

ARITHMETIC SHELLS, FINITE SUPPORT GEOMETRY, AND LORENTZ LIFTS: EXACT IDENTITIES, CERTIFICATES, AND SYMMETRY CONSTRUCTIONS

THE CLANKERS

ABSTRACT. This article constructs exact arithmetic, geometric, and operator-theoretic coordinates for the Erdős–Straus shell problem. It begins with the discriminant-five predecessor identities, the modulus-107 residual data, and a complete calculation of the explicit three-point Fable fibre. For the first occupied 107-shell, the endpoint word is 010: $2E_{21}$ is the outgoing cell C_{0+} , whereas its incoming transpose $C_{-0} = E_{12}$ has coefficient zero at residue 53. Inversive-circle coordinates, the Descartes form, and a finite C_1 – C_2 – C_3 support envelope place this directed datum and the Fable fibre in a common Lorentzian setting.

For every relevant prime p , the divisor-shell construction gives an exact nonnegative integer Q_p and proves the primewise reduction $ES(p) \iff Q_p > 0$. The squarefree carrier refines Q_p to the rank–direction pair (Θ_p, Ξ_p) , while the selected path operator splits its local rank into a commutator-square polar term and an anticommutator-square timelike term. Under the marked support section, the ordered endpoint bits form the exact C_4 full adder $c = (\beta_1 + \beta_2 \bmod 2) + 2\beta_1\beta_2$. Its visible digit and hidden carry are the polar and timelike energies, and the signed commutator retains the order of the two one-sided states. If $\hat{c} = \ell + 2h \in \{0, 1, 2\}$ is the local integer lift, then $\hat{c} = \text{rank } G$ and $3q\hat{c} = \mathcal{E}^{\text{pol}, 123} + 2\mathcal{E}^{\text{tim}, 123}$. Thus classes 0, 1, 2 are exactly the zero, future-null, and future-timelike Lorentz strata. The pair formed by Q_p and the marked rank sum reconstructs the one-sided and timelike multiplicities, with $\mathcal{C}_p^{C_4} \leq \log_2 \text{Vol}_p^{\text{det}} \leq 2\mathcal{C}_p^{C_4}$. Globally, the conjecture is equivalent to $\inf_{p \equiv 1 \pmod{12}} \mathcal{C}_p^{C_4} \geq 3$. The reconstruction matrix belongs to $\text{SL}_2(\mathbb{Z})$, so these two coordinates impose no additional residue law by themselves; the primewise arithmetic pairs below 10000 fill every joint residue class modulo 2 through 8.

The upstream divisor convolution maps through two target coefficients to the complement-orbit rank. Fourier inversion resolves failure into two explicit character-cancellation identities. Averaging the two target identities inserts the exact filter $|1 + \psi(\bar{p})|/2$, so characters odd under the p -rotation make no contribution to simultaneous failure. At the full stabilizer quotient, two distinct missed targets force strict nonprincipal mass beyond the principal mass. Elementary-abelian 2-primary quotients have full support; the sole order-four extremum is a cyclic three-point measure missing its antipode. A pointed-set intertwiner sends that extremum to the proved Fable tetrahedron, where the missing antipode is the fixed apex and the supported face has exact altitude $1/4$. The order-eight and order-sixteen calculations classify the remaining finite 2-primary carrier types, their support volumes, carry characters, and mixer–blade decompositions. Every finite calculation is accompanied by a replayable certificate, with Lean certificates for the discrete cores. The remaining global step is the uniform arithmetic theorem forcing at least one nonzero marked shell class for every relevant prime.

CONTENTS

1.	Scope and attribution	2
1.1.	Road map and relation to complexity geometry	2
1.2.	Mathematical status	10
1.3.	Durable recovery in a long-form research workflow	10
1.4.	Attribution	11
1.5.	Project lineage and supersession	11
2.	The discriminant-five predecessor geometry	11

Date: 7 August 2026.

3.	The modulus-(107) residual shell	13
4.	The explicit constant-Jacobian three-point fibre	15
5.	Withdrawn constructions	17
6.	The central-angle and inversive-circle theorems of u/UmbrellaCorp_HR	17
6.1.	Central-angle field closure and cyclic telescoping	18
6.2.	The Lorentz reflection underlying circle inversion	18
7.	The complete fibre and its canonical marked-factor lift	22
7.1.	Exact classification of the common fibre	22
7.2.	The source-defined invariant lift	23
8.	The singular C_1 – C_2 – C_3 support object	26
9.	Fixed residual shells in the singular support object	33
9.1.	Directed blade transport and the quarter-radius inversion	40
10.	The 12289 orbit-object and its explicit Erdős–Straus certificate	88
10.1.	The Star–Kneser quotient certificate	94
11.	Split-zero support and the exact meaning of the C_1 carrier	110
12.	Rigidified denominator towers and universal Frobenius	131
13.	Mirror completion of the Boolean carrier and the 12289 orbit matrix	133
14.	Sectorwise modular data and the exact C_3 conjugation law	181
15.	The first exact map from a Busy-Beaver extremum to the finite shell carriers	474
	References	486

1. SCOPE AND ATTRIBUTION

1.1. Road map and relation to complexity geometry. The first part of the paper passes from the discriminant-five predecessor identity to exact finite arithmetic carriers, the explicit three-point fibre, and its inversive-circle and Lorentz coordinates. The middle part develops the Erdős–Straus shell criterion, the Star–Kneser support mechanism, the certificate at 12289, and the adjacent $R = 31$ and $R = 27$ shells. The final part studies split-zero support, modular packets, Busy Beaver state spaces, finite operator volumes, and the Coxeter–Fricke chain. This order keeps every later comparison attached to a previously proved finite carrier.

Two geometric results sharpen that route. The centered regular-simplex calculation in theorem 9.35 proves the exact rank–volume identities

$$(n!)^2 \mathcal{V}_n^2 = (n+1) \text{pdet } C_n, \quad ((n-1)!)^2 \mathcal{A}_{n-1}^2 = n \text{pdet } W_n.$$

The Fable collision rays then force the regular 4-simplex completion of theorem 9.43. Its opposite-face centroid is $-E_{21}/4$, so the same quarter is simultaneously the simplex displacement, the coefficient-inversion eigenvalue, and the first coordinate of the common Fable output. The trace-one Gram state and its Hilbert–Schmidt control path further obey

$$\|c_{\text{opp}}\| = \text{tr}(\rho_n^2) = \mathcal{L}_\rho(n)^2 = \frac{1}{n}, \quad \mathcal{V}_n^2 = \frac{(n+1)^{n+1}}{(n!)^2} \mathcal{L}_\rho(n)^{2n};$$

see theorems 9.36 and 9.37. The free-shell pullback gives a second exact finite identity:

$$\dim \mathcal{H}_{R,p}^{\text{simp}} = 3\Omega_R(p), \quad \log \text{Vol}_{R,p}^{\text{det}} = \frac{2 \log 2}{17} L_H^2 = \frac{4 \log 2}{25} L_K^2;$$

see theorem 14.31. The endpoint support cells supply a third exact finite identity:

$$\mathcal{E}_{R,p}^\partial = \frac{1}{4} \sum_{r \in U_R} \|[J_B, C_T(r)]\|_{\text{HS}}^2 = |S \triangle T^{-1}(S)| = \sum_{r \in U_R} \mathcal{N}_T(r).$$

Its two-edge primewise sum isolates one-sided squarefree support and is bounded by the same arithmetic rank:

$$\mathcal{E}_p^{\text{lab}} = 2\Xi_p \leq 2Q_p = \frac{\log \text{Vol}_p^{\det}}{3 \log 2} = \frac{2}{51} L_{H,p}^2 = \frac{4}{75} L_{K,p}^2;$$

see theorems 14.60, 14.73 and 14.246. The endpoint-selected polar lift retains the shell amplitude and the actual endpoint direction in one operator:

$$\mathcal{N}_{0+}^{p,a} = 2H_{p,a}P_{p,a} \otimes \mathcal{J}(C_{0+}), \quad \mathcal{E}_{p,a}^{\text{amp}} = 2\alpha_{p,a}^{\Sigma} \mathcal{E}_{R,p}^{\text{lab}}.$$

Thus the middle-only, exterior-only, and both-channel states select, respectively, the F_{32} , F_{23} , and F_{22} amplitude blocks; see theorem 14.257. The order-forced lift to the three-rotation path carrier satisfies

$$2\mathcal{E}_{p,a}^{\text{amp},123} = 3\mathcal{E}_{p,a}^{\text{amp}}.$$

On a one-sided shell, this path energy lies sharply between the weighted determinant-volume coordinate and both weighted control lengths; see theorems 14.268 and 14.270. The selected source and range modulus spectra coincide under the exact $C_3 : C_2$ stable transfer, while the polar partial isometry retains the signed endpoint direction; see theorem 14.276. Removing the orbit amplitudes from that polar carrier gives the integral rank-direction invariant

$$\mathcal{E}_{p,a}^{\text{pol},123} = 3q_{R,p}(b_1 - b_0)^2 = \frac{3q_{R,p}}{2} \mathcal{E}_{R,p}^{\text{lab}}.$$

Its primewise sum obeys

$$3\Xi_p = \frac{3}{2} \mathcal{E}_p^{\text{lab}} \leq \mathcal{E}_p^{\text{pol},123} \leq 3Q_p;$$

see theorems 14.277 and 14.278. The target-character Lorentz matrix supplies the exact sign refinement: the polar energy is positive precisely on its nonzero null boundary, and there

$$b_1 - b_0 = \frac{\lambda_-}{\lambda_+}.$$

Thus the two one-sided endpoint channels are the two directed null rays; see theorem 14.281. The split-zero target bits recover the endpoint pair entry by entry:

$$(b_0, b_1) = (\beta_1, \beta_2), \quad b_1 - b_0 = \beta_2 - \beta_1.$$

The grading commutator and anticommutator then split every occupied selected path rank as

$$\begin{aligned} \mathcal{E}_{p,a}^{\text{pol},123} &= \frac{1}{4} \|[\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}]\|_{\text{HS}}^2, & \mathcal{E}_{p,a}^{\text{tim},123} &= \frac{1}{4} \|\{\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}\}\|_{\text{HS}}^2, \\ \mathcal{E}_{p,a}^{\text{pol},123} + \mathcal{E}_{p,a}^{\text{tim},123} &= \|V_{\partial,0+}\|_{\text{HS}}^2 = 3q_{R,p}. \end{aligned}$$

Primewise,

$$\mathcal{E}_p^{\text{pol},123} + \mathcal{E}_p^{\text{tim},123} = 3Q_p.$$

The commutator and anticommutator squares are, respectively, the Boolean exclusive-or and conjunction parts; see theorem 14.290. The marked C_4 sum

$$c_{p,a} = (\beta_1 + \beta_2 \bmod 2) + 2\beta_1\beta_2$$

has local integer lift

$$\hat{c}_{p,a} = \text{rank } \mathbf{G}_{p,a}, \quad 3q_{R,p}\hat{c}_{p,a} = \mathcal{E}_{p,a}^{\text{pol},123} + 2\mathcal{E}_{p,a}^{\text{tim},123}.$$

Thus its classes 0, 1, 2 are exactly the zero, future-null, and future-timelike Lorentz strata. Under the primewise shell reduction,

$$\text{ES}(p) \iff \sum_{a \in \mathcal{A}_p} 3q_{R,p}\hat{c}_{p,a} > 0 \iff \exists a \in \mathcal{A}_p : c_{p,a} \neq 0;$$

see theorem 14.325. The full pre-certificate convolution supplies the upstream arithmetic map. With

$$\mathfrak{T}_{p,a}(\mu) = \begin{pmatrix} \mu(T_2(p)) \\ \mu(T_1(p))/2 \end{pmatrix} = \begin{pmatrix} q_{02} \\ q_{11} \end{pmatrix},$$

one has

$$q_{R,p} = q_{02} + q_{11}, \quad \text{rank } V_{\partial,0+} = 3 \text{rank } P_{R,p}^{\text{mod}} = 3(q_{02} + q_{11}).$$

The two C_3 -amplified modular unit lines map to the selected source and range sheets, and their transition map is $V_{\partial,0+}$. Although μ always has positive total mass, the shell is occupied precisely when $\mu \notin \ker \mathfrak{T}_{p,a}$. Hence, for $p \equiv 1 \pmod{4}$,

$$\neg \text{ES}(p) \iff \forall a \in \mathcal{A}_p : \mu_{p,a} \in \ker \mathfrak{T}_{p,a};$$

globally this is the rank gap

$$\text{ES} \iff \inf_{p \equiv 1 \pmod{12}} \text{rank } \mathcal{V}_p^\partial \geq 3;$$

see theorem 14.291. More explicitly,

$$\mathfrak{T}_{p,a}(\mu) = \frac{1}{|H|} \sum_{\chi \in \widehat{H}} \left(\prod_i \sum_{\beta = -e_i}^{e_i} \overline{\chi(\ell_i)}^\beta \right) \mathbf{v}_{p,a}(\chi),$$

where $\mathbf{v}_{p,a}(\chi)$ evaluates the character at the exterior and middle targets and is zero at a target outside H . Hence a target lying in H is missed precisely when its nonprincipal character sum is the negative of the principal mass; see theorem 14.292. The local factors have a second exact consequence: no 2-primary character can be killed by an odd-length kernel. When the binary character is the only nonprincipal survivor, its amplitude is at most one third of the principal mass and both sheets remain positive. For 12289 this is the spectrum $(45, 0, 0, 0, 0, -15, 0, 0, 0, 0)$; see theorem 14.294. If every non-2-primary mode is killed, any missing coefficient must descend to a 2-primary quotient of order at least four; see theorem 14.296. More generally, every elementary-abelian quotient C_2^d has full support and satisfies

$$\nu(x) \geq \frac{M_{p,a}}{3^d};$$

see theorem 14.297. The complete order-four classification therefore leaves one C_4 extremum: a length-three quotient direction supported on $\{-q, 0, q\}$ and missing $2q$; see theorems 14.298 and 14.299. The pointed-set map $2q \mapsto v_*$, $0 \mapsto v_0$, $q \mapsto v_+$, $-q \mapsto v_-$ intertwines both D_3 generators and carries the supported barycentre to the opposite-face centroid at exact spatial altitude $1/4$ from the missing apex; see theorem 14.301. The order-eight support monoid is also completely resolved: C_2^3 is positive; C_8 has the four supports A_3, A_5, A_6, A_7 ; and $C_4 \times C_2$ has the two supports B_6, B_7 ; see theorems 14.303 and 14.304. The inherited eight-cell coordinates carry A_6 and B_6 exactly to the crossed six-path orbit, while their measures land in the phase-zero modular subspace; see theorem 14.305. At order sixteen, the complete finite support monoids contain 36 generating nonfull automorphism types across the four non-elementary groups, and none in C_2^4 . If

$$P \cong \prod_{j=1}^a C_{2^{\lambda_j}} \times C_2^b, \quad \lambda_j \geq 2,$$

then the least generating support volume is

$$m(P) = 3^a 2^b,$$

attained by one automorphism orbit. Thus the order-sixteen volumes are 3, 6, 9, 12, 16. The four non-elementary groups have a common four-sheet-by-four-cell carrier. The C_{16} and $C_8 \times C_2$ laws have non-coboundary carries, the two product laws have zero carry, and the four minimum-support sheet–cell deficits are 3, 2, 0, 0. Carrier inversion retains the nonzero carries as cell displacements and decomposes the minimum supports into 1, 2, 4, 4 exact mixer–blade blocks. Deficit plus block

count is four in all four rows. Cell translation and inversion generate a faithful D_8 -action. Its four cell-Fourier projectors have complex rank four, and its self-conjugate $k = 2$ sector detects the carry by the exact sign $(-1)^{\kappa_P(u, -u)}$. For these four selected carriers, the full irreducible multiplicity row distinguishes all four marked extension classes, while the one-dimensional imbalance detects their non-coboundary rows through

$$\kappa_P \text{ not a coboundary} \iff \text{tr}(R_P U_P) > 0 \iff m_{--}(P) > m_{+-}(P).$$

The binary carry character of the marked extension

$$q_P(u) = \kappa_P(u, u) \pmod{2}, \quad u \in S_P[2],$$

is additive and independent of the section. Its Hamming weight is

$$|\{u : q_P(u) = 1\}| = m_{--}(P) - m_{+-}(P),$$

and a nonzero q_P has weight $|S_P[2]|/2$. For a general finite abelian sheet group whose 2-primary part is $\prod_{i=1}^a C_{2^{\lambda_i}} \times C_2^b$, with $\lambda_i \geq 2$, the map from marked C_4 -extension classes to these characters is onto with kernel $(2C_4)^a$. The character is complete exactly when $a = 0$. In the corresponding visible/hidden bit coordinates, each long cyclic factor obeys

$$(q, h) \star (q', h') = (q + q', h + h' + qq'),$$

and the detection sequence admits an additive section exactly when $a = 0$. For the marked support section, the split-zero endpoint bits form the exact full adder

$$c = (\beta_1 + \beta_2 \bmod 2) + 2\beta_1\beta_2.$$

Its polar and timelike energies recover the visible digit and hidden carry, and the signed commutator coefficient $d = \beta_2 - \beta_1$ restores the ordering that the energy pair forgets. The local lift $\hat{c} = \ell + 2h$ is exactly the rank of the target Lorentz matrix, so the marked classes 0, 1, 2 are the zero, future-null, and future-timelike strata. Its primewise weighted sum is

$$\mathcal{C}_p^{C_4} = \mathcal{E}_p^{\text{pol}, 123} + 2\mathcal{E}_p^{\text{tim}, 123} = 3 \sum_{a \in \mathcal{A}_p} q_{R,p} \text{rank } \mathbf{G}_{p,a},$$

and detects $\text{ES}(p)$ under the primewise shell reduction. If $\mathbf{N}_p^\partial$ and $\mathbf{T}_p^\partial$ are the $q_{R,p}$ -weighted one-sided and timelike shell multiplicities, then

$$Q_p = \mathbf{N}_p^\partial + \mathbf{T}_p^\partial, \quad \frac{\mathcal{C}_p^{C_4}}{3} = \mathbf{N}_p^\partial + 2\mathbf{T}_p^\partial.$$

Hence the marked coordinate reconstructs both multiplicities and lies between $3Q_p$ and $6Q_p$, with exact endpoint strata. In particular, $C_8 \times C_2$ admits a selected binary-sheet carrier of weight two and a distinct cyclic-sheet carrier of weight zero. Writing $\delta_P = m_{--}(P) - m_{+-}(P)$ and e_P for the non-coboundary indicator, the minimum carrier further satisfies

$$d_P + \delta_P = 4e_P, \quad b_P - \delta_P = 4(1 - e_P).$$

The next unclassified purely 2-primary order is 32; see theorems 14.307 to 14.311, 14.313, 14.315, 14.317, 14.325, 14.327, 14.329, 14.331 and 14.333. If a shell is empty and $K = \text{Stab}(B)$, then every target coset $u \in (T + K)/K$ satisfies the sharper quotient identity

$$\sum_{\substack{\psi \in \widehat{H/K} \\ \psi \neq 1}} \hat{\mu}(q^\vee \psi) \psi(u) = - \prod_i (2e_i + 1).$$

The same quotient has nonpositive Star-Kneser surplus. Hence strict domination by the principal mass, checked over the finitely many nonpositive-surplus quotients, is an exact shell-occupancy criterion; see theorem 14.335. Target averaging sharpens that criterion by the factor $|1 + \psi(\bar{p})|/2$. On the full stabilizer quotient it is a strict contraction of the nonprincipal Fourier mass whenever the

two target cosets are distinct; equality would force a forbidden support period. See theorem 14.336. If the target displacement has order m , the contraction gap is at least

$$\left(1 - \cos \frac{\pi}{m}\right) \sum_{\substack{\psi \neq 1 \\ \psi(\bar{p}) \neq 1}} |\widehat{\mu}(\psi)|;$$

see theorem 14.337. Since the shell multiplicity measure is integer-valued, the full support quotient also satisfies

$$W_{\text{rot}}(\nu, d) \geq \sqrt{|Q| \frac{m-1}{m}};$$

see theorem 14.338. Resolving every target-displacement orbit against the actual support gives the sharper support-boundary energy

$$\mathcal{B}_d(C) = \frac{1}{m} \sum_O |C \cap O| |O \setminus C|;$$

see theorem 14.339. Its support-cardinality floor and the Star–Kneser interval refinement are theorem 14.340. For a one-prime-power shell, the support is a proper cyclic interval and both its boundary energy and target-filtered Dirichlet-kernel mass have closed formulas; see theorem 14.344. The sharper support-boundary bound proves occupancy of the shell $(p, R, a) = (13, 7, 5)$, for which the two-target filter in fact removes every active nonprincipal Fourier coefficient; see theorem 14.345. On the remaining prime locus $p \equiv 1 \pmod{24}$, the complete residual-7 classification has the exact Boolean form

$$\mathbf{1}_{\{q_{7,p} > 0\}} = 1 - \prod_{\substack{\ell \mid (p+7)/8 \\ \ell \text{ prime}}} \frac{1 + \left(\frac{\ell}{7}\right)}{2}.$$

Thus the shell is occupied precisely when $(p+7)/8$ has a quadratic-nonresidue prime factor modulo 7. This joins the support bit, the sheet quotient $U_7/Q_7 \cong C_2$, and the internal phase subgroup $Q_7 \cong C_3$ without identifying the three structures; see theorem 14.123. The next residual has a compatible affine coordinate. If $p = 24k + 1$, then

$$c_7 = 3k + 1 = \frac{p+7}{8}, \quad d_{11} = 2k + 1 = \frac{p+11}{12}, \quad 3d_{11} - 2c_7 = 1.$$

Hence c_7 and d_{11} are coprime. The simultaneous residual-7 and residual-11 empty locus is exactly the intersection of the quadratic-residue factor condition on c_7 with the two explicit complement branches of the first residual-11 colon ideal on d_{11} ; see theorem 14.130. The next three affine coordinates are consecutive:

$$(a_{15}, a_{19}, a_{23}) = (6k + 4, 6k + 5, 6k + 6).$$

For the first hard-locus point $p = 1201$, the residual-15 and residual-19 shells are empty, while the residual-23 shell has three orbit classes and gives the explicit decomposition

$$\frac{4}{1201} = \frac{1}{306} + \frac{1}{1082101} + \frac{1}{16218}.$$

Among the 28 residual-7/residual-11 double survivors below 20000, exact finite replay finds that 26 are occupied by at least one of the residual-15, residual-19, or residual-23 shells. The two cases surviving through $R = 23$ are then occupied at $R = 39$ and $R = 31$, respectively:

$$\frac{4}{3361} = \frac{1}{850} + \frac{1}{19769402} + \frac{1}{73525}, \quad \frac{4}{8089} = \frac{1}{2030} + \frac{1}{245371726} + \frac{1}{530845}.$$

Thus every one of the 28 finite hard-locus points is occupied by a shell with $R \leq 39$. See theorems 14.132 to 14.135. The bound $R \leq 39$ does not persist. The least prime empty through

$R = 39$ is 118801; all of its shells through $R = 55$ are empty, and its first occupied shell is $R = 59$. The resulting certificate is

$$\frac{4}{118801} = \frac{1}{29715} + \frac{1}{5077395546660} + \frac{1}{59834124}.$$

See theorems 14.136 and 14.137. The final empty shell at $R = 39$ already reaches both arithmetic targets, but in the opposite exponent channels. Its exact target-incidence matrix is the sheet exchange S , and the established Hadamard map sends it to the mixer grading J . The first occupied shell at $R = 59$ restores both diagonal target incidences. See theorems 14.138 and 14.141. Earlier in the same shell sequence, the first nonzero hidden incidence is $\Xi_{19,29705} = E_{21}$, whose Hadamard image is the ordered blade B_{-+} . Its factor ratio is

$$\frac{13}{5} \equiv -\frac{1}{4} \pmod{19}.$$

Reduction of the complete Fable three-point fibre modulo 19 has the same target coordinate. See theorem 14.140. An exact census of all 9732 primes $p \leq 10^6$, $p \equiv 1 \pmod{24}$, shows that 118801 is the unique case with maximal delay between its first nonzero hidden incidence and its first occupied shell:

$$R_{\text{hid}} = 19, \quad R_{\text{occ}} = 59, \quad R_{\text{occ}} - R_{\text{hid}} = 40.$$

The one-surplus pair $13/5$ occurs once more, at $p = 359041$, but that prime is occupied already at $R = 23$. Thus the prolonged delay is carried by the complete incidence trajectory

$$E_{21} \xrightarrow{R=39} S \xrightarrow{R=59} I_2 + S.$$

See theorem 14.142. The distinction between the two one-surplus primes is already exact at the next residual. At $R = 23$, the exterior and middle reachability sets for 118801 are the quadratic-residue subgroup, whereas the factor $4987 \equiv 19 \pmod{23}$ for 359041 opens its other coset in both channels. The latter shell has one exterior section, two middle sections, and the explicit unit-fraction decomposition in theorem 14.144. In the fixed-target colon ideal, the same distinction is the exact monomial pair

$$X_2 X_{18} \notin J_{23}^{(3)}, \quad X_2 X_{15} X_{16} \in J_{23}^{(3)},$$

with $X_2 X_{15}$ a minimal generator; see theorems 14.210 to 14.212. The forced factor $2 \mid b_\ell$ makes the latter pullback completely explicit. An empty $R = 23$ shell has either only quadratic-residue factor classes modulo 23, or, after factors congruent to 1 are discarded, exactly one of the three log multisets

$$\{1, 2\}, \quad \{2, 21\}, \quad \{1, 2, 21\}.$$

A census through $\ell \leq 10^7$ finds 7328 applicable prime pairs: 5794 are occupied at $R = 23$, while the 1534 survivors split as $1463 + 34 + 35 + 2$ among these four exact factor types. Writing $\ell = 6m + 1$ identifies the shells $R = 23, \dots, 59$ with the ten consecutive coordinates $390m + 66, \dots, 390m + 75$. Their forced divisor skeleton is

$$(6, 1, 2, 3, 10, 1, 6, 1, 2, 15),$$

and two distinct coordinates in this packet can share no prime larger than 7; see theorem 14.214. Every one of those 1534 survivors is occupied by $R = 59$. The three nonresidue types are exhausted by $R = 43$; the only rows reaching 47, 51, 55, 59 belong to the quadratic-residue type. Exactly three rows first occupy at 59, and each has an explicit exterior unit-fraction certificate; see theorem 14.215. For each later residual, the forced divisor skeleton determines an exact set of degree-one cofactor-residue triggers; see theorem 14.216. In the same bounded census, every first occupied row has a support certificate using at most four cofactor prime-factor occurrences, with degree histogram

$$776, 678, 74, 6$$

in degrees one through four; see theorem 14.217. Quotienting the full support carrier by the subgroup generated by the target displacement gives a second exact forcing mechanism. When that

displacement has order two, the two targets form one complete quotient fibre; positive projected Star–Kneser surplus therefore forces an actual target hit. This proves the shell $(37, 19, 14)$, which is missed by the support-boundary criterion, and reconstructs

$$\frac{4}{37} = \frac{1}{14} + \frac{1}{28} + \frac{1}{1036}.$$

See theorems 14.341 and 14.346. More generally, a projected local exponent interval has cardinality $\min(2e+1, n)$, where n is the order of its projected generator. If that generator spans the displacement quotient and $2e+1 \geq n$, one local factor fills the quotient and forces occupancy; see theorems 14.342 and 14.343. For every prime pair

$$q \equiv 7 \text{ or } 11 \pmod{19}, \quad p = 8q - 19,$$

this proves the residual-19 shell $a = 2q$ and gives two closed denominator formulas; see theorem 14.347. This residual-19 family is the first member of an exact power-of-two ladder. If $e \geq 1$, R is prime,

$$\text{ord}_R(2) = 6e + 12, \quad \omega = 2^{2e+4} \pmod{R},$$

and $q \equiv \omega$ or $\omega^2 \pmod{R}$, then every prime $p = 2^{e+2}q - R$ has an occupied residual- R shell at $a = 2^e q$, with a closed unit-fraction formula on each of the two q -branches. The rung $e = 1$ forces $R = 19$; the compact higher-rung instance

$$(e, R, q, p, a) = (3, 331, 31, 661, 248)$$

gives

$$\frac{4}{661} = \frac{1}{248} + \frac{1}{496} + \frac{1}{327856}.$$

See theorems 14.348 and 14.349. The same mechanism is closed under composite cofactors. It is enough that the total residue of b lie in the cubic subgroup $\{1, \omega, \omega^2\}$ modulo R , and that one prime divisor of b have nonidentity cubic residue. All remaining prime factors are unrestricted, and b may contain additional powers of 2. Every prime

$$p = 2^{e+2}b - R$$

therefore has an occupied shell at $a = 2^e b$. An explicit divisor $u \mid a^2$ gives

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{(ap+u)/R} + \frac{1}{(ap+a^2p^2/u)/R}.$$

At $R = 19$, the cofactors 49, 1141, and 2629 realize all three total cubic residue classes. The cofactors 220, 286, and 330 realize the same three classes while admitting prime factors outside the cubic subgroup; see theorems 14.350 to 14.352. The order condition itself imposes a sharp scope boundary:

$$R \pmod{8} \in \{1, 3, 5\}, \quad p \pmod{8} \in \{7, 5, 3\}.$$

Thus the ladder never enters the unresolved $p \equiv 1 \pmod{8}$ locus. Every ladder prime belongs to a congruence class already certified by the elementary $p \equiv 3 \pmod{4}$ or $p \equiv 5 \pmod{12}$ formulas, or by the residual-3 certificate for $p \equiv 13 \pmod{24}$; see theorems 14.353 and 14.354. The shell $(p, R, a) = (37, 11, 12)$ realizes both gains exactly: its unweighted nonprincipal mass equals its principal mass 15, whereas its target-averaged mass is $3 + 4\sqrt{5} < 15$; see theorem 14.355. The marked C_4 coordinates do not supply a further residue obstruction on their own. Their primewise reconstruction matrix has determinant one over $\mathbb{Z}$, and hence over every residue ring; see theorem 14.329. In the exact arithmetic data below 10000, both coordinate pairs fill every joint residue class modulo 2 through 8. A uniform proof must therefore use the shell arithmetic, the target Fourier coefficients, or another invariant beyond the marked rank equations. For each prime

$p \equiv 1 \pmod{4}$, the finite smallest-denominator reduction joins all free shells into one projector of rank Q_p . Its supported carrier satisfies

$$\dim \mathcal{S}_p = 6Q_p, \quad \text{Vol}_p^{\det} = 2^{6Q_p}, \quad L_{H,p}^2 = 51Q_p, \quad L_{K,p}^2 = \frac{75}{2}Q_p.$$

Together with multiplicative inheritance and the explicit decompositions for $p = 2$, $p \equiv 3 \pmod{4}$, and $p \equiv 5 \pmod{12}$, this gives the global equivalence

$$\text{ES} \iff \inf_{p \equiv 1 \pmod{12}} Q_p \geq 1 \iff \inf_{p \equiv 1 \pmod{12}} \text{Vol}_p^{\det} \geq 64;$$

see theorems 14.244 and 14.250. The same orbit multiplicity Q_p carries a canonical four-Time $SU(4)$ factor:

$$\mathcal{V}_p^{\text{Time}} = \mathbb{C}[\mathcal{O}_p] \otimes \mathbb{C}[\mathcal{T}], \quad \dim \mathcal{V}_p^{\text{Time}} = 4Q_p, \quad \log_2 \text{Vol}_p^{\det} = \frac{3}{2} \dim \mathcal{V}_p^{\text{Time}}.$$

Accordingly,

$$\text{ES} \iff \inf_{p \equiv 1 \pmod{12}} \dim \mathcal{V}_p^{\text{Time}} \geq 4;$$

see theorem 14.251. For the arithmetic orbit amplitudes, the same carrier has orbit blocks $\alpha_O \text{diag}(4, -1)$. Its determinant and its two Hilbert–Schmidt lengths are joined by the sharp geometric-mean–root-mean-square bound in theorem 14.269. The label-kernel support, the four equal quarter eigenvalues, the regular tetrahedral ray form, the hidden fourth-coordinate translation, and the exact transport of the Fable face action to Erdős–Straus denominator exchange are shown from complementary viewpoints in figures 1 and 3 to 6; every coordinate used by these images is checked by an exact arithmetic receipt.

The geometric-complexity comparison uses two established formulations. Nielsen’s approach bounds circuit size by the length of a shortest path in a suitable metric geometry on the unitary group [34, 35]; the exact lifted Pauli path constructed below is a one-qubit instance of that framework. Stanford and Susskind propose, up to an order-one factor,

$$\mathcal{C}(t_L, t_R) = \frac{V(t_L, t_R)}{G_N l_{\text{AdS}}},$$

where $V(t_L, t_R)$ is the volume of the maximum-volume codimension-one surface bounded by the two CFT spatial slices [37]. Recent work comparing Bures-metric complexity with holographic subregion volume finds closely tracking behavior and unequal values in the perturbative examples studied [42]. The finite counting, circuit-metric, operator-volume, and simplex rank–volume carriers proved here specify exact comparison data on the arithmetic and finite-geometric side. The affine Fable-to-Sarnak carrier gives a further exact comparison: theorem 9.29 computes the affine-invariant geodesic length of its positive polar factor and its volume Jacobian in one closed identity, whose second term records the anisotropy of the carrier. The order-sixteen support carriers give a second exact comparison: the product of the sheet and cell projection sizes exceeds the occupied support by d_P , group inversion supplies b_P antisymmetric mixer–blade blocks, and

$$d_P + b_P = 4$$

conserves the number of cyclic carrier cells in every non-elementary row. On the full carrier, cell translation and inversion form a faithful dihedral action. Its four rank-four Fourier sectors retain the carry as the phase

$$\mathbf{i}^{k\kappa_P(u, -u)},$$

and the self-conjugate second sector records the two nonzero carry laws by its negative sheet signs. For the selected minimum carriers, the full permutation character distinguishes all four marked extension rows, and the multiplicity difference

$$m_{--}(P) - m_{+-}(P) = \frac{1}{2} \text{tr}(\mathbf{R}_P \mathbf{U}_P) = (1, 2, 0, 0).$$

More intrinsically, the two-torsion carry character

$$q_P : S_P[2] \longrightarrow C_2, \quad q_P(u) = \kappa_P(u, u) \pmod{2},$$

is additive and section-independent, and

$$m_{--}(P) - m_{+-}(P) = |\{u \in S_P[2] : q_P(u) = 1\}|.$$

Thus the spectral defect is the Hamming weight of a finite character intrinsic to the marked extension; if that character is nonzero, its weight is exactly half the two-torsion sheet volume. The general extension-detection sequence has kernel $(2C_4)^a$ when the sheet 2-primary part has a cyclic factors of order at least four. In invariant-factor bit coordinates $c = q + 2h$, its group law contains the forced carry qq' . Thus the hidden kernel is a genuine second layer: the underlying set has a visible/hidden binary chart, while that chart is additive precisely when $a = 0$. The split-zero endpoint bits realize the same quadratic carry law: their polar XOR energy is the visible digit, their timelike AND energy is the hidden digit, and the signed commutator restores the endpoint order. Thus the operator rank decomposition is the digit-by-digit marked C_4 full adder, rather than an independent binary encoding. Its local integer class lift equals the Lorentz rank, and its carry-weighted energy

$$\mathcal{E}_p^{\text{pol},123} + 2\mathcal{E}_p^{\text{tim},123} = 3 \sum_{a \in \mathcal{A}_p} q_{R,p} \text{rank } G_{p,a}$$

is positive exactly when the primewise shell reduction has a supported marked class. Together with Q_p , it reconstructs the weighted one-sided and timelike multiplicities and satisfies

$$\mathcal{C}_p^{C_4} \leq \log_2 \text{Vol}_p^{\text{det}} \leq 2\mathcal{C}_p^{C_4}.$$

Accordingly, the same abstract group $C_8 \times C_2$ carries both the selected binary-sheet model of spectral weight two and a non-split cyclic-sheet model of spectral weight zero. If e_P records whether the selected carry class is nonzero and $\delta_P = m_{--}(P) - m_{+-}(P)$, these same exact data give

$$d_P + \delta_P = 4e_P, \quad b_P - \delta_P = 4(1 - e_P).$$

Thus the two non-split selected rows allocate the four-cell volume between correlation and spectral defects, while the two split selected rows allocate all four cells to the blade sector. A holographic identification additionally requires an explicit map to a boundary state, a bulk dual, and a regularized maximal slice; constructing that map is a research problem in this program. The finite-universe calculation later isolates the exact dimensional datum needed by such a map: an m -label exterior carrier has internal group $SU(m)$, centered spatial rank $m - 1$, and Wick signature $(m - 1, 1)$. Consequently the four-Time carrier is the $SU(4)$, 3+1-dimensional rung, whereas a 4+1-dimensional continuation has five Finger labels and internal group $SU(5)$; see theorems 13.19 and 13.21.

1.2. Mathematical status. The paper uses four statement types.

- (i) A theorem, lemma, proposition, or corollary followed by a complete proof.
- (ii) An exact computation whose input set, algorithm, arithmetic domain, and reproducible certificate are stated explicitly.
- (iii) A theorem imported from a cited source with all hypotheses stated and with either a local proof or a precise proof citation.
- (iv) A withdrawn argument whose invalid inference is identified exactly.

Unproved structural interpretations appear only as source questions or withdrawn arguments.

1.3. Durable recovery in a long-form research workflow. Long calculations in a research programme can outlive the calculation in which they were begun. This project treats each immutable checkpoint as the complete recovery state. Before a long run, the append-only research log records the active objective, the proved and open claims, exact source paths, the current public-release head, immutable artifact hashes, prohibited actions, pending proof attacks, and the first operation to

perform after recovery. Work resumes only after that checkpoint, the active theorem–checker–receipt chain, and the immediately preceding calculation have been reread.

An interrupted write is inspected before it is repeated. Each substantive mathematical change is then propagated through the introduction, theorem dependencies, verification ledger, and bibliography as applicable; the sources are rebuilt, their diagnostics are inspected, and the resulting artifacts receive new hashes. In this way volatile working state is externalized as an auditable sequence of mathematical states. The procedure prevents duplicate edits, loss of ordered coordinate data, and accidental changes in proof status. It also permits a release process to take a read-only snapshot while the maintained research sources continue to develop.

1.4. Attribution. The collective label *The Clankers* identifies the local research corpus and this paper. Each imported theorem, post, diagram, computation, and expository source retains its original copyright and authorship. The text credits every imported result at its first mathematical use and again in the bibliography.

The discriminant-five observation, the instruction to retain its roots, and the predecessor-chain diagrams are attributed to Reddit user u/CivQ17; the exact comment is cited as [2]. The central-angle field-closure and circle-inversion coefficient theorems are attributed to Reddit user u/UmbrellaCorp_HR [3]. The explicit polynomial map is the counterexample publicly announced by Levent Alpöge, whose announcement credits Akhil Mathew for the question and Claude Fable 5 for the construction [7]; it is presented and geometrically digested by Terence Tao [6]. The separate marked-factor pipeline and provenance discussion of Reddit user u/JGPTech, together with Jon Poplett’s exact public implementation, are cited at their use below [4, 5]. The Kneser and Apollonian–Lorentz inputs are cited to their authors at the points where they are used. Kocik’s equal-angle circle-configuration family is cited at its precise use below. Machine-derived coordinate transformations and finite certificates carry their own provenance, separate from the source theorems.

1.5. Project lineage and supersession. This paper consolidates and supersedes Kokuno Yumeto, Miroku Akagi, and Maya Sakuyah, *The Z3 Center Symmetry: Arithmetic Lattices, Automorphic Spectral Triples and QCD* [21]. The earlier finite three-sector and modular constructions are retained as provenance. Their unsupported global identifications are replaced here by the support-indexed statements and explicit hypotheses proved in the following sections.

2. THE DISCRIMINANT-FIVE PREDECESSOR GEOMETRY

The identity and root instruction opening this section come from u/CivQ17’s Reddit comment [2]. The algebraic proofs below verify and extend the attributed observation while retaining that provenance.

Proposition 2.1 (Discriminant-five identity and root involution). *For every $n \in \mathbb{C}$,*

$$\frac{(2n+1)^2 - 5}{4} = n^2 + n - 1. \quad (1)$$

The polynomial $f(n) = n^2 + n - 1$ has discriminant 5, roots

$$r_{\pm} = \frac{-1 \pm \sqrt{5}}{2}, \quad (2)$$

and is invariant under the involution $\tau(n) = -1 - n$.

Proof. Expanding the numerator gives

$$(2n+1)^2 - 5 = 4n^2 + 4n + 1 - 5 = 4(n^2 + n - 1),$$

which proves (1). The discriminant of $n^2 + n - 1$ is

$$1^2 - 4(1)(-1) = 5,$$

and the quadratic formula gives the displayed roots. Finally,

$$f(\tau(n)) = (-1 - n)^2 + (-1 - n) - 1 = n^2 + 2n + 1 - n - 1 - 1 = n^2 + n - 1 = f(n).$$

Since $\tau(\tau(n)) = -1 - (-1 - n) = n$, the map τ is an involution. $\square$

Lemma 2.2 (Odd-root lock). *For every n, μ in a commutative ring in which 4 is invertible,*

$$(2n + 1)^2 + \mu = 4 \left(n^2 + n + \frac{\mu + 1}{4} \right). \quad (3)$$

For $\mu = -5$, this is exactly (1) after division by 4.

Proof. The right-hand side expands to

$$4n^2 + 4n + \mu + 1 = (2n + 1)^2 + \mu.$$

Substitution of $\mu = -5$ yields $(2n + 1)^2 - 5 = 4(n^2 + n - 1)$, as required. $\square$

Proposition 2.3 (Exact C_3 -with-binary-sheet predecessor map). *Let*

$$X = \{1, 3, 5, 7, 9, 11\} \subset \mathbb{Z}/12\mathbb{Z}, \quad R(r) = 4r + 1 \pmod{12}.$$

Then the functional graph of R is

$$1 \longrightarrow 5 \longrightarrow 9 \longrightarrow 1, \quad 3 \longrightarrow 1, \quad 7 \longrightarrow 5, \quad 11 \longrightarrow 9.$$

Under the bijection

$$\begin{array}{lll} (0, 0) \longleftrightarrow 1, & (1, 0) \longleftrightarrow 5, & (2, 0) \longleftrightarrow 9, \\ (0, 1) \longleftrightarrow 7, & (1, 1) \longleftrightarrow 11, & (2, 1) \longleftrightarrow 3, \end{array}$$

from $C_3 \times C_2$ to X , one has

$$R(\alpha, \varepsilon) = (\alpha + 1, 0). \quad (4)$$

Consequently $R^3 = P$ and $R^4 = R$, where $P(\alpha, \varepsilon) = (\alpha, 0)$.

Proof. Direct reduction modulo 12 gives

$$\begin{array}{lll} R(1) = 5, & R(5) = 21 \equiv 9, & R(9) = 37 \equiv 1, \\ R(3) = 13 \equiv 1, & R(7) = 29 \equiv 5, & R(11) = 45 \equiv 9. \end{array}$$

This proves the graph. Reading the six images through the displayed bijection shows that every (α, ε) maps to $(\alpha + 1, 0)$, proving (4). Iterating gives

$$R^2(\alpha, \varepsilon) = (\alpha + 2, 0), \quad R^3(\alpha, \varepsilon) = (\alpha, 0) = P(\alpha, \varepsilon).$$

Therefore $R^4 = R \circ R^3 = R \circ P = R$. $\square$

Corollary 2.4 (Common directed quotient with the six-cycle lift). *Define*

$$\pi : C_3 \times C_2 \rightarrow C_3, \quad \pi(\alpha, \varepsilon) = \alpha, \quad \rho(\alpha) = \alpha + 1,$$

and

$$G(\alpha, \varepsilon) = (\alpha + 1, \varepsilon + 1).$$

Then

$$\pi R = \rho \pi = \pi G, \quad G^3(\alpha, \varepsilon) = (\alpha, \varepsilon + 1), \quad G^6 = I.$$

Proof. For every (α, ε) ,

$$\pi R(\alpha, \varepsilon) = \alpha + 1 = \rho \pi(\alpha, \varepsilon) = \pi G(\alpha, \varepsilon).$$

After three applications of G , the C_3 -coordinate has increased by 3 and hence returned to α , while the C_2 -coordinate has increased by $3 \equiv 1 \pmod{2}$. Thus G^3 is the sheet flip and $G^6 = I$. $\square$

3. THE MODULUS-(107) RESIDUAL SHELL

Definition 3.1 (Busy Beaver data for the residual calculation). The exact numerical inputs are

$$S(1) = 1, \quad S(2) = 6, \quad S(3) = 21, \quad S(4) = 107, \quad S(5) = 47,176,870,$$

$$\Sigma(4) = 13, \quad \text{Space}(3) = 7, \quad \text{Space}(4) = 16, \quad \Sigma(5) = 4098, \quad \text{Space}(5) = 12289.$$

These values use the five-state, two-symbol blank-tape model. The formal enumeration of the bbchallenge Collaboration proves the values of $S(5)$, $\Sigma(5)$, and $\text{Space}(5)$, and proves that one machine realizes all three extrema [15, Theorems 1.1, 6.1, and 6.2]. The residual shell fixed in this section is

$$p_* = 8,803,369, \quad R_* = 107, \quad a_* = \frac{p_* + R_*}{4} = 2,200,869, \quad N_* = p_* a_*.$$

Theorem 3.2 (Exact reconstruction of the (107)-shell residues). *In $\mathbb{F}_{107}$,*

$$S(3)^{-1} = p_*, \tag{5}$$

$$S(5) = N_* = p_* a_*, \tag{6}$$

$$S(3)S(5) = a_*, \tag{7}$$

$$-S(3)S(5) = 11^2. \tag{8}$$

Proof. First reduce the shell parameters modulo (107). Long division gives

$$p_* = 107 \cdot 82,274 + 51,$$

and

$$a_* = 107 \cdot 20,568 + 93.$$

Thus $p_* \equiv 51$ and $a_* \equiv 93$. Since

$$21 \cdot 51 = 1071 = 107 \cdot 10 + 1,$$

we have $S(3)^{-1} = 21^{-1} = 51 = p_*$, proving (5). Next,

$$S(5) = 47,176,870 = 107 \cdot 440,904 + 142 = 107 \cdot 440,905 + 35,$$

so $S(5) \equiv 35$. On the other hand,

$$p_* a_* \equiv 51 \cdot 93 = 4743 = 107 \cdot 44 + 35,$$

which proves (6). Multiplying 35 by 21 gives

$$21 \cdot 35 = 735 = 107 \cdot 6 + 93 \equiv a_*,$$

proving (7). Finally,

$$-93 \equiv 14 \pmod{107}, \quad 11^2 = 121 \equiv 14 \pmod{107},$$

which proves (8). □

Theorem 3.3 (Exact auxiliary identities in the mod-107 report). *The recorded value $\text{Space}(5) = 12289$ satisfies*

$$12289 = 107^2 + 840, \quad 12289 \equiv 91 \pmod{107}, \quad 12289 \equiv 529 \pmod{840}, \tag{9}$$

and

$$289^5 \equiv 529 \pmod{840}. \tag{10}$$

Over $\mathbb{F}_{107}$, let

$$R_B = \begin{pmatrix} 102 & 34 & 19 & 16 & 67 \\ 103 & 56 & 19 & 32 & 6 \\ 104 & 105 & 6 & 96 & 47 \\ 105 & 83 & 77 & 1 & 48 \\ 106 & 62 & 12 & 58 & 58 \end{pmatrix},$$

$$A_1 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 72 & 40 & 4 & 32 & 46 \end{pmatrix}, \quad A_6 = \begin{pmatrix} 62 & 30 & 43 & 97 & 102 \\ 3 & 12 & 64 & 68 & 35 \\ 54 & 74 & 14 & 80 & 85 \\ 13 & 76 & 44 & 102 & 40 \\ 10 & 47 & 88 & 31 & 73 \end{pmatrix}.$$

Then

$$\begin{aligned} \det R_B &= 8, \\ \det A_1 &= 72, & \operatorname{tr}(A_1) &= 46, \\ \det A_6 &= 40, & \operatorname{tr}(A_6) &= 49, \\ \det[A_1, A_6] &= 15, & \operatorname{tr}[A_1, A_6] &= 0 \end{aligned} \tag{11}$$

in $\mathbb{F}_{107}$, where $[A_1, A_6] = A_1 A_6 - A_6 A_1$.

Finally, for

$$\mathbf{h} = (50, 74, 52, 81, 70) \in \mathbb{F}_{107}^5,$$

the elementary symmetric coefficients are

$$(e_0, e_1, e_2, e_3, e_4, e_5) = (1, 6, 22, 53, 98, 93),$$

equivalently

$$\prod_{h \in \{50, 74, 52, 81, 70\}} (X - h) = X^5 - 6X^4 + 22X^3 - 53X^2 + 98X - 93 \quad \text{in } \mathbb{F}_{107}[X]. \tag{12}$$

Proof. The first equality in (9) is $11449 + 840 = 12289$. Division gives

$$12289 = 107 \cdot 114 + 91 = 840 \cdot 14 + 529.$$

For (10),

$$289^2 \equiv 361, \quad 289^4 \equiv 121, \quad 289^5 \equiv 121 \cdot 289 \equiv 529 \pmod{840}.$$

Gaussian elimination in $\mathbb{F}_{107}$, without row rescaling, gives diagonal pivot sequences. The commutator used in the fourth row is

$$[A_1, A_6] = \begin{pmatrix} 42 & 43 & 54 & 78 & 61 \\ 102 & 62 & 76 & 73 & 12 \\ 99 & 46 & 58 & 43 & 9 \\ 19 & 39 & 66 & 98 & 57 \\ 99 & 106 & 22 & 92 & 61 \end{pmatrix}.$$

The resulting pivot data are

matrix	pivots	row swaps
R_B	(102, 93, 57, 9, 10)	0
A_1	(72, 1, 1, 1, 1)	4
A_6	(62, 14, 10, 74, 67)	0
$[A_1, A_6]$	(42, 34, 83, 71, 85)	0.

Their signed products modulo 107 are respectively 8, 72, 40, 15. The traces of A_1 and A_6 are the sums of their displayed diagonal entries. Every commutator has trace zero, and direct summation of the diagonal of the displayed commutator verifies the same identity here. This proves (11).

Direct multiplication of the five linear factors gives coefficient vector

$$(1, 101, 22, 54, 98, 14) = (1, -6, 22, -53, 98, -93) \pmod{107},$$

which is (12) and yields the displayed elementary symmetric coefficients. $\square$

Provenance record 3.4 (Mod-107 verification files). The numerical inputs and matrices are taken from the two matching local reports in the research corpus of The Clankers [8], `BEAVERSHINE/bb107_rigidity_report.txt` and `BEAVERSHINE/13/3/reports/bb107_rigidity_report.txt`. The row labelled `Space5` in `BEAVERSHINE/3/bb107_sweep_full.csv` records the sweep coordinate vector

$$(91, 79, 54, 49, 2, 32).$$

The calculations above use the first coordinate together with the explicitly defined matrix and polynomial identities. Structural interpretations of the remaining sweep-coordinate labels require their separate definitions.

Remark 3.5. Theorem 3.2 establishes the four displayed congruences. A general identification of Busy Beaver extrema with residual-shell parameters would require a separate theorem.

4. THE EXPLICIT CONSTANT-JACOBIAN THREE-POINT FIBRE

Alpöge announced the polynomial map and three-point collision in this section with the credits stated above [7]; Tao reproduces the construction in his technical exposition [6]. The following calculation verifies the displayed formula directly. The marked-factor formulation used later is presented in the separate Reddit pipeline post of u/JGPTech and the accompanying exact implementation [4, 5].

Definition 4.1. Let $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ be the polynomial map

$$\begin{aligned} F_1(x, y, z) &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\ F_2(x, y, z) &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\ F_3(x, y, z) &= 2x - 3x^2 y - x^3 z. \end{aligned} \tag{13}$$

Theorem 4.2 (Constant Jacobian and explicit three-point fibre). *The map F of Theorem 4.1 satisfies*

$$\det JF(x, y, z) = -2 \quad \text{for every } (x, y, z) \in \mathbb{C}^3. \tag{14}$$

Moreover, the three distinct points

$$p_0 = \left(0, 0, -\frac{1}{4}\right), \quad p_+ = \left(1, -\frac{3}{2}, \frac{13}{2}\right), \quad p_- = \left(-1, \frac{3}{2}, \frac{13}{2}\right) \tag{15}$$

have the common image

$$F(p_0) = F(p_+) = F(p_-) = \left(-\frac{1}{4}, 0, 0\right). \tag{16}$$

Proof. We first prove the Jacobian identity. Put

$$u = 1 + xy, \quad h = u^2 z + y^2(1 + 3u). \tag{17}$$

Because $1 + 3u = 4 + 3xy$, the first two coordinates of F become

$$F_1 = uh, \quad F_2 = y + 3xh. \tag{18}$$

On the Zariski-open set $U = \{(x, y, z) : xu \neq 0\}$, one has $y = (u - 1)/x$. The third coordinate also has the rational-chart form

$$F_3 = \frac{x(u + 1 - x^2 h)}{u^2}. \tag{19}$$

Indeed,

$$F_3 = x(2 - 3xy - x^2 z) = x(5 - 3u - x^2 z).$$

Multiplication by u^2/x gives

$$\begin{aligned}\frac{u^2 F_3}{x} &= u^2(5 - 3u) - x^2 u^2 z \\ &= u^2(5 - 3u) - x^2 h + x^2 y^2(1 + 3u) \\ &= u^2(5 - 3u) - x^2 h + (u - 1)^2(1 + 3u).\end{aligned}$$

The purely (u)-dependent terms reduce as follows:

$$\begin{aligned}u^2(5 - 3u) + (u - 1)^2(1 + 3u) &= 5u^2 - 3u^3 + (u^2 - 2u + 1)(1 + 3u) \\ &= 5u^2 - 3u^3 + 3u^3 - 5u^2 + u + 1 \\ &= u + 1.\end{aligned}$$

This proves (19).

Let $\Phi(x, y, z) = (x, u, h)$. Since

$$\frac{\partial(x, u, h)}{\partial(x, y, z)} = \begin{pmatrix} 1 & 0 & 0 \\ y & x & 0 \\ h_x & h_y & u^2 \end{pmatrix},$$

one has

$$\det J\Phi = xu^2. \quad (20)$$

In the (x, u, h) -chart, define

$$G(x, u, h) = \left(uh, \frac{u - 1 + 3x^2 h}{x}, \frac{x(u + 1 - x^2 h)}{u^2} \right).$$

Equations (18) and (19) say that $F = G \circ \Phi$ on U . Differentiation gives

$$JG = \begin{pmatrix} 0 & h & u \\ \frac{3hx^2 - u + 1}{x^2} & \frac{1}{x} & 3x \\ \frac{-3hx^2 + u + 1}{u^2} & \frac{x(2hx^2 - u - 2)}{u^3} & -\frac{x^3}{u^2} \end{pmatrix}. \quad (21)$$

Set $k = hx^2$ and expand the determinant of (21) along its first row. After multiplication by xu^2 , the numerator is

$$\begin{aligned}xu^2 \det JG &= k(-6k + 2u + 4) + (3k - u + 1)(2k - u - 2) \\ &\quad - u(-3k + u + 1).\end{aligned}$$

The middle product is

$$(3k - u + 1)(2k - u - 2) = 6k^2 - 5ku - 4k + u^2 + u - 2.$$

Consequently,

$$\begin{aligned}xu^2 \det JG &= (-6k^2 + 2ku + 4k) + (6k^2 - 5ku - 4k + u^2 + u - 2) \\ &\quad + (3ku - u^2 - u) = -2.\end{aligned}$$

Thus

$$\det JG = -\frac{2}{xu^2} \quad \text{on } U. \quad (22)$$

By the chain rule and (20)–(22),

$$\det JF = (\det JG \circ \Phi) \det J\Phi = -\frac{2}{xu^2}(xu^2) = -2$$

on U . The determinant $\det JF + 2$ is a polynomial in x, y, z that vanishes on the nonempty Zariski-open set U ; hence it is the zero polynomial. This proves (14) on all of $\mathbb{C}^3$.

We now evaluate the three points. At p_0 , $1 + xy = 1$ and $y = x = 0$, so

$$F_1(p_0) = -\frac{1}{4}, \quad F_2(p_0) = 0, \quad F_3(p_0) = 0.$$

At both p_+ and p_- , one has

$$xy = -\frac{3}{2}, \quad 1 + xy = -\frac{1}{2}, \quad 4 + 3xy = -\frac{1}{2}, \quad y^2 = \frac{9}{4}.$$

For p_+ , direct substitution yields

$$\begin{aligned} F_1(p_+) &= \left(-\frac{1}{2}\right)^3 \frac{13}{2} + \frac{9}{4} \left(-\frac{1}{2}\right) \left(-\frac{1}{2}\right) = -\frac{13}{16} + \frac{9}{16} = -\frac{1}{4}, \\ F_2(p_+) &= -\frac{3}{2} + 3 \left(\frac{1}{4}\right) \frac{13}{2} + 3 \left(\frac{9}{4}\right) \left(-\frac{1}{2}\right) = \frac{-12 + 39 - 27}{8} = 0, \\ F_3(p_+) &= 2 - 3 \left(-\frac{3}{2}\right) - \frac{13}{2} = \frac{4 + 9 - 13}{2} = 0. \end{aligned}$$

For p_- , the signs of x and y are reversed while xy , y^2 , and z remain as above. Hence

$$\begin{aligned} F_1(p_-) &= -\frac{13}{16} + \frac{9}{16} = -\frac{1}{4}, \\ F_2(p_-) &= \frac{3}{2} - 3 \left(\frac{1}{4}\right) \frac{13}{2} - 3 \left(\frac{9}{4}\right) \left(-\frac{1}{2}\right) = \frac{12 - 39 + 27}{8} = 0, \\ F_3(p_-) &= -2 - 3 \left(\frac{3}{2}\right) + \frac{13}{2} = \frac{-4 - 9 + 13}{2} = 0. \end{aligned}$$

The first coordinates of p_0, p_+, p_- are $0, 1, -1$, so the three points are distinct. This proves (16). $\square$

5. WITHDRAWN CONSTRUCTIONS

Withdrawn argument 5.1 (Arbitrary cube-root interpolation on the Fable fibre). The polynomial

$$\zeta(x) = 1 - \frac{3}{2}x^2 + i\frac{\sqrt{3}}{2}x$$

sends $x = 0, 1, -1$ to $1, \omega, \omega^2$, respectively. These equalities are correct. The assignment is the unique quadratic interpolant determined by an externally chosen labeling of the three points. Its C_3 -structure therefore depends on that labeling. The later sections return to the original fibre and its involution.

Verification and point of failure. Substitution gives

$$\zeta(0) = 1, \quad \zeta(1) = -\frac{1}{2} + i\frac{\sqrt{3}}{2} = \omega, \quad \zeta(-1) = -\frac{1}{2} - i\frac{\sqrt{3}}{2} = \omega^2.$$

The three numerical evaluations are therefore correct. Let (p_0, p_+, p_-) denote the three source points. Any three prescribed complex values (z_0, z_+, z_-) admit a unique polynomial of degree at most two in the first coordinate $x \in \{0, 1, -1\}$ taking those values. Interpolation alone can therefore realize every labeled three-point configuration. An invariant conclusion would need additional data from the original map. The withdrawn inference treated the external labeling as a canonical invariant. $\square$

6. THE CENTRAL-ANGLE AND INVERSIVE-CIRCLE THEOREMS OF U/UMBRELLACORP_HR

The three source results in this section are attributed to Reddit user u/UmbrellaCorp_HR and to the source post [3].

6.1. Central-angle field closure and cyclic telescoping.

Theorem 6.1 (Central-angle field closure). *Let $K \subseteq \mathbb{R}$ be a field, let $O, P_1, \dots, P_N \in K^2$, and write*

$$u_j = P_j - O \neq 0, \quad u_{N+1} = u_1.$$

Define

$$D_j = \det(u_j, u_{j+1}), \quad E_j = u_j \cdot u_{j+1}.$$

Then $D_j, E_j \in K$. If $E_j \neq 0$, then the tangent of the oriented angle from u_j to u_{j+1} belongs to K and is

$$\tan \theta_j = \frac{D_j}{E_j}.$$

For any continuous choice of the oriented arguments along the prescribed cyclic order, there is an integer w , the winding number, such that

$$\sum_{j=1}^N \operatorname{atan2}(D_j, E_j) = 2\pi w. \quad (23)$$

Equivalently,

$$\prod_{j=1}^N \frac{E_j + iD_j}{\sqrt{E_j^2 + D_j^2}} = 1. \quad (24)$$

Proof. Write $u_j = (x_j, y_j)$ and $z_j = x_j + iy_j$. A direct multiplication gives

$$\begin{aligned} \bar{z}_j z_{j+1} &= (x_j - iy_j)(x_{j+1} + iy_{j+1}) \\ &= x_j x_{j+1} + y_j y_{j+1} + i(x_j y_{j+1} - y_j x_{j+1}) \\ &= E_j + iD_j. \end{aligned}$$

Taking absolute values and squaring yields

$$E_j^2 + D_j^2 = |\bar{z}_j z_{j+1}|^2 = |z_j|^2 |z_{j+1}|^2 = \|u_j\|^2 \|u_{j+1}\|^2.$$

The nonzero-vector hypothesis therefore makes every unit-modulus factor well-defined and gives

$$\frac{E_j + iD_j}{\sqrt{E_j^2 + D_j^2}} = \frac{\bar{z}_j z_{j+1}}{|z_j| |z_{j+1}|}. \quad (25)$$

Multiplying (25) cyclically gives

$$\prod_{j=1}^N \frac{\bar{z}_j z_{j+1}}{|z_j| |z_{j+1}|} = \frac{\prod_{j=1}^N \bar{z}_j z_{j+1}}{\prod_{j=1}^N |z_j| |z_{j+1}|} = \frac{\prod_{j=1}^N |z_j|^2}{\prod_{j=1}^N |z_j|^2} = 1,$$

which proves (24). Taking a continuous argument around the cyclic list proves (23), since a closed loop in S^1 changes its lifted argument by $2\pi w$ for a unique $w \in \mathbb{Z}$. Finally, determinants and dot products use only addition and multiplication of coordinates in K , so $D_j, E_j \in K$; division by nonzero E_j proves $D_j/E_j \in K$. $\square$

6.2. The Lorentz reflection underlying circle inversion. Let $V = K^4$ have coordinates $v = (a, b_0, b_1, c)$ and symmetric bilinear form

$$B(v, w) = b_0 B_0 + b_1 B_1 - \frac{aC + cA}{2}, \quad w = (A, B_0, B_1, C). \quad (26)$$

Its quadratic form is

$$\mathcal{N}(v) = B(v, v) = b_0^2 + b_1^2 - ac. \quad (27)$$

Associate to v the oriented circle or line

$$q_v(x, y) = a(x^2 + y^2) + 2(b_0 x - b_1 y) + c = 0. \quad (28)$$

Theorem 6.2 (Circle-inversion coefficient transformation). *Let $K \subseteq \mathbb{R}$ be a field. Let*

$$w = (A, B_0, B_1, C) \in K^4, \quad A \neq 0, \quad \Delta = \mathcal{N}(w) = B_0^2 + B_1^2 - AC \neq 0.$$

The Euclidean inversion in the nondegenerate circle $q_w = 0$ acts projectively on oriented-circle coefficients by the matrix

$$M_w = \frac{1}{A^2} \begin{pmatrix} B_0^2 + B_1^2 & -2AB_0 & -2AB_1 & A^2 \\ B_0C & B_1^2 - B_0^2 - AC & -2B_0B_1 & AB_0 \\ B_1C & -2B_0B_1 & B_0^2 - B_1^2 - AC & AB_1 \\ C^2 & -2CB_0 & -2CB_1 & B_0^2 + B_1^2 \end{pmatrix}. \quad (29)$$

It satisfies, for every $v \in V$,

$$\mathcal{N}(M_w v) = \frac{\Delta^2}{A^4} \mathcal{N}(v), \quad M_w^2 = \frac{\Delta^2}{A^4} I_4. \quad (30)$$

In particular, it preserves the projective null cone, is a projective involution, and maps K -rational coefficient vectors to K -rational coefficient vectors.

Proof. The Lorentz reflection in w is

$$R_w(v) = v - 2 \frac{B(v, w)}{\Delta} w. \quad (31)$$

Set $\lambda(v) = 2B(v, w)/\Delta$. Since $B(w, w) = \Delta$,

$$\begin{aligned} B(R_w u, R_w v) &= B(u - \lambda(u)w, v - \lambda(v)w) \\ &= B(u, v) - \lambda(v)B(u, w) - \lambda(u)B(w, v) + \lambda(u)\lambda(v)\Delta \\ &= B(u, v) - \frac{4B(u, w)B(v, w)}{\Delta} + \frac{4B(u, w)B(v, w)}{\Delta} \\ &= B(u, v). \end{aligned}$$

Thus R_w is a B -isometry. Moreover,

$$B(R_w v, w) = B(v, w) - 2B(v, w) = -B(v, w),$$

so applying the same reflection twice returns v ; hence $R_w^2 = I_4$.

We next expand $(\Delta/A^2)R_w(v)$. Its first coordinate is

$$\frac{1}{A^2}(\Delta a - 2AB(v, w)) = \frac{1}{A^2}((B_0^2 + B_1^2)a - 2AB_0b_0 - 2AB_1b_1 + A^2c).$$

The second coordinate is

$$\frac{1}{A^2}(\Delta b_0 - 2B_0B(v, w)) = \frac{1}{A^2}(B_0Ca + (B_1^2 - B_0^2 - AC)b_0 - 2B_0B_1b_1 + AB_0c).$$

The third coordinate is

$$\frac{1}{A^2}(\Delta b_1 - 2B_1B(v, w)) = \frac{1}{A^2}(B_1Ca - 2B_0B_1b_0 + (B_0^2 - B_1^2 - AC)b_1 + AB_1c),$$

and the fourth is

$$\frac{1}{A^2}(\Delta c - 2CB(v, w)) = \frac{1}{A^2}(C^2a - 2CB_0b_0 - 2CB_1b_1 + (B_0^2 + B_1^2)c).$$

These are exactly the four rows of (29); therefore

$$M_w = \frac{\Delta}{A^2} R_w. \quad (32)$$

The B -isometry and involution identities for R_w now give

$$\mathcal{N}(M_w v) = \frac{\Delta^2}{A^4} \mathcal{N}(v), \quad M_w^2 = \frac{\Delta^2}{A^4} I_4,$$

which proves (30).

To identify this projective coefficient action with Euclidean inversion, let $P = (x, y)$ and define the null point-circle vector

$$p(P) = (1, -x, y, x^2 + y^2). \quad (33)$$

Then $\mathcal{N}(p(P)) = 0$ and direct substitution into (26) gives

$$-2B(v, p(P)) = q_v(P). \quad (34)$$

Thus point-circle incidence is precisely B -orthogonality.

The inversion circle has center and squared radius

$$O = \left(-\frac{B_0}{A}, \frac{B_1}{A} \right), \quad r^2 = \frac{\Delta}{A^2}, \quad (35)$$

because

$$q_w(P) = A\|P - O\|^2 - \frac{\Delta}{A}. \quad (36)$$

Since $-2B(p(P), w) = q_w(P)$,

$$R_w p(P) = p(P) + \frac{q_w(P)}{\Delta} w. \quad (37)$$

Its first coordinate is

$$1 + \frac{Aq_w(P)}{\Delta} = \frac{\Delta + Aq_w(P)}{\Delta} = \frac{A^2\|P - O\|^2}{\Delta}.$$

Dividing all four homogeneous coordinates by this nonzero first coordinate and recovering the point from the second and third coordinates gives

$$P' = O + \frac{\Delta}{A^2} \frac{P - O}{\|P - O\|^2} = O + r^2 \frac{P - O}{\|P - O\|^2}, \quad (38)$$

the Euclidean inversion of P in the circle with center O and radius r .

Finally, if P lies on the circle represented by v , then $B(v, p(P)) = 0$. The B -invariance of R_w implies

$$B(R_w v, R_w p(P)) = B(v, p(P)) = 0.$$

By (38), $R_w p(P)$ represents the inverted point P' , while by (32), $M_w v$ represents the same projective circle coefficient vector as $R_w v$. Hence $M_w v$ is exactly the image circle under Euclidean inversion. Every entry of (29) lies in K and $A^{-2} \in K$, proving field closure. $\square$

Theorem 6.3 (Full inverse-circle equation and evaluation covariance). *With K, w, A, Δ as in theorem 6.2, let $v = (a, b_0, b_1, c)$, and write*

$$I_w(P) := O + \frac{\Delta}{A^2} \frac{P - O}{\|P - O\|^2}, \quad O = \left(-\frac{B_0}{A}, \frac{B_1}{A} \right).$$

The image of $q_v = 0$ under inversion in $q_w = 0$ is the following explicit equation:

$$\begin{aligned} 0 = & \frac{a(B_0^2 + B_1^2) - 2A(b_0 B_0 + b_1 B_1) + cA^2}{A^2} (x^2 + y^2) \\ & + 2 \left(\frac{b_0(B_1^2 - B_0^2 - AC) + aB_0 C - 2b_1 B_0 B_1 + cAB_0}{A^2} x \right. \\ & \quad \left. - \frac{b_1(B_0^2 - B_1^2 - AC) + aB_1 C - 2b_0 B_0 B_1 + cAB_1}{A^2} y \right) \\ & + \frac{aC^2 - 2b_0 B_0 C - 2b_1 B_1 C + c(B_0^2 + B_1^2)}{A^2}. \end{aligned} \quad (39)$$

With the coefficient scale fixed by (29), its polynomial evaluations obey

$$q_{M_w v}(P) = \|P - O\|^2 q_v(I_w(P)) \quad (40)$$

for every $P \neq O$.

For $\alpha \neq 0$, define the two ordered axial carriers

$$w_\alpha = (4, 0, 0, -4\alpha), \quad w_\alpha^\vee = (-4\alpha, 0, 0, 4). \quad (41)$$

Their full equations are

$$q_{M_{w_\alpha} v}(x, y) = c(x^2 + y^2) + 2\alpha(b_0 x - b_1 y) + \alpha^2 a, \quad (42)$$

$$q_{M_{w_\alpha^\vee} v}(x, y) = c(x^2 + y^2) + 2\alpha^{-1}(b_0 x - b_1 y) + \alpha^{-2} a. \quad (43)$$

Thus the two ordered channels retain distinct coefficient scales at every α except $\alpha = 1$, where the displayed actions agree.

Proof. Multiplying the matrix in (29) by $(a, b_0, b_1, c)^\top$ gives the four coefficients in (39); insertion into (28) gives the displayed equation.

Let $P' = I_w(P)$ and put

$$h(P) := \frac{A^2 \|P - O\|^2}{\Delta}.$$

Equation (37) gives the exact homogeneous identity

$$R_w p(P) = h(P) p(P').$$

Apply the same identity at P' . Since I_w is an involution and

$$\|P' - O\|^2 = \frac{\Delta^2}{A^4 \|P - O\|^2},$$

one obtains

$$R_w p(P') = \frac{\Delta}{A^2 \|P - O\|^2} p(P).$$

Using (34), the B -isometry of R_w , and $M_w = (\Delta/A^2)R_w$, we therefore have

$$\begin{aligned} q_{M_w v}(P) &= -2B \left(\frac{\Delta}{A^2} R_w v, p(P) \right) \\ &= -2\|P - O\|^2 B(R_w v, R_w p(P')) \\ &= \|P - O\|^2 q_v(P'), \end{aligned}$$

which proves (40). Substitution of the two vectors in (41) into (39) proves (42) and (43). $\square$

Corollary 6.4 (Unit-circle coefficient reversal). *For inversion in $x^2 + y^2 - 1 = 0$, one has*

$$(a, b_0, b_1, c) \mapsto (c, b_0, b_1, a).$$

Proof. Substitute $(A, B_0, B_1, C) = (1, 0, 0, -1)$ in (29). $\square$

Provenance record 6.5 (Inversive-circle attribution). The central-angle field-closure theorem, the circle-inversion coefficient matrix, and the full expanded inverse-circle equation are attributed to Reddit user u/UmbrellaCorp_HR. The public source is cited in full as [3]. The surrounding synthesis is credited to The Clankers. The proofs in this section are self-contained reconstructions. The first uses complex edge telescoping. The second identifies the displayed coefficient matrix with a scaled reflection for the bilinear form (26). The third multiplies out that action and derives its exact evaluation covariance.

The coefficient matrix, the full expanded equation, Lorentz-form covariance, projective involution square, pointwise evaluation covariance, unit-circle coefficient reversal, ordered axial formulas, and the three Fable quarter-eigenrays are checked symbolically by the file

verification/verify_umbrella_circle_inversion.py

The captured receipt is

verification/verify_umbrella_circle_inversion_output.txt

The same checker verifies the Hermitian Möbius exchange, its Lorentz action $\text{diag}(1, 1, -1, -1)$, and both ordered branch-circle factorizations of the Fable cubic.

7. THE COMPLETE FIBRE AND ITS CANONICAL MARKED-FACTOR LIFT

Tao's geometric digestion develops the multiplication map $(L, Q) \mapsto LQ$, its resultant-one slice, and the three choices of marked linear factor [6]. The separate Reddit post by u/JGPTEch presents the same marked-factor data as a staged pipeline [4]. Its public implementation gives an exact derivation, three independent language verifiers, and a blind chart search [5]. The coordinate identities below credit each source for its corresponding role.

7.1. Exact classification of the common fibre.

Theorem 7.1 (Complete fibre over $(-1/4, 0, 0)$). *For the polynomial map F in Theorem 4.1,*

$$F^{-1}\left(-\frac{1}{4}, 0, 0\right) = \{p_0, p_+, p_-\},$$

where the three points are those of (15). Every point of this fibre is simple; equivalently, the fibre scheme is reduced.

Proof. Retain the abbreviations

$$u = 1 + xy, \quad h = u^2z + y^2(1 + 3u)$$

from (17). On the specified fibre, $F_1 = uh = -1/4$; therefore $u \neq 0$ and $h \neq 0$. The equation $F_2 = y + 3xh = 0$ gives

$$y = -3xh = \frac{3x}{4u}. \quad (44)$$

Substitution into $u = 1 + xy$ yields

$$4u^2 - 4u - 3x^2 = 0. \quad (45)$$

The rational-chart identity (19), which is valid here because $u \neq 0$, and the equality $h = -1/(4u)$ give

$$0 = F_3 = \frac{x(u + 1 - x^2h)}{u^2} = \frac{x}{4u^3}(4u^2 + 4u + x^2).$$

Thus

$$x(4u^2 + 4u + x^2) = 0. \quad (46)$$

If $x = 0$, then (45) and $u \neq 0$ imply $u = 1$. Equation (44) gives $y = 0$, and $h = u^2z = -1/4$ gives $z = -1/4$. This is p_0 .

Suppose $x \neq 0$. Then the bracket in (46) vanishes. Subtracting (45) from that bracket equation gives

$$8u + 4x^2 = 0, \quad \text{hence} \quad 2u + x^2 = 0.$$

Substituting $x^2 = -2u$ into (45) gives

$$4u^2 - 4u + 6u = 2u(2u + 1) = 0.$$

Since $u \neq 0$, one obtains $u = -1/2$ and $x^2 = 1$. Then (44) gives $y = -3x/2$. Finally,

$$h = -\frac{1}{4u} = \frac{1}{2}, \quad h = u^2z + y^2(1 + 3u) = \frac{z}{4} - \frac{9}{8},$$

so $z = 13/2$. The two signs of x give p_+ and p_- .

By Theorem 4.2, $\det JF = -2$ at every point. In particular, the differential is invertible at each point of the fibre. The Jacobian criterion therefore makes every local fibre algebra a reduced field of length one. Since the preceding argument lists the complete set of fibre points, the entire fibre scheme is reduced. $\square$

Corollary 7.2 (Coordinate algebra and exact elimination certificate). *The reduced coordinate algebra of the fibre is*

$$\mathbb{Q}[x]/(x^3 - x) \cong \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}. \quad (47)$$

In lexicographic order $z > y > x$, a reduced rational Gröbner basis of the fibre ideal is

$$\left\{ z - \frac{27}{4}x^2 + \frac{1}{4}, y + \frac{3}{2}x, x^3 - x \right\}. \quad (48)$$

Proof. The equations in (48) give

$$y = -\frac{3}{2}x, \quad z = \frac{27x^2 - 1}{4}, \quad x(x-1)(x+1) = 0,$$

which are exactly the three solutions proved in Theorem 7.1. Their leading monomials z, y, x^3 are pairwise coprime, so every S -polynomial reduces to zero; Buchberger's criterion proves that the displayed set is a Gröbner basis. Its quotient is $\mathbb{Q}[x]/(x^3 - x)$. Since $x^3 - x = x(x-1)(x+1)$ is squarefree over $\mathbb{Q}$, the Chinese remainder theorem gives (47) and shows that the displayed Gröbner ideal is radical. The actual fibre ideal is also radical by the Jacobian criterion used in Theorem 7.1, and the two radical ideals have the same complex zero set by the classification theorem. Hilbert's Nullstellensatz therefore identifies their scalar extensions to $\mathbb{C}[x, y, z]$. Faithful flatness of $\mathbb{C}/\mathbb{Q}$ identifies the original rational ideals. Hence (48) is a Gröbner basis of the actual fibre ideal itself. $\square$

Proposition 7.3 (Canonical fibre involution). *Define $\sigma(x, y, z) = (-x, -y, z)$. Then*

$$F \circ \sigma = (F_1, -F_2, -F_3). \quad (49)$$

Consequently, σ preserves the common fibre, fixes p_0 , and exchanges p_+ with p_- .

Proof. The quantities xy, y^2 , and z are fixed by σ , so F_1 is fixed. Every term of F_2 contains an odd total occurrence of either x or y , and simultaneous sign reversal changes its sign. Every term of F_3 likewise changes sign. This proves (49). The action on the three displayed points follows by substitution. $\square$

7.2. The source-defined invariant lift.

Theorem 7.4 (Resultant-one linear-quadratic invariant lift). *For $(x, y, z) \in \mathbb{C}^3$, put $t = 1 + xy$ and define*

$$\begin{aligned} a &= x, & b &= t, \\ c &= 1 - \frac{3}{2}xy - \frac{1}{2}x^2z, & d &= \frac{x^2yz + 3xy^2 + xz + y}{2}, \\ e &= x^2y^2z + 3xy^3 + 2xyz + 4y^2 + z. \end{aligned} \quad (50)$$

Then

$$ad + bc = 1, \quad a^2e - abd + b^2c = 1, \quad (51)$$

and

$$F_1 = be, \quad F_2 = 2(ae + bd), \quad F_3 = 2ac. \quad (52)$$

Consequently, for

$$L(U, V) = aU + bV, \quad Q(U, V) = cU^2 + dUV + eV^2, \quad (53)$$

one has

$$LQ = \frac{F_3}{2}U^3 + U^2V + \frac{F_2}{2}UV^2 + F_1V^3, \quad (54)$$

and

$$\text{Res}(L, Q) = 1. \quad (55)$$

Here the homogeneous resultant is defined by

$$\text{Res}(aU + bV, cU^2 + dUV + eV^2) = a^2e - abd + b^2c. \quad (56)$$

Proof. The last expression for e in (50) can be reorganized as

$$e = t^2z + y^2(1 + 3t), \quad (57)$$

because $1 + 3t = 4 + 3xy$. The remaining two nonlinear coefficients become

$$c = \frac{5 - 3t - x^2z}{2}, \quad d = \frac{xzt + y(3t - 2)}{2}. \quad (58)$$

Using $xy = t - 1$, one obtains

$$\begin{aligned} 2(ad + bc) &= x(xzt + y(3t - 2)) + t(5 - 3t - x^2z) \\ &= (t - 1)(3t - 2) + 5t - 3t^2 \\ &= (3t^2 - 5t + 2) + 5t - 3t^2 = 2. \end{aligned}$$

This proves the first identity in (51).

For the second identity, the terms containing z cancel:

$$2(a^2e - abd + b^2c) = 2x^2y^2(1 + 3t) - xyt(3t - 2) + t^2(5 - 3t).$$

Set $r = xy = t - 1$. Then the right-hand side is

$$\begin{aligned} &2(t - 1)^2(1 + 3t) - (t - 1)t(3t - 2) + t^2(5 - 3t) \\ &= (6t^3 - 10t^2 + 2t + 2) + (-3t^3 + 5t^2 - 2t) + (5t^2 - 3t^3) = 2. \end{aligned}$$

Thus $a^2e - abd + b^2c = 1$.

Equation (57) gives

$$be = t^3z + y^2t(1 + 3t) = F_1.$$

Equation (58) gives

$$2ac = x(5 - 3t - x^2z) = 2x - 3x^2y - x^3z = F_3.$$

Finally,

$$\begin{aligned} 2(ae + bd) &= 2x(t^2z + y^2(1 + 3t)) + t(xzt + y(3t - 2)) \\ &= 3xt^2z + 2xy^2(1 + 3t) + ty(3t - 2). \end{aligned}$$

The difference between the desired non- z part $y + 3xy^2(1 + 3t)$ and the last two terms is

$$y(1 + (t - 1)(1 + 3t) - t(3t - 2)) = 0.$$

Hence $F_2 = 2(ae + bd)$.

Multiplication of the two forms in (53) yields

$$LQ = acU^3 + (ad + bc)U^2V + (ae + bd)UV^2 + beV^3.$$

Substituting the identities already proved gives (54). The Sylvester determinant for a homogeneous linear form and a homogeneous quadratic form is the expression in (56); the second identity in (51) therefore proves (55). $\square$

Theorem 7.5 (Exact substitution of the complete fibre). *On the common fibre of Theorem 7.1, the invariant lift in Theorem 7.4 has the following values:*

<i>point</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>L</i>	<i>Q</i>
p_0	0	1	1	0	$-1/4$	V	$U^2 - V^2/4$
p_+	1	$-1/2$	0	1	$1/2$	$U - V/2$	$V(U + V/2)$
p_-	-1	$-1/2$	0	-1	$1/2$	$-(U + V/2)$	$-V(U - V/2)$

All three products are the same squarefree binary cubic,

$$C(U, V) = U^2V - \frac{1}{4}V^3 = V \left(U - \frac{1}{2}V \right) \left(U + \frac{1}{2}V \right), \quad (59)$$

and every row has resultant 1. Thus the three source points are exactly the three resultant-one choices of a marked linear factor of the same squarefree cubic.

Proof. Substitution of p_0, p_+, p_- into (50) gives the five coefficient rows in the table. Multiplying the displayed L and Q in each row gives (59). Alternatively, this follows at once from (54) and the common output $(-1/4, 0, 0)$. The resultant equals 1 in every row by (55).

The cubic in (59) has the three distinct projective linear factors V , $U - V/2$, and $U + V/2$. The table assigns each of these factors to L once, with the complementary product assigned to Q , and the scalar representatives are fixed by $ad + bc = 1$ and resultant 1. Therefore the three points realize precisely the three resultant-one marked linear-factor choices. $\square$

Proposition 7.6 (Blind marked-factor recovery of the Fable map). *The first successful chart returned by the public JGPTech blind search [5] is the coefficient slice $M_2 = 1$ with modulus a . After writing its three chart coordinates as (x, y, z) , the induced map $G = (G_1, G_2, G_3)$ is*

$$\begin{aligned} G_1 &= x - \frac{3}{2}x^2y - \frac{1}{2}x^3z, \\ G_2 &= \frac{1}{2}y + 6xy^2 + \frac{9}{2}x^2y^3 + \frac{3}{2}xz + 3x^2yz + \frac{3}{2}x^3y^2z, \\ G_3 &= z + 4y^2 + 3xyz + 7xy^3 + 3x^2y^2z + 3x^2y^4 + x^3y^3z. \end{aligned}$$

For the map F in theorem 4.1, one has the polynomial identity

$$G = \left(\frac{F_3}{2}, \frac{F_2}{2}, F_1 \right), \quad \det JG = \frac{1}{2}. \quad (60)$$

Consequently F and G have exactly the same fibres. For example, the three distinct points

$$\left(\frac{1}{6}, -4, -288 \right), \quad \left(-\frac{1}{2}, 2, 36 \right), \quad \left(\frac{1}{3}, -4, 0 \right)$$

have common G -image $(1, -2, 0)$ and common F -image $(0, -4, 2)$.

Proof. Expanding the components in (13) gives

$$\begin{aligned} \frac{F_3}{2} &= x - \frac{3}{2}x^2y - \frac{1}{2}x^3z, \\ \frac{F_2}{2} &= \frac{1}{2}y + 6xy^2 + \frac{9}{2}x^2y^3 + \frac{3}{2}xz + 3x^2yz + \frac{3}{2}x^3y^2z, \end{aligned}$$

and

$$F_1 = z + 4y^2 + 3xyz + 7xy^3 + 3x^2y^2z + 3x^2y^4 + x^3y^3z.$$

These are the three displayed components of G . The target-coordinate map $(u, v, w) \mapsto (w/2, v/2, u)$ has determinant $-1/4$; hence

$$\det JG = -\frac{1}{4} \det JF = \frac{1}{2}$$

by (14). Direct substitution of the three displayed points in G gives $(1, -2, 0)$ in every case. Applying the inverse target-coordinate map gives the stated common value of F . $\square$

Provenance record 7.7 (Exact JGPTech pipeline replay). The chart formulas in theorem 7.6 were read from the public `Jacobian/` package at Git revision `2fe1901840ef3aa28232930e1aec8d8bbb61281a` [5]. The repository credits the displayed counterexample to Fable. Its `blind_search.py` begins with a marked linear-quadratic factorization, the visible cubic coefficients, and the resultant equation $R(L, Q) = 1$. It supplies no coefficient slice, final map, or collision point as input. The replay selected $M_2 = 1$ and modulus a , proved the three component differences in (60) to be zero polynomials, and found 24 exact three-point collisions with integral root search bound 4. The independent `aceg.py` manifest verifier passed all five stored maps. Thus the public computation independently recovers the Fable chart once the marked-factor/resultant framework has been supplied.

8. THE SINGULAR C_1 – C_2 – C_3 SUPPORT OBJECT

The three factors in (59) admit a simultaneous idempotent, sheet, and phase envelope. Their phase cycle and sheet reflection generate D_3 , and the occupied sum of all three marked factors is fixed by that action. For the 12289 arithmetic data, the same D_3 acts faithfully on the ambient orbit

$$(3, 6, 3), \quad (3, 3, 6), \quad (6, 3, 3),$$

while the fixed origin-resolved certificate shell carries the divisor- complement C_2 . The equal-fibre criterion in theorem 9.66 proves the obstruction to lifting the phase cycle to that fixed shell. The surviving six-element action and its tetrahedral realization are proved in theorems 9.25 and 10.2.

Let

$$V_F := \mathbb{Q}e_0 \oplus \mathbb{Q}e_1 \oplus \mathbb{Q}e_2,$$

where e_0, e_1, e_2 label, respectively, the marked factors

$$V, \quad U - \frac{1}{2}V, \quad -(U + \frac{1}{2}V).$$

Define operators $r, j \in \text{GL}(V_F)$ by

$$re_0 = e_1, \quad re_1 = e_2, \quad re_2 = e_0,$$

and

$$je_0 = e_0, \quad je_1 = e_2, \quad je_2 = e_1.$$

Then

$$r^3 = 1, \quad j^2 = 1, \quad j r j = r^{-1}, \tag{61}$$

so the generated group is

$$D_3 = C_3 \rtimes C_2 \cong S_3.$$

Here the source involution $\sigma(x, y, z) = (-x, -y, z)$ of theorem 7.3 supplies j , and the displayed cyclic ordering supplies r . The definition of r uses this ordering alone.

Proposition 8.1 (Commuting and reflected couplings of the six-point carrier). *Write the common finite carrier as*

$$\mathcal{C} := C_3 \times C_2 = \{(a, \epsilon) : a \in \mathbb{Z}/3\mathbb{Z}, \epsilon \in \mathbb{Z}/2\mathbb{Z}\}.$$

The two products

$$(a, \epsilon) \circ (b, \delta) = (a + b, \epsilon + \delta), \tag{62}$$

$$(a, \epsilon) \star (b, \delta) = (a + (-1)^\epsilon b, \epsilon + \delta) \tag{63}$$

give

$$(\mathcal{C}, \circ) \cong C_6, \quad (\mathcal{C}, \star) \cong D_3.$$

The first law is the law carried by the diagonal central quotient in the Standard Model gauge group described by Tong [26]. More precisely, the central element

$$\xi = (e^{2\pi i/6}, \eta, \omega), \quad \eta^2 = 1, \quad \omega^3 = 1, \tag{64}$$

generates C_6 ; on a state of integer $U(1)_{\tilde{Y}}$ -charge q , its first component acts as $e^{2\pi i q/6}$. The subgroups generated by ξ^2 and ξ^3 are respectively C_3 and C_2 .

The second law is the law of the singular Fable count orbit: the map

$$\Phi : \mathcal{C} \longrightarrow \langle r, s \rangle, \quad \Phi(a, \epsilon) = r^a s^\epsilon,$$

is an isomorphism from $(\mathcal{C}, \star)$ to D_3 . The two laws exhaust the homomorphisms

$$C_2 \longrightarrow \text{Aut}(C_3) \cong C_2.$$

Under $\circ$, the element $(1, 1)$ has order six. Under $\star$, the element orders are $1^{[1]}, 2^{[3]}, 3^{[2]}$.

Proof. The Chinese-remainder map

$$\mathbb{Z}/6\mathbb{Z} \longrightarrow \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, \quad [n]_6 \longmapsto ([n]_3, [n]_2),$$

transports addition to $\circ$. This proves $(\mathcal{C}, \circ) \cong C_6$ and identifies the powers of ξ with the stated C_3 and C_2 subgroups.

For the reflected law, the relation $srs = r^{-1}$ gives

$$r^a s^\epsilon r^b s^\delta = r^{a+(-1)^\epsilon b} s^{\epsilon+\delta}.$$

Hence Φ preserves multiplication. Both its source and target have six elements, and the displayed expression $r^a s^\epsilon$ is unique, so Φ is bijective. The order assertions follow by direct evaluation of the six pairs. Finally, $\text{Aut}(C_3) \cong C_2$, so a homomorphism from C_2 either acts trivially or sends its generator to inversion. These are exactly $\circ$ and $\star$. $\square$

Corollary 8.2 (Two-sheet phase group and its six-state subgroups). *Let*

$$\mathcal{W} := (C_3^+ \times C_3^-) \rtimes C_2,$$

where the generator Θ of C_2 exchanges the two C_3 -factors. In coordinates $(a, b, \epsilon) \in C_3 \times C_3 \times C_2$, its product is

$$(a, b, \epsilon) \diamond (c, d, \delta) = ((a, b) + \vartheta^\epsilon(c, d), \epsilon + \delta), \quad \vartheta(c, d) = (d, c). \quad (65)$$

The group $\mathcal{W}$ has eighteen elements. It contains the two six-element subgroups

$$\mathcal{W}_{\text{diag}} := \{(a, a, \epsilon) : a \in C_3, \epsilon \in C_2\} \cong C_6, \quad (66)$$

$$\mathcal{W}_{\text{anti}} := \{(a, -a, \epsilon) : a \in C_3, \epsilon \in C_2\} \cong D_3. \quad (67)$$

The diagonal subgroup implements the commuting coupling, while the antidiagonal subgroup implements the inversion coupling.

Proof. Equation (65) is the semidirect-product law, so $|\mathcal{W}| = 3 \cdot 3 \cdot 2 = 18$. The exchange ϑ fixes every pair (a, a) . Hence the law on $\mathcal{W}_{\text{diag}}$ is

$$(a, \epsilon)(c, \delta) = (a + c, \epsilon + \delta),$$

and $(1, 1, 1)$ has order six.

On the antidiagonal,

$$\vartheta(c, -c) = (-c, c) = -(c, -c).$$

The induced law is therefore

$$(a, \epsilon)(c, \delta) = (a + (-1)^\epsilon c, \epsilon + \delta),$$

which is (63). This proves (67). $\square$

Lemma 8.3 (Positivity generated by a two-sheet intertwiner). *Let H_+ and H_- be Hilbert spaces, let*

$$R_\pm \in \mathcal{U}(H_\pm), \quad R_\pm^3 = I,$$

and let $A : H_+ \rightarrow H_-$ be bounded with

$$R_- A = A R_+. \quad (68)$$

On $H_+ \oplus H_-$, define

$$\mathcal{D}_A = \begin{pmatrix} 0 & A^* \\ A & 0 \end{pmatrix}, \quad \mathcal{R} = \begin{pmatrix} R_+ & 0 \\ 0 & R_- \end{pmatrix}. \quad (69)$$

Then $\mathcal{D}_A$ is self-adjoint,

$$\mathcal{D}_A^2 = \begin{pmatrix} A^*A & 0 \\ 0 & AA^* \end{pmatrix} \succeq 0, \quad (70)$$

and $\mathcal{D}_A \mathcal{R} = \mathcal{R} \mathcal{D}_A$.

Proof. The adjoint of the first matrix in (69) is itself. Block multiplication gives (70); for $(x, y) \in H_+ \oplus H_-$,

$$\langle (x, y), \mathcal{D}_A^2(x, y) \rangle = \|Ax\|^2 + \|A^*y\|^2 \geq 0.$$

Iterating (68) gives $R_-^2 A = AR_+^2$. Taking adjoints and using $R_\pm^* = R_\pm^2$ yields

$$A^* R_- = R_+ A^*.$$

The two off-diagonal blocks in $\mathcal{D}_A \mathcal{R}$ and $\mathcal{R} \mathcal{D}_A$ therefore agree. $\square$

Remark 8.4 (Narain lattice test). Narain compactifications provide integral left/right carriers of indefinite signature on which finite-order duality actions can be tested [27]. In the D_4 -containing examples of Beye, Kobayashi, and Kuwakino [28], the right-moving three-cycle

$$s_{\overline{D}_4} \mapsto v_{\overline{D}_4} \mapsto c_{\overline{D}_4} \mapsto s_{\overline{D}_4}$$

must be accompanied by the corresponding left-moving three-cycle in order to preserve the Narain lattice. This is an explicit lattice realization of a coupled $C_3^+ - C_3^-$ action. The packetwise arithmetic intertwiner A is constructed in theorem 14.174: its two sheets are the ordered flag sheets of theorem 14.173, and same-origin sheet exchange intertwines their two three-cycles.

For the support coordinate, put

$$\mathbb{Q}[\varepsilon] := \mathbb{Q}[t]/(t^2 - t), \quad \varepsilon = [t].$$

Thus $\varepsilon^2 = \varepsilon$ and $\varepsilon(1 - \varepsilon) = 0$. The complete finite envelope is the single algebra

$$\mathfrak{S}_{123} := \mathbb{Q}[\varepsilon] \otimes_{\mathbb{Q}} \mathbb{Q}[D_3]. \quad (71)$$

Its acted-on module is

$$\mathfrak{M}_{123} := \mathbb{Q}[\varepsilon] \otimes_{\mathbb{Q}} V_F, \quad (a \otimes g) \cdot (b \otimes v) := ab \otimes gv. \quad (72)$$

The occupied and unoccupied C_1 sectors are the central summands selected by ε and $1 - \varepsilon$; the C_2 and C_3 operators act inside either central summand and cannot exchange them.

Theorem 8.5 (Static Fable element in the singular envelope). *The complete Fable fibre determines the occupied element*

$$\mathbf{f} := \varepsilon \otimes (e_0 + e_1 + e_2). \quad (73)$$

This is an element of the module $\mathfrak{M}_{123}$. Every element of D_3 fixes $\mathbf{f}$. Consequently its orbit in the simultaneous C_1 - C_2 - C_3 module is the singleton $\{\mathbf{f}\}$. Every internal symmetry preserves the occupied ε -sector containing $\mathbf{f}$.

Proof. Each of the three marked factors occurs exactly once in Theorem 7.5; hence its count vector is $(1, 1, 1)$, which is (73). Directly,

$$r(e_0 + e_1 + e_2) = e_1 + e_2 + e_0 = e_0 + e_1 + e_2$$

and

$$j(e_0 + e_1 + e_2) = e_0 + e_2 + e_1 = e_0 + e_1 + e_2.$$

Since r and j generate D_3 , every group element fixes the sum. The idempotent ε is central in (71), so the same is true after tensoring with ε . Multiplication by an element of the D_3 factor preserves the submodule $\varepsilon\mathfrak{M}_{123}$. Moreover,

$$\varepsilon\mathfrak{M}_{123} \cap (1 - \varepsilon)\mathfrak{M}_{123} = \{0\},$$

because multiplication by ε sends an element of the intersection both to itself and to zero. Therefore an internal symmetry preserves occupied support. $\square$

Theorem 8.6 (C_1 -oriented null lift of the marked cubic). *Let the resultant-one coefficient vectors of the marked linear factors in theorem 7.5 be*

$$v_0 = (0, 1), \quad v_+ = \left(1, -\frac{1}{2}\right), \quad v_- = \left(-1, -\frac{1}{2}\right),$$

and adjoin the occupied support coordinate

$$\ell_0 = (1, 0, 1), \quad \ell_+ = \left(1, 1, -\frac{1}{2}\right), \quad \ell_- = \left(1, -1, -\frac{1}{2}\right). \quad (74)$$

Among forms

$$q_\alpha(t, x, y) = t^2 - \alpha x^2 - y^2,$$

there is a unique form for which all three vectors in (74) are null, namely

$$q(t, x, y) = t^2 - \frac{3}{4}x^2 - y^2. \quad (75)$$

The involution $j(t, x, y) = (t, -x, y)$ fixes ℓ_0 and exchanges ℓ_+ and ℓ_- . There is a unique linear operator r cycling ℓ_0, ℓ_+, ℓ_- ; it preserves q and the coordinate t , and it has order three.

For $m = (m_0, m_+, m_-) \in \mathbb{R}_{\geq 0}^3$, put

$$\Lambda(m) = m_0\ell_0 + m_+\ell_+ + m_-\ell_-.$$

Then

$$t(\Lambda(m)) = m_0 + m_+ + m_- =: S(m), \quad (76)$$

$$q(\Lambda(m)) = 3(m_0m_+ + m_+m_- + m_-m_0). \quad (77)$$

Consequently every nonzero nonnegative support vector lies in the component $t > 0$ of the causal cone. The occupied C_1 coordinate therefore fixes the orientation preserved by the C_2 and C_3 operators.

Proof. One has $q_\alpha(\ell_0) = 1 - 1 = 0$, while

$$q_\alpha(\ell_+) = q_\alpha(\ell_-) = 1 - \alpha - \frac{1}{4}.$$

Thus all three vectors are null exactly when $\alpha = 3/4$. This also shows how the coefficient $-1/4$ in the common cubic fixes the spatial coefficient $3/4$ in (75).

Put $X = (\sqrt{3}/2)x$ and $Y = y$. Then

$$q = t^2 - X^2 - Y^2,$$

and the spatial parts of the three vectors are

$$(0, 1), \quad \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right), \quad \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right).$$

They form an equilateral triangle on the unit circle. Rotation through $2\pi/3$ cyclically permutes them, while reflection in the Y -axis fixes the first and exchanges the latter two. Extending both maps by the identity on t proves the assertions about r , j , and q .

The three vectors have pairwise q -inner product $3/2$ and individual q -norm zero. Bilinearity therefore gives

$$q(\Lambda(m)) = 2 \left(\frac{3}{2} \right) (m_0 m_+ + m_+ m_- + m_- m_0),$$

which is (77). Equation (76) follows because every primitive lift has first coordinate one. For $m_i \geq 0$, both displayed quantities are nonnegative, and $S(m) > 0$ whenever $m \neq 0$. Since r and j fix t , this selected component is invariant. $\square$

Remark 8.7 (What is fixed by the C_1 coordinate). Each projective linear factor determines a line, leaving its two vector representatives unordered. The resultant-one identities $\text{Res}(L, Q) = 1$ fixes the representatives in theorem 7.5, and the occupied support mark lifts each representative to first coordinate 1. In particular,

$$\ell_0 + \ell_+ + \ell_- = (3, 0, 0),$$

so the supported lift selects the positive time axis.

Let $\omega = e^{2\pi i/3}$. For a support vector $m = (m_0, m_1, m_2)$ define the unscaled coordinates

$$\Sigma(m) := m_0 + m_1 + m_2, \quad Z(m) := m_0 + \omega m_1 + \omega^2 m_2, \quad (78)$$

and

$$\mathcal{L}(m) := \Sigma(m)^2 - |Z(m)|^2. \quad (79)$$

Expansion using $1 + \omega + \omega^2 = 0$ gives

$$|Z(m)|^2 = m_0^2 + m_1^2 + m_2^2 - m_0 m_1 - m_1 m_2 - m_2 m_0 \quad (80)$$

and therefore

$$\mathcal{L}(m) = 3(m_0 m_1 + m_1 m_2 + m_2 m_0). \quad (81)$$

In particular, $m \in \mathbb{R}_{\geq 0}^3$ is future causal: $\Sigma(m) \geq 0$ and $\mathcal{L}(m) \geq 0$. Both quantities are invariant under D_3 , because that group permutes the three coordinates.

For the Fable vector $m_F = (1, 1, 1)$, one obtains

$$\Sigma(m_F) = 3, \quad Z(m_F) = 0, \quad \mathcal{L}(m_F) = 9. \quad (82)$$

Thus the Fable fibre is occupied, balanced, and future timelike in the unscaled Wick-completed coordinate.

Theorem 8.8 (Fourier, Wick, and Descartes compatibility). *Let $m = (m_0, m_1, m_2)$ and identify $m_+ = m_1$, $m_- = m_2$ in theorem 8.6. If*

$$\Lambda(m) = (S, x, y),$$

then

$$Z(m) = y + i \frac{\sqrt{3}}{2} x, \quad \mathcal{L}(m) = q(\Lambda(m)). \quad (83)$$

For the Euclidean polynomial

$$\mathcal{E}(S, X, Y) = S^2 + X^2 + Y^2,$$

the algebraic Wick substitution in the two phase coordinates satisfies

$$\mathcal{E}(S, iX, iY) = S^2 - X^2 - Y^2. \quad (84)$$

Define the Descartes form of Sarnak [23] by

$$F_D(b_1, b_2, b_3, b_4) = 2 \sum_{j=1}^4 b_j^2 - \left(\sum_{j=1}^4 b_j \right)^2. \quad (85)$$

Let

$$\begin{aligned} e_T &= \frac{1}{2}(1, 1, 1, 1), & e_X &= \frac{1}{\sqrt{2}}(1, -1, 0, 0), \\ e_Y &= \frac{1}{2}(1, 1, -1, -1), & e_W &= \frac{1}{\sqrt{2}}(0, 0, 1, -1). \end{aligned}$$

Then

$$F_D(Te_T + Xe_X + Ye_Y + We_W) = 2(X^2 + Y^2 + W^2 - T^2). \quad (86)$$

The linear embedding

$$\iota(t, x, y) = te_T + \frac{\sqrt{3}}{2}xe_X + ye_Y \quad (87)$$

satisfies

$$F_D(\iota(t, x, y)) = -2q(t, x, y). \quad (88)$$

Thus the three primitive C_1 -oriented Fable vectors map to the Descartes null cone, while every nonzero nonnegative combination has positive $T = t$ and nonpositive Descartes norm.

Proof. From (74),

$$\Lambda(m) = \left(m_0 + m_1 + m_2, m_1 - m_2, m_0 - \frac{1}{2}(m_1 + m_2) \right).$$

Since $\omega = -1/2 + i\sqrt{3}/2$,

$$\begin{aligned} Z(m) &= m_0 + \omega m_1 + \omega^2 m_2 \\ &= m_0 - \frac{1}{2}(m_1 + m_2) + i\frac{\sqrt{3}}{2}(m_1 - m_2), \end{aligned}$$

which proves the first identity in (83). Taking its squared modulus gives

$$|Z(m)|^2 = y^2 + \frac{3}{4}x^2,$$

so

$$\mathcal{L}(m) = S^2 - |Z(m)|^2 = S^2 - \frac{3}{4}x^2 - y^2 = q(\Lambda(m)).$$

Equation (84) follows by direct substitution in the polynomial $\mathcal{E}$. Throughout this construction, “Wick substitution” refers to the displayed formal polynomial operation.

For the Descartes form, the four displayed basis vectors are Euclidean orthonormal. The vector e_T spans the all-ones line, whereas e_X, e_Y, e_W have coordinate sum zero. Hence

$$F_D(e_T) = -2, \quad F_D(e_X) = F_D(e_Y) = F_D(e_W) = 2,$$

and all mixed terms vanish. This proves (86). Setting

$$T = t, \quad X = \frac{\sqrt{3}}{2}x, \quad Y = y, \quad W = 0$$

there gives

$$F_D(\iota(t, x, y)) = 2 \left(\frac{3}{4}x^2 + y^2 - t^2 \right) = -2q(t, x, y),$$

which is (88). The final assertions follow from theorem 8.6. $\square$

Remark 8.9 (Status of the fourth Descartes coordinate). The embedding (87) is an exact isometric identification of the three-dimensional support space with the hyperplane $W = 0$ in the four-dimensional Descartes space. The additional spacelike coordinate W is therefore mathematically present in the ambient Apollonian geometry. Its exact residual-shell interpretation is proved in theorem 9.2.

Provenance record 8.10 (Sarnak source). The Descartes form (85) is equation (1) of Peter Sarnak's *Letter to J. Lagarias about Integral Apollonian Packings*, June 2007 [23]. The orthogonal basis and pullback (88) are proved above.

Theorem 8.11 (Exact inversive-circle–Sarnak isometry). *For $v = (a, b_0, b_1, c)$ define*

$$T = \frac{a+c}{2}, \quad X = b_0, \quad Y = b_1, \quad W = \frac{a-c}{2} \quad (89)$$

and

$$\Psi_{\text{circ}}(v) = Te_T + Xe_X + Ye_Y + We_W. \quad (90)$$

Then

$$F_D(\Psi_{\text{circ}}(v)) = 2\mathcal{N}(v). \quad (91)$$

If $a > 0$ and $\mathcal{N}(v) = 0$, the coefficient-space circle equation is the point-circle

$$q_v(x, y) = a \left[\left(x + \frac{b_0}{a} \right)^2 + \left(y - \frac{b_1}{a} \right)^2 \right] = 0 \quad (92)$$

with center $(-b_0/a, b_1/a)$.

Proof. The change of coordinates gives

$$ac = (T+W)(T-W) = T^2 - W^2.$$

Consequently

$$\mathcal{N}(v) = b_0^2 + b_1^2 - ac = X^2 + Y^2 + W^2 - T^2.$$

Equation (91) now follows from (86). If $\mathcal{N}(v) = 0$, then $c = (b_0^2 + b_1^2)/a$. Completing the two squares in (28) proves (92). $\square$

Corollary 8.12 (The Fable rays as three coefficient-space point-circles). *The coefficient lift*

$$\widehat{\iota}(t, x, y) = \left(t, \frac{\sqrt{3}}{2}x, y, t \right) \quad (93)$$

satisfies

$$\Psi_{\text{circ}} \circ \widehat{\iota} = \iota, \quad \mathcal{N}(\widehat{\iota}(t, x, y)) = -q(t, x, y).$$

The three primitive Fable null rays therefore give

$$\begin{aligned} v_0 &= (1, 0, 1, 1), \\ v_+ &= \left(1, \frac{\sqrt{3}}{2}, -\frac{1}{2}, 1 \right), \\ v_- &= \left(1, -\frac{\sqrt{3}}{2}, -\frac{1}{2}, 1 \right). \end{aligned} \quad (94)$$

They are the point-circles at

$$P_0 = (0, 1), \quad P_+ = \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2} \right), \quad P_- = \left(\frac{\sqrt{3}}{2}, -\frac{1}{2} \right). \quad (95)$$

Writing $\omega = e^{2\pi i/3}$, their complex coordinates are $i, i\omega, i\omega^2$. Thus they form one exact C_3 orbit on the unit circle, while reflection in the vertical axis fixes P_0 and exchanges P_+ and P_- .

Proof. Substitution of (93) into (89) gives $T = t$, $X = \sqrt{3}x/2$, $Y = y$, and $W = 0$; this proves the first two identities. Apply the lift to the three vectors in (74), and then apply (92). The resulting points are exactly (95). Direct multiplication by ω and reflection in the vertical axis prove the orbit and reflection statements. $\square$

Theorem 8.13 (Kocik's equal-angle family in Sarnak coordinates). *Let $p \in \mathbb{R} \setminus \{-1, 1/3\}$. Kocik's general circle formula for four circles having common pairwise configuration product p is*

$$\sum_{j=1}^4 \beta_j^2 = \frac{p}{3p-1} \left(\sum_{j=1}^4 \beta_j \right)^2 \quad (96)$$

[24]. If $\beta = Te_T + Xe_X + Ye_Y + We_W$, then

$$X^2 + Y^2 + W^2 = \frac{p+1}{3p-1} T^2. \quad (97)$$

In particular, $p = 1$ gives the inversive-circle–Sarnak–Descartes null cone, whereas Kocik's Koide specialization $p = 2/3$ gives

$$X^2 + Y^2 + W^2 = \frac{5}{3} T^2. \quad (98)$$

A nonzero Fable null ray under (93) lies on the $p = 1$ member. The Koide member $p = 2/3$ has coefficient $5/3$.

Proof. The basis (86) is Euclidean orthonormal, and only e_T has nonzero coordinate sum. Hence

$$\sum_{j=1}^4 \beta_j^2 = T^2 + X^2 + Y^2 + W^2, \quad \sum_{j=1}^4 \beta_j = 2T.$$

Substitution in (96) and rearrangement give (97). The specializations are direct. A nonzero Fable ray has $X^2 + Y^2 + W^2 = T^2$ and $T > 0$. Since $1 \neq 5/3$, it fails (98). $\square$

Provenance record 8.14 (Kocik source). Equation (96) is equation (5) of Jerzy Kocik's *The Koide Lepton Mass Formula and Geometry of Circles* [24]. The orthogonal-coordinate calculation (97) and its comparison with the coefficient-space and Fable forms are proved above.

The three marked projective fractions themselves are read directly from the linear factors:

$$V = 0 \longleftrightarrow \infty, \quad U - \frac{1}{2}V = 0 \longleftrightarrow \frac{1}{2}, \quad U + \frac{1}{2}V = 0 \longleftrightarrow -\frac{1}{2}. \quad (99)$$

The coefficient $-1/4$ is both the product of the two finite roots and the constant coefficient of the common cubic in these coordinates.

9. FIXED RESIDUAL SHELLS IN THE SINGULAR SUPPORT OBJECT

Let p be an odd prime and let $R \in \mathbb{Z}_{>0}$ satisfy

$$R \equiv -p \pmod{4}, \quad a := \frac{p+R}{4} \in \mathbb{Z}_{>0}.$$

Put $N := pa$ and assume $\gcd(R, N) = 1$.

Theorem 9.1 (Exact divisor-shell equivalence). *For $b, c \in \mathbb{Z}_{>0}$, the identity*

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \quad (100)$$

is equivalent to the existence of a positive divisor $d \mid N^2$ satisfying

$$d \equiv -N \pmod{R}. \quad (101)$$

The two directions are related by

$$d = Rb - N, \quad \frac{N^2}{d} = Rc - N, \quad (102)$$

and

$$b = \frac{N+d}{R}, \quad c = \frac{N+N^2/d}{R}. \quad (103)$$

Proof. Since $4a = p + R$,

$$\frac{4}{p} - \frac{1}{a} = \frac{4a - p}{pa} = \frac{R}{N}.$$

Thus (100) is equivalent to

$$\frac{R}{N} = \frac{1}{b} + \frac{1}{c}. \quad (104)$$

Suppose first that (104) holds. Then $Rbc = N(b + c)$, and hence

$$\begin{aligned} (Rb - N)(Rc - N) &= R^2bc - RN(b + c) + N^2 \\ &= R(Rbc - N(b + c)) + N^2 = N^2. \end{aligned}$$

Therefore $d := Rb - N$ divides N^2 and satisfies $d \equiv -N \pmod{R}$. Both factors in (102) are positive. Indeed, if both were negative, then $b < N/R$ and $c < N/R$, implying $1/b + 1/c > 2R/N$, contrary to (104).

Conversely, let $d > 0$ satisfy (101) and put $e = N^2/d$. Reducing $de = N^2$ modulo R gives

$$(-N)e \equiv N^2 \pmod{R}.$$

Since N is invertible modulo R , this implies $e \equiv -N \pmod{R}$. Consequently the two expressions in (103) are positive integers. They satisfy $Rb - N = d$, $Rc - N = e$, and $de = N^2$. Expanding the last equality and cancelling N^2 gives $Rbc = N(b + c)$, hence (104) and then (100). $\square$

Theorem 9.2 (Fixed Erdős–Straus shells as coefficient-space null point-circles). *For positive integers b, c , put*

$$d = Rb - N, \quad e = Rc - N, \quad v_{b,c} := (d, N, 0, e). \quad (105)$$

Then

$$\mathcal{N}(v_{b,c}) = R(N(b + c) - Rbc). \quad (106)$$

Consequently

$$\frac{R}{N} = \frac{1}{b} + \frac{1}{c} \iff \mathcal{N}(v_{b,c}) = 0. \quad (107)$$

On this null locus, $d, e > 0$, $de = N^2$, and $v_{b,c}$ is the point-circle

$$q_{v_{b,c}}(x, y) = d \left[\left(x + \frac{N}{d} \right)^2 + y^2 \right] = 0. \quad (108)$$

Denominator exchange is the unit-circle coefficient inversion

$$(d, N, 0, e) \mapsto (e, N, 0, d). \quad (109)$$

In the coordinates (89),

$$T = \frac{d + e}{2}, \quad X = N, \quad Y = 0, \quad W = \frac{d - e}{2}, \quad (110)$$

Define $s = b/(b + c)$ and $\zeta = -s(1 - s)$. Then

$$\boxed{\zeta = -\frac{1}{4} + \frac{1}{4} \left(\frac{W}{T + X} \right)^2}. \quad (111)$$

Thus the fourth Sarnak coordinate is exactly the signed divisor-pair asymmetry, and the branch section is $W = 0$.

Proof. Using (105),

$$\begin{aligned}\mathcal{N}(v_{b,c}) &= N^2 - (Rb - N)(Rc - N) \\ &= RN(b + c) - R^2bc,\end{aligned}$$

which proves (106) and (107). Positivity and $de = N^2$ on the shell were proved in theorem 9.1. Completing the square gives (108). Under denominator exchange, d and e are exchanged, so (109) is precisely the coefficient reversal of the unit-circle inversion corollary. The two point coordinates multiply to

$$\left(-\frac{N}{d}\right)\left(-\frac{N}{e}\right) = 1,$$

confirming the same inversion directly.

Equation (110) follows by substitution in (89); its null identity is

$$X^2 + W^2 - T^2 = N^2 - de = 0.$$

Finally,

$$b - c = \frac{d - e}{R} = \frac{2W}{R}, \quad b + c = \frac{2N + d + e}{R} = \frac{2(T + X)}{R}.$$

Therefore $(b - c)/(b + c) = W/(T + X)$. The elementary identity

$$-\frac{bc}{(b + c)^2} = -\frac{1}{4} + \frac{1}{4}\left(\frac{b - c}{b + c}\right)^2$$

proves (111). □

Theorem 9.3 (The complete shell aggregate is timelike in the Fable slice). *Let*

$$\mathcal{D}_{R,N} = \{d > 0 : d \mid N^2, d \equiv -N \pmod{R}\}, \quad M = |\mathcal{D}_{R,N}|,$$

and put

$$V(\mathcal{D}_{R,N}) = \sum_{d \in \mathcal{D}_{R,N}} \left(d, N, 0, \frac{N^2}{d}\right). \quad (112)$$

If $S = \sum_{d \in \mathcal{D}_{R,N}} d$, then

$$V(\mathcal{D}_{R,N}) = (S, MN, 0, S), \quad (113)$$

and its Sarnak coordinates are

$$(T, X, Y, W) = (S, MN, 0, 0). \quad (114)$$

Consequently

$$\mathcal{N}(V(\mathcal{D}_{R,N})) = (MN)^2 - S^2 \leq 0. \quad (115)$$

If $M > 0$, $\gcd(R, N) = 1$, and $R > 2$, the inequality is strict.

Proof. The map $d \mapsto N^2/d$ is an involution of $\mathcal{D}_{R,N}$. Indeed, if $d \equiv -N \pmod{R}$, then reduction of $d(N^2/d) = N^2$ modulo R and cancellation of the unit N give $N^2/d \equiv -N \pmod{R}$. Hence

$$\sum_{d \in \mathcal{D}_{R,N}} \frac{N^2}{d} = \sum_{d \in \mathcal{D}_{R,N}} d = S,$$

which proves (113); (114) follows from (89). Substitution in the coefficient-space norm gives (115) once $S \geq MN$ is known.

Every two-element inversion orbit contributes

$$d + \frac{N^2}{d} \geq 2N,$$

and a fixed orbit contributes N . Summing over the orbits gives $S \geq MN$. Equality holds exactly when every element is fixed, hence equal to N . Such an element would satisfy $N \equiv -N \pmod{R}$,

so $R \mid 2N$. When $\gcd(R, N) = 1$ and $R > 2$, this is impossible. Therefore a nonempty certificate set gives $S > MN$ and strict negativity in (115). $\square$

Proposition 9.4 (Conjugacy of the Erdős–Straus and Fable reflections). *In Sarnak coordinates let*

$$s_W(T, X, Y, W) = (T, X, Y, -W), \quad s_X(T, X, Y, W) = (T, -X, Y, W),$$

and define

$$A(T, X, Y, W) = (T, W, X, Y). \quad (116)$$

Then A preserves $X^2 + Y^2 + W^2 - T^2$, has order three, and

$$As_W A^{-1} = s_X. \quad (117)$$

The geometric Erdős–Straus branch ray $d = e = N$ has projective Sarnak coordinates $(1, 1, 0, 0)$, while the fixed Fable ray v_0 has coordinates $(1, 0, 1, 0)$, and

$$A(1, 1, 0, 0) = (1, 0, 1, 0). \quad (118)$$

Proof. The map A fixes T and cyclically permutes the three spacelike coordinates. It therefore preserves the displayed quadratic form and satisfies $A^3 = I$. Since A sends the W -axis to the X -axis, conjugating the reflection normal to the first axis gives the reflection normal to the second, which is (117). The two ray coordinates follow respectively from (110) with $d = e = N$ and from (94); direct substitution gives (118). $\square$

Theorem 9.5 (Transport of the Fable face action to the Erdős–Straus Lorentz slice). *In the ordered Sarnak coordinates (T, X, Y, W) , let*

$$R_{XY} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

This is the 120° rotation that cycles the three Fable rays. For the spatial three-cycle A in (116), put

$$R_{\text{face}} := A^{-1} R_{XY} A.$$

Then

$$R_{\text{face}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} \\ 0 & 0 & 1 & 0 \\ 0 & \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \end{pmatrix}, \quad A^{-1} s_X A = s_W = \text{diag}(1, 1, 1, -1). \quad (119)$$

These two transformations satisfy

$$R_{\text{face}}^3 = s_W^2 = I, \quad s_W R_{\text{face}} s_W = R_{\text{face}}^{-1}. \quad (120)$$

Write the three Fable rays in Sarnak coordinates as

$$u_0 = (1, 0, 1, 0), \quad u_+ = \left(1, \frac{\sqrt{3}}{2}, -\frac{1}{2}, 0\right), \quad u_- = \left(1, -\frac{\sqrt{3}}{2}, -\frac{1}{2}, 0\right).$$

Their inverse images under A are

$$\begin{aligned} e_0 &= A^{-1} u_0 = (1, 1, 0, 0), \\ e_+ &= A^{-1} u_+ = \left(1, -\frac{1}{2}, 0, \frac{\sqrt{3}}{2}\right), \\ e_- &= A^{-1} u_- = \left(1, -\frac{1}{2}, 0, -\frac{\sqrt{3}}{2}\right). \end{aligned} \quad (121)$$

Each vector is null for

$$q_L(T, X, Y, W) = X^2 + Y^2 + W^2 - T^2$$

and lies on the affine section $T = 1$, $Y = 0$. If $p_a = (X_a, W_a)$ is the spatial point of e_a , then

$$\langle p_a, p_a \rangle = 1, \quad \langle p_a, p_b \rangle = -\frac{1}{2} \quad (a \neq b). \quad (122)$$

Thus the transported face is a regular triangle on the projective null circle, and

$$e_0 \xrightarrow{R_{\text{face}}} e_+ \xrightarrow{R_{\text{face}}} e_- \xrightarrow{R_{\text{face}}} e_0, \quad s_W(e_0) = e_0, \quad s_W(e_+) = e_-. \quad (123)$$

The fixed ray e_0 is exactly the geometric Erdős–Straus branch ray $d = e = N$. Arithmetic membership is established here for e_0 ; e_+ and e_- are the two Lorentz companions supplied by the transported face action.

Proof. The inverse coordinate cycle is

$$A^{-1}(T, X, Y, W) = (T, Y, W, X).$$

Multiplication gives both matrices in (119); the reflection identity is also the inverse form of (117). The planar rotation and reflection satisfy

$$R_{XY}^3 = s_X^2 = I, \quad s_X R_{XY} s_X = R_{XY}^{-1}.$$

Conjugating these identities by A^{-1} proves (120).

Applying the displayed formula for A^{-1} to u_0, u_+, u_- gives (121). Direct substitution gives

$$q_L(e_0) = q_L(e_+) = q_L(e_-) = 0.$$

Their XW -parts have squared length one and pairwise inner product $-1/2$, which proves (122). Conjugating the Fable ray cycle and reflection gives (123). Finally, (118) identifies e_0 with the branch ray $d = e = N$. $\square$

Exact computation 9.6 (Exact check of the Fable-to-Erdős–Straus face transport). The generator, checker, and captured receipt

```
verification/make_fable_es_face_transport_figure.py,
verification/verify_fable_es_face_transport.py,
verification/verify_fable_es_face_transport_output.txt
```

verify the two conjugated matrices, the D_3 relations, all three transported rays, their Lorentz nullity, the regular-triangle Gram data, and the complete cycle and reflection action in theorem 9.5. The maintained vector image is `figures/fable_es_face_transport.pdf`.

The Fable face action transported to denominator exchange

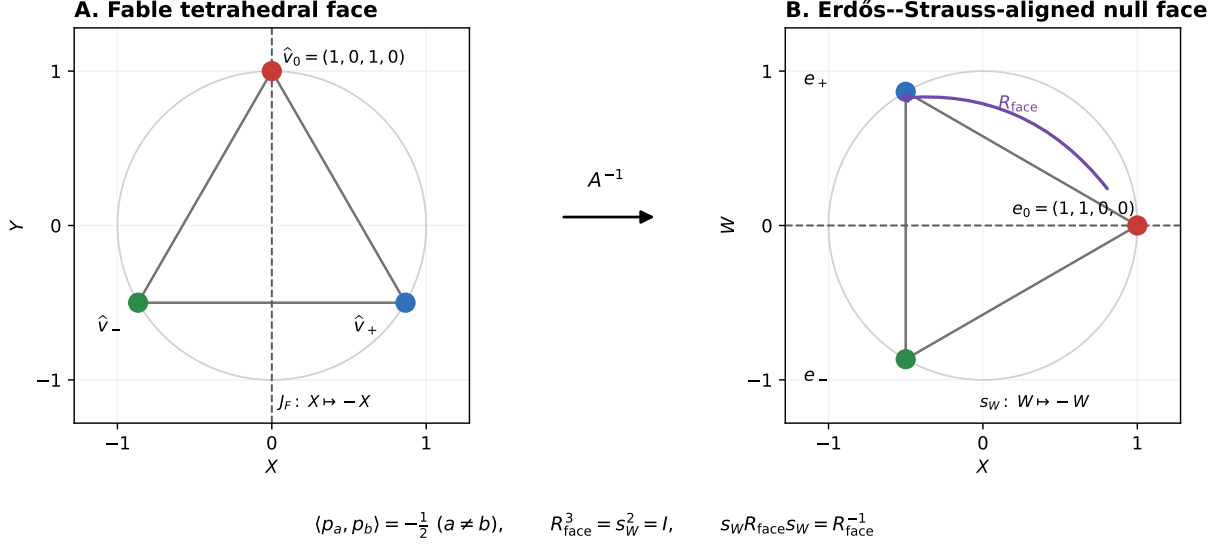


FIGURE 1. **Exact transport of the Fable tetrahedral face to the Erdős–Straus Lorentz slice.** The left panel displays the three Fable rays on the XY unit circle. The map A^{-1} carries them to the XW unit circle in the right panel, preserving the vertex colors and every pairwise inner product. The dashed axis on the right is $W = 0$; reflection across it is precisely denominator exchange s_W . The red fixed point is the Erdős–Straus branch ray $d = e = N$, while the curved arrow shows the transported C_3 face rotation.

Theorem 9.7 (The denominator-exchange quotient and its branch value). *Assume $\gcd(R, N) = 1$, and let (b, c) be a positive solution of (104). Put*

$$s := \frac{b}{b+c}, \quad \zeta := -s(1-s). \quad (124)$$

Then exchange of the two denominators acts by $s \mapsto 1-s$, and

$$\zeta = -\frac{bc}{(b+c)^2} = -\frac{\lambda}{(1+\lambda)^2}, \quad \lambda := \frac{Rb-N}{N}. \quad (125)$$

Over a field k of characteristic different from 2,

$$k[s]^{\langle s \mapsto 1-s \rangle} = k[s(1-s)] = k[\zeta], \quad (126)$$

and s is integral over $k[\zeta]$ with equation

$$s^2 - s - \zeta = 0. \quad (127)$$

Consequently the affine two-sheeted quotient has the unique finite branch value

$$\boxed{\zeta = -\frac{1}{4}}. \quad (128)$$

Equivalently, for $r = (b-c)/(b+c)$,

$$\zeta = -\frac{1}{4} + \frac{r^2}{4}. \quad (129)$$

If $R > 2$, this branch point is absent from the primitive arithmetic certificate set.

Proof. The transposition $(b, c) \mapsto (c, b)$ sends $s = b/(b+c)$ to $1-s$. The first equality in (125) follows directly from (124).

Set $d = Rb - N$ and $e = Rc - N$. By (102), $de = N^2$. Thus, with $\lambda = d/N$, one has $e/N = \lambda^{-1}$, and

$$b = \frac{N+d}{R} = \frac{N(1+\lambda)}{R}, \quad c = \frac{N+e}{R} = \frac{N(1+\lambda^{-1})}{R}.$$

It follows that

$$s = \frac{b}{b+c} = \frac{\lambda}{1+\lambda},$$

which proves the second equality in (125).

Write $u = s - 1/2$. The involution $s \mapsto 1-s$ is $u \mapsto -u$. Hence its invariant polynomial ring is $k[u^2]$. Since

$$u^2 = \frac{1}{4} - s(1-s) = \frac{1}{4} + \zeta,$$

translation by $1/4$ gives (126). Rearranging the definition of ζ gives (127), whose discriminant is $1 + 4\zeta$. It vanishes only at $\zeta = -1/4$, proving (128). Since $r = 2s - 1 = 2u$, the same identity gives (129).

For the final assertion, use the divisor-pair reconstruction

$$N = XYZ, \quad \gcd(X, Y) = 1, \quad R \mid X + Y,$$

under which $d = X^2Z$, $e = Y^2Z$, and

$$s = \frac{X}{X+Y}.$$

The fixed point $s = 1/2$ forces $X = Y$. Coprimality then forces $X = Y = 1$, while the support congruence forces $R \mid 2$. Therefore no primitive certificate has the branch value when $R > 2$. $\square$

Corollary 9.8 (Exact involutive bridge to the Fable fibre). *The involution on the complete Fable fibre has one fixed point p_0 and one transposed pair $\{p_+, p_-\}$. On the marked binary cubic it is induced by*

$$\kappa(U, V) = (-U, V),$$

which fixes the factor V and exchanges the other two marked factors with their resultant-one signs. The common Fable output satisfies

$$F_1(p_0) = F_1(p_+) = F_1(p_-) = -\frac{1}{4},$$

equal to the branch value in (128).

The distinguished scalar and the involution type agree. For $R > 2$, theorem 9.7 excludes the corresponding arithmetic fixed point from the primitive Erdős–Straus certificate set.

Proof. The action on the three fibre points is theorem 7.3. Substitution into the marked factors of theorem 7.5 gives

$$V \mapsto V, \quad U - \frac{1}{2}V \mapsto -\left(U + \frac{1}{2}V\right),$$

and the second application reverses this exchange. The common value is (16); the arithmetic branch value is (128). The final statement is exactly the last assertion of theorem 9.7. $\square$

9.1. Directed blade transport and the quarter-radius inversion. The completed blade calculation for the 107-residual shell supplies the four coefficient polynomials

$$B_1^+ = [0] + [10] + [28] + [78] + [96], \quad B_1^- = [34] + [44] + [62] + [72],$$

and

$$L_{34}^+ = [19] + [87], \quad L_{34}^- = [7] + [99]$$

in the group ring of C_{106} [17]. Order the two scalar signs and the two primitive-leg signs as $+$, $-$, and define the residue-resolved transport matrix

$$\mathcal{M}(t) := \begin{pmatrix} [t](B_1^+ L_{34}^+) & [t](B_1^+ L_{34}^-) \\ [t](B_1^- L_{34}^+) & [t](B_1^- L_{34}^-) \end{pmatrix}. \quad (130)$$

Proposition 9.9 (Directed target coefficient). *The two target residues satisfy*

$$\mathcal{M}(53) = \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix} = 2E_{21}, \quad \mathcal{M}(60) = 0. \quad (131)$$

The two copies of the $(2, 1)$ -entry are the ordered tag decompositions

$$53 = 34 + 19 = 72 + 87 \pmod{106}. \quad (132)$$

The transposed $(1, 2)$ -entry has coefficient zero at residue 53. The endpoint-support calculation in theorem 14.69 identifies the displayed $(2, 1)$ -entry with the outgoing cell C_{0+} of the 107-shell, and the zero $(1, 2)$ -entry with its incoming cell C_{-0} . Equivalently, theorem 14.71 identifies the nonzero E_{21} entry as an exit and E_{12} as an entry.

Proof. The four products have supports

$$\begin{aligned} B_1^+ L_{34}^+ &= 2[9] + [19] + [29] + [47] + [59] + [77] + [87] + 2[97], \\ B_1^+ L_{34}^- &= [3] + [7] + [17] + [21] + [35] + [71] + [85] + [89] + [99] + [103], \\ B_1^- L_{34}^+ &= [15] + [25] + [43] + 2[53] + [63] + [81] + [91], \\ B_1^- L_{34}^- &= [27] + [37] + [41] + [51] + [55] + [65] + [69] + [79]. \end{aligned}$$

The four displayed supports are pairwise disjoint. Only the third contains [53], with coefficient two, and none contains [60]. Its two target terms are $34 + 19$ and $72 + 87$, giving (132) and the stated matrix. $\square$

For a coefficient vector $v = (a, b_0, b_1, c)$, write

$$H(v) := \begin{pmatrix} a & b_0 + ib_1 \\ b_0 - ib_1 & c \end{pmatrix}. \quad (133)$$

Then

$$-\det H(v) = b_0^2 + b_1^2 - ac = \mathcal{N}(v).$$

Proposition 9.10 (Hermitian circle coordinates and the $\mathrm{SO}^+(3, 1)$ action). *For $z = x + iy$, put*

$$\xi_z = \begin{pmatrix} x - iy \\ 1 \end{pmatrix}.$$

Then

$$\xi_z^\dagger H(v) \xi_z = q_v(x, y). \quad (134)$$

For $P \in \mathrm{GL}_2(\mathbb{C})$, define

$$\mathcal{A}_P(H) := (P^{-1})^\dagger H P^{-1}. \quad (135)$$

This is the coefficient transformation of the generalized circle under the Möbius transformation represented by P , and

$$-\det \mathcal{A}_P(H) = \frac{-\det H}{|\det P|^2}. \quad (136)$$

Consequently the restriction $P \in \mathrm{SL}_2(\mathbb{C})$ induces a real-linear action on $\mathrm{Herm}_2(\mathbb{C})$ preserving

$$-\det H = X^2 + Y^2 + W^2 - T^2.$$

Its image belongs to the identity component $\mathrm{SO}^+(3, 1)$. The same matrices preserve the sign-reversed form $T^2 - X^2 - Y^2 - W^2$; for that displayed form the image belongs to $\mathrm{SO}^+(1, 3)$.

Proof. Expanding the left side of (134) gives

$$a(x^2 + y^2) + 2b_0x - 2b_1y + c,$$

which is $q_v(x, y)$. If a homogeneous point ξ is sent to $P\xi$, then

$$\xi^\dagger H \xi = (P\xi)^\dagger (P^{-1})^\dagger H P^{-1} (P\xi),$$

proving (135). Taking determinants gives

$$\det \mathcal{A}_P(H) = \overline{\det(P^{-1})} \det(H) \det(P^{-1}) = \frac{\det H}{|\det P|^2},$$

which proves (136). For $\det P = 1$, $-\det H$ is invariant. The group $\mathrm{SL}_2(\mathbb{C})$ is connected, the action is continuous, and the identity acts as the identity; hence the image lies in the identity component of the real Lorentz group. The generalized-circle congruence law is Theorem 1.52 of [25]; the determinant and component statements are the displayed calculations. $\square$

Theorem 9.11 (Directed idempotent support fixes the quarter). *Let*

$$N = 2E_{21}, \quad P_s = N^\top N = 4E_{11}, \quad P_r = NN^\top = 4E_{22}.$$

There is a unique nonzero scalar μ for which μP_r is idempotent, namely

$$\mu = \frac{1}{4}, \quad Q_r := \mu P_r = E_{22}. \quad (137)$$

Likewise, $Q_s := \frac{1}{4}P_s = E_{11}$ is the unique nonzero idempotent scalar multiple of P_s . The Moore–Penrose inverse and the two support projectors are

$$N^+ = \frac{1}{2}E_{12}, \quad Q_r = NN^+, \quad Q_s = N^+N. \quad (138)$$

Put

$$w_N := (4, 0, 0, -1), \quad w_N^\vee := (-1, 0, 0, 4).$$

Retaining the ordered roles of source multiplicity and target support gives

$$\mathfrak{H}(N) := P_s - Q_r = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix} = H(w_N), \quad (139)$$

$$\mathfrak{H}(N^\top) := P_r - Q_s = \begin{pmatrix} -1 & 0 \\ 0 & 4 \end{pmatrix} = H(w_N^\vee). \quad (140)$$

Their inversive-circle coordinates and Sarnak coordinates are

$$(A, B_0, B_1, C)_N = (4, 0, 0, -1), \quad (T, X, Y, W)_N = \left(\frac{3}{2}, 0, 0, \frac{5}{2}\right), \quad (141)$$

$$(A, B_0, B_1, C)_{N^\top} = (-1, 0, 0, 4), \quad (T, X, Y, W)_{N^\top} = \left(\frac{3}{2}, 0, 0, -\frac{5}{2}\right). \quad (142)$$

Both coefficient vectors have Lorentz norm 4 and Descartes norm 8. The corresponding coefficient-inversion transformations factor as

$$M_{w_N} = \frac{1}{4}R_{w_N}, \quad M_{w_N^\vee} = 4R_{w_N^\vee}. \quad (143)$$

Their simple eigenvalues are therefore

$$\frac{C_N}{A_N} = \frac{T_N - W_N}{T_N + W_N} = -\frac{1}{4}, \quad \frac{C_{N^\top}}{A_{N^\top}} = \frac{T_{N^\top} - W_{N^\top}}{T_{N^\top} + W_{N^\top}} = -4. \quad (144)$$

Coefficient reversal

$$\mathcal{C}(a, b_0, b_1, c) = (c, b_0, b_1, a)$$

exchanges w_N and $w_N^\vee$, fixes T, X, Y , and sends W to $-W$.

Proof. Since $P_r = 4E_{22}$, one has

$$P_r^2 = 16E_{22} = 4P_r.$$

For $\mu \neq 0$,

$$(\mu P_r)^2 = \mu P_r \iff 4\mu^2 P_r = \mu P_r \iff \mu = \frac{1}{4}.$$

The same calculation applies to $P_s = 4E_{11}$. Substitution gives (139) and (140). Direct multiplication by $N^+ = \frac{1}{2}E_{12}$ gives

$$NN^+N = N, \quad N^+NN^+ = N^+, \quad (NN^+)^\top = NN^+, \quad (N^+N)^\top = N^+N,$$

and proves (138).

The coordinate change

$$T = \frac{A+C}{2}, \quad X = B_0, \quad Y = B_1, \quad W = \frac{A-C}{2}$$

gives (141) and (142). In both cases

$$\mathcal{N} = B_0^2 + B_1^2 - AC = 4, \quad F_D = 2\mathcal{N} = 8.$$

The reflection factorization

$$M_w = \frac{\mathcal{N}(w)}{A^2}R_w$$

from (32) gives (143). For $B_0 = B_1 = 0$, $\mathcal{N}(w) = -AC$, so the eigenvalue on the normal line is

$$-\frac{\mathcal{N}(w)}{A^2} = \frac{C}{A} = \frac{T-W}{T+W}.$$

This proves (144). The final assertion follows by applying the displayed coefficient reversal to the two coefficient vectors and to the definitions of T, X, Y, W . $\square$

Theorem 9.12 (Exact restriction of the Fable cubic to the directed branch circle). *Let*

$$C(U, V) = U^2V - \frac{1}{4}V^3$$

be the common binary cubic of (59). For $N = 2E_{21}$, let

$$K_N := \mathfrak{H}(N) = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}, \quad K_{N^\top} := \mathfrak{H}(N^\top) = \begin{pmatrix} -1 & 0 \\ 0 & 4 \end{pmatrix}.$$

For $\mathbf{u} = (U, V)^\top$, the exact polynomial identities are

$$4C(U, V) = V \mathbf{u}^\top K_N \mathbf{u}, \quad (145)$$

$$4C(V, U) = U \mathbf{u}^\top K_{N^\top} \mathbf{u}. \quad (146)$$

The generalized circle represented by K_N is

$$4|z|^2 - 1 = 0. \quad (147)$$

Its intersection with the real projective line has the two finite points

$$\frac{U}{V} = \frac{1}{2}, \quad \frac{U}{V} = -\frac{1}{2},$$

which are exactly the two finite roots of $C(U, V)$. The factor V in (145) retains the third marked root $[U : V] = [1 : 0]$. The transposed circle has finite real intersections

$$\frac{U}{V} = 2, \quad \frac{U}{V} = -2.$$

Let

$$P = i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \in \mathrm{SL}_2(\mathbb{C}).$$

Then

$$\mathcal{A}_P(K_N) = K_{N^\top}. \quad (148)$$

The represented Möbius transformation is $z \mapsto 1/z$. In Sarnak coordinates its Lorentz action is

$$(T, X, Y, W) \mapsto (T, X, -Y, -W), \quad (149)$$

an element of $\mathrm{SO}^+(3, 1)$. It sends

$$\left(\frac{3}{2}, 0, 0, \frac{5}{2}\right) \quad \text{to} \quad \left(\frac{3}{2}, 0, 0, -\frac{5}{2}\right).$$

Proof. Direct multiplication gives

$$\mathbf{u}^\top K_N \mathbf{u} = 4U^2 - V^2,$$

and hence

$$V \mathbf{u}^\top K_N \mathbf{u} = 4U^2V - V^3 = 4C(U, V).$$

Likewise,

$$U \mathbf{u}^\top K_{N^\top} \mathbf{u} = U(-U^2 + 4V^2) = 4C(V, U).$$

This proves (145) and (146), including all three projective factors.

Equation (134) applied to K_N gives (147). Restriction to $y = 0$ gives $4x^2 - 1 = 0$, and the transposed form gives $-x^2 + 4 = 0$.

Put $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Since $P = iS$, one has $\det P = 1$, $P^{-1} = -P$, and

$$(P^{-1})^\dagger K_N P^{-1} = S K_N S = K_{N^\top},$$

proving (148). For a general coefficient matrix,

$$SH(a, b_0, b_1, c)S = H(c, b_0, -b_1, a).$$

Thus T and X are fixed while Y and W change sign, proving (149). The map has determinant 1, fixes the future coordinate T , and preserves $X^2 + Y^2 + W^2 - T^2$. The final two vectors are (141) and (142). $\square$

Theorem 9.13 (The ordered blade signs determine the fourth Lorentz coordinate). *Let*

$$e_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad J = e_+ e_+^\top - e_- e_-^\top = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Use the row and column order $+, -$ from (130). The observed transport is

$$N = 2E_{21} = 2e_- e_+^\top.$$

It satisfies the four exact sign identities

$$JN = -N, \quad NJ = N, \quad JN^\top = N^\top, \quad N^\top J = -N^\top. \quad (150)$$

Thus N has output sign $-$ and input sign $+$, whereas $N^\top$ has output sign $+$ and input sign $-$.

For a diagonal Hermitian coefficient matrix $K = \text{diag}(A, C)$, the Sarnak coordinates are recovered by

$$T(K) = \frac{1}{2} \text{tr } K, \quad W_J(K) = \frac{1}{2} \text{tr}(JK). \quad (151)$$

Consequently

$$(T(\mathfrak{H}(N)), W_J(\mathfrak{H}(N))) = \left(\frac{3}{2}, \frac{5}{2}\right), \quad (152)$$

$$(T(\mathfrak{H}(N^\top)), W_J(\mathfrak{H}(N^\top))) = \left(\frac{3}{2}, -\frac{5}{2}\right). \quad (153)$$

Let

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$SNS = N^\top, \quad SJS = -J, \quad S\mathfrak{H}(N)S = \mathfrak{H}(N^\top). \quad (154)$$

Hence simultaneous exchange of the two ordered sign axes fixes T and reverses W_J . Since the tag-53 coefficient is N , while its transposed matrix unit is absent from that coefficient, the exact blade calculation selects (152) in the stated ordering.

Proof. The relations $Je_+ = e_+$ and $Je_- = -e_-$ give

$$J(2e_- e_+^\top) = -2e_- e_+^\top, \quad (2e_- e_+^\top)J = 2e_- e_+^\top,$$

and transposition gives the remaining two identities in (150). Equations (139) and (140) give

$$\mathfrak{H}(N) = \text{diag}(4, -1), \quad \mathfrak{H}(N^\top) = \text{diag}(-1, 4).$$

Substitution in (151) proves (152) and (153). Direct multiplication proves the first two identities in (154); the third follows either by direct multiplication or from the definitions of the two Gram-support forms. For every diagonal K , cyclicity of trace gives

$$T(SKS) = T(K), \quad W_J(SKS) = \frac{1}{2} \text{tr}(SJSK) = -W_J(K).$$

The final statement now follows from (131). $\square$

Theorem 9.14 (Mixer projectors as the source and range supports of the ordered blade channels).

Let

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (155)$$

Then

$$H^\top H = I_2, \quad HJH^\top = S, \quad HSH^\top = J. \quad (156)$$

Put

$$u_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad u_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad P_{\pm} = u_{\pm} u_{\pm}^{\top} = \frac{1}{2} (I_2 \pm S). \quad (157)$$

Thus the representation of the half-period involution T given by $\rho(T) = S$ satisfies

$$\rho\left(\frac{1}{2}(I \pm T)\right) = P_{\pm}. \quad (158)$$

Conjugation by H sends the ordered raw blades to

$$B_{-+} := H E_{21} H^{\top} = u_- u_+^{\top} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}, \quad (159)$$

$$B_{+-} := H E_{12} H^{\top} = u_+ u_-^{\top} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}. \quad (160)$$

Their source and range projectors are

$$B_{-+}^{\top} B_{-+} = P_+, \quad B_{-+} B_{-+}^{\top} = P_-, \quad (161)$$

$$B_{+-}^{\top} B_{+-} = P_-, \quad B_{+-} B_{+-}^{\top} = P_+. \quad (162)$$

Equivalently,

$$P_- B_{-+} P_+ = B_{-+}, \quad P_+ B_{+-} P_- = B_{+-}. \quad (163)$$

The ordered signs are retained by

$$S B_{-+} = -B_{-+}, \quad B_{-+} S = B_{-+}, \quad (164)$$

$$S B_{+-} = B_{+-}, \quad B_{+-} S = -B_{+-}. \quad (165)$$

Consequently the observed blade $N = 2E_{21}$, identified with the outgoing exit-boundary cell by theorem 14.69, is carried to the directed channel $2B_{-+} : \text{Ran } P_+ \rightarrow \text{Ran } P_-$, while its transpose is carried to the reverse channel $2B_{+-} : \text{Ran } P_- \rightarrow \text{Ran } P_+$.

Proof. The columns of H are u_+ and u_- , which are orthonormal and satisfy $Su_+ = u_+$ and $Su_- = -u_-$. This proves (156) and (157). Equation (158) follows by linearity of ρ .

Since $E_{21} = e_- e_+^{\top}$ and $E_{12} = e_+ e_-^{\top}$, multiplication by H gives (159) and (160). Their source and range products reduce to $u_+ u_+^{\top}$ and $u_- u_-^{\top}$, proving (161)–(163). Finally, applying S to the output vector and input covector in each rank-one operator gives (164) and (165). Multiplication by the scalar two proves the statement for N and $N^{\top}$. $\square$

Exact computation 9.15 (Exact mixer–blade intertwining certificate). The script and captured receipt

verification/verify_mixer_blade_intertwining.py,

verification/verify_mixer_blade_intertwining_output.txt

verify every matrix identity in theorem 9.14 over the exact field $\mathbb{Q}(\sqrt{2})$.

Proposition 9.16 (Location of the Hadamard intertwiner in the globalization hierarchy). *Let $K \subseteq \mathbb{R}$ be a field containing $\sqrt{2}$, and put $S_K = G(K)$. With Φ_2 as in theorem 11.2, define*

$$\mathcal{H} := \text{Ad}_H \circ \text{pr}_1 \circ \Phi_2, \quad \text{Ad}_H(A) = H A H^{\top}, \quad \text{pr}_1(A, S) = A. \quad (166)$$

Then

$$\mathcal{H}(\tilde{N}) = 2B_{-+}. \quad (167)$$

There is no $U \in \text{GL}_2(S_K)$ satisfying $\hat{p}_K(U) = H$. Thus (166) has no realization as conjugation by an invertible split matrix.

The two support calculations occur in distinct codomains:

$$\hat{\chi}_K(\tilde{N}) = E_{21} \in M_2(\mathbb{B}), \quad B_{-+}^{\top} B_{-+} = P_+, \quad B_{-+} B_{-+}^{\top} = P_- \quad \text{in } M_2(K). \quad (168)$$

If $\text{nz}(A)_{ij} = 1$ exactly when $A_{ij} \neq 0$, then

$$\text{nz}(2B_{-+}) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad (169)$$

so the entry-incidence mask after the amplitude change of basis is also different from the retained globalization mask E_{21} .

Proof. Equation (502) gives $\Phi_2(\tilde{N}) = (2E_{21}, E_{21})$. Applying the first-coordinate projection and then Ad_H gives (167) by (159).

Suppose that $U \in \text{GL}_2(S_K)$, with inverse V . Applying $\hat{\chi}_K$ gives mutually inverse Boolean matrices

$$\hat{\chi}_K(U)\hat{\chi}_K(V) = I_2, \quad \hat{\chi}_K(V)\hat{\chi}_K(U) = I_2.$$

An automorphism of the Boolean lattice $\mathbb{B}^2$ permutes its two atoms, so $\hat{\chi}_K(U)$ is a permutation matrix. Consequently $\hat{p}_K(U)$ has one nonzero entry in each row and column. The matrix H has two nonzero entries in each row and column, which rules out $\hat{p}_K(U) = H$.

The first equality in (168) is part of the split pair of $\tilde{N}$; the remaining two were proved in theorem 9.14. Every entry of $2B_{-+}$ is nonzero, proving (169). $\square$

Exact computation 9.17 (Exact layer-location certificate). The script and captured receipt

```
verification/verify_Hadamard_globalization_layer.py,
verification/verify_Hadamard_globalization_layer_output.txt
```

verify the factorization value, the raw and transformed entry masks, both orthogonal support projectors, and the nonmonomial obstruction over $\mathbb{Q}(\sqrt{2})$.

Proposition 9.18 (Sheet-equivariant self-adjoint Hamiltonians). *Let $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and let $P_{\pm} = \frac{1}{2}(I_2 \pm S)$ be the orthogonal projectors in (157). For $K \in M_2(\mathbb{R})$, the following statements are equivalent:*

- (i) $K^T = K$ and $KS = SK$;
- (ii) there are unique $\kappa_+, \kappa_- \in \mathbb{R}$ such that

$$K = \kappa_+ P_+ + \kappa_- P_-; \quad (170)$$

- (iii) there are unique $a, b \in \mathbb{R}$ such that

$$K = \begin{pmatrix} a & b \\ b & a \end{pmatrix}, \quad \kappa_+ = a + b, \quad \kappa_- = a - b. \quad (171)$$

The operator K is positive semidefinite exactly when $\kappa_+, \kappa_- \geq 0$, and it is positive definite exactly when both inequalities are strict.

If Δ is positive definite and commutes with S , then unique $\delta_+, \delta_- > 0$ satisfy

$$\Delta = \delta_+ P_+ + \delta_- P_-; \quad (172)$$

For the explicitly defined logarithmic Hamiltonian

$$K_{\Delta} := -\log \Delta, \quad (173)$$

one has

$$K_{\Delta} = -(\log \delta_+) P_+ - (\log \delta_-) P_-, \quad e^{-K_{\Delta}} = \Delta. \quad (174)$$

Thus P_+ and P_- are the complete spectral projectors for every self-adjoint Hamiltonian respecting the two-sheet exchange.

Proof. Write $K = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$. The equation $KS = SK$ gives $x = w$ and $y = z$; these are also precisely the entries required by self-adjointness, proving the equivalence of (i) and (iii). Substituting $P_{\pm} = \frac{1}{2}(I_2 \pm S)$ gives

$$(a+b)P_+ + (a-b)P_- = aI_2 + bS = \begin{pmatrix} a & b \\ b & a \end{pmatrix},$$

which proves (ii) and uniqueness. The vectors u_+ and u_- form an orthonormal eigenbasis with eigenvalues κ_+, κ_- , proving both positivity criteria.

Apply the same eigenspace decomposition to Δ . Positive definiteness gives $\delta_{\pm} > 0$, so the real functional calculus gives (174). $\square$

Exact computation 9.19 (Sheet-Hamiltonian spectral certificate). The script and captured receipt

`verification/verify_sheet_hamiltonian_spectral_form.py,`
`verification/verify_sheet_hamiltonian_spectral_form_output.txt`

check the projector algebra, both eigenvalues, the commuting spectral form, and the logarithmic reconstruction.

Theorem 9.20 (Exact Lorentz table of the four raw blade cells). *For $\epsilon, \lambda \in \{+, -\}$, put*

$$m_{\epsilon\lambda}(t) := [t](B_1^{\epsilon}L_{34}^{\lambda}). \quad (175)$$

Whenever $m = m_{\epsilon\lambda}(t) > 0$, define

$$N_{\epsilon\lambda}(t) := m e_{\epsilon} e_{\lambda}^{\top}, \quad Q_s := e_{\lambda} e_{\lambda}^{\top}, \quad Q_r := e_{\epsilon} e_{\epsilon}^{\top}, \quad (176)$$

and retain the directed Gram-support order

$$\mathfrak{H}_{\epsilon\lambda}(t) := N_{\epsilon\lambda}(t)^{\top} N_{\epsilon\lambda}(t) - Q_r. \quad (177)$$

Then the four raw cells have the exact data

cell	completed sector	$\mathfrak{H}_{\epsilon\lambda}$	(T, W)	$\mathcal{N}$
$B_1^+ L_{34}^+$	e_+^+	$(m^2 - 1)E_{11}$	$\left(\frac{m^2-1}{2}, \frac{m^2-1}{2}\right)$	0
$B_1^+ L_{34}^-$	e_-^+	$m^2 E_{22} - E_{11}$	$\left(\frac{m^2-1}{2}, -\frac{m^2+1}{2}\right)$	m^2
$B_1^- L_{34}^+$	e_-^-	$m^2 E_{11} - E_{22}$	$\left(\frac{m^2-1}{2}, \frac{m^2+1}{2}\right)$	m^2
$B_1^- L_{34}^-$	e_+^-	$(m^2 - 1)E_{22}$	$\left(\frac{m^2-1}{2}, -\frac{m^2-1}{2}\right)$	0

(178)

In particular, the diagonal raw cells are null, and the two off-diagonal cells form the reciprocal pair exchanged by $W \mapsto -W$. All four entries have the same time coordinate

$$T = \frac{m^2 - 1}{2}.$$

To read the same table as support flow, put

$$\eta_+ := 1, \quad \eta_- := 0, \quad \nabla_{\epsilon\lambda} := \eta_{\epsilon} - \eta_{\lambda}.$$

Under the endpoint carriage of theorem 14.71, $\nabla_{\epsilon\lambda}$ is the discrete support derivative: +1 is entry, -1 is exit, and 0 is stationary. On the two off-diagonal cells,

$$W = -\nabla_{\epsilon\lambda} \frac{m^2 + 1}{2}. \quad (179)$$

For all four cells one has the exact norm identity

$$\mathcal{N} = \frac{1}{4} \|[J_B, N_{\epsilon\lambda}(t)]\|_{\text{HS}}^2 = \frac{1}{144} \|[D, N_{\epsilon\lambda}(t)]\|_{\text{HS}}^2. \quad (180)$$

Thus the Lorentz norm is the squared support-boundary commutator cost.

The map from the two factor signs to the completed signs is

$$\mathbf{u}_{\epsilon,+}\mathbf{u}_{\lambda,-} = \mathbf{u}_{\epsilon\lambda,-}. \quad (181)$$

Consequently the diagonal raw cells lie in e_+^+ , while the off-diagonal raw cells lie in e_-^- . The four raw cells map two-to-one onto these two completed sectors.

At the champion target,

$$\mathcal{M}(53) = 2E_{21} \implies \mathfrak{H}_{-+}(53) = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}, \quad (182)$$

with $(T, W) = (3/2, 5/2)$. Since the coefficient ring is graded by C_{106} , one also has

$$\mathcal{M}(109) = \mathcal{M}(3) = E_{12}, \implies \mathfrak{H}_{+-}(109) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (183)$$

with $(T, W) = (0, -1)$.

Each nonzero cell therefore determines a coefficient vector in the single Lorentz space with quadratic form

$$X^2 + Y^2 + W^2 - T^2.$$

The established cell data determine two null directions and an off-diagonal reciprocal pair. A bijection from the four raw cells to four orthogonal-group components requires an additional quadratic map; the directed Gram-support map gives (178).

Proof. If i is the row index of ϵ and j is the column index of λ , then

$$N_{\epsilon\lambda} = mE_{ij}, \quad N_{\epsilon\lambda}^T N_{\epsilon\lambda} = m^2 E_{jj}.$$

Equation (177) therefore gives

$$\mathfrak{H}_{\epsilon\lambda} = m^2 E_{jj} - E_{ii}.$$

Writing these four matrices in the order $(++)$, $(+-)$, $(-+)$, $(--)$ gives the third column of (178). For a diagonal matrix $\text{diag}(A, C)$,

$$T = \frac{A+C}{2}, \quad W = \frac{A-C}{2}, \quad \mathcal{N} = -AC.$$

Substitution gives the last two columns of the table. For $N_{\epsilon\lambda} = me_\epsilon e_\lambda^T$,

$$[J_B, N_{\epsilon\lambda}] = 2(\eta_\epsilon - \eta_\lambda)N_{\epsilon\lambda}, \quad [D, N_{\epsilon\lambda}] = 12(\eta_\epsilon - \eta_\lambda)N_{\epsilon\lambda}.$$

The Hilbert–Schmidt norm of every matrix unit is 1. The commutators therefore vanish on the two diagonal cells and have squared norms $4m^2$ and $144m^2$, respectively, on the two off-diagonal cells. Comparison with the last column of (178) proves (180). The two off-diagonal rows of that table give (179).

The scalar blade-one factors have lower sign $+$, and the primitive o_{34} -leg factors have lower sign $-$. Multiplication in $\Gamma_\square$ gives (181), including the stated two-to-one map.

Equation (182) follows from (131). For the second specialization, $[109] = [3]$ in C_{106} . The four explicit convolution supports in the proof of theorem 9.9 show that $[3]$ occurs once in $B_1^+ L_{34}^-$ and occurs in none of the other three products. Hence $\mathcal{M}(109) = E_{12}$, and (177) gives (183). $\square$

Theorem 9.21 (A phase-fixed Hermitian Lorentz lift of the ordered blade cells). *Let*

$$\mathcal{O} := \text{span}_{\mathbb{R}}\{E_{12}, E_{21}\}, \quad J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Define

$$\mathfrak{c}(A) := -JA^T, \quad A \in \mathcal{O}. \quad (184)$$

Then

$$\mathfrak{c}(E_{12}) = E_{21}, \quad \mathfrak{c}(E_{21}) = -E_{12}, \quad \mathfrak{c}^2 = -I_{\mathcal{O}}. \quad (185)$$

The operator $-\mathfrak{c}$ also squares to $-I_{\mathcal{O}}$. These two structures are exchanged by the lower-root automorphism $\iota \mapsto -\iota$. Hence the root data determine a conjugate pair of complex structures on the off-diagonal blade plane.

Let $\iota^2 = -1$, with conjugation $\bar{\iota} = -\iota$. The unique real-linear map

$$\chi : \mathcal{O} \longrightarrow \mathbb{R}[\iota]$$

satisfying

$$\chi(E_{12}) = 1, \quad \chi(\mathfrak{c}(A)) = \iota\chi(A)$$

is

$$\chi(bE_{12} + cE_{21}) = b + \iota c. \quad (186)$$

The condition $\chi(E_{12}) = 1$ fixes the complex phase. Replacing $(\mathfrak{c}, \iota)$ by $(-\mathfrak{c}, -\iota)$ gives the conjugate map $b + \iota c \mapsto b - \iota c$. For a raw blade-cell matrix

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

define its Hermitian lift by

$$\mathcal{H}_{\square}(M) := \begin{pmatrix} a & b + \iota c \\ b - \iota c & d \end{pmatrix}. \quad (187)$$

Then

$$-\det \mathcal{H}_{\square}(M) = b^2 + c^2 - ad. \quad (188)$$

The coordinate map

$$(a, b, c, d) \longmapsto (T, X, Y, W) = \left(\frac{a+d}{2}, b, c, \frac{a-d}{2} \right) \quad (189)$$

therefore satisfies

$$-\det \mathcal{H}_{\square}(M) = X^2 + Y^2 + W^2 - T^2. \quad (190)$$

The lower-root automorphism acts on these coordinates by

$$(T, X, Y, W) \longmapsto (T, X, -Y, W).$$

It fixes the quadratic form and its orthogonal-group orbit. The current data therefore fix the Lorentz form and leave the orientation of the Y -axis as a two-element ambiguity. Under this map the four raw cells are

$$\begin{aligned} E_{11} &\longmapsto \left(\frac{1}{2}, 0, 0, \frac{1}{2} \right), & E_{12} &\longmapsto (0, 1, 0, 0), \\ E_{21} &\longmapsto (0, 0, 1, 0), & E_{22} &\longmapsto \left(\frac{1}{2}, 0, 0, -\frac{1}{2} \right). \end{aligned} \quad (191)$$

Thus the diagonal cells are a null pair and the off-diagonal cells are two orthogonal unit spacelike vectors.

The two exact residue specializations are

$$\mathcal{H}_{\square}(\mathcal{M}(53)) = \begin{pmatrix} 0 & 2\iota \\ -2\iota & 0 \end{pmatrix} = H(0, 0, 2, 0), \quad (192)$$

$$\mathcal{H}_{\square}(\mathcal{M}(109)) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = H(0, 1, 0, 0). \quad (193)$$

Their Lorentz norms are 4 and 1, respectively.

Let

$$R_{YW}(T, X, Y, W) = (T, X, -W, Y)$$

and

$$B_{TW}(T, X, Y, W) = \left(\frac{5}{4}T + \frac{3}{4}W, X, Y, \frac{3}{4}T + \frac{5}{4}W \right).$$

The composite

$$L_{\square} := B_{TW} \circ R_{YW}$$

belongs to $\mathrm{SO}^+(3, 1)$ and satisfies

$$L_{\square}(0, 0, 2, 0) = \left(\frac{3}{2}, 0, 0, \frac{5}{2}\right). \quad (194)$$

Accordingly the Hermitian lift of the blade target and the directed Fable branch carrier belong to the same exact $\mathrm{SO}^+(3, 1)$ -orbit.

An explicit $\mathrm{SL}_2(\mathbb{R}[\iota])$ element realizing the same coefficient congruence is obtained by putting

$$q = \frac{1 + \iota}{\sqrt{2}}, \quad \bar{q} = \frac{1 - \iota}{\sqrt{2}}, \quad P_{\square} = \begin{pmatrix} q/2 & -\bar{q}/2 \\ q & \bar{q} \end{pmatrix}.$$

It satisfies

$$\det P_{\square} = 1, \quad \mathcal{A}_{P_{\square}}(\mathcal{H}_{\square}(\mathcal{M}(53))) = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}. \quad (195)$$

The element $P_{\square}$ is one member of the complete transporter set. If G stabilizes $\mathrm{diag}(4, -1)$ under $\mathcal{A}$, then $GP_{\square}$ gives the same target; conversely every transporter has this form.

Proof. Direct multiplication gives

$$-JE_{21} = E_{21}, \quad -JE_{12} = -E_{12},$$

which proves the first two identities in (185). A second application gives $\mathfrak{c}^2 = -I_{\mathcal{O}}$. The two required values of χ determine it on the basis E_{12}, E_{21} , proving (186). Replacing both $\mathfrak{c}$ and ι by their negatives conjugates this coordinate and changes only the sign of $Y = c$.

Taking the determinant in (187) gives

$$\det \mathcal{H}_{\square}(M) = ad - (b + \iota c)(b - \iota c) = ad - b^2 - c^2,$$

which proves (188). Substitution of (189) gives

$$X^2 + Y^2 + W^2 - T^2 = b^2 + c^2 - ad.$$

This proves (190) and the four basis images in (191). Equations (192) and (193) follow from (131) and (183).

The map R_{YW} is a determinant-one rotation of the two spatial coordinates Y, W . In the T, W plane, the matrix of B_{TW} is

$$\begin{pmatrix} 5/4 & 3/4 \\ 3/4 & 5/4 \end{pmatrix}.$$

Its determinant is one and

$$\left(\frac{5}{4}W + \frac{3}{4}T\right)^2 - \left(\frac{5}{4}T + \frac{3}{4}W\right)^2 = W^2 - T^2.$$

It preserves the future sheet. Hence $L_{\square} \in \mathrm{SO}^+(3, 1)$, and direct substitution proves (194).

One has $q\bar{q} = 1$, so direct evaluation gives $\det P_{\square} = 1$. Put

$$K_N = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}.$$

The two columns of $P_{\square}$ are null for K_N , and their K_N -pairing is 2ι . Therefore

$$P_{\square}^{\dagger} K_N P_{\square} = \begin{pmatrix} 0 & 2\iota \\ -2\iota & 0 \end{pmatrix}.$$

Multiplication by $(P_{\square}^{-1})^{\dagger}$ and $P_{\square}^{-1}$ proves (195). Finally, $\mathcal{A}_{GP_{\square}} = \mathcal{A}_G \circ \mathcal{A}_{P_{\square}}$. This proves the stabilizer description of the complete transporter set. $\square$

Corollary 9.22 (The raw blade cells realize the exact quarter $3 + 1$ spectrum). *Let*

$$\mathcal{T}_\square := \mathcal{A}_{P_\square} \circ \mathcal{H}_\square.$$

On the four raw cell basis vectors,

$$\begin{aligned}\mathcal{T}_\square(E_{11}) &= H\left(1, \frac{1}{2}, 0, \frac{1}{4}\right), \\ \mathcal{T}_\square(E_{12}) &= H(0, 0, 1, 0), \\ \mathcal{T}_\square(E_{21}) &= H\left(2, 0, 0, -\frac{1}{2}\right) = \frac{1}{2}\begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}, \\ \mathcal{T}_\square(E_{22}) &= H\left(1, -\frac{1}{2}, 0, \frac{1}{4}\right).\end{aligned}\tag{196}$$

For the quarter-branch coefficient inversion $\mathcal{I}_-$ of (343), these vectors satisfy

$$\mathcal{I}_-\mathcal{T}_\square(E_{21}) = -\frac{1}{4}\mathcal{T}_\square(E_{21}),\tag{197}$$

$$\mathcal{I}_-\mathcal{T}_\square(E_{ij}) = \frac{1}{4}\mathcal{T}_\square(E_{ij}), \quad (ij) \in \{11, 12, 22\}.\tag{198}$$

Hence

$$\mathbb{R}E_{21} \oplus \text{span}_{\mathbb{R}}\{E_{11}, E_{12}, E_{22}\} \xrightarrow{\mathcal{T}_\square} E_{-1/4} \oplus E_{+1/4},\tag{199}$$

where the two target dimensions are one and three.

The residue specializations are

$$\mathcal{T}_\square(\mathcal{M}(53)) = \mathcal{T}_\square(2E_{21}) = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix},\tag{200}$$

$$\mathcal{T}_\square(\mathcal{M}(109)) = \mathcal{T}_\square(E_{12}) = H(0, 0, 1, 0) \in E_{+1/4}.\tag{201}$$

The direct-sum identification (199) is independent of the particular transporter in the stabilizer family described in theorem 9.21. The stabilizer changes the displayed basis inside $E_{+1/4}$ and preserves the one-dimensional normal line and its Lorentz-orthogonal three-space.

Proof. Substitution of the four Hermitian matrices in (191) into $\mathcal{A}_{P_\square}$ gives (196). The first, second, and fourth coefficient vectors satisfy $c = a/4$, so they lie in $E_{+1/4}$ by (345). The third is a nonzero multiple of $(4, 0, 0, -1)$, so it lies in $E_{-1/4}$ by (346). This proves (197) through (199).

Equations (200) and (201) follow from $\mathcal{M}(53) = 2E_{21}$ and $\mathcal{M}(109) = E_{12}$. Every other transporter is $GP_\square$, where G stabilizes the quarter carrier. Its induced Lorentz action fixes the carrier line and preserves its orthogonal complement. Hence it preserves the displayed $1 + 3$ decomposition. $\square$

Corollary 9.23 (The pulled-back quarter reflection changes exactly one raw blade cell). *Define the endomorphism of the raw four-cell space*

$$\mathfrak{Q}_\square := \mathcal{T}_\square^{-1} \circ (4\mathcal{I}_-) \circ \mathcal{T}_\square.\tag{202}$$

Then

$$\begin{aligned}\mathfrak{Q}_\square(E_{11}) &= E_{11}, & \mathfrak{Q}_\square(E_{12}) &= E_{12}, \\ \mathfrak{Q}_\square(E_{21}) &= -E_{21}, & \mathfrak{Q}_\square(E_{22}) &= E_{22}.\end{aligned}\tag{203}$$

Equivalently,

$$\mathfrak{Q}_\square \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ -c & d \end{pmatrix}.\tag{204}$$

Its two spectral projectors are

$$\Pi_{\square,-}(M) = \frac{1}{2}(I - \mathfrak{Q}_{\square})M = cE_{21}, \quad (205)$$

$$\Pi_{\square,+}(M) = \frac{1}{2}(I + \mathfrak{Q}_{\square})M = aE_{11} + bE_{12} + dE_{22}. \quad (206)$$

Thus $B_1^- L_{34}^+$ is the unique raw cell in the negative quarter eigenline. The other three raw cells form the positive quarter three-space. In particular,

$$\Pi_{\square,-}\mathcal{M}(53) = 2E_{21}, \quad \Pi_{\square,+}\mathcal{M}(53) = 0, \quad (207)$$

whereas

$$\Pi_{\square,-}\mathcal{M}(109) = 0, \quad \Pi_{\square,+}\mathcal{M}(109) = E_{12}. \quad (208)$$

Proof. Equations (197) and (198) show that $4\mathcal{I}_-$ has eigenvalue -1 on $\mathcal{T}_{\square}(E_{21})$ and eigenvalue $+1$ on the other three images. Conjugation by $\mathcal{T}_{\square}$ proves (203) and (204). The projector formulas follow by addition and subtraction. Substitution of $\mathcal{M}(53) = 2E_{21}$ and $\mathcal{M}(109) = E_{12}$ proves the final two equations. $\square$

Theorem 9.24 (The Fable D_3 action fixes the unique negative raw cell). *On coefficient vectors (a, b_0, b_1, c) , define*

$$\mathcal{R}_F = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathcal{J}_F = \text{diag}(1, -1, 1, 1).$$

These are the rotation and reflection that act on the three Fable null rays in (347). Pull them back to the raw cell space by

$$\mathfrak{R}_{\square} := \mathcal{T}_{\square}^{-1} \mathcal{R}_F \mathcal{T}_{\square}, \quad \mathfrak{J}_{\square} := \mathcal{T}_{\square}^{-1} \mathcal{J}_F \mathcal{T}_{\square}. \quad (209)$$

In the ordered basis $(E_{11}, E_{12}, E_{21}, E_{22})$, their matrices are

$$\mathfrak{R}_{\square} = \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{2} & 0 & \frac{3}{4} \\ -\frac{\sqrt{3}}{4} & -\frac{1}{2} & 0 & \frac{\sqrt{3}}{4} \\ 0 & 0 & 1 & 0 \\ \frac{3}{4} & -\frac{\sqrt{3}}{2} & 0 & \frac{1}{4} \end{pmatrix}, \quad (210)$$

$$\mathfrak{J}_{\square} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \quad (211)$$

They satisfy

$$\mathfrak{R}_{\square}^3 = I, \quad \mathfrak{J}_{\square}^2 = I, \quad \mathfrak{J}_{\square} \mathfrak{R}_{\square} \mathfrak{J}_{\square} = \mathfrak{R}_{\square}^{-1}. \quad (212)$$

Both operators fix E_{21} , the unique negative quarter cell.

The three Fable null rays have the following exact raw-cell preimages:

$$\begin{aligned} r_0 &= 2E_{11} + 2E_{12} + 2E_{22}, \\ r_+ &= (2 + \sqrt{3})E_{11} - E_{12} + (2 - \sqrt{3})E_{22}, \\ r_- &= (2 - \sqrt{3})E_{11} - E_{12} + (2 + \sqrt{3})E_{22}. \end{aligned} \quad (213)$$

Their actions are

$$r_0 \xrightarrow{\mathfrak{R}_{\square}} r_+ \xrightarrow{\mathfrak{R}_{\square}} r_- \xrightarrow{\mathfrak{R}_{\square}} r_0, \quad \mathfrak{J}_{\square}(r_0) = r_0, \quad \mathfrak{J}_{\square}(r_+) = r_-. \quad (214)$$

Consequently the raw cell space has the invariant direct sum

$$\mathbb{R}E_{21} \oplus \text{span}_{\mathbb{R}}\{E_{11}, E_{12}, E_{22}\}, \quad (215)$$

where D_3 fixes the first summand and acts on the three Fable rays in the second.

The entries of $\mathfrak{R}_{\square}$ involve $\sqrt{3}$. Hence this operator does not preserve the integral raw-cell lattice $\text{span}_{\mathbb{Z}}\{E_{11}, E_{12}, E_{21}, E_{22}\}$. The theorem establishes a Lorentz/Fable D_3 action on the real coefficient space and supplies no cyclic action on the arithmetic group-ring coefficients.

Proof. Conjugating the two displayed coefficient-space matrices by the explicit linear isomorphism $\mathcal{T}_{\square}$ gives (210) and (211). Their products give (212); their third columns show that both fix E_{21} .

Applying $\mathcal{T}_{\square}^{-1}$ to the three vectors in (347) gives (213). Conjugation transports the coefficient-space D_3 action to (214). The displayed matrices preserve the complementary spans in (215). Finally, the irrational entries in (210) show directly that the integer lattice is not invariant. $\square$

Corollary 9.25 (Tetrahedral realization of the Fable D_3 action). *In the ordered raw-cell basis $(E_{11}, E_{12}, E_{21}, E_{22})$, put*

$$v_* = E_{21}, \quad v_0 = r_0, \quad v_+ = r_+, \quad v_- = r_-.$$

The coordinate matrix having these four vectors as columns is

$$P_{\Delta} = \begin{pmatrix} 0 & 2 & 2 + \sqrt{3} & 2 - \sqrt{3} \\ 0 & 2 & -1 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 2 & 2 - \sqrt{3} & 2 + \sqrt{3} \end{pmatrix}, \quad \det P_{\Delta} = -24\sqrt{3}. \quad (216)$$

Hence the four vectors are linearly, and therefore affinely, independent. Their convex hull

$$\Delta_F := \text{conv}\{v_*, v_0, v_+, v_-\}$$

is a nondegenerate tetrahedron in the raw-cell space.

The generators in theorem 9.24 act on its vertices by

$$\mathfrak{R}_{\square} : v_* \mapsto v_*, \quad v_0 \mapsto v_+ \mapsto v_- \mapsto v_0, \quad (217)$$

$$\mathfrak{J}_{\square} : v_* \mapsto v_*, \quad v_0 \mapsto v_0, \quad v_+ \leftrightarrow v_-. \quad (218)$$

Thus the proved $D_3 \cong S_3$ representation is the standard vertex-stabilizer action on a tetrahedron: it fixes v_* and acts as the full symmetry group of the opposite triangular face. The quarter reflection records the transverse sign:

$$\mathfrak{Q}_{\square}(v_*) = -v_*, \quad \mathfrak{Q}_{\square}(v_0) = v_0, \quad \mathfrak{Q}_{\square}(v_+) = v_+, \quad \mathfrak{Q}_{\square}(v_-) = v_-. \quad (219)$$

Proof. Substitution of the vectors in (213) gives the matrix (216); expansion along its third row gives $\det P_{\Delta} = -24\sqrt{3}$. The four columns are therefore linearly independent. Any affine dependence among them would be a linear dependence, so their convex hull is a nondegenerate 3-simplex. Equations (217) and (218) are (214) together with the fact that both generators fix E_{21} . The sign identities follow from (203), because r_0, r_+, r_- lie in $\text{span}_{\mathbb{R}}\{E_{11}, E_{12}, E_{22}\}$. $\square$

Exact computation 9.26 (Coordinate check for the two vector figures). The script and captured receipt

```
verification/verify_visual_tetrahedral_and_c6_figures.py,
verification/verify_visual_tetrahedral_and_c6_figures_output.txt
```

verify the determinant in (216), all vertex actions in (217)–(218), the four quarter signs in (219), and the complete exponent inversion displayed in figure 9.

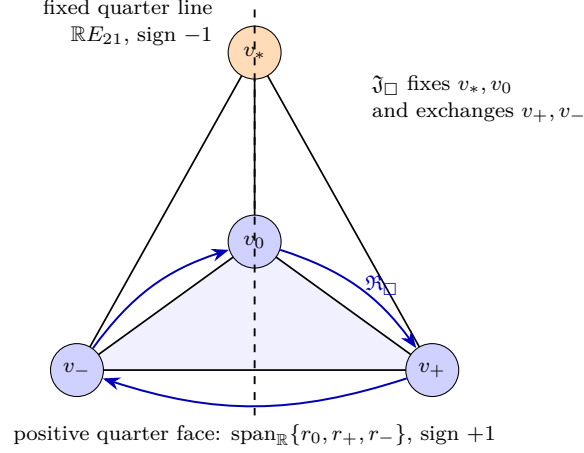


FIGURE 2. Affine tetrahedral realization of the proved Fable action. The apex is the unique negative quarter direction and the opposite face consists of the three Fable rays. The C_3 generator cycles that face, while the reflection fixes v_0 and exchanges v_+ with v_- . The drawing encodes the exact vertex action; no Euclidean edge-length assertion is used.

Theorem 9.27 (Exact spatial quarter altitude of the transported raw-cell tetrahedron). *Define*

$$\Phi_{\Delta} := A^{-1} \circ \Psi_{\text{circ}} \circ \mathcal{T}_{\square} \quad (220)$$

from the raw-cell space to the ordered Sarnak coordinates (T, X, Y, W) . For the four vertices in theorem 9.25, one has

$$\begin{aligned} z_* &:= \Phi_{\Delta}(v_*) = \left(\frac{3}{4}, 0, \frac{5}{4}, 0\right), \\ z_0 &:= \Phi_{\Delta}(v_0) = \left(\frac{5}{2}, 2, \frac{3}{2}, 0\right), \\ z_+ &:= \Phi_{\Delta}(v_+) = \left(\frac{5}{2}, -1, \frac{3}{2}, \sqrt{3}\right), \\ z_- &:= \Phi_{\Delta}(v_-) = \left(\frac{5}{2}, -1, \frac{3}{2}, -\sqrt{3}\right). \end{aligned} \quad (221)$$

The apex is spacelike and the opposite face is null:

$$q_L(z_*) = 1, \quad q_L(z_0) = q_L(z_+) = q_L(z_-) = 0. \quad (222)$$

Moreover, the complete raw-cell action intertwines with the Erdős–Straus-aligned face action:

$$\Phi_{\Delta} \mathfrak{R}_{\square} = R_{\text{face}} \Phi_{\Delta}, \quad \Phi_{\Delta} \mathfrak{J}_{\square} = s_W \Phi_{\Delta}. \quad (223)$$

Let

$$\pi_{\text{sp}}(T, X, Y, W) = (X, Y, W), \quad p_* = \pi_{\text{sp}}(z_*), \quad p_a = \pi_{\text{sp}}(z_a).$$

The centroid of the opposite face and the apex are

$$\bar{p}_{\text{face}} := \frac{1}{3}(p_0 + p_+ + p_-) = \left(0, \frac{3}{2}, 0\right), \quad p_* = \left(0, \frac{5}{4}, 0\right). \quad (224)$$

Consequently,

$$\bar{p}_{\text{face}} - p_* = \left(0, \frac{1}{4}, 0\right). \quad (225)$$

The Euclidean metric data of this spatial tetrahedron are

$$\begin{aligned} \|p_a - p_b\| &= 2\sqrt{3} & (a \neq b, a, b \in \{0, +, -\}), \\ \|p_* - p_a\| &= \frac{\sqrt{65}}{4} & (a \in \{0, +, -\}), \\ \text{Vol conv}\{p_*, p_0, p_+, p_-\} &= \frac{\sqrt{3}}{4}. \end{aligned} \tag{226}$$

Thus the unique negative-quarter raw cell is the fixed spatial apex, the three positive-quarter rays are its null opposite face, and the exact face-to-apex altitude is $1/4$. This is the raw-cell coordinate realization of the tetrahedral quarter in theorem 13.49. The Erdős–Straus input for the four vertices in (221) is exactly the proved alignment of the face reflection with denominator exchange s_W . The later arithmetic intertwiner theorem 14.301 realizes these same four vertices as the exceptional C_4 quotient: its missing antipode is v_* , its three supported classes have equal mass on the opposite face, and their arithmetic barycentre gives the displacement in (225). The later order-eight calculation theorem 14.305 shows that the six-supported C_8 and $C_4 \times C_2$ extrema consist of two such three-cell faces, with their two missing classes occupying the two fixed apex cells of the crossed eight-cell carrier.

Proof. Apply $\mathcal{T}_\square$ from (196) to $v_* = E_{21}$ and to the three raw vectors in (213). The results are respectively

$$\left(2, 0, 0, -\frac{1}{2}\right), \quad (4, 0, 2, 1), \quad (4, \sqrt{3}, -1, 1), \quad (4, -\sqrt{3}, -1, 1).$$

Applying the Sarnak map (89) and then $A^{-1}(T, X, Y, W) = (T, Y, W, X)$ gives (221). Direct substitution into $q_L = X^2 + Y^2 + W^2 - T^2$ proves (222).

By definition of the pulled-back raw-cell generators,

$$\mathcal{T}_\square \mathfrak{R}_\square = R_{XY} \mathcal{T}_\square, \quad \mathcal{T}_\square \mathfrak{J}_\square = s_X \mathcal{T}_\square.$$

The coefficient rotations commute with Ψ_{circ} , and conjugation by A^{-1} gives (119). This proves (223).

Spatial projection of (221) gives

$$p_* = \left(0, \frac{5}{4}, 0\right), \quad p_0 = \left(2, \frac{3}{2}, 0\right), \quad p_+ = \left(-1, \frac{3}{2}, \sqrt{3}\right), \quad p_- = \left(-1, \frac{3}{2}, -\sqrt{3}\right).$$

Their average proves (224) and (225). Direct squared distances give 12 on every face edge and $65/16$ on every apex edge. Finally,

$$\frac{1}{6} |\det(p_0 - p_* \quad p_+ - p_* \quad p_- - p_*)| = \frac{\sqrt{3}}{4},$$

which proves (226). □

Theorem 9.28 (Affine Lorentz spectrum and exact axial origin of the spatial quarter). *Equip the raw-cell space with the Euclidean inner product for which $(E_{11}, E_{12}, E_{21}, E_{22})$ is an orthonormal basis, and let $\eta_L = \text{diag}(-1, 1, 1, 1)$ in the ordered Sarnak coordinates (T, X, Y, W) . The matrix of Φ_Δ is*

$$[\Phi_\Delta] = \begin{pmatrix} \frac{5}{8} & 0 & \frac{3}{4} & \frac{5}{8} \\ 0 & 1 & 0 & 0 \\ \frac{3}{8} & 0 & \frac{5}{4} & \frac{3}{8} \\ \frac{1}{2} & 0 & 0 & -\frac{1}{2} \end{pmatrix}, \quad \det \Phi_\Delta = -\frac{1}{2}. \tag{227}$$

Put

$$\begin{aligned} \mathbf{E}_{\text{raw}} &:= [v_0 - v_* \quad v_+ - v_* \quad v_- - v_*], \\ \mathbf{E}_L &:= [z_0 - z_* \quad z_+ - z_* \quad z_- - z_*], \\ \mathbf{E}_{\text{sp}} &:= [p_0 - p_* \quad p_+ - p_* \quad p_- - p_*] \end{aligned}$$

and define the three Gram matrices

$$G_{\text{raw}} = \mathbf{E}_{\text{raw}}^T \mathbf{E}_{\text{raw}}, \quad G_L = \mathbf{E}_L^T \eta_L \mathbf{E}_L, \quad G_{\text{sp}} = \mathbf{E}_{\text{sp}}^T \mathbf{E}_{\text{sp}}. \quad (228)$$

They are

$$G_{\text{raw}} = \begin{pmatrix} 13 & 7 & 7 \\ 7 & 16 & 4 \\ 7 & 4 & 16 \end{pmatrix}, \quad G_L = \begin{pmatrix} 1 & -5 & -5 \\ -5 & 1 & -5 \\ -5 & -5 & 1 \end{pmatrix}, \quad G_{\text{sp}} = \frac{1}{16} \begin{pmatrix} 65 & -31 & -31 \\ -31 & 65 & -31 \\ -31 & -31 & 65 \end{pmatrix}. \quad (229)$$

Their two generalized characteristic polynomials factor as

$$\det(G_L - \lambda G_{\text{raw}}) = -324(\lambda - 1)(2\lambda - 1)(3\lambda + 1), \quad (230)$$

$$\det(G_{\text{sp}} - \lambda G_{\text{raw}}) = -\frac{27}{4}(\lambda - 1)(2\lambda - 1)(144\lambda - 1). \quad (231)$$

The same three coefficient directions diagonalize both pairs:

$$u_X = (-2, 1, 1)^T, \quad u_W = (0, -1, 1)^T, \quad u_{\text{ax}} = (1, 1, 1)^T. \quad (232)$$

Their generalized values are respectively

$$\begin{array}{c|ccc} & u_X & u_W & u_{\text{ax}} \\ \hline (G_L, G_{\text{raw}}) & 1 & \frac{1}{2} & -\frac{1}{3} \\ (G_{\text{sp}}, G_{\text{raw}}) & 1 & \frac{1}{2} & \frac{1}{144} \end{array}. \quad (233)$$

Thus the Lorentz form on the transported affine tangent space has signature $(2, 1)$. The spatial transport has exact length factors

$$1, \quad \frac{1}{\sqrt{2}}, \quad \frac{1}{12}. \quad (234)$$

The last factor is the one-dimensional face-centroid axis. More precisely,

$$\frac{1}{3} \mathbf{E}_{\text{raw}} u_{\text{ax}} = 2E_{11} - E_{21} + 2E_{22}, \quad \frac{1}{3} \mathbf{E}_{\text{sp}} u_{\text{ax}} = \left(0, \frac{1}{4}, 0\right). \quad (235)$$

The raw vector on the left has length (3) , so the quarter altitude is exactly its image under the unique axial length factor $(1/12)$.

Finally, the raw and spatial affine volumes and their exact ratio are

$$\text{Vol}_{\text{raw}} = 3\sqrt{6}, \quad \text{Vol}_{\text{sp}} = \frac{\sqrt{3}}{4}, \quad \frac{\text{Vol}_{\text{sp}}}{\text{Vol}_{\text{raw}}} = \frac{\sqrt{2}}{24}. \quad (236)$$

Proof. Multiplication of the three matrices in (220) gives (227). Substitution of the four raw vertices and their transported spatial coordinates gives the three matrices in (229). Direct determinants give (230) and (231).

For each vector in (232), direct multiplication proves the corresponding generalized eigenvalue in (233). In particular,

$$\begin{aligned} \mathbf{E}_{\text{raw}} u_X &= -6E_{12}, & \mathbf{E}_{\text{sp}} u_X &= (-6, 0, 0), \\ \mathbf{E}_{\text{raw}} u_W &= 2\sqrt{3}(-E_{11} + E_{22}), & \mathbf{E}_{\text{sp}} u_W &= (0, 0, -2\sqrt{3}). \end{aligned}$$

The equal coefficients of u_{ax} make it the fixed axis of the face permutation, and its third generalized spatial value is distinct from the other two. This proves both its one-dimensionality and (235).

Lastly,

$$\det G_{\text{raw}} = 1944, \quad \det G_{\text{sp}} = \frac{27}{4}.$$

Taking the square root and dividing by $(3!=6)$ gives the two volumes in (236); their quotient is $\sqrt{2}/24$. $\square$

Corollary 9.29 (Exact affine-invariant geodesic length–volume identity). *Let*

$$V_{\text{aff}} := \text{im } E_{\text{raw}}$$

carry the raw Euclidean inner product, and define the spatial transport

$$L_{\text{sp}}: V_{\text{aff}} \longrightarrow \mathbb{R}^3, \quad L_{\text{sp}}(E_{\text{raw}}u) := E_{\text{sp}}u.$$

Write $L_{\text{sp}} = UP$ for its polar decomposition. Thus

$$P = (L_{\text{sp}}^* L_{\text{sp}})^{1/2} \in \text{Sym}^+(V_{\text{aff}}).$$

For the affine-invariant metric

$$g_Q(H, K) := \text{tr}(Q^{-1}H Q^{-1}K) \quad (Q \in \text{Sym}^+(V_{\text{aff}})), \quad (237)$$

let C_{aff} be the geodesic length from I to P , and put

$$J_{\text{vol}} := |\det L_{\text{sp}}| = \det P.$$

Then

$$\text{Spec}(P) = \left\{ 1, \frac{1}{\sqrt{2}}, \frac{1}{12} \right\}, \quad (238)$$

$$C_{\text{aff}}^2 = (\log 12)^2 + \frac{1}{4}(\log 2)^2, \quad J_{\text{vol}} = \frac{1}{12\sqrt{2}} = \frac{\sqrt{2}}{24}, \quad (239)$$

$$2C_{\text{aff}}^2 = \left(\log J_{\text{vol}}^{-1} \right)^2 + \left(\log(6\sqrt{2}) \right)^2. \quad (240)$$

Proof. The spatial generalized eigenvalues in (233) are the eigenvalues of $L_{\text{sp}}^* L_{\text{sp}}$. Their positive square roots prove (238). For the metric (237), the geodesic from I to P is

$$\Gamma(t) = \exp(t \log P), \quad 0 \leq t \leq 1,$$

and its length is $\|\log P\|_{\text{HS}}$ [36]. The three eigenvalues in (238) therefore give

$$C_{\text{aff}}^2 = 0^2 + \left(-\frac{1}{2} \log 2 \right)^2 + (-\log 12)^2.$$

Their product gives $J_{\text{vol}} = 1/(12\sqrt{2})$, agreeing with (236).

Set $a = \log 12$ and $b = \frac{1}{2} \log 2$. Then

$$C_{\text{aff}}^2 = a^2 + b^2, \quad \log J_{\text{vol}}^{-1} = a + b, \quad a - b = \log(6\sqrt{2}).$$

The identity

$$2(a^2 + b^2) = (a + b)^2 + (a - b)^2$$

proves (240). $\square$

Proposition 9.30 (Trace–anisotropy decomposition of affine geodesic length). *Let $d \geq 2$, let $P \in \text{Sym}^+(d)$, and equip $\text{Sym}^+(d)$ with the affine-invariant metric*

$$g_Q(H, K) = \text{tr}(Q^{-1}H Q^{-1}K).$$

Put

$$C(P) := d_g(I, P), \quad J(P) := \det P, \quad X_0 := \log P - \frac{\log J(P)}{d} I.$$

Then

$$\mathcal{C}(P)^2 = \frac{1}{d}(\log J(P))^2 + \|X_0\|_{\text{HS}}^2. \quad (241)$$

For the Fable-to-Sarnak spatial carrier,

$$\|X_0\|_{\text{HS}}^2 = \frac{1}{6} \left(\log(12\sqrt{2}) \right)^2 + \frac{1}{2} \left(\log(6\sqrt{2}) \right)^2. \quad (242)$$

Proof. Set $X = \log P$ and $\mu = d^{-1} \text{tr } X$. Since P is positive definite,

$$\text{tr } X = \log \det P = \log J(P).$$

Moreover $X = X_0 + \mu I$, $\text{tr } X_0 = 0$, and hence

$$\langle X_0, \mu I \rangle_{\text{HS}} = \mu \text{tr } X_0 = 0, \quad \|\mu I\|_{\text{HS}}^2 = d\mu^2.$$

The affine-invariant distance formula $\mathcal{C}(P) = \|\log P\|_{\text{HS}}$ now gives (241).

For the spatial carrier, put

$$s = \log(12\sqrt{2}), \quad r = \log(6\sqrt{2}).$$

The proof of theorem 9.29 gives

$$\mathcal{C}_{\text{aff}}^2 = \frac{1}{2}(s^2 + r^2), \quad \log J_{\text{vol}} = -s.$$

Substitution into (241) with $d = 3$ yields

$$\|X_0\|_{\text{HS}}^2 = \frac{1}{2}(s^2 + r^2) - \frac{1}{3}s^2 = \frac{1}{6}s^2 + \frac{1}{2}r^2,$$

which is (242). □

Corollary 9.31 (Sharp logarithmic volume bound). *Under the conditions of theorem 9.30,*

$$\mathcal{C}(P) \geq \frac{|\log J(P)|}{\sqrt{d}}. \quad (243)$$

Equality holds exactly when

$$P = J(P)^{1/d} I.$$

For the Fable-to-Sarnak spatial carrier the inequality is strict, and its exact squared gap is

$$\mathcal{C}_{\text{aff}}^2 - \frac{1}{3} \left(\log J_{\text{vol}}^{-1} \right)^2 = \frac{1}{6} \left(\log(12\sqrt{2}) \right)^2 + \frac{1}{2} \left(\log(6\sqrt{2}) \right)^2. \quad (244)$$

Proof. Equation (241) is a sum of two nonnegative squared norms, which proves (243). Equality holds exactly when $X_0 = 0$, equivalently when

$$\log P = \frac{\log J(P)}{d} I.$$

Exponentiation gives $P = J(P)^{1/d} I$, and that scalar matrix attains equality. The spectrum in (238) contains three distinct values, so the spatial carrier does not attain equality. Its squared gap is (242). □

Exact computation 9.32 (Exact check of the transported raw-cell tetrahedron). The generator, checker, and captured receipt

```
verification/make_fable_es_raw_tetrahedron_figure.py,
verification/make_fable_12289_count_orbit_figure.py,
verification/verify_fable_es_raw_tetrahedron.py,
verification/verify_fable_es_raw_tetrahedron_output.txt
```

The negative quarter cell is the spatial apex below the null face

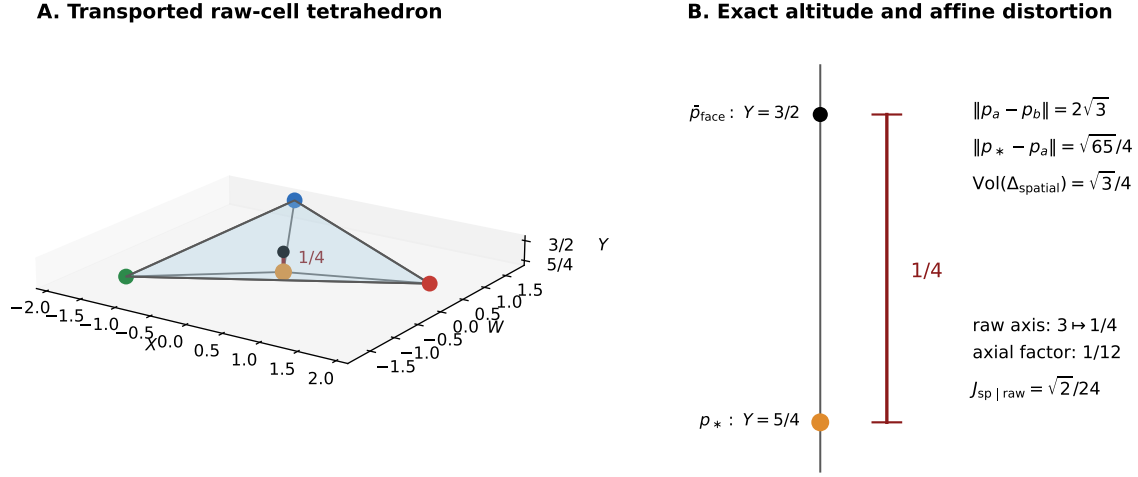


FIGURE 3. Exact spatial realization of the transported raw-cell tetrahedron.

The left panel plots the true spatial coordinates in (221): the colored null face lies at $Y = 3/2$, and the orange spacelike apex lies at $Y = 5/4$. The black point is the face centroid. The right panel reads the two coordinates on their common axis and records the exact altitude, edge lengths, Euclidean volume, axial length factor, and affine-volume factor. Every displayed quantity is checked in theorem 9.32.

verify the complete transport map, the four Lorentz norms, both intertwining identities, the face centroid, the quarter altitude, all six edge lengths, the spatial volume, all three affine Gram matrices, both generalized spectra, the three common eigendirections, and the exact affine-volume factor, positive-polar determinant, quadratic logarithmic identity, and trace–anisotropy decomposition in theorems 9.27 to 9.31 and 10.1. The maintained vector images are

figures/fable_es_raw_tetrahedron.pdf, figures/fable_12289_count_orbit.pdf.

Theorem 9.33 (Vertex projection of a centered regular simplex). *Let $n \geq 2$, let V be a real inner-product space, and suppose $v_0, \dots, v_n \in V$ satisfy*

$$\langle v_i, v_i \rangle = 1, \quad \langle v_i, v_j \rangle = -\frac{1}{n} \quad (i \neq j). \quad (245)$$

Put $W = \text{span}_{\mathbb{R}}\{v_0, \dots, v_n\}$, and let π_0 be the orthogonal projection from W to $W \cap v_0^\perp$. Then

$$\sum_{i=0}^n v_i = 0, \quad c_{\text{opp}} := \frac{1}{n} \sum_{i=1}^n v_i = -\frac{1}{n} v_0. \quad (246)$$

For $1 \leq i \leq n$, put $w_i = \pi_0(v_i)$. Explicitly,

$$w_i = v_i + \frac{1}{n} v_0, \quad \sum_{i=1}^n w_i = 0, \quad (247)$$

and their Gram matrix is

$$\langle w_i, w_j \rangle = \begin{cases} \frac{n^2 - 1}{n^2}, & i = j, \\ -\frac{n+1}{n^2}, & i \neq j. \end{cases} \quad (248)$$

Consequently,

$$u_i := \frac{n}{\sqrt{n^2 - 1}} w_i \implies \langle u_i, u_j \rangle = \begin{cases} 1, & i = j, \\ -\frac{1}{n-1}, & i \neq j. \end{cases} \quad (249)$$

Thus the orthogonal image of the face opposite v_0 is a centered regular $(n-1)$ -simplex. The projection erases the single axial displacement

$$c_{\text{opp}} - 0 = -\frac{1}{n} v_0. \quad (250)$$

Proof. Taking the squared norm of the sum of the $n+1$ vertices gives

$$\left\| \sum_{i=0}^n v_i \right\|^2 = (n+1) + n(n+1) \left(-\frac{1}{n} \right) = 0.$$

This proves the first identity in (246); the second follows by isolating v_0 and dividing by n .

Since $\langle v_i, v_0 \rangle = -1/n$ for $i \geq 1$,

$$\pi_0(v_i) = v_i - \langle v_i, v_0 \rangle v_0 = v_i + \frac{1}{n} v_0.$$

Summing these identities and using $\sum_{i=1}^n v_i = -v_0$ gives $\sum_i w_i = 0$. Direct expansion gives

$$\langle w_i, w_i \rangle = 1 - \frac{2}{n^2} + \frac{1}{n^2} = \frac{n^2 - 1}{n^2},$$

and, for $i \neq j$,

$$\langle w_i, w_j \rangle = -\frac{1}{n} - \frac{2}{n^2} + \frac{1}{n^2} = -\frac{n+1}{n^2}.$$

This proves (248). Multiplication by $n/\sqrt{n^2 - 1}$ sends the off-diagonal entry to

$$-\frac{n+1}{n^2 - 1} = -\frac{1}{n-1},$$

which proves (249). Finally, $\pi_0(c_{\text{opp}}) = 0$ because $c_{\text{opp}} = -v_0/n$, proving (250). $\square$

Corollary 9.34 (The hidden quarter in the regular 4-simplex). *Under the hypotheses of theorem 9.33 with $n = 4$, the face opposite v_0 is the affine translate*

$$\text{conv}\{v_1, v_2, v_3, v_4\} = -\frac{1}{4}v_0 + \text{conv}\{w_1, w_2, w_3, w_4\}. \quad (251)$$

Its centroid is $-v_0/4$, while the centroid of its orthogonal image is 0. The projected Gram matrix has diagonal entry $15/16$ and off-diagonal entry $-5/16$; multiplication of every w_i by $4/\sqrt{15}$ gives a regular tetrahedron with off-diagonal entry $-1/3$.

Every permutation of v_1, v_2, v_3, v_4 extends to an orthogonal transformation of W fixing v_0 . The point $-v_0/4$ is the unique point of the opposite affine face fixed by all these permutations. Therefore the signed quarter displacement is intrinsic to the vertex-stabilized projection.

Proof. Equations (251) and the two Gram entries are the case $n = 4$ of (247)–(248). A permutation of the four opposite vertices preserves every entry of (245), hence induces an orthogonal transformation on W fixing v_0 . A point of the opposite affine face fixed by every permutation must have four equal barycentric coordinates. Their sum is one, so each equals $1/4$; the fixed point is the centroid $-v_0/4$. $\square$

Theorem 9.35 (Exact rank–volume identities for centered regular simplices). *Retain the hypotheses and notation of theorem 9.33. Let*

$$C_n := (\langle v_i, v_j \rangle)_{0 \leq i, j \leq n}, \quad W_n := (\langle w_i, w_j \rangle)_{1 \leq i, j \leq n},$$

and let pdet denote the product of the nonzero eigenvalues. If $\mathbf{1}_m$ is the all-one column of length m , then

$$C_n = \frac{n+1}{n} I_{n+1} - \frac{1}{n} \mathbf{1}_{n+1} \mathbf{1}_{n+1}^\top, \quad \text{Spec}(C_n) = \left\{ 0^{[1]}, \left(\frac{n+1}{n} \right)^{[n]} \right\}, \quad (252)$$

$$W_n = \frac{n+1}{n} \left(I_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top \right), \quad \text{Spec}(W_n) = \left\{ 0^{[1]}, \left(\frac{n+1}{n} \right)^{[n-1]} \right\}. \quad (253)$$

Consequently,

$$\text{rank } C_n = n, \quad \text{rank } W_n = n - 1, \quad \text{pdet } C_n = \frac{n+1}{n} \text{pdet } W_n. \quad (254)$$

Let $\mathcal{V}_n$ be the Euclidean n -volume of $\text{conv}\{v_0, \dots, v_n\}$, and let $\mathcal{A}_{n-1}$ be the Euclidean $(n-1)$ -volume of its face opposite v_0 . Then

$$\mathcal{V}_n^2 = \frac{n+1}{(n!)^2} \text{pdet } C_n = \frac{(n+1)^{n+1}}{(n!)^2 n^n}, \quad (255)$$

$$\mathcal{A}_{n-1}^2 = \frac{n}{((n-1)!)^2} \text{pdet } W_n = \frac{(n+1)^{n-1}}{n^{n-2} ((n-1)!)^2}. \quad (256)$$

The altitude from v_0 to the opposite face is

$$h_n = \|v_0 - c_{\text{opp}}\| = \frac{n+1}{n}, \quad \mathcal{V}_n = \frac{h_n}{n} \mathcal{A}_{n-1}. \quad (257)$$

Thus the loss of one Gram rank under vertex projection, the factor $(n+1)/n$ between the two pseudodeterminants, and the hidden axial displacement $1/n$ are exact parts of one dimension–volume identity.

For $n = 4$, these identities give

$$\text{pdet } C_4 = \frac{625}{256}, \quad \text{pdet } W_4 = \frac{125}{64}, \quad \mathcal{V}_4^2 = \frac{3125}{147456}, \quad \mathcal{A}_3^2 = \frac{125}{576}, \quad h_4 = \frac{5}{4}. \quad (258)$$

Proof. Equations (252) and (253) follow directly from (245) and (248). The matrix $\mathbf{1}_m \mathbf{1}_m^\top$ has eigenvalue m on the all-one line and eigenvalue 0 on its orthogonal complement. The two displayed spectra and (254) follow.

For $1 \leq i \leq n$, put $e_i = v_i - v_0$. Their edge Gram matrix is

$$(\langle e_i, e_j \rangle)_{1 \leq i, j \leq n} = \frac{n+1}{n} (I_n + \mathbf{1}_n \mathbf{1}_n^\top). \quad (259)$$

The second factor has eigenvalue $n+1$ on the all-one line and eigenvalue 1 with multiplicity $n-1$. Hence

$$\det(\langle e_i, e_j \rangle) = (n+1) \left(\frac{n+1}{n} \right)^n.$$

The Gram determinant formula for simplex volume now gives

$$(n!)^2 \mathcal{V}_n^2 = (n+1) \text{pdet } C_n,$$

which is (255).

Using v_n as a base point of the opposite face, the $n-1$ edge vectors $v_i - v_n$, $1 \leq i \leq n-1$, have Gram matrix

$$\frac{n+1}{n} (I_{n-1} + \mathbf{1}_{n-1} \mathbf{1}_{n-1}^\top).$$

Its determinant is

$$n \left(\frac{n+1}{n} \right)^{n-1} = n \text{pdet } W_n.$$

The $(n-1)$ -dimensional Gram determinant formula proves (256).

Finally,

$$v_0 - c_{\text{opp}} = \left(1 + \frac{1}{n} \right) v_0,$$

so $h_n = (n+1)/n$. The pyramid formula $\mathcal{V}_n = h_n \mathcal{A}_{n-1}/n$ proves (257). Substitution of $n=4$ gives (258). $\square$

Corollary 9.36 (Trace-one simplex states and exact entropy–volume identities). *Define*

$$\rho_n := \frac{C_n}{\text{tr } C_n} = \frac{1}{n} \left(I_{n+1} - \frac{1}{n+1} \mathbf{1}_{n+1} \mathbf{1}_{n+1}^\top \right), \quad (260)$$

and

$$\sigma_{n-1} := \frac{W_n}{\text{tr } W_n} = \frac{1}{n-1} \left(I_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top \right). \quad (261)$$

These are positive semidefinite trace-one matrices with

$$\text{Spec}(\rho_n) = \left\{ 0^{[1]}, \left(\frac{1}{n} \right)^{[n]} \right\}, \quad \text{tr}(\rho_n^2) = \frac{1}{n}, \quad S(\rho_n) := -\text{tr}(\rho_n \log \rho_n) = \log n, \quad (262)$$

$$\text{Spec}(\sigma_{n-1}) = \left\{ 0^{[1]}, \left(\frac{1}{n-1} \right)^{[n-1]} \right\}, \quad \text{tr}(\sigma_{n-1}^2) = \frac{1}{n-1}, \quad S(\sigma_{n-1}) = \log(n-1). \quad (263)$$

Here $0 \log 0$ is interpreted by continuity as 0.

The Euclidean volumes are recovered from the support determinants of these two states:

$$\mathcal{V}_n^2 = \frac{(n+1)^{n+1}}{(n!)^2} \text{pdet } \rho_n, \quad \text{pdet } \rho_n = n^{-n}, \quad (264)$$

$$\mathcal{A}_{n-1}^2 = \frac{(n^2-1)^{n-1}}{n^{n-2}((n-1)!)^2} \text{pdet } \sigma_{n-1}, \quad \text{pdet } \sigma_{n-1} = (n-1)^{-(n-1)}. \quad (265)$$

Equivalently,

$$\log \mathcal{V}_n^2 = (n+1) \log(n+1) - 2 \log(n!) - nS(\rho_n), \quad (266)$$

$$\log \mathcal{A}_{n-1}^2 = (n-1) \log(n^2-1) - (n-2) \log n - 2 \log((n-1)!) - (n-1)S(\sigma_{n-1}). \quad (267)$$

For $n = 4$, the four nonzero eigenvalues of ρ_4 are all $1/4$. Their common value equals the magnitude of the directed centroid displacement in (250) and (287).

Proof. The trace of C_n is $n+1$, while

$$\text{tr } W_n = n \frac{n^2-1}{n^2} = \frac{n^2-1}{n}.$$

Dividing (252) and (253) by these trace values gives (262) and (263). The purity, entropy, and pseudodeterminant formulas follow by summing or multiplying the nonzero eigenvalues. Substitution in (255) and (256) proves (264) and (265). Taking logarithms and using the two entropy identities proves (266) and (267). The case $n = 4$ follows from (262). $\square$

Corollary 9.37 (Exact Hilbert–Schmidt control-length–volume carrier). *Let*

$$P_n := I_{n+1} - \frac{1}{n+1} \mathbf{1}_{n+1} \mathbf{1}_{n+1}^\top = n\rho_n$$

be the orthogonal projection onto the support of ρ_n , and put

$$K_n := P_n - \frac{n}{n+1} I_{n+1}. \quad (268)$$

For a Hermitian control H , define the Hilbert–Schmidt cost

$$F_{\text{HS}}(H) := \sqrt{\text{tr}(H^2)}.$$

This cost is bi-invariant. Consider the two paths

$$\gamma_n^\rho(t) := \exp(-it\rho_n) \in U(n+1), \quad \gamma_n^0(t) := \exp(-itK_n) \in SU(n+1), \quad 0 \leq t \leq 1. \quad (269)$$

Their exact lengths are

$$\mathcal{L}_\rho(n) := L_{F_{\text{HS}}}(\gamma_n^\rho) = \frac{1}{\sqrt{n}}, \quad \mathcal{L}_0(n) := L_{F_{\text{HS}}}(\gamma_n^0) = \sqrt{\frac{n}{n+1}}. \quad (270)$$

The hidden axial displacement, state purity, and squared first control length coincide:

$$\|c_{\text{opp}}\| = \text{tr}(\rho_n^2) = \mathcal{L}_\rho(n)^2 = \frac{1}{n}. \quad (271)$$

The full simplex volume has the two exact control-length expressions

$$\boxed{\mathcal{V}_n^2 = \frac{(n+1)^{n+1}}{(n!)^2} \mathcal{L}_\rho(n)^{2n} = \frac{n+1}{(n!)^2} \mathcal{L}_0(n)^{-2n}.} \quad (272)$$

These paths are fully specified instances of the geometric control length used in Nielsen’s circuit geometry [34, 35].

For $n = 4$,

$$\mathcal{L}_\rho(4) = \frac{1}{2}, \quad \mathcal{L}_0(4) = \frac{2}{\sqrt{5}}, \quad \mathcal{V}_4^2 = \frac{5^5}{(4!)^2} \left(\frac{1}{2}\right)^8 = \frac{5}{(4!)^2} \left(\frac{2}{\sqrt{5}}\right)^{-8}. \quad (273)$$

Proof. For a unitary Q ,

$$\text{tr}((QHQ^*)^2) = \text{tr}(QH^2Q^*) = \text{tr}(H^2),$$

which proves bi-invariance. Both paths in (269) have constant adapted Hamiltonians, respectively ρ_n and K_n . Therefore their lengths are the Hilbert–Schmidt norms of those two matrices.

The projection P_n has rank and trace n . Since $\rho_n = P_n/n$,

$$\mathrm{tr}(\rho_n^2) = \frac{1}{n^2} \mathrm{tr}(P_n) = \frac{1}{n}.$$

Equation (246) gives $\|c_{\mathrm{opp}}\| = 1/n$, proving (271).

On the range of P_n , the eigenvalue of K_n is $1/(n+1)$; on its kernel, the eigenvalue is $-n/(n+1)$. Thus

$$\mathrm{tr}(K_n) = 0, \quad \mathrm{tr}(K_n^2) = n \frac{1}{(n+1)^2} + \frac{n^2}{(n+1)^2} = \frac{n}{n+1}.$$

This proves (270) and also shows that $\gamma_n^0(t)$ lies in $SU(n+1)$. Substitution of the two squared lengths into (255) proves (272); the case $n = 4$ is (273). $\square$

Theorem 9.38 (Finite Tomita sector carried by regular-simplex support). *Let V_n be the complexification of the real span of the centered regular-simplex vertices in theorem 9.33, and let*

$$S_n : \mathbb{C}^{n+1} \longrightarrow V_n, \quad S_n e_i = v_i$$

be the synthesis map. Then

$$S_n^* S_n = C_n, \quad \ker S_n = \mathbb{C} \mathbf{1}_{n+1}, \quad P_n = I_{n+1} - \frac{1}{n+1} \mathbf{1}_{n+1} \mathbf{1}_{n+1}^\top \quad (274)$$

is the support projection of C_n and ρ_n .

Define the support corner and its Hilbert–Schmidt standard space by

$$\mathcal{A}_n := P_n M_{n+1}(\mathbb{C}) P_n, \quad \mathcal{H}_n := (\mathcal{A}_n, \langle X, Y \rangle_{\mathrm{HS}} = \mathrm{tr}(X^* Y)), \quad (275)$$

and represent $\mathcal{A}_n$ on $\mathcal{H}_n$ by left multiplication $L_A(X) = AX$. The vector

$$\Omega_n := \frac{1}{\sqrt{n}} P_n \quad (276)$$

is cyclic and separating, and its vector state is

$$\omega_n(A) = \langle \Omega_n, L_A \Omega_n \rangle_{\mathrm{HS}} = \mathrm{tr}(\rho_n A) = \frac{1}{n} \mathrm{tr}_{\mathcal{A}_n}(A). \quad (277)$$

In the Tomita polar decomposition $S_n = J_n \Delta_n^{1/2}$ associated with $(L(\mathcal{A}_n), \Omega_n)$, one has

$$S_n(X) = X^*, \quad J_n(X) = X^*, \quad \Delta_n = I_{\mathcal{H}_n}, \quad J_n L_A J_n = R_{A^*}. \quad (278)$$

Consequently,

$$\Delta_n^{it} L_A \Delta_n^{-it} = L_A \quad (t \in \mathbb{R}), \quad (279)$$

and the logarithmic density generator on the support is

$$H_n^{\mathrm{dens}} := -\log(\rho_n|_{\mathrm{Ran} P_n}) = (\log n) P_n, \quad \mathcal{K}_n^{\mathrm{mod}} := L_{H_n^{\mathrm{dens}}} - R_{H_n^{\mathrm{dens}}} = 0. \quad (280)$$

For $n = 4$, the faithful support density is $\rho_4|_{\mathrm{Ran} P_4} = P_4/4$; its four support eigenvalues are the quarter appearing in (287).

Proof. The first identity in (274) is the definition of the Gram matrix. The vertex sum vanishes, so $\mathbb{C} \mathbf{1}_{n+1} \subseteq \ker S_n$. By theorem 9.35, $\mathrm{rank} S_n = \mathrm{rank} C_n = n$; rank-nullity gives equality of the two one-dimensional spaces. The displayed matrix P_n is the orthogonal projection onto $\mathbf{1}_{n+1}^\perp$, hence onto $(\ker S_n)^\perp$, and $\rho_n = P_n/n$.

For $A \in \mathcal{A}_n$,

$$L_A \Omega_n = \frac{1}{\sqrt{n}} A.$$

Thus $L(\mathcal{A}_n) \Omega_n = \mathcal{H}_n$, proving cyclicity, while $L_A \Omega_n = 0$ implies $A = 0$, proving separation. The Tomita rule on this finite-dimensional core is

$$S_n(L_A \Omega_n) = L_{A^*} \Omega_n,$$

which is exactly $S_n(X) = X^*$. Hilbert–Schmidt adjunction shows that $X \mapsto X^*$ is antiunitary and involutive, so its polar decomposition has $J_n = S_n$ and $\Delta_n = I$. Directly,

$$J_n L_A J_n(X) = (AX^*)^* = XA^*,$$

which proves the last identity in (278) and the flow formula (279). Equation (277) follows from $AP_n = P_n A = A$. Functional calculus on $\text{Ran } P_n$ gives $-\log(P_n/n) = (\log n)P_n$, whose left and right actions coincide. This proves (280). These formulas are the finite-dimensional standard form of Tomita–Takesaki theory [32, 33]; every operator identity used here has been proved directly. $\square$

Remark 9.39 (Status of the label-kernel support). Equation (274) identifies the distinguished label relation exactly: $\ker S_n = \mathbb{C}\mathbf{1}_{n+1}$. This theorem therefore supplies the source relation line for a comparison with the globalization carrier. Identifying that line with the corpus symbol τ requires a specified structure-preserving map and a proof of its compatibility with the respective algebraic operations.

Exact computation 9.40 (Exact check of the finite simplex modular sector). The script and captured receipt

```
verification/verify_regular_simplex_modular_support.py,
verification/verify_regular_simplex_modular_support_output.txt
```

check for every $2 \leq n \leq 9$ that P_n is a self-adjoint idempotent of rank n , its kernel contains the all-one label line, ρ_n is faithful on the support corner, the modular operator fixes every tested support matrix unit, and the density generator is central. They also check the multiplicity-four quarter in the $n = 4$ support state.

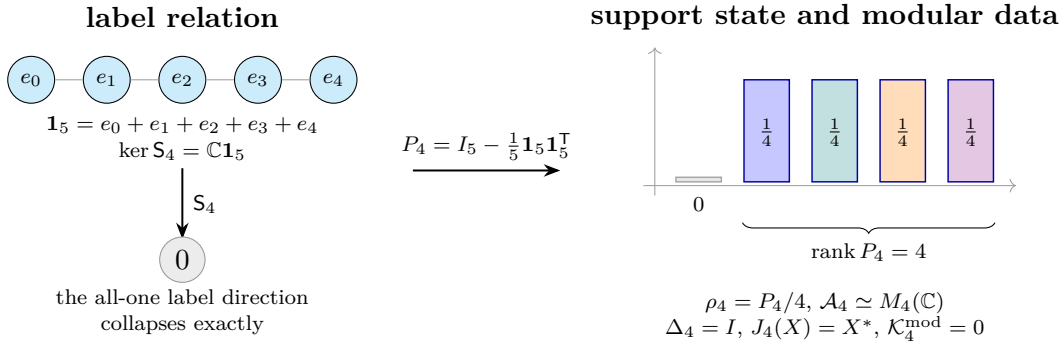


FIGURE 4. The exact support mechanism for the regular 4-simplex. The synthesis map kills precisely the all-one label line. Projection onto its orthogonal complement leaves a rank-four support carrying four equal state eigenvalues $1/4$. On the resulting matrix corner the Tomita operator is adjunction and the modular operator is the identity.

Exact computation 9.41 (Exact check of the simplex rank–volume identities). The script and captured receipt

```
verification/verify_regular_simplex_dimension_volume.py,
verification/verify_regular_simplex_dimension_volume_output.txt
```

check the centered and projected spectra, ranks, pseudodeterminants, edge-Gram determinants, both volume formulas, the altitude identity, the two trace-one state spectra and purities, and both control-length formulas for every $2 \leq n \leq 12$. They also check the exact $n = 4$ values in (258), the four eigenvalues equal to $1/4$ in ρ_4 , and the two lengths in (273).

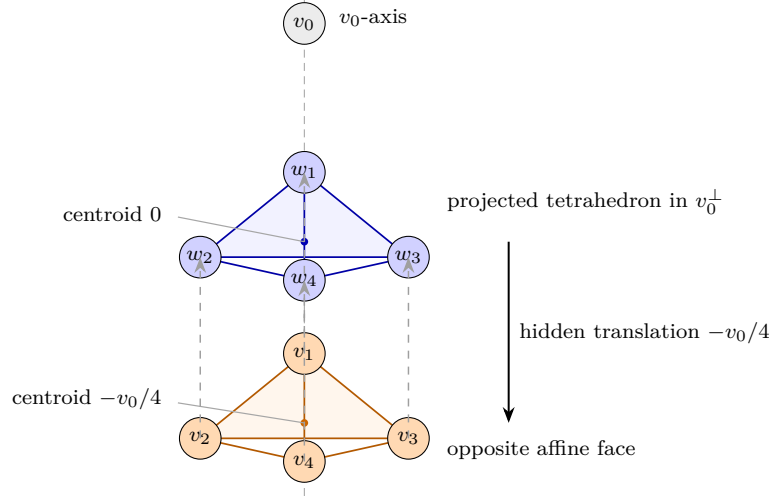


FIGURE 5. The $n = 4$ case of vertex projection. The orange opposite face is the blue projected tetrahedron translated by $-v_0/4$. Orthogonal projection along the v_0 -axis identifies corresponding vertices and sends the orange centroid to 0. The two tetrahedra are drawn in the same oblique image of their internal three coordinates; their vertical separation records the fourth coordinate.

Exact computation 9.42 (Exact check of the simplex-projection identities). The script and captured receipt

```
verification/verify_regular_simplex_projection.py,
verification/verify_regular_simplex_projection_output.txt
```

check the symbolic diagonal and off-diagonal formulas, the unit-rescaled Gram entry, the complete Gram matrices for $2 \leq n \leq 9$, and the exact $n = 4$ entries $15/16$, $-5/16$, and $-v_0/4$.

Theorem 9.43 (The Fable rays force a regular 4-simplex completion). *Let*

$$U_+ := \text{span}_{\mathbb{R}}\{E_{11}, E_{12}, E_{22}\},$$

and retain the three Fable rays r_0, r_+, r_- from (213). Define the fourth centered ray

$$r_\infty := -(r_0 + r_+ + r_-) = -6(E_{11} + E_{22}). \quad (281)$$

There is a unique symmetric bilinear form on U_+ satisfying

$$\langle r_a, r_b \rangle_{G_+} = \begin{cases} 1, & a = b, \\ -\frac{1}{3}, & a \neq b, \end{cases} \quad a, b \in \{\infty, 0, +, -\}. \quad (282)$$

In the ordered basis (E_{11}, E_{12}, E_{22}) , its matrix is

$$G_+ = \begin{pmatrix} \frac{1}{16} & 0 & -\frac{7}{144} \\ 0 & \frac{2}{9} & 0 \\ -\frac{7}{144} & 0 & \frac{1}{16} \end{pmatrix}, \quad \text{Spec}(G_+) = \left\{ \frac{1}{72}, \frac{1}{9}, \frac{2}{9} \right\}. \quad (283)$$

In particular, G_+ is positive definite and the four rays in (282) are the vertices of a centered regular tetrahedron.

Extend G_+ to the full raw-cell space by requiring E_{21} to be unit and orthogonal to U_+ . In the ordered raw-cell basis this gives

$$\widehat{G}_F = \begin{pmatrix} \frac{1}{16} & 0 & 0 & -\frac{7}{144} \\ 0 & \frac{2}{9} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\frac{7}{144} & 0 & 0 & \frac{1}{16} \end{pmatrix}. \quad (284)$$

For $a \in \{\infty, 0, +, -\}$, put

$$s_* := E_{21}, \quad s_a := \frac{\sqrt{15}}{4}r_a - \frac{1}{4}E_{21}. \quad (285)$$

Then

$$\langle s_\alpha, s_\beta \rangle_{\widehat{G}_F} = \begin{cases} 1, & \alpha = \beta, \\ -\frac{1}{4}, & \alpha \neq \beta, \end{cases} \quad \alpha, \beta \in \{*, \infty, 0, +, -\}. \quad (286)$$

Thus the five displayed vectors form a regular 4-simplex. Its face opposite s_* has centroid

$$\frac{1}{4}(s_\infty + s_0 + s_+ + s_-) = -\frac{1}{4}E_{21}, \quad (287)$$

and its orthogonal projection to U_+ is the regular tetrahedron with vertices $(\sqrt{15}/4)r_a$.

The operators $\mathfrak{R}_\square$ and $\mathfrak{J}_\square$ preserve $\widehat{G}_F$, fix s_* and s_∞ , and act on s_0, s_+, s_- by the cycle and reflection in (217)–(218). Hence the maintained Fable D_3 action is an orthogonal vertex symmetry of the completed regular 4-simplex.

Finally, the centroid axis intertwines the regular-simplex quarter with the maintained coefficient inversion:

$$\mathcal{T}_\square\left(-\frac{1}{4}E_{21}\right) = \mathcal{I}_-\mathcal{T}_\square(E_{21}) = -\frac{1}{8}(4, 0, 0, -1). \quad (288)$$

If $\iota_1(t) = (t, 0, 0)$, then the common Fable output is

$$F(p_0) = F(p_+) = F(p_-) = \iota_1\left(-\frac{1}{4}\right). \quad (289)$$

The same scalar $-1/4$ therefore occurs as the opposite-face displacement, the simple inversion eigenvalue, and the first coordinate of the three-point Fable output.

Proof. In the ordered basis (E_{11}, E_{12}, E_{22}) , let

$$P_+ := \begin{pmatrix} 2 & 2 + \sqrt{3} & 2 - \sqrt{3} \\ 2 & -1 & -1 \\ 2 & 2 - \sqrt{3} & 2 + \sqrt{3} \end{pmatrix}, \quad H_3 := \begin{pmatrix} 1 & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & 1 & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & 1 \end{pmatrix}.$$

The columns of P_+ are r_0, r_+, r_- , and $\det P_+ = -24\sqrt{3}$. A symmetric form with Gram matrix H_3 on these three basis vectors must therefore have matrix

$$(P_+^{-1})^\top H_3 P_+^{-1}.$$

Exact multiplication gives the matrix G_+ in (283). Its three eigenvalues are positive. Since $r_\infty = -(r_0 + r_+ + r_-)$, multiplication by the complete 4×4 Gram matrix gives (282). This also proves uniqueness.

The extension conditions determine $\widehat{G}_F$ uniquely and give (284). Substituting (282) into (285) gives

$$\langle s_a, s_a \rangle = \frac{15}{16} + \frac{1}{16} = 1, \quad \langle s_a, s_b \rangle = -\frac{5}{16} + \frac{1}{16} = -\frac{1}{4} \quad (a \neq b),$$

and

$$\langle s_*, s_a \rangle = -\frac{1}{4}.$$

This proves (286). Summing the four opposite vertices and using $\sum_a r_a = 0$ proves (287).

The exact ray action in (214) permutes r_0, r_+, r_- and therefore fixes their negative sum r_∞ . It preserves every entry of (282); hence it preserves G_+ . Both generators fix E_{21} , proving preservation of $\hat{G}_F$ and the stated vertex action.

Equation (196) gives

$$\mathcal{T}_\square(E_{21}) = \frac{1}{2}(4, 0, 0, -1),$$

and (197) gives its $-1/4$ eigenvalue. Linearity proves (288). Equation (289) is the exact common-fibre evaluation in theorem 7.1. $\square$

Regular tetrahedral realization of the four Fable rays

opposite face: $r_0 \mapsto r_+ \mapsto r_- \mapsto r_0$

face reflection: r_0 fixed, $r_+ \leftrightarrow r_-$

forced ray: $r_\infty = -(r_0 + r_+ + r_-)$

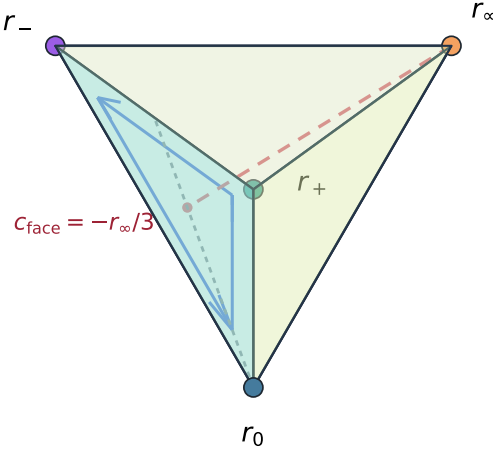


FIGURE 6. A three-dimensional isometric realization of the exact ray Gram form (282). The plotted coordinates are $r_\infty = (1, 1, 1)/\sqrt{3}$, $r_0 = (1, -1, -1)/\sqrt{3}$, $r_+ = (-1, 1, -1)/\sqrt{3}$, and $r_- = (-1, -1, 1)/\sqrt{3}$. Every distinct pair has inner product $-1/3$. The blue arrows realize the C_3 action on the opposite face; its dashed gray axis realizes the reflection fixing r_0 and exchanging r_+ with r_- . The red dashed altitude ends at the exact face centroid $c_{\text{face}} = -r_\infty/3$.

Exact computation 9.44 (Exact check of the regular-tetrahedron vector image). The generator, checker, and captured receipt

```
verification/make_fable_regular_tetrahedron.py,
verification/verify_fable_regular_tetrahedron_figure.py,
verification/verify_fable_regular_tetrahedron_figure_output.txt
```

verify the complete Gram matrix, zero centroid, opposite-face centroid, altitude, three face-edge lengths, and reflection axis used in figure 6. The maintained image is the vector PDF figures/fable_regular_tetrahedron.pdf.

Exact computation 9.45 (Exact verification of the Fable 4-simplex completion). The checker and captured receipt

```
verification/verify_fable_regular_four_simplex.py,
verification/verify_fable_regular_four_simplex_output.txt
```

verify the forced fourth vertex, the positive spectrum of G_+ , the regular-tetrahedron and regular-4-simplex Gram matrices, exact D_3 -invariance and vertex actions, the face centroid, the axis intertwiner, and all three polynomial evaluations in (289).

Corollary 9.46 (The exact $109 \rightarrow 53 \rightarrow$ quarter-circle transport chain). *The single matrix $P_{\square}$ satisfies*

$$2\mathcal{H}_{\square}(\mathcal{M}(109)) \xrightarrow{\mathcal{A}_{P_{\square}}} \mathcal{H}_{\square}(\mathcal{M}(53)) \xrightarrow{\mathcal{A}_{P_{\square}}} \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}. \quad (290)$$

Explicitly,

$$2H(0, 1, 0, 0) \xrightarrow{\mathcal{A}_{P_{\square}}} H(0, 0, 2, 0) \xrightarrow{\mathcal{A}_{P_{\square}}} H(4, 0, 0, -1). \quad (291)$$

In generalized-circle equations, the same chain is

$$4x = 0 \quad \mapsto \quad -4y = 0 \quad \mapsto \quad 4(x^2 + y^2) - 1 = 0. \quad (292)$$

The coefficient two in the first term is required: $\mathcal{M}(109) = E_{12}$, whereas $\mathcal{M}(53) = 2E_{21}$.

Proof. The first two basis images in (196) give

$$\mathcal{A}_{P_{\square}}(H(0, 1, 0, 0)) = H(0, 0, 1, 0).$$

Linearity gives

$$\mathcal{A}_{P_{\square}}(2H(0, 1, 0, 0)) = H(0, 0, 2, 0) = \mathcal{H}_{\square}(\mathcal{M}(53)).$$

Equation (195) gives the second arrow. Finally, the generalized-circle evaluation

$$q_{(a,b_0,b_1,c)}(x, y) = a(x^2 + y^2) + 2b_0x - 2b_1y + c$$

gives the three displayed equations. □

Theorem 9.47 (Multiplicity two identifies the quotient branch with the target carrier). *For $m > 0$, let*

$$N_m = mE_{21}, \quad P_s(m) = N_m^T N_m = m^2 E_{11}, \quad Q_r = E_{22}.$$

For each $\zeta \in \mathbb{R}$, there is a unique diagonal coefficient matrix $j_m(\zeta)$ satisfying

$$j_m(\zeta) - P_s(m) \in \mathbb{R}Q_r, \quad \frac{C(j_m(\zeta))}{A(j_m(\zeta))} = \zeta. \quad (293)$$

It is

$$j_m(\zeta) = P_s(m) + m^2 \zeta Q_r = \begin{pmatrix} m^2 & 0 \\ 0 & m^2 \zeta \end{pmatrix}. \quad (294)$$

For $\zeta < 0$, the coefficient inversion associated with $j_m(\zeta)$ is nondegenerate and has simple normal eigenvalue

$$-\frac{\mathcal{N}(j_m(\zeta))}{A(j_m(\zeta))^2} = \frac{C(j_m(\zeta))}{A(j_m(\zeta))} = \zeta. \quad (295)$$

Its Sarnak coordinates in the $X = Y = 0$ plane are

$$T_m(\zeta) = \frac{m^2}{2}(1 + \zeta), \quad W_m(\zeta) = \frac{m^2}{2}(1 - \zeta). \quad (296)$$

At the denominator-exchange branch value $\zeta_{\text{br}} = -1/4$ of theorem 9.7,

$$j_m(\zeta_{\text{br}}) = m^2 E_{11} - \frac{m^2}{4} E_{22}. \quad (297)$$

The equality

$$j_m(\zeta_{\text{br}}) = P_s(m) - Q_r \quad (298)$$

holds exactly for $m = 2$. Therefore the positive transport multiplicity two is uniquely characterized by the requirement that the universal branch value supply the negative occupied target idempotent. For the observed blade coefficient $N = N_2$,

$$j_2\left(-\frac{1}{4}\right) = 4E_{11} - E_{22} = \mathfrak{H}(N), \quad (299)$$

and (296) gives $(T, W) = (3/2, 5/2)$.

Proof. Any matrix in the affine line specified by the first condition of (293) has the form

$$K = P_s(m) + cQ_r = \text{diag}(m^2, c).$$

Its coefficient ratio is $C(K)/A(K) = c/m^2$. The second condition therefore forces $c = m^2\zeta$, proving existence, uniqueness, and (294).

For $\zeta < 0$,

$$\mathcal{N}(j_m(\zeta)) = -m^4\zeta > 0.$$

The reflection factorization

$$M_w = \frac{\mathcal{N}(w)}{A(w)^2} R_w$$

shows that the eigenvalue on the normal line is

$$-\frac{m^4\zeta}{m^4} = \zeta,$$

which proves (295). Substitution of $A = m^2$ and $C = m^2\zeta$ into $T = (A + C)/2$ and $W = (A - C)/2$ proves (296).

At $\zeta = -1/4$, equation (294) gives (297). Comparing its E_{22} -coefficient with that of $P_s(m) - Q_r$ gives

$$\frac{m^2}{4} = 1.$$

Since $m > 0$, this is equivalent to $m = 2$. The final identities follow by substitution. $\square$

Theorem 9.48 (Kocik parameter and Fable-slice coordinates of the target carrier). *Let $m > 0$ and $-1/4 \leq \zeta < 0$. The Sarnak coordinates of $j_m(\zeta)$ are*

$$(T, X, Y, W) = \left(\frac{m^2}{2}(1 + \zeta), 0, 0, \frac{m^2}{2}(1 - \zeta) \right).$$

Define

$$p(\zeta) := \frac{1 + \zeta^2}{1 - 4\zeta + \zeta^2}. \quad (300)$$

Then

$$\frac{X^2 + Y^2 + W^2}{T^2} = \left(\frac{1 - \zeta}{1 + \zeta} \right)^2 = \frac{p(\zeta) + 1}{3p(\zeta) - 1}. \quad (301)$$

Thus $j_m(\zeta)$ lies on the member with configuration parameter $p(\zeta)$ of Kocik's equal-angle quadratic family (97). Along the full carrier segment,

$$p\left(-\frac{1}{4}\right) = \frac{17}{33}, \quad p(-4 + \sqrt{15}) = \frac{2}{3}, \quad \lim_{\zeta \rightarrow 0^-} p(\zeta) = 1.$$

The function $p(\zeta)$ is strictly increasing on $[-1/4, 0)$.

For the champion coordinate

$$\zeta_* = -\frac{18189}{18190^2},$$

the exact Kocik parameter is

$$p_* = \frac{109478993882049721}{109503067103581321}. \quad (302)$$

The Lorentz isometry

$$L_{XW}(T, X, Y, W) = (T, W, -Y, X) \quad (303)$$

belongs to $\text{SO}^+(3, 1)$ and sends the carrier point to the $W = 0$ Fable slice. More precisely,

$$L_{XW}\Psi_{\text{circ}}(j_m(\zeta)) = \iota \left(\frac{m^2}{2}(1 + \zeta), \frac{m^2}{\sqrt{3}}(1 - \zeta), 0 \right), \quad (304)$$

and the Fable quadratic form has the exact value

$$q \left(\frac{m^2}{2}(1 + \zeta), \frac{m^2}{\sqrt{3}}(1 - \zeta), 0 \right) = m^4 \zeta. \quad (305)$$

The endpoint $\zeta = 0$ is the Descartes–Fable null cone. Relative to the displayed Lorentz form $T^2 - X^2 - Y^2 - W^2$, every point with $\zeta < 0$ is spacelike.

Proof. The coordinate formula is (296). Since $X = Y = 0$,

$$\frac{X^2 + Y^2 + W^2}{T^2} = \frac{(1 - \zeta)^2}{(1 + \zeta)^2}.$$

Solving

$$\frac{(1 - \zeta)^2}{(1 + \zeta)^2} = \frac{p + 1}{3p - 1}$$

for p gives (300), proving (301). Direct substitution gives

$$p(-1/4) = 17/33, \quad p(0) = 1.$$

The equation $p(\zeta) = 2/3$ is

$$\zeta^2 + 8\zeta + 1 = 0.$$

Its root in $[-1/4, 0)$ is $-4 + \sqrt{15}$. Differentiation gives

$$p'(\zeta) = \frac{4(1 - \zeta^2)}{(1 - 4\zeta + \zeta^2)^2} > 0$$

throughout the stated interval.

For $\zeta_* = -n/(n+1)^2$ with $n = 18189$,

$$p(\zeta_*) = \frac{(n+1)^4 + n^2}{(n+1)^4 + 4n(n+1)^2 + n^2}.$$

Exact integer evaluation gives (302); its numerator and denominator are coprime.

The spatial part of L_{XW} exchanges the X - and W -axes and changes the sign of the Y -axis. Its determinant is 1, it fixes T , and it preserves $X^2 + Y^2 + W^2 - T^2$, proving the group assertion. The definition of ι in (87) then gives (304). Finally,

$$\begin{aligned} & q\left(\frac{m^2}{2}(1+\zeta), \frac{m^2}{\sqrt{3}}(1-\zeta), 0\right) \\ &= \frac{m^4}{4}(1+\zeta)^2 - \frac{m^4}{4}(1-\zeta)^2 = m^4\zeta, \end{aligned}$$

which proves (305). $\square$

Proposition 9.49 (The arithmetic cubic family and its Lorentz discriminant). *For $\zeta \in \mathbb{R}$, define*

$$C_\zeta(U, V) := U^2V + \zeta V^3.$$

For $\mathbf{u} = (U, V)^\top$,

$$m^2 C_\zeta(U, V) = V \mathbf{u}^\top J_m(\zeta) \mathbf{u}. \quad (306)$$

Its binary-cubic discriminant is

$$\text{Disc}(C_\zeta) = -4\zeta, \quad (307)$$

and

$$\mathcal{N}(J_m(\zeta)) = \frac{m^4}{4} \text{Disc}(C_\zeta). \quad (308)$$

For $m = 2$,

$$\mathcal{N}(J_2(\zeta)) = 4 \text{Disc}(C_\zeta), \quad F_D(\Psi_{\text{circ}}(J_2(\zeta))) = 8 \text{Disc}(C_\zeta).$$

At $\zeta = -1/4$,

$$C_\zeta = C, \quad \text{Disc}(C) = 1.$$

Thus the common Fable cubic is the discriminant-one endpoint of the arithmetic carrier family.

Proof. Equation (294) gives

$$\mathbf{u}^\top J_m(\zeta) \mathbf{u} = m^2 U^2 + m^2 \zeta V^2.$$

Multiplication by V proves (306). For a binary cubic

$$aU^3 + bU^2V + cUV^2 + dV^3,$$

the discriminant is

$$b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd.$$

Substitution of

$$(a, b, c, d) = (0, 1, 0, \zeta)$$

gives (307). Since

$$\mathcal{N}(J_m(\zeta)) = -m^4\zeta,$$

equation (308) follows. The Descartes relation is (91). Substitution of $\zeta = -1/4$ proves the final statements. $\square$

Theorem 9.50 (Complete Fable fibre over the target-carrier line). *Let $r \in \mathbb{C}^\times$. Define*

$$\begin{aligned} p_0(r) &= (0, 0, -r^2), \\ p_+(r) &= \left(\frac{1}{2r}, -3r, 26r^2\right), \\ p_-(r) &= \left(-\frac{1}{2r}, 3r, 26r^2\right). \end{aligned} \quad (309)$$

Then

$$F(p_0(r)) = F(p_+(r)) = F(p_-(r)) = (-r^2, 0, 0). \quad (310)$$

These are the complete three points of the fibre. Each point is reduced.

The common invariant cubic is

$$C_{-r^2}(U, V) = U^2V - r^2V^3 = V(U - rV)(U + rV), \quad (311)$$

and the three resultant-one marked factorizations supplied by the invariant lift are

$$\begin{aligned} L_0 &= V, & Q_0 &= U^2 - r^2V^2, \\ L_+ &= \frac{U - rV}{2r}, & Q_+ &= 2rV(U + rV), \\ L_- &= -\frac{U + rV}{2r}, & Q_- &= -2rV(U - rV). \end{aligned} \quad (312)$$

Proof. For $p_+(r)$, one has

$$xy = -\frac{3}{2}, \quad 1 + xy = -\frac{1}{2}, \quad 4 + 3xy = -\frac{1}{2}.$$

Substitution in the three displayed coordinates of F gives

$$\begin{aligned} F_1(p_+(r)) &= -\frac{1}{8}(26r^2) + 9r^2 \left(-\frac{1}{2}\right) \left(-\frac{1}{2}\right) = -r^2, \\ F_2(p_+(r)) &= -3r + \frac{39}{4}r - \frac{27}{4}r = 0, \\ F_3(p_+(r)) &= \frac{1}{r} + \frac{9}{4r} - \frac{13}{4r} = 0. \end{aligned}$$

The involution $\sigma(x, y, z) = (-x, -y, z)$ sends $p_+(r)$ to $p_-(r)$ and changes only the signs of F_2, F_3 , so (310) follows for $p_-(r)$. Substitution of $p_0(r)$ is immediate.

The products in (312) are all (311). Under the resultant definition (56), their resultants are all one. Conversely, the resultant-one invariant lift of any point in the fibre gives a resultant-one factorization of C_{-r^2} . Since $r \neq 0$, this cubic has exactly the three distinct projective linear factors

$$V, \quad U - rV, \quad U + rV.$$

Choosing the marked linear factor determines its quadratic complement up to reciprocal rescaling. Under

$$(L, Q) \mapsto (\lambda L, \lambda^{-1}Q),$$

the linear-quadratic resultant is multiplied by λ . Hence the resultant-one requirement fixes the rescaling uniquely, giving exactly (312).

The source coordinates are recovered from the lift coefficients by

$$x = a, \quad 1 + xy = b.$$

When $a \neq 0$,

$$y = \frac{b-1}{a}, \quad z = \frac{2(1 - \frac{3}{2}xy - c)}{x^2}.$$

These formulas recover $p_+(r)$ and $p_-(r)$ from the second and third factorizations. In the first factorization, $a = 0, b = c = 1, d = 0, e = -r^2$; the lift formulas give $x = 0, y = 2d = 0, z = e = -r^2$. Thus the three listed points are the complete fibre. Since $\det JF = -2$, the Jacobian criterion makes every point reduced. $\square$

Corollary 9.51 (Arithmetic divisor orbits as explicit Fable fibres). *For positive N, d , put*

$$r_d := \frac{\sqrt{Nd}}{N+d}. \quad (313)$$

Then

$$\zeta_d = -r_d^2,$$

and the divisor-complement involution satisfies

$$r_{N^2/d} = r_d. \quad (314)$$

Consequently every arithmetic carrier point $\mathcal{J}_{p,a}(d)$ determines the complete Fable fibre

$$\{p_0(r_d), p_+(r_d), p_-(r_d)\}$$

over the target

$$(\zeta_d, 0, 0).$$

The branch value $d = N$ gives $r_d = 1/2$ and recovers the rational fibre in theorem 7.1.

Proof. The first identity is the definition

$$\zeta_d = -\frac{Nd}{(N+d)^2}.$$

Direct calculation gives

$$r_{N^2/d} = \frac{\sqrt{N^3/d}}{N + N^2/d} = \frac{\sqrt{Nd}}{N + d} = r_d.$$

Apply theorem 9.50 with $r = r_d$. □

Theorem 9.52 (The Fable coordinate is the half-rapidity of the Erdős–Straus null vector). *Let $d > 0$, $e = N^2/d$, and use the Sarnak coordinates*

$$T = \frac{d+e}{2}, \quad X = N, \quad Y = 0, \quad W = \frac{d-e}{2}.$$

Put

$$s = \frac{1}{2} \log \frac{d}{N}, \quad x_d = \frac{N+d}{2\sqrt{Nd}}. \quad (315)$$

Then

$$\begin{aligned} \frac{T}{N} &= \cosh(2s) = 2x_d^2 - 1, \\ \frac{W}{N} &= \sinh(2s), \\ x_d &= \cosh s, \\ r_d &= \frac{1}{2x_d}, \quad \zeta_d = -\frac{1}{4x_d^2}. \end{aligned} \quad (316)$$

The two finite points of the corresponding Fable fibre can therefore be written

$$\left(x_d, -\frac{3}{2x_d}, \frac{13}{2x_d^2}\right), \quad \left(-x_d, \frac{3}{2x_d}, \frac{13}{2x_d^2}\right). \quad (317)$$

Moreover,

$$\frac{W}{T+X} = \frac{d-N}{d+N} = \tanh s. \quad (318)$$

The divisor-complement involution $d \leftrightarrow e$ sends

$$s \mapsto -s, \quad W \mapsto -W,$$

and fixes x_d, r_d, ζ_d and the three-point Fable fibre as a set. The Fable involution $\sigma(x, y, z) = (-x, -y, z)$ exchanges the two points in (317) while fixing the arithmetic orientation W . These are two distinct involutions.

Proof. Writing $d/N = e^{2s}$ gives $e/N = e^{-2s}$, so

$$\frac{T}{N} = \frac{e^{2s} + e^{-2s}}{2} = \cosh(2s), \quad \frac{W}{N} = \frac{e^{2s} - e^{-2s}}{2} = \sinh(2s).$$

Also

$$x_d = \frac{1 + d/N}{2\sqrt{d/N}} = \frac{e^s + e^{-s}}{2} = \cosh s.$$

The double-angle identity gives $T/N = 2x_d^2 - 1$. Equation (313) gives

$$r_d = \frac{\sqrt{d/N}}{1 + d/N} = \frac{1}{e^s + e^{-s}} = \frac{1}{2x_d},$$

and hence $\zeta_d = -r_d^2 = -1/(4x_d^2)$. Substitution in (309) proves (317).

Finally,

$$\frac{W}{T+X} = \frac{d-e}{d+e+2N} = \frac{d-N}{d+N} = \frac{e^s - e^{-s}}{e^s + e^{-s}} = \tanh s.$$

The two stated involution actions follow directly from these formulas and from σ . □

Proposition 9.53 (Blade parity equals the Fable splitting squareclass). *Let*

$$N = \prod_{i=1}^k q_i^{\alpha_i}, \quad d = \prod_{i=1}^k q_i^{\delta_i} \mid N^2, \quad \beta_i = \delta_i - \alpha_i,$$

where the q_i are distinct primes. Put

$$\pi(\beta) = \{i : \beta_i \equiv 1 \pmod{2}\}, \quad s_\beta = \prod_{i \in \pi(\beta)} q_i.$$

Then

$$[Nd] = [s_\beta] \quad \text{in} \quad \mathbb{Q}^\times / (\mathbb{Q}^\times)^2. \quad (319)$$

Consequently

$$\mathbb{Q}(r_d) = \mathbb{Q}(\sqrt{s_\beta}), \quad (320)$$

with the understanding that the field is $\mathbb{Q}$ when $s_\beta = 1$. Complementation sends β to $-\beta$ and preserves $\pi(\beta)$, s_β , and the splitting field.

Proof. For every i ,

$$\alpha_i + \delta_i \equiv \delta_i - \alpha_i = \beta_i \pmod{2}.$$

Removing even exponents from

$$Nd = \prod_i q_i^{\alpha_i + \delta_i}$$

gives (319). Since

$$r_d = \frac{\sqrt{Nd}}{N+d}$$

and the denominator is rational, (320) follows. If $d' = N^2/d$, then its centered exponent vector is

$$(2\alpha_i - \delta_i) - \alpha_i = -\beta_i.$$

Parity is unchanged under negation. □

Theorem 9.54 (The shell congruence is a square root of minus one in the blade field). *Use the notation of theorem 9.53. Let*

$$\eta_i \in \{0, 1\}, \quad \eta_i \equiv \beta_i \pmod{2}, \quad \gamma_i = \frac{\beta_i - \eta_i}{2},$$

and put

$$t_\beta = \prod_i q_i^{\gamma_i}, \quad \nu_\beta = t_\beta \sqrt{s_\beta}.$$

Then

$$\frac{d}{N} = t_\beta^2 s_\beta = \nu_\beta^2. \quad (321)$$

Assume that R is an odd prime, $\gcd(R, N) = 1$, and

$$d \equiv -N \pmod{R}.$$

For every prime $\mathfrak{R}$ of $\mathbb{Q}(\sqrt{s_\beta})$ above R ,

$$\bar{\nu}_\beta^2 = -1 \quad \text{in the residue field at } \mathfrak{R}. \quad (322)$$

Consequently

$$\left(\frac{s_\beta}{R}\right) = \left(\frac{-1}{R}\right). \quad (323)$$

If $R \equiv 3 \pmod{4}$, then s_β is a quadratic nonresidue modulo R , the prime R is inert in $\mathbb{Q}(\sqrt{s_\beta})$, and the residue field is $\mathbb{F}_{R^2}$. In that field,

$$\bar{\nu}_\beta^R = -\bar{\nu}_\beta = \bar{\nu}_\beta^{-1}. \quad (324)$$

The divisor-complement eigenvalue is ν_β^{-1} , so its reduction is the Frobenius conjugate of $\bar{\nu}_\beta$.

Proof. Since $\beta_i = 2\gamma_i + \eta_i$,

$$\frac{d}{N} = \prod_i q_i^{\beta_i} = \left(\prod_i q_i^{\gamma_i}\right)^2 \prod_i q_i^{\eta_i} = t_\beta^2 s_\beta,$$

which proves (321). Reduction of $d/N \equiv -1 \pmod{R}$ proves (322). Because t_β is a nonzero square modulo R , Euler's criterion gives (323).

For $R \equiv 3 \pmod{4}$, the right side of (323) is -1 . Hence the quadratic polynomial defining the blade field is irreducible modulo R , so the residue degree is two. The element $\bar{\nu}_\beta$ is a root of $X^2 + 1$ outside $\mathbb{F}_R$. Frobenius permutes the two roots and therefore sends it to $-\bar{\nu}_\beta$. Equation $\bar{\nu}_\beta^2 = -1$ also gives $\bar{\nu}_\beta^{-1} = -\bar{\nu}_\beta$, proving (324). Complementation replaces d/N by N/d and hence replaces the half-eigenvalue by its inverse. $\square$

Theorem 9.55 (The arithmetic half-boost specializes to the Fable inversion matrix on the shell).
Let $N, d > 0$, put $e = N^2/d$, and choose the positive square root $\rho_d = \sqrt{Nd}$. Define

$$S_d := \frac{1}{2\rho_d} \begin{pmatrix} N+d & d-N \\ d-N & N+d \end{pmatrix}, \quad B_d := \frac{1}{N} \begin{pmatrix} T & W \\ W & T \end{pmatrix}, \quad (325)$$

where

$$T = \frac{d+e}{2}, \quad W = \frac{d-e}{2}.$$

Then

$$\det S_d = 1, \quad S_d^2 = B_d. \quad (326)$$

The vectors $(1, 1)^\top$ and $(1, -1)^\top$ are eigenvectors of S_d , with the ordered eigenvalues

$$\nu_d := \frac{\rho_d}{N} = \sqrt{\frac{d}{N}}, \quad \nu_d^{-1} = \frac{\rho_d}{d} = \sqrt{\frac{N}{d}}. \quad (327)$$

With the compatible positive square root for e , divisor complementation satisfies

$$S_e = S_d^{-1}. \quad (328)$$

Put $x_d = \frac{1}{2} \operatorname{tr} S_d$. The two finite points in the Fable fibre attached to d are recovered directly from the two square roots S_d and $-S_d$ of B_d :

$$\begin{aligned} \Phi(S_d) &:= \left(\frac{\operatorname{tr} S_d}{2}, -\frac{3}{\operatorname{tr} S_d}, \frac{26}{(\operatorname{tr} S_d)^2} \right), \\ \Phi(-S_d) &= \sigma(\Phi(S_d)), \quad \sigma(x, y, z) = (-x, -y, z). \end{aligned} \quad (329)$$

The divisor-complement involution $S_d \mapsto S_d^{-1}$ preserves the trace, whereas the Fable-sheet involution $S_d \mapsto -S_d$ changes its sign. Thus these involutions are distinct over the characteristic-zero arithmetic field.

Assume now that R is an odd prime, $\gcd(R, N) = 1$, and $d \equiv -N \pmod{R}$. Let $\mathfrak{R}$ be a prime above R in a field containing ν_d . Reduction at $\mathfrak{R}$ gives

$$\bar{\nu}_d^2 = -1, \quad \bar{S}_d = P_{\bar{\nu}_d} := \begin{pmatrix} 0 & \bar{\nu}_d \\ \bar{\nu}_d & 0 \end{pmatrix}, \quad (330)$$

and therefore

$$\bar{S}_d^2 = -I_2, \quad \bar{S}_d^{-1} = -\bar{S}_d. \quad (331)$$

Consequently the divisor-complement and Fable-sheet operations on the half-boost coincide after reduction on the shell target. At the same target,

$$\operatorname{tr} \bar{S}_d = 0.$$

Hence the affine formula Φ , which divides by $\operatorname{tr} S_d$, has no residue-field value there. The preceding identification is an equality of the two operations on the matrix lift; an interpretation of the corresponding Fable source points requires the projective closure of the affine fibre.

Let the residue-field involution send $\bar{\nu}_d$ to $-\bar{\nu}_d$, and use its transpose-conjugate as $\dagger$. For

$$K_N = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}, \quad K_{N^\top} = \begin{pmatrix} -1 & 0 \\ 0 & 4 \end{pmatrix},$$

one has the exact residual congruence

$$(P_{\bar{\nu}_d}^{-1})^\dagger K_N P_{\bar{\nu}_d}^{-1} = K_{N^\top}. \quad (332)$$

Thus the shell half-boost is the residue-field form of the same SL_2 inversion matrix that exchanges the directed Fable branch forms in theorem 9.12.

Proof. Direct calculation gives

$$\det S_d = \frac{(N+d)^2 - (d-N)^2}{4Nd} = 1.$$

The diagonal and off-diagonal entries of S_d^2 are respectively

$$\frac{(N+d)^2 + (d-N)^2}{4Nd} = \frac{d^2 + N^2}{2Nd} = \frac{T}{N}$$

and

$$\frac{2(N+d)(d-N)}{4Nd} = \frac{d^2 - N^2}{2Nd} = \frac{W}{N},$$

which proves (326). The two displayed eigenvectors give the eigenvalues

$$\frac{(N+d) + (d-N)}{2\rho_d} = \frac{d}{\rho_d} = \frac{\rho_d}{N}$$

and

$$\frac{(N+d) - (d-N)}{2\rho_d} = \frac{N}{\rho_d} = \frac{\rho_d}{d}.$$

This proves (327). Replacing d by $e = N^2/d$ changes the off-diagonal entry of S_d to its negative and preserves its diagonal entry. Since $\det S_d = 1$, the resulting matrix is S_d^{-1} , proving (328).

The trace identity

$$\operatorname{tr} S_d = \frac{N + d}{\rho_d} = 2x_d$$

and $r_d = 1/(2x_d)$ transform (309) into (329). For every 2×2 matrix of determinant one, $\operatorname{tr}(S^{-1}) = \operatorname{tr} S$. This proves the two characteristic-zero involution statements.

At a shell solution, write $\bar{\rho}_d = N\bar{\nu}_d$. Since $d \equiv -N \pmod{R}$, reduction of (325) gives

$$\bar{S}_d = \frac{1}{2N\bar{\nu}_d} \begin{pmatrix} 0 & -2N \\ -2N & 0 \end{pmatrix} = \begin{pmatrix} 0 & \bar{\nu}_d \\ \bar{\nu}_d & 0 \end{pmatrix},$$

where the last equality uses $\bar{\nu}_d^{-1} = -\bar{\nu}_d$. This proves (330) and (331).

Finally, $P_{\bar{\nu}_d}^{-1} = -P_{\bar{\nu}_d}$ and $P_{\bar{\nu}_d}^\dagger = -P_{\bar{\nu}_d}$. Therefore

$$(P_{\bar{\nu}_d}^{-1})^\dagger K_N P_{\bar{\nu}_d}^{-1} = -P_{\bar{\nu}_d} K_N P_{\bar{\nu}_d} = K_{N^\tau},$$

which proves (332). $\square$

Corollary 9.56 (The finite Fable pair is the central two-sheet of the SL_2 Lorentz lift). *Over $K_d = \mathbb{Q}(\sqrt{Nd})$, the matrix S_d of theorem 9.55 represents the projective transformation*

$$m_d(z) = \frac{(N + d)z + (d - N)}{(d - N)z + (N + d)}. \quad (333)$$

The two matrices S_d and $-S_d$ have the same projective action, and their two images under Φ are exactly the two finite points of the Fable fibre. Divisor complementation satisfies

$$m_{N^2/d} = m_d^{-1}. \quad (334)$$

After reduction on the shell $d \equiv -N \pmod{R}$,

$$\bar{m}_d(z) = \frac{1}{z}, \quad [\bar{S}_d]^2 = [I_2] \quad \text{in } \mathrm{PSL}_2. \quad (335)$$

This is the projective inversion represented by $P_{\bar{\nu}_d}$ in (330), and it is the residue-field form of the inversion in (148).

The third Fable point p_0 is not a second lift of this projective transformation. In the resultant-one factorization of theorem 9.50, it is the point associated with the marked factor V ; the other two points are associated with $U - rV$ and $U + rV$. Hence the exact three-point structure is one fixed marked factor together with one central two-sheet.

Proof. The scalar factor $1/(2\sqrt{Nd})$ in S_d cancels from its projective action, giving (333). The matrices S_d and $-S_d$ differ by the central element $-I_2$ and therefore represent the same projective map. Equation (329) identifies their two affine Fable points. Equation (328) gives (334).

On the shell, $N + d \equiv 0$ and $d - N \equiv -2N$, so

$$\bar{m}_d(z) = \frac{-2N}{-2Nz} = \frac{1}{z}.$$

Equation (331) gives $\bar{S}_d^2 = -I_2$, whose projective class is the identity. The last assertion is the three-factor list in (312). $\square$

Corollary 9.57 (The completed lower sign is the residual quadratic character). *Let g be a primitive generator modulo the odd prime R , and write*

$$\Lambda(\beta) = \log_g(d/N) \pmod{R-1}.$$

Then

$$(-1)^{q^\vee(\beta)} = (-1)^{\Lambda(\beta)} = \left(\frac{d/N}{R}\right). \quad (336)$$

On the shell target $d/N \equiv -1 \pmod{R}$,

$$(-1)^{q^\vee(\beta)} = \left(\frac{-1}{R}\right). \quad (337)$$

Proof. Euler's criterion gives

$$\left(\frac{g^k}{R}\right) = (g^{(R-1)/2})^k = (-1)^k.$$

Take $k = \Lambda(\beta)$. □

Theorem 9.58 (The champion blade is the squareclass of the Fable splitting field). *For the champion carrier point,*

$$18189 = 3^2 \cdot 43 \cdot 47, \quad r_* = \frac{\sqrt{18189}}{18190} = \frac{3\sqrt{2021}}{18190}.$$

Hence the corresponding Fable fibre is defined over

$$K_* = \mathbb{Q}(\sqrt{2021}) = \mathbb{Q}(\sqrt{43 \cdot 47})$$

and is

$$\begin{aligned} p_{0,*} &= \left(0, 0, -\frac{18189}{18190^2}\right), \\ p_{+,*} &= \left(\frac{18190}{6\sqrt{2021}}, -\frac{9\sqrt{2021}}{18190}, \frac{26 \cdot 18189}{18190^2}\right), \\ p_{-,*} &= \left(-\frac{18190}{6\sqrt{2021}}, \frac{9\sqrt{2021}}{18190}, \frac{26 \cdot 18189}{18190^2}\right). \end{aligned} \quad (338)$$

The nontrivial element of $\text{Gal}(K_*/\mathbb{Q})$ fixes $p_{0,*}$, exchanges $p_{+,*}$ and $p_{-,*}$, and agrees on this fibre with the Fable involution $\sigma(x, y, z) = (-x, -y, z)$.

Let the blade generators o_1, o_2, o_3, o_4 be indexed by

$$(q_1, q_2, q_3, q_4) = (3, 11, 43, 47),$$

and define the blade-index squareclass map

$$\kappa(I) := \left[\prod_{i \in I} q_i\right] \in \mathbb{Q}^\times / (\mathbb{Q}^\times)^2. \quad (339)$$

For the two champion target vectors,

$$\pi(\beta^\pm) = \{3, 4\},$$

and therefore

$$\kappa(\pi(\beta^\pm)) = [43 \cdot 47] = [2021]. \quad (340)$$

Thus the blade o_{34} records exactly the quadratic squareclass that splits the champion Fable cubic.

Proof. The factorization of 18189 gives the displayed value of r_* . Substitution in (309) proves (338). The Galois action sends $\sqrt{2021}$ to $-\sqrt{2021}$, which changes the signs of the first two coordinates and fixes the third.

The parity computation $\pi(\beta^\pm) = \{3, 4\}$ is (573). Applying (339) gives (340). The asserted identification is the squareclass map on blade indices; no homomorphism from the full sedenion algebra to K_* is used. □

Theorem 9.59 (Exact $3 + 1$ inversion spectrum at the quarter branch). *Let*

$$N := \mathcal{M}(53) = 2E_{21}, \quad P_s := N^\top N, \quad P_r := NN^\top.$$

Then

$$P_s = 4E_{11}, \quad P_r = 4E_{22}. \quad (341)$$

The directed Gram-support difference from theorem 9.11

$$H_- := P_s - \frac{1}{4}P_r = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix} = H(w_N) \quad (342)$$

is the nondegenerate circle $4(x^2 + y^2) - 1 = 0$. Its full coefficient inversion matrix is

$$\mathcal{I}_- = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ \frac{1}{16} & 0 & 0 & 0 \end{pmatrix}. \quad (343)$$

It satisfies

$$\mathcal{I}_-^2 = \frac{1}{16}I_4, \quad \det(\lambda I_4 - \mathcal{I}_-) = \left(\lambda - \frac{1}{4}\right)^3 \left(\lambda + \frac{1}{4}\right). \quad (344)$$

Its eigenspaces are

$$E_{+1/4} = H_-^\perp = \{(a, b_0, b_1, c) : c = a/4\}, \quad \dim E_{+1/4} = 3, \quad (345)$$

and

$$E_{-1/4} = \mathbb{R}(4, 0, 0, -1), \quad \dim E_{-1/4} = 1. \quad (346)$$

The three vectors

$$u_0 = (4, 0, 2, 1), \quad u_+ = (4, \sqrt{3}, -1, 1), \quad u_- = (4, -\sqrt{3}, -1, 1) \quad (347)$$

belong to $E_{+1/4}$, are null for $\mathcal{N}$, and form one C_3 -orbit. The associated point centres are

$$\left(0, \frac{1}{2}\right), \quad \left(-\frac{\sqrt{3}}{4}, -\frac{1}{4}\right), \quad \left(\frac{\sqrt{3}}{4}, -\frac{1}{4}\right). \quad (348)$$

Thus the inversion coefficient space has a $3 + 1$ splitting with spectrum

$$\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}; -\frac{1}{4}\right). \quad (349)$$

For the transposed blade channel $N^\vee = 2E_{12}$, the same ordered construction gives

$$H_+ = (N^\vee)^\top N^\vee - \frac{1}{4}N^\vee(N^\vee)^\top = \begin{pmatrix} -1 & 0 \\ 0 & 4 \end{pmatrix}, \quad (350)$$

whose inversion matrix is

$$\mathcal{I}_+ = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 16 & 0 & 0 & 0 \end{pmatrix} \quad (351)$$

and whose spectrum is

$$(4, 4, 4; -4). \quad (352)$$

The arithmetic coefficient in (131) therefore selects the quarter spectrum; the reciprocal mirror has coefficient zero at residue 53.

Proof. Equation (341) follows by multiplying $N = 2E_{21}$ with its transpose in the two possible orders. Equation (342) follows directly. For $w = (A, 0, 0, C) = (4, 0, 0, -1)$, the full inversion matrix (29) becomes (343). Direct multiplication gives $\mathcal{I}_-^2 = I_4/16$. The b_0 - and b_1 -coordinate vectors are $+1/4$ -eigenvectors. The remaining a, c block is

$$\begin{pmatrix} 0 & 1 \\ 1/16 & 0 \end{pmatrix},$$

with eigenvalues $+1/4$ and $-1/4$. This proves (344).

The bilinear pairing with $w = (4, 0, 0, -1)$ is

$$B((a, b_0, b_1, c), w) = \frac{a - 4c}{2}.$$

Its kernel is $c = a/4$, which is exactly the $+1/4$ -eigenspace. The normal vector w has eigenvalue $-1/4$, proving (345) and (346).

For each vector in (347), one has $b_0^2 + b_1^2 = 4 = ac$, so its norm is zero; its fourth coordinate is one quarter of its first, so it lies in $E_{+1/4}$. On coefficient vectors define

$$\mathcal{R} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathcal{J} = \text{diag}(1, -1, 1, 1).$$

Then

$$\mathcal{R}^3 = I_4, \quad \mathcal{J}^2 = I_4, \quad \mathcal{J}\mathcal{R}\mathcal{J} = \mathcal{R}^{-1},$$

and direct multiplication gives

$$\mathcal{R}u_0 = u_+, \quad \mathcal{R}u_+ = u_-, \quad \mathcal{R}u_- = u_0.$$

Both $\mathcal{R}$ and $\mathcal{J}$ commute with $\mathcal{I}_-$. The point centres in (348) follow from $(-b_0/a, b_1/a)$.

For $N^\vee = 2E_{12}$, the two products in (341) are exchanged. This gives (350). Substitution of $(A, B_0, B_1, C) = (-1, 0, 0, 4)$ into the full coefficient formula gives (351); its a, c block has eigenvalues ± 4 , and its two middle coordinates each have eigenvalue 4. $\square$

Theorem 9.60 (Ordered radius-square family). *Retain the two ordered matrices*

$$N = 2E_{21}, \quad N^\vee = 2E_{12},$$

and let $\alpha > 0$. Define

$$H_\alpha := N^\top N - \alpha N N^\top = \begin{pmatrix} 4 & 0 \\ 0 & -4\alpha \end{pmatrix}, \quad (353)$$

$$H_\alpha^\vee := (N^\vee)^\top N^\vee - \alpha N^\vee (N^\vee)^\top = \begin{pmatrix} -4\alpha & 0 \\ 0 & 4 \end{pmatrix}. \quad (354)$$

The first matrix is the circle

$$4(x^2 + y^2) - 4\alpha = 0$$

of squared radius α . Substitution of its four coefficients $(4, 0, 0, -4\alpha)$ into the complete coefficient-inversion transformation gives

$$\mathcal{I}_\alpha = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & \alpha & 0 & 0 \\ 0 & 0 & \alpha & 0 \\ \alpha^2 & 0 & 0 & 0 \end{pmatrix}. \quad (355)$$

Its exact invariants are

$$\mathcal{I}_\alpha^2 = \alpha^2 I_4, \quad \text{tr}(\mathcal{I}_\alpha) = 2\alpha, \quad \det(\mathcal{I}_\alpha) = -\alpha^4, \quad (356)$$

$$\det(\lambda I_4 - \mathcal{I}_\alpha) = (\lambda - \alpha)^3(\lambda + \alpha). \quad (357)$$

The spectral projectors are

$$\Pi_{\alpha,+} = \frac{1}{2} \left(I_4 + \alpha^{-1} \mathcal{I}_\alpha \right), \quad \Pi_{\alpha,-} = \frac{1}{2} \left(I_4 - \alpha^{-1} \mathcal{I}_\alpha \right), \quad (358)$$

with ranks 3 and 1, respectively. More explicitly,

$$E_{\alpha,+} = \{(a, b_0, b_1, c) : c = \alpha a\} = H_\alpha^\perp, \quad (359)$$

$$E_{\alpha,-} = \mathbb{R}(4, 0, 0, -4\alpha). \quad (360)$$

The mirror matrix $H_\alpha^\vee$ is the circle

$$-4\alpha(x^2 + y^2) + 4 = 0$$

of squared radius α^{-1} . Its complete coefficient-inversion matrix is

$$\mathcal{I}_\alpha^\vee = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & \alpha^{-1} & 0 & 0 \\ 0 & 0 & \alpha^{-1} & 0 \\ \alpha^{-2} & 0 & 0 & 0 \end{pmatrix}, \quad (361)$$

and

$$\det(\lambda I_4 - \mathcal{I}_\alpha^\vee) = (\lambda - \alpha^{-1})^3(\lambda + \alpha^{-1}), \quad (362)$$

$$E_{\alpha,+}^\vee = \{(a, b_0, b_1, c) : a = \alpha c\}, \quad E_{\alpha,-}^\vee = \mathbb{R}(-4\alpha, 0, 0, 4). \quad (363)$$

Consequently, the two ordered channels have simple eigenvalues $-\alpha$ and $-\alpha^{-1}$. Prescribing the simple eigenvalue $-1/4$ in the observed outgoing exit-boundary E_{21} channel of theorem 14.69 determines $\alpha = 1/4$, and then

$$\text{Spec}(\mathcal{I}_{1/4}) = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{4} \right), \quad \text{Spec}(\mathcal{I}_{1/4}^\vee) = (4, 4, 4, -4). \quad (364)$$

Proof. The products $N^\top N = 4E_{11}$ and $NN^\top = 4E_{22}$ give (353). Transposition exchanges these two products and gives (354). The full equation (42), equivalently the coefficient matrix (29), applied first with

$$(A, B_0, B_1, C) = (4, 0, 0, -4\alpha),$$

gives (355). Matrix multiplication proves (356). The b_0 - and b_1 -coordinate axes have eigenvalue α ; the a, c block

$$\begin{pmatrix} 0 & 1 \\ \alpha^2 & 0 \end{pmatrix}$$

has eigenvalues α and $-\alpha$. This proves (357) and (358). For $w_\alpha = (4, 0, 0, -4\alpha)$, the coefficient bilinear form satisfies

$$B((a, b_0, b_1, c), w_\alpha) = 2(\alpha a - c).$$

Its kernel is (359), and $\mathcal{I}_\alpha w_\alpha = -\alpha w_\alpha$, proving (360).

Applying (43) to

$$(A, B_0, B_1, C) = (-4\alpha, 0, 0, 4)$$

gives (361). Its a, c block is

$$\begin{pmatrix} 0 & 1 \\ \alpha^{-2} & 0 \end{pmatrix},$$

and its middle two eigenvalues are both α^{-1} . This proves (362) and (363). Finally, $-\alpha = -1/4$ has the unique positive solution $\alpha = 1/4$, which yields (364). $\square$

The three null vectors are the exact image of the Fable point-circle vectors in (94) under

$$\mathcal{D} = \text{diag}(4, 2, 2, 1). \quad (365)$$

Indeed, $\mathcal{D}v_0 = u_0$, $\mathcal{D}v_+ = u_+$, and $\mathcal{D}v_- = u_-$, while

$$\mathcal{N}(\mathcal{D}v) = 4\mathcal{N}(v). \quad (366)$$

The unit-circle coefficient vector $(1, 0, 0, -1)$ is sent to $(4, 0, 0, -1)$. Hence the Fable C_3 -orbit and its containing circle map to the three null eigenrays and the simple normal eigenline in Theorem 9.59. The simple eigenvalue is the common Fable output and the denominator-exchange branch value:

$$-\frac{1}{4} = F_1(p_0) = F_1(p_+) = F_1(p_-) = \zeta_{\text{branch}}.$$

Provenance record 9.61 (Directed blade source). The four tag polynomials and the target coefficient in Theorem 9.9 are taken from The Clankers, *Integrated Completed Tesserine Sign Algebra, Circular Completion, and Blade-One Positivity for the 107-Residual Erdős–Straus Shell* [17]. The local source is `BEAVERSHINE/11/completed_supersignum_tesserine_blade1_integrated_paper.tex`. The matrix packaging, ordered Gram construction, and the exact $3 + 1$ spectral calculation are proved above.

Remark 9.62 (The codimension-one adelic lamination). Let $S = \{\infty, p, q, r\}$. For the semilocal adele-class construction of Connes and Consani [30], one has

$$\Gamma \cong \mathbb{Z}^3 \times \{\pm 1\}, \quad \mathcal{Y}_S = \Gamma \backslash (\mathbb{A}_S \times \mathbb{R}^3).$$

Their codimension-one lamination theorem gives

$$\dim \mathcal{Y}_S = 4, \quad \dim(\mathbb{R}^3\text{-leaf}) = 3.$$

Set

$$\Gamma^+ = \mathbb{Z}^3 \times \{+1\}, \quad \tilde{\mathcal{Y}}_S = \Gamma^+ \backslash (\mathbb{A}_S \times \mathbb{R}^3).$$

The residual quotient group $\Gamma/\Gamma^+ \cong C_2$ acts by

$$\iota[a, t] = [-a, t], \quad \mathcal{Y}_S = \tilde{\mathcal{Y}}_S / \langle \iota \rangle. \quad (367)$$

The action has two-element orbits when $a \neq 0$. Its fixed locus is the image of $\{0\} \times \mathbb{R}^3$: a fixed class would require $(-a, t) = (\gamma a, t + \ell(\gamma))$ for some $\gamma \in \Gamma^+$, and equality of the $\mathbb{R}^3$ -coordinates forces $\gamma = 1$, after which $2a = 0$ forces $a = 0$. Thus the class-field construction supplies an exact $3 + 1$ -dimensional space together with a generically two-sheeted C_2 -quotient and an explicit fixed stratum. Its coordinates determine the dimensional and sheet structure. A coefficient-space realization consists of a specified map from the three translation coordinates to $H_-^\perp$ and from the transverse coordinate to the line $\mathbb{R}H_-$.

Theorem 9.63 (Cyclic-cubic cover with the quarter $3 + 1$ coefficient fibre). *Let $L/\mathbb{Q}$ be a cyclic cubic extension, let*

$$G = \text{Gal}(L/\mathbb{Q}) = \langle g \rangle \cong C_3,$$

and let

$$\pi^L : X_{\mathbb{Q}}^L \longrightarrow X_{\mathbb{Q}}$$

be the finite abelian cover constructed by Connes and Consani [30]. Define its principal locus by

$$U := \{x \in X_{\mathbb{Q}} : \text{Stab}_G(y) = \{1\} \text{ for } y \in (\pi^L)^{-1}(x)\}$$

and put $P = (\pi^L)^{-1}(U)$. Thus $P \rightarrow U$ is a principal C_3 -cover.

On $V = \mathbb{R}^4$, use the coefficient rotation

$$\mathcal{R} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathcal{R}^3 = I_4, \quad (368)$$

and define the representation $\rho : G \rightarrow \mathrm{GL}(V)$ by $\rho(g) = \mathcal{R}$. The associated coefficient bundle

$$\mathcal{V}_L := P \times_G V, \quad (xg, v) \sim (x, \mathcal{R}v), \quad (369)$$

carries the descended endomorphism induced by $\mathcal{I}_{1/4}$. It splits as

$$\mathcal{V}_L = \mathcal{V}_{L,+} \oplus \mathcal{V}_{L,-}, \quad \mathrm{rank} \mathcal{V}_{L,+} = 3, \quad \mathrm{rank} \mathcal{V}_{L,-} = 1, \quad (370)$$

and the descended endomorphism acts by

$$\mathcal{I}_{1/4}|_{\mathcal{V}_{L,+}} = \frac{1}{4}I, \quad \mathcal{I}_{1/4}|_{\mathcal{V}_{L,-}} = -\frac{1}{4}I. \quad (371)$$

The rank-one summand is the trivial line bundle generated by $(4, 0, 0, -1)$. Replacing g by g^{-1} produces an isomorphic bundle endomorphism.

Proof. Connes and Consani construct

$$X_{\mathbb{Q}}^L = \mathbb{Q}^{\times} \backslash \mathbb{A}_{\mathbb{Q}} / W(L), \quad W(L) = \ker(\widehat{\mathbb{Z}}^* \rightarrow G),$$

with the G -action on each fibre. On the principal locus the action is free and transitive, so $P \rightarrow U$ is a principal G -cover.

The matrices $\mathcal{R}$ and $\mathcal{I}_{1/4}$ commute by direct multiplication. Therefore the map

$$P \times V \rightarrow P \times V, \quad (x, v) \mapsto (x, \mathcal{I}_{1/4}v)$$

respects the equivalence relation in (369) and descends to $\mathcal{V}_L$. The eigenspaces

$$E_{1/4,+} = \{(a, b_0, b_1, c) : c = a/4\}, \quad E_{1/4,-} = \mathbb{R}(4, 0, 0, -1)$$

are $\mathcal{R}$ -invariant. Their associated bundles give (370) and (371). The vector $(4, 0, 0, -1)$ is fixed by $\mathcal{R}$, so its associated line bundle is trivial.

For the other generator, define

$$\mathcal{J} = \mathrm{diag}(1, -1, 1, 1).$$

Then

$$\mathcal{J}\mathcal{R}\mathcal{J} = \mathcal{R}^{-1}, \quad \mathcal{J}\mathcal{I}_{1/4} = \mathcal{I}_{1/4}\mathcal{J}.$$

The fibre map induced by $\mathcal{J}$ intertwines the two associated bundles and their descended endomorphisms. $\square$

Every divisor of $N^2 = p^2a^2$ has a unique expression

$$d = p^j u, \quad j \in \{0, 1, 2\}, \quad u \mid a^2,$$

because the coprimality hypothesis implies $p \nmid a$. Define

$$\lambda_{R,a}(u) := \bar{u} \bar{a}^{-1} \in (\mathbb{Z}/R\mathbb{Z})^{\times}$$

and

$$T_j(p) := -p^{1-j}.$$

The origin-resolved support set is

$$\mathcal{X}_{R,p} := \{(j, u) : j \in \{0, 1, 2\}, \ u \mid a^2, \ \lambda_{R,a}(u) = T_j(p)\}. \quad (372)$$

Let

$$m(R, p) = (m_0, m_1, m_2), \quad m_j := \#\{u : (j, u) \in \mathcal{X}_{R,p}\}. \quad (373)$$

Theorem 9.64 (Arithmetic realization of the singular three-origin object). *The fixed-shell identity (100) holds if and only if*

$$\Sigma(m(R, p)) = m_0 + m_1 + m_2 > 0. \quad (374)$$

Moreover the map

$$\iota(j, u) = \left(2 - j, \frac{a^2}{u}\right) \quad (375)$$

is an involution of $\mathcal{X}_{R,p}$ and therefore

$$m_0 = m_2. \quad (376)$$

The C_2 action (375) is an arithmetic symmetry, and the C_1 idempotent records whether the finite set (372) is occupied.

Proof. The congruence $d \equiv -N \pmod{R}$ for $d = p^j u$ is

$$p^j u \equiv -pa \pmod{R}.$$

Dividing by the units $p^j a$ yields

$$\bar{u} \bar{a}^{-1} = -p^{1-j} = T_j(p).$$

Thus the divisors in Theorem 9.1 are in bijection with $\mathcal{X}_{R,p}$. Its cardinality is exactly $m_0 + m_1 + m_2$, proving (374).

The map (375) squares to the identity. If $\lambda_{R,a}(u) = -p^{1-j}$, then

$$\lambda_{R,a}(a^2/u) = \bar{a} \bar{u}^{-1} = (\bar{u} \bar{a}^{-1})^{-1} = -p^{j-1} = T_{2-j}(p).$$

Hence ι preserves $\mathcal{X}_{R,p}$, exchanges the exterior fibres, and fixes the middle fibre. This proves (376). The proved involution and the nonemptiness of the certificate set give the stated C_2 and C_1 interpretations. $\square$

Proposition 9.65 (Choice and invariance in the Fable three-ray compression). *There are exactly two equivariant bijections from the arithmetic C_2 -set $\{0, 1, 2\}$ to the Fable C_2 -set $\{0, +, -\}$. One is*

$$\rho(1) = 0, \quad \rho(2) = +, \quad \rho(0) = -. \quad (377)$$

The other exchanges $+$ and $-$. The proved arithmetic C_2 -data specify a fixed point and an unordered exchanged pair, hence admit both bijections. Since $m_0 = m_2$, both choices give the same origin-compressed aggregate

$$\Lambda_{\text{orig}}(R, p) = m_1 \ell_0 + m_0(\ell_+ + \ell_-). \quad (378)$$

It satisfies

$$\begin{aligned} t(\Lambda_{\text{orig}}) &= m_0 + m_1 + m_2, \\ q(\Lambda_{\text{orig}}) &= 3(m_0 m_1 + m_1 m_2 + m_2 m_0). \end{aligned} \quad (379)$$

For $p = 12289$ this gives

$$\Lambda_{\text{orig}} = 6\ell_0 + 3\ell_+ + 3\ell_- = (12, 0, 3), \quad q(\Lambda_{\text{orig}}) = 135. \quad (380)$$

Proof. The origin $j = 1$ is fixed by $j \mapsto 2 - j$, while $j = 0$ and $j = 2$ are exchanged. Equivariance therefore forces $1 \mapsto 0$ and permits exactly the two orders of the image pair $\{+, -\}$. The two resulting weighted sums agree because (376) gives $m_0 = m_2$. Equations (379) follow from (76) and (77) after the choice $(m_0, m_+, m_-) = (m_1, m_2, m_0)$. Finally substitute $(m_0, m_1, m_2) = (3, 6, 3)$ and the three vectors (74). $\square$

Theorem 9.66 (Exact origin-cycle criterion and the 12289 obstruction). *For $j \in \{0, 1, 2\}$, put*

$$\mathcal{D}_j(R, p, a) := \left\{u : u \mid a^2, \quad \lambda_{R,a}(u) = T_j(p)\right\}, \quad \mathcal{X}_j := \{j\} \times \mathcal{D}_j(R, p, a). \quad (381)$$

The following conditions are equivalent.

(i) There is a bijection $r : \mathcal{X}_{R,p} \rightarrow \mathcal{X}_{R,p}$ satisfying

$$r(\mathcal{X}_j) = \mathcal{X}_{j+1}, \quad r^3 = \text{id}_{\mathcal{X}_{R,p}}.$$

(ii) The origin fibres have equal cardinality:

$$m_0 = m_1 = m_2. \quad (382)$$

(iii) There are bijections $\varphi_j : \mathcal{D}_j(R, p, a) \rightarrow \mathcal{D}_{j+1}(R, p, a)$ such that

$$\varphi_2 \circ \varphi_1 \circ \varphi_0 = \text{id}_{\mathcal{D}_0(R, p, a)}. \quad (383)$$

Every such divisor cycle obeys the exact residue transport law

$$\lambda_{R,a}(\varphi_j(u)) = \begin{cases} T_{j+1}(p) = p^{-1}\lambda_{R,a}(u), & j = 0, 1, \\ T_0(p) = p^2\lambda_{R,a}(u), & j = 2, \end{cases} \quad \text{in } (\mathbb{Z}/R\mathbb{Z})^\times, \quad (384)$$

The same multiplier p^{-1} occurs on all three edges precisely when

$$p^3 \equiv 1 \pmod{R}. \quad (385)$$

The divisor-complement involution of (375) gives $m_0 = m_2$, so the criterion reduces to $m_0 = m_1$.

For the 12289 shell,

$$(m_0, m_1, m_2) = (3, 6, 3). \quad (386)$$

Moreover $p \equiv 2 \pmod{11}$, so $p^3 \equiv 8 \pmod{11}$. Consequently its origin-resolved certificate set has no cyclic action of the form in (i). The full subgroup of S_3 that stabilizes its multiplicity vector is

$$\{\text{id}, (0\ 2)\} \cong C_2.$$

The transposition $(0\ 2)$ is realized arithmetically by divisor complement. Cyclic coordinate permutation acts instead on the ambient three-vector orbit

$$(3, 6, 3) \mapsto (3, 3, 6) \mapsto (6, 3, 3) \mapsto (3, 6, 3). \quad (387)$$

Proof. Assume (i). Restriction of r gives a bijection $\mathcal{X}_j \rightarrow \mathcal{X}_{j+1}$ for every j , so the three fibres have equal cardinality. This proves (i) $\Rightarrow$ (ii).

Assume (ii). Choose bijections

$$\varphi_0 : \mathcal{D}_0 \rightarrow \mathcal{D}_1, \quad \varphi_1 : \mathcal{D}_1 \rightarrow \mathcal{D}_2,$$

and define

$$\varphi_2 := (\varphi_1 \circ \varphi_0)^{-1} : \mathcal{D}_2 \rightarrow \mathcal{D}_0.$$

They satisfy (383), proving (ii) $\Rightarrow$ (iii). Conversely, the maps in (iii) force equal finite cardinalities, so (iii) $\Rightarrow$ (ii).

Given (iii), define

$$r(j, u) := (j + 1, \varphi_j(u)).$$

Equation (383) gives $r^3 = 1$ on $\mathcal{X}_0$. Since all three φ_j are bijective, its two cyclic conjugates give $r^3 = 1$ on $\mathcal{X}_1$ and $\mathcal{X}_2$ as well. Thus (iii) implies (i).

For $j = 0, 1$, the definition $T_j(p) = -p^{1-j}$ gives $T_{j+1}(p) = p^{-1}T_j(p)$. The closing ratio is $T_0(p)T_2(p)^{-1} = p^2$. These three edge multipliers agree precisely when $p^2 = p^{-1}$ in $(\mathbb{Z}/R\mathbb{Z})^\times$, which is (385). Membership of $\varphi_j(u)$ in $\mathcal{D}_{j+1}$ proves (384). The complement involution gives $m_0 = m_2$ by theorem 9.64. Finally, theorem 10.27 gives $(m_0, m_1, m_2) = (3, 6, 3)$, which violates (382); direct reduction gives $12289^3 \equiv 2^3 \equiv 8 \pmod{11}$. The stabilizer statement follows because exactly the two exterior entries agree. Direct cyclic coordinate permutation gives (387). $\square$

The cyclic structure used later is the product-cover action of theorem 14.33: it rotates an independent three-sheet coordinate and preserves every arithmetic origin label.

Definition 9.67 (Reduced shell coefficient). For a prime p , an integer $a > p/4$, and

$$R := 4a - p, \quad N := pa, \quad g := \gcd(R, N), \quad R_0 := R/g, \quad N_0 := N/g,$$

define

$$\Omega_R^{\text{red}}(p) := \#\{d \in \mathbb{Z}_{>0} : d \mid N_0^2, d \equiv -N_0 \pmod{R_0}\}. \quad (388)$$

Lemma 9.68 (Reduced shell equivalence). *For fixed p and a as in theorem 9.67, there exist $b, c \in \mathbb{Z}_{>0}$ with*

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

if and only if $\Omega_R^{\text{red}}(p) > 0$.

Proof. Since $R = 4a - p$ and $N = pa$,

$$\frac{4}{p} - \frac{1}{a} = \frac{R}{N} = \frac{R_0}{N_0}, \quad \gcd(R_0, N_0) = 1.$$

Suppose first that $R_0/N_0 = 1/b + 1/c$. Then $R_0bc = N_0(b + c)$ and consequently

$$(R_0b - N_0)(R_0c - N_0) = N_0^2. \quad (389)$$

The inequality $R_0/N_0 > 1/b$ implies $R_0b > N_0$; hence $d := R_0b - N_0$ is positive. Equation (389) shows that $d \mid N_0^2$, and clearly $d \equiv -N_0 \pmod{R_0}$. Thus $\Omega_R^{\text{red}}(p) > 0$.

Conversely, let d be counted by (388) and put $e = N_0^2/d$. Reducing $de = N_0^2$ modulo R_0 and using $d \equiv -N_0$ and $\gcd(R_0, N_0) = 1$ gives $e \equiv -N_0 \pmod{R_0}$. Therefore

$$b := \frac{N_0 + d}{R_0}, \quad c := \frac{N_0 + e}{R_0}$$

are positive integers. Substitution into $de = N_0^2$ gives $R_0bc = N_0(b + c)$, hence $R_0/N_0 = 1/b + 1/c$. Adding $1/a$ proves the required identity. $\square$

Lemma 9.69 (Multiplicative inheritance). *If $m, t \in \mathbb{Z}_{>0}$ and*

$$\frac{4}{m} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

for positive integers x, y, z , then

$$\frac{4}{mt} = \frac{1}{tx} + \frac{1}{ty} + \frac{1}{tz}.$$

Proof. Divide the first identity by t . $\square$

Theorem 9.70 (Global positivity equivalence and complete induction). *The following statements are equivalent.*

- (i) *For every integer $n \geq 2$ there are $x, y, z \in \mathbb{Z}_{>0}$ such that $4/n = 1/x + 1/y + 1/z$.*
- (ii) *The same assertion holds for every prime p .*
- (iii) *For every prime p there is an integer $a > p/4$, equivalently a positive admissible shell $R = 4a - p$, such that $\Omega_R^{\text{red}}(p) > 0$.*

Proof. The implication (i) $\Rightarrow$ (ii) is immediate.

Assume (ii) and fix a prime p . In any positive solution

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c},$$

one has $1/a < 4/p$, because $1/b + 1/c > 0$. Hence $a > p/4$ and $R = 4a - p > 0$. By theorem 9.68, $\Omega_R^{\text{red}}(p) > 0$. This proves (ii) $\Rightarrow$ (iii). The reverse implication is the converse direction of theorem 9.68.

For (ii) $\Rightarrow$ (i), use strong induction on n . The base case is

$$\frac{4}{2} = \frac{1}{1} + \frac{1}{2} + \frac{1}{2}.$$

Fix $n > 2$ and assume the assertion for every integer k with $2 \leq k < n$. If n is prime, the assertion is (ii). If n is composite, choose a proper divisor m with $2 \leq m < n$ and write $n = mt$, where $t > 1$. The induction hypothesis supplies a decomposition for $4/m$, and theorem 9.69 supplies one for $4/n$. This completes the induction. $\square$

Corollary 9.71 (Three-origin form of the global positivity statement). *For coprime shells, the global Erdős–Straus assertion is equivalent to*

$$\forall p \in \mathbb{P} \quad \exists R : \Sigma(m(R, p)) = m_0 + m_1 + m_2 > 0. \quad (390)$$

Proof. For a coprime shell, theorem 9.64 identifies $\Sigma(m(R, p))$ with the number of divisors in the fixed-shell certificate set. Apply theorem 9.70. Non-coprime shells are represented by their reduced coefficient $\Omega_R^{\text{red}}(p)$. $\square$

Proposition 9.72 (Exact closure of the simultaneous envelope). *Let $\varepsilon_1 := \varepsilon$ and $\varepsilon_0 := 1 - \varepsilon$. The two central blocks have bases*

$$\{\varepsilon_\delta r^i j^k : i \in \mathbb{Z}/3\mathbb{Z}, \ k \in \mathbb{Z}/2\mathbb{Z}\}, \quad \delta \in \{0, 1\}.$$

Products satisfy

$$(\varepsilon_\delta r^i j^k)(\varepsilon_{\delta'} r^{i'} j^{k'}) = \begin{cases} \varepsilon_\delta r^{i+(-1)^{k'} i'} j^{k+k'}, & \delta = \delta', \\ 0, & \delta \neq \delta', \end{cases} \quad (391)$$

where the exponents are reduced modulo 3 and 2. In particular, every finite product remains in the same singular algebra, and every group operation preserves its support block.

Proof. The central idempotents obey $\varepsilon_\delta^2 = \varepsilon_\delta$ and $\varepsilon_0 \varepsilon_1 = 0$. From $jrj = r^{-1}$ one obtains $j^k r^{i'} = r^{(-1)^{k'} i'} j^k$. Multiplying gives (391). The closure and block-preservation statements follow immediately. $\square$

10. THE 12289 ORBIT-OBJECT AND ITS EXPLICIT ERDŐS–STRAUS CERTIFICATE

For the shell

$$p = 12289, \quad R = 11, \quad a = \frac{p+R}{4} = 3075 = 3 \cdot 5^2 \cdot 41,$$

the exact endpoint counts are

$$c_{11}(3075; -1) = 6, \quad c_{11}(3075; -12289^{-1}) = 3.$$

The divisor involution supplies the equal exterior count, so the complete three-origin vector is

$$m_{12289} = (3, 6, 3). \quad (392)$$

Its full coordinate-permutation D_3 orbit is

$$\mathcal{O}_{12289} = \{(3, 6, 3), (3, 3, 6), (6, 3, 3)\}. \quad (393)$$

This is the orbit of the ambient coordinate action on $\mathbb{Z}_{\geq 0}^3$. Its three points have

$$\Sigma = 12, \quad |Z|^2 = 9, \quad \mathcal{L} = 135 > 0. \quad (394)$$

Theorem 9.66 proves that this coordinate orbit is not induced by a cyclic permutation of the 12289 origin-resolved certificate set.

Theorem 10.1 (Marked D_3 -identification of the Fable face with the 12289 count orbit). *Let*

$$\mathcal{F}_3 := \{e_0, e_+, e_-\}$$

be the three transported Fable rays in (121), and let

$$H_F := \{(1, X, 0, W) : X, W \in \mathbb{R}\}, \quad H_{12} := \{m \in \mathbb{R}^3 : m_0 + m_1 + m_2 = 12\}.$$

Define

$$\mathbf{B} := \begin{pmatrix} 4 & -1 & 0 & -\sqrt{3} \\ 4 & 2 & 0 & 0 \\ 4 & -1 & 0 & \sqrt{3} \end{pmatrix}, \quad \Psi := \mathbf{B}|_{H_{\mathbb{F}}}, \quad (395)$$

and put

$$U_{\text{orb}} := \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad S_{\text{orb}} := \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \quad (396)$$

Then $\Psi : H_{\mathbb{F}} \rightarrow H_{12}$ is an affine bijection, with

$$\Psi(1, X, 0, W) = \begin{pmatrix} 4 - X - \sqrt{3}W \\ 4 + 2X \\ 4 - X + \sqrt{3}W \end{pmatrix} \quad (397)$$

and

$$\Psi^{-1}(m_0, m_1, m_2) = \left(1, \frac{m_1 - 4}{2}, 0, \frac{m_2 - m_0}{2\sqrt{3}}\right). \quad (398)$$

It obeys the exact intertwining equations

$$\mathbf{B}R_{\text{face}} = U_{\text{orb}}\mathbf{B}, \quad \mathbf{B}S_W = S_{\text{orb}}\mathbf{B}. \quad (399)$$

In particular,

$$\begin{aligned} \Psi(e_0) &= (3, 6, 3)^{\top}, \\ \Psi(e_+) &= (3, 3, 6)^{\top}, \quad \Psi(\mathcal{F}_3) = \mathcal{O}_{12289}. \\ \Psi(e_-) &= (6, 3, 3)^{\top}, \end{aligned} \quad (400)$$

The marked fixed point e_0 , the ordered rotations R_{face} and U_{orb} , and affine dependence determine Ψ uniquely.

For $p = (1, X, 0, W)$ and $q = (1, X', 0, W')$,

$$\|\Psi(p) - \Psi(q)\|^2 = 6((X - X')^2 + (W - W')^2). \quad (401)$$

Thus Ψ carries the unit null circle in $H_{\mathbb{F}}$ onto

$$\left\{ m \in H_{12} : \left\| m - (4, 4, 4)^{\top} \right\|^2 = 6 \right\}. \quad (402)$$

If $\omega = e^{2\pi i/3}$ and

$$Z(m) := m_0 + \omega m_1 + \omega^2 m_2,$$

then

$$Z(\Psi(1, X, 0, W)) = 3\omega(X + iW). \quad (403)$$

Consequently the unit null circle has

$$\Sigma = 12, \quad |Z|^2 = 9, \quad \Sigma^2 - |Z|^2 = 135,$$

recovering (394).

The bijection in (400) is an identification of the Fable three-ray carrier with the ambient count-vector orbit. A cyclic lift to the origin-resolved certificate set would be a bijection

$$r : \mathcal{X}_{11,12289} \longrightarrow \mathcal{X}_{11,12289}, \quad r(\mathcal{X}_j) = \mathcal{X}_{j+1}.$$

No such lift exists: its restrictions would force $|\mathcal{X}_0| = |\mathcal{X}_1| = |\mathcal{X}_2|$, whereas

$$(|\mathcal{X}_0|, |\mathcal{X}_1|, |\mathcal{X}_2|) = (3, 6, 3).$$

The divisor-complement involution remains an arithmetic C_2 -action; at the count-vector level it is the reflection $S_{\text{orb}}(3, 6, 3)^{\top} = (3, 6, 3)^{\top}$.

Proof. For $m = \mathbf{B}(1, X, 0, W)^\top$, direct addition gives $m_0 + m_1 + m_2 = 12$. Conversely, if $m \in H_{12}$, substitution of (398) into (397) returns m . This proves the affine bijection and its inverse.

Matrix multiplication using (119) gives (399). Substitution of the three vectors in (121) gives (400). These three points are affinely independent in H_F . An affine map on that plane is determined by their images, which proves uniqueness.

The linear part of Ψ in the ordered coordinates (X, W) is

$$D = \begin{pmatrix} -1 & -\sqrt{3} \\ 2 & 0 \\ -1 & \sqrt{3} \end{pmatrix}, \quad D^\top D = 6I_2.$$

This proves (401) and (402). Using $1 + \omega + \omega^2 = 0$, direct substitution gives

$$(-1 + 2\omega - \omega^2)X + \sqrt{3}(-1 + \omega^2)W = 3\omega(X + iW),$$

which proves (403). The final Lorentz values follow on $X^2 + W^2 = 1$.

The nonexistence of the cyclic arithmetic lift follows from the three fibre cardinalities, equivalently from theorem 9.66. The complement involution and its exterior-fibre exchange are proved in theorem 9.64; direct multiplication gives the displayed fixed-vector identity. $\square$

Corollary 10.2 (Six-element symmetry of the singular count orbit). *Restrict the matrices in (396) to $\mathcal{O}_{12289}$, and write*

$$r := U_{\text{orb}}|_{\mathcal{O}_{12289}}, \quad s := S_{\text{orb}}|_{\mathcal{O}_{12289}}.$$

Then

$$r^3 = 1, \quad s^2 = 1, \quad srs = r^{-1}, \quad (404)$$

and

$$\langle r, s \rangle = \{1, r, r^2, s, sr, sr^2\} \cong D_3. \quad (405)$$

The element orders in this six-element action are

$$1^{[1]}, \quad 2^{[3]}, \quad 3^{[2]}. \quad (406)$$

Thus $\langle r \rangle \cong C_3$ is its orientation-preserving subgroup, and $s\langle r \rangle$ is its orientation-reversing coset. No element generates all six group elements.

The barycenter

$$c = \frac{1}{3}((3, 6, 3) + (3, 3, 6) + (6, 3, 3)) = (4, 4, 4)$$

is fixed by this action. The affine map Ψ transports the complete action to the marked Fable face. Adjoining the central occupied C_1 -idempotent leaves the relations and element orders unchanged. On the fixed origin-resolved certificate shell, the reflection s is realized by divisor complement, whereas r has no arithmetic lift.

Proof. Direct multiplication of the two matrices gives

$$U_{\text{orb}}^3 = I_3, \quad S_{\text{orb}}^2 = I_3, \quad S_{\text{orb}}U_{\text{orb}}S_{\text{orb}} = U_{\text{orb}}^{-1}.$$

The three vectors in $\mathcal{O}_{12289}$ are distinct; U_{orb} cycles them and S_{orb} fixes $(3, 6, 3)$ while exchanging the other two. The induced action is faithful and consists of the six elements in (405). The three reflections have order two, the two nonidentity rotations have order three, and the identity has order one, proving (406).

Coordinate permutations fix c . The intertwining equations (399) transport the action through Ψ . Centrality of the occupied idempotent is proved in theorem 8.5. The final lift statement follows from theorem 9.66. $\square$

Proposition 10.3 (Unique affine count descent and the forced quarter kernel). *Let*

$$p_\star = \left(0, \frac{5}{4}, 0\right), \quad p_0 = \left(2, \frac{3}{2}, 0\right), \quad p_+ = \left(-1, \frac{3}{2}, \sqrt{3}\right), \quad p_- = \left(-1, \frac{3}{2}, -\sqrt{3}\right)$$

be the spatial vertices of the transported raw-cell tetrahedron in (221). Let

$$c = (4, 4, 4)^\top, \quad v_0 = (3, 6, 3)^\top, \quad v_+ = (3, 3, 6)^\top, \quad v_- = (6, 3, 3)^\top.$$

There is a unique affine map

$$\Pi_{\text{cnt}} : \mathbb{R}^3 \longrightarrow H_{12}$$

satisfying

$$\Pi_{\text{cnt}}(p_\star) = c, \quad \Pi_{\text{cnt}}(p_0) = v_0, \quad \Pi_{\text{cnt}}(p_+) = v_+, \quad \Pi_{\text{cnt}}(p_-) = v_-. \quad (407)$$

It is

$$\Pi_{\text{cnt}}(X, Y, W) = \begin{pmatrix} 4 - \frac{X}{2} - \frac{\sqrt{3}W}{2} \\ 4 + X \\ 4 - \frac{X}{2} + \frac{\sqrt{3}W}{2} \end{pmatrix} = \Psi\left(1, \frac{X}{2}, 0, \frac{W}{2}\right). \quad (408)$$

Its linear part is

$$K_{\text{cnt}} = \begin{pmatrix} -\frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} \\ 1 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \end{pmatrix}, \quad K_{\text{cnt}}^\top K_{\text{cnt}} = \begin{pmatrix} \frac{3}{2} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{3}{2} \end{pmatrix}. \quad (409)$$

Consequently,

$$\ker K_{\text{cnt}} = \mathbb{R}(0, 1, 0)^\top. \quad (410)$$

The face centroid

$$\bar{p} := \frac{p_0 + p_+ + p_-}{3} = \left(0, \frac{3}{2}, 0\right)$$

therefore satisfies

$$\Pi_{\text{cnt}}(\bar{p}) = \Pi_{\text{cnt}}(p_\star) = c, \quad \bar{p} - p_\star = \left(0, \frac{1}{4}, 0\right) \in \ker K_{\text{cnt}}. \quad (411)$$

The induced map

$$\mathbb{R}^3 / \mathbb{R}(0, 1, 0) \longrightarrow H_{12}$$

is a similarity with squared length factor 3/2.

Proof. The four points $p_\star, p_0, p_+, p_-$ are affinely independent because their tetrahedral volume is $\sqrt{3}/4 > 0$ by (226). Their four prescribed images therefore determine at most one affine map. Direct substitution in (408) proves all four identities in (407), hence existence and uniqueness.

Matrix multiplication gives (409). Its kernel is the Y -axis, proving (410). Affine dependence gives

$$\Pi_{\text{cnt}}(\bar{p}) = \frac{v_0 + v_+ + v_-}{3} = (4, 4, 4)^\top = c.$$

The coordinate difference in (411) is direct. On the orthogonal complement of the kernel, (409) multiplies every squared length by 3/2, which proves the quotient statement. $\square$

Theorem 10.4 (Canonical axial completion of the count descent). *Let*

$$F_\Delta := \text{aff}\{p_0, p_+, p_-\}, \quad \nu_\Delta := \frac{p_\star - \bar{p}}{\|p_\star - \bar{p}\|} = (0, -1, 0)^\top.$$

The signed Euclidean distance from the marked face, oriented toward the apex, is

$$h_\Delta(X, Y, W) := \langle (X, Y, W)^\top - \bar{p}, \nu_\Delta \rangle = \frac{3}{2} - Y. \quad (412)$$

It is the unique affine scalar function whose restriction to F_Δ is zero, whose gradient has Euclidean length one, and whose value at $p_\star$ is positive. It is invariant under the marked face D_3 -action and

$$h_\Delta(p_0) = h_\Delta(p_+) = h_\Delta(p_-) = 0, \quad h_\Delta(p_\star) = \frac{1}{4}.$$

The completed affine map

$$\widehat{\Pi}_{\text{cnt}} : \mathbb{R}^3 \longrightarrow H_{12} \times \mathbb{R}, \quad q \longmapsto (\Pi_{\text{cnt}}(q), h_\Delta(q)) \quad (413)$$

is bijective, with inverse

$$\widehat{\Pi}_{\text{cnt}}^{-1}((m_0, m_1, m_2)^\top, h) = \left(m_1 - 4, \frac{3}{2} - h, \frac{m_2 - m_0}{\sqrt{3}} \right). \quad (414)$$

For the Euclidean metric on $H_{12} \subset \mathbb{R}^3$ and the standard metric on the last factor, its linear part is

$$\widehat{K}_{\text{cnt}} = \begin{pmatrix} -\frac{1}{2} & 0 & -\frac{\sqrt{3}}{2} \\ 1 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & -1 & 0 \end{pmatrix}, \quad \widehat{K}_{\text{cnt}}^\top \widehat{K}_{\text{cnt}} = \text{diag} \left(\frac{3}{2}, 1, \frac{3}{2} \right). \quad (415)$$

Hence its three singular values are

$$\sqrt{\frac{3}{2}}, \quad 1, \quad \sqrt{\frac{3}{2}},$$

and its affine volume factor is $3/2$.

More generally, if $\ell : \mathbb{R}^3 \rightarrow \mathbb{R}$ is affine, then

$$q \longmapsto (\Pi_{\text{cnt}}(q), \ell(q))$$

is injective exactly when

$$d\ell(0, 1, 0)^\top \neq 0. \quad (416)$$

Thus one scalar coordinate is both necessary and sufficient to restore injectivity, and h_Δ is the unique such coordinate selected by the marked Euclidean tetrahedron with unit gradient.

Proof. The face F_Δ is the plane $Y = 3/2$, and

$$p_\star - \bar{p} = \left(0, -\frac{1}{4}, 0 \right).$$

Therefore $\nu_\Delta = (0, -1, 0)^\top$ and (412) follows. Every affine scalar function vanishing on F_Δ has gradient parallel to ν_Δ . Unit length leaves the two gradients ν_Δ and $-\nu_\Delta$; positivity at the apex selects ν_Δ . The marked D_3 -action fixes the Y -coordinate, so it fixes h_Δ .

For $m \in H_{12}$, one has $m_0 + m_1 + m_2 = 12$. Substitution of

$$X = m_1 - 4, \quad Y = \frac{3}{2} - h, \quad W = \frac{m_2 - m_0}{\sqrt{3}}$$

into (408) returns $(m_0, m_1, m_2)^\top$, proving (414). Direct matrix multiplication proves (415); its determinant is $(3/2)^2$, which gives the stated singular values and volume factor.

Finally,

$$\ker d(\Pi_{\text{cnt}}, \ell) = \ker K_{\text{cnt}} \cap \ker d\ell.$$

Equation (410) identifies the first kernel with $\mathbb{R}(0, 1, 0)^\top$, proving (416). $\square$

The unique split-zero lift of the signed height and its exact restriction to the central count fibre are proved in theorem 11.4.

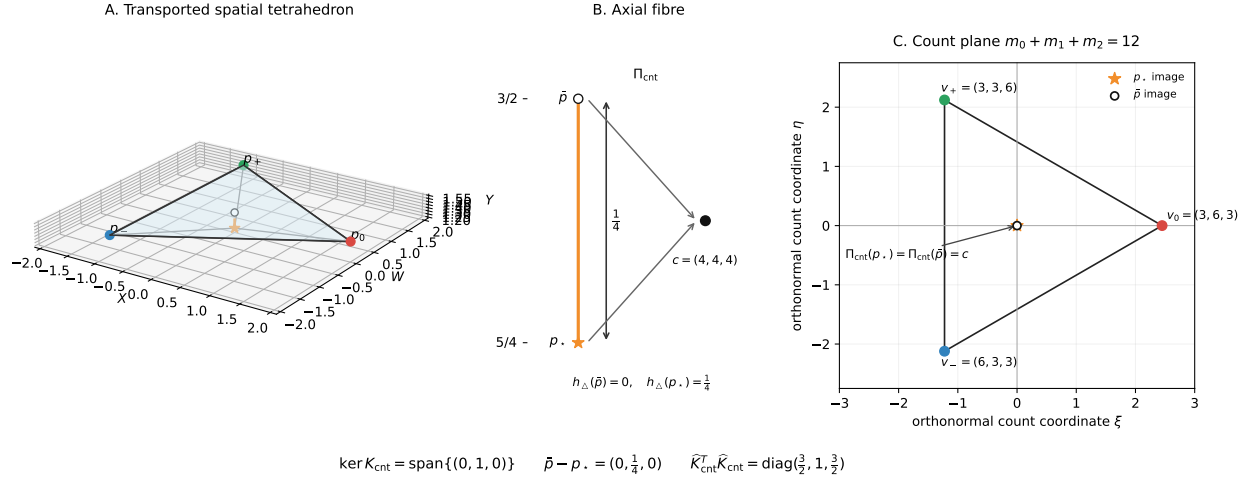
The 12289 count descent erases exactly the quarter-height coordinate

FIGURE 7. **The exact axial coordinate erased by the 12289 count descent.** Panel A shows the transported spatial tetrahedron with its marked D_3 -face and its apex. Panel B isolates the kernel fibre: the face centroid $\bar{p}$ and apex p_* differ by the signed height $1/4$, while both are sent to $c = (4, 4, 4)$. Panel C displays the count orbit in its induced orthonormal chart. The star and open circle at its center are coincident because $\Pi_{\text{cnt}}(p_*) = \Pi_{\text{cnt}}(\bar{p})$. The identities below the panels are (410), (411), and (415).

Theorem 10.5 (Explicit certificate for 12289). *One has*

$$\frac{4}{12289} = \frac{1}{3075} + \frac{1}{3440920} + \frac{1}{2116165800}. \quad (417)$$

Consequently 12289 has the displayed Erdős–Straus decomposition.

Proof. Choose the recorded exponent vector $(-1, -1, -1)$ for the middle target -1 . Its divisor of a^2 is

$$u = a 3^{-1} 5^{-1} 41^{-1} = 5.$$

With

$$N = pa = 37788675, \quad d = pu = 61445,$$

the fixed-shell formulas give

$$b = \frac{N + d}{R} = 3440920$$

and

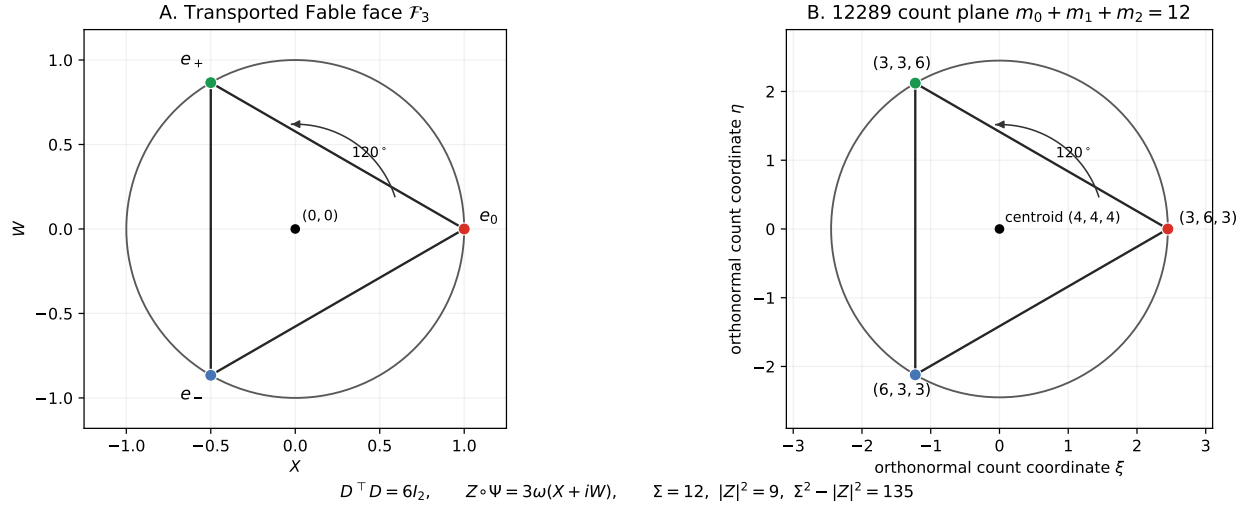
$$c = \frac{N + N^2/d}{R} = 2116165800.$$

Since $d \mid N^2$ and $d \equiv -N \pmod{11}$, the divisor-shell identity gives

$$\frac{11}{N} = \frac{1}{b} + \frac{1}{c}.$$

Also $4/12289 - 1/3075 = 11/N$. Adding $1/3075$ proves (417). $\square$

Exact marked D_3 identification: $\Psi(X, W) = (4 - X - \sqrt{3}W, 4 + 2X, 4 - X + \sqrt{3}W)$



Count-orbit D_3 is exact; a cyclic action on the certificate fibres would require $3 = 6 = 3$ and therefore has no lift.

FIGURE 8. The exact marked D_3 -identification. The left panel is the transported Fable unit face in the (X, W) -plane. The right panel is the plane $m_0 + m_1 + m_2 = 12$, displayed in the orthonormal coordinates induced by the two columns of $D/\sqrt{6}$. Corresponding vertices have the same color. Equations (401) and (403) give every scale and character value displayed below the panels. The final line records the proved obstruction to lifting the count-orbit rotation to the three origin-resolved certificate fibres.

10.1. The Star–Kneser quotient certificate.

Definition 10.6 (Divisor-ratio sumset and target star). Let

$$a = \prod_{i=1}^s \ell_i^{e_i}, \quad H = \langle \bar{\ell}_1, \dots, \bar{\ell}_s \rangle \leq (\mathbb{Z}/R\mathbb{Z})^\times,$$

and write the finite abelian group H additively. Put

$$A_i = \{-e_i \bar{\ell}_i, \dots, 0, \dots, e_i \bar{\ell}_i\} \subseteq H, \quad B = A_1 + \dots + A_s,$$

and

$$T = \{\overline{-1}, \overline{-p^{-1}}\} \cap H.$$

For every subgroup $K \leq H$, define

$$\begin{aligned} a_i(K) &= |(A_i + K)/K|, & t(K) &= |(T + K)/K|, \\ \kappa(K) &= 1 + \sum_{i=1}^s (a_i(K) - 1), & \Sigma_T(K) &= \kappa(K) + t(K) - |H/K|. \end{aligned} \quad (418)$$

Theorem 10.7 (Star–Kneser forcing). *In the notation of theorem 10.6, if*

$$\Sigma_T(K) > 0 \quad \text{for every subgroup } K \leq H,$$

then $B \cap T \neq \emptyset$. Conversely, if $B \cap T = \emptyset$, then

$$\Sigma_T(K) \leq 0$$

for some subgroup $K \leq H$; one may take $K = \text{Stab}(B)$.

Proof. Put $K = \text{Stab}(B)$. The finite form of Kneser's theorem [22] gives

$$|B/K| \geq 1 + \sum_{i=1}^s (|(A_i + K)/K| - 1) = \kappa(K).$$

Assume $B \cap T = \emptyset$. The images of B and T in H/K are disjoint. Indeed, if $b + K = t + K$ for $b \in B$ and $t \in T$, then $t - b \in K$; because K stabilizes B , this implies $t \in b + K \subseteq B$, a contradiction. Therefore

$$|B/K| + |T/K| \leq |H/K|.$$

Using the Kneser bound and $|T/K| = t(K)$ gives

$$\kappa(K) + t(K) \leq |H/K|,$$

or $\Sigma_T(K) \leq 0$. The first assertion is the contrapositive. $\square$

Theorem 10.8 (Three-origin Star-Kneser forcing and exact quotient defect). *Retain the notation of theorem 10.6, and distinguish the two-target and three-origin sets*

$$T^{(2)} := \{\overline{-1}, \overline{-p^{-1}}\} \cap H, \quad T^{(3)} := \{\overline{-p}, \overline{-1}, \overline{-p^{-1}}\} \cap H. \quad (419)$$

Then

$$B \cap T^{(2)} = B \cap T^{(3)}. \quad (420)$$

For every subgroup $K \leq H$, put

$$t^{(3)}(K) := |(T^{(3)} + K)/K|, \quad (421)$$

$$\Sigma^{(3)}(K) := \kappa(K) + t^{(3)}(K) - |H/K|. \quad (422)$$

If

$$\Sigma^{(3)}(K) > 0 \quad \text{for every } K \leq H, \quad (423)$$

then the shell has a divisor certificate.

Suppose the shell has no divisor certificate, and take

$$K = \text{Stab}(B), \quad Q = H/K, \quad C = B/K, \quad U = (T^{(3)} + K)/K. \quad (424)$$

Define the uncovered quotient margin and the Kneser expansion slack by

$$\delta_K := |Q| - |C| - |U|, \quad \epsilon_K := |C| - \kappa(K). \quad (425)$$

Both are nonnegative integers, and

$$-\Sigma^{(3)}(K) = \delta_K + \epsilon_K. \quad (426)$$

Consequently, $\Sigma^{(3)}(K) = 0$ holds precisely when

$$Q = C \dot{\cup} U, \quad |C| = \kappa(K). \quad (427)$$

Finally,

$$\Sigma^{(3)}(K) \geq \Sigma_T(K) \quad (K \leq H). \quad (428)$$

Proof. Every interval A_i is closed under the group inverse, so the sumset $B = A_1 + \cdots + A_s$ is closed under the group inverse. In the original multiplicative residue group,

$$(\overline{-p})^{-1} = \overline{-p^{-1}}.$$

Thus membership of $\overline{-p}$ in B is equivalent to membership of $\overline{-p^{-1}}$ in B . Since $T^{(2)} \subseteq T^{(3)}$, this proves (420).

Apply the proof of theorem 10.7 with $T^{(3)}$ in place of T . Under failure, the sets C and U are disjoint, so $\delta_K \geq 0$. Kneser's theorem gives $|C| \geq \kappa(K)$, so $\epsilon_K \geq 0$. Direct calculation gives

$$\delta_K + \epsilon_K = |Q| - |U| - \kappa(K) = -\Sigma^{(3)}(K),$$

which proves (426). Its two nonnegative summands vanish simultaneously precisely under the two conditions in (427). The inclusion $T^{(2)} \subseteq T^{(3)}$ gives $t^{(3)}(K) \geq t(K)$, proving (428). $\square$

Corollary 10.9 (Exact three-origin surplus data for the two computed shells). *For the 12289 shell of theorem 10.27, the three target logs are*

$$T^{(3)} = \{4, 5, 6\} \subset C_{10},$$

and, for subgroup orders 1, 2, 5, 10,

$$(\Sigma^{(3)}(K))_K = (2, 7, 2, 1). \quad (429)$$

For the supported failed shell $(p, R) = (8803369, 43)$, the three target logs are

$$T^{(3)} = \{6, 7, 8\} \subset C_{14},$$

and, for subgroup orders 1, 2, 7, 14,

$$(\Sigma^{(3)}(K))_K = (-6, 1, 2, 1). \quad (430)$$

Its sumset stabilizer is trivial, and its exact defect pair is

$$(\delta_{\{0\}}, \epsilon_{\{0\}}) = (2, 4). \quad (431)$$

Proof. The logarithms in theorem 10.27 give the first target set; the exterior inverse of 4 is 6 in C_{10} . Quotienting the three displayed target classes by the four subgroups gives target costs $(3, 3, 2, 1)$, and substitution in (422) gives (429).

For $R = 43$, the two recorded target logs are 7, 8, and the exterior inverse is 6. The divisor support is

$$B = \{0, 1, 3, 4, 5, 9, 10, 11, 13\} \subset C_{14}.$$

It has trivial stabilizer. The three-origin target costs in subgroup orders 1, 2, 7, 14 are $(3, 3, 2, 1)$; substitution gives (430). At the trivial stabilizer, $|Q| = 14$, $|C| = 9$, $|U| = 3$, and $\kappa = 5$. Hence

$$\delta_{\{0\}} = 14 - 9 - 3 = 2, \quad \epsilon_{\{0\}} = 9 - 5 = 4,$$

which proves (431). $\square$

Exact computation 10.10 (Three-origin Star–Kneser checker). The script and captured receipt

`verification/verify_three_origin_star_kneser.py,`
`verification/verify_three_origin_star_kneser_output.txt`

verify both surplus vectors, the $R = 43$ defect pair, and the hit equivalence and exact defect decomposition in 89991 finite cyclic test cases.

Theorem 10.11 (Zero-defect three-origin critical family). *Let $m \geq 1$, let*

$$m = e_1 + \cdots + e_s$$

be a composition into positive integers, and put

$$Q = \mathbb{Z}/(2m+4)\mathbb{Z}.$$

For $1 \leq i \leq s$, define

$$A_i = \{-e_i, -e_i + 1, \dots, e_i\} \subset Q,$$

and set

$$C = A_1 + \cdots + A_s, \quad U = \{m+1, m+2, m+3\} \subset Q.$$

Then

$$C = \{-m, -m+1, \dots, m\}, \quad (432)$$

$$Q = C \dot{\cup} U, \quad (433)$$

$$\text{Stab}(C) = \{0\}. \quad (434)$$

Moreover, U has the three-origin form

$$U = \{u, (2m+4)/2, -u\}, \quad u = m+1, \quad (435)$$

and the two defect summands of theorem 10.8 vanish:

$$\delta_{\{0\}} = \epsilon_{\{0\}} = 0, \quad \Sigma^{(3)}(\{0\}) = 0. \quad (436)$$

Proof. Integer interval addition gives

$$[-e_1, e_1] + \cdots + [-e_s, e_s] = [-m, m].$$

The interval on the right has $2m+1$ terms, fewer than the $2m+4$ elements of Q , so its reduction modulo $2m+4$ gives (432) with no repeated residue. Its complement is the three-term interval U , proving (433). A translation preserving C also preserves its complementary proper cyclic interval U . The two endpoints of that interval are the only elements of U having exactly one cyclic neighbour in U . A preserving translation must permute these endpoints and preserve their cyclic difference 2; the exchanged case would require $2 = -2$ in Q , which is impossible because $|Q| = 2m+4 \geq 6$. Hence the translation is zero, proving (434).

In Q ,

$$-u = -(m+1) = m+3, \quad |Q|/2 = m+2,$$

which proves (435). Finally,

$$\kappa(\{0\}) = 1 + \sum_{i=1}^s (2e_i + 1 - 1) = 2m+1 = |C|.$$

Together with $|Q| = 2m+4$, $|U| = 3$, and (433), this gives

$$\delta_{\{0\}} = |Q| - |C| - |U| = 0, \quad \epsilon_{\{0\}} = |C| - \kappa(\{0\}) = 0.$$

The exact defect identity (426) gives the last assertion. $\square$

Corollary 10.12 (Arithmetic realization at the $p = 5$, $R = 7$ shell). *Take*

$$p = 5, \quad R = 7, \quad a = 3, \quad N = 15.$$

Then $H = (\mathbb{Z}/7\mathbb{Z})^\times$, and the divisor-ratio carrier and the three-origin target are

$$B = \{1, 3, 5\}, \quad T^{(3)} = \{2, 4, 6\}.$$

They form a disjoint partition of H , the stabilizer of B is trivial, and

$$\delta_{\{1\}} = \epsilon_{\{1\}} = 0.$$

The shell has no divisor certificate. The same prime has the distinct $R = 3$ certificate

$$a = 2, \quad d = 2, \quad (b, c) = (4, 20),$$

which gives

$$\frac{4}{5} = \frac{1}{2} + \frac{1}{4} + \frac{1}{20}.$$

Proof. The powers $3^{-1}, 3^0, 3^1$ modulo 7 give

$$B = \{5, 1, 3\}.$$

The three target residues are

$$-p = 2, \quad -1 = 6, \quad -p^{-1} = 4 \pmod{7}.$$

This proves the partition and the defect assertions, either directly or by theorem 10.11 with $m = 1$ and additive logarithms to base 3. The positive divisors of $N^2 = 225$ are

$$1, 3, 5, 9, 15, 25, 45, 75, 225.$$

Their residues modulo 7 avoid $-N \equiv 6$, so theorem 9.1 excludes a certificate in this shell.

For $R = 3$, one has $a = (5 + 3)/4 = 2$ and $N = 10$. The divisor $d = 2$ satisfies $d \mid 100$ and $d \equiv -10 \pmod{3}$. The reconstruction formulas give $b = 4$ and $c = 20$, proving the displayed decomposition. $\square$

Corollary 10.13 (Adjacent zero-defect shells for one prime). *For $p = 353$, both adjacent shell parameters*

$$(R, a, N) = (31, 96, 33888), \quad (R, a, N) = (27, 95, 33535)$$

are supported zero-defect failures.

For $R = 31$, let

$$K_{31} = \{1, 2, 4, 8, 16\}.$$

Then $H_{31} = (\mathbb{Z}/31\mathbb{Z})^\times$, the quotient has order six, and

$$B_{31}/K_{31} = \{K_{31}, 3K_{31}, 11K_{31}\}, \quad (437)$$

$$T_{31}^{(3)}K_{31}/K_{31} = \{5K_{31}, 7K_{31}, 15K_{31}\}. \quad (438)$$

For $R = 27$, let

$$K_{27} = \{1, 10, 19\}.$$

Then $H_{27} = (\mathbb{Z}/27\mathbb{Z})^\times$, the quotient again has order six, and

$$B_{27}/K_{27} = \{K_{27}, 2K_{27}, 5K_{27}\}, \quad (439)$$

$$T_{27}^{(3)}K_{27}/K_{27} = \{4K_{27}, 7K_{27}, 8K_{27}\}. \quad (440)$$

In both cases the two displayed three-coset sets partition the quotient, the Kneser lower bound is attained, and

$$\delta_K = \epsilon_K = 0.$$

Proof. The two shell factorizations are

$$96 = 2^5 \cdot 3, \quad 95 = 5 \cdot 19.$$

For $R = 31$, direct exponent reduction gives local quotient-support cardinalities $(1, 3)$, while for $R = 27$ it gives $(3, 1)$. Multiplication of the local support sets gives the displayed carrier cosets. The three-origin residues are

$$T_{31}^{(3)} = \{18, 19, 30\}, \quad T_{27}^{(3)} = \{13, 25, 26\};$$

their cosets are the displayed complementary triples. Hence each carrier misses its target and each shell fails. In either quotient,

$$|Q| = 6, \quad |B/K| = 3, \quad |T^{(3)}K/K| = 3,$$

and

$$\kappa(K) = 1 + (1 - 1) + (3 - 1) = 3.$$

The two defect formulas now give $\delta_K = \epsilon_K = 0$. $\square$

Proposition 10.14 (Ordered-origin inversion across adjacent critical shells). *Put*

$$\mathcal{P}_{\text{adj}} = \{353, 6113, 10433\}.$$

For every $p \in \mathcal{P}_{\text{adj}}$, the two residual shells $R = 31$ and $R = 27$, with $a_R = (p + R)/4$ and $N_R = pa_R$, have the following arithmetic data:

p	a_{31}	N_{31}	a_{27}	N_{27}
353	$2^5 \cdot 3 = 96$	33888	$5 \cdot 19 = 95$	33535
6113	$2^9 \cdot 3 = 1536$	9389568	$5 \cdot 307 = 1535$	9383455
10433	$2^3 \cdot 3 \cdot 109 = 2616$	27292728	$5 \cdot 523 = 2615$	27282295

(441)

Their carrier stabilizers are

$$K_{31} = \{1, 2, 4, 8, 16\}, \quad K_{27} = \{1, 10, 19\}.$$

Writing

$$Q_R = (\mathbb{Z}/R\mathbb{Z})^\times / K_R, \quad g_{31} = 3K_{31}, \quad g_{27} = 5K_{27},$$

both g_{31} and g_{27} have order six. If $\lambda_R(g_R^j) = j \in \mathbb{Z}/6\mathbb{Z}$, then the carrier and the ordered three-origin target satisfy

$$\lambda_R(B_R/K_R) = \{0, 1, 5\}, \quad (442)$$

$$(\lambda_{31}(-pK_{31}), \lambda_{31}(-K_{31}), \lambda_{31}(-p^{-1}K_{31})) = (4, 3, 2), \quad (443)$$

$$(\lambda_{27}(-pK_{27}), \lambda_{27}(-K_{27}), \lambda_{27}(-p^{-1}K_{27})) = (2, 3, 4). \quad (444)$$

The unique group isomorphism $\varphi : Q_{31} \rightarrow Q_{27}$ carrying the three entries of (443) to the corresponding entries of (444) is

$$\varphi(g_{31}^j) = g_{27}^{-j}. \quad (445)$$

It fixes the exponent classes 0 and 3, exchanges 1 with 5, and exchanges 2 with 4.

Each of the six shells is a zero-defect failure:

$$\{d : d \mid N_R^2, d \equiv -N_R \pmod{R}\} = \emptyset, \quad \delta_{K_R} = \epsilon_{K_R} = 0. \quad (446)$$

Let

$$V_R = \mathbb{R}e_{R,+} \oplus \mathbb{R}e_{R,-},$$

where $e_{R,+}$ and $e_{R,-}$ label the carrier endpoints g_R and g_R^{-1} , respectively. The induced endpoint map is

$$\varphi_*e_{31,+} = e_{27,-}, \quad \varphi_*e_{31,-} = e_{27,+}, \quad [\varphi_*] = S := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (447)$$

On the blade carrier

$$W = \mathbb{R}(2E_{21}) \oplus \mathbb{R}(2E_{12}), \quad \sigma_{\text{bl}}(X) = SXS,$$

define

$$\Psi_R(e_{R,+}) = 2E_{21}, \quad \Psi_R(e_{R,-}) = 2E_{12}.$$

Then

$$\Psi_{27} \circ \varphi_* = \sigma_{\text{bl}} \circ \Psi_{31}. \quad (448)$$

Thus the cross-shell endpoint inversion and the ordered blade exchange of theorem 13.53 are isomorphic two-point permutation representations. The arithmetic data above determine the endpoint exchange. A further arithmetic construction is required to select one of the two possible endpoint-to-blade basis bijections.

Proof. For a prime-power factor $\ell^e \parallel a_R$, put

$$A_{\ell,R} = \{\ell^k : -e \leq k \leq e\} \subset (\mathbb{Z}/R\mathbb{Z})^\times.$$

The displayed factorizations in (441) give the following exact quotient-log images, listed in the order of their prime factors:

p	$R = 31$	$R = 27$
353	$(\{0\}, \{0, 1, 5\})$	$(\{0, 1, 5\}, \{0\})$
6113	$(\{0\}, \{0, 1, 5\})$	$(\{0, 1, 5\}, \{0\})$
10433	$(\{0\}, \{0, 1, 5\}, \{0\})$	$(\{0, 1, 5\}, \{0\})$

Multiplying the local sets proves (442); it also gives the displayed stabilizers directly. Reduction of p modulo 31 and 27, followed by discrete logarithms to the displayed generators, gives (443) and (444).

Every isomorphism between the two generated cyclic groups has the form

$$g_{31}^j \mapsto g_{27}^{\eta j}, \quad \eta \in (\mathbb{Z}/6\mathbb{Z})^\times = \{1, -1\}.$$

The first target exponent requires $4\eta \equiv 2 \pmod{6}$, and hence $\eta = -1$. The other two ordered entries then map as $3 \mapsto 3$ and $2 \mapsto 4$. This proves uniqueness and (445).

In every row, the quotient has six elements, the carrier and target each have three elements, the two sets are disjoint, and the local Kneser quantity is

$$\kappa(K_R) = 1 + \sum_{\ell|a_R} (|A_{\ell,R}K_R/K_R| - 1) = 3.$$

This proves both defect equalities. Exact enumeration of the divisors from the prime factorizations of N_R^2 gives the empty sets in (446).

Equation (445) exchanges the two carrier endpoints, which gives (447). Matrix-unit multiplication gives

$$S(2E_{21})S = 2E_{12}, \quad S(2E_{12})S = 2E_{21}.$$

Applying the definitions of Ψ_{31} and Ψ_{27} to the two basis vectors proves (448). $\square$

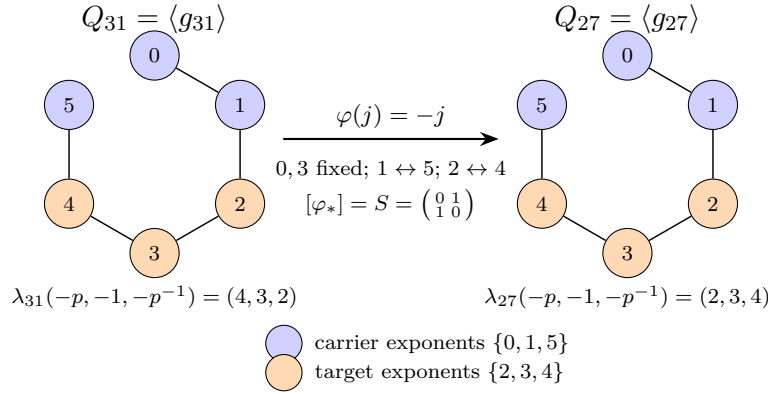


FIGURE 9. The ordered adjacent-shell quotient channel. Exponent inversion is the unique group isomorphism carrying the three ordered origins at $R = 31$ to their corresponding origins at $R = 27$. On the carrier endpoints it is the exchange matrix S , and conjugation by the same matrix exchanges $2E_{21}$ with $2E_{12}$.

Corollary 10.15 (Logarithmic selection of the ordered blade basis). *In the six shells of theorem 10.14, put*

$$q_{31} = 3, \quad q_{27} = 5, \quad u_{R,+} = a_R q_R, \quad u_{R,-} = a_R / q_R.$$

Then

$$u_{R,+} u_{R,-} = a_R^2, \quad \frac{u_{R,+}}{a_R} = q_R, \quad \frac{u_{R,-}}{a_R} = q_R^{-1}. \quad (449)$$

In particular, $u_{R,+}$ and $u_{R,-}$ are complementary divisors of a_R^2 representing the exponent endpoints $+1$ and -1 .

Define the arithmetic logarithmic operator on V_R by

$$\mathcal{L}_R e_{R,+} = (\log q_R) e_{R,+}, \quad \mathcal{L}_R e_{R,-} = -(\log q_R) e_{R,-}. \quad (450)$$

There is a unique bijection from the endpoint basis $\{e_{R,+}, e_{R,-}\}$ to the ordered blade basis $\{2E_{21}, 2E_{12}\}$ whose linear extension intertwines $\mathcal{L}_R$ with the maintained modular Hamiltonian $\widetilde{K}_{q_R}$. It is

$$\Theta_R(e_{R,+}) = 2E_{12}, \quad \Theta_R(e_{R,-}) = 2E_{21}, \quad \Theta_R \mathcal{L}_R = \widetilde{K}_{q_R} \Theta_R. \quad (451)$$

For the cross-shell maps of theorem 10.14,

$$\Theta_{27}\varphi_* = \sigma_{\text{bl}}\Theta_{31}, \quad (452)$$

$$\mathcal{L}_{27}\varphi_* = -\frac{\log 5}{\log 3}\varphi_*\mathcal{L}_{31}, \quad (453)$$

$$\widetilde{K}_5\sigma_{\text{bl}} = -\frac{\log 5}{\log 3}\sigma_{\text{bl}}\widetilde{K}_3. \quad (454)$$

Proof. The local quotient-image table in the proof of theorem 10.14 has exactly one nontrivial factor in each shell: 3 for $R = 31$ and 5 for $R = 27$. Each has valuation one in a_R . Hence the two integers in (449) divide a_R^2 , their product is a_R^2 , and their quotient logs are $+1$ and -1 . This proves (449) and (450).

Equation (878), with $\lambda = q_R$, gives

$$\widetilde{K}_{q_R}(2E_{21}) = -(\log q_R)(2E_{21}), \quad \widetilde{K}_{q_R}(2E_{12}) = (\log q_R)(2E_{12}).$$

The two eigenvalues are distinct. Eigenvalue preservation therefore sends $e_{R,+}$ to $2E_{12}$ and $e_{R,-}$ to $2E_{21}$, proving existence and uniqueness in (451).

Relative to the endpoint basis and the ordered blade basis $(2E_{21}, 2E_{12})$, both $[\varphi_*]$ and $[\sigma_{\text{bl}}]$ are the exchange matrix S , while $[\Theta_R] = S$. This proves (452). Applying the remaining two identities to the two basis vectors gives (453) and (454). $\square$

Theorem 10.16 (CRT and kernel-generation criterion for adjacent inversion). *Let p be a prime with $p \equiv 1 \pmod{4}$, and put*

$$a_R = \frac{p+R}{4} \quad (R \in \{31, 27\}).$$

Retain $K_{31}, K_{27}, g_{31}, g_{27}$ and the logarithms $\lambda_{31}, \lambda_{27}$ from theorem 10.14. Define

$$\rho_R := \lambda_R(pK_R).$$

Then the ordered target has the general exponent formula

$$(\lambda_R(-pK_R), \lambda_R(-K_R), \lambda_R(-p^{-1}K_R)) = (3 + \rho_R, 3, 3 - \rho_R) \quad \text{in } (\mathbb{Z}/6\mathbb{Z})^3. \quad (455)$$

Consequently, the ordered transition

$$(4, 3, 2) \quad \longmapsto \quad (2, 3, 4)$$

occurs exactly when

$$p \bmod 31 \in 3K_{31} = \{3, 6, 12, 17, 24\}, \quad p \bmod 27 \in 2K_{27} = \{2, 11, 20\}. \quad (456)$$

Together with $p \equiv 1 \pmod{4}$, these are the fifteen residue classes

$$\begin{aligned} \mathcal{R}_{31,27} = \{ & 65, 353, 389, 533, 1109, 1181, 1469, 1505, 1649, \\ & 2225, 2297, 2585, 2621, 2765, 3341 \} \pmod{3348}. \end{aligned} \quad (457)$$

Every class in $\mathcal{R}_{31,27}$ is coprime to 3348; hence every class contains infinitely many primes.

For the factorization

$$a_R = \prod_{\ell} \ell^{e_{\ell,R}},$$

put

$$I_{\ell,R} := \{j\lambda_R(\ell K_R) : -e_{\ell,R} \leq j \leq e_{\ell,R}\} \subseteq \mathbb{Z}/6\mathbb{Z}. \quad (458)$$

The shell carrier satisfies

$$\lambda_R(B_R/K_R) = \sum_{\ell|a_R} I_{\ell,R}. \quad (459)$$

Assume (456). The R -shell has stabilizer K_R and is a zero-defect failure exactly when

$$\sum_{\ell|a_R} I_{\ell,R} = \{0, 1, 5\}, \quad K_R B_R = B_R. \quad (460)$$

The exact carrier criterion has the following factor-level formulation. Put

$$q_{31} = 3, \quad q_{27} = 5,$$

and assume

$$e_{q_R,R} = 1, \quad \ell K_R = K_R \quad (\ell \mid a_R, \ell \neq q_R). \quad (461)$$

Define the kernel carrier

$$C_R := \prod_{\substack{\ell^{e_{\ell,R}} \parallel a_R \\ \ell \neq q_R}} \{\ell^j : -e_{\ell,R} \leq j \leq e_{\ell,R}\}. \quad (462)$$

Under (461),

$$(460) \iff C_R = K_R. \quad (463)$$

Proof. Since -1 has exponent 3 in both cyclic quotients, multiplication by -1 , p , and p^{-1} gives (455). The two desired triples are equivalent to

$$\rho_{31} = 1, \quad \rho_{27} = -1.$$

The first equality is $pK_{31} = 3K_{31}$. The second is $pK_{27} = g_{27}^{-1} = 2K_{27}$. This proves (456). The Chinese remainder theorem gives the list in (457). Direct gcd calculation shows that all fifteen representatives are coprime to 3348; the infinitude statement follows from Dirichlet's theorem on primes in arithmetic progressions.

Every divisor $u \mid a_R^2$ has a unique exponent presentation

$$u = \prod_{\ell|a_R} \ell^{e_{\ell,R} + j_{\ell}}, \quad -e_{\ell,R} \leq j_{\ell} \leq e_{\ell,R}.$$

After division by a_R and passage to Q_R , this gives (459). Under the target criterion, the target exponent set is $U = \{2, 3, 4\}$. Suppose first that the two conditions in (460) hold. The carrier image is the complementary set $C = \{0, 1, 5\}$, so the shell fails. The translation stabilizer of C in $\mathbb{Z}/6\mathbb{Z}$ is trivial. Since B_R is K_R -stable, its full stabilizer is therefore exactly K_R .

Write $a_{\ell}(K_R) = |I_{\ell,R}|$. The equality $|C| = 3$ forces

$$\sum_{\ell|a_R} (a_{\ell}(K_R) - 1) \geq 2 :$$

if the sum were at most one, the quotient carrier would have at most two elements. Hence $\kappa(K_R) \geq 3$. Kneser's bound gives $\kappa(K_R) \leq |C| = 3$, and consequently

$$\kappa(K_R) = 3, \quad \delta_{K_R} = 6 - 3 - 3 = 0, \quad \epsilon_{K_R} = 3 - 3 = 0.$$

Thus the failure has zero defect. Conversely, if the shell has stabilizer K_R and is a zero-defect failure, the exact defect theorem gives $Q_R = C \cup U$. Since $U = \{2, 3, 4\}$, one has $C = \{0, 1, 5\}$, while $K_R B_R = B_R$ follows from the stabilizer assumption. This proves (460).

Under (461), every local factor other than q_R lies inside K_R , while $q_R K_R = g_R$ and $e_{q_R,R} = 1$. Therefore

$$B_R = \{q_R^{-1}, 1, q_R\} C_R.$$

The three factors on the right lie in three distinct K_R -cosets. Thus B_R is K_R -stable exactly when the nonempty subset $C_R \subseteq K_R$ is K_R -stable. Left multiplication of K_R on itself is transitive, so this occurs exactly when $C_R = K_R$. The quotient image is then $\{0, 1, 5\}$, proving (463). $\square$

Corollary 10.17 (Prime-order internal kernel weights). *Assume (461). Put*

$$n_{31} = 5, \quad h_{31} = 2, \quad n_{27} = 3, \quad h_{27} = 10,$$

and let

$$\mu_R : K_R \longrightarrow \mathbb{Z}/n_R\mathbb{Z}, \quad \mu_R(h_R^j) = j.$$

Define the internal kernel weight

$$w_R := \sum_{\substack{\ell^{e_{\ell,R}} \parallel a_R, \ell \neq q_R \\ \mu_R(\ell) \neq 0}} e_{\ell,R}. \quad (464)$$

Then

$$C_{31} = K_{31} \iff w_{31} \geq 2, \quad C_{27} = K_{27} \iff w_{27} \geq 1. \quad (465)$$

Consequently, under the ordered-target and single-active-factor conditions, both adjacent shells have full zero-defect support exactly when the two inequalities in (465) hold.

Proof. For each kernel factor, put

$$J_{\ell,R} := \{j\mu_R(\ell) : -e_{\ell,R} \leq j \leq e_{\ell,R}\} \subseteq \mathbb{Z}/n_R\mathbb{Z}.$$

Applying μ_R to (462) gives

$$\mu_R(C_R) = \sum_{\substack{\ell^{e_{\ell,R}} \parallel a_R \\ \ell \neq q_R}} J_{\ell,R}. \quad (466)$$

For $R = 27$, every nonzero $\alpha \in \mathbb{Z}/3\mathbb{Z}$ satisfies

$$\{-\alpha, 0, \alpha\} = \mathbb{Z}/3\mathbb{Z}.$$

Thus the sumset in (466) is the whole group exactly when at least one kernel factor has nonzero internal log. This is $w_{27} \geq 1$.

For $R = 31$, weight zero gives the singleton $\{0\}$, and weight one gives one three-element set $\{-\alpha, 0, \alpha\}$. Suppose $w_{31} \geq 2$. If one active factor has valuation at least two, its local set contains

$$\{-2\alpha, -\alpha, 0, \alpha, 2\alpha\} = \mathbb{Z}/5\mathbb{Z}.$$

Otherwise there are two active valuation-one factors with nonzero logs α and β . After multiplication by α^{-1} , the second log is one of ± 1 or ± 2 . Direct addition gives

$$\{0, \pm 1\} + \{0, \pm 1\} = \mathbb{Z}/5\mathbb{Z}, \quad \{0, \pm 1\} + \{0, \pm 2\} = \mathbb{Z}/5\mathbb{Z}.$$

Additional local sets preserve the full group. Hence $C_{31} = K_{31}$ exactly when $w_{31} \geq 2$. The last assertion follows from theorem 10.16. $\square$

Corollary 10.18 (Residue-factor criterion for full adjacent support). *Assume the ordered-target and single-active-factor conditions of theorem 10.16. Define*

$$\nu_{31}^{\text{nt}}(a_{31}) := \sum_{\substack{\ell^e \parallel a_{31} \\ \ell \bmod 31 \in \{2,4,8,16\}}} e, \quad (467)$$

$$\nu_{27}^{\text{nt}}(a_{27}) := \sum_{\substack{\ell^e \parallel a_{27} \\ \ell \bmod 27 \in \{10,19\}}} e. \quad (468)$$

Then

$$C_{31} = K_{31} \iff \nu_{31}^{\text{nt}}(a_{31}) \geq 2, \quad (469)$$

$$C_{27} = K_{27} \iff \nu_{27}^{\text{nt}}(a_{27}) \geq 1. \quad (470)$$

Consequently, the two adjacent shells have full zero-defect support exactly when both inequalities hold.

Proof. The internal logarithm μ_{31} has kernel $\{1\}$ on

$$K_{31} = \{1, 2, 4, 8, 16\},$$

so its nonzero values occur exactly on the residue classes $2, 4, 8, 16 \pmod{31}$. Likewise, μ_{27} has kernel $\{1\}$ on

$$K_{27} = \{1, 10, 19\},$$

so its nonzero values occur exactly on the residue classes $10, 19 \pmod{27}$. The single-active-factor conditions place every kernel factor in the corresponding subgroup. Therefore

$$\nu_{31}^{\text{nt}}(a_{31}) = w_{31}, \quad \nu_{27}^{\text{nt}}(a_{27}) = w_{27}.$$

The result follows from theorem 10.17. $\square$

Corollary 10.19 (Consecutive-integer form of the adjacent criterion). *Let p satisfy the ordered-target conditions (456), and put*

$$a := a_{31} = \frac{p + 31}{4}.$$

Then

$$a_{27} = a - 1, \quad p = 4a - 31, \quad (471)$$

and the ordered-target conditions are equivalent to

$$a \bmod 31 \in 3K_{31}, \quad a \bmod 27 \in 1 + 5K_{27} = \{6, 15, 24\}. \quad (472)$$

Equivalently,

$$a \bmod 837 \in \mathcal{A}_{31,27}, \quad (473)$$

where

$$\mathcal{A}_{31,27} = \{6, 24, 96, 105, 141, 285, 303, 375, 384, \\ 420, 564, 582, 654, 663, 699\}.$$

Every such a satisfies

$$a \equiv 6 \pmod{9}, \quad v_3(a) = 1. \quad (474)$$

Furthermore,

$$v_5(a - 1) = 1 \iff p \bmod 100 \in \{13, 33, 53, 93\}. \quad (475)$$

On the ordered-target locus, the single-active-factor conditions (461) therefore hold exactly when (475) holds and

$$\ell \mid a/3 \implies \ell \bmod 31 \in \{1, 2, 4, 8, 16\}, \quad (476)$$

$$\ell \mid (a - 1)/5 \implies \ell \bmod 27 \in \{1, 10, 19\} \quad (477)$$

for every prime ℓ . Under these conditions, both shells have full zero-defect support exactly when

$$\sum_{\substack{\ell^e \parallel a/3 \\ \ell \bmod 31 \in \{2, 4, 8, 16\}}} e \geq 2, \quad (478)$$

$$\sum_{\substack{\ell^e \parallel (a-1)/5 \\ \ell \bmod 27 \in \{10, 19\}}} e \geq 1. \quad (479)$$

Proof. The identities in (471) follow directly from the definitions of a_{31} and a_{27} . Modulo 31, one has $p = 4a$ and $4 \in K_{31}$, so

$$pK_{31} = 3K_{31} \iff aK_{31} = 3K_{31}.$$

Modulo 27, one has $p = 4(a - 1)$. Since

$$4(5K_{27}) = 20K_{27} = 2K_{27},$$

the second target condition is equivalent to $(a - 1)K_{27} = 5K_{27}$, which is the second statement in (472). The Chinese remainder theorem then gives the fifteen classes in (473).

The three residues 6, 15, 24 (mod 27) are all congruent to 6 (mod 9). This proves (474). Since

$$4(a - 1) = p + 27,$$

the condition $v_5(a - 1) = 1$ is equivalent to

$$p + 27 \bmod 100 \in \{20, 40, 60, 80\},$$

which gives (475).

Equation (474) supplies the exact valuation of the distinguished factor 3. Equation (475) supplies the exact valuation of the distinguished factor 5. The remaining clauses of (461) say precisely that the prime factors of $a/3$ and $(a - 1)/5$ lie in K_{31} and K_{27} , respectively. This proves (476)–(477). Finally, (478)–(479) are the residue weights from theorem 10.18 after removing the distinguished factors. $\square$

Corollary 10.20 (Exact Diophantine cofactor parameterization). *Let p be a positive prime. The following statements are equivalent.*

- (i) *The prime p satisfies the ordered-target conditions (456) and the single-active-factor conditions (461) for the adjacent $R = 31$ and $R = 27$ shells.*
- (ii) *There is a unique integer $t \geq 1$ such that*

$$p = 60t - 7, \quad b := 5t + 2, \quad c := 3t + 1, \quad (480)$$

and every prime divisor of b belongs modulo 31 to $K_{31} = \{1, 2, 4, 8, 16\}$, while every prime divisor of c belongs modulo 27 to $K_{27} = \{1, 10, 19\}$.

Under these equivalent conditions,

$$a := \frac{p + 31}{4} = 15t + 6 = 3b = 5c + 1, \quad (481)$$

and the two shells have full zero-defect support exactly when

$$\omega_{31}(b) := \sum_{\substack{\ell^e \parallel b \\ \ell \bmod 31 \in \{2, 4, 8, 16\}}} e \geq 2, \quad (482)$$

$$\omega_{27}(c) := \sum_{\substack{\ell^e \parallel c \\ \ell \bmod 27 \in \{10, 19\}}} e \geq 1. \quad (483)$$

For every $t \geq 1$ for which $60t - 7$ is prime, the ordered-target conditions alone are equivalent to

$$t \bmod 93 \in \{0, 6, 9, 57, 81\}. \quad (484)$$

Proof. Assume (i), and put $a = (p + 31)/4$. The ordered-target condition gives $a \equiv 6 \pmod{9}$, hence $3 \mid a$ and $v_3(a) = 1$, by (474). The single-active-factor condition gives $v_5(a - 1) = 1$. Therefore $a \equiv 0 \pmod{3}$ and $a \equiv 1 \pmod{5}$, so there is a unique integer t with

$$a = 15t + 6.$$

Since $p = 4a - 31 > 0$, one has $t \geq 1$. Direct substitution gives (480) and (481). The prime-divisor conditions in (ii) are precisely (476) and (477) with $b = a/3$ and $c = (a - 1)/5$.

Conversely, assume (ii) and define a by (481). Products of elements of K_{31} and K_{27} remain in the respective subgroups, so

$$b \bmod 31 \in K_{31}, \quad c \bmod 27 \in K_{27}.$$

Consequently

$$aK_{31} = 3K_{31}, \quad (a - 1)K_{27} = 5K_{27}.$$

Equation (472) proves the ordered-target conditions. Since $3 \bmod 31 \notin K_{31}$, the stated condition on the prime divisors of b gives $3 \nmid b$, and hence $v_3(a) = 1$. Since $5 \bmod 27 \notin K_{27}$, the stated condition on the prime divisors of c gives $5 \nmid c$, and hence $v_5(a - 1) = 1$. The consecutive-factor characterization (476)–(477) now proves the single-active-factor conditions.

For the final residue assertion, write $a = 15t + 6$, $b = 5t + 2$, and $c = 3t + 1$. The ordered-target conditions are equivalent to

$$b \bmod 31 \in K_{31}, \quad c \bmod 27 \in K_{27}.$$

Since $5^{-1} \equiv 25 \pmod{31}$, the first condition gives

$$t \bmod 31 \in \{0, 6, 9, 19, 26\}.$$

The second condition is equivalent to $t \equiv 0 \pmod{3}$. Applying the Chinese remainder theorem modulo $3 \cdot 31 = 93$ gives exactly

$$t \bmod 93 \in \{0, 6, 9, 57, 81\},$$

which proves (484).

Finally, substituting $b = a/3$ and $c = (a - 1)/5$ into (478)–(479) gives (482)–(483) and proves the full-support criterion. $\square$

Corollary 10.21 (Sixfold parameter and the modulus-360 prime line). *Under the equivalent conditions of theorem 10.20, the parameter t is divisible by 6. Thus there is a unique integer $u \geq 1$ such that*

$$t = 6u, \quad p = 360u - 7, \quad a = 90u + 6, \quad (485)$$

and

$$\frac{a}{3} = 30u + 2 = 2(15u + 1), \quad \frac{a - 1}{5} = 18u + 1. \quad (486)$$

In particular,

$$p \equiv 353 \pmod{360}. \quad (487)$$

For every $u \geq 1$ for which $360u - 7$ is prime, the ordered-target conditions alone are equivalent to

$$u \bmod 31 \in \{0, 1, 17, 25, 29\}. \quad (488)$$

On the ordered single-active locus, full support is equivalent to

$$\sum_{\substack{\ell^e \parallel 2(15u+1) \\ \ell \bmod 31 \in \{2, 4, 8, 16\}}} e \geq 2, \quad (489)$$

$$\sum_{\substack{\ell^e \parallel 18u+1 \\ \ell \bmod 27 \in \{10, 19\}}} e \geq 1. \quad (490)$$

Proof. The single-active-factor condition requires every prime divisor of $c = 3t + 1$ to lie in K_{27} . Since $2 \bmod 27 \notin K_{27}$, the integer c is odd. Hence t is even. Equation (484) gives $3 \mid t$, so $t = 6u$ for a unique integer $u \geq 1$. Substitution proves (485)–(487).

For the target criterion, the residue modulo 27 is

$$a = 90u + 6 \equiv 9u + 6 \pmod{27},$$

which always belongs to $\{6, 15, 24\} = 1 + 5K_{27}$. Modulo 31,

$$a = 3(30u + 2) \equiv 3(2 - u) \pmod{31}.$$

Therefore $a \bmod 31 \in 3K_{31}$ exactly when

$$u \bmod 31 \in 2 - K_{31} = \{0, 1, 17, 25, 29\}.$$

This proves (488). The two weight statements follow by substituting (486) into (482)–(483). $\square$

Corollary 10.22 (Power-of-two/odd-prime cofactor family). *Let $m \geq 1$, let r be either 1 or an odd prime, and put*

$$a = 3 \cdot 2^m r.$$

Assume

$$r \bmod 31 \in K_{31} = \{1, 2, 4, 8, 16\}, \quad (491)$$

$$a - 1 = 5q \quad \text{for a prime } q \text{ with } q \bmod 27 \in \{10, 19\}, \quad (492)$$

$$p = 4a - 31 \quad \text{is prime.} \quad (493)$$

Define

$$\varepsilon_{31}(r) = \begin{cases} 0, & r \equiv 1 \pmod{31}, \\ 1, & r \bmod 31 \in \{2, 4, 8, 16\}. \end{cases}$$

If

$$m + \varepsilon_{31}(r) \geq 2, \quad (494)$$

then p satisfies the ordered adjacent transition

$$(4, 3, 2) \mapsto (2, 3, 4),$$

and both the $R = 31$ and $R = 27$ shells have full zero-defect support.

Proof. Equation (493) gives $p \equiv 1 \pmod{4}$. Modulo 31, equations (491) and $2 \in K_{31}$ give

$$a = 3 \cdot 2^m r \in 3K_{31}.$$

Modulo 27, equation (492) gives

$$a = 1 + 5q \in 1 + 5K_{27}.$$

Hence the ordered-target criterion (472) holds.

The two consecutive factors are

$$\frac{a}{3} = 2^m r, \quad \frac{a-1}{5} = q.$$

Every prime factor of $a/3$ lies in K_{31} , and its nontrivial internal-log weight is exactly $m + \varepsilon_{31}(r)$. Every prime factor of $(a-1)/5$ lies in K_{27} , and its nontrivial internal-log weight is exactly one because $q \bmod 27 \in \{10, 19\}$. Thus (476)–(477) and (478)–(479) all hold. The conclusion follows from theorem 10.19. $\square$

Exact computation 10.23 (Bounded adjacent-inversion criterion). The exact script and captured receipt

```
verification/verify_adjacent_inversion_criterion.py,
verification/verify_adjacent_inversion_criterion_output.txt
```

verify the fifteen CRT classes and all primes $p \leq 500000$. There are 566 target-compatible primes in this range. Fourteen satisfy the quotient-level factor condition, and exactly the following eleven also generate both full kernels:

$$\{353, 6113, 10433, 28433, 55793, 99713, \\ 300593, 340913, 368633, 379433, 497153\}.$$

The internal-weight pairs for the other three primes are

$$(w_{31}, w_{27}) = (5, 0) \text{ at } p = 89633, \quad (1, 1) \text{ at } p = 111593, \quad (2, 0) \text{ at } p = 490313.$$

For all fourteen quotient-factor primes, the checker also evaluates the residue sums in (467) and (468) and verifies their equality with w_{31} and w_{27} , respectively. The same run enumerates the fifteen classes in (473), checks the exact 3- and 5-adic valuation criteria, and verifies that the consecutive-factor predicate and the quotient-factor predicate agree on all 566 target-compatible

primes in the bounded range. It also verifies the equivalent t -parameter criterion in theorem 10.20 on the same 566 primes. The eleven full-support values correspond to

$$(t_1, \dots, t_{11}) = (6, 102, 174, 474, 930, 1662, 5010, 5682, 6144, 6324, 8286).$$

Equivalently, on the sixfold line of theorem 10.21, they are

$$(u_1, \dots, u_{11}) = (1, 17, 29, 79, 155, 277, 835, 947, 1024, 1054, 1381).$$

Ten of the eleven full-support primes belong to the family in theorem 10.22:

$$\{353, 6113, 10433, 28433, 55793, 99713, \\ 300593, 340913, 368633, 497153\}.$$

For the remaining value $p = 379433$, one has

$$a = 94866, \quad \frac{a}{3} = 2 \cdot 97 \cdot 163, \quad \frac{a-1}{5} = 18973,$$

with $97 \equiv 4 \pmod{31}$, $163 \equiv 8 \pmod{31}$, and $18973 \equiv 19 \pmod{27}$. The checker verifies both the ten-member subfamily and this remaining two-odd-prime factorization. For all eleven primes, the checker reconstructs both arithmetic carriers, proves that their stabilizers are K_{31} and K_{27} , obtains the ordered target triples $(4, 3, 2)$ and $(2, 3, 4)$, and exhausts the divisor search in both shells. Hence all twenty-two shells are exact zero-defect failures.

Exact computation 10.24 (Adjacent ordered-origin checker). The script and captured receipt

`verification/verify_adjacent_ordered_quotient_channel.py,`
`verification/verify_adjacent_ordered_quotient_channel_output.txt`

verify the six shell factorizations, their local quotient images and stabilizers, the absence of divisor certificates, the two ordered three-origin log triples, uniqueness of exponent inversion, the complementary divisor-ratio representatives, and the endpoint, blade, and logarithmic selector matrices.

Exact computation 10.25 (Critical-family checker). The script and captured receipt

`verification/verify_three_origin_critical_family.py,`
`verification/verify_three_origin_critical_family_output.txt`

verify 2080 members of the cyclic family, the complete $(p, R, a, N) = (5, 7, 3, 15)$ shell, and the $R = 3$ decomposition of the same prime. They also verify both adjacent $p = 353$ shells, their local quotient-support cardinalities, stabilizers, target cosets, and absence of a divisor certificate.

The critical family and its arithmetic realization show that vanishing of both local defect summands is compatible with an actual failed shell. The adjacent $p = 353$ pair also excludes the inference that the replacement $R \mapsto R - 4$ must leave the zero-defect locus. A global descent requires additional arithmetic data linking more than these two local quotient partitions.

Corollary 10.26 (Residual-shell consequence). *If all star surpluses in theorem 10.7 are positive, then the fixed residual shell has a divisor certificate and hence an Erdős–Straus decomposition.*

Proof. For a divisor

$$u = \prod_i \ell_i^{e_i + \beta_i} \mid a^2, \quad -e_i \leq \beta_i \leq e_i,$$

the quotient residue $\bar{u}\bar{a}^{-1}$ is the element $\sum_i \beta_i \bar{\ell}_i$ of B . A hit at $\overline{-1}$ is the middle origin $j = 1$ in (372); a hit at $\overline{-p^{-1}}$ is the exterior origin $j = 2$. Since every A_i , and hence B , is invariant under inversion, the opposite exterior target $\overline{-p}$ is paired with $\overline{-p^{-1}}$. Thus $B \cap T \neq \emptyset$ produces a divisor counted by theorem 9.1. $\square$

Theorem 10.27 (Complete Star–Kneser certificate at 12289). *For*

$$p = 12289, \quad R = 11, \quad a = 3075 = 3 \cdot 5^2 \cdot 41,$$

the divisor-ratio group is $H = (\mathbb{Z}/11\mathbb{Z})^\times \cong C_{10}$. With 2 as generator and additive logarithmic coordinates,

$$A_3 = \{0, 2, 8\}, \quad A_5 = \{0, 2, 4, 6, 8\}, \quad A_{41} = \{0, 3, 7\},$$

and

$$T = \{4, 5\}.$$

For the four subgroups of C_{10} , the complete quotient table is

$ K $	$ H/K $	$(a_3(K), a_5(K), a_{41}(K))$	$\kappa(K)$	$t(K)$	$\Sigma_T(K)$
1	10	(3, 5, 3)	9	2	1
2	5	(3, 5, 3)	9	2	6
5	2	(1, 1, 2)	2	2	2
10	1	(1, 1, 1)	1	1	1

(495)

In particular,

$$\min_{K \leq H} \Sigma_T(K) = 1,$$

so the shell is forced positive by theorem 10.7. Its exact three-origin count is

$$(m_0, m_1, m_2) = (3, 6, 3), \quad \Omega_{11}(12289) = 12.$$

Proof. The powers of 2 modulo 11 give

$$\log_2(3) = 8, \quad \log_2(5) = 4, \quad \log_2(41) = \log_2(8) = 3.$$

The exponent intervals for 3^1 , 5^2 , and 41^1 therefore give the three displayed subsets. Moreover

$$p \equiv 2 \pmod{11}, \quad p^{-1} \equiv 6 \pmod{11},$$

so

$$\log_2(-1) = 5, \quad \log_2(-p^{-1}) = \log_2(5) = 4.$$

This proves the descriptions of H , the A_i , and T .

The group C_{10} has one subgroup of each order 1, 2, 5, 10. For the trivial subgroup, the quotient sizes are (3, 5, 3) and $t = 2$, hence

$$\kappa = 1 + (3 - 1) + (5 - 1) + (3 - 1) = 9, \quad \Sigma_T = 9 + 2 - 10 = 1.$$

For the subgroup of order 2, reduction to C_5 leaves the three set sizes (3, 5, 3) and the two target classes distinct, giving $\Sigma_T = 9 + 2 - 5 = 6$. For the subgroup of order 5, reduction to parity gives sizes (1, 1, 2) and two target classes, hence

$$\kappa = 1 + 0 + 0 + 1 = 2, \quad \Sigma_T = 2 + 2 - 2 = 2.$$

For $K = H$, all sets have one image, so $\Sigma_T = 1 + 1 - 1 = 1$. This proves (495).

Enumerate the $3 \cdot 5 \cdot 3 = 45$ signed exponent vectors $(\beta_3, \beta_5, \beta_{41})$. The complete fibre over the middle target $5 = \log_2(-1)$ is

$$\begin{aligned} &(-1, -1, -1), \quad (-1, 0, 1), \quad (0, -2, 1), \\ &(0, 2, -1), \quad (1, 0, -1), \quad (1, 1, 1), \end{aligned}$$

while the complete fibre over the exterior target $4 = \log_2(-p^{-1})$ is

$$(-1, -2, 0), \quad (0, 1, 0), \quad (1, -1, 0).$$

Direct reduction of $8\beta_3 + 4\beta_5 + 3\beta_{41}$ modulo 10 verifies every listed vector; inspection of the finite box gives no others. Thus the middle count is 6, one exterior count is 3, and inversion gives the other exterior count 3. Hence

$$\Omega_{11}(12289) = 6 + 2 \cdot 3 = 12.$$

□

Provenance record 10.28 (Source of the quotient certificate). The formulation and the 12289 table are reconstructed from *A Star–Kneser Positivity Criterion for Residual Erdős–Straus Shells*, The Clankers, May 2026 [16], local source `BEAVERSHINE/8/2/star_kneser_erdos_straus.tex`. The present proof restates the finite calculation explicitly.

Exact computation 10.29 (The complete 12289 coefficient-space aggregate). The twelve exponent vectors in theorem 10.27 give six inversion orbits. Writing the origin labels in the first column, the complete divisor set is

$$\begin{array}{l|l}
 0 \leftrightarrow 2 & 615 \longleftrightarrow 2\,321\,925\,135\,375 \\
 0 \leftrightarrow 2 & 5\,125 \longleftrightarrow 278\,631\,016\,245 \\
 0 \leftrightarrow 2 & 230\,625 \longleftrightarrow 6\,191\,800\,361 \\
 1 \leftrightarrow 1 & 61\,445 \longleftrightarrow 23\,240\,035\,125 \\
 1 \leftrightarrow 1 & 2\,765\,025 \longleftrightarrow 516\,445\,225 \\
 1 \leftrightarrow 1 & 23\,041\,875 \longleftrightarrow 61\,973\,427
 \end{array}$$

Here $N = 37\,788\,675$, and the two entries in every row have product N^2 . Thus the fibres have respective cardinalities 3, 6, 3. Direct addition gives

$$\sum_{d \in \mathcal{D}_{11,N}} d = 2630592510468 = 12 \cdot 219216042539.$$

Consequently theorem 9.3 gives

$$V(\mathcal{D}_{11,N}) = 12(219216042539, 37788675, 0, 219216042539). \quad (496)$$

The primitive Sarnak coordinates are

$$(T, X, Y, W) = (219216042539, 37788675, 0, 0),$$

and their exact norm is

$$37788675^2 - 219216042539^2 = -48055671878476699310896. \quad (497)$$

Corollary 10.30 (Outcome of the Fable substitution). *Inside the singular support envelope (71), the Fable triple is an occupied balanced element, while the 12289 table is an occupied closed orbit with the explicit positive certificate (417). The certificate settles the Erdős–Straus instance at 12289 positively.*

11. SPLIT-ZERO SUPPORT AND THE EXACT MEANING OF THE C_1 CARRIER

Let R be a nonzero commutative ring. Define the split-zero semiring

$$G(R) := R \sqcup \{\tau\}, \quad e := 0_R \in R \subset G(R), \quad (498)$$

by

$$r \oplus s = r + s, \quad r \oplus \tau = r, \quad \tau \oplus \tau = \tau,$$

and

$$r \odot s = rs, \quad r \odot \tau = \tau \odot r = \tau, \quad \tau \odot \tau = \tau$$

for $r, s \in R$. Thus τ is the additive zero of $G(R)$, whereas e is the supported copy of the ring zero.

Proposition 11.1 (Exact two-coordinate normal form). *There is a canonical semiring isomorphism*

$$G(R) \cong D_R := \{(r, b) \in R \times \mathbb{B} : b = 0 \implies r = 0_R\} = \{(0_R, 0)\} \cup (R \times \{1\}),$$

where the operations on D_R are componentwise. Under this isomorphism,

$$\tau \longmapsto (0_R, 0), \quad r \longmapsto (r, 1).$$

Consequently the maps

$$p_R : G(R) \rightarrow R, \quad p_R(\tau) = 0_R, \quad p_R(r) = r,$$

and

$$\chi_R : G(R) \rightarrow \mathbb{B}, \quad \chi_R(\tau) = 0, \quad \chi_R(r) = 1,$$

are semiring homomorphisms.

Proof. The displayed map is bijective. Componentwise operations give

$$(r, 1) + (s, 1) = (r + s, 1), \quad (r, 1) + (0, 0) = (r, 1),$$

and

$$(r, 1)(s, 1) = (rs, 1), \quad (r, 1)(0, 0) = (0, 0),$$

which are exactly the defining operations of $G(R)$. The two coordinate projections restricted to D_R are therefore semiring homomorphisms. $\square$

Theorem 11.2 (Entrywise split pair and the full ring-null matrix fibre). *For every $n \geq 1$, entrywise application of the two coordinates in theorem 11.1 gives a semiring isomorphism*

$$\Phi_n : M_n(G(R)) \xrightarrow{\sim} \mathcal{D}_{R,n}, \quad \Phi_n(X) = (\hat{p}_R(X), \hat{\chi}_R(X)), \quad (499)$$

where

$$\mathcal{D}_{R,n} := \{(A, S) \in M_n(R) \times M_n(\mathbb{B}) : S_{ij} = 0 \implies A_{ij} = 0_R\}. \quad (500)$$

In particular, the complete fibre over the zero amplitude matrix is

$$\hat{p}_R^{-1}(0_n) = \{X_S : S \in M_n(\mathbb{B})\}, \quad \Phi_n(X_S) = (0_n, S), \quad (501)$$

and the support coordinate identifies its induced addition and multiplication with those of $M_n(\mathbb{B})$. Thus a ring-null matrix retains an entire Boolean support matrix; it is not determined by its zero amplitude.

For the directed blade matrix of theorem 11.7, the pair is

$$\Phi_2(\tilde{N}) = (2E_{21}, E_{21}). \quad (502)$$

Its support E_{21} precedes the derived source and range projectors E_{11} and E_{22} .

Proof. Apply the isomorphism of theorem 11.1 to every matrix entry. This gives a bijection onto (500). It preserves matrix addition because Boolean addition records the union of entry supports. For multiplication, suppose that the (i, j) -entry of the Boolean product ST is zero. For every k , either $S_{ik} = 0$ or $T_{kj} = 0$; the defining support condition then gives either $A_{ik} = 0$ or $B_{kj} = 0$. Every summand $A_{ik}B_{kj}$ therefore vanishes, so $(AB)_{ij} = 0$. Hence the image is closed under matrix multiplication and (499) is a semiring isomorphism.

Setting the amplitude coordinate equal to 0_n leaves S arbitrary, which proves (501). Finally, $\tilde{2} \mapsto (2, 1)$ and every absent entry maps to $(0, 0)$, so the matrix in (512) maps to $(2E_{21}, E_{21})$. The source and range statements then follow from $E_{12}E_{21} = E_{11}$ and $E_{21}E_{12} = E_{22}$. $\square$

Theorem 11.3 (Uniqueness of the Boolean support character). *For every nonzero commutative ring R , the homomorphism*

$$\chi_R : G(R) \longrightarrow \mathbb{B}, \quad \chi_R(\tau) = 0, \quad \chi_R(r) = 1 \quad (r \in R),$$

is the unique semiring homomorphism from $G(R)$ to $\mathbb{B}$. Consequently its entrywise extension

$$\hat{\chi}_R : M_n(G(R)) \longrightarrow M_n(\mathbb{B}) \quad (503)$$

is the unique matrix-semiring homomorphism induced entrywise by a scalar map $G(R) \rightarrow \mathbb{B}$.

Proof. Let $u : G(R) \rightarrow \mathbb{B}$ be a semiring homomorphism. Since τ is the semiring zero and 1_R is the multiplicative identity,

$$u(\tau) = 0, \quad u(1_R) = 1.$$

Inside the supported copy of R , one has $0_R = 1_R \oplus (-1_R)$. Hence Boolean addition gives

$$u(0_R) = u(1_R) + u(-1_R) = 1 + u(-1_R) = 1.$$

If $u(r) = 0$ for some $r \in R$, then

$$1 = u(0_R) = u(r0_R) = u(r)u(0_R) = 0,$$

a contradiction. Thus $u(r) = 1$ for every supported $r \in R$, so $u = \chi_R$. Entrywise extension preserves matrix addition and matrix multiplication because χ_R preserves finite sums and products. The scalar uniqueness gives the final assertion. This is the Boolean-character theorem of [19] in the notation used here. $\square$

Theorem 11.4 (Unique positive-cone lift of the axial coordinate). *Equip $\mathbb{R}_{\geq 0}$ with its ordinary semiring operations. There is a unique semiring homomorphism*

$$\lambda_+ : \mathbb{R}_{\geq 0} \longrightarrow G(\mathbb{R}) \quad \text{such that} \quad p_{\mathbb{R}} \circ \lambda_+ = \text{id}_{\mathbb{R}_{\geq 0}}. \quad (504)$$

In the two-coordinate form of theorem 11.1, it is

$$\lambda_+(t) = \begin{cases} \tau = (0, 0), & t = 0, \\ (t, 1), & t > 0. \end{cases} \quad (505)$$

Let

$$T_{\Delta} := \text{conv}\{p_{\star}, p_0, p_+, p_{-}\}$$

be the transported spatial tetrahedron of theorem 10.4, and define

$$\sigma_{\Delta} := \lambda_+ \circ h_{\Delta} : T_{\Delta} \longrightarrow G(\mathbb{R}). \quad (506)$$

Then σ_{Δ} is invariant under the marked face D_3 -action and

$$p_{\mathbb{R}}(\sigma_{\Delta}(q)) = h_{\Delta}(q), \quad (507)$$

$$\chi_{\mathbb{R}}(\sigma_{\Delta}(q)) = \begin{cases} 0, & q \in F_{\Delta}, \\ 1, & q \in T_{\Delta} \setminus F_{\Delta}. \end{cases} \quad (508)$$

Thus the C_1 value of the lifted height records exactly whether a point of the tetrahedron has positive signed height above its marked face.

The central count fibre has the exact parametrization

$$\Pi_{\text{cnt}}^{-1}(c) \cap T_{\Delta} = \left\{ q_t := \left(0, \frac{3}{2} - \frac{t}{4}, 0 \right)^{\top} : 0 \leq t \leq 1 \right\}. \quad (509)$$

Along this fibre,

$$\hat{\Pi}_{\text{cnt}}(q_t) = \left(c, \frac{t}{4} \right), \quad \sigma_{\Delta}(q_0) = \tau, \quad \sigma_{\Delta}(q_t) = \frac{t}{4} \quad (0 < t \leq 1). \quad (510)$$

In particular,

$$q_0 = \bar{p}, \quad q_1 = p_{\star}, \quad \sigma_{\Delta}(\bar{p}) = \tau, \quad \sigma_{\Delta}(p_{\star}) = \frac{1}{4}.$$

Proof. For $x, y \geq 0$, positivity satisfies

$$\mathbf{1}_{\{x+y>0\}} = \mathbf{1}_{\{x>0\}} \vee \mathbf{1}_{\{y>0\}}, \quad \mathbf{1}_{\{xy>0\}} = \mathbf{1}_{\{x>0\}} \wedge \mathbf{1}_{\{y>0\}}.$$

The formula

$$\lambda_+(t) = (t, \mathbf{1}_{\{t>0\}})$$

therefore preserves addition and multiplication in the componentwise semiring $D_{\mathbb{R}}$. It sends the additive identity to $(0, 0) = \tau$ and the multiplicative identity to $(1, 1)$, and its first coordinate is t . This proves existence in (504).

Let s be any semiring homomorphism satisfying $p_{\mathbb{R}} \circ s = \text{id}$. Preservation of the additive identity forces $s(0) = \tau$. For $t > 0$, the fibre $p_{\mathbb{R}}^{-1}(t)$ consists of the single supported element t . Hence $s(t) = t$, proving uniqueness.

The affine function h_{Δ} takes the values $1/4, 0, 0, 0$ on the four vertices of T_{Δ} . It consequently takes values in $[0, 1/4]$ throughout the tetrahedron, and its zero locus there is the face F_{Δ} . Equations (507) and (508) now follow from (505). The D_3 -invariance follows from the invariance of h_{Δ} .

It remains to identify the central fibre. Write

$$q = \alpha_{\star} p_{\star} + \alpha_0 p_0 + \alpha_+ p_+ + \alpha_- p_-, \quad \alpha_{\star} + \alpha_0 + \alpha_+ + \alpha_- = 1, \quad \alpha_{\bullet} \geq 0.$$

The equation $\Pi_{\text{cnt}}(q) = c$ is equivalent to $X = W = 0$. The W -coordinate gives $\alpha_+ = \alpha_-$, and the X -coordinate then gives $\alpha_0 = \alpha_+ = \alpha_-$. Therefore

$$q = \alpha_{\star} p_{\star} + (1 - \alpha_{\star}) \bar{p}.$$

Putting $t = \alpha_{\star}$ proves (509). Finally,

$$h_{\Delta}(q_t) = \frac{t}{4}$$

and $\Pi_{\text{cnt}}(q_t) = c$, which proves (510). $\square$

Theorem 11.5 (Carrier retained under cancellation). *For a nonempty finite family $a_1, \dots, a_n \in R$, write $\tilde{a}_i \in G(R)$ for the supported copy of a_i . Then*

$$\tilde{a}_1 \oplus \dots \oplus \tilde{a}_n = a_1 + \dots + a_n. \quad (511)$$

In particular,

$$a_1 + \dots + a_n = 0_R \implies \tilde{a}_1 \oplus \dots \oplus \tilde{a}_n = e \neq \tau,$$

whereas the empty sum equals τ .

Proof. Repeated use of the supported addition law gives (511). If the ring sum is zero, its supported copy is $e = 0_R \in R \subset G(R)$. The empty sum in any semiring is its additive identity, which here is τ . The union in (498) is disjoint, so $e \neq \tau$. $\square$

Corollary 11.6 (Static support under phase and sheet symmetries). *Let a group act on a nonempty finite supported family by permuting its summands. If their ordinary sum is zero, every permuted sum has split value e , never τ . Applied to the C_3 phase permutations and the C_2 sheet involution in $\mathfrak{S}_{123}$, this gives the split-zero form of the C_1 statement: internal phase and sheet symmetries preserve the presence of the carrier even at a cancellation point.*

Proof. A permutation changes neither the cardinality of the nonempty family nor its ordinary sum. Apply theorem 11.5. $\square$

Theorem 11.7 (Split-zero realization of the directed blade supports). *In $G(\mathbb{Z})$, denote the supported copy of $n \in \mathbb{Z}$ by $\tilde{n}$. Lift the tag-53 transport matrix by recording absent entries as τ :*

$$\tilde{N} = \begin{pmatrix} \tau & \tau \\ \tilde{2} & \tau \end{pmatrix} \in M_2(G(\mathbb{Z})). \quad (512)$$

Let $\hat{\chi}$ be the entrywise extension of the support character $\chi_{\mathbb{Z}} : G(\mathbb{Z}) \rightarrow \mathbb{B}$. Then

$$\hat{\chi}(\tilde{N}) = E_{21}, \quad (513)$$

$$\hat{\chi}(\tilde{N}^{\text{T}} \tilde{N}) = E_{11} = Q_{\text{s}}, \quad (514)$$

$$\hat{\chi}(\tilde{N} \tilde{N}^{\text{T}}) = E_{22} = Q_{\text{r}}. \quad (515)$$

Equivalently, with multiplication in $M_2(\mathbb{B})$,

$$\hat{\chi}(\tilde{N})^\top \hat{\chi}(\tilde{N}) = Q_s, \quad \hat{\chi}(\tilde{N}) \hat{\chi}(\tilde{N})^\top = Q_r. \quad (516)$$

Thus the C_1 source and target carriers of the directed blade coefficient are exactly the two real support projectors in (138). In particular, the target carrier is the occupied idempotent E_{22} in the output-sign-— sector. Under the endpoint Boolean carriage of theorem 14.71, the same nonzero matrix cell E_{21} has source state $b = 1$, target state $b = 0$, and is therefore an exit-boundary cell. The coefficient itself remains supported in the split-zero semiring; $b = 0$ records absence of the target divisor fibre, rather than disappearance of the matrix coefficient.

Proof. The character $\chi_{\mathbb{Z}}$ is a semiring homomorphism by theorem 11.1; its entrywise extension therefore commutes with matrix multiplication. From (512),

$$\tilde{N}^\top \tilde{N} = \begin{pmatrix} \tilde{4} & \tau \\ \tau & \tau \end{pmatrix}, \quad \tilde{N} \tilde{N}^\top = \begin{pmatrix} \tau & \tau \\ \tau & \tilde{4} \end{pmatrix}.$$

The entries $\tilde{2}$ and $\tilde{4}$ are supported, so their characters equal 1, while $\chi_{\mathbb{Z}}(\tau) = 0$. This proves (513)–(515). The two identities in (516) follow from

$$E_{12}E_{21} = E_{11}, \quad E_{21}E_{12} = E_{22}.$$

The identifications with Q_s and Q_r are (137) and (138). Finally, (150) places the range of E_{21} in the output-sign-— sector. $\square$

Theorem 11.8 (Arithmetic orbit lift to the directed target carrier). *Fix a prime p , an integer $a > p/4$, and the reduced shell data*

$$R = 4a - p, \quad N = pa, \quad g = \gcd(R, N), \quad R_0 = R/g, \quad N_0 = N/g.$$

Write

$$\mathcal{D}_{p,a}^{\text{red}} = \{d \in \mathbb{Z}_{>0} : d \mid N_0^2, d \equiv -N_0 \pmod{R_0}\}.$$

For $d \in \mathcal{D}_{p,a}^{\text{red}}$, define

$$\zeta_d = -\frac{N_0 d}{(N_0 + d)^2} \quad (517)$$

and

$$\mathcal{J}_{p,a}(d) := \mathcal{J}_2(\zeta_d) = 4E_{11} - \frac{4N_0 d}{(N_0 + d)^2} E_{22}. \quad (518)$$

Then

$$\iota(d) := \frac{N_0^2}{d} \quad (519)$$

is an involution of $\mathcal{D}_{p,a}^{\text{red}}$, and $\mathcal{J}_{p,a}$ induces a bijection

$$\mathcal{D}_{p,a}^{\text{red}} / \langle \iota \rangle \xrightarrow{\sim} \text{im}(\mathcal{J}_{p,a}). \quad (520)$$

Equivalently, for $d_1, d_2 \in \mathcal{D}_{p,a}^{\text{red}}$,

$$\mathcal{J}_{p,a}(d_1) = \mathcal{J}_{p,a}(d_2) \iff d_1 = d_2 \text{ or } d_1 d_2 = N_0^2. \quad (521)$$

Let

$$\Delta_d := \mathcal{J}_{p,a}(d) - 4E_{11} = -\frac{4N_0 d}{(N_0 + d)^2} E_{22}.$$

After lifting its zero entries to τ in $M_2(G(\mathbb{Q}))$, the split-zero support character satisfies

$$\hat{\chi}(\tilde{\Delta}_d) = E_{22} = Q_r. \quad (522)$$

Thus every reduced arithmetic certificate realizes the same directed target carrier as the tag-53 blade coefficient, with its exact amplitude retained in (518).

If $R_0 > 2$, every ι -orbit has two elements and

$$\Omega_R^{\text{red}}(p) = 2 |\text{im}(\mathcal{J}_{p,a})|. \quad (523)$$

In this case

$$\text{im}(\mathcal{J}_{p,a}) \subset \{4E_{11} + cE_{22} : -1 < c < 0\}. \quad (524)$$

The lower endpoint of the corresponding closed segment is

$$4E_{11} - E_{22} = j_2\left(-\frac{1}{4}\right) = \mathfrak{J}(N_2), \quad N_2 = 2E_{21}. \quad (525)$$

Proof. For $d \in \mathcal{D}_{p,a}^{\text{red}}$, put $e = N_0^2/d$. Reduction of $de = N_0^2$ modulo R_0 gives

$$(-N_0)e \equiv N_0^2 \pmod{R_0}.$$

Since $\gcd(R_0, N_0) = 1$, cancellation gives $e \equiv -N_0 \pmod{R_0}$. Hence e lies in the same divisor set, and $\iota^2(d) = d$.

The reconstruction in theorem 9.68 gives

$$b = \frac{N_0 + d}{R_0}, \quad c = \frac{N_0 + N_0^2/d}{R_0} = \frac{N_0(N_0 + d)}{dR_0}.$$

Consequently

$$\frac{b}{b+c} = \frac{d}{N_0+d}, \quad -\frac{bc}{(b+c)^2} = -\frac{N_0d}{(N_0+d)^2}.$$

This proves (517) and, by theorem 9.47, (518).

The section j_2 is injective in ζ , because its E_{22} -coefficient is 4ζ . Set $x = d_1/N_0$ and $y = d_2/N_0$. Direct cross-multiplication gives

$$\begin{aligned} \frac{x}{(1+x)^2} = \frac{y}{(1+y)^2} &\iff x(1+y)^2 - y(1+x)^2 = 0 \\ &\iff (x-y)(1-xy) = 0. \end{aligned}$$

Therefore equality of the two lifted matrices is equivalent to $d_1 = d_2$ or $d_1d_2 = N_0^2$. This proves (521) and the induced bijection (520).

The coefficient of E_{22} in Δ_d is a nonzero rational number. Its split-zero lift has one supported entry, in position $(2, 2)$, and three entries equal to τ . Applying the entrywise support character proves (522).

A fixed point of ι is $d = N_0$. Membership of this divisor in $\mathcal{D}_{p,a}^{\text{red}}$ would give $N_0 \equiv -N_0 \pmod{R_0}$, hence $R_0 \mid 2N_0$. Coprimality gives $R_0 \mid 2$. Thus $R_0 > 2$ makes every orbit two-element, and (523) follows from $|\mathcal{D}_{p,a}^{\text{red}}| = \Omega_R^{\text{red}}(p)$.

For positive N_0, d ,

$$(N_0 + d)^2 - 4N_0d = (N_0 - d)^2 \geq 0.$$

Equality holds exactly at $d = N_0$. When $R_0 > 2$, this divisor is absent, so

$$-1 < -\frac{4N_0d}{(N_0 + d)^2} < 0.$$

This proves (524). Its lower endpoint is (525) by equation (299). $\square$

Theorem 11.9 (Exact C_1 occupancy criterion for a reduced shell). *Retain the reduced shell data and divisor set of theorem 11.8. In $M_2(G(\mathbb{Q}))$, define*

$$\tilde{\Delta}_{p,a}^{\Sigma} := \bigoplus_{d \in \mathcal{D}_{p,a}^{\text{red}}} \tilde{\Delta}_d, \quad (526)$$

where the empty sum is the 2×2 matrix whose four entries are τ . Then

$$\hat{\chi}_{\mathbb{Q}}\left(\tilde{\Delta}_{p,a}^{\Sigma}\right) = \begin{cases} 0, & \Omega_R^{\text{red}}(p) = 0, \\ E_{22}, & \Omega_R^{\text{red}}(p) > 0. \end{cases} \quad (527)$$

Thus

$$\Omega_R^{\text{red}}(p) > 0 \iff \hat{\chi}_{\mathbb{Q}}\left(\tilde{\Delta}_{p,a}^{\Sigma}\right) = E_{22}. \quad (528)$$

If $R_0 > 2$, let $\mathcal{O}_{p,a} := \mathcal{D}_{p,a}^{\text{red}}/\langle \iota \rangle$. For each orbit O , choose either member d_O . The scalar

$$A_{p,a} := 8 \sum_{O \in \mathcal{O}_{p,a}} \frac{N_0 d_O}{(N_0 + d_O)^2} \quad (529)$$

is independent of the chosen orbit members, and, whenever the divisor set is occupied,

$$\tilde{\Delta}_{p,a}^{\Sigma} = -\widetilde{A_{p,a}} E_{22}. \quad (530)$$

In particular, the Boolean carrier in (527) retains a separately specified strictly positive amplitude $A_{p,a}$.

Proof. For every $d \in \mathcal{D}_{p,a}^{\text{red}}$,

$$\Delta_d = -\frac{4N_0 d}{(N_0 + d)^2} E_{22}.$$

Its E_{22} -coefficient is strictly negative, and its other three entries are absent. If the divisor set is empty, the aggregate in (526) is the matrix-semiring zero, so its Boolean image is the zero matrix. If the divisor set is occupied, its E_{22} -entry is the supported copy of the strictly negative number

$$-4 \sum_{d \in \mathcal{D}_{p,a}^{\text{red}}} \frac{N_0 d}{(N_0 + d)^2},$$

and the other entries remain τ . The unique Boolean character from theorem 11.3 therefore sends the aggregate to E_{22} . This proves (527) and (528).

For $e = N_0^2/d$, direct substitution gives

$$\frac{N_0 e}{(N_0 + e)^2} = \frac{N_0 d}{(N_0 + d)^2}.$$

When $R_0 > 2$, every orbit is the pair $\{d, e\}$ by theorem 11.8. Each orbit therefore contributes

$$-\frac{8N_0 d_O}{(N_0 + d_O)^2} E_{22}$$

to the aggregate. This proves the independence assertion, (529), and (530). $\square$

Corollary 11.10 (Global C_1 -carrier form of Erdős–Straus). *The Erdős–Straus conjecture is equivalent to*

$$\forall p \in \mathbb{P} \quad \exists a \in \mathbb{Z}, a > p/4 : \hat{\chi}_{\mathbb{Q}}\left(\tilde{\Delta}_{p,a}^{\Sigma}\right) = E_{22}. \quad (531)$$

The Boolean map appearing here is uniquely determined by the split-zero semiring, and the amplitude attached to every occupied shell is the exact quantity in (529).

Proof. By theorem 9.70, the conjecture is equivalent to the existence, for every prime p , of a fixed shell with $\Omega_R^{\text{red}}(p) > 0$. Apply (528) shell by shell. The uniqueness of the Boolean map is theorem 11.3. $\square$

Theorem 11.11 (Exact amplitude and Lorentz transport of an occupied reduced shell). *Retain the hypotheses of theorem 11.9, assume $R_0 > 2$, and suppose that $\mathcal{D}_{p,a}^{\text{red}} \neq \emptyset$. Put*

$$\alpha := A_{p,a} > 0$$

and define the aggregate coefficient matrix

$$K_{p,a}^\Sigma := 4E_{11} - \alpha E_{22}. \quad (532)$$

Then

$$K_{p,a}^\Sigma = P_s(2) - \alpha Q_r = j_2 \left(-\frac{\alpha}{4} \right). \quad (533)$$

Its inversive-circle and Sarnak coordinates are

$$(A, B_0, B_1, C)_{p,a}^\Sigma = (4, 0, 0, -\alpha), \quad (534)$$

$$(T, X, Y, W)_{p,a}^\Sigma = \left(\frac{4-\alpha}{2}, 0, 0, \frac{4+\alpha}{2} \right). \quad (535)$$

Consequently

$$\mathcal{N}(K_{p,a}^\Sigma) = 4\alpha, \quad F_D(K_{p,a}^\Sigma) = 8\alpha, \quad (536)$$

$$W^2 - T^2 = 4\alpha, \quad \lambda_\perp(K_{p,a}^\Sigma) = -\frac{\alpha}{4}. \quad (537)$$

For $N_2 = 2E_{21}$, the target Gram matrix and its unique nonzero idempotent multiple are

$$N_2 N_2^\top = 4E_{22}, \quad Q_r = \frac{1}{4} N_2 N_2^\top = E_{22}.$$

Thus the simple coefficient-inversion eigenvalue factors exactly as

$$\lambda_\perp(K_{p,a}^\Sigma) = -\alpha \left(\frac{1}{4} \right). \quad (538)$$

Let

$$u_\alpha := \alpha^{1/4}, \quad D_\alpha := \begin{pmatrix} u_\alpha & 0 \\ 0 & u_\alpha^{-1} \end{pmatrix}, \quad P_\alpha := D_\alpha P_\square. \quad (539)$$

Then $D_\alpha, P_\alpha \in \text{SL}_2(\mathbb{R}[\iota])$, and the two ordered off-diagonal channels have the exact transports

$$\mathcal{A}_{P_\alpha}(\mathcal{H}_\square(2\sqrt{\alpha} E_{21})) = K_{p,a}^\Sigma, \quad (540)$$

$$\mathcal{A}_{P_\alpha P_\square}(\mathcal{H}_\square(2\sqrt{\alpha} E_{12})) = K_{p,a}^\Sigma. \quad (541)$$

The intermediate step in the second line is

$$\mathcal{A}_{P_\square}(\mathcal{H}_\square(2\sqrt{\alpha} E_{12})) = \mathcal{H}_\square(2\sqrt{\alpha} E_{21}). \quad (542)$$

For $Q \in \text{SL}_2(\mathbb{C})$,

$$\mathcal{A}_Q(\mathcal{H}_\square(2E_{21})) = K_{p,a}^\Sigma \iff \alpha = 1 \text{ and } Q \text{ belongs to the transporter coset of } P_\square. \quad (543)$$

At $\alpha = 1$, the matrix in (532) is the directed quarter carrier

$$K_{p,a}^\Sigma = 4E_{11} - E_{22} = \mathfrak{H}(N_2), \quad \lambda_\perp = -\frac{1}{4}.$$

Proof. Equation (530) gives the exact transverse coefficient $-\alpha E_{22}$. Adjoining the single fixed source term $4E_{11} = P_s(2)$ gives (532). Formula (294) with $m = 2$ and $\zeta = -\alpha/4$ proves (533).

The coordinate rule

$$T = \frac{A+C}{2}, \quad X = B_0, \quad Y = B_1, \quad W = \frac{A-C}{2}$$

gives (534) and (535). Direct calculation gives

$$-\det K_{p,a}^\Sigma = 4\alpha$$

and

$$\left(\frac{4+\alpha}{2}\right)^2 - \left(\frac{4-\alpha}{2}\right)^2 = 4\alpha.$$

The Descartes value is twice the inversive norm. Equation (295) gives $\lambda_\perp = -\alpha/4$. The idempotent calculation is theorem 9.11, and proves (538).

Write $u = u_\alpha$, so that $\sqrt{\alpha} = u^2$. By (192) and real-linearity,

$$\mathcal{A}_{P_\square}(\mathcal{H}_\square(2u^2E_{21})) = u^2 \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}.$$

Since

$$D_\alpha^{-1} = \begin{pmatrix} u^{-1} & 0 \\ 0 & u \end{pmatrix},$$

one obtains

$$\mathcal{A}_{D_\alpha} \left(u^2 \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix} \right) = \begin{pmatrix} 4 & 0 \\ 0 & -u^4 \end{pmatrix} = K_{p,a}^\Sigma.$$

The identity $\mathcal{A}_D \circ \mathcal{A}_P = \mathcal{A}_{DP}$ proves (540). The first arrow of theorem 9.46, applied by real-linearity, proves (542); composing it with (540) proves (541).

For every $Q \in \mathrm{SL}_2(\mathbb{C})$, the determinant identity (136) gives

$$-\det \mathcal{A}_Q(\mathcal{H}_\square(2E_{21})) = 4.$$

The right side of (543) has inversive norm 4α ; equality of the matrices therefore forces $\alpha = 1$. When $\alpha = 1$, equation (195) supplies $P_\square$, and the complete transporter description in theorem 9.21 gives its stabilizer coset. This proves (543). $\square$

Corollary 11.12 (Champion aggregate amplitude in exact Lorentz coordinates). *For the champion shell of theorem 11.15, the retained aggregate amplitude is*

$$\alpha_* = A_{p_*,a_*} = \frac{36378}{82719025}. \quad (544)$$

Hence

$$K_{p_*,a_*}^\Sigma = 4E_{11} - \frac{36378}{82719025}E_{22}, \quad (545)$$

$$\lambda_\perp(K_{p_*,a_*}^\Sigma) = -\frac{18189}{165438050}, \quad (546)$$

$$(T, X, Y, W)_{p_*,a_*}^\Sigma = \left(\frac{165419861}{82719025}, 0, 0, \frac{165456239}{82719025} \right). \quad (547)$$

In particular, $\alpha_* \neq 1$. The champion arithmetic aggregate and $\mathcal{H}_\square(2E_{21})$ have the common Boolean target support E_{22} and inversive norms

$$\frac{145512}{82719025} \quad \text{and} \quad 4,$$

respectively. The exact directed raw-cell amplitude belonging to the first norm is

$$2\sqrt{\alpha_*} = \frac{6\sqrt{4042}}{9095}. \quad (548)$$

Proof. The champion reduced divisor set is one complementary pair. Taking $d_* = d^-$, one has $d_*/N_* = 1/18189$. Therefore

$$A_{p_*,a_*} = 8 \frac{N_* d_*}{(N_* + d_*)^2} = \frac{8 \cdot 18189}{18190^2} = \frac{36378}{82719025}.$$

Substitution in (532), (535), and (537) gives (545)–(547). The norm is $4\alpha_* = 145512/82719025$. Finally,

$$82719025 = 9095^2, \quad 36378 = 9 \cdot 4042,$$

which proves (548). $\square$

Theorem 11.13 (Every reduced shell lies on one exact Lorentz null-coordinate line). *For arbitrary reduced shell data as in theorem 9.67, define*

$$\alpha_{p,a}^\Sigma := 4 \sum_{d \in \mathcal{D}_{p,a}^{\text{red}}} \frac{N_0 d}{(N_0 + d)^2}, \quad (549)$$

where the empty sum is zero, and put

$$K_{p,a}^\Sigma := 4E_{11} - \alpha_{p,a}^\Sigma E_{22}. \quad (550)$$

Its circle equation and Sarnak coordinates are

$$q_{p,a}^\Sigma(x, y) = 4(x^2 + y^2) - \alpha_{p,a}^\Sigma, \quad (551)$$

$$(T, X, Y, W)_{p,a}^\Sigma = \left(\frac{4 - \alpha_{p,a}^\Sigma}{2}, 0, 0, \frac{4 + \alpha_{p,a}^\Sigma}{2} \right). \quad (552)$$

In null coordinates

$$U := T + W, \quad V := T - W, \quad (553)$$

one has

$$U = 4, \quad V = -\alpha_{p,a}^\Sigma, \quad W^2 - T^2 = -UV = 4\alpha_{p,a}^\Sigma. \quad (554)$$

The following statements are equivalent:

- (i) $\mathcal{D}_{p,a}^{\text{red}} \neq \emptyset$;
- (ii) $\Omega_R^{\text{red}}(p) > 0$;
- (iii) $\alpha_{p,a}^\Sigma > 0$;
- (iv) $-\det K_{p,a}^\Sigma > 0$;
- (v) $V < 0$;
- (vi) $W > |T|$;
- (vii) the zero set of $q_{p,a}^\Sigma$ is the Euclidean circle of radius $\sqrt{\alpha_{p,a}^\Sigma}/2 > 0$.

If the shell is empty, then

$$\alpha_{p,a}^\Sigma = 0, \quad K_{p,a}^\Sigma = 4E_{11}, \quad (U, V) = (4, 0), \quad (T, W) = (2, 2), \quad (555)$$

and (551) has the single real solution $(x, y) = (0, 0)$.

When $R_0 > 2$,

$$\alpha_{p,a}^\Sigma = A_{p,a}, \quad (556)$$

so this theorem extends theorem 11.11 to the empty shell and to the cases in which the divisor involution may have a fixed point.

Proof. Every summand in (549) is strictly positive. Therefore the sum is positive exactly when the divisor set is occupied. This proves the equivalence of (i), (ii), and (iii), using the definition of $\Omega_R^{\text{red}}(p)$.

The matrix in (550) is $H(4, 0, 0, -\alpha_{p,a}^\Sigma)$. Hence

$$-\det K_{p,a}^\Sigma = 4\alpha_{p,a}^\Sigma,$$

which proves (iii) $\iff$ (iv). The coefficient-to-Sarnak map gives (552), and addition and subtraction of its two nonzero coordinates gives (554). Thus (iii) $\iff$ (v). Since

$$W = \frac{4 + \alpha_{p,a}^\Sigma}{2} > 0$$

and

$$W^2 - T^2 = 4\alpha_{p,a}^\Sigma,$$

one has $\alpha_{p,a}^\Sigma > 0$ exactly when $W > |T|$, proving (iii) $\iff$ (vi). Equation (551) proves the equivalence with (vii) and gives the empty-shell statement.

If $R_0 > 2$, every complement orbit is $\{d, N_0^2/d\}$, and

$$\frac{N_0(N_0^2/d)}{(N_0 + N_0^2/d)^2} = \frac{N_0 d}{(N_0 + d)^2}.$$

Each orbit therefore contributes

$$8 \frac{N_0 d}{(N_0 + d)^2}$$

to (549). This is the defining orbit sum for $A_{p,a}$, proving (556). $\square$

Corollary 11.14 (Global Lorentz null-coordinate form of Erdős–Straus). *The Erdős–Straus conjecture is equivalent to each of the following statements:*

$$\forall p \in \mathbb{P} \quad \exists a \in \mathbb{Z}, a > p/4 : -\det K_{p,a}^\Sigma > 0, \quad (557)$$

$$\forall p \in \mathbb{P} \quad \exists a \in \mathbb{Z}, a > p/4 : U = 4, \quad V < 0, \quad (558)$$

$$\forall p \in \mathbb{P} \quad \exists a \in \mathbb{Z}, a > p/4 : q_{p,a}^\Sigma(x, y) = 0 \text{ is a circle of positive radius.} \quad (559)$$

For every shell, the arithmetic family begins at the fixed null point $4E_{11}$ and moves only in the signed direction $-E_{22}$; the displacement is the exact scalar $\alpha_{p,a}^\Sigma$.

Proof. Apply theorem 9.70 and the equivalent statements in theorem 11.13 shell by shell. $\square$

Theorem 11.15 (Exact identification of the champion divisor orbit with the tag-decomposition orbit). *For*

$$p_* = 8,803,369, \quad R_* = 107, \quad a_* = 3^2 11^2 43 47, \quad N_* = p_* a_*,$$

put

$$\mathcal{B} = [-2, 2] \times [-2, 2] \times [-1, 1] \times [-1, 1]$$

and

$$\Lambda(\beta) = 70\beta_1 + 22\beta_2 + 59\beta_3 + 66\beta_4 \quad \text{in } C_{106}. \quad (560)$$

The complete target fibres are

$$\begin{aligned} \{\beta \in \mathcal{B} : \Lambda(\beta) = 53\} &= \{\beta^-, \beta^+\}, \\ \beta^- &= (-2, 0, -1, -1), & \beta^+ &= (2, 0, 1, 1), \end{aligned} \quad (561)$$

$$\{\beta \in \mathcal{B} : \Lambda(\beta) = 60\} = \emptyset. \quad (562)$$

The corresponding complete reduced divisor set is

$$\mathcal{D}_{p_*, a_*} = \{d^-, d^+\}, \quad (563)$$

where

$$\begin{aligned} d^- &= p_* 11^2 = 1,065,207,649, \\ d^+ &= p_* 3^4 11^2 43^2 47^2 = 352,413,001,402,225,929, \end{aligned} \quad (564)$$

and

$$d^- d^+ = N_*^2, \quad d^- < N_* < d^+. \quad (565)$$

Consequently

$$(m_0, m_1, m_2) = (0, 2, 0), \quad \Omega_{107}(p_*) = 2, \quad |\text{im}(\mathcal{J}_{p_*, a_*})| = 1. \quad (566)$$

Split $\Lambda = \Lambda_s + \Lambda_l$ on these two vectors by

$$\Lambda_s(\beta) = 70\beta_1 + 22\beta_2, \quad \Lambda_l(\beta) = 59\beta_3 + 66\beta_4.$$

Then

$$\begin{aligned} (\Lambda_s(\beta^-), \Lambda_l(\beta^-)) &= (72, 87), \\ (\Lambda_s(\beta^+), \Lambda_l(\beta^+)) &= (34, 19) \quad \text{in } C_{106}. \end{aligned} \quad (567)$$

These are exactly the two tag decompositions contributing to $\mathcal{M}(53)$, and hence

$$\sum_{\beta \in \{\beta^-, \beta^+\}} E_{21} = 2E_{21} = \mathcal{M}(53). \quad (568)$$

The endpoint calculation in theorem 14.69 shows that this $(2, 1)$ -entry is $2q_{107, p_*}C_{0+}$, the population lift of the outgoing edge $r_0 \rightarrow r_1$; its incoming transpose has zero coefficient at residue 53. By theorem 14.71, $r_0 \in \partial_T^{\text{out}} S_{107, a_*}$, so the displayed coefficient is the exit-boundary lift. If

$$(53 + \delta, -\delta) \in \{(34, 19), (72, 87)\}, \quad \delta \in \{-19, 19\},$$

then the exact orientation relation is

$$\text{sgn} \left[\frac{1}{2} \left(d(\beta) - \frac{N_*^2}{d(\beta)} \right) \right] = -\text{sgn } \delta(\beta). \quad (569)$$

The blade row and column signs in (568) remain the scalar-sign and primitive-leg-sign coordinates used to define $\mathcal{M}(t)$.

Proof. The logarithms modulo 107, with primitive generator 2, are

$$\log_2(3) = 70, \quad \log_2(11) = 22, \quad \log_2(43) = 59, \quad \log_2(47) = 66.$$

The middle and exterior target logarithms are respectively 53 and 60. To solve the middle target, reduce

$$70\beta_1 + 22\beta_2 + 59\beta_3 + 66\beta_4 \equiv 53 \pmod{106}.$$

Parity forces $\beta_3 = \pm 1$. The complete table of $35\beta_1 + 11\beta_2 \pmod{53}$, with rows $\beta_1 = -2, -1, 0, 1, 2$ and columns $\beta_2 = -2, -1, 0, 1, 2$, is

$$\begin{pmatrix} 14 & 25 & 36 & 47 & 5 \\ 49 & 7 & 18 & 29 & 40 \\ 31 & 42 & 0 & 11 & 22 \\ 13 & 24 & 35 & 46 & 4 \\ 48 & 6 & 17 & 28 & 39 \end{pmatrix}.$$

For $\beta_3 = -1$, the required table entries as $\beta_4 = -1, 0, 1$ are 36, 3, 23. Only 36 occurs, uniquely at $(\beta_1, \beta_2) = (-2, 0)$, and it forces $\beta_4 = -1$. For $\beta_3 = 1$, the required entries are 30, 50, 17. Only 17 occurs, uniquely at $(\beta_1, \beta_2) = (2, 0)$, and it forces $\beta_4 = 1$. This proves (561).

For the exterior target 60, parity forces $\beta_3 = 0$. Division by two reduces the congruence to

$$35\beta_1 + 11\beta_2 \equiv 30 - 33\beta_4 \pmod{53}.$$

For $\beta_4 = -1, 0, 1$, the required residues are 10, 30, 50; none occurs in the displayed table. This proves (562).

For a centered exponent vector β , the middle-origin divisor is

$$d(\beta) = p_* 3^{2+\beta_1} 11^{2+\beta_2} 43^{1+\beta_3} 47^{1+\beta_4}.$$

Substitution of β^- and β^+ gives (564). Their centered exponent vectors are opposites, so their product is $p_*^2 a_*^2 = N_*^2$. Their exponents also give $d^- < N_* < d^+$. The two-target shell theorem and (562) now prove (563) and (566).

Finally,

$$-140 \equiv 72, \quad -125 \equiv 87, \quad 140 \equiv 34, \quad 125 \equiv 19 \pmod{106},$$

which proves (567). Both pairs sum to 53, and theorem 9.9 places both contributions in the $(2, 1)$ -entry, proving (568). The vector β^- gives $d^- < N_*$ and $\delta = 19$, while β^+ gives $d^+ > N_*$ and $\delta = -19$. This proves (569). $\square$

Corollary 11.16 (The champion shell half-boost is the Frobenius Fable inversion over $\mathbb{F}_{107^2}$). *Use the two champion divisors and centered exponent vectors of theorem 11.15. Their half-boost eigenvalues are*

$$\nu_+ = \sqrt{\frac{d^+}{N_*}} = 3\sqrt{2021}, \quad \nu_- = \sqrt{\frac{d^-}{N_*}} = \frac{1}{3\sqrt{2021}} = \nu_+^{-1}. \quad (570)$$

Let

$$\mathbb{F}_{107^2} = \mathbb{F}_{107}(\theta), \quad \theta^2 = 95.$$

Then 95 is a quadratic nonresidue modulo 107, and reduction sends

$$\sqrt{2021} \mapsto \theta.$$

In particular,

$$\bar{\nu}_+ = 3\theta, \quad \bar{\nu}_- = -3\theta, \quad \bar{\nu}_+^{107} = -\bar{\nu}_+ = \bar{\nu}_+^{-1} = \bar{\nu}_-. \quad (571)$$

The reduced half-boosts are

$$\bar{S}_{d^+} = \begin{pmatrix} 0 & 3\theta \\ 3\theta & 0 \end{pmatrix}, \quad \bar{S}_{d^-} = \bar{S}_{d^+}^{-1} = -\bar{S}_{d^+}. \quad (572)$$

Hence the ordered champion divisor pair

$$(d^+, d^-),$$

the ordered tag pair

$$((34, 19), (72, 87)),$$

the Frobenius pair

$$(3\theta, -3\theta),$$

and the two matrix lifts

$$(\bar{S}_{d^+}, -\bar{S}_{d^+})$$

are the same two-element residual orbit under the correspondences already proved. Over the characteristic-zero field $\mathbb{Q}(\sqrt{2021})$, divisor complement and Fable-sheet negation remain distinct operations.

Proof. From (564),

$$\frac{d^+}{N_*} = 3^2 \cdot 43 \cdot 47 = 18189, \quad \frac{d^-}{N_*} = \frac{1}{18189},$$

which proves (570). Reduction gives

$$2021 \equiv 95 \pmod{107}, \quad (3\theta)^2 = 9 \cdot 95 = 855 \equiv -1 \pmod{107}.$$

The exact repeated-squaring computation is

$$95^2 \equiv 37, \quad 95^4 \equiv 85, \quad 95^8 \equiv 56, \quad 95^{16} \equiv 33, \quad 95^{32} \equiv 19 \pmod{107},$$

and therefore

$$95^{53} \equiv 19 \cdot 33 \cdot 85 \cdot 95 \equiv 106 \equiv -1 \pmod{107}.$$

Euler's criterion shows that 95 is a nonresidue. Moreover,

$$\theta^{107} = \theta(\theta^2)^{53} = \theta 95^{53} = -\theta,$$

which proves (571). Apply (330) to d^+ and (331) to its complement d^- . The tag correspondence is (567). This proves (572) and the stated orbit identification. $\square$

Proposition 11.17 (Completed-tesseractine and sedenion refinement of the champion orbit). *Let*

$$\Gamma_{\square} = C_2^{\wedge} \times C_2^{\vee},$$

and write $\mathbf{u}_{\epsilon, \delta}$ for the group basis of $\mathbb{Z}[\Gamma_{\square}]$, so that

$$\mathbf{u}_{\epsilon, \delta} \mathbf{u}_{\epsilon', \delta'} = \mathbf{u}_{\epsilon\epsilon', \delta\delta'}.$$

Put

$$\hat{s} = \mathbf{u}_{-, +}, \quad \check{s} = \mathbf{u}_{+, -}.$$

For $\beta \in \mathcal{B}$, define

$$\begin{aligned} \pi(\beta) &= \{i : \beta_i \equiv 1 \pmod{2}\}, \\ q^{\wedge}(\beta) &= \sum_{i=1}^4 \left\lfloor \frac{\beta_i}{2} \right\rfloor \pmod{2}, \quad q^{\vee}(\beta) = \Lambda(\beta) \pmod{2}. \end{aligned}$$

If o_I denotes the sedenion blade indexed by $I \subseteq \{1, 2, 3, 4\}$, the completed character monomial is

$$m_{\square}^{\text{ch}}(\beta) = \hat{s}^{q^{\wedge}(\beta)} \check{s}^{q^{\vee}(\beta)} o_{\pi(\beta)}[\Lambda(\beta)].$$

The two vectors in (561) satisfy

$$\pi(\beta^{\pm}) = \{3, 4\}, \quad q^{\wedge}(\beta^{\pm}) = q^{\vee}(\beta^{\pm}) = 1, \quad (573)$$

and hence

$$m_{\square}^{\text{ch}}(\beta^{-}) = m_{\square}^{\text{ch}}(\beta^{+}) = \hat{s}\check{s} o_{34}[53]. \quad (574)$$

Equivalently, in the primitive-idempotent sector notation

$$e_{\delta}^{\epsilon} = \frac{1}{4}(1 + \epsilon\hat{s})(1 + \delta\check{s}),$$

each vector contributes

$$e_{-}^{-} o_{34}[53]. \quad (575)$$

The factor decompositions retained by the blade construction are

$$\beta^{\pm} = \beta_s^{\pm} + \beta_1^{\pm},$$

where

$$\begin{aligned} \beta_s^{-} &= (-2, 0, 0, 0), & \beta_1^{-} &= (0, 0, -1, -1), \\ \beta_s^{+} &= (2, 0, 0, 0), & \beta_1^{+} &= (0, 0, 1, 1). \end{aligned}$$

Their complete invariants are

γ	$\Lambda(\gamma)$	$\pi(\gamma)$	$q^{\wedge}(\gamma)$	$q^{\vee}(\gamma)$	$m_{\square}^{\text{ch}}(\gamma)$
β_s^{-}	72	$\emptyset$	1	0	$\hat{s}[72]$
β_1^{-}	87	$\{3, 4\}$	0	1	$\check{s} o_{34}[87]$
β_s^{+}	34	$\emptyset$	1	0	$\hat{s}[34]$
β_1^{+}	19	$\{3, 4\}$	0	1	$\check{s} o_{34}[19]$

(576)

and therefore

$$\begin{aligned} m_{\square}^{\text{ch}}(\beta_s^{-}) m_{\square}^{\text{ch}}(\beta_1^{-}) &= \hat{s}\check{s} o_{34}[72 + 87] = \hat{s}\check{s} o_{34}[53], \\ m_{\square}^{\text{ch}}(\beta_s^{+}) m_{\square}^{\text{ch}}(\beta_1^{+}) &= \hat{s}\check{s} o_{34}[34 + 19] = \hat{s}\check{s} o_{34}[53]. \end{aligned} \quad (577)$$

Thus the completed algebra refines the directed E_{21} -entry by the ordered sign pair

$$((- , +), (+ , -))$$

and by the two ordered tag pairs

$$(72, 87), \quad (34, 19).$$

After multiplication, both completed monomials have the same sector, blade, and target tag. The ordered tag pair and the sign in (569) retain the distinction between the two divisors. No cyclic action on these data is used.

Proof. For $\beta^+ = (2, 0, 1, 1)$,

$$q^\wedge(\beta^+) = 1 + 0 + 0 + 0 \equiv 1 \pmod{2}.$$

For $\beta^- = (-2, 0, -1, -1)$,

$$q^\wedge(\beta^-) = -1 + 0 - 1 - 1 = -3 \equiv 1 \pmod{2}.$$

Both vectors have odd third and fourth coordinates and target tag 53, which proves (573). Equations (574) and (575) follow directly from the two definitions.

For the four factors, direct evaluation gives

$$\begin{aligned} \Lambda(-2, 0, 0, 0) &= -140 \equiv 72, & \Lambda(0, 0, -1, -1) &= -125 \equiv 87, \\ \Lambda(2, 0, 0, 0) &= 140 \equiv 34, & \Lambda(0, 0, 1, 1) &= 125 \equiv 19 \pmod{106}. \end{aligned}$$

The floor sums are respectively 1, 0, 1, 0 modulo 2, and the target parities are respectively 0, 1, 0, 1. This proves (576). Multiplication in $\mathbb{Z}[\Gamma_\square]$, addition of the C_{106} -tags, and $1 \cdot o_{34} = o_{34}$ prove (577). The resulting two contributions are exactly the two copies of the directed entry in (568). The completed algebra formulas are the explicit refinement established in [17]. $\square$

Theorem 11.18 (Exact sixty-four-cell completed sign-blade algebra). *Let A_4 be the real sedenion algebra with binary-subset basis*

$$\{o_I : I \subseteq \{1, 2, 3, 4\}\},$$

and let

$$\Gamma_\square = C_2^\wedge \times C_2^\vee.$$

The completed sign-blade algebra used by the champion-orbit calculation is

$$\mathcal{A}_{64} := \mathbb{R}[\Gamma_\square] \otimes_{\mathbb{R}} A_4. \quad (578)$$

It has the homogeneous basis

$$x_{\epsilon, \delta, I} := \mathbf{u}_{\epsilon, \delta} \otimes o_I, \quad (\epsilon, \delta) \in \{\pm 1\}^2, \quad I \subseteq \{1, 2, 3, 4\}, \quad (579)$$

and therefore

$$\dim_{\mathbb{R}} \mathcal{A}_{64} = 4 \cdot 16 = 64. \quad (580)$$

Writing $b(+) = 0$, $b(-) = 1$, each basis vector has the six-bit label

$$\lambda(x_{\epsilon, \delta, I}) = (b(\epsilon), b(\delta), \mathbf{1}_I) \in (\mathbb{Z}/2\mathbb{Z})^6. \quad (581)$$

For the binary-subset Cayley–Dickson basis, define $\omega_4(I, J) \in \{\pm 1\}$ by

$$o_I o_J = \omega_4(I, J) o_{I \triangle J}.$$

Then the complete homogeneous multiplication is

$$x_{\epsilon, \delta, I} x_{\epsilon', \delta', J} = \omega_4(I, J) x_{\epsilon \epsilon', \delta \delta', I \triangle J}. \quad (582)$$

In particular, the two successive sign adjunctions give the exact dimension tower

$$16 = \dim_{\mathbb{R}} A_4, \quad 32 = \dim_{\mathbb{R}} (\mathbb{R}[C_2] \otimes A_4), \quad 64 = \dim_{\mathbb{R}} \mathcal{A}_{64}. \quad (583)$$

For $\alpha, \beta, \gamma \in \Gamma_\square$ and $I, J, K \subseteq \{1, 2, 3, 4\}$, the two bracketings are

$$(x_{\alpha, I} x_{\beta, J}) x_{\gamma, K} = \omega_4(I, J) \omega_4(I \triangle J, K) x_{\alpha \beta \gamma, I \triangle J \triangle K}, \quad (584)$$

$$x_{\alpha, I} (x_{\beta, J} x_{\gamma, K}) = \omega_4(J, K) \omega_4(I, J \triangle K) x_{\alpha \beta \gamma, I \triangle J \triangle K}. \quad (585)$$

Consequently the support of the associator depends on the four blade bits I, J, K and is independent of the two completed-sign labels α, β, γ . The sign completion retains the sector in which the associator lies while preserving its Cayley–Dickson support.

Proof. The group algebra $\mathbb{R}[\Gamma_\square]$ has basis $\mathbf{u}_{\epsilon,\delta}$ and dimension four. The binary-subset basis of A_4 has sixteen elements. Their elementary tensors form the basis (579), proving (580) and (583).

Multiplication in the first tensor factor gives

$$\mathbf{u}_{\epsilon,\delta}\mathbf{u}_{\epsilon',\delta'} = \mathbf{u}_{\epsilon\epsilon',\delta\delta'},$$

and multiplication in the second gives the defining formula for ω_4 . Tensoring the two equations proves (582). Applying that equation twice with the two bracketings gives (584) and (585). Their common group-algebra label proves the associator assertion. The finite checker `verification/verify_completed_sign_sedenion_64.py` verifies all 64^2 homogeneous products, the displayed source products, the two-sided zero-divisor pair, the completed compressed products, and the completed associator residue. $\square$

Theorem 11.19 (Ordered factor histories above the completed product cells). *Put*

$$\mathcal{P}_4 := \mathcal{P}(\{1, 2, 3, 4\})$$

and retain the completed sign group $\Gamma_\square = C_2^\wedge \times C_2^\vee$. The ordered two-factor history carrier is

$$\mathcal{H}_{\text{all}} := \Gamma_\square \times \Gamma_\square \times \mathcal{P}_4, \quad \#\mathcal{H}_{\text{all}} = 4 \cdot 4 \cdot 16 = 256. \quad (586)$$

Its product projection is

$$\Pi(\alpha, \beta, I) := (\alpha\beta, I) \in \Gamma_\square \times \mathcal{P}_4. \quad (587)$$

Every one of the 64 product cells has exactly four ordered histories above it. The factor-order involution

$$\iota(\alpha, \beta, I) := (\beta, \alpha, I) \quad (588)$$

satisfies

$$\Pi \circ \iota = \Pi. \quad (589)$$

The scalar-then-leg histories used in the champion calculation form

$$\mathcal{H}_{\text{sl}} := \{((\epsilon, +), (\lambda, -), I) : \epsilon, \lambda \in \{\pm 1\}, I \in \mathcal{P}_4\}, \quad (590)$$

and their reversed histories form

$$\mathcal{H}_{\text{ls}} := \iota(\mathcal{H}_{\text{sl}}).$$

They are disjoint 64-element sets, so

$$\#(\mathcal{H}_{\text{sl}} \sqcup \mathcal{H}_{\text{ls}}) = 128. \quad (591)$$

Both project onto the same 32-element lower-minus product carrier:

$$\Pi(\mathcal{H}_{\text{sl}}) = \Pi(\mathcal{H}_{\text{ls}}) = \{((\rho, -), I) : \rho \in \{\pm 1\}, I \in \mathcal{P}_4\}. \quad (592)$$

Over the target product cell $((-, -), \{3, 4\})$, the complete four-element fibre is

$$\begin{aligned} h_{21} &= ((-, +), (+, -), \{3, 4\}), \\ h_{12} &= ((+, +), (-, -), \{3, 4\}), \\ \iota(h_{21}) &= ((+, -), (-, +), \{3, 4\}), \\ \iota(h_{12}) &= ((-, -), (+, +), \{3, 4\}). \end{aligned} \quad (593)$$

Here h_{21} is the observed E_{21} history and h_{12} is the competing E_{12} history in scalar-then-leg order. At residue 53, h_{21} occurs with multiplicity two and h_{12} has coefficient zero. The other two entries in (593) record the reversed factor order.

Proof. The cardinality in (586) follows from $\#\Gamma_\square = 4$ and $\#\mathcal{P}_4 = 16$. For a prescribed product cell (γ, I) , choosing $\alpha \in \Gamma_\square$ uniquely determines $\beta = \alpha^{-1}\gamma$. Hence every fibre of Π has four elements. The group $\Gamma_\square$ is abelian, so $\beta\alpha = \alpha\beta$; this proves (589).

The two lower-sign coordinates in $\mathcal{H}_{\text{sl}}$ are $+$ and $-$, while those in $\mathcal{H}_{\text{ls}}$ are $-$ and $+$. This proves disjointness and (591). Multiplication gives

$$(\epsilon, +)(\lambda, -) = (\epsilon\lambda, -) = (\lambda, -)(\epsilon, +),$$

which proves (592); each restricted fibre has two elements.

Solving $\alpha\beta = (-, -)$ in the four-element group gives the four ordered pairs displayed in (593). The first two are the row-column cells E_{21} and E_{12} in (130). Their residue-53 coefficients are 2 and 0, respectively, by (131). $\square$

Corollary 11.20 (The doubly negative cell is the unique full history orbit). *For*

$$\alpha = (\epsilon, \delta), \quad \beta = (\lambda, \eta) \in \Gamma_\square,$$

define the upper-sign transpose on $\mathcal{H}_{\text{all}}$ by

$$\mathsf{T}(\alpha, \beta, I) := ((\lambda, \delta), (\epsilon, \eta), I). \quad (594)$$

Together with the factor-order involution ι in (588), it satisfies

$$\mathsf{T}^2 = \iota^2 = 1, \quad \mathsf{T}\iota = \iota\mathsf{T}, \quad \Pi\mathsf{T} = \Pi\iota = \Pi. \quad (595)$$

On a fibre over the completed sign (ρ, κ) , the orbit sizes of $\langle \mathsf{T}, \iota \rangle$ are

$$\begin{array}{c|cccc} (\rho, \kappa) & (+, +) & (-, +) & (+, -) & (-, -) \\ \hline \text{orbit size} & 1 & 2 & 2 & 4. \end{array} \quad (596)$$

Thus the doubly negative sector is the unique completed sign sector whose four-history fibres are free transitive $C_2 \times C_2$ -sets. On the champion target fibre,

$$\bar{h}_{21} := \iota(h_{21}), \quad \bar{h}_{12} := \iota(h_{12}),$$

and

$$\mathsf{T}(h_{21}) = h_{12}, \quad \iota(h_{21}) = \bar{h}_{21}, \quad \iota\mathsf{T}(h_{21}) = \bar{h}_{12}, \quad (597)$$

which lists the other three points in the orbit of h_{21} .

Proof. Equation (594) exchanges the two upper coordinates and leaves the two lower coordinates in their original slots. Applying it twice gives the identity. Direct substitution proves that it commutes with ι . Both operations preserve the componentwise product in the abelian group $\Gamma_\square$, proving the last two relations in (595).

Write

$$r = b(\rho), \quad k = b(\kappa).$$

Identify the fibre over (ρ, κ) with $(\mathbb{Z}/2\mathbb{Z})^2$ by the bit pair of its first factor. Under this identification, T adds $(r, 0)$, while ι adds (r, k) . Their spans have dimensions

$$0, \quad 1, \quad 1, \quad 2$$

for $(\rho, \kappa) = (+, +), (-, +), (+, -), (-, -)$, respectively. This proves (596). Substitution of h_{21} from (593) gives the final orbit formulas. $\square$

Remark 11.21 (Test for a further mirror coordinate). Let ρ be a specified involutive automorphism of $\mathcal{A}_{64}$. An external sheet label u with

$$u^2 = 1, \quad ua = \rho(a)u$$

gives the vector-space extension

$$\mathcal{A}_{64} \oplus \mathcal{A}_{64}u, \quad \dim_{\mathbb{R}}(\mathcal{A}_{64} \oplus \mathcal{A}_{64}u) = 128. \quad (598)$$

Theorem 11.19 retains factor order at the history level. Equation (598) is the algebra-level extension when the mirror label is required to survive multiplication. The ordered E_{21}/E_{12} calculation must therefore determine whether its mirror acts on the six existing labels in (581) or requires the external sheet label u . This is an exact coordinate test on the blade basis.

Theorem 11.22 (Residuewise separation of the two ordered e_- blade channels). *For $t \in C_{106}$, write*

$$\mathcal{M}(t) = \begin{pmatrix} a_t & b_t \\ c_t & d_t \end{pmatrix} \quad (599)$$

and define the two completed-sector coefficients

$$\mu_{+,-}(t) := a_t + d_t, \quad \mu_{-,-}(t) := b_t + c_t. \quad (600)$$

Then

$$\#\{(i, j) : \mathcal{M}(t)_{ij} > 0\} \leq 1 \quad (601)$$

for every t . In particular, if $\mu_{-,-}(t) > 0$, exactly one of b_t, c_t is positive. The two possibilities have the exact quarter and split-zero data

<i>active coefficient</i>	$\mathcal{M}(t)$	<i>quarter space</i>	$Q_s(t)$	$Q_r(t)$
$b_t > 0$	$b_t E_{12}$	$E_{+1/4}$	E_{22}	E_{11}
$c_t > 0$	$c_t E_{21}$	$E_{-1/4}$	E_{11}	E_{22}

(602)

where the quarter-space column means $\mathcal{T}_\square(\mathcal{M}(t))$ belongs to the displayed eigenspace, and the last two columns are the Boolean source and range supports.

More precisely, let $\widetilde{\mathcal{M}}(t)$ be the matrix over $G(\mathbb{Z})$ obtained by replacing each positive coefficient by its supported copy and each absent coefficient by τ , and put

$$S(t) := \widehat{\chi}(\widetilde{\mathcal{M}}(t)). \quad (603)$$

For an occupied off-diagonal residue,

$$b_t > 0 \implies S(t) = E_{12}, \quad S(t)^\top S(t) = E_{22}, \quad S(t)S(t)^\top = E_{11}, \quad (604)$$

$$c_t > 0 \implies S(t) = E_{21}, \quad S(t)^\top S(t) = E_{11}, \quad S(t)S(t)^\top = E_{22}. \quad (605)$$

Thus the ordered raw cell determines both the quarter eigenspace and the directed C_1 carrier inside the common completed sector e_- .

At the two explicit residues,

$$(a_{53}, b_{53}, c_{53}, d_{53}) = (0, 0, 2, 0), \quad (606)$$

$$(a_{109}, b_{109}, c_{109}, d_{109}) = (a_3, b_3, c_3, d_3) = (0, 1, 0, 0). \quad (607)$$

For the champion shell this gives the exact equality

$$\Omega_{107}(p_*) = c_{53} = 2, \quad b_{53} = 0, \quad (608)$$

and, for each $d \in \{d^-, d^+\}$,

$$\widehat{\chi}(\widetilde{\Delta}_d) = E_{22} = \widehat{\chi}(\widetilde{\mathcal{M}}(53)\widetilde{\mathcal{M}}(53)^\top). \quad (609)$$

Hence the complete champion divisor orbit, its directed blade sum, its split-zero target carrier, and its simple quarter eigenline select the same ordered $(-, +)$ channel.

Proof. The four raw convolution supports are

$$S_{++} = \{9, 19, 29, 47, 59, 77, 87, 97\},$$

$$S_{+-} = \{3, 7, 17, 21, 35, 71, 85, 89, 99, 103\},$$

$$S_{-+} = \{15, 25, 43, 53, 63, 81, 91\},$$

$$S_{--} = \{27, 37, 41, 51, 55, 65, 69, 79\}.$$

Every displayed residue is taken in C_{106} . Inspection of these four finite sets gives

$$S_{\epsilon\lambda} \cap S_{\epsilon'\lambda'} = \emptyset \quad ((\epsilon, \lambda) \neq (\epsilon', \lambda')).$$

This proves (601). The sign product $\mathbf{u}_{\epsilon,+} \mathbf{u}_{\lambda,-} = \mathbf{u}_{\epsilon\lambda,-}$ sends the two diagonal cells to e_-^+ and the two off-diagonal cells to e_-^- , which proves (600) and the asserted exclusive alternative for b_t, c_t .

By (205) and (206),

$$\Pi_{\square,-}\mathcal{M}(t) = c_t E_{21}, \quad \Pi_{\square,+}\mathcal{M}(t) = a_t E_{11} + b_t E_{12} + d_t E_{22}.$$

Together with (601), these identities give the quarter-space column of (602).

The support character sends a positive coefficient to 1 and an absent coefficient to 0. Hence $b_t > 0$ gives $\mathbf{S}(t) = E_{12}$, while $c_t > 0$ gives $\mathbf{S}(t) = E_{21}$. The four products

$$\begin{aligned} E_{21}^\top E_{21} &= E_{11}, & E_{21} E_{21}^\top &= E_{22}, \\ E_{12}^\top E_{12} &= E_{22}, & E_{12} E_{12}^\top &= E_{11} \end{aligned}$$

prove (604) and (605).

The coefficients at 53 and $3 = [109]$ follow directly from the four support sets, including multiplicity two at 53 in S_{-+} . This proves (606) and (607). The complete champion divisor set has two elements and each contributes one E_{21} cell by (568); therefore $\Omega_{107}(p_*) = c_{53} = 2$. Finally, equation (522) gives $\hat{\chi}(\tilde{\Delta}_d) = E_{22}$, and direct multiplication of the supported matrix $2E_{21}$ gives the same range support. This proves (609). $\square$

Corollary 11.23 (The arithmetic point and the branch scale of the champion orbit). *The two divisors in (563) have the common quotient coordinate*

$$\zeta_* = -\frac{18189}{18190^2} = -\frac{18189}{330876100}. \quad (610)$$

Their common target-carrier image is

$$\mathcal{J}_{p_*,a_*}(d^-) = \mathcal{J}_{p_*,a_*}(d^+) = 4E_{11} - \frac{18189}{82719025}E_{22}. \quad (611)$$

The same two divisors give the blade coefficient

$$\mathcal{M}(53) = 2E_{21}.$$

By theorem 14.69, this coefficient is the outgoing exit-boundary lift $2q_{107,p_}C_{0+}$; it is distinct from the incoming cell $C_{-0} = E_{12}$. Writing $m_* = 2$ for this orbit cardinality and*

$$\begin{aligned} \mathfrak{H}_m &:= P_s(m) - Q_r = m^2 E_{11} - E_{22}, \\ \mathfrak{H}_{m_*} &= j_{m_*} \left(-\frac{1}{m_*^2} \right) = 4E_{11} - E_{22}, \end{aligned} \quad (612)$$

and therefore

$$-\frac{1}{m_*^2} = -\frac{1}{4} = \zeta_{\text{br}}. \quad (613)$$

The actual arithmetic coordinate (610) lies in the open interval $(-1/4, 0)$; equation (613) is the branch scale determined by the two-sheet orbit and its directed blade coefficient.

Proof. Since

$$a_* = 121 \cdot 18189, \quad d^- = p_* \cdot 121, \quad N_* = p_* \cdot 121 \cdot 18189,$$

one has $d^-/N_* = 1/18189$. Formula (517) gives

$$\zeta_* = -\frac{18189}{(18189 + 1)^2},$$

which is (610). Substitution in (518), followed by cancellation of a factor 4, gives (611).

Equation (568) gives $m_* = 2$ and $\mathcal{M}(53) = m_* E_{21}$. For every positive m ,

$$j_m\left(-\frac{1}{m^2}\right) = m^2 E_{11} - E_{22} = \mathfrak{H}_m.$$

Taking $m = m_* = 2$ proves (612); the denominator-exchange branch value in (128) proves (613). Finally, $18189 \neq 1$, so $(18189 - 1)^2 > 0$; the strict inequality in (524) yields $\zeta_* \in (-1/4, 0)$. $\square$

Corollary 11.24 (Global target-carrier formulation). *The Erdős–Straus conjecture is equivalent to*

$$\forall p \in \mathbb{P} \quad \exists a \in \mathbb{Z}, \quad a > p/4 : \text{im}(\mathcal{J}_{p,a}) \neq \emptyset. \quad (614)$$

For every choice with $R_0 > 2$, one arithmetic image point accounts for exactly one denominator-exchange orbit and exactly two divisors counted by $\Omega_R^{\text{red}}(p)$.

Proof. By definition, $\mathcal{D}_{p,a}^{\text{red}} \neq \emptyset$ if and only if $\Omega_R^{\text{red}}(p) > 0$. The map $\mathcal{J}_{p,a}$ has nonempty image exactly when its domain is nonempty. Apply theorem 9.70. The final statement is (523). $\square$

Corollary 11.25 (Origin decomposition of the arithmetic carrier image). *Assume $\gcd(R, pa) = 1$ and $R > 2$. For the origin count vector $m(R, p) = (m_0, m_1, m_2)$ of theorem 9.64, one has*

$$m_0 = m_2, \quad m_1 \equiv 0 \pmod{2}, \quad (615)$$

and

$$|\text{im}(\mathcal{J}_{p,a})| = m_0 + \frac{m_1}{2} = m_2 + \frac{m_1}{2}. \quad (616)$$

Consequently

$$\Omega_R(p) = m_0 + m_1 + m_2 = 2m_0 + m_1 = 2|\text{im}(\mathcal{J}_{p,a})|. \quad (617)$$

For $t \in (\mathbb{Z}/R\mathbb{Z})^\times$, put

$$\mathfrak{c}_R(a; t) := \#\{u > 0 : u \mid a^2, \quad \bar{u} \bar{a}^{-1} = t\}.$$

Then

$$|\text{im}(\mathcal{J}_{p,a})| = \mathfrak{c}_R(a; -p^{-1}) + \frac{1}{2}\mathfrak{c}_R(a; -1). \quad (618)$$

Proof. The involution

$$(j, u) \mapsto \left(2 - j, \frac{a^2}{u}\right)$$

exchanges the origin-0 and origin-2 fibres, proving $m_0 = m_2$. On the middle fibre it sends u to a^2/u . Its only possible fixed point is $u = a$. At that point $\lambda_{R,a}(u) = 1$, while the middle target is -1 ; membership would force $R \mid 2$. Hence the middle fibre is a disjoint union of two-element orbits and m_1 is even.

Each exterior orbit contains one origin-0 member and one origin-2 member, so there are $m_0 = m_2$ exterior orbits. There are $m_1/2$ middle orbits. The bijection (520) now proves (616); summing the three origin counts gives (617). Finally,

$$m_1 = \mathfrak{c}_R(a; -1), \quad m_2 = \mathfrak{c}_R(a; -p^{-1}),$$

by the origin-resolved target equations in equation (372). Substitution proves (618). $\square$

Theorem 11.26 (Exact three-origin, two-target, one-null-coordinate weighted factorization). *Assume the hypotheses of theorem 11.25. For $j \in \{0, 1, 2\}$, put*

$$\mathcal{X}_{R,p}^{(j)} := \{u : (j, u) \in \mathcal{X}_{R,p}\}.$$

For $u \in \mathcal{X}_{R,p}^{(j)}$, define

$$d_j(u) := p^j u, \quad \omega_j(u) := \frac{4N d_j(u)}{(N + d_j(u))^2}, \quad (619)$$

and define the three origin amplitudes by

$$\alpha_j := \sum_{u \in \mathcal{X}_{R,p}^{(j)}} \omega_j(u), \quad (620)$$

with an empty sum equal to zero. Then

$$\alpha_0 = \alpha_2 \quad (621)$$

and

$$\alpha_{p,a}^\Sigma = \alpha_0 + \alpha_1 + \alpha_2 = \alpha_1 + 2\alpha_2. \quad (622)$$

The two-target weighted branch is

$$\mathcal{A}_{R,p} := (\alpha_1, \alpha_2) \in \mathbb{R}_{\geq 0}^2, \quad (623)$$

where the first coordinate belongs to the target -1 and the second belongs to the target $-p^{-1}$. For

$$\varepsilon_2(x, y) := x + 2y, \quad (624)$$

one has the exact factorization

$$(\alpha_0, \alpha_1, \alpha_2) \mapsto \mathcal{A}_{R,p} \xrightarrow{\varepsilon_2} \alpha_{p,a}^\Sigma \mapsto V = -\alpha_{p,a}^\Sigma. \quad (625)$$

In particular,

$$-\det K_{p,a}^\Sigma = 4\varepsilon_2(\mathcal{A}_{R,p}), \quad V = -\varepsilon_2(\mathcal{A}_{R,p}). \quad (626)$$

For $j = 1, 2$, define

$$\gamma_j := \begin{cases} \tau, & m_j = 0, \\ \widetilde{\alpha}_j, & m_j > 0, \end{cases} \quad \widetilde{\mathcal{A}}_{R,p} := (\gamma_1, \gamma_2) \in G(\mathbb{R})^2, \quad (627)$$

where $\widetilde{\alpha}_j$ is the supported copy of α_j . The numerical and Boolean projections satisfy

$$\varepsilon_2(p_{\mathbb{R}}(\gamma_1), p_{\mathbb{R}}(\gamma_2)) = \alpha_{p,a}^\Sigma, \quad (628)$$

$$(\chi_{\mathbb{R}}(\gamma_1), \chi_{\mathbb{R}}(\gamma_2)) = (\mathbf{1}_{m_1 > 0}, \mathbf{1}_{m_2 > 0}). \quad (629)$$

Therefore the following conditions are equivalent:

$$\begin{aligned} \Omega_R(p) &> 0, \\ m_1 &> 0 \text{ or } m_2 > 0, \\ \alpha_1 &> 0 \text{ or } \alpha_2 > 0, \\ \varepsilon_2(\mathcal{A}_{R,p}) &> 0, \\ V &< 0. \end{aligned} \quad (630)$$

The proof uses the arithmetic complement involution $j \leftrightarrow 2 - j$. The cyclic relabelling of the three origins is governed by the equal-fibre criterion in theorem 9.66.

Proof. The map from $\mathcal{X}_{R,p}$ to the divisor certificate set is

$$(j, u) \mapsto d_j(u) = p^j u.$$

It is a bijection by the proof of theorem 9.64. Since $\gcd(R, pa) = 1$, the reduced data satisfy $R_0 = R$ and $N_0 = N$. Summing (619) over all three fibres therefore gives exactly (549).

Under the arithmetic involution

$$(j, u) \mapsto \left(2 - j, \frac{a^2}{u}\right),$$

the divisor changes by

$$d_j(u) \mapsto p^{2-j} \frac{a^2}{u} = \frac{p^2 a^2}{p^j u} = \frac{N^2}{d_j(u)}.$$

For every positive divisor d of N^2 ,

$$\begin{aligned} \frac{4N(N^2/d)}{(N + N^2/d)^2} &= \frac{4N^3/d}{N^2(N + d)^2/d^2} \\ &= \frac{4Nd}{(N + d)^2}. \end{aligned}$$

Thus the involution preserves the weight and exchanges the origin-0 and origin-2 weighted fibres. This proves (621). The bijection with the full divisor set then gives

$$\alpha_{p,a}^\Sigma = \alpha_0 + \alpha_1 + \alpha_2,$$

and substitution of $\alpha_0 = \alpha_2$ proves (622).

The target equations $T_1(p) = -1$ and $T_2(p) = -p^{-1}$ identify the two coordinates of (623). Equations (624) and (622) prove (625). The Lorentz identities are theorem 11.13.

Each weight in (619) is strictly positive. Hence

$$m_j = 0 \iff \alpha_j = 0, \quad m_j > 0 \iff \alpha_j > 0.$$

The two-coordinate model of $G(\mathbb{R})$ in theorem 11.1 now proves (628) and (629). Finally,

$$\Omega_R(p) = m_1 + 2m_2$$

by (617). Positivity of the origin weights proves every equivalence in (630). $\square$

Remark 11.27 (One support datum, two exact realizations). The central idempotent coordinate in $\mathfrak{S}_{123} = \mathbb{Q}[\varepsilon]/(\varepsilon^2 - \varepsilon) \otimes \mathbb{Q}[D_3]$ and the split character χ_R realize the same support datum in two algebraic settings. The linear model displays the two support blocks, and the semiring model records the distinction between an empty carrier and a present carrier whose amplitude cancels. The C_1 carrier and the C_2 divisor involution have direct arithmetic realizations. The shellwise arithmetic C_3 coordinate exists exactly under the equal-fibre condition of theorem 9.66.

Provenance record 11.28 (Globalization sources). The duplicated-zero construction and its componentwise model are taken from The Clankers, *A Note on the Globalization Semiring* $G(\mathbb{Z})$, March 2026 [18], local source `globalization_note.tex`, and from The Clankers, *A Secondary Note on Split-Zero Globalization*, April 2026 [19], local source `globalizationnte/2/split_zero_secondary_note.tex`. The supported-cancellation theorem above is the finite-family specialization needed for the present support object.

12. RIGIDIFIED DENOMINATOR TOWERS AND UNIVERSAL FROBENIUS

For a positive integer Q , put

$$H_Q := \frac{1}{Q}\mathbb{Z} \subset \mathbb{Q}, \quad A_Q := \mathbb{F}_1[T^{H_Q^+}].$$

Every H_Q is abstractly order-isomorphic to $\mathbb{Z}$. We retain the specified inclusion $\iota_Q : \mathbb{Z} \hookrightarrow H_Q$, $\iota_Q(n) = n$. Denote this rigidified object by $\mathfrak{r}_Q = (H_Q, \iota_Q)$.

Theorem 12.1 (Divisibility classifies rigidified morphisms). *For positive integers Q, M ,*

$$\mathrm{Hom}(\mathfrak{r}_Q, \mathfrak{r}_M) = \begin{cases} \{\text{one morphism}\}, & Q \mid M, \\ \emptyset, & Q \nmid M. \end{cases}$$

In particular, $\mathfrak{r}_Q \cong \mathfrak{r}_M$ if and only if $Q = M$.

Proof. Any rigidified morphism f fixes $\mathbb{Z}$, hence $f(1) = 1$. For $a \in \mathbb{Z}$,

$$Qf(a/Q) = f(a) = a.$$

The group $H_M \subset \mathbb{Q}$ is torsion-free, so $f(a/Q) = a/Q$. Thus f , if it exists, is the inclusion $H_Q \hookrightarrow H_M$. This inclusion exists exactly when $1/Q \in M^{-1}\mathbb{Z}$, equivalently $Q \mid M$. The assertion on isomorphisms follows by applying divisibility in both directions. $\square$

Let

$$\phi_Q : \mathbb{F}_1[T] \xrightarrow{\sim} A_Q, \quad \phi_Q(T^n) = T^{n/Q}.$$

If $Q \mid M$, let $j_{Q,M} : A_Q \hookrightarrow A_M$ be induced by inclusion of exponent monoids.

Theorem 12.2 (Exact Frobenius transition). *If $M = mQ$, then*

$$\phi_M^{-1} \circ j_{Q,M} \circ \phi_Q = \text{Frob}_m, \quad \text{Frob}_m(T^n) = T^{mn}.$$

Proof. For $n \in \mathbb{N}$,

$$\phi_Q(T^n) = T^{n/Q} = T^{mn/M},$$

and applying ϕ_M^{-1} gives T^{mn} . $\square$

Theorem 12.3 (LCM join). *Let $Q_1, \dots, Q_r$ be positive integers, and put*

$$L = \text{lcm}(Q_1, \dots, Q_r).$$

Then

$$H_L = H_{Q_1} + \dots + H_{Q_r}. \quad (631)$$

Hence A_L is the least cyclic-root stalk containing every A_{Q_i} .

Proof. Each Q_i divides L , hence $H_{Q_i} \subset H_L$. Put

$$d_i = \frac{L}{Q_i}.$$

Minimality of L implies $\gcd(d_1, \dots, d_r) = 1$, so Bézout gives integers u_i with $\sum_i u_i d_i = 1$. Dividing by L yields

$$\frac{1}{L} = \sum_{i=1}^r \frac{u_i}{Q_i}.$$

Thus the group generated by the H_{Q_i} contains the generator $1/L$ of H_L , proving (631). The stalk assertion follows from the inclusion of exponent monoids. $\square$

Theorem 12.4 (The iterated LCM colimit). *Fix $Q_0 \geq 3$ and set*

$$Q_{j+1} = \text{lcm}(1, 2, \dots, Q_j).$$

Then

$$\bigcup_{j \geq 0} H_{Q_j} = \mathbb{Q}, \quad \varinjlim_j A_{Q_j} = \mathbb{F}_1[T^{\mathbb{Q}_+}].$$

Every Frobenius endomorphism of the limiting stalk is an automorphism.

Proof. Since $\gcd(Q, Q-1) = 1$, for $Q \geq 3$ one has

$$\text{lcm}(1, \dots, Q) \geq Q(Q-1) > Q;$$

hence $Q_j \rightarrow \infty$. Given $a/b \in \mathbb{Q}$, choose j with $Q_j \geq b$. Then $b \mid Q_{j+1}$ and therefore $a/b \in H_{Q_{j+1}}$. This proves the group colimit. Taking the union of the exponent monoids proves the stalk identity. Finally, multiplication by any positive integer m is a bijection of $\mathbb{Q}_+$, with inverse $q \mapsto q/m$, so the corresponding Frobenius is invertible. $\square$

13. MIRROR COMPLETION OF THE BOOLEAN CARRIER AND THE 12289 ORBIT MATRIX

The split-zero carrier distinguishes the unsupported element τ from the supported ring zero. A two-sided support completion must retain that distinction in each oriented copy.

Definition 13.1 (Mirror-completed globalization carrier). Let $S = G(\mathbb{Z}) = \mathbb{Z} \sqcup \{\tau\}$, and put

$$S^{\leftrightarrow} := S_+ \times S_- \quad (632)$$

with componentwise semiring operations. Define

$$\tau_{\leftrightarrow} := (\tau, \tau), \quad 0_+ := (\tilde{0}, \tau), \quad 0_- := (\tau, \tilde{0}), \quad (633)$$

where $\tilde{0}$ denotes the supported copy of the ring zero. The mirror automorphism is

$$\vartheta(x_+, x_-) = (x_-, x_+). \quad (634)$$

The oriented support character is

$$\chi_{\leftrightarrow}(x_+, x_-) := (\chi_{\mathbb{Z}}(x_+), \chi_{\mathbb{Z}}(x_-)) \in \mathbb{B}^2. \quad (635)$$

Proposition 13.2 (Two supported zeros across unsupportedness). *The elements $\tau_{\leftrightarrow}$, 0_+ , 0_- , and $0_+ \oplus 0_- = (\tilde{0}, \tilde{0})$ are distinct and satisfy*

$$0_+^2 = 0_+, \quad 0_-^2 = 0_-, \quad 0_+ 0_- = \tau_{\leftrightarrow}, \quad (636)$$

$$0_+ \oplus 0_- = (\tilde{0}, \tilde{0}). \quad (637)$$

Moreover,

$$\begin{aligned} \chi_{\leftrightarrow}(\tau_{\leftrightarrow}) &= (0, 0), & \chi_{\leftrightarrow}(0_+) &= (1, 0), \\ \chi_{\leftrightarrow}(0_-) &= (0, 1), & \chi_{\leftrightarrow}(0_+ \oplus 0_-) &= (1, 1), \end{aligned} \quad (638)$$

and

$$(S^{\leftrightarrow})^{\vartheta} = \{(x, x) : x \in S\} \cong S. \quad (639)$$

Proof. All identities are componentwise consequences of $\tilde{0}^2 = \tilde{0}$, $\tilde{0} \oplus \tau = \tilde{0}$, and $\tilde{0}\tau = \tau$ in $G(\mathbb{Z})$. The four elements are distinct because $\tilde{0} \neq \tau$. Applying the unique Boolean support character in each coordinate gives (638). The fixed-point equation $(x_+, x_-) = (x_-, x_+)$ is equivalent to $x_+ = x_-$, proving (639). $\square$

For $x \in S$, define the two operation-preserving injections

$$\iota_+(x) := (x, \tau), \quad \iota_-(x) := (\tau, x). \quad (640)$$

Both send τ to $\tau_{\leftrightarrow}$. Their supported ring-zero images are precisely 0_+ and 0_- .

Definition 13.3 (Two-index signed elements). The first sign records the sheet relative to τ , and the second sign records the ordinary sign inside that supported copy of $\mathbb{Z}$. Put

$$\begin{aligned} 1_{++} &:= \iota_+(\tilde{1}), & 1_{+-} &:= \iota_+(\widetilde{-1}), \\ 1_{-+} &:= \iota_-(\tilde{1}), & 1_{--} &:= \iota_-(\widetilde{-1}), \end{aligned} \quad (641)$$

and retain the four entries in the indexed matrix

$$\mathcal{U}_2 := \begin{pmatrix} 1_{++} & 1_{+-} \\ 1_{-+} & 1_{--} \end{pmatrix}. \quad (642)$$

Proposition 13.4 (Exact dual signed-line calculus). *The two local additive cancellations are*

$$1_{++} \oplus 1_{+-} = 0_+, \quad 1_{-+} \oplus 1_{--} = 0_-. \quad (643)$$

The mirror satisfies

$$\vartheta(0_+) = 0_-, \quad \vartheta(1_{++}) = 1_{-+}, \quad \vartheta(1_{+-}) = 1_{--}. \quad (644)$$

Products of elements belonging to opposite single-sheet images equal $\tau_{\leftrightarrow}$. Products within one sheet are the ordinary integer products in that sheet.

Proof. The first identity in (643) is

$$(\widetilde{1}, \tau) \oplus (\widetilde{-1}, \tau) = (\widetilde{0}, \tau) = 0_+,$$

and the second is its mirror. The remaining identities follow from (634). If the sheet indices differ, every coordinate of the product contains a factor τ ; hence the product is $\tau_{\leftrightarrow}$. $\square$

Definition 13.5 (Finite family of supported universes). Let I be a nonempty finite index set and define

$$S^{(I)} := \prod_{a \in I} S_a, \quad \tau_I := (\tau)_{a \in I}. \quad (645)$$

For $x \in S$, let $\iota_a(x)$ have coordinate x at a and coordinate τ elsewhere. Set

$$0_a := \iota_a(\widetilde{0}), \quad 1_{a,+} := \iota_a(\widetilde{1}), \quad 1_{a,-} := \iota_a(\widetilde{-1}). \quad (646)$$

For $x = (x_a)_{a \in I}$, define

$$\text{Supp}(x) := \{a \in I : x_a \neq \tau\}. \quad (647)$$

The symbols

$$Z_0 := \tau_I, \quad Z_{1,a} := 0_a \cong \mathbb{Z}/1\mathbb{Z} \quad (648)$$

record, respectively, the absence of every supported stalk and the supported trivial ring in sheet a . Here Z_0 is a support-state symbol; it is not the quotient ring $\mathbb{Z}/0\mathbb{Z}$.

Theorem 13.6 (Union–intersection calculus for supported universes). *For all $x, y \in S^{(I)}$,*

$$\text{Supp}(x \oplus y) = \text{Supp}(x) \cup \text{Supp}(y), \quad \text{Supp}(xy) = \text{Supp}(x) \cap \text{Supp}(y). \quad (649)$$

The product support character

$$\chi_I : S^{(I)} \longrightarrow \mathbb{B}^I, \quad \chi_I(x)_a := \chi_{\mathbb{Z}}(x_a), \quad (650)$$

is the indicator vector of $\text{Supp}(x)$. In particular,

$$\chi_I(\tau_I) = 0, \quad \chi_I(0_a) = e_a. \quad (651)$$

Each 0_a is therefore supported, even though its integer coordinate is zero. An element is single-sheet when its support has cardinality one and mixed-sheet when its support has cardinality at least two.

Proof. At coordinate a , a sum is τ exactly when both summands are τ , while a product is supported exactly when both factors are supported. This proves (649) and the indicator-vector assertion. Equation (651) follows because $\chi_{\mathbb{Z}}(\tau) = 0$ and $\chi_{\mathbb{Z}}(\widetilde{0}) = 1$. $\square$

Theorem 13.7 (Power-set branches and directed null fibres). *Let I be a nonempty finite set, let R be a commutative ring, and regard $\mathcal{P}(I)$ as a semiring under union and intersection. Put*

$$G_{\mathcal{P}(I)}(R) = \{(0, A) : A \subseteq I\} \cup \{(r, I) : r \in R\}, \quad z_A := (0, A). \quad (652)$$

For every $i \in I$, the membership map

$$\epsilon_i(A) := \mathbf{1}_{\{i \in A\}} \quad (653)$$

is a Boolean branch. Write ϵ_i also for its semiring extension to $G_{\mathcal{P}(I)}(R)$, given by

$$\epsilon_i(z_A) = \mathbf{1}_{\{i \in A\}}, \quad \epsilon_i(r, I) = 1. \quad (654)$$

The categorical product of all branches is the isomorphism

$$\epsilon_I := \prod_{i \in I} \epsilon_i : \mathcal{P}(I) \xrightarrow{\sim} \mathbb{B}^I, \quad A \longmapsto (\mathbf{1}_{\{i \in A\}})_{i \in I}. \quad (655)$$

In particular, the complete branch tuple recovers A .

Write $\tau_R = (0, 0)$ and $e_R = (0, 1)$ in $G(R)$. There is an injective semiring map

$$\begin{aligned} \Delta_I : G_{\mathcal{P}(I)}(R) &\longrightarrow G(R)^I, \\ \Delta_I(z_A)_i &= \begin{cases} e_R, & i \in A, \\ \tau_R, & i \notin A, \end{cases} \\ \Delta_I(r, I)_i &= (r, 1) \quad (i \in I). \end{aligned} \tag{656}$$

Its image consists exactly of the zero-amplitude support vectors with coordinates in $\{\tau_R, e_R\}$, together with the constant supported amplitude vectors. The i -th branch in (655) equals the i -th Boolean support coordinate of Δ_I .

Assume now that $2_R \neq 0_R$. In $M_2(G_{\mathcal{P}(I)}(R))$, let $\tilde{N}_I$ have entry $(2_R, I)$ at position $(2, 1)$ and $\tau_{\mathcal{P}(I)} := z_\emptyset$ elsewhere. For $A \subseteq I$, let $Z_{A,21}$ have entry z_A at position $(2, 1)$ and $\tau_{\mathcal{P}(I)}$ elsewhere. Entrywise amplitude and branch maps satisfy

$$\hat{p}(\tilde{N}_I) = 2E_{21}, \quad \hat{\epsilon}_i(\tilde{N}_I) = E_{21} \quad (i \in I), \tag{657}$$

$$\hat{p}(Z_{A,21}) = 0_2, \quad \hat{\epsilon}_i(Z_{A,21}) = \begin{cases} E_{21}, & i \in A, \\ 0_2, & i \notin A. \end{cases} \tag{658}$$

Moreover,

$$Z_{A,21} + Z_{B,21} = Z_{A \cup B, 21}, \quad Z_{A,21} Z_{B,21} = 0_2^{\text{amb}}, \tag{659}$$

where 0_2^{amb} is the matrix all of whose entries are $\tau_{\mathcal{P}(I)}$. Thus each scalar z_A is multiplicatively idempotent, while its placement in the directed E_{21} cell is square-zero under matrix composition.

Proof. Union and intersection are detected coordinatewise by membership, so each ϵ_i preserves both semiring operations. A subset is uniquely determined by its membership vector, proving (655).

For $A, B \subseteq I$, the i -th coordinates of $\Delta_I(z_{A \cup B})$ and $\Delta_I(z_{A \cap B})$ are respectively the join and meet of the corresponding coordinates of $\Delta_I(z_A)$ and $\Delta_I(z_B)$. A supported amplitude has support in every coordinate; addition and multiplication with a zero-amplitude support vector therefore reproduce union and intersection with I . This proves that Δ_I preserves both operations. Its amplitude and membership coordinates recover its source, so it is injective and has the displayed image.

The matrix identities in (657)–(658) follow entrywise. Addition in the distinguished cell is union. Every summand in the matrix product of two $Z_{A,21}$ contains a zero entry because $E_{21}^2 = 0_2$, proving (659). $\square$

Corollary 13.8 (Colour atoms and Time atoms). *For $\mathcal{C} = \{r, g, b\}$, put*

$$0_c := z_{\{c\}} \quad (c \in \mathcal{C}), \quad \tau := z_\emptyset, \quad e := z_{\mathcal{C}}. \tag{660}$$

Then

$$0_c^2 = 0_c, \quad 0_c 0_d = \tau \quad (c \neq d), \quad 0_r + 0_g + 0_b = e. \tag{661}$$

The three Boolean branches recover every colour-support subset. Under $\Delta_{\mathcal{C}}$, these atoms are precisely the three local supported zeros already denoted $0_r, 0_g, 0_b$ in $G(R)^{\mathcal{C}}$.

For $\mathcal{T} = \{M, N, T, E\}$, the four atoms $0_t := z_{\{t\}}$ satisfy

$$0_t^2 = 0_t, \quad 0_t 0_u = \tau \quad (t \neq u), \quad \sum_{t \in \mathcal{T}} 0_t = e. \tag{662}$$

Their joint branch coordinate is the four-cube $\mathbb{B}^{\mathcal{T}}$. The centered real permutation carrier of these same four labels is the three-dimensional tetrahedral Times space of theorem 13.42; its metric is supplied by that centered representation.

Proof. Singletons are the atoms of the two power-set lattices. Their self-meets, pairwise meets, and total joins give the displayed identities. The final statement invokes the centered permutation-space construction, whose Gram form is proved in theorem 13.42. $\square$

Theorem 13.9 (Centred branch projection from support cubes to simplex carriers). *Let I be a finite set of cardinality $n \geq 2$. Embed the branch cube $\mathbb{B}^I$ in $\mathbb{R}^I$ by $0 \mapsto 0$ and $1 \mapsto 1$, and write*

$$\mathbf{j}_I := \sum_{i \in I} e_i, \quad \Pi_I := I_n - \frac{1}{n} \mathbf{j}_I \mathbf{j}_I^\top. \quad (663)$$

For $A \subseteq I$, put

$$q_I(A) := \Pi_I \mathbf{1}_A = \mathbf{1}_A - \frac{|A|}{n} \mathbf{j}_I. \quad (664)$$

Then $q_I(A) \in \mathbf{j}_I^\perp$, and

$$q_I(\emptyset) = q_I(I) = 0, \quad q_I(I \setminus A) = -q_I(A), \quad (665)$$

$$\langle q_I(A), q_I(B) \rangle = |A \cap B| - \frac{|A||B|}{n}. \quad (666)$$

Every permutation $\sigma \in S_I$ satisfies

$$q_I(\sigma A) = \sigma q_I(A). \quad (667)$$

For each atom $\{i\}$, define its unit direction

$$\nu_i := \sqrt{\frac{n}{n-1}} q_I(\{i\}). \quad (668)$$

The atom directions obey

$$\sum_{i \in I} \nu_i = 0, \quad \langle \nu_i, \nu_j \rangle = \begin{cases} 1, & i = j, \\ -\frac{1}{n-1}, & i \neq j. \end{cases} \quad (669)$$

They span the $(n-1)$ -dimensional space $\mathbf{j}_I^\perp$. Consequently, the three Colour branches give the 120° Colour triangle, while the four Time branches give the tetrahedral Times carrier.

Proof. The matrix Π_I is the orthogonal projector onto $\mathbf{j}_I^\perp$, which gives (664). The empty and full subsets map to zero. Since $\mathbf{1}_{I \setminus A} = \mathbf{j}_I - \mathbf{1}_A$ and $\Pi_I \mathbf{j}_I = 0$, complements map to opposite vectors. Expanding the Euclidean inner product gives (666). Coordinate permutation commutes with Π_I , proving (667).

For atoms,

$$\langle q_I(\{i\}), q_I(\{j\}) \rangle = \delta_{ij} - \frac{1}{n}.$$

Multiplication by $n/(n-1)$ gives (669). Its Gram matrix has one-dimensional kernel $\mathbb{R}\mathbf{j}_I$ and rank $n-1$. $\square$

Exact computation 13.10 (Centred branch/simplex certificate). The script and captured receipt

```
verification/verify_power_set_simplex_bridge.py,
verification/verify_power_set_simplex_bridge_output.txt
```

exhaust the Colour and Time support cubes, verify the complement law and the subset Gram formula, and compute the atom Gram ranks and angles exactly.

Exact computation 13.11 (Power-set branch and directed-fibre certificate). The script and captured receipt

```
verification/verify_power_set_branch_directed_fibre.py,
verification/verify_power_set_branch_directed_fibre_output.txt \quad (670)
```

exhaust all colour-support subsets, check the semiring embedding, verify the colour and Time atom identities, and check every directed branch mask and square-zero matrix product.

Proposition 13.12 (Colour support and the $SU(3)$ baryonic Wilson junction). *Let $V = \mathbb{C}[\mathcal{C}]$ with ordered Colour basis (e_r, e_g, e_b) and dual basis $(e_r^\vee, e_g^\vee, e_b^\vee)$. Let*

$$\varepsilon_{\mathcal{C}} := e_r \wedge e_g \wedge e_b \in \Lambda^3 V, \quad \varepsilon_{\mathcal{C}}^\vee := e_r^\vee \wedge e_g^\vee \wedge e_b^\vee \in \Lambda^3 V^*. \quad (671)$$

Write their mutually dual components as ε^{abc} and ε_{abc} , respectively. Either component is nonzero exactly when (a, b, c) is a permutation of (r, g, b) . Equivalently, every nonzero component has the exact power-set support certificate

$$0_a + 0_b + 0_c = e, \quad 0_a 0_b = 0_b 0_c = 0_c 0_a = \tau. \quad (672)$$

The coefficient $\text{sgn}(a, b, c)$ belongs to the alternating tensor; the power-set branches retain the three-colour cover.

Let $U_1, U_2, U_3 : V_x \rightarrow V_y$ be three Colour holonomies, and define

$$W_{\mathcal{C}}(U_1, U_2, U_3) := \frac{1}{3!} \varepsilon_{abc} \varepsilon^{a'b'c'} (U_1)^a_{a'} (U_2)^b_{b'} (U_3)^c_{c'}. \quad (673)$$

For $g_x, g_y \in SU(3)$, set

$$U_j^g := g_y U_j g_x^{-1}. \quad (674)$$

Then

$$W_{\mathcal{C}}(U_1^g, U_2^g, U_3^g) = W_{\mathcal{C}}(U_1, U_2, U_3). \quad (675)$$

Thus the three atomic support branches give an exact colour-incidence coordinate for the gauge-invariant three-arm Wilson junction.

Proof. The alternating tensor is supported on the six permutations of the three Colour labels. Those triples are exactly the triples of atoms with pairwise meet τ and total join e , proving (672).

For every $g \in GL(V)$,

$$\varepsilon_{abc} g^a_d g^b_e g^c_f = (\det g) \varepsilon_{def}. \quad (676)$$

Apply this identity to g_y at the upper indices of (673) and to g_x^{-1} at the lower indices. Their determinants are both one, so all endpoint factors cancel and (675) follows. $\square$

Remark 13.13 (Three-arm geometry). When the three holonomies run along paths meeting at a common junction, (673) is the standard three-quark Wilson observable. Lattice calculations extract the static three-quark potential from this contraction and compare it with the minimal flux-tube length [45]. The associated area-law geometry admits a separate comparison of three-bladed Y surfaces and pairwise Δ surfaces [46].

Exact computation 13.14 (Colour Wilson-junction certificate). The script and captured receipt

```
verification/verify_colour_baryonic_wilson_junction.py,
verification/verify_colour_baryonic_wilson_junction_output.txt
```

exhaust the alternating Colour support, verify (676) for a generic 3×3 matrix, and check endpoint gauge invariance over exact integer data.

Corollary 13.15 (The four-Time $SU(4)$ volume junction). *Let $V_{\mathcal{T}} = \mathbb{C}[\mathcal{T}]$, with basis e_t and dual basis $e_t^\vee$, and put*

$$\varepsilon_{\mathcal{T}} := e_M \wedge e_N \wedge e_T \wedge e_E \in \Lambda^4 V_{\mathcal{T}}, \quad \varepsilon_{\mathcal{T}}^\vee := e_M^\vee \wedge e_N^\vee \wedge e_T^\vee \wedge e_E^\vee \in \Lambda^4 V_{\mathcal{T}}^*. \quad (677)$$

Its nonzero components are the twenty-four orderings (t_1, t_2, t_3, t_4) of $\mathcal{T}$. Each has the support certificate

$$\sum_{j=1}^4 0_{t_j} = e, \quad 0_{t_j} 0_{t_k} = \tau \quad (j \neq k). \quad (678)$$

For four holonomies $U_1, \dots, U_4$ on $V_{\mathcal{T}}$, the contraction

$$W_{\mathcal{T}}(U_1, \dots, U_4) := \frac{1}{4!} \varepsilon_{a_1 a_2 a_3 a_4} \varepsilon^{b_1 b_2 b_3 b_4} \prod_{j=1}^4 (U_j)^{a_j}_{b_j} \quad (679)$$

is invariant under

$$U_j \mapsto g_y U_j g_x^{-1}, \quad g_x, g_y \in SU(4). \quad (680)$$

Moreover,

$$q_{\mathcal{T}}(\{t\}) = e_t - \frac{1}{4} \mathbf{j}_{\mathcal{T}} = \lambda_t, \quad (681)$$

so the same four support atoms give the A_3 weights and, after the factor $2/\sqrt{3}$, the tetrahedral unit directions μ_t .

Proof. The support statement is the four-label version of (672). The rank-four determinant identity

$$\varepsilon_{a_1 a_2 a_3 a_4} g^{a_1}_{c_1} g^{a_2}_{c_2} g^{a_3}_{c_3} g^{a_4}_{c_4} = (\det g) \varepsilon_{c_1 c_2 c_3 c_4}$$

applied at both endpoints proves the $SU(4)$ invariance. Equation (681) follows by substituting $n = 4$ and $A = \{t\}$ in (664); the remaining assertion is (796). $\square$

Exact computation 13.16 (Four-Time volume-junction certificate). The script and captured receipt

verification/verify_time_SU4_volume_junction.py,

verification/verify_time_SU4_volume_junction_output.txt

exhaust the twenty-four Time orderings and verify the four-arm endpoint gauge invariance over exact integer data.

Theorem 13.17 (Finite-universe exterior-junction law). *Let $I = (i_1, \dots, i_n)$ be an ordered finite label set with $n \geq 2$, let $V_I = \mathbb{C}[I]$, and write $e_i^\vee$ for the basis dual to e_i . Put*

$$\varepsilon_I := e_{i_1} \wedge \dots \wedge e_{i_n} \in \Lambda^n V_I, \quad \varepsilon_I^\vee := e_{i_1}^\vee \wedge \dots \wedge e_{i_n}^\vee \in \Lambda^n V_I^*. \quad (682)$$

The following statements hold.

- (i) *A component $\varepsilon_I^\vee(e_{a_1}, \dots, e_{a_n})$ is nonzero exactly when $(a_1, \dots, a_n)$ is an ordering of I . In the power-set globalization this is equivalent to*

$$\sum_{j=1}^n 0_{a_j} = e, \quad 0_{a_j} 0_{a_k} = \tau \quad (j \neq k). \quad (683)$$

- (ii) *For linear maps $U_1, \dots, U_n : V_x \rightarrow V_y$, define*

$$W_I(U_1, \dots, U_n) := \frac{1}{n!} \varepsilon_{a_1 \dots a_n} \varepsilon^{b_1 \dots b_n} \prod_{j=1}^n (U_j)^{a_j}_{b_j}. \quad (684)$$

For $g_y, g_x \in GL(V_I)$,

$$W_I(g_y U_1 g_x^{-1}, \dots, g_y U_n g_x^{-1}) = \frac{\det g_y}{\det g_x} W_I(U_1, \dots, U_n). \quad (685)$$

It is therefore invariant under independent $SL(V_I)$, hence $SU(n)$, endpoint actions.

- (iii) *The centred singleton branch is*

$$q_I(\{i\}) = e_i - \frac{1}{n} \mathbf{j}_I. \quad (686)$$

The n vectors in (686) are the defining-weight orbit of A_{n-1} ; after multiplication by $\sqrt{n/(n-1)}$, they are the unit vertices of the centred regular $(n-1)$ -simplex.

Thus the Colour $SU(3)$ Wilson junction and the four-Time $SU(4)$ volume junction are the cases $n = 3$ and $n = 4$ of one exterior-support construction.

Proof. The alternating tensor is nonzero on a basis n -tuple exactly when no basis vector repeats. Since I has n elements, this says precisely that the tuple uses every label once. Singleton supports are pairwise disjoint exactly when their products are τ , and their union is all of I exactly when their sum is e . This proves (i).

At the output endpoint the alternating tensor obeys

$$\varepsilon_{a_1 \dots a_n} (g_y)^{a_1}_{c_1} \dots (g_y)^{a_n}_{c_n} = (\det g_y) \varepsilon_{c_1 \dots c_n}.$$

At the input endpoint the corresponding identity for g_x^{-1} contributes $\det(g_x^{-1}) = (\det g_x)^{-1}$. Substitution in (684) proves (685).

Finally, (686) is (664) with $A = \{i\}$. Its Gram matrix is given by (669), which is the Gram matrix of the defining A_{n-1} weight orbit and of the centred regular $(n-1)$ -simplex after the stated scalar multiplication. $\square$

Exact computation 13.18 (Finite-universe exterior-junction certificate). The script and receipt

`verification/verify_finite_universe_exterior_junction.py,`
`verification/verify_finite_universe_exterior_junction_output.txt`

exhaust the alternating components and evaluate the endpoint action over exact integer matrices for $2 \leq n \leq 5$.

Theorem 13.19 (Exterior–Wick rung and dimension count). *Let $I = (i_1, \dots, i_m)$ be an ordered label set with $m \geq 2$, and equip $E_I := \mathbb{R}[I]$ with the Euclidean inner product for which the label basis is orthonormal. Define*

$$\mathbf{j}_I := \sum_{i \in I} e_i, \quad u_I := \frac{1}{\sqrt{m}} \mathbf{j}_I, \quad H_I := u_I^\perp. \quad (687)$$

Let $u_I^\vee$ be the metric dual of u_I , let $\Omega_I^\vee := e_{i_1}^\vee \wedge \dots \wedge e_{i_m}^\vee$, and put

$$\widehat{\omega}_I := \iota_{u_I} \Omega_I^\vee \in \Lambda^{m-1} E_I^*, \quad \omega_I := \widehat{\omega}_I|_{H_I} \in \Lambda^{m-1} H_I^*. \quad (688)$$

Then:

(i) *The decomposition*

$$E_I = \mathbb{R}u_I \oplus H_I \quad (689)$$

is orthogonal, $\dim H_I = m-1$, and ω_I is the unit oriented covolume form of H_I . In particular,

$$u_I^\vee \wedge \widehat{\omega}_I = \Omega_I^\vee. \quad (690)$$

(ii) *The symmetric form*

$$\eta_I(x, y) := -u_I^\vee(x)u_I^\vee(y) + \langle \Pi_I x, \Pi_I y \rangle, \quad \Pi_I := I_m - u_I u_I^\top, \quad (691)$$

has matrix

$$[\eta_I] = I_m - \frac{2}{m} \mathbf{j}_I \mathbf{j}_I^\top \quad (692)$$

in the label basis and signature $(m-1, 1)$.

(iii) *On $\mathbb{R}_t \times H_I$, for any real order-zero term R_0 , the lifted operator*

$$P_I := \partial_t^2 - 4\Delta_{H_I} + R_0 \quad (693)$$

has principal symbol

$$\sigma_{P_I}(\sigma, \xi) = -\sigma^2 + 4\|\xi\|^2, \quad (694)$$

and hence the same signature $(m-1, 1)$. The identification $\partial_t \mapsto u_I$ realizes its one temporal plus $m-1$ spatial directions on the splitting in (689). This is the finite-label instance of the corrected Wick-lift principal-symbol theorem in [41].

- (iv) *Together with theorem 13.17, the same ordered label system therefore carries the following exact ladder. For the five-label row write*

$$\mathcal{F} := \{t, i, m, r, p\}, \quad (695)$$

standing for thumb, index, middle (also called long), ring, and pinky:

m	named labels	internal determinant group	Wick signature
3	three Colours	$SU(3)$	$(2, 1)$
4	four Times	$SU(4)$	$(3, 1)$
5	five Fingers	$SU(5)$	$(4, 1)$.

(696)

The exact maps joining the support and volume data are

$$0_i \mapsto q_I(\{i\}) \in H_I, \quad \Omega_I^\vee \mapsto \widehat{\omega}_I = \iota_{u_I} \Omega_I^\vee \mapsto \omega_I = \widehat{\omega}_I|_{H_I}. \quad (697)$$

Proof. The vector u_I has unit length. Orthogonal projection along its span is $\Pi_I = I_m - u_I u_I^\top$, so its image H_I has dimension $m - 1$, proving the decomposition. Extend u_I to an oriented orthonormal basis $u_I, h_1, \dots, h_{m-1}$ of E_I . Interior contraction gives

$$\omega_I(h_1, \dots, h_{m-1}) = \Omega_I^\vee(u_I, h_1, \dots, h_{m-1}) = 1,$$

and the exterior identity $u_I^\vee \wedge \widehat{\omega}_I = \Omega_I^\vee$ proves (690).

The form η_I takes the value -1 on u_I , equals the Euclidean form on H_I , and makes the two summands orthogonal. Since $u_I u_I^\top = m^{-1} \mathbf{j}_I \mathbf{j}_I^\top$, its matrix is (692); its eigenvalues are -1 once and $+1$ $m - 1$ times. The principal symbol in (694) follows directly from (693); the term R_0 has order zero. Finally, the determinant-group column is theorem 13.17, and substitution of $m = 3, 4, 5$ gives the table. The two maps in (697) are respectively the centred branch map and the contraction-restriction map. $\square$

Definition 13.20 (Fuzzionic carrier datum). Let $(\mathcal{M}^m, g)$ be a time-oriented Lorentzian manifold and let $\Sigma^{m-1} \subset \mathcal{M}$ be an oriented spacelike hypersurface. An m -label Fuzzionic carrier datum over an open set $U \subseteq \Sigma$ consists of:

- (i) a rank- m Hermitian bundle $\mathcal{E} \rightarrow U$ with a unit determinant covolume section $\epsilon_\mathcal{E}^\vee \in \Gamma(U, \Lambda^m \mathcal{E}^*)$;
- (ii) a Hermitian connection $\nabla^\mathcal{E}$ satisfying $\nabla^\mathcal{E} \epsilon_\mathcal{E}^\vee = 0$, so its endpoint holonomies lie in $SU(m)$;
- (iii) a pointwise oriented isometry

$$\Theta_x : H_I \longrightarrow T_x \Sigma, \quad x \in U. \quad (698)$$

The induced spatial volume form is

$$\text{vol}_{\Theta, I} := (\Theta^{-1})^* \omega_I. \quad (699)$$

For $m = 4$ and $I = \mathcal{T} = \{M, N, T, E\}$, this is a *four-Time Fuzzionic carrier*. A physical matter state equipped with such a carrier is called a *Fuzzionic matter state*.

The ER=EPR proposal associates wormhole connectivity with entanglement [38]. The complexity-volume proposal uses the volume of a maximal codimension-one bulk slice [37]. Fuzzball and microstate-geometry constructions supply horizon-scale bulk degrees of freedom in string theory [40]. These results motivate the following precise physical statement; its carrier and dimension count are supplied by the preceding theorem.

Conjecture 13.21 (Four-Time Fuzzionic matter-state conjecture). *For a $3 + 1$ -dimensional holographic bulk state whose interior description contains a maximal spacelike Einstein–Rosen-bridge slice Σ^3 , the bulk microstate degrees of freedom admit a global four-Time Fuzzionic carrier datum. Under its soldering map, the volume entering the complexity-volume prescription is*

$$\mathcal{C}_V \propto \frac{1}{G_4 \ell} \int_{\Sigma^3} \text{vol}_{\Theta, \mathcal{T}}. \quad (700)$$

Thus the proposed internal coding is four-Time $SU(4)$, in direct analogy with three-Colour $SU(3)$ coding of the baryonic determinant carrier in theorem 13.12.

For a genuinely 4+1-dimensional bulk, the same dimension rule selects the five-Finger continuation, with spatial rank four, Wick signature $(4, 1)$, and internal determinant group $SU(5)$ [39].

Remark 13.22 (Mathematical and physical scope). The algebraic statements proved here are the exterior-support law, the spatial covolume, the Wick signature, and the $SU(m)$ determinant action. The Fuzzionic assertion is the existence of the bundle, connection, and soldering map in a holographic microstate geometry. In particular, the three-Colour, four-Time, and five-label rows are fixed by the dimension count rather than by a choice between $SU(4)$ and $SU(5)$.

Exact computation 13.23 (Exterior–Wick rung certificate). The script and receipt

verification/verify_exterior_Wick_rung.py,
verification/verify_exterior_Wick_rung_output.txt

check the singleton Gram matrices, contracted covolumes, Lorentz spectra, and the $m = 3, 4, 5$ dimension table over exact symbolic data.

Corollary 13.24 (The supported zero in each universe). *The single-sheet subsemiring $\iota_a(S)$ is isomorphic to S . Under this isomorphism, 0_a is prime in the divisibility sense and*

$$P_{0,a} := \{\tau_I, 0_a\} \quad (701)$$

is a prime ideal of $\iota_a(S)$. Thus every distinct supported zero lies on the prime spectrum of its own sheet.

Proof. The map $\iota_a : S \rightarrow \iota_a(S)$ is a semiring isomorphism. It therefore suffices to prove the two assertions for the supported integer zero $\tilde{0} \in S$.

The multiples of $\tilde{0}$ are exactly $\{\tau, \tilde{0}\}$: multiplication by τ gives τ , while multiplication by an integer gives the ordinary integer product zero. If $\tilde{0}$ divides xy , then $xy \in \{\tau, \tilde{0}\}$. When one factor is τ , that factor is divisible by $\tilde{0}$. Otherwise $x, y \in \mathbb{Z}$, and $xy = 0$ forces $x = 0$ or $y = 0$. Hence $\tilde{0}$ is prime in the divisibility sense.

The set $\{\tau, \tilde{0}\}$ contains the additive identity, is closed under addition, and is stable under multiplication by every element of S . It is therefore an ideal. If xy belongs to this ideal, the preceding case distinction gives $x \in \{\tau, \tilde{0}\}$ or $y \in \{\tau, \tilde{0}\}$. Thus it is a prime ideal. Applying ι_a gives the stated assertions for 0_a and $P_{0,a}$. $\square$

Theorem 13.25 (Cyclic triality and its exact integral plane). *Take $I = C_3 = \{r, g, b\}$. Cyclic coordinate permutation fixes τ_I and acts transitively on $0_r, 0_g, 0_b$. Let*

$$\mathbf{j} := e_r + e_g + e_b, \quad v_a := 3e_a - \mathbf{j}. \quad (702)$$

Then

$$v_r + v_g + v_b = 0, \quad \langle v_a, v_a \rangle = 6, \quad \langle v_a, v_b \rangle = -3 \quad (a \neq b). \quad (703)$$

Consequently the three support directions are pairwise separated by 120° .

Proof. The coordinate permutation assertions are immediate. In the ordered basis (e_r, e_g, e_b) , the three vectors are

$$(2, -1, -1), \quad (-1, 2, -1), \quad (-1, -1, 2).$$

Their sum and inner products are those in (703). The ratio of an off-diagonal inner product to the common squared length is $-3/6 = -1/2$, whose angle is 120° . $\square$

Definition 13.26 (Handed Wick pair over the two supported zeros). Let V be a real Euclidean plane, let $e \in V$ be a unit support direction, and let $I_+ \in \text{End}_{\mathbb{R}}(V)$ be an orthogonal complex structure:

$$I_+^2 = -\text{id}_V. \quad (704)$$

Define

$$I_- := -I_+, \quad \mathcal{V}_+ := \{0_+\} \times V, \quad \mathcal{V}_- := \{0_-\} \times V. \quad (705)$$

Write $0_\epsilon + v$ for $(0_\epsilon, v) \in \mathcal{V}_\epsilon$ and $i_\epsilon v$ for $I_\epsilon v$. Define the sheet exchange

$$\mathcal{A} : \mathcal{V}_+ \longrightarrow \mathcal{V}_-, \quad \mathcal{A}(0_+ + v) := 0_- + v. \quad (706)$$

The datum $(\mathcal{V}_+, I_+; \mathcal{V}_-, I_-; \mathcal{A})$ is called the *handed Wick pair* determined by (V, e, I_+) .

Theorem 13.27 (The same normal locale has opposite helicity labels). *The two complex structures in theorem 13.26 satisfy*

$$I_-^2 = -\text{id}_V, \quad I_- = -I_+ = I_+^{-1}. \quad (707)$$

The sheet exchange is an isometry whose linear part is anti-complex-linear:

$$\mathcal{A}I_+ = -I_- \mathcal{A}. \quad (708)$$

If $n := I_+e$, then

$$n = I_+e = -I_-e, \quad -n = -I_+e = I_-e. \quad (709)$$

Equivalently, in the two sheetwise complex notations,

$$+i_+ = -i_-, \quad -i_+ = +i_- \quad (710)$$

as labels of the same underlying real displacements. Affinely,

$$\mathcal{A}(0_+ + i_+e) = 0_- - i_-e, \quad \mathcal{A}(0_+ - i_+e) = 0_- + i_-e. \quad (711)$$

Thus the quotient of $\mathcal{V}_+ \sqcup \mathcal{V}_-$ by the pairing generated by $\mathcal{A}$ obeys

$$[0_+ + i_+e] = [0_- - i_-e], \quad [0_+ - i_+e] = [0_- + i_-e]. \quad (712)$$

The ordered frames (e, I_+e) and (e, I_-e) have opposite orientations.

Proof. Equation (707) follows from $I_- = -I_+$ and $I_+^2 = -\text{id}_V$. The linear part of $\mathcal{A}$ is the identity on V , so

$$\mathcal{A}I_+ = I_+ = -I_- \mathcal{A},$$

which proves anti-complex-linearity. Substitution of $n = I_+e$ gives (709); the affine and quotient identities follow by applying $\mathcal{A}$. Finally, $(e, I_-e) = (e, -I_+e)$, so the two ordered frames have opposite orientations. $\square$

Proposition 13.28 (Frame-free carrier of a localized point). *Let L be a real affine line and let*

$$\mathcal{F}(L) := \text{Iso}_{\text{aff}}(\mathbb{R}, L) \quad (713)$$

be its affine-frame torsor. The affine group $G = \text{Aff}_1(\mathbb{R})$ acts on the right by

$$p \cdot g := p \circ g. \quad (714)$$

Form the associated carrier

$$\mathcal{L} := \mathcal{F}(L) \times_G \mathbb{R} = (\mathcal{F}(L) \times \mathbb{R}) / \text{sim}, \quad (p, t) \sim (p \circ g, g^{-1}t). \quad (715)$$

There is a canonical bijection

$$\kappa_L : L \longrightarrow \mathcal{L}, \quad \kappa_L(\ell) := [p, p^{-1}(\ell)], \quad (716)$$

and the displayed class is independent of the frame p .

If $q = p \circ g$, the two scalar readings of the same localized point are

$$t_p := p^{-1}(\ell), \quad t_q := q^{-1}(\ell) = g^{-1}t_p. \quad (717)$$

They are the same literal scalar precisely on the fixed locus

$$t_p \in \text{Fix}(g). \quad (718)$$

Thus localization is carried by the equivalence class $[p, t_p]$; a scalar assigned to that class is a reading in a selected frame.

For the reflection $s(t) = -t$, take the paired frames $p_+ = p$ and $p_- = p \circ s$. Their common zero locale has two retained support labels

$$0_+ := (p(0), +), \quad 0_- := (p(0), -) \quad (719)$$

in the doubled carrier $L \times \{+, -\}$. For every $\ell \in L$,

$$(t_+(\ell), t_-(\ell)) = (t, -t), \quad t = p^{-1}(\ell), \quad (720)$$

and equality of the two readings occurs exactly at the common zero locale. The paired reading lies in the sign line

$$\mathbb{R}_{\text{sgn}} = \{(t, -t) : t \in \mathbb{R}\} \subset \mathbb{R}^2. \quad (721)$$

The labels 0_+ and 0_- remain distinct after the forgetful map to L identifies their underlying locale.

Proof. If $q = p \circ g$, then

$$[q, q^{-1}(\ell)] = [p \circ g, g^{-1}p^{-1}(\ell)] = [p, p^{-1}(\ell)],$$

so κ_L is well defined. The map

$$[p, t] \mapsto p(t)$$

is well defined because $(p \circ g)(g^{-1}t) = p(t)$, and it is inverse to κ_L . Equation (717) follows from $(p \circ g)^{-1} = g^{-1} \circ p^{-1}$; equality of its two sides is equivalent to (718).

For $s(t) = -t$, one has $s^{-1} = s$, so $t_- = s^{-1}t_+ = -t_+$. Its fixed locus is $\{0\}$, proving the final claims. The two support labels are different elements of the doubled carrier even though the forgetful map sends both to $p(0)$. $\square$

Theorem 13.29 (Associated support–geometry carrier for finite universe frames). *Let I be a finite set of cardinality $n \geq 2$, let $H \leq \text{Sym}(I)$, and let $\mathcal{F}$ be a right H -torsor. Put*

$$S_I := S^I, \quad W_I := \left\{ (t_i)_{i \in I} \in \mathbb{R}^I : \sum_{i \in I} t_i = 0 \right\}. \quad (722)$$

The group H acts on both factors by

$$(h \cdot x)_i = x_{h^{-1}i}, \quad (h \cdot t)_i = t_{h^{-1}i}. \quad (723)$$

Define the associated carrier

$$\mathcal{X}(\mathcal{F}, I) := \mathcal{F} \times_H (S_I \times W_I), \quad (f; x, t) \sim (fh; h^{-1} \cdot x, h^{-1} \cdot t). \quad (724)$$

Its support component

$$\mathcal{S}(\mathcal{F}, I) := \mathcal{F} \times_H S_I \quad (725)$$

is a semiring. In any common frame f , its operations are

$$[f, x] \oplus [f, y] = [f, x \oplus y], \quad [f, x] \odot [f, y] = [f, x \odot y]. \quad (726)$$

For $\mathcal{P}(I)$, endowed with its induced H -action, the formula

$$\text{Supp}_{\mathcal{F}}[f, x] := [f, \text{Supp}(x)] \in \mathcal{F} \times_H \mathcal{P}(I) \quad (727)$$

is well defined. Its cardinality is independent of the frame. Addition and multiplication obey

$$\text{Supp}_{\mathcal{F}}(X \oplus Y) = \text{Supp}_{\mathcal{F}}(X) \cup_{\mathcal{F}} \text{Supp}_{\mathcal{F}}(Y), \quad (728)$$

$$\text{Supp}_{\mathcal{F}}(X \odot Y) = \text{Supp}_{\mathcal{F}}(X) \cap_{\mathcal{F}} \text{Supp}_{\mathcal{F}}(Y). \quad (729)$$

Here the fibrewise union and intersection are computed after representing both classes in one frame.

Let

$$\mu_i := \sqrt{\frac{n}{n-1}} \left(e_i - \frac{1}{n} \sum_{j \in I} e_j \right) \in W_I. \quad (730)$$

The unsupported point

$$\tau_{\mathcal{F}} := [f; (\tau_I, 0)] \quad (731)$$

is independent of f . For a fixed f , the set

$$\mathcal{Z}_f := \{[f; (0_i, \mu_i)] : i \in I\} \quad (732)$$

is independent of the selected frame as an unlabelled subset of $\mathcal{X}(\mathcal{F}, I)$. Every member has support cardinality one, whereas $\tau_{\mathcal{F}}$ has support cardinality zero. The geometric directions satisfy

$$\sum_{i \in I} \mu_i = 0, \quad \langle \mu_i, \mu_j \rangle = \begin{cases} 1, & i = j, \\ -\frac{1}{n-1}, & i \neq j. \end{cases} \quad (733)$$

Thus $n = 2$, $n = 3$, and $n = 4$ recover, within one construction, the signed line, the Colour plane, and the tetrahedral Times space. Their dimensions and their group actions remain distinct.

Proof. Coordinate permutation is a semiring automorphism of S_I and an orthogonal linear automorphism of W_I . If the common frame in (726) is changed from f to fh , both operands change by h^{-1} ; equivariance of the two operations proves that their classes are unchanged. Any two classes have representatives in a common frame because $\mathcal{F}$ is a torsor.

For $x \in S_I$,

$$\text{Supp}(h^{-1} \cdot x) = h^{-1} \text{Supp}(x). \quad (734)$$

This proves that (727) is well defined and that its cardinality is frame-independent. Equations (728) and (729) follow fibrewise from theorem 13.6.

Both τ_I and $0 \in W_I$ are fixed by every permutation, which proves the independence asserted in (731). The relations

$$h \cdot 0_i = 0_{hi}, \quad h \cdot \mu_i = \mu_{hi}$$

show that a change of frame only permutes the set in (732). Its support-cardinality statements follow from (651). Finally, direct calculation from (730) gives (733). The augmentation spaces for $n = 2, 3, 4$ have dimensions 1, 2, 3, respectively. $\square$

The finite cases $2 \leq n \leq 6$, including support equivariance, fibrewise union and intersection, the local-zero orbit, and the exact simplex Gram matrices, are checked in `verification/verify_associated_support_geometry.py`. The captured output is `verification/verify_associated_support_geometry_output.txt`.

Theorem 13.30 (Support purity of fibrewise reciprocal certificates). *Replace $S = G(\mathbb{Z})$ in theorem 13.29 by $S_{\mathbb{Q}} := G(\mathbb{Q})$, and write*

$$\mathcal{S}_{\mathbb{Q}}(\mathcal{F}, I) := \mathcal{F} \times_H S_{\mathbb{Q}}^I. \quad (735)$$

For a universe class $A = [f, a] \in \mathcal{F} \times_H I$, write A also for the associated singleton $[f, \{a\}]$, and define

$$j_A : G(\mathbb{Q}) \longrightarrow \mathcal{S}_{\mathbb{Q}}(\mathcal{F}, I), \quad j_A(q) := [f, \iota_a(q)]. \quad (736)$$

This map is well defined, injective, and operation-preserving. Its support is

$$\text{Supp}_{\mathcal{F}}(j_A(q)) = \begin{cases} \emptyset, & q = \tau, \\ A, & q \in \mathbb{Q}. \end{cases} \quad (737)$$

Suppose

$$X_1 \oplus \cdots \oplus X_r = Y \quad (738)$$

in $\mathcal{S}_{\mathbb{Q}}(\mathcal{F}, I)$, with $\text{Supp}_{\mathcal{F}}(Y) = A$. Then

$$\text{Supp}_{\mathcal{F}}(X_j) \subseteq A \quad (1 \leq j \leq r). \quad (739)$$

Every non-unsupported summand in (738) therefore has support exactly A .

In particular, for positive integers n, x, y, z ,

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \iff j_A\left(\frac{\tilde{4}}{n}\right) = j_A\left(\frac{\tilde{1}}{x}\right) \oplus j_A\left(\frac{\tilde{1}}{y}\right) \oplus j_A\left(\frac{\tilde{1}}{z}\right). \quad (740)$$

Thus the classical Erdős–Straus identity occupies one associated universe class. A mixed-stalk model of certificate failure requires a further map from the absence of a fibrewise certificate to a mixed support class.

Proof. If $[f, a] = [fh, h^{-1}a]$, then

$$[fh, \iota_{h^{-1}a}(q)] = [f, h \cdot \iota_{h^{-1}a}(q)] = [f, \iota_a(q)],$$

so j_A is well defined. The coordinate injection $\iota_a : G(\mathbb{Q}) \rightarrow G(\mathbb{Q})^I$ is operation-preserving and injective; these properties pass to the associated carrier. Formula (737) follows directly from the definition of ι_a .

Apply (728) repeatedly to (738). It gives

$$A = \bigcup_{j=1}^r \text{Supp}_{\mathcal{F}}(X_j),$$

which proves (739). Finally, j_A is injective and restricts to ordinary rational addition on its supported sheet, proving (740). $\square$

Definition 13.31 (The Colour plane). Let X be a C_3 -torsor and let

$$E_X := \mathbb{R}[X] \quad (741)$$

be the real permutation space with its invariant inner product for which the vectors e_x , $x \in X$, are orthonormal. Define

$$\varepsilon_X\left(\sum_{x \in X} a_x e_x\right) := \sum_{x \in X} a_x, \quad \mathbf{j}_X := \sum_{x \in X} e_x. \quad (742)$$

The *Colour plane* of X is

$$\text{Col}(X) := \ker \varepsilon_X = \mathbf{j}_X^\perp. \quad (743)$$

The term Colour plane is introduced here for the construction in (743).

Theorem 13.32 (Construction and universal property of the Colour plane). *The orthogonal map*

$$q_X(v) := v - \frac{\langle v, \mathbf{j}_X \rangle}{3} \mathbf{j}_X \quad (744)$$

is C_3 -equivariant, has kernel $\mathbb{R}\mathbf{j}_X$, and induces a C_3 -equivariant isomorphism

$$\bar{q}_X : E_X / \mathbb{R}\mathbf{j}_X \xrightarrow{\sim} \text{Col}(X). \quad (745)$$

Every linear map $f : E_X \rightarrow W$ satisfying $f(\mathbf{j}_X) = 0$ factors uniquely through the quotient in (745).

For $x \in X$, define

$$\nu_x := \sqrt{\frac{3}{2}} q_X(e_x) = \sqrt{\frac{3}{2}} \left(e_x - \frac{1}{3} \mathbf{j}_X \right). \quad (746)$$

Then

$$\sum_{x \in X} \nu_x = 0, \quad \langle \nu_x, \nu_y \rangle = \begin{cases} 1, & x = y, \\ -\frac{1}{2}, & x \neq y. \end{cases} \quad (747)$$

The three distinguished directions therefore have pairwise angle $2\pi/3$. If R is the generator of the C_3 -action, then

$$R\nu_x = \nu_{Rx}. \quad (748)$$

Every equivariant bijection of C_3 -torsors induces an orthogonal isomorphism of their Colour planes carrying the corresponding three distinguished directions to one another.

Proof. Since $\langle \mathbf{j}_X, \mathbf{j}_X \rangle = 3$, (744) is the orthogonal projection onto $\mathbf{j}_X^\perp$. The vector $\mathbf{j}_X$ is C_3 -fixed, so q_X is equivariant. Its kernel and image give (745). The factorization statement is the universal property of the cokernel of $\mathbb{R}\mathbf{j}_X \hookrightarrow E_X$.

For $x, y \in X$, direct calculation gives

$$\left\langle e_x - \frac{1}{3}\mathbf{j}_X, e_y - \frac{1}{3}\mathbf{j}_X \right\rangle = \delta_{xy} - \frac{1}{3}.$$

Multiplication by $3/2$ yields (747). Their sum vanishes because $\sum_x e_x = \mathbf{j}_X$. The off-diagonal inner product is $-1/2 = \cos(2\pi/3)$. Equivariance gives (748); functoriality under equivariant bijections follows from the permutation-space construction. $\square$

Theorem 13.33 (Fixed loci in cyclic augmentation carriers). *For $n \geq 2$, let C_n act regularly on*

$$E_n := \mathbb{R}[C_n] \tag{749}$$

and let

$$W_n := \left\{ (x_0, \dots, x_{n-1}) \in \mathbb{R}^n : \sum_{j=0}^{n-1} x_j = 0 \right\} \tag{750}$$

be its augmentation carrier. Write R_n for the cyclic shift. For $0 \leq k < n$,

$$\dim \text{Fix} \left(R_n^k \mid W_n \right) = \gcd(n, k) - 1. \tag{751}$$

More generally, if a permutation $\pi \in S_n$ has $c(\pi)$ cycles, then

$$\dim \text{Fix}(\pi \mid W_n) = c(\pi) - 1. \tag{752}$$

Consequently, the nonidentity shifts of W_2 , W_3 , and W_5 have zero fixed locus. On the cyclically ordered Times carrier W_4 , the quarter-turns also have zero fixed locus, while the half-turn has the fixed line

$$\mathbb{R}(1, -1, 1, -1) \tag{753}$$

in the order (M, N, T, E).

Proof. The permutation R_n^k has $d = \gcd(n, k)$ cycles. A vector fixed by that permutation is constant on each cycle, so its fixed space in E_n has dimension d . The augmentation equation imposes one nonzero linear relation on those d cycle constants. This proves (751). The same argument with the cycles of π proves (752). Substitution of $n = 2, 3, 4, 5$ gives the listed cases; for the half-turn of W_4 , the fixed vectors have the form (a, b, a, b) , and the augmentation equation gives $b = -a$. $\square$

Definition 13.34 (Value direction and universe index). Fix the cyclically ordered colour set

$$\mathcal{C} := \{g, r, b\}, \quad \sigma : g \mapsto r \mapsto b \mapsto g. \tag{754}$$

For each universe $u \in \mathcal{C}$, let $\mathcal{V}_u = 0_u + \text{Col}(\mathcal{C})$ be an affine copy of the Colour plane with supported origin 0_u . For $v \in \mathcal{C}$, define

$${}^v 1_u := 0_u + \nu_v, \quad [{}^v 1_u] := {}^v 1_u - 0_u = \nu_v. \tag{755}$$

The left superscript v records the value direction. The subscript u records the universe in which that value is read.

Theorem 13.35 (Colour replacement for binary negation). *For every fixed $u \in \mathcal{C}$,*

$$[{}^g 1_u] + [{}^r 1_u] + [{}^b 1_u] = 0, \tag{756}$$

and

$$\langle [{}^v 1_u], [{}^w 1_u] \rangle = \begin{cases} 1, & v = w, \\ -\frac{1}{2}, & v \neq w. \end{cases} \tag{757}$$

Consequently,

$$-[g1_u] = [r1_u] + [b1_u], \quad (758)$$

$$-[r1_u] = [b1_u] + [g1_u], \quad (759)$$

$$-[b1_u] = [g1_u] + [r1_u]. \quad (760)$$

Thus the vector opposite to one colour direction is the resultant of the other two colour directions. It is not represented by an individual colour label.

Proof. Equations (756) and (757) are (747) written in the value–universe notation. Moving one term of (756) to the other side gives the three resultant identities. $\square$

Theorem 13.36 (Colour-by-colour $C_3 \times C_3$ carrier). *Let*

$$\mathcal{P}_{\text{colour}} := \{v1_u : u, v \in \mathcal{C}\}. \quad (761)$$

Define the universe and value rotations by

$$R_{\text{universe}}(v1_u) := v1_{\sigma(u)}, \quad R_{\text{value}}(v1_u) := \sigma(v)1_u. \quad (762)$$

They commute, each has order three, and they generate a faithful $C_3 \times C_3$ -action on $\mathcal{P}_{\text{colour}}$.

The bilinear mixed coordinate of the point $v1_u$ is

$$\Phi_{u,v} := \nu_u \otimes \nu_v \in \text{Col}(\mathcal{C}) \otimes \text{Col}(\mathcal{C}). \quad (763)$$

These nine coordinates span a four-dimensional real space and satisfy

$$\sum_{v \in \mathcal{C}} \Phi_{u,v} = 0, \quad \sum_{u \in \mathcal{C}} \Phi_{u,v} = 0. \quad (764)$$

Their complete Gram rule is

$$\langle \Phi_{u,v}, \Phi_{u',v'} \rangle = \gamma(u, u') \gamma(v, v'), \quad \gamma(a, c) := \begin{cases} 1, & a = c, \\ -\frac{1}{2}, & a \neq c. \end{cases} \quad (765)$$

In particular, distinct tensor coordinates have inner product $-1/2$ when exactly one colour index changes and $1/4$ when both colour indices change.

For every $u, v \in \mathcal{C}$,

$$-\Phi_{u,v} = \Phi_{\sigma(u),v} + \Phi_{\sigma^2(u),v} = \Phi_{u,\sigma(v)} + \Phi_{u,\sigma^2(v)}. \quad (766)$$

Hence the two independent colour indices provide two exact two-colour descriptions of the direction opposite to $\Phi_{u,v}$.

Proof. The two maps in (762) act on different indices, so they commute. Their action on the ordered pair (u, v) is the regular translation action of $C_3 \times C_3$, proving faithfulness.

The span of the ν_u is the two-dimensional Colour plane. Therefore the tensors $\nu_u \otimes \nu_v$ span its four-dimensional tensor square. The row and column relations follow from $\nu_g + \nu_r + \nu_b = 0$. The tensor-product inner product gives

$$\langle \nu_u \otimes \nu_v, \nu_{u'} \otimes \nu_{v'} \rangle = \langle \nu_u, \nu_{u'} \rangle \langle \nu_v, \nu_{v'} \rangle,$$

which is (765). Finally,

$$-\nu_u = \nu_{\sigma(u)} + \nu_{\sigma^2(u)}$$

and the corresponding identity in the second tensor factor give (766). $\square$

Theorem 13.37 (Co-rotating and counter-rotating Colour planes). *Put $V = \text{Col}(\mathcal{C})$, choose the oriented orthonormal basis in which*

$$\nu_g = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \nu_r = \begin{pmatrix} -1/2 \\ \sqrt{3}/2 \end{pmatrix}, \quad \nu_b = \begin{pmatrix} -1/2 \\ -\sqrt{3}/2 \end{pmatrix}, \quad (767)$$

and let

$$R = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}. \quad (768)$$

Thus $R\nu_g = \nu_r$, $R\nu_r = \nu_b$, and $R\nu_b = \nu_g$. Identify

$$V \otimes V \xrightarrow{\sim} M_2(\mathbb{R}), \quad x \otimes y \mapsto xy^\top. \quad (769)$$

The simultaneous and counter-rotating actions become

$$\mathcal{D}(X) := RXR^\top, \quad \mathcal{A}(X) := RXR. \quad (770)$$

Their fixed spaces are the distinct orthogonal planes

$$\text{Fix}(\mathcal{D}) = \text{span}_{\mathbb{R}}\{I, J\}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad (771)$$

$$\text{Fix}(\mathcal{A}) = \text{span}_{\mathbb{R}}\{K, L\}, \quad K = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad L = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (772)$$

Write $\Phi_{u,v} = \nu_u \nu_v^\top$ under (769). The three co-rotating orbit sums are

$$D_g := \Phi_{g,g} + \Phi_{r,r} + \Phi_{b,b} = \frac{3}{2}I, \quad (773)$$

$$D_r := \Phi_{g,r} + \Phi_{r,b} + \Phi_{b,g} = \begin{pmatrix} -\frac{3}{4} & \frac{3\sqrt{3}}{4} \\ -\frac{3\sqrt{3}}{4} & -\frac{3}{4} \end{pmatrix}, \quad (774)$$

$$D_b := \Phi_{g,b} + \Phi_{r,g} + \Phi_{b,r} = \begin{pmatrix} -\frac{3}{4} & -\frac{3\sqrt{3}}{4} \\ \frac{3\sqrt{3}}{4} & -\frac{3}{4} \end{pmatrix}. \quad (775)$$

The three counter-rotating orbit sums are

$$A_g := \Phi_{g,g} + \Phi_{r,b} + \Phi_{b,r} = \frac{3}{2}K, \quad (776)$$

$$A_r := \Phi_{g,r} + \Phi_{r,g} + \Phi_{b,b} = \begin{pmatrix} -\frac{3}{4} & \frac{3\sqrt{3}}{4} \\ \frac{3\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}, \quad (777)$$

$$A_b := \Phi_{g,b} + \Phi_{r,r} + \Phi_{b,g} = \begin{pmatrix} -\frac{3}{4} & -\frac{3\sqrt{3}}{4} \\ -\frac{3\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}. \quad (778)$$

Each ordered triple is a centred equilateral triple:

$$D_g + D_r + D_b = 0, \quad A_g + A_r + A_b = 0, \quad (779)$$

and, for the Frobenius inner product,

$$(\langle D_u, D_v \rangle)_{u,v} = (\langle A_u, A_v \rangle)_{u,v} = \frac{9}{4} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \quad \langle D_u, A_v \rangle = 0. \quad (780)$$

Thus the two ways of entwining the value and universe trialities retain two separate two-dimensional carriers inside the same four-dimensional tensor space.

Proof. Under (769), $(R \otimes R)(x \otimes y)$ maps to $RXR^\top$, while $(R \otimes R^{-1})(x \otimes y)$ maps to RXR . A matrix fixed by $X \mapsto RXR^\top$ commutes with R . Since the minimal polynomial of R is $t^2 + t + 1$, its real matrix centralizer is $\text{span}\{I, J\}$. Substitution of a general matrix $X = (x_{ij})$ into $RXR = X$ gives $x_{22} = -x_{11}$ and $x_{21} = x_{12}$, proving (772). The four displayed basis matrices are pairwise orthogonal across the two planes.

Substitution of the three vectors in (767) gives (773)–(778). Adding those matrices proves (779). Their Frobenius products give (780). $\square$

Proposition 13.38 (Exact C_2 -to- C_3 representation comparison). *Let Y be a C_2 -torsor with basis e_0, e_1 , and let s exchange the two basis vectors. Put*

$$\mathbf{j}_Y = e_0 + e_1, \quad \mathbf{u}_Y = e_0 - e_1. \quad (781)$$

Then

$$\mathbb{R}[Y] = \mathbb{R}\mathbf{j}_Y \oplus \mathbb{R}\mathbf{u}_Y, \quad s\mathbf{j}_Y = \mathbf{j}_Y, \quad s\mathbf{u}_Y = -\mathbf{u}_Y. \quad (782)$$

The orthogonal map

$$\frac{\mathbf{j}_Y}{\sqrt{2}} \mapsto i, \quad \frac{\mathbf{u}_Y}{\sqrt{2}} \mapsto 1 \quad (783)$$

intertwines s with $\rho(\zeta) = -\bar{\zeta}$ and hence realizes

$$\mathbb{C}_\rho = \mathbb{R}_{\text{sgn}} \oplus i\mathbb{R}_{\text{triv}}. \quad (784)$$

For a C_3 -torsor X , the corresponding decomposition is

$$\mathbb{R}[X] = \mathbb{R}\mathbf{j}_X \oplus \text{Col}(X) \cong \mathbb{R}_{\text{triv}} \oplus \text{Col}(X). \quad (785)$$

Thus the full C_3 permutation datum has one invariant support direction and the two-dimensional Colour plane constructed in theorem 13.31.

Proof. The vectors $\mathbf{j}_Y, \mathbf{u}_Y$ are orthogonal and have squared length 2. Their transformation laws prove (782). Since

$$\rho(i) = i, \quad \rho(1) = -1,$$

the map in (783) is an orthogonal intertwiner. Equation (785) is the orthogonal decomposition by the fixed line and its augmentation plane. $\square$

Proposition 13.39 (Anti-unitary implementation of sheet reflections). *Let $I = C_N$, let $\mathcal{H}_N = \mathbb{C}^{C_N}$ with basis (e_a) , and set*

$$Ue_a = e_{a+1}, \quad J_c \left(\sum_{a \in C_N} z_a e_a \right) := \sum_{a \in C_N} \bar{z}_a e_{c-a}. \quad (786)$$

Then U is unitary, every J_c is anti-unitary, and

$$U^N = I, \quad J_c^2 = I, \quad J_c U J_c^{-1} = U^{-1}. \quad (787)$$

If P_a is the projection onto $\mathbb{C}e_a$, then

$$J_c P_a J_c^{-1} = P_{c-a}. \quad (788)$$

For $N = 2$, J_1 exchanges the support directions of 0_+ and 0_- . For $N = 3$, the three operators J_c give the three reflections of the red–green–blue triality plane.

Proof. The displayed formula for J_c is conjugate-linear and preserves the standard Hermitian norm, so it is anti-unitary. Applying it twice sends e_a to $e_{c-(c-a)} = e_a$. On a basis vector,

$$J_c U J_c^{-1} e_a = e_{a-1} = U^{-1} e_a,$$

which proves (787). Conjugating the rank-one basis projection gives (788). $\square$

Theorem 13.40 (Triality of trialities). *Let $I = C_3 \times C_3$, with indices (a, b) , and define*

$$0_{a,b} := \iota_{(a,b)}(\tilde{0}), \quad 0_a^{(2)} := \bigoplus_{b \in C_3} 0_{a,b}. \quad (789)$$

Then

$$\text{Supp}(0_{a,b}) = \{(a, b)\}, \quad \text{Supp}(0_a^{(2)}) = \{a\} \times C_3. \quad (790)$$

Let U rotate the first coordinate and, for each a , let V_a rotate the second coordinate in the fibre over a . These permutations satisfy

$$U^3 = V_a^3 = I, \quad V_a V_c = V_c V_a, \quad UV_a U^{-1} = V_{a+1}. \quad (791)$$

Their permutation group is

$$C_3^3 \rtimes C_3 = C_3 \wr C_3. \quad (792)$$

In $\mathbb{R}^{C_3 \times C_3}$, write

$$j_a := \sum_{b \in C_3} e_{a,b}, \quad J := \sum_{a \in C_3} j_a, \quad v_{a,b} := 3e_{a,b} - j_a, \quad w_a := 3j_a - J. \quad (793)$$

For fixed a , the three vectors $v_{a,b}$ have pairwise angle 120° ; the three vectors w_a also have pairwise angle 120° . Thus the nine supported zeros carry an inner triality in each fibre and an outer triality of the three fibres.

Proof. The support statements follow from theorem 13.6. The permutation relations are direct, and they give the stated semidirect product because the three commuting inner rotations form C_3^3 and the outer rotation permutes those three factors transitively.

For fixed a , direct calculation gives

$$\sum_b v_{a,b} = 0, \quad \langle v_{a,b}, v_{a,b} \rangle = 6, \quad \langle v_{a,b}, v_{a,c} \rangle = -3 \quad (b \neq c).$$

Since $\langle j_a, j_a \rangle = 3$, $\langle j_a, j_c \rangle = 0$ for $a \neq c$, and $\langle J, J \rangle = 9$, one also obtains

$$\sum_a w_a = 0, \quad \langle w_a, w_a \rangle = 18, \quad \langle w_a, w_c \rangle = -9 \quad (a \neq c).$$

Both off-diagonal-to-diagonal ratios are $-1/2$, proving the two 120° assertions. $\square$

Definition 13.41 (The Times space). Let

$$\mathcal{T} := \{M, N, T, E\} \quad (794)$$

denote the ordered Time labels Morning, Noon, Twilight, and Evening. In the real permutation space $\mathbb{R}[\mathcal{T}]$, put

$$\mathbf{j}_{\mathcal{T}} := e_M + e_N + e_T + e_E, \quad \text{Time}(\mathcal{T}) := \mathbf{j}_{\mathcal{T}}^\perp. \quad (795)$$

For $a \in \mathcal{T}$, define

$$\lambda_a := e_a - \frac{1}{4}\mathbf{j}_{\mathcal{T}}, \quad \mu_a := \frac{2}{\sqrt{3}}\lambda_a. \quad (796)$$

The term *Times space* is introduced here for the three-dimensional space in (795).

Theorem 13.42 (The tetrahedral four-Time carrier). *The four vectors in (796) satisfy*

$$\mu_M + \mu_N + \mu_T + \mu_E = 0 \quad (797)$$

and

$$\langle \mu_a, \mu_c \rangle = \begin{cases} 1, & a = c, \\ -\frac{1}{3}, & a \neq c. \end{cases} \quad (798)$$

They span $\text{Time}(\mathcal{T})$, which has dimension three. Every pair of distinct Time directions has angle

$$\theta_{\mathcal{T}} = \arccos\left(-\frac{1}{3}\right). \quad (799)$$

For each Time universe $u \in \mathcal{T}$, define

$$v1_u := 0_u + \mu_v, \quad [v1_u] := v1_u - 0_u = \mu_v \quad (v \in \mathcal{T}). \quad (800)$$

Then

$$\sum_{v \in \mathcal{T}} [v1_u] = 0 \quad (801)$$

and, for each $v \in \mathcal{T}$,

$$-[v1_u] = \sum_{\substack{c \in \mathcal{T} \\ c \neq v}} [c1_u]. \quad (802)$$

Thus the direction opposite to one Time is the resultant of the other three Time directions.

Proof. Since $\sum_a e_a = \mathbf{j}_{\mathcal{T}}$, the vectors λ_a sum to zero. Direct calculation gives

$$\langle \lambda_a, \lambda_c \rangle = \delta_{ac} - \frac{1}{4}.$$

Multiplication by $4/3$ gives (798). Its Gram matrix is

$$\frac{4}{3}I_4 - \frac{1}{3}J_4.$$

It has eigenvalue zero on $\mathbf{j}_{\mathcal{T}}$ and eigenvalue $4/3$ with multiplicity three on $\mathbf{j}_{\mathcal{T}}^\perp$. Hence the four vectors span the three-dimensional Times space and form a regular tetrahedron. Equations (801) and (802) are (797) in value–universe notation. $\square$

Theorem 13.43 (A_3 , tetrahedral symmetry, and the Time cycle). *The unscaled vectors λ_a are the four weights of the defining representation in the standard real A_3 weight space, and*

$$\lambda_a - \lambda_c = e_a - e_c \quad (a \neq c) \quad (803)$$

are the twelve A_3 roots.

The full orthogonal symmetry group of the set $\{\mu_M, \mu_N, \mu_T, \mu_E\}$ is

$$W(A_3) \cong S_4. \quad (804)$$

Its orientation-preserving subgroup is A_4 . The subgroup

$$V_4 = \{1, (MN)(TE), (MT)(NE), (ME)(NT)\} \quad (805)$$

is a normal subgroup of A_4 .

Let

$$\Omega : M \mapsto N \mapsto T \mapsto E \mapsto M \quad (806)$$

be the cyclic Time relabeling. In the orthonormal basis

$$u = \frac{e_M - e_T}{\sqrt{2}}, \quad v = \frac{e_N - e_E}{\sqrt{2}}, \quad w = \frac{e_M - e_N + e_T - e_E}{2} \quad (807)$$

of $\text{Time}(\mathcal{T})$, one has

$$[\Omega]_{(u,v,w)} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (808)$$

Consequently $\Omega^4 = I$, $\det \Omega = -1$, and the Time cycle is the orthogonal sum of a quarter-turn plane and an alternating line.

Under the standard $SU(4)$ interpretation, (795) is the real Cartan weight space and Ω is an order-four Weyl-group element. The central subgroup $Z(SU(4)) \cong C_4$ acts by scalar fourth roots of unity on the defining complex representation. Its adjoint action on the Cartan space is the identity.

Proof. Equation (803) is immediate from (796). Every permutation of the four coordinates preserves the centered hyperplane and permutes the four weights. Conversely, an orthogonal symmetry of a regular tetrahedron permutes its four vertices, proving (804). A permutation matrix has determinant equal to the sign of its permutation. Its restriction to the fixed line $\mathbb{R}\mathbf{j}_{\mathcal{T}}$ has determinant one, so its restriction to the Times space has the same determinant. This identifies the proper subgroup with A_4 . The three double transpositions together with the identity form the displayed normal Klein subgroup.

The vectors in (807) are orthonormal and lie in the centered hyperplane. Direct application of the four-cycle gives

$$\Omega u = v, \quad \Omega v = -u, \quad \Omega w = -w,$$

which proves (808). The final statements are the standard weight-space, Weyl-group, and central actions of $SU(4)$. $\square$

Proposition 13.44 (Local threefold tangent structure at each Time). *For $a \in \mathcal{T}$, let*

$$T_a := \mu_a^\perp \cap \text{Time}(\mathcal{T}). \quad (809)$$

Then $\dim T_a = 2$. The stabilizer of μ_a in A_4 is the cyclic group of order three that rotates the remaining three Time vertices. Its restriction to T_a is a rotation through $2\pi/3$.

Proof. The vector μ_a is nonzero in the three-dimensional Times space, so its orthogonal complement there has dimension two. An even permutation fixing a is an even permutation of the other three labels; these are the three powers of their three-cycle. The two nontrivial powers act orthogonally on T_a , have order three, and have no nonzero fixed vector there. Their rotation angles are therefore $2\pi/3$ and $4\pi/3$. $\square$

Theorem 13.45 (Orthogonal flag from the binary line through Colours to Times). *In the ordered Time coordinates (M, N, T, E) , define*

$$h_1 := \frac{e_M - e_N}{\sqrt{2}}, \quad h_2 := \frac{e_M + e_N - 2e_T}{\sqrt{6}}, \quad h_3 := \frac{e_M + e_N + e_T - 3e_E}{\sqrt{12}}. \quad (810)$$

These vectors are orthonormal, and

$$\mathbb{R}h_1 \subset \mathbb{R}h_1 \oplus \mathbb{R}h_2 \subset \mathbb{R}h_1 \oplus \mathbb{R}h_2 \oplus \mathbb{R}h_3 = \text{Time}(\mathcal{T}). \quad (811)$$

The first member is the centered binary line on $\{M, N\}$. The second is the Colour direction space on $\{M, N, T\}$. The third is the Times space. Moreover,

$$\mu_E = -h_3. \quad (812)$$

Every $x \in \text{Time}(\mathcal{T})$ has the unique expansion

$$x = x_1 h_1 + x_2 h_2 + x_3 h_3. \quad (813)$$

Its projection to the selected Colour plane is

$$\text{pr}_{\text{Colour}}(x) = x_1 h_1 + x_2 h_2, \quad (814)$$

and its new Time component is $x_3 h_3$. In particular,

$${}^E 20_u = 0_u + 20\mu_E = 0_u - 20h_3. \quad (815)$$

No orthogonal transformation preserving the selected Colour plane can send a Colour-plane vector to the nonzero displacement $20\mu_E$.

Proof. Direct inner-product calculation proves that the three vectors in (810) are orthonormal and have coordinate sum zero. The first two have vanishing E coordinate, so they span

$$\{x \in \mathbb{R}^{\mathcal{T}} : x_E = 0, x_M + x_N + x_T = 0\},$$

the centered direction space of the selected three-label face. Adding h_3 gives the complete three-dimensional centered Time space. Substitution in (796) gives

$$\mu_E = \frac{2}{\sqrt{3}} \left(-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, \frac{3}{4} \right) = -h_3.$$

The coordinate and projection formulas follow from orthonormality. An orthogonal transformation preserving the Colour plane maps that plane to itself, while $20\mu_E$ is a nonzero vector in its orthogonal complement. $\square$

Remark 13.46 (Relative choice of a binary–Colour–Time flag). The flag in (811) uses the selected ordered labels

$$(M, N) \subset (M, N, T) \subset (M, N, T, E).$$

The set of all such complete flags is preserved transitively by S_4 . Hence the family of bridges is intrinsic to the tetrahedral carrier; one specific bridge records a relative choice of edge, face, and omitted Time.

Theorem 13.47 (Colour correlation plane and the Fable complementary resultant). *Read $x \otimes y$ as the rank-one operator $z \mapsto x\langle y, z \rangle$. With $\Phi_{u,v} = \nu_u \otimes \nu_v$, define*

$$\Gamma_g := \Phi_{g,g} + \Phi_{r,r} + \Phi_{b,b}, \quad (816)$$

$$\Gamma_r := \Phi_{g,r} + \Phi_{r,b} + \Phi_{b,g}, \quad (817)$$

$$\Gamma_b := \Phi_{g,b} + \Phi_{r,g} + \Phi_{b,r}. \quad (818)$$

Then

$$\Gamma_g + \Gamma_r + \Gamma_b = 0, \quad (819)$$

and, for $c, d \in \{g, r, b\}$,

$$\langle \Gamma_c, \Gamma_d \rangle_{\text{HS}} = \begin{cases} \frac{9}{2}, & c = d, \\ -\frac{9}{4}, & c \neq d. \end{cases} \quad (820)$$

Thus

$$\hat{\Gamma}_c := \frac{\sqrt{2}}{3} \Gamma_c \quad (821)$$

are three unit directions separated by $2\pi/3$.

If R is the colour rotation $g \mapsto r \mapsto b \mapsto g$, then

$$\Gamma_g = \frac{3}{2} I_{\text{Col}}, \quad \Gamma_r = \frac{3}{2} R^{-1}, \quad \Gamma_b = \frac{3}{2} R^{-2}. \quad (822)$$

The Fable marked-factor occupancy vector

$$m_F = (1, 1, 1) \quad (823)$$

has centered Colour reading zero and support reading three. Its diagonal-correlation reading is

$$\text{Corr}_{\mathcal{C}}(m_F) := \Phi_{g,g} + \Phi_{r,r} + \Phi_{b,b} = \Gamma_g = \frac{3}{2} I_{\text{Col}}. \quad (824)$$

The Fable common target is $(-1/4, 0, 0)$ in its original affine coefficient chart. Its intrinsic Colour-correlation representative is the two-colour quarter resultant

$$\mathbf{q}_{rb} := \frac{1}{4} \Gamma_r + \frac{1}{4} \Gamma_b. \quad (825)$$

It is characterized inside the Colour correlation plane by

$$\frac{1}{4} \Gamma_g + \mathbf{q}_{rb} = 0_{\text{Corr}(\mathcal{C})}. \quad (826)$$

Thus the signed scalar appearing in the affine chart is replaced by the resultant of the two Colour directions complementary to (g) .

Proof. The tight-frame operator of the three simplex vertices is

$$\sum_{a \in \mathcal{C}} \nu_a \otimes \nu_a = \frac{3}{2} I_{\text{Col}},$$

which follows directly from $\nu_a = \sqrt{3/2} q_{\mathcal{C}}(e_a)$ and $\sum_a e_a \otimes e_a = I_{\mathbb{R}[\mathcal{C}]}$. Since $\nu_{\sigma(a)} = R\nu_a$, the three correlation operators are those in (822). On the Colour plane,

$$I + R + R^2 = 0, \quad \text{tr}(I) = 2, \quad \text{tr}(R) = \text{tr}(R^2) = -1.$$

These identities give (819) and the Hilbert–Schmidt Gram matrix (820).

The centered reading of $(1, 1, 1)$ vanishes because it is the diagonal vector $\mathbf{j}_{\mathcal{C}}$, while its augmentation is three. The diagonal-correlation formula is (816). Finally, (819) gives (826), whose second summand is exactly the resultant in (825). $\square$

Theorem 13.48 (Time correlation tetrahedron). *Let Ω be the Time cycle in (806), and put $\Psi_{u,v} := \mu_u \otimes \mu_v$. Define*

$$\Theta_{\text{M}} := \Psi_{\text{M},\text{M}} + \Psi_{\text{N},\text{N}} + \Psi_{\text{T},\text{T}} + \Psi_{\text{E},\text{E}}, \quad (827)$$

$$\Theta_{\text{N}} := \Psi_{\text{M},\text{N}} + \Psi_{\text{N},\text{T}} + \Psi_{\text{T},\text{E}} + \Psi_{\text{E},\text{M}}, \quad (828)$$

$$\Theta_{\text{T}} := \Psi_{\text{M},\text{T}} + \Psi_{\text{N},\text{E}} + \Psi_{\text{T},\text{M}} + \Psi_{\text{E},\text{N}}, \quad (829)$$

$$\Theta_{\text{E}} := \Psi_{\text{M},\text{E}} + \Psi_{\text{N},\text{M}} + \Psi_{\text{T},\text{N}} + \Psi_{\text{E},\text{T}}. \quad (830)$$

Then

$$\Theta_{\text{M}} + \Theta_{\text{N}} + \Theta_{\text{T}} + \Theta_{\text{E}} = 0, \quad (831)$$

and

$$\langle \Theta_a, \Theta_c \rangle_{\text{HS}} = \begin{cases} \frac{16}{3}, & a = c, \\ -\frac{16}{9}, & a \neq c. \end{cases} \quad (832)$$

Therefore

$$\hat{\Theta}_a := \frac{\sqrt{3}}{4} \Theta_a \quad (833)$$

form a regular tetrahedron in a three-dimensional correlation subspace. As operators on the Times space,

$$\Theta_{\text{M}} = \frac{4}{3} I_{\text{Time}}, \quad \Theta_{\text{N}} = \frac{4}{3} \Omega^{-1}, \quad \Theta_{\text{T}} = \frac{4}{3} \Omega^{-2}, \quad \Theta_{\text{E}} = \frac{4}{3} \Omega^{-3}. \quad (834)$$

The balanced Time count $(1, 1, 1, 1)$ has centered Times reading zero, support reading four, and diagonal-correlation reading

$$\text{Corr}_{\mathcal{T}}(1, 1, 1, 1) = \Theta_{\text{M}} = \frac{4}{3} I_{\text{Time}}. \quad (835)$$

Its three-Time quarter resultant is

$$\mathfrak{q}_{\text{NTE}} := \frac{1}{4} \Theta_{\text{N}} + \frac{1}{4} \Theta_{\text{T}} + \frac{1}{4} \Theta_{\text{E}}. \quad (836)$$

It is characterized by

$$\frac{1}{4} \Theta_{\text{M}} + \mathfrak{q}_{\text{NTE}} = 0_{\text{Corr}(\mathcal{T})}. \quad (837)$$

Proof. The four tetrahedral unit vectors form the tight frame

$$\sum_{a \in \mathcal{T}} \mu_a \otimes \mu_a = \frac{4}{3} I_{\text{Time}}.$$

Because $\mu_{\Omega(a)} = \Omega \mu_a$, this gives (834). The centered permutation representation of the four-cycle satisfies

$$I + \Omega + \Omega^2 + \Omega^3 = 0.$$

Its dimension is three, and

$$\mathrm{tr}(\Omega) = \mathrm{tr}(\Omega^2) = \mathrm{tr}(\Omega^3) = -1.$$

Taking Hilbert–Schmidt products of the four operators in (834) proves (832). The remaining assertions follow from (831) and the tight-frame identity. $\square$

Theorem 13.49 (Time-to-Colour projection and the quarter altitude). *For $a \in \mathcal{T}$, let*

$$c_a^\wedge := \frac{1}{3} \sum_{b \in \mathcal{T} \setminus \{a\}} \mu_b, \quad h_a := \mu_a - c_a^\wedge. \quad (838)$$

Then

$$0_{\mathcal{T}} = \frac{1}{4}\mu_a + \frac{3}{4}c_a^\wedge, \quad 0_{\mathcal{T}} - c_a^\wedge = \frac{1}{4}h_a. \quad (839)$$

Consequently,

$$\frac{\mathrm{dist}(0_{\mathcal{T}}, c_a^\wedge)}{\mathrm{dist}(\mu_a, c_a^\wedge)} = \frac{1}{4}. \quad (840)$$

Let π_a be orthogonal projection onto $\mu_a^\perp$. The three vectors

$$\pi_a(\mu_b) = \mu_b + \frac{1}{3}\mu_a, \quad b \neq a, \quad (841)$$

have equal norm and pairwise angle $2\pi/3$. Moreover,

$$\pi_a(c_a^\wedge) = \pi_a(0_{\mathcal{T}}) = 0_{\mathcal{C}}. \quad (842)$$

Thus the global Time centre and the centre of the opposite Time face have the same Colour projection and differ by one quarter of the full tetrahedral altitude. If the altitude from $c_a^\wedge$ to μ_a is assigned coordinate length one and the global Time centre is assigned coordinate zero, then $c_a^\wedge$ has coordinate $-1/4$.

Proof. The simplex identity $\sum_{b \in \mathcal{T}} \mu_b = 0$ gives

$$c_a^\wedge = -\frac{1}{3}\mu_a.$$

Substitution proves both identities in (839) and the distance ratio (840). For $b \neq a$, $\langle \mu_b, \mu_a \rangle = -1/3$, so orthogonal projection gives (841). For distinct $b, c \in \mathcal{T} \setminus \{a\}$,

$$\|\pi_a(\mu_b)\|^2 = \frac{8}{9}, \quad \langle \pi_a(\mu_b), \pi_a(\mu_c) \rangle = -\frac{4}{9}.$$

Their unit directions therefore have inner product $-1/2$, hence angle $2\pi/3$. Finally, $c_a^\wedge = -\mu_a/3$ proves (842). The point in (842) is the supported zero of the projected Colour carrier; it is distinct from the unsupported symbol τ . $\square$

Theorem 13.50 (Two-layer quarter invariant and the complete Fable fibre). *Fix $a \in \mathcal{T}$, put*

$$K_{\mathcal{T}} := \mathrm{conv}\{\mu_c : c \in \mathcal{T}\}, \quad \pi_a := I - \mu_a \otimes \mu_a,$$

and write $B_a := \mathcal{T} \setminus \{a\}$. For $b \in B_a$, define

$$\nu_b^{(a)} := \frac{3}{2\sqrt{2}} \pi_a(\mu_b). \quad (843)$$

The three vectors in (843) are unit vectors with

$$\sum_{b \in B_a} \nu_b^{(a)} = 0, \quad \langle \nu_b^{(a)}, \nu_c^{(a)} \rangle = -\frac{1}{2} \quad (b \neq c). \quad (844)$$

The fibre of the projected Colour zero inside the Time tetrahedron is

$$\pi_a^{-1}(0_{\mathcal{C}}) \cap K_{\mathcal{T}} = [c_a^\wedge, \mu_a]. \quad (845)$$

On its affine line, the functional

$$q_a(x) := \frac{3}{4} \langle x, \mu_a \rangle \quad (846)$$

satisfies

$$q_a(c_a^\wedge) = -\frac{1}{4}, \quad q_a(0_T) = 0, \quad q_a(\mu_a) = \frac{3}{4}, \quad q_a(h_a) = 1. \quad (847)$$

Thus $c_a^\wedge$ is the unique point of the opposite Time face in the zero fibre, and its coordinate relative to the global centre and the full face-to-vertex altitude is $-1/4$.

Select $b_0 \in B_a$. For $b \in B_a$, put

$$u_b := \langle \nu_{b_0}^{(a)}, \nu_b^{(a)} \rangle, \quad h_b := \frac{1}{4} - \frac{1}{2}u_b. \quad (848)$$

In the order (b_0, b_1, b_2) , the exact values are

$$\begin{array}{c|ccc} b & b_0 & b_1 & b_2 \\ \hline u_b & 1 & -\frac{1}{2} & -\frac{1}{2} \\ h_b & -\frac{1}{4} & \frac{1}{2} & \frac{1}{2} \\ u_b h_b & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{array} \quad (849)$$

The product reading of the three labelled Colour states is therefore constant, although the states are distinct.

Let

$$A_F := \mathbb{Q}[x]/(x^3 - x)$$

be the reduced coordinate algebra of the complete Fable fibre in theorem 7.2, and let $e := x^2 \in A_F$. The restrictions of the two factors $u = 1 + xy$ and $h = u^2 z + y^2(1 + 3u)$ are

$$e^2 = e, \quad u_F = 1 - \frac{3}{2}e, \quad h_F = -\frac{1}{4} + \frac{3}{4}e = \frac{1}{4} - \frac{1}{2}u_F. \quad (850)$$

Consequently,

$$u_F h_F = -\frac{1}{4} \quad \text{in } A_F. \quad (851)$$

Under the ordered identification

$$(b_0, b_1, b_2) \longleftrightarrow (x = 0, x = 1, x = -1), \quad (852)$$

the three evaluations of (u_F, h_F) are exactly the first two rows of (849). Hence the first coordinate of the Fable common target is the two-layer product

$$F_1 = u_F h_F = -\frac{1}{4}. \quad (853)$$

Together with $F_2 = F_3 = 0$, this recovers the complete target $(-1/4, 0, 0)$ of theorem 7.1.

Finally, let

$$w_N = (4, 0, 0, -1), \quad L_a = \mathbb{R}\mu_a,$$

and define

$$\Phi_a : L_a \longrightarrow \mathbb{R}w_N, \quad \Phi_a(x) := q_a(x)w_N. \quad (854)$$

Then

$$\Phi_a(h_a) = w_N, \quad \Phi_a(c_a^\wedge) = -\frac{1}{4}w_N = \mathcal{I}_- w_N. \quad (855)$$

The scalar $1/4$ in this formula is also the unique nonzero scalar λ for which

$$\lambda N N^\top = 4\lambda E_{22} \quad (856)$$

is idempotent.

Proof. For $b \neq a$, the tetrahedral Gram matrix gives

$$\pi_a(\mu_b) = \mu_b + \frac{1}{3}\mu_a.$$

Its squared norm is $8/9$, while two distinct projected face vertices have inner product $-4/9$. Equation (844) follows after multiplication by $3/(2\sqrt{2})$.

Let $x = \sum_{c \in \mathcal{T}} t_c \mu_c \in K_{\mathcal{T}}$, where $t_c \geq 0$ and $\sum_c t_c = 1$. If $\pi_a(x) = 0$, then

$$\sum_{b \in B_a} t_b \pi_a(\mu_b) = 0.$$

The only linear relation among the three vertices of the projected regular triangle is their sum-zero relation. Hence the three coefficients t_b , for $b \in B_a$, are equal. Writing their common value as s gives $t_a + 3s = 1$, and therefore

$$x = t_a \mu_a + s \sum_{b \in B_a} \mu_b = (t_a - s) \mu_a.$$

As t_a ranges from zero to one, this is exactly the segment from $-\mu_a/3 = c_a^-$ to μ_a . This proves (845). Substitution of $c_a^- = -\mu_a/3$ and $h_a = 4\mu_a/3$ into (846) proves (847).

Equation (844) gives the row $(1, -1/2, -1/2)$. Applying the affine expression in (848) gives the second row of (849); multiplication gives its third row.

In A_F , one has $e^2 = e$. The Gröbner relations (48) give

$$y = -\frac{3}{2}x, \quad z = \frac{27}{4}e - \frac{1}{4}.$$

Substitution into the definitions of u and h , followed only by the relation $e^2 = e$, gives (850). It follows that

$$\left(1 - \frac{3}{2}e\right) \left(-\frac{1}{4} + \frac{3}{4}e\right) = -\frac{1}{4} + \frac{9}{8}(e - e^2) = -\frac{1}{4},$$

which proves (851). Evaluation at the three roots $(0,1,-1)$ proves (852) and (853).

The map Φ_a is the unique linear map from L_a to $\mathbb{R}w_N$ sending h_a to w_N . Equations (847) and (346) prove (855). Since $NN^T = 4E_{22}$, the idempotency equation for a nonzero λ is $16\lambda^2 = 4\lambda$, whose unique solution is $\lambda = 1/4$. $\square$

Corollary 13.51 (Regular five-cell realization of the quarter spectrum). *Let n be a unit vector orthogonal to $\text{Time}(\mathcal{T})$, and define*

$$\eta_a := \frac{\sqrt{15}}{4}\mu_a - \frac{1}{4}n \quad (a \in \mathcal{T}), \quad \eta_* := n. \quad (857)$$

Then

$$\sum_{v \in \mathcal{T} \cup \{*\}} \eta_v = 0, \quad \langle \eta_v, \eta_w \rangle = -\frac{1}{4} \quad (v \neq w). \quad (858)$$

Thus these five vectors are the unit vertices of a regular four-simplex.

Let R_5 act as the identity on $\text{Time}(\mathcal{T})$ and send n to $-n$, and put $Q_5 := R_5/4$. Relative to the coefficient order (a, b_0, b_1, c) and the target basis (h_1, h_2, h_3, n) , define

$$\Lambda = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{1}{8} & 0 & 0 & \frac{1}{2} \\ \frac{1}{8} & 0 & 0 & -\frac{1}{2} \end{pmatrix}. \quad (859)$$

Then

$$\Lambda \mathcal{I}_- = Q_5 \Lambda, \quad \Lambda^T \Lambda = \text{diag}\left(\frac{1}{32}, 1, 1, \frac{1}{2}\right). \quad (860)$$

In particular, $\mathcal{I}_-$ is self-adjoint for the positive inner product with Gram matrix $\Lambda^\top \Lambda$, and its $(3+1)$ -spectrum is the regular-five-cell reflection spectrum $(1/4, 1/4, 1/4; -1/4)$.

Proof. The tetrahedral relations give

$$\|\eta_a\|^2 = \frac{15}{16} + \frac{1}{16} = 1, \quad \langle \eta_a, \eta_b \rangle = \frac{15}{16} \left(-\frac{1}{3}\right) + \frac{1}{16} = -\frac{1}{4}$$

for distinct $a, b \in \mathcal{T}$, while $\langle \eta_a, n \rangle = -1/4$. Their sum vanishes by $\sum_a \mu_a = 0$, proving (858).

The vectors

$$(0, 1, 0, 0), \quad (0, 0, 1, 0), \quad (4, 0, 0, 1), \quad (4, 0, 0, -1)$$

are an eigenbasis of $\mathcal{I}_-$, with the first three carrying eigenvalue $1/4$ and the fourth carrying eigenvalue $-1/4$. The map Λ sends this ordered eigenbasis to (h_1, h_2, h_3, n) . This proves the first identity in (860). Direct multiplication gives its second identity and hence

$$\mathcal{I}_-^\top (\Lambda^\top \Lambda) = (\Lambda^\top \Lambda) \mathcal{I}_-.$$

□

Corollary 13.52 (Exact pointed isometry from the Time altitude to the ordered blade carrier). *Fix $a \in \mathcal{T}$, retain the ordered blade*

$$N = 2E_{21},$$

and write

$$L_a := \mathbb{R}h_a = \mathbb{R}\mu_a, \quad L_N := \mathbb{R}w_N.$$

Orient L_a by h_a and L_N by

$$w_N = (4, 0, 0, -1), \quad H(w_N) = \mathcal{T}_\square(N) = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}. \quad (861)$$

Let

$$G_N := \Lambda^\top \Lambda = \text{diag}\left(\frac{1}{32}, 1, 1, \frac{1}{2}\right), \quad (862)$$

and equip the two lines with

$$g_{\mathcal{T},a}(x, y) := q_a(x)q_a(y), \quad g_N(v, w) := v^\top G_N w. \quad (863)$$

Then

$$\Phi_a : L_a \longrightarrow L_N, \quad \Phi_a(x) = q_a(x)w_N, \quad (864)$$

is the unique orientation-preserving linear isometry satisfying $\Phi_a(h_a) = w_N$. Its distinguished-point values are

$$\Phi_a(c_a^\wedge) = -\frac{1}{4}w_N = \mathcal{I}_- w_N, \quad \Phi_a(0\mathcal{T}) = 0, \quad \Phi_a(\mu_a) = \frac{3}{4}w_N. \quad (865)$$

Thus the selected Time altitude and the ordered E_{21} carrier have the same pointed metric coordinate: face centre, global centre, and selected vertex occur at $-1/4, 0, 3/4$, respectively. The selection of a fixes the Time orientation, while the order E_{21} fixes the blade orientation.

Proof. Equations (847) give

$$q_a(h_a) = 1, \quad c_a^\wedge = -\frac{1}{4}h_a, \quad \mu_a = \frac{3}{4}h_a.$$

The metric in (860) gives

$$w_N^\top G_N w_N = 16 \left(\frac{1}{32}\right) + \frac{1}{2} = 1.$$

For $r, s \in \mathbb{R}$, one therefore has

$$g_{\mathcal{T},a}(rh_a, sh_a) = rs = g_N(rw_N, sw_N).$$

This proves $\Phi_a^* g_N = g_{\mathcal{T},a}$. It also proves preservation of the stated orientations. The three point values follow from the displayed relations and $\mathcal{I}_- w_N = -w_N/4$. Equation (200) proves

$H(w_N) = \mathcal{T}_\square(N)$. An orientation-preserving isometry between one-dimensional metric spaces is determined by the image of its positive unit vector, proving uniqueness. $\square$

Corollary 13.53 (Ordered E_{21}/E_{12} sheets and reciprocal inversion). *Retain the selected Time vertex a , the directed carrier vector w_N , and the metric G_N from theorem 13.52. Let*

$$\mathcal{C} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad w_N^\vee := \mathcal{C}w_N = (-1, 0, 0, 4), \quad (866)$$

and define

$$G_N^\vee := \mathcal{C}^\top G_N \mathcal{C} = \text{diag}\left(\frac{1}{2}, 1, 1, \frac{1}{32}\right). \quad (867)$$

Then

$$\mathcal{C} : (\mathbb{R}w_N, G_N) \longrightarrow (\mathbb{R}w_N^\vee, G_N^\vee) \quad (868)$$

is an orientation-preserving isometry of the two oriented lines, where w_N and $w_N^\vee$ are their positive unit vectors. Its extension to the full four-dimensional coefficient carrier has determinant -1 .

The two Time maps

$$\Phi_{a,-}(x) := q_a(x)w_N, \quad \Phi_{a,+}(x) := q_a(x)w_N^\vee \quad (869)$$

satisfy

$$\Phi_{a,+} = \mathcal{C}\Phi_{a,-}. \quad (870)$$

For the two raw coefficient-inversion operators,

$$\mathcal{C}\mathcal{I}_-\mathcal{C} = \mathcal{I}_+^{-1} = \frac{1}{16}\mathcal{I}_+. \quad (871)$$

Consequently,

$$\Phi_{a,-}(c_a^\wedge) = \mathcal{I}_-w_N = -\frac{1}{4}w_N, \quad (872)$$

$$\Phi_{a,+}(c_a^\wedge) = \mathcal{I}_+^{-1}w_N^\vee = -\frac{1}{4}w_N^\vee, \quad (873)$$

$$\mathcal{I}_+w_N^\vee = -4w_N. \quad (874)$$

Thus the Time face point uses the raw inversion on the directed sheet and the inverse raw inversion on the transposed sheet.

After retaining only the reflection parts,

$$R_- := 4\mathcal{I}_-, \quad R_+ := \frac{1}{4}\mathcal{I}_+, \quad (875)$$

one has

$$R_+ = \mathcal{C}R_-\mathcal{C}, \quad R_-^2 = R_+^2 = I_4, \quad R_-w_N = -w_N, \quad R_+w_N^\vee = -w_N^\vee. \quad (876)$$

On the raw 2×2 blade carrier, put

$$S = E_{12} + E_{21}.$$

Then

$$S(2E_{21})S = 2E_{12}. \quad (877)$$

For the transported modular Hamiltonian of theorem 14.20, with $\ell = \log \lambda$, the ordered generators retain opposite eigenvalues:

$$\widetilde{K}_\lambda(2E_{21}) = -\ell(2E_{21}), \quad \widetilde{K}_\lambda(2E_{12}) = \ell(2E_{12}). \quad (878)$$

The reciprocal raw inversion scales and the opposite modular signs are therefore preserved by the sheet labels.

Proof. Direct multiplication gives

$$\mathcal{C}w_N = w_N^\vee, \quad \det \mathcal{C} = -1, \quad (w_N^\vee)^\top G_N^\vee w_N^\vee = 1.$$

For $r, s \in \mathbb{R}$,

$$(\mathcal{C}(rw_N))^\top G_N^\vee (\mathcal{C}(sw_N)) = rs = (rw_N)^\top G_N (sw_N),$$

which proves (868); the chosen positive vectors prove the line-orientation statement. Equations (869) and (866) give (870).

The explicit matrices (343) and (351) satisfy

$$\mathcal{I}_+^2 = 16I_4, \quad \mathcal{C}\mathcal{I}_-\mathcal{C} = \frac{1}{16}\mathcal{I}_+.$$

This proves (871). Applying it to $w_N^\vee = \mathcal{C}w_N$, together with (865), proves (872)–(874). Multiplying (871) by the displayed scalar factors gives (876).

The raw exchange identity is a matrix-unit product. The two modular eigenvalue equations are (1159) and (1158), transported through (1161). $\square$

Theorem 13.54 (Regular-simplex projection ladder). *Let $m \geq 2$, and let $v_0, \dots, v_m$ be unit vertices of a centred regular m -simplex:*

$$\sum_{i=0}^m v_i = 0, \quad \langle v_i, v_j \rangle = -\frac{1}{m} \quad (i \neq j). \quad (879)$$

Fix k , and put

$$c_k := \frac{1}{m} \sum_{i \neq k} v_i = -\frac{1}{m} v_k, \quad P_k := I - v_k \otimes v_k. \quad (880)$$

For every $i \neq k$,

$$P_k v_i = v_i + \frac{1}{m} v_k = v_i - c_k. \quad (881)$$

Hence the opposite facet and its orthogonal projection differ by the single hidden displacement c_k , and the projected vertices form a centred regular $(m-1)$ -simplex. More precisely,

$$\hat{v}_i^{(k)} := \frac{m}{\sqrt{m^2 - 1}} P_k v_i \quad (i \neq k) \quad (882)$$

satisfy

$$\sum_{i \neq k} \hat{v}_i^{(k)} = 0, \quad \langle \hat{v}_i^{(k)}, \hat{v}_j^{(k)} \rangle = -\frac{1}{m-1} \quad (i \neq j; i, j \neq k). \quad (883)$$

If $K_m := \text{conv}\{v_0, \dots, v_m\}$, then

$$P_k^{-1}(0) \cap K_m = [c_k, v_k]. \quad (884)$$

The full-altitude coordinate

$$q_k^{(m)}(x) := \frac{m}{m+1} \langle x, v_k \rangle \quad (885)$$

has the values

$$q_k^{(m)}(c_k) = -\frac{1}{m+1}, \quad q_k^{(m)}(0) = 0, \quad q_k^{(m)}(v_k) = \frac{m}{m+1}. \quad (886)$$

For a specified opposite facet, the line $\mathbb{R}v_k$ is the unique orthogonal-projection direction that sends its centroid to the global centre. Therefore a regular m -simplex has exactly $m+1$ such vertex-axis projection lines. In particular, a regular four-simplex has five symmetry-equivalent projections onto centred regular tetrahedra.

Proof. The sum-zero identity gives the formula for c_k . Since $\langle v_i, v_k \rangle = -1/m$, orthogonal projection gives (881). For $i, j \neq k$, direct calculation yields

$$\|P_k v_i\|^2 = 1 - \frac{1}{m^2}, \quad \langle P_k v_i, P_k v_j \rangle = -\frac{1}{m} - \frac{1}{m^2}.$$

Their ratio is $-1/(m-1)$, proving (883).

The barycentric argument in the proof of theorem 13.50 applies with m opposite-face vertices. The unique relation among their projected vertices is their sum-zero relation, so the zero fibre in K_m is the segment $[c_k, v_k]$. The altitude vector is

$$v_k - c_k = \frac{m+1}{m} v_k.$$

Equation (886) follows immediately.

Suppose orthogonal projection along a line $\mathbb{R}r$ sends the centroid of the specified opposite facet to the global centre. Then $c_k \in \mathbb{R}r$. Since $c_k = -v_k/m$, the projection line is $\mathbb{R}v_k$. Each of the $m+1$ facets has a distinct opposite vertex line, completing the proof. $\square$

Corollary 13.55 (The quarter shared by the three- and four-dimensional steps). *For the tetrahedron-to-Colour projection ($m=3$), the hidden face centre has full-altitude coordinate*

$$q_k^{(3)}(c_k) = -\frac{1}{4}. \quad (887)$$

For the four-simplex-to-tetrahedron projection ($m=4$), every vertex of the selected tetrahedral facet has the same hidden displacement

$$v_i - P_k v_i = c_k = -\frac{1}{4} v_k \quad (i \neq k), \quad (888)$$

while the facet centre has full-altitude coordinate $-1/5$. Thus the same quarter occurs in the two adjacent layers: first as the relative centre-to-face coordinate of the tetrahedron, then as the common hidden coordinate of the tetrahedral facet inside the regular four-simplex.

Proof. Set $m=3$ and $m=4$, respectively, in (881) and (886). $\square$

Theorem 13.56 (Reciprocal invariant of adjacent simplex layers). *Let $m \geq 3$. In a centred regular $(m-1)$ -simplex, select a vertex u_k and let d_k be the centroid of its opposite face. In a centred regular m -simplex, select a vertex v_ℓ , let P_ℓ be orthogonal projection onto $v_\ell^\perp$, and take any $i \neq \ell$. Then*

$$q_k^{(m-1)}(d_k) = -\frac{1}{m}, \quad v_i - P_\ell v_i = -\frac{1}{m} v_\ell, \quad (889)$$

where $q_k^{(m-1)}$ measures the lower face-to-vertex altitude with total length one. Thus the scalar $-1/m$ occurs successively as

- (i) *the lower simplex's centre-to-opposite-face altitude coordinate; and*
- (ii) *the upper simplex facet's common hidden displacement along the omitted vertex axis.*

Both quantities are ratios of collinear vectors and are preserved by every isometry of the corresponding regular simplex. No identification of the two ambient Euclidean spaces is required.

Proof. Apply theorem 13.54 first in dimension $m-1$. Its full-altitude formula gives

$$q_k^{(m-1)}(d_k) = -\frac{1}{(m-1)+1} = -\frac{1}{m}.$$

Apply the same theorem in dimension m . Its projection formula gives

$$P_\ell v_i = v_i + \frac{1}{m} v_\ell \quad (i \neq \ell),$$

and hence the second identity in (889). Isometries preserve centroids, orthogonal projections, collinearity, and vector ratios, proving the invariance statement. $\square$

Theorem 13.57 (Colour-weighted reconstruction of the Fable Jacobian). *Let $\gamma \in \mathbb{C}$. In the marked linear-quadratic factor space of theorem 7.4, consider the chart*

$$a = x, \quad b = 1 + xy, \quad c = 1 + \gamma(3xy + x^2z). \quad (890)$$

The coefficient 3 is the support reading of the three projected Colour states in the proposed interpretation. Algebraically, the displayed chart is an input: its coefficient 3 matches the cardinality of the Colour orbit, and the common scalar γ weights both the three-state term and the transverse term. The Colour carrier alone has not been shown to select this chart. Impose the two exact equations

$$ad + bc = 1, \quad a^2e - abd + b^2c = 1. \quad (891)$$

On $x \neq 0$, these equations have the unique solution

$$d_\gamma = -(3\gamma + 1)y - 3\gamma xy^2 - \gamma xz - \gamma x^2yz, \quad (892)$$

$$e_\gamma = -\frac{(6\gamma + 3)y}{x} - (12\gamma + 2)y^2 - 6\gamma xy^3 - 2\gamma z - 4\gamma xyz - 2\gamma x^2y^2z. \quad (893)$$

The pair (d_γ, e_γ) extends polynomially through $x = 0$ for all y, z precisely when

$$\gamma = -\frac{1}{2}. \quad (894)$$

The forced scalar equals the off-diagonal coefficient in the Colour Gram matrix (844). At this value,

$$d = \frac{x^2yz + 3xy^2 + xz + y}{2}, \quad (895)$$

$$e = x^2y^2z + 3xy^3 + 2xyz + 4y^2 + z. \quad (896)$$

The three visible coefficients of the product LQ , in the ordered target coordinates used by F , are

$$\tilde{F} := (be, 2(ae + bd), 2ac). \quad (897)$$

They satisfy the polynomial identity

$$\tilde{F} = F \quad (898)$$

and

$$\det J\tilde{F} = 4\gamma = -2. \quad (899)$$

Thus, within the displayed chart class, polynomial extension through the hidden factor $a = 0$ recovers the full constant-Jacobian map from the marked-factor incidence equations. The equality with the Colour Gram coefficient is an exact scalar comparison; a derivation of the chart from the Colour carrier is a separate problem.

Proof. The first equation in (891) gives

$$d_\gamma = \frac{1 - bc}{a}.$$

Substitution of (890) and collection in x, y, z gives (892). The resultant equation then gives

$$e_\gamma = \frac{1 + abd - b^2c}{a^2},$$

which expands to (893). Its only possible pole along $x = 0$ has coefficient

$$-(6\gamma + 3)y.$$

Polynomial extension for every y is therefore equivalent to $6\gamma + 3 = 0$, proving (894) and its uniqueness. Substitution gives (895) and (896).

Multiplication gives

$$LQ = acU^3 + (ad + bc)U^2V + (ae + bd)UV^2 + beV^3.$$

Insert (890), (895), and (896). The three unfixed coefficients become

$$\begin{aligned} be &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\ 2(ae + bd) &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\ 2ac &= 2x - 3x^2 y - x^3 z. \end{aligned}$$

These are the three components of (13), proving (898). Before substituting the forced value of γ , direct differentiation on $x \neq 0$ gives

$$\det \frac{\partial(be, 2(ae + bd), 2ac)}{\partial(x, y, z)} = 4\gamma.$$

At $\gamma = -1/2$, the map is polynomial on all of $\mathbb{C}^3$, and the identity extends to every point. This proves (899). $\square$

Remark 13.58 (Scope of the chart reconstruction). The uniqueness in theorem 13.57 belongs to the displayed chart class $c = 1 + \gamma(3xy + x^2 z)$, whose coefficient pattern is supplied by one balanced three-state contribution and one transverse contribution. A classification of every polynomial chart on the resultant-one incidence space is a separate problem.

Proposition 13.59 (Centered zeta endpoints and their scalar Fable values). *Put*

$$\rho(s) := s - \frac{1}{2} \tag{900}$$

and

$$\xi_R(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s). \tag{901}$$

Then

$$\rho(1-s) = -\rho(s), \quad \rho(0) = -\frac{1}{2}, \quad \rho(1) = \frac{1}{2}, \quad \rho(0)\rho(1) = -\frac{1}{4}. \tag{902}$$

The analytically continued zeta function and the completed function satisfy

$$\zeta(0) = -\frac{1}{2}, \quad \zeta(-1) = -\frac{1}{12}, \quad \xi_R(0) = \xi_R(1) = \frac{1}{2}. \tag{903}$$

For the marked-factor chart of theorem 13.57, the Laurent coefficient along the hidden divisor is

$$[x^{-1}]e_\gamma = -3(2\gamma + 1)y. \tag{904}$$

Consequently its unique polynomial value is

$$\gamma_* = \zeta(0) = \rho(0) = -\frac{1}{2}. \tag{905}$$

At this value,

$$\det JF = 4\zeta(0) = -2, \quad F_1|_{\text{complete fibre}} = \rho(0)\rho(1) = -\frac{1}{4}. \tag{906}$$

Equations (905) and (906) compare equal scalar values in three separately defined constructions. Their projective and scheme-level morphisms are constructed in theorem 13.60.

Proof. The first four identities follow directly from $\rho(s) = s - \frac{1}{2}$. The zeta functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s)$$

gives, as $s \rightarrow 0$,

$$\zeta(s) = (1 + O(s)) \pi^{-1} \left(\frac{\pi s}{2} + O(s^3) \right) (1 + O(s)) \left(-\frac{1}{s} + O(1) \right),$$

and hence $\zeta(0) = -1/2$. Substitution of $s = -1$, together with $\zeta(2) = \pi^2/6$, gives

$$\zeta(-1) = 2^{-1}\pi^{-2}\sin\left(-\frac{\pi}{2}\right)\Gamma(2)\zeta(2) = -\frac{1}{12}.$$

At $s = 0$, the expansions

$$\Gamma\left(\frac{s}{2}\right) = \frac{2}{s} + O(1), \quad s(s-1) = -s + O(s^2), \quad \zeta(s) = -\frac{1}{2} + O(s)$$

give $\xi_R(0) = 1/2$. The functional equation $\xi_R(s) = \xi_R(1-s)$ then gives $\xi_R(1) = 1/2$.

Equation (893) has exactly one negative power of x , whose coefficient is the right-hand side of (904). It vanishes for every y precisely when $2\gamma + 1 = 0$, proving (905). Equations (899) and (853), together with (902), prove (906). $\square$

Theorem 13.60 (Centered zeta–Fable projective isomorphism). *Let $\mathbb{P}_{U,V}^1$ and $\mathbb{P}_{S,T}^1$ be projective lines over $\mathbb{C}$, and put*

$$C_F(U, V) := V \left(U - \frac{1}{2}V \right) \left(U + \frac{1}{2}V \right), \quad E_\zeta(S, T) := TS(S - T). \quad (907)$$

The projective automorphism

$$\varphi : \mathbb{P}_{U,V}^1 \longrightarrow \mathbb{P}_{S,T}^1, \quad [U : V] \longmapsto \left[U + \frac{1}{2}V : V \right] \quad (908)$$

has inverse

$$[S : T] \longmapsto \left[S - \frac{1}{2}T : T \right] \quad (909)$$

and satisfies

$$\varphi^* E_\zeta = C_F. \quad (910)$$

Consequently,

$$\operatorname{div}(C_F) = \varphi^{-1}([0] + [1] + [\infty]), \quad (911)$$

with the three marked points carried by

$$\left[-\frac{1}{2} : 1 \right] \longmapsto [0 : 1], \quad \left[\frac{1}{2} : 1 \right] \longmapsto [1 : 1], \quad [1 : 0] \longmapsto [1 : 0]. \quad (912)$$

Thus the common Fable binary cubic is the pullback of the projective completed-endpoint divisor, including the point at infinity.

Let

$$\mathcal{Z}_0 := \operatorname{Spec} \frac{\mathbb{C}[q]}{(2q+1)}, \quad \mathcal{E}_{\text{poly}} := \operatorname{Spec} \frac{\mathbb{C}[\gamma]}{(2\gamma+1)}. \quad (913)$$

The coordinate-ring map

$$\psi^\# : \frac{\mathbb{C}[\gamma]}{(2\gamma+1)} \longrightarrow \frac{\mathbb{C}[q]}{(2q+1)}, \quad \gamma \longmapsto q, \quad (914)$$

is an isomorphism. It identifies the endpoint value $q = \zeta(0) = -1/2$ with the unique polynomial-extension value $\gamma_ = -1/2$. On this scalar locus,*

$$4q = q^{-1} = -2. \quad (915)$$

Therefore

$$\det JF = 4\zeta(0) = \zeta(0)^{-1} = -2. \quad (916)$$

Proof. The displayed inverse proves that φ is a projective automorphism. Substitution gives

$$\varphi^* E_\zeta = V \left(U + \frac{1}{2}V \right) \left(U - \frac{1}{2}V \right) = C_F(U, V),$$

which proves (910)–(912). On the affine chart $V \neq 0$, write $t = U/V$ and $s = t + 1/2$. Then

$$\frac{C_F(U, V)}{V^3} = t^2 - \frac{1}{4} = s(s - 1),$$

so this automorphism is exactly the centered coordinate $\rho(s) = s - 1/2$ in projective form.

Both coordinate rings in (913) are canonically isomorphic to $\mathbb{C}$, and $\psi^\#$ respects their defining relations. Its inverse sends q to γ . Hence $\psi^\#$ is an isomorphism. In its target ring, $q = -1/2$, so $4q = -2 = q^{-1}$. The Jacobian formula $\det JF = 4\gamma$ from (899), transported by $\gamma = q = \zeta(0)$, proves (916). $\square$

The construction order at this point is fixed by the split-zero coordinates of theorem 11.1:

$$\tau \longmapsto (0_R, 0), \quad e = 0_R \longmapsto (0_R, 1).$$

Entrywise, the directed blade lift is the matrix

$$\tilde{N} = \begin{pmatrix} \tau & \tau \\ \tilde{2} & \tau \end{pmatrix}, \quad \hat{\chi}(\tilde{N}) = E_{21},$$

as proved in theorem 11.7. These are the primary zero and support coordinates. The affine equalizer below begins after passage to the globalized two-sheet zeta deformation and records two scalar images of that prior support geometry.

Theorem 13.61 (Globalization equalizer and the two Fable scalar fibres). *Put $q := \zeta(0) = -1/2$. On the affine parameter line $\mathbb{A}_u^1$, define*

$$Z_q(u) := (1 + u)q - u, \quad D(u) := \delta(1 + u) = -u(1 + u), \quad (917)$$

where $Z_q(u) = Z_{1,u}(0)$ is the origin value of the globalized two-sheet zeta deformation and $\delta(\lambda) = \lambda - \lambda^2$ is its Stone idempotence defect. Their difference factors as

$$Z_q(u) - D(u) = (u - 1) \left(u + \frac{1}{2} \right). \quad (918)$$

Consequently their equalizer is the reduced two-point scheme

$$\mathcal{J} := \text{Eq}(Z_q, D) = \text{Spec} \frac{\mathbb{C}[u]}{((u - 1)(2u + 1))}. \quad (919)$$

Let

$$\mathcal{W} := \text{Spec} \frac{\mathbb{C}[w]}{((4w - 1)(w + 2))}. \quad (920)$$

The common-value morphism

$$\vartheta : \mathcal{J} \longrightarrow \mathcal{W}, \quad \vartheta^\#([w]) = [-u(1 + u)] \quad (921)$$

is an isomorphism. Its inverse on coordinate rings sends

$$[u] \longmapsto -\frac{2[w] + 1}{3}. \quad (922)$$

On the two components it is

$$u = -\frac{1}{2} \longmapsto w = \frac{1}{4}, \quad u = 1 \longmapsto w = -2. \quad (923)$$

The first component is $u = q$ and has common value $q^2 = 1/4$. The centered endpoint reflection $\rho(1 - s) = -\rho(s)$ gives the oriented product

$$\rho(0)\rho(1) = q(-q) = -\frac{1}{4}. \quad (924)$$

The second component has common value

$$-2 = 4q = q^{-1} = \det JF. \quad (925)$$

Thus the Fable fibre value $-1/4$ and the Fable Jacobian scalar -2 arise from the two components of one explicit globalization equalizer, with the quarter sign supplied by centered reflection. Composing the second common-value point with the analytic zeta map gives $\zeta(-2) = 0$.

Proof. Substitution of $q = -1/2$ into (917) gives

$$Z_q(u) = -\frac{1}{2} - \frac{3}{2}u, \quad D(u) = -u - u^2.$$

Their difference is

$$u^2 - \frac{1}{2}u - \frac{1}{2} = (u - 1) \left(u + \frac{1}{2} \right),$$

which proves (918) and (919). The two factors are coprime over $\mathbb{C}$, so the equalizer is reduced.

In the coordinate ring of $\mathcal{S}$, the class $w = -u(1 + u)$ satisfies $(4w - 1)(w + 2) = 0$. Conversely, in the coordinate ring of $\mathcal{W}$, the class $u = -(2w + 1)/3$ satisfies $(u - 1)(2u + 1) = 0$. The two substitutions compose to the identity because

$$-\frac{2(-u(1 + u)) + 1}{3} - u = \frac{(2u + 1)(u - 1)}{3}$$

and

$$-\left(-\frac{2w + 1}{3}\right) \left(1 - \frac{2w + 1}{3}\right) - w = -\frac{(4w - 1)(w + 2)}{9}.$$

This proves (921)–(923). The remaining identities follow from $u = q = -1/2$, the centered reflection formula, and (916). $\square$

Theorem 13.62 (Full split-pair lift of the globalization equalizer). *For $a \in \mathbb{C}$, let $\tilde{a}$ denote the supported copy of a in $G(\mathbb{C})$, and put*

$$\mathbf{e}_{21}(a) := \begin{pmatrix} \tau & \tau \\ \tilde{a} & \tau \end{pmatrix} \in M_2(G(\mathbb{C})). \quad (926)$$

The full split-pair map of theorem 11.2 gives

$$\Phi_2(\mathbf{e}_{21}(a)) = (aE_{21}, E_{21}). \quad (927)$$

Hence

$$\mathcal{L}_{21}^{\text{spl}} := \{(aE_{21}, E_{21}) : a \in \mathbb{C}\} \subset \mathcal{D}_{\mathbb{C},2} \quad (928)$$

is an affine line, with isomorphism

$$\iota_{21} : \mathbb{A}_a^1 \xrightarrow{\sim} \mathcal{L}_{21}^{\text{spl}}, \quad a \mapsto (aE_{21}, E_{21}). \quad (929)$$

Let

$$\mathfrak{R} : \mathbb{A}_w^1 \longrightarrow \mathbb{A}_a^1, \quad w \mapsto -w,$$

and let $\mathfrak{r} = \mathfrak{R}|_{\mathcal{W}}$. The ambient affine morphism

$$\mathfrak{B}_{21} := \iota_{21} \circ \mathfrak{R} : \mathbb{A}_w^1 \longrightarrow \mathcal{L}_{21}^{\text{spl}}, \quad w \mapsto (-wE_{21}, E_{21}) \quad (930)$$

has restriction

$$\mathfrak{b}_{21} := \mathfrak{B}_{21}|_{\mathcal{W}} = \iota_{21} \circ \mathfrak{r} : \mathcal{W} \longrightarrow \mathcal{L}_{21}^{\text{spl}}, \quad w \mapsto (-wE_{21}, E_{21}) \quad (931)$$

lifts the common-value coordinate to the full directed split pair. Its pullback through the equalizer is

$$\kappa_{21} : \mathcal{J} \xrightarrow{\vartheta} \mathcal{W} \xrightarrow{\mathbf{b}_{21}} \mathcal{L}_{21}^{\text{spl}} \quad (932)$$

has the coordinate formula

$$\kappa_{21}(u) = (u(1+u)E_{21}, E_{21}). \quad (933)$$

On the two components of the equalizer,

$$u = -\frac{1}{2} \mapsto \left(-\frac{1}{4}E_{21}, E_{21}\right), \quad (934)$$

$$u = 1 \mapsto (2E_{21}, E_{21}) = \Phi_2(\tilde{N}). \quad (935)$$

The first-coordinate Fable target line has the explicit isomorphism

$$j_{F,21} : \{(X, 0, 0) : X \in \mathbb{C}\} \xrightarrow{\sim} \mathcal{L}_{21}^{\text{spl}}, \quad (X, 0, 0) \mapsto (XE_{21}, E_{21}). \quad (936)$$

Consequently,

$$\kappa_{21}\left(-\frac{1}{2}\right) = j_{F,21}(F(p_0)) = j_{F,21}(F(p_+)) = j_{F,21}(F(p_-)). \quad (937)$$

The second pair is the complete directed-blade datum of (502).

The involution $a = -w$ is the unique affine map $\mathbb{A}_w^1 \rightarrow \mathbb{A}_a^1$ sending

$$\frac{1}{4} \mapsto -\frac{1}{4}, \quad -2 \mapsto 2.$$

Moreover,

$$\Phi_2(\mathbf{e}_{21}(0)) = (0_2, E_{21}) \neq (0_2, 0_2). \quad (938)$$

Thus the lift retains the complete matrix support coordinate at zero amplitude.

Proof. Equation (927) follows directly from (499): the $(2, 1)$ -entry is supported for every a , including $a = 0$, and the other three entries are absent. Extraction of the $(2, 1)$ -amplitude is the inverse of (929).

On $\mathcal{J}$, the common-value coordinate is $w = -u(1+u)$ by (921). Applying $\mathbf{r}$ gives $a = u(1+u)$, which proves (933). Evaluation at $u = -1/2$ and $u = 1$ proves (934) and (935).

If $a = \alpha w + \beta$ has the two displayed values, then

$$\frac{\alpha}{4} + \beta = -\frac{1}{4}, \quad -2\alpha + \beta = 2.$$

Their difference gives $\alpha = -1$, and then $\beta = 0$. This proves the asserted uniqueness. Finally, substituting $a = 0$ into (927) proves (938). $\square$

Theorem 13.63 (Endpoint-divisor realization of the two Fable scalars). *Let $X = \mathbf{P}_{\mathbb{C}}^1$ have affine coordinate s and marked point ∞ . Put*

$$D = [0] + [1], \quad f(s) = s(s-1), \quad \rho(s) = s - \frac{1}{2}, \quad q = \zeta(0) = -\frac{1}{2}.$$

Then the centered endpoint product and the projective logarithmic residue are

$$\rho(0)\rho(1) = q(-q) = -\frac{1}{4}, \quad \text{Res}_{\infty}(d \log f) = -2. \quad (939)$$

More generally, the marked-infinity map

$$\mathbf{r}_{\infty} : \text{Div}^+(\mathbb{A}_{\mathbb{C}}^1) \longrightarrow \mathbb{Z}, \quad \sum_a n_a[a] \mapsto -\sum_a n_a \quad (940)$$

is a homomorphism of additive commutative monoids, and

$$\mathbf{r}_{\infty}\left(\sum_a n_a[a]\right) = \text{Res}_{\infty} d \log \left(\prod_a (s-a)^{n_a}\right).$$

For the divisor D , its value agrees with the second component of the globalization equalizer and with the Fable Jacobian scalar:

$$\mathfrak{r}_\infty(D) = \vartheta_{\mathbb{C}}(u = 1) = q^{-1} = 4q = \det JF = -2. \quad (941)$$

Let $\iota_{\mathbb{C}} : \mathbb{C} \rightarrow G(\mathbb{C})$ be the supported-amplitude inclusion, let $e = 0_{\mathbb{C}} \in G(\mathbb{C})$, and let $\chi : G(\mathbb{C}) \rightarrow \mathbf{B}$ be the support character. The analytic continuation of the second endpoint invariant is the typed composite

$$D \xrightarrow{\mathfrak{r}_\infty} -2 \xrightarrow{\zeta} 0_{\mathbb{C}} \xrightarrow{\iota_{\mathbb{C}}} e \xrightarrow{\chi} 1. \quad (942)$$

In particular, this composite ends at the supported zero e , rather than at the absent element $\tau = 0_{G(\mathbb{C})}$.

Proof. The centered coordinates of the two finite marked points are

$$\rho(0) = -\frac{1}{2} = q, \quad \rho(1) = \frac{1}{2} = -q,$$

which proves the first identity in (939). Moreover,

$$d \log f = \frac{ds}{s} + \frac{ds}{s-1}.$$

Its residues at 0 and 1 are both 1. The residue theorem on $\mathbf{P}_{\mathbb{C}}^1$ therefore gives residue -2 at ∞ . For an arbitrary effective affine divisor, the same calculation gives the sum of its multiplicities at the finite points and its negative at ∞ , proving (940).

The value $\vartheta_{\mathbb{C}}(1) = -2$ is (923); the remaining equalities in (941) are (915) and (916). Finally, $\zeta(-2) = 0_{\mathbb{C}}$, the inclusion $\iota_{\mathbb{C}}$ sends the ring zero to the supported zero e , and $\chi(e) = 1$. This proves (942). $\square$

Theorem 13.64 (Endpoint-equalizer isomorphism and transported reflection). *Let*

$$\mathcal{E} := \operatorname{Spec} \frac{\mathbb{C}[s]}{(s(s-1))}$$

be the reduced scheme of the functional-equation endpoints. The morphism

$$\alpha : \mathcal{E} \longrightarrow \mathcal{J}, \quad \alpha^\#([u]) = \left[\frac{3s-1}{2} \right], \quad (943)$$

is an isomorphism, with inverse on coordinate rings

$$[s] \longmapsto \left[\frac{2u+1}{3} \right]. \quad (944)$$

The composite

$$\omega := \vartheta \circ \alpha : \mathcal{E} \xrightarrow{\sim} \mathcal{W}$$

is given on coordinate rings by

$$\omega^\#([w]) = \left[\frac{1-9s^2}{4} \right] = \left[\frac{1-9s}{4} \right]. \quad (945)$$

In particular,

$$s = 0 \longmapsto u = -\frac{1}{2} \longmapsto w = \frac{1}{4}, \quad s = 1 \longmapsto u = 1 \longmapsto w = -2. \quad (946)$$

Let $\sigma_{\mathcal{E}}$ be the endpoint reflection $\sigma_{\mathcal{E}}^\#([s]) = [1-s]$. Its transports through α and ω are

$$\sigma_{\mathcal{J}}^\#([u]) = \left[\frac{1}{2} - u \right], \quad \sigma_{\mathcal{W}}^\#([w]) = \left[-w - \frac{7}{4} \right]. \quad (947)$$

Consequently the three involutions exchange

$$0 \longleftrightarrow 1, \quad -\frac{1}{2} \longleftrightarrow 1, \quad \frac{1}{4} \longleftrightarrow -2,$$

respectively. For the centered coordinate $\rho = s - \frac{1}{2}$,

$$\rho \circ \sigma_{\mathcal{E}} = -\rho, \quad \rho(0)\rho(1) = -\omega(0) = -\frac{1}{4}. \quad (948)$$

Proof. Substituting $u = (3s - 1)/2$ gives

$$(u - 1)(2u + 1) = \frac{9}{2}s(s - 1),$$

so (943) defines a morphism. Conversely, substituting $s = (2u + 1)/3$ gives

$$s(s - 1) = \frac{2}{9}(2u + 1)(u - 1),$$

so (944) defines its inverse. The two substitutions compose identically on the coordinate generators.

The common-value coordinate pulls back as

$$-u(1 + u) = -\frac{3s - 1}{2} \frac{3s + 1}{2} = \frac{1 - 9s^2}{4}.$$

The relation $s^2 = s$ in the coordinate ring of $\mathcal{E}$ gives the second representative in (945) and the component chain (946).

Under $s \mapsto 1 - s$,

$$\frac{3s - 1}{2} \mapsto \frac{3(1 - s) - 1}{2} = \frac{1}{2} - \frac{3s - 1}{2}.$$

Using $w = (1 - 9s)/4$, the same reflection gives

$$w \mapsto \frac{1 - 9(1 - s)}{4} = -w - \frac{7}{4}.$$

This proves (947). Finally, $\rho(1 - s) = -\rho(s)$, while $\rho(0) = -1/2$, $\rho(1) = 1/2$, and $\omega(0) = 1/4$, proving (948). $\square$

Theorem 13.65 (Endpoint algebra norm transported from the globalization equalizer). *Let*

$$A_{\mathcal{E}} = \frac{\mathbb{C}[s]}{(s(s - 1))}, \quad \rho = s - \frac{1}{2} \in A_{\mathcal{E}}.$$

Under the scheme isomorphism $\alpha : \mathcal{E} \xrightarrow{\sim} \mathcal{J}$ of theorem 13.64, the centered endpoint section is the class

$$\rho = \frac{4u - 1}{6} \quad \text{in} \quad \frac{\mathbb{C}[u]}{((u - 1)(2u + 1))}. \quad (949)$$

Then

$$\rho^2 = \frac{1}{4}, \quad \text{Tr}_{A_{\mathcal{E}}/\mathbb{C}}(\rho) = 0, \quad N_{A_{\mathcal{E}}/\mathbb{C}}(\rho) = -\frac{1}{4}. \quad (950)$$

The binary cubic obtained from this intrinsic norm is exactly the common Fable cubic:

$$C_{\rho}(U, V) := U^2V + N_{A_{\mathcal{E}}/\mathbb{C}}(\rho)V^3 = U^2V - \frac{1}{4}V^3 = C_F(U, V). \quad (951)$$

It satisfies

$$\text{Disc}(C_{\rho}) = -4N_{A_{\mathcal{E}}/\mathbb{C}}(\rho) = 1. \quad (952)$$

The oriented endpoint invariant and the Fable Jacobian scalar are recovered from the same section by

$$\rho(0)\rho(1) = N_{A_{\mathcal{E}}/\mathbb{C}}(\rho) = -\frac{1}{4}, \quad \det JF = 4\rho(0) = 8N_{A_{\mathcal{E}}/\mathbb{C}}(\rho) = -2. \quad (953)$$

Proof. The inverse of $\alpha^\#([u]) = [(3s - 1)/2]$ sends s to $(2u + 1)/3$. Therefore

$$s - \frac{1}{2} = \frac{2u + 1}{3} - \frac{1}{2} = \frac{4u - 1}{6},$$

which proves (949). Its values on the two equalizer components $u = -1/2$ and $u = 1$ are $-1/2$ and $1/2$, respectively.

The defining relation $s^2 = s$ gives

$$\rho^2 = s^2 - s + \frac{1}{4} = \frac{1}{4}.$$

In the basis $(1, s)$ of the rank-two algebra $A_{\mathcal{E}}$, multiplication by ρ has matrix

$$m_\rho = \begin{pmatrix} -\frac{1}{2} & 0 \\ 1 & \frac{1}{2} \end{pmatrix}. \quad (954)$$

Its trace is zero and its determinant is $-1/4$, proving (950). Equivalently, under $A_{\mathcal{E}} \cong \mathbb{C}|_{s=0} \times \mathbb{C}|_{s=1}$, the element ρ has values $(-1/2, 1/2)$, whose product is $-1/4$.

Equation (951) is (59). Thus the Fable cubic is obtained after transporting the centered endpoint section through the globalization equalizer and taking its rank-two algebra norm. The binary-cubic invariant

$$\text{Disc}(U^2V + \lambda V^3) = -4\lambda$$

was proved in theorem 9.49; applying it to $\lambda = N(\rho)$ proves (952). Finally, $\rho(0) = -1/2$, and theorem 13.59 gives $\det JF = 4\rho(0)$. The equality $4\rho(0) = 8N(\rho)$ follows from the two values in (950). $\square$

Theorem 13.66 (Restricted zeta projector and the trivial-zero first jet). *In the coordinate ring*

$$A_{\mathcal{W}} = \frac{\mathbb{C}[w]}{((4w - 1)(w + 2))},$$

put

$$e_{1/4} = \frac{4(w + 2)}{9}, \quad e_{-2} = \frac{1 - 4w}{9}. \quad (955)$$

These are orthogonal idempotents with sum 1, taking the respective values

$$e_{1/4}(\tfrac{1}{4}) = 1, \quad e_{1/4}(-2) = 0, \quad e_{-2}(\tfrac{1}{4}) = 0, \quad e_{-2}(-2) = 1.$$

Let $c = \zeta(\frac{1}{4})$. Then $c < 0$, and the regular function on $\mathcal{W}$ whose values agree with the analytic zeta function is

$$Z_{\mathcal{W}} = c e_{1/4} = \frac{4c}{9}(w + 2). \quad (956)$$

Its pullbacks through the two scheme isomorphisms are

$$\vartheta^\#(Z_{\mathcal{W}}) = \frac{2c}{3}(1 - u), \quad \omega^\#(Z_{\mathcal{W}}) = c(1 - s). \quad (957)$$

Consequently,

$$V_{\mathcal{E}}(\omega^\# Z_{\mathcal{W}}) = \{s = 1\} \xrightarrow{\alpha} \{u = 1\} \xrightarrow{\vartheta} \{w = -2\} = V_{\mathcal{W}}(Z_{\mathcal{W}}). \quad (958)$$

Composing with analytic evaluation and split-zero support gives

$$\{s = 1\} \longrightarrow \{u = 1\} \longrightarrow \{w = -2\} \xrightarrow{\zeta} \{0_{\mathbb{C}}\} \xrightarrow{\iota_{\mathbb{C}}} \{e\} \xrightarrow{\chi} \{1\}. \quad (959)$$

The corresponding full split-pair morphism is

$$\mathcal{Z}_{21}^{\text{spl}} : \mathcal{W} \longrightarrow \mathcal{L}_{21}^{\text{spl}}, \quad w \longmapsto (Z_{\mathcal{W}}(w)E_{21}, E_{21}). \quad (960)$$

Its component values are

$$\mathcal{Z}_{21}^{\text{spl}}\left(\frac{1}{4}\right) = (cE_{21}, E_{21}), \quad \mathcal{Z}_{21}^{\text{spl}}(-2) = (0_2, E_{21}) = \Phi_2(\mathbf{e}_{21}(0)). \quad (961)$$

Let

$$\mathcal{N}_{-2} = \operatorname{Spec} \frac{\mathbb{C}[\varepsilon]}{(\varepsilon^2)}, \quad w = -2 + \varepsilon, \quad \mathcal{N}_0 = \operatorname{Spec} \frac{\mathbb{C}[t]}{(t^2)}.$$

The first jet of zeta induces an isomorphism

$$j_\zeta : \mathcal{N}_{-2} \xrightarrow{\sim} \mathcal{N}_0, \quad j_\zeta^\#([t]) = \zeta'(-2)[\varepsilon] = -\frac{\zeta(3)}{4\pi^2}[\varepsilon]. \quad (962)$$

On the fixed-support stratum, this first jet is

$$\mathcal{Z}_{21}^{\text{spl}}(-2 + \varepsilon) = \left(-\frac{\zeta(3)}{4\pi^2} \varepsilon E_{21}, E_{21} \right) \pmod{\varepsilon^2}. \quad (963)$$

For the endpoint divisor $D = [0] + [1]$ of theorem 13.63, the two maps out of its residue coordinate have the exact values

$$\mathcal{Z}_{21}^{\text{spl}}(\mathfrak{r}_\infty(D)) = (0_2, E_{21}), \quad (964)$$

$$\mathfrak{b}_{21}(\mathfrak{r}_\infty(D)) = (2E_{21}, E_{21}) = \Phi_2(\tilde{N}). \quad (965)$$

Proof. The Chinese-remainder decomposition

$$A_{\mathcal{W}} \cong \mathbb{C}|_{w=1/4} \times \mathbb{C}|_{w=-2}$$

sends $e_{1/4}$ and e_{-2} to $(1, 0)$ and $(0, 1)$, respectively. This proves their idempotent relations.

For $0 < s < 1$, the Dirichlet eta series satisfies

$$\eta(s) = \sum_{k \geq 1} ((2k-1)^{-s} - (2k)^{-s}) > 0,$$

while $\eta(s) = (1 - 2^{1-s})\zeta(s)$ and $1 - 2^{1-s} < 0$. Hence $c = \zeta(1/4) < 0$. A regular function on the reduced two-point scheme $\mathcal{W}$ is determined by its two values. Since $\zeta(-2) = 0$, its restriction is

$$c e_{1/4} + 0 e_{-2},$$

which is (956). Composing this scalar function with the fixed-support isomorphism ι_{21} proves (960) and (961).

Using $w = -u(1+u)$ and $(u-1)(2u+1) = 0$, one obtains

$$\vartheta^\#(e_{1/4}) = \frac{4}{9}(2 - u - u^2) = \frac{2}{3}(1 - u).$$

Using $u = (3s-1)/2$ then gives

$$\alpha^\# \vartheta^\#(e_{1/4}) = 1 - s.$$

Because $c \neq 0$, their vanishing ideals are generated by $w+2$, $u-1$, and $s-1$, respectively. This proves (957)–(959).

The functional equation in the form

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s)$$

gives

$$\zeta'(-2) = 2^{-2} \pi^{-3} \left(-\frac{\pi}{2}\right) \Gamma(3) \zeta(3) = -\frac{\zeta(3)}{4\pi^2}.$$

This scalar is nonzero. Taylor expansion modulo ε^2 gives

$$\zeta(-2 + \varepsilon) = \zeta'(-2)\varepsilon.$$

Retaining the support coordinate E_{21} gives (963). Therefore (962) has inverse $[\varepsilon] \mapsto [t]/\zeta'(-2)$. Finally, $\mathfrak{r}_\infty(D) = -2$ by (941). Evaluation of $\mathcal{Z}_{21}^{\text{spl}}$ and $\mathfrak{b}_{21}$ at this value proves (964) and (965). $\square$

Theorem 13.67 (Globalization dilation orbit and the trivial-zero Fable family). *Retain*

$$q = \zeta(0) = -\frac{1}{2}$$

and the common-value scheme $\mathcal{W}$ of theorem 13.61. Its second component is

$$w = q^{-1} = 4q = -2.$$

Let

$$g_{\text{triv}}(z) := \frac{1}{z\Gamma(z/2)}, \quad (966)$$

where the value at $z = 0$ is supplied by removable continuation, and let

$$\mathcal{T}_{\text{triv}} := V(g_{\text{triv}}) \subset \mathbb{A}_{\mathbb{C}}^{1,\text{an}}.$$

Then $\mathcal{T}_{\text{triv}}$ is reduced, and the map from the reduced discrete analytic space on the positive integers

$$\mu : \mathbb{N}_{>0} \xrightarrow{\sim} \mathcal{T}_{\text{triv}}, \quad \mu(n) = nq^{-1} = -2n, \quad (967)$$

is an isomorphism. Its image is exactly the trivial-zero divisor of ζ .

Every point of this divisor has a nonzero first jet. More precisely,

$$\zeta'(-2n) = (-1)^n \frac{(2n)!}{2(2\pi)^{2n}} \zeta(2n+1) \neq 0 \quad (n \geq 1). \quad (968)$$

Consequently, for

$$\mathcal{N}_{-2n} = \text{Spec } \frac{\mathbb{C}[\varepsilon]}{(\varepsilon^2)}, \quad \mathcal{N}_0 = \text{Spec } \frac{\mathbb{C}[t]}{(t^2)},$$

the first jet of zeta is the isomorphism

$$j_{\zeta,n} : \mathcal{N}_{-2n} \xrightarrow{\sim} \mathcal{N}_0, \quad j_{\zeta,n}^{\#}([t]) = \zeta'(-2n)[\varepsilon]. \quad (969)$$

Put

$$\Omega_{\zeta} := \mathbb{A}_{\mathbb{C}}^{1,\text{an}} \setminus \{1\}.$$

The analytic split-pair morphism

$$\hat{\mathcal{Z}}_{21} : \Omega_{\zeta} \longrightarrow (\mathcal{L}_{21}^{\text{spl}})^{\text{an}}, \quad z \longmapsto (\zeta(z)E_{21}, E_{21}) \quad (970)$$

restricts on $\mathcal{W}$ to $\mathcal{Z}_{21}^{\text{spl}}$. Define the synchronized target

$$\mathcal{P}_{21} := \{((aE_{21}, E_{21}), (bE_{21}, E_{21})) : a, b \in \mathbb{C}\}.$$

The two morphisms form the joint analytic map

$$\mathfrak{J}_{21} := (\hat{\mathcal{Z}}_{21}, \mathfrak{B}_{21}) : \Omega_{\zeta} \longrightarrow \mathcal{P}_{21}, \quad z \longmapsto ((\zeta(z)E_{21}, E_{21}), (-zE_{21}, E_{21})). \quad (971)$$

At every trivial zero and on its first infinitesimal neighborhood,

$$\mathfrak{J}_{21}(-2n) = ((0_2, E_{21}), (2nE_{21}, E_{21})), \quad (972)$$

$$\mathfrak{J}_{21}(-2n + \varepsilon) = ((\zeta'(-2n)\varepsilon E_{21}, E_{21}), ((2n - \varepsilon)E_{21}, E_{21})) \pmod{\varepsilon^2}. \quad (973)$$

Both entries retain the Boolean coordinate E_{21} . The first tangent coefficient is the odd-zeta value in (968); the second tangent coefficient is -1 .

In the displayed target coordinates of the Fable map, put

$$L_{\lambda}(X, Y, Z) := (\lambda X, Y, Z), \quad F^{(\lambda)} := L_{\lambda} \circ F \quad (\lambda \in \mathbb{C}^{\times}). \quad (974)$$

The orbit

$$\mathcal{K}_F := \{F^{(\lambda)} : \lambda \in \mathbb{C}^{\times}\}$$

is an algebraic copy of $\mathbb{G}_m$, and

$$\det JF^{(\lambda)} = -2\lambda, \quad F^{(\lambda)}(p_0) = F^{(\lambda)}(p_+) = F^{(\lambda)}(p_-) = \left(-\frac{\lambda}{4}, 0, 0\right). \quad (975)$$

For $\lambda = n \in \mathbb{N}_{>0}$, these identities become

$$\det JF^{(n)} = nq^{-1} = -2n, \quad F^{(n)}(p_\bullet) = \left(n N_{A_{\mathcal{E}}/\mathbb{C}}(\rho), 0, 0\right) = \left(-\frac{n}{4}, 0, 0\right). \quad (976)$$

The ambient split-pair morphism $\mathfrak{B}_{21}$ of theorem 13.62 restricts to an isomorphism from $\mathcal{T}_{\text{triv}}$ onto the reduced discrete subspace

$$\mathcal{R}_{21} := \{(2nE_{21}, E_{21}) : n \in \mathbb{N}_{>0}\} \subset \mathcal{L}_{21}^{\text{spl}}.$$

For every $n \geq 1$ and every $p_\bullet \in \{p_0, p_+, p_-\}$, its exact coordinate formula is

$$\begin{aligned} \mathfrak{B}_{21}(\mu(n)) &= \Phi_2(\mathbf{e}_{21}(2n)) = (2nE_{21}, E_{21}) \\ &= \left((- \det JF^{(n)})E_{21}, E_{21}\right) = \left((-8F_1^{(n)}(p_\bullet))E_{21}, E_{21}\right). \end{aligned} \quad (977)$$

Consequently,

$$\mathfrak{J}_{21}(\mu(n)) = \left((0_2, E_{21}), \left((- \det JF^{(n)})E_{21}, E_{21}\right)\right). \quad (978)$$

Its Boolean source and range supports are independent of n :

$$\hat{\chi}(\mathbf{e}_{21}(2n)) = E_{21}, \quad E_{21}^\top E_{21} = E_{11}, \quad E_{21} E_{21}^\top = E_{22}. \quad (979)$$

Thus the diagram

$$\begin{array}{ccc} \mathbb{N}_{>0} & \xrightarrow{n \mapsto F^{(n)}} & \mathcal{K}_F \\ \downarrow n \mapsto -2n & & \downarrow H \mapsto \det JH \\ \mathcal{T}_{\text{triv}} & \hookrightarrow & \mathbb{A}_{\mathbb{C}}^{1, \text{an}} \end{array} \quad (980)$$

commutes. Each $F^{(n)}$ is noninjective with nonzero constant Jacobian.

Proof. The reciprocal identity $q^{-1} = 4q = -2$ follows from $q = -1/2$. The entire function $1/\Gamma(z/2)$ has simple zeros at $0, -2, -4, \dots$. Its zero at 0 is removed by division by z , since

$$\frac{1}{\Gamma(z/2)} = \frac{z}{2} + O(z^2).$$

Hence (966) is entire with simple zero set $\{-2n : n \geq 1\}$. This proves that $\mathcal{T}_{\text{triv}}$ is reduced and that μ is an isomorphism of reduced discrete analytic spaces.

At $z = -2n$, the only vanishing factor in the functional equation

$$\zeta(z) = 2^z \pi^{z-1} \sin\left(\frac{\pi z}{2}\right) \Gamma(1-z) \zeta(1-z)$$

is the sine factor. Its derivative there is

$$\frac{\pi}{2} \cos(-n\pi) = (-1)^n \frac{\pi}{2}.$$

Substitution of $\Gamma(1+2n) = (2n)!$ gives (968). The factor $\zeta(2n+1)$ is greater than 1, so the derivative is nonzero. Taylor expansion modulo ε^2 now proves (969). The function ζ is holomorphic on Ω_ζ , so (970) is an analytic morphism. Its restriction to the two points of $\mathcal{W}$ has amplitudes $\zeta(1/4)$ and $\zeta(-2) = 0$; hence it is $\mathcal{Z}_{21}^{\text{spl}}$. Combining its Taylor expansion with

$$\mathfrak{B}_{21}(-2n + \varepsilon) = ((2n - \varepsilon)E_{21}, E_{21})$$

proves (972) and (973).

The linear map L_λ has determinant λ . The chain rule and $\det JF = -2$ therefore give

$$\det JF^{(\lambda)} = \lambda \det JF = -2\lambda.$$

The coefficient of z in the first component of F is 1, so the corresponding coefficient in $F^{(\lambda)}$ is λ . Coefficient extraction is therefore the inverse of $\lambda \mapsto F^{(\lambda)}$, proving $\mathcal{K}_F \cong \mathbb{G}_m$. Applying L_λ to the

common target $(-1/4, 0, 0)$ proves the second identity in (975). Since L_λ is an automorphism for $\lambda \neq 0$, it preserves the three-point collision. Finally,

$$N_{A_{\mathcal{E}}/\mathbb{C}}(\rho) = -\frac{1}{4}, \quad 8N_{A_{\mathcal{E}}/\mathbb{C}}(\rho) = q^{-1} = -2,$$

so restriction to $\lambda = n$ proves (976). Since $\mu(n) = -2n$ and $\mathfrak{B}_{21}(w) = (-wE_{21}, E_{21})$,

$$\mathfrak{B}_{21}(\mu(n)) = (2nE_{21}, E_{21}).$$

The identities $-\det JF^{(n)} = 2n$ and $-8F_1^{(n)}(p_\bullet) = 2n$ prove (977) and (978). Boolean matrix multiplication proves (979). These calculations also prove the commutative square. $\square$

Proposition 13.68 (Endpoint-reciprocal-spectral three-cycle on the directed stratum). *Put*

$$q = \zeta(0) = -\frac{1}{2}, \quad \mathcal{C}_q := \{1, q, q^{-1}\} \subset \mathbb{P}_{\mathbb{C}}^1.$$

The unique projective automorphism carrying the displayed ordered triple cyclically is

$$\Theta([A : B]) = [-B : A + B], \quad M_\Theta = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}. \quad (981)$$

On the affine chart it is

$$\Theta(a) = -\frac{1}{a+1},$$

and

$$1 \xrightarrow{\Theta} q \xrightarrow{\Theta} q^{-1} \xrightarrow{\Theta} 1. \quad (982)$$

The matrix satisfies $M_\Theta^3 = -I_2$; hence Θ has order three in $\mathrm{PGL}_2(\mathbb{C})$.

Let

$$\mathcal{C}_{q,21}^{\mathrm{spl}} := \{(aE_{21}, E_{21}) : a \in \mathcal{C}_q\} \subset \mathcal{L}_{21}^{\mathrm{spl}}.$$

The fixed-support lift

$$\Theta_{21}(aE_{21}, E_{21}) := (\Theta(a)E_{21}, E_{21}) \quad (983)$$

is an order-three automorphism of $\mathcal{C}_{q,21}^{\mathrm{spl}}$, with

$$(E_{21}, E_{21}) \mapsto \left(-\frac{1}{2}E_{21}, E_{21}\right) \mapsto (-2E_{21}, E_{21}) \mapsto (E_{21}, E_{21}). \quad (984)$$

Every point retains Boolean coordinate E_{21} .

The twisted spectral function

$$\mathcal{L}_U(s) := q^{-1}(4\pi^2)^{-s}\zeta(2s) \quad (985)$$

satisfies

$$\mathcal{L}_U(0) = q^{-1}\zeta(0) = 1. \quad (986)$$

Thus the three points in (982) are respectively the twisted spectral value, the zeta endpoint, and the reciprocal endpoint, where

$$q^{-1} = \det JF = -2.$$

At the reciprocal endpoint, the two ambient morphisms have the exact values

$$\widehat{\mathcal{Z}}_{21}(q^{-1}) = (0_2, E_{21}), \quad \mathfrak{B}_{21}(q^{-1}) = (2E_{21}, E_{21}). \quad (987)$$

The zeta image at the reciprocal endpoint is the zero-amplitude point of the fixed E_{21} -support fibre; it is distinct from the ambient additive zero $(0_2, 0_2)$. The second value is the directed blade pair. Their amplitudes and their common Boolean coordinate are both retained in $\mathfrak{J}_{21}(q^{-1})$.

Proof. The three points $1, -1/2, -2$ are distinct and none is the pole -1 of the affine formula for Θ . Direct evaluation gives

$$\Theta(1) = -\frac{1}{2}, \quad \Theta\left(-\frac{1}{2}\right) = -2, \quad \Theta(-2) = 1.$$

A projective automorphism is determined by its values on an ordered triple of distinct points, proving uniqueness. Matrix multiplication gives

$$M_{\Theta}^2 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \quad M_{\Theta}^3 = -I_2.$$

The embedding $a \mapsto (aE_{21}, E_{21})$ is injective and retains the second coordinate, so it transports the scalar action to (983).

Equation (986) follows by substituting $s = 0$, since $(4\pi^2)^0 = 1$ and $q^{-1}q = 1$. The identity $q^{-1} = \det JF = -2$ is (916). Finally, $\zeta(q^{-1}) = \zeta(-2) = 0$, while $\mathfrak{B}_{21}(w) = (-wE_{21}, E_{21})$. These substitutions prove (987). $\square$

The following degree-four construction extends the marked linear-factor pipeline used for the explicit three-dimensional map [5, 6].

Theorem 13.69 (Quartic marked-factor carrier and its residual binary sheet). *Let*

$$L = aU + bV, \quad Q_3 = cU^3 + dU^2V + eUV^2 + fV^3,$$

and let $M_4(L, Q_3) = LQ_3$. In coefficient coordinates,

$$M_4(a, b, c, d, e, f) = (ac, ad + bc, ae + bd, af + be, bf). \quad (988)$$

The resultant is

$$R_4(L, Q_3) = Q_3(-b, a) = -b^3c + ab^2d - a^2be + a^3f. \quad (989)$$

Under

$$(L, Q_3) \mapsto (\lambda L, \lambda^{-1}Q_3), \quad (990)$$

one has

$$R_4(\lambda L, \lambda^{-1}Q_3) = \lambda^2 R_4(L, Q_3). \quad (991)$$

On the resultant-one hypersurface

$$Y_4 := \{R_4 = 1\} \subset \mathbb{C}^6,$$

the differential of $M_4|_{Y_4} : Y_4 \rightarrow \mathbb{C}^5$ is an isomorphism at every point. Each square-free binary quartic has four marked linear factors, while the equation $R_4 = 1$ leaves the residual action

$$(L, Q_3) \mapsto (-L, -Q_3). \quad (992)$$

Hence its resultant-one marked fibre is canonically the four-element root set times μ_2 . A selected cyclic ordering of the root set identifies this fibre with $C_4 \times C_2$.

On the coefficient slice

$$X_4 := \{R_4 = 1, ad + bc = 1\} \subset \mathbb{C}^6, \quad (993)$$

put

$$K := 2b^2d - abe + a^2f. \quad (994)$$

Then

$$b^2 + 1 = aK. \quad (995)$$

For either $\iota \in \mathbb{C}$ with $\iota^2 = -1$, this becomes

$$(b - \iota)(b + \iota) = aK. \quad (996)$$

The boundary $a = 0$ therefore has the two components $b = \iota$ and $b = -\iota$, exchanged by the residual action (992).

Proof. Equation (988) follows by multiplication, and evaluation at $(-b, a)$ gives (989). The resultant of a linear form and a cubic is cubic in the coefficients of the linear form and linear in the coefficients of the cubic. This gives the weight two identity (991).

At a point of Y_4 , the forms L and Q_3 are coprime. If $(\dot{L}, \dot{Q}_3) \in T_{(L, Q_3)}Y_4$ lies in the kernel of dM_4 , then

$$\dot{L}Q_3 + L\dot{Q}_3 = 0.$$

Unique factorization gives $\dot{L} = tL$ and $\dot{Q}_3 = -tQ_3$. Differentiating (991) gives

$$dR_4(tL, -tQ_3) = 2tR_4 = 2t.$$

Tangency to Y_4 gives $t = 0$. The hypersurface Y_4 is smooth because Euler's identity for the degree-four polynomial R_4 gives

$$aR_a + bR_b + cR_c + dR_d + eR_e + fR_f = 4R_4 = 4$$

on Y_4 . Its tangent dimension is five, equal to the target dimension, so the differential is an isomorphism.

A square-free quartic has four choices for its marked linear factor. For each choice, equation $\lambda^2 R_4 = 1$ has two solutions, interchanged by $\lambda = -1$. This proves the fibre statement.

On X_4 , the equation $bc = 1 - ad$ transforms (989) into

$$1 = -b^2 + 2ab^2d - a^2be + a^3f.$$

Rearranging gives (995), and $b^2 + 1 = (b - \iota)(b + \iota)$ gives (996). $\square$

Theorem 13.70 (Erdős–Strauss as a quartic marked-factor fourfold). *Retain the linear and cubic forms*

$$L = aU + bV, \quad Q_3 = cU^3 + dU^2V + eUV^2 + fV^3$$

and the resultant-one carrier Y_4 from theorem 13.69. Define

$$\mathcal{X}_{\text{ES}} := \{(a, b, c, d, e, f) \in Y_4 : 4af - be = 0\}. \quad (997)$$

The second equation is invariant under the factor gauge (990), and

$$4af - be = \det \begin{pmatrix} 2a & b \\ e & 2f \end{pmatrix}. \quad (998)$$

The variety $\mathcal{X}_{\text{ES}}$ is a smooth affine fourfold, and the restriction of M_4 to it has injective differential everywhere.

On the open set $ac \neq 0$, put

$$p = -\frac{b}{a}, \quad \frac{Q_3}{c} = U^3 - s_1U^2V + s_2UV^2 - s_3V^3. \quad (999)$$

Then

$$4af - be = 0 \iff 4s_3 = ps_2. \quad (1000)$$

If the cubic in (999) splits as

$$(U - xV)(U - yV)(U - zV),$$

then (1000) is equivalent to

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \quad (1001)$$

whenever $pxyz \neq 0$.

At the projective boundary $a = 0$, one has

$$e = 0, \quad -b^3c = 1, \quad (1002)$$

with d and f free. Thus the finite-prime locus and its projective boundary belong to one four-dimensional marked-factor carrier.

Proof. Under (990), both af and be are fixed. Equation (998) follows by taking the displayed determinant.

No point of $\mathcal{X}_{\text{ES}}$ has $b = 0$. Indeed, at $b = 0$ the resultant-one equation becomes $a^3f = 1$, while the second equation becomes $4af = 0$. These two identities are incompatible. The Jacobian minor of R_4 and $4af - be$, taken in the variables (c, e) , is

$$\det \begin{pmatrix} -b^3 & -a^2b \\ 0 & -b \end{pmatrix} = b^4.$$

The two defining differentials are therefore independent on their common locus. Since Y_4 has dimension five, $\mathcal{X}_{\text{ES}}$ is smooth of dimension four. The differential of $M_4|_{Y_4}$ is an isomorphism by theorem 13.69; its restriction to the four-dimensional tangent space is injective.

From (999),

$$\frac{e}{c} = s_2, \quad \frac{f}{c} = -s_3, \quad \frac{b}{a} = -p.$$

Division of $4af = be$ by ac gives

$$-4s_3 = -ps_2,$$

which proves (1000). For the split cubic, $s_2 = xy + xz + yz$ and $s_3 = xyz$, proving (1001).

Finally, if $a = 0$, then $b \neq 0$ and $4af - be = 0$ gives $e = 0$. The resultant equation reduces to $-b^3c = 1$, proving (1002). $\square$

Corollary 13.71 (Four-to-five-dimensional simplex coordinate). *The dimension pair in theorem 13.70 is the adjacent pair*

$$\dim \mathcal{X}_{\text{ES}} = 4, \quad \dim Y_4 = 5.$$

For the regular-simplex carriers of these dimensions, the reciprocal shared by the two adjacent projection layers is

$$-\frac{1}{5}. \tag{1003}$$

More explicitly, it is the full-altitude coordinate of an opposite-face centroid in a regular four-simplex and the common hidden displacement of the opposite facet under a vertex-axis projection of a regular five-simplex.

Proof. Apply theorem 13.56 with $m = 5$. $\square$

Remark 13.72 (Metric problem attached to the dimension pair). The determinantal equation (998) proves the four-dimensional carrier inside the five-dimensional factor space. An identification of (1003) with a scalar intrinsic to that carrier requires an explicit metric or tight frame on the tangent and normal data of $\mathcal{X}_{\text{ES}} \subset Y_4$. This is the next intertwining calculation.

Exact computation 13.73 (Quartic Erdős–Strauss lift and simplex checker). The script

`verification/verify_es_quartic_marked_factor_lift.py`

checks the product and resultant formulas, gauge weights, determinant carrier, smoothness minor, dimension count, and the adjacent four-to-five-dimensional reciprocal.

Theorem 13.74 (Explicit four-Time constant-Jacobian collision). *Fix $\iota^2 = -1$. Define a polynomial map $\Phi_\iota : \mathbb{C}^4 \rightarrow X_4$ in coordinates (a, y, z, w) by*

$$b = \iota + ay, \quad c = -\iota + 2ay + a^2z, \tag{1004}$$

$$d = -\iota y - a(\iota z + 2y^2) - a^2yz, \quad e = 2z - 7\iota y^2 + aw, \tag{1005}$$

$$f = \iota w + 3\iota y^3 - 4yz + a(6\iota y^2z + wy + 4y^4) + 2a^2y^3z. \tag{1006}$$

Delete the fixed U^3V -coefficient from $M_4 \circ \Phi_\iota$, and write

$$P_\iota = (P_0, P_2, P_3, P_4) : \mathbb{C}^4 \longrightarrow \mathbb{C}^4.$$

Then

$$P_0 = a^3z + 2a^2y - \iota a, \quad (1007)$$

$$P_2 = -a^3y^2z - 2\iota a^2yz + a^2w - 2a^2y^3 - 10\iota ay^2 + 3az + y, \quad (1008)$$

$$P_3 = 2a^3y^3z + 6\iota a^2y^2z + 2a^2wy + 4a^2y^4 + 2\iota aw - 4\iota ay^3 - 2ayz + 2\iota z + 7y^2, \quad (1009)$$

$$P_4 = 2a^3y^4z + 8\iota a^2y^3z + a^2wy^2 + 4a^2y^5 + 2\iota awy + 7\iota ay^4 - 10ay^2z - 4\iota yz - w - 3y^3. \quad (1010)$$

Its Jacobian determinant is

$$\det DP_\iota = -2. \quad (1011)$$

The four distinct points

$$q_M = \left(0, 0, \frac{\iota}{2}, 0\right), \quad (1012)$$

$$q_N = (\iota, -1, -3\iota, 13), \quad (1013)$$

$$q_T = \left(\frac{1}{\sqrt{2}}, -1 - \sqrt{2}\iota, 2\sqrt{2} + 6\iota, -34 - 19\sqrt{2}\iota\right), \quad (1014)$$

$$q_E = \left(\frac{1}{\sqrt{2}}, 1 - \sqrt{2}\iota, -2\sqrt{2} + 6\iota, 34 - 19\sqrt{2}\iota\right) \quad (1015)$$

satisfy

$$P_\iota(q_M) = P_\iota(q_N) = P_\iota(q_T) = P_\iota(q_E) = (0, 0, -1, 0). \quad (1016)$$

They are the four resultant-one marked factorizations of

$$H(U, V) = U^3V - UV^3 = UV(U - V)(U + V). \quad (1017)$$

Thus the collision occupancy is exactly $(1, 1, 1, 1)$, indexed by the four Times.

Proof. Substitution of (1004)–(1006) into (993) gives

$$ad + bc = 1, \quad R_4 = 1.$$

Substitution into the four unfixed coefficients $(ac, ae + bd, af + be, bf)$ gives (1007)–(1010). Direct differentiation and collection in the basis $1, \iota$ of $\mathbb{Q}(\iota)$ gives (1011).

The four marked states in (a, b, c, d, e, f) -coordinates are

	a	b	c	d	e	f
M	0	ι	$-\iota$	0	ι	0
N	ι	0	0	$-\iota$	0	ι
T	$1/\sqrt{2}$	$-1/\sqrt{2}$	0	$\sqrt{2}$	$\sqrt{2}$	0
E	$1/\sqrt{2}$	$1/\sqrt{2}$	0	$\sqrt{2}$	$-\sqrt{2}$	0

(1018)

Each row has resultant one and product H . Equations (1004)–(1006) send the four points (1012)–(1015) to the four rows of (1018). Their common product has coefficient vector $(0, 1, 0, -1, 0)$; deleting the fixed second entry gives (1016). $\square$

The incidence equations, constant Jacobian, four source evaluations, resultant-one conditions, and common quartic product are independently checked over $\mathbb{Q}(i, \sqrt{2})$ in `verification/verify_quartic_time_collision.py`. The captured output is `verification/verify_quartic_time_collision_output.txt`.

Corollary 13.75 (Intrinsic Time coordinate of the quartic common target). *Label the four unfixed coefficient directions*

$$(U^4, U^2V^2, UV^3, V^4) \longleftrightarrow (M, N, T, E). \quad (1019)$$

The common target in (1016) is represented in the Time carrier by the three-Time face resultant

$$\mathfrak{h}_{MNE} := \mu_M + \mu_N + \mu_E. \quad (1020)$$

If A_{MNE} is the affine face plane through these three Time vertices and $\pi_{\text{T}}^{\text{face}}$ is orthogonal projection onto that plane, then

$$\pi_{\text{T}}^{\text{face}}(\mu_{\text{T}}) = \pi_{\text{T}}^{\text{face}}(\mathfrak{h}_{\text{MNE}}) = \frac{1}{3}\mathfrak{h}_{\text{MNE}}. \quad (1021)$$

Thus the selected Time vertex and the common-target resultant have the same flat three-Time projection and distinct altitude coordinates.

Proof. The centred simplex relation gives $\mu_{\text{M}} + \mu_{\text{N}} + \mu_{\text{T}} + \mu_{\text{E}} = 0$. Hence the coefficient appearing in the third component of (1016) is represented by (1020). The affine face plane is

$$A_{\text{MNE}} = \{x : \langle x, \mu_{\text{T}} \rangle = -1/3\}.$$

Orthogonal projection of both μ_{T} and $\mathfrak{h}_{\text{MNE}}$ onto this plane gives the face centroid $\mathfrak{h}_{\text{MNE}}/3$. $\square$

Theorem 13.76 (The tetrahedral S_4 inside the Klein–Fano carrier). *Let*

$$G := \text{PSL}(2, \mathbb{F}_7) \cong \text{GL}(3, \mathbb{F}_2), \quad |G| = 168. \quad (1022)$$

The group G acts transitively on the seven points of the Fano plane. For every Fano point p , its stabilizer has order 24 and acts faithfully on the four Fano lines not incident with p . Hence

$$G_p \cong S_4. \quad (1023)$$

Choosing p and a bijection from those four lines to $\{\text{M}, \text{N}, \text{T}, \text{E}\}$ identifies this action with the tetrahedral Time action in theorem 13.43.

The seven subgroups G_p form a single conjugacy orbit under G . Thus the family of localized tetrahedral carriers is G -invariant, while selecting one carrier requires selecting a Fano point.

In the tetrahedral picture of Klein’s quartic, the twelve visible proper tetrahedral rotations give A_4 . The inside-out symmetry acts as an odd permutation of the four tetrahedral corners. Together these symmetries give the corresponding S_4 subgroup of the full order-168 orientation-preserving automorphism group [29].

Proof. Orbit–stabilizer gives

$$|G_p| = \frac{168}{7} = 24.$$

An element of G_p that fixes each of the four nonincident lines fixes their pairwise incidence points and hence every point of the Fano plane. The action on the four lines is therefore faithful, so the order-24 stabilizer is S_4 . Transitivity makes the seven stabilizers conjugate. The final Klein-quartic statement is the tetrahedral subgroup description recorded in [29]. $\square$

Theorem 13.77 (Support-to-Toeplitz transform). *Let $N \geq 2$, let $\omega_N = e^{2\pi i/N}$, and let $A \subseteq C_N$. For $a \in C_N$, define*

$$v_a := (1, \omega_N^a, \omega_N^{2a}, \dots, \omega_N^{(N-1)a})^{\text{T}}, \quad (1024)$$

and put

$$T_A := \sum_{a \in A} v_a v_a^*, \quad H_A := N I_N - T_A. \quad (1025)$$

Then H_A is Hermitian, positive semidefinite, and Toeplitz, and

$$\ker H_A = \text{span}\{v_a : a \in A\}, \quad (1026)$$

$$\text{corank } H_A = |A|, \quad (1027)$$

$$\text{tr } H_A = N(N - |A|). \quad (1028)$$

If $A = \{a\}$, the kernel is generated by v_a , and its polynomial is

$$P_a(z) := \sum_{j=0}^{N-1} \omega_N^{aj} z^j = \frac{1 - z^N}{1 - \omega_N^a z}. \quad (1029)$$

Its zeros are

$$z = \omega_N^{k-a}, \quad k = 1, \dots, N-1, \quad (1030)$$

and hence lie on the unit circle.

Proof. The finite geometric-series identity gives

$$v_a^* v_b = \sum_{j=0}^{N-1} \omega_N^{(b-a)j} = N \delta_{ab}. \quad (1031)$$

Thus the v_a form an orthogonal basis and

$$H_A v_b = \begin{cases} 0, & b \in A, \\ N v_b, & b \notin A. \end{cases}$$

This proves positivity, (1026), and (1027). Each $v_a v_a^*$ has trace N , which proves (1028). Its (j, k) -entry is $\omega_N^{a(j-k)}$, so every entry of H_A depends only on $j - k$; hence H_A is Toeplitz.

For singleton A , summing the geometric progression gives (1029). Its numerator vanishes at all N -th roots of unity, and the denominator removes precisely the root $z = \omega_N^{-a}$. The remaining roots are exactly those in (1030). $\square$

Corollary 13.78 (Pure and mixed support regimes). *Let $x \in S^{\langle C_N \rangle}$ and take $A = \text{Supp}(x)$ in theorem 13.77. Then*

A	corank H_A	kernel regime
$\emptyset$	0	unsupported, no kernel
$ A = 1$	1	single universe, circle-supported kernel zeros
$ A \geq 2$	$ A $	mixed universes, higher-corank kernel

(1032)

The singleton case supplies a positive self-adjoint Toeplitz Hamiltonian with a one-dimensional kernel. The mixed case is exactly the higher-corank locus in which individual kernel polynomials require the full kernel module for their spectral interpretation.

Proof. Combine (1027) with theorem 13.6. The singleton root statement is (1030); the higher-corank statement records the complete kernel in (1026). $\square$

Let U be the cyclic coordinate operator

$$U = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad U^3 = I_3. \quad (1033)$$

Theorem 13.79 (The complete 12289 coordinate orbit as one matrix). *Place the ordered coordinate-orbit vectors from (393) in cyclic order as columns:*

$$M_{12289} := \begin{pmatrix} 3 & 3 & 6 \\ 6 & 3 & 3 \\ 3 & 6 & 3 \end{pmatrix}. \quad (1034)$$

Then

$$M_{12289} = 3I_3 + 6U + 3U^2 = 3(I_3 + U)^2, \quad (1035)$$

and its opposite orientation is $M_{12289}^\top$. The ordered blade difference and the mirror-even product are

$$M_{12289} - M_{12289}^\top = 3 \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}, \quad (1036)$$

$$M_{12289} M_{12289}^\top = 9I_3 + 45J_3 = \begin{pmatrix} 54 & 45 & 45 \\ 45 & 54 & 45 \\ 45 & 45 & 54 \end{pmatrix}, \quad (1037)$$

where J_3 is the all-ones matrix. If $\omega = e^{2\pi i/3}$, the eigenvalues of M_{12289} are

$$12, \quad 3\omega, \quad 3\omega^2, \quad (1038)$$

and the eigenvalues of $M_{12289}M_{12289}^\top$ are

$$144, \quad 9, \quad 9. \quad (1039)$$

In particular,

$$M_{12289}(1, 1, 1)^\top = 12(1, 1, 1)^\top, \quad 144 - 9 = 135, \quad (1040)$$

recovering (394) from the single orbit matrix.

Proof. The columns of $3I_3 + 6U + 3U^2$ are respectively $(3, 6, 3)^\top$, $(3, 3, 6)^\top$, and $(6, 3, 3)^\top$, proving (1035). Transposition sends U to U^2 , which gives (1036). Since

$$(I_3 + U)(I_3 + U^2) = 2I_3 + U + U^2 = I_3 + J_3$$

and $J_3^2 = 3J_3$, one obtains

$$M_{12289}M_{12289}^\top = 9(I_3 + J_3)^2 = 9I_3 + 45J_3.$$

The eigenvalues of U are $1, \omega, \omega^2$. Evaluating $3(1 + z)^2$ at these three values gives $12, 3\omega, 3\omega^2$. Their squared absolute values give (1039). The final identities follow directly. $\square$

Definition 13.80 (Mirror-completed orbit matrix). For an integer n , write $\tilde{n} \in G(\mathbb{Z})$ for its supported copy. Define $\mathbf{M}_{12289} \in M_3(S^{\leftrightarrow})$ by

$$\mathbf{M}_{12289} := \begin{pmatrix} (\tilde{3}, \tilde{3}) & (\tilde{3}, \tilde{6}) & (\tilde{6}, \tilde{3}) \\ (\tilde{6}, \tilde{3}) & (\tilde{3}, \tilde{3}) & (\tilde{3}, \tilde{6}) \\ (\tilde{3}, \tilde{6}) & (\tilde{6}, \tilde{3}) & (\tilde{3}, \tilde{3}) \end{pmatrix}. \quad (1041)$$

Equivalently, its two coordinate matrices are $(M_{12289}, M_{12289}^\top)$. On $M_3(S^{\leftrightarrow})$, put

$$\Theta_3(A_+, A_-) := (A_-^\top, A_+^\top). \quad (1042)$$

Proposition 13.81 (One object with two retained orientations). *The map Θ_3 is an involutive semiring anti-automorphism and*

$$\Theta_3(\mathbf{M}_{12289}) = \mathbf{M}_{12289}. \quad (1043)$$

The two coordinate projections recover M_{12289} and $M_{12289}^\top$ separately. Hence the mirror completion retains the ordered blade difference (1036) while placing both orientations in one matrix-semiring element.

Proof. The component swap is a semiring automorphism and transposition reverses matrix products. Their composite is therefore an anti-automorphism, and applying it twice gives the identity. Since the two coordinate matrices of $\mathbf{M}_{12289}$ are mutual transposes, (1043) follows. Coordinate projection gives the final assertion. $\square$

14. SECTORWISE MODULAR DATA AND THE EXACT C_3 CONJUGATION LAW

The earlier sectorial construction in the local corpus contains a sound multiple-modular starting point together with an incorrect calculation of the action of an antiunitary on the two nontrivial C_3 -character projectors. This section gives the closed-operator statement and separates the two possible antiunitary actions. The source audit used the local sectorial drafts, Summers's account of Tomita–Takesaki theory, and Takesaki's left-Hilbert-algebra treatment [31, 32, 33].

Theorem 14.1 (Antiunitary transport of a standard pair). *For $j = 1, 2$, let $M_j \subset B(H_j)$ be a von Neumann algebra and let $\Omega_j \in H_j$ be cyclic and separating for M_j . Let $C : H_1 \rightarrow H_2$ be antiunitary and suppose*

$$C\Omega_1 = \Omega_2, \quad CM_1C^{-1} = M_2. \quad (1044)$$

Write

$$S_j = J_j\Delta_j^{1/2}$$

for the polar decomposition of the closed Tomita operator associated with (M_j, Ω_j) . Then

$$C\text{Dom}(S_1) = \text{Dom}(S_2), \quad S_2C = CS_1, \quad (1045)$$

$$J_2 = CJ_1C^{-1}, \quad \Delta_2 = C\Delta_1C^{-1}. \quad (1046)$$

For every $t \in \mathbb{R}$,

$$C\Delta_1^{it}C^{-1} = \Delta_2^{-it}. \quad (1047)$$

Consequently, if $\sigma_t^{(j)}(A) = \Delta_j^{it}A\Delta_j^{-it}$, then

$$C\sigma_t^{(1)}(A)C^{-1} = \sigma_{-t}^{(2)}(CAC^{-1}) \quad (A \in M_1). \quad (1048)$$

Proof. Let $S_{j,0}$ be the Tomita operator on the core $M_j\Omega_j$: $S_{j,0}(A\Omega_j) = A^*\Omega_j$. For $A \in M_1$, put $B = CAC^{-1} \in M_2$. The hypotheses give

$$CS_{1,0}C^{-1}(B\Omega_2) = CA^*\Omega_1 = (CAC^{-1})^*\Omega_2 = B^*\Omega_2.$$

Thus $CS_{1,0}C^{-1} = S_{2,0}$ on their corresponding cores. Conjugation by a bounded antiunitary preserves graph closure, which proves (1045).

The operator CJ_1C^{-1} is antiunitary, and $C\Delta_1^{1/2}C^{-1}$ is positive and self-adjoint. Hence

$$S_2 = (CJ_1C^{-1})(C\Delta_1^{1/2}C^{-1})$$

is a polar decomposition. Uniqueness of the polar decomposition gives (1046). Real Borel functional calculus is preserved by antiunitary conjugation, while scalar coefficients are complex conjugated. Applying this to $\lambda^{it} = \exp(it \log \lambda)$ gives (1047). Substitution into the definition of the modular automorphism groups gives (1048). $\square$

The map $A \mapsto CAC^{-1}$ in Theorem 14.1 is conjugate-linear, multiplicative, and star-preserving:

$$C(AB)C^{-1} = (CAC^{-1})(CBC^{-1}), \quad CA^*C^{-1} = (CAC^{-1})^*. \quad (1049)$$

In particular, it preserves the order of multiplication.

Proposition 14.2 (Character projectors under linear and antiunitary maps). *Let U be unitary with $U^3 = I$, let $\omega = e^{2\pi i/3}$, and put*

$$P_k = \frac{1}{3} \sum_{m=0}^2 \omega^{-km} U^m, \quad k \in \mathbb{Z}/3\mathbb{Z}. \quad (1050)$$

If C is antiunitary and $CUC^{-1} = U^\varepsilon$, where $\varepsilon \in \{1, -1\}$, then

$$CP_kC^{-1} = P_{-\varepsilon k}. \quad (1051)$$

If V is linear unitary and $VUV^{-1} = U^\varepsilon$, then

$$VP_kV^{-1} = P_{\varepsilon k}. \quad (1052)$$

Thus an antiunitary commuting with U exchanges P_1 and P_2 , whereas an antiunitary sending U to U^{-1} fixes all three projectors. A linear unitary sending U to U^{-1} exchanges P_1 and P_2 .

Proof. Antiunitarity conjugates the coefficients in (1050), so

$$CP_k C^{-1} = \frac{1}{3} \sum_{m=0}^2 \omega^{km} U^{\varepsilon m}.$$

For $\varepsilon = 1$ this is P_{-k} . For $\varepsilon = -1$, replacing m by $-m$ in $\mathbb{Z}/3\mathbb{Z}$ gives P_k . This proves (1051). The same calculation without complex conjugation of the coefficients gives (1052). $\square$

An explicit three-dimensional test displays all three cases. Take

$$U = \text{diag}(1, \omega, \omega^2), \quad Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

and let K_0 denote entrywise complex conjugation. Then

$$\begin{aligned} K_0 U K_0^{-1} &= U^{-1}, & K_0 P_k K_0^{-1} &= P_k, \\ (Q K_0) U (Q K_0)^{-1} &= U, & (Q K_0) P_1 (Q K_0)^{-1} &= P_2, \\ Q U Q^{-1} &= U^{-1}, & Q P_1 Q^{-1} &= P_2. \end{aligned}$$

Lemma 14.3 (The two algebra maps induced by an antiunitary). *Let $C : \mathcal{H}_1 \rightarrow \mathcal{H}_2$ be an antiunitary bijection and let $\mathcal{A}_i \subseteq \mathcal{B}(\mathcal{H}_i)$ be unital $*$ -algebras such that*

$$C \mathcal{A}_1 C^{-1} = \mathcal{A}_2.$$

For $A \in \mathcal{A}_1$, define

$$\Phi_C(A) := C A C^{-1}, \quad \Theta_C(A) := C A^* C^{-1}. \quad (1053)$$

Then Φ_C is a conjugate-linear multiplicative $*$ -isomorphism:

$$\Phi_C(\lambda A + \mu B) = \bar{\lambda} \Phi_C(A) + \bar{\mu} \Phi_C(B), \quad (1054)$$

$$\Phi_C(AB) = \Phi_C(A) \Phi_C(B), \quad \Phi_C(A^*) = \Phi_C(A)^*. \quad (1055)$$

The map Θ_C is a complex-linear order-reversing $*$ -isomorphism:

$$\Theta_C(\lambda A + \mu B) = \lambda \Theta_C(A) + \mu \Theta_C(B), \quad (1056)$$

$$\Theta_C(AB) = \Theta_C(B) \Theta_C(A), \quad \Theta_C(A^*) = \Theta_C(A)^*. \quad (1057)$$

Proof. Anti-linearity of C gives (1054). Inserting $C^{-1}C = I$ between two operators gives

$$C A B C^{-1} = (C A C^{-1})(C B C^{-1}),$$

and taking adjoints gives the second equality in (1055). Since $(\lambda A + \mu B)^* = \bar{\lambda} A^* + \bar{\mu} B^*$, the two scalar conjugations in Θ_C cancel, proving (1056). Finally,

$$C(AB)^* C^{-1} = C B^* A^* C^{-1} = (C B^* C^{-1})(C A^* C^{-1}),$$

and another adjoint calculation proves (1057). $\square$

Theorem 14.4 (Finite Hilbert–Schmidt modular blocks). *Let $n \geq 1$, let $H_n = M_n(\mathbb{C})$ with Hilbert–Schmidt inner product, and represent $M_n(\mathbb{C})$ on H_n by left multiplication. Given a positive definite matrix ρ_n , set $\Omega_n = \rho_n^{1/2}$. Then Ω_n is cyclic and separating, and the modular objects are*

$$S_n(Y) = \rho_n^{-1/2} Y^* \rho_n^{1/2}, \quad J_n(Y) = Y^*, \quad (1058)$$

$$\Delta_n(Y) = \rho_n Y \rho_n^{-1}. \quad (1059)$$

If $\rho_n = \text{diag}(r_{n,1}, \dots, r_{n,n})$, then

$$\Delta_n(E_{ij}) = \frac{r_{n,i}}{r_{n,j}} E_{ij}. \quad (1060)$$

For $K_n = -\log \Delta_n$,

$$K_n(E_{ij}) = \log \left(\frac{r_{n,j}}{r_{n,i}} \right) E_{ij}. \quad (1061)$$

The direct sum of the blocks with $n = 1, 2, 3$ is again a standard pair, with Tomita and modular objects equal to the corresponding direct sums.

Proof. Every $Y \in H_n$ has the form $Y = A\rho_n^{1/2}$, where $A = Y\rho_n^{-1/2}$, so Ω_n is cyclic. The equality $A\rho_n^{1/2} = 0$ implies $A = 0$, so it is separating. Applying the defining Tomita map to $Y = A\rho_n^{1/2}$ gives the first formula in (1058). The map $Y \mapsto Y^*$ is antiunitary, and direct calculation gives

$$J_n \Delta_n^{1/2}(Y) = (\rho_n^{1/2} Y \rho_n^{-1/2})^* = \rho_n^{-1/2} Y^* \rho_n^{1/2}.$$

This proves (1058)–(1059). Equations (1060) and (1061) follow by applying the diagonal matrices on the left and right. Cyclicity, separation, graph closure, and polar decomposition are componentwise under finite direct sums. $\square$

Theorem 14.5 (Simplex-synthesis realization of the 1–2–3 modular linking algebra). *For $n = 1$, put $V_1 = \mathbb{C}$, $v_0 = 1$, and $v_1 = -1$. For $n = 2, 3$, use the centered regular n -simplex in theorem 9.33. In every case let*

$$S_n : \mathbb{C}^{n+1} \longrightarrow V_n, \quad S_n e_i = v_i, \quad P_n = I_{n+1} - \frac{1}{n+1} \mathbf{1}_{n+1} \mathbf{1}_{n+1}^\top. \quad (1062)$$

The operator

$$U_n := \sqrt{\frac{n}{n+1}} S_n P_n : \mathbb{C}^{n+1} \longrightarrow V_n \quad (1063)$$

has initial projection P_n and final projection I_{V_n} :

$$U_n^* U_n = P_n, \quad U_n U_n^* = I_{V_n}. \quad (1064)$$

Consequently,

$$\Phi_n : P_n M_{n+1}(\mathbb{C}) P_n \longrightarrow B(V_n), \quad \Phi_n(A) = U_n A U_n^*, \quad (1065)$$

is a unital $*$ -isomorphism.

Put

$$P = P_1 \oplus P_2 \oplus P_3 \in M_9(\mathbb{C}), \quad U = U_1 \oplus U_2 \oplus U_3, \quad V = V_1 \oplus V_2 \oplus V_3. \quad (1066)$$

Then

$$\Phi : P M_9(\mathbb{C}) P \longrightarrow B(V) \cong M_6(\mathbb{C}), \quad \Phi(X) = U X U^*, \quad (1067)$$

is a unital $*$ -isomorphism. For $1 \leq a, b \leq 3$, it restricts to

$$\Phi_{ab} : P_a M_{a+1,b+1}(\mathbb{C}) P_b \xrightarrow{\cong} B(V_b, V_a) = \mathcal{L}_{ab}, \quad \dim_{\mathbb{C}} \mathcal{L}_{ab} = ab. \quad (1068)$$

For composable support-corner matrices X and Y ,

$$\Phi_{ab}(X) \Phi_{bc}(Y) = \Phi_{ac}(XY), \quad \Phi_{ab}(X)^* = \Phi_{ba}(X^*). \quad (1069)$$

Let $a_n \in P_n M_{n+1}(\mathbb{C}) P_n$ be positive and invertible in that support corner, and put

$$a = a_1 \oplus a_2 \oplus a_3, \quad d_n = \Phi_n(a_n), \quad d = d_1 \oplus d_2 \oplus d_3. \quad (1070)$$

Write a_n^{-1, P_n} for the inverse of a_n in the corner with unit P_n . On the Hilbert–Schmidt spaces, define

$$\begin{aligned} \Delta_a(X) &= a X a^{-1, P}, & J_P(X) &= X^*, \\ \Delta_d(Y) &= d Y d^{-1}, & J_V(Y) &= Y^*, \end{aligned} \quad (1071)$$

where $a^{-1, P} = a_1^{-1, P_1} \oplus a_2^{-1, P_2} \oplus a_3^{-1, P_3}$. The Hilbert–Schmidt unitary induced by Φ intertwines the full modular data:

$$\Phi \Delta_a = \Delta_d \Phi, \quad \Phi J_P = J_V \Phi, \quad \Phi(-\log \Delta_a) = (-\log \Delta_d) \Phi. \quad (1072)$$

On the rectangular corner in (1068), this reads

$$\Phi_{ab}(a_a X a_b^{-1, P_b}) = d_a \Phi_{ab}(X) d_b^{-1}. \quad (1073)$$

For the regular-simplex trace state $a_n = P_n/n$, one has

$$d_n = \frac{1}{n} I_{V_n}, \quad \Delta_{a_n} = I, \quad -\log \Delta_{a_n} = 0. \quad (1074)$$

For the positive matrices

$$d_1 = (1), \quad d_2 = \text{diag}(2, 3), \quad d_3 = \text{diag}(5, 7, 11), \quad (1075)$$

the first modular Hamiltonian is zero and the second and third are nonzero. Under Φ , their nine modular corners and the six oriented closed paths are exactly those of theorems 14.7 and 14.8. Thus the three simplex-supported modular Hamiltonians occur as three specified diagonal sectors of one full linking algebra, with all inter-sector maps retained.

Proof. For $n = 2, 3$, theorem 9.38 gives

$$S_n^* S_n = \frac{n+1}{n} P_n, \quad \ker S_n = \mathbb{C} \mathbf{1}_{n+1}.$$

The same identities follow directly for $n = 1$ from $S_1 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$. Hence

$$U_n^* U_n = \frac{n}{n+1} P_n S_n^* S_n P_n = P_n.$$

The rank of P_n and the dimension of V_n are both n , so U_n maps $\text{Ran } P_n$ onto V_n ; this proves $U_n U_n^* = I_{V_n}$.

If $A, B \in P_n M_{n+1}(\mathbb{C}) P_n$, then

$$\Phi_n(A) \Phi_n(B) = U_n A (U_n^* U_n) B U_n^* = U_n A P_n B U_n^* = \Phi_n(AB).$$

The adjoint identity is immediate, and $Y \mapsto U_n^* Y U_n$ is the inverse map. This proves (1065). Taking orthogonal direct sums gives $U^* U = P$ and $U U^* = I_V$, and proves (1067). Its restriction to the (a, b) -corner is (1068); the source and target dimensions are both ab . The same multiplication calculation with three indices and the adjoint calculation give (1069).

Conjugation by U preserves the Hilbert–Schmidt inner product:

$$\text{tr}_V(\Phi(X)^* \Phi(Y)) = \text{tr}_{\text{Ran } P}(X^* Y).$$

Moreover,

$$\Phi(a X a^{-1, P}) = (U a U^*)(U X U^*)(U a^{-1, P} U^*) = d \Phi(X) d^{-1}.$$

This proves the first identity in (1072); the second follows from $\Phi(X^*) = \Phi(X)^*$, and finite-dimensional functional calculus gives the third. Restriction to one rectangular corner gives (1073).

For $a_n = P_n/n$, equations (1065) and (1071) give (1074). Substitution of (1075) yields the stated three Hamiltonians. Finally, (1069) sends every product of three composable support-corner matrix units to the corresponding product in $M_6(\mathbb{C})$. The six paths and their modular weights are therefore the paths and weights proved in theorems 14.7 and 14.8. $\square$

Exact computation 14.6 (Exact check of the simplex 1–2–3 modular intertwiner). The script and captured receipt

```
verification/verify_simplex_123_modular_intertwiner.py,
verification/verify_simplex_123_modular_intertwiner_output.txt
```

verify exact ordered Helmert realizations of (1064), the pulled-back Gram forms, the displayed segment, triangle, and tetrahedron, the transported positive matrices and their support inverses, every matrix-unit modular eigenvalue, the complete 3×3 dimension matrix, all six three-sector weight products, and the stated behavior of the three modular Hamiltonians.

Regular-simplex support carried into the full 1-2-3 modular linking algebra

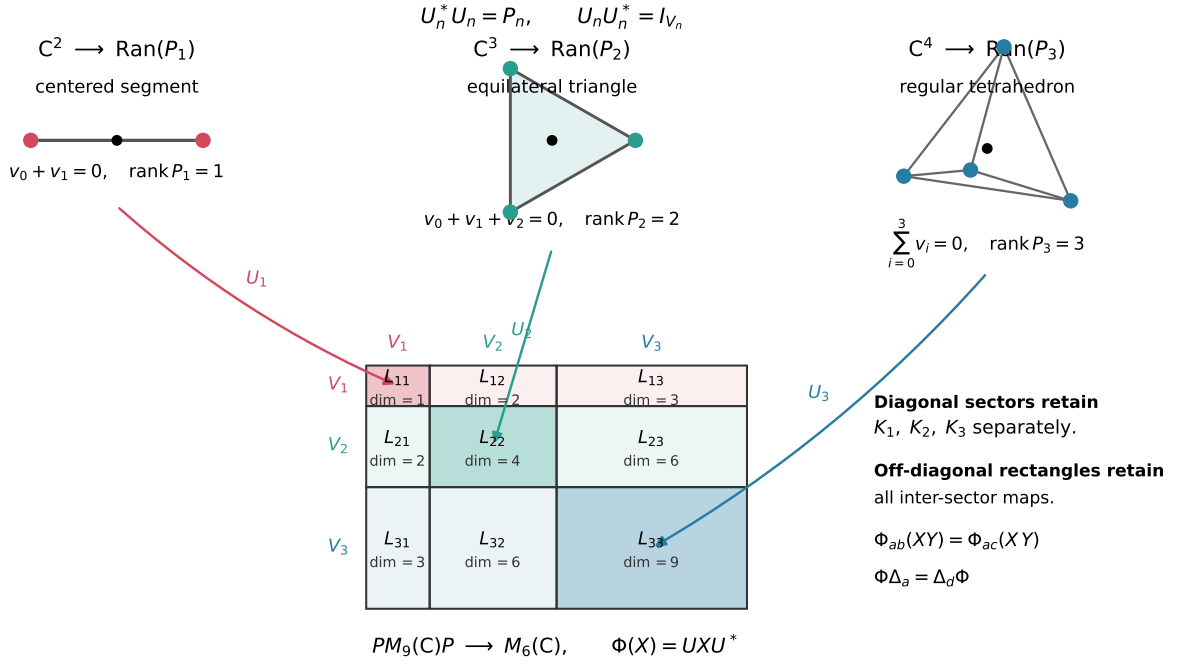


FIGURE 10. **Exact visual realization of theorem 14.5.** The top row shows the centered regular 1-, 2-, and 3-simplex label geometries. Their all-one vertex relations leave support ranks 1, 2, and 3. The three support unitaries carry those spaces to the diagonal sectors V_1, V_2, V_3 . The lower block image displays the entire linking algebra: the diagonal rectangles retain the three modular Hamiltonians, and the six off-diagonal rectangles retain every typed inter-sector map. Rectangle side lengths encode the sector dimensions, so the displayed cell areas are exactly $\dim_{\mathcal{C}} \mathcal{L}_{ab} = ab$.

Theorem 14.7 (A single linking algebra for the 1, 2, and 3 modular blocks). *Put*

$$(n_1, n_2, n_3) = (1, 2, 3), \quad \mathcal{V} = \mathbb{C}^{n_1} \oplus \mathbb{C}^{n_2} \oplus \mathbb{C}^{n_3} \cong \mathbb{C}^6,$$

and let E_a be the orthogonal projection onto the a -th summand. For positive diagonal matrices

$$\rho_a = \text{diag}(r_{a,1}, \dots, r_{a,n_a}), \quad \rho = \rho_1 \oplus \rho_2 \oplus \rho_3,$$

define the nine rectangular corners

$$\mathcal{L}_{ab} := E_a M_6(\mathbb{C}) E_b \cong M_{n_a, n_b}(\mathbb{C}), \quad 1 \leq a, b \leq 3. \quad (1076)$$

Represent $M_6(\mathbb{C})$ on its Hilbert–Schmidt space by left multiplication and use $\Omega = \rho^{1/2}$. Its modular objects are

$$J_\rho(X) = X^*, \quad \Delta_\rho(X) = \rho X \rho^{-1}, \quad K_\rho = -\log \Delta_\rho. \quad (1077)$$

Write $E_{a,i;b,j}$ for the matrix unit from the j -th coordinate of the b -th summand to the i -th coordinate of the a -th summand. Then

$$\Delta_\rho(E_{a,i;b,j}) = \frac{r_{a,i}}{r_{b,j}} E_{a,i;b,j}, \quad (1078)$$

$$K_\rho(E_{a,i;b,j}) = \log\left(\frac{r_{b,j}}{r_{a,i}}\right) E_{a,i;b,j}. \quad (1079)$$

The Hilbert–Schmidt orthogonal decomposition

$$M_6(\mathbb{C}) = \bigoplus_{a,b=1}^3 \mathcal{L}_{ab} \quad (1080)$$

is reduced by Δ_ρ and K_ρ . Its dimension matrix is

$$(\dim_{\mathbb{C}} \mathcal{L}_{ab})_{a,b=1}^3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}. \quad (1081)$$

Multiplication and adjunction give the typed composition laws

$$\mathcal{L}_{ab} \mathcal{L}_{bc} \subseteq \mathcal{L}_{ac}, \quad \mathcal{L}_{ab} \mathcal{L}_{cd} = \{0\} \quad (b \neq c), \quad \mathcal{L}_{ab}^* = \mathcal{L}_{ba}. \quad (1082)$$

More precisely, multiplication induces a bimodule isomorphism

$$m_{abc} : \mathcal{L}_{ab} \otimes_{M_{n_b}(\mathbb{C})} \mathcal{L}_{bc} \longrightarrow \mathcal{L}_{ac}, \quad X \otimes Y \longmapsto XY. \quad (1083)$$

For every $1 \leq j \leq 2$ and $1 \leq k \leq 3$, the oriented three-sector path closes exactly:

$$E_{1,1;2,j} E_{2,j;3,k} E_{3,k;1,1} = E_{1,1;1,1}. \quad (1084)$$

Its modular weights and Hamiltonian weights obey

$$\frac{r_{1,1}}{r_{2,j}} \frac{r_{2,j}}{r_{3,k}} \frac{r_{3,k}}{r_{1,1}} = 1, \quad (1085)$$

$$\log \frac{r_{2,j}}{r_{1,1}} + \log \frac{r_{3,k}}{r_{2,j}} + \log \frac{r_{1,1}}{r_{3,k}} = 0. \quad (1086)$$

The six reversed paths are obtained by J_ρ , and

$$J_\rho \Delta_\rho J_\rho = \Delta_\rho^{-1}, \quad J_\rho K_\rho J_\rho = -K_\rho. \quad (1087)$$

Let

$$\mathfrak{D} = \mathcal{L}_{11} \oplus \mathcal{L}_{22} \oplus \mathcal{L}_{33} \cong M_1(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C}).$$

Every $$ -automorphism of $\mathfrak{D}$ fixes its three minimal central projections. In particular, no $*$ -automorphism of $\mathfrak{D}$ acts as the cycle $E_1 \mapsto E_2 \mapsto E_3 \mapsto E_1$. Equations (1082) and (1083) nevertheless provide typed compositions among all three diagonal blocks inside $M_6(\mathbb{C})$.*

Proof. Equation (1077) is theorem 14.4 with $n = 6$. Left and right multiplication by the block-diagonal matrices ρ and ρ^{-1} preserve each corner. Applying them to a matrix unit proves (1078) and (1079). The matrix units in distinct corners are Hilbert–Schmidt orthogonal, and each corner has $n_a n_b$ matrix units. This proves (1080) and (1081).

The identities $E_b E_c = \delta_{bc} E_b$ prove (1082). The map m_{abc} is surjective because

$$E_{a,i;c,k} = E_{a,i;b,1} E_{b,1;c,k}.$$

On matrix units, its inverse is

$$E_{a,i;c,k} \mapsto \frac{1}{n_b} \sum_{q=1}^{n_b} E_{a,i;b,q} \otimes E_{b,q;c,k}. \quad (1088)$$

The balancing relation over $M_{n_b}(\mathbb{C})$ shows that all nonzero summands on the right represent the same tensor and that tensors with unmatched middle indices vanish. Thus (1088) is a two-sided inverse.

Matrix-unit multiplication proves (1084). Multiplication of the three eigenvalues in (1078) telescopes to (1085); taking the real logarithms gives (1086). Since $J_\rho(X) = X^*$, it reverses every path. Direct calculation gives

$$J_\rho \Delta_\rho J_\rho(X) = \rho^{-1} X \rho = \Delta_\rho^{-1}(X),$$

and real functional calculus gives the second identity in (1087).

The center of $\mathfrak{D}$ is $\mathbb{C}E_1 \oplus \mathbb{C}E_2 \oplus \mathbb{C}E_3$. A $*$ -automorphism permutes its minimal central projections. If it sends E_a to E_b , it restricts to a complex-linear bijection

$$E_a \mathfrak{D} E_a \longrightarrow E_b \mathfrak{D} E_b.$$

The two spaces have dimensions n_a^2 and n_b^2 . Since $(n_1^2, n_2^2, n_3^2) = (1, 4, 9)$, equality forces $a = b$. Hence every minimal central projection is fixed. $\square$

Corollary 14.8 (The explicit 1–2–3 modular cycle). *For*

$$\rho_1 = (1), \quad \rho_2 = \text{diag}(2, 3), \quad \rho_3 = \text{diag}(5, 7, 11),$$

the three forward corner spectra are

$$\text{Spec}(\Delta_\rho|_{\mathcal{L}_{12}}) = \left\{ \frac{1}{2}, \frac{1}{3} \right\}, \quad (1089)$$

$$\text{Spec}(\Delta_\rho|_{\mathcal{L}_{23}}) = \left\{ \frac{2}{5}, \frac{2}{7}, \frac{2}{11}, \frac{3}{5}, \frac{3}{7}, \frac{3}{11} \right\}, \quad (1090)$$

$$\text{Spec}(\Delta_\rho|_{\mathcal{L}_{31}}) = \{5, 7, 11\}. \quad (1091)$$

For each of the six choices $(j, k) \in \{1, 2\} \times \{1, 2, 3\}$, the corresponding three entries multiply to 1, and their K_ρ -eigenvalues add to 0. The spectra on $\mathcal{L}_{21}, \mathcal{L}_{32}, \mathcal{L}_{13}$ are the termwise reciprocals.

Proof. Substitution in (1078) gives (1089)–(1091). The remaining assertions are (1085)–(1087). $\square$

Theorem 14.9 (Support-line factorization and the six-Hamiltonian D_3 orbit). *Use the ordered orthonormal basis*

$$e_\star; \quad f_0, f_1; \quad g_0, g_1, g_2$$

of $\mathcal{V} = \mathbb{C}^1 \oplus \mathbb{C}^2 \oplus \mathbb{C}^3$, and put

$$\begin{aligned} u_\epsilon &= |e_\star\rangle\langle f_\epsilon| \in \mathcal{L}_{12}, & v_\epsilon &= u_\epsilon^* \in \mathcal{L}_{21}, \\ p_m &= |e_\star\rangle\langle g_m| \in \mathcal{L}_{13}, & q_m &= p_m^* \in \mathcal{L}_{31}, \end{aligned} \quad (1092)$$

where $\epsilon \in C_2$ and $m \in C_3$. The matrix units in the second and third diagonal blocks and in the two mixed corners factor through the one-dimensional block as

$$v_\epsilon u_{\epsilon'} = |f_\epsilon\rangle\langle f_{\epsilon'}|, \quad q_m p_{m'} = |g_m\rangle\langle g_{m'}|, \quad (1093)$$

$$v_\epsilon p_m = |f_\epsilon\rangle\langle g_m|, \quad q_m u_\epsilon = |g_m\rangle\langle f_\epsilon|. \quad (1094)$$

In particular, with the row and column order $(0, 1)$, the ordered blade cells satisfy

$$2E_{21}^{(2)} = 2v_1 u_0, \quad E_{12}^{(2)} = v_0 u_1. \quad (1095)$$

For $\rho_2 = \text{diag}(2, 3)$, their exact modular data are

$$\Delta_\rho(2v_1 u_0) = \frac{3}{2}(2v_1 u_0), \quad K_\rho(2v_1 u_0) = -\log\left(\frac{3}{2}\right)(2v_1 u_0), \quad (1096)$$

$$\Delta_\rho(v_0 u_1) = \frac{2}{3}(v_0 u_1), \quad K_\rho(v_0 u_1) = \log\left(\frac{3}{2}\right)(v_0 u_1). \quad (1097)$$

Let

$$\gamma_{\epsilon,m} := u_\epsilon \otimes (v_\epsilon p_m) \otimes q_m, \quad \Gamma := \bigoplus_{\epsilon \in C_2, m \in C_3} \mathbb{C} \gamma_{\epsilon,m}. \quad (1098)$$

Here Γ is a subspace of $\mathcal{L}_{12} \otimes_{\mathbb{C}} \mathcal{L}_{23} \otimes_{\mathbb{C}} \mathcal{L}_{31}$. Multiplication restricts to

$$\mu : \Gamma \longrightarrow \mathcal{L}_{11}, \quad \mu(\gamma_{\epsilon,m}) = E_{1,1;1,1}. \quad (1099)$$

Consequently

$$\text{rank } \mu = 1, \quad \ker \mu = \left\{ \sum_{\epsilon,m} c_{\epsilon,m} \gamma_{\epsilon,m} : \sum_{\epsilon,m} c_{\epsilon,m} = 0 \right\}, \quad \dim \ker \mu = 5. \quad (1100)$$

Define permutation matrices by

$$S_2 f_\epsilon = f_{\epsilon+1}, \quad R_3 g_m = g_{m+1}, \quad F_3 g_m = g_{-m},$$

and set

$$W_r = 1 \oplus I_2 \oplus R_3, \quad W_f = 1 \oplus S_2 \oplus F_3. \quad (1101)$$

They obey

$$W_r^3 = W_f^2 = I_6, \quad W_f W_r W_f = W_r^{-1}. \quad (1102)$$

Their conjugation action preserves Γ and is

$$\text{Ad}_{W_r}(\gamma_{\epsilon,m}) = \gamma_{\epsilon,m+1}, \quad \text{Ad}_{W_f}(\gamma_{\epsilon,m}) = \gamma_{\epsilon+1,-m}. \quad (1103)$$

Thus the six basis paths form the regular D_3 -set, and μ is D_3 -invariant.

For the density

$$\rho = \text{diag}(1; 2, 3; 5, 7, 11), \quad (1104)$$

put

$$\rho^{(m,\epsilon)} := W_r^m W_f^\epsilon \rho W_f^\epsilon W_r^{-m}, \quad \Delta^{(m,\epsilon)} = \text{Ad}_{\rho^{(m,\epsilon)}}, \quad K^{(m,\epsilon)} = -\log \Delta^{(m,\epsilon)}. \quad (1105)$$

The six densities, listed in the order $(0, 0), (0, 1), (1, 0), (1, 1), (2, 0), (2, 1)$, are

$$\begin{aligned} (1, 2, 3, 5, 7, 11), & \quad (1, 3, 2, 5, 11, 7), \\ (1, 2, 3, 11, 5, 7), & \quad (1, 3, 2, 7, 5, 11), \\ (1, 2, 3, 7, 11, 5), & \quad (1, 3, 2, 11, 7, 5). \end{aligned} \quad (1106)$$

They are pairwise distinct. Their span has dimension four. In the same ordering, its two independent relations are

$$-3\rho^{(0,0)} - 7\rho^{(0,1)} - 5\rho^{(1,0)} + 7\rho^{(1,1)} + 8\rho^{(2,0)} = 0, \quad (1107)$$

$$7\rho^{(0,0)} - 5\rho^{(0,1)} - 7\rho^{(1,0)} - 3\rho^{(1,1)} + 8\rho^{(2,1)} = 0. \quad (1108)$$

All six modular operators, and hence all six modular Hamiltonians, are distinct. If $\mathcal{R} = \text{Ad}_{W_r}$ and $\mathcal{F} = \text{Ad}_{W_f}$ on the Hilbert–Schmidt space, then

$$\mathcal{R}\Delta^{(m,\epsilon)}\mathcal{R}^{-1} = \Delta^{(m+1,\epsilon)}, \quad \mathcal{R}K^{(m,\epsilon)}\mathcal{R}^{-1} = K^{(m+1,\epsilon)}, \quad (1109)$$

$$\mathcal{F}\Delta^{(m,\epsilon)}\mathcal{F}^{-1} = \Delta^{(-m,\epsilon+1)}, \quad \mathcal{F}K^{(m,\epsilon)}\mathcal{F}^{-1} = K^{(-m,\epsilon+1)}. \quad (1110)$$

Finally, the antiunitary

$$\mathcal{C}_f := \mathcal{F}J_\rho \quad (1111)$$

is an involution and satisfies

$$\mathcal{C}_f\Delta^{(m,\epsilon)}\mathcal{C}_f^{-1} = (\Delta^{(-m,\epsilon+1)})^{-1}, \quad \mathcal{C}_fK^{(m,\epsilon)}\mathcal{C}_f^{-1} = -K^{(-m,\epsilon+1)}. \quad (1112)$$

Proof. All identities in (1093)–(1094) are products of rank-one operators. Equation (1095) is the specialization $(\epsilon, \epsilon') = (1, 0)$ and $(0, 1)$. Applying (1078) and (1079) with $(r_{2,1}, r_{2,2}) = (2, 3)$ proves (1096) and (1097).

The six tensors $\gamma_{\epsilon,m}$ use six different triples of matrix units, so they are linearly independent. Their products are

$$u_\epsilon(v_\epsilon p_m)q_m = |e_\star\rangle\langle f_\epsilon||f_\epsilon\rangle\langle g_m||g_m\rangle\langle e_\star| = |e_\star\rangle\langle e_\star|.$$

This proves (1099) and (1100).

The definitions of S_2, R_3, F_3 give $S_2^2 = I_2$, $R_3^3 = F_3^2 = I_3$, and $F_3R_3F_3 = R_3^{-1}$. Hence (1102) holds. Conjugation sends a rank-one operator $|x\rangle\langle y|$ to $|Wx\rangle\langle Wy|$, which proves (1103). This action is transitive, and the stabilizer of $\gamma_{0,0}$ is the identity; it is therefore regular. Equation (1099) makes the invariance of μ immediate.

Conjugating the diagonal matrix in (1104) gives (1106). The entries 2, 3 determine ϵ , and the ordered positions of 5, 7, 11 determine m , so the six matrices are pairwise distinct. Direct substitution proves (1107) and (1108), giving rank at most four. The minor formed from coordinates 1, 2, 4, 5 and densities $(0, 0), (0, 1), (1, 0), (1, 1)$ has determinant -32 , giving rank at least four.

If two of the inner automorphisms in (1105) were equal, the quotient of their density matrices would commute with every element of $M_6(\mathbb{C})$ and would therefore be scalar. Their first diagonal entries are all one, so that scalar would be one. Pairwise distinction of the densities now proves pairwise distinction of the six modular operators. Functional calculus then proves the corresponding assertion for the generators. Conjugating $\rho^{(m,\epsilon)}$ by W_r and W_f , using (1102), proves (1109) and (1110). The matrices W_f and $\rho^{(m,\epsilon)}$ are real, while $W_f = W_f^* = W_f^{-1}$; hence $\mathcal{F}$ commutes with J_ρ and $\mathcal{C}_f^2 = I$. Combining the reflection formula with (1087) proves (1112). $\square$

Corollary 14.10 (Ordered blade pullback and the simplex-support D_3 action). *Use the ordered Helmert vectors*

$$\eta_0 = \frac{1}{\sqrt{2}}(1, -1, 0)^\top, \quad \eta_1 = \frac{1}{\sqrt{6}}(1, 1, -2)^\top \quad \text{in } \text{Ran } P_2, \quad (1113)$$

and

$$\begin{aligned} \theta_0 &= \frac{1}{\sqrt{2}}(1, -1, 0, 0)^\top, \\ \theta_1 &= \frac{1}{\sqrt{6}}(1, 1, -2, 0)^\top, \\ \theta_2 &= \frac{1}{\sqrt{12}}(1, 1, 1, -3)^\top \quad \text{in } \text{Ran } P_3. \end{aligned} \quad (1114)$$

For the support unitaries in theorem 14.5,

$$U_2^* f_\epsilon = \eta_\epsilon, \quad U_3^* g_m = \theta_m.$$

The two ordered blade cells in (1095) therefore pull back to the distinct support matrices

$$B_{-,+} := \Phi_2^{-1}(2E_{21}^{(2)}) = 2|\eta_1\rangle\langle\eta_0| = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 0 \\ -2 & 2 & 0 \end{pmatrix}, \quad (1115)$$

$$B_{+,-} := \Phi_2^{-1}(E_{12}^{(2)}) = |\eta_0\rangle\langle\eta_1| = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 & 1 & -2 \\ -1 & -1 & 2 \\ 0 & 0 & 0 \end{pmatrix}. \quad (1116)$$

Their ordered relation and products are

$$B_{-,+} = 2B_{+,-}^*, \quad (1117)$$

$$B_{-,+}B_{+,-} = 2|\eta_1\rangle\langle\eta_1|, \quad B_{+,-}B_{-,+} = 2|\eta_0\rangle\langle\eta_0|. \quad (1118)$$

The pulled positive matrix is

$$\tilde{\rho}_2 := \Phi_2^{-1}(\text{diag}(2, 3)) = \frac{1}{2} \begin{pmatrix} 3 & -1 & -2 \\ -1 & 3 & -2 \\ -2 & -2 & 4 \end{pmatrix}, \quad (1119)$$

and its modular operator satisfies

$$\Delta_{\tilde{\rho}_2}(B_{-,+}) = \frac{3}{2}B_{-,+}, \quad \Delta_{\tilde{\rho}_2}(B_{+,-}) = \frac{2}{3}B_{+,-}. \quad (1120)$$

Inside the support corners, define

$$\begin{aligned} \hat{S}_2 &:= \sum_{\epsilon \in C_2} |\eta_{\epsilon+1}\rangle\langle\eta_\epsilon|, \\ \hat{R}_3 &:= \sum_{m \in C_3} |\theta_{m+1}\rangle\langle\theta_m|, & \hat{F}_3 &:= \sum_{m \in C_3} |\theta_{-m}\rangle\langle\theta_m|. \end{aligned} \quad (1121)$$

They obey

$$\hat{S}_2^2 = P_2, \quad \hat{R}_3^3 = \hat{F}_3^2 = P_3, \quad \hat{F}_3\hat{R}_3\hat{F}_3 = \hat{R}_3^*. \quad (1122)$$

With $P = P_1 \oplus P_2 \oplus P_3$, put

$$\widehat{W}_r = P_1 \oplus P_2 \oplus \hat{R}_3, \quad \widehat{W}_f = P_1 \oplus \hat{S}_2 \oplus \hat{F}_3. \quad (1123)$$

Then

$$\widehat{W}_r^3 = \widehat{W}_f^2 = P, \quad \widehat{W}_f\widehat{W}_r\widehat{W}_f = \widehat{W}_r^*, \quad \Phi(\widehat{W}_r) = W_r, \quad \Phi(\widehat{W}_f) = W_f. \quad (1124)$$

Hence the regular six-element D_3 path orbit and its six modular Hamiltonians in theorem 14.9 have an exact realization on the three simplex-support corners.

Proof. The vectors in (1113)–(1114) are orthonormal bases of the all-one orthogonal complements in $\mathbb{C}^3$ and $\mathbb{C}^4$. In those bases, the support unitaries are the ordered Helmert matrices. Therefore

$$\Phi_2^{-1}(E_{ij}^{(2)}) = U_2^* E_{ij}^{(2)} U_2 = |\eta_i\rangle\langle\eta_j|.$$

Substitution gives (1115)–(1116). Taking adjoints and multiplying the two rank-one operators proves (1117)–(1118).

Conjugating $\text{diag}(2, 3)$ by U_2^* and U_2 gives (1119). The modular intertwining identity (1073), together with (1096)–(1097), gives (1120).

The operators in (1121) act by

$$\eta_\epsilon \longmapsto \eta_{\epsilon+1}, \quad \theta_m \longmapsto \theta_{m+1}, \quad \theta_m \longmapsto \theta_{-m},$$

respectively, and vanish on the all-one kernel lines. Their products are therefore the support projections and the support inverse displayed in (1122). Direct sums give the first three identities in (1124). The last two follow from $U_2\hat{S}_2U_2^* = S_2$, $U_3\hat{R}_3U_3^* = R_3$, and $U_3\hat{F}_3U_3^* = F_3$. Conjugation

by the $*$ -isomorphism Φ now transports the complete path action and the six conjugated modular densities from theorem 14.9. $\square$

Exact computation 14.11 (Exact check of the ordered blade and D_3 support pullback). The script and captured receipt

```
verification/verify_simplex_blade_d3_pullback.py,
verification/verify_simplex_blade_d3_pullback_output.txt
```

verify the two displayed 3×3 blade matrices, their adjoint relation, their two ordered rank-one products, the pulled positive matrix, the modular weights $3/2$ and $2/3$, the support-corner C_2 and C_3 relations, and the full 1–2–3 D_3 intertwining relation.

Corollary 14.12 (Exact simplex-support pullback of the two arithmetic raw blade carriers). *Use the ordered support unitary U_2 of theorem 14.5 and the two arithmetic raw blade coefficients*

$$N_{-,+} = 2E_{21} \quad \text{at residue 53}, \quad N_{+,-} = E_{12} \quad \text{at residue 109}. \quad (1125)$$

For the directed Gram-support carrier of (177), one has

$$\mathfrak{H}_{-,+} = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathfrak{H}_{+,-} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (1126)$$

Their inverse images in the simplex-support corner are

$$\tilde{\mathfrak{H}}_{-,+} := \Phi_2^{-1}(\mathfrak{H}_{-,+}) = \frac{1}{6} \begin{pmatrix} 11 & -13 & 2 \\ -13 & 11 & 2 \\ 2 & 2 & -4 \end{pmatrix}, \quad (1127)$$

$$\tilde{\mathfrak{H}}_{+,-} := \Phi_2^{-1}(\mathfrak{H}_{+,-}) = \frac{1}{3} \begin{pmatrix} -1 & 2 & -1 \\ 2 & -1 & -1 \\ -1 & -1 & 2 \end{pmatrix}. \quad (1128)$$

Both matrices belong to $P_2 M_3(\mathbb{R}) P_2$; equivalently,

$$\tilde{\mathfrak{H}}_{-,+} \mathbf{1}_3 = 0, \quad \tilde{\mathfrak{H}}_{+,-} \mathbf{1}_3 = 0. \quad (1129)$$

Under the Hermitian circle and Sarnak coordinate map of theorem 9.10, the complete data are

N	$q_N(x, y)$	(T, X, Y, W)	$\mathcal{N}$	c/a	$\Phi_2^{-1}(\mathfrak{H}_N)$
$2E_{21}$	$4(x^2 + y^2) - 1$	$(\frac{3}{2}, 0, 0, \frac{5}{2})$	4	$-\frac{1}{4}$	$\tilde{\mathfrak{H}}_{-,+}$
E_{12}	$1 - (x^2 + y^2)$	$(0, 0, 0, -1)$	1	-1	$\tilde{\mathfrak{H}}_{+,-}$

(1130)

Thus the ordered $(-, +)$ and raw transposed $(+, -)$ channels are carried into the same simplex-support corner while retaining different circle coefficients, Lorentz vectors, norms, and simple ratios. In particular,

$$\mathfrak{H}(2E_{12}) = \begin{pmatrix} -1 & 0 \\ 0 & 4 \end{pmatrix} \neq \mathfrak{H}(E_{12}) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (1131)$$

The first matrix belongs to the transpose of the multiplicity-two champion coefficient; the second belongs to the actual multiplicity-one raw cell at residue 109.

Proof. In the ordered Helmholtz basis of (1113),

$$U_2 = \begin{pmatrix} \eta_0^\top \\ \eta_1^\top \end{pmatrix}, \quad \Phi_2^{-1}(A) = U_2^* A U_2.$$

The two raw carrier matrices in (1126) are the specializations (182) and (183). Direct multiplication by U_2^* and U_2 gives (1127) and (1128). Since $U_2 \mathbf{1}_3 = 0$, (1129) follows.

For a diagonal carrier

$$H = \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix},$$

theorem 9.10 gives

$$q_H(x, y) = a(x^2 + y^2) + c, \quad (T, X, Y, W) = \left(\frac{a+c}{2}, 0, 0, \frac{a-c}{2} \right), \quad \mathcal{N} = -ac.$$

Substitution of $(a, c) = (4, -1)$ and $(-1, 1)$ proves (1130). Finally, applying (177) to $2E_{12}$ and E_{12} gives the two matrices in (1131). $\square$

Exact computation 14.13 (Exact check of the simplex-support Lorentz composition). The script and captured receipt

```
verification/verify_simplex_blade_lorentz_composition.py,
verification/verify_simplex_blade_lorentz_composition_output.txt
```

verify the two pulled 3×3 carriers, their all-one kernel line, the two circle equations, the two Sarnak vectors and norms, and the exact simple ratios $-1/4$ and -1 .

Corollary 14.14 (Unit-circle inversion distinguishes the multiplicity-two transpose from the raw transposed cell). *Let*

$$u = (1, 0, 0, -1), \quad q_u(x, y) = x^2 + y^2 - 1,$$

and let $\mathcal{J} = M_u$ be the coefficient action of inversion in $q_u = 0$ from theorem 6.2. Then

$$\mathcal{J} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad \mathcal{J}^2 = I_4. \quad (1132)$$

For the two arithmetic raw cells in (1125),

$$\mathcal{J}(4, 0, 0, -1) = (-1, 0, 0, 4), \quad (1133)$$

$$\mathcal{J}(-1, 0, 0, 1) = (1, 0, 0, -1) = -(-1, 0, 0, 1). \quad (1134)$$

Consequently inversion maps the circle of radius $1/2$,

$$4(x^2 + y^2) - 1 = 0,$$

to the circle of radius 2,

$$4 - (x^2 + y^2) = 0.$$

The unit circle represented by the raw multiplicity-one cell is fixed as a point set and its oriented coefficient vector changes sign.

In Sarnak coordinates the same action is

$$(T, X, Y, W) \mapsto (T, X, Y, -W). \quad (1135)$$

In particular,

$$\left(\frac{3}{2}, 0, 0, \frac{5}{2} \right) \mapsto \left(\frac{3}{2}, 0, 0, -\frac{5}{2} \right), \quad (0, 0, 0, -1) \mapsto (0, 0, 0, 1). \quad (1136)$$

Let

$$S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{S}_2 := \Phi_2^{-1}(S_2) = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{pmatrix}. \quad (1137)$$

Then $\hat{S}_2^2 = P_2$. If

$$\tilde{\mathfrak{H}}_{+,-}^{(2)} := \Phi_2^{-1} \begin{pmatrix} -1 & 0 \\ 0 & 4 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 1 & 7 & -8 \\ 7 & 1 & -8 \\ -8 & -8 & 16 \end{pmatrix}, \quad (1138)$$

then the exact support-corner actions are

$$\widehat{S}_2 \widetilde{\mathfrak{H}}_{-,+} \widehat{S}_2 = \widetilde{\mathfrak{H}}_{+,-}^{(2)}, \quad (1139)$$

$$\widehat{S}_2 \widetilde{\mathfrak{H}}_{+,-} \widehat{S}_2 = -\widetilde{\mathfrak{H}}_{+,-}. \quad (1140)$$

Proof. For $u = (1, 0, 0, -1)$, the quantities in theorem 6.2 are $A = 1$ and $\Delta = 1$. Substitution in (29) gives (1132). Applying that matrix to the two vectors in (1130) proves (1133) and (1134). The circle equations follow directly from the first and fourth coefficients.

The Sarnak coordinates satisfy

$$T = \frac{a+c}{2}, \quad X = b_0, \quad Y = b_1, \quad W = \frac{a-c}{2}.$$

Exchanging a and c proves (1135) and (1136).

In the ordered Helmert basis,

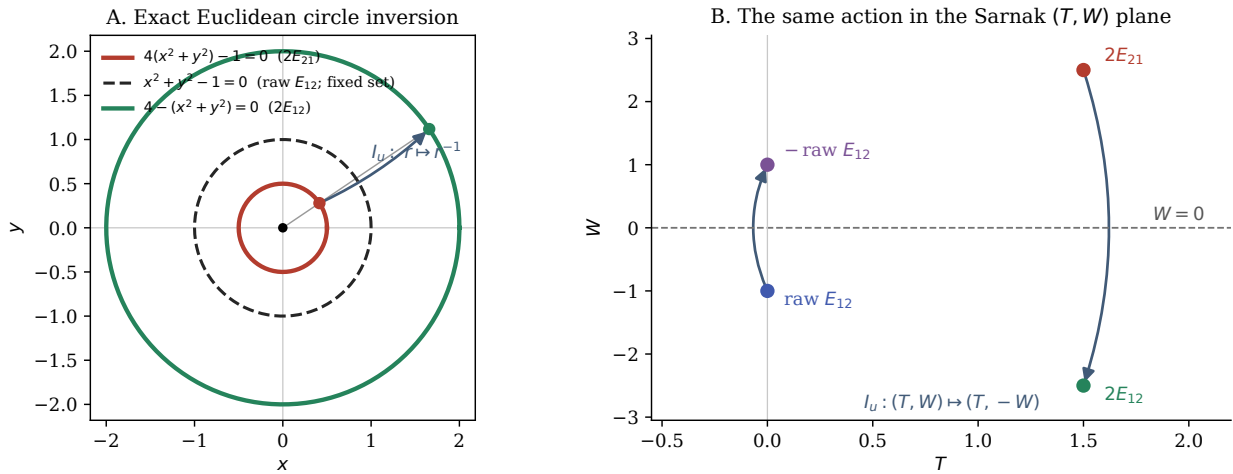
$$\widehat{S}_2 = |\eta_0\rangle\langle\eta_1| + |\eta_1\rangle\langle\eta_0|.$$

This gives the matrix in (1137) and $\widehat{S}_2^2 = P_2$. Since Φ_2 is a unital $*$ -isomorphism on the support corner,

$$\widehat{S}_2 \Phi_2^{-1}(H) \widehat{S}_2 = \Phi_2^{-1}(S_2 H S_2)$$

for every $H \in M_2(\mathbb{R})$. Applying this identity to $\text{diag}(4, -1)$ and $\text{diag}(-1, 1)$, followed by direct multiplication with U_2^* and U_2 , proves (1138)–(1140). $\square$

Unit-circle inversion retains the arithmetic multiplicity distinction



$M_U(a, b_0, b_1, c) = (c, b_0, b_1, a)$; the raw E_{12} circle is projectively fixed, whereas $2E_{21}$ maps to $2E_{12}$.

FIGURE 11. The exact unit-circle inversion of theorem 14.14. In the Euclidean plane, the radius-1/2 champion circle maps to the radius-2 multiplicity-two transposed circle, while the raw multiplicity-one unit circle is fixed as a point set and changes coefficient orientation. In Sarnak coordinates, both actions are the reflection $(T, W) \mapsto (T, -W)$.

Exact computation 14.15 (Exact check of unit-circle inversion in the simplex-support corner). The script and captured receipt

```
verification/verify_simplex_unit_circle_inversion.py,
verification/verify_simplex_unit_circle_inversion_output.txt
```

verify the coefficient involution, both arithmetic images, the Sarnak reflection, the pulled 3×3 sheet exchange, and both support-corner conjugation identities.

Theorem 14.16 (Support-indexed modular family). *Let $N \geq 1$, let $A \subseteq C_N$, and, with the Fourier vectors of theorem 13.77, put*

$$\Pi_A := \frac{1}{N} T_A, \quad Q_A := I_N - \Pi_A. \quad (1141)$$

For $N = 1$, the same formula uses $v_0 = (1)$. Let $\lambda > 0$, $\lambda \neq 1$, and define

$$\rho_{A,\lambda} := \Pi_A + \lambda Q_A, \quad \Omega_{A,\lambda} := \rho_{A,\lambda}^{1/2}. \quad (1142)$$

Then Π_A and Q_A are orthogonal projections, $\rho_{A,\lambda}$ is positive and invertible, and

$$\rho_{A,\lambda}^{-1} = \Pi_A + \lambda^{-1} Q_A. \quad (1143)$$

Represent $M_N(\mathbb{C})$ on its Hilbert–Schmidt space by left multiplication and use the standard vector $\Omega_{A,\lambda}$. Its modular operator and modular Hamiltonian are

$$\Delta_{A,\lambda}(X) = \rho_{A,\lambda} X \rho_{A,\lambda}^{-1}, \quad K_{A,\lambda} := -\log \Delta_{A,\lambda}. \quad (1144)$$

The Hilbert–Schmidt orthogonal decomposition

$$M_N(\mathbb{C}) = \Pi_A M_N \Pi_A \hat{\oplus} \Pi_A M_N Q_A \hat{\oplus} Q_A M_N \Pi_A \hat{\oplus} Q_A M_N Q_A \quad (1145)$$

reduces both operators. Their restrictions are

$\mathcal{E}$	$\Pi_A M_N \Pi_A$	$\Pi_A M_N Q_A$	$Q_A M_N \Pi_A$	$Q_A M_N Q_A$
$\Delta_{A,\lambda} \mathcal{E}$	1	λ^{-1}	λ	1
$K_{A,\lambda} \mathcal{E}$	0	$\log \lambda$	$-\log \lambda$	0.

(1146)

If $r = |A|$, the multiplicities of $0, \log \lambda, -\log \lambda$ in $K_{A,\lambda}$ are

$$r^2 + (N - r)^2, \quad r(N - r), \quad r(N - r), \quad (1147)$$

respectively.

Suppose W is unitary or antiunitary on $\mathbb{C}^N$ and

$$W \Pi_A W^{-1} = \Pi_B. \quad (1148)$$

The induced map $\widehat{W}(X) := W X W^{-1}$ satisfies

$$\widehat{W} K_{A,\lambda} \widehat{W}^{-1} = K_{B,\lambda}. \quad (1149)$$

For $t \in \mathbb{R}$,

$$\widehat{W} \Delta_{A,\lambda}^{it} \widehat{W}^{-1} = \begin{cases} \Delta_{B,\lambda}^{it}, & W \text{ unitary,} \\ \Delta_{B,\lambda}^{-it}, & W \text{ antiunitary.} \end{cases} \quad (1150)$$

For singleton support, the multiplicity triples for $N = 1, 2, 3$ are

N	mult(0)	mult($\log \lambda$)	mult($-\log \lambda$)
1	1	0	0
2	2	1	1
3	5	2	2.

(1151)

At $A = \emptyset$ and $A = C_N$, one has $\Delta_{A,\lambda} = I$ and $K_{A,\lambda} = 0$. The support class therefore remains part of the datum; the modular operator alone identifies these two extreme cases.

Proof. Fourier orthogonality (1031) gives

$$\Pi_A^2 = \Pi_A = \Pi_A^*, \quad Q_A^2 = Q_A = Q_A^*, \quad \Pi_A Q_A = 0.$$

Equations (1142) and (1143) follow by restricting to the two orthogonal ranges. The finite Hilbert–Schmidt standard form is theorem 14.4.

For $X \in \Pi_A M_N Q_A$,

$$\rho_{A,\lambda} X \rho_{A,\lambda}^{-1} = \Pi_A X Q_A \lambda^{-1} = \lambda^{-1} X.$$

The other three channels give $1, \lambda, 1$ in the displayed order. Applying $-\log$ proves (1146). Their dimensions are

$$r^2, \quad r(N-r), \quad (N-r)r, \quad (N-r)^2,$$

which proves (1147) and (1151).

Hypothesis (1148) gives

$$W \rho_{A,\lambda} W^{-1} = \rho_{B,\lambda}.$$

Real functional calculus gives (1149). Complex functional calculus preserves it for linear W and conjugates it to $-it$ for antiunitary W , proving (1150). The two extreme cases follow directly from $\Pi_\emptyset = 0$ and $\Pi_{C_N} = I_N$. $\square$

Corollary 14.17 (C_3 -transport of the support-indexed blocks). *Let $N = 3$, let $Ue_j = e_{j+1}$, and retain the character projectors P_k from (1050). Then*

$$\Pi_{\{a\}} = P_{-a}, \quad \Pi_A = \sum_{a \in A} P_{-a}. \quad (1152)$$

If C is antiunitary and $CUC^{-1} = U^\varepsilon$, $\varepsilon \in \{1, -1\}$, then

$$C\Pi_A C^{-1} = \Pi_{-\varepsilon A}, \quad \widehat{C} K_{A,\lambda} \widehat{C}^{-1} = K_{-\varepsilon A, \lambda}, \quad (1153)$$

and

$$\widehat{C} \Delta_{A,\lambda}^{it} \widehat{C}^{-1} = \Delta_{-\varepsilon A, \lambda}^{-it}. \quad (1154)$$

Thus an antiunitary commuting with U sends A to $-A$, while an antiunitary sending U to U^{-1} retains A .

Proof. The cyclic shift satisfies

$$Uv_a = \omega_3^{-a} v_a.$$

Hence $N^{-1}v_a v_a^* = P_{-a}$, proving (1152). Proposition 14.2 gives

$$CP_{-a}C^{-1} = P_{\varepsilon a} = \Pi_{\{-\varepsilon a\}}.$$

Summing over $a \in A$ and applying theorem 14.16 proves the remaining formulas. $\square$

Theorem 14.18 (Exact $N = 2$ Fourier realization of the ordered blade channels). *Let $a \in C_2$, write $\bar{a} = a + 1 \pmod{2}$, and set*

$$f_a := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ (-1)^a \end{pmatrix}, \quad f_{\bar{a}} := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -(-1)^a \end{pmatrix}, \quad U_a := (f_a \ f_{\bar{a}}). \quad (1155)$$

For the singleton support $A = \{a\}$, one has

$$\Pi_A = f_a f_a^*, \quad Q_A = f_{\bar{a}} f_{\bar{a}}^*, \quad U_a^* \Pi_A U_a = E_{11}, \quad U_a^* Q_A U_a = E_{22}. \quad (1156)$$

Consequently, the unitary carriage $\Phi_a(X) := U_a^ X U_a$ gives the ordered channel identities*

$$\begin{aligned} \Phi_a(\Pi_A M_2(\mathbb{C}) \Pi_A) &= \mathbb{C} E_{11}, & \Phi_a(\Pi_A M_2(\mathbb{C}) Q_A) &= \mathbb{C} E_{12}, \\ \Phi_a(Q_A M_2(\mathbb{C}) \Pi_A) &= \mathbb{C} E_{21}, & \Phi_a(Q_A M_2(\mathbb{C}) Q_A) &= \mathbb{C} E_{22}. \end{aligned} \quad (1157)$$

In particular,

$$K_{A,\lambda}(f_a f_{\bar{a}}^*) = (\log \lambda) f_a f_{\bar{a}}^*, \quad \Phi_a(f_a f_{\bar{a}}^*) = E_{12}, \quad (1158)$$

$$K_{A,\lambda}(f_{\bar{a}} f_a^*) = -(\log \lambda) f_{\bar{a}} f_a^*, \quad \Phi_a(f_{\bar{a}} f_a^*) = E_{21}. \quad (1159)$$

After the ordered source–range identification

$$\Phi_a(\Pi_A) = Q_s = E_{11}, \quad \Phi_a(Q_A) = Q_r = E_{22}, \quad (1160)$$

the observed arithmetic coefficient and its transpose satisfy

$$N = 2E_{21} \in Q_r M_2 Q_s, \quad N^\top = 2E_{12} \in Q_s M_2 Q_r. \quad (1161)$$

Thus the ordered $(-, +)$ blade is the $-\log \lambda$ modular channel and the ordered $(+, -)$ blade is the $+\log \lambda$ modular channel.

The ordered channel lines in (1157) depend only on the ordered pair (Π_A, Q_A) . More precisely, if V is unitary and

$$V^* \Pi_A V = E_{11}, \quad V^* Q_A V = E_{22},$$

then

$$V = U_a D, \quad D = \text{diag}(\eta_s, \eta_r), \quad |\eta_s| = |\eta_r| = 1, \quad (1162)$$

and

$$D^* E_{21} D = \bar{\eta}_r \eta_s E_{21}, \quad D^* E_{12} D = \bar{\eta}_s \eta_r E_{12}. \quad (1163)$$

Hence the two one-dimensional subspaces and their order are intrinsic to (Π_A, Q_A) ; an individual generator carries reciprocal phase freedom. The coefficient computation $\mathcal{M}(53) = 2E_{21}$ in theorem 9.9 fixes the displayed arithmetic generator in the ordered blade basis.

Proof. For $N = 2$, the vectors in (1024) are $\sqrt{2}f_0$ and $\sqrt{2}f_1$. They are orthogonal, so U_a is unitary and

$$\Pi_A = \frac{1}{2} v_a v_a^* = f_a f_a^*, \quad Q_A = f_{\bar{a}} f_{\bar{a}}^*.$$

This proves (1156). For $i, j \in \{a, \bar{a}\}$,

$$U_a^*(f_i f_j^*) U_a = (U_a^* f_i)(U_a^* f_j)^*.$$

Taking the ordered pairs $(i, j) = (a, a), (a, \bar{a}), (\bar{a}, a), (\bar{a}, \bar{a})$ gives $E_{11}, E_{12}, E_{21}, E_{22}$, respectively, proving (1157). Equations (1158) and (1159) now follow from (1146). The blade source and range projectors are (341), which proves (1160) and (1161).

The first column of V is a unit vector in $\text{ran } \Pi_A = \mathbb{C}f_a$, and its second column is a unit vector in $\text{ran } Q_A = \mathbb{C}f_{\bar{a}}$. This gives (1162). Direct multiplication by D proves (1163). $\square$

Corollary 14.19 (The $N = 2$ support exchange reverses the directed channels). *Let $A = \{a\}$ and $A^c = \{\bar{a}\}$. Then*

$$\rho_{A^c, \lambda} = \lambda \rho_{A, \lambda}^{-1}, \quad \Delta_{A^c, \lambda} = \Delta_{A, \lambda}^{-1}, \quad K_{A^c, \lambda} = -K_{A, \lambda}. \quad (1164)$$

If $S = E_{12} + E_{21}$, then

$$SE_{21}S = E_{12}, \quad SE_{12}S = E_{21}.$$

Thus exchange of the ordered source and range interchanges the two blade lines and reverses their modular eigenvalues.

Proof. For singleton support in C_2 , $\Pi_{A^c} = Q_A$ and $Q_{A^c} = \Pi_A$. Therefore

$$\rho_{A^c, \lambda} = Q_A + \lambda \Pi_A = \lambda(\Pi_A + \lambda^{-1} Q_A) = \lambda \rho_{A, \lambda}^{-1}.$$

The scalar factor cancels from left–right modular conjugation, giving the last two identities in (1164). The two matrix-unit identities are direct products. $\square$

Corollary 14.20 (Exact modular–Lorentz orientation identity). *Retain the ordered source–range identification (1160), and put*

$$J = E_{11} - E_{22}, \quad h_\lambda := (\log \lambda) E_{22}.$$

The transported modular Hamiltonian on $M_2(\mathbb{C})$ is

$$\widetilde{K}_\lambda := \Phi_a K_{A,\lambda} \Phi_a^{-1} = -\operatorname{ad}_{h_\lambda} = \frac{\log \lambda}{2} \operatorname{ad}_J. \quad (1165)$$

For $m > 0$, define the two ordered blade cells

$$X_- = mE_{21}, \quad X_+ = mE_{12},$$

where the subscript records the sign of the modular eigenvalue. Their source–Gram minus occupied–range forms are

$$\mathfrak{H}_-(m) := X_-^\top X_- - \frac{1}{m^2} X_- X_-^\top = m^2 E_{11} - E_{22}, \quad (1166)$$

$$\mathfrak{H}_+(m) := X_+^\top X_+ - \frac{1}{m^2} X_+ X_+^\top = m^2 E_{22} - E_{11}. \quad (1167)$$

Writing

$$T_m = \frac{m^2 - 1}{2}, \quad W_m = \frac{m^2 + 1}{2},$$

these identities become

$$\mathfrak{H}_-(m) = T_m I_2 + W_m J, \quad \mathfrak{H}_+(m) = T_m I_2 - W_m J. \quad (1168)$$

If

$$\widetilde{K}_\lambda(X_\pm) = \kappa_\pm X_\pm, \quad \kappa_- = -\log \lambda, \quad \kappa_+ = \log \lambda,$$

then the fourth Sarnak coordinate satisfies the single exact formula

$$W_J(\mathfrak{H}_\pm(m)) = -\frac{m^2 + 1}{2 \log \lambda} \kappa_\pm. \quad (1169)$$

The corresponding coefficient ratios are

$$\frac{T_m - W_J(\mathfrak{H}_-(m))}{T_m + W_J(\mathfrak{H}_-(m))} = -\frac{1}{m^2}, \quad \frac{T_m - W_J(\mathfrak{H}_+(m))}{T_m + W_J(\mathfrak{H}_+(m))} = -m^2. \quad (1170)$$

At $m = 2$, the $(-, +)$ arithmetic cell $2E_{21}$ gives

$$(\kappa, T, W_J) = \left(-\log \lambda, \frac{3}{2}, \frac{5}{2} \right), \quad \frac{C}{A} = -\frac{1}{4},$$

whereas the transposed $(+, -)$ cell $2E_{12}$ gives

$$(\kappa, T, W_J) = \left(\log \lambda, \frac{3}{2}, -\frac{5}{2} \right), \quad \frac{C}{A} = -4.$$

Proof. In the ordered blade basis, $\rho_{A,\lambda} = E_{11} + \lambda E_{22}$. Therefore

$$\widetilde{K}_\lambda(X) = -[(\log \lambda) E_{22}, X].$$

Since $E_{22} = (I_2 - J)/2$, this is (1165). The four identities

$$J E_{21} = -E_{21}, \quad E_{21} J = E_{21}, \quad J E_{12} = E_{12}, \quad E_{12} J = -E_{12}$$

give $\kappa_- = -\log \lambda$ and $\kappa_+ = \log \lambda$.

Direct multiplication gives

$$X_-^\top X_- = m^2 E_{11}, \quad \frac{1}{m^2} X_- X_-^\top = E_{22},$$

and transposition exchanges E_{11} and E_{22} . This proves (1166) and (1167). Expanding the two diagonal matrices in the basis I_2, J proves (1168). Substitution of $\kappa_{\pm}$ gives (1169). The two fractions in (1170) follow from $A = T + W_J$ and $C = T - W_J$. Setting $m = 2$ gives the final two rows. $\square$

Theorem 14.21 (The quarter reflection is a quadratic polynomial in the modular Hamiltonian). *Let $\ell = \log \lambda \neq 0$, and retain the transported modular Hamiltonian $\widetilde{K}_\lambda$ from (1165). On the raw four-cell space,*

$$\widetilde{K}_\lambda \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & \ell b \\ -\ell c & 0 \end{pmatrix}. \quad (1171)$$

Its three spectral projectors are

$$\mathcal{P}_-^K := \frac{\widetilde{K}_\lambda(\widetilde{K}_\lambda - \ell I_{\text{End}})}{2\ell^2}, \quad \mathcal{P}_-^K(M) = cE_{21}, \quad (1172)$$

$$\mathcal{P}_+^K := \frac{\widetilde{K}_\lambda(\widetilde{K}_\lambda + \ell I_{\text{End}})}{2\ell^2}, \quad \mathcal{P}_+^K(M) = bE_{12}, \quad (1173)$$

$$\mathcal{P}_0^K := I_{\text{End}} - \frac{\widetilde{K}_\lambda^2}{\ell^2}, \quad \mathcal{P}_0^K(M) = aE_{11} + dE_{22}. \quad (1174)$$

The negative modular projector equals the negative raw quarter projector, and the positive raw quarter projector is the sum of the other two:

$$\mathcal{P}_-^K = \Pi_{\square,-}, \quad \Pi_{\square,+} = \mathcal{P}_0^K + \mathcal{P}_+^K. \quad (1175)$$

Equivalently, the pulled-back quarter reflection satisfies

$$\mathfrak{Q}_{\square} = I_{\text{End}} + \frac{\widetilde{K}_\lambda}{\ell} - \left(\frac{\widetilde{K}_\lambda}{\ell} \right)^2. \quad (1176)$$

Transport to the Hermitian circle coefficient space gives

$$4\mathcal{I}_- = \mathcal{T}_{\square} \left[I_{\text{End}} + \frac{\widetilde{K}_\lambda}{\ell} - \left(\frac{\widetilde{K}_\lambda}{\ell} \right)^2 \right] \mathcal{T}_{\square}^{-1}. \quad (1177)$$

The same transporter sends the source and range idempotents to the null Hermitian pair

$$\mathcal{T}_{\square}(E_{11}) = H \left(1, \frac{1}{2}, 0, \frac{1}{4} \right), \quad u_{\text{s}} := \left(\frac{5}{8}, \frac{1}{2}, 0, \frac{3}{8} \right), \quad (1178)$$

$$\mathcal{T}_{\square}(E_{22}) = H \left(1, -\frac{1}{2}, 0, \frac{1}{4} \right), \quad u_{\text{r}} := \left(\frac{5}{8}, -\frac{1}{2}, 0, \frac{3}{8} \right). \quad (1179)$$

For the Lorentz pairing

$$\langle u, v \rangle_L := X_u X_v + Y_u Y_v + W_u W_v - T_u T_v,$$

one has

$$\langle u_{\text{s}}, u_{\text{s}} \rangle_L = \langle u_{\text{r}}, u_{\text{r}} \rangle_L = 0, \quad \langle u_{\text{s}}, u_{\text{r}} \rangle_L = -\frac{1}{2}, \quad (1180)$$

and

$$\langle u_{\text{s}} + u_{\text{r}}, u_{\text{s}} + u_{\text{r}} \rangle_L = -1, \quad \langle u_{\text{s}} - u_{\text{r}}, u_{\text{s}} - u_{\text{r}} \rangle_L = 1. \quad (1181)$$

Proof. Equation (1171) follows by substituting $J = \text{diag}(1, -1)$ into (1165). The eigenvalues on $(E_{11}, E_{12}, E_{21}, E_{22})$ are $(0, \ell, -\ell, 0)$. Evaluating the three displayed polynomials at these four eigenvalues proves (1172)–(1174). Comparison with (205) and (206) gives (1175). Since

$$\mathfrak{Q}_{\square} = I_{\text{End}} - 2\Pi_{\square,-},$$

substitution of (1172) proves (1176). Conjugating by $\mathcal{T}_{\square}$ and using (202) proves (1177).

The two transported matrices are the first and fourth formulas in (196). Applying (189) gives u_s and u_r . Direct substitution in the Lorentz pairing proves (1180). Their sum and difference are

$$u_s + u_r = \left(\frac{5}{4}, 0, 0, \frac{3}{4}\right), \quad u_s - u_r = (0, 1, 0, 0),$$

which proves (1181). $\square$

Corollary 14.22 (The support-line blade plane intertwines the modular and quarter operators). *Let*

$$\mathcal{E}_2 := E_2 M_6(\mathbb{C}) E_2$$

and define the block coordinate map

$$\beta_2 : \mathcal{E}_2 \longrightarrow M_2(\mathbb{C}), \quad \beta_2(|f_i\rangle\langle f_j|) = E_{i+1,j+1}, \quad i, j \in \{0, 1\}. \quad (1182)$$

It is a $*$ -algebra isomorphism. Put

$$\mathcal{O}_2 := \text{span}_{\mathbb{R}}\{v_0 u_1, v_1 u_0\} \subset \mathcal{E}_2, \quad \mathcal{V}_{\pm} := \text{span}_{\mathbb{R}}\{\mathcal{T}_{\square}(E_{12}), \mathcal{T}_{\square}(E_{21})\}, \quad (1183)$$

$$\Phi_{\square} := \mathcal{T}_{\square} \circ \beta_2|_{\mathcal{O}_2} : \mathcal{O}_2 \longrightarrow \mathcal{V}_{\pm}, \quad \ell := \log\left(\frac{3}{2}\right). \quad (1184)$$

Then $\Phi_{\square}$ is a real-linear isomorphism and

$$\Phi_{\square}(K_{\rho} X) = 4\ell \mathcal{I}_{-}(\Phi_{\square} X), \quad X \in \mathcal{O}_2. \quad (1185)$$

Equivalently,

$$\Phi_{\square}(\Delta_{\rho} X) = \exp(-4\ell \mathcal{I}_{-})(\Phi_{\square} X), \quad X \in \mathcal{O}_2. \quad (1186)$$

The two arithmetic generators retain their ordered coefficients:

$$\Phi_{\square}(2v_1 u_0) = \mathcal{T}_{\square}(2E_{21}) = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathcal{I}_{-}\Phi_{\square}(2v_1 u_0) = -\frac{1}{4}\Phi_{\square}(2v_1 u_0), \quad (1187)$$

$$\Phi_{\square}(v_0 u_1) = \mathcal{T}_{\square}(E_{12}) = H(0, 0, 1, 0), \quad \mathcal{I}_{-}\Phi_{\square}(v_0 u_1) = \frac{1}{4}\Phi_{\square}(v_0 u_1). \quad (1188)$$

Thus (1185) is a typed morphism of real operator modules from the second linking block to the ordered quarter carrier.

Proof. Products and adjoints of the rank-one matrices $|f_i\rangle\langle f_j|$ show directly that

$$\beta_2(XY) = \beta_2(X)\beta_2(Y), \quad \beta_2(X^*) = \beta_2(X)^*. \quad (1189)$$

The inverse sends $E_{i+1,j+1}$ to $|f_i\rangle\langle f_j|$, proving the asserted $*$ -algebra isomorphism. Equations (1095), (1096), and (1097) give

$$K_{\rho}(v_1 u_0) = -\ell v_1 u_0, \quad K_{\rho}(v_0 u_1) = \ell v_0 u_1. \quad (1190)$$

On the target lines, (197) and (198) give

$$4\mathcal{I}_{-}\mathcal{T}_{\square}(E_{21}) = -\mathcal{T}_{\square}(E_{21}), \quad 4\mathcal{I}_{-}\mathcal{T}_{\square}(E_{12}) = \mathcal{T}_{\square}(E_{12}). \quad (1191)$$

The two target vectors are linearly independent by (196). Hence $\Phi_{\square}$ is a real-linear isomorphism. Comparing (1190) and (1191) on the displayed basis proves (1185). Applying the exponential functional calculus proves (1186). The final two formulas are the coefficient-preserving specializations of (196). $\square$

Theorem 14.23 (Maximal D_3 -equivariant transport of the six support paths). *Let*

$$\mathcal{R}_{\square} := \text{span}_{\mathbb{R}}\{E_{11}, E_{12}, E_{21}, E_{22}\}, \quad \mathcal{R}_{\square,+} := \text{span}_{\mathbb{R}}\{E_{11}, E_{12}, E_{22}\}. \quad (1192)$$

For $(m, \epsilon) \in C_3 \times C_2$, define

$$x_{m,\epsilon} := \mathfrak{R}_{\square}^m \mathfrak{J}_{\square}^{\epsilon}(E_{11}). \quad (1193)$$

The six vectors are

$$x_{0,0} = E_{11}, \quad x_{0,1} = E_{22}, \quad (1194)$$

$$x_{1,0} = \frac{1}{4}E_{11} - \frac{\sqrt{3}}{4}E_{12} + \frac{3}{4}E_{22}, \quad x_{1,1} = \frac{3}{4}E_{11} + \frac{\sqrt{3}}{4}E_{12} + \frac{1}{4}E_{22}, \quad (1195)$$

$$x_{2,0} = \frac{1}{4}E_{11} + \frac{\sqrt{3}}{4}E_{12} + \frac{3}{4}E_{22}, \quad x_{2,1} = \frac{3}{4}E_{11} - \frac{\sqrt{3}}{4}E_{12} + \frac{1}{4}E_{22}. \quad (1196)$$

They are pairwise distinct and span $\mathcal{R}_{\square,+}$.

The map

$$\Psi : \Gamma \longrightarrow \mathcal{R}_{\square,+}, \quad \Psi(\gamma_{\epsilon,m}) = x_{m,\epsilon}, \quad (1197)$$

is a surjective real D_3 -module morphism. Explicitly,

$$\Psi \operatorname{Ad}_{W_r} = \mathfrak{R}_{\square} \Psi, \quad \Psi \operatorname{Ad}_{W_f} = \mathfrak{J}_{\square} \Psi. \quad (1198)$$

Its kernel has the basis

$$k_1 = \gamma_{0,0} + \gamma_{1,0} - \gamma_{0,1} - \gamma_{1,1}, \quad (1199)$$

$$k_2 = \gamma_{0,0} + \gamma_{1,0} - \gamma_{0,2} - \gamma_{1,2}, \quad (1200)$$

$$k_3 = \sum_{m=0}^2 (\gamma_{0,m} - \gamma_{1,m}). \quad (1201)$$

Consequently

$$\operatorname{rank} \Psi = 3, \quad \dim \ker \Psi = 3. \quad (1202)$$

Let $s_{\star} := |e_{\star}\rangle\langle e_{\star}| \in \mathcal{L}_{11}$ and define

$$\alpha : \mathcal{R}_{\square,+} \longrightarrow \mathcal{L}_{11}, \quad \alpha(M) := \operatorname{tr}(M)s_{\star}. \quad (1203)$$

Then α is D_3 -invariant and the support-collapse triangle commutes:

$$\mu = \alpha \circ \Psi. \quad (1204)$$

The quarter action is also preserved:

$$\mathfrak{Q}_{\square} \Psi = \Psi, \quad 4\mathcal{I}_{-}\mathcal{T}_{\square} \Psi = \mathcal{T}_{\square} \Psi. \quad (1205)$$

Every real D_3 -module morphism $L : \Gamma \rightarrow \mathcal{R}_{\square}$ has rank at most three. Hence Ψ has the maximal rank available in this morphism class.

Proof. Substitution of the matrices in (210) and (211) into (1193) gives (1196). The first two vectors give E_{11} , E_{22} , and

$$E_{12} = \frac{2}{\sqrt{3}}(x_{2,0} - x_{1,0}). \quad (1206)$$

Thus the image is $\mathcal{R}_{\square,+}$ and has dimension three.

The source action in (1103) and the target relation $\mathfrak{J}_{\square}\mathfrak{R}_{\square}\mathfrak{J}_{\square} = \mathfrak{R}_{\square}^{-1}$ give

$$\Psi(\gamma_{\epsilon,m+1}) = x_{m+1,\epsilon} = \mathfrak{R}_{\square} x_{m,\epsilon},$$

$$\Psi(\gamma_{\epsilon+1,-m}) = x_{-m,\epsilon+1} = \mathfrak{J}_{\square} x_{m,\epsilon}.$$

This proves (1198). Direct substitution of (1196) shows that each vector in (1201) lies in $\ker \Psi$. The three vectors are linearly independent. Rank-nullity and $\operatorname{rank} \Psi = 3$ prove (1202) and show that they form a kernel basis.

Every vector in (1196) has trace one. The first and fourth rows of $\mathfrak{R}_{\square}$ sum to $(1, 0, 0, 1)$, while $\mathfrak{J}_{\square}$ exchanges the first and fourth coordinates. Therefore tr is invariant under both generators. Since $\mu(\gamma_{\epsilon,m}) = s_{\star}$, equation (1204) follows on the path basis. The raw positive carrier is the $+1$ -eigenspace of $\mathfrak{Q}_{\square}$; conjugation by $\mathcal{T}_{\square}$ gives (1205).

It remains to determine the maximal possible rank. Regularity of the action on the six path labels gives

$$\Gamma^{D_3} = \mathbb{R} \sum_{\epsilon, m} \gamma_{\epsilon, m}, \quad \dim \Gamma^{D_3} = 1. \quad (1207)$$

Solving the two fixed-point equations for the target matrices gives

$$\mathcal{R}_{\square}^{D_3} = \text{span}_{\mathbb{R}}\{E_{21}, E_{11} + E_{22}\}, \quad \dim \mathcal{R}_{\square}^{D_3} = 2. \quad (1208)$$

Suppose a real D_3 -module morphism $L : \Gamma \rightarrow \mathcal{R}_{\square}$ were surjective. For any $y \in \mathcal{R}_{\square}^{D_3}$, choose $x \in \Gamma$ with $Lx = y$ and put

$$\bar{x} = \frac{1}{6} \sum_{g \in D_3} gx.$$

Then $\bar{x} \in \Gamma^{D_3}$ and $L\bar{x} = y$. Surjectivity would therefore give a surjection $\Gamma^{D_3} \rightarrow \mathcal{R}_{\square}^{D_3}$, contradicting (1207) and (1208). A rank-four map into the four-dimensional target would be surjective. Hence every equivariant map has rank at most three, and Ψ attains that rank. $\square$

Theorem 14.24 (The six support paths form an exact Lorentz-null hexagon). *For a Hermitian circle matrix, write*

$$\text{coeff}_{\square}(H(A, B_0, B_1, C)) = (A, B_0, B_1, C),$$

and define

$$\Lambda_{\square} := \Psi_{\text{circ}} \circ \text{coeff}_{\square} \circ \mathcal{T}_{\square} : \mathcal{R}_{\square, +} \longrightarrow \mathbb{R}^{3,1}. \quad (1209)$$

For $M = aE_{11} + bE_{12} + dE_{22}$, its circle coefficients and Lorentz coordinates are

$$\text{coeff}_{\square} \mathcal{T}_{\square}(M) = \left(a + d, \frac{a - d}{2}, b, \frac{a + d}{4} \right), \quad (1210)$$

$$\Lambda_{\square}(M) = \left(\frac{5(a + d)}{8}, \frac{a - d}{2}, b, \frac{3(a + d)}{8} \right). \quad (1211)$$

Put

$$v_{m, \epsilon} := \text{coeff}_{\square} \mathcal{T}_{\square}(x_{m, \epsilon}), \quad u_{m, \epsilon} := \Lambda_{\square}(x_{m, \epsilon}). \quad (1212)$$

In the order $(0, 0), (0, 1), (1, 0), (1, 1), (2, 0), (2, 1)$, the six pairs are

(m, ϵ)	$v_{m, \epsilon}$	$u_{m, \epsilon}$
$(0, 0)$	$(1, \frac{1}{2}, 0, \frac{1}{4})$	$(\frac{5}{8}, \frac{1}{2}, 0, \frac{3}{8})$
$(0, 1)$	$(1, -\frac{1}{2}, 0, \frac{1}{4})$	$(\frac{5}{8}, -\frac{1}{2}, 0, \frac{3}{8})$
$(1, 0)$	$(1, -\frac{1}{4}, -\frac{\sqrt{3}}{4}, \frac{1}{4})$	$(\frac{5}{8}, -\frac{1}{4}, -\frac{\sqrt{3}}{4}, \frac{3}{8})$
$(1, 1)$	$(1, \frac{1}{4}, \frac{\sqrt{3}}{4}, \frac{1}{4})$	$(\frac{5}{8}, \frac{1}{4}, \frac{\sqrt{3}}{4}, \frac{3}{8})$
$(2, 0)$	$(1, -\frac{1}{4}, \frac{\sqrt{3}}{4}, \frac{1}{4})$	$(\frac{5}{8}, -\frac{1}{4}, \frac{\sqrt{3}}{4}, \frac{3}{8})$
$(2, 1)$	$(1, \frac{1}{4}, -\frac{\sqrt{3}}{4}, \frac{1}{4})$	$(\frac{5}{8}, \frac{1}{4}, -\frac{\sqrt{3}}{4}, \frac{3}{8})$

(1213)

Every coefficient vector has circle norm zero, and every Lorentz vector is null.

Let $\eta = \text{diag}(-1, 1, 1, 1)$. The complete Lorentz Gram matrix of the ordered six-tuple is

$$\mathcal{G}_6 := (u_i^{\top} \eta u_j)_{i, j=1}^6 = -\frac{1}{8} \begin{pmatrix} 0 & 4 & 3 & 1 & 3 & 1 \\ 4 & 0 & 1 & 3 & 1 & 3 \\ 3 & 1 & 0 & 4 & 3 & 1 \\ 1 & 3 & 4 & 0 & 1 & 3 \\ 3 & 1 & 3 & 1 & 0 & 4 \\ 1 & 3 & 1 & 3 & 4 & 0 \end{pmatrix}. \quad (1214)$$

It has eigenvalues

$$-\frac{3}{2}, \quad \frac{3}{4}, \quad \frac{3}{4}, \quad 0, \quad 0, \quad 0. \quad (1215)$$

Hence the six vectors span a three-dimensional Lorentz subspace with two positive directions and one negative direction.

The Lorentz operators

$$\mathcal{R}_L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathcal{J}_L = \text{diag}(1, -1, 1, 1) \quad (1216)$$

satisfy

$$\mathcal{R}_L^\top \eta \mathcal{R}_L = \eta, \quad \mathcal{J}_L^\top \eta \mathcal{J}_L = \eta, \quad \mathcal{R}_L^3 = \mathcal{J}_L^2 = I, \quad \mathcal{J}_L \mathcal{R}_L \mathcal{J}_L = \mathcal{R}_L^{-1}. \quad (1217)$$

They give the exact transport equations

$$\Lambda_\square \mathfrak{R}_\square = \mathcal{R}_L \Lambda_\square, \quad \Lambda_\square \mathfrak{J}_\square = \mathcal{J}_L \Lambda_\square. \quad (1218)$$

Thus, for $\Gamma_\mathbb{R} := \text{span}_\mathbb{R}\{\gamma_{\epsilon,m}\}$ and $\Xi := \Lambda_\square \circ \Psi$,

$$\Xi \text{Ad}_{W_r} = \mathcal{R}_L \Xi, \quad \Xi \text{Ad}_{W_f} = \mathcal{J}_L \Xi. \quad (1219)$$

Their barycentre is

$$q_6 := \frac{1}{6} \sum_{m,\epsilon} u_{m,\epsilon} = \left(\frac{5}{8}, 0, 0, \frac{3}{8}\right) = \Lambda_\square \left(\frac{E_{11} + E_{22}}{2}\right). \quad (1220)$$

It is fixed by $\mathcal{R}_L$ and $\mathcal{J}_L$, and

$$\langle q_6, q_6 \rangle_L = -\frac{1}{4}. \quad (1221)$$

Each vertex has the form

$$u_{m,\epsilon} = q_6 + s_{m,\epsilon}, \quad (s_{m,\epsilon})_T = (s_{m,\epsilon})_W = 0, \quad (s_{m,\epsilon})_X^2 + (s_{m,\epsilon})_Y^2 = \frac{1}{4}. \quad (1222)$$

The six transverse vectors are the vertices of a regular hexagon in the affine plane $q_6 + \text{span}_\mathbb{R}\{e_X, e_Y\}$.

The support-collapse functional has the exact Lorentz expression

$$\alpha(M) = (T(\Lambda_\square M) + W(\Lambda_\square M))_{s_\star}, \quad M \in \mathcal{R}_{\square,+}, \quad (1223)$$

and therefore

$$\mu(z) = (T(\Xi z) + W(\Xi z))_{s_\star}, \quad z \in \Gamma_\mathbb{R}. \quad (1224)$$

Finally, every $u_{m,\epsilon}$ lies on the $p = 1$ member of (2384), the barycentre q_6 lies on its $p = 17$ member, and every nonzero chord $u_{m,\epsilon} - u_{m',\epsilon'}$ lies on its $p = 1/3$ boundary member.

Proof. Linearity and the three basis images $\mathcal{T}_\square(E_{11})$, $\mathcal{T}_\square(E_{12})$, and $\mathcal{T}_\square(E_{22})$ in (196) give (1210). Applying (89) gives (1211). Substitution of (1196) proves the table (1213).

For every row, one has

$$B_0^2 + B_1^2 = \frac{1}{4} = AC,$$

which proves both null assertions. Pairwise substitution in the Lorentz form gives (1214). Its characteristic polynomial is

$$t^3 \left(t + \frac{3}{2}\right) \left(t - \frac{3}{4}\right)^2,$$

proving (1215) and the assertion about the spanned subspace.

Direct multiplication proves (1217). Equations (209) and (89) give (1218); composition with (1198) gives (1219).

Summing the six rows of (1213) gives (1220). Its Lorentz square is

$$\left(\frac{3}{8}\right)^2 - \left(\frac{5}{8}\right)^2 = -\frac{1}{4}.$$

Subtracting it from each row proves (1222). After ordering the transverse coordinates by angle, consecutive coordinates are separated by $\pi/3$, proving the hexagon assertion.

For $M = aE_{11} + bE_{12} + dE_{22}$, (1211) gives

$$T(\Lambda_{\square}M) + W(\Lambda_{\square}M) = a + d = \text{tr}(M).$$

This proves (1223); composition with (1204) proves (1224).

For each vertex, $X^2 + Y^2 + W^2 = T^2$, which gives $p = 1$. For q_6 ,

$$\frac{X^2 + Y^2 + W^2}{T^2} = \frac{9}{25},$$

which gives $p = 17$. Every nonzero chord has $T = W = 0$, so the Kocik equation gives $3p - 1 = 0$, hence $p = 1/3$. $\square$

Theorem 14.25 (Individual reduced divisors form an ordered Lorentz-null arc). *Retain the reduced shell data and divisor set of theorem 11.8. For $d \in \mathcal{D}_{p,a}^{\text{red}}$, put*

$$e_d := \frac{N_0^2}{d}, \quad b_d := \frac{N_0 + d}{R_0}, \quad c_d := \frac{N_0 + e_d}{R_0}. \quad (1225)$$

Then

$$R_0 b_d - N_0 = d, \quad R_0 c_d - N_0 = e_d, \quad (1226)$$

so both members of the complementary pair satisfy the reduced-shell congruence. Define their ordered denominator weights by

$$x_d := \frac{b_d}{b_d + c_d} = \frac{d}{N_0 + d}, \quad y_d := \frac{c_d}{b_d + c_d} = \frac{N_0}{N_0 + d}. \quad (1227)$$

They satisfy $x_d + y_d = 1$. The associated raw certificate cell is

$$\mathcal{C}_{p,a}(d) := x_d^2 E_{11} + x_d y_d E_{12} + y_d^2 E_{22} \in \mathcal{R}_{\square,+}. \quad (1228)$$

Its Hermitian lift is the rank-one matrix

$$\mathcal{H}_{\square}(\mathcal{C}_{p,a}(d)) = \begin{pmatrix} x_d^2 & x_d y_d \\ x_d y_d & y_d^2 \end{pmatrix} = \begin{pmatrix} x_d \\ y_d \end{pmatrix} \begin{pmatrix} x_d & y_d \end{pmatrix}. \quad (1229)$$

Put

$$\delta_d := x_d - y_d = \frac{d - N_0}{d + N_0} = \frac{b_d - c_d}{b_d + c_d}, \quad \alpha_d := 4x_d y_d = \frac{4N_0 d}{(N_0 + d)^2} = \frac{4b_d c_d}{(b_d + c_d)^2}. \quad (1230)$$

The two coordinates obey

$$\delta_d^2 + \alpha_d = 1. \quad (1231)$$

For the divisor involution $\iota(d) = e_d$, one has

$$x_{e_d} = y_d, \quad y_{e_d} = x_d, \quad \delta_{e_d} = -\delta_d, \quad \alpha_{e_d} = \alpha_d, \quad \mathcal{C}_{p,a}(e_d) = \mathfrak{J}_{\square} \mathcal{C}_{p,a}(d). \quad (1232)$$

The fixed and anti-fixed parts are therefore

$$\mathcal{C}_{p,a}^+(d) := \frac{\mathcal{C}_{p,a}(d) + \mathcal{C}_{p,a}(e_d)}{2} = \frac{2 - \alpha_d}{4}(E_{11} + E_{22}) + \frac{\alpha_d}{4}E_{12}, \quad (1233)$$

$$\mathcal{C}_{p,a}^-(d) := \frac{\mathcal{C}_{p,a}(d) - \mathcal{C}_{p,a}(e_d)}{2} = \frac{\delta_d}{2}(E_{11} - E_{22}). \quad (1234)$$

On $\mathcal{R}_{\square,+}$, the fixed and anti-fixed spaces of $\mathfrak{J}_{\square}$ are

$$\text{Fix}(\mathfrak{J}_{\square}) = \text{span}_{\mathbb{R}}\{E_{11} + E_{22}, E_{12}\}, \quad \text{Anti}(\mathfrak{J}_{\square}) = \mathbb{R}(E_{11} - E_{22}). \quad (1235)$$

Let

$$u_d := \Lambda_{\square}(\mathcal{C}_{p,a}(d)). \quad (1236)$$

Its circle coefficients and Lorentz coordinates are

$$\text{coeff}_\square \mathcal{T}_\square(\mathcal{C}_{p,a}(d)) = \left(\frac{2 - \alpha_d}{2}, \frac{\delta_d}{2}, \frac{\alpha_d}{4}, \frac{2 - \alpha_d}{8} \right), \quad (1237)$$

$$u_d = \left(\frac{5(2 - \alpha_d)}{16}, \frac{\delta_d}{2}, \frac{\alpha_d}{4}, \frac{3(2 - \alpha_d)}{16} \right). \quad (1238)$$

Every u_d is Lorentz null, and its two transverse coordinates satisfy the exact parabolic equation

$$X(u_d)^2 + Y(u_d) = \frac{1}{4}, \quad -\frac{1}{2} < X(u_d) < \frac{1}{2}, \quad 0 < Y(u_d) \leq \frac{1}{4}. \quad (1239)$$

Complementation is transported without changing the remaining coordinates:

$$u_{e_d} = \mathcal{J}_L u_d. \quad (1240)$$

Consequently

$$\text{Fix}(\mathcal{J}_L) = \text{span}_{\mathbb{R}}\{e_T, e_Y, e_W\}, \quad \text{Anti}(\mathcal{J}_L) = \mathbb{R}e_X. \quad (1241)$$

The reflection has an arithmetic fixed point precisely when

$$d = N_0 \iff R_0 \mid 2. \quad (1242)$$

At this point $(\delta_d, \alpha_d) = (0, 1)$. Hence shells with $R_0 > 2$ meet the two open halves of (1239) in complementary pairs and do not contain its reflection-fixed vertex.

Define the affine maps

$$\kappa_\square(rE_{11} + sE_{12} + tE_{22}) := 4E_{11} - 4sE_{22}, \quad (1243)$$

$$\mathcal{B}_\square(T, X, Y, W) := (2 - 2Y, 0, 0, 2 + 2Y). \quad (1244)$$

They satisfy

$$\kappa_\square(\mathcal{C}_{p,a}(d)) = 4E_{11} - \alpha_d E_{22} = \mathcal{J}_{p,a}(d), \quad (1245)$$

$$\mathcal{B}_\square(u_d) = \Psi_{\text{circ}}(\text{coeff}_\square \mathcal{J}_{p,a}(d)), \quad (1246)$$

$$\kappa_\square \mathfrak{J}_\square = \kappa_\square, \quad \mathcal{B}_\square \mathcal{J}_L = \mathcal{B}_\square. \quad (1247)$$

The translation kernel of the affine descent is

$$\ker_{\text{tr}}(\kappa_\square) = \text{span}_{\mathbb{R}}\{E_{11}, E_{22}\}, \quad (1248)$$

which contains the anti-fixed line in (1235). Restricted to the rank-one barycentric certificate cells, it induces the bijection

$$\mathcal{D}_{p,a}^{\text{red}} / \langle \iota \rangle \xrightarrow{\sim} \text{im}(\mathcal{J}_{p,a}) \quad (1249)$$

of (520).

Finally, retain the two ordered blade channels by putting

$$N_d^{-,+} := 2\sqrt{\alpha_d} E_{21}, \quad N_d^{+,-} := 2\sqrt{\alpha_d} E_{12}. \quad (1250)$$

With P_{α_d} defined by (539), their exact transport words are

$$\mathcal{A}_{P_{\alpha_d}}(\mathcal{H}_\square(N_d^{-,+})) = \mathcal{J}_{p,a}(d), \quad (1251)$$

$$\mathcal{A}_{P_{\alpha_d} P_\square}(\mathcal{H}_\square(N_d^{+,-})) = \mathcal{J}_{p,a}(d), \quad (1252)$$

$$\mathcal{A}_{P_\square}(\mathcal{H}_\square(N_d^{+,-})) = \mathcal{H}_\square(N_d^{-,+}). \quad (1253)$$

Thus the ordered certificate is retained in δ_d , the complementary divisor reflection is retained in $\mathfrak{J}_\square$ and $\mathcal{J}_L$, and the exact diagonal carrier is their reflection quotient through (1243).

Proof. The reduced-shell congruence for d , the identity $de_d = N_0^2$, and $\gcd(R_0, N_0) = 1$ give $e_d \equiv -N_0 \pmod{R_0}$, as in theorem 9.68. This proves (1225) and (1226). Substitution of the two reconstructed denominators gives

$$b_d + c_d = \frac{(N_0 + d)^2}{dR_0},$$

and hence (1227). The outer-product identity (1229) follows from the definition of $\mathcal{H}_\square$.

Equations (1230) and (1231) follow from $x_d + y_d = 1$. Replacing d by N_0^2/d exchanges x_d and y_d . The matrix in (211) exchanges the E_{11} - and E_{22} -coefficients and fixes the E_{12} -coefficient. This proves (1232). Taking the half-sum and half-difference gives (1233) and (1234), and also proves the two raw eigenspace formulas.

Apply (1210) and (1211) to (1228). Since

$$x_d^2 + y_d^2 = 1 - 2x_dy_d = \frac{2 - \alpha_d}{2}, \quad x_d^2 - y_d^2 = \delta_d,$$

the results are (1237) and (1238). The circle norm vanishes because

$$\left(\frac{\delta_d}{2}\right)^2 + \left(\frac{\alpha_d}{4}\right)^2 = \frac{(2 - \alpha_d)^2}{16} = \frac{2 - \alpha_d}{2} \frac{2 - \alpha_d}{8}.$$

This is equivalent to Lorentz nullity. The identity $\delta_d^2 + \alpha_d = 1$ gives (1239). Positivity of d, N_0 gives the displayed open bounds for X and the stated bounds for Y . The intertwining equation (1218) gives (1240), and the diagonal matrix for $\mathcal{J}_L$ gives its two eigenspaces.

The only fixed point of $d \mapsto N_0^2/d$ in positive divisors is $d = N_0$. It belongs to the reduced divisor set exactly when $N_0 \equiv -N_0 \pmod{R_0}$. Coprimality converts this condition to $R_0 \mid 2$, proving (1242).

The E_{12} -coefficient of $\mathcal{C}_{p,a}(d)$ is $x_dy_d = \alpha_d/4$. This proves (1245). The circle–Sarnak image of $4E_{11} - \alpha_d E_{22}$ is

$$\left(\frac{4 - \alpha_d}{2}, 0, 0, \frac{4 + \alpha_d}{2}\right),$$

which is $\mathcal{B}_\square(u_d)$. Both coequalizer equations follow from the displayed matrices for $\mathfrak{J}_\square$ and $\mathcal{J}_L$. The map $\kappa_\square$ depends only on the E_{12} -coefficient, so its translation kernel is $\text{span}_{\mathbb{R}}\{E_{11}, E_{22}\}$. The exact-fibre statement (521) proves (1249).

The last three equations are the individual-divisor specializations of the coefficient congruences (540)–(542), with $\alpha = \alpha_d$. Their source matrices retain the distinct $(-, +)$ and $(+, -)$ blade positions, and the intermediate equation records the transposition step explicitly. $\square$

Theorem 14.26 (The reduced-divisor arc has an exact three-arc D_3 -closure). *Let*

$$R_\perp := \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, \quad R_\perp^3 = I, \tag{1254}$$

the transverse block of $\mathcal{R}_L$. For $-1 < \delta < 1$ and $m \in \mathbb{Z}/3\mathbb{Z}$, define

$$U_m(\delta) := \left(\frac{5(1 + \delta^2)}{16}, R_\perp^m \begin{pmatrix} \delta/2 \\ (1 - \delta^2)/4 \end{pmatrix}, \frac{3(1 + \delta^2)}{16} \right), \tag{1255}$$

where the middle column supplies the (X, Y) -coordinates. Then

$$U_m(\delta_d) = \mathcal{R}_L^m u_d, \quad \delta_d = \frac{d - N_0}{d + N_0}, \tag{1256}$$

for every $d \in \mathcal{D}_{p,a}^{\text{red}}$. Each $U_m(\delta)$ is Lorentz null.

Let $\mathcal{A}_m$ denote the image of U_m . For $z = (X, Y)^\top$, put

$$\xi_m := (R_\perp^m e_X)^\top z, \quad \eta_m := (R_\perp^m e_Y)^\top z. \tag{1257}$$

The arc $\mathcal{A}_m$ is characterized by

$$\xi_m^2 + \eta_m = \frac{1}{4}, \quad -\frac{1}{2} < \xi_m < \frac{1}{2}, \quad 0 < \eta_m \leq \frac{1}{4}, \quad (1258)$$

together with

$$T = \frac{5(1 + 4\xi_m^2)}{16}, \quad W = \frac{3(1 + 4\xi_m^2)}{16}. \quad (1259)$$

In the common (X, Y) -chart, the three parabolic equations are

$$X^2 + Y = \frac{1}{4}, \quad (1260)$$

$$(X + \sqrt{3}Y)^2 + 2\sqrt{3}X - 2Y = 1, \quad (1261)$$

$$(X - \sqrt{3}Y)^2 - 2\sqrt{3}X - 2Y = 1. \quad (1262)$$

For fixed δ , the three cyclic points have barycentre

$$c_\delta := \frac{1}{3} \sum_{m=0}^2 U_m(\delta) = \left(\frac{5(1 + \delta^2)}{16}, 0, 0, \frac{3(1 + \delta^2)}{16} \right) = \frac{1 + \delta^2}{2} q_6. \quad (1263)$$

Their transverse radius is

$$r_\delta = \frac{1 + \delta^2}{4}, \quad \langle c_\delta, c_\delta \rangle_L = -r_\delta^2. \quad (1264)$$

Consequently their Lorentz Gram matrix has zero diagonal and constant off-diagonal entry

$$-\frac{3}{2}r_\delta^2 = -\frac{3(1 + \delta^2)^2}{32}. \quad (1265)$$

The barycentre lies on the $p = 17$ Kocik–Sarnak ray for every δ .

The three open arcs have exactly three pairwise intersections:

$$\mathcal{A}_0 \cap \mathcal{A}_1 = \left\{ w_{01} := \left(\frac{5}{12}, \frac{\sqrt{3}}{6}, \frac{1}{6}, \frac{1}{4} \right) \right\}, \quad (1266)$$

$$\mathcal{A}_0 \cap \mathcal{A}_2 = \left\{ w_{02} := \left(\frac{5}{12}, -\frac{\sqrt{3}}{6}, \frac{1}{6}, \frac{1}{4} \right) \right\}, \quad (1267)$$

$$\mathcal{A}_1 \cap \mathcal{A}_2 = \left\{ w_{12} := \left(\frac{5}{12}, 0, -\frac{1}{3}, \frac{1}{4} \right) \right\}. \quad (1268)$$

There is no triple intersection. Each w_{ij} is null, their pairwise Lorentz inner products equal $-1/6$, and

$$\frac{w_{01} + w_{02} + w_{12}}{3} = \left(\frac{5}{12}, 0, 0, \frac{1}{4} \right) = \frac{2}{3} q_6. \quad (1269)$$

The intersection parameters and quotient amplitude are

$$\delta = \pm \frac{1}{\sqrt{3}}, \quad \frac{d}{N_0} = 2 \pm \sqrt{3}, \quad \alpha = \frac{2}{3}. \quad (1270)$$

The closures of the arcs terminate at the six Lorentz-null vertices in (1213):

$$U_m(1) = u_{m,0}, \quad U_m(-1) = u_{m,1}. \quad (1271)$$

Thus their six endpoints are exactly the previously proved Lorentz-null hexagon.

The arithmetic map

$$(\mathbb{Z}/3\mathbb{Z}) \times \mathcal{D}_{p,a}^{\text{red}} \longrightarrow \mathbb{R}^{3,1}, \quad (m, d) \longmapsto U_m(\delta_d), \quad (1272)$$

is injective. In particular, none of the three points w_{01}, w_{02}, w_{12} and none of the six endpoints is an arithmetic certificate point. The reflection acts by

$$\mathcal{J}_L U_m(\delta_d) = U_{-m}(-\delta_d) = U_{-m}(\delta_{e_d}). \quad (1273)$$

Every complementary pair $\{d, e_d\}$ with $d \neq e_d$ therefore generates a six-point D_3 -orbit. A fixed divisor $d = e_d = N_0$, which occurs exactly when $R_0 \mid 2$, generates a three-point orbit.

Proof. Substitution of $\alpha = 1 - \delta^2$ into (1238) gives $U_0(\delta)$. The intertwining equation (1218) gives the remaining two parametrizations and proves (1256). The Lorentz square of $U_0(\delta)$ is zero by theorem 14.25; the Lorentz isometry $\mathcal{R}_L$ proves the assertion for every m .

Since $R_\perp$ is orthogonal, $R_\perp^{-m}z = (\xi_m, \eta_m)^\top$. Applying the base equation $X^2 + Y = 1/4$ in this rotated coordinate pair proves (1258). The identities $\delta = 2\xi_m$ and $1 - \delta^2 = 4\eta_m$ give (1259). Substitution of the three powers of $R_\perp$ gives (1260)–(1262).

For the transverse vector

$$z_0(\delta) = \left(\frac{\delta}{2}, \frac{1 - \delta^2}{4} \right),$$

one has

$$\sum_{m=0}^2 R_\perp^m z_0(\delta) = 0, \quad \|z_0(\delta)\|^2 = \frac{(1 + \delta^2)^2}{16} = r_\delta^2. \quad (1274)$$

This proves the barycentre and radius formulas. Distinct rotations have transverse inner product $-r_\delta^2/2$. Adding the barycentre square $-r_\delta^2$ gives (1265). The ratio $W(c_\delta)^2/T(c_\delta)^2 = 9/25$ gives the $p = 17$ statement by (2384).

Suppose $U_0(\delta) = U_k(\varepsilon)$ for $k = 1$ or 2 . Equality of the T -coordinates gives $\varepsilon^2 = \delta^2$. The case $\varepsilon = \delta$ has no solution because a nonzero transverse vector cannot be fixed by a rotation through $2\pi/3$. For $\varepsilon = -\delta$, the two coordinate equations for $k = 1$ are

$$\sqrt{3}\delta^2 + 2\delta - \sqrt{3} = 0, \quad 3\delta^2 + 2\sqrt{3}\delta - 3 = 0. \quad (1275)$$

Their common roots are $1/\sqrt{3}$ and $-\sqrt{3}$; only the first lies in $(-1, 1)$. For $k = 2$, the admissible root is $-1/\sqrt{3}$. Rotation of the first intersection supplies the third. This proves the three displayed intersection formulas and excludes a triple intersection. Direct inner products give the null, Gram, and barycentre assertions. Solving $\delta = (d - N_0)/(d + N_0)$ for d/N_0 and using $\alpha = 1 - \delta^2$ proves (1270).

Taking $\delta \rightarrow \pm 1$ in (1255) gives (1271). Positive finite d, N_0 give $-1 < \delta_d < 1$, so no endpoint is attained.

Finally, every δ_d is rational. The preceding intersection calculation shows that points on distinct arcs can agree only at $\delta = \pm 1/\sqrt{3}$, which is irrational. Points on one arc have unique parameter δ , and $d = N_0(1 + \delta)/(1 - \delta)$ recovers the divisor. This proves injectivity. The relation $\mathcal{J}_L \mathcal{R}_L^m = \mathcal{R}_L^{-m} \mathcal{J}_L$, together with $\delta_{e_d} = -\delta_d$, proves (1273). The orbit sizes follow from (1242). $\square$

Theorem 14.27 (Star–Kneser certificates admit an exact free D_3 Lorentz cover). *Assume*

$$R = 4a - p > 2, \quad \gcd(R, pa) = 1, \quad N = pa, \quad (1276)$$

and retain the origin-resolved certificate set $\mathcal{X}_{R,p}$ of (372). Put

$$\tilde{\mathcal{X}}_{R,p} := (\mathbb{Z}/3\mathbb{Z}) \times \mathcal{X}_{R,p}. \quad (1277)$$

For $(m, j, u) \in \tilde{\mathcal{X}}_{R,p}$, let

$$d_j(u) := p^j u, \quad \delta_j(u) := \frac{d_j(u) - N}{d_j(u) + N}. \quad (1278)$$

Define two permutations of $\tilde{\mathcal{X}}_{R,p}$ by

$$r(m, j, u) = (m + 1, j, u), \quad (1279)$$

$$s(m, j, u) = \left(-m, 2 - j, \frac{a^2}{u} \right). \quad (1280)$$

They satisfy

$$r^3 = s^2 = 1, \quad srs = r^{-1}, \quad (1281)$$

and define a free D_3 -action.

The raw and Lorentz maps

$$\Phi_{\text{raw}}(m, j, u) := \mathfrak{R}_{\square}^m \mathcal{C}_{p,a}(d_j(u)), \quad (1282)$$

$$\Phi_L(m, j, u) := \Lambda_{\square} \Phi_{\text{raw}}(m, j, u) = U_m(\delta_j(u)) \quad (1283)$$

are injective and D_3 -equivariant:

$$\Phi_{\text{raw}} \circ r = \mathfrak{R}_{\square} \circ \Phi_{\text{raw}}, \quad \Phi_{\text{raw}} \circ s = \mathfrak{J}_{\square} \circ \Phi_{\text{raw}}, \quad (1284)$$

$$\Phi_L \circ r = \mathcal{R}_L \circ \Phi_L, \quad \Phi_L \circ s = \mathcal{J}_L \circ \Phi_L. \quad (1285)$$

For each $m \in \mathbb{Z}/3\mathbb{Z}$, define the sheetwise affine descents

$$\kappa_m := \kappa_{\square} \circ \mathfrak{R}_{\square}^{-m}, \quad \mathcal{B}_m := \mathcal{B}_{\square} \circ \mathcal{R}_L^{-m}. \quad (1286)$$

They obey

$$\kappa_{m+1} \mathfrak{R}_{\square} = \kappa_m, \quad \kappa_{-m} \mathfrak{J}_{\square} = \kappa_m, \quad (1287)$$

$$\mathcal{B}_{m+1} \mathcal{R}_L = \mathcal{B}_m, \quad \mathcal{B}_{-m} \mathcal{J}_L = \mathcal{B}_m. \quad (1288)$$

For every cover point,

$$\kappa_m \Phi_{\text{raw}}(m, j, u) = \mathcal{J}_{p,a}(d_j(u)), \quad (1289)$$

$$\mathcal{B}_m \Phi_L(m, j, u) = \Psi_{\text{circ}}(\text{coeff}_{\square} \mathcal{J}_{p,a}(d_j(u))). \quad (1290)$$

The common descended value induces a bijection

$$\tilde{\mathcal{X}}_{R,p}/D_3 \xrightarrow{\sim} \text{im}(\mathcal{J}_{p,a}). \quad (1291)$$

Consequently

$$|\Phi_L(\tilde{\mathcal{X}}_{R,p})| = 3\Omega_R(p), \quad |\tilde{\mathcal{X}}_{R,p}/D_3| = \frac{\Omega_R(p)}{2} = |\text{im}(\mathcal{J}_{p,a})|. \quad (1292)$$

Thus the positive-surplus hypothesis in theorem 10.7 produces a nonempty Lorentz-null D_3 -cover and a nonempty arithmetic carrier image.

For the complete 12289 shell, the six complementary pairs in theorem 10.29 give the following exact parameters, where $d < N$ is the displayed representative:

origin pair	d	δ_d	α_d
$0 \leftrightarrow 2$	615	-30722/30723	61445/943902729
$0 \leftrightarrow 2$	5125	-18431/18436	184335/339886096
$0 \leftrightarrow 2$	230625	-6107/6182	921675/38217124
$1 \leftrightarrow 1$	61445	-307/308	615/94864
$1 \leftrightarrow 1$	2765025	-19/22	123/484
$1 \leftrightarrow 1$	23041875	-8/33	1025/1089

(1293)

The cover has 36 distinct Lorentz-null points and six free D_3 -orbits. Its origin multiplicities are

$$3(m_0, m_1, m_2) = (9, 18, 9). \quad (1294)$$

No permutation of $\mathcal{X}_{11,12289}$ cycles its three origin fibres, because their cardinalities are $(3, 6, 3)$. The C_3 -action in the cover acts on the first coordinate m and preserves each arithmetic origin label.

Proof. The arithmetic involution in (375) preserves $\mathcal{X}_{R,p}$, so s is well defined. Direct substitution proves (1281). A nonidentity rotation changes the first coordinate. A fixed point of a reflection would require

$$j = 1, \quad u = \frac{a^2}{u},$$

hence $u = a$. Its origin target equation would then be $1 = -1 \pmod{R}$, which contradicts $R > 2$. The action is therefore free.

The map $(j, u) \mapsto d_j(u)$ is the bijection from theorem 9.64. Under the present coprimality hypothesis, $R_0 = R$ and $N_0 = N$. Hence theorem 14.26 applies and proves the injectivity of both cover maps. Rotation equivariance follows from their definitions. For reflection, use

$$d_{2-j}(a^2/u) = \frac{N^2}{d_j(u)}, \quad (1295)$$

together with $\mathcal{C}_{p,a}(N^2/d) = \mathfrak{J}_\square \mathcal{C}_{p,a}(d)$ and $\mathfrak{J}_\square \mathfrak{R}_\square^m = \mathfrak{R}_\square^{-m} \mathfrak{J}_\square$. This proves the raw equivariance; the Lorentz intertwining gives (1285).

The two indexed rotation identities in (1287) and (1288) follow by cancelling adjacent inverse powers. For reflection,

$$\kappa_{-m} \mathfrak{J}_\square = \kappa_\square \mathfrak{R}_\square^m \mathfrak{J}_\square = \kappa_\square \mathfrak{J}_\square \mathfrak{R}_\square^{-m} = \kappa_\square \mathfrak{R}_\square^{-m} = \kappa_m,$$

using $\kappa_\square \mathfrak{J}_\square = \kappa_\square$. The Lorentz identity is identical with $(\mathcal{B}_\square, \mathcal{R}_L, \mathcal{J}_L)$ in place of $(\kappa_\square, \mathfrak{R}_\square, \mathfrak{J}_\square)$. Equations (1289) and (1290) now follow from (1245) and (1246).

The descended value is constant on every D_3 -orbit. Conversely, if two cover points have the same descended value, the exact-fibre theorem (521) says that their divisors are equal or complementary. A power of r connects any two sheet indices over one divisor, and a reflection followed by a power of r connects the two complementary divisors. Thus every fibre is one D_3 -orbit, proving (1291). Freeness and $|\mathcal{X}_{R,p}| = \Omega_R(p)$ give (1292). The Star–Kneser implication then follows from theorem 10.26.

For $p = 12289$, substitute the six representatives in theorem 10.29 into

$$\delta_d = \frac{d - N}{d + N}, \quad \alpha_d = \frac{4Nd}{(N + d)^2}, \quad N = 37788675.$$

This gives (1293). The cardinality and orbit assertions follow from $\Omega_{11}(12289) = 12$. Multiplication of the origin counts by the three sheet indices gives (1294). A cyclic permutation of the three arithmetic fibres would preserve their cardinalities, which would force $m_0 = m_1 = m_2$. The vector $(3, 6, 3)$ excludes such a permutation. $\square$

Theorem 14.28 (Order-determined C_1 – C_2 – C_3 linking decomposition of every free shell). *Assume (1276). Let*

$$\mathcal{O}_{R,p} := \mathcal{X}_{R,p}/\langle \iota \rangle, \quad q_{R,p} := |\mathcal{O}_{R,p}| = \frac{\Omega_R(p)}{2}. \quad (1296)$$

For every $O \in \mathcal{O}_{R,p}$, there is a unique ordered pair

$$x_O^- \in O, \quad x_O^+ := \iota(x_O^-), \quad d(x_O^-) < N < d(x_O^+). \quad (1297)$$

Put $x_O^0 = x_O^-$ and $x_O^1 = x_O^+$. The map

$$\theta_{123} : \tilde{\mathcal{X}}_{R,p} \longrightarrow \mathcal{O}_{R,p} \times C_2 \times C_3, \quad \theta_{123}(m, x_O^\epsilon) = (O, \epsilon, m) \quad (1298)$$

is a bijection. Its linearization

$$\Theta_{123} : \mathbb{C}[\tilde{\mathcal{X}}_{R,p}] \xrightarrow{\cong} \mathbb{C}[\mathcal{O}_{R,p}] \otimes \Gamma, \quad e_{(m, x_O^\epsilon)} \longmapsto e_O \otimes \gamma_{\epsilon, m} \quad (1299)$$

intertwines both generators:

$$\Theta_{123} \mathbf{R} = (I \otimes \text{Ad}_{W_r} |_\Gamma) \Theta_{123}, \quad (1300)$$

$$\Theta_{123} \mathbf{S} = (I \otimes \text{Ad}_{W_f} |_\Gamma) \Theta_{123}, \quad (1301)$$

where $\mathbf{R}$ and $\mathbf{S}$ are the permutation operators induced by (1279) and (1280).

The multiplication map of the linking algebra gives the orbitwise support map

$$\Pi_{123} := (I \otimes \mu) \Theta_{123} : \mathbb{C}[\tilde{\mathcal{X}}_{R,p}] \longrightarrow \mathbb{C}[\mathcal{O}_{R,p}] \otimes \mathcal{L}_{11}, \quad (1302)$$

with basis action

$$\Pi_{123}e_{(m,x_O^\epsilon)} = e_O \otimes E_{1,1;1,1}. \quad (1303)$$

It is D_3 -invariant. On the distinguished basis in (1303), the induced label map has the free D_3 -orbits as its fibres, and

$$\text{rank } \Pi_{123} = q_{R,p}, \quad \dim \ker \Pi_{123} = 5q_{R,p}. \quad (1304)$$

Let

$$\mathcal{A}_{-,+} := 2 \sum_{O \in \mathcal{O}_{R,p}} \sum_{m \in C_3} |e_{(m,x_O^+)}\rangle \langle e_{(m,x_O^-)}|, \quad (1305)$$

$$\mathcal{A}_{+,-} := \mathcal{A}_{-,+}^*. \quad (1306)$$

After the basis identification $\gamma_{\epsilon,m} \leftrightarrow f_\epsilon \otimes g_m$, their exact transported forms are

$$\Theta_{123}\mathcal{A}_{-,+}\Theta_{123}^{-1} = I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes 2E_{21}^{(2)} \otimes I_3, \quad (1307)$$

$$\Theta_{123}\mathcal{A}_{+,-}\Theta_{123}^{-1} = I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes 2E_{12}^{(2)} \otimes I_3. \quad (1308)$$

Define the ordered sheet projectors

$$Q_{<} := \sum_{O,m} |e_{(m,x_O^-)}\rangle \langle e_{(m,x_O^-)}|, \quad Q_{>} := \sum_{O,m} |e_{(m,x_O^+)}\rangle \langle e_{(m,x_O^+)}|. \quad (1309)$$

Then

$$\frac{1}{4}\mathcal{A}_{+,-}\mathcal{A}_{-,+} = Q_{<}, \quad \frac{1}{4}\mathcal{A}_{-,+}\mathcal{A}_{+,-} = Q_{>}, \quad (1310)$$

and

$$\mathbf{S}\mathcal{A}_{-,+}\mathbf{S}^{-1} = \mathcal{A}_{+,-}, \quad \mathbf{R}\mathcal{A}_{-,+}\mathbf{R}^{-1} = \mathcal{A}_{-,+}. \quad (1311)$$

Consequently, for every shell under the stated hypotheses,

$$\Omega_R(p) > 0 \iff \Pi_{123} \neq 0 \iff \mathcal{A}_{-,+} \neq 0. \quad (1312)$$

The three factors retain, respectively, the occupied support line, the two ordered complement sheets, and the three Lorentz-cover rotations.

Proof. The complement identity $d(x)d(\iota x) = N^2$ and freeness of ι imply that every orbit has two unequal positive entries with product N^2 . Exactly one lies below N , proving (1297). The labels (O, ϵ, m) therefore recover every cover point uniquely, so θ_{123} and Θ_{123} are bijective.

The source generators act on the ordered labels by

$$r(O, \epsilon, m) = (O, \epsilon, m+1), \quad s(O, \epsilon, m) = (O, \epsilon+1, -m). \quad (1313)$$

This is exactly the path action in (1103), proving (1300) and (1301).

Equation (1099) gives (1303). For one fixed orbit O , the image is the one-dimensional line generated by $e_O \otimes E_{1,1;1,1}$, and the kernel consists of the coefficient vectors whose six entries sum to zero. Different orbit labels give independent image lines. This proves (1304). The regular path action is transitive on the six labels over each O , so these six-element sets are both the fibres and the free D_3 -orbits.

The operator $\mathcal{A}_{-,+}$ sends each low-sheet basis vector to twice the corresponding high-sheet basis vector and kills the high sheet. Under Θ_{123} , this is precisely $I \otimes 2E_{21}^{(2)} \otimes I_3$; adjunction proves the transposed formula. The two products are four times the low and high projectors, which proves (1310). Reflection exchanges the sheet labels and reverses m , while rotation changes only m . This proves (1311). Finally, $|\mathcal{X}_{R,p}| = 2q_{R,p} = \Omega_R(p)$, and all three objects in (1312) vanish exactly when $q_{R,p} = 0$. $\square$

Corollary 14.29 (Exact simplex-support and Lorentz pullback of every free shell). *Under the conditions of theorem 14.28, let*

$$\mathcal{U}_{R,p} := I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes \mathbf{U}_2^* \otimes I_3, \quad \tilde{\Theta}_{123} := \mathcal{U}_{R,p} \Theta_{123}. \quad (1314)$$

This is a unitary map onto

$$\mathcal{H}_{R,p}^{\text{simp}} := \mathbb{C}[\mathcal{O}_{R,p}] \otimes \text{Ran } P_2 \otimes \mathbb{C}^3. \quad (1315)$$

Define the multiplicity-two transposed simplex blade by

$$\mathbf{B}_{+,-}^{(2)} := \Phi_2^{-1}(2E_{12}) = \mathbf{B}_{-,+}^* = 2\mathbf{B}_{+,-}. \quad (1316)$$

Then the two global low/high blade operators satisfy

$$\tilde{\Theta}_{123} \mathcal{A}_{-,+} \tilde{\Theta}_{123}^{-1} = I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes \mathbf{B}_{-,+} \otimes I_3, \quad (1317)$$

$$\tilde{\Theta}_{123} \mathcal{A}_{+,-} \tilde{\Theta}_{123}^{-1} = I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes \mathbf{B}_{+,-}^{(2)} \otimes I_3. \quad (1318)$$

Let

$$E_0 := |\eta_0\rangle\langle\eta_0|, \quad E_1 := |\eta_1\rangle\langle\eta_1|.$$

The ordered sheet projectors become

$$\tilde{\Theta}_{123} Q_{<} \tilde{\Theta}_{123}^{-1} = I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes E_0 \otimes I_3, \quad (1319)$$

$$\tilde{\Theta}_{123} Q_{>} \tilde{\Theta}_{123}^{-1} = I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes E_1 \otimes I_3. \quad (1320)$$

Accordingly,

$$\frac{1}{4} \mathbf{B}_{+,-}^{(2)} \mathbf{B}_{-,+} = E_0, \quad \frac{1}{4} \mathbf{B}_{-,+} \mathbf{B}_{+,-}^{(2)} = E_1. \quad (1321)$$

The global directed Gram-support carrier

$$\mathcal{H}_{R,p} := \mathcal{A}_{+,-} \mathcal{A}_{-,+} - Q_{>} \quad (1322)$$

has the exact pullback

$$\tilde{\Theta}_{123} \mathcal{H}_{R,p} \tilde{\Theta}_{123}^{-1} = I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes \tilde{\mathfrak{H}}_{-,+} \otimes I_3. \quad (1323)$$

Every compression to one orbit label O and one rotation label m therefore has the circle and Sarnak data

$$q(x, y) = 4(x^2 + y^2) - 1, \quad (T, X, Y, W) = \left(\frac{3}{2}, 0, 0, \frac{5}{2}\right), \quad \mathcal{N} = 4. \quad (1324)$$

The pulled rotation and reflection operators are

$$\tilde{\mathbf{R}} = I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes P_2 \otimes R_3, \quad \tilde{\mathbf{S}} = I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes \hat{S}_2 \otimes F_3. \quad (1325)$$

Writing

$$P_{R,p}^{\text{simp}} = I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes P_2 \otimes I_3,$$

one has

$$\tilde{\mathbf{R}}^3 = \tilde{\mathbf{S}}^2 = P_{R,p}^{\text{simp}}, \quad \tilde{\mathbf{S}} \tilde{\mathbf{R}} \tilde{\mathbf{S}} = \tilde{\mathbf{R}}^*. \quad (1326)$$

Finally,

$$\Omega_R(p) > 0 \iff \mathcal{H}_{R,p}^{\text{simp}} \neq 0 \iff I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes \mathbf{B}_{-,+} \otimes I_3 \neq 0. \quad (1327)$$

Proof. The map $\mathbf{U}_2^* : \mathbb{C}^2 \rightarrow \text{Ran } P_2$ is unitary, so $\mathcal{U}_{R,p}$ and $\tilde{\Theta}_{123}$ have the stated source and range. Conjugating (1307)–(1308) by $\mathcal{U}_{R,p}$, and using

$$\mathbf{U}_2^*(2E_{21})\mathbf{U}_2 = \mathbf{B}_{-,+}, \quad \mathbf{U}_2^*(2E_{12})\mathbf{U}_2 = \mathbf{B}_{+,-}^{(2)},$$

proves (1317)–(1318).

The same conjugation sends E_{11} to E_0 and E_{22} to E_1 . This proves (1319)–(1320); the two products in (1321) follow either from these identities or by multiplying the two rank-one blades.

In the (f_0, f_1) basis,

$$(2E_{12})(2E_{21}) - E_{22} = \text{diag}(4, -1).$$

Conjugating this identity by U_2^* and U_2 gives $\tilde{\mathfrak{H}}_{-,+}$, proving (1323). The fibrewise data in (1324) are those of (1130).

The source action in (1313) is

$$I \otimes I_2 \otimes R_3, \quad I \otimes S_2 \otimes F_3.$$

Conjugation by $\mathcal{U}_{R,p}$ sends I_2 to P_2 and S_2 to $\hat{S}_2$, proving (1325). The identities $R_3^3 = F_3^2 = I_3$, $F_3 R_3 F_3 = R_3^*$, and $\hat{S}_2^2 = P_2$ prove (1326). The last equivalence is (1312) transported by the unitary $\mathcal{U}_{R,p}$. $\square$

Exact computation 14.30 (Exact tensor check of the free-shell simplex and Lorentz pullback). The script and captured receipt

```
verification/verify_general_shell_simplex_lorentz_pullback.py,
verification/verify_general_shell_simplex_lorentz_pullback_output.txt
```

verify the pulled blade operators, quarter projectors, Gram-support carrier, and D_3 generators for one through four complement-orbit factors.

Theorem 14.31 (Exact free-shell control-length-determinant-volume identity). *Retain the conditions and notation of theorem 14.29, and assume $q_{R,p} \geq 1$. Write*

$$q := q_{R,p} = |\mathcal{O}_{R,p}|, \quad \Omega := \Omega_R(p) = 2q,$$

and let

$$H_{R,p}^{\text{sh}} := (\tilde{\Theta}_{123} \mathcal{H}_{R,p} \tilde{\Theta}_{123}^{-1}) \Big|_{\mathcal{H}_{R,p}^{\text{simp}}}. \quad (1328)$$

Then

$$\dim \mathcal{H}_{R,p}^{\text{simp}} = 6q = 3\Omega, \quad \text{Spec}(H_{R,p}^{\text{sh}}) = \{4^{[3q]}, (-1)^{[3q]}\}. \quad (1329)$$

The determinant volume of this carrier is

$$\text{Vol}_{R,p}^{\det} := \left| \det_{\mathcal{H}_{R,p}^{\text{simp}}} H_{R,p}^{\text{sh}} \right| = 4^{3q} = 2^{3\Omega}. \quad (1330)$$

For a Hermitian control A , put

$$F_{\text{HS}}(A) := \sqrt{\text{tr}(A^2)}.$$

Let $I_{R,p}$ denote the identity on $\mathcal{H}_{R,p}^{\text{simp}}$, and define

$$K_{R,p}^{\text{sh}} := H_{R,p}^{\text{sh}} - \frac{3}{2} I_{R,p}. \quad (1331)$$

The paths

$$\gamma_H(t) := \exp(-itH_{R,p}^{\text{sh}}), \quad \gamma_K(t) := \exp(-itK_{R,p}^{\text{sh}}), \quad 0 \leq t \leq 1, \quad (1332)$$

lie in $U(6q)$ and $SU(6q)$, respectively. Their F_{HS} -lengths satisfy

$$L_H^2 := L_{F_{\text{HS}}}(\gamma_H)^2 = 51q = \frac{51}{2}\Omega, \quad (1333)$$

$$L_K^2 := L_{F_{\text{HS}}}(\gamma_K)^2 = \frac{75}{2}q = \frac{75}{4}\Omega. \quad (1334)$$

Consequently,

$$\boxed{\log \text{Vol}_{R,p}^{\det} = \frac{2 \log 2}{17} L_H^2 = \frac{4 \log 2}{25} L_K^2.} \quad (1335)$$

Free-shell support: one spectrum, two exact length--volume coordinates

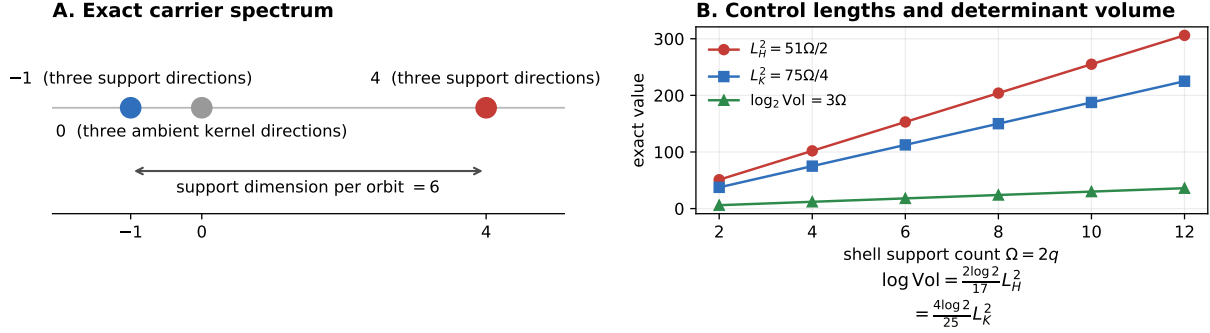


FIGURE 12. The exact spectrum of one orbit block and the three linear laws relating shell support, Hilbert–Schmidt control length, and determinant volume. The zero eigenvalue belongs to the ambient all-one kernel; the determinant and both control paths are taken on the six-dimensional support block.

Proof. The support factor $\text{Ran } P_2$ has dimension two, so

$$\dim \mathcal{H}_{R,p}^{\text{simp}} = q \cdot 2 \cdot 3 = 6q.$$

Equation (1296) gives $\Omega = 2q$, proving the dimension formula in (1329).

In the orthonormal basis transported from (f_0, f_1) , the middle factor of $H_{R,p}^{\text{sh}}$ is $\text{diag}(4, -1)$. The orbit factor has dimension q , and the rotation factor has dimension three. Thus each of the two middle-factor eigenvalues has multiplicity $3q$. This proves the spectrum in (1329), and multiplication of the absolute eigenvalues gives

$$\text{Vol}_{R,p}^{\text{det}} = 4^{3q} = 2^{6q} = 2^{3\Omega}.$$

The Hilbert–Schmidt cost is bi-invariant, and both paths have constant adapted Hamiltonians. Therefore

$$L_H^2 = \text{tr}((H_{R,p}^{\text{sh}})^2) = 3q(4^2 + (-1)^2) = 51q.$$

The trace of $H_{R,p}^{\text{sh}}$ is $9q$, while its dimension is $6q$. Hence the scalar part is exactly $3I_{R,p}/2$. The two eigenvalues of $K_{R,p}^{\text{sh}}$ are $5/2$ and $-5/2$, each with multiplicity $3q$. It follows that

$$\text{tr } K_{R,p}^{\text{sh}} = 0, \quad L_K^2 = 6q \left(\frac{5}{2}\right)^2 = \frac{75}{2}q.$$

The zero trace places γ_K in $SU(6q)$. Finally,

$$\log \text{Vol}_{R,p}^{\text{det}} = 3\Omega \log 2,$$

and substitution of the two length formulas proves (1335). $\square$

Exact computation 14.32 (Exact free-shell control-length and determinant-volume check). The script and captured receipt

```
verification/verify_general_shell_control_volume.py,
verification/verify_general_shell_control_volume_output.txt
```

verify the dimension, spectrum, determinant, two squared lengths, and both length–volume identities for one through five complement-orbit factors. The vector image is generated by

```
verification/make_general_shell_control_volume_figure.py.
```

Theorem 14.33 (Exact product factorization of the 12289 synthetic D_3 -cover). *Let*

$$\mathbf{v} := \begin{pmatrix} 3 \\ 6 \\ 3 \end{pmatrix}, \quad S_0 := \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \quad (1336)$$

With U as in (1033), one has

$$U^3 = S_0^2 = I_3, \quad S_0 U S_0 = U^{-1}, \quad S_0 \mathbf{v} = \mathbf{v}. \quad (1337)$$

Thus U and S_0 act on

$$\mathcal{O}_{12289} = \{\mathbf{v}, U\mathbf{v}, U^2\mathbf{v}\} \quad (1338)$$

as the transitive three-point D_3 -action.

Put

$$\mathcal{X} := \mathcal{X}_{11,12289}, \quad \tilde{\mathcal{X}} := (\mathbb{Z}/3\mathbb{Z}) \times \mathcal{X}, \quad (1339)$$

and retain the arithmetic involution

$$\iota(j, u) = \left(2 - j, \frac{3075^2}{u} \right). \quad (1340)$$

Define

$$\pi_{\mathcal{X}}(m, j, u) := (j, u), \quad \pi_{\mathcal{O}}(m, j, u) := U^m \mathbf{v}, \quad (1341)$$

$$\Theta(m, j, u) := ((j, u), U^m \mathbf{v}). \quad (1342)$$

Then

$$\Theta : \tilde{\mathcal{X}} \xrightarrow{\sim} \mathcal{X} \times \mathcal{O}_{12289} \quad (1343)$$

is a bijection. Under this bijection, the generators in (1279) and (1280) act by

$$r(x, w) = (x, Uw), \quad s(x, w) = (\iota(x), S_0 w). \quad (1344)$$

Equivalently,

$$\pi_{\mathcal{X}} \circ r = \pi_{\mathcal{X}}, \quad \pi_{\mathcal{X}} \circ s = \iota \circ \pi_{\mathcal{X}}, \quad (1345)$$

$$\pi_{\mathcal{O}} \circ r = U \circ \pi_{\mathcal{O}}, \quad \pi_{\mathcal{O}} \circ s = S_0 \circ \pi_{\mathcal{O}}. \quad (1346)$$

The two fibre systems have the exact cardinalities

$$\left| \pi_{\mathcal{X}}^{-1}(x) \right| = 3 \quad (x \in \mathcal{X}), \quad \left| \pi_{\mathcal{O}}^{-1}(w) \right| = 12 \quad (w \in \mathcal{O}_{12289}). \quad (1347)$$

Finally, the quotient maps give bijections

$$\tilde{\mathcal{X}}/\langle r \rangle \xrightarrow{\sim} \mathcal{X}, \quad \tilde{\mathcal{X}}/D_3 \xrightarrow{\sim} \mathcal{X}/\langle \iota \rangle \xrightarrow{\sim} \text{im}(\mathcal{J}_{12289,3075}). \quad (1348)$$

Their cardinalities are 12, 6, and 6, respectively.

Proof. Direct matrix multiplication proves (1337). The three columns $\mathbf{v}, U\mathbf{v}, U^2\mathbf{v}$ are the three distinct vectors in (393), so

$$\mathbb{Z}/3\mathbb{Z} \longrightarrow \mathcal{O}_{12289}, \quad m \longmapsto U^m \mathbf{v}$$

is a bijection. Consequently $\Theta = (\pi_{\mathcal{X}}, \pi_{\mathcal{O}})$ is the product of the identity map on $\mathcal{X}$ with this bijection, proving (1343) and the fibre counts in (1347).

For rotation,

$$\Theta(r(m, x)) = \Theta(m + 1, x) = (x, U^{m+1}\mathbf{v}).$$

For reflection, (1337) gives

$$S_0 U^m \mathbf{v} = U^{-m} S_0 \mathbf{v} = U^{-m} \mathbf{v},$$

and hence

$$\Theta(s(m, x)) = \Theta(-m, \iota(x)) = (\iota(x), S_0 U^m \mathbf{v}).$$

This proves (1344) and the four projection identities.

The subgroup $\langle r \rangle$ is free and transitive on the first factor of $\tilde{\mathcal{X}}$, so its quotient is identified with $\mathcal{X}$ by $\pi_{\mathcal{X}}$. The remaining order-two action is exactly ι . Therefore the second quotient in (1348) is $\mathcal{X}/\langle \iota \rangle$. The involution has no fixed point: such a point would have $j = 1$ and $u = 3075$, whose target equation is $1 = -1 \pmod{11}$. Since $|\mathcal{X}| = 12$, its quotient has six elements. The last bijection is the exact descended-fibre identification (1291). $\square$

Theorem 14.34 (Exact permutation representation of the 12289 synthetic cover). *Let*

$$K_0 := \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (1349)$$

The coordinate-orbit matrix M_{12289} satisfies

$$UM_{12289} = M_{12289}U, \quad S_0M_{12289} = M_{12289}K_0. \quad (1350)$$

Since $\det M_{12289} = 108$, these identities give the matrices of the coordinate rotation and reflection in the ordered orbit basis

$$(\mathbf{v}, U\mathbf{v}, U^2\mathbf{v})$$

as U and K_0 , respectively.

Let

$$V_{\mathcal{X}} := \mathbb{C}[\mathcal{X}], \quad V_{\mathcal{O}} := \mathbb{C}[\mathcal{O}_{12289}], \quad V_{\tilde{\mathcal{X}}} := \mathbb{C}[\tilde{\mathcal{X}}],$$

and define $P_{\iota} \in \text{End}(V_{\mathcal{X}})$ by

$$P_{\iota}e_x = e_{\iota(x)} \quad (x \in \mathcal{X}).$$

The factor actions induced by (1344) are

$$r_{\mathcal{X}} = I_{12}, \quad s_{\mathcal{X}} = P_{\iota}, \quad r_{\mathcal{O}} = U, \quad s_{\mathcal{O}} = K_0. \quad (1351)$$

The linearization of Θ in (1343) is the vector-space isomorphism

$$\hat{\Theta} : V_{\tilde{\mathcal{X}}} \xrightarrow{\sim} V_{\mathcal{X}} \otimes V_{\mathcal{O}}, \quad e_{(m,x)} \mapsto e_x \otimes e_{U^m \mathbf{v}}. \quad (1352)$$

If $R_{\tilde{\mathcal{X}}}$ and $S_{\tilde{\mathcal{X}}}$ denote the permutation operators induced by r and s , then

$$\hat{\Theta}R_{\tilde{\mathcal{X}}}\hat{\Theta}^{-1} = I_{12} \otimes U, \quad (1353)$$

$$\hat{\Theta}S_{\tilde{\mathcal{X}}}\hat{\Theta}^{-1} = P_{\iota} \otimes K_0. \quad (1354)$$

Their characteristic polynomials are

$$\det(tI_{36} - R_{\tilde{\mathcal{X}}}) = (t^3 - 1)^{12}, \quad (1355)$$

$$\det(tI_{36} - S_{\tilde{\mathcal{X}}}) = (t^2 - 1)^{18}. \quad (1356)$$

Write $\mathbf{1}$, ε , and V_{std} for the trivial, sign, and two-dimensional irreducible representations of D_3 . Then

$$V_{\mathcal{X}} \cong 6(\mathbf{1} \oplus \varepsilon), \quad (1357)$$

$$V_{\mathcal{O}} \cong \mathbf{1} \oplus V_{\text{std}}, \quad (1358)$$

$$V_{\tilde{\mathcal{X}}} \cong 6\mathbf{1} \oplus 6\varepsilon \oplus 12V_{\text{std}} \cong 6\mathbb{C}[D_3]. \quad (1359)$$

In the class order

$$1, \quad \{r, r^2\}, \quad \{s, rs, r^2s\},$$

the three permutation characters are

$$\chi_{\mathcal{X}} = (12, 12, 0), \quad \chi_{\mathcal{O}} = (3, 0, 1), \quad \chi_{\tilde{\mathcal{X}}} = (36, 0, 0). \quad (1360)$$

Proof. The first identity in (1350) follows because $M_{12289} = 3I_3 + 6U + 3U^2$. The columns of $S_0 M_{12289}$ are

$$S_0 \mathbf{v} = \mathbf{v}, \quad S_0 U \mathbf{v} = U^2 \mathbf{v}, \quad S_0 U^2 \mathbf{v} = U \mathbf{v},$$

which are the columns of $M_{12289} K_0$. The determinant follows from the three eigenvalues in (1038).

Equations (1353) and (1354) are the linearizations of (1344). The matrix U is one three-cycle. The involution ι is the product of six disjoint transpositions, so P_i has eigenvalues $+1$ and -1 , each with multiplicity six. The matrix K_0 has eigenvalue $+1$ with multiplicity two and eigenvalue -1 with multiplicity one. The Kronecker formulas therefore give (1355) and (1356).

On $\mathcal{X}$, the rotation acts identically and every reflection acts as the fixed-point-free involution ι . Thus $V_{\mathcal{X}} \cong 6(\mathbf{1} \oplus \varepsilon)$. The transitive three-point action has one constant line and the standard two-dimensional summand, proving (1358). Tensoring these two decompositions and using $\varepsilon \otimes V_{\text{std}} \cong V_{\text{std}}$ gives (1359). The displayed character values follow either from these decompositions or by counting fixed basis vectors. Their pointwise product is

$$(12, 12, 0)(3, 0, 1) = (36, 0, 0),$$

as required by the product action. $\square$

Theorem 14.35 (Exact directed-blade factorization of the two reflection factors). *Retain $\mathcal{X} = \mathcal{X}_{11,12289}$ and the involution ι from (1340). Put*

$$N_{\text{ar}} := 12289 \cdot 3075, \quad d(j, u) := 12289^j u,$$

and define

$$\mathcal{X}_{<} := \{x \in \mathcal{X} : d(x) < N_{\text{ar}}\}.$$

Then

$$d(x)d(\iota(x)) = N_{\text{ar}}^2, \quad |\mathcal{X}_{<}| = 6, \quad \mathcal{X} = \coprod_{x \in \mathcal{X}_{<}} \{x, \iota(x)\}. \quad (1361)$$

Thus $d < N_{\text{ar}}$ orders every arithmetic complement pair.

On $V_{\mathcal{X}} = \mathbb{C}[\mathcal{X}]$, define

$$A_{-} := 2 \sum_{x \in \mathcal{X}_{<}} e_{\iota(x)} e_x^*, \quad A_{+} := A_{-}^{\top} = 2 \sum_{x \in \mathcal{X}_{<}} e_x e_{\iota(x)}^*, \quad (1362)$$

$$Q_{<} := \sum_{x \in \mathcal{X}_{<}} e_x e_x^*, \quad Q_{>} := \sum_{x \in \mathcal{X}_{<}} e_{\iota(x)} e_{\iota(x)}^*. \quad (1363)$$

In the ordered basis $(e_x, e_{\iota(x)})$ of each summand, the restrictions of A_{-} and A_{+} are respectively $2E_{21}$ and $2E_{12}$. Consequently

$$P_i = \frac{1}{2}(A_{-} + A_{+}), \quad A_{+}A_{-} = 4Q_{<}, \quad A_{-}A_{+} = 4Q_{>}, \quad (1364)$$

$$A_{-}^2 = A_{+}^2 = 0, \quad P_i A_{-} P_i = A_{+}. \quad (1365)$$

Let F_{ij} be the matrix units of $M_3(\mathbb{C})$, and let

$$j : M_2(\mathbb{C}) \longrightarrow M_3(\mathbb{C}), \quad j(E_{ij}) = F_{i+1, j+1}. \quad (1366)$$

For the ordered blade coefficient $N = 2E_{21}$ from theorem 9.9, set

$$B_{-} := j(N) = 2F_{32}, \quad B_{+} := j(N^{\top}) = 2F_{23}. \quad (1367)$$

The orbit-basis reflection in (1349) has the exact factorization

$$K_0 = F_{11} + \frac{1}{2}(B_{-} + B_{+}), \quad K_0 B_{-} K_0 = B_{+}. \quad (1368)$$

For $m \in \mathbb{Z}/3\mathbb{Z}$, write

$$B_{\pm, m} := U^m B_{\pm} U^{-m}, \quad K_m := U^m K_0 U^{-m}. \quad (1369)$$

Then

$$K_m = U^m F_{11} U^{-m} + \frac{1}{2}(B_{-,m} + B_{+,m}), \quad K_m B_{-,m} K_m = B_{+,m}, \quad (1370)$$

$$\sum_{m=0}^2 B_{-,m} = 2U, \quad \sum_{m=0}^2 B_{+,m} = 2U^2, \quad \sum_{m=0}^2 K_m = J_3. \quad (1371)$$

The oriented coordinate-orbit matrix is therefore

$$M_{12289} - M_{12289}^T = \frac{3}{2} \sum_{m=0}^2 (B_{-,m} - B_{+,m}). \quad (1372)$$

Under the product isomorphism $\hat{\Theta}$, the complete reflection in (1354) has the matrix-entry expansion

$$\begin{aligned} \hat{\Theta} S_{\hat{\chi}} \hat{\Theta}^{-1} &= \frac{1}{2}(A_- + A_+) \otimes F_{11} \\ &+ \frac{1}{4}(A_- \otimes B_- + A_- \otimes B_+ + A_+ \otimes B_- + A_+ \otimes B_+). \end{aligned} \quad (1373)$$

Thus both factors of the (36)-point reflection are assembled from the same ordered $2E_{21}$, $2E_{12}$ channel pair. The fixed one-dimensional coordinate of K_0 , the six arithmetic pair orderings, and all four tensor-product channel positions remain explicit in (1373).

Proof. For $x = (j, u)$, direct substitution in (1340) gives

$$d(\iota(x)) = 12289^{2-j} \frac{3075^2}{u} = \frac{N_{\text{ar}}^2}{d(x)}.$$

The involution has no fixed point by the proof of theorem 14.33. Hence exactly one member of each of its six orbits lies below N_{ar} , proving (1361). Restriction to each ordered pair gives

$$A_- = 2E_{21}, \quad A_+ = 2E_{12}, \quad P_\ell = E_{12} + E_{21}.$$

Multiplication on the six direct summands proves (1364) and (1365).

The embedding j gives

$$B_- = 2F_{32}, \quad B_+ = 2F_{23}, \quad K_0 = F_{11} + F_{23} + F_{32},$$

which proves (1368). Conjugation by U^m proves (1370). Entrywise,

$$\begin{aligned} \frac{1}{2}(B_{-,0} + B_{-,1} + B_{-,2}) &= F_{32} + F_{13} + F_{21} = U, \\ \frac{1}{2}(B_{+,0} + B_{+,1} + B_{+,2}) &= F_{23} + F_{31} + F_{12} = U^2. \end{aligned}$$

The three fixed matrix units sum to I_3 , so the three reflections sum to $I_3 + U + U^2 = J_3$. Combining these identities with $M_{12289} - M_{12289}^T = 3(U - U^2)$ proves (1372).

Finally, substitute

$$P_\ell = \frac{1}{2}(A_- + A_+), \quad K_0 = F_{11} + \frac{1}{2}(B_- + B_+)$$

into $P_\ell \otimes K_0$ and distribute the product. This gives (1373) with every coefficient displayed. $\square$

Theorem 14.36 (Exact source and range partition of the four directed channels). *Retain the operators of theorem 14.35. For $\sigma, v \in \{-, +\}$, put*

$$C_{\sigma v} := A_\sigma \otimes B_v, \quad V_{\sigma v} := \frac{1}{4} C_{\sigma v}. \quad (1374)$$

Every $V_{\sigma v}$ is a rank-six square-zero partial isometry. Its source and range projectors

$$S_{\sigma v} := V_{\sigma v}^T V_{\sigma v}, \quad R_{\sigma v} := V_{\sigma v} V_{\sigma v}^T \quad (1375)$$

are given entry by entry by

<i>channel</i>	$S_{\sigma v}$	$R_{\sigma v}$
V_{--}	$Q_{<} \otimes F_{22}$	$Q_{>} \otimes F_{33}$
V_{-+}	$Q_{<} \otimes F_{33}$	$Q_{>} \otimes F_{22}$
V_{+-}	$Q_{>} \otimes F_{22}$	$Q_{<} \otimes F_{33}$
V_{++}	$Q_{>} \otimes F_{33}$	$Q_{<} \otimes F_{22}$

(1376)

The four source projectors are pairwise orthogonal, as are the four range projectors, and both families add to the full off-axis carrier:

$$\sum_{\sigma, v} S_{\sigma v} = \sum_{\sigma, v} R_{\sigma v} = I_{V_{\mathcal{X}}} \otimes (F_{22} + F_{33}). \quad (1377)$$

Transposition and reflection preserve the ordered signs by

$$V_{--}^T = V_{++}, \quad V_{-+}^T = V_{+-}, \quad (1378)$$

$$(P_l \otimes K_0)V_{\sigma v}(P_l \otimes K_0) = V_{\bar{\sigma}\bar{v}}, \quad \bar{-} = +, \quad \bar{+} = -. \quad (1379)$$

The two channels carrying the directed coordinate blade B_- and the two carrying its transpose combine respectively to

$$V_{21}^{(12289)} := V_{--} + V_{+-} = P_l \otimes F_{32}, \quad (1380)$$

$$V_{12}^{(12289)} := V_{-+} + V_{++} = P_l \otimes F_{23} = (V_{21}^{(12289)})^T. \quad (1381)$$

Consequently

$$(V_{21}^{(12289)})^T V_{21}^{(12289)} = I_{V_{\mathcal{X}}} \otimes F_{22} = I_{V_{\mathcal{X}}} \otimes j(E_{11}), \quad (1382)$$

$$V_{21}^{(12289)} (V_{21}^{(12289)})^T = I_{V_{\mathcal{X}}} \otimes F_{33} = I_{V_{\mathcal{X}}} \otimes j(E_{22}). \quad (1383)$$

Thus the source-to-target carrier $E_{11} \rightarrow E_{22}$ of theorem 11.7 occurs over every arithmetic basis vector in $\mathcal{X}_{11,12289}$, with its direction retained.

For completeness, let $\tilde{V}_{\sigma v}$ be the matrix over $G(\mathbb{Z})$ obtained by replacing the six unit entries of $V_{\sigma v}$ by $\tilde{1}$ and every absent entry by τ . The unique entrywise Boolean character satisfies

$$\hat{\chi}_{\mathbb{Z}}(\tilde{V}_{\sigma v}) = V_{\sigma v}, \quad (1384)$$

where the right-hand side is read as a Boolean incidence matrix. Its Boolean source and range projectors are precisely those in (1376). The theorem therefore establishes the carrier statement on the finite $\mathcal{X}_{11,12289}$ cover. The global Erdős–Strauss carrier is the quantified condition in (531).

Proof. From (1364) and (1367),

$$\begin{aligned} A_+ A_- &= 4Q_{<}, & A_- A_+ &= 4Q_{>}, \\ B_+ B_- &= 4F_{22}, & B_- B_+ &= 4F_{33}. \end{aligned}$$

For example,

$$V_{--}^T V_{--} = \frac{1}{16} (A_+ A_-) \otimes (B_+ B_-) = Q_{<} \otimes F_{22},$$

whereas

$$V_{--} V_{--}^T = \frac{1}{16} (A_- A_+) \otimes (B_- B_+) = Q_{>} \otimes F_{33}.$$

The same multiplication with the other three ordered sign pairs proves every entry of (1376). Each A_{σ} has rank six and each B_v has rank one, so every $V_{\sigma v}$ has rank six. Its nonzero entries equal one. The identities $A_{\sigma}^2 = B_v^2 = 0$ give $V_{\sigma v}^2 = 0$. The source and range identities in (1376) prove the partial-isometry assertion.

The relations

$$Q_{<} Q_{>} = 0, \quad Q_{<} + Q_{>} = I_{V_{\mathcal{X}}}, \quad F_{22} F_{33} = 0$$

prove the orthogonality and the two sums in (1377). Transposition reverses both signs. Conjugation by P_ι exchanges A_- and A_+ , while conjugation by K_0 exchanges B_- and B_+ ; this proves (1378) and (1379).

Using $A_- + A_+ = 2P_\iota$, $B_- = 2F_{32}$, and $B_+ = 2F_{23}$, one obtains

$$\begin{aligned} V_{--} + V_{+-} &= \frac{1}{4}(A_- + A_+) \otimes B_- = P_\iota \otimes F_{32}, \\ V_{-+} + V_{++} &= \frac{1}{4}(A_- + A_+) \otimes B_+ = P_\iota \otimes F_{23}. \end{aligned}$$

Since $P_\iota^2 = I_{V_\chi}$, multiplication proves (1382) and (1383). The identities $F_{22} = j(E_{11})$ and $F_{33} = j(E_{22})$ follow directly from (1366).

Finally, every $V_{\sigma v}$ is a partial permutation matrix. Its split-zero lift has one supported unit in each occupied row and column. The unique Boolean character sends each supported unit to one and each τ to zero, proving (1384); Boolean multiplication then gives the same source and range projectors as the displayed table. $\square$

Theorem 14.37 (Basepointed torsor isomorphism between completed histories and the 12289 four-channel carrier). *Let*

$$\mathcal{F}_{--,34} := \Pi^{-1}(((-, -), \{3, 4\}))$$

be the four-history fibre of theorem 11.20, and put

$$\mathcal{V}_{12289} := \{V_{--}, V_{-+}, V_{+-}, V_{++}\}.$$

On $\mathcal{V}_{12289}$, define the two partial conjugations

$$\mathfrak{a}(X) := (P_\iota \otimes I_3)X(P_\iota \otimes I_3), \quad (1385)$$

$$\mathfrak{b}(X) := (I_{V_\chi} \otimes K_0)X(I_{V_\chi} \otimes K_0). \quad (1386)$$

They act by

$$\mathfrak{a}(V_{\sigma v}) = V_{\bar{\sigma}, v}, \quad \mathfrak{b}(V_{\sigma v}) = V_{\sigma, \bar{v}}. \quad (1387)$$

Consequently, $\mathfrak{a}$ and $\mathfrak{b}$ are commuting involutions whose action on $\mathcal{V}_{12289}$ is free and transitive. Their composite has the two equal descriptions

$$\mathfrak{ab}(V_{\sigma v}) = V_{\sigma v}^\top = (P_\iota \otimes K_0)V_{\sigma v}(P_\iota \otimes K_0). \quad (1388)$$

There is a unique bijection

$$\Phi : \mathcal{F}_{--,34} \longrightarrow \mathcal{V}_{12289}$$

satisfying

$$\Phi(h_{21}) = V_{--}, \quad \Phi \circ \mathsf{T} = \mathfrak{b} \circ \Phi, \quad \Phi \circ \iota = \mathfrak{a} \circ \Phi. \quad (1389)$$

It is given explicitly by

completed ordered history	12289 channel	
h_{21}	V_{--}	
$\mathsf{T}(h_{21}) = h_{12}$	V_{-+}	
$\iota(h_{21}) = \bar{h}_{21}$	V_{+-}	
$\iota\mathsf{T}(h_{21}) = \bar{h}_{12}$	V_{++}	

(1390)

The two factor-order orbits recover the complete directed and transposed coordinate carriers:

$$\sum_{h \in \{h_{21}, \iota(h_{21})\}} \Phi(h) = V_{--} + V_{+-} = V_{21}^{(12289)}, \quad (1391)$$

$$\sum_{h \in \{h_{12}, \iota(h_{12})\}} \Phi(h) = V_{-+} + V_{++} = V_{12}^{(12289)}. \quad (1392)$$

Proof. The relations

$$P_\iota A_\sigma P_\iota = A_{\bar{\sigma}}, \quad K_0 B_v K_0 = B_{\bar{v}}$$

and the tensor factorization

$$V_{\sigma v} = \frac{1}{4} A_\sigma \otimes B_v$$

give (1387). Each partial conjugation flips one sign and preserves the other. They therefore commute, and their joint orbit through V_{--} contains all four channels exactly once. Applying both flips both signs. The transpose identities in (1378) and the reflection identity in (1379) give (1388).

On $\mathcal{F}_{--,34}$, the commuting involutions T and ι act freely and transitively by theorem 11.20. Hence the basepoint equation $\Phi(h_{21}) = V_{--}$ and the two intertwining equations determine Φ on all four histories. They give the four rows of (1390), which are distinct, so Φ is a bijection.

The ι -orbit of h_{21} maps to the two channels carrying B_- , while the ι -orbit of h_{12} maps to the two channels carrying B_+ . Equations (1380) and (1381) prove the two orbit-sum identities. $\square$

Corollary 14.38 (Affine C_3 crosswalk and tetrahedral closure). *Put $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ and retain $b(+) = 0$, $b(-) = 1$. Identify $\mathcal{F}_{--,34}$ with $\mathbb{F}_2^2$ by the first factor:*

$$((\epsilon, \delta), (\lambda, \eta), \{3, 4\}) \mapsto x = \begin{pmatrix} b(\epsilon) \\ b(\delta) \end{pmatrix}.$$

Identify $\mathcal{V}_{12289}$ with $\mathbb{F}_2^2$ by

$$V_{\sigma v} \mapsto y = \begin{pmatrix} b(\sigma) \\ b(v) \end{pmatrix}.$$

In these coordinates, the map Φ of theorem 14.37 is

$$y = \phi(x) := Lx + c, \quad L = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (1393)$$

The two history involutions are translations by

$$t_{\mathsf{T}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad t_{\iota} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad (1394)$$

and the two channel involutions are translations by

$$e_{\mathsf{b}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad e_{\mathsf{a}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The linear part of ϕ intertwines them exactly:

$$Lt_{\mathsf{T}} = e_{\mathsf{b}}, \quad Lt_{\iota} = e_{\mathsf{a}}. \quad (1395)$$

Moreover,

$$L^2 + L + I_2 = 0, \quad L^3 = I_2, \quad (I_2 + L + L^2)c = 0. \quad (1396)$$

Hence the affine map ϕ has order three. It fixes

$$x_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and cycles the other three points of $\mathbb{F}_2^2$. The translation group and this order-three map generate

$$\langle \tau_v \ (v \in \mathbb{F}_2^2), \phi \rangle = \mathbb{F}_2^2 \rtimes \langle L \rangle \cong \text{Alt}(4). \quad (1397)$$

Thus the four-point carrier has one fixed point and one three-point orbit under the crosswalk C_3 , while its history translations form the normal $C_2 \times C_2$ subgroup of the tetrahedral rotation group.

Finally, the linear involution

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (1398)$$

satisfies

$$R^2 = I_2, \quad RLR = L^{-1}.$$

It enlarges the linear group to $\langle L, R \rangle = \text{GL}(2, \mathbb{F}_2)$, and

$$\mathbb{F}_2^2 \rtimes \text{GL}(2, \mathbb{F}_2) = \text{AGL}(2, \mathbb{F}_2) \cong S_4. \quad (1399)$$

Proof. Substitution of the four rows in (1390) gives

x	$\phi(x)$
$(1, 0)^\top$	$(1, 1)^\top$
$(0, 0)^\top$	$(1, 0)^\top$
$(0, 1)^\top$	$(0, 1)^\top$
$(1, 1)^\top$	$(0, 0)^\top$,

which is precisely (1393). The translations in (1394) are the bit formulas proved in theorem 11.20. Flipping the first, respectively second, channel sign adds $(1, 0)^\top$, respectively $(0, 1)^\top$. Direct multiplication by L proves (1395).

Matrix multiplication over $\mathbb{F}_2$ gives

$$L^2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad I_2 + L + L^2 = 0.$$

This proves (1396) and $\phi^3(x) = x$. Solving $x = Lx + c$ gives the unique fixed point $(0, 1)^\top$; the remaining three points form one orbit.

Translations by $\mathbb{F}_2^2$ form the regular Klein four group. Since L cyclically permutes its three nonidentity elements, the generated semidirect product has order 12 and is the tetrahedral rotation group $\text{Alt}(4)$. Finally, direct multiplication gives $RLR = L^2 = L^{-1}$. Thus L and R generate the six-element group $\text{GL}(2, \mathbb{F}_2)$. Its affine extension has order $4 \cdot 6 = 24$ and acts faithfully on the four affine points, which identifies it with S_4 . $\square$

Corollary 14.39 (Tetrahedral actions on the 64-cell completed sign-sedenion carrier). *Write the homogeneous basis of $\mathcal{A}_{64} = \mathbb{R}[\mathbb{F}_2^2] \otimes_{\mathbb{R}} A_4$ as*

$$x_{g,I} := \mathbf{u}_g \otimes o_I, \quad g \in \mathbb{F}_2^2, \quad I \subseteq \{1, 2, 3, 4\}.$$

For $M \in \text{GL}(2, \mathbb{F}_2)$ and $v \in \mathbb{F}_2^2$, define

$$\rho_M(x_{g,I}) := x_{Mg,I}, \quad (1400)$$

$$\ell_v(x_{g,I}) := x_{v+g,I}. \quad (1401)$$

Every ρ_M is an algebra automorphism of $\mathcal{A}_{64}$. Every ℓ_v is an invertible linear operator, and

$$\rho_M \ell_v \rho_M^{-1} = \ell_{Mv}. \quad (1402)$$

Consequently these operators give a faithful action

$$\mathbb{F}_2^2 \rtimes \text{GL}(2, \mathbb{F}_2) \cong S_4 \curvearrowright \mathcal{A}_{64}. \quad (1403)$$

For every blade I , the four-dimensional subspace

$$\mathcal{Q}_I := \text{span}_{\mathbb{R}}\{x_{g,I} : g \in \mathbb{F}_2^2\} \quad (1404)$$

is invariant. Hence the 64 homogeneous cells are the disjoint union of 16 four-cell tetrahedral blocks. Restricting the linear part to $\langle L \rangle$ gives the $\text{Alt}(4)$ subgroup in (1397).

For the mirror matrix R in (1398),

$$\rho_R(x_{(\alpha,\beta),I}) = x_{(\beta,\alpha),I}. \quad (1405)$$

Thus this mirror acts on the existing six-bit labels. Retaining the mirror as an additional multiplicative label gives the extension

$$\mathcal{A}_{64} \oplus \mathcal{A}_{64}u, \quad u^2 = 1, \quad ua = \rho_R(a)u,$$

of real dimension 128, as in (598).

Proof. The homogeneous multiplication law from theorem 11.18 is

$$x_{g,I}x_{h,J} = \omega_4(I, J)x_{g+h, I\Delta J}.$$

Since $M(g+h) = Mg + Mh$,

$$\rho_M(x_{g,I}x_{h,J}) = \omega_4(I, J)x_{Mg+Mh, I\Delta J} = \rho_M(x_{g,I})\rho_M(x_{h,J}).$$

This proves the algebra-automorphism assertion. Direct substitution proves (1402). Faithfulness is already visible on each four-element sign basis $\{x_{g,I} : g \in \mathbb{F}_2^2\}$, where the affine group realizes all 24 permutations. The blade index is unchanged, proving the 16-block decomposition. Equation (1405) follows by multiplying the sign-bit column by R . The final dimension is $64 + 64 = 128$. $\square$

Theorem 14.40 (Crossed two-sheet refinement of the tetrahedral carrier). *Let*

$$G := \mathbb{F}_2^2, \quad X := G \times \mathbb{F}_2,$$

where the second coordinate is a sheet label. For $v, w \in G$, define the independent sheet translations

$$\tau_{v,w}(g, \epsilon) := \begin{cases} (g+v, 0), & \epsilon = 0, \\ (g+w, 1), & \epsilon = 1. \end{cases} \quad (1406)$$

Let

$$\Theta(g, \epsilon) := (g, \epsilon + 1), \quad (1407)$$

$$J_R(g, \epsilon) := (Rg, \epsilon + 1), \quad (1408)$$

$$\hat{R}(g, \epsilon) := (Rg, \epsilon), \quad (1409)$$

where R is the matrix in (1398). Then

$$\Theta J_R = J_R \Theta = \hat{R} \quad (1410)$$

and

$$\Theta \tau_{v,w} \Theta^{-1} = \tau_{w,v}, \quad (1411)$$

$$J_R \tau_{v,w} J_R^{-1} = \tau_{Rw, Rv}. \quad (1412)$$

The resulting finite closures have orders

$$|\langle \tau_{v,w}, \Theta \rangle| = 32, \quad (1413)$$

$$|\langle \tau_{v,w}, \Theta, J_R \rangle| = 64. \quad (1414)$$

If R_+ and R_- apply R independently on the two sheets, then

$$|\langle \tau_{v,w}, \Theta, R_+, R_- \rangle| = 128. \quad (1415)$$

Let L_+ and L_- apply the order-three matrix L from (1393) independently on the two sheets. Put

$$T := G \rtimes \langle L \rangle \cong \text{Alt}(4), \quad S := G \rtimes \langle L, R \rangle \cong S_4.$$

Then

$$|\langle \tau_{v,w}, L_+, L_-, \Theta \rangle| = |T \wr C_2| = 288, \quad (1416)$$

$$|\langle \tau_{v,w}, L_+, L_-, \Theta, J_R \rangle| = 576, \quad (1417)$$

$$|\langle \tau_{v,w}, L_+, L_-, R_+, R_-, \Theta \rangle| = |S \wr C_2| = 1152. \quad (1418)$$

On the phase subgroup $\langle L_+ \rangle \times \langle L_- \rangle \cong C_3^2$, the pure exchange and crossed mirror induce

$$\vartheta(a, b) = (b, a), \quad \vartheta_R(a, b) = (-b, -a). \quad (1419)$$

Consequently

$$J_R L_+ J_R^{-1} = L_-^{-1}, \quad J_R L_- J_R^{-1} = L_+^{-1}. \quad (1420)$$

The crossed phase group has product

$$(a, b, \epsilon) \diamond_R (c, d, \delta) = ((a, b) + \vartheta_R^\epsilon(c, d), \epsilon + \delta). \quad (1421)$$

Its diagonal and antidiagonal six-element subgroups are

$$\{(a, a, \epsilon)\} \cong D_3, \quad (1422)$$

$$\{(a, -a, \epsilon)\} \cong C_6. \quad (1423)$$

Thus the crossed mirror exchanges the two subgroup types appearing for the pure sheet exchange in (66)–(67). Retaining both Θ and J_R gives a 36-element phase closure.

Finally, the external-mirror algebra in (598) has the homogeneous product

$$(x_{g,I}u^\epsilon)(x_{h,J}u^\delta) = \omega_4(I, J)x_{g+R^\epsilon h, I \triangle J}u^{\epsilon+\delta}. \quad (1424)$$

Hence

$$\mathcal{A}_{64} \oplus \mathcal{A}_{64}u \cong \mathbb{R}[G \rtimes_R C_2] \otimes_{\mathbb{R}} A_4. \quad (1425)$$

The successive sign and sheet carriers therefore give the exact sedenion-row dimension sequence

$$16, \quad 32, \quad 64, \quad 128. \quad (1426)$$

More generally, for a finite sign-sheet group H and the n -fold Cayley–Dickson algebra A_n ,

$$\dim_{\mathbb{R}}(\mathbb{R}[H] \otimes_{\mathbb{R}} A_n) = |H| 2^n. \quad (1427)$$

Proof. The maps $\tau_{v,w}$ form $G \times G$, of order 16. Substitution in (1407)–(1408) proves (1410)–(1412). Adjoining either sheet exchange gives a semidirect product of order $16 \cdot 2 = 32$. Their product is the simultaneous internal reflection $\hat{R}$. It fixes the zero bit vector and exchanges two nonzero vectors, whereas every nonidentity translation has no fixed bit vector. It therefore supplies a further factor of two, proving (1414). Independent internal reflections form

$$(\langle R \rangle \times \langle R \rangle) \rtimes C_2,$$

of order eight, over the 16-element translation group; this proves (1415).

The one-sheet affine groups T and S have orders 12 and 24. Pure sheet exchange gives their wreath products, whose orders are $12^2 \cdot 2 = 288$ and $24^2 \cdot 2 = 1152$. The simultaneous R -reflection lies outside $T \times T$ and commutes with sheet exchange, so it doubles 288 to 576.

The identity $RLR = L^{-1}$ proves (1419) and (1420). On the diagonal, $\vartheta_R(a, a) = (-a, -a)$, which gives the reflected $C_3 \rtimes C_2$ law and hence D_3 . On the antidiagonal, $\vartheta_R(a, -a) = (a, -a)$, which gives $C_3 \times C_2 \cong C_6$. The two commuting exchange involutions give a four-element outer action on C_3^2 , hence $9 \cdot 4 = 36$.

Equation (1424) follows from $ux_{h,J} = \rho_R(x_{h,J})u$ and the homogeneous product in (582). Its sign-sheet labels multiply by

$$(g, \epsilon)(h, \delta) = (g + R^\epsilon h, \epsilon + \delta),$$

the group law of $G \rtimes_R C_2$. This proves (1425). The final two dimension formulas follow from $\dim_{\mathbb{R}} A_n = 2^n$. Every finite closure and every displayed subgroup type is checked by `verification/verify_two_sheet_tetrahedral_granularity.py`. $\square$

Corollary 14.41 (Explicit crossed-mirror matrices on the eight sign-sheet cells). *Order the four sign cells as*

$$(+, +), (+, -), (-, +), (-, -).$$

The matrices induced by L and R are

$$P_L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad (1428)$$

$$P_R = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1429)$$

They satisfy

$$P_L^3 = I_4, \quad P_R^2 = I_4, \quad P_R P_L P_R = P_L^{-1}.$$

On the ordered pair of sheets, define

$$\Theta_8 = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}, \quad J_8 = \begin{pmatrix} 0 & P_R \\ P_R & 0 \end{pmatrix}, \quad \hat{R}_8 = \begin{pmatrix} P_R & 0 \\ 0 & P_R \end{pmatrix}. \quad (1430)$$

Then

$$\Theta_8 J_8 = J_8 \Theta_8 = \hat{R}_8.$$

The diagonal and antidiagonal phase matrices are

$$U_\Delta = \begin{pmatrix} P_L & 0 \\ 0 & P_L \end{pmatrix}, \quad U_\nabla = \begin{pmatrix} P_L & 0 \\ 0 & P_L^{-1} \end{pmatrix}. \quad (1431)$$

Their exchange relations are

$$\begin{aligned} \Theta_8 U_\Delta &= U_\Delta \Theta_8, & \Theta_8 U_\nabla \Theta_8 &= U_\nabla^{-1}, \\ J_8 U_\Delta J_8 &= U_\Delta^{-1}, & J_8 U_\nabla &= U_\nabla J_8. \end{aligned} \quad (1432)$$

Consequently

$$\begin{aligned} \langle U_\Delta, \Theta_8 \rangle &\cong C_6, & \langle U_\nabla, \Theta_8 \rangle &\cong D_3, \\ \langle U_\Delta, J_8 \rangle &\cong D_3, & \langle U_\nabla, J_8 \rangle &\cong C_6. \end{aligned}$$

Moreover,

$$J_8^\top = J_8, \quad J_8^2 = I_8 \succeq 0. \quad (1433)$$

Thus J_8 is the block coupling $\mathcal{D}_A$ from (69) with $A = P_R$, and its commuting order-three phase is the antidiagonal phase U_∇ .

Proof. The first two matrices are the permutations of $\mathbb{F}_2^2$ induced by L and R . Block multiplication, together with $RLR = L^{-1}$, gives every relation in (1432). The four subgroup identifications follow from the commuting and inversion actions of their order-two generators on their order-three generators. The last equation follows from $P_R^\top = P_R = P_R^{-1}$. The exact matrix products are checked in `verification/verify_two_sheet_tetrahedral_granularity.py`. $\square$

Theorem 14.42 (Crossed-mirror $2 \otimes (1 \oplus 3)$ modular spectral entwinement). *In the four-cell basis of theorem 14.41, put*

$$C_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (1434)$$

Then

$$P_L = (1) \oplus C_3, \quad P_R = (1) \oplus S_3, \quad S_3 C_3 S_3 = C_3^{-1}. \quad (1435)$$

Consequently the eight-cell carrier has the exact factorization

$$\mathcal{H}_8 = \mathbb{C}_{\text{sh}}^2 \otimes (\mathbb{C}_{\text{fix}} \oplus \mathbb{C}_{\text{orb}}^3), \quad (1436)$$

under which

$$\Theta_8 = \mathbf{X}_2 \otimes I_4, \quad J_8 = \mathbf{X}_2 \otimes ((1) \oplus \mathbf{S}_3), \quad \mathbf{X}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (1437)$$

The three explicit block generators

$$\mathbf{A}_1 = (0), \quad \mathbf{A}_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{A}_3 = \frac{\mathbf{C}_3 - \mathbf{C}_3^{-1}}{i\sqrt{3}} \quad (1438)$$

are self-adjoint and satisfy

$$\text{Spec}(\mathbf{A}_1) = \{0\}, \quad \text{Spec}(\mathbf{A}_2) = \{-1, 1\}, \quad \text{Spec}(\mathbf{A}_3) = \{-1, 0, 1\}. \quad (1439)$$

In particular,

$$\mathbf{A}_\oplus := \mathbf{A}_1 \oplus \mathbf{A}_2 \oplus \mathbf{A}_3 \quad \text{has spectrum} \quad \{-1, -1, 0, 0, 1, 1\}. \quad (1440)$$

On $\mathcal{H}_8$, define

$$K_{\text{sh}} := \mathbf{A}_2 \otimes I_4, \quad K_{\text{ph}} := \mathbf{A}_2 \otimes (\mathbf{A}_1 \oplus \mathbf{A}_3). \quad (1441)$$

These are commuting self-adjoint operators, and

$$K_{\text{ph}} = \frac{U_\nabla - U_\nabla^{-1}}{i\sqrt{3}}. \quad (1442)$$

Let κ_8 be coordinatewise complex conjugation and set

$$\mathcal{C}_8 := J_8 \kappa_8. \quad (1443)$$

Then $\mathcal{C}_8$ is an antiunitary involution and

$$\mathcal{C}_8 K_{\text{sh}} \mathcal{C}_8^{-1} = -K_{\text{sh}}, \quad \mathcal{C}_8 K_{\text{ph}} \mathcal{C}_8^{-1} = -K_{\text{ph}}. \quad (1444)$$

For $s, t \in \mathbb{R}$, the operator

$$K_8(s, t) := sK_{\text{sh}} + tK_{\text{ph}}, \quad \Delta_8(s, t) := \exp(K_8(s, t)) \quad (1445)$$

is positive and invertible and obeys

$$\mathcal{C}_8 \Delta_8(s, t) \mathcal{C}_8^{-1} = \Delta_8(s, t)^{-1}. \quad (1446)$$

Its generator has characteristic polynomial

$$\begin{aligned} \det(\lambda I_8 - K_8(s, t)) &= (\lambda - s)^2 (\lambda + s)^2 (\lambda - s - t)(\lambda - s + t) \\ &\quad \cdot (\lambda + s + t)(\lambda + s - t). \end{aligned} \quad (1447)$$

Thus the spectrum of $\Delta_8(s, t)$ consists of four reciprocal pairs:

$$\{e^s, e^s, e^{-s}, e^{-s}, e^{s+t}, e^{s-t}, e^{-s-t}, e^{-s+t}\}. \quad (1448)$$

Proof. The lower-right 3-by-3 blocks of P_L and P_R are $\mathbf{C}_3$ and $\mathbf{S}_3$, respectively. This proves (1435); tensoring with the two sheet labels gives (1436) and (1437).

The eigenvalues of $\mathbf{C}_3$ are $1, \omega, \omega^2$, where $\omega = e^{2\pi i/3}$. Applying

$$z \mapsto \frac{z - z^{-1}}{i\sqrt{3}}$$

gives $0, 1, -1$. This proves self-adjointness and (1439); the direct-sum statement follows.

Writing

$$U_\nabla = \text{diag}(P_L, P_L^{-1})$$

shows directly that

$$\frac{U_\nabla - U_\nabla^{-1}}{i\sqrt{3}} = \mathbf{A}_2 \otimes (\mathbf{A}_1 \oplus \mathbf{A}_3).$$

This proves (1442); it also proves that the two operators in (1441) commute.

All matrices in J_8 are real. Moreover,

$$X_2 A_2 X_2 = -A_2, \quad \overline{A_3} = -A_3, \quad S_3 A_3 S_3 = -A_3.$$

Substitution in $C_8 A C_8^{-1} = J_8 \overline{A} J_8$ proves (1444). Real functional calculus then proves (1446). Finally, the spectra in (1439) give (1447) and (1448). Every matrix identity and multiplicity is checked in `verification/verify_crossed_mirror_modular_spectral_entwinement.py`. $\square$

Theorem 14.43 (Exact finite standard-pair locus of the eight-cell family). *Let $\mathcal{M} \subseteq \mathcal{B}(\mathcal{H}_8)$ be a finite-dimensional von Neumann algebra and let $\Omega \in \mathcal{H}_8$ be cyclic and separating for $\mathcal{M}$. If the modular operator of $(\mathcal{M}, \Omega)$ is unitarily equivalent to $\Delta_8(s, t)$, then*

$$s = 0. \quad (1449)$$

Conversely, every $t \in \mathbb{R}$ admits a cyclic-and-separating standard pair on $\mathcal{H}_8$ whose modular conjugation and modular operator are exactly

$$C_8, \quad \Delta_8(0, t). \quad (1450)$$

The converse has the following explicit Hilbert–Schmidt realization. Put

$$\mathcal{B} := M_2(\mathbb{C}) \oplus M_2(\mathbb{C}), \quad \mathcal{K} := \text{HS}_2 \oplus \text{HS}_2, \quad (1451)$$

and let $\mathcal{B}$ act on $\mathcal{K}$ by left multiplication. For

$$\rho_t := \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}, \quad \Omega_t := (\rho_t^{1/2}, \rho_t^{1/2}), \quad (1452)$$

the Tomita polar data on $\mathcal{K}$ are

$$J_{\text{HS}}(X_1, X_2) := (X_1^*, X_2^*), \quad (1453)$$

$$\Delta_{\text{HS},t}(X_1, X_2) := (\rho_t X_1 \rho_t^{-1}, \rho_t X_2 \rho_t^{-1}). \quad (1454)$$

To state the intertwiner, let e_+, e_- be the sheet basis, let $\omega = e^{2\pi i/3}$, and in $\mathbb{C}_{\text{fix}} \oplus \mathbb{C}_{\text{orb}}^3$ put

$$f = (1, 0, 0, 0)^T, \quad v_0 = \frac{1}{\sqrt{3}}(0, 1, 1, 1)^T, \quad v_+ = \frac{1}{\sqrt{3}}(0, 1, \omega, \omega^2)^T, \quad v_- = \overline{v_+}. \quad (1455)$$

Define eight vectors in $\mathcal{H}_8$ by

$$\begin{aligned} q_{1,11} &:= \frac{e_+ \otimes f + e_- \otimes f}{\sqrt{2}}, & q_{1,12} &:= e_+ \otimes v_+, \\ q_{1,21} &:= \omega^2 e_- \otimes v_+, & q_{1,22} &:= \frac{i(e_+ \otimes f - e_- \otimes f)}{\sqrt{2}}, \end{aligned} \quad (1456)$$

$$\begin{aligned} q_{2,11} &:= \frac{e_+ \otimes v_0 + e_- \otimes v_0}{\sqrt{2}}, & q_{2,12} &:= e_- \otimes v_-, \\ q_{2,21} &:= \omega e_+ \otimes v_-, & q_{2,22} &:= \frac{i(e_+ \otimes v_0 - e_- \otimes v_0)}{\sqrt{2}}. \end{aligned} \quad (1457)$$

There is a unique linear map $Q : \mathcal{K} \rightarrow \mathcal{H}_8$ satisfying

$$Q(E_{ab}, 0) = q_{1,ab}, \quad Q(0, E_{ab}) = q_{2,ab} \quad (1 \leq a, b \leq 2). \quad (1458)$$

It is unitary and obeys

$$Q^{-1} C_8 Q = J_{\text{HS}}, \quad Q^{-1} \Delta_8(0, t) Q = \Delta_{\text{HS},t}. \quad (1459)$$

Thus the required standard pair on $\mathcal{H}_8$ is

$$\mathcal{M}_8 := Q \mathcal{B} Q^{-1}, \quad \Omega_8(t) := Q \Omega_t. \quad (1460)$$

Proof. Write the finite-dimensional algebra as

$$\mathcal{M} \cong \bigoplus_j M_{n_j}(\mathbb{C}).$$

Every faithful representation has the form

$$\mathcal{H}_8 \cong \bigoplus_j (\mathbb{C}^{n_j} \otimes \mathbb{C}^{m_j}),$$

with M_{n_j} acting on the first tensor factor. In one summand, a vector is an n_j -by- m_j matrix. Cyclicity requires full column rank, so $m_j \leq n_j$; separation requires full row rank, so $m_j \geq n_j$. Hence

$$m_j = n_j, \quad \sum_j n_j^2 = 8. \quad (1461)$$

The modular spectrum in the j -th block consists of the ratios $\lambda_{j,a}/\lambda_{j,b}$ of the positive eigenvalues of its density matrix. The $a = b$ terms contribute the eigenvalue 1 with multiplicity at least n_j . The three square partitions of 8 are

$$8 = 1 + \cdots + 1, \quad 8 = 4 + 1 + 1 + 1 + 1, \quad 8 = 4 + 4, \quad (1462)$$

and their forced unit-eigenvalue multiplicities are 8, 6, 4, respectively. Every modular operator of such a standard pair therefore has the eigenvalue 1 with multiplicity at least 4.

From (1448), the multiplicity of 1 in $\Delta_8(s, t)$ is

$$4\mathbf{1}_{\{s=0\}} + 2\mathbf{1}_{\{t=s\}} + 2\mathbf{1}_{\{t=-s\}}. \quad (1463)$$

When $s \neq 0$, at most one of $t = s$ and $t = -s$ can hold, so this multiplicity is at most 2. This proves (1449).

It remains to verify the displayed construction on $s = 0$. The vectors in (1456) and (1457) form an orthonormal basis. In the ordered matrix-unit basis

$$E_{11}, E_{12}, E_{21}, E_{22}$$

of each Hilbert–Schmidt block, direct multiplication gives

$$Q^{-1}K_{\text{ph}}Q = \text{diag}(0, 1, -1, 0, 0, 1, -1, 0), \quad (1464)$$

$$Q^{-1}C_8Q = J_{\text{HS}}. \quad (1465)$$

On the other hand,

$$\log \rho_t = \text{diag}(t/2, -t/2),$$

so left multiplication by $\log \rho_t$ minus right multiplication by $\log \rho_t$ has eigenvalues

$$0, t, -t, 0$$

on those four matrix units. Equations (1454) and (1464) now give the second identity in (1459); the first is (1465).

Each $\rho_t^{1/2}$ is invertible, so Ω_t is cyclic and separating for the two left matrix blocks. Transport through Q proves (1450) and (1460). All basis products, antiunitary phases, square partitions, and multiplicities are checked in `verification/verify_eight_cell_standard_pair_locus.py`. $\square$

Theorem 14.44 (Arithmetic six-path carrier inside the eight-cell standard pair). *Let Γ be the six-dimensional arithmetic path space in (1098), with basis $\gamma_{\epsilon, m}$ for $\epsilon \in C_2$ and $m \in C_3$. Let*

$$\mathcal{H}_{8, \text{orb}} := \mathbb{C}_{\text{sh}}^2 \otimes \mathbb{C}_{\text{orb}}^3 \subset \mathcal{H}_8,$$

and write g_0, g_1, g_2 for the orbit basis in which C_3 and S_3 have the matrices in (1434). The map

$$\Xi_\Gamma : \Gamma \longrightarrow \mathcal{H}_{8, \text{orb}}, \quad \Xi_\Gamma(\gamma_{\epsilon, m}) := e_\epsilon \otimes g_{2-m} \quad (1466)$$

is unitary, where the phase index is read modulo 3. It intertwines the two exact D_3 generators:

$$\Xi_\Gamma \text{Ad}_{W_r}|_\Gamma = U_\Delta|_{\mathcal{H}_{8,\text{orb}}} \Xi_\Gamma, \quad (1467)$$

$$\Xi_\Gamma \text{Ad}_{W_f}|_\Gamma = J_8|_{\mathcal{H}_{8,\text{orb}}} \Xi_\Gamma. \quad (1468)$$

Define the six path Fourier modes by

$$\begin{aligned} \hat{\gamma}_{\epsilon,0} &:= \frac{1}{\sqrt{3}} \sum_{m=0}^2 \gamma_{\epsilon,m}, \\ \hat{\gamma}_{\epsilon,+} &:= \frac{\omega^2}{\sqrt{3}} \sum_{m=0}^2 \omega^{-m} \gamma_{\epsilon,m}, \\ \hat{\gamma}_{\epsilon,-} &:= \frac{\omega}{\sqrt{3}} \sum_{m=0}^2 \omega^m \gamma_{\epsilon,m}. \end{aligned} \quad (1469)$$

Then

$$\Xi_\Gamma \hat{\gamma}_{\epsilon,0} = e_\epsilon \otimes v_0, \quad \Xi_\Gamma \hat{\gamma}_{\epsilon,+} = e_\epsilon \otimes v_+, \quad \Xi_\Gamma \hat{\gamma}_{\epsilon,-} = e_\epsilon \otimes v_-. \quad (1470)$$

Under the standard-pair unitary Q in (1458), these six modes occupy the following exact Hilbert–Schmidt matrix units:

$$\begin{aligned} Q^{-1} \Xi_\Gamma \hat{\gamma}_{0,0} &= \left(0, \frac{E_{11} - iE_{22}}{\sqrt{2}}\right), & Q^{-1} \Xi_\Gamma \hat{\gamma}_{1,0} &= \left(0, \frac{E_{11} + iE_{22}}{\sqrt{2}}\right), \\ Q^{-1} \Xi_\Gamma \hat{\gamma}_{0,+} &= (E_{12}, 0), & Q^{-1} \Xi_\Gamma \hat{\gamma}_{1,+} &= (\omega E_{21}, 0), \\ Q^{-1} \Xi_\Gamma \hat{\gamma}_{0,-} &= (0, \omega^2 E_{21}), & Q^{-1} \Xi_\Gamma \hat{\gamma}_{1,-} &= (0, E_{12}). \end{aligned} \quad (1471)$$

The unit vector spanning the D_3 -invariant arithmetic support line is

$$\gamma_{\text{inv}} := \frac{1}{\sqrt{6}} \sum_{\epsilon \in C_2} \sum_{m \in C_3} \gamma_{\epsilon,m}. \quad (1472)$$

It satisfies

$$Q^{-1} \Xi_\Gamma \gamma_{\text{inv}} = (0, E_{11}), \quad K_{\text{ph}} \Xi_\Gamma \gamma_{\text{inv}} = 0, \quad \mathcal{C}_8 \Xi_\Gamma \gamma_{\text{inv}} = \Xi_\Gamma \gamma_{\text{inv}}. \quad (1473)$$

Moreover,

$$\Gamma = \mathbb{C} \gamma_{\text{inv}} \hat{\oplus} \ker \mu, \quad (1474)$$

where μ is the arithmetic path-collapse map in (1099). Thus its unique invariant support line is exactly a modular unit line in the second Hilbert–Schmidt block.

The later classification theorem 14.303 identifies this same six-path space with both six-supported order-eight extrema:

$$A_6(q) \quad \text{and} \quad B_6(q, r).$$

Under the explicit map in theorem 14.305, their two missing classes occupy the two fixed cells $\mathbb{C}_{\text{sh}}^2 \otimes \mathbb{C}_{\text{fix}}$, while their six supported classes occupy $\mathcal{H}_{8,\text{orb}}$. Their local-interval coefficient vectors lie in the phase-zero subspace.

Proof. Both bases in (1466) are orthonormal, so Ξ_Γ is unitary. The arithmetic rotation sends (ϵ, m) to $(\epsilon, m+1)$, while $\mathcal{C}_3$ sends g_k to g_{k-1} . Hence

$$g_{2-(m+1)} = \mathcal{C}_3 g_{2-m},$$

which proves (1467). The arithmetic reflection sends (ϵ, m) to $(\epsilon+1, -m)$, while $\mathcal{S}_3 g_k = g_{1-k}$. The identity

$$2 - (-m) = 1 - (2 - m) \pmod{3}$$

proves (1468).

Substitution in (1469) gives (1470). Applying the eight defining columns of Q in (1456) and (1457) then gives every equality in (1471).

The invariant vector is

$$\gamma_{\text{inv}} = \frac{\widehat{\gamma}_{0,0} + \widehat{\gamma}_{1,0}}{\sqrt{2}}.$$

The first line of (1471) therefore sends it to $(0, E_{11})$. Equations (1464) and (1465) prove (1473). Finally, μ sends each of the six orthonormal path vectors to the same scalar matrix unit. Its kernel is the orthogonal complement of their equal-weight sum, which proves (1474). The complete six-vector calculation is checked in `verification/verify_six_path_eight_cell_standard_bridge.py`. $\square$

Theorem 14.45 (Shellwise modular-support factorization of the free Star–Kneser carrier). *Assume the conditions of theorem 14.28, and retain the unitaries Ξ_Γ and Q from theorems 14.43 and 14.44. Put*

$$\mathcal{K}_\Gamma := Q^{-1} \mathcal{H}_{8,\text{orb}} \subset \text{HS}_2 \oplus \text{HS}_2 \quad (1475)$$

and define

$$\mathcal{V}_{R,p} := (I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes Q^{-1} \Xi_\Gamma) \Theta_{123}. \quad (1476)$$

Then $\mathcal{V}_{R,p}$ is a unitary map

$$\mathbb{C}[\widetilde{\mathcal{X}}_{R,p}] \longrightarrow \mathbb{C}[\mathcal{O}_{R,p}] \otimes \mathcal{K}_\Gamma.$$

Give each Hilbert–Schmidt block the inner product $\langle X, Y \rangle_{\text{HS}} = \text{Tr}(X^* Y)$, and define

$$\lambda_{11}(X_1, X_2) := \text{Tr}(E_{11}^* X_2), \quad (1477)$$

$$\mathbf{m}_{11}(X_1, X_2) := \sqrt{6} \lambda_{11}(X_1, X_2) E_{1,1;1,1}. \quad (1478)$$

The free-shell support map has the exact factorization

$$\Pi_{123} = (I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes \mathbf{m}_{11}) \mathcal{V}_{R,p}. \quad (1479)$$

Let

$$P_{11}^{(2)} := |(0, E_{11})\rangle \langle (0, E_{11})|, \quad P_{R,p}^{\text{mod}} := I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes P_{11}^{(2)}. \quad (1480)$$

Its range lies in $\mathbb{C}[\mathcal{O}_{R,p}] \otimes \mathcal{K}_\Gamma$, and

$$\mathcal{V}_{R,p}^* P_{R,p}^{\text{mod}} \mathcal{V}_{R,p} = \Theta_{123}^* \left(I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes |\gamma_{\text{inv}}\rangle \langle \gamma_{\text{inv}}| \right) \Theta_{123} = \frac{1}{6} \Theta_{123}^* \left(I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes \mu^* \mu \right) \Theta_{123}. \quad (1481)$$

Consequently,

$$\text{rank } P_{R,p}^{\text{mod}} = q_{R,p}, \quad \dim \ker \Pi_{123} = 5q_{R,p}, \quad (1482)$$

$$\mathcal{V}_{R,p}(\ker \Pi_{123}) = (\mathbb{C}[\mathcal{O}_{R,p}] \otimes \mathcal{K}_\Gamma) \cap \ker P_{R,p}^{\text{mod}}. \quad (1483)$$

Write

$$\widehat{K}_{\text{ph}} := Q^{-1} K_{\text{ph}} Q, \quad \mathbf{K}_{R,p} := I_{\mathbb{C}[\mathcal{O}_{R,p}]} \otimes \widehat{K}_{\text{ph}}, \quad \mathbf{J}_{R,p} := \kappa_{\mathcal{O}_{R,p}} \otimes J_{\text{HS}},$$

where $\kappa_{\mathcal{O}_{R,p}}$ is coefficient conjugation in the orbit basis. The modular support projector obeys

$$\mathbf{K}_{R,p} P_{R,p}^{\text{mod}} = P_{R,p}^{\text{mod}} \mathbf{K}_{R,p} = 0, \quad \mathbf{J}_{R,p} P_{R,p}^{\text{mod}} \mathbf{J}_{R,p}^{-1} = P_{R,p}^{\text{mod}}. \quad (1484)$$

In particular,

$$\Omega_R(p) > 0 \iff P_{R,p}^{\text{mod}} \neq 0 \iff \Pi_{123} \neq 0. \quad (1485)$$

Equivalently, the fixed residual shell has an Erdős–Straus decomposition precisely when the modular support projector is occupied:

$$\left(\exists b, c \in \mathbb{Z}_{>0} : \frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \iff P_{R,p}^{\text{mod}} \neq 0. \quad (1486)$$

Thus every occupied complement orbit contributes exactly one zero-generator modular unit line, while its five transverse arithmetic modes form the kernel of the support map.

Proof. For

$$x = \sum_{\epsilon \in C_2} \sum_{m \in C_3} c_{\epsilon, m} \gamma_{\epsilon, m},$$

the path-collapse identity (1099) gives

$$\mu(x) = \left(\sum_{\epsilon, m} c_{\epsilon, m} \right) E_{1,1;1,1}. \quad (1487)$$

By (1473),

$$Q^{-1} \Xi_{\Gamma} \gamma_{\text{inv}} = (0, E_{11}).$$

The map $Q^{-1} \Xi_{\Gamma}$ is unitary onto $\mathcal{H}_{\Gamma}$; hence

$$\begin{aligned} \lambda_{11}(Q^{-1} \Xi_{\Gamma} x) &= \left\langle (0, E_{11}), Q^{-1} \Xi_{\Gamma} x \right\rangle_{\text{HS} \oplus \text{HS}} \\ &= \langle \gamma_{\text{inv}}, x \rangle_{\Gamma} = \frac{1}{\sqrt{6}} \sum_{\epsilon, m} c_{\epsilon, m}. \end{aligned} \quad (1488)$$

Equations (1487) and (1488) prove the one-orbit factorization. Tensoring it with $\mathbb{C}[\mathcal{O}_{R,p}]$ and using (1302) proves (1479).

The unitary $Q^{-1} \Xi_{\Gamma}$ sends $|\gamma_{\text{inv}}\rangle \langle \gamma_{\text{inv}}|$ to $P_{11}^{(2)}$. Moreover, μ sends all six orthonormal path basis vectors to the unit matrix $E_{1,1;1,1}$, so

$$|\gamma_{\text{inv}}\rangle \langle \gamma_{\text{inv}}| = \frac{1}{6} \mu^* \mu.$$

Tensoring by the orbit-label space and conjugating by Θ_{123} proves (1481). The projection has one-dimensional range over each of the $q_{R,p}$ orbit labels. Its orthogonal complement has dimension five over each label and is carried by (1479) to zero. This proves (1482) and (1483).

The vector $(0, E_{11})$ lies in the zero eigenspace of $\widehat{K}_{\text{ph}}$ and is fixed by J_{HS} , by (1473). The associated rank-one projection therefore satisfies (1484). Finally, $\text{rank } P_{R,p}^{\text{mod}} = q_{R,p}$ and (1296) give

$$P_{R,p}^{\text{mod}} \neq 0 \iff q_{R,p} > 0 \iff \Omega_R(p) > 0.$$

The last equivalence with $\Pi_{123} \neq 0$ is also (1312). Finally, $\Omega_R(p) = |\mathcal{X}_{R,p}| = m_0 + m_1 + m_2$, so theorem 9.64 proves (1486). The complete matrix calculation is checked in `verification/verify_shellwise_modular_support_map.py`. $\square$

Exact computation 14.46 (Exact shellwise modular-support checker). The script and captured receipt

`verification/verify_shellwise_modular_support_map.py,`

`verification/verify_shellwise_modular_support_map_output.txt`

verify the coefficient factorization, the equal-weight projection, the rank and nullity for one through six orbit factors, and the fixedness under the modular generator and conjugation.

Corollary 14.47 (Modular Star–Kneser forcing). *Retain the conditions of theorem 14.45 and the finite additive group H , sumset B , target T , and surpluses $\Sigma_T(K)$ of theorem 10.6. If*

$$\Sigma_T(K) > 0 \quad \text{for every subgroup } K \leq H, \quad (1489)$$

then

$$P_{R,p}^{\text{mod}} \neq 0, \quad (1490)$$

and the fixed residual shell has an Erdős–Straus decomposition. Conversely, if $P_{R,p}^{\text{mod}} = 0$, then

$$K = \text{Stab}(B) \quad \text{satisfies} \quad \Sigma_T(K) \leq 0. \quad (1491)$$

For the complete 12289 shell with $R = 11$ and $a = 3075$,

$$\text{rank } P_{11,12289}^{\text{mod}} = 6, \quad \dim \ker \Pi_{123} = 30. \quad (1492)$$

Proof. The exact convolution theorem gives

$$B \cap T \neq \emptyset \iff \mu(T_1(p)) + \mu(T_2(p)) > 0, \quad (1493)$$

$$\Omega_R(p) = \mu(T_1(p)) + 2\mu(T_2(p)). \quad (1494)$$

Both coefficients are nonnegative integers, so

$$B \cap T \neq \emptyset \iff \Omega_R(p) > 0 \iff P_{R,p}^{\text{mod}} \neq 0, \quad (1495)$$

where the last equivalence is (1485). The forward implication from (1489) and the obstruction (1491) now follow directly from theorem 10.7. The fixed-shell Erdős–Straus conclusion is (1486).

For $p = 12289$, theorem 10.27 gives $\Omega_{11}(12289) = 12$. Hence

$$q_{11,12289} = \frac{\Omega_{11}(12289)}{2} = 6.$$

Substitution in (1482) proves (1492). $\square$

Theorem 14.48 (Finite primewise modular-projector criterion). *Let p be an odd prime.*

If $p \equiv 3 \pmod{4}$, put

$$a = \frac{p+1}{4}, \quad N = pa, \quad b = N+1, \quad c = N(N+1). \quad (1496)$$

Then

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}. \quad (1497)$$

If $p \equiv 5 \pmod{12}$, put

$$a = \frac{p+3}{4}, \quad N = pa, \quad d = a, \quad b = \frac{N+d}{3} = \frac{(p+1)(p+3)}{12}, \quad c = \frac{N+N^2/d}{3} = pb. \quad (1498)$$

Then $R = 4a - p = 3$, and

$$d \mid N^2, \quad d \equiv -N \pmod{3}, \quad (1499)$$

so

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}. \quad (1500)$$

Suppose $p \equiv 1 \pmod{4}$, and define the finite denominator set

$$\mathcal{A}_p := \left\{ a \in \mathbb{Z}_{>0} : \frac{p}{4} < a \leq \frac{3p}{4} \right\}, \quad R_a := 4a - p. \quad (1501)$$

For every $a \in \mathcal{A}_p$, the conditions

$$R_a > 2, \quad \gcd(R_a, pa) = 1 \quad (1502)$$

hold. Hence theorem 14.45 supplies a finite standard-pair support projector $P_{R_a,p}^{\text{mod}}$. On the finite orthogonal sum

$$\mathcal{H}_p^{\text{mod}} := \bigoplus_{a \in \mathcal{A}_p} (\mathbb{C}[\mathcal{O}_{R_a,p}] \otimes \mathcal{H}_\Gamma), \quad \mathcal{P}_p^{\text{mod}} := \bigoplus_{a \in \mathcal{A}_p} P_{R_a,p}^{\text{mod}}, \quad (1503)$$

one has the exact equivalence

$$\left(\exists x, y, z \in \mathbb{Z}_{>0} : \frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right) \iff \mathcal{P}_p^{\text{mod}} \neq 0. \quad (1504)$$

Moreover,

$$|\mathcal{A}_p| = \frac{p-1}{2}, \quad \{R_a : a \in \mathcal{A}_p\} = \{3, 7, 11, \dots, 2p-3\}, \quad (1505)$$

and

$$\text{rank } \mathcal{P}_p^{\text{mod}} = \sum_{a \in \mathcal{A}_p} q_{R_a,p} = \frac{1}{2} \sum_{a \in \mathcal{A}_p} \Omega_{R_a}(p). \quad (1506)$$

Proof. For $p \equiv 3 \pmod{4}$, the data in (1496) satisfy

$$\frac{4}{p} - \frac{1}{a} = \frac{4a - p}{pa} = \frac{1}{N}$$

and

$$\frac{1}{b} + \frac{1}{c} = \frac{1}{N+1} + \frac{1}{N(N+1)} = \frac{1}{N}.$$

This proves (1497).

If $p \equiv 5 \pmod{12}$, write $p = 12m + 5$. Then

$$a = 3m + 2, \quad R = 4a - p = 3.$$

Modulo 3, both p and $a = d$ equal -1 , so

$$N = pa \equiv 1 \pmod{3}, \quad d \equiv -N \pmod{3}.$$

Also $d = a$ divides $N^2 = p^2 a^2$, proving (1499). The two reconstruction formulas give exactly the values of b and c in (1498). Directly,

$$\frac{4}{p} - \frac{1}{a} = \frac{4a - p}{pa} = \frac{3}{N}$$

and

$$\frac{1}{b} + \frac{1}{c} = \frac{1}{b} + \frac{1}{pb} = \frac{p+1}{pb} = \frac{3}{pa} = \frac{3}{N}.$$

This proves (1500).

Now let $p \equiv 1 \pmod{4}$. If

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z},$$

choose $a = \min\{x, y, z\}$. The summand $1/a$ is at least the average of the three positive summands, whereas the other two summands are nonzero. Therefore

$$\frac{1}{a} \geq \frac{4}{3p}, \quad \frac{1}{a} < \frac{4}{p}, \tag{1507}$$

which is equivalent to $a \in \mathcal{A}_p$. In particular, $a < p$, so $p \nmid a$. Since p is odd,

$$\begin{aligned} \gcd(R_a, a) &= \gcd(4a - p, a) = \gcd(p, a) = 1, \\ \gcd(R_a, p) &= \gcd(4a - p, p) = \gcd(4a, p) = 1. \end{aligned} \tag{1508}$$

Also $R_a > 0$ and $R_a \equiv -p \equiv 3 \pmod{4}$, hence $R_a \geq 3$. This proves (1502).

After relabelling the remaining two denominators as b, c , (1486) gives

$$P_{R_a, p}^{\text{mod}} \neq 0.$$

Thus a decomposition makes the finite direct-sum projector nonzero. Conversely, if $\mathcal{P}_p^{\text{mod}} \neq 0$, then one summand $P_{R_a, p}^{\text{mod}}$ is nonzero, and (1486) supplies positive integers b, c with

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}.$$

This proves (1504).

Write $p = 4k + 1$. Then

$$\mathcal{A}_p = \{k+1, k+2, \dots, 3k\},$$

which has $2k = (p-1)/2$ elements, and $a \mapsto 4a - p$ sends this set bijectively to

$$\{3, 7, 11, \dots, 8k-1\} = \{3, 7, 11, \dots, 2p-3\}.$$

This proves (1505). Finally, $\text{rank } P_{R_a, p}^{\text{mod}} = q_{R_a, p}$ and $q_{R_a, p} = \Omega_{R_a}(p)/2$ prove (1506). $\square$

Exact computation 14.49 (Finite-primewise reduction checker). The script and captured receipt

verification/verify_primewise_finite_modular_projector_reduction.py,
verification/verify_primewise_finite_modular_projector_reduction_output.txt

verify the denominator interval, residual-modulus list, coprimality, shell count, the explicit $R = 1$ identity, and the $R = 3$ endpoint certificate for every applicable odd prime up to 5000.

Theorem 14.50 (Endpoint divisor-orbit lower bound). *Let $p \equiv 1 \pmod{4}$ be prime, and define*

$$\nu_a(p) := \#\{R \in \mathbb{Z}_{>0} : R \mid p+1, R \equiv 3 \pmod{4}\}, \quad (1509)$$

$$\nu_p(p) := \#\{R \in \mathbb{Z}_{>0} : R \mid p+4, R \equiv 3 \pmod{4}\}. \quad (1510)$$

For every modulus counted by either function, put

$$a_R := \frac{p+R}{4}, \quad N_R := pa_R.$$

Then $a_R \in \mathcal{A}_p$.

If $R \mid p+1$, the divisor $d = a_R$ gives the certificate orbit

$$\{a_R, p^2 a_R\} \quad (1511)$$

and reconstructs

$$b = \frac{a_R(p+1)}{R}, \quad c = pb. \quad (1512)$$

Conversely, $d = a_R$ is a certificate in the R -shell only if $R \mid p+1$.

If $R \mid p+4$, the divisor $d = p$ gives the certificate orbit

$$\{p, pa_R^2\} \quad (1513)$$

and reconstructs

$$b = \frac{p(a_R+1)}{R}, \quad c = a_R b. \quad (1514)$$

Conversely, $d = p$ is a certificate in the R -shell only if $R \mid p+4$.

All the orbits in (1511) and (1513) are distinct. Consequently,

$$\boxed{Q_p \geq \nu_a(p) + \nu_p(p)}. \quad (1515)$$

Every modulus counted by either endpoint function also supplies an explicit nonzero image under the two-target extraction:

$$\boxed{R \mid (p+1) \vee R \mid (p+4) \implies \mu_{p,a_R} \notin \ker \mathfrak{T}_{p,a_R}}. \quad (1516)$$

For the primewise polar direct sum and the two operator energies of theorems 14.290 and 14.291, this gives

$$\boxed{\text{rank } \mathcal{V}_p^\partial = \mathcal{E}_p^{\text{pol},123} + \mathcal{E}_p^{\text{tim},123} \geq 3(\nu_a(p) + \nu_p(p))}. \quad (1517)$$

In particular,

$$\text{Vol}_p^{\det} \geq 2^{6(\nu_a(p) + \nu_p(p))}, \quad (1518)$$

$$L_{H,p}^2 \geq 51(\nu_a(p) + \nu_p(p)), \quad (1519)$$

$$L_{K,p}^2 \geq \frac{75}{2}(\nu_a(p) + \nu_p(p)). \quad (1520)$$

If the Erdős–Straus identity fails at p , every odd prime factor of $(p+1)/2$ and every prime factor of $p+4$ is congruent to 1 modulo 4.

Proof. If R is counted by either (1509) or (1510), then $R \equiv 3 \pmod{4}$, so a_R is an integer and $a_R > p/4$. Moreover $R \leq p+4 \leq 2p$ for $p \geq 5$, hence $a_R \leq 3p/4$. Thus $a_R \in \mathcal{A}_p$.

The integer a_R is a unit modulo R . Indeed, a common divisor of a_R and R divides $4a_R - R = p$, while R is coprime to p . Therefore

$$\begin{aligned} a_R \equiv -N_R \pmod{R} &\iff a_R \equiv -pa_R \pmod{R} \\ &\iff p+1 \equiv 0 \pmod{R}. \end{aligned}$$

Since $a_R \mid N_R^2$, this proves the certificate criterion for $d = a_R$. Its complementary divisor is $N_R^2/a_R = p^2a_R$. The reconstruction formulas give

$$\frac{N_R + a_R}{R} = \frac{a_R(p+1)}{R} = b, \quad \frac{N_R + p^2a_R}{R} = pb = c,$$

proving (1511) and (1512).

Likewise, p is a unit modulo R , and

$$\begin{aligned} p \equiv -N_R \pmod{R} &\iff p \equiv -pa_R \pmod{R} \\ &\iff a_R + 1 \equiv 0 \pmod{R} \\ &\iff p+4 \equiv 0 \pmod{R}, \end{aligned}$$

where the last equivalence uses $4(a_R + 1) = p + R + 4$. The complementary divisor is $N_R^2/p = pa_R^2$, and reconstruction gives

$$\frac{N_R + p}{R} = \frac{p(a_R + 1)}{R} = b, \quad \frac{N_R + pa_R^2}{R} = a_Rb = c.$$

This proves the second criterion and (1513)–(1514).

Orbits belonging to different R -shells are distinct summands of $\mathcal{P}_p^{\text{mod}}$. Within one shell, $a_R < p$; the orbit in (1511) has a_R as its unique member not exceeding p , while the orbit in (1513) has p . Hence these two orbits are distinct as well. Each orbit contributes one to $q_{R,p}$, proving (1515). The three analytic lower bounds follow from theorem 14.244. Each counted shell has $q_{R,p} > 0$, so (2656) proves (1516). The identities $\text{rank } \mathcal{V}_p^\partial = 3Q_p$ and $\mathcal{E}_p^{\text{pol},123} + \mathcal{E}_p^{\text{tim},123} = 3Q_p$ then prove (1517).

Finally, an odd positive integer has a divisor congruent to 3 modulo 4 exactly when at least one of its prime factors is congruent to 3 modulo 4. If the identity fails, then $Q_p = 0$, so (1515) forces $\nu_a(p) = \nu_p(p) = 0$. Applying the preceding divisor criterion to $(p+1)/2$ and $p+4$ proves the final assertion. $\square$

Exact computation 14.51 (Endpoint divisor-orbit checker). The script and captured receipt

```
verification/verify_endpoint_divisor_orbit_families.py,
verification/verify_endpoint_divisor_orbit_families_output.txt
```

verify both certificate equivalences, both reconstruction formulas, pairwise distinctness of the resulting complement orbits, survival under the two-target extraction, the C_3 -amplified polar rank bound, and the prime-factor obstruction for every prime $p \equiv 1 \pmod{4}$ up to 5000.

Proposition 14.52 (Endpoint three-target base grid). *Let $p \equiv 1 \pmod{4}$ be prime, let*

$$R \in \{3, 7, 11, \dots, 2p-3\}, \quad a := \frac{p+R}{4}, \quad N := pa,$$

and retain the origin-resolved support set $\mathcal{X}_{R,p}$ of (372). For every $j \in \{0, 1, 2\}$ and every $u \mid a^2$,

$$(j, u) \in \mathcal{X}_{R,p} \iff 4u + p^{2-j} \equiv 0 \pmod{R}. \quad (1521)$$

On the nine-cell base grid $\{0, 1, 2\} \times \{1, a, a^2\}$, the complete membership table is

$$\begin{array}{c|ccc}
 & u = 1 & u = a & u = a^2 \\
 \hline
 j = 0 & \emptyset & R \mid p + 1 & \emptyset \\
 j = 1 & R \mid p + 4 & \emptyset & R \mid p + 4 \\
 j = 2 & \emptyset & R \mid p + 1 & \emptyset
 \end{array} \tag{1522}$$

Here each divisibility entry is the necessary and sufficient condition for that cell to belong to $\mathcal{X}_{R,p}$, while $\emptyset$ records a cell that is empty for every displayed (p, R) .

If a is prime, then the base grid contains every divisor $u \mid a^2$, and therefore

$$P_{R,p}^{\text{mod}} \neq 0 \iff R \mid p + 1 \text{ or } R \mid p + 4. \tag{1523}$$

More precisely,

$$\text{rank } P_{R,p}^{\text{mod}} = \mathbf{1}_{R \mid p+1} + \mathbf{1}_{R \mid p+4}. \tag{1524}$$

Proof. The defining condition for $\mathcal{X}_{R,p}$ is

$$p^j u \equiv -pa \pmod{R}.$$

Both p and 4 are units modulo R , and $4a \equiv p \pmod{R}$. Dividing by p^j and multiplying by 4 therefore proves (1521).

For $u = a$, direct substitution into (1521) gives

$$\begin{aligned}
 (0, a) \in \mathcal{X}_{R,p} &\iff p + p^2 \equiv 0 \pmod{R} \iff R \mid p + 1, \\
 (1, a) \in \mathcal{X}_{R,p} &\iff 2p \equiv 0 \pmod{R}, \\
 (2, a) \in \mathcal{X}_{R,p} &\iff p + 1 \equiv 0 \pmod{R}.
 \end{aligned}$$

The middle congruence is impossible because $R \geq 3$ is odd and $\gcd(R, p) = 1$.

For $u = 1$, the three conditions are

$$p^2 + 4 \equiv 0 \pmod{R}, \quad p + 4 \equiv 0 \pmod{R}, \quad 5 \equiv 0 \pmod{R}.$$

The last is impossible because no positive divisor of 5 is congruent to 3 modulo 4 . The first is also impossible: since $R \equiv 3 \pmod{4}$, some prime $q \equiv 3 \pmod{4}$ divides R . If $R \mid p^2 + 4$, then

$$(p 2^{-1})^2 \equiv -1 \pmod{q},$$

contradicting the quadratic nonresiduosity of -1 modulo $q \equiv 3 \pmod{4}$.

For $u = a^2$, multiplication by the unit 4 and the congruence $16a^2 \equiv p^2 \pmod{R}$ transform the three conditions into

$$5p^2 \equiv 0 \pmod{R}, \quad p(p + 4) \equiv 0 \pmod{R}, \quad p^2 + 4 \equiv 0 \pmod{R}.$$

The first and third are impossible by the preceding arguments, while the middle condition is equivalent to $R \mid p + 4$. This proves (1522).

If a is prime, its squared divisor set is exactly $\{1, a, a^2\}$. The table then exhausts $\mathcal{X}_{R,p}$. Each satisfied endpoint condition contributes two cells, and

$$\text{rank } P_{R,p}^{\text{mod}} = \frac{1}{2} |\mathcal{X}_{R,p}|.$$

Equations (1523) and (1524) follow. $\square$

Exact computation 14.53 (Endpoint three-target-grid checker). The script and captured receipt

```
verification/verify_endpoint_three_target_grid.py,
verification/verify_endpoint_three_target_grid_output.txt
```

verify (1521), all nine entries of (1522), and the prime- a occupancy criterion for every applicable prime $p \leq 5000$.

Corollary 14.54 (Complete endpoint-coordinate divisor count). *Retain the conditions and notation of theorem 14.52. Put*

$$\mathcal{D}(a^2) := \{u \in \mathbb{Z}_{>0} : u \mid a^2\}$$

and define

$$\mathcal{E}_{R,a} := \{u \in \mathcal{D}(a^2) : R \mid 4u + 1\}, \quad (1525)$$

$$\mathcal{M}_{R,a} := \{u \in \mathcal{D}(a^2) : R \mid u + a\}. \quad (1526)$$

If

$$\mathcal{X}_{R,p}^{(j)} := \{u \in \mathcal{D}(a^2) : (j, u) \in \mathcal{X}_{R,p}\},$$

then

$$\mathcal{X}_{R,p}^{(2)} = \mathcal{E}_{R,a}, \quad \mathcal{X}_{R,p}^{(1)} = \mathcal{M}_{R,a}, \quad \mathcal{X}_{R,p}^{(0)} = \left\{ \frac{a^2}{u} : u \in \mathcal{E}_{R,a} \right\}. \quad (1527)$$

The complement involution $u \mapsto a^2/u$ preserves $\mathcal{M}_{R,a}$ and has no fixed point there. Consequently,

$$q_{R,p} = |\mathcal{E}_{R,a}| + \frac{1}{2}|\mathcal{M}_{R,a}|. \quad (1528)$$

The two divisor sets also have the exact integer descriptions

$$\mathcal{E}_{R,a} = \left\{ \frac{kR-1}{4} : \begin{array}{l} k \in \mathbb{Z}_{>0}, \ k \equiv 3 \pmod{4}, \\ (kR-1)/4 \mid a^2 \end{array} \right\}, \quad (1529)$$

$$\mathcal{M}_{R,a} = \left\{ kR-a : \begin{array}{l} k \in \mathbb{Z}_{>0}, \ kR > a, \\ kR-a \mid a^2 \end{array} \right\}. \quad (1530)$$

For $p \equiv 1 \pmod{4}$, let

$$R_a := 4a - p \quad (a \in \mathcal{A}_p).$$

Then the complete primewise projector rank is

$$Q_p = \sum_{a \in \mathcal{A}_p} \left(|\mathcal{E}_{R_a,a}| + \frac{1}{2}|\mathcal{M}_{R_a,a}| \right). \quad (1531)$$

Hence the Erdős–Straus identity at p is equivalent to the existence of $a \in \mathcal{A}_p$ for which

$$\mathcal{E}_{R_a,a} \neq \emptyset \quad \text{or} \quad \mathcal{M}_{R_a,a} \neq \emptyset. \quad (1532)$$

Proof. Equation (1521) at $j = 2$ gives

$$\mathcal{X}_{R,p}^{(2)} = \mathcal{E}_{R,a}.$$

At $j = 1$, the relation $p \equiv 4a \pmod{R}$ gives

$$4u + p \equiv 4(u + a) \pmod{R},$$

and 4 is a unit modulo R . This proves $\mathcal{X}_{R,p}^{(1)} = \mathcal{M}_{R,a}$. At $j = 0$, the same relation gives

$$4u + p^2 \equiv 4(u + 4a^2) \pmod{R}.$$

For $u \mid a^2$, both u and a are units modulo R , and

$$u + 4a^2 \equiv 0 \pmod{R} \iff 4\frac{a^2}{u} + 1 \equiv 0 \pmod{R}.$$

This proves the third equality in (1527).

If $u \in \mathcal{M}_{R,a}$, then

$$u + a \equiv 0 \pmod{R}.$$

Multiplication of $a^2/u + a$ by the unit u gives

$$a^2 + au = a(a + u) \equiv 0 \pmod{R}.$$

Thus $a^2/u \in \mathcal{M}_{R,a}$, and the converse follows by applying the same involution again. A fixed point would be $u = a$, but $R \mid u + a$ would then imply $R \mid 2a$, which is impossible because $R \geq 3$ and $\gcd(R, a) = 1$. Therefore $|\mathcal{M}_{R,a}|$ is even. The exterior fibres have equal cardinality $|\mathcal{E}_{R,a}|$, so

$$q_{R,p} = \frac{1}{2}|\mathcal{X}_{R,p}| = |\mathcal{E}_{R,a}| + \frac{1}{2}|\mathcal{M}_{R,a}|,$$

proving (1528).

If $u \in \mathcal{E}_{R,a}$, write $4u + 1 = kR$. Since $R \equiv 3 \pmod{4}$, reduction modulo 4 forces $k \equiv 3 \pmod{4}$, and $u = (kR - 1)/4$. The converse is immediate. Likewise, $R \mid u + a$ is equivalent to $u = kR - a > 0$ for an integer $k > 0$. This proves (1529) and (1530).

Finally, sum (1528) over $a \in \mathcal{A}_p$ and use (2221). This gives (1531). The equivalence (1532) follows from theorem 14.244. $\square$

Exact computation 14.55 (Complete endpoint-coordinate divisor checker). The script and captured receipt

```
verification/verify_complete_endpoint_divisor_formula.py,
verification/verify_complete_endpoint_divisor_formula_output.txt
```

verify the three origin formulas, both complement pairings, the absence of middle fixed points, the orbit count (1528), and both integer parametrizations for every applicable prime $p \leq 5000$.

Theorem 14.56 (Middle coprime-factor orbit classification). *Let $p \equiv 1 \pmod{4}$ be prime, let $a \in \mathcal{A}_p$, and put $R = 4a - p$. Retain $\mathcal{E}_{R,a}$ and $\mathcal{M}_{R,a}$ from (1525) and (1526). Define*

$$\mathcal{C}_{R,a} := \left\{ \{x, y\} : \begin{array}{l} x, y \in \mathbb{Z}_{>0}, \quad \gcd(x, y) = 1, \\ xy \mid a, \quad R \mid x + y \end{array} \right\}, \quad (1533)$$

where $\{x, y\}$ is unordered. For $\{x, y\} \in \mathcal{C}_{R,a}$, put $h = a/(xy)$. Then

$$\Phi_{R,a} : \mathcal{C}_{R,a} \longrightarrow \mathcal{M}_{R,a} / (u \sim a^2/u), \quad \{x, y\} \longmapsto \{hx^2, hy^2\}, \quad (1534)$$

is a bijection. Consequently,

$$q_{R,p} = |\mathcal{E}_{R,a}| + |\mathcal{C}_{R,a}|. \quad (1535)$$

If $u \in \mathcal{M}_{R,a}$ is written as

$$u = kR - a, \quad k \in \mathbb{Z}_{>0}, \quad (1536)$$

then

$$\gcd(k, a) = \gcd(u, a). \quad (1537)$$

In particular,

$$\gcd(k, a) = 1 \iff u = 1. \quad (1538)$$

The endpoint orbit occurs precisely when

$$\{1, a^2\} \subset \mathcal{M}_{R,a} \iff R \mid p + 4. \quad (1539)$$

Finally, the map

$$r \longmapsto \{a/r, ar\} \quad (1540)$$

injects

$$\mathcal{B}_{R,a} := \{r : r \mid a, R \mid r + 1\} \quad (1541)$$

into the middle complement orbits. Hence

$$q_{R,p} \geq |\mathcal{E}_{R,a}| + |\mathcal{B}_{R,a}|. \quad (1542)$$

Proof. Since $0 < a < p$ and p is prime, $\gcd(p, a) = 1$. The identity $R = 4a - p$ therefore gives

$$\gcd(R, a) = 1. \quad (1543)$$

Take $u \in \mathcal{M}_{R,a}$, put $v = a^2/u$, and define

$$g := \gcd(u, a), \quad x := \frac{u}{g}, \quad y := \frac{a}{g}.$$

Then $\gcd(x, y) = 1$. The divisibility $u \mid a^2$ gives

$$gx \mid g^2 y^2, \quad \text{hence} \quad x \mid gy^2.$$

As $\gcd(x, y) = 1$, one has $x \mid g$. Writing $g = hx$ gives

$$u = hx^2, \quad a = hxy, \quad v = hy^2. \quad (1544)$$

The middle congruence is

$$R \mid u + a = hx(x + y).$$

Because $hx \mid a$, (1543) gives $\gcd(R, hx) = 1$. Thus

$$R \mid x + y.$$

This constructs the preimage in $\mathcal{C}_{R,a}$.

Conversely, let $\{x, y\} \in \mathcal{C}_{R,a}$ and $h = a/(xy)$. Then

$$(hx^2)(hy^2) = a^2$$

and

$$hx^2 + a = hx(x + y), \quad hy^2 + a = hy(x + y).$$

Both displayed divisors therefore lie in $\mathcal{M}_{R,a}$. Exchanging complementary divisors exchanges x and y , so the unordered pair is uniquely recovered. This proves (1534). Combining it with (1528) proves (1535).

For (1537), substitute $R = 4a - p$ into (1536):

$$u = (4k - 1)a - kp.$$

Using $\gcd(p, a) = 1$, one obtains

$$\gcd(u, a) = \gcd(kp, a) = \gcd(k, a).$$

If this common divisor is 1, the conditions $u \mid a^2$ and $\gcd(u, a) = 1$ force $u = 1$. The converse is immediate. Moreover,

$$1 \in \mathcal{M}_{R,a} \iff R \mid a + 1 \iff R \mid p + 4,$$

where the last step uses $p = 4a - R$ and the oddness of R . Complement closure now gives (1539).

For $r \mid a$, the two divisors in (1540) are complementary, and

$$R \mid a/r + a \iff R \mid r + 1 \iff R \mid ar + a,$$

because $\gcd(R, a) = \gcd(R, r) = 1$. The value $r = 1$ cannot occur, since $R \geq 3$. Distinct admissible values of r give distinct unordered pairs, proving the injection and (1542). $\square$

Theorem 14.57 (Middle residue-lift family). *Let $R, x, y \in \mathbb{Z}_{>0}$ satisfy*

$$R \equiv 3 \pmod{4}, \quad \gcd(x, y) = 1, \quad R \mid x + y. \quad (1545)$$

Let p be prime and suppose

$$p \equiv -R \pmod{4xy}, \quad R \leq 2p - 3. \quad (1546)$$

Put

$$h := \frac{p + R}{4xy}, \quad a := hxy, \quad u := hx^2, \quad v := hy^2. \quad (1547)$$

Then $p \equiv 1 \pmod{4}$, $a \in \mathcal{A}_p$, and

$$\{u, v\} \subset \mathcal{M}_{R,a}. \quad (1548)$$

In particular $q_{R,p} \geq 1$. The divisor certificate $d = pu$ reconstructs

$$b = \frac{phx(x+y)}{R}, \quad c = \frac{phy(x+y)}{R}, \quad (1549)$$

and

$$\frac{4}{p} = \frac{1}{hxy} + \frac{1}{b} + \frac{1}{c}. \quad (1550)$$

Proof. The congruence in (1546) makes h a positive integer and gives $p \equiv -R \equiv 1 \pmod{4}$. Since

$$a = \frac{p+R}{4}$$

and $0 < R \leq 2p-3$, one has $a \in \mathcal{A}_p$ and $4a-p=R$. The data in (1547) satisfy

$$uv = a^2, \quad u+a = hx(x+y), \quad v+a = hy(x+y).$$

The seed condition therefore proves (1548). For $d = pu$, the reconstruction formulas of theorem 9.1 give

$$b = \frac{p(a+u)}{R}, \quad c = \frac{p(a+v)}{R}.$$

Substitution yields (1549) and (1550). $\square$

Exact computation 14.58 (Middle coprime-factor orbit checker). The script and captured receipt

`verification/verify_middle_coprime_factor_orbits.py,`

`verification/verify_middle_coprime_factor_orbits_output.txt`

verify the orbit bijection, the exact count (1535), the k -gcd identity, the balanced-divisor injection, and the middle residue lifts for all tested shells through $p = 5000$ and lifted primes through $p = 20000$.

Theorem 14.59 (Squarefree coordinates for both shell channels). *Retain the conditions and notation of theorem 14.56. Define*

$$\mathcal{S}_{R,a}^{\text{ext}} := \left\{ (s, t, m) : \begin{array}{l} s, t, m \in \mathbb{Z}_{>0}, \ s \text{ squarefree}, \\ a = stm, \ R \mid 4st^2 + 1 \end{array} \right\}, \quad (1551)$$

$$\mathcal{S}_{R,a}^{\text{mid}} := \left\{ (s, \{t, m\}) : \begin{array}{l} s, t, m \in \mathbb{Z}_{>0}, \ s \text{ squarefree}, \\ a = stm, \ R \mid t+m \end{array} \right\}, \quad (1552)$$

where $\{t, m\}$ is unordered. The maps

$$\Psi_{R,a}^{\text{ext}} : \mathcal{S}_{R,a}^{\text{ext}} \longrightarrow \mathcal{E}_{R,a}, \quad (s, t, m) \longmapsto st^2, \quad (1553)$$

$$\Psi_{R,a}^{\text{mid}} : \mathcal{S}_{R,a}^{\text{mid}} \longrightarrow \mathcal{M}_{R,a}/(u \sim a^2/u), \quad (s, \{t, m\}) \longmapsto \{st^2, sm^2\} \quad (1554)$$

are bijections. In particular,

$$q_{R,p} = |\mathcal{S}_{R,a}^{\text{ext}}| + |\mathcal{S}_{R,a}^{\text{mid}}|. \quad (1555)$$

For an exterior coordinate (s, t, m) , the certificate $d = p^2 st^2$ gives

$$b = \frac{pst(m+pt)}{R}, \quad c = \frac{sm(pt+m)}{R}. \quad (1556)$$

For a middle coordinate $(s, \{t, m\})$, the certificate $d = pst^2$ gives

$$b = \frac{pst(t+m)}{R}, \quad c = \frac{psm(t+m)}{R}. \quad (1557)$$

Proof. Every positive integer u has a unique expression

$$u = st^2$$

with s squarefree. If $u \mid a^2$, prime valuations give

$$st \mid a.$$

Thus there is a unique $m \in \mathbb{Z}_{>0}$ such that

$$a = stm, \quad \frac{a^2}{u} = sm^2. \quad (1558)$$

Conversely, every triple in (1558) produces a divisor pair of a^2 . Complementation exchanges t and m .

The exterior condition is

$$R \mid 4u + 1 \iff R \mid 4st^2 + 1,$$

which proves (1553). The middle condition is

$$R \mid u + a \iff R \mid st(t + m).$$

Since $st \mid a$ and $\gcd(R, a) = 1$, it is equivalent to

$$R \mid t + m.$$

The same criterion holds for the complementary divisor sm^2 . Moreover, $t = m$ would imply $R \mid 2t$, which is impossible because $\gcd(R, t) = 1$ and $R \geq 3$. This proves (1554). Equation (1555) follows from (1528).

In these coordinates the square-divisor hull from (2211) is

$$\mathfrak{h}(st^2) = st. \quad (1559)$$

Substituting $a = stm$, $u = st^2$, and $a^2/u = sm^2$ into the reconstruction formulas of theorem 9.1 gives (1556) and (1557). $\square$

Corollary 14.60 (Squarefree-coordinate rank–direction–volume law). *For each prime $p \equiv 1 \pmod{4}$, define*

$$\Theta_p := \sum_{a \in \mathcal{A}_p} \left(|\mathcal{S}_{R_a, a}^{\text{ext}}| + |\mathcal{S}_{R_a, a}^{\text{mid}}| \right). \quad (1560)$$

Define the two occupied-channel index sets and their directional symmetric-difference count by

$$\begin{aligned} \mathcal{A}_p^{\text{ext}} &:= \left\{ a \in \mathcal{A}_p : \mathcal{S}_{R_a, a}^{\text{ext}} \neq \emptyset \right\}, \\ \mathcal{A}_p^{\text{mid}} &:= \left\{ a \in \mathcal{A}_p : \mathcal{S}_{R_a, a}^{\text{mid}} \neq \emptyset \right\}, \\ \Xi_p &:= \left| \mathcal{A}_p^{\text{ext}} \triangle \mathcal{A}_p^{\text{mid}} \right|. \end{aligned} \quad (1561)$$

Then

$$\Theta_p = Q_p. \quad (1562)$$

Moreover,

$$\boxed{0 \leq \Xi_p \leq \Theta_p = Q_p.} \quad (1563)$$

Equality $\Xi_p = \Theta_p$ holds exactly when every occupied shell has one nonempty channel, that channel contains one squarefree coordinate, and the other channel is empty. Consequently,

$$\dim \mathcal{S}_p = 6\Theta_p, \quad (1564)$$

$$\text{Spec}(H_p^{\text{sh}}) = \{4^{[3\Theta_p]}, (-1)^{[3\Theta_p]}\}, \quad (1565)$$

$$\text{Vol}_p^{\det} = 2^{6\Theta_p}, \quad (1566)$$

$$L_{H,p}^2 = 51\Theta_p, \quad L_{K,p}^2 = \frac{75}{2}\Theta_p. \quad (1567)$$

Proof. Sum (1555) over $a \in \mathcal{A}_p$ and compare with (2221). This proves (1562).

For fixed a , put

$$x_a := |\mathcal{S}_{R,a,a}^{\text{ext}}|, \quad y_a := |\mathcal{S}_{R,a,a}^{\text{mid}}|.$$

Its contribution to Ξ_p is one exactly when precisely one of x_a, y_a is positive, and is zero otherwise. Therefore

$$\mathbf{1}_{\{x_a > 0, y_a = 0\} \cup \{x_a = 0, y_a > 0\}} \leq x_a + y_a.$$

Equality holds for an occupied shell exactly when one of x_a, y_a equals one and the other equals zero. Summing this pointwise inequality proves (1563) and its equality statement. The remaining identities follow from theorem 14.244. $\square$

Exact computation 14.61 (Squarefree two-channel coordinate checker). The script and captured receipt

```
verification/verify_squarefree_shell_coordinates.py,
verification/verify_squarefree_shell_coordinates_output.txt
```

verify the unique squarefree factor coordinates, complement exchange, both channel congruences, and (1555) for every tested shell through $p = 5000$. They also verify (1563) and its equality criterion for all 329 tested primes; the aggregate directional count is 6364, the aggregate orbit rank is 16229, and none of the tested primewise carriers is an equality case. The only tested primes with $\Xi_p = 0$ are 5 and 17; no tested prime congruent to 1 (mod 12) has zero directional coordinate.

Theorem 14.62 (Exact displacement of the endpoint divisor fibres). *Let $p \equiv 1 \pmod{4}$ be prime, let $a \in \mathcal{A}_p$, and put $R = 4a - p$. Write*

$$\rho_{R,a} : \{u \in \mathbb{Z}_{>0} : u \mid a^2\} \longrightarrow \mathbb{Z}/R\mathbb{Z}, \quad \rho_{R,a}(u) = \bar{u},$$

and define the exterior target and endpoint displacement by

$$e_R := -\bar{4}^{-1}, \quad \delta_{R,p} := \bar{4}^{-1} - \bar{a} = -\overline{(p-1)}\bar{4}^{-1}. \quad (1568)$$

Then the complete endpoint sections are the two exact fibres

$$\boxed{\mathcal{E}_{R,a} = \rho_{R,a}^{-1}(e_R), \quad \mathcal{M}_{R,a} = \rho_{R,a}^{-1}(e_R + \delta_{R,p}).} \quad (1569)$$

The ambient affine translation

$$\bar{u} \longmapsto \bar{u} + \delta_{R,p}$$

carries the exterior target to the middle target. Its displacement vanishes exactly on the coupled locus:

$$\delta_{R,p} = 0 \iff R \mid p-1. \quad (1570)$$

In particular, for every $u_E \in \mathcal{E}_{R,a}$ and $u_M \in \mathcal{M}_{R,a}$,

$$u_M - u_E \equiv \delta_{R,p} \equiv -\overline{(p-1)}\bar{4}^{-1} \pmod{R}. \quad (1571)$$

Proof. The identity $4a = p + R$ gives

$$\bar{a} = \bar{p}\bar{4}^{-1} \quad \text{in } \mathbb{Z}/R\mathbb{Z}.$$

Therefore

$$\bar{4}^{-1} - \bar{a} = \overline{(1-p)}\bar{4}^{-1},$$

which proves (1568). The defining congruences for the complete exterior and middle sections are, respectively,

$$\bar{u} = -\bar{4}^{-1} = e_R, \quad \bar{u} = -\bar{a} = e_R + \delta_{R,p}.$$

This proves (1569). Since R is odd, $\bar{4}$ is a unit; hence $\delta_{R,p} = 0$ exactly when $p \equiv 1 \pmod{R}$, proving (1570). Subtracting the two target congruences proves (1571). $\square$

Theorem 14.63 (Dihedral endpoint-orbit law). *Retain the conditions of theorem 14.62. For $r \in U_R$, put*

$$\mathcal{F}_{R,a}(r) := \{u \in \mathbb{Z}_{>0} : u \mid a^2, \bar{u} = r\}.$$

Define two bijections of U_R by

$$T_{R,p}(r) := \bar{p}r, \quad J_{R,a}(r) := \bar{a}^2 r^{-1}. \quad (1572)$$

Then $J_{R,a}$ is an involution and

$$J_{R,a}T_{R,p}J_{R,a} = T_{R,p}^{-1}. \quad (1573)$$

Let $r_0 := -\bar{a}$ and, for $j \in \mathbb{Z}$, set $r_j := \bar{p}^j r_0$. The three labelled endpoint targets are

$$\boxed{r_{-1} = -\bar{4}^{-1} \quad (\text{exterior}), \quad r_0 = -\bar{a} \quad (\text{middle}), \quad r_1 = -\bar{p}\bar{a} \quad (\text{opposite exterior}).} \quad (1574)$$

Moreover,

$$J_{R,a}(r_j) = r_{-j}.$$

The divisor complement $u \mapsto a^2/u$ induces the fibre bijections

$$\mathcal{F}_{R,a}(r_j) \xrightarrow{\sim} \mathcal{F}_{R,a}(r_{-j}). \quad (1575)$$

Consequently,

$$\boxed{2q_{R,p} = |\mathcal{F}_{R,a}(r_{-1})| + |\mathcal{F}_{R,a}(r_0)| + |\mathcal{F}_{R,a}(r_1)|.} \quad (1576)$$

If $n = \text{ord}_{U_R}(\bar{p}) \geq 3$, the restrictions of $T_{R,p}$ and $J_{R,a}$ to $\{r_j : j \in \mathbb{Z}/n\mathbb{Z}\}$ give the faithful dihedral action of order $2n$. On a coupled shell, $n = 1$, and the three labelled targets lie over one residue.

Proof. Both a and p are units modulo R . Direct evaluation gives

$$J_{R,a}^2(r) = \bar{a}^2 (\bar{a}^2 r^{-1})^{-1} = r$$

and

$$J_{R,a}T_{R,p}J_{R,a}(r) = \bar{a}^2 (\bar{p}\bar{a}^2 r^{-1})^{-1} = \bar{p}^{-1}r.$$

This proves (1573).

Since $4a \equiv p \pmod{R}$,

$$r_{-1} = -\bar{p}^{-1}\bar{a} = -\bar{4}^{-1}.$$

The remaining two descriptions in (1574) follow from the definition of r_j . For every j ,

$$J_{R,a}(r_j) = \bar{a}^2 (-\bar{p}^j \bar{a})^{-1} = -\bar{p}^{-j} \bar{a} = r_{-j}.$$

Residue reduction of a^2/u proves (1575). In particular,

$$|\mathcal{F}_{R,a}(r_{-1})| = |\mathcal{F}_{R,a}(r_1)|.$$

The first two fibres are $\mathcal{E}_{R,a}$ and $\mathcal{M}_{R,a}$, respectively. Thus (1528) gives

$$2q_{R,p} = 2|\mathcal{E}_{R,a}| + |\mathcal{M}_{R,a}|,$$

which is (1576). Finally, $T_{R,p}$ acts by $j \mapsto j + 1$ and $J_{R,a}$ by $j \mapsto -j$. For $n \geq 3$, these are the standard faithful dihedral permutations of $\mathbb{Z}/n\mathbb{Z}$. $\square$

Corollary 14.64 (Exact C_6 endpoint carrier at $p = 73$). *For $p = 73$, $R = 7$, and $a = 20$, multiplication by $\bar{p} = \bar{3}$ generates the target cycle*

$$(r_0, r_1, r_2, r_3, r_4, r_5) = (1, 3, 2, 6, 4, 5). \quad (1577)$$

The middle target is $r_0 = 1$, the exterior target is $r_{-1} = r_5 = 5$, and its complementary exterior target is $r_1 = 3$. Under the canonical map

$$\Phi_7(u) = (\chi_7(u), \chi_7(u)u) : U_7 \longrightarrow C_2 \times C_3,$$

the cycle is

j	0	1	2	3	4	5
r_j	1	3	2	6	4	5
$\Phi_7(r_j)$	(1, 1)	(-1, 4)	(1, 2)	(-1, 1)	(1, 4)	(-1, 2)

(1578)

Hence one step of the C_6 rotation flips the C_2 sheet and rotates the C_3 phase.

Proof. The class $3 \in U_7$ has order 6, and repeated multiplication of $r_0 = -20 \equiv 1 \pmod{7}$ gives (1577). Since

$$\Phi_7(3) = (-1, 4),$$

multiplicativity of Φ_7 gives every entry of (1578). $\square$

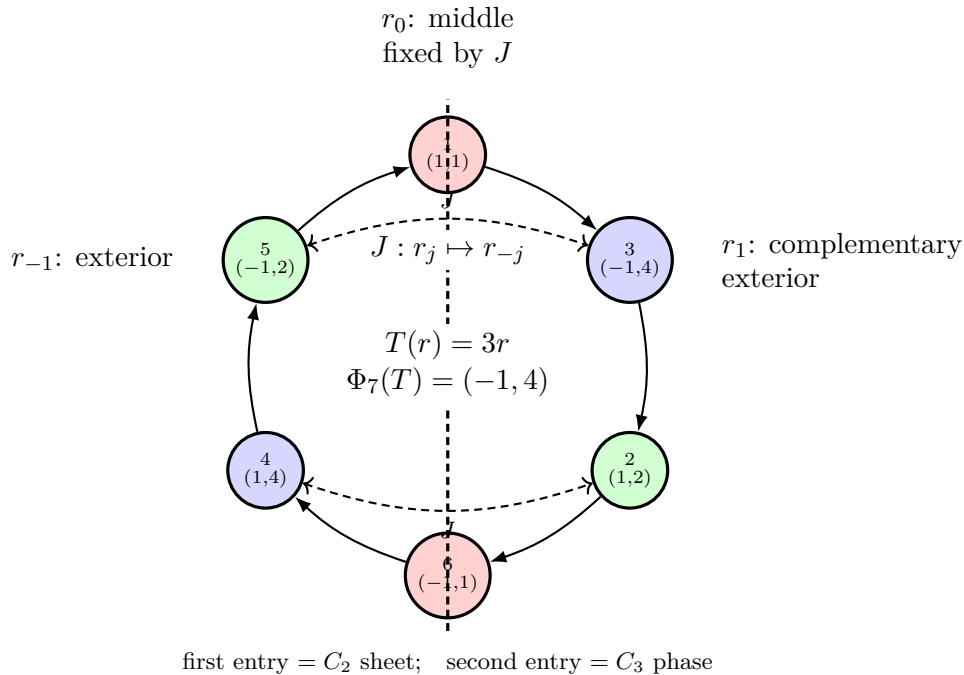


FIGURE 13. The exact endpoint target orbit for $p = 73$, $R = 7$. Each node displays r_j above $\Phi_7(r_j)$. Multiplication by $p \equiv 3$ advances the six-cycle, simultaneously flipping the C_2 sheet and rotating the C_3 phase. Divisor complement is the reflection J ; it fixes the middle target and exchanges the two labelled exterior directions.

Exact computation 14.65 (Endpoint dihedral-orbit checker). The script and captured receipt

`verification/verify_endpoint_dihedral_orbit.py,`
`verification/verify_endpoint_dihedral_orbit_output.txt`

verify the multiplicative target transport, complement reflection, dihedral relation, and three-fibre rank identity in all 67810 admissible shells through $p = 2000$, comprising 1666950 divisors with multiplicity. They also verify the complete C_2 -sheet/ C_3 -phase cycle in (1578).

Theorem 14.66 (Support-orbit classification of the endpoint channels). *Retain the conditions of theorem 14.63, and put*

$$S_{R,a} := \{r \in U_R : \mathcal{F}_{R,a}(r) \neq \emptyset\}, \quad H_{R,a} := \{h \in U_R : hS_{R,a} = S_{R,a}\}. \quad (1579)$$

Then

$$J_{R,a}(S_{R,a}) = S_{R,a}, \quad H_{R,a} \leq U_R. \quad (1580)$$

The endpoint-channel status is determined exactly by the intersection of $S_{R,a}$ with the three labelled targets:

<i>channel status</i>	<i>target support</i>	
<i>empty</i>	$S_{R,a} \cap \{r_{-1}, r_0, r_1\} = \emptyset$	
<i>exterior only</i>	$r_{-1}, r_1 \in S_{R,a}, \quad r_0 \notin S_{R,a}$	
<i>middle only</i>	$r_0 \in S_{R,a}, \quad r_{-1}, r_1 \notin S_{R,a}$	
<i>both</i>	$r_{-1}, r_0, r_1 \in S_{R,a}$	(1581)

In particular,

$$q_{R,p} > 0 \iff S_{R,a} \cap \{r_{-1}, r_0, r_1\} \neq \emptyset. \quad (1582)$$

If $\bar{p} \in H_{R,a}$, membership in $S_{R,a}$ is constant on the complete target orbit

$$\mathcal{O}_{R,p,a} := \{r_j : j \in \mathbb{Z}\}.$$

Hence an exterior-only or middle-only shell necessarily satisfies

$$\bar{p} \notin H_{R,a}. \quad (1583)$$

The reverse implication fails. At $(p, R, a) = (73, 11, 21)$,

$$S_{11,21} = \{1, 3, 4, 5, 7, 8, 9, 10\}, \quad (r_{-1}, r_0, r_1) = (8, 1, 7), \quad (1584)$$

so both channels are occupied, while $\bar{p} = \bar{7} \notin H_{11,21}$ because $4 \in S_{11,21}$ and $7 \cdot 4 \equiv 6 \notin S_{11,21} \pmod{11}$.

Proof. The divisor complement $u \mapsto a^2/u$ is a bijection of the divisors of a^2 , and its residue action is $J_{R,a}$. This proves $J_{R,a}(S_{R,a}) = S_{R,a}$.

The identity fixes $S_{R,a}$. If h and k fix it, then so does hk ; if $hS_{R,a} = S_{R,a}$, applying h^{-1} gives $h^{-1}S_{R,a} = S_{R,a}$. Thus $H_{R,a}$ is a subgroup.

The exterior and middle fibres are $\mathcal{F}_{R,a}(r_{-1})$ and $\mathcal{F}_{R,a}(r_0)$. Reflection stability gives

$$r_{-1} \in S_{R,a} \iff r_1 \in S_{R,a}.$$

This proves the four rows of (1581); the three-fibre identity (1576) then proves (1582).

If $\bar{p} \in H_{R,a}$, then $T_{R,p}(S_{R,a}) = S_{R,a}$. Since $r_j = T_{R,p}^j(r_0)$, every point of $\mathcal{O}_{R,p,a}$ has the same membership status. This proves (1583). Finally, the divisors of 21^2 , reduced modulo 11, give the support set in (1584); the displayed multiplication by 7 proves the last assertion. $\square$

Exact computation 14.67 (Endpoint-support and rotational-stabilizer audit). The script and captured receipt

`verification/verify_endpoint_support_stabilizers.py,`
`verification/verify_endpoint_support_stabilizers_output.txt`

verify theorem 14.66 in all 67810 admissible shells through $p = 2000$, comprising 1666950 divisors with multiplicity. The exact occupancy counts are

status	empty	exterior only	middle only	both
shells	65253	1642	314	601.

Among these shells, $\bar{p}S_{R,a} = S_{R,a}$ holds in 316 empty cases and 455 both-channel cases, and in no one-sided case. There are 146 both-channel shells without this rotational support-stability, beginning with (73, 11, 21).

Theorem 14.68 (Complete Boolean endpoint-to-blade cell bridge). *Retain the conditions of theorem 14.66. Define*

$$b_j := \mathbf{1}_{S_{R,a}}(r_j) \in \{0, 1\}, \quad e_1 := e_+, \quad e_0 := e_-, \quad J_B := e_1 e_1^\top - e_0 e_0^\top. \quad (1585)$$

The two consecutive directed target edges have Boolean cells

$$C_{-0} := e_{b_0} e_{b_{-1}}^\top, \quad C_{0+} := e_{b_1} e_{b_0}^\top. \quad (1586)$$

Then

$$b_{-1} = b_1, \quad C_{0+} = C_{-0}^\top, \quad (1587)$$

and

$$J_B C_{-0} = (2b_0 - 1)C_{-0}, \quad C_{-0} J_B = (2b_{-1} - 1)C_{-0}. \quad (1588)$$

Put

$$\beta_{R,p} := (b_0 - b_{-1}, b_1 - b_0). \quad (1589)$$

The four support patterns and both directed Boolean cells are

status	(b_{-1}, b_0, b_1)	$\beta_{R,p}$	C_{-0}	C_{0+}
empty	(0, 0, 0)	(0, 0)	E_{22}	E_{22}
exterior only	(1, 0, 1)	(-1, +1)	E_{21}	E_{12}
middle only	(0, 1, 0)	(+1, -1)	E_{12}	E_{21}
both	(1, 1, 1)	(0, 0)	E_{11}	E_{11}

(1590)

Under the exact Hadamard intertwiner of theorem 9.14, the ordered pair (C_{-0}, C_{0+}) becomes

status	$(HC_{-0}H^\top, HC_{0+}H^\top)$
empty	(P_-, P_-)
exterior only	(B_{-+}, B_{+-})
middle only	(B_{+-}, B_{-+})
both	(P_+, P_+)

(1591)

Thus every one-sided pattern contains both ordered off-diagonal masks, one on each reflected edge; their order distinguishes the two patterns. The empty and both-channel patterns give the two diagonal mixer projectors. The Boolean cells record the support masks; the integer fibre populations remain in (1576).

Proof. The unique Boolean support character of theorem 11.3 assigns 1 to a supported fibre and 0 to an unsupported fibre. The reflection stability in (1580) gives $b_{-1} = b_1$, and hence

$$e_{b_1} e_{b_0}^\top = \left(e_{b_0} e_{b_{-1}}^\top \right)^\top.$$

Since

$$J_B e_b = (2b - 1)e_b \quad (b \in \{0, 1\}),$$

left and right multiplication prove (1588).

The equality $b_{-1} = b_1$ leaves the four bit strings displayed in (1590). Substitution in (1586) gives the four matrix units on each edge and the four boundary vectors. Finally, theorem 9.14 gives

$$HE_{22}H^\top = P_-, \quad HE_{21}H^\top = B_{-+}, \quad HE_{12}H^\top = B_{+-}, \quad HE_{11}H^\top = P_+.$$

Applying these four identities to both transpose-related columns of (1590) proves (1591). $\square$

Corollary 14.69 (The $R = 107$ shell selects the outgoing blade edge). *For the shell*

$$(p, R, a) = (8803369, 107, 2200869),$$

one has

$$(b_{-1}, b_0, b_1) = (0, 1, 0), \quad C_{-0} = E_{12}, \quad C_{0+} = E_{21}, \quad q_{107,8803369} = 1. \quad (1592)$$

Consequently the population lift of the outgoing edge is

$$2q_{107,8803369}C_{0+} = 2E_{21} = \mathcal{M}(53). \quad (1593)$$

The incoming edge is the transposed mask E_{12} , whose coefficient at residue 53 is zero. Hence the exact coefficient calculation selects the outgoing direction $r_0 \rightarrow r_1$.

Proof. Theorem 14.226 gives $\mathcal{E}_{107,2200869} = \emptyset$, $|\mathcal{M}_{107,2200869}| = 2$, and $q_{107,8803369} = 1$. The middle-only row of (1590) gives the two cells. The directed coefficient and the absence of its transpose are (131). $\square$

Exact computation 14.70 (Boolean endpoint-to-blade certificate). The script and captured receipt

```
verification/verify_endpoint_Boolean_blade_bridge.py,
verification/verify_endpoint_Boolean_blade_bridge_output.txt
```

verify the reflection transpose, both bimodule signs, source and range projectors, boundary vectors, all eight directed Hadamard images, and the $R = 107$ outgoing-edge selection over $\mathbb{Q}(\sqrt{2})$.

Theorem 14.71 (Support-boundary involution and discrete blade commutator). *Retain the conditions and notation of theorem 14.66. Write $T = T_{R,p}$, $J = J_{R,a}$, and $S = S_{R,a}$, and define the outgoing and incoming boundary-source sets*

$$\partial_T^{\text{out}} S := S \setminus T^{-1}(S), \quad \partial_T^{\text{in}} S := T^{-1}(S) \setminus S. \quad (1594)$$

The crossed reflection

$$K := J \circ T \quad (1595)$$

is an involution and restricts to mutually inverse bijections

$$\partial_T^{\text{out}} S \xleftrightarrow[K]{K} \partial_T^{\text{in}} S. \quad (1596)$$

For $r \in U_R$, put

$$b(r) := \mathbf{1}_S(r), \quad C_T(r) := e_{b(Tr)} e_{b(r)}^\top, \quad \nabla_T b(r) := b(Tr) - b(r). \quad (1597)$$

Then the four transitions are

$b(r) \longrightarrow b(Tr)$	$\nabla_T b(r)$	$C_T(r)$	boundary status
$0 \longrightarrow 0$	0	E_{22}	neither
$0 \longrightarrow 1$	+1	E_{12}	$r \in \partial_T^{\text{in}} S$
$1 \longrightarrow 0$	-1	E_{21}	$r \in \partial_T^{\text{out}} S$
$1 \longrightarrow 1$	0	E_{11}	neither.

(1598)

Their exact commutator is

$$[J_B, C_T(r)] = 2\nabla_T b(r) C_T(r). \quad (1599)$$

For

$$D := 3I_2 + 6J_B = \text{diag}(9, -3), \quad (1600)$$

one consequently has

$$[D, C_T(r)] = 12\nabla_T b(r) C_T(r). \quad (1601)$$

Thus E_{12} is the +12 entry-boundary line and E_{21} is the -12 exit-boundary line.

On the two labelled endpoint edges, exterior-only support gives

$$r_{-1} \in \partial_T^{\text{out}} S, \quad r_0 \in \partial_T^{\text{in}} S, \quad (C_{-0}, C_{0+}) = (E_{21}, E_{12}), \quad (1602)$$

whereas middle-only support gives

$$r_{-1} \in \partial_T^{\text{in}} S, \quad r_0 \in \partial_T^{\text{out}} S, \quad (C_{-0}, C_{0+}) = (E_{12}, E_{21}). \quad (1603)$$

In particular, the $R = 107$ coefficient satisfies

$$\mathcal{M}(53) = 2q_{107,8803369} C_T(r_0) = 2E_{21}, \quad r_0 \in \partial_T^{\text{out}} S_{107,2200869}. \quad (1604)$$

Proof. The dihedral relation $J \circ T \circ J = T^{-1}$ gives

$$K^2 = J \circ T \circ J \circ T = I.$$

If $r \in \partial_T^{\text{out}} S$, then $r \in S$ and $Tr \notin S$. Since $J(S) = S$, one has $J(Tr) \notin S$, while

$$T(J(Tr)) = Jr \in S$$

because $T \circ J \circ T = J$. Hence $Kr = J(Tr)$ lies in $\partial_T^{\text{in}} S$. Applying the involution K again proves (1596).

The four possibilities for $(b(r), b(Tr))$ give (1598) directly from $C_T(r) = e_{b(Tr)} e_{b(r)}^T$. Left and right multiplication by J_B give

$$J_B C_T(r) = (2b(Tr) - 1) C_T(r), \quad C_T(r) J_B = (2b(r) - 1) C_T(r).$$

Their difference is (1599). Since scalar matrices have zero commutator,

$$[D, C_T(r)] = 6[J_B, C_T(r)],$$

which proves (1601).

The exterior-only support word 101 gives an exit on $r_{-1} \rightarrow r_0$ and an entry on $r_0 \rightarrow r_1$. The middle-only word 010 reverses those two transitions. This proves (1602) and (1603). Finally, theorem 14.69 gives the middle-only word, $q_{107,8803369} = 1$, and $\mathcal{M}(53) = 2C_{0+} = 2E_{21}$, proving (1604). $\square$

Exact computation 14.72 (Endpoint-boundary and commutator replay). The script and captured receipt

```
verification/verify_endpoint_boundary_commutator.py,
verification/verify_endpoint_boundary_commutator_output.txt
```

verify the crossed-reflection boundary bijection in all 67810 admissible shells through $p = 2000$. The scan finds 1419667 outgoing boundary sources and the same number of incoming boundary sources. On the 135620 labelled endpoint edges, the transition counts are

transition	$0 \rightarrow 0$	$0 \rightarrow 1$	$1 \rightarrow 0$	$1 \rightarrow 1$
count	130506	1956	1956	1202.

The checker also verifies both commutator identities for all four Boolean cells and replays the $R = 107$ exit-boundary coefficient. Its total support-boundary energy is 2839334; exactly 771 scanned shells have zero energy, and the labelled endpoint energy is 3912.

Corollary 14.73 (Support-boundary Dirichlet energy and Lorentz sum). *Retain the conditions and notation of theorem 14.71. Define*

$$\mathcal{E}_{R,p}^{\partial} := \frac{1}{4} \sum_{r \in U_R} \|[J_B, C_T(r)]\|_{\text{HS}}^2. \quad (1605)$$

Then

$$\begin{aligned}
 \mathcal{E}_{R,p}^{\partial} &= \frac{1}{144} \sum_{r \in U_R} \|[D, C_T(r)]\|_{\text{HS}}^2 \\
 &= \sum_{r \in U_R} (\nabla_T b(r))^2 \\
 &= |S \triangle T^{-1}(S)| \\
 &= 2|\partial_T^{\text{out}} S| = 2|\partial_T^{\text{in}} S|.
 \end{aligned} \tag{1606}$$

In particular, $\mathcal{E}_{R,p}^{\partial}$ is a nonnegative even integer, and

$$\mathcal{E}_{R,p}^{\partial} = 0 \iff T(S) = S \iff \bar{p} \in H_{R,a}. \tag{1607}$$

For each $r \in U_R$, put

$$\mathfrak{H}_T(r) := C_T(r)^{\top} C_T(r) - e_{b(Tr)} e_{b(Tr)}^{\top}, \quad \mathcal{N}_T(r) := -\det \mathfrak{H}_T(r). \tag{1608}$$

Then

$$\mathcal{N}_T(r) = (\nabla_T b(r))^2, \quad \mathcal{E}_{R,p}^{\partial} = \sum_{r \in U_R} \mathcal{N}_T(r). \tag{1609}$$

Thus the complete finite Dirichlet energy is the sum of the Lorentz norms of the unit-mass endpoint cells.

The two labelled edges have energy

$$\begin{aligned}
 \mathcal{E}_{R,p}^{\text{lab}} &:= \frac{1}{4} \left(\|[J_B, C_{-0}]\|_{\text{HS}}^2 + \|[J_B, C_{0+}]\|_{\text{HS}}^2 \right) \\
 &= (b_0 - b_{-1})^2 + (b_1 - b_0)^2 = 2(b_0 - b_{-1})^2.
 \end{aligned} \tag{1610}$$

Consequently,

$$\mathcal{E}_{R,p}^{\text{lab}} = \begin{cases} 2, & \text{for an exterior-only or middle-only shell,} \\ 0, & \text{for an empty or both-channel shell.} \end{cases} \tag{1611}$$

For the first occupied $R = 107$ shell,

$$\mathcal{E}_{107,8803369}^{\partial} = 20, \quad \mathcal{E}_{107,8803369}^{\text{lab}} = 2. \tag{1612}$$

Proof. The commutator formulas in theorem 14.71 and the unit Hilbert–Schmidt norm of every matrix unit give

$$\frac{1}{4} \|[J_B, C_T(r)]\|_{\text{HS}}^2 = \frac{1}{144} \|[D, C_T(r)]\|_{\text{HS}}^2 = (\nabla_T b(r))^2.$$

The last quantity equals one exactly when membership of r in S differs from membership of r in $T^{-1}(S)$. Summation therefore gives the symmetric-difference formula in (1606). The symmetric difference is the disjoint union of the outgoing and incoming boundary-source sets. Their cardinalities agree by (1596), proving the remaining equalities.

The energy vanishes exactly when $T^{-1}(S) = S$, equivalently $T(S) = S$. The definition of $H_{R,a}$ in (1579) gives (1607).

Equation (1608) is the unit-mass case of the directed Gram–support carrier in (177). The four rows of (178) give $\mathcal{N}_T(r) = 0$ on stationary cells and $\mathcal{N}_T(r) = 1$ on entry or exit cells. This proves (1609).

Finally, $b_{-1} = b_1$ by (1587). Substitution into the two-edge commutator sum proves (1610). The four support words in (1590) give (1611). The exact $R = 107$ values are replayed by theorem 14.72. $\square$

Theorem 14.74 (Coupled-shell three-orbit balance). *Let $p \equiv 1 \pmod{4}$ be prime, let $a \in \mathcal{A}_p$, and put $R = 4a - p$. For every squarefree coordinate*

$$a = stm, \quad s \text{ squarefree},$$

one has

$$(R \mid 4st^2 + 1 \text{ and } R \mid t + m) \implies R \mid p - 1. \quad (1613)$$

If $R \mid p - 1$, then for every such coordinate,

$$R \mid 4st^2 + 1 \iff R \mid t + m. \quad (1614)$$

Suppose $R \mid p - 1$, and define

$$\mathcal{T}_{R,a} := \left\{ (s, \{t, m\}) : \begin{array}{l} s, t, m \in \mathbb{Z}_{>0}, \text{ } s \text{ squarefree,} \\ a = stm, \text{ } R \mid 4st^2 + 1 \end{array} \right\}. \quad (1615)$$

The exterior condition in (1615) is invariant under $t \leftrightarrow m$. Every packet contributes exactly the three distinct complement orbits

$$\{(2, st^2), (0, sm^2)\}, \quad \{(2, sm^2), (0, st^2)\}, \quad \{(1, st^2), (1, sm^2)\}. \quad (1616)$$

These are all occupied orbits in the shell. Hence

$$q_{R,p} = 3|\mathcal{T}_{R,a}|. \quad (1617)$$

In particular,

$$Q_p \geq 3|\mathcal{T}_{R,a}|, \quad (1618)$$

$$\text{Vol}_p^{\det} \geq 2^{18|\mathcal{T}_{R,a}|}, \quad (1619)$$

$$L_{H,p}^2 \geq 153|\mathcal{T}_{R,a}|, \quad L_{K,p}^2 \geq \frac{225}{2}|\mathcal{T}_{R,a}|. \quad (1620)$$

The coupled shells for p are indexed exactly by

$$\{R : R \mid p - 1, \text{ } R \equiv 3 \pmod{4}\}. \quad (1621)$$

Proof. Suppose both coordinate congruences in (1613) hold. Then

$$m \equiv -t \pmod{R},$$

so

$$a = stm \equiv -st^2 \pmod{R}.$$

The exterior congruence now gives

$$p = 4a - R \equiv -4st^2 \equiv 1 \pmod{R},$$

proving (1613).

Now suppose $R \mid p - 1$. Since $p = 4a - R$,

$$4a \equiv 1 \pmod{R}. \quad (1622)$$

If $R \mid 4st^2 + 1$, then

$$(-t)(4st) \equiv 1 \pmod{R}.$$

On the other hand, $a = stm$ and (1622) give

$$m(4st) \equiv 1 \pmod{R}.$$

Because $\gcd(R, 4st) = 1$, these two inverse expressions imply

$$m \equiv -t \pmod{R}.$$

Thus $R \mid t + m$. Conversely, if $R \mid t + m$, then

$$4a = 4stm \equiv -4st^2 \pmod{R}.$$

Equation (1622) yields $R \mid 4st^2 + 1$, proving (1614).

The equivalence is unchanged by exchanging t and m , so the exterior coordinate set is complement-stable. A fixed point $t = m$ would give $R \mid 2t$; this is impossible because $\gcd(R, t) = 1$ and $R \geq 3$. Each packet therefore has two ordered exterior coordinates and one unordered middle coordinate. Theorem 14.59 identifies these with the three orbits in (1616) and proves (1617). The quantitative bounds follow from theorem 14.60.

Finally, a coupled shell necessarily has $R \mid p - 1$ and $R \equiv 3 \pmod{4}$. Conversely, if R is a positive divisor of $p - 1$ with $R \equiv 3 \pmod{4}$, then

$$a = \frac{p + R}{4}$$

is an integer satisfying

$$\frac{p}{4} < a < \frac{p}{2} < \frac{3p}{4}.$$

Thus $a \in \mathcal{A}_p$ and $4a - p = R$, proving (1621). $\square$

Corollary 14.75 (Coupled-shell equality of divisor support sections). *Retain the conditions of theorem 14.74, and assume $R \mid p - 1$. Then the complete exterior and middle divisor sets coincide:*

$$\mathcal{E}_{R,a} = \mathcal{M}_{R,a} = \left\{ u \in \mathbb{Z}_{>0} : u \mid a^2, u \equiv -a \equiv -\bar{4}^{-1} \pmod{R} \right\}. \quad (1623)$$

Consequently,

$$q_{R,p} = \frac{3}{2} |\mathcal{E}_{R,a}| = \frac{3}{2} |\mathcal{M}_{R,a}|, \quad q_{R,p} > 0 \iff \mathcal{E}_{R,a} = \mathcal{M}_{R,a} \neq \emptyset. \quad (1624)$$

The complement involution $u \mapsto a^2/u$ acts without fixed points on this common section, so its cardinality is even.

Proof. The coupling condition and $p = 4a - R$ give

$$4a \equiv p \equiv 1 \pmod{R}, \quad a \equiv \bar{4}^{-1} \pmod{R}.$$

For every $u \mid a^2$,

$$R \mid 4u + 1 \iff u \equiv -\bar{4}^{-1} \iff u \equiv -a \iff R \mid u + a.$$

The definitions (1525) and (1526) now prove (1623). Substitution in (1528) proves (1624). The final assertion is the fixed-point-free middle complement theorem in theorem 14.54. $\square$

Corollary 14.76 (First failure of universal coupled-shell coverage). *For $p = 73$, every coupled shell is empty, whereas the noncoupled shell $R = 7$ has rank 4.*

Proof. The divisors of $p - 1 = 72$ that are $3 \pmod{4}$ consist only of $R = 3$. Its denominator is

$$a = \frac{73 + 3}{4} = 19,$$

and

$$\{u : u \mid 19^2\} = \{1, 19, 361\} \equiv \{1\} \pmod{3}.$$

Both endpoint targets equal $-a \equiv -4^{-1} \equiv 2 \pmod{3}$. Hence

$$\mathcal{E}_{3,19} = \mathcal{M}_{3,19} = \emptyset. \quad (1625)$$

For $R = 7$,

$$a = \frac{73 + 7}{4} = 20, \quad \delta_{7,73} = 4^{-1} - 20 \equiv 3 \pmod{7}.$$

Direct divisor evaluation gives

$$\mathcal{E}_{7,20} = \{5, 40\}, \quad \mathcal{M}_{7,20} = \{1, 8, 50, 400\}. \quad (1626)$$

The exterior target 5 (mod 7) is carried by the displacement 3 to the middle target 1 (mod 7), exactly as in theorem 14.62. Therefore

$$q_{7,73} = |\mathcal{E}_{7,20}| + \frac{1}{2}|\mathcal{M}_{7,20}| = 2 + 2 = 4.$$

Since $7 \nmid 72$, this occupied shell lies outside the coupled family. $\square$

Exact computation 14.77 (Finite coupled-shell coverage and displacement scan). The script and captured receipt

verification/scan_coupled_shell_coverage.py,
verification/scan_coupled_shell_coverage_output.txt

enumerate every prime $p \leq 10^6$ with $p \equiv 1 \pmod{12}$, every coupled residual $R \mid p-1$ with $R \equiv 3 \pmod{4}$, and every divisor of the associated a^2 . Among the 19564 relevant primes, 16757 have an occupied coupled shell and 2807 have none. Every one of those 2807 primes has an occupied general shell with first residual at most 59. At that first general shell, 2750 primes meet both endpoint fibres, 36 meet only the exterior fibre, and 21 meet only the middle fibre. The receipt records the complete residual distributions and the first witnesses in each one-sided class.

Exact computation 14.78 (Coupled-shell three-orbit checker). The script and captured receipt

verification/verify_coupled_shell_three_orbit_balance.py,
verification/verify_coupled_shell_three_orbit_balance_output.txt

verify the forcing implication, the channel equivalence, literal equality of the complete divisor sections, complement stability, and the three-orbit count for every admissible shell through $p = 5000$. The replay checks the divisor-set equality in all 625 coupled shells in that range.

Theorem 14.79 (All-shell unit-group coefficient and modular rank). *Let $p \equiv 1 \pmod{4}$ be prime, let $a \in \mathcal{A}_p$, put $R = 4a - p$, and factor*

$$a = \prod_{\ell^e \parallel a} \ell^e.$$

Retain the divisor sets $\mathcal{E}_{R,a}$ and $\mathcal{M}_{R,a}$ from (1525) and (1526). Let

$$U_R = (\mathbb{Z}/R\mathbb{Z})^\times$$

and write $[\bar{v}]$ for the basis element of $\mathbb{Z}[U_R]$ indexed by the residue of a unit v . Define

$$\mathfrak{P}_{R,a} := \prod_{\ell^e \parallel a} \left([\bar{1}] + \sum_{j=1}^e ([\bar{\ell}^j] + [\bar{\ell}^{-j}]) \right) \in \mathbb{Z}[U_R] \quad (1627)$$

and let

$$c_{R,a} := \text{coeff}_{[\bar{-1}]} \mathfrak{P}_{R,a}.$$

Then

$$\boxed{c_{R,a} = |\mathcal{M}_{R,a}|}, \quad c_{R,a} \equiv 0 \pmod{2}, \quad (1628)$$

and

$$\boxed{q_{R,p} = |\mathcal{E}_{R,a}| + \frac{1}{2}c_{R,a} = \text{rank } P_{R,p}^{\text{mod}}}. \quad (1629)$$

When $q_{R,p} > 0$, the exact shell complexity quantities are

$$\text{Vol}_{R,p}^{\text{det}} = 2^{6|\mathcal{E}_{R,a}|+3c_{R,a}}, \quad (1630)$$

$$L_H^2 = 51|\mathcal{E}_{R,a}| + \frac{51}{2}c_{R,a}, \quad (1631)$$

$$L_K^2 = \frac{75}{2}|\mathcal{E}_{R,a}| + \frac{75}{4}c_{R,a}. \quad (1632)$$

Finite Fourier inversion on U_R gives

$$c_{R,a} = \frac{1}{\varphi(R)} \sum_{\chi \in \widehat{U}_R} \chi(-1) \prod_{\ell^e \parallel a} \left(1 + \sum_{j=1}^e (\chi(\ell)^j + \chi(\ell)^{-j}) \right). \quad (1633)$$

Proof. The identities $0 < a < p$ and $R = 4a - p$ give $\gcd(R, a) = 1$. Thus every divisor of a represents a unit modulo R .

Consider the ordered coprime-factor set

$$\widetilde{\mathcal{C}}_{R,a} = \{(x, y) : x, y > 0, \gcd(x, y) = 1, xy \mid a, R \mid x + y\}.$$

At a prime power $\ell^e \parallel a$, an ordered pair in this set chooses one of

$$\ell \nmid xy, \quad \ell^j \parallel x \quad (1 \leq j \leq e), \quad \ell^j \parallel y \quad (1 \leq j \leq e).$$

These three types contribute

$$\overline{1}, \quad \overline{\ell}^j, \quad \overline{\ell}^{-j}$$

to the unit ratio $\overline{x} \overline{y}^{-1}$. Expansion of (1627) therefore gives

$$c_{R,a} = \#\{(x, y) : x, y > 0, \gcd(x, y) = 1, xy \mid a, \overline{x} \overline{y}^{-1} = \overline{-1}\}.$$

Since y is a unit modulo R , the final equality is equivalent to $R \mid x + y$. Hence

$$c_{R,a} = |\widetilde{\mathcal{C}}_{R,a}|.$$

The exchange $(x, y) \mapsto (y, x)$ is free: a fixed point would have $x = y = 1$, contrary to $R \nmid 2$ for the odd integer $R \geq 3$. It follows that

$$c_{R,a} = 2|\mathcal{C}_{R,a}|.$$

The bijection (1534) identifies $\mathcal{C}_{R,a}$ with the complement orbits in $\mathcal{M}_{R,a}$. The complement action on $\mathcal{M}_{R,a}$ is free by theorem 14.54. Therefore

$$c_{R,a} = 2|\mathcal{C}_{R,a}| = |\mathcal{M}_{R,a}|,$$

which proves (1628).

Equation (1528) now gives the first equality in (1629), and (1482) gives the second. Substitution in (1330), (1333), and (1334) proves (1630)–(1632).

Finally, evaluate $\mathfrak{P}_{R,a}$ under each character of U_R and apply finite Fourier inversion. The element $\overline{-1}$ has order two, so $\overline{\chi(-1)} = \chi(-1)$, which proves (1633). $\square$

Theorem 14.80 (Finite factor-residue support ideal). *Retain the conditions of theorem 14.79. For $g \in U_R \setminus \{\overline{1}\}$, define*

$$\nu_g(a) := \sum_{\substack{\ell^e \parallel a \\ \ell = g}} e$$

and write

$$\boldsymbol{\nu}_R(a) = (\nu_g(a))_{g \neq \overline{1}}.$$

For an arbitrary vector $\boldsymbol{\nu} \in \mathbb{Z}_{\geq 0}^{U_R \setminus \{\overline{1}\}}$, put

$$\mathcal{R}_R^{\mathbf{E}}(\boldsymbol{\nu}) := \left\{ \prod_{g \neq \overline{1}} g^{e_g} : 0 \leq e_g \leq 2\nu_g \right\}, \quad (1634)$$

$$\mathcal{R}_R^{\mathbf{M}}(\boldsymbol{\nu}) := \left\{ \prod_{g \neq \overline{1}} g^{d_g} : -\nu_g \leq d_g \leq \nu_g \right\}. \quad (1635)$$

Let

$$t_R := \overline{-4^{-1}} \in U_R, \quad s_R := \overline{-1} \in U_R,$$

and define the occupied valuation up-set

$$\mathcal{U}_R := \left\{ \boldsymbol{\nu} : t_R \in \mathcal{R}_R^{\mathbf{E}}(\boldsymbol{\nu}) \text{ or } s_R \in \mathcal{R}_R^{\mathbf{M}}(\boldsymbol{\nu}) \right\}.$$

Its coordinatewise-minimal subset

$$\text{Min}(\mathcal{U}_R)$$

is finite. In the polynomial ring with variables $\{X_g : g \in U_R \setminus \{\bar{1}\}\}$, define

$$I_R := \left\langle \prod_{g \neq \bar{1}} X_g^{\mu_g} : \boldsymbol{\mu} \in \text{Min}(\mathcal{U}_R) \right\rangle \quad (1636)$$

and

$$\mathbf{m}_R(a) := \prod_{g \neq \bar{1}} X_g^{\nu_g(a)}.$$

Then

$$q_{R,p} > 0 \iff \mathbf{m}_R(a) \in I_R. \quad (1637)$$

Let $D(U_R)$ denote the least positive integer such that every sequence of $D(U_R)$ elements of U_R contains a nonempty subsequence with product $\bar{1}$. Every minimal generator of I_R has total degree at most

$$D(U_R) - 1. \quad (1638)$$

Proof. A divisor $u \mid a^2$ chooses, for every residue class g , an exponent between 0 and $2\nu_g(a)$. Hence

$$\mathcal{E}_{R,a} \neq \emptyset \iff t_R \in \mathcal{R}_R^{\mathbf{E}}(\boldsymbol{\nu}_R(a)).$$

For the middle channel, collect all prime powers of a having the same residue g . Their local exponent intervals add to the full integer interval $[-\nu_g(a), \nu_g(a)]$. Expansion of (1627) therefore gives

$$c_{R,a} > 0 \iff s_R \in \mathcal{R}_R^{\mathbf{M}}(\boldsymbol{\nu}_R(a)).$$

Factors congruent to 1 (mod R) may change positive multiplicities, but they do not change either support set.

The set $\mathcal{U}_R$ is closed under coordinatewise increase. Consequently,

$$\boldsymbol{\nu}_R(a) \in \mathcal{U}_R \iff \boldsymbol{\nu}_R(a) \text{ dominates an element of } \text{Min}(\mathcal{U}_R),$$

which is equivalent to $\mathbf{m}_R(a) \in I_R$. Equation (1629) now proves (1637).

It remains to prove finiteness and the degree bound. Suppose first that t_R has a representation in $\mathcal{R}_R^{\mathbf{E}}(\boldsymbol{\nu})$. Write this representation as a sequence containing e_g copies of g . Whenever its length is at least $D(U_R)$, remove a nonempty product-one subsequence. Iteration leaves a representation of t_R of length at most $D(U_R) - 1$. If e'_g is the remaining multiplicity, then

$$\mu_g := \left\lceil \frac{e'_g}{2} \right\rceil$$

defines an occupied vector dominated by $\boldsymbol{\nu}$, and

$$\sum_g \mu_g \leq \sum_g e'_g \leq D(U_R) - 1.$$

For a middle representation, use $|d_g|$ labelled copies of g or g^{-1} , according to the sign of d_g . The same subsequence removal leaves a target representation of length at most $D(U_R) - 1$. Its

remaining absolute multiplicities define an occupied vector dominated by ν and of total degree at most $D(U_R) - 1$. Thus every occupied vector dominates one in the finite box

$$\left\{ \mu : \sum_g \mu_g \leq D(U_R) - 1 \right\}.$$

All coordinatewise-minimal occupied vectors lie in this box, proving finiteness and (1638). $\square$

Theorem 14.81 (Safe-residual parity-sheet reduction). *Retain the conditions of theorem 14.80. Suppose*

$$R = 2q + 1$$

with q an odd prime. Choose a primitive root $\gamma \in U_R$, and let

$$\lambda_R : U_R \longrightarrow \mathbb{Z}/(2q)\mathbb{Z}, \quad \lambda_R(\gamma) = 1$$

be the corresponding additive logarithm. Put

$$r_R := \lambda_R(\overline{2}).$$

Write the prime factorization of a , with each prime repeated according to its valuation, as

$$a = \ell_1 \cdots \ell_m, \quad \kappa_i := \lambda_R(\overline{\ell_i}).$$

Indices with $\kappa_i = 0$ may be omitted. Define

$$\mathcal{B}_R^M(\kappa) := \left\{ \sum_{i=1}^m \delta_i \kappa_i : \delta_i \in \{-1, 0, 1\} \right\}, \quad (1639)$$

$$\mathcal{B}_R^E(\kappa) := \left\{ \sum_{i=1}^m \epsilon_i \kappa_i : \epsilon_i \in \{0, 1, 2\} \right\} \quad (1640)$$

in $\mathbb{Z}/(2q)\mathbb{Z}$. Then

$$\lambda_R(s_R) = q, \quad \lambda_R(t_R) = q - 2r_R, \quad (1641)$$

and

$$q_{R,p} > 0 \iff q \in \mathcal{B}_R^M(\kappa) \text{ or } q - 2r_R \in \mathcal{B}_R^E(\kappa). \quad (1642)$$

Both targets in (1641) are odd. Consequently, every middle representation satisfies

$$\#\{i : \kappa_i \text{ is odd and } \delta_i \neq 0\} \equiv 1 \pmod{2}, \quad (1643)$$

and every exterior representation satisfies

$$\#\{i : \kappa_i \text{ is odd and } \epsilon_i = 1\} \equiv 1 \pmod{2}. \quad (1644)$$

In particular, if every nonzero factor logarithm of a is even, then $q_{R,p} = 0$.

Suppose now that exactly one factor-logarithm occurrence is odd; denote it by k . Write every remaining occurrence as $\kappa_i = 2h_i$ in $\mathbb{Z}/(2q)\mathbb{Z}$, and define

$$\mathcal{A}_R^M := \left\{ \sum_i \delta_i h_i : \delta_i \in \{-1, 0, 1\} \right\}, \quad (1645)$$

$$\mathcal{A}_R^E := \left\{ \sum_i \epsilon_i h_i : \epsilon_i \in \{0, 1, 2\} \right\} \quad (1646)$$

in $\mathbb{Z}/q\mathbb{Z}$. The complete occupation criterion is

$$q_{R,p} > 0 \iff \frac{q-k}{2} \in \mathcal{A}_R^M \text{ or } \frac{q+k}{2} \in \mathcal{A}_R^M \text{ or } \frac{q-2r_R-k}{2} \in \mathcal{A}_R^E. \quad (1647)$$

Each displayed half is the class modulo q of the indicated even integer divided by 2.

Proof. Since γ has order $2q$,

$$\lambda_R(\overline{-1}) = q.$$

Moreover,

$$\lambda_R(\overline{-4^{-1}}) = \lambda_R(\overline{-1}) - \lambda_R(\overline{4}) = q - 2r_R,$$

which proves (1641). Repeating each prime factor according to its valuation converts the intervals in (1634) and (1635) into independent occurrence coefficients $\epsilon_i \in \{0, 1, 2\}$ and $\delta_i \in \{-1, 0, 1\}$, respectively. Applying λ_R to theorem 14.80 proves (1642).

Reduction modulo 2 proves (1643) and (1644). If all κ_i are even, neither odd target can occur.

Assume that k is the unique odd occurrence. In the middle channel its coefficient must be 1 or -1 . Subtracting k , or adding k , and then dividing the resulting congruence by 2 gives the first two membership tests in (1647). In the exterior channel the coefficient of k must be 1. Subtracting k and dividing by 2 gives the third membership test. These steps are reversible, proving the equivalence. $\square$

Corollary 14.82 (Canonical Legendre-sheet form). *Under the conditions of theorem 14.81, let*

$$Q_R := \{u^2 : u \in U_R\}$$

be the quadratic-residue subgroup and let $\chi_R : U_R \rightarrow \{\pm 1\}$ be the quadratic character. The map

$$\Phi_R : U_R \longrightarrow \{\pm 1\} \times Q_R, \quad \Phi_R(u) = (\chi_R(u), \chi_R(u)u) \quad (1648)$$

is a group isomorphism. Its target values are

$$\Phi_R(s_R) = (-1, 1), \quad \Phi_R(t_R) = (-1, \overline{4}^{-1}). \quad (1649)$$

For each repeated factor occurrence, write

$$\Phi_R(\overline{\ell}_i) = (\sigma_i, v_i).$$

Then the shell is occupied exactly when at least one of the following two systems has a solution:

$$\prod_i \sigma_i^{\delta_i} = -1, \quad \prod_i v_i^{\delta_i} = 1, \quad \delta_i \in \{-1, 0, 1\}, \quad (1650)$$

$$\prod_i \sigma_i^{\epsilon_i} = -1, \quad \prod_i v_i^{\epsilon_i} = \overline{4}^{-1}, \quad \epsilon_i \in \{0, 1, 2\}. \quad (1651)$$

If exactly one occurrence, indexed by 0, is a quadratic nonresidue, then $v_0 = -\overline{\ell}_0 \in Q_R$ and the criterion becomes

$$\begin{aligned} q_{R,p} > 0 &\iff v_0^{-1} \in \left\{ \prod_{i \neq 0} v_i^{\delta_i} : \delta_i \in \{-1, 0, 1\} \right\} \\ &\text{or } v_0 \in \left\{ \prod_{i \neq 0} v_i^{\delta_i} : \delta_i \in \{-1, 0, 1\} \right\} \\ &\text{or } \overline{4}^{-1} v_0^{-1} \in \left\{ \prod_{i \neq 0} v_i^{\epsilon_i} : \epsilon_i \in \{0, 1, 2\} \right\}. \end{aligned} \quad (1652)$$

For $R = 7$,

$$Q_7 = \{\overline{1}, \overline{2}, \overline{4}\} = \langle \overline{2} \rangle \cong C_3,$$

and

$$U_7 \cong C_2 \times C_3, \quad s_7 \longmapsto (-1, \overline{1}), \quad t_7 \longmapsto (-1, \overline{2}). \quad (1653)$$

Thus the $R = 7$ shell has an exact binary sheet and three-phase carrier; the Boolean coordinate

$$\text{Supp}_{7,p}(a) := \mathbf{1}_{\{q_{7,p} > 0\}}$$

records whether either distinguished point on the nonresidue sheet is reached.

Proof. Because q is odd, $R \equiv 3 \pmod{4}$, so $\chi_R(-1) = -1$. Hence $\chi_R(u)u \in Q_R$ for every $u \in U_R$, and (1648) is a homomorphism. Its inverse is

$$(\sigma, v) \mapsto \sigma v.$$

This proves the isomorphism. Since 4^{-1} is a square, (1649) follows immediately. Applying Φ_R to the two occurrence boxes in (1642) gives (1650) and (1651).

With one nonresidue occurrence, the sheet equations force $\delta_0 = 1$ or -1 in the middle channel and $\epsilon_0 = 1$ in the exterior channel. Solving the corresponding Q_R -equations proves (1652). Finally, $Q_7 = \langle \bar{2} \rangle$, while

$$s_7 = \bar{6}, \quad t_7 = \bar{5}, \quad -\bar{5} = \bar{2}.$$

This proves (1653). $\square$

Corollary 14.83 (The $R = 107$ carrier has exactly two parity sheets). *Retain the notation of theorem 11.15. For $R_* = 107$,*

$$q = 53, \quad \gamma = 2, \quad r_{107} = 1,$$

and the factor logarithms of

$$a_* = 3^2 11^2 43 47$$

are

$$70, 70, 22, 22, 59, 66.$$

The unique odd occurrence is $k = 59$, arising from 43, and the multiset of even half-logarithms is

$$(35, 35, 11, 11, 33).$$

The three reduced targets in (1647) are

$$\frac{53 - 59}{2} = 50, \quad \frac{53 + 59}{2} = 3, \quad \frac{53 - 2 - 59}{2} = 49 \quad \text{in } \mathbb{Z}/53\mathbb{Z}. \quad (1654)$$

In the aggregate coefficient box

$$-2 \leq \beta_3, \beta_{11} \leq 2, \quad -1 \leq \beta_{47} \leq 1,$$

the first two target fibres are

$$\{(\beta_3, \beta_{11}, \beta_{47}) : 35\beta_3 + 11\beta_{11} + 33\beta_{47} = 50\} = \{(2, 0, 1)\}, \quad (1655)$$

$$\{(\beta_3, \beta_{11}, \beta_{47}) : 35\beta_3 + 11\beta_{11} + 33\beta_{47} = 3\} = \{(-2, 0, -1)\}, \quad (1656)$$

where the equations are in $\mathbb{Z}/53\mathbb{Z}$. In the exterior coefficient box

$$0 \leq \epsilon_3, \epsilon_{11} \leq 4, \quad 0 \leq \epsilon_{47} \leq 2,$$

the target 49 has no representative. Hence the two centered middle vectors and the exterior fibre are exactly

$$\beta^- = (-2, 0, -1, -1), \quad \beta^+ = (2, 0, 1, 1), \quad \mathcal{E}_{107, a_*} = \emptyset.$$

Furthermore,

$$\lambda_{107}(a_*) = 97, \quad \lambda_{107}(t_{107}) = 51, \quad 51 - 97 = 60 \pmod{106}.$$

Thus 51 is the unshifted exterior target and 60 is its centered coordinate in theorem 11.15. Under the exact blade identification

$$2C_{0+} = 2E_{21} = \mathcal{M}(53),$$

the two summands are precisely the two parity sheets distinguished by the coefficient $+1$ or -1 on the unique odd factor logarithm. Here $r_0 \in \partial_T^{\text{out}} S_{107, a_}$, so this is the exit-boundary cell by theorem 14.71. The incoming cell is $C_{-0} = E_{12}$ and has zero coefficient at residue 53, as proved in theorem 14.69.*

Proof. The logarithms and the two centered fibres were proved in theorem 11.15. Applying theorem 14.81 gives (1654). The finite table in the proof of theorem 11.15 gives the unique memberships (1655) and (1656) and excludes 49 from the exterior box. Finally,

$$2(70) + 2(22) + 59 + 66 = 309 \equiv 97 \pmod{106},$$

which proves (14.83). The blade statement is then (568). $\square$

Exact computation 14.84 (Safe-residual parity replay). The script and captured receipt

```
verification/verify_safe_residual_parity_bridge.py,
verification/verify_safe_residual_parity_bridge_output.txt
```

check the canonical Legendre-sheet isomorphism and both target images for all 24 safe residual primes through 1000, verify the exact $C_2 \times C_3$ target carrier at $R = 7$, compare the full $2q$ -modulus boxes with the quotient criterion on 255837 one-odd profiles through $R = 107$, and replay the complete $R = 107$ calculation. The latter check returns the reduced targets

$$(50, 3, 49),$$

the two centered vectors

$$(-2, 0, -1, -1), \quad (2, 0, 1, 1),$$

an empty exterior fibre, and the target transport

$$51 - 97 = 60 \pmod{106}.$$

Corollary 14.85 (Consecutive factor-ideal formulation). *Let $p = 4n + 1$ be prime. For*

$$1 \leq j \leq 2n, \quad a_j := n + j, \quad R_j := 4j - 1,$$

one has

$$a_j \in \mathcal{A}_p, \quad 4a_j - p = R_j.$$

Moreover,

$$\boxed{\text{ES}(p) \iff \exists j \in \{1, \dots, 2n\} : \mathfrak{m}_{R_j}(n + j) \in I_{R_j}.} \quad (1657)$$

Thus a counterexample at p is exactly a length- $2n$ affine sequence satisfying

$$\mathfrak{m}_{4j-1}(n + j) \notin I_{4j-1} \quad (1 \leq j \leq 2n). \quad (1658)$$

Proof. From $p = 4n + 1$ and (1501),

$$\mathcal{A}_p = \{n + 1, n + 2, \dots, 3n\}.$$

Writing $a = n + j$ gives $1 \leq j \leq 2n$ and

$$R_a = 4(n + j) - (4n + 1) = 4j - 1.$$

The primewise divisor criterion (1532), together with theorem 14.80, proves (1657). Negation gives (1658). $\square$

Theorem 14.86 (Prime-factor colon reduction). *Retain the conditions of theorem 14.85. Fix $1 \leq j \leq 2n$, and write*

$$R = 4j - 1, \quad a = n + j.$$

Let ℓ be a prime divisor of a , let $1 \leq e \leq v_\ell(a)$, and suppose that the class $\bar{\ell} \in U_R$ is not $\bar{1}$. Then

$$\mathfrak{m}_R(a) = X_\ell^e \mathfrak{m}_R\left(\frac{a}{\ell^e}\right), \quad (1659)$$

and

$$\boxed{q_{R,p} > 0 \iff \mathfrak{m}_R\left(\frac{a}{\ell^e}\right) \in (I_R : X_\ell^e).} \quad (1660)$$

Proof. The factor-residue valuation of a/ℓ^e is obtained from that of a by subtracting e in the $\bar{\ell}$ -coordinate. This proves (1659). For every monomial M ,

$$X_{\bar{\ell}}^e M \in I_R \iff M \in (I_R : X_{\bar{\ell}}^e).$$

Apply this identity to (1659) and use (1637). $\square$

Theorem 14.87 (Three-position target packets after factor extraction). *Retain the conditions of theorem 14.86 with $e = 1$, and write $a = \ell b$. Define two subsets of U_R :*

$$\mathcal{T}_{R,\ell}^E := \{t_R, t_R \bar{\ell}^{-1}, t_R \bar{\ell}^{-2}\}, \quad (1661)$$

$$\mathcal{T}_{R,\ell}^M := \{s_R \bar{\ell}, s_R, s_R \bar{\ell}^{-1}\}. \quad (1662)$$

Then

$$q_{R,p} > 0 \iff \mathcal{R}_R^E(\nu_R(b)) \cap \mathcal{T}_{R,\ell}^E \neq \emptyset \text{ or } \mathcal{R}_R^M(\nu_R(b)) \cap \mathcal{T}_{R,\ell}^M \neq \emptyset. \quad (1663)$$

Proof. The extracted prime factor ℓ may occur in an exterior divisor with local exponent 0, 1, or 2. Hence

$$\mathcal{R}_R^E(\nu_R(a)) = \bigcup_{j=0}^2 \bar{\ell}^j \mathcal{R}_R^E(\nu_R(b)).$$

The exterior target t_R belongs to this union exactly when the reduced reachability set meets $\mathcal{T}_{R,\ell}^E$.

In the middle channel the extracted factor has local signed exponent $-1, 0$, or 1 , so

$$\mathcal{R}_R^M(\nu_R(a)) = \bigcup_{d=-1}^1 \bar{\ell}^d \mathcal{R}_R^M(\nu_R(b)).$$

Solving $s_R = \bar{\ell}^d u$ for the reduced target u gives $\mathcal{T}_{R,\ell}^M$. The general reachability criterion in theorem 14.80 proves (1663). $\square$

Corollary 14.88 (Exact homothety between the two extracted-factor packets). *Under the conditions of theorem 14.87, put*

$$\alpha_{R,\ell} := (4\bar{\ell})^{-1} \in U_R.$$

Then

$$\mathcal{T}_{R,\ell}^E = \alpha_{R,\ell} \mathcal{T}_{R,\ell}^M. \quad (1664)$$

Thus the exterior and middle packets are two scalar-related ordered length-three orbit segments. This statement uses no order-three action.

Proof. Put

$$\mathcal{B}_{R,\ell} := \{1, \bar{\ell}^{-1}, \bar{\ell}^{-2}\}.$$

Since $t_R = -4^{-1}$ and $s_R = -1$, the definitions give

$$\mathcal{T}_{R,\ell}^E = t_R \mathcal{B}_{R,\ell}, \quad \mathcal{T}_{R,\ell}^M = s_R \bar{\ell} \mathcal{B}_{R,\ell}.$$

Their scalar ratio is

$$\frac{t_R}{s_R \bar{\ell}} = \frac{-4^{-1}}{-\bar{\ell}} = (4\bar{\ell})^{-1}.$$

This proves (1664). $\square$

Theorem 14.89 (Weighted packet exchange and ordered-blade amplitude ratio). *Retain the conditions of theorem 14.88, set*

$$A_R := \mathbb{Z}/R\mathbb{Z}, \quad \alpha := \alpha_{R,\ell},$$

and form the ordered packet-channel module

$$V_{R,\ell} := A_R e_E \oplus A_R e_M.$$

The map

$$\mathcal{J}_{R,\ell} \begin{pmatrix} x \\ y \end{pmatrix} := \begin{pmatrix} 0 & \alpha \\ \alpha^{-1} & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad (1665)$$

is an involution and exchanges the two tagged packets:

$$\begin{aligned} \mathcal{J}_{R,\ell}(\mathcal{T}_{R,\ell}^E e_E) &= \mathcal{T}_{R,\ell}^M e_M, \\ \mathcal{J}_{R,\ell}(\mathcal{T}_{R,\ell}^M e_M) &= \mathcal{T}_{R,\ell}^E e_E. \end{aligned} \quad (1666)$$

Let

$$W_R := A_R E_{21} \oplus A_R E_{12} \subset M_2(A_R), \quad S := E_{12} + E_{21},$$

and let $\mathcal{S}(X) := SXS$. Every line-preserving A_R -module isomorphism

$$\Phi : V_{R,\ell} \longrightarrow W_R$$

that satisfies

$$\Phi \mathcal{J}_{R,\ell} = \mathcal{S} \Phi$$

has the form

$$\Phi_u(e_E) = uE_{21}, \quad \Phi_u(e_M) = \alpha uE_{12} \quad (u \in U_R). \quad (1667)$$

Consequently an equal-amplitude carriage,

$$\Phi(e_E) = uE_{21}, \quad \Phi(e_M) = uE_{12},$$

exists if and only if $\alpha = 1$.

Finally, define the packet and blade gradings by

$$\begin{aligned} \Gamma_P e_E &= -e_E, & \Gamma_P e_M &= e_M, \\ \Gamma_B E_{21} &= -E_{21}, & \Gamma_B E_{12} &= E_{12}. \end{aligned}$$

Then

$$\Phi_u \Gamma_P = \Gamma_B \Phi_u, \quad \mathcal{J}_{R,\ell} \Gamma_P = -\Gamma_P \mathcal{J}_{R,\ell}, \quad \mathcal{S} \Gamma_B = -\Gamma_B \mathcal{S}. \quad (1668)$$

The blade grading is the sign grading of the two modular channels in theorem 14.18.

Proof. The matrix in (1665) has square I_2 . Equation (1664) gives

$$\alpha^{-1} \mathcal{T}_{R,\ell}^E = \mathcal{T}_{R,\ell}^M, \quad \alpha \mathcal{T}_{R,\ell}^M = \mathcal{T}_{R,\ell}^E,$$

which proves (1666).

Write

$$\Phi(e_E) = u_E E_{21}, \quad \Phi(e_M) = u_M E_{12},$$

where both coefficients are units. Since

$$\mathcal{S}(E_{21}) = E_{12}, \quad \mathcal{S}(E_{12}) = E_{21},$$

the intertwining equation evaluated at e_E is

$$\alpha^{-1} u_M E_{12} = u_E E_{12}.$$

Equivalently,

$$u_M = \alpha u_E.$$

The equation at e_M gives the same condition. This proves (1667) and its converse. The two amplitudes are equal exactly when $\alpha = 1$. The grading identities follow on the two displayed bases. $\square$

Corollary 14.90 (The first two weighted packet-blade carriages). *For the universal extracted factor 3, the first two exchange matrices and blade amplitude ratios are*

$$R = 11 : \quad [\mathcal{J}_{11,3}] = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \Phi_u(e_E) = uE_{21}, \quad \Phi_u(e_M) = uE_{12}, \quad (1669)$$

$$R = 23 : \quad [\mathcal{J}_{23,3}] = \begin{pmatrix} 0 & 2 \\ 12 & 0 \end{pmatrix}, \quad \Phi_u(e_E) = uE_{21}, \quad \Phi_u(e_M) = 2uE_{12}. \quad (1670)$$

Thus residual 11 has an equal-amplitude ordered-blade carriage, whereas residual 23 retains the factor 2 between its two ordered channel lines.

Proof. Modulo 11, $12^{-1} = 1$. Modulo 23, $12^{-1} = 2$ and $2^{-1} = 12$. Substitute these values into (1665) and (1667). $\square$

Theorem 14.91 (One-box form of the two divisor channels). *Retain the conditions of theorem 14.87, write $a = \ell b$, and let $\bar{b} \in U_R$ be the residue class of b . Then*

$$\boxed{\mathcal{R}_R^E(\nu_R(b)) = \bar{b} \mathcal{R}_R^M(\nu_R(b))}. \quad (1671)$$

Consequently the two-channel packet test is the single signed-box test

$$\begin{aligned} q_{R,p} > 0 &\iff \mathcal{R}_R^M(\nu_R(b)) \cap (\mathcal{T}_{R,\ell}^M \cup \bar{b}^{-1} \mathcal{T}_{R,\ell}^E) \neq \emptyset \\ &\iff \mathcal{R}_R^M(\nu_R(b)) \cap (\mathcal{T}_{R,\ell}^M \cup (4\bar{\ell}\bar{b})^{-1} \mathcal{T}_{R,\ell}^M) \neq \emptyset. \end{aligned} \quad (1672)$$

Proof. Write

$$\nu_R(b) = (\nu_g)_{g \in U_R \setminus \{\bar{1}\}}.$$

The residue of b is

$$\bar{b} = \prod_g g^{\nu_g}.$$

For each exterior exponent e_g with $0 \leq e_g \leq 2\nu_g$, put $d_g = e_g - \nu_g$. This gives a bijection

$$\prod_g \{0, \dots, 2\nu_g\} \longleftrightarrow \prod_g \{-\nu_g, \dots, \nu_g\},$$

and

$$\prod_g g^{e_g} = \bar{b} \prod_g g^{d_g}.$$

This proves (1671). Pulling the exterior target packet back through multiplication by $\bar{b}$ gives the first line of (1672). Equation (1664) gives the second. $\square$

Theorem 14.92 (Universal every-second-shell binary packet). *Let $p = 12k + 1$ be prime, and let $\eta_k \in \{0, 1\}$ be determined by $\eta_k \equiv k \pmod{2}$. For $1 \leq m \leq 3k$, put*

$$\begin{aligned} j_m^{(2)} &:= 2m - \eta_k, \\ R_m^{(2)} &:= 8m - 4\eta_k - 1, \\ b_m^{(2)} &:= \frac{3k - \eta_k}{2} + m. \end{aligned} \quad (1673)$$

Then

$$1 \leq j_m^{(2)} \leq 6k, \quad 3k + j_m^{(2)} = 2b_m^{(2)}, \quad R_m^{(2)} = 4j_m^{(2)} - 1,$$

and

$$8b_m^{(2)} = p + R_m^{(2)}. \quad (1674)$$

Define

$$\mathcal{T}_m^{(2)} := \mathcal{T}_{R_m^{(2)}, 2}^M = \{-2, -1, -2^{-1}\} \subset U_{R_m^{(2)}},$$

$$\mathcal{D}_{m,p}^{(2)} := \mathcal{T}_m^{(2)} \cup \bar{p}^{-1} \mathcal{T}_m^{(2)}.$$

Then

$$q_{R_m^{(2)},p} > 0 \iff \mathcal{R}_{R_m^{(2)}}^M(\nu_{R_m^{(2)}}(b_m^{(2)})) \cap \mathcal{D}_{m,p}^{(2)} \neq \emptyset. \quad (1675)$$

The only members of this residual family for which 2 has order less than five are $R = 3, 7, 15$. Their target volumes are

$$R = 3: \quad |\mathcal{D}_{m,p}^{(2)}| = 2, \quad (1676)$$

$$R = 7: \quad |\mathcal{D}_{m,p}^{(2)}| = \begin{cases} 3, & \bar{p} \in \langle \bar{2} \rangle, \\ 6, & \bar{p} \notin \langle \bar{2} \rangle, \end{cases} \quad (1677)$$

$$R = 15: \quad |\mathcal{D}_{m,p}^{(2)}| = \begin{cases} 3, & \bar{p} = \bar{1}, \\ 4, & \bar{p} \in \langle \bar{2} \rangle \setminus \{\bar{1}\}, \\ 6, & \bar{p} \notin \langle \bar{2} \rangle. \end{cases} \quad (1678)$$

For every other $R_m^{(2)}$,

$$|\mathcal{D}_{m,p}^{(2)}| = \begin{cases} 3, & \bar{p} = \bar{1}, \\ 4, & \bar{p} \in \{\bar{2}, \bar{2}^{-1}\}, \\ 5, & \bar{p} \in \{\bar{2}^2, \bar{2}^{-2}\}, \\ 6, & \text{otherwise.} \end{cases} \quad (1679)$$

Proof. The parity condition on η_k makes $(3k - \eta_k)/2$ integral. Direct substitution gives

$$3k + j_m^{(2)} = 3k + 2m - \eta_k = 2b_m^{(2)}$$

and

$$4j_m^{(2)} - 1 = 8m - 4\eta_k - 1 = R_m^{(2)}.$$

It also gives

$$8b_m^{(2)} = 12k - 4\eta_k + 8m = p + R_m^{(2)}.$$

The displayed range of m yields the asserted range of $j_m^{(2)}$.

One has $0 < R_m^{(2)} < 2p$, and $R_m^{(2)} \equiv 3 \pmod{4}$, whereas $p \equiv 1 \pmod{4}$. Thus $R_m^{(2)} \neq p$, and primality of p gives $\gcd(p, R_m^{(2)}) = 1$. The affine relation then gives $\gcd(b_m^{(2)}, R_m^{(2)}) = 1$. Applying theorem 14.91 with $\ell = 2$ proves (1675).

Let d be the order of 2 modulo $R_m^{(2)}$. If $d \leq 4$, then $R_m^{(2)}$ divides one of

$$2 - 1 = 1, \quad 2^2 - 1 = 3, \quad 2^3 - 1 = 7, \quad 2^4 - 1 = 15.$$

Together with $R_m^{(2)} \geq 3$ and $R_m^{(2)} \equiv 3 \pmod{4}$, this gives precisely

$$R_m^{(2)} \in \{3, 7, 15\}.$$

Modulo 3, the packet $\mathcal{T}_m^{(2)}$ is U_3 , proving (1676). Modulo 7,

$$\langle \bar{2} \rangle = \{1, 2, 4\}, \quad \mathcal{T}_m^{(2)} = -\langle \bar{2} \rangle.$$

Translation by $\bar{p}^{-1}$ either fixes this coset or sends it to the other coset of $\langle \bar{2} \rangle$, proving (1677).

Modulo 15,

$$\langle \bar{2} \rangle = \{1, 2, 4, 8\}, \quad \mathcal{T}_m^{(2)} = \{-2, -1, -2^{-1}\} = \{13, 14, 7\}.$$

The identity translate has three points. Translation by any of 2, 4, 8 has a two-point intersection with this packet and hence a four-point union. A translating element outside $\langle \bar{2} \rangle$ places the translated packet in the other coset and gives a six-point union. This proves (1678).

For every remaining residual, $d \geq 5$. After multiplication by -1 , the packet is the three-term segment

$$\{2^{-1}, 1, 2\}$$

inside $\langle \bar{2} \rangle$. Its translate overlaps in three, two, or one points exactly for translating exponents $0, \pm 1$, or ± 2 , and is otherwise disjoint. This proves (1679). $\square$

Corollary 14.93 (Automatic first-shell binary families). *Every prime in any of the congruence classes*

$$p \equiv 13 \pmod{24} \quad \text{or} \quad p \equiv 73, 97, 145 \pmod{168} \quad (1680)$$

satisfies the Erdős–Straus conjecture.

Proof. Write $p = 12k + 1$. If k is odd, then $\eta_k = 1$, and the first every-second shell has

$$j_1^{(2)} = 1, \quad R_1^{(2)} = 3.$$

Its target packet is all of U_3 . The identity belongs to the signed reachability set, so (1675) gives $q_{3,p} > 0$. The condition that k be odd is exactly $p \equiv 13 \pmod{24}$.

If k is even, then $\eta_k = 0$, and the first every-second shell has

$$j_1^{(2)} = 2, \quad R_1^{(2)} = 7.$$

By (1677), its target is all of U_7 whenever

$$\bar{p} \notin \langle \bar{2} \rangle = \{\bar{1}, \bar{2}, \bar{4}\},$$

equivalently whenever $p \equiv 3, 5, 6 \pmod{7}$. Again the identity in the signed reachability set proves $q_{7,p} > 0$. Solving these three conditions together with $p \equiv 1 \pmod{24}$ gives respectively

$$p \equiv 73, 145, 97 \pmod{168},$$

which is the second set of classes in (1680). $\square$

Corollary 14.94 (Automatic second-shell binary families). *Every prime in either congruence class*

$$p \equiv 73, 97 \pmod{120} \quad (1681)$$

satisfies the Erdős–Straus conjecture.

Proof. Write $p = 12k + 1$. The two classes in (1681) have $p \equiv 1 \pmod{24}$, so k is even and $\eta_k = 0$. The second every-second shell therefore has

$$j_2^{(2)} = 4, \quad R_2^{(2)} = 15, \quad b_2^{(2)} = \frac{3k}{2} + 2.$$

Its middle target packet is

$$\mathcal{T}_{15,2}^M = \{-2, -1, -2^{-1}\} = \{\bar{7}, \bar{13}, \bar{14}\} \subset U_{15}.$$

The two stated classes are exactly the simultaneous solutions of

$$p \equiv 1 \pmod{24}, \quad \bar{p} \in \{\bar{7}, \bar{13}\} \subset U_{15}.$$

Thus $\bar{p} \in \mathcal{T}_{15,2}^M$, whence $\bar{1} \in \bar{p}^{-1} \mathcal{T}_{15,2}^M$. The identity also belongs to $\mathcal{R}_{15}^M(\nu_{15}(b_2^{(2)}))$, by taking every signed exponent equal to zero. The one-box test (1675) now gives $q_{15,p} > 0$. $\square$

Corollary 14.95 (Universal every-third-shell reduction). *Let $p = 12k + 1$ be prime. For every integer m with $1 \leq m \leq 2k$, put*

$$j = 3m, \quad R_m = 12m - 1, \quad a_m = 3(k + m).$$

Then $a_m = n + j$, $4a_m - p = R_m$, and

$$\boxed{q_{R_m,p} > 0 \iff \mathbf{m}_{R_m}(k + m) \in (I_{R_m} : X_{\bar{3}}).} \quad (1682)$$

More generally, for every h satisfying $0 \leq h \leq v_3(k+m)$,

$$q_{R_m, p} > 0 \iff \mathfrak{m}_{R_m} \left(\frac{k+m}{3^h} \right) \in (I_{R_m} : X_{\bar{3}}^{h+1}). \quad (1683)$$

Proof. Since $p = 4n + 1 = 12k + 1$, one has $n = 3k$. Substitution of $j = 3m$ gives

$$n + j = 3(k + m), \quad 4j - 1 = 12m - 1.$$

The class of 3 modulo $12m - 1$ is a nonidentity unit. Apply theorem 14.86 first with $e = 1$, and then with $e = h + 1$. $\square$

Corollary 14.96 (Prime-twisted packet and its exact target volume). *Retain the conditions of theorem 14.95, and put*

$$b_m := k + m, \quad \mathcal{T}_m := \mathcal{T}_{R_m, 3}^M, \quad \mathcal{D}_{m, p} := \mathcal{T}_m \cup \bar{p}^{-1} \mathcal{T}_m.$$

Then

$$q_{R_m, p} > 0 \iff \mathcal{R}_{R_m}^M(\nu_{R_m}(b_m)) \cap \mathcal{D}_{m, p} \neq \emptyset. \quad (1684)$$

Moreover,

$$|\mathcal{D}_{m, p}| = \begin{cases} 3, & \bar{p} = \bar{1}, \\ 4, & \bar{p} \in \{\bar{3}, \bar{3}^{-1}\}, \\ 5, & \bar{p} \in \{\bar{3}^2, \bar{3}^{-2}\}, \\ 6, & \text{otherwise} \end{cases} \quad \text{in } U_{R_m}. \quad (1685)$$

Proof. The affine relation gives

$$12b_m = 12(k + m) = p + R_m.$$

Since $m \leq 2k$, one has

$$0 < R_m = 12m - 1 < 24k + 2 = 2p.$$

Also $R_m \neq p$, because their difference is congruent to $-2 \pmod{12}$. As p is prime, a common divisor of p and R_m is therefore 1. The displayed affine relation and $\gcd(12, R_m) = 1$ then also give $\gcd(b_m, R_m) = 1$. Thus both $\bar{p}$ and $\bar{b}_m$ belong to U_{R_m} . Thus

$$(12\bar{b}_m)^{-1} = \bar{p}^{-1} \quad \text{in } U_{R_m},$$

and (1684) follows from theorem 14.91.

Let d be the order of $\bar{3}$ in U_{R_m} . One has $d \geq 5$. Indeed, an order at most four would force R_m to divide one of

$$3 - 1 = 2, \quad 3^2 - 1 = 8, \quad 3^3 - 1 = 26, \quad 3^4 - 1 = 80.$$

No divisor $R_m \geq 11$ congruent to 11 $\pmod{12}$ occurs in this list.

Scalar multiplication by -1 does not change intersection cardinality, and

$$-\mathcal{T}_m = \{\bar{3}^{-1}, \bar{1}, \bar{3}\}.$$

Inside the cyclic subgroup generated by $\bar{3}$, this is the three-term segment with exponents $-1, 0, 1$. Its translate by $\bar{p}^{-1}$ overlaps in three, two, or one points precisely when the translating exponent is $0, \pm 1$, or ± 2 , respectively. Every other translate is disjoint. Subtracting the overlap from six proves (1685). $\square$

Corollary 14.97 (Automatic first-shell ternary families). *Every prime in any of the congruence classes*

$$p \equiv 73, 85, 109 \pmod{132} \quad (1686)$$

satisfies the Erdős–Straus conjecture.

Proof. Write $p = 12k + 1$. At $m = 1$, the every-third family gives

$$R_1 = 11, \quad b_1 = k + 1, \quad \mathcal{T}_{11,3}^M = \{-3, -1, -3^{-1}\} = \{\bar{8}, \bar{10}, \bar{7}\}.$$

The three classes in (1686) are exactly the simultaneous solutions of

$$p \equiv 1 \pmod{12}, \quad \bar{p} \in \{\bar{7}, \bar{8}, \bar{10}\} \subset U_{11}.$$

Consequently $\bar{1} \in \bar{p}^{-1} \mathcal{T}_{11,3}^M$. The identity belongs to $\mathcal{R}_{11}^M(\nu_{11}(b_1))$, so (1684) gives $q_{11,p} > 0$. $\square$

Theorem 14.98 (Exact four-carrier residue sieve). *Let*

$$\mathcal{A} := \{r \in \mathbb{Z}/9240\mathbb{Z} : r \equiv 1 \pmod{12}, \gcd(r, 9240) = 1\}.$$

Let $\mathcal{C} \subset \mathcal{A}$ be the union of the residue classes in (1680), (1681), and (1686). Then

$$|\mathcal{A}| = 480, \quad |\mathcal{C}| = 438, \quad \frac{|\mathcal{C}|}{|\mathcal{A}|} = \frac{73}{80}. \quad (1687)$$

Every prime $p \equiv 1 \pmod{12}$ whose class modulo 9240 lies in $\mathcal{C}$ satisfies the Erdős–Straus conjecture. The remaining 42 classes are exactly those satisfying

$$\begin{aligned} r &\equiv 1 \pmod{24}, \\ r &\equiv 1, 4 \pmod{5}, \\ r &\equiv 1, 2, 4 \pmod{7}, \\ r &\equiv 1, 2, 3, 4, 5, 6, 9 \pmod{11}. \end{aligned} \quad (1688)$$

Proof. Since

$$9240 = \text{lcm}(24, 5, 7, 11),$$

the Chinese remainder theorem identifies $\mathcal{A}$ with

$$\{\bar{1}, \bar{13}\} \times U_5 \times U_7 \times U_{11}.$$

Hence

$$|\mathcal{A}| = 2 \cdot 4 \cdot 6 \cdot 10 = 480.$$

Under this product description, the four automatic carriers are:

$$\begin{array}{l|l} R = 3 & r \equiv 13 \pmod{24}, \\ R = 7 & r \equiv 1 \pmod{24}, \bar{r} \in \{\bar{3}, \bar{5}, \bar{6}\} \subset U_7, \\ R = 15 & r \equiv 1 \pmod{24}, \bar{r} \in \{\bar{2}, \bar{3}\} \subset U_5, \\ R = 11 & \bar{r} \in \{\bar{7}, \bar{8}, \bar{10}\} \subset U_{11}. \end{array}$$

A class avoids all four carriers precisely when it obeys (1688). The four remaining coordinate sets have cardinalities 1, 2, 3, 7, so the complement has

$$1 \cdot 2 \cdot 3 \cdot 7 = 42$$

elements. Therefore

$$|\mathcal{C}| = 480 - 42 = 438, \quad \frac{438}{480} = \frac{73}{80}.$$

The final assertion follows from theorems 14.93, 14.94 and 14.97. $\square$

Theorem 14.99 (The $R = 23$ signed-two-power carrier). *Let $p = 12k + 1$ be prime and put*

$$b := k + 2 = \frac{p + 23}{12}, \quad \mathcal{T} := \mathcal{T}_{23,3}^M = \{\bar{15}, \bar{20}, \bar{22}\} \subset U_{23}.$$

For $e = 1, 2$, define

$$\mathcal{P}_e := \bigcup_{d=-e}^e \bar{2}^{-d} \mathcal{T}.$$

Then

$$\mathcal{P}_1 = \{\overline{7}, \overline{10}, \overline{11}, \overline{15}, \overline{17}, \overline{19}, \overline{20}, \overline{21}, \overline{22}\}, \quad (1689)$$

$$\begin{aligned} \mathcal{P}_2 &= \{\overline{5}, \overline{7}, \overline{10}, \overline{11}, \overline{14}, \overline{15}, \overline{17}, \overline{19}, \overline{20}, \overline{21}, \overline{22}\} \\ &= U_{23} \setminus \langle \overline{2} \rangle. \end{aligned} \quad (1690)$$

In particular,

$$p \equiv 1 \pmod{48}, \quad \bar{p} \in \mathcal{P}_1 \implies q_{23,p} > 0, \quad (1691)$$

$$p \equiv 25 \pmod{48}, \quad \bar{p} \in \mathcal{P}_2 \implies q_{23,p} > 0. \quad (1692)$$

The second line says that every quadratic nonresidue $p \bmod 23$ is certified when $p \equiv 25 \pmod{48}$.

Proof. Direct multiplication in U_{23} gives (1689) and (1690). The powers of 2 modulo 23 form the order-eleven subgroup

$$\langle \overline{2} \rangle = \{\overline{1}, \overline{2}, \overline{3}, \overline{4}, \overline{6}, \overline{8}, \overline{9}, \overline{12}, \overline{13}, \overline{16}, \overline{18}\}.$$

It is the quadratic-residue subgroup, so its complement is the set of quadratic nonresidues displayed in (1690).

If $2^e \mid b$, then the signed reachability box contains

$$\{\overline{2}^d : -e \leq d \leq e\} \subset \mathcal{R}_{23}^M(\nu_{23}(b)).$$

For $\bar{p} \in \mathcal{P}_e$, there are $d \in \{-e, \dots, e\}$ and $t \in \mathcal{T}$ such that $\bar{p} = \overline{2}^{-d}t$. Therefore

$$\overline{2}^d = \bar{p}^{-1}t \in \bar{p}^{-1}\mathcal{T}.$$

The one-box test (1684) proves occupation.

Finally,

$$p \equiv 1 \pmod{48} \implies b = \frac{p+23}{12} \equiv 2 \pmod{4},$$

whereas

$$p \equiv 25 \pmod{48} \implies 4 \mid b.$$

Thus the first class supplies $e = 1$ and the second supplies $e = 2$, which proves (1691) and (1692). $\square$

Exact computation 14.100 (The signed-length-two certificate for $p = 2521$). For $p = 2521$,

$$p \equiv 25 \pmod{48}, \quad \bar{p} = \overline{14} \in U_{23}, \quad b = \frac{2521+23}{12} = 212 = 4 \cdot 53.$$

The signed exponent -2 of the factor 2 gives

$$\overline{2}^{-2} = \overline{6}, \quad \bar{p}\overline{6} = \overline{14}\overline{6} = \overline{15} \in \mathcal{T}.$$

Hence

$$\overline{6} \in \mathcal{R}_{23}^M(\nu_{23}(212)) \cap \bar{p}^{-1}\mathcal{T},$$

which is an explicit $R = 23$ witness.

Corollary 14.101 (The $R = 23$ refinement of the automatic residue sieve). *Let*

$$M^\sharp := 425040 = \text{lcm}(9240, 48, 23)$$

and

$$\mathcal{A}^\sharp := \left\{ r \in \mathbb{Z}/M^\sharp\mathbb{Z} : r \equiv 1 \pmod{12}, \gcd(r, M^\sharp) = 1 \right\}.$$

Let $\mathcal{C}^\sharp$ contain the lifts of the four carriers in theorem 14.98, together with the $R = 23$ classes in (1691) and (1692). Then

$$|\mathcal{A}^\sharp| = 21120, \quad |\mathcal{C}^\sharp| = 20112, \quad \frac{|\mathcal{C}^\sharp|}{|\mathcal{A}^\sharp|} = \frac{419}{440}. \quad (1693)$$

Every prime represented by $\mathcal{C}^\sharp$ satisfies the Erdős–Straus conjecture.

Proof. Each of the 480 classes in $\mathcal{A}$ has two lifts from modulus 8 to modulus 16 and 22 unit lifts modulo 23. Thus

$$|\mathcal{A}^\#| = 480 \cdot 2 \cdot 22 = 21120.$$

The previous four carriers certify $438 \cdot 44 = 19272$ lifts.

Each of the remaining 42 classes satisfies $r \equiv 1 \pmod{24}$. Its two lifts modulo 48 are $\bar{1}$ and $\bar{25}$. On the first lift, (1691) certifies $|\mathcal{P}_1| = 9$ of the 22 unit classes modulo 23. On the second lift, (1692) certifies $|\mathcal{P}_2| = 11$. The $R = 23$ carrier therefore adds

$$42(9 + 11) = 840$$

classes. Consequently

$$|\mathcal{C}^\#| = 19272 + 840 = 20112.$$

The complement has

$$42((22 - 9) + (22 - 11)) = 42(13 + 11) = 1008$$

classes, and division by 48 gives

$$\frac{20112}{21120} = \frac{419}{440}.$$

□

Proposition 14.102 (Fixed-factor carrier at $R = 23$). *Retain*

$$b = \frac{p + 23}{12}, \quad \mathcal{T} = \mathcal{T}_{23,3}^M.$$

Let $\ell \neq 23$ be prime and let $e \geq 1$. Define

$$\mathcal{P}_{\ell,e} := \bigcup_{d=-e}^e \bar{\ell}^{-d} \mathcal{T} \subset U_{23}. \quad (1694)$$

If

$$\ell^e \mid b \quad \text{and} \quad \bar{p} \in \mathcal{P}_{\ell,e},$$

then $q_{23,p} > 0$. The divisibility condition is equivalent to

$$p \equiv -23 \pmod{12\ell^e}.$$

Proof. The condition $\ell^e \mid b$ places every residue

$$\bar{\ell}^d, \quad -e \leq d \leq e,$$

in $\mathcal{R}_{23}^M(\nu_{23}(b))$. If $\bar{p} = \bar{\ell}^{-d} t$ for some $t \in \mathcal{T}$, then

$$\bar{\ell}^d = \bar{p}^{-1} t \in \bar{p}^{-1} \mathcal{T}.$$

The one-box test proves $q_{23,p} > 0$. Since $12b = p + 23$, the final equivalence follows directly. □

Theorem 14.103 (Additive signed-box criterion at $R = 23$). *Let $p = 12k + 1$ be prime and put*

$$b := k + 2 = \frac{p + 23}{12}, \quad \mathcal{T} := \mathcal{T}_{23,3}^M = \{\bar{15}, \bar{20}, \bar{22}\}.$$

Let

$$\lambda : U_{23} \longrightarrow \mathbb{Z}/22\mathbb{Z}$$

be the group isomorphism determined by $\lambda(\bar{5}) = 1$. For the prime factorization

$$b = \prod_{r \mid b} r^{\nu_r(b)},$$

define the signed logarithmic box

$$\mathcal{B}_{23}(b) := \left\{ \sum_{r|b} d_r \lambda(\bar{r}) : -\nu_r(b) \leq d_r \leq \nu_r(b) \right\} \subset \mathbb{Z}/22\mathbb{Z}. \quad (1695)$$

Set

$$\rho := \lambda(\bar{p}), \quad \mathcal{S} := \{5, 11, 17\} \subset \mathbb{Z}/22\mathbb{Z}, \quad \mathcal{S} - \rho := \{s - \rho : s \in \mathcal{S}\}.$$

Then

$$q_{23,p} > 0 \iff \mathcal{B}_{23}(b) \cap (\mathcal{S} \cup (\mathcal{S} - \rho)) \neq \emptyset. \quad (1696)$$

Proof. The identities

$$5^{11} \equiv -1 \pmod{23}, \quad 5 \not\equiv -1 \pmod{23}$$

show that 5 has order 22, so λ is an isomorphism. Directly,

$$5^{17} \equiv 15, \quad 5^5 \equiv 20, \quad 5^{11} \equiv 22 \pmod{23},$$

and hence

$$\lambda(\mathcal{T}) = \{17, 5, 11\} = \mathcal{S}. \quad (1697)$$

The middle reachability set consists of the residues

$$\prod_{r|b} \bar{r}^{d_r}, \quad -\nu_r(b) \leq d_r \leq \nu_r(b).$$

Applying λ therefore gives

$$\lambda(\mathcal{R}_{23}^M(\nu_{23}(b))) = \mathcal{B}_{23}(b).$$

Moreover,

$$\lambda(\bar{p}^{-1}\mathcal{T}) = \mathcal{S} - \rho.$$

At this shell, $12b = p + 23$, so (1672) reads

$$q_{23,p} > 0 \iff \mathcal{R}_{23}^M(\nu_{23}(b)) \cap (\mathcal{T} \cup \bar{p}^{-1}\mathcal{T}) \neq \emptyset.$$

Applying the bijection λ proves (1696). $\square$

Corollary 14.104 (Parity obstruction in the $R = 23$ logarithmic box). *Assume that $\bar{p}$ is a quadratic residue modulo 23. If $q_{23,p} > 0$, then $b = (p + 23)/12$ has a prime divisor whose residue modulo 23 is a quadratic nonresidue.*

Proof. The quadratic residues in $U_{23} = \langle \bar{5} \rangle$ have even λ -coordinate. Thus ρ is even, while every member of $\mathcal{S} \cup (\mathcal{S} - \rho)$ is odd. If every prime divisor of b were a quadratic residue, every summand in (1695) would be even, and hence $\mathcal{B}_{23}(b)$ would contain only even classes. Its intersection with the two target triples would be empty, contradicting (1696). $\square$

Corollary 14.105 (Quadratic-character balance at $R = 23$). *For a positive integer n not divisible by 23, put*

$$\Omega_{23}^-(n) := \sum_{\substack{r|n \\ r \text{ prime} \\ \left(\frac{r}{23}\right) = -1}} \nu_r(n).$$

Then

$$\left(\frac{b}{23}\right) = \left(\frac{p}{23}\right), \quad \Omega_{23}^-(b) \equiv \frac{1 - \left(\frac{p}{23}\right)}{2} \pmod{2}. \quad (1698)$$

In particular, if $\bar{p}$ is a quadratic residue and $q_{23,p} > 0$, then

$$\Omega_{23}^-(b) \geq 2. \quad (1699)$$

Proof. The affine shell identity gives

$$b = \frac{p+23}{12} \equiv 2p \pmod{23}.$$

Since $2 \equiv 5^2 \pmod{23}$, multiplication by 2 preserves quadratic character. This proves the first equality in (1698). Multiplicativity of the Legendre symbol gives

$$\left(\frac{b}{23}\right) = (-1)^{\Omega_{23}^-(b)},$$

which proves the parity formula. If p is a quadratic residue and the shell is occupied, theorem 14.104 gives $\Omega_{23}^-(b) > 0$, while the parity formula makes this integer even. Therefore it is at least 2. $\square$

Theorem 14.106 (Complete edge-log classification at $R = 23$). *Retain the notation of theorem 14.103. From the prime factorization of $b = (p+23)/12$, form the multiset*

$$\mathfrak{L}_{23}^\times(b) := \prod_{\substack{r|b, \, r \text{ prime} \\ \lambda(\bar{r}) \neq 0}} \{\lambda(\bar{r})\}^{\nu_r(b)}. \quad (1700)$$

If

$$\rho = \lambda(\bar{p}) \in \{1, 21\},$$

then the $R = 23$ shell is empty precisely in the following cases:

$\bar{p}$	ρ	$\mathfrak{L}_{23}^\times(b)$
$\bar{5}$	1	$\{3\}, \{1, 2\}, \{1, 1, 1\}$
$\bar{14}$	21	$\{1\}, \{2, 21\}, \{8, 15\}, \{1, 1, 21\}, \{15, 15, 15\}.$

(1701)

Equivalently, after prime factors congruent to 1 (mod 23) are deleted, the multisets of prime-factor residues are

$\bar{p}$	$\{\bar{r} : r \mid b \text{ prime}\}_{\text{multiset}}$
$\bar{5}$	$\{10\}, \{5, 2\}, \{5, 5, 5\}$
$\bar{14}$	$\{5\}, \{2, 14\}, \{16, 19\}, \{5, 5, 14\}, \{19, 19, 19\}.$

(1702)

Proof. Put

$$A_\rho := \mathcal{S} \cup (\mathcal{S} - \rho).$$

The signed box $\mathcal{B}_{23}(b)$ is stable under negation. For $\rho = 1$ and $\rho = 21 = -1$, respectively,

$$A_1 = \{4, 5, 10, 11, 16, 17\}, \quad A_{21} = \{5, 6, 11, 12, 17, 18\}.$$

It follows that, for either value of ρ ,

$$\mathcal{B}_{23}(b) \cap A_\rho = \emptyset \iff \mathcal{B}_{23}(b) \cap \mathcal{F} = \emptyset, \quad (1703)$$

where

$$\mathcal{F} = A_1 \cup (-A_1) = A_{21} \cup (-A_{21}) = \{4, 5, 6, 10, 11, 12, 16, 17, 18\}.$$

The allowed one-term classes are therefore

$$\mathcal{C}^\times = \{\pm 1, \pm 2, \pm 3, \pm 7, \pm 8, \pm 9\} \subset \mathbb{Z}/22\mathbb{Z}. \quad (1704)$$

For completeness, the unordered two-term multisets in $\mathcal{C}^\times$ whose signed boxes avoid $\mathcal{F}$ are

(1, 1)	(1, 2)	(1, 8)	(1, -8)	(1, -2)	(1, -1)
(2, -1)	(7, 7)	(7, 8)	(7, -8)	(7, -7)	(8, -7)
(8, -1)	(-8, -7)	(-8, -1)	(-7, -7)	(-2, -1)	(-1, -1).

This table is obtained by testing $x+y$ and $x-y$ for the twelve classes in (1704); every omitted pair has a signed sum in $\mathcal{F}$. Testing a third entry against the displayed pairs leaves exactly

$$\{\epsilon_1, \epsilon_2, \epsilon_3\}, \quad \epsilon_j \in \{1, -1\}, \quad \text{or} \quad \{7\epsilon_1, 7\epsilon_2, 7\epsilon_3\}, \quad \epsilon_j \in \{1, -1\}. \quad (1705)$$

Every box in the first family contains

$$D_1 := \{0, \pm 1, \pm 2, \pm 3\},$$

and every box in the second contains

$$D_7 := \{0, \pm 1, \pm 7, \pm 8\}.$$

By negation symmetry, a possible fourth entry may be represented by $x \in \{1, 2, 3, 7, 8, 9\}$. The following table gives, for each x , an element of D_1 and an element of D_7 whose sum with x lies in $\mathcal{F}$:

x	1	2	3	7	8	9
$d_1 \in D_1$	3	2	1	3	2	1
$x + d_1$	4	4	4	10	10	10
$d_7 \in D_7$	-7	8	1	-1	8	1
$x + d_7$	16	10	4	6	16	10.

Hence no four-term multiset of nonzero logarithms has a signed box disjoint from $\mathcal{F}$. The same conclusion holds for every longer multiset because its signed box contains the box of each four-term submultiset.

Finally,

$$\sum_{x \in \mathfrak{L}_{23}^\times(b)} x = \lambda(\bar{b}) = \lambda(\overline{2p}) = \rho + 2 \pmod{22}.$$

Imposing total 3 on the safe one-, two-, and three-term lists gives

$$\{3\}, \quad \{1, 2\}, \quad \{1, 1, 1\};$$

imposing total 1 gives

$$\{1\}, \quad \{2, 21\}, \quad \{8, 15\}, \quad \{1, 1, 21\}, \quad \{15, 15, 15\}.$$

Logarithm-zero prime factors make no contribution to either the total or the signed box. This proves (1701) by theorem 14.103. The residue table follows from

$$5^1 \equiv 5, \quad 5^2 \equiv 2, \quad 5^3 \equiv 10, \quad 5^8 \equiv 16, \quad 5^{15} \equiv 19, \quad 5^{21} \equiv 14 \pmod{23}.$$

□

Corollary 14.107 (Four nontrivial residue occurrences force occupation). *Suppose $p \equiv 5$ or $14 \pmod{23}$. If*

$$\sum_{\substack{r|b, r \text{ prime} \\ r \not\equiv 1 \pmod{23}}} \nu_r(b) \geq 4,$$

then $q_{23,p} > 0$, and the Erdős–Straus equation has a positive integer solution for p .

Proof. The sum is the cardinality of $\mathfrak{L}_{23}^\times(b)$, counted with multiplicity. Every empty-shell multiset in (1701) has cardinality at most three. □

Corollary 14.108 (Exact two-adic survivor form on the edge branches). *Let p be a prime satisfying*

$$p \equiv 1 \pmod{48}, \quad p \equiv 5 \text{ or } 14 \pmod{23},$$

and put $b = (p + 23)/12$. Then

$$q_{23,p} = 0 \iff b = 2\ell u, \tag{1706}$$

where ℓ is a prime satisfying

$$\ell \equiv p \pmod{23}, \quad \nu_\ell(b) = 1,$$

and every prime divisor of u is congruent to 1 $\pmod{23}$. Consequently, failure of the factorization on the right-hand side of (1706) implies $q_{23,p} > 0$ and gives a positive integer solution of the Erdős–Straus equation for p .

Proof. Write $p = 48k + 1$. Then

$$b = \frac{48k + 24}{12} = 4k + 2,$$

so $\nu_2(b) = 1$. Since $5^2 \equiv 2 \pmod{23}$, the nonzero prime-log multiset $\mathfrak{L}_{23}^\times(b)$ contains exactly one occurrence of 2.

Put $\rho = \lambda(\bar{p})$. If $p \equiv 5 \pmod{23}$, then $\rho = 1$, and the only multiset in the first row of (1701) that contains 2 is $\{1, 2\}$. If $p \equiv 14 \pmod{23}$, then $\rho = 21$, and the only multiset in the second row that contains 2 is $\{2, 21\}$. Thus an empty shell has exactly one further nonzero prime logarithm, it occurs once, and it equals ρ . The corresponding prime ℓ satisfies $\ell \equiv p \pmod{23}$; all remaining prime factors have logarithm zero and hence are congruent to 1 $\pmod{23}$. This proves the forward implication in (1706).

Conversely, the displayed factorization gives

$$\mathfrak{L}_{23}^\times(b) = \{2, \rho\}.$$

This is $\{1, 2\}$ when $\rho = 1$ and $\{2, 21\}$ when $\rho = 21$. Both occur in (1701), so the $R = 23$ shell is empty. $\square$

Corollary 14.109 (Adjacent-shell isolation of an edge survivor). *Retain the conditions of theorem 14.108, and assume $q_{23,p} = 0$. Write*

$$b = 2\ell u$$

as in (1706), and put $c = b - 1$. Then the first six reduced affine coordinates are

$$\left(a_3, a_7, \frac{a_{11}}{3}, a_{15}, a_{19}, \frac{a_{23}}{3}\right) = (6\ell u - 5, 2(3\ell u - 2), 2\ell u - 1, 2(3\ell u - 1), 6\ell u - 1, 2\ell u). \quad (1707)$$

Their gcds with b are

$$\begin{array}{c|ccccc} x & a_3 & a_7 & a_{11}/3 & a_{15} & a_{19} \\ \hline \gcd(b, x) & \gcd(b, 5) & 2 & 1 & 2 & 1. \end{array} \quad (1708)$$

Moreover,

$$\gcd\left(u, a_3 a_7 \frac{a_{11}}{3} a_{15} a_{19}\right) = 1, \quad (1709)$$

$$\gcd\left(\ell, a_7 \frac{a_{11}}{3} a_{15} a_{19}\right) = 1, \quad \gcd(\ell, a_3) = \gcd(\ell, 5). \quad (1710)$$

Thus the distinguished prime ℓ can occur in one of the five preceding coordinates only when $\ell = 5$, and then only in a_3 .

Proof. Substitute $c = 2\ell u - 1$ into (2075) to obtain (1707). The Euclidean algorithm gives

$$\begin{aligned} \gcd(b, 3b - 5) &= \gcd(b, 5), & \gcd(b, 3b - 4) &= \gcd(b, 4), \\ \gcd(b, b - 1) &= 1, & \gcd(b, 3b - 2) &= \gcd(b, 2), \\ \gcd(b, 3b - 1) &= 1. \end{aligned}$$

Since $\nu_2(b) = 1$, the two gcds on the right involving 4 and 2 both equal 2. This proves (1708).

Every prime divisor of u is 1 $\pmod{23}$, so none equals 2 or 5. The gcd table therefore proves (1709). The prime ℓ is odd and is not 1 $\pmod{23}$. Applying the same table to ℓ proves (1710). $\square$

Corollary 14.110 (The first factor-three refinement). *Every prime satisfying*

$$p \equiv 49 \pmod{144}, \quad p \equiv 5, 14 \pmod{23} \quad (1711)$$

satisfies the Erdős–Straus conjecture.

Proof. At $\ell = 3$ and $e = 1$,

$$\mathcal{P}_{3,1} = \{\bar{5}, \bar{14}, \bar{15}, \bar{20}, \bar{22}\}.$$

If $p \equiv 49 \pmod{144}$, then

$$b = \frac{p+23}{12} \equiv 6 \pmod{12},$$

so $3 \mid b$. The two stated residue classes modulo 23 belong to $\mathcal{P}_{3,1}$, and theorem 14.102 applies. $\square$

Corollary 14.111 (The factor-three refinement of the residue sieve). *Let*

$$M^b := 1275120 = \text{lcm}(425040, 144)$$

and let $\mathcal{A}^b$ be the reduced classes modulo M^b that are congruent to 1 (mod 12). Adjoin the family in (1711) to the carriers of theorem 14.101. The resulting certified set $\mathcal{C}^b$ satisfies

$$|\mathcal{A}^b| = 63360, \quad |\mathcal{C}^b| = 60420, \quad \frac{|\mathcal{C}^b|}{|\mathcal{A}^b|} = \frac{1007}{1056}. \quad (1712)$$

Proof. Passing from $M^\sharp$ to M^b adds the three lifts modulo 9 above each class already congruent to 1 (mod 3). Hence

$$|\mathcal{A}^b| = 3 \cdot 21120 = 63360$$

and the earlier carriers certify

$$3 \cdot 20112 = 60336$$

classes.

For each of the 42 base classes in (1688), there is exactly one lift with $p \equiv 49 \pmod{144}$. On that lift, the two classes $\bar{5}, \bar{14} \pmod{23}$ were absent from $\mathcal{P}_1$ and are supplied by $\mathcal{P}_{3,1}$. Thus the factor-three carrier adds

$$42 \cdot 2 = 84$$

classes. Therefore

$$|\mathcal{C}^b| = 60336 + 84 = 60420,$$

and

$$\frac{60420}{63360} = \frac{1007}{1056}.$$

$\square$

Corollary 14.112 (The first factor-seventeen refinement). *Every prime satisfying*

$$p \equiv 181 \pmod{204}, \quad p \equiv 2, 4, 6, 9, 12, 15, 18, 20, 22 \pmod{23} \quad (1713)$$

satisfies the Erdős–Straus conjecture.

Proof. At $\ell = 17$ and $e = 1$,

$$\mathcal{P}_{17,1} = \{\bar{2}, \bar{4}, \bar{6}, \bar{9}, \bar{12}, \bar{15}, \bar{18}, \bar{20}, \bar{22}\}.$$

The congruence $p \equiv 181 \pmod{204}$ is

$$p \equiv -23 \pmod{12 \cdot 17},$$

and hence $17 \mid b = (p+23)/12$. The assertion follows from theorem 14.102. $\square$

Corollary 14.113 (The factor-seventeen refinement of the residue sieve). *Let*

$$M^{(17)} := 21677040 = 17M^b.$$

Lift the carrier set of theorem 14.111 to the reduced classes modulo $M^{(17)}$, and adjoin the family in (1713). If $\mathcal{A}^{(17)}$ and $\mathcal{C}^{(17)}$ denote the resulting admissible and certified sets, then

$$|\mathcal{A}^{(17)}| = 1013760, \quad |\mathcal{C}^{(17)}| = 968232, \quad \frac{|\mathcal{C}^{(17)}|}{|\mathcal{A}^{(17)}|} = \frac{40343}{42240}. \quad (1714)$$

The remaining class count is 45528.

Proof. The prime 17 does not divide M^b . Every reduced class modulo M^b therefore has 16 reduced lifts modulo $M^{(17)}$. Hence

$$|\mathcal{A}^{(17)}| = 16 \cdot 63360 = 1013760,$$

and the preceding certified set contributes

$$16 \cdot 60420 = 966720$$

classes.

The 2940-class remainder of theorem 14.111 has 252 members above each of the six residues

$$2, 4, 6, 9, 12, 18 \pmod{23}.$$

These are precisely the members of $\mathcal{P}_{17,1}$ occurring in that remainder. Above each such base class, exactly one of the 16 reduced lifts has

$$p \equiv -23 \equiv 11 \pmod{17}.$$

The factor-seventeen carrier therefore contributes

$$6 \cdot 252 = 1512$$

new classes. It follows that

$$|\mathcal{C}^{(17)}| = 966720 + 1512 = 968232,$$

and division gives

$$\frac{968232}{1013760} = \frac{40343}{42240}.$$

The complement has $1013760 - 968232 = 45528$ classes. $\square$

Theorem 14.114 (Divisor-sparse submaximal prime-twisted targets). *Let $p = 12k + 1$ be prime, and retain the every-third-shell data*

$$R_m = 12m - 1, \quad 1 \leq m \leq 2k.$$

For positive N, X , define

$$\delta_{11}(N; X) := \#\{d \mid N : d \leq X, d \equiv 11 \pmod{12}\}.$$

Let

$$\mathcal{N}_j(k) := \#\{1 \leq m \leq 2k : |\mathcal{D}_{m,p}| = j\}.$$

Then, with $X_k = 24k - 1$,

$$\mathcal{N}_3(k) = \delta_{11}(k; X_k), \tag{1715}$$

$$\mathcal{N}_4(k) = \delta_{11}(6k - 1; X_k) + \delta_{11}(18k + 1; X_k), \tag{1716}$$

$$\mathcal{N}_5(k) = \delta_{11}(3k - 2; X_k) + \delta_{11}(27k + 2; X_k), \tag{1717}$$

$$\mathcal{N}_6(k) = 2k - \mathcal{N}_3(k) - \mathcal{N}_4(k) - \mathcal{N}_5(k). \tag{1718}$$

In particular, for every $\varepsilon > 0$,

$$\mathcal{N}_3(k) + \mathcal{N}_4(k) + \mathcal{N}_5(k) = O_\varepsilon(k^\varepsilon), \quad \frac{\mathcal{N}_6(k)}{2k} \longrightarrow 1. \tag{1719}$$

Proof. By (1685), target volume three is equivalent to $p \equiv 1 \pmod{R_m}$. Since

$$p - 1 = 12k, \quad \gcd(R_m, 12) = 1,$$

this is equivalent to $R_m \mid k$.

Target volume four is equivalent to $p \equiv 3$ or $p \equiv 3^{-1} \pmod{R_m}$. These two congruences are respectively equivalent to

$$R_m \mid p - 3 = 2(6k - 1), \quad R_m \mid 3p - 1 = 2(18k + 1).$$

Because R_m is odd, they are equivalent to

$$R_m \mid 6k - 1, \quad R_m \mid 18k + 1.$$

Likewise, target volume five is equivalent to $p \equiv 3^2$ or $p \equiv 3^{-2} \pmod{R_m}$. The corresponding integer divisibilities are

$$R_m \mid p - 9 = 4(3k - 2), \quad R_m \mid 9p - 1 = 4(27k + 2).$$

Since $\gcd(R_m, 4) = 1$, these are equivalent to

$$R_m \mid 3k - 2, \quad R_m \mid 27k + 2.$$

The map $m \mapsto R_m$ is a bijection from $\{1, \dots, 2k\}$ onto the positive integers

$$d \equiv 11 \pmod{12}, \quad d \leq 24k - 1.$$

The five residue classes

$$1, \quad 3, \quad 3^{-1}, \quad 3^2, \quad 3^{-2}$$

are distinct because the order of 3 modulo R_m is at least five. The five divisor conditions above are therefore disjoint and give (1715)–(1718).

Each restricted divisor count is at most the ordinary divisor function of an integer of size $O(k)$. The bound $\tau(N) = O_\varepsilon(N^\varepsilon)$ gives the first assertion in (1719); division by $2k$ gives the second. $\square$

Corollary 14.115 (The $R = 11$ packet collision and $R = 23$ packet separation). *In the every-third-shell reduction, the first two pairs of target packets are*

$$\mathcal{T}_{11,3}^E = \mathcal{T}_{11,3}^M = \{7, 8, 10\} \subset U_{11}, \quad (1720)$$

$$\mathcal{T}_{23,3}^E = \{7, 17, 21\}, \quad \mathcal{T}_{23,3}^M = \{15, 20, 22\} \subset U_{23}. \quad (1721)$$

Moreover,

$$\mathcal{T}_{11,3}^E = \mathcal{T}_{11,3}^M, \quad \mathcal{T}_{23,3}^E = 2\mathcal{T}_{23,3}^M. \quad (1722)$$

The corresponding prime-twisted target volumes are

R	$ \mathcal{D}_{m,p} = 3$	$ \mathcal{D}_{m,p} = 4$	$ \mathcal{D}_{m,p} = 5$	$ \mathcal{D}_{m,p} = 6$
11	$p \equiv 1$	$p \equiv 3, 4$	$p \equiv 5, 9$	$p \equiv 2, 6, 7, 8, 10$
23	$p \equiv 1$	$p \equiv 3, 8$	$p \equiv 9, 18$	$p \equiv 2, 4, 5, 6, 7, 10, 11, 12, 13$ $p \equiv 14, 15, 16, 17, 19, 20, 21, 22$

(1723)

Every congruence in a row is taken modulo the residual R . Thus the two three-position packets coincide at residual 11, while they are disjoint at residual 23.

Proof. Modulo 11,

$$t_{11} = 8, \quad s_{11} = 10, \quad 3^{-1} = 4.$$

Substitution into (1661) and (1662) gives

$$\begin{aligned} \{8, 8 \cdot 4, 8 \cdot 4^2\} &= \{8, 10, 7\}, \\ \{10 \cdot 3, 10, 10 \cdot 4\} &= \{8, 10, 7\}. \end{aligned}$$

Modulo 23,

$$t_{23} = 17, \quad s_{23} = 22, \quad 3^{-1} = 8.$$

The same substitution gives

$$\begin{aligned} \{17, 17 \cdot 8, 17 \cdot 8^2\} &= \{17, 21, 7\}, \\ \{22 \cdot 3, 22, 22 \cdot 8\} &= \{20, 22, 15\}. \end{aligned}$$

The two latter sets have empty intersection. Furthermore,

$$3^{-1} \equiv 4, \quad 3^{-2} \equiv 5 \pmod{11},$$

and

$$3^{-1} \equiv 8, \quad 3^{-2} \equiv 18 \pmod{23}.$$

The four cases of (1685) therefore give the two rows of (1723). Finally, $12^{-1} \equiv 1 \pmod{11}$ and $12^{-1} \equiv 2 \pmod{23}$. Equation (1664) proves (1722). $\square$

Theorem 14.116 (Uniqueness of the universal packet collision). *For $m \geq 1$, put $R_m = 12m - 1$. Then*

$$\mathcal{T}_{R_m,3}^E = \mathcal{T}_{R_m,3}^M \iff m = 1. \quad (1724)$$

Equivalently, residual 11 is the unique member of the universal every-third-shell family at which the exterior and middle three-position target packets coincide.

Proof. All of 3, 4, 12, 36 are units modulo R_m . In fractional notation inside U_{R_m} , the packets are

$$\mathcal{T}_{R_m,3}^E = \left\{ -\frac{1}{4}, -\frac{1}{12}, -\frac{1}{36} \right\}, \quad \mathcal{T}_{R_m,3}^M = \left\{ -3, -1, -\frac{1}{3} \right\}.$$

Each displayed packet has three distinct entries. Indeed, a repeated entry would imply that the order of 3 modulo R_m is at most two, whereas $3 \not\equiv 1 \pmod{R_m}$ and $3^2 \not\equiv 1 \pmod{R_m}$ for $R_m \geq 11$. If the two sets coincide, their sums coincide. Since

$$-\frac{13}{36} - \left(-\frac{13}{3} \right) = \frac{143}{36},$$

one obtains $R_m \mid 143$. Their products also coincide, and

$$-\frac{1}{1728} - (-1) = \frac{1727}{1728},$$

so $R_m \mid 1727$. But

$$\gcd(143, 1727) = 11.$$

As $R_m \geq 11$, it follows that $R_m = 11$, hence $m = 1$. The converse is (1720). $\square$

Exact computation 14.117 (Forced-factor target-packet certificate). The script and captured receipt

verification/verify_forced_factor_target_packets.py,

verification/verify_forced_factor_target_packets_output.txt

verify the exact packet sets at residuals 11 and 23, compare the packet criterion with direct factor insertion on all reduced valuation states of total degree at most four, and compare it with the original divisor definitions on 555 affine prime shells for each residual. It also checks the packet homothety and the unique collision over the first 1000 members of the universal residual family; the preceding theorem proves the uniqueness claim for all members.

Corollary 14.118 (Two-target divisor forcing). *Retain the conditions of theorem 14.79. Each of the following conditions forces $q_{R,p} > 0$:*

- (i) *there is a divisor $u \mid a^2$ such that $4u + 1 \equiv 0 \pmod{R}$;*
- (ii) *there are coprime positive integers x, y such that $xy \mid a$ and $x + y \equiv 0 \pmod{R}$.*

Under (i), one has

$$|\mathcal{E}_{R,a}| \geq 1.$$

Under (ii), both (x, y) and (y, x) contribute to $c_{R,a}$, and hence

$$c_{R,a} \geq 2.$$

In particular, a divisor $d \mid a$ with

$$d \equiv -1 \pmod{R} \quad (1725)$$

forces $c_{R,a} \geq 2$ through the ordered pairs $(d, 1)$ and $(1, d)$.

Proof. Condition (i) says exactly that $u \in \mathcal{E}_{R,a}$. Under (ii), the two ordered pairs belong to the set counted in (1627); they are distinct because $R \geq 3$. The rank identity (1629) proves the assertion. Taking $x = d$ and $y = 1$ proves the final statement. $\square$

Corollary 14.119 (Exact first-residual classification). *Let $p \equiv 1 \pmod{12}$ be prime and put*

$$a_3 := \frac{p+3}{4}.$$

Then $a_3 \in \mathcal{A}_p$, $4a_3 - p = 3$, and the following are equivalent:

- (i) $q_{3,p} > 0$;
- (ii) $\mathcal{E}_{3,a_3} \neq \emptyset$;
- (iii) $c_{3,a_3} > 0$;
- (iv) a_3 has a prime divisor $\ell \equiv 2 \pmod{3}$.

Whenever these conditions hold, both support channels are occupied:

$$|\mathcal{E}_{3,a_3}| \geq 1, \quad c_{3,a_3} \geq 2, \quad q_{3,p} \geq 2. \quad (1726)$$

Proof. Write $p = 12k + 1$. Then $a_3 = 3k + 1$, so $3 \nmid a_3$; the inequalities

$$\frac{p}{4} < a_3 \leq \frac{3p}{4}$$

show that $a_3 \in \mathcal{A}_p$.

Suppose that $\ell \mid a_3$ and $\ell \equiv 2 \pmod{3}$. The divisor $u = \ell$ satisfies

$$4u + 1 \equiv u + 1 \equiv 0 \pmod{3},$$

so $\ell \in \mathcal{E}_{3,a_3}$. The coprime ordered pairs $(\ell, 1)$ and $(1, \ell)$ also satisfy $3 \mid \ell + 1$, so theorem 14.118 gives (1726).

Conversely, suppose that a_3 has no prime divisor congruent to 2 (mod 3). Since $3 \nmid a_3$, every prime divisor of a_3 is congruent to 1 (mod 3). Consequently every divisor $u \mid a_3^2$ is congruent to 1 (mod 3), and hence

$$4u + 1 \equiv 2 \pmod{3}.$$

Thus $\mathcal{E}_{3,a_3} = \emptyset$. Likewise, every pair of positive integers x, y with $xy \mid a_3$ satisfies $x \equiv y \equiv 1 \pmod{3}$, so $x + y \equiv 2 \pmod{3}$. The counting formula (1627) gives $c_{3,a_3} = 0$. The rank identity (1629) completes all four equivalences. $\square$

Exact computation 14.120 (Two-target divisor-forcing certificate). The script and captured receipt

```
verification/verify_two_target_divisor_forcing.py,
verification/verify_two_target_divisor_forcing_output.txt
```

check theorem 14.119 for all 159 primes $p \equiv 1 \pmod{12}$ below 5000. They also check (1725) in 441 admissible-shell instances below $p = 1000$.

Remark 14.121 (Relation to residue-class methods). Elsholtz and Tao describe the established parametric approach in terms of solution families occupying specified residue classes, with those classes feeding sieve arguments and computational verification [1]. The first-residual criterion above acts on the prime factors of the affine shell coordinate

$$a_3 = \frac{p+3}{4}.$$

It gives a complete decision for the $R = 3$ shell. When this shell is empty, occupancy of a later shell remains to be established.

Theorem 14.122 (Exact second-residual factor classification). *Let $p \equiv 1 \pmod{12}$ be prime and put*

$$a_7 := \frac{p+7}{4}.$$

For $r \in U_7$, define

$$\nu_r(a_7) := \sum_{\substack{\ell|a_7 \\ \ell \text{ prime} \\ \ell \equiv r \pmod{7}}} v_\ell(a_7). \quad (1727)$$

Then $a_7 \in \mathcal{A}_p$, $4a_7 - p = 7$, and $q_{7,p} > 0$ if and only if at least one of the following conditions holds:

$$\nu_5(a_7) + \nu_6(a_7) > 0, \quad (1728)$$

$$\nu_3(a_7) \geq 3, \quad (1729)$$

$$\nu_3(a_7) \geq 1 \quad \text{and} \quad \nu_2(a_7) + \nu_4(a_7) \geq 1. \quad (1730)$$

Equivalently, the $R = 7$ shell is empty exactly in the union of the following two cases:

- (a) every prime divisor of a_7 lies in $\{1, 2, 4\} \pmod{7}$;
- (b) every prime divisor lies in $\{1, 3\} \pmod{7}$ and $1 \leq \nu_3(a_7) \leq 2$.

Proof. The integer 3 generates U_7 , with

$$\begin{array}{c|cccccc} j & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline 3^j \pmod{7} & 1 & 3 & 2 & 6 & 4 & 5. \end{array}$$

The congruence $7 \nmid a_7$ follows from $4a_7 = p + 7$ and $p \neq 7$.

If $\nu_5(a_7) > 0$, choose a prime divisor $\ell \equiv 5 \pmod{7}$. Then

$$4\ell + 1 \equiv 0 \pmod{7},$$

so $\ell \in \mathcal{E}_{7,a_7}$. If $\nu_6(a_7) > 0$, a prime divisor $\ell \equiv -1 \pmod{7}$ forces the middle channel by theorem 14.118.

If $\nu_3(a_7) \geq 3$, the product d of three prime factors congruent to 3 (mod 7), counted with multiplicity, is a divisor of a_7 and satisfies

$$d \equiv 3^3 \equiv -1 \pmod{7}.$$

This again forces the middle channel. Under (1730), a 3-residue prime and a 2-residue prime have product $-1 \pmod{7}$. A 3-residue prime x and a 4-residue prime y are distinct and satisfy

$$\gcd(x, y) = 1, \quad x + y \equiv 0 \pmod{7}.$$

Thus every listed condition gives $q_{7,p} > 0$.

Suppose that none of (1728)–(1730) holds. If $\nu_3(a_7) = 0$, every divisor of a_7 lies in the subgroup

$$H = \{1, 2, 4\} \leq U_7.$$

Hence every divisor of a_7^2 lies in H , while the exterior target is $5 \notin H$. Every unit ratio $\bar{x}\bar{y}^{-1}$ arising from (1627) also lies in H , while the middle target is $-1 \equiv 6 \notin H$. Both channels are empty.

It remains to consider $1 \leq \nu_3(a_7) \leq 2$. Failure of the other conditions says that all prime divisors lie in $\{1, 3\} \pmod{7}$. Writing $e = \nu_3(a_7)$, the residue of an exterior divisor has the form 3^j with $0 \leq j \leq 2e \leq 4$; it cannot equal $5 = 3^5$. A middle ratio has the form 3^j with $-e \leq j \leq e$; for $e \leq 2$, it cannot equal $-1 = 3^3$. Thus both channels are empty in this case as well. The two cases are exactly the negation of the three forcing conditions. $\square$

Corollary 14.123 (Hard-locus Boolean sheet formula at $R = 7$). *Let $p \equiv 1 \pmod{24}$ be prime and put*

$$c := \frac{p+7}{8}, \quad a_7 := \frac{p+7}{4} = 2c. \quad (1731)$$

Then

$$\begin{aligned} q_{7,p} > 0 & \iff \text{some prime } \ell \mid c \text{ satisfies } \left(\frac{\ell}{7}\right) = -1 \\ & \iff \text{some prime } \ell \mid c \text{ satisfies } \ell \equiv 3, 5, \text{ or } 6 \pmod{7}. \end{aligned} \quad (1732)$$

Equivalently,

$$\text{Supp}_{7,p}(a_7) = 1 - \prod_{\substack{\ell|c \\ \ell \text{ prime}}} \frac{1 + \left(\frac{\ell}{7}\right)}{2}. \quad (1733)$$

Thus the exact support bit is the Boolean disjunction of the quadratic-nonresidue sheet flags of the distinct prime divisors of c . The three structural components are

$$\text{Supp}_{7,p}(a_7) \in \{0, 1\}, \quad U_7/Q_7 \cong C_2, \quad Q_7 \cong C_3. \quad (1734)$$

Proof. Write $p = 24k + 1$. Then

$$a_7 = 6k + 2 = 2(3k + 1), \quad c = 3k + 1,$$

which proves (1731). Also $7 \nmid c$: otherwise $p = 8c - 7$ would be divisible by 7, contrary to primality and $p \equiv 1 \pmod{24}$.

The marked factor $2 \mid a_7$ satisfies $2 \in Q_7 = \{1, 2, 4\}$, so

$$\nu_2(a_7) + \nu_4(a_7) \geq 1.$$

Apply theorem 14.122. A prime factor of c congruent to 5 or 6 (mod 7) activates (1728). A prime factor congruent to 3 (mod 7) activates (1730) together with the marked factor 2. Hence a quadratic-nonresidue prime factor of c forces occupancy.

Conversely, if every prime divisor of c lies in $Q_7 = \{1, 2, 4\}$, then the same is true of every prime divisor of $a_7 = 2c$. This is the first empty case in theorem 14.122, so $q_{7,p} = 0$. This proves (1732). For every prime $\ell \mid c$, the factor

$$\frac{1 + \left(\frac{\ell}{7}\right)}{2}$$

is 1 on Q_7 and 0 on its other coset. Its product is therefore 1 exactly when all prime factors remain on the residue sheet. Subtracting from 1 proves (1733). The group identifications in (1734) are (1653). $\square$

Exact computation 14.124 (Second-residual factor certificate). The script and captured receipt

`verification/verify_residual_seven_factor_classification.py`,

`verification/verify_residual_seven_factor_classification_output.txt`

check the classification for every $a \leq 3000$ coprime to 7, and for all 555 primes $p \equiv 1 \pmod{12}$ below 20000. Among the 267 tested primes $p \equiv 1 \pmod{24}$, the exact Boolean formula (1733) marks 201 occupied shells and 66 empty shells. The first empty pairs (p, c) are

$$(337, 43), \quad (457, 58), \quad (1009, 127), \quad (1129, 142).$$

Theorem 14.125 (Exact $R = 11$ factor-ideal classification). *Let $p \equiv 1 \pmod{12}$ be prime and put*

$$a_{11} := \frac{p + 11}{4}.$$

For $r \in \{2, 3, \dots, 10\}$, define

$$\nu_r(a_{11}) := \sum_{\substack{\ell|a_{11} \\ \ell \text{ prime} \\ \ell \equiv r \pmod{11}}} v_\ell(a_{11})$$

and form the factor-residue monomial

$$\mathbf{m}_{11}(a_{11}) := \prod_{r=2}^{10} X_r^{\nu_r(a_{11})}.$$

Let I_{11} be the monomial ideal generated by the following twenty-one monomials:

$$\mathcal{G}_{11}^{(1)} := \{X_8, X_{10}\}, \quad (1735)$$

$$\mathcal{G}_{11}^{(2)} := \{X_6X_7, X_3X_7, X_6X_9, X_7X_9, X_5X_6, \\ X_4X_6, X_4X_7, X_2X_9, X_2X_5, X_2X_4, X_2^2\}, \quad (1736)$$

$$\mathcal{G}_{11}^{(3)} := \{X_3X_6^2, X_3^2X_6, X_5^2X_7, X_2X_3^2, X_2X_7^2\}, \quad (1737)$$

$$\mathcal{G}_{11}^{(4)} := \{X_6^4, X_5X_7^3\}, \quad (1738)$$

$$\mathcal{G}_{11}^{(5)} := \{X_7^5\}, \quad (1739)$$

$$I_{11} := \left\langle \bigcup_{d=1}^5 \mathcal{G}_{11}^{(d)} \right\rangle. \quad (1740)$$

Then $a_{11} \in \mathcal{A}_p$, $4a_{11} - p = 11$, and

$$\boxed{q_{11,p} > 0 \iff \mathbf{m}_{11}(a_{11}) \in I_{11}.} \quad (1741)$$

The displayed twenty-one generators are the coordinatewise-minimal occupied factor-residue states.

Proof. The integer 2 generates U_{11} . Its nonidentity powers are

k	1	2	3	4	5	6	7	8	9
$2^k \bmod 11$	2	4	8	5	10	9	7	3	6.

Write $\lambda(r)$ for the exponent in this table. The exterior target is

$$-4^{-1} \equiv 8 \equiv 2^3 \pmod{11},$$

whereas the middle target is

$$-1 \equiv 10 \equiv 2^5 \pmod{11}.$$

Let

$$\boldsymbol{\nu} = (\nu_2, \nu_4, \nu_8, \nu_5, \nu_{10}, \nu_9, \nu_7, \nu_3, \nu_6),$$

so the coordinate order agrees with the powers $2^1, \dots, 2^9$. A divisor of a_{11}^2 can use an exponent between 0 and $2\nu_{2^k}$ in the 2^k -residue coordinate. Hence the exterior channel is occupied exactly when

$$3 \in \sum_{k=1}^9 k\{0, 1, \dots, 2\nu_{2^k}\} \pmod{10}. \quad (1742)$$

Expansion of $\mathfrak{P}_{11,a_{11}}$ in (1627) shows that the middle channel is occupied exactly when

$$5 \in \sum_{k=1}^9 k\{-\nu_{2^k}, \dots, \nu_{2^k}\} \pmod{10}. \quad (1743)$$

Prime factors congruent to 1 (mod 11) multiply the relevant coefficients by positive integers and do not alter either support set.

Both reachability conditions are increasing under coordinatewise increase of $\boldsymbol{\nu}$. Every occupied vector dominates an occupied vector whose coordinate sum is at most nine. To see this, write a target representation as a sequence in C_{10} , using the selected positive exponent contributions for (1742), and the selected signed contributions for (1743). If the sequence has at least ten terms, order ten of them and inspect their successive partial sums, together with zero. Either one partial sum is zero or two are equal. In either event a nonempty subsequence has sum zero. Removing that subsequence preserves the target and does not increase any coordinate. Iteration leaves at most nine terms.

There are

$$\sum_{s=1}^9 \binom{s+8}{8} = \binom{18}{9} - 1 = 48619$$

nonzero nine-coordinate vectors of total degree at most nine. For each such vector, finite addition in C_{10} evaluates (1742) and (1743). Removing every occupied vector that strictly dominates another occupied vector leaves exactly the twenty-one exponent vectors encoded by (1735)–(1739). The finite calculation is recorded in theorem 14.127. Therefore an arbitrary valuation vector is occupied exactly when it dominates one of those twenty-one vectors, which is precisely monomial membership in I_{11} . Finally, (1629) identifies occupancy with $q_{11,p} > 0$. $\square$

Remark 14.126 (Direct target certificates for the generators). In exponent coordinates modulo 10, the degree-one certificates are

$$3 \equiv 3, \quad 5 \equiv -5.$$

The remaining generators are certified by the following target representations:

$$\begin{aligned} 3 &\equiv 2 \cdot 7 + 9 \equiv 6 + 7 \equiv 2 \cdot 2 + 9 \equiv 1 + 2 \equiv 3 \cdot 1, \\ 5 &\equiv -7 - 8 \equiv -6 - 9 \equiv -4 + 9 \equiv -2 + 7 \equiv -1 + 6 \equiv -1 - 4, \\ 3 &\equiv 2 \cdot 8 + 3 \cdot 9 \equiv 7 \cdot 9, \\ 5 &\equiv -2 \cdot 8 - 9 \equiv -2 \cdot 4 - 7 \equiv -1 + 2 \cdot 8 \equiv -1 - 2 \cdot 7 \\ &\equiv -4 - 3 \cdot 7 \equiv -5 \cdot 7 \pmod{10}. \end{aligned}$$

Each summand uses no more than the factor valuation encoded by its generator; positive exterior coefficients may use twice that valuation. These congruences give a direct witness for every generator of I_{11} .

Exact computation 14.127 ($R = 11$ factor-ideal certificate). The script and captured receipt

```
verification/verify_residual_eleven_factor_classification.py,
verification/verify_residual_eleven_factor_classification_output.txt
```

perform the 48619-state antichain calculation, compare the resulting twenty-one generators with the stated ideal, and then compare ideal membership with the original exterior and middle divisor definitions for every $a \leq 3000$ coprime to 11 and every applicable prime $p < 20000$. It also derives the two colon ideals in theorem 14.128 and checks their prime-shell consequences over the same prime range. Finally, it checks the two explicit complement branches below for all 2728 integers $c \leq 3000$ coprime to 11 and for all fourteen applicable $R = 23$ edge primes below 20000.

Theorem 14.128 (Forced factor and colon filtration at $R = 11$). *Let $p = 4n + 1$ be prime with $p \equiv 1 \pmod{12}$, and retain $a_{11} = n + 3$ and I_{11} from theorem 14.125. Then*

$$3 \mid a_{11}.$$

For every integer e satisfying $1 \leq e \leq v_3(a_{11})$, put

$$\tilde{\mathbf{m}}_{11}^{(e)}(a_{11}) := \frac{\mathbf{m}_{11}(a_{11})}{X_3^e}, \quad J_{11}^{(e)} := (I_{11} : X_3^e).$$

Then

$$q_{11,p} > 0 \iff \tilde{\mathbf{m}}_{11}^{(e)}(a_{11}) \in J_{11}^{(e)}. \quad (1744)$$

The first colon ideal has the irredundant presentation

$$\begin{aligned} J_{11}^{(1)} = \langle X_7, X_8, X_{10}, X_6 X_9, X_6^2, X_5 X_6, X_4 X_6, X_3 X_6, \\ X_2 X_9, X_2 X_5, X_2 X_4, X_2 X_3, X_2^2 \rangle. \end{aligned} \quad (1745)$$

For every $e \geq 2$,

$$J_{11}^{(e)} = \langle X_2, X_6, X_7, X_8, X_{10} \rangle. \quad (1746)$$

Moreover,

$$v_3(a_{11}) \geq 2 \iff p \equiv 25 \pmod{36}. \quad (1747)$$

In this residue class define

$$b_{11} := \frac{a_{11}}{9} = \frac{p+11}{36}.$$

Then the empty-shell condition has the exact factor-class form

$$q_{11,p} = 0 \iff \ell \bmod 11 \in \{1, 3, 4, 5, 9\} \text{ for every prime } \ell \mid b_{11}. \quad (1748)$$

Proof. The congruence $p = 4n + 1 \equiv 1 \pmod{12}$ gives $n \equiv 0 \pmod{3}$, hence $3 \mid n + 3 = a_{11}$. The prime 3 has residue 3 $\pmod{11}$, so removing e copies of the actual prime factor 3 divides the factor-residue monomial by X_3^e .

For a monomial M divisible by X_3^e , the definition of a colon ideal gives

$$M \in I_{11} \iff \frac{M}{X_3^e} \in (I_{11} : X_3^e).$$

Combining this identity with (1741) proves (1744).

For a monomial ideal generated by monomials G , the colon by X_3^e is generated by

$$\left\{ \frac{G}{\gcd(G, X_3^e)} \right\}.$$

Apply this rule to the twenty-one monomials in (1735)–(1739), and remove a monomial whenever it is divisible by another member of the resulting list. The thirteen remaining monomials are exactly those in (1745). Removing two copies of X_3 leaves the five divisibility-minimal monomials

$$X_2, X_6, X_7, X_8, X_{10}.$$

No minimal generator of I_{11} contains a power of X_3 greater than two. The colon ideal therefore remains equal to this five-generator ideal for every $e \geq 2$.

Since $a_{11} = n + 3$, divisibility by 9 is equivalent to $n \equiv 6 \pmod{9}$, and hence to

$$p = 4n + 1 \equiv 25 \pmod{36}.$$

When this holds,

$$\mathfrak{m}_{11}(b_{11}) = \frac{\mathfrak{m}_{11}(a_{11})}{X_3^2}.$$

Membership in the ideal $\langle X_2, X_6, X_7, X_8, X_{10} \rangle$ means precisely that b_{11} has a prime factor in one of the residue classes 2, 6, 7, 8, 10 $\pmod{11}$. Its negation is (1748). $\square$

Corollary 14.129 (Exact complement of the first $R = 11$ colon ideal). *Let c be a positive integer coprime to 11. For $r \in U_{11} \setminus \{1\}$, let*

$$\nu_r(c) := \sum_{\substack{\ell^e \parallel c \\ \ell \equiv r \pmod{11}}} e.$$

Then

$$\mathfrak{m}_{11}(c) \notin J_{11}^{(1)} \quad (1749)$$

if and only if exactly one of the following two branches holds:

(A) Every prime divisor of c lies in

$$\{1, 3, 4, 5, 9\} \pmod{11}.$$

(B) *One has*

$$c = vq_2^{\epsilon_2}q_6^{\epsilon_6}, \quad \epsilon_2, \epsilon_6 \in \{0, 1\}, \quad 1 \leq \epsilon_2 + \epsilon_6 \leq 2, \quad (1750)$$

where every prime divisor of v is $1 \pmod{11}$, and, whenever $\epsilon_j = 1$, the number q_j is prime and

$$q_j \equiv j \pmod{11}.$$

A factor whose exponent is zero is omitted.

Proof. By (1745), nonmembership first forces

$$\nu_7(c) = \nu_8(c) = \nu_{10}(c) = 0.$$

The generators X_2^2 and X_6^2 give

$$\nu_2(c) \leq 1, \quad \nu_6(c) \leq 1.$$

The remaining eight quadratic generators are

$$X_jX_k, \quad j \in \{2, 6\}, \quad k \in \{3, 4, 5, 9\}.$$

Consequently, if $\nu_2(c) = \nu_6(c) = 0$, the remaining prime-factor classes are precisely those in branch (A). If either $\nu_2(c)$ or $\nu_6(c)$ is positive, all four valuations $\nu_3(c), \nu_4(c), \nu_5(c), \nu_9(c)$ vanish, and the two surviving valuations $\nu_2(c), \nu_6(c)$ are at most one. This is exactly the factorization in branch (B). Conversely, neither branch is divisible by a generator of $J_{11}^{(1)}$, so both imply (1749). The two branches are disjoint because branch (A) has $\nu_2(c) = \nu_6(c) = 0$, whereas branch (B) does not. $\square$

Theorem 14.130 (Coprime $R = 7/R = 11$ hard-locus coordinates). *Let $p \equiv 1 \pmod{24}$ be prime. Write $p = 24k + 1$, and define*

$$c_7 := \frac{p+7}{8} = 3k+1, \quad d_{11} := \frac{p+11}{12} = 2k+1. \quad (1751)$$

Then

$$p = 8c_7 - 7 = 12d_{11} - 11, \quad 3d_{11} - 2c_7 = 1, \quad \gcd(c_7, d_{11}) = 1, \quad (1752)$$

and the two shell denominators are

$$a_7 = 2c_7, \quad a_{11} = 3d_{11}. \quad (1753)$$

The two shells are simultaneously empty if and only if both of the following conditions hold:

(i) *every prime divisor of c_7 lies in*

$$\{1, 2, 4\} \pmod{7};$$

(ii)

$$\mathbf{m}_{11}(d_{11}) \notin J_{11}^{(1)}.$$

Equivalently, condition (ii) says that exactly one of the following two disjoint alternatives holds:

(A) *every prime divisor of d_{11} lies in*

$$\{1, 3, 4, 5, 9\} \pmod{11};$$

(B)

$$d_{11} = vq_2^{\epsilon_2}q_6^{\epsilon_6}, \quad \epsilon_2, \epsilon_6 \in \{0, 1\}, \quad 1 \leq \epsilon_2 + \epsilon_6 \leq 2,$$

where every prime divisor of v is $1 \pmod{11}$, and each present q_j is prime with $q_j \equiv j \pmod{11}$.

Proof. The identities in (1751) follow from $p = 24k + 1$. They give

$$8c_7 - 7 = 24k + 1 = 12d_{11} - 11$$

and

$$3d_{11} - 2c_7 = 3(2k + 1) - 2(3k + 1) = 1.$$

Every common divisor of c_7 and d_{11} therefore divides 1, which proves the coprimality assertion. Direct substitution in $a_R = (p + R)/4$ gives (1753).

By theorem 14.123, the residual-7 shell is empty exactly when every prime divisor of c_7 lies in $\{1, 2, 4\} \pmod{7}$. Apply theorem 14.128 with $e = 1$. Since

$$\frac{\mathbf{m}_{11}(a_{11})}{X_3} = \mathbf{m}_{11}(d_{11}),$$

the residual-11 shell is empty exactly when $\mathbf{m}_{11}(d_{11}) \notin J_{11}^{(1)}$. The two explicit alternatives are precisely theorem 14.129. $\square$

Exact computation 14.131 (Hard-locus $R = 7/R = 11$ certificate). The script and captured receipt

```
verification/verify_hard_locus_R7_R11_pair.py,
verification/verify_hard_locus_R7_R11_pair_output.txt,
verification/HardLocusR7R11CoprimeCertificate.lean,
verification/HardLocusR7R11CoprimeCertificate_output.txt
```

compare the two factor criteria with the original exterior and middle divisor definitions for every prime $p \equiv 1 \pmod{24}$ below 20000. The Lean source independently checks the two reduced coordinate identities, the forced-factor identities, the Bézout relation, and coprimality for every $k \in \mathbb{N}$. There are 267 primes in the finite range. Exactly 28 leave both shells empty. The first twelve triples (p, c_7, d_{11}) are

$$\begin{aligned} &(1201, 151, 101), (2137, 268, 179), (2377, 298, 199), \\ &(2521, 316, 211), (2689, 337, 225), (3361, 421, 281), \\ &(3697, 463, 309), (4057, 508, 339), (4657, 583, 389), \\ &(5737, 718, 479), (7177, 898, 599), (7417, 928, 619). \end{aligned}$$

Corollary 14.132 (Next three affine coordinates on the hard locus). *Retain the notation and conditions of theorem 14.130. Then*

$$a_{15} = \frac{p + 15}{4} = 6k + 4 = 2(c_7 + 1), \tag{1754}$$

$$a_{19} = \frac{p + 19}{4} = 6k + 5 = 2c_7 + 3 = 3d_{11} + 2, \tag{1755}$$

$$b_{23} := \frac{p + 23}{12} = 2k + 2 = d_{11} + 1, \quad a_{23} = 3b_{23} = 6k + 6. \tag{1756}$$

In particular, a_{15}, a_{19}, a_{23} are three consecutive integers.

Proof. Insert $p = 24k + 1$, $c_7 = 3k + 1$, and $d_{11} = 2k + 1$ into the four displayed quotients. $\square$

Theorem 14.133 (Resolution of the first $R = 7/R = 11$ hard point). *For the first prime in the finite hard-locus list,*

$$p = 1201, \quad (k, c_7, d_{11}) = (50, 151, 101),$$

the next three affine shells are

$$(a_{15}, a_{19}, b_{23}, a_{23}) = (304, 305, 102, 306).$$

Their exact occupation numbers are

$$q_{15,1201} = 0, \quad q_{19,1201} = 0, \quad q_{23,1201} = 3. \quad (1757)$$

The residual-23 exterior and ordered middle witnesses are

$$\mathcal{E}_{23,306} = \{17, 2754\}, \quad \mathcal{M}_{23,306} = \{(6, 17), (17, 6)\}. \quad (1758)$$

The exterior witness $u = 17$ reconstructs

$$\boxed{\frac{4}{1201} = \frac{1}{306} + \frac{1}{1082101} + \frac{1}{16218}}. \quad (1759)$$

Proof. The three relevant factorizations are

$$304 = 2^4 \cdot 19, \quad 305 = 5 \cdot 61, \quad 102 = 2 \cdot 3 \cdot 17.$$

Modulo 15, the first factorization occupies only classes 2 and 4. Both alternatives in (2070) therefore fail, so $q_{15,1201} = 0$.

For $R = 19$, use the primitive logarithm with $\lambda_{19}(2) = 1$. Since

$$\lambda_{19}(61) = \lambda_{19}(4) = 2, \quad \lambda_{19}(5) = 16,$$

the log multiset is $(2, 16)$. Both of its reachability sets are

$$\Sigma_E(2, 16) = \Sigma_M(2, 16) = \{0, 2, 4, 14, 16\}.$$

This set contains neither the exterior target 7 nor the middle target 9, proving $q_{19,1201} = 0$.

For $R = 23$, use $\lambda_{23}(5) = 1$. The reduced factor $b_{23} = 102$ has log multiset

$$(\lambda_{23}(2), \lambda_{23}(17), \lambda_{23}(3)) = (2, 7, 16).$$

The singleton (7) is a degree-one generator of $J_{23}^{(3)}$ by (2093), so the shell is occupied. Direct enumeration of the divisors of

$$306 = 2 \cdot 3^2 \cdot 17$$

gives exactly the two sets in (1758). Hence

$$q_{23,1201} = |\mathcal{E}_{23,306}| + \frac{1}{2}|\mathcal{M}_{23,306}| = 2 + 1 = 3.$$

Finally,

$$23 \mid 4 \cdot 17 + 1, \quad 306 + 1201 \cdot 17 = 23 \cdot 901.$$

Apply theorem 14.241 with $u = 17$. Its two reconstructed denominators are

$$1201 \cdot 901 = 1082101, \quad \frac{306 \cdot 901}{17} = 16218,$$

which proves (1759). □

Exact computation 14.134 (First hard-locus continuation certificate). The script and captured receipts

```
verification/verify_first_hard_locus_R15_R19_R23.py,
verification/verify_first_hard_locus_R15_R19_R23_output.txt,
verification/FirstHardLocusR15R19R23Certificate.lean,
verification/FirstHardLocusR15R19R23Certificate_output.txt
```

verify the affine identities, the three support-ideal evaluations, the complete exterior and middle witness sets, and (1759). The Lean source independently checks the general affine formulas, the exterior divisibility data, and the unit-fraction identity.

The same Python calculation compares all three ideal tests with the original divisor definitions on the 28 residual-7/residual-11 double survivors below 20000. Their occupation-pattern counts are

$(q_{15} > 0, q_{19} > 0, q_{23} > 0)$	$(0, 0, 0)$	$(0, 0, 1)$	$(0, 1, 0)$	$(0, 1, 1)$	$(1, 0, 1)$	$(1, 1, 0)$	$(1, 1, 1)$
count	2	3	3	2	15	1	2.

Thus 26 of the 28 points are occupied by at least one of the next three shells in this finite range. The two points empty through $R = 23$ are

$$p = 3361, \quad p = 8089.$$

Continuing the same exact replay through $R = 39$ gives

p	preceding empty shells	first occupied shell
3361	$R = 27, 31, 35$	$R = 39, q_{39} = 3,$
8089	$R = 27$	$R = 31, q_{31} = 4.$

The extended Lean source independently proves the two exterior reconstruction chains and the resulting unit-fraction identities. Hence all 28 finite hard-locus points are occupied by a shell $R \leq 39$.

Theorem 14.135 (Exact resolution of the finite hard-locus tail). *For the two primes left empty by the five shells $R = 7, 11, 15, 19, 23$, the subsequent shell data are as follows.*

(i) For $p = 3361$,

$$q_{27,3361} = q_{31,3361} = q_{35,3361} = 0, \quad q_{39,3361} = 3.$$

At the occupied shell $a_{39} = 850$,

$$\mathcal{E}_{39,850} = \{68, 42500\}, \quad \mathcal{M}_{39,850} = \{(5, 34), (34, 5)\}.$$

The exterior witness $u = 68$ gives

$$\frac{4}{3361} = \frac{1}{850} + \frac{1}{19769402} + \frac{1}{73525}. \quad (1760)$$

(ii) For $p = 8089$,

$$q_{27,8089} = 0, \quad q_{31,8089} = 4.$$

At the occupied shell $a_{31} = 2030$,

$$\mathcal{E}_{31,2030} = \{116, 1015\},$$

$$\mathcal{M}_{31,2030} = \{(2, 29), (29, 2), (35, 58), (58, 35)\}.$$

The exterior witness $u = 116$ gives

$$\frac{4}{8089} = \frac{1}{2030} + \frac{1}{245371726} + \frac{1}{530845}. \quad (1761)$$

Consequently, every residual-7/residual-11 double survivor below 20000 is occupied by at least one shell $R \leq 39$.

Proof. The complete factorisations needed for the empty shells are

$$\begin{array}{l|l} p = 3361 & a_{27} = 847 = 7 \cdot 11^2 \quad a_{31} = 848 = 2^4 \cdot 53 \quad a_{35} = 849 = 3 \cdot 283, \\ p = 8089 & a_{27} = 2029 \end{array}$$

with 2029 prime. Substitution of their complete divisor lattices into the exterior and middle definitions gives

$$\mathcal{E}_{27,847} = \mathcal{M}_{27,847} = \emptyset,$$

$$\mathcal{E}_{31,848} = \mathcal{M}_{31,848} = \emptyset,$$

$$\mathcal{E}_{35,849} = \mathcal{M}_{35,849} = \emptyset,$$

$$\mathcal{E}_{27,2029} = \mathcal{M}_{27,2029} = \emptyset.$$

For the occupied shells,

$$850 = 2 \cdot 5^2 \cdot 17, \quad 2030 = 2 \cdot 5 \cdot 7 \cdot 29,$$

and the same exhaustive divisor calculation gives exactly the displayed sets. Therefore

$$q_{39,3361} = 2 + \frac{1}{2} \cdot 2 = 3, \quad q_{31,8089} = 2 + \frac{1}{2} \cdot 4 = 4.$$

For $p = 3361$, the exterior reconstruction is

$$\begin{aligned} 4 \cdot 68 + 1 &= 39 \cdot 7, & 850 + 3361 \cdot 68 &= 39 \cdot 5882, \\ 3361 \cdot 5882 &= 19769402, & \frac{850 \cdot 5882}{68} &= 73525, \end{aligned}$$

which proves (1760). For $p = 8089$,

$$\begin{aligned} 4 \cdot 116 + 1 &= 31 \cdot 15, & 2030 + 8089 \cdot 116 &= 31 \cdot 30334, \\ 8089 \cdot 30334 &= 245371726, & \frac{2030 \cdot 30334}{116} &= 530845, \end{aligned}$$

which proves (1761). The finite conclusion then follows from theorem 14.134. $\square$

Theorem 14.136 (The first escape through $R = 39$ and its $R = 59$ exit). *The least prime $p \equiv 1 \pmod{24}$ for which*

$$q_{R,p} = 0 \quad (R = 3, 7, \dots, 39)$$

is

$$p = 118801 = 24 \cdot 4950 + 1.$$

For this prime, all fourteen shells

$$R = 3, 7, \dots, 55$$

are empty. Their affine coordinates, together with the first occupied coordinate, are the consecutive integers

$$a_{4j-1} = 29700 + j \quad (1 \leq j \leq 15).$$

The complete factorisations are

R	a_R	factorisation	R	a_R	factorisation
3	29701	$7 \cdot 4243$	7	29702	$2 \cdot 14851$
11	29703	$3 \cdot 9901$	15	29704	$2^3 \cdot 47 \cdot 79$
19	29705	$5 \cdot 13 \cdot 457$	23	29706	$2 \cdot 3 \cdot 4951$
27	29707	$61 \cdot 487$	31	29708	$2^2 \cdot 7 \cdot 1061$
35	29709	$3^2 \cdot 3301$	39	29710	$2 \cdot 5 \cdot 2971$
43	29711	$11 \cdot 37 \cdot 73$	47	29712	$2^4 \cdot 3 \cdot 619$
51	29713	$43 \cdot 691$	55	29714	$2 \cdot 83 \cdot 179$
59	29715	$3 \cdot 5 \cdot 7 \cdot 283$			

At the $R = 59$ shell,

$$\mathcal{E}_{59,29715} = \{21225\}, \quad \mathcal{M}_{59,29715} = \{(1, 1415), (1415, 1)\},$$

and therefore

$$q_{59,118801} = 1 + \frac{1}{2} \cdot 2 = 2.$$

The exterior witness $u = 21225$ gives

$$\frac{4}{118801} = \frac{1}{29715} + \frac{1}{5077395546660} + \frac{1}{59834124}. \quad (1762)$$

Proof. The affine identity is the specialization of

$$a_{4j-1} = 6k + j$$

at $k = 4950$. Trial division through the square root proves that 118801 is prime; the independent Lean certificate proves the same statement. The displayed table is the complete prime factorisation of each affine coordinate. Substitution of the resulting complete divisor lattices into the exterior and middle definitions gives

$$\mathcal{E}_{R,a_R} = \mathcal{M}_{R,a_R} = \emptyset \quad (R = 3, 7, \dots, 55),$$

and gives exactly the two displayed sets at $R = 59$. This proves all shell counts.

For the exterior reconstruction,

$$\begin{aligned} 21225 \mid 29715^2, \quad 4 \cdot 21225 + 1 &= 59 \cdot 1439, \\ 29715 + 118801 \cdot 21225 &= 59 \cdot 42738660. \end{aligned}$$

The remaining denominators are

$$118801 \cdot 42738660 = 5077395546660, \quad \frac{29715 \cdot 42738660}{21225} = 59834124,$$

which proves (1762).

Finally, exhaustive evaluation of the original exterior and middle definitions on all 1372 primes $p \equiv 1 \pmod{24}$ with $p \leq 118801$ finds

$$\{p : q_{R,p} = 0 \text{ for } R = 3, 7, \dots, 39\} = \{118801\}.$$

This proves the least-prime assertion. $\square$

Exact computation 14.137 (First $R = 39$ escape and $R = 59$ exit certificate). The independent sources and captured receipts

```
verification/verify_first_escape_through_R39.py,
verification/verify_first_escape_through_R39_output.txt,
verification/FirstEscapeThroughR39Certificate.lean,
verification/FirstEscapeThroughR39Certificate_output.txt
```

check the least-prime enumeration, all fifteen coordinate factorisations, every empty exterior and middle set through $R = 55$, the complete $R = 59$ witness sets, the shell count, primality, the exterior reconstruction chain, and (1762). The Python replay evaluates the original divisor definitions; the Lean source proves the affine formula, primality, the exterior and middle arithmetic, and the rational identity without a proof placeholder or added axiom.

Theorem 14.138 (Target transposition at the first $R = 39$ escape). *For reduced shell data, retain the reachability sets and targets of theorem 14.80, and define the target-incidence matrix*

$$\Xi_{R,a} := \begin{pmatrix} \mathbf{1}_{\{t_R \in \mathcal{R}_R^E(\nu_R(a))\}} & \mathbf{1}_{\{s_R \in \mathcal{R}_R^E(\nu_R(a))\}} \\ \mathbf{1}_{\{t_R \in \mathcal{R}_R^M(\nu_R(a))\}} & \mathbf{1}_{\{s_R \in \mathcal{R}_R^M(\nu_R(a))\}} \end{pmatrix}. \quad (1763)$$

The diagonal entries are the exterior and middle occupancy indicators:

$$q_{R,p} > 0 \iff (\Xi_{R,a})_{11} + (\Xi_{R,a})_{22} > 0. \quad (1764)$$

At the last empty shell of theorem 14.136,

$$(p, R, a) = (118801, 39, 29710), \quad a = 2 \cdot 5 \cdot 2971,$$

the factor residues are $(2, 5, 7)$, the two targets are

$$t_{39} = 29, \quad s_{39} = 38,$$

and the complete reachability sets are

$$\mathcal{R}_{39}^E(\boldsymbol{\nu}_{39}(29710)) = \{1, 2, 4, 5, 7, 10, 11, 14, 16, 19, 20, 22, 23, 25, 28, 31, 32, 35, 37, 38\}, \quad (1765)$$

$$\mathcal{R}_{39}^M(\boldsymbol{\nu}_{39}(29710)) = \{1, 2, 4, 5, 7, 8, 10, 14, 16, 17, 19, 20, 22, 23, 28, 29, 31, 34, 35, 37\}. \quad (1766)$$

Consequently

$$\Xi_{39,29710} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = S. \quad (1767)$$

The two off-diagonal entries have the explicit exponent witnesses

$$2 \cdot 5^2 \cdot 7 \equiv 38 \pmod{39}, \quad 5^{-1}7^{-1} \equiv 29 \pmod{39}. \quad (1768)$$

On the integer divisor lattice, these are represented by

$$u_{\times} = 2 \cdot 5^2 \cdot 2971 = 148550 \equiv -1 \pmod{39}$$

and by the signed ratio

$$\frac{x_{\times}}{y_{\times}} = \frac{1}{5 \cdot 2971}, \quad y_{\times} \equiv 35 \pmod{39}, \quad y_{\times}^{-1} \equiv 29 \pmod{39}.$$

The second pair has

$$x_{\times} + y_{\times} = 14856 \equiv 36 \pmod{39},$$

so it reaches the exterior target in the middle exponent alphabet without satisfying the middle gluing equation.

Let H and J be the exact matrices in (155). Then

$$H\Xi_{39,29710}H^{\top} = HSH^{\top} = J. \quad (1769)$$

The established endpoint-to-blade bridge assigns this empty shell the pair (P_-, P_-) . Equation (1769) records its crossed pre-occupancy incidence.

At the first occupied shell

$$(p, R, a) = (118801, 59, 29715),$$

the factor residues are $(3, 5, 7, 47)$ and

$$\Xi_{59,29715} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (1770)$$

The diagonal entries are witnessed by

$$21225 \equiv 44 = t_{59} \pmod{59}, \quad 1415 \equiv 58 = s_{59} \pmod{59}, \quad 1 + 1415 = 24 \cdot 59.$$

Thus the passage from $R = 39$ to $R = 59$ changes the diagonal of Ξ from $(0, 0)$ to $(1, 1)$, exactly where the shell rank changes from zero to two. At the occupied comparison shell

$$(p, R, a) = (3361, 39, 850),$$

one also has $\Xi_{39,850} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$; hence the transposition matrix in (1767) depends on the factor-residue datum and not on the modulus alone.

Proof. For the $R = 39$ shell, each of the three nontrivial factor residues occurs once. Therefore the exterior exponents range independently over $\{0, 1, 2\}$, while the middle exponents range independently over $\{-1, 0, 1\}$. Exhausting the resulting 27 exponent vectors gives (1765) and (1766). Since

$$4 \cdot 29 + 1 = 3 \cdot 39, \quad 38 \equiv -1 \pmod{39},$$

the two lists give (1767). The exponent vectors $(1, 2, 1)$ and $(0, -1, -1)$ give (1768); the displayed integer representatives verify their lifts to the divisor lattice.

Equation (1764) is precisely the two-target criterion in theorem 14.80. The matrix identity in (1769) is (156). The empty-shell projector pair follows from (1591).

At $R = 59$, exhaustion of the $3^4 = 81$ exterior and middle exponent vectors gives 54 residue classes in each reachability set and gives (1770). The displayed congruences are the correct exterior and middle witnesses. The same finite exponent calculation for $850 = 2 \cdot 5^2 \cdot 17$ gives the final comparison. $\square$

Corollary 14.139 (Identity-support increment from $R = 39$ to $R = 59$). *For a target-incidence matrix, write*

$$D_{R,a} := \text{diag}((\Xi_{R,a})_{11}, (\Xi_{R,a})_{22}), \quad C_{R,a} := \Xi_{R,a} - D_{R,a}.$$

For the two shells of theorem 14.138,

$$(D_{39,29710}, C_{39,29710}) = (0, S), \quad (D_{59,29715}, C_{59,29715}) = (I_2, S). \quad (1771)$$

Equivalently,

$$\Xi_{59,29715} - \Xi_{39,29710} = I_2. \quad (1772)$$

Hadamard transport gives

$$\begin{aligned} HD_{39,29710}H^\top &= 0, & HC_{39,29710}H^\top &= J, \\ HD_{59,29715}H^\top &= I_2, & HC_{59,29715}H^\top &= J, \end{aligned} \quad (1773)$$

and hence

$$H\Xi_{39,29710}H^\top = J, \quad H\Xi_{59,29715}H^\top = I_2 + J = 2E_{11}.$$

For every shell,

$$\text{tr } \Xi_{R,a} = \mathbf{1}_{\{\mathcal{E}_{R,a} \neq \emptyset\}} + \mathbf{1}_{\{\mathcal{M}_{R,a} \neq \emptyset\}}, \quad q_{R,p} > 0 \iff \text{tr } \Xi_{R,a} > 0. \quad (1774)$$

Thus $D_{R,a}$ is the binary occupied-channel carrier. At the empty $R = 39$ shell,

$$q_{39,118801} = 0, \quad \|C_{39,29710}\|_{\text{HS}}^2 = \|S\|_{\text{HS}}^2 = 2.$$

The shell rank and its determinant-volume carrier vanish there, while the crossed pre-occupancy grading remains nonzero.

Proof. Equations (1767) and (1770) give

$$\Xi_{39,29710} = S, \quad \Xi_{59,29715} = I_2 + S.$$

Taking diagonal and off-diagonal parts proves (1771) and (1772). Conjugation by H , using $HI_2H^\top = I_2$ and (156), proves (1773).

The two diagonal entries of $\Xi_{R,a}$ are the two target conditions in theorem 14.80; this proves (1774). Finally, $\text{tr}(S^\top S) = 2$. $\square$

Theorem 14.140 (The first hidden blade and the mod-19 Fable quarter). *For the fifteen shells*

$$R = 3, 7, \dots, 59$$

of $p = 118801$, the complete target-incidence trajectory is

$$\Xi_{R,a_R} = 0 \quad (R \in \{3, 7, 11, 15, 23, 27, 31, 35, 43, 47, 51, 55\}), \quad (1775)$$

$$\Xi_{19,29705} = E_{21}, \quad \Xi_{39,29710} = E_{21} + E_{12} = S, \quad \Xi_{59,29715} = I_2 + S. \quad (1776)$$

Hence the first nonzero incidence matrix in increasing residual order is the raw ordered blade E_{21} . Under the exact Hadamard intertwiner,

$$H\Xi_{19,29705}H^\top = HE_{21}H^\top = B_{-+}. \quad (1777)$$

The $R = 19$ affine coordinate factors as

$$a_{19} = 29705 = 5 \cdot 13 \cdot 457,$$

with factor residues $(1, 5, 13)$. Its complete reachability sets are

$$\mathcal{R}_{19}^E(\nu_{19}(29705)) = \{1, 2, 5, 6, 7, 8, 9, 13, 17\}, \quad (1778)$$

$$\mathcal{R}_{19}^M(\nu_{19}(29705)) = \{1, 3, 4, 5, 8, 12, 13, 14, 15\}. \quad (1779)$$

The two targets are

$$t_{19} = 14, \quad s_{19} = 18,$$

so only the crossed entry

$$t_{19} \in \mathcal{R}_{19}^M$$

occurs. A primitive factor-pair witness is

$$\frac{13}{5} \equiv 14 \equiv -\frac{1}{4} \pmod{19}, \quad 4 \cdot 13 + 5 = 3 \cdot 19, \quad 13 + 5 + 1 = 19. \quad (1780)$$

Let

$$\rho_{19} : \mathbb{Z}[1/2] \longrightarrow \mathbb{F}_{19} \quad (1781)$$

be the localization morphism induced by reduction modulo 19. The map F of theorem 4.1 has integral coefficients, so it base changes to a polynomial map

$$F_{19} : \mathbb{F}_{19}^3 \longrightarrow \mathbb{F}_{19}^3$$

and satisfies

$$F_{19} \circ \rho_{19}^3 = \rho_{19}^3 \circ F \quad \text{on } \mathbb{Z}[1/2]^3. \quad (1782)$$

The three marked inputs and their common image reduce to

$$\rho_{19}^3(p_0) = (0, 0, 14), \quad \rho_{19}^3(p_+) = (1, 8, 16), \quad (1783)$$

$$\rho_{19}^3(p_-) = (18, 11, 16), \quad (1784)$$

$$F_{19}(0, 0, 14) = F_{19}(1, 8, 16) = F_{19}(18, 11, 16) = (14, 0, 0). \quad (1785)$$

Consequently the marked Fable target and the first hidden arithmetic blade meet under explicit morphisms at the same element:

$$\text{pr}_1 \rho_{19}^3 F(p_0) = \rho_{19} \left(-\frac{1}{4} \right) = 14 = 13 \cdot 5^{-1} \quad \text{in } \mathbb{F}_{19}. \quad (1786)$$

Proof. The exact factorisations in theorem 14.136 determine each factor-residue multiset. Exhaustion of the exterior alphabet $\{0, 1, 2\}$ and the middle alphabet $\{-1, 0, 1\}$ gives (1775) and (1776). The $R = 19$ exhaustion gives (1778) and (1779). Since

$$4 \cdot 14 + 1 = 3 \cdot 19, \quad 18 \equiv -1 \pmod{19},$$

the two lists give $\Xi_{19,29705} = E_{21}$. Equation (1777) is (159). Direct integer arithmetic proves (1780).

Because 2 is invertible modulo 19, reduction modulo 19 extends uniquely from $\mathbb{Z}$ to the localization $\mathbb{Z}[1/2]$, proving (1781). Polynomial evaluation commutes with every ring morphism, which proves (1782). Applying this identity to the three equalities in (16) gives the common image in (1785); direct reduction of the three coordinates gives (1784). Finally, $\rho_{19}(-1/4) = 14$, and (1780) proves (1786). $\square$

Exact computation 14.141 (First-escape target-transposition certificate). The independent sources and captured receipts

```
verification/verify_first_escape_target_transposition.py,
verification/verify_first_escape_target_transposition_output.txt,
verification/FirstEscapeTargetTranspositionCertificate.lean,
verification/FirstEscapeTargetTranspositionCertificate_output.txt
```

check all fifteen affine factorisations, both exponent alphabets and every incidence matrix through $R = 59$, the complete $R = 19$ and $R = 39$ reachability sets, the crossed witnesses, the integral

forms of the E_{21} and S Hadamard identities, the $R = 59$ correction, the $p = 3361$ comparison, and all three reduced Fable collisions. The Lean source independently proves the displayed integer congruences, the finite-field factor ratio, and the three polynomial evaluations.

Theorem 14.142 (Bounded hidden-incidence census and unique maximal delay). *Let*

$$\mathcal{P}_{10^6} := \{p \leq 10^6 : p \text{ is prime and } p \equiv 1 \pmod{24}\}, \quad \mathcal{R} := \{3, 7, \dots, 107\},$$

and put $a_{p,R} := (p + R)/4$. For $p \in \mathcal{P}_{10^6}$, define

$$R_{\text{occ}}(p) := \min\{R \in \mathcal{R} : \text{tr } \Xi_{R,a_{p,R}} > 0\}.$$

When the set

$$\mathcal{H}_p := \{R < R_{\text{occ}}(p) : \Xi_{R,a_{p,R}} \neq 0\}$$

is nonempty, put $R_{\text{hid}}(p) := \min \mathcal{H}_p$.

There are 9732 primes in $\mathcal{P}_{10^6}$. The first occupied residual exists for every one of them and satisfies

$$R_{\text{occ}}(p) \leq 59.$$

The shells preceding first occupation comprise 6463 empty shells. Their incidence matrices have the exact multiplicities

$$\begin{array}{c|cccc} \Xi & 0 & E_{12} & E_{21} & S \\ \hline \# & 6381 & 3 & 40 & 39. \end{array} \quad (1787)$$

Exactly 75 primes have $\mathcal{H}_p \neq \emptyset$. Their first hidden matrices have multiplicities

$$\begin{array}{c|cccc} \Xi_{R_{\text{hid}},a} & E_{12} & E_{21} & S \\ \hline \# & 2 & 38 & 35, \end{array} \quad (1788)$$

and the delay

$$g(p) := R_{\text{occ}}(p) - R_{\text{hid}}(p)$$

has distribution

$$\begin{array}{c|cccccccc} g & 4 & 8 & 12 & 16 & 20 & 28 & 36 & 40 \\ \hline \# & 27 & 23 & 15 & 5 & 2 & 1 & 1 & 1. \end{array} \quad (1789)$$

The unique case with $g(p) = 40$ is

$$(p, R_{\text{hid}}, R_{\text{occ}}) = (118801, 19, 59).$$

Precisely seven primes have two nonzero hidden matrices before first occupation. Their complete nonzero hidden paths are

$$\begin{array}{c|cc} p & (R, \Xi_{R,a_{p,R}}) & R_{\text{occ}} \\ \hline 2521 & (11, E_{21}), (19, E_{21}) & 23 \\ 28921 & (11, E_{21}), (19, S) & 23 \\ 104161 & (11, E_{21}), (19, S) & 23 \\ 118801 & (19, E_{21}), (39, S) & 59 \\ 214561 & (11, S), (19, E_{21}) & 23 \\ 246241 & (11, S), (27, S) & 39 \\ 415969 & (23, S), (27, E_{12}) & 31. \end{array} \quad (1790)$$

In all seven rows the first occupied incidence matrix is $I_2 + S$.

For a hidden shell whose lower-left entry is 1, define its primitive quarter-pair set

$$\mathcal{Q}_{p,R} := \left\{ (u, v) \in \mathbb{N}^2 : \gcd(u, v) = 1, \quad uv \mid a_{p,R}, \quad uv^{-1} \equiv -\frac{1}{4} \pmod{R} \right\}.$$

Across every such pair before first occupation, the strict inequality

$$R - u - v > 0$$

holds exactly twice:

p	R	u	v	$(4u + v)/R$	$R - u - v$	R_{occ}
118801	19	13	5	3	1	59
359041	19	13	5	3	1	23.

(1791)

No primitive quarter pair in the census has $R - u - v = 0$.

Proof. For each prime and residual, complete factorisation of $a_{p,R}$ gives the factor-residue multiset. Iterated multiplication by the three local values associated with the exterior alphabet $\{0, 1, 2\}$ and the middle alphabet $\{-1, 0, 1\}$ gives every entry of $\Xi_{R,a_{p,R}}$. The diagonal criterion in theorem 14.138 identifies the first occupied residual.

An independent replay constructs every divisor of $a_{p,R}^2$. Its exterior residues and its relative middle residues

$$\{d \bmod R : d \mid a_{p,R}^2\}, \quad \{da_{p,R}^{-1} \bmod R : d \mid a_{p,R}^2\}$$

reproduce all 16195 matrices through first occupation. Counting the resulting finite lists gives (1787)–(1790).

For each nonzero lower-left entry, complete enumeration of coprime pairs u, v with $uv \mid a_{p,R}$ gives $\mathcal{Q}_{p,R}$. Testing the integer

$$R - u - v$$

over the full list gives exactly the two rows in (1791) and no zero case. □

Exact computation 14.143 (Bounded hidden-incidence census certificate). The independent checker and stored receipt

```
verification/scan_hidden_target_incidence.py,
verification/scan_hidden_target_incidence_output.txt,
verification/HiddenIncidenceMatrixStructure.lean,
verification/HiddenIncidenceMatrixStructure_output.txt
```

enumerate the 9732 primes, factor every affine shell coordinate, compute all target-incidence matrices through $R = 107$, replay every prefix matrix from the complete divisor lattice, enumerate every primitive quarter pair, and verify the stated counts, paths, delays, and surplus assertions. The Lean source proves that a Boolean 2×2 matrix with zero diagonal is exactly one of $0, E_{12}, E_{21}, S$, together with the two integer differences used by the unique maximal-delay and one-surplus conclusions. Both sources compile, and a fresh Python run agrees exactly with the stored receipt.

Theorem 14.144 (Prime-cofactor $R = 19$ incidence table and quadratic-coset $R = 23$ branch). *Let $\ell \notin \{5, 13, 19\}$ be prime, assume that*

$$p = 260\ell - 19$$

is prime, and write $a_R = (p + R)/4$. Let

$$\lambda_{19} : U_{19} \longrightarrow \mathbb{Z}/18\mathbb{Z}$$

be the discrete logarithm determined by $\lambda_{19}(2) = 1$, and put $k = \lambda_{19}(\bar{\ell})$. Then

$$a_{19} = 5 \cdot 13 \cdot \ell,$$

and the complete target-incidence table is

$$\begin{array}{c|c} \Xi_{19,a_{19}} & k \pmod{18} \\ \hline E_{21} & 0, 5 \\ S & 8, 13 \\ I_2 + S & 2, 4, 6, 9, 11, 12 \\ \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} & 14, 16 \\ \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} & 10, 15 \\ \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} & 7 \\ \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} & 1, 3, 17. \end{array} \tag{1792}$$

The two positive-surplus cases in (1791) correspond to

$$\begin{array}{c|c|c|c} \ell & p & \ell \bmod 19 & \lambda_{19}(\ell) \\ \hline 457 & 118801 & 1 & 0 \\ 1381 & 359041 & 13 & 5. \end{array}$$

Consequently both have

$$\Xi_{19,a_{19}} = E_{21}, \quad (u, v) = (13, 5), \quad 4u + v = 3 \cdot 19, \quad 19 - u - v = 1.$$

At $R = 23$, let

$$Q_{23} := \{1, 2, 3, 4, 6, 8, 9, 12, 13, 16, 18\} \subset U_{23}$$

be the quadratic-residue subgroup. For $p = 118801$,

$$a_{23} = 29706 = 2 \cdot 3 \cdot 4951, \quad (2, 3, 4951) \equiv (2, 3, 6) \pmod{23},$$

and

$$\mathcal{R}_{23}^E(\nu_{23}(a_{23})) = \mathcal{R}_{23}^M(\nu_{23}(a_{23})) = Q_{23}. \tag{1793}$$

Since

$$t_{23} = 17, \quad s_{23} = 22$$

are not in Q_{23} , this gives

$$\Xi_{23,29706} = 0.$$

For $p = 359041$,

$$a_{23} = 89766 = 2 \cdot 3^2 \cdot 4987, \quad (2, 3, 4987) \equiv (2, 3, 19) \pmod{23},$$

where $19 \notin Q_{23}$. In this case

$$\mathcal{R}_{23}^E(\nu_{23}(a_{23})) = \mathcal{R}_{23}^M(\nu_{23}(a_{23})) = U_{23}, \tag{1794}$$

so

$$\Xi_{23,89766} = I_2 + S.$$

The complete divisor sections are

$$\mathcal{E}_{23,89766} = \{179532\}, \quad \mathcal{M}_{23,89766} = \{3, 2685978252\}, \quad q_{23,359041} = 2. \tag{1795}$$

The exterior section $u = 179532$ reconstructs

$$\boxed{\frac{4}{359041} = \frac{1}{89766} + \frac{1}{1006242664629726} + \frac{1}{1401292143}}. \tag{1796}$$

Proof. The element 2 generates U_{19} , and direct exponentiation gives

$$\lambda_{19}(5) = 16, \quad \lambda_{19}(13) = 5, \quad \lambda_{19}(t_{19}) = 7, \quad \lambda_{19}(s_{19}) = 9.$$

For $k = \lambda_{19}(\ell)$, the logarithmic exterior and middle reachability sets are therefore

$$\begin{aligned} L_E(k) &:= \{16e + 5f + kg : e, f, g \in \{0, 1, 2\}\}, \\ L_M(k) &:= \{16e + 5f + kg : e, f, g \in \{-1, 0, 1\}\}, \end{aligned}$$

with all entries taken modulo 18. By (1763),

$$\Xi_{19,a_{19}} = \begin{pmatrix} \mathbf{1}_{\{7 \in L_E(k)\}} & \mathbf{1}_{\{9 \in L_E(k)\}} \\ \mathbf{1}_{\{7 \in L_M(k)\}} & \mathbf{1}_{\{9 \in L_M(k)\}} \end{pmatrix}.$$

Evaluation for $k = 0, \dots, 17$ gives (1792). The two displayed values of ℓ then give the common E_{21} matrix and the stated quarter-pair identities.

Modulo 23, the residues 2, 3, 6 are quadratic residues. Multiplying their exterior powers 0, 1, 2, or their middle powers $-1, 0, 1$, produces exactly the eleven elements of Q_{23} . This proves (1793). For the second prime, the powers of 2 and 3 permitted by $\nu_{23}(89766)$ again fill Q_{23} in each alphabet. The residue 19 is a nonresidue, and its permitted powers meet both cosets of Q_{23} . Each product set is therefore all of U_{23} , proving (1794).

The factorisation

$$89766^2 = 2^2 3^4 4987^2$$

has 45 divisors. Testing them in the exact congruences

$$23 \mid 4u + 1, \quad 23 \mid 89766 + u$$

gives precisely the three divisors in (1795). Finally,

$$4 \cdot 179532 + 1 = 31223 \cdot 23.$$

Substitution of $p = 359041$, $R = 23$, $a = 89766$, and $u = 179532$ into (1835) gives the two remaining denominators in (1796). $\square$

Exact computation 14.145 (One-surplus quarter-branch certificate). The independent sources and stored receipts

```
verification/compare_one_surplus_quarter_branches.py,
verification/compare_one_surplus_quarter_branches_output.txt,
verification/OneSurplusQuarterBranchCertificate.lean,
verification/OneSurplusQuarterBranchCertificate_output.txt
```

verify the full 18-row incidence table, both shell trajectories, the two $R = 23$ product sets, all divisor sections, primality, and the rational identity. A fresh Python run agrees exactly with its stored receipt. The Lean source independently checks the two relevant $R = 19$ matrices, the four $R = 23$ reachability equalities, exclusion of both targets from Q_{23} , the fixed-target pair distinction (2, 15) versus (2, 18), the divisor arithmetic, primality, and (1796).

Corollary 14.146 (Explicit $R = 11$ split on an $R = 23$ edge). *Retain the conditions of theorem 14.108, assume $q_{23,p} = 0$, and write*

$$b = 2\ell u, \quad x := \ell u, \quad c := 2x - 1.$$

Then $q_{11,p} = 0$ if and only if one of the following disjoint cases holds:

(i) $x \equiv 2 \pmod{3}$, and every prime divisor of

$$\frac{2x - 1}{3}$$

lies in $\{1, 3, 4, 5, 9\} \pmod{11}$;

(ii) $x \not\equiv 2 \pmod{3}$, and $c = 2x - 1$ satisfies branch (A) or branch (B) of theorem 14.129.

Proof. The affine coordinates on this edge satisfy

$$a_{11} = 3c = 3(2x - 1).$$

If $x \equiv 2 \pmod{3}$, then $3 \mid c$, so $v_3(a_{11}) \geq 2$. Use $e = 2$ in theorem 14.128. The remaining factor-residue monomial is that of $c/3$, and

$$J_{11}^{(2)} = \langle X_2, X_6, X_7, X_8, X_{10} \rangle.$$

Its complement consists exactly of integers whose prime factors lie in $\{1, 3, 4, 5, 9\} \pmod{11}$, proving case (i).

If $x \not\equiv 2 \pmod{3}$, then $3 \nmid c$ and $v_3(a_{11}) = 1$. Use $e = 1$ in the same theorem and apply theorem 14.129. This gives case (ii). $\square$

Corollary 14.147 (Pairwise-coprime five-Finger carrier on the three-adic edge). *Retain the conditions and notation of theorem 14.146, and suppose*

$$x \equiv 2 \pmod{3}.$$

Put

$$d := \frac{2x - 1}{3}.$$

Then d is odd,

$$p = 36d - 11, \quad (a_3, a_7, a_{11}, a_{15}, a_{19}) = (9d - 2, 9d - 1, 9d, 9d + 1, 9d + 2), \quad (1797)$$

and the ordered Finger coordinates

$$\begin{aligned} (F_t, F_i, F_m, F_r, F_p) &:= \left(a_3, \frac{a_7}{2}, \frac{a_{11}}{9}, \frac{a_{15}}{2}, a_{19} \right) \\ &= \left(9d - 2, \frac{9d - 1}{2}, d, \frac{9d + 1}{2}, 9d + 2 \right) \end{aligned} \quad (1798)$$

are pairwise coprime. The forced-factor reassembly is

$$\text{diag}(1, 2, 9, 2, 1) \begin{pmatrix} F_t \\ F_i \\ F_m \\ F_r \\ F_p \end{pmatrix} = \begin{pmatrix} 9d - 2 \\ 9d - 1 \\ 9d \\ 9d + 1 \\ 9d + 2 \end{pmatrix}. \quad (1799)$$

Proof. The identity $2x - 1 = 3d$ gives $x = (3d + 1)/2$. Since x is odd, d is odd. Using $p = 24x - 23$ gives

$$p = 12(3d + 1) - 23 = 36d - 11.$$

Substitution in $a_R = (p + R)/4$, for $R = 3, 7, 11, 15, 19$, proves (1797) and (1798).

For adjacent Finger coordinates one has

$$\begin{aligned} 2F_i - F_t &= 1, & 9F_m - 2F_i &= 1, \\ 2F_r - 9F_m &= 1, & F_p - 2F_r &= 1. \end{aligned}$$

The remaining pairs are controlled by

$$\begin{aligned} 9F_m - F_t &= 2, & 2F_r - F_t &= 3, & F_p - F_t &= 4, \\ F_r - F_i &= 1, & F_p - 2F_i &= 3, & F_p - 9F_m &= 2. \end{aligned}$$

The three coordinates F_t, F_m, F_p are odd. Moreover

$$F_t \equiv F_i \equiv 1 \pmod{3}.$$

Each possible common divisor in the second display therefore divides 1, proving pairwise coprimality. The diagonal identity is the definition of the five forced-factor quotients. $\square$

Corollary 14.148 (Exact five-Finger factor intersection). *Retain the conditions and notation of theorem 14.147. Then*

$$q_{3,p} = q_{7,p} = q_{11,p} = q_{15,p} = q_{19,p} = q_{23,p} = 0 \quad (1800)$$

if and only if all five conditions below hold:

- (i) *every prime divisor of F_t is congruent to 1 (mod 3);*
- (ii) *every prime divisor of F_i lies in $\{1, 2, 4\}$ (mod 7);*
- (iii) *every prime divisor of F_m lies in $\{1, 3, 4, 5, 9\}$ (mod 11);*
- (iv) *every prime divisor of F_r lies in $\{1, 2, 4, 8\}$ (mod 15);*
- (v)

$$\mathbf{m}_{19}^{\log}(F_p) \notin I_{19}^{\log}.$$

Proof. The retained edge conditions already give $q_{23,p} = 0$. Substitute

$$a_3 = F_t, \quad a_7 = 2F_i, \quad a_{11} = 9F_m, \quad a_{15} = 2F_r, \quad a_{19} = F_p$$

into theorem 14.221. Its residual- 3, residual-7, residual-15, and residual-19 clauses give conditions (i), (ii), (iv), and (v). On the three-adic branch, theorem 14.146 says that the residual- 11 shell is empty exactly when every prime divisor of

$$\frac{2x-1}{3} = d = F_m$$

lies in the five displayed residue classes. This is condition (iii). □

Corollary 14.149 (Fixed Vandermonde volume of the five shell coordinates). *Let*

$$A_j := 9d - 2 + j, \quad 0 \leq j \leq 4,$$

and form the evaluation matrix

$$\mathcal{V}_d := (A_j^k)_{0 \leq k, j \leq 4}.$$

Then

$$\det \mathcal{V}_d = 288 = 2^5 3^2, \quad \text{disc} \left(\prod_{j=0}^4 (T - A_j) \right) = 82944 = 2^{10} 3^4. \quad (1801)$$

In particular, a prime at least 5 divides at most one of the five unreduced shell coordinates.

Proof. The coordinates are consecutive and ordered increasingly, so the Vandermonde determinant is

$$\prod_{0 \leq i < j \leq 4} (A_j - A_i) = \prod_{0 \leq i < j \leq 4} (j - i) = 1^4 2^3 3^2 4 = 288.$$

The discriminant is its square. A prime dividing two coordinates divides their difference, which belongs to $\{1, 2, 3, 4\}$; the last assertion follows. □

Corollary 14.150 (Sixty-class congruence shadow of the five-Finger intersection). *Retain the conditions of theorem 14.148, and assume that the six shells in (1800) are empty. Then the center parameter d belongs to the set $\mathcal{D}$ of residue classes determined by*

$$\begin{aligned} d &\equiv 3 \pmod{4}, \\ d &\equiv 3 \text{ or } 9 \pmod{23}, \\ d &\equiv 1, 5, \text{ or } 6 \pmod{7}, \\ d &\equiv 1, 3, 4, 5, \text{ or } 9 \pmod{11}, \\ d &\equiv 0 \text{ or } 2 \pmod{5}. \end{aligned} \quad (1802)$$

The set $\mathcal{D}$ has exactly 60 elements modulo

$$4 \cdot 23 \cdot 7 \cdot 11 \cdot 5 = 35420.$$

Thus its density in the integers is

$$\frac{60}{35420} = \frac{3}{1771}.$$

Equivalently, the corresponding primes $p = 36d - 11$ occupy exactly 60 classes modulo 1275120. For every $r \in \mathcal{D}$, the map

$$\mathbb{Z}/19\mathbb{Z} \longrightarrow \mathbb{Z}/19\mathbb{Z}, \quad k \longmapsto r + 35420k \quad (1803)$$

is a bijection. Hence $F_p = 9d + 2$ assumes every residue class modulo 19 along each progression $d \equiv r \pmod{35420}$.

Proof. The edge conditions give $p \equiv 1 \pmod{48}$, and $p = 36d - 11$; hence $3d \equiv 1 \pmod{4}$, so $d \equiv 3 \pmod{4}$. Likewise

$$p \equiv 5 \text{ or } 14 \pmod{23}$$

and $36 \equiv 13 \pmod{23}$ give $d \equiv 3 \text{ or } 9 \pmod{23}$.

The permitted residual-7 factor classes form the subgroup $\{1, 2, 4\}$ of U_7 . Therefore

$$F_i = \frac{9d - 1}{2} \equiv d + 3 \pmod{7}$$

lies in that subgroup, giving $d \equiv 1, 5, 6 \pmod{7}$. The permitted residual-11 classes are the quadratic-residue subgroup of U_{11} , so $F_m = d$ lies in $\{1, 3, 4, 5, 9\} \pmod{11}$.

Finally, the permitted residual-15 classes form the subgroup $\{1, 2, 4, 8\}$ of U_{15} . Since

$$F_r = \frac{9d + 1}{2} \equiv 12d + 8 \pmod{15},$$

membership in this subgroup is equivalent to $d \equiv 0 \text{ or } 2 \pmod{5}$. The five moduli in (1802) are pairwise coprime, so the Chinese remainder theorem gives

$$1 \cdot 2 \cdot 3 \cdot 5 \cdot 2 = 60$$

classes modulo 35420. The affine map $d \mapsto 36d - 11$ carries distinct classes modulo 35420 to distinct classes modulo $36 \cdot 35420$. Finally,

$$\gcd(35420, 19) = 1,$$

so (1803) is a bijection. Multiplication by 9 and translation by 2 preserve bijectivity modulo 19, proving the assertion for F_p . $\square$

Theorem 14.151 (Quadratic split-support lift of the first four Fingers). *For $R \in \{3, 7, 11, 15\}$, put*

$$K_R := \mathbb{Q}(\sqrt{-R})$$

and define the split-support monoid

$$\mathcal{S}_R := \left\{ n \geq 1 : q \mid n, \text{ } q \text{ prime} \implies \left(\frac{-R}{q} \right) = 1 \right\},$$

where the symbol is the Kronecker symbol. Under the conditions of theorem 14.147, the first four factor clauses in theorem 14.148 are equivalent to

$$F_t \in \mathcal{S}_3, \quad F_i \in \mathcal{S}_7, \quad F_m \in \mathcal{S}_{11}, \quad F_r \in \mathcal{S}_{15}. \quad (1804)$$

For $n \in \mathcal{S}_R$, let

$$\mathcal{L}_R(n) := \left\{ \prod_{q^e \parallel n} \mathfrak{q}_q^e : \mathfrak{q}_q \mid q\mathcal{O}_{K_R}, \text{ } N_{K_R/\mathbb{Q}}(\mathfrak{q}_q) = q \right\}. \quad (1805)$$

Then every ideal in $\mathcal{L}_R(n)$ has norm n , and $\mathcal{L}_R(n)$ is a torsor under

$$C_2^{\omega(n)},$$

where the factor indexed by q exchanges the two conjugate prime ideals above q . Consequently,

$$\mathcal{L}_3(F_t) \times \mathcal{L}_7(F_i) \times \mathcal{L}_{11}(F_m) \times \mathcal{L}_{15}(F_r) \quad (1806)$$

is a torsor under

$$C_2^{\omega(F_t F_i F_m F_r)}.$$

Proof. Direct evaluation of the four quadratic characters gives

R	$\{a \in U_R : (\frac{-R}{a}) = 1\}$
3	{1}
7	{1, 2, 4}
11	{1, 3, 4, 5, 9}
15	{1, 2, 4, 8}.

The retained affine conditions make each Finger coprime to the corresponding R , so the standard quadratic splitting criterion identifies these kernels with the rational primes that split in K_R . This proves (1804).

If $q^e \parallel n$ and $n \in \mathcal{S}_R$, then

$$q\mathcal{O}_{K_R} = \mathfrak{q}\bar{\mathfrak{q}}, \quad N(\mathfrak{q}) = N(\bar{\mathfrak{q}}) = q.$$

Selecting either member of this conjugate pair independently for each distinct $q \mid n$ gives (1805). Ideal norms are multiplicative, so every resulting ideal has norm n . Independent conjugate exchange at each rational prime acts freely and transitively, proving the torsor assertion. Pairwise coprimality of the four Fingers gives

$$\sum_{X \in \{t, i, m, r\}} \omega(F_X) = \omega(F_t F_i F_m F_r),$$

which proves the final statement. □

Corollary 14.152 (Residual-19 split-support safe face). *Put*

$$K_{19} := \mathbb{Q}(\sqrt{-19}), \quad \mathcal{S}_{19} := \left\{ n \geq 1 : q \mid n \implies \left(\frac{-19}{q} \right) = 1 \right\}.$$

If

$$F_p \in \mathcal{S}_{19},$$

then

$$\mathfrak{m}_{19}^{\log}(F_p) \notin I_{19}^{\log}, \quad q_{19,p} = 0.$$

Equivalently, the simultaneous split-support system

$$(F_t, F_i, F_m, F_r, F_p) \in \mathcal{S}_3 \times \mathcal{S}_7 \times \mathcal{S}_{11} \times \mathcal{S}_{15} \times \mathcal{S}_{19} \tag{1807}$$

forces the first five residual shells to be empty.

Proof. With primitive logarithm $\lambda_{19}(2) = 1$, the split residues are exactly the even logarithms:

$$\{1, 4, 5, 6, 7, 9, 11, 16, 17\} = \{2^{2j} : 0 \leq j \leq 8\} \pmod{19}.$$

Thus every entry of the factor-log multiset of F_p is even. Every element of both reachability sets

$$\Sigma_E(M) = \left\{ \sum e_j k_j : e_j \in \{0, 1, 2\} \right\}, \quad \Sigma_M(M) = \left\{ \sum d_j k_j : d_j \in \{-1, 0, 1\} \right\}$$

is therefore even modulo 18. The exterior and middle targets are 7 and 9, respectively, and both are odd. Neither target is reachable, proving the residual-19 assertion. The first four claims in (1807) imply the preceding four empty-shell conditions by theorem 14.151. □

Theorem 14.153 (Exact five-Finger six-shell survivor and its $R = 31$ closure). *Let*

$$p = 415969, \quad d = 11555, \quad p = 36d - 11. \tag{1808}$$

The five Fingers are

$$(F_t, F_i, F_m, F_r, F_p) = (103993, 51997, 11555, 51998, 103997). \tag{1809}$$

Every residual shell through $R = 27$ is empty:

$$q_{R,p} = 0 \quad \text{for } R \in \{3, 7, 11, 15, 19, 23, 27\}. \quad (1810)$$

The next shell, $R = 31$, has

$$a_{31} = 104000 = 2^6 \cdot 5^3 \cdot 13$$

and endpoint sets

$$\mathcal{E}_{31,104000} = \{5200, 166400, 650000, 20800000\}, \quad (1811)$$

$$\mathcal{M}_{31,104000} = \{5, 160, 625, 5120, 16900, 20000, 540800, 640000, 2112500, 17305600, 67600000, 2163200000\}. \quad (1812)$$

Consequently,

$$q_{31,p} = |\mathcal{E}_{31,104000}| + \frac{1}{2}|\mathcal{M}_{31,104000}| = 10. \quad (1813)$$

The least exterior seed $u = 5200$ gives the origin-2 divisor

$$\begin{aligned} N &= pa = 43260776000, \\ D &= p^2u = 899757086597200, \\ \frac{N^2}{D} &= \frac{a^2}{u} = 2080000. \end{aligned} \quad (1814)$$

The divisor-shell reconstruction therefore gives

$$\boxed{\frac{4}{415969} = \frac{1}{104000} + \frac{1}{29025817657200} + \frac{1}{1395576000}}. \quad (1815)$$

Proof. The complete factorization

$$p - 1 = 2^5 \cdot 3 \cdot 7 \cdot 619$$

and the base 13 give

$$13^{p-1} \equiv 1 \pmod{p}$$

and

$$\left(13^{(p-1)/2}, 13^{(p-1)/3}, 13^{(p-1)/7}, 13^{(p-1)/619}\right) \equiv (415968, 32421, 190698, 191070) \pmod{p}.$$

The gcd of one less than each displayed residue with p is 1. The integer 619 has no prime divisor at most 24, so it is prime. The Pocklington–Lucas criterion proves that p is prime.

Substitution of $d = 11555$ into

$$\left(9d - 2, \frac{9d - 1}{2}, d, \frac{9d + 1}{2}, 9d + 2\right)$$

gives (1809). The relevant shell denominators and their complete factorizations are

R	$a_R = (p + R)/4$	factorization of a_R
3	103993	103993
7	103994	$2 \cdot 11 \cdot 29 \cdot 163$
11	103995	$3^2 \cdot 5 \cdot 2311$
15	103996	$2^2 \cdot 25999$
19	103997	103997
23	103998	$2 \cdot 3 \cdot 17333$
27	103999	$7 \cdot 83 \cdot 179$
31	104000	$2^6 \cdot 5^3 \cdot 13$.

(1816)

Enumerating the divisors of a_R^2 in the defining congruences

$$\mathcal{E}_{R,a_R} = \{u \mid a_R^2 : R \mid 4u + 1\}, \quad \mathcal{M}_{R,a_R} = \{u \mid a_R^2 : R \mid u + a_R\}$$

gives empty sets for the first seven rows of (1816). The final row gives exactly (1811) and (1812). This proves (1810) and (1813).

For $u = 5200$,

$$4u + 1 = 20801 = 31 \cdot 671, \quad u \mid a^2.$$

Thus $(2, u) \in \mathcal{X}_{31,p}$, and the exact origin map

$$(j, u) \mapsto p^j u$$

gives $D = p^2 u$. Direct calculation gives

$$D \mid N^2, \quad D \equiv -N \pmod{31},$$

and the values in (1814). Applying theorem 9.1 yields

$$b = \frac{N + D}{31} = 29025817657200, \quad c = \frac{N + N^2/D}{31} = 1395576000.$$

Finally,

$$4abc = p(ab + ac + bc) = 16851217553150051635200000000,$$

which verifies (1815). $\square$

Corollary 14.154 (The $R = 31$ exterior progression of the five-Finger carrier). *Let p be prime and suppose*

$$p \equiv 14209 \pmod{22320}. \quad (1817)$$

Then

$$d := \frac{p + 11}{36}$$

is an integer satisfying $d \equiv 395 \pmod{620}$. The residual $R = 31$ shell contains the exterior seed

$$a = \frac{p + 31}{4}, \quad u = \frac{a}{20} = \frac{p + 31}{80}. \quad (1818)$$

In particular,

$$\boxed{\frac{4}{p} = \frac{1}{(p + 31)/4} + \frac{1}{p(p + 31)(p + 20)/2480} + \frac{1}{(p + 31)(p + 20)/124}}. \quad (1819)$$

Proof. The congruence (1817) gives

$$p \equiv -11 \pmod{36}, \quad p + 31 \equiv 0 \pmod{80}, \quad p + 20 \equiv 0 \pmod{31}.$$

Hence d , a , and u in (1818) are integers. Moreover,

$$p = 36d - 11, \quad d \equiv 395 \pmod{620}.$$

The final congruence in the first display gives

$$4u + 1 = \frac{p + 31}{20} + 1 = \frac{p + 51}{20} \equiv 0 \pmod{31}.$$

Thus $u \in \mathcal{E}_{31,a}$. Substitution of $u = a/20$ into the exterior reconstruction formulas gives

$$b = \frac{p(a + pu)}{31} = \frac{p(p + 31)(p + 20)}{2480}, \quad c = \frac{pa + a^2/u}{31} = \frac{(p + 31)(p + 20)}{124}.$$

Their integrality follows from the divisibilities above. Finally, the numerator of the three right-hand terms over the common denominator $p(p + 31)(p + 20)$ is

$$4p(p + 20) + 2480 + 124p = 4(p + 20)(p + 31),$$

which proves (1819). $\square$

Theorem 14.155 (Certified all-five split face below twenty million). *Let $d > 0$ satisfy*

$$\begin{aligned} d \equiv 3 \pmod{4}, \quad d \equiv 3, 9 \pmod{23}, \quad d \equiv 1, 5, 6 \pmod{7}, \\ d \equiv 1, 3, 4, 5, 9 \pmod{11}, \quad d \equiv 0, 2 \pmod{5}, \end{aligned} \quad (1820)$$

put $p = 36d - 11$, and retain the rows for which $p < 20000000$ is prime and $q_{23,p} = 0$. There are exactly 147 retained rows. Exactly 25 have their first four Fingers split-supported at residuals 3, 7, 11, 15, and exactly 10 have all five Fingers split-supported at residuals 3, 7, 11, 15, 19. The ten primes in the latter set, together with the first occupied residual after 23, are

p	d	$\min\{R \geq 27 : R \equiv 3 \pmod{4}, q_{R,p} > 0\}$
3368401	93567	27
5153569	143155	39
5719921	158887	27
7273249	202035	31
9527281	264647	31
11996161	333227	31
13503121	375087	31
15866449	440735	31
17722609	492295	27
17904769	497355	31

(1821)

In particular, $p = 3368401$ is the least such prime in the stated bounded class.

Proof. The congruences in (1820) define exactly 60 classes modulo

$$4 \cdot 23 \cdot 7 \cdot 11 \cdot 5 = 35420.$$

For every representative d in those classes for which $36d - 11 < 20000000$, the certificate enumerates $p = 36d - 11$, tests primality by trial division through $\lfloor \sqrt{p} \rfloor$, and computes the two endpoint sets of the $R = 23$ shell from every divisor of a_{23}^2 . This leaves 147 rows. For each retained row it factors

$$(F_t, F_i, F_m, F_r, F_p) = \left(9d - 2, \frac{9d - 1}{2}, d, \frac{9d + 1}{2}, 9d + 2\right)$$

and checks that every prime divisor lies respectively in

residual	allowed prime residues
3	$\{1\}$
7	$\{1, 2, 4\}$
11	$\{1, 3, 4, 5, 9\}$
15	$\{1, 2, 4, 8\}$
19	$\{1, 4, 5, 6, 7, 9, 11, 16, 17\}$

(1822)

The resulting counts are 25 through the first four columns and 10 through all five. Direct divisor enumeration in the successive shells gives the final column of (1821). The script and its captured receipt are identified in Computation 14.199. $\square$

Theorem 14.156 (Least bounded all-five row and its unique $R = 27$ endpoint). *For the least row in Theorem 14.155,*

$$p = 3368401, \quad d = 93567, \quad p = 36d - 11, \quad (1823)$$

the five Fingers are

$$(F_t, F_i, F_m, F_r, F_p) = (842101, 421051, 93567, 421052, 842105). \quad (1824)$$

All six shells through $R = 23$ are empty. The next shell has

$$a_{27} = 842107 = 7 \cdot 59 \cdot 2039, \quad (1825)$$

and

$$\mathcal{E}_{27,842107} = \{49684313\} = \{59a_{27}\}, \quad \mathcal{M}_{27,842107} = \emptyset. \quad (1826)$$

Thus $q_{27,p} = 1$. Its unique endpoint gives

$$\boxed{\frac{4}{3368401} = \frac{1}{842107} + \frac{1}{20878683089630846060} + \frac{1}{105057558340}}. \quad (1827)$$

Proof. The Finger factorizations are

$$\begin{aligned} 842101 &= 13 \cdot 211 \cdot 307, & 421051 &= 29 \cdot 14519, & 93567 &= 3 \cdot 31189, \\ 421052 &= 2^2 \cdot 105263, & 842105 &= 5 \cdot 11 \cdot 61 \cdot 251. \end{aligned} \quad (1828)$$

Their residues lie in the five sets of (1822), and the direct $R = 23$ endpoint calculation is empty. This proves the first six empty-shell assertions by the split-support criteria already proved for the first five residuals.

For completeness, primality of p has the Pocklington certificate

$$p - 1 = 2^4 \cdot 3 \cdot 5^2 \cdot 7 \cdot 401.$$

The integer 401 is prime by trial division through 20, and base 31 satisfies

$$31^{p-1} \equiv 1 \pmod{p}$$

and

$$\frac{q}{31^{(p-1)/q} \bmod p} \mid \begin{array}{ccccc} 2 & 3 & 5 & 7 & 401 \\ 3368400 & 2938148 & 263135 & 1806350 & 1223878 \end{array} \quad (1829)$$

with

$$\gcd(31^{(p-1)/q} - 1, p) = 1 \quad (q = 2, 3, 5, 7, 401).$$

Every divisor of a_{27}^2 has the form

$$u = 7^i 5^j 14^k, \quad 0 \leq i, j, k \leq 2.$$

Modulo 27, the exterior and middle targets are respectively 20 and $-a_{27} \equiv 23$. Since

$$\begin{aligned} 7^i 5^j 14^k &\equiv 20 \pmod{27} \iff (i, j, k) = (1, 2, 1), \\ 7^i 5^j 14^k &\equiv 23 \pmod{27} \text{ has no solution for } 0 \leq i, j, k \leq 2, \end{aligned}$$

the endpoint sets are exactly those in (1826).

For $u = 59a_{27}$, put $N = pa_{27}$ and use the exterior origin $j = 2$. Then

$$D = p^2 u = 563724440583478782713, \quad \frac{N^2}{D} = \frac{a_{27}}{59} = 14273, \quad (1830)$$

$$\frac{N + D}{27} = 20878683089630846060, \quad \frac{N + N^2/D}{27} = 105057558340. \quad (1831)$$

The shell reconstruction theorem gives (1827). Equivalently, both sides of the integer cross multiplication are

$$7388523758382026891769703924779291200.$$

□

Corollary 14.157 (The $R = 27$ exterior progression of the all-five split carrier). *Let p be prime and suppose*

$$p \equiv 3985 \pmod{6372}. \quad (1832)$$

Then the $R = 27$ shell contains the exterior seed

$$a = \frac{p + 27}{4}, \quad u = 59a. \quad (1833)$$

In particular,

$$\boxed{\frac{4}{p} = \frac{1}{(p+27)/4} + \frac{1}{p(p+27)(59p+1)/108} + \frac{1}{(p+27)(59p+1)/6372}}. \quad (1834)$$

Proof. The congruence in (1832) is the simultaneous system

$$p \equiv -27 \pmod{236}, \quad p \equiv 16 \pmod{27}.$$

Hence $59 \mid a$, so $u = 59a \mid a^2$, and

$$4u + 1 = 59(p + 27) + 1 \equiv 0 \pmod{27}.$$

Thus $u \in \mathcal{E}_{27,a}$. With $N = pa$ and $D = p^2u$,

$$\frac{N^2}{D} = \frac{a}{59},$$

and shell reconstruction gives the three denominators in (1834). Their equality can also be read directly from

$$4p(59p + 1) + 108 + 6372p = 4(p + 27)(59p + 1).$$

□

Lemma 14.158 (Exact endpoint-family reconstruction). *Let p be an odd prime, let $R > 0$ satisfy*

$$R \equiv -p \pmod{4}, \quad a = \frac{p+R}{4}, \quad \gcd(R, pa) = 1,$$

and let $u \mid a^2$.

(i) *If $R \mid 4u + 1$, then*

$$(a, b, c) = \left(a, \frac{p(a+pu)}{R}, \frac{pa+a^2/u}{R} \right) \quad (1835)$$

is a positive integral solution of $4/p = 1/a + 1/b + 1/c$.

(ii) *If $R \mid a + u$, then*

$$(a, b, c) = \left(a, \frac{p(a+u)}{R}, \frac{p(a+a^2/u)}{R} \right) \quad (1836)$$

is a positive integral solution of $4/p = 1/a + 1/b + 1/c$.

Proof. Put $N = pa$. In the first case take $D = p^2u$. Since

$$p \equiv 4a \pmod{R}, \quad pu + a \equiv a(4u + 1) \equiv 0 \pmod{R},$$

one has $D \equiv -N \pmod{R}$. Moreover

$$\frac{N^2}{D} = \frac{a^2}{u}.$$

The denominator formulas in theorem 9.1 are therefore exactly (1835).

In the second case take $D = pu$. Then

$$D + N = p(u + a) \equiv 0 \pmod{R}, \quad \frac{N^2}{D} = \frac{pa^2}{u},$$

and the same reconstruction theorem gives (1836). □

Proposition 14.159 (Fixed-seed CRT carriers). *For a positive integer*

$$u = \prod_{\ell} \ell^{e_{\ell}},$$

put

$$\kappa(u) := \prod_{\ell} \ell^{\lceil e_{\ell}/2 \rceil}. \quad (1837)$$

Let $R \equiv 3 \pmod{4}$.

If $R \mid 4u + 1$, consider the exterior congruence system

$$p \equiv 25 \pmod{36}, \quad p \equiv -R \pmod{4\kappa(u)}. \quad (1838)$$

If instead a middle endpoint is required, consider

$$\begin{aligned} p &\equiv 25 \pmod{36}, & p &\equiv -R \pmod{4\kappa(u)}, \\ p &\equiv -R - 4u \pmod{4R}. \end{aligned} \quad (1839)$$

Each compatible system is one residue class modulo the least common multiple of its displayed moduli. Every prime $p > R$ in that class has $a = (p + R)/4$, satisfies $u \mid a^2$, and yields respectively an exterior or middle Erdős–Straus certificate through theorem 14.158.

Conversely, if a five-Finger prime p has a fixed exterior endpoint u at residual R , then it satisfies (1838); if it has a fixed middle endpoint u , then it satisfies (1839).

Proof. Prime-adic valuations give the exact equivalence

$$u \mid a^2 \iff \kappa(u) \mid a.$$

Since $4a = p + R$, the condition on the right is equivalent to

$$p \equiv -R \pmod{4\kappa(u)}.$$

For a middle endpoint,

$$R \mid a + u \iff p + R + 4u \equiv 0 \pmod{4R}.$$

The exterior endpoint condition is the fixed divisibility $R \mid 4u + 1$. The Chinese remainder theorem now gives the asserted description whenever the displayed congruences are compatible.

For a prime $p > R$, one also has $\gcd(R, pa) = 1$. Indeed, a prime divisor common to R and a would divide $p = 4a - R$, but would be smaller than p . The reconstruction lemma therefore applies. Reading the same equivalences in reverse proves the converse. $\square$

Theorem 14.160 (Exterior endpoint–origin transpose). *Retain the hypotheses of theorem 14.158, and let $u \in \mathcal{E}_{R,a}$. Write its unique squarefree coordinates as*

$$u = st^2, \quad a = stm, \quad s \text{ squarefree}, \quad (1840)$$

and put

$$v := \frac{a^2}{u} = sm^2. \quad (1841)$$

Then u occupies the $j = 2$ fibre and v occupies the $j = 0$ fibre:

$$(2, u), (0, v) \in \mathcal{X}_{R,p}. \quad (1842)$$

The associated divisors

$$D_2 = p^2u, \quad D_0 = v$$

are complementary divisors of $N^2 = (pa)^2$. Their ordered reconstructions are

$$(b_2, c_2) = \left(\frac{pst(m + pt)}{R}, \frac{sm(pt + m)}{R} \right), \quad (1843)$$

$$(b_0, c_0) = \left(\frac{sm(pt + m)}{R}, \frac{pst(m + pt)}{R} \right). \quad (1844)$$

Thus the endpoint and origin fibres reconstruct the same unordered Erdős–Straus certificate and exchange its two residual denominators:

$$(b_0, c_0) = (c_2, b_2). \quad (1845)$$

Proof. The squarefree-coordinate bijection in theorem 14.59 gives (1840) and (1841). The endpoint-fibre identity (1527) then proves (1842). Moreover,

$$D_2 D_0 = p^2 uv = p^2 a^2 = N^2.$$

The $j = 2$ reconstruction is (1556). For $D_0 = v = sm^2$, the general shell formulas give

$$\frac{N + D_0}{R} = \frac{pstm + sm^2}{R} = \frac{sm(pt + m)}{R}$$

and

$$\frac{N + N^2/D_0}{R} = \frac{pstm + p^2 st^2}{R} = \frac{pst(m + pt)}{R}.$$

These are (1844) and (1845). $\square$

Theorem 14.161 (Fixed origin-carrier progressions). *Let $R \equiv 3 \pmod{4}$, let $v \in \mathbb{Z}_{>0}$, and write*

$$v = sm^2 \tag{1846}$$

with s squarefree. Put

$$L_R := \text{lcm}(R, 9), \quad B_R := \frac{R + 25}{4},$$

and define

$$\mathcal{T}_{R,v} := \left\{ \bar{t} \in \mathbb{Z}/L_R\mathbb{Z} : \begin{array}{l} 4st^2 + 1 \equiv 0 \pmod{R}, \\ smt \equiv B_R \pmod{9} \end{array} \right\}. \tag{1847}$$

For a positive representative t_0 of an element of $\mathcal{T}_{R,v}$, let

$$t = t_0 + L_R n, \quad p = 4smt - R, \quad n \in \mathbb{Z}_{\geq 0}. \tag{1848}$$

Whenever $p > R$ is prime, one has

$$p \equiv 25 \pmod{36}, \quad a = \frac{p + R}{4} = smt,$$

and the residual- R exterior orbit has endpoint and origin carrier

$$u = st^2 \in \mathcal{E}_{R,a}, \quad v = sm^2 = \frac{a^2}{u}. \tag{1849}$$

Its two ordered divisor reconstructions are exactly (1843) and (1844).

Conversely, every prime $p \equiv 25 \pmod{36}$ whose residual- R exterior orbit has $j = 0$ carrier v occurs in exactly one residue class in $\mathcal{T}_{R,v}$. The quadratic residue condition has the intrinsic form

$$4st^2 \equiv -1 \pmod{R} \iff (2st)^2 \equiv -s \pmod{R}. \tag{1850}$$

Hence a root modulo R exists precisely when $-s$ is a square in $(\mathbb{Z}/R\mathbb{Z})^\times$; compatibility with the congruence modulo 9 is recorded without loss by $\mathcal{T}_{R,v}$.

Proof. The first congruence in (1847) gives $R \mid 4u + 1$ for $u = st^2$. The second gives

$$a = smt \equiv B_R \pmod{9},$$

and therefore

$$p = 4a - R \equiv 25 \pmod{36}.$$

Equation (1846) gives

$$uv = (st^2)(sm^2) = a^2.$$

Thus $u \in \mathcal{E}_{R,a}$, $v = a^2/u$, and theorem 14.160 supplies both reconstructions.

For the converse, the unique squarefree decomposition of v , and then the squarefree-coordinate bijection of theorem 14.59, recover a unique positive integer t with

$$a = smt, \quad u = st^2.$$

The exterior condition gives the first congruence in (1847). The identity $p = 4a - R$, together with $p \equiv 25 \pmod{36}$, gives the second. Reducing t modulo L_R proves the asserted progression description. Finally, the exterior congruence makes s and t units modulo R . Multiplication by s , and its inverse operation, prove the two implications in (1850). $\square$

Theorem 14.162 (Residual-free carrier incidence). *Let $p \equiv 1 \pmod{4}$ be prime.*

- (i) *An exterior endpoint occurs in the primewise shell space if and only if there exist positive integers k, s, t, h , with s squarefree, such that*

$$kp + 1 = 4sth, \quad k \mid t + h, \quad t < 2h. \quad (1851)$$

The associated shell coordinates are

$$m = \frac{t+h}{k}, \quad a = stm, \quad R = 4stm - p. \quad (1852)$$

Necessarily $k \equiv 3 \pmod{4}$, and the prime lies in the exact interval

$$4st(m-t) - 1 \leq p < 4smt \leq 3p. \quad (1853)$$

- (ii) *A middle orbit occurs in the primewise shell space if and only if there exist positive integers ℓ, s, t, m , with s squarefree, such that*

$$(4\ell st - 1)(4\ell sm - 1) = 4\ell^2 sp + 1. \quad (1854)$$

The associated shell coordinates are

$$a = stm, \quad R = 4stm - p, \quad t + m = \ell R. \quad (1855)$$

The prime lies in the exact interval

$$4smt - (t + m) \leq p < 4smt \leq 2p. \quad (1856)$$

Exchanging t and m exchanges the two factors in (1854) and preserves the orbit.

Consequently, the Erdős–Strauss identity at p holds if and only if at least one of (1851) and (1854) has a solution.

Proof. Suppose first that an exterior endpoint occurs. Its unique squarefree coordinates have the form

$$u = st^2, \quad a = stm, \quad R \mid 4st^2 + 1.$$

Put

$$k = \frac{4st^2 + 1}{R}, \quad h = km - t.$$

Using $R = 4stm - p$ gives

$$kp + 1 = 4st(km - t) = 4sth.$$

The left side is positive, so $h > 0$, and

$$t + h = km.$$

The denominator condition $a \in \mathcal{A}_p$ gives

$$\begin{aligned} 0 &\geq 4a - 3p \\ &= \frac{4st(t+h) - 3kp}{k} \\ &= \frac{4st(t-2h) + 3}{k}. \end{aligned}$$

If $t \geq 2h$, the last numerator is positive. Hence $t < 2h$. Reduction of

$$kR = 4st^2 + 1$$

modulo 4 gives $k \equiv 3 \pmod{4}$. Since $0 < R \leq 4st^2 + 1$, one also obtains (1853).

Conversely, assume (1851), and define m, a, R by (1852). Then

$$\begin{aligned} kR &= 4kst \frac{t+h}{k} - kp \\ &= 4st(t+h) - (4sth - 1) = 4st^2 + 1. \end{aligned}$$

Thus $R > 0$, so $p < 4a$. Since $t < 2h$,

$$4st(t-2h) + 3 \leq -4st + 3 < 0,$$

and the preceding identity for $4a - 3p$ gives $4a < 3p$. Consequently $a \in \mathcal{A}_p$, and $u = st^2$ belongs to $\mathcal{E}_{R,a}$.

Now suppose that a middle orbit occurs. Write its squarefree coordinates as

$$a = stm, \quad R \mid t + m,$$

and put $\ell = (t + m)/R$. Direct expansion gives

$$\begin{aligned} (4\ell st - 1)(4\ell sm - 1) &= 16\ell^2 s^2 tm - 4\ell s(t + m) + 1 \\ &= 16\ell^2 s^2 tm - 4\ell^2 s(4stm - p) + 1 \\ &= 4\ell^2 sp + 1. \end{aligned}$$

This proves (1854).

Conversely, expansion of (1854) and division by $4\ell s$ yield

$$4\ell stm - (t + m) = \ell p.$$

Hence

$$t + m = \ell(4stm - p) = \ell R.$$

In particular, $R > 0$. Moreover,

$$t + m \leq 2tm$$

and

$$\begin{aligned} \ell p &= 4\ell stm - (t + m) \\ &\geq (4\ell s - 2)tm \\ &\geq 2tm \\ &\geq t + m. \end{aligned}$$

Therefore $R = (t + m)/\ell \leq p$, and hence

$$p < 4a = p + R \leq 2p < 3p.$$

Thus $a = stm$ lies in $\mathcal{A}_p$, and $R \mid t + m$ makes $\{st^2, sm^2\}$ a middle orbit. The equality $p = 4smt - (t + m)/\ell$, together with $\ell \geq 1$, gives the first inequality in (1856); the other two have just been proved. The final equivalence follows from theorem 14.54. $\square$

Remark 14.163 (Exact carrier arities). The two alternatives in theorem 14.162 can be displayed as

$$\frac{kp + 1}{4} = sth, \quad k \mid t + h, \quad (1857)$$

$$4\ell^2 sp + 1 = X_t X_m, \quad X_t = 4\ell st - 1, \quad X_m = 4\ell sm - 1. \quad (1858)$$

Thus the exterior channel has three positive carrier factors and a directed gluing equation, whereas the middle channel has two exchangeable factors. The scalar surplus is +1 in both formulas.

Theorem 14.164 (Exterior divisor-carrier parametrization). *Let $p \equiv 1 \pmod{4}$ be prime. An exterior endpoint occurs in the primewise shell space if and only if there exist positive integers k, s, t , with s squarefree, such that*

$$k \mid 4st^2 + 1, \quad 4st \mid kp + 1, \quad (1859)$$

and, on writing

$$h = \frac{kp + 1}{4st}, \quad (1860)$$

one has $t < 2h$. The inverse map is

$$R = \frac{4st^2 + 1}{k}, \quad m = \frac{t + h}{k}, \quad a = stm. \quad (1861)$$

All four reconstructed quantities are positive integers, one has

$$k \equiv 3 \pmod{4}, \quad p < 4a < 3p, \quad st^2 \in \mathcal{E}_{R,a}, \quad (1862)$$

and the associated ordered denominator triple is

$$\frac{4}{p} = \frac{1}{stm} + \frac{1}{psth} + \frac{1}{smh}. \quad (1863)$$

Proof. Suppose first that an exterior endpoint occurs. The exterior part of theorem 14.162 supplies positive k, s, t, h , with s squarefree, such that

$$kp + 1 = 4sth, \quad k \mid t + h, \quad t < 2h.$$

It also gives

$$kR = 4st^2 + 1.$$

These identities prove both divisibilities in (1859) and recover all three coordinates in (1861).

Conversely, assume (1859) and define h by (1860). Since $k \mid 4st^2 + 1$, one has $\gcd(k, 4st) = 1$. Modulo k ,

$$4sth \equiv 1, \quad 4st(-t) \equiv 1.$$

Cancellation of $4st$ gives

$$h \equiv -t \pmod{k}.$$

Thus $m = (t + h)/k$ is a positive integer. Define R, a by (1861). Then

$$\begin{aligned} k(4a - p) &= 4st(t + h) - kp \\ &= 4st^2 + 1 = kR, \end{aligned}$$

so $R = 4a - p > 0$. The strict inequality $t < 2h$ and the calculation in the proof of theorem 14.162 give $4a < 3p$. Moreover,

$$kR = 4st^2 + 1,$$

so $R \mid 4st^2 + 1$ and $st^2 \in \mathcal{E}_{R,a}$. Reduction of $kp + 1 = 4sth$ modulo (4) gives $k \equiv 3 \pmod{4}$.

Finally,

$$p(t + h) + m = pkm + m = m(kp + 1) = 4sthm.$$

After division by $psth$, this is exactly (1863). $\square$

Theorem 14.165 (One-leg factor-residue criterion). *Let $p \equiv 1 \pmod{4}$ be prime, let $k \equiv 3 \pmod{4}$ be a positive integer, and put*

$$N_k(p) := \frac{kp + 1}{4}.$$

Suppose that $N_k(p)$ has a squarefree divisor s satisfying

$$k \mid 4s + 1. \quad (1864)$$

Define

$$h = \frac{N_k(p)}{s}, \quad m = \frac{h + 1}{k}, \quad R = \frac{4s + 1}{k}, \quad a = sm. \quad (1865)$$

Then all four quantities are positive integers,

$$u = s \in \mathcal{E}_{R,a}, \quad p < 4a \leq 2p, \quad (1866)$$

and the Erdős–Strauss identity holds at p .

Consequently, an Erdős–Strauss counterexample $p \equiv 1 \pmod{4}$ would have to satisfy

$$k \nmid 4s + 1 \quad (1867)$$

for every positive integer $k \equiv 3 \pmod{4}$ and every squarefree divisor $s \mid N_k(p)$.

Proof. The identities

$$4N_k(p) = kp + 1, \quad N_k(p) = sh$$

give

$$kp + 1 = 4sh.$$

Modulo k , hypothesis (1864) therefore gives

$$-h \equiv 1 \pmod{k}.$$

Thus $h \equiv -1 \pmod{k}$, so $m = (h + 1)/k$ is a positive integer. With $t = 1$, the three integers k, s, t satisfy

$$k \mid 4st^2 + 1, \quad 4st \mid kp + 1, \quad t < 2h.$$

Theorem 14.164 gives

$$u = st^2 = s \in \mathcal{E}_{R,a},$$

and its reconstructed residual is

$$\begin{aligned} 4sm - p &= \frac{4s(h + 1) - (4sh - 1)}{k} \\ &= \frac{4s + 1}{k} = R. \end{aligned}$$

Finally, $s \leq N_k(p)$ gives

$$R \leq \frac{4N_k(p) + 1}{k} = p + \frac{2}{k} < p + 1.$$

Since R is integral, $R \leq p$, and hence

$$p < 4a = p + R \leq 2p.$$

The last assertion is the contrapositive of the proved construction. $\square$

Theorem 14.166 (Middle divisor-carrier parametrization). *Let $p \equiv 1 \pmod{4}$ be prime. A middle orbit occurs in the primewise shell space if and only if there exist positive integers ℓ, s, t , with s squarefree, such that, on writing*

$$q = 4\ell st - 1, \quad (1868)$$

one has

$$q \mid p + 4st^2. \quad (1869)$$

The inverse map is

$$R = \frac{p + 4st^2}{q}, \quad m = \ell R - t, \quad a = stm. \quad (1870)$$

These are positive integers satisfying

$$0 < R \leq p, \quad \{st^2, sm^2\} \subset \mathcal{M}_{R,a}, \quad (1871)$$

and the associated denominator triple is

$$\frac{4}{p} = \frac{1}{stm} + \frac{1}{p\ell st} + \frac{1}{p\ell sm}. \quad (1872)$$

Equivalently,

$$q \mid \ell p + t, \quad m = \frac{\ell p + t}{q}. \quad (1873)$$

In the special case $t = 1$, the criterion becomes

$$4\ell s - 1 \mid \ell p + 1 \iff 4\ell s - 1 \mid p + 4s. \quad (1874)$$

Proof. Suppose first that a middle orbit occurs. Its squarefree coordinates satisfy

$$a = stm, \quad t + m = \ell R$$

for a positive integer ℓ . Therefore

$$\begin{aligned} qR &= (4\ell st - 1)R \\ &= 4st(t + m) - R \\ &= 4st^2 + p. \end{aligned}$$

This proves (1869) and the first formula in (1870); the remaining two formulas recover the original coordinates.

Conversely, assume (1869) and define R, m, a by (1870). Then

$$\begin{aligned} p &= qR - 4st^2 \\ &= (4\ell st - 1)R - 4st^2 \\ &= 4st(\ell R - t) - R \\ &= 4stm - R. \end{aligned}$$

Since $p, R > 0$, this identity forces $m > 0$. It also gives

$$t + m = \ell R.$$

As in the proof of theorem 14.162,

$$\begin{aligned} \ell p &= 4\ell stm - (t + m) \\ &\geq (4\ell s - 2)tm \\ &\geq 2tm \\ &\geq t + m, \end{aligned}$$

so $R = (t + m)/\ell \leq p$. Thus $a \in \mathcal{A}_p$, and $R \mid t + m$ gives the middle orbit in (1871).

The sum on the right of (1872) has numerator

$$\ell p + t + m = 4\ell stm$$

over the common denominator $p\ell stm$, proving the identity.

Finally, $q = 4\ell st - 1$ is coprime to $4st$, and modulo q ,

$$4st(\ell p + t) \equiv p + 4st^2.$$

This proves the equivalence in (1873). The quotient identity follows from

$$qm = q(\ell R - t) = \ell(p + 4st^2) - qt = \ell p + t.$$

Taking $t = 1$ proves (1874). □

Theorem 14.167 (Integral determinant flags of the two carriers). *Let $p \equiv 1 \pmod{4}$ be prime.*

- (i) *Positive exterior-carrier data are equivalent to positive integers k, s, t, h, R , with s squarefree and $t < 2h$, for which*

$$E_p = \begin{pmatrix} 4st & k \\ p & h \end{pmatrix}, \quad E_R = \begin{pmatrix} k & 4st \\ t & R \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}). \quad (1875)$$

The shell coordinates are

$$m = \frac{t+h}{k}, \quad a = stm.$$

- (ii) Positive middle-carrier data are equivalent to positive integers ℓ, s, t, q, R , with s squarefree, for which

$$M_1 = \begin{pmatrix} 4st & q \\ 1 & \ell \end{pmatrix}, \quad \det M_1 = 1, \quad (1876)$$

and

$$M_p = \begin{pmatrix} q & 4st \\ t & R \end{pmatrix}, \quad \det M_p = p. \quad (1877)$$

The remaining coordinates are

$$m = \ell R - t, \quad a = stm,$$

and the third matrix

$$M_t = \begin{pmatrix} q & \ell \\ p & m \end{pmatrix} \quad (1878)$$

satisfies $\det M_t = t$.

Proof. For the exterior channel, the two determinant equations in (1875) are

$$4sth - kp = 1, \quad kR - 4st^2 = 1.$$

They are precisely the two quotient equations in theorem 14.164. Modulo k they give

$$4sth \equiv 1, \quad 4st(-t) \equiv 1.$$

Since $4st$ is a unit modulo k , one obtains $h \equiv -t \pmod{k}$. Thus $m = (t+h)/k$ is integral, and theorem 14.164 proves both directions.

For the middle channel,

$$\det M_1 = 4\ell st - q,$$

so $\det M_1 = 1$ is exactly $q = 4\ell st - 1$. The second determinant equation is

$$qR - 4st^2 = p,$$

which is the quotient equation in theorem 14.166. It gives

$$p = 4st(\ell R - t) - R.$$

Positivity of p, R forces $m = \ell R - t > 0$, and the middle divisor-carrier theorem proves both directions. Finally,

$$\begin{aligned} \det M_t &= qm - \ell p \\ &= q(\ell R - t) - \ell(qR - 4st^2) \\ &= t(4\ell st - q) = t. \end{aligned}$$

□

Theorem 14.168 (Determinant flags transport the two ordered blade channels). *Let*

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R}), \quad J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and retain the ordered basis e_+, e_- . Define

$$\mathcal{N}_+(A) := AE_{21}A^{-1}, \quad \mathcal{N}_-(A) := AE_{12}A^{-1}, \quad (1879)$$

and

$$P_{+,A} := AE_{11}A^{-1}, \quad P_{-,A} := AE_{22}A^{-1}, \quad J_A := AJA^{-1}. \quad (1880)$$

Then

$$J_A \mathcal{N}_{-+}(A) = -\mathcal{N}_{-+}(A), \quad \mathcal{N}_{-+}(A) J_A = \mathcal{N}_{-+}(A), \quad (1881)$$

$$J_A \mathcal{N}_{+-}(A) = \mathcal{N}_{+-}(A), \quad \mathcal{N}_{+-}(A) J_A = -\mathcal{N}_{+-}(A), \quad (1882)$$

and

$$P_{-,A} \mathcal{N}_{-+}(A) P_{+,A} = \mathcal{N}_{-+}(A), \quad P_{+,A} \mathcal{N}_{+-}(A) P_{-,A} = \mathcal{N}_{+-}(A). \quad (1883)$$

Thus the first transported channel has source Ae_+ and range Ae_- , while the second has source Ae_- and range Ae_+ .

Suppose now that $A \in M_2(\mathbb{Z})$ and $\Delta := \det A > 0$. The integral blades

$$\mathcal{B}_{-+}(A) := \Delta \mathcal{N}_{-+}(A), \quad \mathcal{B}_{+-}(A) := \Delta \mathcal{N}_{+-}(A) \quad (1884)$$

are

$$\mathcal{B}_{-+}(A) = \begin{pmatrix} bd & -b^2 \\ d^2 & -bd \end{pmatrix}, \quad (1885)$$

$$\mathcal{B}_{+-}(A) = \begin{pmatrix} -ac & a^2 \\ -c^2 & ac \end{pmatrix}. \quad (1886)$$

In particular, the first matrix depends only on the second column of A , and the second depends only on the first column. Their actions on the displayed columns are

$$\mathcal{B}_{-+}(A) Ae_+ = \Delta Ae_-, \quad \mathcal{B}_{-+}(A) Ae_- = 0, \quad (1887)$$

$$\mathcal{B}_{+-}(A) Ae_- = \Delta Ae_+, \quad \mathcal{B}_{+-}(A) Ae_+ = 0. \quad (1888)$$

Put

$$G_A := (A^{-1})^\top A^{-1}, \quad X^{\dagger A} := G_A^{-1} X^\top G_A. \quad (1889)$$

Then the two channels are exact adjoints:

$$\mathcal{B}_{-+}(A)^{\dagger A} = \mathcal{B}_{+-}(A). \quad (1890)$$

By contrast,

$$\mathcal{N}_{-+}(A)^\top = \mathcal{N}_{+-}(A) \iff A^\top A = |\det A| I_2. \quad (1891)$$

Every determinant matrix in theorem 14.167 has positive entries, so its ordinary transpose is not its reverse ordered channel.

Finally, if $a, b, c, d > 0$, then

$$\frac{d}{b} - \frac{c}{a} = \frac{\Delta}{ab} > 0. \quad (1892)$$

Hence the second carrier column lies counterclockwise from the first in the standard oriented plane, and the transported $(-, +)$ blade runs from the first column to the second.

Proof. Conjugating the four raw identities

$$J E_{21} = -E_{21}, \quad E_{21} J = E_{21}, \quad J E_{12} = E_{12}, \quad E_{12} J = -E_{12}$$

by A proves (1881) and (1882). Conjugation of $E_{22} E_{21} E_{11} = E_{21}$ and $E_{11} E_{12} E_{22} = E_{12}$ proves (1883).

The inverse matrix is

$$A^{-1} = \frac{1}{\Delta} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Direct multiplication gives (1885) and (1886), and their actions on the two columns give (1887) and (1888).

Since $G_A^{-1} = A A^\top$, one has

$$\begin{aligned} G_A^{-1} \mathcal{B}_{-+}(A)^\top G_A &= \Delta A A^\top (A^{-1})^\top E_{12} A^\top (A^{-1})^\top A^{-1} \\ &= \Delta A E_{12} A^{-1} = \mathcal{B}_{+-}(A), \end{aligned}$$

which proves (1890).

For the ordinary-transpose assertion, put $Q = A^\top A$. Multiplying $\mathcal{N}_{-+}(A)^\top = \mathcal{N}_{+-}(A)$ on the left by $A^\top$ and on the right by A gives

$$E_{12}Q = QE_{12}.$$

Writing $Q = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}$ shows that this equality holds exactly when $\beta = 0$ and $\alpha = \gamma$. Positivity of Q and $\det Q = (\det A)^2$ then give $Q = |\det A|I_2$. The converse is immediate. For every positive carrier matrix the off-diagonal entry of $A^\top A$ is $ab + cd > 0$, so the criterion fails. Equation (1892) is the determinant identity $ad - bc = \Delta$ divided by ab . $\square$

Corollary 14.169 (Explicit blade matrices of the two arithmetic carriers). *For the exterior determinant flags,*

$$\mathcal{B}_{-+}(E_p) = \begin{pmatrix} kh & -k^2 \\ h^2 & -kh \end{pmatrix}, \quad \mathcal{B}_{+-}(E_p) = \begin{pmatrix} -4pst & (4st)^2 \\ -p^2 & 4pst \end{pmatrix}, \quad (1893)$$

$$\mathcal{B}_{-+}(E_R) = \begin{pmatrix} 4stR & -(4st)^2 \\ R^2 & -4stR \end{pmatrix}, \quad \mathcal{B}_{+-}(E_R) = \begin{pmatrix} -kt & k^2 \\ -t^2 & kt \end{pmatrix}. \quad (1894)$$

For the middle determinant flags,

$$\mathcal{B}_{-+}(M_1) = \begin{pmatrix} q\ell & -q^2 \\ \ell^2 & -q\ell \end{pmatrix}, \quad \mathcal{B}_{+-}(M_1) = \begin{pmatrix} -4st & (4st)^2 \\ -1 & 4st \end{pmatrix}, \quad (1895)$$

$$\mathcal{B}_{-+}(M_p) = \begin{pmatrix} 4stR & -(4st)^2 \\ R^2 & -4stR \end{pmatrix}, \quad \mathcal{B}_{+-}(M_p) = \begin{pmatrix} -qt & q^2 \\ -t^2 & qt \end{pmatrix}, \quad (1896)$$

$$\mathcal{B}_{-+}(M_t) = \begin{pmatrix} \ell m & -\ell^2 \\ m^2 & -\ell m \end{pmatrix}, \quad \mathcal{B}_{+-}(M_t) = \begin{pmatrix} -qp & q^2 \\ -p^2 & qp \end{pmatrix}. \quad (1897)$$

Their five oriented slope gaps are

$$\frac{h}{k} - \frac{p}{4st} = \frac{1}{4stk}, \quad \frac{R}{4st} - \frac{t}{k} = \frac{1}{4stk}, \quad (1898)$$

$$\frac{\ell}{q} - \frac{1}{4st} = \frac{1}{4stq}, \quad \frac{R}{4st} - \frac{t}{q} = \frac{p}{4stq}, \quad \frac{m}{\ell} - \frac{p}{q} = \frac{t}{q\ell}. \quad (1899)$$

Whenever two determinant flags have the same ordered second column, their directed integral blades agree. Equality of first columns gives the corresponding statement for the reverse integral blades.

Proof. Substitution of the five matrices from theorem 14.167 into (1885) and (1886) gives all ten displayed matrices. Dividing each positive determinant identity by the product of the two first-row entries gives (1898) and (1899). The final assertions follow from the column dependence in (1885) and (1886). $\square$

Theorem 14.170 (Shared-column equality and split reverse defect). *Let*

$$A_x = \begin{pmatrix} x & b \\ c & d \end{pmatrix}, \quad A_y = \begin{pmatrix} y & b \\ c & d \end{pmatrix}$$

be invertible real matrices with $c \neq 0$, and put $\delta := y - x$. Their directed integral blades agree:

$$\mathcal{B}_{-+}(A_x) = \mathcal{B}_{-+}(A_y) = \begin{pmatrix} bd & -b^2 \\ d^2 & -bd \end{pmatrix}. \quad (1900)$$

The difference of their reverse integral blades is

$$\mathcal{D}_{x,y} := \mathcal{B}_{+-}(A_y) - \mathcal{B}_{+-}(A_x) = \delta \begin{pmatrix} -c & x+y \\ 0 & c \end{pmatrix}. \quad (1901)$$

It satisfies

$$\operatorname{tr} \mathcal{D}_{x,y} = 0, \quad \det \mathcal{D}_{x,y} = -\delta^2 c^2, \quad \mathcal{D}_{x,y}^2 = \delta^2 c^2 I_2. \quad (1902)$$

More precisely, for

$$U_{x,y,c} := \begin{pmatrix} 1 & \frac{x+y}{2c} \\ 0 & 1 \end{pmatrix},$$

one has

$$\mathcal{D}_{x,y} = \delta U_{x,y,c} \begin{pmatrix} -c & 0 \\ 0 & c \end{pmatrix} U_{x,y,c}^{-1}. \quad (1903)$$

For $x \neq y$, its eigenvalues are $-\delta c$ and δc , with respective eigenvectors

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} x+y \\ 2c \end{pmatrix}.$$

For $x \neq y$, set

$$\Theta_{x,y} := \frac{\mathcal{D}_{x,y}}{\delta c} = \begin{pmatrix} -1 & \frac{x+y}{c} \\ 0 & 1 \end{pmatrix}. \quad (1904)$$

Then $\Theta_{x,y}^2 = I_2$, and its two spectral support idempotents are

$$\Pi_{-,x,y} := \frac{1}{2}(I_2 - \Theta_{x,y}) = \begin{pmatrix} 1 & -\frac{x+y}{2c} \\ 0 & 0 \end{pmatrix}, \quad (1905)$$

$$\Pi_{+,x,y} := \frac{1}{2}(I_2 + \Theta_{x,y}) = \begin{pmatrix} 0 & \frac{x+y}{2c} \\ 0 & 1 \end{pmatrix}. \quad (1906)$$

They obey

$$\Pi_{\pm,x,y}^2 = \Pi_{\pm,x,y}, \quad \Pi_{-,x,y}\Pi_{+,x,y} = \Pi_{+,x,y}\Pi_{-,x,y} = 0, \quad \Pi_{-,x,y} + \Pi_{+,x,y} = I_2, \quad (1907)$$

and

$$\mathcal{D}_{x,y} = \delta c(\Pi_{+,x,y} - \Pi_{-,x,y}). \quad (1908)$$

Thus variation of the first carrier column leaves the directed $(-, +)$ blade fixed and is detected exactly by the split reverse $(+, -)$ defect.

Proof. Equation (1900) follows immediately from (1885), which depends only on the second column. Equation (1886) gives

$$\begin{aligned} \mathcal{D}_{x,y} &= \begin{pmatrix} -yc & y^2 \\ -c^2 & yc \end{pmatrix} - \begin{pmatrix} -xc & x^2 \\ -c^2 & xc \end{pmatrix} \\ &= (y-x) \begin{pmatrix} -c & x+y \\ 0 & c \end{pmatrix}. \end{aligned}$$

The trace and determinant follow from this triangular matrix. Its square is $c^2 I_2$ before multiplication by δ^2 , because its upper-right entry in the square is

$$-c(x+y) + (x+y)c = 0.$$

Direct multiplication proves (1903) and the two displayed eigenvector identities. Dividing (1902) by $\delta^2 c^2$ proves $\Theta_{x,y}^2 = I_2$. The formulas (1905) and (1906) then give (1907) and (1908) by direct multiplication. $\square$

Corollary 14.171 (Arithmetic overlap has one common blade and one split defect). *Let p be an odd prime and let*

$$a = stm, \quad R = 4a - p, \quad u = st^2$$

be positive integers. Suppose that

$$R \mid 4u + 1. \quad (1909)$$

Then

$$R \mid u + a \iff R \mid p - 1. \quad (1910)$$

When these equivalent divisibilities hold, put

$$k := \frac{4st^2 + 1}{R}, \quad q := \frac{p + 4st^2}{R}.$$

The exterior residual flag and middle prime flag

$$E_R = \begin{pmatrix} k & 4st \\ t & R \end{pmatrix}, \quad M_p = \begin{pmatrix} q & 4st \\ t & R \end{pmatrix}$$

satisfy

$$\det E_R = 1, \quad \det M_p = p, \quad q - k = \frac{p - 1}{R}.$$

Their directed blades coincide:

$$\mathcal{B}_{-+}(E_R) = \mathcal{B}_{-+}(M_p) = \begin{pmatrix} 4stR & -(4st)^2 \\ R^2 & -4stR \end{pmatrix}. \quad (1911)$$

Their reverse-channel defect is

$$\begin{aligned} \mathcal{D}_{R,p,s,t} &:= \mathcal{B}_{+-}(M_p) - \mathcal{B}_{+-}(E_R) \\ &= \frac{p - 1}{R} \begin{pmatrix} -t & q + k \\ 0 & t \end{pmatrix}, \end{aligned} \quad (1912)$$

and therefore

$$\mathcal{D}_{R,p,s,t}^2 = \left(\frac{t(p - 1)}{R} \right)^2 I_2, \quad \det \mathcal{D}_{R,p,s,t} = - \left(\frac{t(p - 1)}{R} \right)^2. \quad (1913)$$

Proof. The integer $R = 4a - p$ is odd. Under (1909),

$$4(u + a) = (4u + 1) + (4a - 1) \equiv p - 1 \pmod{R}.$$

Since 4 is a unit modulo R , this proves (1910). The determinant identities are

$$kR - 4st^2 = 1, \quad qR - 4st^2 = p,$$

and their difference gives

$$(q - k)R = p - 1.$$

Apply theorem 14.170 with

$$(x, y, b, c, d) = (k, q, 4st, t, R).$$

□

Theorem 14.172 (Half-period and mixer realization of the split defect). *Retain the data of theorem 14.170, with $x \neq y$. On the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, put*

$$g_0 := \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}, \quad (Tf)(r) := f\left(r + \frac{1}{2}\right),$$

and let

$$V_0 := \text{span}_{\mathbb{R}}\{\mathbf{1}, g_0\}.$$

Then V_0 is T -invariant,

$$T\mathbf{1} = \mathbf{1}, \quad Tg_0 = -g_0.$$

Define

$$\begin{aligned} \Phi_{x,y,c}: \mathbb{R}^2 &\longrightarrow V_0, \\ \Phi_{x,y,c} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} &:= \frac{z_2}{2c} \mathbf{1} + \left(z_1 - \frac{x + y}{2c} z_2 \right) g_0. \end{aligned} \quad (1914)$$

This is a linear isomorphism and

$$\Phi_{x,y,c}\Theta_{x,y} = T\Phi_{x,y,c}. \quad (1915)$$

For

$$P_-^{\text{hp}} := \frac{1}{2}(I - T), \quad P_+^{\text{hp}} := \frac{1}{2}(I + T),$$

one has

$$\Phi_{x,y,c}\Pi_{-,x,y} = P_-^{\text{hp}}\Phi_{x,y,c}, \quad \Phi_{x,y,c}\Pi_{+,x,y} = P_+^{\text{hp}}\Phi_{x,y,c}. \quad (1916)$$

In particular,

$$\Phi_{x,y,c}(\text{Ran } \Pi_{-,x,y}) = \mathbb{R}g_0, \quad \Phi_{x,y,c}(\text{Ran } \Pi_{+,x,y}) = \mathbb{R}\mathbf{1}.$$

In the bases (e_1, e_2) and $(\mathbf{1}, g_0)$, the matrix of $\Phi_{x,y,c}$ is

$$F_{x,y,c} = \begin{pmatrix} 0 & \frac{1}{2c} \\ 1 & -\frac{x+y}{2c} \end{pmatrix}, \quad \det F_{x,y,c} = -\frac{1}{2c}. \quad (1917)$$

Let $H, S, P_{\pm}$ be the Hadamard, sheet-exchange, and mixer-support matrices of theorem 9.14, and put

$$C_{x,y,c} := HF_{x,y,c}. \quad (1918)$$

Then

$$C_{x,y,c}\Theta_{x,y} = SC_{x,y,c}, \quad (1919)$$

$$C_{x,y,c}\Pi_{\pm,x,y} = P_{\pm}C_{x,y,c}, \quad (1920)$$

$$C_{x,y,c}\mathcal{D}_{x,y}C_{x,y,c}^{-1} = \delta c S. \quad (1921)$$

Proof. The definition of g_0 gives

$$g_0\left(r + \frac{1}{2}\right) = -g_0(r),$$

so the matrix of $T|_{V_0}$ in the ordered basis $(\mathbf{1}, g_0)$ is $J = \text{diag}(1, -1)$. Equation (1917) follows from (1914); its nonzero determinant proves bijectivity. Direct multiplication gives

$$F_{x,y,c}\Theta_{x,y} = JF_{x,y,c}.$$

This is (1915). Applying the two polynomials $\frac{1}{2}(1 - X)$ and $\frac{1}{2}(1 + X)$ to the intertwined involutions proves (1916).

The existing Hadamard identities are

$$HJ = SH, \quad HE_{11} = P_+H, \quad HE_{22} = P_-H.$$

Multiplying the finite-dimensional intertwining identities by H proves (1919) and (1920). Finally, $\mathcal{D}_{x,y} = \delta c \Theta_{x,y}$ gives (1921). $\square$

Corollary 14.173 (Coupled packets are two-sheeted flag families). *Retain the conditions of theorem 14.74, and assume $R \mid p - 1$. The map*

$$\begin{aligned} \pi_{R,a} : \mathcal{S}_{R,a}^{\text{ext}} &\longrightarrow \mathcal{T}_{R,a}, \\ (s, t, m) &\longmapsto (s, \{t, m\}) \end{aligned} \quad (1922)$$

is a two-sheeted surjection. Its deck involution is

$$\iota(s, t, m) := (s, m, t), \quad (1923)$$

and it acts freely.

For $\sigma = (s, t, m) \in \mathcal{S}_{R,a}^{\text{ext}}$, put

$$u_{\sigma} := st^2, \quad k_{\sigma} := \frac{4st^2 + 1}{R}, \quad q_{\sigma} := \frac{p + 4st^2}{R}, \quad (1924)$$

and define

$$E_\sigma := \begin{pmatrix} k_\sigma & 4st \\ t & R \end{pmatrix}, \quad M_\sigma := \begin{pmatrix} q_\sigma & 4st \\ t & R \end{pmatrix}. \quad (1925)$$

Then

$$\mathcal{B}_{-+}(E_\sigma) = \mathcal{B}_{-+}(M_\sigma), \quad (1926)$$

whereas the reverse-channel defect is

$$\mathcal{D}_\sigma := \mathcal{B}_{+-}(M_\sigma) - \mathcal{B}_{+-}(E_\sigma) = \frac{p-1}{R} \begin{pmatrix} -t & q_\sigma + k_\sigma \\ 0 & t \end{pmatrix}. \quad (1927)$$

The complementary sheet $\iota\sigma = (s, m, t)$ carries the same formulas with t and m exchanged. For each of the two sheets, the maps

$$\Theta_\sigma := \frac{R}{t(p-1)} \mathcal{D}_\sigma, \quad \Phi_\sigma := \Phi_{k_\sigma, q_\sigma, t}, \quad C_\sigma := C_{k_\sigma, q_\sigma, t} \quad (1928)$$

identify its split involution with the half-period action and the mixer exchange:

$$\Phi_\sigma \Theta_\sigma = T \Phi_\sigma, \quad C_\sigma \Theta_\sigma = S C_\sigma. \quad (1929)$$

Every fiber $\{\sigma, \iota\sigma\}$ supplies exactly the three complement orbits

$$\begin{aligned} \{(2, st^2), (0, sm^2)\}, \quad \{(2, sm^2), (0, st^2)\}, \\ \{(1, st^2), (1, sm^2)\}. \end{aligned} \quad (1930)$$

Consequently,

$$\boxed{\begin{aligned} |\mathcal{S}_{R,a}^{\text{ext}}| &= |\mathcal{E}_{R,a}| = c_{R,a} = 2|\mathcal{T}_{R,a}|, \\ q_{R,p} &= 3|\mathcal{T}_{R,a}| = \frac{3}{2}|\mathcal{S}_{R,a}^{\text{ext}}|. \end{aligned}} \quad (1931)$$

In particular, one shared-column arithmetic carrier forces

$$q_{R,p} \geq 3, \quad Q_p \geq 3, \quad (1932)$$

and therefore proves the Erdős–Strauss identity at p .

Proof. Under $R \mid p-1$, (1614) identifies the exterior condition with $R \mid t+m$, and it is unchanged by $t \leftrightarrow m$. A fixed point would have $t = m$, hence $R \mid 2t$; this is impossible because $R \geq 3$ is odd and $\gcd(R, t) = 1$. Thus (1922) has the asserted fibers.

The flag and defect identities follow from theorem 14.171; their half-period and mixer realizations follow from theorem 14.172. The squarefree coordinate bijections in theorem 14.59 identify the two ordered sheets with the two exterior coordinates and their quotient with the middle complement orbit. The three-orbit description in theorem 14.74 now gives (1930). Finally, Theorem 14.79 gives $c_{R,a} = |\mathcal{M}_{R,a}|$. The middle complement action is free, and the middle-coordinate bijection identifies its orbit set with $\mathcal{T}_{R,a}$. Hence

$$c_{R,a} = |\mathcal{M}_{R,a}| = 2|\mathcal{T}_{R,a}|, \quad q_{R,p} = |\mathcal{E}_{R,a}| + \frac{1}{2}c_{R,a},$$

which proves (1931). A nonempty fiber has one packet, proving (1932) and the final assertion. $\square$

Theorem 14.174 (A coupled packet carries both six-state laws). *Retain the conditions of theorem 14.173. Fix*

$$P = (s, \{t, m\}) \in \mathcal{T}_{R,a},$$

and let $\pi_{R,a}^{-1}(P) = \{\sigma, \iota\sigma\}$. For $\rho = (s, x, y)$ in this two-element fiber, put

$$u_\rho := sx^2, \quad u_{\iota\rho} := sy^2 = \frac{a^2}{u_\rho}. \quad (1933)$$

The six arithmetic cells supplied by P are exactly

$$\mathcal{X}(P) := \left\{ (j, u_\rho) : j \in \mathbb{Z}/3\mathbb{Z}, \rho \in \pi_{R,a}^{-1}(P) \right\} \subset \mathcal{X}_{R,P}. \quad (1934)$$

Define permutations of $\mathcal{X}(P)$ by

$$r_P(j, u_\rho) := (j + 1, u_\rho), \quad (1935)$$

$$\alpha_P(j, u_\rho) := (j, u_{\iota\rho}), \quad (1936)$$

$$\kappa_P(j, u_\rho) := (2 - j, u_{\iota\rho}). \quad (1937)$$

Then

$$r_P^3 = \alpha_P^2 = \kappa_P^2 = 1, \quad \alpha_P r_P = r_P \alpha_P, \quad \kappa_P r_P \kappa_P = r_P^{-1}. \quad (1938)$$

Consequently $g_P := r_P \alpha_P$ generates a regular C_6 -action on $\mathcal{X}(P)$, while r_P, κ_P generate a regular D_3 -action on the same six cells.

Let

$$\mathcal{H}_P := \mathbb{C}[\mathcal{X}(P)] = \mathcal{H}_\sigma \oplus \mathcal{H}_{\iota\sigma}, \quad \mathcal{H}_\rho := \text{span}\{e_{0,\rho}, e_{1,\rho}, e_{2,\rho}\}.$$

In the ordered basis

$$(e_{0,\sigma}, e_{1,\sigma}, e_{2,\sigma}, e_{0,\iota\sigma}, e_{1,\iota\sigma}, e_{2,\iota\sigma}),$$

the three permutations have the matrices

$$\mathbf{R}_P = \begin{pmatrix} R_3 & 0 \\ 0 & R_3 \end{pmatrix}, \quad \mathbf{A}_P = \begin{pmatrix} 0 & I_3 \\ I_3 & 0 \end{pmatrix}, \quad \mathbf{K}_P = \begin{pmatrix} 0 & F_3 \\ F_3 & 0 \end{pmatrix}, \quad (1939)$$

where

$$R_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad F_3 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \quad (1940)$$

The unitary

$$A_\sigma : \mathcal{H}_\sigma \longrightarrow \mathcal{H}_{\iota\sigma}, \quad A_\sigma e_{j,\sigma} := e_{j,\iota\sigma} \quad (1941)$$

satisfies

$$R_{\iota\sigma} A_\sigma = A_\sigma R_\sigma. \quad (1942)$$

Its two-sheet coupling operator from theorem 8.3 is precisely $\mathbf{A}_P$. In particular,

$$\mathbf{A}_P = \mathbf{A}_P^*, \quad \mathbf{A}_P^2 = I_{\mathcal{H}_P} \succeq 0, \quad \mathbf{A}_P \mathbf{R}_P = \mathbf{R}_P \mathbf{A}_P. \quad (1943)$$

Proof. Write

$$u := st^2, \quad v := sm^2.$$

The three complement orbits in (1930) contain respectively

$$\{(2, u), (0, v)\}, \quad \{(2, v), (0, u)\}, \quad \{(1, u), (1, v)\}.$$

Their union is

$$\{(j, u) : j \in \mathbb{Z}/3\mathbb{Z}\} \sqcup \{(j, v) : j \in \mathbb{Z}/3\mathbb{Z}\},$$

which proves (1934). Equation (1933) follows from

$$\frac{a^2}{st^2} = \frac{s^2 t^2 m^2}{st^2} = sm^2.$$

Hence κ_P is the restriction of the arithmetic complement (375).

The first two relations in (1938) follow directly from addition modulo 3 and the free deck involution. The pure sheet exchange leaves the origin coordinate fixed, so it commutes with r_P . Finally,

$$\kappa_P r_P \kappa_P(j, u_\rho) = (j - 1, u_\rho) = r_P^{-1}(j, u_\rho).$$

For $0 \leq n < 6$,

$$g_P^n(j, u_\rho) = (j + n, u_{\iota^n \rho}). \quad (1944)$$

The residue of n modulo 3 and its parity determine n modulo 6, so the six displayed points are distinct. Thus the C_6 -action is regular. The six elements

$$r_P^b \kappa_P^\epsilon, \quad b \in \mathbb{Z}/3\mathbb{Z}, \quad \epsilon \in \{0, 1\},$$

send a fixed cell to three cells on its own sheet when $\epsilon = 0$, and to three cells on the other sheet when $\epsilon = 1$. Their images are distinct, proving regularity of the D_3 -action.

The matrices in (1939) are the permutation matrices of (1935)–(1937). The identities

$$R_3^3 = F_3^2 = I_3, \quad F_3 R_3 F_3 = R_3^{-1}$$

give the same relations by matrix multiplication. Equation (1942) follows because A_σ preserves j . Since A_σ is unitary, the block operator in theorem 8.3 is $\mathbf{A}_P$, and its square is the identity. This proves (1943). $\square$

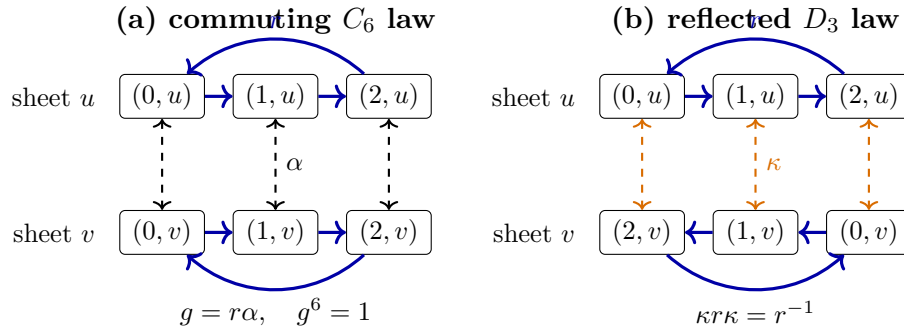


FIGURE 14. The same six arithmetic cells support two different exact couplings. Blue arrows are the origin rotation r . In (a), same-origin sheet exchange α commutes with r , and $r\alpha$ generates the regular C_6 -action. In (b), arithmetic complement κ exchanges the sheets while reflecting the origin coordinate, producing the regular D_3 -action.

Corollary 14.175 (Arithmetic exhaustion and spectral splitting of the packet couplings). *Retain the conditions of theorem 14.174, choose one ordered sheet σ , and put*

$$\mathcal{C} = C_3 \times C_2.$$

The map

$$\Psi_\sigma : \mathcal{C} \longrightarrow \mathcal{X}(P), \quad \Psi_\sigma(a, \epsilon) := (a, u_{\epsilon\sigma}) \quad (1945)$$

is a bijection. Let $\circ$ and $\star$ be the commuting and reflected products from (62) and (63). If

$$x = (a, \epsilon), \quad y = (b, \delta), \quad \beta_P := r_P \kappa_P,$$

then

$$\Psi_\sigma(x \circ y) = r_P^a \alpha_P^\epsilon \Psi_\sigma(y), \quad (1946)$$

$$\Psi_\sigma(x \star y) = r_P^a \beta_P^\epsilon \Psi_\sigma(y). \quad (1947)$$

Consequently these two arithmetic actions realize, and exhaust, the two homomorphisms

$$C_2 \longrightarrow \text{Aut}(C_3).$$

On $\mathcal{H}_P$, put

$$\mathbf{P}_{P,\pm} := \frac{1}{2}(I_{\mathcal{H}_P} \pm \mathbf{A}_P), \quad \mathcal{H}_P^\pm := \text{im } \mathbf{P}_{P,\pm}. \quad (1948)$$

These are orthogonal projections of rank three. They commute with $\mathbf{R}_P$ and $\mathbf{K}_P$. For

$$\omega := e^{2\pi i/3}, \quad \mathbf{G}_P := \mathbf{R}_P \mathbf{A}_P,$$

one has

$$\text{Spec}(\mathbf{G}_P) = \{1, -1, \omega, -\omega, \omega^2, -\omega^2\}, \quad \det(XI_6 - \mathbf{G}_P) = X^6 - 1. \quad (1949)$$

Let V denote the two-dimensional irreducible complex representation of D_3 . Under the reflected packet action,

$$\mathcal{H}_P^+ \cong \mathbf{1} \oplus V, \quad \mathcal{H}_P^- \cong \text{sgn} \oplus V, \quad \mathcal{H}_P \cong \mathbf{1} \oplus \text{sgn} \oplus V^{\oplus 2}. \quad (1950)$$

The trivial and sign lines are spanned respectively by

$$c_P := \sum_{j=0}^2 (e_{j,\sigma} + e_{j,\iota\sigma}), \quad s_P := \sum_{j=0}^2 (e_{j,\sigma} - e_{j,\iota\sigma}). \quad (1951)$$

Proof. The two sheet endpoints in (1933) are distinct, and each occurs once at every origin. Hence (1945) is a bijection. For the commuting product,

$$r_P^a \alpha_P^\epsilon(b, u_{\iota^\epsilon \sigma}) = (a + b, u_{\iota^\epsilon + \delta \sigma}),$$

which is (1946). Since

$$\beta_P(j, u_\rho) = r_P(2 - j, u_{\iota\rho}) = (-j, u_{\iota\rho}),$$

one likewise has

$$r_P^a \beta_P^\epsilon(b, u_{\iota^\epsilon \sigma}) = (a + (-1)^\epsilon b, u_{\iota^\epsilon + \delta \sigma}),$$

which proves (1947). Theorem 8.1 proves exhaustion.

Equation (1943) makes $\mathbf{A}_P$ a self-adjoint involution, so $\mathbf{P}_{P,\pm}$ are orthogonal projections. The matrix $\mathbf{A}_P$ has zero diagonal and hence zero trace. Therefore

$$\text{tr } \mathbf{P}_{P,+} = \text{tr } \mathbf{P}_{P,-} = 3,$$

which gives their ranks. The first commutation follows from (1943); the second follows from

$$\mathbf{A}_P \mathbf{K}_P = \mathbf{K}_P \mathbf{A}_P = \begin{pmatrix} F_3 & 0 \\ 0 & F_3 \end{pmatrix}. \quad (1952)$$

The eigenvalues of R_3 are $1, \omega, \omega^2$. For each such eigenvalue, the corresponding two-dimensional eigenspace of $\mathbf{R}_P$ contains one $+1$ and one -1 eigenline of $\mathbf{A}_P$, because $\mathbf{A}_P$ exchanges the two sheets. Thus $\mathbf{G}_P = \mathbf{R}_P \mathbf{A}_P$ has the six eigenvalues in (1949), all with multiplicity one. Their product polynomial is $X^6 - 1$.

The origin rotation fixes both vectors in (1951). Arithmetic complement fixes c_P and sends s_P to $-s_P$. Thus these vectors span the trivial and sign representations. The orthogonal complement of $\mathbb{C}c_P$ in $\mathcal{H}_P^+$ has dimension two, as does the orthogonal complement of $\mathbb{C}s_P$ in $\mathcal{H}_P^-$. On each complement, $\mathbf{R}_P$ has eigenvalues ω, ω^2 . Every one-dimensional representation of D_3 sends the order-three generator to 1, so neither complement contains a one-dimensional subrepresentation. Each is therefore isomorphic to V , proving (1950). $\square$

Theorem 14.176 (Packet parity is the tensor lift of the ordered blade supports). *Retain the conditions of theorem 14.174 and choose one ordered sheet σ . Let*

$$\mathcal{S}_P := \text{span}_{\mathbb{C}}\{\eta_\sigma, \eta_{\iota\sigma}\}, \quad \mathcal{O}_3 := \text{span}_{\mathbb{C}}\{f_0, f_1, f_2\},$$

and define the unitary

$$\Upsilon_\sigma : \mathcal{S}_P \otimes \mathcal{O}_3 \longrightarrow \mathcal{H}_P, \quad \Upsilon_\sigma(\eta_\rho \otimes f_j) := e_{j,\rho}. \quad (1953)$$

Then

$$\begin{aligned} \Upsilon_\sigma^{-1} \mathbf{A}_P \Upsilon_\sigma &= S \otimes I_3, \\ \Upsilon_\sigma^{-1} \mathbf{R}_P \Upsilon_\sigma &= I_2 \otimes R_3, \\ \Upsilon_\sigma^{-1} \mathbf{K}_P \Upsilon_\sigma &= S \otimes F_3. \end{aligned} \quad (1954)$$

Put

$$\eta_{\pm} := \frac{\eta_{\sigma} \pm \eta_{\iota\sigma}}{\sqrt{2}},$$

and define the lifted ordered blades

$$\mathbb{B}_{P,-+} := \Upsilon_{\sigma}((\eta_{-}\eta_{+}^{*}) \otimes I_3) \Upsilon_{\sigma}^{-1}, \quad (1955)$$

$$\mathbb{B}_{P,+-} := \Upsilon_{\sigma}((\eta_{+}\eta_{-}^{*}) \otimes I_3) \Upsilon_{\sigma}^{-1}. \quad (1956)$$

They are rank-three partial isometries with exact source and range supports

$$\mathbb{B}_{P,-+}^{*} \mathbb{B}_{P,-+} = \mathbf{P}_{P,+}, \quad \mathbb{B}_{P,-+} \mathbb{B}_{P,-+}^{*} = \mathbf{P}_{P,-}, \quad (1957)$$

$$\mathbb{B}_{P,+-}^{*} \mathbb{B}_{P,+-} = \mathbf{P}_{P,-}, \quad \mathbb{B}_{P,+-} \mathbb{B}_{P,+-}^{*} = \mathbf{P}_{P,+}. \quad (1958)$$

Their ordered signs and origin covariance are

$$\mathbf{A}_P \mathbb{B}_{P,-+} = -\mathbb{B}_{P,-+}, \quad \mathbb{B}_{P,-+} \mathbf{A}_P = \mathbb{B}_{P,-+}, \quad (1959)$$

$$\mathbf{A}_P \mathbb{B}_{P,+-} = \mathbb{B}_{P,+-}, \quad \mathbb{B}_{P,+-} \mathbf{A}_P = -\mathbb{B}_{P,+-}, \quad (1960)$$

$$\mathbf{R}_P \mathbb{B}_{P,-+} = \mathbb{B}_{P,-+} \mathbf{R}_P, \quad \mathbf{R}_P \mathbb{B}_{P,+-} = \mathbb{B}_{P,+-} \mathbf{R}_P. \quad (1961)$$

Let $U_{\sigma} : \mathbb{C}^2 \rightarrow \mathcal{S}_P$ send the standard basis to $(\eta_{\sigma}, \eta_{\iota\sigma})$, and let C_{σ} and Θ_{σ} be the exact flag maps in (1928). Define

$$\mathfrak{C}_{\sigma} := \Upsilon_{\sigma} \circ ((U_{\sigma} C_{\sigma}) \otimes I_3). \quad (1962)$$

Writing

$$\Pi_{\sigma,\pm} := \Pi_{\pm,k_{\sigma},q_{\sigma}},$$

with the third parameter $c = t$, one has

$$\mathfrak{C}_{\sigma}(\Theta_{\sigma} \otimes I_3) = \mathbf{A}_P \mathfrak{C}_{\sigma}, \quad (1963)$$

$$\mathfrak{C}_{\sigma}(\Pi_{\sigma,\pm} \otimes I_3) = \mathbf{P}_{P,\pm} \mathfrak{C}_{\sigma}, \quad (1964)$$

$$\mathfrak{C}_{\sigma}(\mathcal{D}_{\sigma} \otimes I_3) \mathfrak{C}_{\sigma}^{-1} = \frac{(p-1)t}{R} \mathbf{A}_P. \quad (1965)$$

For the complementary sheet $\iota\sigma = (s, m, t)$, the corresponding map obeys

$$\mathfrak{C}_{\iota\sigma}(\mathcal{D}_{\iota\sigma} \otimes I_3) \mathfrak{C}_{\iota\sigma}^{-1} = \frac{(p-1)m}{R} \mathbf{A}_P. \quad (1966)$$

Thus both flag defects determine the same packet involution, with their positive arithmetic scales retained.

Finally, reversing the chosen sheet order fixes $\mathbf{P}_{P,+}$ and $\mathbf{P}_{P,-}$, but sends

$$\mathbb{B}_{P,-+} \longmapsto -\mathbb{B}_{P,-+}, \quad \mathbb{B}_{P,+-} \longmapsto -\mathbb{B}_{P,+-}. \quad (1967)$$

Proof. Equation (1954) follows directly from the three permutation matrices in (1939). Under Υ_{σ} , the packet parity projections are

$$\mathbf{P}_{P,\pm} = (\eta_{\pm}\eta_{\pm}^{*}) \otimes I_3.$$

The source and range products in (1957) and (1958) are therefore the rank-one identities from (161) and (162), tensored with I_3 . The bimodule signs follow in the same way from (164) and (165). Both lifted blades act as the identity on $\mathcal{O}_3$, which proves (1961).

The sheet-basis map satisfies

$$U_{\sigma} S U_{\sigma}^{-1} = S_{\mathcal{S}_P},$$

where $S_{\mathcal{S}_P}$ exchanges η_{σ} and $\eta_{\iota\sigma}$. Equation (1929) consequently gives

$$(U_{\sigma} C_{\sigma}) \Theta_{\sigma} = S_{\mathcal{S}_P} (U_{\sigma} C_{\sigma}).$$

Tensoring with I_3 and applying Υ_σ proves (1963). Applying $\frac{1}{2}(1 \pm X)$ proves (1964). From (1927) and (1928),

$$\mathcal{D}_\sigma = \frac{(p-1)t}{R} \Theta_\sigma.$$

This proves (1965); exchanging t and m proves (1966).

Under $\sigma \leftrightarrow \iota\sigma$, the vector η_+ is fixed and η_- changes sign. The two parity projectors are therefore fixed, while each rank-one ordered blade changes sign. This proves (1967). $\square$

Theorem 14.177 (Exact transition between the complementary packet flags). *Retain the conditions of theorem 14.176. Write*

$$h_\sigma := k_\sigma + q_\sigma, \quad h_{\iota\sigma} := k_{\iota\sigma} + q_{\iota\sigma}, \quad h_P := h_\sigma + h_{\iota\sigma},$$

where $\sigma = (s, t, m)$ and $\iota\sigma = (s, m, t)$. The transition from the σ -flag coordinates to the $\iota\sigma$ -flag coordinates is

$$\begin{aligned} \mathcal{T}_{\sigma \rightarrow \iota\sigma} &:= \mathfrak{C}_{\iota\sigma}^{-1} \mathfrak{C}_\sigma = T_{\sigma \rightarrow \iota\sigma} \otimes I_3, \\ T_{\sigma \rightarrow \iota\sigma} &:= C_{\iota\sigma}^{-1} S C_\sigma = \begin{pmatrix} -1 & \frac{h_P}{2m} \\ 0 & \frac{2t}{m} \end{pmatrix}. \end{aligned} \quad (1968)$$

The reverse transition is

$$T_{\iota\sigma \rightarrow \sigma} = \begin{pmatrix} -1 & \frac{h_P}{2m} \\ 0 & \frac{2t}{m} \end{pmatrix} = T_{\sigma \rightarrow \iota\sigma}^{-1}. \quad (1969)$$

In particular,

$$\det T_{\sigma \rightarrow \iota\sigma} = -\frac{m}{t}, \quad \text{Spec}(T_{\sigma \rightarrow \iota\sigma}) = \left\{ -1, \frac{m}{t} \right\}. \quad (1970)$$

The two eigenlines are generated by

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} \frac{h_P}{2(t+m)} \\ 1 \end{pmatrix}, \quad (1971)$$

with eigenvalues -1 and m/t , respectively. Since the deck action is free, $t \neq m$. On the six-dimensional flag carrier, -1 and m/t each have multiplicity three and

$$\det \mathcal{T}_{\sigma \rightarrow \iota\sigma} = -\left(\frac{m}{t}\right)^3.$$

Put

$$\Pi_{\rho, \pm} := \Pi_{\pm, k_\rho, q_\rho} \quad (\rho \in \{\sigma, \iota\sigma\}).$$

The transition preserves the ordered support labels:

$$\mathcal{T}_{\sigma \rightarrow \iota\sigma}(\Theta_\sigma \otimes I_3) = (\Theta_{\iota\sigma} \otimes I_3) \mathcal{T}_{\sigma \rightarrow \iota\sigma}, \quad (1972)$$

$$\mathcal{T}_{\sigma \rightarrow \iota\sigma}(\Pi_{\sigma, \pm} \otimes I_3) = (\Pi_{\iota\sigma, \pm} \otimes I_3) \mathcal{T}_{\sigma \rightarrow \iota\sigma}. \quad (1973)$$

The reverse-channel defects retain their unequal arithmetic scales:

$$\mathcal{T}_{\sigma \rightarrow \iota\sigma}(\mathcal{D}_\sigma \otimes I_3) = \frac{t}{m}(\mathcal{D}_{\iota\sigma} \otimes I_3) \mathcal{T}_{\sigma \rightarrow \iota\sigma}. \quad (1974)$$

The corresponding transport on packet space is

$$\mathcal{P}_{\sigma \rightarrow \iota\sigma} := \mathfrak{C}_{\iota\sigma} \mathfrak{C}_\sigma^{-1}.$$

Put

$$\widehat{\Upsilon}_\sigma := \Upsilon_\sigma \circ (U_\sigma \otimes I_3).$$

In the ordered sheet basis $(\eta_\sigma, \eta_{\iota\sigma})$, put

$$r := \frac{t}{m}, \quad d := h_\sigma - rh_{\iota\sigma}.$$

Then

$$\hat{\Upsilon}_\sigma^{-1} \mathcal{P}_{\sigma \rightarrow \iota\sigma} \hat{\Upsilon}_\sigma = L_{\sigma \rightarrow \iota\sigma} \otimes I_3, \quad L_{\sigma \rightarrow \iota\sigma} = \frac{1}{2} \begin{pmatrix} r-d-1 & r-d+1 \\ r+d+1 & r+d-1 \end{pmatrix}. \quad (1975)$$

Consequently,

$$\det L_{\sigma \rightarrow \iota\sigma} = -\frac{t}{m}, \quad \text{Spec}(L_{\sigma \rightarrow \iota\sigma}) = \left\{ -1, \frac{t}{m} \right\}. \quad (1976)$$

Thus both the flag-coordinate transition and the packet-space transport have one negative direction and one positive scale direction; the scale is reciprocal between the two realizations.

Proof. For $h = x + y$, the matrices in (1917) obey

$$F_{x,y,c} = \begin{pmatrix} 0 & \frac{1}{2c} \\ 1 & -\frac{h}{2c} \end{pmatrix}, \quad F_{x,y,c}^{-1} = \begin{pmatrix} h & 1 \\ 2c & 0 \end{pmatrix}.$$

The unitary in (1953) is unchanged as an actual map when the two sheet labels are reversed, while $U_{\iota\sigma} = U_\sigma S$. Hence

$$\mathfrak{C}_{\iota\sigma}^{-1} \mathfrak{C}_\sigma = (C_{\iota\sigma}^{-1} S C_\sigma) \otimes I_3.$$

Using $C_\rho = H F_\rho$ and $H S H = J$ gives

$$C_{\iota\sigma}^{-1} S C_\sigma = F_{\iota\sigma}^{-1} J F_\sigma.$$

Substitution of the two displayed inverse matrices gives (1968). Exchanging t and m gives (1969); direct multiplication gives both inverse identities. The matrix is upper triangular, which proves (1970). Applying it to the two vectors in (1971) proves the eigenline statement. Tensoring with I_3 gives the multiplicities and determinant on the full flag carrier.

From (1929),

$$T_{\sigma \rightarrow \iota\sigma} \Theta_\sigma = C_{\iota\sigma}^{-1} S C_\sigma \Theta_\sigma = C_{\iota\sigma}^{-1} C_\sigma = \Theta_{\iota\sigma} T_{\sigma \rightarrow \iota\sigma}.$$

This proves (1972). Applying $\frac{1}{2}(1 \pm X)$ proves (1973). Finally,

$$\mathcal{D}_\sigma = \frac{(p-1)t}{R} \Theta_\sigma, \quad \mathcal{D}_{\iota\sigma} = \frac{(p-1)m}{R} \Theta_{\iota\sigma},$$

so the involution intertwining gives (1974).

For the packet-space transport,

$$\hat{\Upsilon}_\sigma^{-1} \mathcal{P}_{\sigma \rightarrow \iota\sigma} \hat{\Upsilon}_\sigma = (S C_{\iota\sigma} C_\sigma^{-1}) \otimes I_3.$$

The intermediate flag product is

$$F_{\iota\sigma} F_\sigma^{-1} = \begin{pmatrix} r & 0 \\ d & 1 \end{pmatrix}.$$

Since $S H = H J$,

$$S C_{\iota\sigma} C_\sigma^{-1} = H J \begin{pmatrix} r & 0 \\ d & 1 \end{pmatrix} H,$$

and multiplication gives the matrix in (1975). Its trace is $r - 1$ and its determinant is $-r$, proving (1976). $\square$

Theorem 14.178 (Canonical fixed line of an unordered arithmetic packet). *Retain the conditions of theorem 14.177. Put*

$$\delta_P := \frac{p-1}{R}, \quad \ell_P := \frac{t+m}{R}. \quad (1977)$$

These are positive integers. The positive transition eigenline has the order-independent coordinate

$$\gamma_P := \frac{h_P}{2(t+m)} = 4s\ell_P - \frac{\delta_P + 2}{\ell_P R}, \quad w_P := \begin{pmatrix} \gamma_P \\ 1 \end{pmatrix}. \quad (1978)$$

In particular,

$$0 < \gamma_P < 4s\ell_P,$$

and

$$T_{\sigma \rightarrow \iota\sigma} w_P = \frac{m}{t} w_P, \quad T_{\iota\sigma \rightarrow \sigma} w_P = \frac{t}{m} w_P. \quad (1979)$$

Define the oriented coordinate

$$\mu_\sigma := \frac{(m-t)(\delta_P + 2)}{t+m}. \quad (1980)$$

Then $\mu_{\iota\sigma} = -\mu_\sigma$, and the vector

$$v_P := (1 + \mu_\sigma)\eta_\sigma + (1 - \mu_\sigma)\eta_{\iota\sigma} \in \mathcal{S}_P \quad (1981)$$

is unchanged when the ordered sheet σ is replaced by $\iota\sigma$. The two flag charts map w_P to the same line in $\mathcal{S}_P$, with exact scale factors

$$U_\sigma C_\sigma w_P = \frac{1}{2\sqrt{2}t} v_P, \quad U_{\iota\sigma} C_{\iota\sigma} w_P = \frac{1}{2\sqrt{2}m} v_P. \quad (1982)$$

Let Q_P^{fix} be the rank-one orthogonal projection onto $\mathbb{C}v_P$, and lift it to the six-dimensional packet carrier by

$$\mathbb{Q}_P^{\text{fix}} := \Upsilon_\sigma(Q_P^{\text{fix}} \otimes I_3)\Upsilon_\sigma^{-1}.$$

This rank-three orthogonal projection is independent of the chosen ordered sheet and has the exact ordered-blade expansion

$$\mathbb{Q}_P^{\text{fix}} = \frac{1}{1 + \mu_\sigma^2} \left(\mathbf{P}_{P,+} + \mu_\sigma(\mathbb{B}_{P,-+} + \mathbb{B}_{P,+-}) + \mu_\sigma^2 \mathbf{P}_{P,-} \right). \quad (1983)$$

It commutes with the arithmetic-origin rotation. Its commutator with sheet exchange is

$$[\mathbf{A}_P, \mathbb{Q}_P^{\text{fix}}] = \frac{2\mu_\sigma}{1 + \mu_\sigma^2} (\mathbb{B}_{P,+-} - \mathbb{B}_{P,-+}) \neq 0. \quad (1984)$$

Thus the order-independent packet line is neither pure positive parity nor pure negative parity: its exact off-diagonal part is carried by the two ordered blade channels.

Proof. The coupled-shell conditions give

$$R \mid p-1, \quad R \mid t+m,$$

which proves the integrality in (1977). Since

$$p = 4stm - R, \quad t+m = \ell_P R,$$

the sheet data in (1924) satisfy

$$q_\sigma = \frac{p + 4st^2}{R} = 4s\ell_P t - 1,$$

$$q_{\iota\sigma} = \frac{p + 4sm^2}{R} = 4s\ell_P m - 1,$$

and

$$q_\rho - k_\rho = \frac{p-1}{R} = \delta_P \quad (\rho \in \{\sigma, \iota\sigma\}).$$

Therefore

$$h_\rho = 2q_\rho - \delta_P$$

and

$$h_P = 2(4s\ell_P^2 R - \delta_P - 2). \quad (1985)$$

Division by $2(t + m) = 2\ell_P R$ proves (1978). Its positivity follows from $h_P > 0$; the displayed correction to $4s\ell_P$ is positive, which proves the strict upper bound. Equation (1979) follows from (1968) and (1969).

Apply the two matrices $F_\sigma = F_{k_\sigma, q_\sigma, t}$ and $F_{l\sigma} = F_{k_{l\sigma}, q_{l\sigma}, m}$ to w_P . Using $h_P = h_\sigma + h_{l\sigma}$ gives

$$\begin{aligned} F_\sigma w_P &= \frac{1}{2t} \begin{pmatrix} 1 \\ \frac{th_{l\sigma} - mh_\sigma}{t + m} \end{pmatrix}, \\ F_{l\sigma} w_P &= \frac{1}{2m} \begin{pmatrix} 1 \\ \frac{mh_\sigma - th_{l\sigma}}{t + m} \end{pmatrix}. \end{aligned}$$

The formulas just proved for the two heights give

$$th_{l\sigma} - mh_\sigma = (m - t)(\delta_P + 2).$$

Hence

$$F_\sigma w_P = \frac{1}{2t} \begin{pmatrix} 1 \\ \mu_\sigma \end{pmatrix}, \quad F_{l\sigma} w_P = \frac{1}{2m} \begin{pmatrix} 1 \\ -\mu_\sigma \end{pmatrix}. \quad (1986)$$

Applying the Hadamard matrix H , followed by the two ordered sheet maps, proves (1982). Exchanging t and m changes the sign of μ_σ and exchanges $\eta_\sigma, \eta_{l\sigma}$, so (1981) is unchanged.

In the parity basis,

$$v_P = \sqrt{2}(\eta_+ + \mu_\sigma \eta_-), \quad \langle v_P, v_P \rangle = 2(1 + \mu_\sigma^2).$$

The rank-one projection onto this line is therefore

$$Q_P^{\text{fix}} = \frac{1}{1 + \mu_\sigma^2} (\eta_+ \eta_+^* + \mu_\sigma (\eta_- \eta_+^* + \eta_+ \eta_-^*) + \mu_\sigma^2 \eta_- \eta_-^*).$$

Tensoring with I_3 and applying Υ_σ proves (1983). Each term commutes with $\mathbf{R}_P$. Reversing the sheet order changes the signs of both μ_σ and the two directed blades, and hence fixes the displayed projection.

In the parity basis, sheet exchange is $\text{diag}(1, -1)$. Direct multiplication gives

$$[\mathbf{A}_P, \mathbb{Q}_P^{\text{fix}}] = \frac{2\mu_\sigma}{1 + \mu_\sigma^2} (\mathbb{B}_{P,+-} - \mathbb{B}_{P,-+}).$$

The packet action is free, so $t \neq m$, and therefore $\mu_\sigma \neq 0$. This proves (1984). $\square$

Theorem 14.179 (Square-discriminant reconstruction and the packet quarter). *Retain the conditions of theorem 14.173. Define*

$$\mathcal{D}_{R,a} := \left\{ (s, \ell) : \begin{array}{l} s \mid a, \quad s \text{ squarefree}, \quad \ell \in \mathbb{Z}_{>0}, \\ D_{s,\ell} := \ell^2 R^2 - \frac{4a}{s} \text{ is a positive square} \end{array} \right\}. \quad (1987)$$

The map

$$\begin{aligned} \Phi_{R,a} : \mathcal{T}_{R,a} &\longrightarrow \mathcal{D}_{R,a}, \\ (s, \{t, m\}) &\longmapsto \left(s, \frac{t + m}{R} \right) \end{aligned} \quad (1988)$$

is a bijection. If $d_{s,\ell} := \sqrt{D_{s,\ell}} > 0$, its inverse is

$$(s, \ell) \mapsto \left(s, \left\{ \frac{\ell R - d_{s,\ell}}{2}, \frac{\ell R + d_{s,\ell}}{2} \right\} \right). \quad (1989)$$

Every pair in $\mathcal{D}_{R,a}$ satisfies

$$\ell R \leq \frac{a}{s} + 1. \quad (1990)$$

Consequently,

$$|\mathcal{T}_{R,a}| = \sum_{\substack{s|a \\ s \text{ squarefree}}} \# \left\{ \ell \in \mathbb{Z}_{>0} : \begin{array}{l} \ell R \leq a/s + 1, \\ \ell^2 R^2 - 4a/s \text{ is a positive square} \end{array} \right\}. \quad (1991)$$

For an ordered sheet $\sigma = (s, t, m)$ over $P = (s, \{t, m\})$, put

$$\vartheta_\sigma := \frac{t}{t+m}, \quad r_\sigma := \frac{t-m}{t+m}, \quad \zeta_P := -\frac{tm}{(t+m)^2}. \quad (1992)$$

Then

$$D_{s,\ell_P} = (t-m)^2, \quad r_\sigma^2 = \frac{D_{s,\ell_P}}{\ell_P^2 R^2}, \quad (1993)$$

$$\mu_\sigma = -(\delta_P + 2)r_\sigma, \quad \zeta_P = -\frac{1}{4} + \frac{r_\sigma^2}{4}. \quad (1994)$$

In particular, the fixed-line coordinate of theorem 14.178 determines the exact distance from the branch value in theorem 9.7:

$$\boxed{\zeta_P + \frac{1}{4} = \frac{D_{s,\ell_P}}{4\ell_P^2 R^2} = \frac{\mu_\sigma^2}{4(\delta_P + 2)^2}}. \quad (1995)$$

Moreover,

$$-\frac{1}{4} < \zeta_P < 0. \quad (1996)$$

Proof. Let $P = (s, \{t, m\}) \in \mathcal{T}_{R,a}$. By equation (1614),

$$t+m = \ell_P R, \quad tm = \frac{a}{s}.$$

Thus t and m are the roots of

$$X^2 - \ell_P R X + \frac{a}{s},$$

whose discriminant is

$$\ell_P^2 R^2 - \frac{4a}{s} = (t-m)^2 > 0.$$

This proves that $\Phi_{R,a}$ maps into $\mathcal{D}_{R,a}$.

Conversely, let $(s, \ell) \in \mathcal{D}_{R,a}$, and write $d = \sqrt{D_{s,\ell}}$. Since

$$d^2 \equiv \ell^2 R^2 \pmod{4},$$

the integers d and ℓR have the same parity. Also

$$0 < d^2 = \ell^2 R^2 - \frac{4a}{s} < \ell^2 R^2,$$

so the two numbers in (1989) are distinct positive integers. Their sum is ℓR , and their product is

$$\frac{(\ell R - d)(\ell R + d)}{4} = \frac{\ell^2 R^2 - d^2}{4} = \frac{a}{s}.$$

Hence they produce a squarefree coordinate $a = stm$ with $R \mid t+m$. The equivalence (1614) places the resulting unordered triple in $\mathcal{T}_{R,a}$. The sum and product recover the same quadratic polynomial, so the two constructions are inverse.

For positive integers t, m ,

$$t + m \leq tm + 1,$$

because $(t - 1)(m - 1) \geq 0$. This proves (1990), and the bijection gives (1991).

Apply theorem 9.7 with $b = t$ and $c = m$. Its centered branch coordinate is r_σ , and direct expansion gives

$$-\vartheta_\sigma(1 - \vartheta_\sigma) = -\frac{tm}{(t + m)^2} = -\frac{1}{4} + \frac{(t - m)^2}{4(t + m)^2}.$$

The discriminant identity and $t + m = \ell_P R$ prove (1993). The definition (1980) gives

$$\mu_\sigma = \frac{(m - t)(\delta_P + 2)}{t + m} = -(\delta_P + 2)r_\sigma.$$

Substitution proves (1994) and (1995). Finally $tm > 0$ gives $\zeta_P < 0$, while $t \neq m$ makes the final two terms in (1995) positive. This proves (1996). $\square$

Theorem 14.180 (Cartesian packet cover and the Fable quarter endpoints). *Retain the conditions of theorem 14.179. Regard the two finite sets as finite reduced $\mathbb{Q}$ -schemes*

$$\mathbf{S}_{R,a} := \coprod_{\sigma \in \mathcal{S}_{R,a}^{\text{ext}}} \text{Spec } \mathbb{Q}, \quad \mathbf{T}_{R,a} := \coprod_{P \in \mathcal{T}_{R,a}} \text{Spec } \mathbb{Q}.$$

Let

$$q: \mathbb{A}_y^1 \longrightarrow \mathbb{A}_\zeta^1, \quad q(y) := -\frac{1}{4} + \frac{y^2}{4}, \quad (1997)$$

and define

$$y_{R,a}(\sigma) := \frac{\mu_\sigma}{\delta_{\pi_{R,a}(\sigma)} + 2} = \frac{m - t}{m + t}, \quad \zeta_{R,a}(P) := \zeta_P. \quad (1998)$$

Then the square

$$\begin{array}{ccc} \mathbf{S}_{R,a} & \xrightarrow{y_{R,a}} & \mathbb{A}_y^1 \\ \pi_{R,a} \downarrow & & \downarrow q \\ \mathbf{T}_{R,a} & \xrightarrow{\zeta_{R,a}} & \mathbb{A}_\zeta^1 \end{array} \quad (1999)$$

is cartesian. Equivalently,

$$\mathbf{S}_{R,a} \cong \mathbf{T}_{R,a} \times_{\mathbb{A}_\zeta^1} \mathbb{A}_y^1. \quad (2000)$$

Under this isomorphism, the deck involution ι is $y \mapsto -y$. The pullback is finite étale: its two points over P have coordinates

$$y = \pm \frac{d_{s,\ell_P}}{\ell_P R}, \quad (2001)$$

and neither coordinate is zero.

Put

$$h(y) := \frac{y^2}{4}. \quad (2002)$$

Every ordered arithmetic sheet satisfies

$$0 < h(y_{R,a}(\sigma)) < \frac{1}{4}, \quad \zeta_{\pi_{R,a}(\sigma)} = -\frac{1}{4} + h(y_{R,a}(\sigma)). \quad (2003)$$

On the complete Fable fibre of theorem 4.2, let x denote the first source coordinate. Its exact values under the same two morphisms are

$$\begin{array}{c|ccc} p & p_0 & p_+ & p_- \\ \hline x(p) & 0 & 1 & -1 \\ h(x(p)) & 0 & \frac{1}{4} & \frac{1}{4} \\ q(x(p)) & -\frac{1}{4} & 0 & 0 \\ F_1(p) & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{array} \quad (2004)$$

In particular,

$$q(x(p)) - F_1(p) = h(x(p)) \quad (p \in \{p_0, p_+, p_-\}). \quad (2005)$$

Thus the fixed Fable point realizes height 0, its transposed pair realizes height $1/4$, and every primitive coupled arithmetic packet lies strictly between these two endpoint heights.

Proof. The relation between μ_σ and the centered packet coordinate in (1994) gives

$$y_{R,a}(\sigma) = -r_\sigma = \frac{m-t}{m+t}.$$

Consequently,

$$q(y_{R,a}(\sigma)) = -\frac{1}{4} + \frac{(m-t)^2}{4(m+t)^2} = -\frac{tm}{(t+m)^2} = \zeta_{\pi_{R,a}(\sigma)}.$$

This proves commutativity of (1999), and exchanging t with m sends y to $-y$.

Fix $P = (s, \{t, m\})$. The coordinate algebra of the fibre product over this component is

$$\begin{aligned} \mathbb{Q}[Y]/(Y^2 - (1 + 4\zeta_P)) &= \mathbb{Q}[Y]/(Y^2 - y_{R,a}(\sigma)^2) \\ &= \mathbb{Q}[Y]/((Y - y_{R,a}(\sigma))(Y + y_{R,a}(\sigma))). \end{aligned}$$

The coordinate $y_{R,a}(\sigma)$ is nonzero because $t \neq m$. The two linear factors are therefore coprime, so the Chinese remainder theorem identifies this algebra with $\mathbb{Q} \oplus \mathbb{Q}$. Its two rational points are exactly σ and $\iota\sigma$. Applying this on every component proves (2000). Moreover,

$$y_{R,a}(\sigma)^2 = \frac{D_{s,\ell_P}}{\ell_P^2 R^2},$$

which proves (2001). Since $q'(y) = y/2$, the cover is ramified only at $y = 0$; the arithmetic pullback avoids that point and is finite étale.

Positivity of t and m , together with $t \neq m$, gives

$$0 < |m - t| < m + t.$$

This proves (2003). Finally, the source coordinates in (15) give $x(p_0) = 0$, $x(p_+) = 1$, and $x(p_-) = -1$, while (16) gives $F_1(p) = -1/4$ at all three points. Substitution into (1997) and (2002) proves (2004) and (2005). $\square$

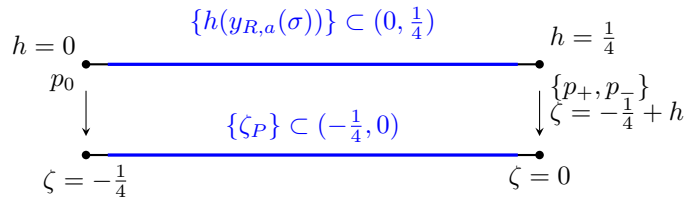


FIGURE 15. The universal quarter height. The Fable fibre supplies the two endpoint heights; primitive coupled arithmetic packets occupy the open height interval and map to the open ζ -interval.

Theorem 14.181 (Blade-coupling formula for the packet quarter height). *Retain the conditions of theorem 14.178 and theorem 14.179. On the two-dimensional parity space $\mathcal{S}_P$, put*

$$E_+ := \eta_+ \eta_+^*, \quad E_- := \eta_- \eta_-^*, \quad Q := Q_P^{\text{fix}}, \quad S := E_+ - E_-,$$

and define

$$a_\pm(P) := \text{tr}(E_\pm Q). \quad (2006)$$

Then

$$a_+(P) = \frac{1}{1 + \mu_\sigma^2}, \quad a_-(P) = \frac{\mu_\sigma^2}{1 + \mu_\sigma^2}, \quad a_+(P) + a_-(P) = 1, \quad (2007)$$

and, for the Hilbert–Schmidt norm $\|X\|_{\text{HS}}^2 = \text{tr}(X^*X)$,

$$\|[S, Q]\|_{\text{HS}}^2 = 8a_+(P)a_-(P). \quad (2008)$$

The packet quarter height is therefore

$$\zeta_P + \frac{1}{4} = \frac{1}{4(\delta_P + 2)^2} \frac{a_-(P)}{a_+(P)} = \frac{\|[S, Q]\|_{\text{HS}}^2}{32(\delta_P + 2)^2 a_+(P)^2}. \quad (2009)$$

On the six-dimensional packet carrier, put

$$A_\pm(P) := \text{tr}(\mathbf{P}_{P,\pm} \mathbb{Q}_P^{\text{fix}}), \quad \mathcal{C}_P := [\mathbf{A}_P, \mathbb{Q}_P^{\text{fix}}]. \quad (2010)$$

Then

$$A_\pm(P) = 3a_\pm(P), \quad \|\mathcal{C}_P\|_{\text{HS}}^2 = \frac{8}{3} A_+(P) A_-(P), \quad (2011)$$

and the same height is recovered intrinsically from the lifted ordered blades:

$$\zeta_P + \frac{1}{4} = \frac{1}{4(\delta_P + 2)^2} \frac{A_-(P)}{A_+(P)} = \frac{3\|\mathcal{C}_P\|_{\text{HS}}^2}{32(\delta_P + 2)^2 A_+(P)^2}. \quad (2012)$$

For every arithmetic packet, $A_-(P) > 0$ and $\mathcal{C}_P \neq 0$. In the algebraic one-parameter family obtained by allowing $\mu_\sigma = 0$, the following three conditions are equivalent:

$$\zeta_P + \frac{1}{4} = 0, \quad Q = E_+, \quad [S, Q] = 0. \quad (2013)$$

Proof. In the ordered parity basis (η_+, η_-) ,

$$Q = \frac{1}{1 + \mu_\sigma^2} \begin{pmatrix} 1 & \mu_\sigma \\ \mu_\sigma & \mu_\sigma^2 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Taking the two diagonal traces proves (2007). Direct multiplication gives

$$[S, Q] = \frac{2\mu_\sigma}{1 + \mu_\sigma^2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and hence

$$\|[S, Q]\|_{\text{HS}}^2 = \frac{8\mu_\sigma^2}{(1 + \mu_\sigma^2)^2} = 8a_+(P)a_-(P).$$

Since

$$\frac{a_-(P)}{a_+(P)} = \mu_\sigma^2,$$

substitution into (1995) proves (2009).

The lift from $\mathcal{S}_P$ to $\mathcal{H}_P$ tensors every operator with I_3 and then applies the unitary Υ_σ . It therefore multiplies each trace and each squared Hilbert–Schmidt norm by 3. This proves (2011) and (2012). Arithmetic packets have $\mu_\sigma \neq 0$, so their negative-parity weight and commutator norm are positive. Finally, in the extended one-parameter family the displayed matrix shows that each condition in (2013) is equivalent to $\mu_\sigma = 0$. $\square$

Theorem 14.182 (Exact packet control complexity). *For orthogonal projections E, Q of equal rank on a finite-dimensional Hilbert space, define*

$$\mathfrak{C}_{\text{HS}}(E, Q) := \inf_U \int_0^1 \sqrt{\text{tr}(H_U(t)^2)} dt, \quad H_U(t) := i\dot{U}(t)U(t)^*, \quad (2014)$$

where the infimum ranges over piecewise C^1 unitary paths satisfying

$$U(0) = I, \quad U(1)EU(1)^* = Q.$$

For an arithmetic packet P , let

$$\theta_\sigma := \arctan(\mu_\sigma) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad \mathbf{Y} := \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (2015)$$

Then

$$\mathfrak{C}_{\text{HS}}^{(2)}(P) := \mathfrak{C}_{\text{HS}}(E_+, Q_P^{\text{fix}}) = \sqrt{2}|\theta_\sigma|, \quad (2016)$$

$$\mathfrak{C}_{\text{HS}}^{(6)}(P) := \mathfrak{C}_{\text{HS}}(\mathbf{P}_{P,+}, Q_P^{\text{fix}}) = \sqrt{6}|\theta_\sigma| = \sqrt{3}\mathfrak{C}_{\text{HS}}^{(2)}(P). \quad (2017)$$

A minimizing path in each case is

$$U_\sigma(t) := \exp(-it\theta_\sigma \mathbf{Y}), \quad (2018)$$

$$\mathbb{U}_\sigma(t) := \Upsilon_\sigma(U_\sigma(t) \otimes I_3) \Upsilon_\sigma^{-1}. \quad (2019)$$

The parity weights and ordered-blade coupling are the exact trigonometric data of this control:

$$a_+(P) = \cos^2 \theta_\sigma, \quad a_-(P) = \sin^2 \theta_\sigma, \quad (2020)$$

$$\| [S, Q_P^{\text{fix}}] \|_{\text{HS}}^2 = 2 \sin^2(2\theta_\sigma), \quad \| [\mathbf{A}_P, Q_P^{\text{fix}}] \|_{\text{HS}}^2 = 6 \sin^2(2\theta_\sigma). \quad (2021)$$

Consequently,

$$\zeta_P + \frac{1}{4} = \frac{1}{4(\delta_P + 2)^2} \tan^2 \left(\frac{\mathfrak{C}_{\text{HS}}^{(2)}(P)}{\sqrt{2}} \right) = \frac{1}{4(\delta_P + 2)^2} \tan^2 \left(\frac{\mathfrak{C}_{\text{HS}}^{(6)}(P)}{\sqrt{6}} \right). \quad (2022)$$

Proof. Because $\tan \theta_\sigma = \mu_\sigma$,

$$U_\sigma(1)\eta_+ = \cos \theta_\sigma \eta_+ + \sin \theta_\sigma \eta_-.$$

The rank-one projection onto this vector is

$$\frac{1}{1 + \mu_\sigma^2} \begin{pmatrix} 1 & \mu_\sigma \\ \mu_\sigma & \mu_\sigma^2 \end{pmatrix} = Q_P^{\text{fix}}.$$

The adapted Hamiltonian of (2018) is the constant matrix $\theta_\sigma \mathbf{Y}$. Since $\text{tr}(\mathbf{Y}^2) = 2$, its path length is $\sqrt{2}|\theta_\sigma|$. Tensoring with I_3 multiplies the squared Hilbert–Schmidt norm by 3, so (2019) has length $\sqrt{6}|\theta_\sigma|$.

It remains to prove that these lengths are least. Let $U(t)$ be any admissible path and put $Q(t) = U(t)EU(t)^*$. Relative to $\text{ran } Q(t) \oplus \ker Q(t)$, write

$$H_U(t) = \begin{pmatrix} A(t) & B(t)^* \\ B(t) & D(t) \end{pmatrix}.$$

Then

$$\dot{Q}(t) = -i[H_U(t), Q(t)], \quad \|\dot{Q}(t)\|_{\text{HS}}^2 = 2 \text{tr}(B(t)^* B(t)) \leq \|H_U(t)\|_{\text{HS}}^2.$$

Thus every unitary control path is at least as long as its projection path. The required endpoint bound can be read directly from reflections. Put

$$\mathcal{R}(t) := 2Q(t) - I.$$

This is a unitary path and

$$\|\dot{\mathcal{R}}(t)\mathcal{R}(t)^*\|_{\text{HS}} = 2\|\dot{Q}(t)\|_{\text{HS}}.$$

For either endpoint pair used here, let $\theta_1, \dots, \theta_r \in (0, \pi/2)$ be its principal angles. The cosine–sine decomposition shows that

$$(2Q(1) - I)(2Q(0) - I)$$

has eigenvalues $e^{2i\theta_j}$ and $e^{-2i\theta_j}$ for $1 \leq j \leq r$. The bi-invariant Hilbert–Schmidt distance between the endpoint reflections is therefore

$$\sqrt{\sum_{j=1}^r ((2\theta_j)^2 + (-2\theta_j)^2)} = \sqrt{8 \sum_{j=1}^r \theta_j^2}.$$

Since the reflection path has twice the length of the projection path, every admissible unitary path has length at least

$$\sqrt{2 \sum_{j=1}^r \theta_j^2}.$$

For (E_+, Q_P^{fix}) , the sole principal angle is $|\theta_\sigma|$. For $(\mathbf{P}_{P,+}, \mathbb{Q}_P^{\text{fix}})$, the three principal angles are all $|\theta_\sigma|$. The lower bounds are consequently $\sqrt{2}|\theta_\sigma|$ and $\sqrt{6}|\theta_\sigma|$, respectively, and the displayed paths attain them.

Equations (2020) and (2021) follow from $\mu_\sigma = \tan \theta_\sigma$ and equation (2007), equation (2008), and equation (2011). Substitution into (2009) proves (2022). $\square$

Corollary 14.183 (Packet interval volume from control complexity). *Let*

$$I_P := \left[0, \frac{|y_{R,a}(\sigma)|}{2}\right] \subset \mathbb{R}, \quad \mathcal{V}_P := \text{vol}_1(I_P). \quad (2023)$$

Then $\mathcal{V}_P$ is independent of the ordered sheet and

$$\mathcal{V}_P^2 = \zeta_P + \frac{1}{4}, \quad \mathcal{V}_P = \frac{1}{2(\delta_P + 2)} \tan \left(\frac{\mathfrak{e}_{\text{HS}}^{(2)}(P)}{\sqrt{2}} \right) = \frac{1}{2(\delta_P + 2)} \tan \left(\frac{\mathfrak{e}_{\text{HS}}^{(6)}(P)}{\sqrt{6}} \right). \quad (2024)$$

Proof. The deck involution changes the sign of $y_{R,a}(\sigma)$, so it fixes the interval length. Equations (2002) and (1998) give

$$\mathcal{V}_P^2 = \frac{y_{R,a}(\sigma)^2}{4} = \zeta_P + \frac{1}{4}.$$

Taking the positive square root in (2022) proves the two control-length formulas. $\square$

Corollary 14.184 (Arithmetic reconstruction from packet control volume). *Retain the conditions of theorem 14.183, and put*

$$L_P := t + m = \ell_P R, \quad d_P := |m - t|. \quad (2025)$$

Then

$$d_P = 2L_P \mathcal{V}_P = \frac{L_P}{\delta_P + 2} \tan \left(\frac{\mathfrak{e}_{\text{HS}}^{(2)}(P)}{\sqrt{2}} \right) = \frac{L_P}{\delta_P + 2} \tan \left(\frac{\mathfrak{e}_{\text{HS}}^{(6)}(P)}{\sqrt{6}} \right). \quad (2026)$$

The unordered divisor pair is recovered by

$$\{t, m\} = \left\{ \frac{L_P - d_P}{2}, \frac{L_P + d_P}{2} \right\} = \left\{ L_P \left(\frac{1}{2} - \mathcal{V}_P \right), L_P \left(\frac{1}{2} + \mathcal{V}_P \right) \right\}. \quad (2027)$$

Moreover,

$$tm = L_P^2 \left(\frac{1}{4} - \mathcal{V}_P^2 \right), \quad D_{s, \ell_P} = d_P^2 = 4L_P^2 \mathcal{V}_P^2. \quad (2028)$$

In particular,

$$2L_P \mathcal{V}_P \in \mathbb{Z}_{>0}, \quad 2L_P \mathcal{V}_P \equiv L_P \pmod{2}, \quad 2L_P \mathcal{V}_P \leq L_P - 2. \quad (2029)$$

Thus the packet volume belongs to the finite lattice

$$\mathcal{V}_P \in \left\{ \frac{d}{2L_P} : 1 \leq d \leq L_P - 2, \quad d \equiv L_P \pmod{2} \right\}, \quad (2030)$$

and obeys

$$\mathcal{V}_P \geq \begin{cases} \frac{1}{2L_P}, & L_P \text{ odd}, \\ \frac{1}{L_P}, & L_P \text{ even}, \end{cases} \quad \mathcal{V}_P \leq \frac{1}{2} - \frac{1}{L_P}. \quad (2031)$$

Proof. The cover coordinate in (1998) gives

$$\mathcal{V}_P = \frac{|m-t|}{2(m+t)} = \frac{d_P}{2L_P}.$$

Combining this equality with (2024) proves (2026). Solving the two equations

$$t+m = L_P, \quad |m-t| = d_P$$

proves (2027). Multiplying its two entries gives the product formula in (2028); the discriminant formula follows from (1993).

The packet action is free, so $d_P > 0$. Since t, m are positive integers, $d_P \leq L_P - 2$, and the sum and difference have the same parity. This proves (2029) and (2030). The least positive integer of the parity of L_P is 1 when L_P is odd and 2 when it is even. Combining these least values with $d_P \leq L_P - 2$ proves (2031). $\square$

Theorem 14.185 (Packet sum-gap polynomial and exact moment carrier). *Let $p \equiv 1 \pmod{4}$ be prime, let $a \in \mathcal{A}_p$, put $R = 4a - p$, and suppose $R \mid p-1$. For*

$$P = (s, \{t, m\}) \in \mathcal{T}_{R,a}, \quad L_P := t+m, \quad d_P := |m-t|,$$

define

$$\mathcal{G}_{R,a}(X, Z) := \sum_{P \in \mathcal{T}_{R,a}} X^{L_P} Z^{d_P} \in \mathbb{Z}_{\geq 0}[X, Z]. \quad (2032)$$

Every coefficient of $\mathcal{G}_{R,a}$ belongs to $\{0, 1\}$. More precisely, for positive integers L, d ,

$$[X^L Z^d] \mathcal{G}_{R,a} = 1 \quad (2033)$$

if and only if

$$\begin{aligned} L > d > 0, \quad L &\equiv d \pmod{2}, \\ s_{L,d} &:= \frac{4a}{L^2 - d^2} \text{ is a positive squarefree integer,} \\ R &\mid s_{L,d}(L-d)^2 + 1. \end{aligned} \quad (2034)$$

Every occupied monomial also satisfies $R \mid L$.

The value at $(1, 1)$ recovers all coupled-shell counts:

$$\boxed{\mathcal{G}_{R,a}(1, 1) = |\mathcal{T}_{R,a}| = \frac{1}{2} c_{R,a} = \frac{1}{2} |\mathcal{E}_{R,a}| = \frac{1}{3} q_{R,p}.} \quad (2035)$$

Let

$$\mathcal{D}_X := X \frac{\partial}{\partial X}, \quad \mathcal{D}_Z := Z \frac{\partial}{\partial Z}.$$

For all $j, k \in \mathbb{Z}_{\geq 0}$,

$$\mathcal{D}_X^j \mathcal{D}_Z^k \mathcal{G}_{R,a}(X, Z) \Big|_{(X,Z)=(1,1)} = \sum_{P \in \mathcal{T}_{R,a}} L_P^j d_P^k \quad (2036)$$

$$= 2^k \sum_{P \in \mathcal{T}_{R,a}} L_P^{j+k} \mathcal{V}_P^k \quad (2037)$$

$$= \sum_{P \in \mathcal{T}_{R,a}} \frac{L_P^{j+k}}{(\delta_P + 2)^k} \tan^k \left(\frac{\mathfrak{C}_{\text{HS}}^{(2)}(P)}{\sqrt{2}} \right) \quad (2038)$$

$$= \sum_{P \in \mathcal{T}_{R,a}} \frac{L_P^{j+k}}{(\delta_P + 2)^k} \tan^k \left(\frac{\mathfrak{C}_{\text{HS}}^{(6)}(P)}{\sqrt{6}} \right). \quad (2039)$$

In particular,

$$\mathcal{D}_Z \mathcal{G}_{R,a} \Big|_{(1,1)} = 2 \sum_{P \in \mathcal{T}_{R,a}} L_P \mathcal{V}_P, \quad (2040)$$

$$\mathcal{D}_Z^2 \mathcal{G}_{R,a} \Big|_{(1,1)} = \sum_{P \in \mathcal{T}_{R,a}} D_{s,\ell_P} = 4 \sum_{P \in \mathcal{T}_{R,a}} L_P^2 \left(\zeta_P + \frac{1}{4} \right). \quad (2041)$$

Thus $\mathcal{G}_{R,a} \neq 0$ if and only if the coupled shell is occupied; in that case it certifies the Erdős–Strauss identity at p .

Proof. For a packet $P = (s, \{t, m\})$, the sum and absolute difference recover the unordered pair by

$$\{t, m\} = \left\{ \frac{L_P - d_P}{2}, \frac{L_P + d_P}{2} \right\}.$$

They then recover the remaining coordinate because

$$L_P^2 - d_P^2 = 4tm, \quad s = \frac{a}{tm} = \frac{4a}{L_P^2 - d_P^2}.$$

Two packets contributing the same monomial are therefore equal, which proves the 0-1 coefficient assertion.

Conversely, suppose (L, d) satisfies (2034). Put

$$t = \frac{L - d}{2}, \quad m = \frac{L + d}{2}, \quad s = s_{L,d}.$$

The parity and inequality conditions make t, m positive integers, and

$$stm = \frac{4a}{L^2 - d^2} \frac{L^2 - d^2}{4} = a.$$

The last congruence in (2034) is precisely

$$R \mid 4st^2 + 1.$$

Hence $(s, \{t, m\}) \in \mathcal{T}_{R,a}$, proving (2033). The coupled-channel equivalence (1614) gives $R \mid L$.

Evaluation at $(1, 1)$, followed by (1931), proves (2035). Since

$$\mathcal{D}_X^j \mathcal{D}_Z^k (X^L Z^d) = L^j d^k X^L Z^d,$$

termwise evaluation proves (2036). Substitution of (2026) proves (2037) and (2038) and (2039). The cases $(j, k) = (0, 1)$ and $(0, 2)$, together with (2028) and (2024), give (2040) and (2041). The final assertion follows from (1617). $\square$

Example 14.186 (The first shared-column arithmetic carrier). Take

$$(p, a, R, s, t, m, u) = (13, 4, 3, 2, 1, 2, 2).$$

Then

$$3 \mid 4u + 1 = 9, \quad 3 \mid u + a = 6, \quad (k, q) = (3, 7),$$

and

$$E_R = \begin{pmatrix} 3 & 8 \\ 1 & 3 \end{pmatrix}, \quad M_p = \begin{pmatrix} 7 & 8 \\ 1 & 3 \end{pmatrix}.$$

The common directed blade is

$$\mathcal{B}_{-+}(E_R) = \mathcal{B}_{-+}(M_p) = \begin{pmatrix} 24 & -64 \\ 9 & -24 \end{pmatrix}.$$

The reverse blades and their defect are

$$\mathcal{B}_{+-}(E_R) = \begin{pmatrix} -3 & 9 \\ -1 & 3 \end{pmatrix}, \quad \mathcal{B}_{+-}(M_p) = \begin{pmatrix} -7 & 49 \\ -1 & 7 \end{pmatrix},$$

$$\mathcal{D}_{3,13,2,1} = \begin{pmatrix} -4 & 40 \\ 0 & 4 \end{pmatrix}, \quad \mathcal{D}_{3,13,2,1}^2 = 16I_2.$$

Corollary 14.187 ($k = 3$ split-prime obstruction). *Let $p \equiv 1 \pmod{4}$ be prime, and put*

$$N_3(p) := \frac{3p+1}{4}.$$

If $N_3(p)$ has a prime divisor $q \equiv 2 \pmod{3}$, then the Erdős–Strauss identity holds at p through an exterior endpoint. More precisely, with

$$s = q, \quad t = 1, \quad h = \frac{N_3(p)}{q}, \quad m = \frac{h+1}{3}, \quad (2042)$$

one has

$$R = \frac{4q+1}{3}, \quad a = \frac{q(h+1)}{3}, \quad u = q \in \mathcal{E}_{R,a}. \quad (2043)$$

Consequently, an Erdős–Strauss counterexample $p \equiv 1 \pmod{4}$ would have to satisfy

$$q \mid N_3(p) \implies q \equiv 1 \pmod{3} \quad (2044)$$

for every prime q .

Proof. Since $N_3(p) \equiv 1 \pmod{3}$, the cofactor

$$h = \frac{N_3(p)}{q}$$

is also congruent to 2 $\pmod{3}$. Hence $m = (h+1)/3$ is a positive integer. The four integers

$$k = 3, \quad s = q, \quad t = 1, \quad h$$

satisfy

$$3p + 1 = 4qh, \quad 3 \mid 1 + h.$$

The resulting residual is

$$R = 4qm - p = \frac{4q+1}{3} > 0.$$

Moreover $q \leq N_3(p)$ gives $R \leq p$, and therefore

$$p < 4a = p + R \leq 2p < 3p.$$

The exterior alternative of theorem 14.162 now proves (2043). The final assertion is its contrapositive, since $3 \nmid N_3(p)$. $\square$

Example 14.188 (The first resistant prime after 60). For $p = 73$,

$$N_3(73) = 55 = 5 \cdot 11.$$

Taking $q = 5$ gives

$$(s, t, h, m, R, a) = (5, 1, 11, 4, 7, 20)$$

and the exact decomposition

$$\frac{4}{73} = \frac{1}{20} + \frac{1}{220} + \frac{1}{4015}.$$

Exact computation 14.189 (Residual-free carrier-incidence certificate). The script and captured receipt

```
verification/verify_residual_free_carrier_incidence.py,
verification/verify_residual_free_carrier_incidence_output.txt
```

verify all 1504 exterior coordinates and all 1366 middle divisor coordinates occurring for primes $p \leq 1000$. For both exterior flags and all three middle flags, the checker also verifies the transported blade signs, source and range corners, column actions, pulled-back adjoint, ordinary-transpose inequality, and oriented slope gap. Among these data, 236 coordinates lie in both carriers. For every one of them the checker verifies the common directed blade, the split reverse defect, its square and determinant, both eigenlines, and the two complementary spectral support idempotents, together with the half-period and mixer intertwining identities. It also verifies the free two-sheet quotient, the exact three-orbit rank, and both regular six-state actions on all 35 nonempty coupled shells in the same range. For every packet it further verifies the complementary flag transitions, the canonical fixed line, the square-discriminant reconstruction, the fixed-line quarter identity, and the control-volume arithmetic reconstruction. For every coupled shell it also verifies the 0-1 support criterion for $\mathcal{G}_{R,a}$, its count at $(1, 1)$, and its first two exact gap moments. The first five records $(p, a, R, |\mathcal{S}_{R,a}^{\text{ext}}|, |\mathcal{T}_{R,a}|, q_{R,p})$ are

$$(13, 4, 3, 2, 1, 3), \quad (37, 10, 3, 4, 2, 6), \quad (61, 16, 3, 4, 2, 6), \quad (97, 25, 3, 2, 1, 3), \quad (109, 28, 3, 6, 3, 9).$$

The same receipt verifies the two converse reconstructions

$$(p, k, s, t, h, m, a, R) = (349, 7, 47, 1, 13, 2, 94, 27)$$

and

$$(p, \ell, s, t, m, a, R) = (13, 1, 2, 1, 2, 4, 3).$$

For the 2374 primes $p < 100000$ with $p \equiv 1 \pmod{12}$, the same checker tests theorem 14.165 for all $k \equiv 3 \pmod{4}$ with $k \leq 255$, records the first successful (k, s) , and verifies the resulting unit-fraction identity. It constructs certificates for 2369 of these primes. The five primes not reached by this bounded one-leg search are

$$241, \quad 1249, \quad 6481, \quad 21169, \quad 43969.$$

Each of these five primes is reached by the $t = 1$ case of theorem 14.166. The exact data are

p	ℓ	q	s	m	R	a
241	1	11	3	22	23	66
1249	2	7	1	357	179	357
6481	1	7	2	926	927	1852
21169	5	39	2	2714	543	5428
43969	24	1631	17	647	27	10999

The checker verifies the corresponding five middle-orbit factorizations and all five unit-fraction identities.

Corollary 14.190 (Infinitely many five-Finger primes on every residual). *For every positive integer $R \equiv 3 \pmod{4}$, there are infinitely many primes*

$$p \equiv 25 \pmod{36}$$

whose residual- R shell contains an exterior endpoint. More precisely, there exist a prime s , a positive integer m , and a primitive arithmetic progression

$$p_n = 4sm(1 + L_R n) - R, \quad L_R = \text{lcm}(R, 9), \quad (2045)$$

such that every prime value $p_n > R$ has

$$a_n = sm(1 + L_R n), \quad u_n = s(1 + L_R n)^2 \in \mathcal{E}_{R, a_n}. \quad (2046)$$

Proof. Choose a reduced residue class $s_0 \pmod{R}$ satisfying

$$4s_0 + 1 \equiv 0 \pmod{R}.$$

Dirichlet's theorem on primes in arithmetic progressions supplies a prime $s \neq 3$ in this class. In particular,

$$\gcd(s, 9R) = 1.$$

Choose m so that

$$sm \equiv B_R \pmod{9}, \quad \gcd(m, R) = 1. \quad (2047)$$

Such an m exists. For prime divisors $q \neq 3$ of R , impose $m \equiv 1 \pmod{q}$ and use the Chinese remainder theorem with the displayed congruence modulo 9. If $3 \mid R$, then $R \equiv 3 \pmod{12}$, so

$$B_R = \frac{R + 25}{4} \equiv 1 \pmod{3};$$

hence the required residue of m modulo 9 is already nonzero modulo 3.

Put $t_n = 1 + L_R n$. Since $t_n \equiv 1 \pmod{R}$ and modulo 9,

$$4st_n^2 + 1 \equiv 0 \pmod{R}, \quad smt_n \equiv B_R \pmod{9}.$$

Thus t_n lies in the residue set of theorem 14.161 for the fixed carrier $v = sm^2$, and (2046) follows for every prime value $p_n > R$.

It remains to prove that the progression in (2045) is primitive. Write

$$p_0 = 4sm - R, \quad M = 4smL_R.$$

The integer p_0 is odd. If a prime $q \mid sm$, then

$$p_0 \equiv -R \pmod{q},$$

which is nonzero because $\gcd(sm, R) = 1$. If $q \mid R$, then

$$p_0 \equiv 4sm \pmod{q},$$

which is again nonzero. Finally,

$$p_0 \equiv 25 \equiv 1 \pmod{3}.$$

These cases prove

$$\gcd(p_0, M) = 1.$$

Dirichlet's theorem therefore supplies infinitely many prime values of $p_0 + Mn$, and all but finitely many exceed R . $\square$

Corollary 14.191 (Two residual-27 origin-carrier families). *For $R = 27$, the carrier $v = 251$ has*

$$s = 251, \quad m = 1, \quad \mathcal{T}_{27;251} = \{\overline{23}\} \subset \mathbb{Z}/27\mathbb{Z}.$$

Consequently,

$$p = 23065 + 27108n \quad (2048)$$

is the complete five-Finger progression with this fixed origin carrier. Writing $h = 23 + 27n$, every prime member has

$$\frac{4}{p} = \frac{1}{251h} + \frac{1}{251(ph+1)/27} + \frac{1}{251ph(ph+1)/27}. \quad (2049)$$

The member $n = 942$ is

$$p = 25558801, \quad h = 25457,$$

with exterior endpoint and origin carrier

$$u = 162662771099, \quad v = 251.$$

Its exact certificate is

$$\frac{4}{25558801} = \frac{1}{6389707} + \frac{1}{6048638876354} + \frac{1}{3935549286554136428490178}. \quad (2050)$$

The carrier

$$v = 2878829 = 149 \cdot 139^2$$

has

$$s = 149, \quad m = 139, \quad \mathcal{T}_{27;2878829} = \{\overline{11}\}.$$

Its complete five-Finger progression is

$$p = 911257 + 2236788n. \quad (2051)$$

The member $n = 18$ is

$$p = 41173441, \quad t = 497,$$

with

$$u = 36804341, \quad v = 2878829,$$

and exact certificate

$$\frac{4}{41173441} = \frac{1}{10293367} + \frac{1}{15696790434988} + \frac{1}{2310838595737973730884}. \quad (2052)$$

Proof. For $v = 251$, the two congruences in (1847) reduce to

$$t^2 \equiv 16 \pmod{27}, \quad t \equiv 5 \pmod{9}.$$

The first has roots 4 and 23 modulo 27, and the second selects 23. Substitution in (1848) gives (2048). The transpose reconstruction in theorem 14.160 gives (2049). Integer substitution at $n = 942$ gives (2050).

For $v = 149 \cdot 139^2$, the two congruences reduce to

$$t^2 \equiv 13 \pmod{27}, \quad t \equiv 2 \pmod{9}.$$

The roots of the first are 11 and 16 modulo 27; the second selects 11. Equations (2051) and (2052) now follow from the same reconstruction and direct integer substitution. $\square$

Exact computation 14.192 (Endpoint–origin transpose certificate). The script and captured receipt

`verification/verify_five_finger_origin_carrier_progressions.py,`

`verification/verify_five_finger_origin_carrier_progressions_output.txt`

verify both squarefree decompositions, both complete residue sets, five exact parameters in each progression, the quadratic-character identity, both ordered transposes, and both displayed unit-fraction certificates. They also verify the universal-residual constructor at

$$R \in \{27, 31, 35, 39, 55\},$$

including the root condition, the modulo-9 branch condition, the coprimality condition, and the primitive progression constants.

Remark 14.193 (What a finite progression cover records). Every finite set of certified rows admits a finite progression cover by fixing one endpoint seed in each row and applying theorem 14.159. The additional content of a bounded cover is therefore the reuse and arithmetic size of its seeds. A global finite-cover theorem requires a finite seed set, or a finite collection of parameterized seed laws, chosen independently of the search bound.

Theorem 14.194 (Eight endpoint progressions covering the certified all-five face). *Let $t \in \mathbb{Z}_{\geq 0}$. The following table defines eight families, with $a = (p + R)/4$.*

X	R	endpoint	u	p
A	27	$\mathcal{E}$	$59a$	$3985 + 6372t$
B	39	$\mathcal{M}$	2	$889 + 936t$
C	27	$\mathcal{E}$	317	$2509 + 11412t$
D	31	$\mathcal{M}$	16	$1393 + 4464t$
E	31	$\mathcal{E}$	116	$1825 + 2088t$
F	31	$\mathcal{M}$	1352	$39697 + 58032t$
G	27	$\mathcal{E}$	$a^2/4913$	$25405 + 31212t$
H	31	$\mathcal{M}$	200	$4129 + 22320t$

(2053)

Whenever p in one of these rows is prime, its endpoint belongs to $\mathcal{E}_{R,a}$ or $\mathcal{M}_{R,a}$ as displayed, and hence gives an Erdős–Straus decomposition. Every row has

$$p \equiv 25 \pmod{36},$$

so all eight families remain in the five-Finger prime class.

The denominator triples are as follows. Family A is

$$A: \quad a = \frac{p+27}{4}, \quad b = \frac{p(p+27)(59p+1)}{108}, \quad c = \frac{(p+27)(59p+1)}{6372}. \quad (2054)$$

For the other seven families,

$$B: \quad a = \frac{p+39}{4}, \quad b = \frac{p(p+47)}{156}, \quad c = \frac{p(p+39)(p+47)}{1248}, \quad (2055)$$

$$C: \quad a = \frac{p+27}{4}, \quad b = \frac{p(47p+1)}{4}, \quad c = \frac{(p+27)(47p+1)}{5072}, \quad (2056)$$

$$D: \quad a = \frac{p+31}{4}, \quad b = \frac{p(p+95)}{124}, \quad c = \frac{p(p+31)(p+95)}{7936}, \quad (2057)$$

$$E: \quad a = \frac{p+31}{4}, \quad b = \frac{p(15p+1)}{4}, \quad c = \frac{(p+31)(15p+1)}{1856}, \quad (2058)$$

$$F: \quad a = \frac{p+31}{4}, \quad b = \frac{p(p+5439)}{124}, \quad c = \frac{p(p+31)(p+5439)}{670592}, \quad (2059)$$

and, on putting

$$K_p := p(p+27) + 19652, \\ G: \quad a = \frac{p+27}{4}, \quad b = \frac{p(p+27)K_p}{2122416}, \quad c = \frac{K_p}{108}, \quad (2060)$$

$$H: \quad a = \frac{p+31}{4}, \quad b = \frac{p(p+831)}{124}, \quad c = \frac{p(p+31)(p+831)}{99200}. \quad (2061)$$

The union of the eight progressions contains all ten primes in (1821). More precisely, if $p_X(t)$ denotes the affine expression in row X of (2053), then

$$\begin{aligned} p_A(528) &= 3368401, & p_B(5505) &= 5153569, & p_C(501) &= 5719921, \\ p_D(1629) &= 7273249, & p_D(2687) &= 11996161, & p_D(3554) &= 15866449, \\ p_E(4562) &= 9527281, & p_E(7598) &= 15866449, & p_F(232) &= 13503121, \\ p_G(567) &= 17722609, & p_H(802) &= 17904769. \end{aligned} \quad (2062)$$

Thus $p = 15866449$ lies in both D and E , while every other prime in the certified bounded list lies in at least one displayed family.

Proof. All eight moduli in (2053) are divisible by 36, and all eight initial residues are congruent to 25 modulo 36. This proves the five-Finger class assertion.

For family A ,

$$a = 59(17 + 27t), \quad u = 59a, \quad 4u + 1 = 27(8767 + 13924t).$$

Thus $u \mid a^2$ and $u \in \mathcal{E}_{27,a}$. For family B ,

$$a = 2(116 + 117t), \quad a + u = 39(6 + 6t),$$

so $u = 2$ belongs to $\mathcal{M}_{39,a}$. For family C ,

$$a = 317(2 + 9t), \quad 4u + 1 = 4 \cdot 317 + 1 = 27 \cdot 47,$$

so $u = 317$ belongs to $\mathcal{E}_{27,a}$. For family D ,

$$a = 4(89 + 279t), \quad a + u = 31(12 + 36t),$$

so $u = 16$ belongs to $\mathcal{M}_{31,a}$. For family E ,

$$a = 58(8 + 9t), \quad 4u + 1 = 4 \cdot 116 + 1 = 31 \cdot 15,$$

so $u = 116$ belongs to $\mathcal{E}_{31,a}$. For family F ,

$$a = 52(191 + 279t), \quad a + u = 31(364 + 468t),$$

so $u = 1352 = 2^3 \cdot 13^2$ belongs to $\mathcal{M}_{31,a}$.

For family G , put $h = 22 + 27t$. Then

$$a = 289h, \quad u = \frac{a^2}{4913} = 17h^2,$$

and

$$4u + 1 = 68h^2 + 1 = 27(1219 + 2992t + 1836t^2).$$

Thus $u \in \mathcal{E}_{27,a}$. Finally, family H has

$$a = 20(52 + 279t), \quad a + u = 31(40 + 180t),$$

so $u = 200$ belongs to $\mathcal{M}_{31,a}$.

In every row $u \mid a^2$. Direct reduction also gives $\gcd(R, pa) = 1$ whenever p is prime. In family E , this uses only $p > 31$: a prime p cannot be divisible by 31, and then $a = (p + 31)/4$ is also a unit modulo 31. The remaining seven rows have the unit residues exposed by the displayed affine formulas. The endpoint-family reconstruction lemma now proves all eight decompositions. Substitution in (1835) and (1836) gives (2054)–(2061).

The identities in (2062) follow by integer multiplication and addition. They contain the ten entries of (1821), proving the bounded cover assertion. $\square$

Theorem 14.195 (First all-five row outside the eight bounded families). *Apply the residue-class and endpoint predicates of theorem 14.155 with the extended bound $p < 100000000$. There are exactly*

$$519 \text{ empty } R = 23 \text{ edge rows}$$

and exactly 29 all-five split-supported rows. Thirteen of those rows lie in at least one of the progressions A–H in (2053); sixteen lie in none.

The least row outside A–H is

$$p = 25371889, \quad d = 704775. \tag{2063}$$

Its first occupied residual is $R = 27$, where

$$a = \frac{p + 27}{4} = 6342979 = 19 \cdot 47 \cdot 7103, \tag{2064}$$

and

$$\mathcal{E}_{27,a} = \{47, 2564183\}, \quad \mathcal{M}_{27,a} = \emptyset. \quad (2065)$$

In particular, the eight bounded families do not cover the all-five face below one hundred million.

Proof. The sixty residue classes in (1820) are enumerated up to the stated bound. Trial division by the complete prime table through $\sqrt{100000000}$ proves primality or compositeness of every candidate and supplies every factorization used in the split-support and endpoint tests. This gives the three asserted counts and the least row in (2063).

For a separate primality certificate,

$$p - 1 = 2^4 \cdot 3 \cdot 17^2 \cdot 31 \cdot 59.$$

Pocklington base 46 has

$$\begin{array}{c|ccccc} q & 2 & 3 & 17 & 31 & 59 \\ \hline 46^{(p-1)/q} \bmod p & 25371888 & 14684078 & 7058867 & 22442215 & 19278722 \end{array} \quad (2066)$$

and

$$\gcd(46^{(p-1)/q} - 1, p) = 1 \quad (q = 2, 3, 17, 31, 59).$$

Every divisor of a^2 has the form

$$u = 19^i 47^j 7103^k, \quad 0 \leq i, j, k \leq 2.$$

The exterior and middle targets modulo 27 are 20 and $-a \equiv 23$, respectively. Since

$$19^i 20^j 2^k \equiv 20 \pmod{27} \iff (i, j, k) \in \{(0, 1, 0), (2, 0, 1)\},$$

whereas

$$19^i 20^j 2^k \equiv 23 \pmod{27}$$

has no solution in the exponent box, the endpoint sets are precisely those in (2065). $\square$

Corollary 14.196 (The new residual-27 exterior-47 family). *Let p be prime and suppose*

$$p \equiv 349 \pmod{1692}. \quad (2067)$$

Then $p \equiv 25 \pmod{36}$, and the residual-27 shell has

$$a = \frac{p+27}{4}, \quad u = 47 \in \mathcal{E}_{27,a}.$$

The resulting identity is

$$\boxed{\frac{4}{p} = \frac{1}{(p+27)/4} + \frac{1}{p(7p+1)/4} + \frac{1}{(p+27)(7p+1)/752}}. \quad (2068)$$

The prime in (2063) is the member with $t = 14995$, and hence

$$\frac{4}{25371889} = \frac{1}{6342979} + \frac{1}{1126532321342534} + \frac{1}{5992199575342}. \quad (2069)$$

Proof. The congruence (2067) is the Chinese remainder solution of

$$p \equiv 25 \pmod{36}, \quad p \equiv -27 \pmod{188}.$$

Consequently

$$a = 47(2 + 9t), \quad 4u + 1 = 4 \cdot 47 + 1 = 27 \cdot 7.$$

The exterior reconstruction formula gives (2068). Finally,

$$25371889 = 349 + 1692 \cdot 14995,$$

and substitution gives (2069). $\square$

Exact computation 14.197 (Eight-progression bounded-cover certificate). The script and captured receipt

```
verification/verify_five_finger_all_split_progression_cover.py,
verification/verify_five_finger_all_split_progression_cover_output.txt
```

check the bounded membership of every family, the union of the eight families, the five-Finger residue class, and five exact parameter values in each progression. At every tested value they verify $u \mid a^2$, the displayed endpoint congruence, equality between shell reconstruction and the closed denominator formulas, and the integer cross multiplication of the resulting unit-fraction identity.

Exact computation 14.198 (Extended all-five progression scan). The script and captured receipt

```
verification/explore_five_finger_all_split_progression_coverage.py,
verification/explore_five_finger_all_split_progression_coverage_output.txt
```

verify the extended counts, list all 29 rows and all 16 rows outside $A-H$, reproduce the Pocklington certificate and both endpoint sets of the least uncovered row, and check five exact parameters of the exterior-47 progression.

Exact computation 14.199 (All-five split face and first $R = 27$ certificate). The script and captured receipt

```
verification/verify_five_finger_all_split_first_row.py,
verification/verify_five_finger_all_split_first_row_output.txt
```

enumerate the 60 residue classes, reproduce all bounded counts and all ten rows, verify the Pocklington certificate, factor every Finger of the least row, enumerate both $R = 27$ endpoint sets, and check the final integer identity and its prime progression.

Exact computation 14.200 (Five-Finger survivor and $R = 31$ certificate). The script and captured receipt

```
verification/verify_five_finger_first_six_survivor.py,
verification/verify_five_finger_first_six_survivor_output.txt,
verification/verify_five_finger_split_face_scan.py,
verification/verify_five_finger_split_face_scan_output.txt
```

independently check the primality certificate, all Finger coordinates and factorizations, the seven empty shells, both exact $R = 31$ endpoint sets, the origin-2 divisor congruence, and the final integer cross multiplication. The bounded scan independently enumerates all 96 three-adic empty $R = 23$ edge rows below two million and verifies the three first-four split-supported rows and their residual-19 status.

Exact computation 14.201 (Three-adic five-Finger certificate). The script and captured receipt

```
verification/verify_r23_three_adic_five_finger.py,
verification/verify_r23_three_adic_five_finger_output.txt
```

check the consecutive-shell identity, forced-factor reassembly, and all ten pairwise gcds for the first 5000 odd values of d , the fixed Vandermonde determinant and discriminant, and every applicable affine edge prime below 20000. It also enumerates the congruence shadow, checks that it has exactly 60 classes modulo 35420, checks the residual-19 bijection on every one of them, verifies the four quadratic character kernels, and verifies the parity obstruction on the residual-19 split-support face.

Theorem 14.202 (Exact $R = 15$ parity-coupling classification). *Let $p \equiv 1 \pmod{12}$ be prime and put*

$$a_{15} := \frac{p+15}{4}.$$

For $r \in U_{15} \setminus \{1\}$, define

$$\nu_r(a_{15}) := \sum_{\substack{\ell|a_{15} \\ \ell \text{ prime} \\ \ell \equiv r \pmod{15}}} v_\ell(a_{15}).$$

Then $a_{15} \in \mathcal{A}_p$, $4a_{15} - p = 15$, and

$$q_{15,p} > 0 \iff \nu_{11}(a_{15}) + \nu_{14}(a_{15}) > 0 \quad \text{or} \quad (\nu_2(a_{15}) + \nu_8(a_{15}))(\nu_7(a_{15}) + \nu_{13}(a_{15})) > 0. \quad (2070)$$

Equivalently, if

$$\mathbf{m}_{15}(a_{15}) = \prod_{r \in U_{15} \setminus \{1\}} X_r^{\nu_r(a_{15})},$$

then the exact support ideal is

$$\begin{aligned} I_{15} &:= \langle X_{11}, X_{14}, X_2X_7, X_2X_{13}, X_8X_7, X_8X_{13} \rangle \\ &= \langle X_{11}, X_{14} \rangle + (X_2, X_8)(X_7, X_{13}), \end{aligned} \quad (2071)$$

and

$$q_{15,p} > 0 \iff \mathbf{m}_{15}(a_{15}) \in I_{15}.$$

Proof. The map

$$\phi : U_{15} \longrightarrow C_4 \times C_2, \quad \phi(2^i(-1)^j) = (i, j),$$

is an isomorphism. Its complete table is

r	1	2	4	8	14	13	11	7
$\phi(r)$	(0, 0)	(1, 0)	(2, 0)	(3, 0)	(0, 1)	(1, 1)	(2, 1)	(3, 1).

The exterior target is

$$-4^{-1} \equiv 11 \pmod{15}, \quad \phi(11) = (2, 1),$$

and the middle target is

$$-1 \equiv 14 \pmod{15}, \quad \phi(14) = (0, 1).$$

A prime factor congruent to 11 (mod 15) supplies an exterior divisor because

$$4 \cdot 11 + 1 = 45.$$

A prime factor congruent to 14 (mod 15) supplies the middle target. The four cross-products are

$$2 \cdot 13 \equiv 8 \cdot 7 \equiv 11 \pmod{15}, \quad 2 \cdot 7 \equiv 8 \cdot 13 \equiv 14 \pmod{15}.$$

Thus each generator of I_{15} gives an exterior or middle witness, proving that either displayed condition implies $q_{15,p} > 0$.

For the converse, suppose that neither displayed condition holds. There are no prime factors in the direct target classes 11 and 14. If no prime factor lies in $\{7, 13\}$, every available factor has second coordinate zero under ϕ , so neither target can occur.

Suppose instead that a factor in $\{7, 13\}$ occurs. Failure of the coupling condition then excludes factors in $\{2, 8\}$. Every available factor with second coordinate one has odd first coordinate, while every available factor with second coordinate zero has even first coordinate. Any exterior divisor or signed middle factorization whose second coordinate is one uses an odd total coefficient on the first collection. Its first coordinate is consequently odd, even after all contributions from residues 1 and 4 are included. Both targets have even first coordinate. Neither target is reachable, so

$$\mathcal{E}_{15,a_{15}} = \emptyset, \quad c_{15,a_{15}} = 0.$$

Equation (1629) gives $q_{15,p} = 0$. This proves the converse and the ideal formulation. $\square$

Exact computation 14.203 ($R = 15$ parity-coupling certificate). The script and captured receipt

verification/verify_residual_fifteen_factor_classification.py,
verification/verify_residual_fifteen_factor_classification_output.txt

recover the six minimal generators from all 329 nonzero seven-coordinate states of total degree at most four. They also compare the criterion with the original divisor definitions for all 1600 integers $a \leq 3000$ coprime to 15 and all 555 applicable prime shells below 20000.

Corollary 14.204 (First four consecutive support ideals). *Let $p = 4n + 1$ be a prime with $p \equiv 1 \pmod{12}$. The first four affine shell coordinates are*

$$(a_3, a_7, a_{11}, a_{15}) = (n + 1, n + 2, n + 3, n + 4).$$

The first two support ideals are

$$I_3 = \langle X_2 \rangle, \tag{2072}$$

$$I_7 = \langle X_5, X_6, X_3^3, X_2X_3, X_3X_4 \rangle. \tag{2073}$$

Together with I_{11} from (1740) and I_{15} from (2071), one has

$$\begin{aligned} q_{3,p} + q_{7,p} + q_{11,p} + q_{15,p} > 0 \quad &\Longleftrightarrow \quad \mathbf{m}_3(n+1) \in I_3 \text{ or } \mathbf{m}_7(n+2) \in I_7 \\ &\text{or } \mathbf{m}_{11}(n+3) \in I_{11} \text{ or } \mathbf{m}_{15}(n+4) \in I_{15}. \end{aligned} \tag{2074}$$

Proof. The affine-coordinate identity follows from theorem 14.85. The exact $R = 3$ classification gives I_3 . The three alternatives in theorem 14.122 give the five generators of I_7 . The $R = 11$ and $R = 15$ assertions are theorems 14.125 and 14.202. Taking the union of the four shell occupancies proves (2074). $\square$

Corollary 14.205 (Exact first-four survivor branches). *Let $p \equiv 1 \pmod{12}$ be prime and put*

$$c := \frac{p+11}{12}.$$

Then

$$\left(a_3, a_7, \frac{a_{11}}{3}, a_{15}, a_{19}, \frac{a_{23}}{3} \right) = (3c-2, 3c-1, c, 3c+1, 3c+2, c+1). \tag{2075}$$

In particular, the first four shells are all empty,

$$q_{3,p} = q_{7,p} = q_{11,p} = q_{15,p} = 0,$$

if and only if all four conditions below hold:

- (i) *every prime divisor of $3c-2$ is congruent to 1 (mod 3);*
- (ii) *either every prime divisor of $3c-1$ lies in $\{1, 2, 4\} \pmod{7}$, or every such prime divisor lies in $\{1, 3\} \pmod{7}$ and its total 3-residue valuation is one or two;*
- (iii)

$$\mathbf{m}_{11}(c) \notin J_{11}^{(1)};$$

- (iv) *writing*

$$S_{15}^A := \{1, 2, 4, 8\}, \quad S_{15}^B := \{1, 4, 7, 13\},$$

one has either

$$\{\ell \bmod 15 : \ell \text{ prime}, \ell \mid 3c+1\} \subseteq S_{15}^A$$

or

$$\{\ell \bmod 15 : \ell \text{ prime}, \ell \mid 3c+1\} \subseteq S_{15}^B.$$

The four reduced coordinates occurring in these conditions satisfy

$$\begin{aligned} \gcd(3c-2, 3c-1) &= \gcd(3c-1, c) = \gcd(c, 3c+1) \\ &= \gcd(3c-2, 3c+1) = 1, \end{aligned} \quad (2076)$$

$$\gcd(3c-2, c) = \gcd(c, 2), \quad \gcd(3c-1, 3c+1) = \gcd(3c-1, 2). \quad (2077)$$

Proof. Since $p = 12c - 11$, one has

$$n = \frac{p-1}{4} = 3c-3.$$

Substitution into $a_{4j-1} = n + j$ proves (2075).

Condition (i) is the empty-shell side of theorem 14.119. Condition (ii) is the two-case empty-shell classification in theorem 14.122. Since

$$a_{11} = 3c, \quad \mathfrak{m}_{11}(c) = \frac{\mathfrak{m}_{11}(a_{11})}{X_3},$$

theorem 14.128 gives condition (iii).

For the last shell, the negation of (2070) excludes the direct target classes 11, 14 (mod 15) and prevents simultaneous occurrence of a factor from $\{2, 8\}$ and a factor from $\{7, 13\}$. This is equivalent to condition (iv). The four rank terms are nonnegative, so their sum vanishes exactly when all four individual empty-shell conditions hold.

The gcd identities follow from the Euclidean algorithm. For example,

$$\gcd(3c-1, c) = 1, \quad \gcd(3c-2, 3c+1) = \gcd(3c-2, 3) = 1.$$

The remaining identities are obtained in the same way. □

Theorem 14.206 (Exact $R = 19$ factor-log ideal). *Let $p \equiv 1 \pmod{12}$ be prime and put*

$$a_{19} := \frac{p+19}{4}.$$

Let

$$\lambda_{19} : U_{19} \longrightarrow \mathbb{Z}/18\mathbb{Z}$$

be the primitive logarithm determined by $\lambda_{19}(2) = 1$. For a finite multiset

$$M = (k_1, \dots, k_d), \quad 1 \leq k_1 \leq \dots \leq k_d \leq 17,$$

define

$$\Sigma_E(M) := \left\{ \sum_{j=1}^d e_j k_j : e_j \in \{0, 1, 2\} \right\} \subseteq \mathbb{Z}/18\mathbb{Z}, \quad (2078)$$

$$\Sigma_M(M) := \left\{ \sum_{j=1}^d d_j k_j : d_j \in \{-1, 0, 1\} \right\} \subseteq \mathbb{Z}/18\mathbb{Z}. \quad (2079)$$

Let $\mathcal{G}_{19}$ be the submultiset-minimal collection of multisets satisfying

$$7 \in \Sigma_E(M) \quad \text{or} \quad 9 \in \Sigma_M(M). \quad (2080)$$

For $1 \leq k \leq 17$, put

$$\nu_k(a) := \sum_{\substack{\ell^e \parallel a \\ \lambda_{19}(\ell) = k}} e, \quad \mathfrak{m}_{19}^{\log}(a) := \prod_{k=1}^{17} X_k^{\nu_k(a)},$$

and define

$$I_{19}^{\log} := \left\langle \prod_{k \in M} X_k : M \in \mathcal{G}_{19} \right\rangle. \quad (2081)$$

Then

$$\boxed{q_{19,p} > 0 \iff \mathfrak{m}_{19}^{\log}(a_{19}) \in I_{19}^{\log}.} \quad (2082)$$

The ideal has 174 minimal generators. Their degree distribution is

$$\begin{array}{c|cccccccc} d & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline |\mathcal{G}_{19}^{(d)}| & 2 & 32 & 91 & 30 & 9 & 5 & 3 & 1 & 1. \end{array} \quad (2083)$$

The degree-one log multisets are

$$\mathcal{G}_{19}^{(1)} = \{(7), (9)\}, \quad (2084)$$

corresponding to prime-factor residues 14 and 18 (mod 19). The degree-two log multisets are

$$\begin{aligned} \mathcal{G}_{19}^{(2)} = \{ & (1, 3), (1, 5), (1, 6), (1, 8), \\ & (1, 10), (1, 12), (2, 3), (2, 5), \\ & (2, 11), (3, 4), (3, 6), (3, 11), \\ & (3, 12), (4, 5), (4, 13), (4, 17), \\ & (5, 10), (5, 14), (5, 15), (6, 13), \\ & (6, 15), (8, 17), (10, 15), (10, 17), \\ & (11, 14), (11, 16), (12, 13), (12, 15), \\ & (13, 14), (13, 15), (13, 17), (14, 15)\}. \end{aligned} \quad (2085)$$

Proof. The element 2 has order 18 modulo 19. Moreover,

$$-4^{-1} \equiv 14 \equiv 2^7 \pmod{19}, \quad -1 \equiv 18 \equiv 2^9 \pmod{19}.$$

Thus (2078) is the exterior reachability set, (2079) is the middle reachability set, and (2080) is precisely the shell-occupation test of theorem 14.80 in primitive-log coordinates. This proves (2082) once the minimal multisets are determined.

Since $U_{19} \cong C_{18}$, its Davenport constant is 18. Consequently every minimal generator has degree at most 17 by (1638). Exhaustive finite addition in $\mathbb{Z}/18\mathbb{Z}$ over all sorted multisets of degrees $1, \dots, 17$, followed by removal of every occupied multiset having an occupied proper submultiset, gives 174 rows with the degree profile (2083). The canonical newline-separated list of those rows has SHA-256

d2b4644e85c20abe36ad943b8e5b5bbd659d758f3ed1bbd9dc7307f418ae4d01.

Direct extraction of its first two degrees gives (2084) and (2085). The complete finite calculation is recorded in theorem 14.207. $\square$

Exact computation 14.207 ($R = 19$ factor-log certificate). The script and captured receipt

```
verification/verify_residual_nineteen_factor_classification.py,
verification/verify_residual_nineteen_factor_classification_output.txt
```

perform the complete antichain calculation through degree 17, check the generator count, degree profile, canonical hash, and the two displayed low-degree tables, and compare ideal membership with the original exterior and middle divisor definitions for all 2843 integers $a \leq 3000$ coprime to 19 and all 555 applicable prime shells below 20000.

Theorem 14.208 (Fixed-target $R = 23$ colon ideal). *Let $p \equiv 1 \pmod{12}$ be prime and put*

$$b := \frac{p+23}{12}, \quad a_{23} = 3b.$$

Let

$$\lambda_{23} : U_{23} \longrightarrow \mathbb{Z}/22\mathbb{Z}$$

be the primitive logarithm determined by $\lambda_{23}(\overline{5}) = 1$. If

$$M = (k_1, \dots, k_d), \quad 1 \leq k_1 \leq \dots \leq k_d \leq 21,$$

is a finite multiset, define

$$\Sigma_E^{23}(M) := \left\{ \sum_{j=1}^d e_j k_j : e_j \in \{0, 1, 2\} \right\}, \quad (2086)$$

$$\Sigma_M^{23}(M) := \left\{ \sum_{j=1}^d d_j k_j : d_j \in \{-1, 0, 1\} \right\} \quad (2087)$$

in $\mathbb{Z}/22\mathbb{Z}$, and put

$$E_{23} := \{7, 13, 19\}, \quad M_{23} := \{5, 11, 17\}. \quad (2088)$$

Let $\mathcal{G}_{23}^{(3)}$ be the submultiset-minimal collection of multisets M satisfying

$$\Sigma_E^{23}(M) \cap E_{23} \neq \emptyset \quad \text{or} \quad \Sigma_M^{23}(M) \cap M_{23} \neq \emptyset. \quad (2089)$$

In $\mathbb{Z}[X_1, \dots, X_{21}]$, define

$$J_{23}^{(3)} := \left\langle \prod_{k \in M} X_k : M \in \mathcal{G}_{23}^{(3)} \right\rangle \quad (2090)$$

and

$$\nu_k^{23}(b) := \sum_{\substack{\ell^e \parallel b \\ \lambda_{23}(\overline{\ell})=k}} e, \quad \mathbf{m}_{23}^{\log}(b) := \prod_{k=1}^{21} X_k^{\nu_k^{23}(b)}.$$

Then

$$\boxed{q_{23,p} > 0 \iff \mathbf{m}_{23}^{\log}(b) \in J_{23}^{(3)}}. \quad (2091)$$

Let $I_{23}^{\log}$ denote the ideal of theorem 14.80 after transport through λ_{23} . Since $\lambda_{23}(\overline{3}) = 16$, one has the exact colon identity

$$J_{23}^{(3)} = \left(I_{23}^{\log} : X_{16} \right). \quad (2092)$$

Every minimal generator of $J_{23}^{(3)}$ has degree at most 21. Its degree-one part is

$$\mathcal{G}_{23}^{(3,1)} = \{(5), (7), (11), (13), (17), (19)\}, \quad (2093)$$

and its degree-two part consists of the following 44 pairs:

$$\begin{aligned} \mathcal{G}_{23}^{(3,2)} = \{ & (1, 3), (1, 4), (1, 6), (1, 9), (1, 10), (1, 12), (1, 14), (1, 16), \\ & (1, 18), (1, 20), (2, 3), (2, 9), (2, 15), (3, 4), (3, 8), (3, 10), \\ & (3, 14), (3, 16), (3, 20), (4, 9), (4, 15), (4, 21), (6, 21), (8, 9), \\ & (8, 21), (9, 10), (9, 14), (9, 16), (9, 18), (9, 20), (9, 21), (10, 15), \\ & (10, 21), (12, 15), (12, 21), (14, 15), (14, 21), (15, 18), (15, 20), \\ & (15, 21), (16, 21), (18, 21), (20, 21), (21, 21) \}. \end{aligned} \quad (2094)$$

Proof. Direct calculation in U_{23} gives

$$\lambda_{23}(\overline{2}) = 2, \quad \lambda_{23}(\overline{3}) = 16, \quad \lambda_{23}(\overline{-4^{-1}}) = 7, \quad \lambda_{23}(\overline{-1}) = 11.$$

In the exterior channel, the forced factor 3 in $a_{23} = 3b$ occurs with exponent $e \in \{0, 1, 2\}$. The target for the remaining factors of b is therefore

$$\{7 - 16e : e = 0, 1, 2\} = \{7, 13, 19\} = E_{23}.$$

In the middle channel its signed exponent is $d \in \{-1, 0, 1\}$, and the corresponding target set is

$$\{11 - 16d : d = -1, 0, 1\} = \{5, 11, 17\} = M_{23}.$$

Equations (2086) and (2087) are precisely the two remaining factor-reachability sets. This proves (2091). It also proves (2092), because

$$\mathbf{m}_{23}^{\log}(a_{23}) = X_{16}\mathbf{m}_{23}^{\log}(b).$$

This fixed-target form also recovers the moving-target criterion in theorem 14.103. Write

$$\rho := \lambda_{23}(\bar{p}), \quad S := \{5, 11, 17\}.$$

The affine identity $b \equiv 2p \pmod{23}$ gives

$$\lambda_{23}(\bar{b}) = \rho + 2.$$

Translation by the total logarithm of b maps the middle box bijectively onto the exterior box. Thus, for $v \in \Sigma_M^{23}(M)$,

$$v \in S - \rho \iff v + \lambda_{23}(\bar{b}) \in S + 2 = E_{23}. \quad (2095)$$

The other moving target S is already M_{23} , proving equality of the two criteria.

The occupied multiset set is upward closed. Applying the product-one subsequence deletion argument from theorem 14.80 in $U_{23} \cong C_{22}$ gives the degree bound

$$D(C_{22}) - 1 = 21.$$

Testing the 21 one-term multisets and the 231 two-term multisets in the two fixed boxes gives (2093) and (2094). The independent finite replay is recorded below. $\square$

Exact computation 14.209 ($R = 23$ fixed-target and low-degree certificate). The script and captured receipt

```
verification/verify_residual_twenty_three_factor_classification.py,
verification/verify_residual_twenty_three_factor_classification_output.txt
```

check the primitive logarithm, the equivalence between the fixed and moving targets, the 50 minimal rows of degrees one and two, and the canonical low-degree SHA-256

3f9c363f28a56953fe396adb70d2fb17fdb03f70d0117c8e5bda6570bfe13bc2.

They compare (2091) with the original exterior and middle divisor definitions for all 2870 integers $b \leq 3000$ coprime to 23 and all 555 applicable prime shells below 20000. They also enumerate the 39 safe odd-sector valuation vectors, their nine maximal rows, and the first forbidden even-log multiplicities used below. The complete odd-bearing calculation contains 82 safe profiles and 25 maximal profiles, with SHA-256

c854a92934205485e9ff4ff440ff2cb107f1fc0666ded9d8a554fead9e3393a8.

Corollary 14.210 (Colon-ideal explanation of the one-surplus split). *For the two primes in (1791), put*

$$b_p := \frac{p + 23}{12}.$$

In the fixed-target coordinates of theorem 14.208,

$$\mathbf{m}_{23}^{\log}(b_{118801}) = X_2 X_{18} \notin J_{23}^{(3)}, \quad (2096)$$

$$\mathbf{m}_{23}^{\log}(b_{359041}) = X_2 X_{15} X_{16} \in J_{23}^{(3)}. \quad (2097)$$

More precisely, X_2X_{15} is a coordinatewise-minimal generator of $J_{23}^{(3)}$, whereas X_2X_{18} is an empty degree-two profile. Hence the exact transition from the common hidden matrix E_{21} is

p	$\Xi_{19,a_{19}}$	$\Xi_{23,a_{23}}$
118801	E_{21}	0
359041	E_{21}	$I_2 + S$.

Proof. The fixed cofactors are

$$b_{118801} = 9902 = 2 \cdot 4951, \quad b_{359041} = 29922 = 2 \cdot 3 \cdot 4987.$$

Modulo 23,

$$4951 \equiv 6, \quad 4987 \equiv 19.$$

The primitive logarithm of theorem 14.208 satisfies

$$\lambda_{23}(2) = 2, \quad \lambda_{23}(3) = 16, \quad \lambda_{23}(6) = 18, \quad \lambda_{23}(19) = 15.$$

This proves the two monomial identities.

Neither 2 nor 18 occurs in the degree-one list (2093), and the pair (2, 18) does not occur in the complete degree-two list (2094). Since X_2X_{18} has degree two, no generator of $J_{23}^{(3)}$ divides it. Thus it is outside the ideal. The pair (2, 15) does occur in (2094); consequently

$$X_2X_{15} \mid X_2X_{15}X_{16}$$

and the second monomial lies in the ideal. The incidence matrices now follow from theorem 14.144. $\square$

Corollary 14.211 (Prime-cofactor E_{21} -survivor criterion through $R = 23$). *Let $\ell \notin \{5, 13, 19, 23\}$ be prime, assume that*

$$p = 260\ell - 19$$

is prime with $p \equiv 1 \pmod{12}$, and put

$$b_\ell := \frac{p + 23}{12} = \frac{65\ell + 1}{3}.$$

Then

$$\boxed{\Xi_{19,a_{19}} = E_{21} \text{ and } q_{23,p} = 0 \iff \lambda_{19}(\ell) \in \{0, 5\} \text{ and } \mathbf{m}_{23}^{\log}(b_\ell) \notin J_{23}^{(3)}}. \quad (2098)$$

Equivalently, among the prime-cofactor shells carrying the ordered $R = 19$ blade E_{21} , escape at $R = 23$ is exactly membership of the pulled-back factor monomial in $J_{23}^{(3)}$.

Proof. The E_{21} rows of (1792) are exactly $\lambda_{19}(\ell) = 0$ and 5. The fixed-target classification (2091) gives

$$q_{23,p} = 0 \iff \mathbf{m}_{23}^{\log}(b_\ell) \notin J_{23}^{(3)}.$$

Taking the conjunction proves (2098). $\square$

Corollary 14.212 (Forced-log classification of the prime-cofactor pullback). *Under the conditions of theorem 14.211, suppose $\lambda_{19}(\ell) \in \{0, 5\}$, so that $\Xi_{19,a_{19}} = E_{21}$. Then b_ℓ is even, coprime to 23, and*

$$\boxed{q_{23,p} = 0 \iff \left[\nu_k^{23}(b_\ell) = 0 \text{ for every odd } k \right] \text{ or } \left[\mathbf{m}_{23}^{\log}(b_\ell) \in \{X_1X_2, X_2X_{21}, X_1X_2X_{21}\} \right]}. \quad (2099)$$

Equivalently, the first alternative says that every prime divisor of b_ℓ is a quadratic residue modulo 23. In each of the other three alternatives there is an odd integer u , every prime divisor of which is congruent to 1 (mod 23), and primes

$$q_5 \equiv 5 \pmod{23}, \quad q_{14} \equiv 14 \pmod{23},$$

such that, respectively,

$$b_\ell = 2uq_5, \quad b_\ell = 2uq_{14}, \quad b_\ell = 2uq_5q_{14}. \quad (2100)$$

The displayed nonresidue factors occur to exponent one.

Proof. Because $p \equiv 1 \pmod{12}$, the integer $b_\ell = (65\ell + 1)/3$ is defined. The prime ℓ is odd, so $65\ell + 1$ is even; division by 3 preserves parity. Hence

$$\nu_2^{23}(b_\ell) \geq 1,$$

because $\lambda_{23}(2) = 2$. If $23 \mid b_\ell$, then $23 \mid p = 12b_\ell - 23$; this is impossible because the displayed prime family does not contain $p = 23$. Thus b_ℓ is coprime to 23.

If every nonzero factor logarithm is even, both sumsets in (2086) and (2087) lie in $2\mathbb{Z}/22\mathbb{Z}$, whereas every target in (2088) is odd. The shell is therefore empty.

Suppose an odd factor logarithm occurs. The complete odd-bearing down-set in theorem 14.219 has twenty-five coordinatewise-maximal profiles. Exactly one of them contains the forced even logarithm 2: its odd multiset is $(1, 21)$ and its even multiset is (2) . Thus every empty odd-bearing profile containing 2 is one of

$$(1, 2), \quad (2, 21), \quad (1, 2, 21).$$

Conversely, direct substitution in (2086) and (2087) shows that all three multisets miss the two target triples. This proves (2099).

Finally,

$$\lambda_{23}(5) = 1, \quad \lambda_{23}(2) = 2, \quad \lambda_{23}(14) = 21.$$

Logarithm zero is exactly residue 1 (mod 23), and the even logarithms are exactly the nonzero quadratic residues. Translating the three exceptional monomials into prime-factor language gives (2100). $\square$

Exact computation 14.213 (Bounded prime-cofactor pullback census). The files

verification/scan_prime_cofactor_r23_pullback.py,
verification/scan_prime_cofactor_r23_pullback_output.txt

enumerate every prime $\ell \leq 10^7$, retain the two $R = 19$ logarithmic classes and primality of $p = 260\ell - 19$, factor b_ℓ , and evaluate the two fixed-target sumsets. There are 7328 applicable prime pairs. Exactly 5794 are occupied at $R = 23$, and the 1534 empty shells have the factor-type counts

type	even logs only	(1, 2)	(2, 21)	(1, 2, 21)
count	1463	34	35	2.

(2101)

The serialized survivor rows have SHA-256

$$6f318e7183729c5cbe35655674865e65bc5380b896a6ec935d487a547314d465.$$

Proposition 14.214 (Consecutive affine packet after $R = 23$). *Under the conditions of theorem 14.211, there is a unique integer $m \geq 0$ such that*

$$\ell = 6m + 1, \quad p = 1560m + 241, \quad b_\ell = 130m + 22. \quad (2102)$$

For $t \geq 0$, put

$$R_t := 23 + 4t, \quad a_t := \frac{p + R_t}{4}.$$

Then

$$a_t = 390m + 66 + t = 3b_\ell + t. \quad (2103)$$

In particular, the shells $R = 23, 27, \dots, 59$ have the ten consecutive coordinates

$$(a_0, a_1, \dots, a_9) = (390m + 66, 390m + 67, \dots, 390m + 75). \quad (2104)$$

Their forced divisor skeleton is

$$(6, 1, 2, 3, 10, 1, 6, 1, 2, 15), \quad (2105)$$

where the t -th displayed integer divides a_t . More explicitly,

$$\begin{aligned} a_0 &= 6(65m + 11), & a_2 &= 2(195m + 34), & a_3 &= 3(130m + 23), \\ a_4 &= 10(39m + 7), & a_6 &= 6(65m + 12), & a_8 &= 2(195m + 37), \\ a_9 &= 15(26m + 5). \end{aligned} \tag{2106}$$

Finally,

$$\gcd(a_s, a_t) \mid |t - s|. \tag{2107}$$

Consequently, for $0 \leq s < t \leq 9$, a prime dividing both a_s and a_t belongs to $\{2, 3, 5, 7\}$.

Proof. Reduction of $p = 260\ell - 19$ modulo 12 gives

$$1 \equiv 8\ell + 5 \pmod{12},$$

and hence $\ell \equiv 1 \pmod{3}$. The prime ℓ is odd, so $\ell = 6m + 1$ for a unique $m \geq 0$. Substitution gives (2102). For $R_t = 23 + 4t$,

$$4(390m + 66 + t) = (1560m + 241) + (23 + 4t),$$

which proves (2103). The displayed factorizations prove (2105).

For any s, t ,

$$a_t - a_s = t - s.$$

Every common divisor of a_s and a_t therefore divides $|t - s|$, proving (2107). When $0 < t - s \leq 9$, every prime divisor of $t - s$ is one of 2, 3, 5, 7.

The identities and common-divisor implication are independently checked in

verification/PrimeCofactorAffinePacketCertificate.lean,
verification/PrimeCofactorAffinePacketCertificate_output.txt.

□

Exact computation 14.215 (First post-23 escape in the bounded prime-cofactor family). For each of the 1534 empty rows in theorem 14.213, the files

verification/scan_prime_cofactor_post23_escape.py,
verification/scan_prime_cofactor_post23_escape_output.txt

factor $a_R = (p + R)/4$, evaluate both multiplicative reachability sets, and stop at the first occupied residual $R > 23$. Every row is occupied by $R = 59$, with distribution

R	27	31	35	39	43	47	51	55	59
count	556	574	234	144	10	10	1	2	3.

(2108)

Writing Q for the even-log factor type and P for the paired $(1, 2, 21)$ type, the full stratification is

type \ R	27	31	35	39	43	47	51	55	59
Q	523	561	219	136	8	10	1	2	3
$(1, 2)$	17	4	11	2	0	0	0	0	0
$(2, 21)$	16	9	3	5	2	0	0	0	0
P	0	0	1	1	0	0	0	0	0.

(2109)

The three maximal-delay rows all have $\Xi_{59, a_{59}} = I_2 + S$:

ℓ	p	a_{59}	$\text{fac}(a_{59})$
457	118801	29715	$3 \cdot 5 \cdot 7 \cdot 283$
163477	42504001	10626015	$3 \cdot 5 \cdot 281 \cdot 2521$
1945297	505777201	126444315	$3 \cdot 5 \cdot 8429621.$

(2110)

Least exterior witnesses $u = 21225, 53130075, 1896664725$, respectively, give

$$\frac{4}{118801} = \frac{1}{29715} + \frac{1}{5077395546660} + \frac{1}{59834124}, \quad (2111)$$

$$\frac{4}{42504001} = \frac{1}{10626015} + \frac{1}{1626852000211200354510} + \frac{1}{7655053462902}, \quad (2112)$$

$$\frac{4}{505777201} = \frac{1}{126444315} + \frac{1}{8223506743037744477326560} + \frac{1}{1083943249686304}. \quad (2113)$$

The canonical escape-row hash is

$$40ff82510a860198c07b31c5aaa898b87b679128de483d57851de543514afc70.$$

Proposition 14.216 (Degree-one colon triggers for the affine packet). *For*

$$\mathcal{R} := \{27, 31, 35, 39, 43, 47, 51, 55, 59\},$$

let d_R be the forced divisor from (2105); thus

$$(d_{27}, d_{31}, d_{35}, d_{39}, d_{43}, d_{47}, d_{51}, d_{55}, d_{59}) = (1, 2, 3, 10, 1, 6, 1, 2, 15). \quad (2114)$$

Write

$$\delta_R := \nu_R(d_R)$$

for its factor-residue valuation vector, and for $u \in U_R \setminus \{\bar{1}\}$ let $\mathbf{e}_u$ denote the corresponding unit coordinate vector. Define the degree-one trigger set

$$T_R^{(1)} := \{u \in U_R \setminus \{\bar{1}\} : \delta_R + \mathbf{e}_u \in \mathcal{U}_R\}. \quad (2115)$$

Then

$$\begin{aligned} T_{27}^{(1)} &= \{20, 26\}, \\ T_{31}^{(1)} &= \{15, 23, 27, 29, 30\}, \\ T_{35}^{(1)} &= \{23, 26, 32, 34\}, \\ T_{39}^{(1)} &= \{17, 19, 23, 29, 31, 34, 35, 37, 38\}, \\ T_{43}^{(1)} &= \{32, 42\}, \\ T_{47}^{(1)} &= \{15, 22, 23, 30, 31, 35, 39, 41, 43, 44, 45, 46\}, \\ T_{51}^{(1)} &= \{38, 50\}, \\ T_{55}^{(1)} &= \{24, 27, 41, 48, 53, 54\}, \\ T_{59}^{(1)} &= \{18, 23, 39, 44, 47, 54, 55, 56, 58\}. \end{aligned} \quad (2116)$$

If $c_R := a_R/d_R$ has a prime divisor q whose unit residue $\bar{q}$ belongs to $T_R^{(1)}$, then $q_{R,p} > 0$. Equivalently,

$$q_{R,p} = 0 \implies \bar{q} \notin T_R^{(1)} \text{ for every prime } q \mid c_R. \quad (2117)$$

Proof. For each R , insert $\delta_R + \mathbf{e}_u$ into (1634) and (1635), for every $u \in U_R \setminus \{\bar{1}\}$. Testing membership of t_R and s_R gives exactly (2116).

If $q \mid c_R$, then

$$\nu_R(a_R) \geq \delta_R + \mathbf{e}_{\bar{q}}.$$

The occupied set $\mathcal{U}_R$ is an up-set by theorem 14.80. Membership of $\bar{q}$ in $T_R^{(1)}$ therefore implies $\nu_R(a_R) \in \mathcal{U}_R$, which is equivalent to $q_{R,p} > 0$.

The complete finite evaluation is replayed by

$$\text{verification/analyze_prime_cofactor_escape_complexity.py.}$$

□

Exact computation 14.217 (Minimal escape-certificate complexity). For each first-occupied row in theorem 14.215, put

$$\gamma_R(a_R) := \nu_R(a_R) - \delta_R$$

and define

$$\kappa_R(a_R) := \min \left\{ \sum_{u \neq \bar{1}} \eta_u : 0 \leq \eta \leq \gamma_R(a_R), \delta_R + \eta \in \mathcal{U}_R \right\}. \quad (2118)$$

This is the least number of cofactor prime-factor occurrences whose residue multiset, together with the forced divisor, already occupies the shell. Exhaustion of every submultiset gives

R	$\kappa = 1$	$\kappa = 2$	$\kappa = 3$	$\kappa = 4$	total
27	261	251	42	2	556
31	280	281	13	0	574
35	108	108	14	4	234
39	118	26	0	0	144
43	2	5	3	0	10
47	5	5	0	0	10
51	0	1	0	0	1
55	0	0	2	0	2
59	2	1	0	0	3
total	776	678	74	6	1534.

(2119)

Thus every bounded escape has a certificate supported on at most four cofactor prime-factor occurrences. The serialized minimal certificates have SHA-256

28f6c4432217a0df53d2f1c0b13433cdbff60d853f80407b00c3324b09f736fe.

The exact source and captured receipt are

verification/analyze_prime_cofactor_escape_complexity.py,
verification/analyze_prime_cofactor_escape_complexity_output.txt.

Corollary 14.218 (Parity-split $R = 23$ survivor bounds). *Retain the conditions and notation of theorem 14.208, and suppose $q_{23,p} = 0$. For $1 \leq k \leq 21$, write*

$$\alpha_k := \nu_k^{23}(b).$$

Then

$$\alpha_5 = \alpha_7 = \alpha_{11} = \alpha_{13} = \alpha_{17} = \alpha_{19} = 0. \quad (2120)$$

In the coordinate order

$$(\alpha_1, \alpha_3, \alpha_9, \alpha_{15}, \alpha_{21}),$$

the remaining odd-log valuation vector lies in the coordinatewise down-set generated by the following nine maximal vectors:

$$\begin{aligned} \mathcal{V}_{23}^{\text{odd}} = \{ & (0, 0, 0, 6, 0), (0, 0, 1, 1, 0), (0, 1, 0, 0, 1), \\ & (0, 2, 2, 0, 0), (0, 3, 0, 1, 0), (0, 5, 1, 0, 0), \\ & (0, 8, 0, 0, 0), (1, 0, 0, 3, 0), (3, 0, 0, 0, 1) \}. \end{aligned} \quad (2121)$$

Let

$$\eta_{23}(b) := \sum_{j=1}^{10} \nu_{2j}^{23}(b) \quad (2122)$$

be the number of nonidentity even-log factor occurrences. If the odd sector is nonzero, then

$$\begin{aligned} \alpha_1 + \alpha_9 + \alpha_{15} + \alpha_{21} > 0 & \implies \eta_{23}(b) \leq 2, \\ \alpha_1 = \alpha_9 = \alpha_{15} = \alpha_{21} = 0, \quad \alpha_3 > 0 & \implies \eta_{23}(b) \leq 3. \end{aligned} \quad (2123)$$

If every nonzero factor logarithm is even, then $q_{23,p} = 0$.

Proof. Equation (2120) is the degree-one part of (2093). For the remaining five odd logarithms, the least pure multiplicities that reach a fixed target are

$$\frac{k}{\min\{e \geq 1 : (k, \dots, k) \text{ is occupied}\}} \mid \begin{array}{ccccc} 1 & 3 & 9 & 15 & 21 \\ 4 & 9 & 3 & 7 & 2. \end{array} \quad (2124)$$

Hence an empty odd sector lies in a box of

$$4 \cdot 9 \cdot 3 \cdot 7 \cdot 2 = 1512$$

valuation vectors. Exact evaluation of (2086) and (2087) leaves 39 vectors. Removing every row dominated by another safe row gives exactly the nine rows in (2121). This finite calculation is replayed in theorem 14.209.

It remains to prove the even-log bounds. Divide the even-log subgroup

$$2\mathbb{Z}/22\mathbb{Z}$$

by 2 and identify it with $\mathbb{Z}/11\mathbb{Z}$. If $m = \eta_{23}(b)$, the middle contributions of the m nonzero even logarithms form a sumset $A \subseteq \mathbb{Z}/11\mathbb{Z}$. Each factor occurrence contributes a three-element set $\{-h, 0, h\}$. Repeated application of Cauchy–Davenport gives

$$|A| \geq \min(2m + 1, 11). \quad (2125)$$

Fix one occurring odd logarithm k and set

$$B_k := \left\{ \frac{s-k}{2}, \frac{s+k}{2} : s \in M_{23} \right\} \subseteq \mathbb{Z}/11\mathbb{Z}.$$

The cardinalities are

$$\frac{k}{|B_k|} \mid \begin{array}{ccccc} 1 & 3 & 9 & 15 & 21 \\ 6 & 4 & 6 & 6 & 6. \end{array} \quad (2126)$$

If $A \cap B_k \neq \emptyset$, choosing coefficient 1 or -1 on the odd occurrence reaches M_{23} , so the shell is occupied. An empty shell therefore satisfies

$$|A| + |B_k| \leq 11.$$

For $k \in \{1, 9, 15, 21\}$, equations (2125) and (2126) force $m \leq 2$. For $k = 3$, they force $m \leq 3$. This proves (2123). Finally, if all nonzero logs are even, both fixed reachability boxes contain only even classes, whereas every member of $E_{23} \cup M_{23}$ is odd. The shell is then empty. $\square$

Corollary 14.219 (Complete odd-bearing $R = 23$ survivor down-set). *Retain the conditions and notation of theorem 14.208. Let $O(b)$ and $V(b)$ be the multisets of odd and nonzero even factor logarithms of b , respectively; factors with logarithm 0 are omitted. For multisets A, B , write $A \preceq B$ when each multiplicity in A is at most the corresponding multiplicity in B . In the table below, k^e denotes e copies of k .*

Then $q_{23,p} = 0$ if and only if either $O(b) = \emptyset$, or there is a row (O_i, V_i) in the following table such that

$$O(b) \preceq O_i, \quad V(b) \preceq V_i.$$

O_i	V_i	O_i	V_i
15^2	16	$3^2, 9^2$	$\emptyset$
15^2	8^2	$3^3, 15$	$\emptyset$
15^4	8	$3^3, 9$	6
15^6	$\emptyset$	3^4	12
9, 15	$\emptyset$	3^4	6^2
9^2	6	$3^5, 9$	$\emptyset$
3, 21	$\emptyset$	3^6	6
3, 15	6	3^8	$\emptyset$
3, 9	12	1, 21	2
3, 9	6^2	1, 15	8
3^2	18	1, 15^3	$\emptyset$
3^2	6, 12	$1^3, 21$	$\emptyset$
3^2	6^3		

(2127)

Consequently, the support of any odd-bearing empty shell uses only the logarithm classes

$$1, 2, 3, 6, 8, 9, 12, 15, 16, 18, 21.$$

In the prime-cofactor family of theorem 14.211, the forced logarithm 2 selects the unique maximal row

$$(O_i, V_i) = ((1, 21), (2)),$$

and hence gives the three exceptional monomials in theorem 14.212.

Proof. If $O(b) = \emptyset$, the shell is empty by theorem 14.218. Suppose $O(b) \neq \emptyset$. The same corollary places the odd valuation vector among 38 nonzero safe vectors and bounds $|V(b)|$ by 3. There are

$$1 + 10 + \binom{11}{2} + \binom{12}{3} = 286$$

even-log multisets of cardinality at most 3. Direct evaluation of the two fixed reachability boxes on the resulting finite product leaves 82 safe profiles. Their coordinatewise-maximal profiles are exactly the 25 rows in (2127).

The occupied profiles form an upward-closed multiset ideal. Hence every profile below one of the displayed safe maximal rows is safe, and every odd-bearing safe profile lies below one of those rows. This proves both implications. The finite calculation and the canonical row hash are replayed in theorem 14.209. $\square$

Corollary 14.220 (Unrestricted exact first-six survivor criterion). *Let $p \equiv 1 \pmod{12}$ be prime and put*

$$c := \frac{p+11}{12}.$$

Then

$$q_{3,p} = q_{7,p} = q_{11,p} = q_{15,p} = q_{19,p} = q_{23,p} = 0 \quad (2128)$$

if and only if the following three conditions hold simultaneously:

- (i) conditions (i)–(iv) of theorem 14.205;
- (ii)

$$\mathfrak{m}_{19}^{\log}(3c+2) \notin I_{19}^{\log};$$

- (iii)

$$\mathfrak{m}_{23}^{\log}(c+1) \notin J_{23}^{(3)}.$$

Every such survivor satisfies the complete $R = 23$ classification of theorem 14.219 at $b = c + 1$. The six reduced coordinates are

$$(3c - 2, 3c - 1, c, 3c + 1, 3c + 2, c + 1).$$

The only pairs among them whose gcd can exceed 1 are

$$\begin{aligned} \gcd(3c - 2, c) &= \gcd(c, 2), & \gcd(3c - 2, 3c + 2) &= \gcd(3c - 2, 4), \\ \gcd(3c - 2, c + 1) &= \gcd(c + 1, 5), & \gcd(3c - 1, 3c + 1) &= \gcd(3c - 1, 2), \\ \gcd(3c - 1, c + 1) &= \gcd(c + 1, 4), & \gcd(c, 3c + 2) &= \gcd(c, 2), \\ \gcd(3c + 1, c + 1) &= \gcd(c + 1, 2). \end{aligned} \quad (2129)$$

Hence distinct reduced coordinates can share no prime factor other than 2 or 5.

Proof. Equation (2075) identifies the six shell coordinates. The first four vanish exactly under condition (i) by theorem 14.205. The $R = 19$ shell vanishes exactly under condition (ii) by theorem 14.206, and the $R = 23$ shell vanishes exactly under condition (iii) by theorem 14.208. This proves the equivalence.

Every identity in (2129) follows from one Euclidean step. For example,

$$\gcd(3c - 2, c + 1) = \gcd(c + 1, 5), \quad \gcd(3c - 1, c + 1) = \gcd(c + 1, 4).$$

The omitted pairs differ by 1, by 3 while both are prime to 3, or reduce to $\gcd(c, c + 1)$. Their gcd is therefore 1. $\square$

Corollary 14.221 (Exact first-six intersection on an $R = 23$ edge branch). *Let p be a prime satisfying*

$$p \equiv 1 \pmod{48}, \quad p \equiv 5 \text{ or } 14 \pmod{23},$$

and put $b = (p + 23)/12$. Then

$$q_{3,p} = q_{7,p} = q_{11,p} = q_{15,p} = q_{19,p} = q_{23,p} = 0 \quad (2130)$$

if and only if there are an odd prime ℓ and an odd positive integer u for which all six conditions below hold:

(i)

$$b = 2\ell u, \quad \ell \equiv p \pmod{23}, \quad \nu_\ell(b) = 1,$$

and every prime divisor of u is congruent to 1 (mod 23);

(ii) *every prime divisor of $6\ell u - 5$ is congruent to 1 (mod 3);*

(iii) *every prime divisor of $3\ell u - 2$ lies in $\{1, 2, 4\}$ (mod 7);*

(iv)

$$\mathfrak{m}_{11}(2\ell u - 1) \notin J_{11}^{(1)};$$

(v) *every prime divisor of $3\ell u - 1$ lies in $\{1, 2, 4, 8\}$ (mod 15);*

(vi)

$$\mathfrak{m}_{19}^{\log}(6\ell u - 1) \notin I_{19}^{\log}.$$

Here $J_{11}^{(1)}$ is the residual-eleven colon ideal from theorem 14.128, and $I_{19}^{\log}$ is the exact residual-nineteen ideal from theorem 14.206. Condition (iv) has the explicit disjoint factor branches in theorem 14.146.

Proof. By theorem 14.108, the $R = 23$ shell is empty exactly when condition (i) holds. In this case

$$c := \frac{p + 11}{12} = b - 1 = 2\ell u - 1,$$

and (2075) becomes

$$\left(a_3, a_7, \frac{a_{11}}{3}, a_{15}, a_{19}, \frac{a_{23}}{3}\right) = (6\ell u - 5, 2(3\ell u - 2), 2\ell u - 1, 2(3\ell u - 1), 6\ell u - 1, 2\ell u).$$

Condition (ii) is the $R = 3$ empty-shell criterion in theorem 14.205. Since ℓu is odd, $3\ell u - 2$ is odd, so a_7 contains the prime factor 2. The second $R = 7$ branch in theorem 14.205 requires every prime factor of a_7 to lie in $\{1, 3\} \pmod{7}$, and therefore cannot contain 2. The first branch remains and is exactly condition (iii).

Condition (iv) is the residual-eleven criterion applied to $a_{11}/3 = 2\ell u - 1$. Likewise $a_{15} = 2(3\ell u - 1)$ contains the prime factor 2. Of the two residual-fifteen sets

$$S_{15}^A = \{1, 2, 4, 8\}, \quad S_{15}^B = \{1, 4, 7, 13\},$$

only the first contains 2. Hence the $R = 15$ empty-shell criterion is exactly condition (v).

Finally, theorem 14.206 gives

$$q_{19,p} = 0 \iff \mathfrak{m}_{19}^{\log}(a_{19}) \notin I_{19}^{\log},$$

and $a_{19} = 6\ell u - 1$, which is condition (vi). These six equivalences prove the assertion. $\square$

Corollary 14.222 (Primewise all-shell coefficient criterion). *Let $p \equiv 1 \pmod{4}$ be prime. For each $a \in \mathcal{A}_p$, put $R_a = 4a - p$, and define*

$$\mathfrak{e}_p := \sum_{a \in \mathcal{A}_p} |\mathcal{E}_{R_a, a}|, \quad \mathfrak{c}_p := \sum_{a \in \mathcal{A}_p} c_{R_a, a}. \quad (2131)$$

Then $\mathfrak{c}_p$ is even and

$$Q_p = \text{rank } \mathcal{P}_p^{\text{mod}} = \mathfrak{e}_p + \frac{1}{2}\mathfrak{c}_p. \quad (2132)$$

The aggregate complexity quantities satisfy

$$\text{Vol}_p^{\det} = 2^{6\mathfrak{e}_p + 3\mathfrak{c}_p}, \quad (2133)$$

$$L_{H,p}^2 = 51\mathfrak{e}_p + \frac{51}{2}\mathfrak{c}_p, \quad (2134)$$

$$L_{K,p}^2 = \frac{75}{2}\mathfrak{e}_p + \frac{75}{4}\mathfrak{c}_p. \quad (2135)$$

In particular,

$$\text{ES}(p) \iff \mathfrak{e}_p > 0 \vee \mathfrak{c}_p > 0. \quad (2136)$$

Equivalently, failure at p requires, for every $a \in \mathcal{A}_p$,

$$\mathcal{E}_{R_a, a} = \emptyset, \quad c_{R_a, a} = 0. \quad (2137)$$

Proof. Sum (1629) over $a \in \mathcal{A}_p$. This gives

$$Q_p = \mathfrak{e}_p + \frac{1}{2}\mathfrak{c}_p.$$

Each summand of $\mathfrak{c}_p$ is even by (1628). The aggregate modular rank and complexity formulas in theorem 14.244 now give (2132)–(2135). The same corollary identifies $\text{ES}(p)$ with $Q_p > 0$. Both totals in (2131) are nonnegative, which proves (2136). Their sum vanishes exactly when every displayed summand vanishes, proving (2137). $\square$

Corollary 14.223 (Intrinsic two-factor coordinates at prime residual moduli). *Retain the conditions of theorem 14.79, and assume that R is prime. Put*

$$n := \frac{R-1}{2}, \quad U_R[2] := \{u \in U_R : u^2 = 1\}, \quad U_R[n] := \{u \in U_R : u^n = 1\}.$$

The integer n is odd, and

$$\Theta_R : U_R \longrightarrow U_R[2] \times U_R[n], \quad u \longmapsto (u^n, u^{n+1}) \quad (2138)$$

is a group isomorphism with inverse

$$(\varepsilon, \omega) \longmapsto \varepsilon\omega.$$

Its distinguished coefficient target is

$$\Theta_R(-1) = (-1, 1). \quad (2139)$$

The image of $\mathfrak{P}_{R,a}$ is

$$\mathfrak{Q}_{R,a} := \prod_{\ell^e \| a} \left([(1, 1)] + \sum_{j=1}^e \left([(\ell^{nj}, \ell^{(n+1)j})] + [(\ell^{-nj}, \ell^{-(n+1)j})] \right) \right), \quad (2140)$$

where both entries are reduced modulo R . Thus

$$c_{R,a} = \text{coeff}_{[(-1,1)]} \mathfrak{Q}_{R,a}, \quad \text{rank } P_{R,p}^{\text{mod}} = |\mathcal{E}_{R,a}| + \frac{1}{2} \text{coeff}_{[(-1,1)]} \mathfrak{Q}_{R,a}. \quad (2141)$$

Proof. The prime satisfies $R \equiv 3 \pmod{4}$, so $R - 1 = 2n$ with n odd. For every $u \in U_R$,

$$(u^n)^2 = 1, \quad (u^{n+1})^n = 1, \quad u^n u^{n+1} = u^{2n+1} = u^R = u.$$

Hence Θ_R takes values in the displayed product and multiplication is a left inverse. Conversely, if $\varepsilon^2 = \omega^n = 1$, then

$$(\varepsilon\omega)^n = \varepsilon, \quad (\varepsilon\omega)^{n+1} = \omega,$$

because n is odd. This proves the inverse formula. The same parity gives

$$(-1)^n = -1, \quad (-1)^{n+1} = 1.$$

Applying the induced group-ring isomorphism to (1627), followed by (1629), proves the remaining claims. $\square$

Corollary 14.224 (Exact 12289 shell and its six modular unit lines). *For*

$$p = 12289, \quad R = 11, \quad a = \frac{p+R}{4} = 3075 = 3 \cdot 5^2 \cdot 41,$$

the higher-order factor is

$$U_{11}[5] = \{1, 3, 4, 5, 9\}.$$

Under $\Theta_{11}(u) = (u^5, u^6)$, one has

$$\Theta_{11}(3) = (1, 3), \quad \Theta_{11}(5) = (1, 5), \quad \Theta_{11}(41) = (-1, 3), \quad (2142)$$

and the group-ring element factors as

$$\begin{aligned} \mathfrak{Q}_{11,3075} = & ([(1, 1)] + [(1, 3)] + [(1, 4)]) \left(\sum_{\omega \in U_{11}[5]} [(1, \omega)] \right) \\ & \cdot ([(1, 1)] + [(-1, 3)] + [(-1, 4)]). \end{aligned} \quad (2143)$$

Its coefficient at $[(-1, 1)]$ has the exact count

$$\text{coeff}_{[(-1,1)]} \mathfrak{Q}_{11,3075} = 3 \cdot 2 = 6. \quad (2144)$$

Accordingly,

$$c_{11,3075} = 6, \quad \mathcal{E}_{11,3075} = \{41, 1845, 15375\}, \quad (2145)$$

and the middle complement orbits are

$$\{5, 1891125\}, \quad \{225, 42025\}, \quad \{1875, 5043\}. \quad (2146)$$

Consequently,

$$q_{11,12289} = \text{rank } P_{11,12289}^{\text{mod}} = 3 + \frac{6}{2} = 6. \quad (2147)$$

The six orbit labels are

$$\begin{aligned} \{(2, 41), (0, 230625)\}, & \quad \{(2, 1845), (0, 5125)\}, & \quad \{(2, 15375), (0, 615)\}, \\ \{(1, 5), (1, 1891125)\}, & \quad \{(1, 225), (1, 42025)\}, & \quad \{(1, 1875), (1, 5043)\}. \end{aligned} \quad (2148)$$

Over each displayed orbit, the unique D_3 -invariant line in its six-path fiber is $\mathbb{C}\gamma_{\text{inv}}$, and the shellwise unitary sends it to

$$\mathbb{C}(0, E_{11}).$$

Thus $P_{11,12289}^{\text{mod}}$ is the orthogonal sum of six orbit-indexed copies of the modular E_{11} unit line.

Proof. Direct exponentiation modulo 11 gives (2142). The local factor for 3^1 is

$$[(1, 1)] + [(1, 3)] + [(1, 4)].$$

The five terms belonging to 5^2 range once over the entire subgroup $U_{11}[5]$, and the local factor for 41^1 is

$$[(1, 1)] + [(-1, 3)] + [(-1, 4)].$$

This proves (2143). A product contributing to $[(-1, 1)]$ must choose one of the two C_2 -odd terms in the 41-factor. After choosing one of the three terms in the 3-factor, exactly one term in the full $U_{11}[5]$ -sum makes the second component equal to 1. Therefore the target coefficient is $3 \cdot 2 = 6$.

The ordered coprime-factor pairs contributing to the coefficient of -1 modulo 11 are

$$(1, 615), (3, 41), (25, 41), (41, 3), (41, 25), (615, 1).$$

They give the same six coefficient terms in coprime-factor coordinates. Direct evaluation of the two divisor conditions gives

$$\mathcal{E}_{11,3075} = \{41, 1845, 15375\}$$

and the three pairs in (2146). Equations (1629) and (1481) prove (2147). Complementing the three exterior divisors and adjoining the three middle pairs gives (2148). The final assertion follows from (1473) in each orbit fiber. $\square$

Exact computation 14.225 (All-shell unit-group and primewise-rank checker). The script and captured receipt

```
verification/verify_all_shell_unit_group_modular_rank.py,
verification/verify_all_shell_unit_group_modular_rank_output.txt
```

verify the identity $c_{R,a} = |\mathcal{M}_{R,a}|$, the shellwise and primewise modular-rank formulas, and the associated determinant-volume exponents for every admissible shell through $p = 5000$. They also verify the intrinsic product for every prime residual modulus through $R = 199$, together with the exact $C_2 \times C_5$ factorization, coefficient, divisor sets, and six complement orbits at $p = 12289$.

Exact computation 14.226 (A first occupied shell supplied only by the unit-group coefficient). The prime

$$p = 8803369$$

has no occupied residual shell for

$$R = 3, 7, 11, \dots, 103.$$

Its first occupied shell occurs at

$$R = 107, \quad a = \frac{p+R}{4} = 2200869 = 3^2 \cdot 11^2 \cdot 43 \cdot 47. \quad (2149)$$

For this shell,

$$\mathcal{E}_{107,2200869} = \emptyset. \quad (2150)$$

The two ordered coprime-factor pairs contributing to the coefficient of $[-1]$ are

$$(1, 18189), \quad (18189, 1), \quad 18189 = 3^2 \cdot 43 \cdot 47 \equiv -1 \pmod{107}. \quad (2151)$$

Equivalently,

$$c_{107,2200869} = \text{coeff}_{[-1]} \mathfrak{P}_{107,2200869} = 2. \quad (2152)$$

The corresponding middle divisors form the single complement orbit

$$\mathcal{M}_{107,2200869} = \{121, 40031606241\}. \quad (2153)$$

Therefore

$$q_{107,8803369} = \text{rank } P_{107,8803369}^{\text{mod}} = 0 + \frac{2}{2} = 1. \quad (2154)$$

Its endpoint word and directed cells are therefore

$$(b_{-1}, b_0, b_1) = (0, 1, 0), \quad C_{-0} = E_{12}, \quad C_{0+} = E_{21},$$

and

$$2q_{107,8803369}C_{0+} = 2E_{21} = \mathcal{M}(53).$$

Moreover,

$$r_{-1} \in \partial_T^{\text{in}} S_{107,a}, \quad r_0 \in \partial_T^{\text{out}} S_{107,a}.$$

Thus $r_{-1} \rightarrow r_0$ is the incoming entry-boundary edge and $r_0 \rightarrow r_1$ is the outgoing exit-boundary edge. This is the outgoing-edge selection of theorem 14.69.

The script and captured receipt

```
verification/verify_least_residual_shell_scan.py,
verification/verify_least_residual_shell_scan_output.txt
```

verify these exact data. They also determine the first occupied residual shell for every one of the 166011 primes $p \equiv 1 \pmod{12}$ below 10^7 . Every such shell has $R \leq 107$, and the displayed prime is the unique member of the tested range whose first occupied shell has $R = 107$.

Theorem 14.227 (Coupled-shell unit-group coefficient refinement). *Retain the conditions of theorem 14.74, assume $R \mid p - 1$, and retain $\mathfrak{P}_{R,a}$ and $c_{R,a}$ from theorem 14.79. Then*

$$|\mathcal{T}_{R,a}| = \frac{c_{R,a}}{2}, \quad |\mathcal{E}_{R,a}| = c_{R,a}, \quad q_{R,p} = \frac{3}{2}c_{R,a}. \quad (2155)$$

Consequently,

$$Q_p \geq \frac{3}{2}c_{R,a}, \quad (2156)$$

$$\text{Vol}_p^{\text{det}} \geq 2^{9c_{R,a}}, \quad (2157)$$

$$L_{H,p}^2 \geq \frac{153}{2}c_{R,a}, \quad L_{K,p}^2 \geq \frac{225}{4}c_{R,a}. \quad (2158)$$

Proof. The all-shell coefficient theorem gives

$$c_{R,a} = 2|\mathcal{C}_{R,a}| = 2|\mathcal{T}_{R,a}|.$$

Theorem 14.75 gives

$$\mathcal{E}_{R,a} = \mathcal{M}_{R,a}, \quad |\mathcal{E}_{R,a}| = c_{R,a}.$$

Substitution in the all-shell rank formula gives

$$q_{R,p} = \frac{3}{2}c_{R,a} = 3|\mathcal{T}_{R,a}|,$$

proving all three equalities in (2155). The bounds follow from theorem 14.60. $\square$

Corollary 14.228 ($R = 7$ and the exact C_6 coefficient). *Let $p \equiv 1 \pmod{28}$ be prime and put*

$$a = \frac{p+7}{4}.$$

For every prime $\ell \mid a$, let $\kappa(\ell) \in \mathbb{Z}/6\mathbb{Z}$ be determined by

$$\ell \equiv 3^{\kappa(\ell)} \pmod{7}.$$

In $\mathbb{Z}[z]/(z^6 - 1)$, define

$$\mathfrak{P}_{7,a}(z) = \prod_{\ell^e \parallel a} \left(1 + \sum_{j=1}^e \left(z^{j\kappa(\ell)} + z^{-j\kappa(\ell)} \right) \right). \quad (2159)$$

Then

$$q_{7,p} = \frac{3}{2} \text{coeff}_{z^3} \mathfrak{P}_{7,a}(z). \quad (2160)$$

Proof. The residue 3 generates $(\mathbb{Z}/7\mathbb{Z})^\times \cong C_6$, and

$$3^3 \equiv -1 \pmod{7}.$$

Under this identification, (1627) becomes (2159), while the coefficient at $\overline{-1}$ becomes the coefficient at z^3 . Equation (2160) follows from (2155). $\square$

Corollary 14.229 (Intrinsic $C_2 \times C_3$ coordinates at $R = 7$). *Let $p \equiv 1 \pmod{28}$ be prime, put $a = (p + 7)/4$, and define*

$$U_7[2] := \{u \in U_7 : u^2 = 1\}, \quad U_7[3] := \{u \in U_7 : u^3 = 1\}.$$

The map

$$\Theta : U_7 \longrightarrow U_7[2] \times U_7[3], \quad u \longmapsto (u^3, u^4) \quad (2161)$$

is a group isomorphism with inverse

$$(\varepsilon, \omega) \longmapsto \varepsilon\omega.$$

It sends

$$-1 \longmapsto (-1, 1). \quad (2162)$$

The group-ring image of $\mathfrak{P}_{7,a}$ is

$$\mathfrak{Q}_{7,a} := \prod_{\ell^e \parallel a} \left([(1, 1)] + \sum_{j=1}^e \left([(\ell^{3j}, \ell^{4j})] + [(\ell^{-3j}, \ell^{-4j})] \right) \right) \in \mathbb{Z}[U_7[2] \times U_7[3]], \quad (2163)$$

where every component is reduced modulo 7. Then

$$q_{7,p} = \frac{3}{2} \text{coeff}_{[(-1,1)]} \mathfrak{Q}_{7,a}. \quad (2164)$$

Proof. Every $u \in U_7$ satisfies $u^6 = 1$. Hence

$$(u^3)^2 = 1, \quad (u^4)^3 = 1, \quad u^3 u^4 = u^7 = u.$$

Thus Θ takes values in the displayed product and is injective. Both its domain and codomain have order 6, so it is bijective. If $\varepsilon^2 = \omega^3 = 1$, then

$$(\varepsilon\omega)^3 = \varepsilon, \quad (\varepsilon\omega)^4 = \omega,$$

which proves the inverse formula. Equation (2162) follows from

$$(-1)^3 = -1, \quad (-1)^4 = 1.$$

Applying the induced group-ring isomorphism to (1627) gives (2163). Coefficient extraction at the image of -1 , followed by (2155), proves (2164). $\square$

Corollary 14.230 (Exact $R = 7$ occupancy criterion). *Let $p \equiv 1 \pmod{28}$ be prime and put $a = (p + 7)/4$. Define*

$$\mathcal{P}_2(a) := \{\ell : \ell \mid a, \ell \text{ prime}, \ell \equiv 6 \pmod{7}\}, \quad (2165)$$

$$\mathcal{P}_3(a) := \{\ell : \ell \mid a, \ell \text{ prime}, \ell \equiv 2 \text{ or } 4 \pmod{7}\}, \quad (2166)$$

$$E_{\text{mix}}(a) := \sum_{\substack{\ell^e \parallel a \\ \ell \equiv 3 \text{ or } 5 \pmod{7}}} e. \quad (2167)$$

Then

$$q_{7,p} > 0 \iff \mathcal{P}_2(a) \neq \emptyset \vee E_{\text{mix}}(a) \geq 3 \vee (E_{\text{mix}}(a) \geq 1 \wedge \mathcal{P}_3(a) \neq \emptyset). \quad (2168)$$

Consequently, if $p \equiv 1 \pmod{28}$ fails the Erdős–Straus identity, then

$$\mathcal{P}_2(a) = \emptyset, \quad E_{\text{mix}}(a) \leq 2, \quad E_{\text{mix}}(a) > 0 \implies \mathcal{P}_3(a) = \emptyset. \quad (2169)$$

Proof. Use the exponent coordinate $\ell \equiv 3^{\kappa(\ell)} \pmod{7}$ from theorem 14.228. The six nonzero residue classes have the table

$\ell \pmod{7}$	1	2	3	4	5	6
$\kappa(\ell)$	0	2	1	4	5	3.

For a prime power $\ell^e \parallel a$, the support of its local factor in $\mathbb{Z}[C_6]$ is

$$S_{\ell,e} = \{0, \pm\kappa(\ell), \pm 2\kappa(\ell), \dots, \pm e\kappa(\ell)\} \subset \mathbb{Z}/6\mathbb{Z}.$$

The coefficient of z^3 is positive exactly when

$$3 \in \sum_{\ell^e \parallel a} S_{\ell,e}. \quad (2170)$$

A prime in $\mathcal{P}_2(a)$ has $\kappa(\ell) = 3$, so its local support already contains the target. Assume henceforth that $\mathcal{P}_2(a) = \emptyset$. Every prime congruent to 1 modulo 7 contributes only 0. Every prime represented in $\mathcal{P}_3(a)$ contributes the even subgroup

$$\{0, 2, 4\} \subset \mathbb{Z}/6\mathbb{Z}.$$

For $\ell \equiv 3$ or $5 \pmod{7}$, one has $\kappa(\ell) = \pm 1$. Combining all such local supports therefore gives precisely the residue classes represented by the integer interval

$$[-E_{\text{mix}}(a), E_{\text{mix}}(a)].$$

If $E_{\text{mix}}(a) \geq 3$, this interval contains 3. If $1 \leq E_{\text{mix}}(a) \leq 2$ and $\mathcal{P}_3(a) \neq \emptyset$, the sum $1 + 2 = 3$ reaches the target. In all remaining cases the reachable classes are even, or are represented by the interval $[-2, 2]$; neither set contains 3 modulo 6. This proves (2168).

A positive $R = 7$ shell contributes $q_{7,p} > 0$ to Q_p , so theorem 14.244 supplies the Erdős–Straus identity. Negating the three alternatives in (2168) gives (2169). $\square$

Exact computation 14.231 (Exact $R = 7$ occupancy checker). The script and captured receipt

verification/verify_r7_exact_occupancy_criterion.py,
verification/verify_r7_exact_occupancy_criterion_output.txt

compare (2168) with the coefficient of z^3 for every $a \leq 20000$ satisfying $\gcd(a, 7) = 1$.

Corollary 14.232 (Balanced divisors in a coupled shell). *Retain the hypotheses of theorem 14.227, and retain $\mathcal{B}_{R,a}$ from (1541). Then*

$$c_{R,a} \geq 2|\mathcal{B}_{R,a}|, \quad |\mathcal{T}_{R,a}| \geq |\mathcal{B}_{R,a}|, \quad q_{R,p} \geq 3|\mathcal{B}_{R,a}|. \quad (2171)$$

For each $r \in \mathcal{B}_{R,a}$, the corresponding packet consists of the three complement orbits

$$\{(2, ar), (0, a/r)\}, \quad \{(2, a/r), (0, ar)\}, \quad \{(1, ar), (1, a/r)\}. \quad (2172)$$

Consequently,

$$Q_p \geq 3|\mathcal{B}_{R,a}|, \quad (2173)$$

$$\text{Vol}_p^{\text{det}} \geq 2^{18|\mathcal{B}_{R,a}|}, \quad (2174)$$

$$L_{H,p}^2 \geq 153|\mathcal{B}_{R,a}|, \quad L_{K,p}^2 \geq \frac{225}{2}|\mathcal{B}_{R,a}|. \quad (2175)$$

Proof. For $r \in \mathcal{B}_{R,a}$, the two ordered coprime-factor pairs

$$(r, 1), \quad (1, r)$$

occur in the coefficient expansion of $\mathfrak{P}_{R,a}$. Their unit ratios are

$$\bar{r} = \overline{-1}, \quad \bar{r}^{-1} = \overline{-1}.$$

Distinct balanced divisors give distinct ordered pairs, proving the first inequality in (2171). The coefficient and rank identities in theorem 14.227 prove the other two.

Under the middle coprime-factor bijection (1534), the unordered pair $\{r, 1\}$ maps to

$$\{a/r, ar\}.$$

The coupled-shell theorem supplies the two exterior orientations and the middle orbit on this complement pair, giving (2172). The final bounds follow from theorem 14.244. $\square$

Exact computation 14.233 (Intrinsic $C_2 \times C_3$ and balanced-divisor checker). The script and captured receipt

```
verification/verify_intrinsic_c2_c3_coupled_certificate.py,
verification/verify_intrinsic_c2_c3_coupled_certificate_output.txt
```

verify the intrinsic product coordinates, the target $(-1, 1)$, the balanced-divisor coefficient bound, and all three orbit representatives in every coupled shell through $p = 5000$.

Exact computation 14.234 (Coupled-shell unit-group coefficient checker). The script and captured receipt

```
verification/verify_coupled_shell_unit_group_formula.py,
verification/verify_coupled_shell_unit_group_formula_output.txt
```

verify the group-ring coefficient, the free factor exchange, the three-halves rank formula, the $R = 3$ divisor specialization, and the $R = 7$ C_6 coefficient for every coupled shell through $p = 5000$.

Theorem 14.235 ($R = 3$ three-orbit balance). *The $R = 3$ case of theorem 14.74 admits an explicit prime-factor criterion. Let $p \equiv 1 \pmod{12}$ be prime and put*

$$a := \frac{p+3}{4}. \quad (2176)$$

Then $a \equiv 1 \pmod{3}$ and $4a - p = 3$. Define

$$\mathcal{T}(a) := \left\{ (s, \{t, m\}) : \begin{array}{l} s, t, m \in \mathbb{Z}_{>0}, \ s \text{ squarefree,} \\ a = stm, \ s \equiv 2 \pmod{3} \end{array} \right\}. \quad (2177)$$

Every element $(s, \{t, m\}) \in \mathcal{T}(a)$ determines exactly the three distinct complement orbits represented by

$$\mathcal{O}_{s,t,m}^{(1)} := \{(2, st^2), (0, sm^2)\}, \quad (2178)$$

$$\mathcal{O}_{s,t,m}^{(2)} := \{(2, sm^2), (0, st^2)\}, \quad (2179)$$

$$\mathcal{O}_{s,t,m}^{(3)} := \{(1, st^2), (1, sm^2)\}. \quad (2180)$$

These are all occupied orbits in the $R = 3$ shell. Therefore

$$q_{3,p} = 3|\mathcal{T}(a)|. \quad (2181)$$

In particular,

$$q_{3,p} > 0 \iff a \text{ has a prime divisor } q \equiv 2 \pmod{3}. \quad (2182)$$

Proof. Write $p = 12n + 1$. Then $a = 3n + 1$, proving $a \equiv 1 \pmod{3}$. In any squarefree coordinate factorization

$$a = stm,$$

none of s, t, m is divisible by 3. Hence $t^2 \equiv 1 \pmod{3}$, and the exterior condition becomes

$$3 \mid 4st^2 + 1 \iff s \equiv 2 \pmod{3}.$$

If $s \equiv 2 \pmod{3}$, the congruence $a \equiv 1 \pmod{3}$ gives

$$tm \equiv 2 \pmod{3}.$$

Thus t and m have opposite nonzero residue classes, so

$$3 \mid t + m.$$

Conversely, $3 \mid t + m$ makes t and m opposite modulo 3; then $tm \equiv 2 \pmod{3}$, and $a = stm \equiv 1 \pmod{3}$ forces $s \equiv 2 \pmod{3}$. The exterior and middle coordinate criteria of theorem 14.59 therefore select the same packet set $\mathcal{T}(a)$.

Within every packet, $t \neq m$, since their residues modulo 3 are different. The two ordered exterior coordinates give (2178) and (2179); the unordered middle coordinate gives (2180). This proves (2181).

If $\mathcal{T}(a) \neq \emptyset$, its squarefree factor $s \equiv 2 \pmod{3}$ contains a prime divisor congruent to 2 modulo 3. Conversely, if $q \mid a$ is prime and $q \equiv 2 \pmod{3}$, then

$$s = q, \quad t = 1, \quad m = a/q$$

defines an element of $\mathcal{T}(a)$. This proves (2182). $\square$

Corollary 14.236 (Multiplicative $R = 3$ rank formula). *Under the conditions of theorem 14.235, write*

$$a = \prod_{\ell} \ell^{e_{\ell}}, \quad a_+ := \prod_{\ell \equiv 1 \pmod{3}} \ell^{e_{\ell}}. \quad (2183)$$

Let

$$d(n) := \#\{r \in \mathbb{Z}_{>0} : r \mid n\}.$$

Then

$$|\mathcal{T}(a)| = \frac{d(a^2) - d(a_+^2)}{4}, \quad (2184)$$

$$q_{3,p} = \frac{3}{4} \left(d(a^2) - d(a_+^2) \right). \quad (2185)$$

In particular, the difference $d(a^2) - d(a_+^2)$ is divisible by 4. If

$$\Delta_3(a) := d(a^2) - d(a_+^2),$$

then

$$Q_p \geq \frac{3}{4} \Delta_3(a), \quad (2186)$$

$$\text{Vol}_p^{\det} \geq 2^{\frac{9}{2} \Delta_3(a)}, \quad (2187)$$

$$L_{H,p}^2 \geq \frac{153}{4} \Delta_3(a), \quad L_{K,p}^2 \geq \frac{225}{8} \Delta_3(a). \quad (2188)$$

Proof. Let χ_3 be the nontrivial real character modulo 3. Since $a \equiv 1 \pmod{3}$, no prime divisor of a equals 3. Consider ordered triples (s, t, m) satisfying

$$a = stm, \quad s \text{ squarefree.}$$

For $\ell^e \parallel a$, omission of ℓ from s leaves $e + 1$ allocations of the exponent e between t and m . Inclusion of ℓ in s leaves e allocations of the remaining exponent $e - 1$. Hence the total number of ordered triples is

$$\prod_{\ell^e \parallel a} (2e + 1) = d(a^2).$$

Weighting each triple by $\chi_3(s)$ changes the local factor to

$$(e + 1) + e\chi_3(\ell).$$

For $\ell \equiv 1 \pmod{3}$, this is $2e + 1$; for $\ell \equiv 2 \pmod{3}$, it is 1. Therefore the character-weighted sum is

$$\prod_{\substack{\ell^e \parallel a \\ \ell \equiv 1 \pmod{3}}} (2e + 1) = d(a_+^2).$$

The number of ordered triples with $s \equiv 2 \pmod{3}$ is consequently

$$\frac{d(a^2) - d(a_+^2)}{2}.$$

For these triples, t and m have different residues modulo 3, so the exchange $t \leftrightarrow m$ has no fixed point. Passing to unordered pairs divides the count by 2 and proves (2184). Equation (2185) follows from (2181). The $R = 3$ shell is one summand of Q_p ; substitution into theorem 14.60 gives the remaining bounds. $\square$

Corollary 14.237 (Exact residual-three packet polynomial and divisor moments). *Under the conditions of theorem 14.235, let*

$$\sigma_r(n) := \sum_{d|n} d^r \quad (r \in \mathbb{Z}_{\geq 0}),$$

and define

$$A_r(a) := \prod_{\ell^e \parallel a} \left(\sigma_r(\ell^e) + \sigma_r(\ell^{e-1}) \right), \quad (2189)$$

$$B_r(a) := \prod_{\ell^e \parallel a} \left(\sigma_r(\ell^e) + \chi_3(\ell) \sigma_r(\ell^{e-1}) \right), \quad (2190)$$

$$H(a) := \prod_{\ell^e \parallel a} \left((e + 1)\ell^e + e\ell^{e-1} \right), \quad (2191)$$

$$K(a) := \prod_{\ell^e \parallel a} \left((e + 1)\ell^e + e\chi_3(\ell)\ell^{e-1} \right), \quad (2192)$$

where χ_3 is the nontrivial real character modulo 3. Then the complete residual-three packet polynomial is

$$\boxed{\mathcal{G}_{3,a}(X, Z) = \sum_{\substack{s|a \text{ squarefree} \\ s \equiv 2 \pmod{3}}} \sum_{\substack{t|a/s \\ t < \sqrt{a/s}}} X^{t+a/(st)} Z^{a/(st)-t}.} \quad (2193)$$

Its value and its first two sum-gap moments are

$$\mathcal{G}_{3,a}(1, 1) = \frac{A_0(a) - B_0(a)}{4}, \quad (2194)$$

$$\mathcal{D}_X \mathcal{G}_{3,a}|_{(1,1)} = \frac{A_1(a) - B_1(a)}{2}, \quad (2195)$$

$$\mathcal{D}_X^2 \mathcal{G}_{3,a}|_{(1,1)} = \frac{A_2(a) - B_2(a) + H(a) - K(a)}{2}, \quad (2196)$$

$$\mathcal{D}_Z^2 \mathcal{G}_{3,a}|_{(1,1)} = \frac{A_2(a) - B_2(a) - H(a) + K(a)}{2}. \quad (2197)$$

In particular,

$$\sum_{P \in \mathcal{T}_{3,a}} D_{s,\ell_P} = 4 \sum_{P \in \mathcal{T}_{3,a}} L_P^2 \left(\zeta_P + \frac{1}{4} \right) = \frac{A_2(a) - B_2(a) - H(a) + K(a)}{2}. \quad (2198)$$

Proof. By theorem 14.235, a packet is specified by a squarefree divisor $s \mid a$ with $s \equiv 2 \pmod{3}$, together with an unordered factor pair of

$$n := \frac{a}{s}.$$

Since $a \equiv 1 \pmod{3}$, one has $n \equiv 2 \pmod{3}$. A square not divisible by 3 is congruent to 1 modulo 3, so n is not a square. Its unordered factor pairs are therefore indexed exactly once by

$$t \mid n, \quad t < \sqrt{n}, \quad m = \frac{n}{t}.$$

Substitution of $L_P = t + m$ and $d_P = m - t$ into (2032) proves (2193).

Pairing complementary divisors t and n/t gives

$$\sum_{\substack{t \mid n \\ t < \sqrt{n}}} \left(t + \frac{n}{t} \right) = \sigma_1(n), \quad (2199)$$

$$\sum_{\substack{t \mid n \\ t < \sqrt{n}}} \left(t + \frac{n}{t} \right)^2 = \sigma_2(n) + nd(n), \quad (2200)$$

$$\sum_{\substack{t \mid n \\ t < \sqrt{n}}} \left(\frac{n}{t} - t \right)^2 = \sigma_2(n) - nd(n). \quad (2201)$$

Indeed, the last two identities are half of the corresponding sums over all divisors of n .

Because $3 \nmid a$, the indicator of the required squarefree residue class is

$$\mathbf{1}_{s \equiv 2 \pmod{3}} = \frac{1 - \chi_3(s)}{2}.$$

For every prime power $\ell^e \parallel a$, excluding ℓ from the squarefree divisor s contributes $\sigma_r(\ell^e)$, while including it contributes $\sigma_r(\ell^{e-1})$. Consequently,

$$\sum_{\substack{s \mid a \\ s \text{ squarefree}}} \sigma_r(a/s) = A_r(a), \quad (2202)$$

$$\sum_{\substack{s \mid a \\ s \text{ squarefree}}} \chi_3(s) \sigma_r(a/s) = B_r(a). \quad (2203)$$

The same local calculation applied to $(a/s)d(a/s)$ gives

$$\sum_{\substack{s|a \\ s \text{ squarefree}}} \frac{a}{s} d(a/s) = H(a), \quad (2204)$$

$$\sum_{\substack{s|a \\ s \text{ squarefree}}} \chi_3(s) \frac{a}{s} d(a/s) = K(a). \quad (2205)$$

Applying the residue-class indicator to (2199)–(2201) proves (2195)–(2197). The case $r = 0$, followed by the free exchange of the two unequal factors, proves (2194); it is the count in (2184). Finally, (2041) proves (2198). $\square$

Exact computation 14.238 (Residual-three packet-polynomial certificate). The script and captured receipt

```
verification/verify_residual_three_packet_polynomial.py,
verification/verify_residual_three_packet_polynomial_output.txt
```

independently enumerate the packet polynomial for all 159 primes $p \equiv 1 \pmod{12}$ below 5000. They verify coefficient uniqueness, the multiplicative rank, and (2195)–(2197). The 113 occupied shells contain 540 packets in total.

Corollary 14.239 (Paired factor identities and quantitative bounds). *Under the conditions of theorem 14.235, suppose $q \mid a$ is prime with $q \equiv 2 \pmod{3}$, and put $m = a/q$. Then*

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{pq(m+1)/3} + \frac{1}{pa(m+1)/3}, \quad (2206)$$

and

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{pq(m+p)/3} + \frac{1}{a(p+m)/3}. \quad (2207)$$

Moreover,

$$Q_p \geq 3|\mathcal{T}(a)|, \quad (2208)$$

$$\text{Vol}_p^{\det} \geq 2^{18|\mathcal{T}(a)|}, \quad (2209)$$

$$L_{H,p}^2 \geq 153|\mathcal{T}(a)|, \quad L_{K,p}^2 \geq \frac{225}{2}|\mathcal{T}(a)|. \quad (2210)$$

Proof. The factor q gives the packet

$$(s, t, m) = (q, 1, a/q).$$

Equations (2206) and (2207) are respectively (1557) and (1556) with $R = 3$. The divisibility of the displayed denominators also follows directly from

$$m \equiv 2 \pmod{3}, \quad p \equiv 1 \pmod{3}.$$

The $R = 3$ summand contributes $3|\mathcal{T}(a)|$ to Q_p by (2181). The remaining inequalities follow from theorem 14.60. $\square$

Exact computation 14.240 ($R = 3$ three-orbit checker). The script and captured receipt

```
verification/verify_r3_three_orbit_balance.py,
verification/verify_r3_three_orbit_balance_output.txt
```

verify the exact packet criterion, the two exterior orientations, the middle orbit, (2181), the prime-factor criterion, and both identities for every tested prime $p \leq 20000$.

Theorem 14.241 (Exterior residue-lift family). *For $u \in \mathbb{Z}_{>0}$, write*

$$u = \prod_{\ell} \ell^{e_{\ell}}$$

and define its square-divisor hull by

$$\mathfrak{h}(u) := \prod_{\ell} \ell^{\lceil e_{\ell}/2 \rceil}. \quad (2211)$$

For every positive integer a ,

$$u \mid a^2 \iff \mathfrak{h}(u) \mid a. \quad (2212)$$

Let $R, u \in \mathbb{Z}_{>0}$ satisfy

$$R \equiv 3 \pmod{4}, \quad R \mid 4u + 1, \quad (2213)$$

and let p be a prime satisfying

$$p \equiv -R \pmod{4\mathfrak{h}(u)}, \quad R \leq 2p - 3. \quad (2214)$$

Put

$$a := \frac{p + R}{4}. \quad (2215)$$

Then $p \equiv 1 \pmod{4}$, $a \in \mathcal{A}_p$, $u \mid a^2$, and

$$(2, u) \in \mathcal{X}_{R,p}. \quad (2216)$$

In particular $q_{R,p} \geq 1$. The divisor certificate $d = p^2 u$ reconstructs the positive integers

$$b = \frac{p(a + pu)}{R}, \quad c = \frac{pa + a^2/u}{R}, \quad (2217)$$

and they satisfy

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}. \quad (2218)$$

Proof. For each prime ℓ ,

$$u \mid a^2 \iff e_{\ell} \leq 2v_{\ell}(a) \iff \left\lceil \frac{e_{\ell}}{2} \right\rceil \leq v_{\ell}(a).$$

Taking all prime valuations proves (2212).

The congruence in (2214) implies $p \equiv -R \equiv 1 \pmod{4}$, and it also gives

$$\mathfrak{h}(u) \mid \frac{p + R}{4} = a.$$

Thus $u \mid a^2$ by (2212). Since $R > 0$, $a > p/4$; since $R \leq 2p - 3 < 2p$, one has $a < 3p/4$. Therefore $a \in \mathcal{A}_p$, and $4a - p = R$.

The seed condition $R \mid 4u + 1$ says precisely that $u \in \mathcal{E}_{R,a}$. Equation (1527) proves (2216), and (1528) gives $q_{R,p} \geq 1$. The corresponding divisor is $d = p^2 u$. Substitution into the reconstruction formulas of theorem 9.1 gives (2217) and (2218). $\square$

Corollary 14.242 (Two explicit modulo-20 interior families). *If p is prime and $p \equiv 13 \pmod{20}$, then*

$$\frac{4}{p} = \frac{1}{(p+7)/4} + \frac{1}{p(3p+1)/4} + \frac{1}{(p+7)(3p+1)/80}. \quad (2219)$$

If p is prime and $p \equiv 17 \pmod{20}$, then

$$\frac{4}{p} = \frac{1}{(p+3)/4} + \frac{1}{p(7p+1)/4} + \frac{1}{(p+3)(7p+1)/80}. \quad (2220)$$

Proof. Take $u = 5$, for which $\mathfrak{h}(5) = 5$ and

$$4u + 1 = 21.$$

For $R = 7$, the prime congruence in (2214) is $p \equiv 13 \pmod{20}$. Equations (2215) and (2217) give the three denominators in (2219). For $R = 3$, the same congruence is $p \equiv 17 \pmod{20}$, and the reconstruction gives (2220). $\square$

Exact computation 14.243 (Exterior residue-lift checker). The script and captured receipt

`verification/verify_exterior_residue_lift_families.py,`

`verification/verify_exterior_residue_lift_families_output.txt`

verify the hull criterion for $u \leq 300$, all residue lifts generated by these u for primes $p \leq 20000$, their reconstructed unit-fraction identities, and both modulo-20 formulas.

Corollary 14.244 (Primewise aggregate control-length–determinant–volume identity). *Let $p \equiv 1 \pmod{4}$ be prime, retain $\mathcal{A}_p$ from (1501), and put*

$$Q_p := \text{rank } \mathcal{P}_p^{\text{mod}} = \sum_{a \in \mathcal{A}_p} q_{R_a, p}. \quad (2221)$$

Form the finite supported carrier and its shell operator by

$$\mathcal{S}_p := \bigoplus_{\substack{a \in \mathcal{A}_p \\ q_{R_a, p} > 0}} \mathcal{H}_{R_a, p}^{\text{simp}}, \quad H_p^{\text{sh}} := \bigoplus_{\substack{a \in \mathcal{A}_p \\ q_{R_a, p} > 0}} H_{R_a, p}^{\text{sh}}. \quad (2222)$$

Then

$$\dim \mathcal{S}_p = 6Q_p, \quad \text{Spec}(H_p^{\text{sh}}) = \{4^{[3Q_p]}, (-1)^{[3Q_p]}\}, \quad (2223)$$

and

$$\text{Vol}_p^{\text{det}} := \left| \det_{\mathcal{S}_p} H_p^{\text{sh}} \right| = 4^{3Q_p} = 2^{6Q_p}. \quad (2224)$$

When $Q_p = 0$, the direct sums in (2222) are the zero vector space and its unique endomorphism. Its action on $\Lambda^0 \mathcal{S}_p$ is the identity, so the determinant in (2224) equals 1.

Let I_p be the identity on $\mathcal{S}_p$, set

$$K_p^{\text{sh}} := H_p^{\text{sh}} - \frac{3}{2}I_p, \quad (2225)$$

and define the two control paths

$$\gamma_{H, p}(t) := \exp(-itH_p^{\text{sh}}), \quad \gamma_{K, p}(t) := \exp(-itK_p^{\text{sh}}), \quad 0 \leq t \leq 1. \quad (2226)$$

For the Hilbert–Schmidt cost $F_{\text{HS}}(A) = \sqrt{\text{tr}(A^2)}$, their squared lengths are

$$L_{H, p}^2 = 51Q_p, \quad L_{K, p}^2 = \frac{75}{2}Q_p. \quad (2227)$$

Consequently,

$$\boxed{\log \text{Vol}_p^{\text{det}} = \frac{2 \log 2}{17} L_{H, p}^2 = \frac{4 \log 2}{25} L_{K, p}^2} \quad (2228)$$

The following five statements are equivalent:

$$\exists x, y, z \in \mathbb{Z}_{>0} : \frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}; \quad (2229)$$

$$Q_p > 0; \quad (2230)$$

$$\text{Vol}_p^{\text{det}} > 1; \quad (2231)$$

$$L_{H, p}^2 > 0; \quad (2232)$$

$$L_{K, p}^2 > 0. \quad (2233)$$

Proof. Apply theorem 14.31 to each $a \in \mathcal{A}_p$ for which $q_{R_a,p} > 0$. The direct sum has

$$\sum_{a \in \mathcal{A}_p} 6q_{R_a,p} = 6Q_p$$

dimensions. Its two eigenvalue multiplicities are both

$$\sum_{a \in \mathcal{A}_p} 3q_{R_a,p} = 3Q_p,$$

which proves (2223). Multiplicativity of the determinant under finite direct sums gives (2224).

The two squared Hilbert–Schmidt lengths add over orthogonal direct sums. Summing (1333) and (1334) over $\mathcal{A}_p$ proves (2227). Substitution gives (2228).

Finally, (1504) and (2221) identify (2229) with (2230). Equations (2224) and (2227) prove the remaining equivalences. $\square$

Remark 14.245 (Target incidence at the zero-volume boundary). The determinant-volume carrier in theorem 14.244 is indexed by actual complement orbits, and its positivity is therefore controlled by the diagonal carrier $D_{R,a}$ in theorem 14.139. The shell (118801, 39, 29710) has $D_{39,29710} = 0$, so its contribution to Q_p , the control lengths, and the determinant volume is zero. Its off-diagonal carrier is nevertheless the exact matrix

$$C_{39,29710} = S, \quad HC_{39,29710}H^\top = J.$$

The present complexity–volume identity is therefore an occupied-support identity. The crossed incidence is an additional finite grading on its zero-support boundary. At $R = 59$, the addition of $D_{59,29715} = I_2$ creates the first nonzero shell rank while retaining the same crossed grading.

Corollary 14.246 (Primewise directional-energy bound). *Retain the conditions and notation of theorem 14.244. Define the one-sided shell set*

$$\mathcal{A}_p^{\text{one}} := \left\{ a \in \mathcal{A}_p : \begin{array}{l} \text{the endpoint status of the } R_a\text{-shell} \\ \text{is exterior-only or middle-only} \end{array} \right\}, \quad (2234)$$

The squarefree-coordinate bijections give

$$\mathcal{A}_p^{\text{one}} = \mathcal{A}_p^{\text{ext}} \triangle \mathcal{A}_p^{\text{mid}}, \quad |\mathcal{A}_p^{\text{one}}| = \Xi_p. \quad (2235)$$

Define the primewise labelled boundary energy

$$\mathcal{E}_p^{\text{lab}} := \sum_{a \in \mathcal{A}_p} \mathcal{E}_{R_a,p}^{\text{lab}}. \quad (2236)$$

Then

$$\boxed{\begin{aligned} \mathcal{E}_p^{\text{lab}} &= 2\Xi_p \\ &= 2|\mathcal{A}_p^{\text{one}}| \\ &\leq 2Q_p \\ &= \frac{1}{3 \log 2} \log \text{Vol}_p^{\text{det}} \\ &= \frac{2}{51} L_{H,p}^2 = \frac{4}{75} L_{K,p}^2. \end{aligned}} \quad (2237)$$

Equality in the inequality of (2237) holds exactly when every occupied shell is one-sided and

$$q_{R_a,p} = 1 \quad \text{for every } a \in \mathcal{A}_p \text{ with } q_{R_a,p} > 0. \quad (2238)$$

Moreover,

$$\mathcal{E}_p^{\text{lab}} > 0 \quad \implies \quad \exists x, y, z \in \mathbb{Z}_{>0} : \frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}. \quad (2239)$$

Proof. The labelled-energy classification in (1611), together with (2235), gives

$$\mathcal{E}_p^{\text{lab}} = 2\Xi_p = 2|\mathcal{A}_p^{\text{one}}|.$$

Every one-sided shell is occupied by (1582), so its positive integer orbit count satisfies $q_{R_a,p} \geq 1$. Hence

$$|\mathcal{A}_p^{\text{one}}| \leq \sum_{a \in \mathcal{A}_p^{\text{one}}} q_{R_a,p} \leq \sum_{a \in \mathcal{A}_p} q_{R_a,p} = Q_p.$$

The first inequality is an equality exactly when every one-sided shell has orbit count one. The second is an equality exactly when no both-channel shell is occupied. This proves the stated equality case. Substitution of (2224) and (2227) proves the remaining equalities in (2237). Finally, positive labelled energy forces $Q_p > 0$, and (2229)–(2230) prove (2239). $\square$

Exact computation 14.247 (Primewise aggregate complexity–volume checker). The script and captured receipt

```
verification/verify_primewise_aggregate_complexity_volume.py,
verification/verify_primewise_aggregate_complexity_volume_output.txt
```

verify the aggregate dimension, eigenvalue multiplicities, determinant volume, both squared control lengths, and both length–volume identities for finite test lists containing zero through six complement orbits.

Corollary 14.248 (Primewise coupled-coefficient decomposition). *Let $p \equiv 1 \pmod{4}$ be prime. For $a \in \mathcal{A}_p$, write*

$$R_a = 4a - p$$

and define

$$\mathcal{A}_p^{\text{rem}} := \{a \in \mathcal{A}_p : R_a \nmid p-1\}, \quad (2240)$$

$$Q_p^{\text{rem}} := \sum_{a \in \mathcal{A}_p^{\text{rem}}} q_{R_a,p}. \quad (2241)$$

Let

$$\mathcal{R}_p^{\text{coup}} := \{R : R \mid p-1, R \equiv 3 \pmod{4}\}, \quad a_R := \frac{p+R}{4}, \quad (2242)$$

and set

$$C_p^{\text{coup}} := \sum_{R \in \mathcal{R}_p^{\text{coup}}} c_{R,a_R}. \quad (2243)$$

Then C_p^{coup} is even and

$$\boxed{Q_p = Q_p^{\text{rem}} + \frac{3}{2}C_p^{\text{coup}}.} \quad (2244)$$

The modular–polar amplification of the same decomposition is

$$\boxed{\text{rank } \mathcal{V}_p^\partial = \mathcal{E}_p^{\text{pol},123} + \mathcal{E}_p^{\text{tim},123} = 3Q_p^{\text{rem}} + \frac{9}{2}C_p^{\text{coup}}.} \quad (2245)$$

Consequently, the primewise determinant volume and the two control lengths admit the exact arithmetic decompositions

$$\log \text{Vol}_p^{\text{det}} = 6(\log 2)Q_p^{\text{rem}} + 9(\log 2)C_p^{\text{coup}}, \quad (2246)$$

$$L_{H,p}^2 = 51Q_p^{\text{rem}} + \frac{153}{2}C_p^{\text{coup}}, \quad (2247)$$

$$L_{K,p}^2 = \frac{75}{2}Q_p^{\text{rem}} + \frac{225}{4}C_p^{\text{coup}}. \quad (2248)$$

In particular,

$$\exists x, y, z \in \mathbb{Z}_{>0} : \frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \iff Q_p^{\text{rem}} > 0 \vee C_p^{\text{coup}} > 0. \quad (2249)$$

Equivalently, p fails the Erdős–Straus identity precisely when

$$Q_p^{\text{rem}} = 0 \quad \text{and} \quad c_{R,(p+R)/4} = 0 \quad \text{for every } R \in \mathcal{R}_p^{\text{coup}}. \quad (2250)$$

Equivalently, the remainder shells and coupled shells separate the global two-target obstruction as

$$\neg \text{ES}(p) \iff \left[\forall a \in \mathcal{A}_p^{\text{rem}} : \mu_{p,a} \in \ker \mathfrak{T}_{p,a} \right] \wedge \left[\forall R \in \mathcal{R}_p^{\text{coup}} : c_{R,(p+R)/4} = 0 \right]. \quad (2251)$$

Proof. The shell set $\mathcal{A}_p$ is the disjoint union of $\mathcal{A}_p^{\text{rem}}$ and the shells satisfying $R_a \mid p-1$. The coupled-shell indexing in theorem 14.74 identifies the latter shells bijectively with $\mathcal{R}_p^{\text{coup}}$ through

$$R \mapsto a_R = \frac{p+R}{4}.$$

For every member of this set, theorem 14.227 gives

$$q_{R,p} = \frac{3}{2} c_{R,a_R}.$$

Summing over the two parts of $\mathcal{A}_p$ proves (2244). Every coefficient c_{R,a_R} is an even nonnegative integer, so their sum has the same properties. Multiplying (2244) by 3 and applying (2658) and (2636) proves (2245).

Substitute (2244) into the identities

$$\log \text{Vol}_p^{\text{det}} = 6(\log 2)Q_p, \quad L_{H,p}^2 = 51Q_p, \quad L_{K,p}^2 = \frac{75}{2}Q_p$$

from theorem 14.244. This proves (2246)–(2248).

The same corollary identifies the unit-fraction assertion with $Q_p > 0$. Both summands on the right side of (2244) are nonnegative. This proves (2249). Finally, $C_p^{\text{coup}} = 0$ holds exactly when each of its nonnegative group-ring coefficients vanishes, which proves (2250). The nonnegative remainder sum vanishes exactly when every remainder-shell rank vanishes. By (2656), this is equivalent to $\mu_{p,a} \in \ker \mathfrak{T}_{p,a}$ for every $a \in \mathcal{A}_p^{\text{rem}}$. Combining this statement with the coefficient condition proves (2251). $\square$

Exact computation 14.249 (Primewise coupled-coefficient checker). The script and captured receipt

`verification/verify_primewise_coupled_coefficient_decomposition.py,`

`verification/verify_primewise_coupled_coefficient_decomposition_output.txt`

verify the rank decomposition, its C_3 -amplified modular–polar form, the determinant-volume exponent, both control-length identities, and the separated remainder-kernel/coupled-coefficient zero criterion for every primewise shell through $p = 5000$.

Corollary 14.250 (Uniform primewise complexity–volume gap). *Let*

$$\mathbb{P}_{1,12} := \{p \in \mathbb{P} : p \equiv 1 \pmod{12}\}.$$

For $p \in \mathbb{P}_{1,12}$, retain the quantities

$$Q_p, \quad \Xi_p, \quad \mathcal{E}_p^{\text{lab}}, \quad \text{Vol}_p^{\text{det}}, \quad L_{H,p}^2, \quad L_{K,p}^2$$

from theorem 14.244. The following statements are equivalent:

(i) For every integer $n \geq 2$, there are $x, y, z \in \mathbb{Z}_{>0}$ such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

(ii)

$$\inf_{p \in \mathbb{P}_{1,12}} Q_p \geq 1.$$

(iii)

$$\inf_{p \in \mathbb{P}_{1,12}} \text{Vol}_p^{\det} \geq 64.$$

(iv)

$$\inf_{p \in \mathbb{P}_{1,12}} L_{H,p}^2 \geq 51.$$

(v)

$$\inf_{p \in \mathbb{P}_{1,12}} L_{K,p}^2 \geq \frac{75}{2}.$$

The directional squarefree coordinate gives the further exact implication

$$\inf_{p \in \mathbb{P}_{1,12}} \Xi_p \geq 1 \iff \inf_{p \in \mathbb{P}_{1,12}} \mathcal{E}_p^{\text{lab}} \geq 2 \implies \text{ES}. \quad (2252)$$

Proof. By theorem 9.70, it is enough and necessary to prove the decomposition for every prime. The prime 2 has

$$\frac{4}{2} = \frac{1}{1} + \frac{1}{2} + \frac{1}{2}.$$

Every prime $p \equiv 3 \pmod{4}$ has the explicit decomposition (1497). Hence the global assertion remains only for primes congruent to 1 or 5 modulo 12. Every prime $p \equiv 5 \pmod{12}$ has the endpoint-shell decomposition (1500). Consequently the global assertion is equivalent to the local assertion for every $p \in \mathbb{P}_{1,12}$.

Corollary 14.244 identifies that local assertion with $Q_p > 0$. Since $Q_p \in \mathbb{Z}_{\geq 0}$, positivity for every $p \in \mathbb{P}_{1,12}$ is equivalent to

$$\inf_{p \in \mathbb{P}_{1,12}} Q_p \geq 1.$$

The exact identities

$$\text{Vol}_p^{\det} = 2^{6Q_p}, \quad L_{H,p}^2 = 51Q_p, \quad L_{K,p}^2 = \frac{75}{2}Q_p$$

then give the remaining three equivalences. Equation (2237) gives $\mathcal{E}_p^{\text{lab}} = 2\Xi_p$, proving the equivalence in (2252). The bound $\Xi_p \leq Q_p$ from (1563) proves its final implication. $\square$

Theorem 14.251 (Arithmetic four-Time multiplicity carrier). *Let $p \in \mathbb{P}_{1,12}$. For $a \in \mathcal{A}_p$, retain the complement-orbit set $\mathcal{O}_{R_a,p}$ from (1296), and form the finite disjoint union*

$$\mathcal{O}_p := \bigsqcup_{a \in \mathcal{A}_p} \mathcal{O}_{R_a,p}. \quad (2253)$$

Then

$$|\mathcal{O}_p| = Q_p = \sum_{a \in \mathcal{A}_p} \left(|\mathcal{S}_{R_a,a}^{\text{ext}}| + |\mathcal{S}_{R_a,a}^{\text{mid}}| \right). \quad (2254)$$

Let $\mathcal{T} = (\text{M}, \text{N}, \text{T}, \text{E})$ be the ordered four-Time label set and define

$$\mathcal{V}_p^{\text{Time}} := \mathbb{C}[\mathcal{O}_p] \otimes_{\mathbb{C}} \mathbb{C}[\mathcal{T}]. \quad (2255)$$

This carrier has the following exact properties.

(i) Its dimension is

$$\dim_{\mathbb{C}} \mathcal{V}_p^{\text{Time}} = 4Q_p. \quad (2256)$$

- (ii) The standard $SU(4)$ -action on $\mathbb{C}[\mathcal{T}]$ induces the action $I_{\mathbb{C}[\mathcal{O}_p]} \otimes g$ on $\mathcal{V}_p^{\text{Time}}$. It fixes the top covolume

$$\epsilon_{p,\mathcal{T}}^\vee := \bigwedge_{O \in \mathcal{O}_p} \left(e_{O,M}^\vee \wedge e_{O,N}^\vee \wedge e_{O,T}^\vee \wedge e_{O,E}^\vee \right). \quad (2257)$$

The product is independent of the ordering of $\mathcal{O}_p$.

- (iii) The existing six-path carrier and the four-Time carrier have the common arithmetic multiplicity factor

$$\mathcal{S}_p \cong \mathbb{C}[\mathcal{O}_p] \otimes \text{Ran } P_2 \otimes \mathbb{C}^3, \quad (2258)$$

$$\mathcal{V}_p^{\text{Time}} = \mathbb{C}[\mathcal{O}_p] \otimes \mathbb{C}[\mathcal{T}]. \quad (2259)$$

- (iv) The primewise Erdős–Straus assertion is equivalent to each of

$$Q_p > 0, \quad \mathcal{V}_p^{\text{Time}} \neq 0, \quad \dim_{\mathbb{C}} \mathcal{V}_p^{\text{Time}} \geq 4. \quad (2260)$$

Moreover,

$$\log_2 \text{Vol}_p^{\det} = \frac{3}{2} \dim_{\mathbb{C}} \mathcal{V}_p^{\text{Time}}, \quad (2261)$$

$$L_{H,p}^2 = \frac{51}{4} \dim_{\mathbb{C}} \mathcal{V}_p^{\text{Time}}, \quad L_{K,p}^2 = \frac{75}{8} \dim_{\mathbb{C}} \mathcal{V}_p^{\text{Time}}. \quad (2262)$$

The directional squarefree coordinate satisfies

$$\boxed{\mathcal{E}_p^{\text{lab}} = 2\Xi_p \leq \frac{1}{2} \dim_{\mathbb{C}} \mathcal{V}_p^{\text{Time}} = \frac{1}{3} \log_2 \text{Vol}_p^{\det}}. \quad (2263)$$

Equality in the inequality holds exactly when every occupied shell is one-sided and has $q_{R_a,p} = 1$.

Consequently,

$$\text{ES} \iff \inf_{p \in \mathbb{P}_{1,12}} \dim_{\mathbb{C}} \mathcal{V}_p^{\text{Time}} \geq 4 \iff \inf_{p \in \mathbb{P}_{1,12}} \text{Vol}_p^{\det} \geq 64. \quad (2264)$$

Proof. The disjoint union in (2253) has cardinality

$$\sum_{a \in \mathcal{A}_p} q_{R_a,p} = Q_p$$

by (2221). The squarefree-coordinate identity is (1555) summed over $\mathcal{A}_p$, equivalently (1562). Tensor dimensions now give (2256).

In the basis indexed by $\mathcal{O}_p \times \mathcal{T}$, the action of $I \otimes g$ is a direct sum of Q_p copies of g . Its determinant is therefore

$$(\det g)^{Q_p} = 1.$$

The displayed covolume is fixed by the same calculation. Interchanging two orbit blocks interchanges two degree-four factors, producing the sign $(-1)^{4 \cdot 4} = 1$; hence (2257) does not require an ordering of $\mathcal{O}_p$.

Equations (1315) and (2222) identify the direct sum of shell spaces with the first tensor product in (iii). The second is the definition in (2255).

The equivalence in (2260) follows from (2229)–(2230) and (2256). Substitution of $\dim \mathcal{V}_p^{\text{Time}} = 4Q_p$ into (2224) and (2227) proves (2261) and (2262). Combining (2237) with $\dim \mathcal{V}_p^{\text{Time}} = 4Q_p$ proves (2263); its equality statement is (2238). Finally, (2264) is theorem 14.250 with the same substitution. $\square$

Exact computation 14.252 (Arithmetic four-Time carrier certificate). The script and receipt

verification/verify_arithmetic_four_time_carrier.py,
verification/verify_arithmetic_four_time_carrier_output.txt

check the common orbit-factor dimensions, block-order sign, $SU(4)$ determinant action, determinant volume, both control lengths, and the nonvanishing equivalence for multiplicities $0 \leq Q_p \leq 8$. They also check (2263) for every integer pair $0 \leq \Xi_p \leq Q_p \leq 8$, including its algebraic equality criterion.

Remark 14.253 (Meaning of the four-Time extension). The theorem identifies an exact common arithmetic multiplicity space. The $C_2 \times C_3$ shell action remains the proved six-path action on $\text{Ran } P_2 \otimes \mathbb{C}^3$. The $SU(4)$ action occurs on a separate four-dimensional factor once for every arithmetic complement orbit. Their common object is precisely $\mathbb{C}[\mathcal{O}_p]$.

Remark 14.254 (Exact scope of the arithmetic bridge). Theorems 14.43 and 14.44 identify the arithmetic six-path carrier, its full D_3 action, and its invariant support line inside the eight-cell standard pair. The full linking algebra in theorem 14.7 is $M_6(\mathbb{C})$, of complex dimension 36, whereas the finite standard algebra in (1460) is $M_2(\mathbb{C}) \oplus M_2(\mathbb{C})$, of complex dimension 8. Their dimensions exclude a unital $*$ -isomorphism. The exact bridge proved here is the path-space intertwiner Ξ_Γ , followed by the Hilbert–Schmidt unitary Q^{-1} .

Remark 14.255 (Exact scope of the crosswalk). The blade orientation in (1389) is fixed by the observed E_{21} history and the embedding $B_- = j(2E_{21})$. The arithmetic orientation is fixed by $d < N_{\text{ar}}$, which defines A_- . The construction uses the basepoint equation $\Phi(h_{21}) = V_{--}$. A derivation of this basepoint equation from a common ambient multiplication law remains open. Independently, the 107-shell endpoint calculation in theorem 14.69 identifies the same E_{21} matrix line with the outgoing exit-boundary edge $r_0 \rightarrow r_1$. This fixes the direction of that input line; it does not supply the still-open ambient multiplication law.

Theorem 14.256 (Amplitude-retaining polar carrier of every occupied reduced shell). *Fix reduced shell data as in theorem 11.8, and assume $\mathcal{D}_{p,a}^{\text{red}} \neq \emptyset$. For $d \in \mathcal{D}_{p,a}^{\text{red}}$, put*

$$\alpha_d := \frac{4N_0 d}{(N_0 + d)^2}, \quad \iota(d) := \frac{N_0^2}{d}. \quad (2265)$$

Let

$$\mathcal{H}_{p,a} := \mathbb{C}[\mathcal{D}_{p,a}^{\text{red}}] \quad (2266)$$

with orthonormal basis $(e_d)_d$, and define

$$P_{p,a} e_d := e_{\iota(d)}, \quad (2267)$$

$$H_{p,a} := \sum_d \sqrt{\alpha_d} e_d e_d^*, \quad D_{p,a} := H_{p,a}^2 = \sum_d \alpha_d e_d e_d^*. \quad (2268)$$

Then

$$P_{p,a}^\top = P_{p,a}, \quad P_{p,a}^2 = I_{\mathcal{H}_{p,a}}, \quad P_{p,a} H_{p,a} = H_{p,a} P_{p,a}. \quad (2269)$$

Let (f_1, f_2, f_3) be the standard basis of $\mathbb{C}^3$. The amplitude-retaining directed and transposed channels are

$$\mathcal{N}_{21}^{p,a} := 2H_{p,a} P_{p,a} \otimes F_{32}, \quad (2270)$$

$$\mathcal{N}_{12}^{p,a} := (\mathcal{N}_{21}^{p,a})^\top = 2H_{p,a} P_{p,a} \otimes F_{23}. \quad (2271)$$

Their basis actions are

$$\mathcal{N}_{21}^{p,a}(e_d \otimes f_2) = 2\sqrt{\alpha_d} e_{\iota(d)} \otimes f_3, \quad (2272)$$

$$\mathcal{N}_{12}^{p,a}(e_d \otimes f_3) = 2\sqrt{\alpha_d} e_{\iota(d)} \otimes f_2, \quad (2273)$$

and both operators vanish on the other two coordinate summands. In particular, the coordinate coefficient on every displayed arrow is $2\sqrt{\alpha_d} F_{32} = j(N_d^{-,+})$, respectively $2\sqrt{\alpha_d} F_{23} = j(N_d^{+,-})$, with the ordered blade coefficients of theorem 14.25.

The exact weighted source and range products are

$$(\mathcal{N}_{21}^{p,a})^\top \mathcal{N}_{21}^{p,a} = 4D_{p,a} \otimes F_{22}, \quad (2274)$$

$$\mathcal{N}_{21}^{p,a} (\mathcal{N}_{21}^{p,a})^\top = 4D_{p,a} \otimes F_{33}. \quad (2275)$$

Since every α_d is positive, $H_{p,a}$ is invertible. The Moore–Penrose inverse is

$$(\mathcal{N}_{21}^{p,a})^\dagger = \frac{1}{2} H_{p,a}^{-1} P_{p,a} \otimes F_{23}. \quad (2276)$$

It recovers the unweighted source and range carriers exactly:

$$(\mathcal{N}_{21}^{p,a})^\dagger \mathcal{N}_{21}^{p,a} = I_{\mathcal{H}_{p,a}} \otimes F_{22} = I_{\mathcal{H}_{p,a}} \otimes \mathcal{J}(E_{11}), \quad (2277)$$

$$\mathcal{N}_{21}^{p,a} (\mathcal{N}_{21}^{p,a})^\dagger = I_{\mathcal{H}_{p,a}} \otimes F_{33} = I_{\mathcal{H}_{p,a}} \otimes \mathcal{J}(E_{22}). \quad (2278)$$

The polar partial-isometry factor is

$$U_{21}^{p,a} := P_{p,a} \otimes F_{32}, \quad \mathcal{N}_{21}^{p,a} = U_{21}^{p,a} (2H_{p,a} \otimes F_{22}). \quad (2279)$$

If every positive entry of $\mathcal{N}_{21}^{p,a}$ is lifted to its supported copy in $G(\mathbb{R})$ and every absent entry is lifted to τ , the unique entrywise Boolean character returns $U_{21}^{p,a}$. Thus the numerical operator retains all shell amplitudes, while its Boolean image retains exactly the complement incidence and the directed $E_{11} \rightarrow E_{22}$ carrier.

The all-shell amplitude is the trace of $D_{p,a}$, and the Hilbert–Schmidt norm of the directed channel is

$$\mathrm{tr} D_{p,a} = \alpha_{p,a}^\Sigma, \quad \|\mathcal{N}_{21}^{p,a}\|_{\mathrm{HS}}^2 = 4\alpha_{p,a}^\Sigma. \quad (2280)$$

For $(p, a) = (12289, 3075)$, one has $R_0 = 11$ and $N_0 = 12289 \cdot 3075$. The established certificate map

$$\theta_d : \mathcal{X}_{11,12289} \xrightarrow{\sim} \mathcal{D}_{12289,3075}^{\mathrm{red}}, \quad \theta_d(j, u) = 12289^j u, \quad (2281)$$

intertwines the two complement involutions. Its linearization $\hat{\theta}_d : V_{\mathcal{X}} \rightarrow \mathcal{H}_{12289,3075}$ therefore satisfies

$$(\hat{\theta}_d^{-1} \otimes I_3) U_{21}^{12289,3075} (\hat{\theta}_d \otimes I_3) = P_\iota \otimes F_{32} = V_{21}^{(12289)}. \quad (2282)$$

Hence the complete four-channel carrier of theorem 14.36 is the exact polar carrier of the amplitude-retaining reduced-shell operator.

Proof. The complement identity in theorem 11.8 proves that $P_{p,a}$ is an involutive permutation matrix. Direct substitution gives

$$\alpha_{\iota(d)} = \frac{4N_0(N_0^2/d)}{(N_0 + N_0^2/d)^2} = \frac{4N_0d}{(N_0 + d)^2} = \alpha_d.$$

Therefore $H_{p,a} e_{\iota(d)} = \sqrt{\alpha_d} e_{\iota(d)}$, proving (2269) and the two basis actions.

The matrix-unit products

$$F_{23} F_{32} = F_{22}, \quad F_{32} F_{23} = F_{33}$$

together with

$$(H_{p,a} P_{p,a})^\top = H_{p,a} P_{p,a}, \quad (H_{p,a} P_{p,a})^2 = D_{p,a}$$

prove (2274) and (2275). Multiplying (2276) on the two sides of $\mathcal{N}_{21}^{p,a}$ gives the projectors in (2277) and (2278). These two products are self-adjoint, and the two remaining Penrose identities follow by one further multiplication. This proves the Moore–Penrose assertion.

Since $F_{32} F_{22} = F_{32}$ and $P_{p,a} H_{p,a} = H_{p,a} P_{p,a}$, the product in (2279) is exactly $\mathcal{N}_{21}^{p,a}$. The source and range projectors of $U_{21}^{p,a}$ are the right-hand sides of (2277) and (2278). Its occupied entries are precisely the occupied entries of $\mathcal{N}_{21}^{p,a}$, proving the Boolean statement.

By (549),

$$\mathrm{tr} D_{p,a} = \sum_d \alpha_d = \alpha_{p,a}^\Sigma.$$

Taking the trace in (2274) proves (2280).

For the final specialization, $4 \cdot 3075 - 12289 = 11$ and $\gcd(11, 12289 \cdot 3075) = 1$. The bijection in the proof of theorem 9.64 is therefore exactly (2281). Moreover,

$$\theta_d \left(2 - j, \frac{3075^2}{u} \right) = 12289^{2-j} \frac{3075^2}{u} = \frac{N_0^2}{12289^j u} = \iota(\theta_d(j, u)).$$

Thus $\hat{\theta}_d^{-1} P_{12289, 3075} \hat{\theta}_d = P_\iota$, which proves (2282). $\square$

Corollary 14.257 (Endpoint-selected polar carrier and amplitude-weighted direction). *Let $p \equiv 1 \pmod{4}$ be prime, let $a \in \mathcal{A}_p$, and put $R = 4a - p$. Retain the endpoint bits b_j , the cell*

$$C_{0+} = e_{b_1} e_{b_0}^\top,$$

and, for an occupied shell, the operators $\mathcal{H}_{p,a}$, $H_{p,a}$, and $P_{p,a}$ of theorem 14.256. For an empty shell, read these as the zero-dimensional space and its zero operators, and put $\alpha_{p,a}^\Sigma = 0$. Define

$$\mathcal{N}_{0+}^{p,a} := 2H_{p,a}P_{p,a} \otimes \mathcal{J}(C_{0+}), \quad U_{0+}^{p,a} := P_{p,a} \otimes \mathcal{J}(C_{0+}), \quad (2283)$$

$$\mathcal{J}_\partial^{p,a} := I_{\mathcal{H}_{p,a}} \otimes (F_{22} - F_{33}). \quad (2284)$$

The complete status table is

status	C_{0+}	$\mathcal{N}_{0+}^{p,a}$
empty	E_{22}	0
exterior only	E_{12}	$\mathcal{N}_{12}^{p,a}$
middle only	E_{21}	$\mathcal{N}_{21}^{p,a}$
both	E_{11}	$2H_{p,a}P_{p,a} \otimes F_{22}$.

(2285)

Its Moore–Penrose source and range are the actual endpoint projectors:

$$(\mathcal{N}_{0+}^{p,a})^\dagger \mathcal{N}_{0+}^{p,a} = I_{\mathcal{H}_{p,a}} \otimes \mathcal{J}(e_{b_0} e_{b_0}^\top), \quad (2286)$$

$$\mathcal{N}_{0+}^{p,a} (\mathcal{N}_{0+}^{p,a})^\dagger = I_{\mathcal{H}_{p,a}} \otimes \mathcal{J}(e_{b_1} e_{b_1}^\top). \quad (2287)$$

Moreover,

$$\text{rank } \mathcal{N}_{0+}^{p,a} = |\mathcal{D}_{p,a}^{\text{red}}|, \quad \|\mathcal{N}_{0+}^{p,a}\|_{\text{HS}}^2 = 4\alpha_{p,a}^\Sigma. \quad (2288)$$

The grading commutator retains both the sign and the full amplitude:

$$\boxed{[\mathcal{J}_\partial^{p,a}, \mathcal{N}_{0+}^{p,a}] = 2(b_1 - b_0) \mathcal{N}_{0+}^{p,a}.} \quad (2289)$$

Consequently the amplitude-weighted directional energy is

$$\boxed{\begin{aligned} \mathcal{E}_{p,a}^{\text{amp}} &:= \frac{1}{4} \|[\mathcal{J}_\partial^{p,a}, \mathcal{N}_{0+}^{p,a}]\|_{\text{HS}}^2 \\ &= 4\alpha_{p,a}^\Sigma (b_1 - b_0)^2 \\ &= 2\alpha_{p,a}^\Sigma \mathcal{E}_{R,p}^{\text{lab}}. \end{aligned}} \quad (2290)$$

Thus this energy is positive exactly on a one-sided shell and vanishes on an empty or both-channel shell.

Proof. The endpoint table (1590) and the embedding $\mathcal{J}(E_{ij}) = F_{i+1, j+1}$ give (2285). On an occupied shell, $H_{p,a}P_{p,a}$ is invertible and $\mathcal{J}(C_{0+})$ is a rank-one matrix unit. Its Moore–Penrose inverse is therefore

$$(\mathcal{N}_{0+}^{p,a})^\dagger = \frac{1}{2} H_{p,a}^{-1} P_{p,a} \otimes \mathcal{J}(C_{0+}^\top).$$

Multiplication gives (2286) and (2287). The same calculation gives the rank formula, while

$$\text{tr} \left((\mathcal{N}_{0+}^{p,a})^\top \mathcal{N}_{0+}^{p,a} \right) = 4 \text{tr } D_{p,a} = 4\alpha_{p,a}^\Sigma$$

proves the norm formula. Every identity also holds on the zero-dimensional empty-shell space.

The lower-coordinate embedding sends J_B to $F_{22} - F_{33}$. Applying (1599) at r_0 gives

$$[F_{22} - F_{33}, j(C_{0+})] = 2(b_1 - b_0)j(C_{0+}),$$

which proves (2289). Taking the squared Hilbert–Schmidt norm and using (1610) gives (2290). $\square$

Corollary 14.258 (Global directed-carrier form of Erdős–Straus). *For an empty reduced shell, read $\mathcal{H}_{p,a}$, $\mathcal{N}_{21}^{p,a}$, and $U_{21}^{p,a}$ as the zero-dimensional space and its zero operators. For every shell,*

$$\text{rank } \mathcal{N}_{21}^{p,a} = \text{rank } U_{21}^{p,a} = |\mathcal{D}_{p,a}^{\text{red}}| = \Omega_R^{\text{red}}(p). \quad (2291)$$

Consequently the following conditions are equivalent:

$$\text{ES} \iff \forall p \in \mathbb{P} \exists a \in \mathbb{Z}, a > p/4 : \mathcal{N}_{21}^{p,a} \neq 0 \quad (2292)$$

$$\iff \forall p \in \mathbb{P} \exists a \in \mathbb{Z}, a > p/4 : U_{21}^{p,a} \neq 0 \quad (2293)$$

$$\iff \forall p \in \mathbb{P} \exists a \in \mathbb{Z}, a > p/4 : \begin{cases} (\mathcal{N}_{21}^{p,a})^\dagger \mathcal{N}_{21}^{p,a} = I_{\mathcal{H}_{p,a}} \otimes F_{22}, \\ \mathcal{N}_{21}^{p,a} (\mathcal{N}_{21}^{p,a})^\dagger = I_{\mathcal{H}_{p,a}} \otimes F_{33}, \\ I_{\mathcal{H}_{p,a}} \neq 0. \end{cases} \quad (2294)$$

The endpoint-selected carrier gives the further exact free-shell form

$$\text{ES} \iff \forall p \in \mathbb{P}, p \equiv 1 \pmod{4} \exists a \in \mathcal{A}_p : \mathcal{N}_{0+}^{p,a} \neq 0. \quad (2295)$$

For every witnessing shell, the entrywise split-zero character satisfies

$$\widehat{\chi}_{\mathbb{R}}(\widetilde{\mathcal{N}}_{21}^{p,a}) = U_{21}^{p,a}, \quad (2296)$$

and its source-to-range direction is

$$I_{\mathcal{H}_{p,a}} \otimes j(E_{11}) \longrightarrow I_{\mathcal{H}_{p,a}} \otimes j(E_{22}).$$

Thus (531) and the directed channel criterion above have the same shell witnesses, with the full divisor amplitudes retained by $\mathcal{N}_{21}^{p,a}$.

Proof. If the shell is occupied, $H_{p,a}P_{p,a}$ is invertible on $\mathcal{H}_{p,a}$, while F_{32} has rank one. Hence

$$\text{rank } \mathcal{N}_{21}^{p,a} = \dim \mathcal{H}_{p,a} = |\mathcal{D}_{p,a}^{\text{red}}|.$$

The same rank calculation applies to $U_{21}^{p,a}$. The empty-shell case gives zero on every term, proving (2291).

The global positivity criterion in theorem 9.70 states that Erdős–Straus is equivalent to an occupied reduced shell for every prime. Equation (2291) proves (2292) and (2293). By theorem 14.257, $\mathcal{N}_{0+}^{p,a} \neq 0$ exactly when the free shell is occupied. The explicit decompositions for $p = 2$ and $p \equiv 3 \pmod{4}$ then prove (2295). The source and range identities in theorem 14.256 prove (2294). Its entrywise split-zero calculation proves (2296). $\square$

Theorem 14.259 (Exact complement-orbit spectral descent). *Retain an occupied reduced shell and assume $R_0 > 2$. Let*

$$\mathcal{O}_{p,a} := \mathcal{D}_{p,a}^{\text{red}} / \langle \iota \rangle.$$

Every orbit $O \in \mathcal{O}_{p,a}$ contains two divisors, and

$$\alpha_O := \alpha_d \quad (d \in O) \quad (2297)$$

is well defined. The complement eigenspaces

$$\mathcal{H}_{p,a}^+ := \ker(P_{p,a} - I), \quad \mathcal{H}_{p,a}^- := \ker(P_{p,a} + I) \quad (2298)$$

are orthogonal, have dimension $|\mathcal{O}_{p,a}|$, and reduce $H_{p,a}$ and $D_{p,a}$. There is a canonical isometry

$$J_+ : \mathbb{C}[\mathcal{O}_{p,a}] \longrightarrow \mathcal{H}_{p,a}^+, \quad J_+ e_O = \frac{e_d + e_{\iota(d)}}{\sqrt{2}}, \quad d \in O. \quad (2299)$$

It carries the quotient amplitude operator to

$$J_+^\top D_{p,a} J_+ = \sum_{O \in \mathcal{O}_{p,a}} \alpha_O e_O e_O^*. \quad (2300)$$

Put

$$\Lambda_{p,a} := H_{p,a} P_{p,a}. \quad (2301)$$

Its restrictions to the complement eigenspaces have the exact spectra

$$\text{Spec}(\Lambda_{p,a}|_{\mathcal{H}_{p,a}^+}) = \{\sqrt{\alpha_O} : O \in \mathcal{O}_{p,a}\}, \quad (2302)$$

$$\text{Spec}(\Lambda_{p,a}|_{\mathcal{H}_{p,a}^-}) = \{-\sqrt{\alpha_O} : O \in \mathcal{O}_{p,a}\}. \quad (2303)$$

Consequently

$$\det(tI - \Lambda_{p,a}) = \prod_{O \in \mathcal{O}_{p,a}} (t^2 - \alpha_O), \quad (2304)$$

$$\det(tI - D_{p,a}) = \prod_{O \in \mathcal{O}_{p,a}} (t - \alpha_O)^2, \quad (2305)$$

$$\text{tr}(\Lambda_{p,a}^{2k+1}) = 0, \quad k \geq 0, \quad (2306)$$

$$\text{tr}(\Lambda_{p,a}^{2k}) = 2 \sum_{O \in \mathcal{O}_{p,a}} \alpha_O^k, \quad k \geq 1, \quad (2307)$$

$$\text{tr}(D_{p,a}^k) = 2 \sum_{O \in \mathcal{O}_{p,a}} \alpha_O^k, \quad k \geq 1. \quad (2308)$$

In particular,

$$\alpha_{p,a}^\Sigma = 2 \sum_{O \in \mathcal{O}_{p,a}} \alpha_O, \quad \|\mathcal{N}_{21}^{p,a}\|_{\text{HS}}^2 = 8 \sum_{O \in \mathcal{O}_{p,a}} \alpha_O. \quad (2309)$$

Under the arithmetic hypotheses and notation of theorem 14.260, define

$$\mathcal{O}_{02} := \{O : O \text{ meets the origin-0 and origin-2 fibres}\},$$

$$\mathcal{O}_{11} := \{O : O \text{ lies in the origin-1 fibre}\}.$$

Then

$$\mathcal{O}_{p,a} = \mathcal{O}_{02} \sqcup \mathcal{O}_{11}, \quad (2310)$$

$$|\mathcal{O}_{02}| = \mu(T_2(p)), \quad (2311)$$

$$|\mathcal{O}_{11}| = \frac{1}{2}\mu(T_1(p)), \quad (2312)$$

$$|\mathcal{O}_{p,a}| = \mu(T_2(p)) + \frac{1}{2}\mu(T_1(p)). \quad (2313)$$

The corresponding amplitude sums are

$$\sum_{O \in \mathcal{O}_{02}} \alpha_O = \sum_{d \text{ in origin } 0} \alpha_d = \sum_{d \text{ in origin } 2} \alpha_d, \quad (2314)$$

$$\sum_{O \in \mathcal{O}_{11}} \alpha_O = \frac{1}{2} \sum_{d \text{ in origin } 1} \alpha_d. \quad (2315)$$

For $(p, a) = (12289, 3075)$, ordering the six orbits by their member $d < N_0$ gives the exact quotient spectrum

$$\{\alpha_O : O \in \mathcal{O}_{12289,3075}\} = \left\{ \frac{61445}{943902729}, \frac{184335}{339886096}, \frac{615}{94864}, \frac{921675}{38217124}, \frac{123}{484}, \frac{1025}{1089} \right\}. \quad (2316)$$

These six rational numbers are pairwise distinct. The three exterior orbits and three middle orbits therefore retain data beyond the two coefficient values of the parity quotient $\mu = q^*(3, 6)^\top$.

Proof. The hypotheses $R_0 > 2$ and theorem 11.8 give two elements in every complement orbit. Equation (2269) gives $\alpha_{\iota(d)} = \alpha_d$, proving (2297). The spectral projectors

$$Q_{\pm} := \frac{1}{2}(I \pm P_{p,a})$$

have ranges $\mathcal{H}_{p,a}^{\pm}$. Since $D_{p,a}$ and $H_{p,a}$ commute with $P_{p,a}$, both eigenspaces are reducing. The right-hand side of (2299) is independent of the member $d \in O$; different orbits have disjoint supports. This proves the isometry and (2300).

On the two-dimensional coordinate subspace of an orbit $O = \{d, \iota(d)\}$,

$$H_{p,a} = \sqrt{\alpha_O} I_2, \quad P_{p,a} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \Lambda_{p,a} = \sqrt{\alpha_O} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The invariant and anti-invariant lines therefore have eigenvalues $\sqrt{\alpha_O}$ and $-\sqrt{\alpha_O}$. Taking the product over the complement orbits proves (2302)–(2305). Summing the powers of the two eigenvalues on each orbit proves (2306)–(2308). The case $k = 1$, together with (2280), proves (2309).

The arithmetic involution exchanges the origin-0 and origin-2 fibres and preserves the origin-1 fibre. It has no fixed divisor when $R_0 > 2$. The coefficient identities $m_j = \mu(T_j(p))$ and $m_0 = m_2$ from theorem 14.260 therefore give (2310)–(2313). Complement invariance of α_d gives (2314) and (2315).

For 12289, the members below $N_0 = 37788675$ are

$$615, \quad 5125, \quad 61445, \quad 230625, \quad 2765025, \quad 23041875.$$

Substitution into $\alpha_d = 4N_0d/(N_0 + d)^2$ gives the six fractions in (2316). Cross multiplication of each pair proves that they are distinct. The orbit-type counts are $\mu(T_2) = 3$ and $\mu(T_1)/2 = 3$, while the displayed spectrum has six distinct entries, proving the final assertion. $\square$

Theorem 14.260 (The Star–Kneser support sets are the supports of the exact divisor convolution). *Retain the factorization, finite additive group, and subsets of theorem 10.6. For*

$$I_i := \{-e_i, -e_i + 1, \dots, e_i\},$$

define the nonnegative integer-valued function

$$\mu_i(h) := \#\{\beta_i \in I_i : \beta_i \bar{\ell}_i = h\}, \quad h \in H, \quad (2317)$$

and let

$$\mu := \mu_1 * \dots * \mu_s, \quad (f * g)(h) := \sum_{x+y=h} f(x)g(y). \quad (2318)$$

Then

$$\mu(h) = \#\left\{(\beta_1, \dots, \beta_s) \in \prod_{i=1}^s I_i : \sum_{i=1}^s \beta_i \bar{\ell}_i = h\right\}, \quad (2319)$$

$$\text{supp}(\mu_i) = A_i, \quad \text{supp}(\mu) = B, \quad (2320)$$

$$\mu(-h) = \mu(h). \quad (2321)$$

Here $-h$ denotes the inverse of h in the additive notation for H .

For a subgroup $K \leq H$, let $\pi_K : H \rightarrow H/K$ be the quotient map and define

$$(\pi_{K*} f)(\bar{h}) := \sum_{x \in \pi_K^{-1}(\bar{h})} f(x). \quad (2322)$$

Then

$$\pi_{K*} \mu = (\pi_{K*} \mu_1) * \dots * (\pi_{K*} \mu_s), \quad (2323)$$

$$\text{supp}(\pi_{K*} \mu_i) = (A_i + K)/K, \quad \text{supp}(\pi_{K*} \mu) = (B + K)/K. \quad (2324)$$

Consequently the subgroup surplus has the exact form

$$\Sigma_T(K) = 1 + \sum_{i=1}^s (|\text{supp}(\pi_{K*}\mu_i)| - 1) + |\pi_K(T)| - |H/K|. \quad (2325)$$

Extend μ by zero from H to $(\mathbb{Z}/R\mathbb{Z})^\times$. The origin counts are the three exact target coefficients

$$m_j = \mu(T_j(p)), \quad j = 0, 1, 2. \quad (2326)$$

In particular,

$$B \cap T \neq \emptyset \iff \mu(T_1(p)) + \mu(T_2(p)) > 0. \quad (2327)$$

Since $T_0(p)$ is the group inverse of $T_2(p)$,

$$m_0 = m_2, \quad \Omega_R(p) = \mu(T_1(p)) + 2\mu(T_2(p)). \quad (2328)$$

If $R > 2$, the fibre over $T_1(p) = -1$ has no fixed point under $(\beta_i)_i \mapsto (-\beta_i)_i$, so $\mu(T_1(p))$ is even. The cover of theorem 14.27 therefore satisfies

$$|\Phi_L(\tilde{\mathcal{X}}_{R,p})| = 3(\mu(T_1(p)) + 2\mu(T_2(p))), \quad (2329)$$

$$|\tilde{\mathcal{X}}_{R,p}/D_3| = \mu(T_2(p)) + \frac{1}{2}\mu(T_1(p)) = |\text{im}(\mathcal{J}_{p,a})|. \quad (2330)$$

For $(p, R, a) = (12289, 11, 3075)$, use the additive logarithm to base 2 on $H \cong C_{10}$. The full coefficient vector is

$$(\mu(0), \mu(1), \dots, \mu(9)) = (3, 6, 3, 6, 3, 6, 3, 6, 3, 6). \quad (2331)$$

For the four subgroups of orders 1, 2, 5, 10, the pushed coefficient vectors, with quotient cosets in increasing representative order, are

$ K $	$\pi_{K*}\mu$
1	(3, 6, 3, 6, 3, 6, 3, 6, 3, 6)
2	(9, 9, 9, 9, 9)
5	(15, 30)
10	(45)

(2332)

The target coordinates $T_1 = 5$ and $T_2 = 4$ give $m_1 = 6$, $m_2 = 3$, $\Omega_{11}(12289) = 12$, and six carrier orbits.

Proof. For an exponent vector $\beta = (\beta_1, \dots, \beta_s)$, put

$$u(\beta) := \prod_{i=1}^s \ell_i^{e_i + \beta_i}. \quad (2333)$$

Unique prime factorization makes $\beta \mapsto u(\beta)$ a bijection from $\prod_i I_i$ to the positive divisors of a^2 , and

$$\overline{u(\beta)} \bar{a}^{-1} = \sum_{i=1}^s \beta_i \bar{\ell}_i \quad (2334)$$

in the additive notation for H . Expanding the convolution in (2318) now gives (2319).

The support of μ_i is A_i by its definition. Every summand in the convolution is a nonnegative integer. A coefficient $\mu(h)$ is positive exactly when $h = h_1 + \dots + h_s$ with $h_i \in A_i$. This is the defining membership condition for $B = A_1 + \dots + A_s$, proving (2320). Negating every exponent is a bijection between the fibres over h and $-h$, proving (2321).

For two functions $f, g : H \rightarrow \mathbb{Z}_{\geq 0}$, finite rearrangement of the sums gives

$$\begin{aligned} (\pi_{K*}(f * g))(\bar{h}) &= \sum_{\pi_K(z)=\bar{h}} \sum_{x+y=z} f(x)g(y) \\ &= \sum_{\bar{x}+\bar{y}=\bar{h}} (\pi_{K*}f)(\bar{x})(\pi_{K*}g)(\bar{y}). \end{aligned}$$

Iteration proves (2323). A quotient coefficient is positive exactly when its coset meets the original support, which proves (2324). Substitution in (418) gives (2325).

The bijection $\beta \mapsto u(\beta)$, followed by $(j, u) \mapsto p^j u$, identifies the fibre over $T_j(p)$ with the origin- j certificate fibre. This proves (2326). The definition of T gives (2327). Inversion sends $T_2(p) = -p^{-1}$ to $T_0(p) = -p$; apply (2321) to obtain (2328).

The exponent involution has the unique fixed vector $\beta = 0$, whose residue is the identity of H . For $R > 2$, the identity differs from $-1 = T_1(p)$. Hence the middle target fibre is a disjoint union of two-element involution orbits. The cardinality formulas in theorem 14.27 now give (2329) and (2330).

For the final calculation, the three local exponent intervals and their additive logarithmic steps are

$$(I_3, I_5, I_{41}) = ([-1, 1], [-2, 2], [-1, 1]), \quad (\log_2 3, \log_2 5, \log_2 41) = (8, 4, 3) \pmod{10}.$$

Enumeration of the $3 \cdot 5 \cdot 3 = 45$ exponent vectors gives (2331). Summing its entries over the cosets of the subgroups of orders 2, 5, 10 gives (2332). Reading entries five and four gives the stated target counts and the remaining numerical conclusions. $\square$

Theorem 14.261 (Exact character transform and blade intertwiner of the 12289 shell). *Retain the additive coordinate $H = C_{10}$, the coefficient function μ , and the local exponent data in theorem 14.260. Put*

$$\zeta_{10} := e^{2\pi i/10}, \quad \chi_k(h) := \zeta_{10}^{kh}, \quad \hat{\mu}(k) := \sum_{h=0}^9 \mu(h) \overline{\chi_k(h)}.$$

Then the character transform factors through the three local exponent intervals as

$$\hat{\mu}(k) = \prod_{(e,d) \in \{(1,8), (2,4), (1,3)\}} \left(\sum_{\beta=-e}^e \zeta_{10}^{-kd\beta} \right), \quad (2335)$$

and its complete coefficient vector is

$$(\hat{\mu}(0), \dots, \hat{\mu}(9)) = (45, 0, 0, 0, 0, -15, 0, 0, 0, 0). \quad (2336)$$

Let

$$q : C_{10} \longrightarrow C_2, \quad q(h) = h \pmod{2}, \quad \nu = \begin{pmatrix} 3 \\ 6 \end{pmatrix},$$

and let $q^ : \mathbb{C}^{C_2} \rightarrow \mathbb{C}^{C_{10}}$ denote pullback of functions. Then*

$$\mu = q^* \nu. \quad (2337)$$

Define the unitary Fourier matrix W_{10} and the character embedding ι by

$$(W_{10})_{h,k} = \frac{1}{\sqrt{10}} \zeta_{10}^{hk}, \quad \iota e_0 = e_0, \quad \iota e_1 = e_5. \quad (2338)$$

For the C_2 Fourier frame U_0 of (1155), one has the exact intertwining equations

$$W_{10}^* q^* U_0 = \sqrt{5} \iota, \quad (2339)$$

$$W_{10}^* \mu = \frac{1}{\sqrt{10}} (45e_0 - 15e_5) = \sqrt{5} \iota U_0^* \nu. \quad (2340)$$

Let L_ν be left convolution by ν on $\mathbb{C}[C_2]$. In the ordered point basis and the ordered Fourier frame, respectively,

$$L_\nu = \begin{pmatrix} 3 & 6 \\ 6 & 3 \end{pmatrix}, \quad (2341)$$

$$U_0^* L_\nu U_0 = \begin{pmatrix} 9 & 0 \\ 0 & -3 \end{pmatrix}. \quad (2342)$$

With $\Pi_{\{0\}} = f_0 f_0^*$ and $Q_{\{0\}} = f_1 f_1^*$, this is equivalently

$$L_\nu = 9\Pi_{\{0\}} - 3Q_{\{0\}} = 12\Pi_{\{0\}} - 3I_2. \quad (2343)$$

Consequently the support-indexed modular Hamiltonian of theorem 14.16 satisfies

$$\text{ad}_{L_\nu} = 12 \text{ad}_{\Pi_{\{0\}}} = K_{\{0\}, e^{12}}. \quad (2344)$$

Under the blade carriage Φ_0 ,

$$\Phi_0 \text{ad}_{L_\nu} \Phi_0^{-1} = \text{ad}_{\text{diag}(9, -3)} = 6 \text{ad}_J, \quad (2345)$$

and hence

$$\text{ad}_{\text{diag}(9, -3)}(E_{12}) = 12E_{12}, \quad \text{ad}_{\text{diag}(9, -3)}(E_{21}) = -12E_{21}. \quad (2346)$$

The arithmetic blade $2E_{21}$ therefore occupies the -12 channel. By theorem 14.71, this is the exit-boundary eigenline; the transposed E_{12} line with eigenvalue $+12$ is the entry-boundary line. The eigenvalue gap is

$$9 - (-3) = 12 = \Omega_{11}(12289). \quad (2347)$$

Thus the divisor multiplicity, its parity quotient, and the ordered blade carrier are joined by the explicit maps q^* , ι , U_0 , and ad_{L_ν} . The general survivor theorem theorem 14.294 shows why this parity quotient retains support: the binary coefficient has magnitude $15 = 45/3$, so Fourier inversion is strictly positive on both parity sheets. The binary mode records multiplicity and does not remove a residue class.

Proof. The exponent-box formula in (2319) and multiplicativity of the character in each exponent give (2335). Its middle factor is

$$\sum_{\beta=-2}^2 \zeta_{10}^{-4k\beta}.$$

The exponents $4\beta \pmod{10}$ run through the five even residue classes. This sum is zero unless $k \in \{0, 5\}$, and it equals 5 at those two characters. At $k = 0$, the other factors both equal 3, giving $\hat{\mu}(0) = 45$. At $k = 5$, the step-8 factor equals 3, while the step-3 factor is

$$\sum_{\beta=-1}^1 (-1)^\beta = -1.$$

Therefore $\hat{\mu}(5) = -15$, proving (2336).

Equation (2331) says that $\mu(h) = 3$ on the even class and $\mu(h) = 6$ on the odd class. This is exactly (2337). If $v_r^{(2)}$ and $v_k^{(10)}$ denote the Fourier vectors of (1024), then

$$q^* v_r^{(2)} = v_{5r}^{(10)}, \quad r = 0, 1.$$

The column lengths are $\sqrt{2}$ and $\sqrt{10}$, respectively, which gives (2339). Application to ν , together with (2336), proves (2340).

Direct convolution on C_2 gives (2341). The vectors f_0, f_1 are its eigenvectors with eigenvalues $9, -3$, proving (2342) and (2343). For $\rho_{\{0\}, e^{12}} = \Pi_{\{0\}} + e^{12} Q_{\{0\}}$,

$$\log \rho_{\{0\}, e^{12}} = 12 Q_{\{0\}}.$$

The definition $K = -\log \Delta$ in (1144) therefore yields

$$K_{\{0\}, e^{12}} = -12 \text{ad}_{Q_{\{0\}}} = 12 \text{ad}_{\Pi_{\{0\}}} = \text{ad}_{L_\nu},$$

which is (2344). Conjugating by U_0 proves (2345). The two matrix commutators give (2346). Finally, $2E_{21}$ is the arithmetic blade by (1161). Equations (2328) and (2331) give

$$\Omega_{11}(12289) = \mu(5) + 2\mu(4) = 6 + 2 \cdot 3 = 12.$$

This proves (2347). □

Theorem 14.262 (Finite-abelian divisor transform and two-character gap criterion). *Let H be a finite abelian group, let $\widehat{H} = \text{Hom}(H, S^1)$, and let the functions μ_i, μ be as in theorem 14.260. For $f : H \rightarrow \mathbb{C}$, define*

$$\widehat{f}(\chi) := \sum_{h \in H} f(h) \overline{\chi(h)}. \quad (2348)$$

Then

$$\widehat{\mu}(\chi) = \prod_{i=1}^s \left(\sum_{\beta=-e_i}^{e_i} \overline{\chi(\bar{\ell}_i)}^\beta \right), \quad (2349)$$

$$\mu(h) = \frac{1}{|H|} \sum_{\chi \in \widehat{H}} \widehat{\mu}(\chi) \chi(h). \quad (2350)$$

Let $K \leq H$, put $Q = H/K$, and write

$$q : H \longrightarrow Q, \quad q^\vee : \widehat{Q} \longrightarrow \widehat{H}, \quad q^\vee(\psi) = \psi \circ q.$$

For a function $f : H \rightarrow \mathbb{C}$, the following conditions are equivalent:

- (1) $f = q^*g$ for a unique function $g : Q \rightarrow \mathbb{C}$;
- (2) $f(h+k) = f(h)$ for all $h \in H$ and $k \in K$;
- (3) $\text{supp } \widehat{f} \subseteq K^\perp := \text{im}(q^\vee)$.

When they hold,

$$\widehat{f}(q^\vee \psi) = |K| \widehat{g}(\psi), \quad \psi \in \widehat{Q}. \quad (2351)$$

For a finite abelian group G , let W_G be the unitary Fourier matrix with entries

$$(W_G)_{x,\gamma} = |G|^{-1/2} \gamma(x).$$

Define

$$\iota_q : \mathbb{C}^{\widehat{Q}} \longrightarrow \mathbb{C}^{\widehat{H}}, \quad \iota_q e_\psi = e_{q^\vee \psi}.$$

Then the quotient pullback has the exact unitary-Fourier form

$$W_H^* q^* W_Q = \sqrt{|K|} \iota_q. \quad (2352)$$

Suppose now that $Q = C_2$, that

$$g = \nu = \begin{pmatrix} a \\ b \end{pmatrix}, \quad a \geq 0, \quad b > 0,$$

and that the three target classes satisfy

$$q(T_0(p)) = q(T_2(p)) = 0, \quad q(T_1(p)) = 1.$$

Left convolution by ν has the exact forms

$$\mathbf{L}_\nu = \begin{pmatrix} a & b \\ b & a \end{pmatrix}, \quad (2353)$$

$$U_0^* \mathbf{L}_\nu U_0 = \text{diag}(a+b, a-b), \quad (2354)$$

$$\mathbf{L}_\nu = (a+b)\Pi_{\{0\}} + (a-b)Q_{\{0\}} = (a-b)I_2 + 2b\Pi_{\{0\}}. \quad (2355)$$

Therefore

$$\text{ad}_{\mathbf{L}_\nu} = 2b \text{ad}_{\Pi_{\{0\}}} = K_{\{0\}, e^{2b}}, \quad (2356)$$

and the two ordered blade lines have eigenvalues $+2b$ and $-2b$. Moreover,

$$(m_0, m_1, m_2) = (a, b, a), \quad \Omega_R(p) = b + 2a. \quad (2357)$$

The Fourier eigenvalue gap agrees with the shell count precisely when

$$2b = \Omega_R(p) \iff b = 2a. \quad (2358)$$

Proof. For finite functions on H , rearrangement of the finite sums gives

$$\widehat{f * g}(\chi) = \widehat{f}(\chi)\widehat{g}(\chi).$$

The definition of μ_i yields

$$\widehat{\mu}_i(\chi) = \sum_{\beta=-e_i}^{e_i} \overline{\chi(\bar{\ell}_i)}^\beta.$$

Iteration proves (2349). Character orthogonality gives

$$\frac{1}{|H|} \sum_{\chi \in \widehat{H}} \chi(h-x) = \begin{cases} 1, & h=x, \\ 0, & h \neq x, \end{cases}$$

and substitution into (2348) proves (2350).

A pullback q^*g is constant on every K -coset. Conversely, a K -translation-invariant function has one value on each coset and hence descends uniquely to Q . If f is K -translation invariant and $\chi|_K \neq 1$, choose $k \in K$ with $\chi(k) \neq 1$. Translation of the Fourier sum gives

$$\widehat{f}(\chi) = \overline{\chi(k)} \widehat{f}(\chi),$$

so $\widehat{f}(\chi) = 0$. If the transform is supported on $K^\perp$, the inversion formula contains only characters trivial on K , and thus gives $f(h+k) = f(h)$. This proves the three-way equivalence.

For $\psi \in \widehat{Q}$, summation coset by coset gives

$$\widehat{q^*g}(q^\vee\psi) = \sum_{\bar{h} \in Q} \sum_{x \in q^{-1}(\bar{h})} g(\bar{h}) \overline{\psi(\bar{h})} = |K| \widehat{g}(\psi),$$

which proves (2351). The character columns satisfy

$$q^*v_\psi^Q = v_{q^\vee\psi}^H.$$

Their lengths are $\sqrt{|Q|}$ and $\sqrt{|H|}$, so their unit columns differ by $\sqrt{|K|}$. This proves (2352).

The C_2 convolution table gives (2353). Its Fourier eigenvectors are f_0, f_1 , with eigenvalues $a+b, a-b$, proving (2354) and (2355). Hence

$$\text{ad}_{L_\nu} = 2b \text{ad}_{\Pi_{\{0\}}}.$$

For $\rho_{\{0\}, e^{2b}} = \Pi_{\{0\}} + e^{2b}Q_{\{0\}}$, the definition (1144) gives

$$K_{\{0\}, e^{2b}} = -2b \text{ad}_{Q_{\{0\}}} = 2b \text{ad}_{\Pi_{\{0\}}},$$

proving (2356). The ordered Fourier carriage of theorem 14.18 gives the two blade eigenvalues.

The assumed target classes and $f = q^*\nu$ give $(m_0, m_1, m_2) = (a, b, a)$. Equation (2328) gives $\Omega_R(p) = b + 2a$. The two convolution eigenvalues differ by $2b$. Equating that gap with $\Omega_R(p)$ yields

$$2b = b + 2a \iff b = 2a,$$

which proves (2358). □

Theorem 14.263 (Amplitude pushforward from complement orbits to the ordered two-character blade). *Assume the arithmetic hypotheses of theorem 11.26 and theorem 14.259. On $\mathbb{C}[\mathcal{O}_{p,a}]$, put*

$$D_{p,a}^{\text{orb}} := J_+^\top D_{p,a} J_+, \quad E_{02} := \sum_{O \in \mathcal{O}_{02}} e_O e_O^*, \quad E_{11} := \sum_{O \in \mathcal{O}_{11}} e_O e_O^*. \quad (2359)$$

Let

$$r : \mathcal{O}_{p,a} \longrightarrow C_2, \quad r(O) = \begin{cases} 0, & O \in \mathcal{O}_{02}, \\ 1, & O \in \mathcal{O}_{11}. \end{cases}$$

The pushforwards of the counting measure and the exact orbit-amplitude measure are

$$c_{p,a}(0) := \text{tr}(E_{02}) = m_2, \quad c_{p,a}(1) := \text{tr}(E_{11}) = \frac{m_1}{2}, \quad (2360)$$

$$\eta_{p,a}(0) := \text{tr}(E_{02}D_{p,a}^{\text{orb}}) = \alpha_2, \quad \eta_{p,a}(1) := \text{tr}(E_{11}D_{p,a}^{\text{orb}}) = \frac{\alpha_1}{2}. \quad (2361)$$

Thus

$$\begin{pmatrix} m_2 \\ m_1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} c_{p,a}(0) \\ c_{p,a}(1) \end{pmatrix}, \quad \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \eta_{p,a}(0) \\ \eta_{p,a}(1) \end{pmatrix}. \quad (2362)$$

Left convolution by $\eta_{p,a}$ on $\mathbb{C}[C_2]$ has the point-basis, Fourier, and projector forms

$$\mathbf{L}_\eta = \begin{pmatrix} \alpha_2 & \alpha_1/2 \\ \alpha_1/2 & \alpha_2 \end{pmatrix}, \quad (2363)$$

$$U_0^* \mathbf{L}_\eta U_0 = \text{diag} \left(\alpha_2 + \frac{\alpha_1}{2}, \alpha_2 - \frac{\alpha_1}{2} \right), \quad (2364)$$

$$\mathbf{L}_\eta = \left(\alpha_2 - \frac{\alpha_1}{2} \right) I_2 + \alpha_1 \Pi_{\{0\}}. \quad (2365)$$

Consequently,

$$\text{ad}_{\mathbf{L}_\eta} = \alpha_1 \text{ad}_{\Pi_{\{0\}}}, \quad (2366)$$

and, after carriage by U_0 , the ordered blade lines E_{12} and E_{21} have eigenvalues $+\alpha_1$ and $-\alpha_1$, respectively. The Fourier eigenvalue gap and the total quotient amplitude are

$$\text{gap}(\mathbf{L}_\eta) = \alpha_1, \quad \sum_{\epsilon \in C_2} \eta_{p,a}(\epsilon) = \frac{\alpha_{p,a}^\Sigma}{2}. \quad (2367)$$

They agree exactly under the amplitude balance equation

$$\text{gap}(\mathbf{L}_\eta) = \frac{\alpha_{p,a}^\Sigma}{2} \iff \alpha_1 = 2\alpha_2. \quad (2368)$$

Likewise,

$$\text{gap}(\mathbf{L}_c) = |\mathcal{O}_{p,a}| \iff m_1 = 2m_2. \quad (2369)$$

For $(p, a) = (12289, 3075)$, one has

$$c_{12289,3075} = (3, 3), \quad (2370)$$

$$\eta_{12289,3075}(0) = \frac{5176199252561022635}{209357204251173091344}, \quad (2371)$$

$$\eta_{12289,3075}(1) = \frac{1026107}{853776}. \quad (2372)$$

The count convolution has Fourier spectrum $(6, 0)$, so its gap equals the six-orbit count. The amplitude quantities are

$$\text{gap}(\mathbf{L}_\eta) = \alpha_1 = \frac{1026107}{426888}, \quad (2373)$$

$$\frac{\alpha_{12289,3075}^\Sigma}{2} = \frac{128395625711903946059}{104678602125586545672}, \quad (2374)$$

$$\alpha_1 - 2\alpha_2 = \frac{28287287984238504}{12015450198070081} > 0. \quad (2375)$$

Hence the 12289 count identity $m_1 = 2m_2$ holds, and its exact orbit-amplitude measure has the positive defect in (2375).

Proof. The quotient amplitude operator is diagonal in the orbit basis:

$$D_{p,a}^{\text{orb}} = \sum_{O \in \mathcal{O}_{p,a}} \alpha_O e_O e_O^*.$$

Taking the traces against E_{02} and E_{11} gives the two amplitude sums in theorem 14.259. Those sums are α_2 and $\alpha_1/2$ by (620) and (621). The corresponding projector ranks are m_2 and $m_1/2$. This proves (2360), (2361), and the two coordinate maps in (2362).

The multiplication table of C_2 gives (2363). The two Fourier vectors f_0, f_1 have eigenvalues $\alpha_2 + \alpha_1/2$ and $\alpha_2 - \alpha_1/2$. Since $\Pi_{\{0\}} = f_0 f_0^*$ and $Q_{\{0\}} = f_1 f_1^*$, this proves (2364) and (2365). Scalar matrices vanish under the commutator, proving (2366). In the ordered Fourier frame, evaluation of the commutator with $\text{diag}(\alpha_2 + \alpha_1/2, \alpha_2 - \alpha_1/2)$ on E_{12} and E_{21} gives the two ordered eigenvalues.

The difference of the Fourier eigenvalues is α_1 . Equations (622) and (2361) give

$$\eta_{p,a}(0) + \eta_{p,a}(1) = \alpha_2 + \frac{\alpha_1}{2} = \frac{\alpha_{p,a}^\Sigma}{2}.$$

Equating these two quantities gives $\alpha_1 = \alpha_2 + \alpha_1/2$, which is equivalent to $\alpha_1 = 2\alpha_2$. Replacing $(\alpha_2, \alpha_1/2)$ by $(m_2, m_1/2)$ proves (2369).

For the 12289 shell, the exterior orbit amplitudes are

$$\frac{61445}{943902729}, \quad \frac{184335}{339886096}, \quad \frac{921675}{38217124},$$

and the middle orbit amplitudes are

$$\frac{615}{94864}, \quad \frac{123}{484}, \quad \frac{1025}{1089}.$$

Adding each triple gives (2371) and (2372). The established origin counts give $c = (3, 3)$. The displayed sums, followed by exact rational subtraction, give (2373)–(2375). $\square$

Corollary 14.264 (Global two-character spectral form of shell positivity). *Read an empty reduced shell as the zero-dimensional orbit space and set $\eta_{p,a} = 0$ there. For every coprime shell in the preceding theorem, write*

$$\lambda_+(p, a) := \alpha_2 + \frac{\alpha_1}{2}, \quad \lambda_-(p, a) := \alpha_2 - \frac{\alpha_1}{2}. \quad (2376)$$

The two origin amplitudes are reconstructed exactly by

$$\alpha_1 = \lambda_+ - \lambda_-, \quad \alpha_2 = \frac{\lambda_+ + \lambda_-}{2}, \quad \alpha_{p,a}^\Sigma = 2\lambda_+. \quad (2377)$$

Consequently,

$$\Omega_R(p) > 0 \iff \lambda_+(p, a) > 0 \iff \mathbf{L}_{\eta_{p,a}} \neq 0. \quad (2378)$$

In the coprime-shell formulation, the Erdős–Straus assertion is therefore equivalent to

$$\forall p \in \mathbb{P} \quad \exists R = 4a - p > 0 : \lambda_+(p, a) > 0. \quad (2379)$$

The character difference $\lambda_+ - \lambda_- = \alpha_1$ detects the middle target, and the character average $(\lambda_+ + \lambda_-)/2 = \alpha_2$ detects the exterior target.

Proof. The reconstruction equations follow by adding and subtracting the two entries in (2376). Every occupied orbit amplitude is strictly positive, so

$$\lambda_+ = \sum_{O \in \mathcal{O}_{p,a}} \alpha_O = \frac{\alpha_{p,a}^\Sigma}{2}$$

is positive exactly when the reduced shell is occupied. This also holds for the declared zero-dimensional empty-shell value. The convolution matrix vanishes exactly when its two nonnegative coefficient values α_2 and $\alpha_1/2$ vanish. This proves (2378). Apply theorem 9.71 to obtain (2379). $\square$

Theorem 14.265 (Kocik–Sarnak realization of the transported support dyad). *Let*

$$v_s := \left(1, \frac{1}{2}, 0, \frac{1}{4}\right), \quad v_r := \left(1, -\frac{1}{2}, 0, \frac{1}{4}\right) \quad (2380)$$

be the two circle-coefficient vectors in (1178) and (1179). Their generalized-circle equations are the two point-circles

$$\left(x + \frac{1}{2}\right)^2 + y^2 = 0, \quad \left(x - \frac{1}{2}\right)^2 + y^2 = 0. \quad (2381)$$

The restriction of the circle–Sarnak map to their ordered span has matrix

$$D_{sr} := \begin{pmatrix} 5/8 & 5/8 \\ 1/2 & -1/2 \\ 0 & 0 \\ 3/8 & 3/8 \end{pmatrix} \quad (2382)$$

in the coefficient basis (v_s, v_r) and the Lorentz coordinate order (T, X, Y, W) . Put

$$\eta = \text{diag}(-1, 1, 1, 1).$$

Then

$$D_{sr}^T \eta D_{sr} = \begin{pmatrix} 0 & -1/2 \\ -1/2 & 0 \end{pmatrix}. \quad (2383)$$

The cleared Kocik family is

$$(3p - 1)(X^2 + Y^2 + W^2) = (p + 1)T^2. \quad (2384)$$

The two columns of D_{sr} satisfy (2384) at $p = 1$. Their sum and difference are

$$u_\Sigma = u_s + u_r = \left(\frac{5}{4}, 0, 0, \frac{3}{4}\right), \quad v_\Sigma = v_s + v_r = \left(2, 0, 0, \frac{1}{2}\right), \quad (2385)$$

$$u_\Delta = u_s - u_r = (0, 1, 0, 0), \quad v_\Delta = v_s - v_r = (0, 1, 0, 0). \quad (2386)$$

The sum satisfies the Kocik family at

$$p_\Sigma = 17. \quad (2387)$$

The difference satisfies its boundary member at

$$p_\Delta = \frac{1}{3}, \quad (2388)$$

and its generalized-circle equation is the oriented line

$$2x = 0. \quad (2389)$$

Thus the transported source and range idempotents form a nondegenerate Lorentz null basis whose symmetric and antisymmetric directions are the $p = 17$ timelike carrier and the $p = 1/3$ spacelike boundary carrier, respectively.

Proof. Substitution of the two coefficient vectors in (28) gives

$$x^2 + y^2 + x + \frac{1}{4} = \left(x + \frac{1}{2}\right)^2 + y^2$$

and its reflected equation, proving (2381). Applying (89) to the two columns gives (2382). Matrix multiplication with η proves (2383).

For either column,

$$X^2 + Y^2 + W^2 = \frac{1}{4} + \frac{9}{64} = \frac{25}{64} = T^2,$$

so (2384) holds at $p = 1$. For u_Σ ,

$$\frac{X^2 + Y^2 + W^2}{T^2} = \frac{9}{25}.$$

Solving

$$\frac{p+1}{3p-1} = \frac{9}{25}$$

gives $p = 17$. For u_Δ , one has $T = 0$ and $X^2 + Y^2 + W^2 = 1$. The cleared equation then forces $3p - 1 = 0$, giving $p = 1/3$. Finally, (28) applied to $v_\Delta = (0, 1, 0, 0)$ gives $2x = 0$. $\square$

Theorem 14.266 (Two-character Lorentz resolution and reciprocal Kocik sheets). *Retain the hypotheses and notation of theorem 14.263. In coefficient space and Lorentz space, respectively, put*

$$\epsilon_{\text{tar}} := -E_{22} = (0, 0, 0, -1), \quad n_{\text{tar}} := \Psi_{\text{circ}}(\epsilon_{\text{tar}}) = \left(-\frac{1}{2}, 0, 0, \frac{1}{2}\right). \quad (2390)$$

Let $L_{\text{tar}} := \mathbb{R}\epsilon_{\text{tar}}$ and define the ordered target-resolved lift

$$\mathcal{K}_{p,a}^{\text{res}} := ((\lambda_+ - \lambda_-)\epsilon_{\text{tar}}, (\lambda_+ + \lambda_-)\epsilon_{\text{tar}}) = (\alpha_1\epsilon_{\text{tar}}, 2\alpha_2\epsilon_{\text{tar}}) \in L_{\text{tar}}^{\oplus 2}. \quad (2391)$$

For

$$\Sigma(A, B) := A + B, \quad \Delta(A, B) := B - A,$$

one has

$$\Sigma\mathcal{K}_{p,a}^{\text{res}} = 2\lambda_+\epsilon_{\text{tar}}, \quad \Delta\mathcal{K}_{p,a}^{\text{res}} = 2\lambda_-\epsilon_{\text{tar}}, \quad (2392)$$

$$\Psi_{\text{circ}}^{\oplus 2}(\mathcal{K}_{p,a}^{\text{res}}) = ((\lambda_+ - \lambda_-)n_{\text{tar}}, (\lambda_+ + \lambda_-)n_{\text{tar}}). \quad (2393)$$

The coordinate matrix

$$\begin{pmatrix} \alpha_1 \\ 2\alpha_2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_+ \\ \lambda_- \end{pmatrix} \quad (2394)$$

has determinant 2. Hence the ordered lift retains both character coordinates. Its decomposition

$$\mathcal{K}_{p,a}^{\text{res}} = \lambda_+(\epsilon_{\text{tar}}, \epsilon_{\text{tar}}) + \lambda_-(-\epsilon_{\text{tar}}, \epsilon_{\text{tar}}) \quad (2395)$$

places the λ_- -term in the kernel of Σ .

The aggregate shell matrix and its Sarnak coordinates are therefore

$$K_{p,a}^\Sigma = 4E_{11} + \Sigma\mathcal{K}_{p,a}^{\text{res}} = 4E_{11} - 2\lambda_+E_{22}, \quad (2396)$$

$$(T, X, Y, W)_{p,a}^\Sigma = (2 - \lambda_+, 0, 0, 2 + \lambda_+). \quad (2397)$$

In the null coordinates of (553), the two resolved target components, their sum, and their ordered difference have values

$$(U_1, V_1) = (0, -(\lambda_+ - \lambda_-)), \quad (U_2, V_2) = (0, -(\lambda_+ + \lambda_-)), \quad (2398)$$

$$(U_\Sigma, V_\Sigma) = (0, -2\lambda_+), \quad (U_\Delta, V_\Delta) = (0, -2\lambda_-). \quad (2399)$$

After addition of the fixed source $4E_{11}$, the full shell has

$$U = 4, \quad V = -2\lambda_+. \quad (2400)$$

Moreover,

$$\lambda_+ \geq 0, \quad |\lambda_-| \leq \lambda_+. \quad (2401)$$

Write $\ell := \lambda_+$. The unique finite parameter for which the point in (2397) satisfies the cleared Kocik family (2384) is

$$\kappa(\ell) := \frac{\ell^2 + 4}{\ell^2 + 8\ell + 4}. \quad (2402)$$

For $\ell \geq 0$,

$$1 - \kappa(\ell) = \frac{8\ell}{\ell^2 + 8\ell + 4}, \quad (2403)$$

$$\kappa(\ell) - \frac{1}{3} = \frac{2(\ell - 2)^2}{3(\ell^2 + 8\ell + 4)}. \quad (2404)$$

Thus $1/3 \leq \kappa(\ell) \leq 1$; the lower endpoint occurs exactly at $\ell = 2$, and the finite upper endpoint occurs exactly at $\ell = 0$. For $\ell > 0$,

$$\kappa\left(\frac{4}{\ell}\right) = \kappa(\ell). \quad (2405)$$

The coefficient-inversion coordinate from (538) is

$$q_{\perp} := \lambda_{\perp}(K_{p,a}^{\Sigma}) = -\frac{\alpha_{p,a}^{\Sigma}}{4} = -\frac{\lambda_{+}}{2}. \quad (2406)$$

In this coordinate,

$$\kappa(q_{\perp}) = \frac{q_{\perp}^2 + 1}{q_{\perp}^2 - 4q_{\perp} + 1}, \quad \kappa(q_{\perp}^{-1}) = \kappa(q_{\perp}). \quad (2407)$$

In particular,

$$\kappa\left(-\frac{1}{4}\right) = \kappa(-4) = \frac{17}{33}. \quad (2408)$$

Let

$$\mathcal{S}(T, X, Y, W) := (W, X, Y, T). \quad (2409)$$

The $q_{\perp} = -1/4$ shell point and the $p_{\Sigma} = 17$ support-dyad sum in (2385) obey

$$\left(\frac{3}{2}, 0, 0, \frac{5}{2}\right) = 2\mathcal{S}\left(\frac{5}{4}, 0, 0, \frac{3}{4}\right). \quad (2410)$$

On the $X = Y = 0$ meridian, transposition sends a finite Kocik parameter p to

$$p^{\mathcal{S}} = \frac{p}{2p-1}. \quad (2411)$$

Consequently 17 is sent to $17/33$, exactly the value in (2408).

Proof. The circle-Sarnak map in (89) sends ϵ_{tar} to n_{tar} . The reconstruction formulas in (2377) give (2391). Addition and ordered subtraction prove (2392) and (2393). The determinant in (2394) is 2, and direct expansion gives (2395).

Since $\alpha^{\Sigma} = 2\lambda_{+}$, the aggregate formulas are the coefficient and Sarnak formulas in theorem 11.13. The vector n_{tar} has null coordinates $(U, V) = (0, -1)$, proving (2398)–(2400). Finally, $\lambda_{+} - \lambda_{-} = \alpha_1 \geq 0$ and $\lambda_{+} + \lambda_{-} = 2\alpha_2 \geq 0$, which proves (2401).

Substitution of $T = 2 - \ell$, $W = 2 + \ell$, and $X = Y = 0$ into (2384) gives

$$(3\kappa - 1)(2 + \ell)^2 = (\kappa + 1)(2 - \ell)^2.$$

Solving for κ proves (2402). Subtraction from 1 and $1/3$ gives (2403) and (2404). Replacing ℓ by $4/\ell$ proves (2405).

Equation (2406) follows from $\alpha^{\Sigma} = 2\lambda_{+}$. Substitution $\ell = -2q_{\perp}$ gives the first identity in (2407); replacing $q_{\perp}$ by its reciprocal gives the second. Evaluation at $-1/4$ and -4 gives (2408).

At $q_{\perp} = -1/4$, one has $\alpha^{\Sigma} = 1$, so (2397) gives $(3/2, 0, 0, 5/2)$. Equation (2385) gives the second vector in (2410). This proves that identity. If $X = Y = 0$, the Kocik equation gives

$$\frac{W^2}{T^2} = \frac{p+1}{3p-1}.$$

Transposition replaces this ratio by its reciprocal. Solving

$$\frac{p^{\mathcal{S}} + 1}{3p^{\mathcal{S}} - 1} = \frac{3p - 1}{p + 1}$$

gives (2411). Its value at $p = 17$ is $17/33$. □

Theorem 14.267 (Amplitude-weighted C_1 – C_2 – C_3 path carrier and exact Lorentz pushforward). Assume (1276). Use the bijection

$$(j, u) \mapsto d_j(u) = p^j u$$

to identify $\mathcal{O}_{R,p}$ with the complement-orbit set $\mathcal{O}_{p,a}$ of theorem 14.259. Read an empty shell as the zero-dimensional orbit space and every displayed sum over it as zero. For every such shell, put

$$D_{p,a}^{\text{orb}} = \sum_{O \in \mathcal{O}_{p,a}} \alpha_O e_O e_O^*, \quad H_{p,a}^{\text{orb}} := (D_{p,a}^{\text{orb}})^{1/2} = \sum_O \sqrt{\alpha_O} e_O e_O^*. \quad (2412)$$

The amplitude-weighted support map is

$$\Pi_{123}^\alpha := (H_{p,a}^{\text{orb}} \otimes I_{\mathcal{L}_{11}}) \Pi_{123}, \quad (2413)$$

and its basis action is

$$\Pi_{123}^\alpha e_{(m, x_O^\epsilon)} = \sqrt{\alpha_O} e_O \otimes E_{1,1;1,1}. \quad (2414)$$

It is D_3 -invariant and satisfies

$$\text{rank } \Pi_{123}^\alpha = q_{R,p}, \quad \dim \ker \Pi_{123}^\alpha = 5q_{R,p}. \quad (2415)$$

Define the two ordered weighted blades by

$$\mathcal{W}_{-,+} := 2 \sum_{O \in \mathcal{O}_{p,a}} \sum_{m \in C_3} \sqrt{\alpha_O} |e_{(m, x_O^+)}\rangle \langle e_{(m, x_O^-)}|, \quad (2416)$$

$$\mathcal{W}_{+,-} := \mathcal{W}_{-,+}^*. \quad (2417)$$

Their transported forms retain the full orbit-amplitude factor:

$$\Theta_{123} \mathcal{W}_{-,+} \Theta_{123}^{-1} = 2H_{p,a}^{\text{orb}} \otimes E_{21}^{(2)} \otimes I_3, \quad (2418)$$

$$\Theta_{123} \mathcal{W}_{+,-} \Theta_{123}^{-1} = 2H_{p,a}^{\text{orb}} \otimes E_{12}^{(2)} \otimes I_3. \quad (2419)$$

Put

$$\mathcal{D}_< := \sum_{O,m} \alpha_O |e_{(m, x_O^-)}\rangle \langle e_{(m, x_O^-)}|, \quad \mathcal{D}_> := \sum_{O,m} \alpha_O |e_{(m, x_O^+)}\rangle \langle e_{(m, x_O^+)}|. \quad (2420)$$

Then

$$\frac{1}{4} \mathcal{W}_{+,-} \mathcal{W}_{-,+} = \mathcal{D}_<, \quad \frac{1}{4} \mathcal{W}_{-,+} \mathcal{W}_{+,-} = \mathcal{D}_>. \quad (2421)$$

The polar partial isometries are exactly

$$V_{-,+} := \frac{1}{2} \mathcal{A}_{-,+}, \quad V_{+,-} := \frac{1}{2} \mathcal{A}_{+,-}, \quad (2422)$$

and the two polar decompositions are

$$\mathcal{W}_{-,+} = V_{-,+} 2\mathcal{D}_<^{1/2}, \quad \mathcal{W}_{+,-} = V_{+,-} 2\mathcal{D}_>^{1/2}. \quad (2423)$$

Reflection exchanges the two weighted blades and rotation preserves each:

$$\mathbf{S} \mathcal{W}_{-,+} \mathbf{S}^{-1} = \mathcal{W}_{+,-}, \quad \mathbf{R} \mathcal{W}_{-,+} \mathbf{R}^{-1} = \mathcal{W}_{-,+}. \quad (2424)$$

The orbit trace, ranks, and Hilbert–Schmidt norms are

$$\text{tr } D_{p,a}^{\text{orb}} = \sum_O \alpha_O = \lambda_+ = \frac{\alpha_{p,a}^\Sigma}{2}, \quad (2425)$$

$$\|\Pi_{123}^\alpha\|_{\text{HS}}^2 = 6\lambda_+ = 3\alpha_{p,a}^\Sigma, \quad (2426)$$

$$\text{rank } \mathcal{W}_{-,+} = 3q_{R,p}, \quad \|\mathcal{W}_{-,+}\|_{\text{HS}}^2 = 12\lambda_+ = 6\alpha_{p,a}^\Sigma. \quad (2427)$$

Compared with the divisor-space channel $\mathcal{N}_{21}^{p,a}$ of theorem 14.256, one has

$$2 \text{ rank } \mathcal{W}_{-,+} = 3 \text{ rank } \mathcal{N}_{21}^{p,a}, \quad 2\|\mathcal{W}_{-,+}\|_{\text{HS}}^2 = 3\|\mathcal{N}_{21}^{p,a}\|_{\text{HS}}^2. \quad (2428)$$

The exact two-type pushforward on the diagonal orbit algebra is the linear map

$$\mathfrak{P}_{02|11} : \mathbb{R}^{\mathcal{O}_{p,a}} \longrightarrow \mathbb{R}^2, \quad (z_O)_O \longmapsto \left(\sum_{O \in \mathcal{O}_{02}} z_O, \sum_{O \in \mathcal{O}_{11}} z_O \right). \quad (2429)$$

It sends the complete amplitude vector to

$$\mathfrak{P}_{02|11}((\alpha_O)_O) = \left(\alpha_2, \frac{\alpha_1}{2} \right). \quad (2430)$$

Its kernel is

$$\ker \mathfrak{P}_{02|11} = \left\{ (z_O)_O : \sum_{O \in \mathcal{O}_{02}} z_O = 0, \quad \sum_{O \in \mathcal{O}_{11}} z_O = 0 \right\}. \quad (2431)$$

If

$$\rho_{p,a} := \#\{\nu \in \{02, 11\} : \mathcal{O}_\nu \neq \emptyset\},$$

then

$$\text{rank } \mathfrak{P}_{02|11} = \rho_{p,a}, \quad \dim \ker \mathfrak{P}_{02|11} = q_{R,p} - \rho_{p,a}. \quad (2432)$$

Thus the weighted path carrier retains every α_O , while (2431) records exactly the within-type information omitted by the two-character image.

The subsequent coordinate maps are

$$\begin{pmatrix} \alpha_2 \\ \alpha_1/2 \end{pmatrix} \xrightarrow{\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}} \begin{pmatrix} \lambda_+ \\ \lambda_- \end{pmatrix} \xrightarrow{\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}} \begin{pmatrix} \alpha_1 \\ 2\alpha_2 \end{pmatrix}. \quad (2433)$$

In the target Lorentz line of theorem 14.266, this gives

$$((\lambda_+ - \lambda_-)n_{\text{tar}}, (\lambda_+ + \lambda_-)n_{\text{tar}}) = (\alpha_1 n_{\text{tar}}, 2\alpha_2 n_{\text{tar}}), \quad (2434)$$

$$\alpha_1 n_{\text{tar}} + 2\alpha_2 n_{\text{tar}} = \alpha_{p,a}^\Sigma n_{\text{tar}}. \quad (2435)$$

After the fixed source $4E_{11}$ is added, the Sarnak point and its two null coordinates are

$$(T, X, Y, W) = (2 - \lambda_+, 0, 0, 2 + \lambda_+), \quad (U, V) = \left(4, -\alpha_{p,a}^\Sigma \right). \quad (2436)$$

For $(p, a) = (12289, 3075)$, both orbit types contain three elements, so

$$\text{rank } \mathfrak{P}_{02|11} = 2, \quad \dim \ker \mathfrak{P}_{02|11} = 4. \quad (2437)$$

The exact norm values are

$$\|\Pi_{123}^\alpha\|_{\text{HS}}^2 = \frac{128395625711903946059}{17446433687597757612}, \quad (2438)$$

$$\|\mathcal{W}_{-,+}\|_{\text{HS}}^2 = \frac{128395625711903946059}{8723216843798878806}. \quad (2439)$$

Proof. Under the coprimality hypothesis, the reduced data are $R_0 = R$ and $N_0 = N$. The certificate bijection $(j, u) \mapsto p^j u$ intertwines the two complement involutions, so it induces the stated identification of orbit sets. Complement invariance of the amplitude gives a single strictly positive value α_O on each occupied orbit.

Equations (2413) and (1303) give (2414). The orbit factor is invertible on every occupied orbit line, so the image and kernel dimensions are those of Π_{123} . The D_3 -action changes only (ϵ, m) , proving invariance.

On the six-dimensional block over O , the two weighted blades are

$$2\sqrt{\alpha_O} E_{21}^{(2)} \otimes I_3, \quad 2\sqrt{\alpha_O} E_{12}^{(2)} \otimes I_3.$$

This proves the transported forms. Their two products are

$$4\alpha_O E_{11}^{(2)} \otimes I_3, \quad 4\alpha_O E_{22}^{(2)} \otimes I_3,$$

which proves (2421). Removing the strictly positive coefficient $2\sqrt{\alpha_O}$ from each occupied arrow leaves the low-to-high partial isometry $\mathcal{A}_{-,+}/2$ and its adjoint. This proves the polar formulas. The rotation and reflection equations follow blockwise from (1311).

Each orbit contributes six equal squared coefficients to Π_{123}^α , and three squared coefficients $(2\sqrt{\alpha_O})^2$ to either weighted blade. Hence

$$\|\Pi_{123}^\alpha\|_{\text{HS}}^2 = 6 \sum_O \alpha_O, \quad \|\mathcal{W}_{-,+}\|_{\text{HS}}^2 = 12 \sum_O \alpha_O.$$

Now apply (2309) and (2377). The rank statement follows from the three nonzero arrows over every orbit. The earlier divisor-space carrier has rank $2q_{R,p}$ and squared Hilbert–Schmidt norm $4\alpha_{p,a}^\Sigma$, proving (2428).

The two rows of $\mathfrak{P}_{02|11}$ have disjoint supports. Each nonempty row is nonzero, so its rank is the number $\rho_{p,a}$ of nonempty orbit types. Its displayed kernel follows directly, and rank–nullity proves (2432). The amplitude values in (2430) are (2361). Applying the Hadamard matrix and its second coordinate map proves (2433). Equations (2393), (622), and (2400) then prove the Lorentz and Sarnak formulas.

For the 12289 shell, the three exterior and three middle amplitudes listed in the proof of theorem 14.263 give $\rho_{12289,3075} = 2$ and $q_{11,12289} = 6$. Substitution of (2374) into (2426) and (2427) gives the two stated rational norms. $\square$

Corollary 14.268 (Order-forced endpoint lift to the weighted C_1 – C_2 – C_3 path carrier). *Under the conditions of theorem 14.267, retain the endpoint bits b_0, b_1 of theorem 14.257. The endpoint-support set and ordered complement sheet are*

$$\mathcal{B} = \{0 < 1\}, \quad \mathcal{X}_O = \{x_O^- < x_O^+\}.$$

There is exactly one order isomorphism

$$\varpi_O : \mathcal{B} \longrightarrow \mathcal{X}_O, \quad \varpi_O(0) = x_O^-, \quad \varpi_O(1) = x_O^+. \quad (2440)$$

Using $x_O^0 = x_O^-$ and $x_O^1 = x_O^+$, define

$$\mathcal{W}_{\partial,0+}^{p,a} := 2 \sum_{O \in \mathcal{O}_{p,a}} \sum_{m \in C_3} \sqrt{\alpha_O} |e_{(m, x_O^{b_1})}\rangle \langle e_{(m, x_O^{b_0})}|. \quad (2441)$$

For an empty shell, the sum and its carrier space are zero-dimensional. In the ordered basis (f_0, f_1) of the complement-sheet factor,

$$\Theta_{123} \mathcal{W}_{\partial,0+}^{p,a} \Theta_{123}^{-1} = 2H_{p,a}^{\text{orb}} \otimes E_{b_1+1, b_0+1}^{(2)} \otimes I_3. \quad (2442)$$

Consequently,

<i>status</i>	(b_0, b_1)	$\mathcal{W}_{\partial,0+}^{p,a}$
<i>empty</i>	$(0, 0)$	0
<i>exterior only</i>	$(0, 1)$	$\mathcal{W}_{-,+}$
<i>middle only</i>	$(1, 0)$	$\mathcal{W}_{+,-}$
<i>both</i>	$(1, 1)$	$2\mathcal{D}_{>}^{1/2}$.

(2443)

On an occupied shell, its Moore–Penrose source and range are

$$(\mathcal{W}_{\partial,0+}^{p,a})^\dagger \mathcal{W}_{\partial,0+}^{p,a} = \Theta_{123}^{-1} \left(I_{\mathbb{C}[\mathcal{O}_{p,a}]} \otimes E_{b_0+1, b_0+1}^{(2)} \otimes I_3 \right) \Theta_{123}, \quad (2444)$$

$$\mathcal{W}_{\partial,0+}^{p,a} (\mathcal{W}_{\partial,0+}^{p,a})^\dagger = \Theta_{123}^{-1} \left(I_{\mathbb{C}[\mathcal{O}_{p,a}]} \otimes E_{b_1+1, b_1+1}^{(2)} \otimes I_3 \right) \Theta_{123}. \quad (2445)$$

Put

$$\mathcal{J}_{\partial,123}^{p,a} := \Theta_{123}^{-1} \left(I_{\mathbb{C}[\mathcal{O}_{p,a}]} \otimes (E_{22}^{(2)} - E_{11}^{(2)}) \otimes I_3 \right) \Theta_{123}. \quad (2446)$$

Then

$$\text{rank } \mathcal{W}_{\partial,0+}^{p,a} = 3q_{R,p}, \quad \|\mathcal{W}_{\partial,0+}^{p,a}\|_{\text{HS}}^2 = 6\alpha_{p,a}^\Sigma, \quad (2447)$$

$$[\mathcal{J}_{\partial,123}^{p,a}, \mathcal{W}_{\partial,0+}^{p,a}] = 2(b_1 - b_0)\mathcal{W}_{\partial,0+}^{p,a}. \quad (2448)$$

The selected path energy is therefore

$$\begin{aligned} \mathcal{E}_{p,a}^{\text{amp},123} &:= \frac{1}{4} \|[\mathcal{J}_{\partial,123}^{p,a}, \mathcal{W}_{\partial,0+}^{p,a}]\|_{\text{HS}}^2 \\ &= 6\alpha_{p,a}^\Sigma (b_1 - b_0)^2 \\ &= 3\alpha_{p,a}^\Sigma \mathcal{E}_{R,p}^{\text{lab}} \\ &= \frac{3}{2} \mathcal{E}_{p,a}^{\text{amp}}. \end{aligned} \quad (2449)$$

The divisor and path carriers obey the three exact multiplicity relations

$$\begin{aligned} 2 \text{rank } \mathcal{W}_{\partial,0+}^{p,a} &= 3 \text{rank } \mathcal{N}_{0+}^{p,a}, \\ 2\|\mathcal{W}_{\partial,0+}^{p,a}\|_{\text{HS}}^2 &= 3\|\mathcal{N}_{0+}^{p,a}\|_{\text{HS}}^2, \\ 2\mathcal{E}_{p,a}^{\text{amp},123} &= 3\mathcal{E}_{p,a}^{\text{amp}}. \end{aligned} \quad (2450)$$

In particular,

$$\mathcal{W}_{\partial,0+}^{p,a} \neq 0 \iff \mathcal{N}_{0+}^{p,a} \neq 0 \iff q_{R,p} > 0.$$

Proof. The endpoint-support order is $0 < 1$. The divisor inequalities in (1297) give $x_O^- < x_O^+$. A map between two ordered two-element sets is an order isomorphism exactly when it sends the least element to the least element; this proves existence and uniqueness in (2440). Applying the linearized map Θ_{123} to (2441) proves the block formula (2442). The four bit pairs then give (2443).

On an occupied shell, $H_{p,a}^{\text{orb}}$ is invertible and $E_{b_1+1,b_0+1}^{(2)}$ has rank one. Its Moore–Penrose inverse is

$$\frac{1}{2}(H_{p,a}^{\text{orb}})^{-1} \otimes E_{b_0+1,b_1+1}^{(2)} \otimes I_3.$$

Multiplying on both sides proves the source and range formulas. Since $\text{tr } D_{p,a}^{\text{orb}} = \alpha_{p,a}^\Sigma/2$,

$$\left\| 2H_{p,a}^{\text{orb}} \otimes E_{b_1+1,b_0+1}^{(2)} \otimes I_3 \right\|_{\text{HS}}^2 = 12 \text{tr } D_{p,a}^{\text{orb}} = 6\alpha_{p,a}^\Sigma.$$

This also gives the rank statement. The grading has eigenvalue $2\epsilon - 1$ on f_ϵ , so its commutator with the selected matrix unit is $2(b_1 - b_0)$ times that matrix unit. This proves the commutator and energy identities.

Finally, an occupied complement orbit has two divisor points and three rotation labels. Thus $\text{rank } \mathcal{N}_{0+}^{p,a} = 2q_{R,p}$ and $\text{rank } \mathcal{W}_{\partial,0+}^{p,a} = 3q_{R,p}$. The norm and energy identities in theorem 14.257 give the remaining two relations in (2450). All quantities vanish on an empty shell. $\square$

Theorem 14.269 (Amplitude-weighted shell control and determinant volume). *Assume (1276), suppose $\mathcal{O}_{p,a} \neq \emptyset$, and retain the orbit amplitudes, weighted blades, and weighted sheet operators of theorem 14.267. Write*

$$q := q_{R,p} = |\mathcal{O}_{p,a}|, \quad M_2(\alpha) := \sum_{O \in \mathcal{O}_{p,a}} \alpha_O^2.$$

Define

$$\mathcal{H}_{p,a}^\alpha := \mathcal{W}_{+,-} \mathcal{W}_{-,+} - \mathcal{D}_>, \quad (2451)$$

and transport it to the simplex-support space:

$$H_{p,a}^{\text{sh},\alpha} := \tilde{\Theta}_{123} \mathcal{H}_{p,a}^\alpha \tilde{\Theta}_{123}^{-1}. \quad (2452)$$

Then

$$H_{p,a}^{\text{sh},\alpha} = D_{p,a}^{\text{orb}} \otimes \tilde{\mathfrak{H}}_{-,+} \otimes I_3, \quad (2453)$$

and its spectral multiset is

$$\text{Spec}(H_{p,a}^{\text{sh},\alpha}) = \bigsqcup_{O \in \mathcal{O}_{p,a}} \left\{ (4\alpha_O)^{[3]}, (-\alpha_O)^{[3]} \right\}. \quad (2454)$$

In particular, this carrier is invertible on the $6q$ -dimensional simplex-support space, and its determinant volume is

$$\text{Vol}_{p,a}^{\text{det},\alpha} := \left| \det H_{p,a}^{\text{sh},\alpha} \right| = 4^{3q} \prod_{O \in \mathcal{O}_{p,a}} \alpha_O^6, \quad (2455)$$

$$\log \text{Vol}_{p,a}^{\text{det},\alpha} = 3q \log 4 + 6 \sum_{O \in \mathcal{O}_{p,a}} \log \alpha_O. \quad (2456)$$

On $\mathcal{H}_{R,p}^{\text{simp}}$, put

$$\hat{D}_{p,a}^\alpha := D_{p,a}^{\text{orb}} \otimes I_{\text{Ran } P_2} \otimes I_3, \quad K_{p,a}^{\text{sh},\alpha} := H_{p,a}^{\text{sh},\alpha} - \frac{3}{2} \hat{D}_{p,a}^\alpha. \quad (2457)$$

The two constant-control paths

$$\gamma_{H,\alpha}(t) := \exp(-itH_{p,a}^{\text{sh},\alpha}), \quad \gamma_{K,\alpha}(t) := \exp(-itK_{p,a}^{\text{sh},\alpha}), \quad 0 \leq t \leq 1, \quad (2458)$$

lie in $U(6q)$ and $SU(6q)$, respectively. Their F_{HS} -lengths are

$$L_{H,\alpha}^2 = 51M_2(\alpha), \quad (2459)$$

$$L_{K,\alpha}^2 = \frac{75}{2}M_2(\alpha). \quad (2460)$$

The determinant volume and both lengths satisfy the sharp inequality

$$\left(\text{Vol}_{p,a}^{\text{det},\alpha} \right)^{1/(6q)} = 2 \left(\prod_O \alpha_O \right)^{1/q} \leq 2 \sqrt{\frac{M_2(\alpha)}{q}} = \frac{2L_{H,\alpha}}{\sqrt{51q}} = 2 \sqrt{\frac{2}{75q}} L_{K,\alpha}. \quad (2461)$$

Equality holds precisely when all orbit amplitudes are equal. The choice $\alpha_O = 1$ for every O recovers theorem 14.31.

Proof. Equations (2418) and (2419) give

$$\Theta_{123} \mathcal{W}_{+,-} \Theta_{123}^{-1} = 4D_{p,a}^{\text{orb}} \otimes E_{11}^{(2)} \otimes I_3.$$

The basis definition of $\mathcal{D}_>$ gives

$$\Theta_{123} \mathcal{D}_> \Theta_{123}^{-1} = D_{p,a}^{\text{orb}} \otimes E_{22}^{(2)} \otimes I_3.$$

Applying $I \otimes \mathbf{U}_2^* \otimes I_3$ and its inverse proves (2453). The middle factor $\tilde{\mathfrak{H}}_{-,+}$ has eigenvalues 4 and -1 . Each orbit label therefore contributes $4\alpha_O$ and $-\alpha_O$, each repeated for the three rotation labels. This proves (2454). Positivity of every α_O proves invertibility. Multiplication of the six absolute eigenvalues over each orbit gives

$$(4\alpha_O)^3 \alpha_O^3 = 4^3 \alpha_O^6,$$

which proves (2455) and (2456).

The spectrum of the orbitwise scalar operator $(3/2)\hat{D}_{p,a}^\alpha$ consists of $(3/2)\alpha_O$ with multiplicity six over O . Hence the two eigenvalues of $K_{p,a}^{\text{sh},\alpha}$ over O are $(5/2)\alpha_O$ and $-(5/2)\alpha_O$, each with

multiplicity three. Their sum vanishes orbit by orbit, placing $\gamma_{K,\alpha}$ in $SU(6q)$. Direct summation gives

$$\begin{aligned} L_{H,\alpha}^2 &= 3 \sum_O \left((4\alpha_O)^2 + (-\alpha_O)^2 \right) = 51 \sum_O \alpha_O^2, \\ L_{K,\alpha}^2 &= 6 \sum_O \left(\frac{5}{2} \alpha_O \right)^2 = \frac{75}{2} \sum_O \alpha_O^2. \end{aligned}$$

Apply the arithmetic–geometric mean inequality to the positive numbers α_O^2 :

$$\left(\prod_O \alpha_O^2 \right)^{1/q} \leq \frac{1}{q} \sum_O \alpha_O^2.$$

Taking positive square roots and using (2455) and the two length formulas proves (2461). Equality in the arithmetic–geometric mean inequality holds precisely when the α_O^2 are equal. Their positivity makes this equivalent to equality of all α_O . $\square$

Corollary 14.270 (Endpoint direction between determinant volume and control length). *Under the conditions of theorem 14.269, suppose the shell is one-sided. Then*

$$6q \left(\text{Vol}_{p,a}^{\det,\alpha} \right)^{1/(6q)} \leq \mathcal{E}_{p,a}^{\text{amp},123} \leq \frac{12\sqrt{q}}{\sqrt{51}} L_{H,\alpha} = 12\sqrt{\frac{2q}{75}} L_{K,\alpha}. \quad (2462)$$

Both inequalities are equalities simultaneously exactly when all orbit amplitudes α_O are equal.

Proof. On a one-sided shell, $\mathcal{E}_{R,p}^{\text{lab}} = 2$, so

$$\mathcal{E}_{p,a}^{\text{amp},123} = 6\alpha_{p,a}^\Sigma = 12 \sum_O \alpha_O.$$

The arithmetic–geometric mean inequality and the determinant formula give

$$\sum_O \alpha_O \geq q \left(\prod_O \alpha_O \right)^{1/q} = \frac{q}{2} \left(\text{Vol}_{p,a}^{\det,\alpha} \right)^{1/(6q)}.$$

The Cauchy–Schwarz inequality gives

$$\sum_O \alpha_O \leq \sqrt{qM_2(\alpha)}.$$

Multiplying by 12, then using $L_{H,\alpha}^2 = 51M_2(\alpha)$ and $L_{K,\alpha}^2 = (75/2)M_2(\alpha)$, proves (2462). Equality in either finite mean inequality holds exactly when all positive α_O are equal. $\square$

Exact computation 14.271 (Exact endpoint-selected weighted path check). The script and captured receipt

```
verification/verify_endpoint_selected_weighted_123_carrier.py,
verification/verify_endpoint_selected_weighted_123_carrier_output.txt
```

verify all four endpoint states on the exact 12289 orbit amplitudes. They check the order-forced path cell, Moore–Penrose source and range, signed commutator, the three 3 : 2 multiplicity identities, and the two finite mean inequalities in (2462). They also restrict the source and range moduli to their selected sheets, recover the identical orbit amplitudes there, verify polar transport with its signed grading, implement the stable tensor flip on both sheets, and check the squared-path versus cubed-divisor characteristic-product identity of theorem 14.276.

Amplitude-weighted shell: orbitwise spectrum and sharp volume bound

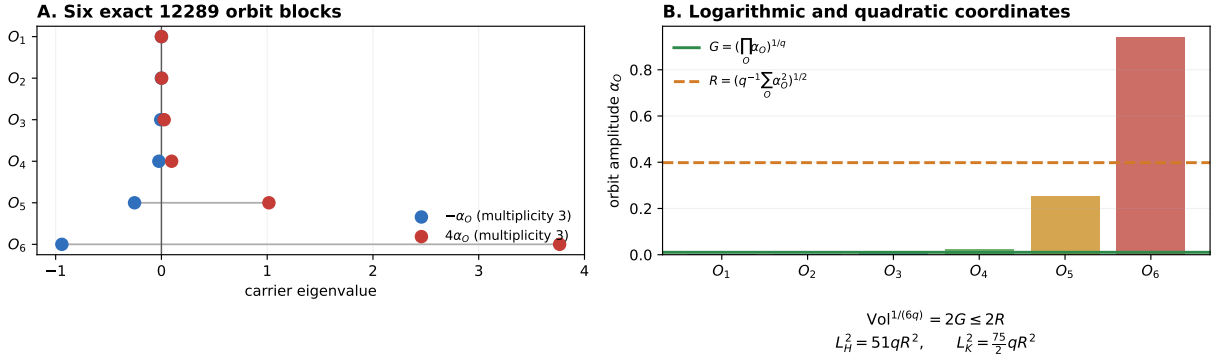


FIGURE 16. The amplitude-weighted carrier for the six exact 12289 complement orbits. Each orbit contributes the two eigenvalues $-\alpha_O$ and $4\alpha_O$, both with multiplicity three. The right panel shows the same six amplitudes together with their geometric mean G and root mean square R . Thus the determinant-volume coordinate is $2G$, while the two Hilbert–Schmidt lengths are fixed multiples of R ; the sharp relation is $2G \leq 2R$.

Exact computation 14.272 (Exact amplitude-weighted control and volume check). The script and captured receipt

```
verification/verify_weighted_shell_control_volume.py,
verification/verify_weighted_shell_control_volume_output.txt
```

verify the orbitwise spectrum, determinant, two squared lengths, and the sharp mean inequality for exact rational weight systems. They also check the six amplitudes in (2316). The vector image is generated by

```
verification/make_weighted_shell_control_volume_figure.py.
```

Theorem 14.273 (Individual Lorentz points recover the ordered-blade range coefficient). *Assume (1276) and retain the ordered complement orbits, Lorentz arcs, and weighted blades of theorems 14.26, 14.28 and 14.267. For $O \in \mathcal{O}_{p,a}$, put*

$$d_O^- := d(x_O^-), \quad d_O^+ := d(x_O^+) = \frac{N^2}{d_O^-}, \quad \delta_O := \frac{d_O^- - N}{d_O^- + N}, \quad \alpha_O := 1 - \delta_O^2. \quad (2463)$$

For $m \in C_3$, let

$$Q_{O,m}^- := |e_{(m,x_O^-)}\rangle \langle e_{(m,x_O^-)}|, \quad Q_{O,m}^+ := |e_{(m,x_O^+)}\rangle \langle e_{(m,x_O^+)}|, \quad (2464)$$

$$W_{O,m}^{-,+} := Q_{O,m}^+ \mathcal{W}_{-,+} Q_{O,m}^-, \quad W_{O,m}^{+,-} := (W_{O,m}^{-,+})^*. \quad (2465)$$

Then

$$W_{O,m}^{-,+} = 2\sqrt{\alpha_O} |e_{(m,x_O^+)}\rangle \langle e_{(m,x_O^-)}|, \quad (2466)$$

$$W_{O,m}^{+,-} = 2\sqrt{\alpha_O} |e_{(m,x_O^-)}\rangle \langle e_{(m,x_O^+)}|, \quad (2467)$$

and their ordered products are

$$\frac{1}{4}W_{O,m}^{+,-}W_{O,m}^{-,+} = \alpha_O Q_{O,m}^{\leq}, \quad (2468)$$

$$\frac{1}{4}W_{O,m}^{-,+}W_{O,m}^{+,-} = \alpha_O Q_{O,m}^{\geq}. \quad (2469)$$

Let $\mathcal{H}_{O,m}$ be the span of the two displayed basis vectors, and define the order-determined coordinate isomorphism

$$\vartheta_{O,m} : \mathcal{H}_{O,m} \longrightarrow \mathbb{C}^2, \quad e_{(m,x_O^-)} \longmapsto e_1, \quad e_{(m,x_O^+)} \longmapsto e_2. \quad (2470)$$

The signed range product is precisely the arithmetic target coefficient:

$$\vartheta_{O,m} \left(-\frac{1}{4}W_{O,m}^{-,+}W_{O,m}^{+,-} \right) \vartheta_{O,m}^{-1} = -\alpha_O E_{22} = \Delta_{d_O^-} = \Delta_{d_O^+}. \quad (2471)$$

The corresponding signed source product is

$$\vartheta_{O,m} \left(-\frac{1}{4}W_{O,m}^{+,-}W_{O,m}^{-,+} \right) \vartheta_{O,m}^{-1} = -\alpha_O E_{11}. \quad (2472)$$

At the individual Lorentz point $U_m(\delta_O)$, the rotated transverse coordinates satisfy

$$\xi_m(U_m(\delta_O)) = \frac{\delta_O}{2}, \quad \eta_m(U_m(\delta_O)) = \frac{\alpha_O}{4}. \quad (2473)$$

Consequently

$$W_{O,m}^{-,+} = 4\sqrt{\eta_m(U_m(\delta_O))} |e_{(m,x_O^+)}\rangle \langle e_{(m,x_O^-)}|, \quad (2474)$$

and

$$\frac{1}{16}W_{O,m}^{+,-}W_{O,m}^{-,+} = \eta_m(U_m(\delta_O)) Q_{O,m}^{\leq}, \quad (2475)$$

$$\frac{1}{16}W_{O,m}^{-,+}W_{O,m}^{+,-} = \eta_m(U_m(\delta_O)) Q_{O,m}^{\geq}. \quad (2476)$$

The sheetwise Lorentz descent therefore has the pointwise form

$$\mathcal{B}_m U_m(\delta_O) = \Psi_{\text{circ}}(4E_{11} + \Delta_{d_O^-}) = \Psi_{\text{circ}}(4E_{11}) + \alpha_O n_{\text{tar}} = \Psi_{\text{circ}}(4E_{11}) + 4\eta_m(U_m(\delta_O)) n_{\text{tar}}. \quad (2477)$$

Thus each of the three Lorentz-cover points over one complement orbit retains the same amplitude in its rotated arc height, blade modulus, ordered range product, arithmetic target coefficient, and target null ray. The source and range projectors remain distinguished by (2468) and (2469).

Proof. The complement relation and the strict order in (1297) give $d_O^+ = N^2/d_O^-$ and $d_O^- < N < d_O^+$. Equations (1230) and (1232) give $\alpha_{d_O^-} = \alpha_{d_O^+} = \alpha_O$.

Compressing (2416) to the two-dimensional (O, m) -block proves (2466); taking the adjoint proves (2467). The two matrix-unit products give (2468) and (2469). The map $\vartheta_{O,m}$ sends these two projectors to E_{11} and E_{22} , respectively. Since $\Delta_d = -\alpha_d E_{22}$ by (518), this proves (2471) and (2472).

The definition of U_m and the rotated coordinates in (1257) give (2473). Substitution of $\alpha_O = 4\eta_m(U_m(\delta_O))$ gives the blade and product formulas in arc coordinates. Finally, (1290) gives the first equality in (2477). Linearity of Ψ_{circ} , together with $\Delta_{d_O^-} = \alpha_O \epsilon_{\text{tar}}$ and $n_{\text{tar}} = \Psi_{\text{circ}}(\epsilon_{\text{tar}})$, proves the remaining equalities. $\square$

Lemma 14.274 (Finite-fibre stable spectral exchange). *Let D be an operator on a finite-dimensional Hilbert space $\mathcal{H}$, and let A and B be nonempty finite sets. Put*

$$D_A := D \otimes I_{\mathbb{C}[A]}, \quad D_B := D \otimes I_{\mathbb{C}[B]}.$$

The tensor-factor flip

$$\Sigma_{A,B} : (\mathcal{H} \otimes \mathbb{C}[A]) \otimes \mathbb{C}[B] \longrightarrow (\mathcal{H} \otimes \mathbb{C}[B]) \otimes \mathbb{C}[A], \quad u \otimes e_a \otimes e_b \longmapsto u \otimes e_b \otimes e_a \quad (2478)$$

is unitary and obeys

$$\Sigma_{A,B}(D_A \otimes I_{\mathbb{C}[B]})\Sigma_{A,B}^{-1} = D_B \otimes I_{\mathbb{C}[A]}. \quad (2479)$$

If D is self-adjoint, its finite spectral counting measure μ_D therefore gives

$$\mu_{D_A} = |A|\mu_D, \quad \mu_{D_B} = |B|\mu_D, \quad |B|\mu_{D_A} = |A|\mu_{D_B}. \quad (2480)$$

For every function φ on $\text{Spec } D$,

$$|B| \text{tr } \varphi(D_A) = |A| \text{tr } \varphi(D_B) = |A||B| \text{tr } \varphi(D), \quad (2481)$$

and

$$\det(tI - D_A)^{|B|} = \det(tI - D_B)^{|A|} = \det(tI - D)^{|A||B|}. \quad (2482)$$

Proof. The map $\Sigma_{A,B}$ permutes an orthonormal tensor-product basis. Both sides of (2479) send $u \otimes e_b \otimes e_a$ to $Du \otimes e_b \otimes e_a$. Every eigenvalue of D acquires multiplicity $|A|$ in D_A and $|B|$ in D_B . This proves the three remaining identities. $\square$

Theorem 14.275 (Exact $C_3 : C_2$ stable spectral transfer). *Assume (1276) and $\mathcal{O}_{p,a} \neq \emptyset$. Retain the ordered complement data of theorem 14.273. Put*

$$\mathcal{H}_{>}^{\text{path}} := Q_{>}\mathbb{C}[\tilde{\mathcal{X}}_{R,p}], \quad \mathcal{D}_{>}^{\circ} := \mathcal{D}_{>}|_{\mathcal{H}_{>}^{\text{path}}}. \quad (2483)$$

Let (f_0, f_1) and $(g_m)_{m \in C_3}$ be the distinguished bases of $\mathbb{C}[C_2]$ and $\mathbb{C}[C_3]$. With $d_O^0 = d_O^-$ and $d_O^1 = d_O^+$, the order-determined maps

$$T_{>} : \mathcal{H}_{>}^{\text{path}} \longrightarrow \mathbb{C}[\mathcal{O}_{p,a}] \otimes \mathbb{C}[C_3], \quad T_{>}e_{(m,x_O^+)} = e_O \otimes g_m, \quad (2484)$$

$$T_{\text{div}} : \mathcal{H}_{p,a} \longrightarrow \mathbb{C}[\mathcal{O}_{p,a}] \otimes \mathbb{C}[C_2], \quad T_{\text{div}}e_{d_O^\epsilon} = e_O \otimes f_\epsilon \quad (2485)$$

are unitary. They give the exact factorizations

$$T_{>}\mathcal{D}_{>}^{\circ}T_{>}^{-1} = D_{p,a}^{\text{orb}} \otimes I_{\mathbb{C}[C_3]}, \quad (2486)$$

$$T_{\text{div}}D_{p,a}T_{\text{div}}^{-1} = D_{p,a}^{\text{orb}} \otimes I_{\mathbb{C}[C_2]}. \quad (2487)$$

Consequently the basis map

$$\mathfrak{F}_{3:2} : \mathcal{H}_{>}^{\text{path}} \otimes \mathbb{C}[C_2] \longrightarrow \mathcal{H}_{p,a} \otimes \mathbb{C}[C_3], \quad e_{(m,x_O^+)} \otimes f_\epsilon \longmapsto e_{d_O^\epsilon} \otimes g_m \quad (2488)$$

is unitary and satisfies

$$\mathfrak{F}_{3:2}(\mathcal{D}_{>}^{\circ} \otimes I_{\mathbb{C}[C_2]})\mathfrak{F}_{3:2}^{-1} = D_{p,a} \otimes I_{\mathbb{C}[C_3]}. \quad (2489)$$

Applying theorem 14.274 with $A = C_3$, $B = C_2$, and $D = D_{p,a}^{\text{orb}}$, the two spectral counting measures are

$$\mu_{>}^{\text{path}} = 3 \sum_{O \in \mathcal{O}_{p,a}} \delta_{\alpha_O}, \quad \mu^{\text{div}} = 2 \sum_{O \in \mathcal{O}_{p,a}} \delta_{\alpha_O}, \quad 2\mu_{>}^{\text{path}} = 3\mu^{\text{div}}. \quad (2490)$$

For every function φ on the finite amplitude set $\{\alpha_O : O \in \mathcal{O}_{p,a}\}$, finite functional calculus gives

$$2 \text{tr } \varphi(\mathcal{D}_{>}^{\circ}) = 3 \text{tr } \varphi(D_{p,a}) = 6 \sum_{O \in \mathcal{O}_{p,a}} \varphi(\alpha_O). \quad (2491)$$

In particular, for every integer $k \geq 0$,

$$2 \text{tr } ((\mathcal{D}_{>}^{\circ})^k) = 3 \text{tr } (D_{p,a}^k) = 6 \sum_O \alpha_O^k, \quad (2492)$$

where the $k = 0$ case records the two multiplicities. The characteristic polynomials obey

$$\det(tI - \mathcal{D}_{>}^{\circ})^2 = \det(tI - D_{p,a})^3 = \prod_{O \in \mathcal{O}_{p,a}} (t - \alpha_O)^6, \quad (2493)$$

and the two operator radii agree:

$$\|\mathcal{D}_{>}^{\circ}\| = \|D_{p,a}\| = \max_{O \in \mathcal{O}_{p,a}} \alpha_O. \quad (2494)$$

Thus the rank and Hilbert–Schmidt relations in (2428) belong to one stable spectral identity; the explicit intertwiner (2488) retains the orbit label, the ordered complement sheet, and the Lorentz rotation separately.

Proof. Equations (2420) and (2412) give

$$\mathcal{D}_{>}^{\circ} e_{(m, x_O^+)} = \alpha_O e_{(m, x_O^+)}, \quad D_{p,a}^{\text{orb}} e_O = \alpha_O e_O.$$

This proves (2486). The two divisors in O have the same amplitude by (2297); hence

$$D_{p,a} e_{d_O^{\epsilon}} = \alpha_O e_{d_O^{\epsilon}},$$

which proves (2487). The displayed basis maps send orthonormal bases to orthonormal bases. Applying both sides of (2489) to $e_{d_O^{\epsilon}} \otimes g_m$ proves the intertwining identity.

Each orbit amplitude consequently occurs three times on $\mathcal{H}_{>}^{\text{path}}$ and twice on $\mathcal{H}_{p,a}$. This proves the counting-measure identity. Summation against φ proves (2491); powers and characteristic polynomials give (2492) and (2493). The largest diagonal entry gives (2494). $\square$

Corollary 14.276 (Endpoint-selected stable spectral transfer and directed polar carrier). *Under the conditions of theorem 14.268, suppose $\mathcal{O}_{p,a} \neq \emptyset$, and abbreviate $W = \mathcal{W}_{\partial,0+}^{p,a}$. Put*

$$Q_0 := Q_{<}, \quad Q_1 := Q_{>},$$

and define the selected source and range spaces by

$$\mathcal{H}_{\partial,0}^{\text{path}} := \text{Ran}(W^*W) = Q_{b_0} \mathbb{C}[\tilde{\mathcal{X}}_{R,p}], \quad (2495)$$

$$\mathcal{H}_{\partial,1}^{\text{path}} := \text{Ran}(WW^*) = Q_{b_1} \mathbb{C}[\tilde{\mathcal{X}}_{R,p}]. \quad (2496)$$

The two positive amplitude operators

$$\mathcal{D}_{\partial,0}^{\circ} := \frac{1}{4} W^*W|_{\mathcal{H}_{\partial,0}^{\text{path}}}, \quad (2497)$$

$$\mathcal{D}_{\partial,1}^{\circ} := \frac{1}{4} WW^*|_{\mathcal{H}_{\partial,1}^{\text{path}}} \quad (2498)$$

satisfy, for $j \in \{0, 1\}$,

$$\mathcal{D}_{\partial,j}^{\circ} e_{(m, x_O^{b_j})} = \alpha_O e_{(m, x_O^{b_j})}. \quad (2499)$$

Consequently the maps

$$T_{\partial,j} : \mathcal{H}_{\partial,j}^{\text{path}} \longrightarrow \mathbb{C}[\mathcal{O}_{p,a}] \otimes \mathbb{C}[C_3], \quad T_{\partial,j} e_{(m, x_O^{b_j})} = e_O \otimes g_m \quad (2500)$$

are unitary and give

$$T_{\partial,j} \mathcal{D}_{\partial,j}^{\circ} T_{\partial,j}^{-1} = D_{p,a}^{\text{orb}} \otimes I_{\mathbb{C}[C_3]}. \quad (2501)$$

For each $j \in \{0, 1\}$, the basis map

$$\mathfrak{F}_{\partial,j}^{3:2} : \mathcal{H}_{\partial,j}^{\text{path}} \otimes \mathbb{C}[C_2] \longrightarrow \mathcal{H}_{p,a} \otimes \mathbb{C}[C_3], \quad e_{(m, x_O^{b_j})} \otimes f_{\epsilon} \longmapsto e_{d_O^{\epsilon}} \otimes g_m \quad (2502)$$

is unitary and obeys

$$\mathfrak{F}_{\partial,j}^{3:2} (\mathcal{D}_{\partial,j}^{\circ} \otimes I_{\mathbb{C}[C_2]}) (\mathfrak{F}_{\partial,j}^{3:2})^{-1} = D_{p,a} \otimes I_{\mathbb{C}[C_3]}. \quad (2503)$$

The polar partial isometry of the selected weighted carrier is

$$V_{\partial,0+} := \frac{1}{2} W (\mathcal{D}_{\partial,0}^{\circ})^{-1/2}, \quad V_{\partial,0+} e_{(m, x_O^{b_0})} = e_{(m, x_O^{b_1})}, \quad (2504)$$

extended by zero on the orthogonal complement of its source. It satisfies

$$V_{\partial,0+}^* V_{\partial,0+} = I_{\mathcal{H}_{\partial,0}^{\text{path}}}, \quad V_{\partial,0+} V_{\partial,0+}^* = I_{\mathcal{H}_{\partial,1}^{\text{path}}}, \quad (2505)$$

$$\mathcal{D}_{\partial,1}^{\circ} = V_{\partial,0+} \mathcal{D}_{\partial,0}^{\circ} V_{\partial,0+}^*, \quad [\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}] = 2(b_1 - b_0) V_{\partial,0+}. \quad (2506)$$

Writing $\mu_{\partial,j}^{\text{path}}$ for the spectral counting measure of $\mathcal{D}_{\partial,j}^\circ$, one has

$$\mu_{\partial,j}^{\text{path}} = 3 \sum_{O \in \mathcal{O}_{p,a}} \delta_{\alpha_O}, \quad 2\mu_{\partial,j}^{\text{path}} = 3\mu^{\text{div}}. \quad (2507)$$

For every function φ on the finite amplitude set,

$$2 \operatorname{tr} \varphi(\mathcal{D}_{\partial,j}^\circ) = 3 \operatorname{tr} \varphi(D_{p,a}) = 6 \sum_{O \in \mathcal{O}_{p,a}} \varphi(\alpha_O), \quad (2508)$$

and

$$\det(tI - \mathcal{D}_{\partial,j}^\circ)^2 = \det(tI - D_{p,a})^3 = \prod_{O \in \mathcal{O}_{p,a}} (t - \alpha_O)^6. \quad (2509)$$

Thus the selected source and range have identical positive amplitude data. Their ordered endpoint direction is retained by $V_{\partial,0+}$ through the signed commutator in (2506).

Proof. Equation (2442) gives

$$\begin{aligned} \frac{1}{4} W^* W &= \Theta_{123}^{-1} (D_{p,a}^{\text{orb}} \otimes E_{b_0+1, b_0+1}^{(2)} \otimes I_3) \Theta_{123}, \\ \frac{1}{4} W W^* &= \Theta_{123}^{-1} (D_{p,a}^{\text{orb}} \otimes E_{b_1+1, b_1+1}^{(2)} \otimes I_3) \Theta_{123}. \end{aligned}$$

Every α_O is positive on an occupied complement orbit. The two ranges are therefore exactly the selected sheets, and restriction gives (2499). The displayed basis maps then prove (2501). Combining that factorization with (2487), or applying theorem 14.274, proves (2503).

On every selected basis vector, the operator in (2504) divides the coefficient $2\sqrt{\alpha_O}$ by the same coefficient. Hence it is the stated basis transport, which proves the support and modulus-transport formulas. The grading eigenvalues on its source and range are $2b_0 - 1$ and $2b_1 - 1$, respectively; their difference proves the commutator identity. Finally each α_O occurs three times on each selected path sheet and twice on the divisor carrier. This proves the counting-measure identity, and finite functional calculus and products over the amplitude multiset prove the remaining formulas. $\square$

Corollary 14.277 (Polar rank–direction invariant and exact endpoint classification). *Under the conditions of theorem 14.276, put $q = q_{R,p}$. On an occupied shell define*

$$\mathcal{E}_{p,a}^{\text{pol},123} := \frac{1}{4} \|[\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}]\|_{\text{HS}}^2, \quad (2510)$$

and set this quantity equal to zero on an empty shell. Then

$$\boxed{\mathcal{E}_{p,a}^{\text{pol},123} = 3q(b_1 - b_0)^2 = \frac{3q}{2} \mathcal{E}_{R,p}^{\text{lab}}.} \quad (2511)$$

If

$$\rho_{p,a}^{\text{path}} := 3q_{R,p}, \quad (2512)$$

then, on every occupied shell,

$$\rho_{p,a}^{\text{path}} = \operatorname{rank} \mathcal{D}_{\partial,j}^\circ \quad (j = 0, 1), \quad (2513)$$

and the exact status table is

endpoint status	$(\rho_{p,a}^{\text{path}}, \mathcal{E}_{p,a}^{\text{pol},123})$
empty	(0, 0)
exterior only or middle only	(3q, 3q)
both	(3q, 0).

(2514)

Consequently

$$0 \leq \mathcal{E}_{p,a}^{\text{pol},123} \leq \rho_{p,a}^{\text{path}}, \quad (2515)$$

with equality on the right exactly for a one-sided shell. The ordered pair in (2514) therefore recovers all three endpoint statuses without using the orbit amplitudes.

Proof. On an occupied shell, (2505) gives

$$\|V_{\partial,0+}\|_{\text{HS}}^2 = \text{tr}(V_{\partial,0+}^* V_{\partial,0+}) = \dim \mathcal{H}_{\partial,0}^{\text{path}} = 3q.$$

Combining this with (2506) yields

$$\mathcal{E}_{p,a}^{\text{pol},123} = (b_1 - b_0)^2 \|V_{\partial,0+}\|_{\text{HS}}^2 = 3q(b_1 - b_0)^2.$$

Equation (1610) gives $\mathcal{E}_{R,p}^{\text{lab}} = 2(b_1 - b_0)^2$, proving (2511). The multiplicity in (2499) proves (2513). Substitution of the four bit pairs in (2443) proves the table and the sharp bound. $\square$

Corollary 14.278 (Primewise polar rank–direction bound). *Retain the conditions and notation of theorem 14.246. Define*

$$\mathcal{E}_p^{\text{pol},123} := \sum_{a \in \mathcal{A}_p} \mathcal{E}_{p,a}^{\text{pol},123}. \quad (2516)$$

Then

$$\mathcal{E}_p^{\text{pol},123} = 3 \sum_{a \in \mathcal{A}_p^{\text{one}}} q_{R_a,p}, \quad (2517)$$

and

$$\begin{aligned} \frac{3}{2} \mathcal{E}_p^{\text{lab}} &= 3\Xi_p \leq \mathcal{E}_p^{\text{pol},123} \leq 3Q_p \\ &= \frac{3}{4} \dim_{\mathbb{C}} \mathcal{V}_p^{\text{Time}} = \frac{1}{2} \log_2 \text{Vol}_p^{\text{det}} \\ &= \frac{1}{2 \log 2} \log \text{Vol}_p^{\text{det}} = \frac{1}{17} L_{H,p}^2 = \frac{2}{25} L_{K,p}^2. \end{aligned} \quad (2518)$$

The lower inequality is an equality exactly when every one-sided shell has $q_{R_a,p} = 1$. The upper inequality is an equality exactly when every occupied shell is one-sided. Moreover,

$$\mathcal{E}_p^{\text{pol},123} > 0 \iff \Xi_p > 0 \iff \mathcal{E}_p^{\text{lab}} > 0, \quad (2519)$$

and any of these conditions implies the Erdős–Straus equation for p .

Proof. Summing (2511) over the shells and using (2235) proves (2517). Each one-sided occupied shell has $q_{R_a,p} \geq 1$, while

$$\sum_{a \in \mathcal{A}_p^{\text{one}}} q_{R_a,p} \leq \sum_{a \in \mathcal{A}_p} q_{R_a,p} = Q_p.$$

This proves both inequalities and their equality cases. The remaining equalities in (2518) follow by multiplying (2237) by $3/2$. The positivity equivalence follows from the positive integer orbit count on every one-sided shell. Its Erdős–Straus consequence is (2239). $\square$

Corollary 14.279 (Stable transfer of target coefficients and target-null mass). *Under the conditions of theorem 14.276, for $j \in \{0, 1\}$ define the aggregate target coefficients and target-null vectors by*

$$\Delta_{\partial,j}^{\text{path}} := \sum_{O \in \mathcal{O}_{p,a}} \sum_{m \in C_3} (-\alpha_O E_{22}), \quad \Delta^{\text{div}} := \sum_{d \in \mathcal{D}_{p,a}^{\text{red}}} \Delta_d, \quad (2520)$$

$$\nu_{\partial,j}^{\text{path}} := \sum_{O \in \mathcal{O}_{p,a}} \sum_{m \in C_3} \alpha_O n_{\text{tar}}, \quad \nu^{\text{div}} := \sum_{d \in \mathcal{D}_{p,a}^{\text{red}}} \alpha_d n_{\text{tar}}. \quad (2521)$$

Then

$$2\Delta_{\partial,j}^{\text{path}} = 3\Delta^{\text{div}} = -3\alpha_{p,a}^{\Sigma} E_{22}, \quad (2522)$$

$$2\nu_{\partial,j}^{\text{path}} = 3\nu^{\text{div}} = 3\alpha_{p,a}^{\Sigma} n_{\text{tar}}. \quad (2523)$$

Tensoring $\mathfrak{F}_{\partial,j}^{3:2}$ with the identity on the two-coordinate target space gives the corresponding stable equivalence before these aggregates are taken. Thus both the selected source and the selected range carry the same aggregate target coefficient and target-null mass.

Proof. There are three Lorentz rotations over each orbit and two complementary divisors in each orbit. Moreover

$$\Delta_{d_O^-} = \Delta_{d_O^+} = -\alpha_O E_{22}, \quad \alpha_{p,a}^\Sigma = 2 \sum_O \alpha_O.$$

Summing these identities proves both formulas for either selected sheet. The final assertion follows from (2503) after multiplication by $-E_{22}$ on the target-coordinate factor. $\square$

Corollary 14.280 (Target-character Lorentz cone and global carrier rank). *Let*

$$H_2 := \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \eta := \begin{pmatrix} \alpha_2 \\ \alpha_1/2 \end{pmatrix}, \quad \lambda := \begin{pmatrix} \lambda_+ \\ \lambda_- \end{pmatrix}. \quad (2524)$$

Then

$$\lambda = H_2 \eta, \quad H_2 \lambda = 2\eta = \begin{pmatrix} 2\alpha_2 \\ \alpha_1 \end{pmatrix}. \quad (2525)$$

The symmetric matrix

$$\mathbf{G}_{p,a} := \begin{pmatrix} \lambda_+ & \lambda_- \\ \lambda_- & \lambda_+ \end{pmatrix} \quad (2526)$$

is positive semidefinite and has eigenvalues $2\alpha_2$ and α_1 . Its determinant is the target-character Lorentz norm

$$\det \mathbf{G}_{p,a} = \lambda_+^2 - \lambda_-^2 = 2\alpha_1 \alpha_2. \quad (2527)$$

Consequently:

- (i) $\mathbf{G}_{p,a} = 0$ exactly when the shell is empty;
- (ii) $\mathbf{G}_{p,a}$ has rank one exactly when one target class is occupied and the other is empty;
- (iii) $\mathbf{G}_{p,a}$ is positive definite exactly when both target classes are occupied.

Equivalently, (λ_+, λ_-) lies in the closed future cone

$$\lambda_+ \geq 0, \quad \lambda_+^2 - \lambda_-^2 \geq 0, \quad (2528)$$

whose two null coordinates are α_1 and $2\alpha_2$. The global coprime-shell criterion is

$$\text{ES} \iff \forall p \in \mathbb{P} \quad \exists a > p/4 : \mathbf{G}_{p,a} \in \text{PSD}_2(\mathbb{R}) \setminus \{0\}. \quad (2529)$$

For the 12289 shell,

$$\lambda_+ = \frac{128395625711903946059}{104678602125586545672}, \quad \lambda_- = -\frac{14143643992119252}{12015450198070081}, \quad (2530)$$

$$\lambda_+^2 - \lambda_-^2 = \frac{5311334286447633252931945}{44686039104187389308828736} > 0. \quad (2531)$$

Thus its target-character vector is timelike and its skew character is negative.

Proof. The first identity in (2525) is (2376). Since $H_2^2 = 2I_2$, the second follows. Carriage of $\mathbf{G}_{p,a}$ by $U_0 = H_2/\sqrt{2}$ gives

$$U_0^* \mathbf{G}_{p,a} U_0 = \text{diag}(\lambda_+ + \lambda_-, \lambda_+ - \lambda_-) = \text{diag}(2\alpha_2, \alpha_1).$$

Both eigenvalues are nonnegative, and their product proves (2527). Strict positivity of each occupied target amplitude gives the three rank statements and (2528). Equation (2529) follows from (2379).

Substitution of (2371) and (2372) into (2376) gives (2530). Exact multiplication in (2527) gives (2531). $\square$

Corollary 14.281 (Polar direction as the nonzero Lorentz-boundary detector). *Under the conditions of theorem 14.280, put*

$$r_{p,a} := \text{rank } \mathbf{G}_{p,a}. \quad (2532)$$

Then

$$\boxed{\mathcal{E}_{p,a}^{\text{pol},123} = 3q_{R,p} r_{p,a}(2 - r_{p,a}).} \quad (2533)$$

In particular,

$$\mathcal{E}_{p,a}^{\text{pol},123} > 0 \iff \mathbf{G}_{p,a} \neq 0 \text{ and } \det \mathbf{G}_{p,a} = 0 \iff \lambda_+ > 0, \lambda_+^2 - \lambda_-^2 = 0. \quad (2534)$$

Thus the amplitude-free polar energy detects precisely the nonzero null boundary of the target-character Lorentz cone.

On this boundary,

$$b_1 - b_0 = \frac{\lambda_-}{\lambda_+}, \quad (2535)$$

and therefore

$$[\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}] = 2 \frac{\lambda_-}{\lambda_+} V_{\partial,0+}. \quad (2536)$$

The exterior-only shell lies on the ray $\lambda_- = \lambda_+ > 0$ and has $b_1 - b_0 = 1$; the middle-only shell lies on the ray $\lambda_- = -\lambda_+ < 0$ and has $b_1 - b_0 = -1$.

Proof. By theorem 14.280, the ranks 0, 1, 2 correspond respectively to an empty shell, a shell with exactly one occupied target class, and a shell with both target classes occupied. The target reconstruction in (2377) identifies those three alternatives with the empty, one-sided, and both-channel endpoint statuses. Comparing this rank table with (2514) proves (2533) and (2534).

For an exterior-only shell, $\alpha_1 = 0$ and $\alpha_2 > 0$, so (2376) gives $\lambda_- = \lambda_+ > 0$. Its endpoint bits are $(b_0, b_1) = (0, 1)$. For a middle-only shell, $\alpha_2 = 0$ and $\alpha_1 > 0$, so $\lambda_- = -\lambda_+ < 0$, while $(b_0, b_1) = (1, 0)$. This proves (2535); substitution into (2506) proves the commutator formula. $\square$

Theorem 14.282 (Typewise stable spectral calculus and positive-function Lorentz cones). *Under the conditions of theorem 14.276, retain the orbit partition $\mathcal{O}_{p,a} = \mathcal{O}_{02} \sqcup \mathcal{O}_{11}$ from (2310). For $j \in \{0, 1\}$ and $\nu \in \{02, 11\}$, put*

$$\mathcal{H}_{\partial,j,\nu}^{\text{path}} := \text{span}\{e_{(m,x_O^{b_j})} : O \in \mathcal{O}_\nu, m \in C_3\}, \quad \mathcal{D}_{\partial,j,\nu}^\circ := \mathcal{D}_{\partial,j}^\circ|_{\mathcal{H}_{\partial,j,\nu}^{\text{path}}}, \quad (2537)$$

$$\mathcal{H}_\nu^{\text{div}} := \text{span}\{e_{d_\epsilon} : O \in \mathcal{O}_\nu, \epsilon \in C_2\}, \quad D_{p,a,\nu} := D_{p,a}|_{\mathcal{H}_\nu^{\text{div}}}. \quad (2538)$$

The unitary $\mathfrak{F}_{\partial,j}^{3:2}$ restricts for each (j, ν) to a unitary

$$\mathfrak{F}_{\partial,j,\nu}^{3:2} : \mathcal{H}_{\partial,j,\nu}^{\text{path}} \otimes \mathbb{C}[C_2] \longrightarrow \mathcal{H}_\nu^{\text{div}} \otimes \mathbb{C}[C_3] \quad (2539)$$

satisfying

$$\mathfrak{F}_{\partial,j,\nu}^{3:2} (\mathcal{D}_{\partial,j,\nu}^\circ \otimes I_{\mathbb{C}[C_2]}) (\mathfrak{F}_{\partial,j,\nu}^{3:2})^{-1} = D_{p,a,\nu} \otimes I_{\mathbb{C}[C_3]}. \quad (2540)$$

Hence the typewise spectral counting measures are

$$\mu_{\partial,j,\nu}^{\text{path}} = 3 \sum_{O \in \mathcal{O}_\nu} \delta_{\alpha_O}, \quad \mu_\nu^{\text{div}} = 2 \sum_{O \in \mathcal{O}_\nu} \delta_{\alpha_O}, \quad 2\mu_{\partial,j,\nu}^{\text{path}} = 3\mu_\nu^{\text{div}}. \quad (2541)$$

For every function φ on the finite amplitude set,

$$2 \text{tr } \varphi(\mathcal{D}_{\partial,j,\nu}^\circ) = 3 \text{tr } \varphi(D_{p,a,\nu}) = 6 \sum_{O \in \mathcal{O}_\nu} \varphi(\alpha_O). \quad (2542)$$

The typewise characteristic polynomials satisfy

$$\det(tI - \mathcal{D}_{\partial,j,\nu}^\circ)^2 = \det(tI - D_{p,a,\nu})^3 = \prod_{O \in \mathcal{O}_\nu} (t - \alpha_O)^6. \quad (2543)$$

Now let φ take nonnegative real values on the amplitude set and define

$$\eta_\varphi := \begin{pmatrix} \eta_{\varphi,02} \\ \eta_{\varphi,11} \end{pmatrix} := \begin{pmatrix} \sum_{O \in \mathcal{O}_{02}} \varphi(\alpha_O) \\ \sum_{O \in \mathcal{O}_{11}} \varphi(\alpha_O) \end{pmatrix}, \quad \lambda_\varphi := H_2 \eta_\varphi = \begin{pmatrix} \lambda_{\varphi,+} \\ \lambda_{\varphi,-} \end{pmatrix}. \quad (2544)$$

The two masses are recovered from either carrier by

$$\eta_{\varphi,\nu} = \frac{1}{3} \operatorname{tr} \varphi(\mathcal{D}_{\partial,j,\nu}^\circ) = \frac{1}{2} \operatorname{tr} \varphi(D_{p,a,\nu}). \quad (2545)$$

The symmetric matrix

$$\mathbf{G}_\varphi := \begin{pmatrix} \lambda_{\varphi,+} & \lambda_{\varphi,-} \\ \lambda_{\varphi,-} & \lambda_{\varphi,+} \end{pmatrix} \quad (2546)$$

obeys

$$U_0^* \mathbf{G}_\varphi U_0 = \operatorname{diag}(2\eta_{\varphi,02}, 2\eta_{\varphi,11}), \quad U_0 = \frac{1}{\sqrt{2}} H_2. \quad (2547)$$

It is positive semidefinite, and

$$\det \mathbf{G}_\varphi = \lambda_{\varphi,+}^2 - \lambda_{\varphi,-}^2 = 4\eta_{\varphi,02}\eta_{\varphi,11}. \quad (2548)$$

Its rank is zero when both typewise masses vanish, one when exactly one is positive, and two when both are positive. Moreover, for nonnegative scalars r, s and nonnegative spectral functions φ, ψ ,

$$\mathbf{G}_{r\varphi+s\psi} = r\mathbf{G}_\varphi + s\mathbf{G}_\psi. \quad (2549)$$

Thus the nonnegative finite spectral calculus maps linearly into the closed positive-semidefinite Lorentz cone.

For $\varphi(t) = t$,

$$\eta_{\varphi,02} = \alpha_2, \quad \eta_{\varphi,11} = \frac{\alpha_1}{2}, \quad \mathbf{G}_\varphi = \mathbf{G}_{p,a}. \quad (2550)$$

In particular, (2548) becomes $\det \mathbf{G}_{p,a} = 2\alpha_1\alpha_2$.

Proof. The basis formula (2502) leaves the orbit label O unchanged. It therefore maps the stable subspace belonging to $\mathcal{O}_\nu$ onto the stable divisor subspace belonging to the same orbit class for either selected sheet. This proves (2539) and (2540). Every orbit in that class occurs three times on each selected path carrier and twice on the divisor carrier. This proves (2541); integration against φ and formation of characteristic polynomials give (2542) and (2543).

Equation (2545) is (2542) divided by six. Direct multiplication by U_0 proves (2547). The two diagonal entries are nonnegative; their product is (2548), and their vanishing patterns give the rank assertions. Every construction in (2544) and (2546) is linear in φ , which proves (2549). Finally, (2314) and (2315) give (2550). $\square$

Corollary 14.283 (Strictly positive spectral preservation of the polar null direction). *Under the conditions of theorem 14.282, suppose*

$$\varphi(\alpha_O) > 0 \quad \text{for every } O \in \mathcal{O}_{p,a}. \quad (2551)$$

Put

$$r_\varphi := \operatorname{rank} \mathbf{G}_\varphi. \quad (2552)$$

Then

$$r_\varphi = \operatorname{rank} \mathbf{G}_{p,a} = r_{p,a}, \quad (2553)$$

and

$$\mathcal{E}_{p,a}^{\text{pol},123} = 3q_{R,p} r_\varphi (2 - r_\varphi). \quad (2554)$$

If this energy is positive, then

$$b_1 - b_0 = \frac{\lambda_{\varphi,-}}{\lambda_{\varphi,+}}, \quad [\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}] = 2 \frac{\lambda_{\varphi,-}}{\lambda_{\varphi,+}} V_{\partial,0+}. \quad (2555)$$

Consequently every strictly positive spectral function preserves the empty, one-sided, or both-channel status and, in the one-sided case, the directed Lorentz-null ray.

Proof. Condition (2551) makes

$$\eta_{\varphi,\nu} = \sum_{O \in \mathcal{O}_\nu} \varphi(\alpha_O)$$

positive exactly when $\mathcal{O}_\nu$ is nonempty. Hence the diagonal form in (2547) has the same zero pattern as $\text{diag}(2\alpha_2, \alpha_1)$, proving (2553). Equation (2554) follows from (2533). If the rank is one, exactly one typewise mass is positive. Thus $\lambda_{\varphi,-} = \lambda_{\varphi,+} > 0$ for the 02-only class and $\lambda_{\varphi,-} = -\lambda_{\varphi,+} < 0$ for the 11-only class. The endpoint status table proves the first identity in (2555); the polar commutator proves the second. $\square$

Corollary 14.284 (Moment, heat, and resolvent Lorentz families). *Under the conditions of theorem 14.282, define for $\nu \in \{02, 11\}$*

$$M_{\nu,k} := \sum_{O \in \mathcal{O}_\nu} \alpha_O^k, \quad k \in \mathbb{Z}_{\geq 0}, \quad (2556)$$

$$\Theta_\nu(\beta) := \sum_{O \in \mathcal{O}_\nu} e^{-\beta \alpha_O}, \quad \beta \geq 0, \quad (2557)$$

$$R_\nu(z) := \sum_{O \in \mathcal{O}_\nu} \frac{1}{z - \alpha_O}, \quad z > \max_{O \in \mathcal{O}_{p,a}} \alpha_O. \quad (2558)$$

The corresponding character vectors and Lorentz matrices are obtained from theorem 14.282 using

$$\varphi_k(t) = t^k, \quad \varphi_\beta(t) = e^{-\beta t}, \quad \varphi_z(t) = \frac{1}{z - t}. \quad (2559)$$

All three Lorentz-matrix families are positive semidefnite. For either $j \in \{0, 1\}$, their typewise stable trace identities are

$$2 \operatorname{tr}((\mathcal{D}_{\partial,j,\nu}^\circ)^k) = 3 \operatorname{tr}(D_{p,a,\nu}^k) = 6M_{\nu,k}, \quad (2560)$$

$$2 \operatorname{tr} e^{-\beta \mathcal{D}_{\partial,j,\nu}^\circ} = 3 \operatorname{tr} e^{-\beta D_{p,a,\nu}} = 6\Theta_\nu(\beta), \quad (2561)$$

$$2 \operatorname{tr}(zI - \mathcal{D}_{\partial,j,\nu}^\circ)^{-1} = 3 \operatorname{tr}(zI - D_{p,a,\nu})^{-1} = 6R_\nu(z). \quad (2562)$$

The moment sequence determines the other two families through the convergent expansions

$$\Theta_\nu(\beta) = \sum_{k=0}^{\infty} \frac{(-\beta)^k}{k!} M_{\nu,k}, \quad (2563)$$

$$R_\nu(z) = \sum_{k=0}^{\infty} \frac{M_{\nu,k}}{z^{k+1}}. \quad (2564)$$

For each of the three choices of φ , the Lorentz determinant is four times the product of the two displayed typewise masses. All three functions in (2559) are strictly positive on the occupied amplitude

set. Hence their Lorentz matrices have rank $r_{p,a}$. On a one-sided shell,

$$\boxed{\begin{aligned} b_1 - b_0 &= \frac{M_{02,k} - M_{11,k}}{M_{02,k} + M_{11,k}} \\ &= \frac{\Theta_{02}(\beta) - \Theta_{11}(\beta)}{\Theta_{02}(\beta) + \Theta_{11}(\beta)} \\ &= \frac{R_{02}(z) - R_{11}(z)}{R_{02}(z) + R_{11}(z)}. \end{aligned}} \quad (2565)$$

This holds for every $k \geq 0$, $\beta \geq 0$, and $z > \max_O \alpha_O$. Thus these strictly positive spectral families transport the endpoint direction but do not alter the shellwise nonvanishing criterion.

Proof. Every amplitude is nonnegative, so all functions in (2559) are nonnegative on the stated domains. Apply (2542) to each function. The exponential series is absolutely convergent for every β , and the geometric series for $(z - \alpha_O)^{-1}$ converges because $z > \max_O \alpha_O$. There are finitely many orbits, so termwise summation gives (2563) and (2564). The determinant statement is (2548). Strict positivity allows application of theorem 14.283. On a one-sided shell one typewise sum is zero and the other is positive, so direct substitution proves (2565). $\square$

Theorem 14.285 (Signed orbit trace, arithmetic numerator, and target rapidity). *Assume the hypotheses of theorems 14.259 and 14.263, and assume $R_0 > 2$. For every $O \in \mathcal{O}_{p,a}$, let $d_O < N$ be the unique lower member of the complement orbit $O = \{d_O, N^2/d_O\}$, and put*

$$\chi_{\text{tar}}(O) := \begin{cases} +1, & O \in \mathcal{O}_{02}, \\ -1, & O \in \mathcal{O}_{11}. \end{cases} \quad (2566)$$

Then the skew character is the exact signed orbit trace

$$\lambda_- = \sum_{O \in \mathcal{O}_{p,a}} \chi_{\text{tar}}(O) \frac{4Nd_O}{(N + d_O)^2} \quad (2567)$$

$$= \sum_{O \in \mathcal{O}_{02}} \alpha_O - \sum_{O \in \mathcal{O}_{11}} \alpha_O = \alpha_2 - \frac{\alpha_1}{2} = \frac{1}{2}(\alpha_0 - \alpha_1 + \alpha_2). \quad (2568)$$

Equivalently,

$$2\lambda_- = \sum_{j=0}^2 (-1)^j \sum_{u \in \mathcal{X}_{R,p}^{(j)}} \frac{4N p^j u}{(N + p^j u)^2}. \quad (2569)$$

Every divisor weight has the logarithmic-radius form

$$\frac{4Nd}{(N + d)^2} = \text{sech}^2\left(\frac{1}{2} \log \frac{d}{N}\right). \quad (2570)$$

Define the positive integer

$$\mathcal{D}_{p,a} := \prod_{O \in \mathcal{O}_{p,a}} (N + d_O)^2 \quad (2571)$$

and the signed integer

$$\mathcal{B}_{p,a} := 4N \sum_{O \in \mathcal{O}_{p,a}} \chi_{\text{tar}}(O) d_O \prod_{\substack{O' \in \mathcal{O}_{p,a} \\ O' \neq O}} (N + d_{O'})^2. \quad (2572)$$

The empty product is 1. Then

$$\lambda_- = \frac{\mathcal{B}_{p,a}}{\mathcal{D}_{p,a}}, \quad \text{sgn}(\lambda_-) = \text{sgn}(\mathcal{B}_{p,a}). \quad (2573)$$

In particular,

$$\lambda_- = 0 \iff \mathcal{B}_{p,a} = 0 \iff \alpha_1 = 2\alpha_2. \quad (2574)$$

Suppose both target classes are occupied. The Lorentz mass and rapidity of the target-character vector are

$$\mathfrak{m}_{p,a} := \sqrt{\lambda_+^2 - \lambda_-^2} = \sqrt{2\alpha_1\alpha_2}, \quad (2575)$$

$$\vartheta_{p,a} := \operatorname{arctanh}\left(\frac{\lambda_-}{\lambda_+}\right) = \frac{1}{2} \log\left(\frac{2\alpha_2}{\alpha_1}\right). \quad (2576)$$

Thus

$$\lambda_+ = \mathfrak{m}_{p,a} \cosh \vartheta_{p,a}, \quad \lambda_- = \mathfrak{m}_{p,a} \sinh \vartheta_{p,a}. \quad (2577)$$

The balance locus in (2368) is the zero-rapidity locus.

If the two orbit types have equal cardinality m , order their lower representatives as

$$e_1 \leq \cdots \leq e_m < N \quad \text{for } \mathcal{O}_{02}, \quad h_1 \leq \cdots \leq h_m < N \quad \text{for } \mathcal{O}_{11}.$$

The componentwise inequalities $e_k \leq h_k$ imply $\lambda_- \leq 0$, with strict inequality when at least one comparison is strict. Reversing every comparison reverses the conclusion.

For the 12289 shell, the ordered lower representatives are

$$(e_1, e_2, e_3) = (615, 5125, 230625), \quad (h_1, h_2, h_3) = (61445, 2765025, 23041875). \quad (2578)$$

Hence $\lambda_- < 0$ follows directly from componentwise radial comparison. Its exact rapidity satisfies

$$\exp(2\vartheta_{12289,3075}) = \frac{5176199252561022635}{251615052171246869483} < 1. \quad (2579)$$

Proof. The complement involution sends d to N^2/d , preserves $4Nd/(N+d)^2$, and has no fixed divisor under the stated hypotheses. Every orbit therefore has one member below N , and $\alpha_O = 4Nd_O/(N+d_O)^2$. Equations (2314), (2315), and (2376) give (2567) and (2568). Substitution of (620) proves (2569).

For $x = \frac{1}{2} \log(d/N)$,

$$\cosh^2 x = \frac{1}{4} \left(\frac{d}{N} + 2 + \frac{N}{d} \right) = \frac{(N+d)^2}{4Nd},$$

which proves (2570). Passage to the common denominator $\mathcal{D}_{p,a}$ gives (2572) and (2573). Its denominator is positive. The final equivalence in (2574) is $\lambda_- = \alpha_2 - \alpha_1/2$.

On the two-target interior,

$$\lambda_+ + \lambda_- = 2\alpha_2 > 0, \quad \lambda_+ - \lambda_- = \alpha_1 > 0.$$

Hence

$$\exp(2\vartheta_{p,a}) = \frac{\lambda_+ + \lambda_-}{\lambda_+ - \lambda_-} = \frac{2\alpha_2}{\alpha_1}.$$

Together with $\lambda_+^2 - \lambda_-^2 = 2\alpha_1\alpha_2$, this proves (2575)–(2577).

For $0 < d < N$, differentiation gives

$$\frac{d}{dd} \frac{4Nd}{(N+d)^2} = \frac{4N(N-d)}{(N+d)^3} > 0.$$

The componentwise comparison statement follows by summation. The representatives in (2578) are read from their p -adic origins and are componentwise strictly ordered. Finally, substitution of (2371) and (2372) into $\exp(2\vartheta) = 2\alpha_2/\alpha_1$ gives (2579). $\square$

Theorem 14.286 (Signed resolvent character and pointwise blade–circle realization). *Assume the hypotheses of theorems 14.273 and 14.282. For $\nu \in \{02, 11\}$, define the monic type polynomial and its logarithmic derivative by*

$$P_\nu(z) := \prod_{O \in \mathcal{O}_\nu} (z - \alpha_O), \quad R_\nu(z) := \sum_{O \in \mathcal{O}_\nu} \frac{1}{z - \alpha_O} = \frac{P'_\nu(z)}{P_\nu(z)}. \quad (2580)$$

The empty product is 1. On the complement of the amplitude spectrum, put

$$\Pi(z) := R_{02}(z) + R_{11}(z), \quad \Xi(z) := R_{02}(z) - R_{11}(z). \quad (2581)$$

For either $j \in \{0, 1\}$, the stable carriers give

$$2 \left[\operatorname{tr}(zI - \mathcal{D}_{\partial,j,02}^\circ)^{-1} + \operatorname{tr}(zI - \mathcal{D}_{\partial,j,11}^\circ)^{-1} \right] = 6\Pi(z), \quad (2582)$$

$$2 \left[\operatorname{tr}(zI - \mathcal{D}_{\partial,j,02}^\circ)^{-1} - \operatorname{tr}(zI - \mathcal{D}_{\partial,j,11}^\circ)^{-1} \right] = 6\Xi(z), \quad (2583)$$

and the same left sides with the factor 2 replaced by 3 after $\mathcal{D}_{\partial,j,\nu}^\circ$ is replaced by $D_{p,a,\nu}$.

The signed character has the exact rational form

$$\Xi(z) = \frac{\mathcal{W}_{p,a}(z)}{P_{02}(z)P_{11}(z)}, \quad \mathcal{W}_{p,a}(z) := P'_{02}(z)P_{11}(z) - P'_{11}(z)P_{02}(z). \quad (2584)$$

Let $G(z)$ be the monic greatest common divisor of $P_{02}(z)$ and $P_{11}(z)$, and write

$$P_{02} = GA, \quad P_{11} = GB, \quad \gcd(A, B) = 1. \quad (2585)$$

Then

$$\Xi(z) = \frac{A'(z)}{A(z)} - \frac{B'(z)}{B(z)} = \frac{A'(z)B(z) - B'(z)A(z)}{A(z)B(z)}. \quad (2586)$$

For any amplitude value c , let $m_{02}(c)$ and $m_{11}(c)$ be its orbit multiplicities in the two classes. Its signed residue is

$$\operatorname{Res}_{z=c} \Xi(z) = m_{02}(c) - m_{11}(c). \quad (2587)$$

Thus a common amplitude disappears from the pole divisor precisely to the extent that its two type multiplicities agree.

For $|z| > \max_O \alpha_O$, the Laurent expansions are

$$\Pi(z) = \sum_{k=0}^{\infty} \frac{M_{02,k} + M_{11,k}}{z^{k+1}}, \quad (2588)$$

$$\Xi(z) = \sum_{k=0}^{\infty} \frac{M_{02,k} - M_{11,k}}{z^{k+1}}. \quad (2589)$$

In particular,

$$\lim_{z \rightarrow \infty} z\Xi(z) = |\mathcal{O}_{02}| - |\mathcal{O}_{11}|, \quad (2590)$$

$$\lim_{z \rightarrow \infty} z^2 \left(\Xi(z) - \frac{|\mathcal{O}_{02}| - |\mathcal{O}_{11}|}{z} \right) = \lambda_-. \quad (2591)$$

The symmetric resolvent-character matrix

$$\mathbf{G}(z) := \begin{pmatrix} \Pi(z) & \Xi(z) \\ \Xi(z) & \Pi(z) \end{pmatrix} \quad (2592)$$

has eigenvalues $2R_{02}(z)$ and $2R_{11}(z)$. For real $z > \max_O \alpha_O$, it is positive semidefinite and

$$\det \mathbf{G}(z) = 4R_{02}(z)R_{11}(z). \quad (2593)$$

On a one-sided shell, for real $z > \max_O \alpha_O$,

$$\boxed{\Xi(z) = (b_1 - b_0)\Pi(z)}, \quad \frac{\Xi(z)}{\Pi(z)} = b_1 - b_0. \quad (2594)$$

Thus the signed resolvent character lies on the positive null ray for an exterior-only shell and on the negative null ray for a middle-only shell.

Fix $m \in C_3$. The individual ordered blade and rotated Lorentz arc give the pointwise formulas

$$R_\nu(z) = \sum_{O \in \mathcal{O}_\nu} \operatorname{tr} \mathcal{H}_{O,m} \left[Q_{O,m}^> \left(z I_{\mathcal{H}_{O,m}} - \frac{1}{4} W_{O,m}^{-,+} W_{O,m}^{+,-} \right)^{-1} Q_{O,m}^> \right] \quad (2595)$$

$$= \sum_{O \in \mathcal{O}_\nu} \frac{1}{z - 4\eta_m(U_m(\delta_O))}. \quad (2596)$$

On the arithmetic target line,

$$\left(z E_{22} + \Delta_{d_O^-} \right)_{E_{22}\mathbb{C}^2}^{-1} = \frac{1}{z - \alpha_O} E_{22}. \quad (2597)$$

The pointwise circle descent also gives

$$R_\nu(z) n_{\text{tar}} = \sum_{O \in \mathcal{O}_\nu} \frac{\mathcal{B}_m U_m(\delta_O) - \Psi_{\text{circ}}(4E_{11})}{\alpha_O(z - \alpha_O)}. \quad (2598)$$

Consequently every entry and the determinant of $\mathbf{G}(z)$ can be computed from the typewise ordered-blade resolvents, from the rotated arc heights, or from the target-null circle descent.

For the 12289 shell, the two exact polynomials are

$$P_{02}(z) = \left(z - \frac{61445}{943902729} \right) \left(z - \frac{184335}{339886096} \right) \left(z - \frac{921675}{38217124} \right), \quad (2599)$$

$$P_{11}(z) = \left(z - \frac{615}{94864} \right) \left(z - \frac{123}{484} \right) \left(z - \frac{1025}{1089} \right). \quad (2600)$$

All six roots are distinct. Hence the residues of Ξ are $+1$ at the three roots of P_{02} and -1 at the three roots of P_{11} . Its numerator is

$$\mathcal{W}_{12289,3075}(z) = -\frac{347}{12794188948921747974357647576064} \widehat{\mathcal{W}}_{12289}(z), \quad (2601)$$

where

$$\begin{aligned} \widehat{\mathcal{W}}_{12289}(z) := & 43401481466401148947095343104z^4 \\ & - 18209177424538142005968238080z^3 \\ & + 395994243119213681673328400z^2 \\ & - 2827184709924732899115000z \\ & + 831864914962023065625. \end{aligned} \quad (2602)$$

The coefficient of z^4 in $\mathcal{W}_{12289,3075}(z)$ is

$$\lambda_- = -\frac{14143643992119252}{12015450198070081}, \quad (2603)$$

as required by (2591) and the equal cardinality of the two orbit classes.

Proof. The logarithmic-derivative identity in (2580) follows by differentiating the finite product. The stable trace identities follow from (2562); addition and subtraction give (2582) and (2583). Bringing the two logarithmic derivatives to a common denominator proves (2584). Substitution of (2585) cancels the two copies of G'/G and proves (2586). The residue of a logarithmic derivative is the root multiplicity, proving (2587).

The geometric series for each $(z - \alpha_O)^{-1}$ gives the two Laurent expansions. Their first two coefficients prove (2590) and (2591). Hadamard carriage of $G(z)$ gives $\text{diag}(2R_{02}(z), 2R_{11}(z))$, which proves (2593) and positivity on the stated real domain. On a one-sided shell one type polynomial is 1, so one logarithmic derivative vanishes. Equations (2581) then prove (2594).

Equations (2469) and (2473) give (2595) and (2596). Equation (2471) gives the inverse on the one-dimensional target line. Dividing (2477) by $\alpha_O(z - \alpha_O)$ and summing over one orbit class proves (2598).

For the 12289 shell, evaluate $4Nd/(N + d)^2$ at the ordered representatives in (2578). This gives (2599) and (2600). Exact differentiation and integer denominator clearing give (2601) and (2602). The leading coefficient is the difference of the two first moments, hence (2603). $\square$

Theorem 14.287 (Signed-moment defect filtration). *Retain the notation of theorem 14.286, and put*

$$S_k := M_{02,k} - M_{11,k} \quad (k \geq 0). \quad (2604)$$

Let $P_{02} = GA$ and $P_{11} = GB$ be the coprime factorization in (2585), and define

$$a := \deg A, \quad b := \deg B, \quad d := a + b, \quad V(z) := A'(z)B(z) - B'(z)A(z). \quad (2605)$$

Suppose that Ξ is nonzero, and let

$$r := \min\{k \geq 0 : S_k \neq 0\}. \quad (2606)$$

Then $0 \leq r \leq d - 1$, and

$$\deg V = d - r - 1, \quad \text{lc}(V) = S_r. \quad (2607)$$

The unreduced numerator satisfies

$$\mathcal{W}_{p,a} = G^2V, \quad \deg \mathcal{W}_{p,a} = \deg P_{02} + \deg P_{11} - r - 1, \quad \text{lc}(\mathcal{W}_{p,a}) = S_r. \quad (2608)$$

The following conditions are equivalent:

- (i) $P_{02} = P_{11}$;
- (ii) Ξ vanishes identically;
- (iii) $S_k = 0$ for every $k \geq 0$;
- (iv) $S_k = 0$ for $0 \leq k \leq d - 1$.

In particular, each successive vanishing signed moment lowers the degree of the signed-resolvent numerator by one, and the first nonzero signed moment is retained as its leading coefficient.

For completeness, write

$$P_\nu(z) = z^{n_\nu} + c_{\nu,1}z^{n_\nu-1} + \cdots + c_{\nu,n_\nu}, \quad \nu \in \{02, 11\}. \quad (2609)$$

The typewise moments and coefficients determine one another through

$$M_{\nu,k} + \sum_{j=1}^{k-1} c_{\nu,j}M_{\nu,k-j} + kc_{\nu,k} = 0, \quad 1 \leq k \leq n_\nu, \quad (2610)$$

$$M_{\nu,k} + \sum_{j=1}^{n_\nu} c_{\nu,j}M_{\nu,k-j} = 0, \quad k > n_\nu. \quad (2611)$$

If $n_{02} = n_{11} = m$, denote the two coefficient arrays by

$$P_{02}(z) = z^m + a_1z^{m-1} + \cdots + a_m, \quad P_{11}(z) = z^m + b_1z^{m-1} + \cdots + b_m. \quad (2612)$$

When the polynomials differ, the defect order obeys $1 \leq r \leq m$, and

$$a_j = b_j \quad (1 \leq j < r), \quad S_r = r(b_r - a_r). \quad (2613)$$

Proof. Equation (2589) gives

$$\Xi(z) = \frac{S_r}{z^{r+1}} + O(z^{-r-2}) \quad (z \rightarrow \infty). \quad (2614)$$

The denominator $A(z)B(z)$ is monic of degree d . Comparing (2614) with $\Xi = V/(AB)$ gives (2607). Since G is monic and

$$(GA)'(GB) - (GB)'(GA) = G^2(A'B - B'A),$$

equation (2608) follows.

Equality of the type polynomials gives $\Xi = 0$ by their logarithmic derivatives. If $\Xi = 0$, then $A'B - B'A = 0$, so the rational function A/B has zero derivative. It is constant; monicity makes the constant equal to 1. Coprimality then gives $A = B = 1$ and $P_{02} = P_{11} = G$. The Laurent expansion proves the equivalence with the vanishing of every S_k . Finally, if $S_0, \dots, S_{d-1}$ all vanish while $V \neq 0$, then (2607) would give $\deg V < 0$, a contradiction. This proves the four-way equivalence.

Equations (2610) and (2611) are Newton's identities for the roots of each monic type polynomial. In the equal-degree case, $S_0 = 0$. Applying the initial identities successively shows that $S_1 = \dots = S_{r-1} = 0$ is equivalent to $a_j = b_j$ for $1 \leq j < r$. Subtracting the two identities at index r then leaves

$$S_r + r(a_r - b_r) = 0,$$

which is (2613). Two distinct monic degree- m polynomials differ in one of their first m coefficients, so $r \leq m$. $\square$

Corollary 14.288 (Lorentz splitting order and the balanced (12289) stratum). *Under the hypotheses of theorem 14.287, the resolvent-character matrix has the asymptotic decomposition*

$$G(z) = \Pi(z)I_2 + \left(\frac{S_r}{z^{r+1}} + O(z^{-r-2}) \right) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (2615)$$

Thus the two type eigenvalues first separate at order z^{-r-1} , with difference $2S_r z^{-r-1} + O(z^{-r-2})$.

For the 12289 shell, both type polynomials have degree 3,

$$S_0 = 0, \quad S_1 = \lambda_- = -\frac{14143643992119252}{12015450198070081} \neq 0. \quad (2616)$$

Consequently $r = 1$, the signed numerator has degree 4, and its leading coefficient is exactly λ_- . Equivalently, if a_1 and b_1 are the first coefficients of (2599) and (2600), then

$$b_1 - a_1 = \lambda_-. \quad (2617)$$

Proof. Equation (2615) follows by substituting (2614) into (2592). The exact (12289) claims follow from equal type cardinality, (2603), and (2607). The final coefficient identity is the case $r = 1$ of (2613). $\square$

Corollary 14.289 (Split-zero Boolean strata of the target Lorentz cone). *Retain the split target coordinates $(\gamma_1, \gamma_2) \in G(\mathbb{R})^2$ from (627), and write*

$$b_j := \chi_{\mathbb{R}}(\gamma_j) \in \mathbb{B}, \quad x := p_{\mathbb{R}}(\gamma_1) = \alpha_1, \quad y := p_{\mathbb{R}}(\gamma_2) = \alpha_2. \quad (2618)$$

The numerical cone map

$$\mathfrak{C}(\gamma_1, \gamma_2) := \begin{pmatrix} y + x/2 & y - x/2 \\ y - x/2 & y + x/2 \end{pmatrix} = G_{p,a} \quad (2619)$$

has the following exact support strata:

(b_1, b_2)	(λ_+, λ_-)	$\text{rank } G_{p,a}$	<i>Lorentz stratum</i>
$(0, 0)$	$(0, 0)$	0	zero
$(1, 0)$	$(x/2, -x/2)$	1	future-null
$(0, 1)$	(y, y)	1	future-null
$(1, 1)$	$(y + x/2, y - x/2)$	2	future-timelike.

(2620)

In particular,

$$\Omega_R(p) > 0 \iff b_1 \vee b_2 = 1 \iff \mathbf{G}_{p,a} \neq 0, \quad (2621)$$

$$\det \mathbf{G}_{p,a} > 0 \iff b_1 \wedge b_2 = 1. \quad (2622)$$

Consequently,

$$\forall p \in \mathbb{P} \quad \exists a > p/4 : b_1 \wedge b_2 = 1 \implies \text{ES}. \quad (2623)$$

Both rank-one strata occur in coprime shells. For $(p, a, R, N) = (3, 2, 5, 6)$, the reduced divisor set is $\{4, 9\}$, its origin vector is $(1, 0, 1)$, and

$$\alpha_1 = 0, \quad \alpha_2 = \frac{24}{25}, \quad (\lambda_+, \lambda_-) = \left(\frac{24}{25}, \frac{24}{25} \right). \quad (2624)$$

It yields

$$\frac{4}{3} = \frac{1}{2} + \frac{1}{2} + \frac{1}{3}. \quad (2625)$$

For $(p, a, R, N) = (29, 10, 11, 290)$, the reduced divisor set is $\{29, 2900\}$, its origin vector is $(0, 2, 0)$, and

$$\alpha_1 = \frac{80}{121}, \quad \alpha_2 = 0, \quad (\lambda_+, \lambda_-) = \left(\frac{40}{121}, -\frac{40}{121} \right). \quad (2626)$$

It yields

$$\frac{4}{29} = \frac{1}{10} + \frac{1}{29} + \frac{1}{290}. \quad (2627)$$

Thus the Boolean conjunction in (2622) is a proper strengthening of the shell predicate in (2621).

Proof. Equation (628) gives $x = \alpha_1$ and $y = \alpha_2$, including the value 0 on an empty target class. Substitution into (2376) proves (2619). Its eigenvalues are x and $2y$. The Boolean-coordinate identity (629) now gives every row of (2620), as well as (2621) and (2622). The global implication follows from (2529).

For $p = 3$ and $a = 2$, direct divisor enumeration gives

$$d \mid 36, \quad d \equiv -6 \pmod{5} \implies d \in \{4, 9\}.$$

The two divisors form one exterior complement orbit, and its weight is

$$\frac{4 \cdot 6 \cdot 4}{(6 + 4)^2} = \frac{24}{25}.$$

This proves (2624); direct addition proves (2625).

For $p = 29$ and $a = 10$, direct divisor enumeration gives

$$d \mid 290^2, \quad d \equiv -290 \pmod{11} \implies d \in \{29, 2900\}.$$

Both divisors lie in the origin-1 fibre and form one complement orbit. Their common weight is

$$\frac{4 \cdot 290 \cdot 29}{(290 + 29)^2} = \frac{40}{121}.$$

Hence $\alpha_1 = 80/121$, proving (2626); direct addition proves (2627). Each example has $b_1 \vee b_2 = 1$ and $b_1 \wedge b_2 = 0$, proving the final assertion. $\square$

Corollary 14.290 (Split-zero XOR-AND decomposition of path rank and complexity volume). *Retain the split-zero target coordinates of theorem 14.289, but write their Boolean values as*

$$\beta_j := \chi_{\mathbb{R}}(\gamma_j) \in \{0, 1\} \quad (j = 1, 2) \quad (2628)$$

to distinguish them from the endpoint bits. Put

$$d_{\partial,p,a} := b_1 - b_0 \quad (2629)$$

for the signed endpoint direction of theorem 14.268. Then

$$(b_0, b_1) = (\beta_1, \beta_2), \quad d_{\partial, p, a} = \beta_2 - \beta_1. \quad (2630)$$

For operators A, B , write $\{A, B\} := AB + BA$. On an occupied shell define the timelike both-channel rank by

$$\mathcal{E}_{p, a}^{\text{tim}, 123} := \frac{1}{4} \|\{\mathcal{J}_{\partial, 123}^{p, a}, V_{\partial, 0+}\}\|_{\text{HS}}^2 = 3q_{R, p} \beta_1 \beta_2. \quad (2631)$$

Set this quantity equal to zero on an empty shell. The four Boolean strata give the exact coefficient table

(β_1, β_2)	$d_{\partial, p, a}$	$(\beta_1 - \beta_2)^2$	$\beta_1 \beta_2$
(0, 0)	0	0	0
(1, 0)	-1	1	0
(0, 1)	1	1	0
(1, 1)	0	0	1.

(2632)

On every occupied shell the commutator and anticommutator exhaust the selected path rank:

$$\frac{1}{4} \left(\|\mathcal{J}_{\partial, 123}^{p, a}, V_{\partial, 0+}\|_{\text{HS}}^2 + \|\{\mathcal{J}_{\partial, 123}^{p, a}, V_{\partial, 0+}\}\|_{\text{HS}}^2 \right) = \|V_{\partial, 0+}\|_{\text{HS}}^2 = 3q_{R, p} = \rho_{p, a}^{\text{path}}. \quad (2633)$$

Consequently,

$$\rho_{p, a}^{\text{path}} = \mathcal{E}_{p, a}^{\text{pol}, 123} + \mathcal{E}_{p, a}^{\text{tim}, 123}. \quad (2634)$$

The first summand is supported exactly on the two Boolean exclusive-or strata; the second is supported exactly on the Boolean conjunction stratum. By theorem 14.323, these two coefficients are the visible digit and hidden carry of the marked C_4 sum

$$\iota_+(\beta_1) + \iota_+(\beta_2) = (\beta_1 + \beta_2 \bmod 2) + 2\beta_1 \beta_2.$$

The energy pair therefore reads the two C_4 digits of the endpoint support class. Its equality on the two one-sided strata loses their order; the signed commutator coefficient $\beta_2 - \beta_1$ restores it. The local integer lift of that marked class is the Lorentz rank:

$$\hat{c}_{p, a} = \text{rank } \mathbf{G}_{p, a}.$$

Consequently the carry-weighted operator energy gives a second exact complexity coordinate,

$$3q_{R, p} \text{rank } \mathbf{G}_{p, a} = \mathcal{E}_{p, a}^{\text{pol}, 123} + 2\mathcal{E}_{p, a}^{\text{tim}, 123},$$

whose local zero set is exactly the unsupported stratum; see theorem 14.325.

Define

$$\mathcal{E}_p^{\text{tim}, 123} := \sum_{a \in \mathcal{A}_p} \mathcal{E}_{p, a}^{\text{tim}, 123}. \quad (2635)$$

Then the primewise complexity-volume coordinate has the exact decomposition

$$\begin{aligned} \mathcal{E}_p^{\text{pol}, 123} + \mathcal{E}_p^{\text{tim}, 123} &= 3Q_p \\ &= \frac{3}{4} \dim_{\mathbb{C}} \mathcal{V}_p^{\text{Time}} = \frac{1}{2} \log_2 \text{Vol}_p^{\text{det}} \\ &= \frac{1}{17} L_{H, p}^2 = \frac{2}{25} L_{K, p}^2. \end{aligned} \quad (2636)$$

The carry-weighted marked coordinate refines this equality by recovering the $q_{R, p}$ -weighted one-sided and timelike multiplicities separately. It also places the determinant volume between one and two copies of the marked rank sum; see theorem 14.327. In particular,

$$\text{ES}(p) \iff \mathcal{E}_p^{\text{pol}, 123} + \mathcal{E}_p^{\text{tim}, 123} > 0 \quad (2637)$$

under the primewise shell reduction. Equivalently,

$$\neg \text{ES}(p) \iff \forall a \in \mathcal{A}_p : \mathcal{E}_{p,a}^{\text{pol},123} = \mathcal{E}_{p,a}^{\text{tim},123} = 0. \quad (2638)$$

Thus a proof may force either a one-sided polar shell or a both-channel timelike shell; a counterexample must exclude both kinds simultaneously.

Proof. The target bits (β_1, β_2) record the middle and exterior classes, respectively. Reading all four rows of (2443) gives

$$(\beta_1, \beta_2) = (b_0, b_1) \in \{(0, 0), (1, 0), (0, 1), (1, 1)\}.$$

This proves (2630) and the coefficient table.

On an occupied shell, the grading eigenvalues on the source and range of $V_{\partial,0+}$ are

$$s_0 = 2b_0 - 1, \quad s_1 = 2b_1 - 1.$$

Therefore

$$[\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}] = 2(b_1 - b_0)V_{\partial,0+}, \quad (2639)$$

$$\{\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}\} = 2(b_0 + b_1 - 1)V_{\partial,0+}. \quad (2640)$$

Since $\|V_{\partial,0+}\|_{\text{HS}}^2 = 3q_{R,p}$, the second identity gives

$$\frac{1}{4} \|\{\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}\}\|_{\text{HS}}^2 = 3q_{R,p}(b_0 + b_1 - 1)^2.$$

For the three occupied bit pairs,

$$(b_0 + b_1 - 1)^2 = \beta_1 \beta_2.$$

This proves (2631). The binary identity

$$(b_1 - b_0)^2 + (b_0 + b_1 - 1)^2 = 1$$

and (2510) prove (2633) and (2634) on every occupied shell. Both terms are zero on an empty shell. Summing over shells gives

$$\mathcal{E}_p^{\text{pol},123} + \mathcal{E}_p^{\text{tim},123} = 3 \sum_{a \in \mathcal{A}_p} q_{R,a,p} = 3Q_p.$$

The remaining identities are (2518); the Erdős–Straus equivalence is (2229). $\square$

Theorem 14.291 (Exact pre-certificate convolution, modular amplification, and polar transport). *Retain the conditions of theorems 14.45, 14.260 and 14.276. Write $\tilde{f}$ for the extension by zero of a function $f : H \rightarrow \mathbb{C}$ to $(\mathbb{Z}/R\mathbb{Z})^\times$, and define the two-target extraction map*

$$\mathfrak{T}_{p,a} : \mathbb{C}^H \longrightarrow \mathbb{C}^2, \quad \mathfrak{T}_{p,a}(f) := \begin{pmatrix} \tilde{f}(T_2(p)) \\ \frac{1}{2}\tilde{f}(T_1(p)) \end{pmatrix}. \quad (2641)$$

For the exact divisor convolution $\mu = \mu_1 * \cdots * \mu_s$, put

$$q_{02} := \mu(T_2(p)), \quad q_{11} := \frac{1}{2}\mu(T_1(p)). \quad (2642)$$

Then q_{02}, q_{11} are nonnegative integers and

$$\boxed{\mathfrak{T}_{p,a}(\mu) = \begin{pmatrix} q_{02} \\ q_{11} \end{pmatrix}, \quad q_{R,p} = q_{02} + q_{11}.} \quad (2643)$$

The two target coefficients also recover the ordered endpoint pair:

$$\boxed{b_0 = \beta_1 = \mathbf{1}_{\{q_{11} > 0\}}, \quad b_1 = \beta_2 = \mathbf{1}_{\{q_{02} > 0\}}.} \quad (2644)$$

The divisor map $(j, u) \mapsto p^j u$ induces the canonical complement-orbit bijection

$$\mathcal{O}_{R,p} \xrightarrow{\sim} \mathcal{O}_{p,a}; \quad (2645)$$

use this bijection to write O for an orbit in either model. Put

$$u_{\text{mod}} := (0, E_{11}) \in \mathcal{K}_\Gamma, \quad \mathcal{K}_{p,a}^{\text{mod}} := \mathbb{C}[\mathcal{O}_{R,p}] \otimes \mathcal{K}_\Gamma. \quad (2646)$$

For $j \in \{0, 1\}$, define

$$\mathcal{U}_{\partial,j}^{p,a} : \mathcal{K}_{p,a}^{\text{mod}} \otimes \mathbb{C}[C_3] \longrightarrow \mathbb{C}[\tilde{\mathcal{X}}_{R,p}] \quad (2647)$$

by

$$\mathcal{U}_{\partial,j}^{p,a}((e_O \otimes u_{\text{mod}}) \otimes g_m) = e_{(m, x_O^{b_j})}, \quad (2648)$$

and let it vanish on $(\ker P_{R,p}^{\text{mod}}) \otimes \mathbb{C}[C_3]$. These maps are partial isometries with

$$(\mathcal{U}_{\partial,j}^{p,a})^* \mathcal{U}_{\partial,j}^{p,a} = P_{R,p}^{\text{mod}} \otimes I_{\mathbb{C}[C_3]}, \quad (2649)$$

$$\mathcal{U}_{\partial,j}^{p,a} (\mathcal{U}_{\partial,j}^{p,a})^* = Q_{b_j}, \quad (2650)$$

and the selected polar transport is their exact transition map:

$$\boxed{V_{\partial,0+} = \mathcal{U}_{\partial,1}^{p,a} (\mathcal{U}_{\partial,0}^{p,a})^*}. \quad (2651)$$

Consequently,

$$\boxed{\text{rank } V_{\partial,0+} = 3 \text{rank } P_{R,p}^{\text{mod}} = 3(q_{02} + q_{11})}. \quad (2652)$$

The two operator energies are the following functions of the same extracted target vector:

$$\mathcal{E}_{p,a}^{\text{pol},123} = 3(q_{02} + q_{11}) \left(\mathbf{1}_{\{q_{02} > 0\}} - \mathbf{1}_{\{q_{11} > 0\}} \right)^2, \quad (2653)$$

$$\mathcal{E}_{p,a}^{\text{tim},123} = 3(q_{02} + q_{11}) \mathbf{1}_{\{q_{02} > 0\}} \mathbf{1}_{\{q_{11} > 0\}}. \quad (2654)$$

The full convolution is nonzero on every shell, since

$$\sum_{h \in H} \mu(h) = \prod_{i=1}^s (2e_i + 1) > 0. \quad (2655)$$

Shell occupancy is exactly the assertion that this nonzero vector survives the target extraction:

$$\boxed{\mu \notin \ker \mathfrak{T}_{p,a} \iff P_{R,p}^{\text{mod}} \neq 0 \iff V_{\partial,0+} \neq 0}. \quad (2656)$$

For a prime $p \equiv 1 \pmod{4}$, form the direct sum

$$\mathcal{V}_p^\partial := \bigoplus_{a \in \mathcal{A}_p} V_{\partial,0+}^{p,a}. \quad (2657)$$

Then

$$\boxed{\text{rank } \mathcal{V}_p^\partial = 3 \text{rank } \mathcal{P}_p^{\text{mod}} = 3Q_p}, \quad (2658)$$

and the exact counterexample condition is

$$\boxed{\neg \text{ES}(p) \iff \forall a \in \mathcal{A}_p : \mu_{p,a} \in \ker \mathfrak{T}_{p,a}}. \quad (2659)$$

Writing $\mathbb{P}_{1,12} = \{p \in \mathbb{P} : p \equiv 1 \pmod{12}\}$, the global statement is equivalently

$$\boxed{\text{ES} \iff \forall p \in \mathbb{P}_{1,12} \exists a \in \mathcal{A}_p : \mu_{p,a} \notin \ker \mathfrak{T}_{p,a} \iff \inf_{p \in \mathbb{P}_{1,12}} \text{rank } \mathcal{V}_p^\partial \geq 3}. \quad (2660)$$

Proof. The coefficient identities in (2326) give $m_2 = q_{02}$ and $m_1 = 2q_{11}$. The middle fibre is a disjoint union of two-element complement orbits, so q_{11} is an integer. Exterior orbits pair origins 0 and 2, while middle orbits pair two distinct origin-1 points. Hence

$$q_{R,p} = m_2 + \frac{1}{2}m_1 = q_{02} + q_{11},$$

proving (2643). Middle support is present exactly when $q_{11} > 0$, and exterior support is present exactly when $q_{02} > 0$. Apply (2630) to obtain (2644).

The divisor bijection is the map used in the proof of (2326); it intertwines the two complement involutions. This proves (2645). The vectors $(e_O \otimes u_{\text{mod}}) \otimes g_m$ form an orthonormal basis of $\text{Ran}(P_{R,p}^{\text{mod}}) \otimes \mathbb{C}[C_3]$. Their images in (2648) form the orthonormal basis of $\mathcal{H}_{\partial,j}^{\text{path}}$. This proves the two projection identities. On a selected source vector,

$$\mathcal{U}_{\partial,1}^{p,a}(\mathcal{U}_{\partial,0}^{p,a})^* e_{(m,x_{O_0}^{b_0})} = e_{(m,x_{O_0}^{b_1})};$$

on its orthogonal complement the composite vanishes. Comparison with (2504) proves (2651). The modular projector has rank $q_{02} + q_{11}$, and the amplified initial space has three copies of every modular unit line, proving (2652).

Substitution of (2644) in the commutator and anticommutator formulas of theorem 14.290 proves (2653) and (2654).

Each local multiplicity function has total mass $2e_i + 1$, and total mass is multiplicative under finite convolution. This proves (2655). Since both coordinates of $\mathfrak{T}_{p,a}(\mu)$ are nonnegative, its image is zero exactly when $q_{02} = q_{11} = 0$. The shellwise equivalence in (1485) and the polar rank formula prove (2656). Finally, summation over the finite shell set gives (2658); combine it with (1504) to obtain (2659). The explicit decompositions outside $\mathbb{P}_{1,12}$, together with theorem 14.250, reduce the global statement to these primes. Since $\text{rank } \mathcal{V}_p^\partial = 3Q_p$ and $Q_p \in \mathbb{Z}_{\geq 0}$, this proves (2660). $\square$

Theorem 14.292 (Exact Fourier coordinates of the two-target obstruction). *Retain the conditions of theorem 14.291. For $t \in (\mathbb{Z}/R\mathbb{Z})^\times$ and $\chi \in \hat{H}$, define*

$$\text{ev}_{H,t}(\chi) := \begin{cases} \chi(t), & t \in H, \\ 0, & t \notin H, \end{cases} \quad (2661)$$

and put

$$D_i(\chi) := \sum_{\beta=-e_i}^{e_i} \overline{\chi(\bar{\ell}_i)}^\beta, \quad (2662)$$

$$\mathbf{v}_{p,a}(\chi) := \begin{pmatrix} \text{ev}_{H,T_2(p)}(\chi) \\ \frac{1}{2} \text{ev}_{H,T_1(p)}(\chi) \end{pmatrix}. \quad (2663)$$

Then

$$\boxed{\mathfrak{T}_{p,a}(\mu) = \frac{1}{|H|} \sum_{\chi \in \hat{H}} \left(\prod_{i=1}^s D_i(\chi) \right) \mathbf{v}_{p,a}(\chi).} \quad (2664)$$

Every factor $D_i(\chi)$ is real, and

$$\hat{\mu}(\chi) = \prod_{i=1}^s D_i(\chi) = \hat{\mu}(\bar{\chi}) \in \mathbb{R}. \quad (2665)$$

Let

$$\hat{H}[2] := \{\chi \in \hat{H} : \chi^2 = 1\},$$

and let $\mathcal{R}_{\hat{H}}$ contain one member of every pair $\{\chi, \bar{\chi}\}$ in $\hat{H} \setminus \hat{H}[2]$. The same target vector therefore has the real form

$$\boxed{\mathfrak{T}_{p,a}(\mu) = \frac{1}{|H|} \left(\sum_{\chi \in \hat{H}[2]} \hat{\mu}(\chi) \mathbf{v}_{p,a}(\chi) + 2 \sum_{\chi \in \mathcal{R}_{\hat{H}}} \hat{\mu}(\chi) \Re \mathbf{v}_{p,a}(\chi) \right).} \quad (2666)$$

Write

$$M_{p,a} := \hat{\mu}(1) = \prod_{i=1}^s (2e_i + 1) > 0. \quad (2667)$$

For $j \in \{1, 2\}$, the vanishing of the extended target coefficient is equivalent to the following exact alternative:

$$\boxed{\tilde{\mu}(T_j(p)) = 0 \iff [T_j(p) \notin H] \vee \left[\sum_{\substack{\chi \in \widehat{H} \\ \chi \neq 1}} \hat{\mu}(\chi) \chi(T_j(p)) = -M_{p,a} \right]}. \quad (2668)$$

In the second alternative $T_j(p) \in H$. Consequently, $\mu \in \ker \mathfrak{T}_{p,a}$ precisely when (2668) holds for both $j = 1$ and $j = 2$. Thus the arithmetic obstruction consists of subgroup exclusion or exact cancellation of the positive principal mass at each of the two targets.

For the 12289 shell, only the characters 0 and 5 occur, $\hat{\mu}(0) = 45$, $\hat{\mu}(5) = -15$, $\chi_5(T_2) = 1$, and $\chi_5(T_1) = -1$. Hence

$$\mathfrak{T}_{12289,3075}(\mu) = \frac{1}{10} \left[45 \begin{pmatrix} 1 \\ 1/2 \end{pmatrix} - 15 \begin{pmatrix} 1 \\ -1/2 \end{pmatrix} \right] = \begin{pmatrix} 3 \\ 3 \end{pmatrix}. \quad (2669)$$

Proof. The finite Fourier inversion formula (2350) evaluates $\mu(t)$ for $t \in H$. Both that value and $\text{ev}_{H,t}$ are zero after extension when $t \notin H$. Applying the inversion formula to $T_2(p)$ and $T_1(p)$, with the factor $1/2$ in the second coordinate, gives (2664). The local factorization is (2349).

Because each character value lies on the unit circle,

$$D_i(\chi) = 1 + 2 \sum_{\beta=1}^{e_i} \Re(\chi(\bar{\ell}_i)^\beta).$$

This proves that $D_i(\chi)$ is real and unchanged by $\chi \mapsto \bar{\chi}$, giving (2665). Pairing every non-real character with its conjugate proves (2666).

At the principal character, every local factor is $2e_i + 1$, which proves (2667). If $T_j(p)$ does not belong to H , its extended coefficient is zero. If it belongs to H , isolate the principal-character term in Fourier inversion:

$$|H| \mu(T_j(p)) = M_{p,a} + \sum_{\substack{\chi \in \widehat{H} \\ \chi \neq 1}} \hat{\mu}(\chi) \chi(T_j(p)).$$

The coefficient is nonnegative, and it vanishes exactly when the displayed nonprincipal sum equals $-M_{p,a}$. This proves (2668) and the simultaneous kernel statement. Substitution of the two nonzero transform coefficients from (2336) proves (2669). $\square$

Exact computation 14.293 (Two-target Fourier factorization checker). The script and captured receipt

```
verification/verify_two_target_fourier_factorization.py,
verification/verify_two_target_fourier_factorization_output.txt
```

verify the local character products, real character coefficients, Fourier inversion, principal-mass cancellation criterion, and the exact (3, 3) target extraction for the 12289 shell. They also verify, on two empty cyclic shells, the stabilizer-defect decomposition, the quotient/annihilator Fourier identity, exact cancellation of the principal mass at every missed target coset, and failure of the strict Parseval survival inequality.

Theorem 14.294 (Cyclotomic survivors and exact binary-sheet positivity). *Retain the conditions and notation of theorem 14.292, and put*

$$m_i := 2e_i + 1, \quad z_i(\chi) := \overline{\chi(\bar{\ell}_i)}.$$

Then

$$D_i(\chi) = \begin{cases} m_i, & z_i(\chi) = 1, \\ z_i(\chi)^{-e_i} \frac{z_i(\chi)^{m_i} - 1}{z_i(\chi) - 1}, & z_i(\chi) \neq 1. \end{cases} \quad (2670)$$

Consequently,

$$D_i(\chi) = 0 \iff z_i(\chi) \neq 1 \text{ and } z_i(\chi)^{m_i} = 1. \quad (2671)$$

The exact survivor spectrum is therefore

$$\mathcal{S}_\mu := \{\chi \in \hat{H} : \hat{\mu}(\chi) \neq 0\} = \{\chi : \text{no local factor satisfies (2671)}\}. \quad (2672)$$

Every character of 2-power order belongs to $\mathcal{S}_\mu$.

If $\Gamma \leq \hat{H}$ contains $\mathcal{S}_\mu$, then

$$\mu(h + k) = \mu(h) \quad (h \in H, k \in \Gamma^\perp). \quad (2673)$$

In particular, $\mathcal{S}_\mu = \{1\}$ implies

$$\mu(h) = \frac{M_{p,a}}{|H|} > 0 \quad (h \in H);$$

this case can occur only when $|H|$ is odd.

Suppose now that the 2-primary character subgroup is $\{1, \eta\}$, where η is nontrivial, and every character outside this subgroup is killed by at least one local factor. Then

$$\mathcal{S}_\mu = \{1, \eta\}, \quad \mu(h) = \frac{M_{p,a} + A_{p,a}\eta(h)}{|H|}, \quad (2674)$$

where

$$A_{p,a} := \hat{\mu}(\eta) = \prod_{\eta(\bar{\ell}_i)=1} m_i \prod_{\eta(\bar{\ell}_i)=-1} (-1)^{e_i}. \quad (2675)$$

Because the $\bar{\ell}_i$ generate H , at least one factor occurs in the second product. Hence

$$\frac{|A_{p,a}|}{M_{p,a}} = \prod_{\eta(\bar{\ell}_i)=-1} \frac{1}{m_i} \leq \frac{1}{3}, \quad \mu(h) \geq \frac{2M_{p,a}}{3|H|} > 0. \quad (2676)$$

Thus the surviving binary character changes multiplicities on the two sheets and cannot create unsupportedness.

For the 12289 shell, with $H = C_{10}$ and $\eta(h) = (-1)^h$,

$$(\hat{\mu}(0), \dots, \hat{\mu}(9)) = (45, 0, 0, 0, 0, -15, 0, 0, 0, 0), \quad (2677)$$

and

$$\mu(h) = \frac{45 - 15(-1)^h}{10} = \begin{cases} 3, & h \text{ even}, \\ 6, & h \text{ odd}. \end{cases} \quad (2678)$$

Proof. Equation (2670) is the finite geometric-series identity applied to $\sum_{\beta=-e_i}^{e_i} z_i(\chi)^\beta$. Since every $z_i(\chi)$ is a root of unity, (2671) follows. The product formula for $\hat{\mu}$ gives (2672).

If χ has 2-power order, every $z_i(\chi)$ has 2-power order. The odd integer m_i cannot be divisible by a nontrivial 2-power order, so no local factor vanishes. Thus every such character survives.

For $k \in \Gamma^\perp$, Fourier inversion gives

$$\mu(h+k) = \frac{1}{|H|} \sum_{\chi \in \mathcal{S}_\mu} \widehat{\mu}(\chi) \chi(h) \chi(k) = \mu(h),$$

which proves (2673). If only the principal character survives, the same formula gives the stated constant positive value. An even-order finite abelian group has a nontrivial 2-primary character, so this case forces $|H|$ odd.

For the binary case, the preceding survivor statement and the assumed annihilation of every other character give $\mathcal{S}_\mu = \{1, \eta\}$. Since η has order two, $\eta(\bar{\ell}_i) \in \{1, -1\}$, and direct summation gives

$$D_i(\eta) = \begin{cases} m_i, & \eta(\bar{\ell}_i) = 1, \\ (-1)^{e_i}, & \eta(\bar{\ell}_i) = -1. \end{cases}$$

This proves (2675). The character η is nontrivial on at least one generator. Since every $e_i \geq 1$, every corresponding $m_i \geq 3$, proving (2676). Fourier inversion then gives (2674) and positivity on both sheets.

For 12289, the length-five factor with step 4 kills precisely the characters outside $\{1, \eta\}$. The local binary factors are

$$3, \quad 5, \quad -1,$$

so $A_{12289,3075} = -15$, while $M_{12289,3075} = 45$. Equations (2677) and (2678) follow. $\square$

Exact computation 14.295 (Cyclotomic survivor and binary-sheet checker). The script and captured receipt

```
verification/verify_cyclotomic_two_sheet_positivity.py,
verification/verify_cyclotomic_two_sheet_positivity_output.txt
```

check the root-order zero criterion on 2-groups through order 16, five exact C_{2m} coupling families, the strict binary-amplitude bound, and the complete 12289 spectrum and positive two-sheet reconstruction. The quantitative implication from the one-third amplitude bound to strict positivity on both sheets is also certified by theorem 14.360.

Corollary 14.296 (First possible purely 2-primary obstruction). *Retain the conditions and notation of theorem 14.294, and let*

$$\Gamma := \{\chi \in \widehat{H} : \text{ord}(\chi) \text{ is a power of } 2\}, \quad K := \Gamma^\perp.$$

Suppose every character in $\widehat{H} \setminus \Gamma$ is killed by at least one local factor D_i . Then μ is constant on the cosets of K , so there is a unique function

$$\bar{\mu} : H/K \longrightarrow \mathbb{Z}_{\geq 0}, \quad \bar{\mu}(h+K) = \mu(h).$$

The quotient pullback identifies

$$q^\vee(\widehat{H/K}) = \Gamma, \quad |H/K| = |\Gamma|.$$

If $\mu(h_0) = 0$ for some $h_0 \in H$, then

$$\boxed{|H/K| = |\Gamma| \geq 4.} \tag{2679}$$

Equivalently, complete annihilation of the odd-order spectral modes proves full support whenever the 2-primary quotient is trivial or binary.

Proof. Every character in Γ survives by theorem 14.294, while the stated condition kills every character outside Γ . Hence $\mathcal{S}_\mu = \Gamma$. Applying (2673) with this subgroup shows that μ is K -periodic and therefore defines $\bar{\mu}$.

Finite character duality identifies the characters of H/K with the characters of H that are trivial on K . Since $K = \Gamma^\perp$, the double-annihilator identity gives $q^\vee(\widehat{H/K}) = \Gamma$, and therefore $|H/K| = |\Gamma|$.

If $|\Gamma| = 1$, only the principal character survives and theorem 14.294 gives $\mu(h) = M_{p,a}/|H| > 0$ for every h . If $|\Gamma| = 2$, the binary part of the same theorem gives $\mu(h) \geq 2M_{p,a}/(3|H|) > 0$ for every h . A zero coefficient therefore forces $|\Gamma| > 2$. The order of the 2-primary subgroup Γ is a power of two, so $|\Gamma| \geq 4$. $\square$

Theorem 14.297 (Elementary-abelian 2-primary quotients have full support). *Retain the conditions of theorem 14.296, and suppose*

$$P := H/K \cong C_2^d.$$

Let q_i be the image of $\bar{\ell}_i$ in P , choose a set I of d indices for which $\{q_i : i \in I\}$ is a basis, and put

$$M_I := \prod_{i \in I} (2e_i + 1), \quad M_{\text{rest}} := \frac{M_{p,a}}{M_I}.$$

For $\nu = \pi_\mu$, every $x \in P$ satisfies*

$$\nu(x) \geq M_{\text{rest}} \prod_{i \in I} e_i \geq \frac{M_{p,a}}{3^d} > 0. \quad (2680)$$

Equivalently,

$$\mu(h) \geq \frac{2^d M_{p,a}}{3^d |H|} > 0 \quad (h \in H). \quad (2681)$$

Thus no elementary-abelian 2-primary quotient can carry a missing coefficient.

Proof. Push the i -th local interval kernel to P . Because q_i has order two, even values of β contribute to 0 and odd values contribute to q_i . Among the $2e_i + 1$ integers $-e_i, \dots, e_i$, the two parity counts are e_i and $e_i + 1$ in some order. Hence the local measure has the form

$$\lambda_i = u_i \delta_0 + v_i \delta_{q_i}, \quad \{u_i, v_i\} = \{e_i, e_i + 1\}.$$

The convolution $\lambda_I := *_{i \in I} \lambda_i$ has one term for each basis expansion of $x \in P$, so

$$\lambda_I(x) \geq \prod_{i \in I} e_i.$$

Let λ_{rest} be the convolution of the remaining local measures. It has nonnegative coefficients and total mass M_{rest} . Therefore

$$\nu(x) = \sum_{y \in P} \lambda_I(x - y) \lambda_{\text{rest}}(y) \geq \left(\prod_{i \in I} e_i \right) M_{\text{rest}}.$$

Since $e_i \geq 1$,

$$\frac{e_i}{2e_i + 1} \geq \frac{1}{3},$$

which proves the second inequality in (2680). Finally, $\nu(\pi(h)) = |K|\mu(h)$ and $|H| = 2^d |K|$, proving (2681). $\square$

Theorem 14.298 (Complete order-four 2-primary classification). *Retain the conditions of theorem 14.296, assume $|\Gamma| = 4$, and write*

$$P := H/K, \quad \pi : H \longrightarrow P, \quad \nu := \pi_*\mu.$$

Thus $\nu(\pi(h)) = |K|\mu(h)$, its total mass is $M_{p,a}$, and its Fourier transform is the restriction of $\hat{\mu}$ to Γ .

(1) If $P \cong C_2 \times C_2$, then

$$\boxed{\nu(x) \geq \frac{M_{p,a}}{9} > 0, \quad \mu(h) \geq \frac{4M_{p,a}}{9|H|} > 0.} \quad (2682)$$

(2) Suppose $P \cong C_4$. Choose $\chi \in \hat{P}$ of order four, put

$$q_i := \pi(\bar{\ell}_i) \in C_4, \quad L_{\text{odd}} := \prod_{q_i \in \{1,3\}} m_i, \quad L_2 := \prod_{q_i=2} m_i,$$

and use the value 1 for an empty product. Then

$$\frac{|\hat{\nu}(\chi)|}{M_{p,a}} = \frac{|\hat{\nu}(\chi^3)|}{M_{p,a}} = \frac{1}{L_{\text{odd}}L_2}, \quad \frac{|\hat{\nu}(\chi^2)|}{M_{p,a}} = \frac{1}{L_{\text{odd}}}. \quad (2683)$$

In particular,

$$\frac{|\hat{\nu}(\chi)| + |\hat{\nu}(\chi^2)| + |\hat{\nu}(\chi^3)|}{M_{p,a}} = \frac{1 + 2/L_2}{L_{\text{odd}}} \leq 1. \quad (2684)$$

Every coefficient is positive unless

$$L_{\text{odd}} = 3, \quad L_2 = 1. \quad (2685)$$

These equalities mean that there is exactly one index i_0 with q_{i_0} odd and $e_{i_0} = 1$, while every other q_i is zero. Writing $q := q_{i_0}$, one then has

$$\boxed{\nu = \frac{M_{p,a}}{3}(\delta_{-q} + \delta_0 + \delta_q), \quad \nu(2q) = 0.} \quad (2686)$$

Thus the unique order-four zero is the antipode of the unique length-three quotient direction.

Proof. For a character $\psi \in \Gamma$, each local amplitude satisfies

$$\frac{|D_i(\psi)|}{m_i} = \begin{cases} 1, & \psi(q_i) = 1, \\ 1/m_i, & \psi(q_i) \neq 1, \end{cases}$$

because the nontrivial values have order two or four and m_i is odd.

Suppose first that $P \cong C_2 \times C_2$. Since the q_i generate P , two of them form a basis. Apply theorem 14.297 with $d = 2$. This gives both inequalities in (2682).

Now let $P \cong C_4$. A generator χ and its inverse are nontrivial on the indices with q_i odd or $q_i = 2$, whereas χ^2 is nontrivial precisely when q_i is odd. Multiplying the local amplitude ratios proves (2683). The q_i generate C_4 , so at least one is odd and $L_{\text{odd}} \geq 3$; also $L_2 \geq 1$. This proves (2684). The bound is strict unless $L_{\text{odd}} = 3$ and $L_2 = 1$, and strictness gives $\nu(x) > 0$ by Fourier inversion.

In the equality case, the product defining L_{odd} contains one factor, that factor is $m_{i_0} = 3$, and no index has image 2. Every remaining image is zero. Choose $\chi(q) = i$. The single length-three factor gives

$$\hat{\nu}(\chi) = \frac{M_{p,a}}{3}, \quad \hat{\nu}(\chi^2) = -\frac{M_{p,a}}{3}, \quad \hat{\nu}(\chi^3) = \frac{M_{p,a}}{3}.$$

Fourier inversion on C_4 is exactly (2686). □

Corollary 14.299 (Purely 2-primary obstruction dichotomy). *Under complete annihilation of the characters outside Γ , a missing coefficient has exactly two possible locations:*

- (1) $|\Gamma| \geq 8$ and $H/\Gamma^\perp \not\cong C_2^d$ for every d ; or
- (2) $H/\Gamma^\perp \cong C_4$ and the quotient pushforward is the three-point measure in (2686).

In the second case the sole missing quotient class is the antipode $2q$. The order-at-least-eight branch is resolved through order eight in theorem 14.303.

Proof. Theorem 14.296 gives $|\Gamma| \geq 4$. If $|\Gamma| = 4$, the quotient is isomorphic to C_4 or $C_2 \times C_2$. The first part of theorem 14.298 excludes the second group, and its second part gives the asserted C_4 measure. If $|\Gamma| > 4$, its power-of-two order is at least eight. Theorem 14.297 excludes every elementary-abelian quotient. $\square$

Exact computation 14.300 (Exact order-four 2-primary classification). The script and captured receipt

```
verification/verify_order_four_two_primary.py,
verification/verify_order_four_two_primary_output.txt
```

exhaust finite products of up to four local kernels with $1 \leq e_i \leq 3$. On C_4 it checks the amplitude formulas and proves that a zero occurs exactly for the three-point antipodal configuration in (2686). On $C_2 \times C_2$ it checks the stronger coefficient bound (2682). The local one-third inequality, its finite-product form, and preservation of a uniform lower bound under convolution with a nonnegative residual measure are also certified by theorem 14.360.

Theorem 14.301 (Exceptional C_4 shell to the Fable tetrahedron). *Retain the exceptional cyclic case (2686). On the pointed set $C_4 = \{0, q, 2q, -q\}$, define*

$$\rho_q : 0 \mapsto q \mapsto -q \mapsto 0, \quad 2q \mapsto 2q, \quad (2687)$$

$$\sigma_q(x) := -x. \quad (2688)$$

Then

$$\rho_q^3 = \sigma_q^2 = 1, \quad \sigma_q \rho_q \sigma_q = \rho_q^{-1}.$$

There is a unique pointed-set bijection

$$\Theta_q : C_4 \longrightarrow \{v_*, v_0, v_+, v_-\}, \quad \begin{array}{ll} 2q \mapsto v_*, & 0 \mapsto v_0, \\ q \mapsto v_+, & -q \mapsto v_-, \end{array} \quad (2689)$$

with the intertwining

$$\Theta_q \rho_q = \Re_{\square} \Theta_q, \quad \Theta_q \sigma_q = \Im_{\square} \Theta_q. \quad (2690)$$

The arithmetic measure and the quarter reflection satisfy

$$(\Theta_q)_* \nu = \frac{M_{p,a}}{3} (\delta_{v_0} + \delta_{v_+} + \delta_{v_-}), \quad (2691)$$

$$\Re_{\square} \Theta_q(x) = \begin{cases} -\Theta_q(x), & x = 2q, \\ \Theta_q(x), & x \in \{0, q, -q\}. \end{cases} \quad (2692)$$

Thus the sole unsupported quotient class is carried to the unique negative quarter vertex, and the three supported classes are carried to the positive opposite face.

Finally, let

$$b_\nu := \frac{1}{M_{p,a}} \sum_{x \in C_4} \nu(x) \Theta_q(x) = \frac{1}{3} (v_0 + v_+ + v_-).$$

For the raw-cell-Lorentz map Φ_Δ and spatial projection π_{sp} from theorem 9.27,

$$\boxed{\pi_{\text{sp}} \Phi_\Delta(b_\nu) - \pi_{\text{sp}} \Phi_\Delta(\Theta_q(2q)) = \left(0, \frac{1}{4}, 0\right)}. \quad (2693)$$

Proof. The two permutations in (2687)–(2688) fix $2q$ and act on $\{0, q, -q\}$ as a three-cycle and a reflection fixing 0. Their displayed relations follow immediately. Equations (217) and (218) give the same action on $\{v_*, v_0, v_+, v_-\}$, proving (2690).

Equation (2686) gives (2691). The quarter signs in (219) give (2692). The pushed-forward probability measure has barycentre $(v_0 + v_+ + v_-)/3$. Linearity of Φ_Δ , followed by (224)–(225), proves (2693). $\square$

Exact computation 14.302 (Exact exceptional- C_4 /Fable intertwiner). The script and captured receipt

```
verification/verify_exceptional_c4_fable_bridge.py,
verification/verify_exceptional_c4_fable_bridge_output.txt
```

check the two pointed-set permutations, their D_3 relations, every vertex intertwining in (2690), the measure pushforward, all four quarter signs, and the exact spatial altitude (2693). The last rational coordinate calculation is also certified by theorem 14.360.

Theorem 14.303 (Complete order-eight 2-primary support classification). *Retain the conditions of theorem 14.296, put $P = H/K$, and suppose $|P| = 8$. If the quotient pushforward $\nu = \pi_*\mu$ has a zero coefficient, then exactly one of the following cases occurs.*

(1) *If $P \cong C_8$, there is a generator q for which $\text{supp } \nu$ is one of the four sets*

$$\begin{aligned} A_3(q) &:= \{0, q, -q\}, \\ A_5(q) &:= \{0, q, -q, 2q, -2q\}, \\ A_6(q) &:= P \setminus \{2q, -2q\}, \\ A_7(q) &:= P \setminus \{4q\}. \end{aligned} \tag{2694}$$

Every displayed type occurs as a finite sum of local interval supports.

(2) *If $P \cong C_4 \times C_2$, there are an order-four element q and an order-two element $r \notin \langle q \rangle$ for which $\text{supp } \nu$ is one of*

$$\begin{aligned} B_6(q, r) &:= \{0, q, -q, r, r+q, r-q\} = P \setminus \{2q, 2q+r\}, \\ B_7(q, r) &:= P \setminus \{2q\}. \end{aligned} \tag{2695}$$

Both displayed types occur as finite sums of local interval supports.

(3) *If $P \cong C_2^3$, every coefficient of ν is positive.*

Proof. For $u \in P$ and $e \geq 1$, write

$$I_e(u) := \{\beta u : -e \leq \beta \leq e\}.$$

The support of ν is the sumset of the local interval supports $I_{e_i}(q_i)$, and the q_i generate P .

First suppose $P = C_8 = \langle q \rangle$. Its primitive elements form the two antipodal direction pairs $\{q, -q\}$ and $\{3q, -3q\}$. If both pairs occur among the local directions, then already

$$I_1(q) + I_1(3q) = C_8.$$

Thus a nonfull support uses only one primitive direction pair. After replacing q by another generator, the sum of all primitive local supports is

$$I_E(q),$$

where E is the sum of their radii. The cases $E = 1, 2, 3$ give $A_3(q)$, $A_5(q)$, and $A_7(q)$, respectively, while $E \geq 4$ fills C_8 .

The remaining nonzero directions have orders four and two. A short order-four factor is

$$I_1(2q) = \{0, 2q, -2q\};$$

two such factors fill the even subgroup, and a radius-at-least-two order-four factor already equals that subgroup. Adding either to a primitive interval of radius one fills C_8 , except that one short order-four factor gives

$$I_1(q) + I_1(2q) = A_7(q).$$

Every order-two factor has support $\{0, 4q\}$. Such factors do not enlarge one another, and

$$I_1(q) + \{0, 4q\} = A_6(q).$$

Adding $\{0, 4q\}$ to a primitive interval of radius at least two fills C_8 . This proves the cyclic list, including its completeness.

Now suppose $P = C_4 \times C_2$. Choose an order-four element q . The order-four directions form the two antipodal pairs represented by q and $q + r$, where $r \notin \langle q \rangle$ has order two; both pairs have midpoint $2q$. If only one order-four direction occurs, generation of P requires an external order-two direction. A nonfull sum then has radius one in the order-four direction and only one external binary direction:

$$I_1(q) + \{0, r\} = B_6(q, r).$$

A second external direction, the midpoint direction, or order-four radius at least two fills P .

If both order-four directions occur, their smallest possible sum is

$$I_1(q) + I_1(q + r) = B_7(q, r).$$

Any further nonzero local support fills its sole missing class $2q$. This proves the $C_4 \times C_2$ list.

Finally, *theorem 14.297*, with $d = 3$, gives

$$\nu(x) \geq \frac{M_{p,a}}{27} > 0 \quad (x \in C_2^3).$$

□

Corollary 14.304 (Purely 2-primary obstructions through order eight). *Under complete annihilation of the characters outside the 2-primary subgroup, every missing coefficient has one of the following exact locations:*

- (1) the exceptional C_4 support $\{0, q, -q\}$;
- (2) one of the four cyclic order-eight supports in (2694);
- (3) one of the two $C_4 \times C_2$ supports in (2695); or
- (4) a non-elementary 2-primary quotient of order at least 16.

In particular, no elementary-abelian quotient occurs at any order. The order-sixteen branch is classified in theorem 14.307. Its four non-elementary groups occupy the exact four-sheet carriers of theorem 14.308. Their transported inversion operators decompose every minimum-support two-cycle into the exact mixer-blade block, and the missing sheet-cell deficit plus the block count is four by theorem 14.311. Cell translation and inversion further produce the faithful dihedral carrier action of theorem 14.313; its second Fourier sector detects the nonzero carry sheets. The irreducible character calculation of theorem 14.315 recovers the non-coboundary carry exactly from the strict inequality $m_{--}(P) > m_{+-}(P)$. The two-torsion character of theorem 14.317 identifies that multiplicity difference with the Hamming weight of $u \mapsto \kappa_P(u, u) \pmod{2}$, and gives half of $|S_P[2]|$ whenever the character is nonzero. The marked-extension calculation in theorem 14.319 proves that this character map is onto with kernel $(2C_4)^a$, and exhibits the alternative cyclic-sheet $C_8 \times C_2$ carrier whose non-split class has zero character. The visible/hidden coordinate calculation of theorem 14.321 proves that every long cyclic factor carries the quadratic bit law $(q, h) \star (q', h') = (q + q', h + h' + qq')$, and that the detection sequence splits exactly in the elementary-abelian case. The split-zero operator calculation of theorem 14.323 identifies its XOR and AND energies with the visible digit and hidden carry of that same law; the signed commutator retains the endpoint ordering, one pair realizes the classes 0, 1, 2, and Baer closure supplies class 3. Theorem 14.325 further identifies the local integer lift of those classes with the Lorentz rank and turns their carry-weighted primewise energy into the exact nonzero-marked-class support criterion. The spectral defect, the correlation deficit, and the blade count of the selected minimum carriers then satisfy the two exact balance laws of theorem 14.331. After that classification the next unclassified purely 2-primary order is 32.

Proof. The order-four case is theorem 14.298. Theorem 14.303 gives the complete order-eight list and excludes C_2^3 . Theorem 14.297 excludes every C_2^d . A remaining 2-primary quotient has power-of-two order strictly larger than eight, hence order at least 16. □

Theorem 14.305 (Order-eight supports in the inherited eight-cell carrier). *Let P be either $C_8 = \langle q \rangle$, with $r = 4q$, or $C_4 \times C_2$, with q of order four and $r \notin \langle q \rangle$ of order two. In both cases there is a disjoint pointed-set decomposition*

$$P = \coprod_{\epsilon \in C_2} (\epsilon r + \{0, q, 2q, -q\}). \quad (2696)$$

Put $e_0 = e_+$ and $e_1 = e_-$, using the sheet basis of theorem 14.43. The map

$$\Theta_{q,r} : P \longrightarrow \{e_\epsilon \otimes f, e_\epsilon \otimes g_0, e_\epsilon \otimes g_1, e_\epsilon \otimes g_2 : \epsilon \in C_2\} \quad (2697)$$

defined by

$$\begin{aligned} \Theta_{q,r}(\epsilon r + 2q) &= e_\epsilon \otimes f, & \Theta_{q,r}(\epsilon r - q) &= e_\epsilon \otimes g_0, \\ \Theta_{q,r}(\epsilon r + q) &= e_\epsilon \otimes g_1, & \Theta_{q,r}(\epsilon r) &= e_\epsilon \otimes g_2 \end{aligned} \quad (2698)$$

is a bijection.

On each four-cell sheet, let ρ fix $\epsilon r + 2q$ and cycle

$$\epsilon r \longmapsto \epsilon r + q \longmapsto \epsilon r - q \longmapsto \epsilon r.$$

Let σ fix 0 and $2q$ and exchange q with $-q$, and define

$$\begin{aligned} d(\epsilon r + x) &:= \epsilon r + \sigma(x), \\ j(\epsilon r + x) &:= (\epsilon + 1)r + \sigma(x). \end{aligned} \quad (2699)$$

Then

$$\boxed{\Theta_{q,r}\rho = U_\Delta \Theta_{q,r}, \quad \Theta_{q,r}d = \widehat{R}_8 \Theta_{q,r}, \quad \Theta_{q,r}j = J_8 \Theta_{q,r}.} \quad (2700)$$

Both $\langle \rho, d \rangle$ and $\langle \rho, j \rangle$ are D_3 actions.

Write

$$\mathcal{O}_\epsilon := \{e_\epsilon \otimes g_0, e_\epsilon \otimes g_1, e_\epsilon \otimes g_2\}, \quad a_\epsilon := e_\epsilon \otimes f.$$

The six order-eight support types occupy the inherited cells as follows:

support	image under $\Theta_{q,r}$	inherited action
$A_3(q)$	$\mathcal{O}_0$	diagonal D_3
$A_5(q)$	$\mathcal{O}_0 \cup \{a_0, a_1\}$	diagonal D_3
$A_6(q)$	$\mathcal{O}_0 \cup \mathcal{O}_1$	crossed D_3
$A_7(q)$	all cells except $e_1 \otimes g_2$	not phase-invariant
$B_6(q, r)$	$\mathcal{O}_0 \cup \mathcal{O}_1$	crossed D_3
$B_7(q, r)$	all cells except a_0	diagonal D_3

Here the first four rows apply to C_8 , and the last two to $C_4 \times C_2$. Thus $A_7(q)$ is the unique order-eight support which does not inherit the existing phase cycle through (2698).

For either six-point support $A_6(q)$ or $B_6(q, r)$, restriction and linear extension give a unitary map

$$\mathbb{C}[\text{supp } \nu] \xrightarrow{\Theta_{q,r}} \mathcal{H}_{8,\text{orb}}. \quad (2701)$$

Under

$$\delta_{\epsilon r} \mapsto \gamma_{\epsilon,0}, \quad \delta_{\epsilon r+q} \mapsto \gamma_{\epsilon,1}, \quad \delta_{\epsilon r-q} \mapsto \gamma_{\epsilon,2},$$

this unitary is exactly the map Ξ_Γ from (1466).

Moreover, every local-interval measure with either six-point support has the factorization

$$\nu = c(b_0\delta_0 + b_1\delta_r) * (\delta_{-q} + \delta_0 + \delta_q), \quad c, b_0, b_1 > 0. \quad (2702)$$

Consequently its coefficient vector satisfies

$$\boxed{\Theta_{q,r} \left(\sum_{x \in P} \nu(x) \delta_x \right) = c\sqrt{3} (b_0 e_0 + b_1 e_1) \otimes v_0 \in \ker(K_{\text{ph}}|_{\mathcal{H}_{8,\text{orb}}}).} \quad (2703)$$

Proof. The two decompositions in (2696) are immediate: for C_8 , the two sheets are

$$\{0, q, 2q, -q\}, \quad 4q + \{0, q, 2q, -q\},$$

and for $C_4 \times C_2$ they are the two cosets of $\langle q \rangle$. Hence (2698) lists each source and target cell exactly once.

The lower three cells are ordered so that P_L sends

$$g_2 \mapsto g_1 \mapsto g_0 \mapsto g_2,$$

while P_R fixes g_2 and exchanges g_1 with g_0 . Equations (1430) and (1431) therefore give all three intertwining in (2700). They also give the two D_3 relations.

Applying $\Theta_{q,r}$ to the explicit support sets in (2694) and (2695) gives the table. In the A_7 row the missing class is r , which maps to $e_1 \otimes g_2$; the phase cycle moves g_2 , proving the stated failure of inherited phase invariance.

For the six-point cases, the proof of theorem 14.303 shows that there is exactly one primitive radius-one factor. Every remaining nontrivial factor lies in the same order-two sheet direction. Their convolution is $b_0\delta_0 + b_1\delta_r$, up to the positive scalar contributed by zero quotient directions. This proves (2702). The three face coefficients are equal on each sheet. Since

$$v_0 = \frac{1}{\sqrt{3}}(g_0 + g_1 + g_2) \quad \text{and} \quad A_3 v_0 = 0,$$

the coefficient vector maps exactly as in (2703). Its identification with Ξ_Γ follows from (1466). $\square$

Exact computation 14.306 (Exact order-eight support and eight-cell certificates). The scripts and captured receipts

```
verification/verify_order_eight_two_primary_support.py,
verification/verify_order_eight_two_primary_support_output.txt,
verification/verify_order_eight_eight_cell_bridge.py,
verification/verify_order_eight_eight_cell_bridge_output.txt
```

close the full finite support monoid for each group of order eight. They find 11 reachable cyclic supports with six generating nonfull labeled supports, 13 reachable $C_4 \times C_2$ supports with three generating nonfull labeled supports, and no generating nonfull support among the 16 reachable C_2^3 supports. The second checker verifies the three cell intertwining, every row of the carrier table, and the unique failure of inherited phase invariance.

The Lean 4.32 source and receipt

```
verification/OrderEightCarrierCertificate.lean,
verification/OrderEightCarrierCertificate_output.txt
```

independently certify the order-three phase, both order-two reflections, both D_3 relations, and preservation of the face locus by the phase and the diagonal and crossed reflections.

Theorem 14.307 (Complete order-sixteen 2-primary support classification). *Retain the conditions of theorem 14.296, put $P = H/K$, and suppose $|P| = 16$. For $u \in P$ and $e \geq 1$, write*

$$I_e(u) := \{\beta u : -e \leq \beta \leq e\},$$

and, when h has order two, write

$$B(h) := I_1(h) = \{0, h\}.$$

If the quotient pushforward $\nu = \pi_\mu$ has a zero coefficient, then, up to $\text{Aut}(P)$, its support occurs in exactly one row of the following four tables.*

If $P = C_{16} = \langle q \rangle$, put $B_2 := B(8q)$. The complete list is

$ \text{supp } \nu $	$ \text{Aut}(P) \text{supp } \nu $	$\text{supp } \nu$
3	4	$I_1(q)$
5	4	$I_2(q)$
6	2	$I_1(q) + B_2$
7	4	$I_3(q)$
8	2	$I_1(q) + I_1(7q)$
9	4	$I_4(q)$
9	4	$I_1(q) + I_1(4q)$
9	4	$I_1(q) + I_1(6q)$
10	2	$I_2(q) + B_2$
11	4	$I_5(q)$
12	4	$I_2(q) + I_1(7q)$
12	1	$I_1(q) + I_2(4q)$
13	4	$I_6(q)$
13	2	$I_1(q) + I_2(6q)$
14	4	$I_2(q) + I_1(6q)$
14	1	$I_3(q) + B_2$
15	1	$I_7(q)$.

If $P = C_8 \times C_2$, let q have order eight and let $r \notin \langle q \rangle$ have order two. The complete list is

$ \text{supp } \nu $	$ \text{Aut}(P) \text{supp } \nu $	$\text{supp } \nu$
6	4	$I_1(q) + B(r)$
8	4	$I_1(q) + I_1(q + r)$
9	4	$I_1(q) + I_1(2q + r)$
10	4	$I_2(q) + B(r)$
12	8	$I_1(q) + I_2(q + r)$
12	1	$I_1(q) + B(4q) + B(r)$
12	1	$I_1(q) + I_2(2q + r)$
13	4	$I_2(q) + I_1(2q + r)$
14	1	$I_1(q) + I_1(q + r) + B(4q)$
14	2	$I_3(q) + B(r)$
15	2	$I_2(q) + I_2(q + r)$
15	1	$I_1(q) + I_3(q + r)$.

If $P = C_4 \times C_4$, let q, r be an invariant-factor basis. The complete list is

$ \text{supp } \nu $	$ \text{Aut}(P) \text{supp } \nu $	$\text{supp } \nu$
9	12	$I_1(q) + I_1(r)$
12	6	$I_2(q) + I_1(r)$
14	3	$I_1(q) + I_1(r) + B(2q + 2r)$
15	3	$I_1(q) + I_1(r) + I_1(2q + r)$.

If $P = C_4 \times C_2 \times C_2$, let q generate the order-four factor and let r, s be a basis of the binary factor. The complete list is

$ \text{supp } \nu $	$ \text{Aut}(P) \text{supp } \nu $	$\text{supp } \nu$
12	4	$I_1(q) + B(r) + B(s)$
14	6	$I_1(q) + I_1(q + s) + B(r)$
15	1	$I_1(q) + I_1(q + r) + I_1(q + s)$.

If $P = C_2^4$, every coefficient of ν is positive. Every row in the four displayed tables occurs in the finite monoid generated by the local interval supports.

Proof. The support of ν is a sumset of local intervals $I_e(u)$, and the local directions generate P . For each of the five abelian groups of order sixteen, enumerate the distinct $I_e(u)$ with $u \neq 0$ and $1 \leq e \leq \text{ord}(u)$. Their respective counts are

$$35, \quad 19, \quad 15, \quad 15, \quad 15.$$

Starting with $\{0\}$, repeated addition of these local supports reaches a fixed finite monoid after respectively two, three, three, three, and four factors. The monoid cardinalities are

$$63, \quad 64, \quad 54, \quad 60, \quad 67.$$

Filter these finite lists by two exact tests: the support generates P , and the support is not all of P . This leaves respectively

$$51, \quad 36, \quad 24, \quad 11, \quad 0$$

labeled supports. Automorphisms are enumerated by the possible images of an invariant-factor basis; their group orders are

$$8, \quad 16, \quad 96, \quad 192, \quad 20160.$$

Orbit closure gives

$$17, \quad 12, \quad 4, \quad 3, \quad 0$$

orbits. Direct sumset evaluation of every displayed template gives one representative of every orbit, and the middle column of each table is its orbit cardinality. The orbit cardinalities sum to the labeled-support counts above. This proves exhaustion, mutual inequivalence, and occurrence inside the finite local-support monoid. The C_2^4 assertion also follows from theorem 14.297. $\square$

Theorem 14.308 (Exact four-sheet carrier and carry cocycle at order sixteen). *Retain the generators used in theorem 14.307, and suppose P is non-elementary. Define an order-four subgroup $A_P = \langle a_P \rangle$, a sheet group S_P , and a section $\sigma_P : S_P \rightarrow P$ by*

P	S_P	a_P	σ_P
C_{16}	C_4	$4q$	$t \mapsto tq$
$C_8 \times C_2$	C_2^2	$2q$	$(\epsilon, \delta) \mapsto \epsilon q + \delta r$
$C_4 \times C_4$	C_4	q	$t \mapsto tr$
$C_4 \times C_2^2$	C_2^2	q	$(\epsilon, \delta) \mapsto \epsilon r + \delta s.$

In every row, the map

$$\Theta_P : S_P \times C_4 \longrightarrow P, \quad \Theta_P(u, j) = \sigma_P(u) + ja_P \quad (2704)$$

is a bijection. Addition transported through Θ_P is

$$(u, j) \boxplus_P (v, k) = (u + v, j + k + \kappa_P(u, v)), \quad (2705)$$

where the second coordinate is read in C_4 , and

$$\kappa_{C_{16}}(t, t') = \left\lfloor \frac{\tilde{t} + \tilde{t}'}{4} \right\rfloor, \quad 0 \leq \tilde{t}, \tilde{t}' \leq 3, \quad (2706)$$

$$\kappa_{C_8 \times C_2}((\epsilon, \delta), (\epsilon', \delta')) = \epsilon\epsilon', \quad \epsilon, \delta, \epsilon', \delta' \in \{0, 1\}, \quad (2707)$$

$$\kappa_{C_4 \times C_4} = 0, \quad \kappa_{C_4 \times C_2^2} = 0. \quad (2708)$$

Each κ_P vanishes when either argument is zero and satisfies

$$\kappa_P(u, v) + \kappa_P(u + v, w) = \kappa_P(v, w) + \kappa_P(u, v + w) \quad \text{in } C_4. \quad (2709)$$

The first two cocycles are not coboundaries. The latter two are zero. Equivalently, the extensions

$$0 \longrightarrow A_P \longrightarrow P \longrightarrow S_P \longrightarrow 0$$

split exactly for $C_4 \times C_4$ and $C_4 \times C_2^2$. The two nonzero carries are fixed explicitly by

$$4\sigma_{C_{16}}(1) = a_{C_{16}}, \quad 2\sigma_{C_8 \times C_2}(1, 0) = a_{C_8 \times C_2}, \quad 2\sigma_{C_8 \times C_2}(0, 1) = 0. \quad (2710)$$

Proof. In each row, σ_P contains exactly one representative of every coset of A_P . Every element of a coset has a unique expression $\sigma_P(u) + ja_P$, proving bijectivity of Θ_P .

For C_{16} , write the sheet representatives as $\tilde{t}, \tilde{t}' \in \{0, 1, 2, 3\}$. Euclidean division gives

$$\tilde{t} + \tilde{t}' = (\tilde{t} + \tilde{t}' \bmod 4) + 4 \left\lfloor \frac{\tilde{t} + \tilde{t}'}{4} \right\rfloor.$$

The multiple of $4q = a_{C_{16}}$ is exactly the cell carry in (2706). For $C_8 \times C_2$, addition of the first binary sheet representatives contributes $2q = a_P$ precisely when $\epsilon = \epsilon' = 1$; the r -coordinate contributes no carry. The remaining two rows are direct products. This proves (2705). Associativity on P , written in these coordinates, gives (2709).

Changing a section by a function $f : S_P \rightarrow C_4$ changes its carry by

$$f(u) + f(v) - f(u + v).$$

Thus the carry can be killed exactly when the quotient map has a subgroup section, or equivalently when P has a subgroup complementary to A_P . In C_{16} , the unique subgroup of order four is A_P , so no complement exists. In $C_8 \times C_2$, a complement would be a copy of C_2^2 inside

$$P[2] = \langle 4q, r \rangle.$$

This is the only subgroup of $P[2]$ having order four, and it meets A_P in $\langle 4q \rangle$. Hence it is not a complement. In the remaining rows, $\langle r \rangle$ and $\langle r, s \rangle$ are complements. The lift relations follow directly from the displayed sections. $\square$

Corollary 14.309 (Exact order-sixteen minimum support volume). *Write an abelian group P of order sixteen as*

$$P \cong \prod_{j=1}^a C_{2^{\lambda_j}} \times C_2^b, \quad \lambda_j \geq 2.$$

Among the supports in the local-interval monoid which generate P , the least cardinality is

$$\boxed{m(P) = 3^a 2^b}. \quad (2711)$$

There is exactly one $\text{Aut}(P)$ -orbit attaining this cardinality. It is represented by the direct interval box

$$\sum_{j=1}^a I_1(q_j) + \sum_{k=1}^b B(r_k), \quad (2712)$$

where q_j, r_k are an invariant-factor basis. Thus the five order-sixteen types have minimum support volumes

P	C_{16}	$C_8 \times C_2$	$C_4 \times C_4$	$C_4 \times C_2^2$	C_2^4
$m(P)$	3	6	9	12	16.

Proof. For the four non-elementary groups, read the least row of the corresponding complete table in theorem 14.307. Each least row is exactly (2712), its factors occupy independent invariant-factor coordinates, and hence its cardinality is $3^a 2^b$. The orbit column shows uniqueness. For C_2^4 , the elementary-abelian theorem forces full support, and four independent binary factors attain all sixteen points. $\square$

Corollary 14.310 (Minimum-support locations in the four-sheet carrier). *For the four non-elementary groups, let*

P	M_P
C_{16}	$I_1(q)$
$C_8 \times C_2$	$I_1(q) + B(r)$
$C_4 \times C_4$	$I_1(q) + I_1(r)$
$C_4 \times C_2^2$	$I_1(q) + B(r) + B(s).$

These are the unique minimum-support orbits from theorem 14.309. Their exact inverse images under Θ_P are

$$\Theta_{C_{16}}^{-1}(M_{C_{16}}) = \{(0, 0), (1, 0), (3, 3)\}, \quad (2713)$$

$$\Theta_{C_8 \times C_2}^{-1}(M_{C_8 \times C_2}) = \{(0, \delta, 0) : \delta \in C_2\} \cup \{(1, \delta, j) : \delta \in C_2, j \in \{0, 3\}\}, \quad (2714)$$

$$\Theta_{C_4 \times C_4}^{-1}(M_{C_4 \times C_4}) = \{0, 1, 3\} \times \{0, 1, 3\}, \quad (2715)$$

$$\Theta_{C_4 \times C_2^2}^{-1}(M_{C_4 \times C_2^2}) = C_2^2 \times \{0, 1, 3\}. \quad (2716)$$

For $X_P = \Theta_P^{-1}(M_P)$, let

$$d_P := |\text{pr}_{S_P} X_P| |\text{pr}_{C_4} X_P| - |X_P|.$$

Then

P	C_{16}	$C_8 \times C_2$	$C_4 \times C_4$	$C_4 \times C_2^2$
$ X_P $	3	6	9	12
$ \text{pr}_{S_P} X_P $	3	4	3	4
$ \text{pr}_{C_4} X_P $	2	2	3	3
d_P	3	2	0	0.

(2717)

In particular,

$$|\text{pr}_{S_P} X_P| |\text{pr}_{C_4} X_P| = m(P) + d_P. \quad (2718)$$

Thus the minimum support is a sheet-cell rectangle exactly in the two split selected carriers. In the two non-split selected carriers, the positive value of d_P records the exact missing sheet-cell pairs forced by the carry.

Proof. Apply the four coordinate maps in (2704) to the four displayed minimum supports. This gives (2713)–(2716) term by term. Counting each set and its two coordinate projections gives (2717). The last identity is the definition of d_P together with theorem 14.309. The last assertion follows from the split classification in theorem 14.308. $\square$

Theorem 14.311 (Carrier inversion decomposes into the exact mixer-blade blocks). *For a non-elementary abelian group P of order sixteen, retain $\Theta_P : S_P \times C_4 \rightarrow P$, κ_P , and $X_P = \Theta_P^{-1}(M_P)$ from theorems 14.308 and 14.310. Group inversion on P transports to the involution*

$$\iota_P(u, j) = (-u, -j - \kappa_P(u, -u)) \quad \text{on } S_P \times C_4. \quad (2719)$$

Its fixed loci on the full carriers are

P	$(S_P \times C_4)^{\iota_P}$
C_{16}	$\{(0, 0), (0, 2)\}$
$C_8 \times C_2$	$\{((0, \delta), j) : \delta \in C_2, j \in \{0, 2\}\}$
$C_4 \times C_4$	$\{(t, j) : t, j \in \{0, 2\}\}$
$C_4 \times C_2^2$	$C_2^2 \times \{0, 2\}.$

(2720)

Let

$$\mathcal{K}_P := \mathbb{R}^{S_P \times C_4}, \quad U_P \delta_\omega := \delta_{\iota_P(\omega)}, \quad \Pi_{P, \pm} := \frac{1}{2}(I \pm U_P).$$

Then U_P is an orthogonal self-adjoint involution, and $\Pi_{P, +}, \Pi_{P, -}$ are its two orthogonal spectral projectors. Their full-carrier ranks are

P	C_{16}	$C_8 \times C_2$	$C_4 \times C_4$	$C_4 \times C_2^2$
$\text{rank } \Pi_{P, +}$	9	10	10	12
$\text{rank } \Pi_{P, -}$	7	6	6	4.

(2721)

Every X_P is ι_P -invariant. Its fixed locus and the ranks on $\mathcal{K}_{X_P} := \text{span}\{\delta_x : x \in X_P\}$ are

P	$X_P^{\iota_P}$	$\text{rank } \Pi_{P,+} _{\mathcal{K}_{X_P}}$	$\text{rank } \Pi_{P,-} _{\mathcal{K}_{X_P}}$
C_{16}	$\{(0, 0)\}$	2	1
$C_8 \times C_2$	$\{((0, \delta), 0) : \delta \in C_2\}$	4	2
$C_4 \times C_4$	$\{(0, 0)\}$	5	4
$C_4 \times C_2^2$	$C_2^2 \times \{0\}$	8	4.

(2722)

For every two-point orbit

$$\mathcal{O} = \{\omega, \iota_P(\omega)\} \subseteq X_P,$$

the restriction of U_P to $\mathbb{R}^{\mathcal{O}}$ has matrix

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

in either point order. Its restrictions of $\Pi_{P,\pm}$ are therefore the mixer projectors $P_{\pm} = \frac{1}{2}(I \pm S)$ of theorem 9.14. Moreover,

$$\Pi_{P,-} \text{End}(\mathbb{R}^{\mathcal{O}}) \Pi_{P,+} \quad \text{and} \quad \Pi_{P,+} \text{End}(\mathbb{R}^{\mathcal{O}}) \Pi_{P,-} \quad (2723)$$

are one-dimensional. In either point order they are generated by B_{-+} and B_{+-} , respectively. Reversing the point order negates both displayed generators and leaves their two lines fixed.

Let

$$b_P := \frac{|X_P| - |X_P^{\iota_P}|}{2} = \text{rank } \Pi_{P,-}|_{\mathcal{K}_{X_P}}. \quad (2724)$$

Thus b_P is simultaneously the number of two-point inversion orbits and the number of mixer-blade blocks in the minimum support. Together with the correlation deficit d_P of (2717), one has

P	C_{16}	$C_8 \times C_2$	$C_4 \times C_4$	$C_4 \times C_2^2$
d_P	3	2	0	0
b_P	1	2	4	4
$d_P + b_P$	4	4	4	4.

(2725)

In particular, the four carrier cells obey the exact conservation law

$$\boxed{d_P + b_P = |C_4| = 4.} \quad (2726)$$

Proof. Apply the transported law (2705) to $(u, 0)$ and $(-u, 0)$. Since their sum maps to zero in P ,

$$\sigma_P(u) + \sigma_P(-u) = \kappa_P(u, -u)a_P.$$

Negating $\Theta_P(u, j) = \sigma_P(u) + ja_P$ therefore gives (2719).

For C_{16} , one has

$$\kappa(t, -t) = \begin{cases} 0, & t = 0, \\ 1, & t \neq 0. \end{cases}$$

Solving $\iota_P(t, j) = (t, j)$ gives $(0, 0)$ and $(0, 2)$. For $C_8 \times C_2$, inversion fixes every sheet and the cell equation is

$$2j \equiv -\epsilon \pmod{4}.$$

It has the two solutions $j = 0, 2$ precisely when $\epsilon = 0$. In the split $C_4 \times C_4$ row the fixed equations are $2t = 2j = 0$ in C_4 . In the split $C_4 \times C_2^2$ row every sheet is fixed and $j = 0, 2$. This proves (2720).

The operator U_P permutes an orthonormal basis by an involution, so it is orthogonal and self-adjoint. Every fixed point contributes one positive eigenvector. Every two-point orbit contributes one positive vector and one negative vector. Counting the fixed loci in (2720) proves (2721).

Each M_P is a sum of symmetric intervals and order-two binary factors, and hence is closed under group inversion. Substitution in the four exact coordinate sets of theorem 14.310 gives (2722). On a two-point orbit, inversion exchanges the two point masses, so its matrix is S in either point order.

The projector and corner calculations are exactly (157)–(163). Reversing the point order conjugates each corner generator by S , which negates it and preserves its line.

The negative rank is one for every two-point orbit and zero for every fixed point, proving (2724). The fixed counts in (2722) give

$$b_P = (1, 2, 4, 4).$$

Adding this row to $d_P = (3, 2, 0, 0)$ from (2717) proves (2725) and (2726). $\square$

Exact computation 14.312 (Exact carrier-inversion and blade-block certificates). The script and captured receipt

```
verification/verify_order_sixteen_carrier_inversion_blade.py,
verification/verify_order_sixteen_carrier_inversion_blade_output.txt
```

verify the inversion formula on all sixty-four displayed carrier coordinates, the full and minimum fixed loci, the exact rational 2-by-2 mixer and corner identities, and the four deficit-conservation equalities. The independent Lean 4.32 source and receipt

```
verification/OrderSixteenCarrierInversionBladeCertificate.lean,
verification/OrderSixteenCarrierInversionBladeCertificate_output.txt
```

certify the transported inversions, full spectra, invariant minimum supports, fixed loci, two-cycle blocks, and conservation identities in twenty-four theorem statements, without `sorry`, `admit`, or an added axiom.

Theorem 14.313 (Carry-phased dihedral Fourier sectors on the four-cell carrier). *Retain the conditions and notation of theorem 14.311. Define the cell rotation*

$$R_P(u, j) := (u, j + 1) \quad (2727)$$

and let U_P denote the linear inversion operator on $\mathcal{K}_P$. Then

$$R_P^4 = I, \quad U_P^2 = I, \quad U_P R_P U_P = R_P^{-1}. \quad (2728)$$

The resulting action of the order-eight group

$$D_8 := \langle r, s : r^4 = s^2 = 1, \, sr s = r^{-1} \rangle, \quad r \mapsto R_P, \quad s \mapsto U_P, \quad (2729)$$

is faithful. Its orbit sizes on the full carrier are

P	C_{16}	$C_8 \times C_2$	$C_4 \times C_4$	$C_4 \times C_2^2$
orbit sizes	$(4, 4, 8)$	$(4, 4, 4, 4)$	$(4, 4, 8)$	$(4, 4, 4, 4)$

(2730)

Complexify $\mathcal{K}_P$, put $\mathbf{i}^2 = -1$, and for $u \in S_P$, $k \in \mathbb{Z}/4\mathbb{Z}$, define

$$v_{u,k} := \sum_{j=0}^3 \mathbf{i}^{-kj} \delta_{(u,j)}. \quad (2731)$$

Then

$$R_P v_{u,k} = \mathbf{i}^k v_{u,k}, \quad U_P v_{u,k} = \mathbf{i}^{k\kappa_P(u,-u)} v_{-u,-k}. \quad (2732)$$

Accordingly the four cell spectral projectors are

$$Q_{P,k} := \frac{1}{4} \sum_{m=0}^3 \mathbf{i}^{-km} R_P^m, \quad \text{rank}_{\mathbb{C}} Q_{P,k} = 4, \quad U_P Q_{P,k} U_P = Q_{P,-k}. \quad (2733)$$

The $k = 0$ and $k = 2$ sectors are self-conjugate, while $k = 1$ and $k = 3$ are exchanged. On the self-conjugate $k = 2$ sector, the inversion phase on sheet u is

$$(-1)^{\kappa_P(u,-u)}. \quad (2734)$$

With the sheet orders

$$(0, 1, 2, 3) \quad \text{for } S_P = C_4, \quad ((0, 0), (0, 1), (1, 0), (1, 1)) \quad \text{for } S_P = C_2^2,$$

these signs are

$$\begin{array}{c|cccc} P & C_{16} & C_8 \times C_2 & C_4 \times C_4 & C_4 \times C_2^2 \\ \hline k = 2 \text{ signs} & (1, -1, -1, -1) & (1, 1, -1, -1) & (1, 1, 1, 1) & (1, 1, 1, 1). \end{array} \quad (2735)$$

Thus the two nonzero selected carry classes are detected by the second cell-Fourier sector, whereas both split selected carriers have positive second-sector phase on every sheet.

Proof. The map R_P is translation by the order-four element a_P , so it has order four. Equation (2719) gives

$$\iota_P R_P \iota_P(u, j) = (u, j - 1),$$

which proves (2728). The presentation in (2729) therefore acts. Its rotation has order four. If a reflected group element acted trivially, it would force $R_P = R_P^{-1}$, contrary to that order. Hence the action is faithful.

Cell rotation is transitive on the four cells of a fixed sheet. Inversion sends sheet u to sheet $-u$. The inversion orbits on C_4 have sizes 1, 1, 2, while inversion fixes every element of C_2^2 . Multiplication by the four cell positions gives (2730).

From (2731),

$$R_P v_{u,k} = \sum_{j=0}^3 \mathbf{i}^{-kj} \delta_{(u,j+1)} = \mathbf{i}^k v_{u,k}.$$

Put $c_u = \kappa_P(u, -u)$. Under inversion the target cell is $\ell = -j - c_u$, so $j = -\ell - c_u$ in C_4 . Therefore

$$\begin{aligned} U_P v_{u,k} &= \sum_{j=0}^3 \mathbf{i}^{-kj} \delta_{(-u, -j - c_u)} \\ &= \mathbf{i}^{kc_u} \sum_{\ell=0}^3 \mathbf{i}^{k\ell} \delta_{(-u, \ell)} = \mathbf{i}^{kc_u} v_{-u, -k}. \end{aligned}$$

This proves (2732). The character orthogonality identity on C_4 proves the projector formula and rank statement in (2733); conjugation by U_P then sends k to $-k$.

For $k = 2$, the phase is $\mathbf{i}^{2c_u} = (-1)^{c_u}$. The carry values are

$$\begin{array}{c|c} P & (\kappa_P(u, -u))_{u \in S_P} \\ \hline C_{16} & (0, 1, 1, 1) \\ C_8 \times C_2 & (0, 0, 1, 1) \\ C_4 \times C_4 & (0, 0, 0, 0) \\ C_4 \times C_2^2 & (0, 0, 0, 0). \end{array}$$

Taking their parities proves (2735). □

Exact computation 14.314 (Exact dihedral and carry-phased Fourier certificates). The exact Gaussian-rational script and receipt

`verification/verify_order_sixteen_carrier_dihedral_fourier.py,`

`verification/verify_order_sixteen_carrier_dihedral_fourier_output.txt`

check all dihedral relations, orbit sizes, four Fourier eigenvectors on every sheet, carry phases, Fourier ranks, and the second-sector sign table. The independent Lean 4.32 source and receipt

`verification/OrderSixteenCarrierDihedralFourierCertificate.lean,`

`verification/OrderSixteenCarrierDihedralFourierCertificate_output.txt`

certify four faithful dihedral relation checks, four exact orbit partitions, four Fourier-exponent transports, and four second-sector sign tables in sixteen theorem statements, without `sorry`, `admit`, or an added axiom.

Theorem 14.315 (The irreducible dihedral spectrum recovers the selected carrier extension class). For $\alpha, \beta \in \{1, -1\}$, let $\chi_{\alpha\beta}$ be the one-dimensional representation of D_8 defined by

$$\chi_{\alpha\beta}(r) = \alpha, \quad \chi_{\alpha\beta}(s) = \beta,$$

and let ρ be the irreducible two-dimensional representation. Define the two reflection-orbit modules

$$\begin{aligned} V_0 &:= \text{Ind}_{\langle s \rangle}^{D_8} \mathbf{1} = \chi_{++} \oplus \chi_{-+} \oplus \rho, \\ V_1 &:= \text{Ind}_{\langle rs \rangle}^{D_8} \mathbf{1} = \chi_{++} \oplus \chi_{--} \oplus \rho. \end{aligned} \quad (2736)$$

The four complexified carrier representations of theorem 14.313 decompose as

$$\begin{array}{c|c} P & \mathcal{K}_P^{\mathbb{C}} \\ \hline C_{16} & V_0 \oplus V_1 \oplus \mathbb{C}[D_8] \\ C_8 \times C_2 & 2V_0 \oplus 2V_1 \\ C_4 \times C_4 & 2V_0 \oplus \mathbb{C}[D_8] \\ C_4 \times C_2^2 & 4V_0. \end{array} \quad (2737)$$

In the element order

$$(1, r, r^2, r^3, s, rs, r^2s, r^3s),$$

their permutation characters are

$$\begin{array}{c|c} P & \chi_{\mathcal{K}_P} \\ \hline C_{16} & (16, 0, 0, 0, 2, 2, 2, 2) \\ C_8 \times C_2 & (16, 0, 0, 0, 4, 4, 4, 4) \\ C_4 \times C_4 & (16, 0, 0, 0, 4, 0, 4, 0) \\ C_4 \times C_2^2 & (16, 0, 0, 0, 8, 0, 8, 0). \end{array} \quad (2738)$$

Write $m_{\alpha\beta}(P)$ and $m_{\rho}(P)$ for the corresponding irreducible multiplicities. Then

$$\begin{array}{c|ccccc} P & m_{++} & m_{+-} & m_{-+} & m_{--} & m_{\rho} \\ \hline C_{16} & 3 & 1 & 2 & 2 & 4 \\ C_8 \times C_2 & 4 & 0 & 2 & 2 & 4 \\ C_4 \times C_4 & 3 & 1 & 3 & 1 & 4 \\ C_4 \times C_2^2 & 4 & 0 & 4 & 0 & 4. \end{array} \quad (2739)$$

The reflection traces recover the two relevant multiplicity differences:

$$\frac{1}{2} \text{tr}(U_P) = m_{-+}(P) - m_{+-}(P), \quad \frac{1}{2} \text{tr}(R_P U_P) = m_{--}(P) - m_{+-}(P). \quad (2740)$$

For the four carrier laws, one therefore has the exact spectral criterion

$$\boxed{\kappa_P \text{ is not a coboundary} \iff \text{tr}(R_P U_P) > 0 \iff m_{--}(P) > m_{+-}(P).} \quad (2741)$$

The full multiplicity row distinguishes the four selected marked extension classes. The common two-dimensional multiplicity $m_{\rho}(P) = 4$ records the four cyclic cell modes, while the one-dimensional imbalance in (2741) detects precisely the two non-coboundary selected classes.

Proof. Consider one orbit of the dihedral action on the carrier. If $u \neq -u$, the orbit contains the four cells over u and the four cells over $-u$. Its cardinality equals $|D_8|$, so its permutation module is the regular representation.

Suppose $u = -u$. The four cells over u form one transitive orbit. Put $c_u = \kappa_P(u, -u)$. On that orbit,

$$U_P(u, j) = (u, -j - c_u).$$

If c_u is even, a rotation of the cell coordinate carries its point stabilizer to $\langle s \rangle$; if c_u is odd, it carries the stabilizer to $\langle rs \rangle$. The two orbit modules are therefore V_0 and V_1 , respectively.

For the four sheet involutions and carry rows, the orbit-module lists are

P	orbit modules
C_{16}	$V_0, V_1, \mathbb{C}[\mathbf{D}_8]$
$C_8 \times C_2$	V_0, V_0, V_1, V_1
$C_4 \times C_4$	$V_0, V_0, \mathbb{C}[\mathbf{D}_8]$
$C_4 \times C_2^2$	$V_0, V_0, V_0, V_0.$

This proves (2737).

The four one-dimensional characters satisfy

$$\chi_{\alpha\beta}(r^a s^b) = \alpha^a \beta^b,$$

and the character of ρ is

$$(2, 0, -2, 0, 0, 0, 0, 0).$$

The induced modules in (2736) follow either from their fixed-vector dimensions under $\langle s \rangle$ and $\langle rs \rangle$, or by taking their inner products with this character table. The regular character is

$$(8, 0, 0, 0, 0, 0, 0, 0)$$

and decomposes as

$$\mathbb{C}[\mathbf{D}_8] = \chi_{++} \oplus \chi_{+-} \oplus \chi_{-+} \oplus \chi_{--} \oplus 2\rho.$$

Adding the orbit characters proves (2738), and taking the character inner products

$$\langle \chi, \psi \rangle_{\mathbf{D}_8} = \frac{1}{8} \sum_{g \in \mathbf{D}_8} \chi(g) \psi(g)$$

proves (2739).

Reading the columns for s and rs in the same table gives (2740). In particular,

$$\frac{1}{2} \operatorname{tr}(\mathbf{R}_P U_P) = m_{--}(P) - m_{+-}(P) = (1, 2, 0, 0)$$

in the four displayed rows. The carry calculations in theorem 14.308 prove that precisely the first two cocycles are not coboundaries. This proves (2741). $\square$

Exact computation 14.316 (Exact dihedral irreducible-spectrum certificates). The exact character calculation and captured receipt

```
verification/verify_order_sixteen_carrier_dihedral_irreducibles.py,
verification/verify_order_sixteen_carrier_dihedral_irreducibles_output.txt
```

enumerate every fixed point of the eight group elements, compute the five character inner products over $\mathbb{Q}$, recover the four orbit-module lists, and check both reflection-trace identities and the spectral carry criterion. The independent Lean 4.32 source and receipt

```
verification/OrderSixteenCarrierDihedralIrreducibleCertificate.lean,
verification/OrderSixteenCarrierDihedralIrreducibleCertificate_output.txt
```

certify four permutation-character tables, four irreducible-multiplicity tables, four orbit-module decompositions, and four spectral carry criteria in sixteen theorem statements, without `sorry`, `admit`, or an added axiom.

Theorem 14.317 (Two-torsion carry character and spectral weight). *Let S be a finite abelian group, and let $\kappa : S \times S \rightarrow C_4$ be a symmetric 2-cocycle satisfying $\kappa(0, u) = \kappa(u, 0) = 0$. Give $E = S \times C_4$ the carrier law*

$$(u, j) + (v, k) = (u + v, j + k + \kappa(u, v)).$$

On

$$S[2] := \{u \in S : 2u = 0\}$$

define

$$q_\kappa(u) := \kappa(u, u) \pmod{2}. \quad (2742)$$

Then $q_\kappa : S[2] \rightarrow C_2$ is a homomorphism. It is unchanged when κ is replaced by the cocycle obtained from any other section of the same extension.

Let cell translation and carrier inversion act on $\mathbb{C}[E]$ as the dihedral generators

$$R(u, j) = (u, j + 1), \quad U(u, j) = (-u, -j - \kappa(u, -u)).$$

If $m_{\alpha\beta}$ are the four one-dimensional irreducible multiplicities, then

$$\boxed{m_{--} - m_{+-} = \frac{1}{2} \text{tr}(RU) = |\{u \in S[2] : q_\kappa(u) = 1\}|.} \quad (2743)$$

Consequently, if $q_\kappa \neq 0$, then

$$m_{--} - m_{+-} = \frac{|S[2]|}{2}, \quad (2744)$$

and the carrier extension is non-split.

For the four order-sixteen carriers, the exact characters are

P	$S_P[2]$	q_{κ_P}	$ \{u : q_{\kappa_P}(u) = 1\} $
C_{16}	$\{0, 2\} \subset C_4$	$(0, 1)$	1
$C_8 \times C_2$	C_2^2	$q(\epsilon, \delta) = \epsilon$	2
$C_4 \times C_4$	$\{0, 2\} \subset C_4$	0	0
$C_4 \times C_2^2$	C_2^2	0	0.

(2745)

Thus the two nonzero spectral defects are the half-cardinalities of the two corresponding two-torsion sheet groups.

Proof. For $u, v \in S[2]$, additivity of doubling in E gives

$$2((u, 0) + (v, 0)) = 2(u, 0) + 2(v, 0).$$

Comparing the C_4 -coordinates yields

$$\kappa(u + v, u + v) + 2\kappa(u, v) = \kappa(u, u) + \kappa(v, v) \pmod{4}.$$

Reduction modulo two proves that q_κ is additive.

If a section is changed by $f : S \rightarrow C_4$, with $f(0) = 0$, then its cocycle is

$$\kappa'(u, v) = \kappa(u, v) + f(u) + f(v) - f(u + v).$$

For $u \in S[2]$,

$$\kappa'(u, u) - \kappa(u, u) = 2f(u) \pmod{4},$$

so $q_{\kappa'} = q_\kappa$. In particular, a split extension has $q_\kappa = 0$; hence $q_\kappa \neq 0$ forces the extension to be non-split.

If $u \neq -u$, the eight cells over u and $-u$ form the regular dihedral module, which contributes equally to m_{--} and m_{+-} . If $u = -u$, the four cells over u form V_0 when $q_\kappa(u) = 0$ and V_1 when $q_\kappa(u) = 1$. The module V_0 contains neither χ_{--} nor χ_{+-} , whereas V_1 contains χ_{--} once and does not contain χ_{+-} . Summing the orbit contributions proves (2743).

If $q_\kappa \neq 0$, its kernel has index two in $S[2]$, so its two fibers have equal cardinality. This proves (2744). Finally, substituting the four cocycles of theorem 14.308 gives (2745). $\square$

Exact computation 14.318 (Two-torsion carry-character certificates). The script and captured receipt

```
verification/verify_order_sixteen_two_torsion_carry_character.py,
verification/verify_order_sixteen_two_torsion_carry_character_output.txt
```

derive $S_P[2]$ and q_{κ_P} from the four carrier laws, verify additivity, test invariance under all 64 section changes in every row, count the V_1 orbit modules, and recompute the spectral multiplicity difference. The independent Lean 4.32 source and receipt

```
verification/OrderSixteenTwoTorsionCarryCharacterCertificate.lean,
verification/OrderSixteenTwoTorsionCarryCharacterCertificate_output.txt
```

certify the general translation equivalence between the zero and one fibers of a nonzero binary carry character, the resulting equality of their cardinalities, all four explicit additivity laws, all four weights, and both nonzero equal-fiber instances in eleven theorem statements, without `sorry`, `admit`, or an added axiom.

Theorem 14.319 (Marked C_4 -extension detection and its exact kernel). *Let $\mathcal{E}_4(S)$ be the abelian group of equivalence classes of abelian extensions*

$$0 \longrightarrow C_4 \longrightarrow E \longrightarrow S \longrightarrow 0$$

with the kernel identified with C_4 . For a cocycle representative κ , define

$$Q_S : \mathcal{E}_4(S) \longrightarrow \text{Hom}(S[2], C_2), \quad Q_S([\kappa]) = q_\kappa. \quad (2746)$$

Then Q_S is a well-defined surjective homomorphism.

Choose an invariant-factor decomposition of the 2-primary part

$$S_{(2)} \cong \prod_{i=1}^a C_{2^{\lambda_i}} \times C_2^b, \quad \lambda_i \geq 2.$$

Under the resulting coordinates,

$$\mathcal{E}_4(S) \cong C_4^a \times C_2^b, \quad \text{Hom}(S[2], C_2) \cong C_2^{a+b},$$

and

$$Q_S(c_1, \dots, c_a; \epsilon_1, \dots, \epsilon_b) = (\bar{c}_1, \dots, \bar{c}_a; \epsilon_1, \dots, \epsilon_b), \quad (2747)$$

where $\bar{c}_i$ is reduction modulo two. Hence

$$0 \longrightarrow (2C_4)^a \longrightarrow C_4^a \times C_2^b \xrightarrow{Q_S} C_2^{a+b} \longrightarrow 0 \quad (2748)$$

is exact. Every binary carry character has exactly 2^a marked extension classes above it. In particular, q_κ determines the marked extension class precisely when $S_{(2)}$ is elementary abelian.

For sheet groups of order four, all classes are given by the exact table

S	$[\kappa]$	E	$ q_\kappa^{-1}(1) $	$split$
C_4	0	$C_4 \times C_4$	0	yes
C_4	1	C_{16}	1	no
C_4	2	$C_8 \times C_2$	0	no
C_4	3	C_{16}	1	no
C_2^2	(0, 0)	$C_4 \times C_2^2$	0	yes
C_2^2	$(a, b) \neq (0, 0)$	$C_8 \times C_2$	2	no.

(2749)

Consequently, $P = C_8\langle q \rangle \times C_2\langle r \rangle$ has two marked C_4 -carrier presentations with different spectral weights:

$$\begin{aligned} A &= \langle 2q \rangle, & S &= C_2^2, & \kappa((\epsilon, \delta), (\epsilon', \delta')) &= \epsilon\epsilon', & |q_\kappa^{-1}(1)| &= 2, \\ A' &= \langle 2q + r \rangle, & S' &= C_4, & \kappa'(t, t') &= 2 \left\lfloor \frac{\tilde{t} + \tilde{t}'}{4} \right\rfloor, & |q_{\kappa'}^{-1}(1)| &= 0. \end{aligned} \quad (2750)$$

The second presentation is non-split because

$$4q = 2(2q + r) \neq 0,$$

although its binary carry character vanishes. Thus the multiplicity difference in (2743) is intrinsic to the marked carrier extension, rather than to the abstract group P alone. The minimum-support construction in theorem 14.308 selects the first $C_8 \times C_2$ presentation.

Proof. For a cyclic summand $C_n = \langle u \rangle$, choose a lift $x \in E$ of u . Then $nx = c \in C_4$. Replacing x by $x + d$, with $d \in C_4$, replaces c by $c + nd$. Conversely, the presentation

$$\langle x, z : 4z = 0, \quad nx = cz, \quad [x, z] = 0 \rangle$$

gives an abelian extension with class $c \in C_4/nC_4$. Therefore

$$\mathcal{E}_4(C_n) \cong C_4/nC_4.$$

This group is zero for odd n , is C_2 for $n = 2$, and is C_4 when $4 \mid n$. Additivity over direct products gives $\mathcal{E}_4(S) \cong C_4^a \times C_2^b$.

For $n = 2^\lambda$, the nonzero element of $C_n[2]$ is $2^{\lambda-1}u$. Doubling its chosen lift gives

$$2(2^{\lambda-1}x) = 2^\lambda x = c.$$

Its image in $C_4/2C_4 = C_2$ is therefore $c \bmod 2$. This proves (2747). Addition of cocycles proves that Q_S is a homomorphism, and theorem 14.317 proves independence of the section. The coordinate map is onto, with kernel $(2C_4)^a$, proving (2748).

For $S = C_4$, use

$$\kappa_c(t, t') = c \left\lfloor \frac{\tilde{t} + \tilde{t}'}{4} \right\rfloor.$$

The relation matrix for E_c is

$$\begin{pmatrix} 4 & -c \\ 0 & 4 \end{pmatrix}.$$

Its Smith invariant factors are

$$(4, 4), \quad (1, 16), \quad (2, 8), \quad (1, 16)$$

for $c = 0, 1, 2, 3$, respectively. The carry character has values $(0, c \bmod 2)$ on $C_4[2] = \{0, 2\}$.

For $S = C_2^2$, write the class as $(a, b) \in C_2^2$, using the lifts $a, b \in \{0, 1\} \subset C_4$. The cocycle is

$$\kappa_{a,b}((\epsilon, \delta), (\epsilon', \delta')) = a\epsilon\epsilon' + b\delta\delta',$$

and the relation matrix is

$$\begin{pmatrix} 2 & 0 & -a \\ 0 & 2 & -b \\ 0 & 0 & 4 \end{pmatrix}.$$

Its Smith factors are $(2, 2, 4)$ for $(a, b) = (0, 0)$ and $(1, 2, 8)$ for every nonzero pair. Moreover,

$$q_{\kappa_{a,b}}(\epsilon, \delta) = a\epsilon + b\delta,$$

so every nonzero pair has a two-element one-fiber. This proves (2749).

For the first $C_8 \times C_2$ presentation, the bijection is

$$((\epsilon, \delta), j) \longmapsto \epsilon q + \delta r + 2jq.$$

For the second it is

$$(t, j) \longmapsto tq + j(2q + r).$$

Direct addition gives the two cocycles in (2750). The relation $4q = 2(2q + r) \neq 0$ completes the claimed distinction. $\square$

Exact computation 14.320 (Marked C_4 -extension detection certificates). The exact script and captured receipt

```
verification/verify_marked_C4_extension_detection.py,
verification/verify_marked_C4_extension_detection_output.txt
```

compute the Smith factors, every binary carry weight, and every coboundary status in (2749). They also build both $C_8 \times C_2$ carrier bijections and recover their cocycles by transporting the group law. The independent Lean 4.32 source and receipt

```
verification/MarkedC4ExtensionDetectionCertificate.lean,
verification/MarkedC4ExtensionDetectionCertificate_output.txt
```

certify associativity of all eight finite carrier laws, their exact torsion-kernel counts, all carry weights, the invisible non-split cyclic class 2, and the binary coboundary table in fourteen theorem statements, without `sorry`, `admit`, or an added axiom.

Corollary 14.321 (Forced coupling of visible and hidden extension bits). *Use the invariant-factor coordinates of theorem 14.319. Every long cyclic-sheet coordinate $c_i \in C_4$ has a unique expression*

$$c_i = q_i + 2h_i, \quad q_i, h_i \in \mathbb{F}_2, \quad q_i = c_i \bmod 2. \quad (2751)$$

Consequently, the displayed invariant-factor decomposition gives a bijection

$$\mathcal{E}_4(S) \longleftrightarrow \mathbb{F}_2^a \times \mathbb{F}_2^a \times \mathbb{F}_2^b, \quad (c; \epsilon) \longleftrightarrow (q, h; \epsilon).$$

Under Baer addition, this bijection transports the group law to

$$(q, h; \epsilon) \star (q', h'; \epsilon') = (q + q', h + h' + q \odot q'; \epsilon + \epsilon'), \quad (2752)$$

where all entries lie in $\mathbb{F}_2$ and $(q \odot q')_i = q_i q'_i$. In these coordinates,

$$Q_S(q, h; \epsilon) = (q; \epsilon), \quad \ker Q_S = \{(0, h; 0) : h \in \mathbb{F}_2^a\}.$$

The detection exact sequence has a group-homomorphic section if and only if

$$a = 0. \quad (2753)$$

Thus a set-theoretic visible/hidden bit chart always exists, while every long cyclic sheet forces the quadratic carry $q \odot q'$ and prevents that chart from being an additive product chart.

For the cyclic order-four sheet $S = C_4$, the complete bit table is

c	q	h	E_c	marked extension
0	0	0	$C_4 \times C_4$	<i>split</i>
1	1	0	C_{16}	<i>non-split</i>
2	0	1	$C_8 \times C_2$	<i>non-split</i>
3	1	1	C_{16}	<i>non-split.</i>

(2754)

In particular, $c = 2$ is the unique nonzero class invisible to Q_{C_4} .

Proof. For $c \in C_4$, take its representative in $\{0, 1, 2, 3\}$. The two quantities

$$q = c \bmod 2, \quad h = \frac{c - q}{2} \bmod 2$$

give the unique expression (2751). For bits $q, q' \in \{0, 1\}$, the integer identity

$$q + q' = (q + q' \bmod 2) + 2qq'$$

shows that addition in C_4 becomes

$$(q, h) \star (q', h') = (q + q', h + h' + qq').$$

Applying this identity independently to the a long cyclic factors and using ordinary addition on the C_2^b factors proves (2752). The formulas for Q_S and its kernel follow at once from (2747).

Suppose $a > 0$ and a group-homomorphic section of Q_S exists. Every element of C_2^{a+b} has order at most two, so its image under the section has order at most two. Each C_4 -coordinate of such an image is therefore 0 or 2, and hence has visible bit zero. The composite with Q_S cannot be the identity on any of the first a target coordinates, a contradiction. If $a = 0$, the map $Q_S : C_2^b \rightarrow C_2^b$ is the identity. This proves (2753). Finally, substituting $c = 0, 1, 2, 3$ into (2749) gives (2754). $\square$

Exact computation 14.322 (Visible/hidden bit certificates). The exact finite script and captured receipt

```
verification/verify_marked_C4_visible_hidden_bits.py,
verification/verify_marked_C4_visible_hidden_bits_output.txt
```

check all sixteen sums in C_4 , all sixty-four associativity triples in the transported four-element law, every candidate function $C_2 \rightarrow C_4$, and the four rows of (2754). The independent Lean 4.32 source and receipt

```
verification/MarkedC4VisibleHiddenBitsCertificate.lean,
verification/MarkedC4VisibleHiddenBitsCertificate_output.txt
```

certify reconstruction, the coupled addition law, associativity, the nonzero double of every visible-odd coordinate, the absence of an additive section, and the invisible kernel generator in six theorem statements, without `sorry`, `admit`, or an added axiom.

Corollary 14.323 (Split-zero full-adder realization of the marked C_4 bits). *Retain the split-zero endpoint bits (β_1, β_2) and operator energies of theorem 14.290. Let*

$$\iota_+ : C_2 \longrightarrow C_4, \quad \iota_+(0) = 0, \quad \iota_+(1) = 1$$

be the marked support section, and define

$$\ell := \beta_1 + \beta_2 \pmod{2}, \quad h := \beta_1 \beta_2, \quad c := \iota_+(\beta_1) + \iota_+(\beta_2) \in C_4. \quad (2755)$$

Then

$$\boxed{c = \ell + 2h = \beta_1 + \beta_2,} \quad (2756)$$

where the final sum is the integer sum in $\{0, 1, 2\}$. Thus ℓ and h are precisely the visible and hidden bits of the marked C_4 -coordinate c . Equivalently, in the coupled law of theorem 14.321,

$$(\beta_1, 0) \star (\beta_2, 0) = (\ell, h). \quad (2757)$$

On every occupied shell, the two operator energies recover these two bits:

$$\boxed{\mathcal{E}_{p,a}^{\text{pol},123} = 3q_{R,p}\ell, \quad \mathcal{E}_{p,a}^{\text{tim},123} = 3q_{R,p}h.} \quad (2758)$$

The ordered endpoint direction

$$d := \beta_2 - \beta_1$$

is retained by the signed commutator. More precisely,

$$\boxed{[\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}] = 2dV_{\partial,0+}, \quad \{\mathcal{J}_{\partial,123}^{p,a}, V_{\partial,0+}\} = 2(c-1)V_{\partial,0+}.} \quad (2759)$$

The map

$$(\beta_1, \beta_2) \longmapsto (c, d) \quad (2760)$$

is injective, with inverse

$$\beta_1 = \frac{c-d}{2}, \quad \beta_2 = \frac{c+d}{2} \quad (2761)$$

on its four-element image. The complete table is

$$\begin{array}{c|ccc}
 (\beta_1, \beta_2) & \ell & h & c & d \\
 \hline
 (0, 0) & 0 & 0 & 0 & 0 \\
 (1, 0) & 1 & 0 & 1 & -1 \\
 (0, 1) & 1 & 0 & 1 & 1 \\
 (1, 1) & 0 & 1 & 2 & 0.
 \end{array} \tag{2762}$$

Hence the energy pair (ℓ, h) identifies the two one-sided strata, whereas d distinguishes their order. A single endpoint pair has marked class in

$$\{0, 1, 2\} \subset C_4, \quad 3 \notin \text{im}(c). \tag{2763}$$

The subgroup generated by the local image is all of C_4 , since

$$1 + 2 = 3 \pmod{4}. \tag{2764}$$

Proof. For bits β_1, β_2 , the elementary full-adder identity is

$$\beta_1 + \beta_2 = (\beta_1 + \beta_2 \bmod 2) + 2\beta_1\beta_2.$$

This is (2756). Applying (2752) to the two visible-only inputs proves (2757).

The Boolean identity

$$(\beta_2 - \beta_1)^2 = \ell$$

and (2632) give the first formula in (2758); the second is (2631). Equations (2639) and (2640), together with $c = \beta_1 + \beta_2$, prove (2759).

Solving the two integer equations

$$c = \beta_1 + \beta_2, \quad d = \beta_2 - \beta_1$$

gives (2761), hence injectivity. Evaluation on the four bit pairs gives (2762) and (2763). Finally, $1 + 2 = 3$ in C_4 , proving (2764). $\square$

Exact computation 14.324 (Split-zero C_4 full-adder certificates). The exact finite script and receipt

`verification/verify_split_zero_C4_full_adder_bridge.py,`

`verification/verify_split_zero_C4_full_adder_bridge_output.txt`

check the four endpoint rows, their bit decomposition, the coupled visible-only sum, both energy coefficients, the inverse ordered map, the three-class local image, and its four-class generated subgroup. The independent Lean 4.32 source and receipt

`verification/SplitZeroC4FullAdderBridgeCertificate.lean,`

`verification/SplitZeroC4FullAdderBridgeCertificate_output.txt`

certify the digit identity, the coupled sum, both energy-bit recoveries, injectivity after adjoining direction, one-sided energy degeneracy with opposite directions, absence of class 3 from one pair, and occurrence of class 3 in the generated closure in eight theorem statements, without `sorry`, `admit`, or an added axiom.

Corollary 14.325 (Marked C_4 rank coordinate and primewise support criterion). *For a shell with $R = 4a - p$, retain the data of theorem 14.323 and let*

$$\widehat{c}_{p,a} := \ell + 2h \in \{0, 1, 2\} \subset \mathbb{Z} \tag{2765}$$

be the local integer lift of the marked class $c_{p,a} \in C_4$. Then

$$\boxed{\widehat{c}_{p,a} = \text{rank } G_{p,a}.} \tag{2766}$$

The operator energies recover this rank coordinate with its carry weight:

$$\boxed{3q_{R,p}\widehat{c}_{p,a} = \mathcal{E}_{p,a}^{\text{pol},123} + 2\mathcal{E}_{p,a}^{\text{tim},123}.} \tag{2767}$$

The three possible local classes are exactly the three Lorentz strata:

$\hat{c}_{p,a}$	$c_{p,a}$	$\text{rank } \mathbf{G}_{p,a}$	<i>stratum</i>
0	0	0	<i>zero</i>
1	1	1	<i>future-null</i>
2	2	2	<i>future-timelike.</i>

(2768)

In particular,

$$\Omega_R(p) > 0 \iff \hat{c}_{p,a} > 0 \iff c_{p,a} \neq 0 \iff \text{rank } \mathbf{G}_{p,a} > 0, \quad (2769)$$

and

$$\det \mathbf{G}_{p,a} > 0 \iff \hat{c}_{p,a} = 2. \quad (2770)$$

Define the primewise marked C_4 rank coordinate

$$\mathcal{C}_p^{C_4} := \sum_{a \in \mathcal{A}_p} 3q_{R,p} \hat{c}_{p,a}. \quad (2771)$$

Then

$$\mathcal{C}_p^{C_4} = \mathcal{E}_p^{\text{pol},123} + 2\mathcal{E}_p^{\text{tim},123} = 3 \sum_{a \in \mathcal{A}_p} q_{R,p} \text{rank } \mathbf{G}_{p,a}, \quad (2772)$$

and, under the primewise shell reduction,

$$\text{ES}(p) \iff \mathcal{C}_p^{C_4} > 0 \iff \exists a \in \mathcal{A}_p : c_{p,a} \neq 0. \quad (2773)$$

Proof. Compare the four rows of (2620) with (2762). In every row,

$$\text{rank } \mathbf{G}_{p,a} = \beta_1 + \beta_2 = \ell + 2h = \hat{c}_{p,a}.$$

This proves (2766) and (2768). Equation (2758) gives

$$\mathcal{E}_{p,a}^{\text{pol},123} + 2\mathcal{E}_{p,a}^{\text{tim},123} = 3q_{R,p}(\ell + 2h),$$

which is (2767).

The local support equivalence (2621) and the three-row table give (2769). The determinant criterion (2622) gives (2770). Summing the weighted energy identity over a proves (2772). Both primewise energy coordinates

$$\mathcal{E}_p^{\text{pol},123} + \mathcal{E}_p^{\text{tim},123}, \quad \mathcal{E}_p^{\text{pol},123} + 2\mathcal{E}_p^{\text{tim},123}$$

have nonnegative local summands with the same shellwise zero set. Their primewise sums therefore have the same zero set. Hence (2637) proves (2773). $\square$

Exact computation 14.326 (Marked C_4 rank-support certificates). The exact rational-coordinate script and captured receipt

```
verification/verify_split_zero_C4_rank_support.py,
verification/verify_split_zero_C4_rank_support_output.txt
```

construct all four Lorentz matrices, compute their determinants and ranks, and compare them with the marked classes and the weighted energy coefficients. The independent Lean 4.32 source and receipt

```
verification/SplitZeroC4RankSupportCertificate.lean,
verification/SplitZeroC4RankSupportCertificate_output.txt
```

prove class-lift/rank equality, weighted energy recovery, the nonzero-class support test, the rank-one and rank-two characterizations, and the positive-rank/nonzero-class equivalence in six theorem statements, without `sorry`, `admit`, or an added axiom.

Corollary 14.327 (Primewise null–timelike reconstruction and marked C_4 volume sandwich). *For $p \in \mathbb{P}_{1,12}$, let $\ell_{p,a}$ and $h_{p,a}$ be the visible and hidden bits of the marked shell class, and define the weighted one-sided and timelike multiplicities*

$$\mathbf{N}_p^\partial := \sum_{a \in \mathcal{A}_p} q_{R_{a,p}} \ell_{p,a}, \quad \mathbf{T}_p^\partial := \sum_{a \in \mathcal{A}_p} q_{R_{a,p}} h_{p,a}. \quad (2774)$$

Then

$$Q_p = \mathbf{N}_p^\partial + \mathbf{T}_p^\partial, \quad \frac{\mathcal{C}_p^{C_4}}{3} = \mathbf{N}_p^\partial + 2\mathbf{T}_p^\partial. \quad (2775)$$

In particular, the two multiplicities are recovered by

$$\mathbf{N}_p^\partial = 2Q_p - \frac{\mathcal{C}_p^{C_4}}{3}, \quad \mathbf{T}_p^\partial = \frac{\mathcal{C}_p^{C_4}}{3} - Q_p. \quad (2776)$$

The marked rank sum obeys the sharp bounds

$$3Q_p \leq \mathcal{C}_p^{C_4} \leq 6Q_p. \quad (2777)$$

Equality on the left holds exactly when $\mathbf{T}_p^\partial = 0$, and equality on the right holds exactly when $\mathbf{N}_p^\partial = 0$.

The determinant volume therefore satisfies

$$\mathcal{C}_p^{C_4} \leq \log_2 \text{Vol}_p^{\det} \leq 2\mathcal{C}_p^{C_4}. \quad (2778)$$

The left equality holds exactly when every occupied shell is timelike; the right equality holds exactly when every occupied shell is one-sided. If $Q_p > 0$, the weighted Lorentz-rank mean is

$$\bar{r}_p := \frac{\mathcal{C}_p^{C_4}}{3Q_p} = \frac{\sum_{a \in \mathcal{A}_p} q_{R_{a,p}} \text{rank } \mathbf{G}_{p,a}}{\sum_{a \in \mathcal{A}_p} q_{R_{a,p}}} = 1 + \frac{\mathbf{T}_p^\partial}{Q_p} \in [1, 2]. \quad (2779)$$

Finally, under the primewise shell reduction,

$$\text{ES} \iff \inf_{p \in \mathbb{P}_{1,12}} \mathcal{C}_p^{C_4} \geq 3. \quad (2780)$$

Proof. Equation (2758), summed over the shells, gives

$$\mathcal{E}_p^{\text{pol},123} = 3\mathbf{N}_p^\partial, \quad \mathcal{E}_p^{\text{tim},123} = 3\mathbf{T}_p^\partial.$$

The identity $\mathcal{E}_p^{\text{pol},123} + \mathcal{E}_p^{\text{tim},123} = 3Q_p$ from (2636) proves the first formula in (2775). Equation (2772) proves the second. Solving these two integer equations gives (2776).

Both multiplicities in (2774) are nonnegative. Substitution into (2775) proves (2777) and its equality statements. Since

$$\log_2 \text{Vol}_p^{\det} = 6Q_p,$$

the same bounds give (2778), with the displayed equality cases. Division of (2772) by $3Q_p$ gives (2779).

Each $\mathcal{C}_p^{C_4}$ is a nonnegative multiple of 3. By (2773), it is nonzero exactly when the primewise Erdős–Straus assertion holds. The reduction to $\mathbb{P}_{1,12}$ in theorem 14.250 now proves (2780). $\square$

Exact computation 14.328 (Marked C_4 primewise rank–volume certificates). The exact finite script and stored receipt

```
verification/verify_marked_C4_primewise_rank_volume.py,
verification/verify_marked_C4_primewise_rank_volume_output.txt
```

check 2197 three-shell packets, the two multiplicity identities, their inverse formulas, both sharp bounds and equality cases, the volume sandwich, and the support gap. The independent Lean 4.32 source and receipt

```
verification/MarkedC4PrimewiseRankVolumeCertificate.lean,
verification/MarkedC4PrimewiseRankVolumeCertificate_output.txt
```

prove the same arithmetic core in ten theorem statements, without `sorry`, `admit`, or an added axiom.

Corollary 14.329 (Residue freedom of the marked C_4 reconstruction). *Put*

$$\mathcal{K}_p := \frac{\mathcal{C}_p^{C_4}}{3}.$$

The reconstruction in (2775) is the integral linear map

$$\underbrace{\begin{pmatrix} Q_p \\ \mathcal{K}_p \end{pmatrix}}_{\mathbf{U}} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \mathbf{N}_p^\partial \\ \mathbf{T}_p^\partial \end{pmatrix}, \quad \mathbf{U}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}. \quad (2781)$$

In particular, $\mathbf{U} \in \mathrm{SL}_2(\mathbb{Z})$. For every commutative ring A , the induced map

$$\mathbf{U}_A : A^2 \longrightarrow A^2$$

is an automorphism. Thus, for every $m \geq 1$ and every $(q, k) \in (\mathbb{Z}/m\mathbb{Z})^2$, there is a unique pair $(n, t) \in (\mathbb{Z}/m\mathbb{Z})^2$ satisfying

$$q = n + t, \quad k = n + 2t, \quad n = 2q - k, \quad t = k - q. \quad (2782)$$

Consequently, no proper congruence condition on $(Q_p, \mathcal{K}_p)$ follows from the two marked reconstruction equations alone. Any arithmetic congruence restriction on the primewise image must enter through the shell data.

Proof. Direct multiplication gives

$$\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$

This identity remains valid after applying the canonical map $\mathbb{Z} \rightarrow A$, proving the assertion over every commutative ring. Taking $A = \mathbb{Z}/m\mathbb{Z}$ gives (2782). The final assertion follows because every residue pair has a unique preimage under these equations. $\square$

Exact computation 14.330 (Primewise marked- C_4 residue saturation below 10000). The exact arithmetic script and stored receipt

```
verification/verify_marked_C4_primewise_residue_freedom.py,
verification/verify_marked_C4_primewise_residue_freedom_output.txt
```

evaluate every residual shell for the 300 primes $p < 10000$, $p \equiv 1 \pmod{12}$. For each modulus $2 \leq m \leq 8$, both sets

$$\left\{ (\mathbf{N}_p^\partial, \mathbf{T}_p^\partial) \bmod m \right\}_p, \quad \left\{ (Q_p, \mathcal{K}_p) \bmod m \right\}_p$$

equal $(\mathbb{Z}/m\mathbb{Z})^2$. All four joint parity strata occur, and both $\mathbf{N}_p^\partial$ and $\mathbf{T}_p^\partial$ are positive for every prime in this finite range. The residue saturation and positivity assertions here are bounded exact calculations.

The independent Lean 4.32 source and receipt

```
verification/MarkedC4ResidueFreedomCertificate.lean,
verification/MarkedC4ResidueFreedomCertificate_output.txt
```

prove the all-ring inverse and the unique-preimage theorem, including its specialization to every $\text{ZMod}(m)$, without `sorry`, `admit`, or an added axiom.

Corollary 14.331 (Spectral–support balance of the minimum carriers). *Retain the correlation deficit d_P from (2717), the mixer–blade count b_P from (2724), and put*

$$\delta_P := m_{--}(P) - m_{+-}(P) = \frac{1}{2} \text{tr}(\mathbf{R}_P U_P).$$

Let

$$e_P := \begin{cases} 1, & \kappa_P \text{ is not a coboundary,} \\ 0, & \kappa_P \text{ is a coboundary.} \end{cases}$$

Then the four exact rows are

P	C_{16}	$C_8 \times C_2$	$C_4 \times C_4$	$C_4 \times C_2^2$
d_P	3	2	0	0
b_P	1	2	4	4
δ_P	1	2	0	0
e_P	1	1	0	0.

(2783)

They satisfy the two exact balance laws

$$\boxed{d_P + \delta_P = 4e_P, \quad b_P - \delta_P = 4(1 - e_P).} \quad (2784)$$

Equivalently,

$$\delta_P = e_P b_P, \quad d_P = e_P(4 - b_P), \quad b_P = \delta_P + 4(1 - e_P). \quad (2785)$$

Thus, on either non-split selected minimum carrier, the irreducible spectral defect equals the number of mixer–blade blocks and completes the correlation deficit to the four-cell volume. On either split selected minimum carrier, both the correlation and spectral defects vanish, and all four cells occur as blade blocks.

Proof. The rows for d_P and b_P are (2717) and (2725). The row for δ_P is (2745), and the spectral criterion in theorem 14.315 gives the row for e_P . Adding and subtracting the four columns proves (2784). Combining that law with $d_P + b_P = 4$ from (2726) proves (2785). $\square$

Exact computation 14.332 (Exact spectral–support balance certificates). The script and captured receipt

```
verification/verify_order_sixteen_spectral_support_balance.py,  
verification/verify_order_sixteen_spectral_support_balance_output.txt
```

recompute d_P from the two coordinate projections of the minimum support, b_P from its inversion orbits, e_P by exhaustive coboundary search, and δ_P from the five dihedral character inner products. They verify (2784) and (2785) in every row. The independent Lean 4.32 source and receipt

```
verification/OrderSixteenSpectralSupportBalanceCertificate.lean,  
verification/OrderSixteenSpectralSupportBalanceCertificate_output.txt
```

certify four data rows, each carrying the deficit–blade conservation law, both identities in (2784), and the equality $\delta_P = e_P b_P$, without `sorry`, `admit`, or an added axiom.

Corollary 14.333 (Purely 2-primary obstructions through order sixteen). *Under complete annihilation of the characters outside the 2-primary subgroup, every missing coefficient lies in one of the order-four, order-eight, or order-sixteen supports classified in theorems 14.298, 14.303 and 14.307, or in a non-elementary 2-primary quotient of order at least 32. No elementary-abelian quotient occurs at any order.*

Proof. The cited theorems exhaust orders four, eight, and sixteen. Theorem 14.297 excludes every elementary-abelian quotient. Any remaining finite 2-group quotient has power-of-two order strictly larger than sixteen. $\square$

Exact computation 14.334 (Exact order-sixteen support certificates). The script and captured receipt

```
verification/verify_order_sixteen_two_primary_support.py,
verification/verify_order_sixteen_two_primary_support_output.txt
```

close all five finite support monoids, enumerate their automorphism groups, verify every structural template and orbit, and prove the five minimum support cardinalities.

The independent Lean 4.32 source and receipt

```
verification/OrderSixteenSupportCertificate.lean,
verification/OrderSixteenSupportCertificate_output.txt
```

certify the five local-support counts, five complete monoid counts, five generating-nonfull counts, five minima, and five explicit attaining boxes. The certificate contains no `sorry`, `admit`, or added axiom.

The four-sheet carrier is checked independently by

```
verification/verify_order_sixteen_four_sheet_carrier.py,
verification/verify_order_sixteen_four_sheet_carrier_output.txt.
```

It verifies all four bijections and transported group laws, both nonzero cocycle identities, the absence or presence of a coboundary in each row, and every minimum-support coordinate and projection count. The Lean 4.32 source and receipt

```
verification/OrderSixteenFourSheetCarrierCertificate.lean,
verification/OrderSixteenFourSheetCarrierCertificate_output.txt
```

certify the same carrier claims in twenty-two theorem statements, without `sorry`, `admit`, or an added axiom.

Theorem 14.335 (Stabilizer-quotient character certificate for target failure). *Retain the conditions and notation of theorems 14.260 and 14.292. Suppose $\mu \in \ker \mathfrak{T}_{p,a}$, equivalently $B \cap T = \emptyset$, and put*

$$K := \text{Stab}(B), \quad Q := H/K, \quad C := B/K, \quad U := (T + K)/K. \quad (2786)$$

Define

$$\delta_K := |Q| - |C| - |U|, \quad \epsilon_K := |C| - \kappa(K). \quad (2787)$$

Then

$$\boxed{\delta_K, \epsilon_K \in \mathbb{Z}_{\geq 0}, \quad -\Sigma_T(K) = \delta_K + \epsilon_K.} \quad (2788)$$

Let

$$\bar{\mu}_K := \pi_{K*} \mu : Q \longrightarrow \mathbb{Z}_{\geq 0},$$

and let

$$q^\vee : \hat{Q} \longrightarrow \hat{H}, \quad q^\vee(\psi) = \psi \circ \pi_K$$

be the dual injection. Its Fourier coefficients are

$$\boxed{\widehat{\bar{\mu}_K}(\psi) = \hat{\mu}(q^\vee \psi) = \prod_{i=1}^s D_i(q^\vee \psi).} \quad (2789)$$

For every target coset $u \in U$,

$$\boxed{\sum_{\substack{\psi \in \hat{Q} \\ \psi \neq 1}} \hat{\mu}(q^\vee \psi) \psi(u) = -M_{p,a}.} \quad (2790)$$

In particular, whenever $U \neq \emptyset$,

$$M_{p,a} \leq \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} \prod_{i=1}^s |D_i(q^\vee \psi)|. \quad (2791)$$

Consequently, if $T \neq \emptyset$ and every subgroup $K \leq H$ with $\Sigma_T(K) \leq 0$ satisfies

$$\sum_{\substack{\psi \in \widehat{H/K} \\ \psi \neq 1}} \prod_{i=1}^s |D_i(q^\vee \psi)| < M_{p,a}, \quad (2792)$$

then the shell is occupied. A second sufficient condition is obtained from Parseval: it is enough that, for every such K ,

$$(|H/K| - 1) \left(|H/K| \sum_{\bar{h} \in H/K} (\pi_{K*} \mu(\bar{h}))^2 - M_{p,a}^2 \right) < M_{p,a}^2. \quad (2793)$$

Proof. Since K stabilizes B , the sets C and U are disjoint in Q . Hence $\delta_K \geq 0$. Kneser's theorem gives $|C| \geq \kappa(K)$, so $\epsilon_K \geq 0$. Directly,

$$\delta_K + \epsilon_K = |Q| - |U| - \kappa(K) = -\Sigma_T(K),$$

which proves (2788).

Finite rearrangement of the quotient pushforward gives

$$\widehat{\bar{\mu}_K}(\psi) = \sum_{\bar{h} \in Q} \sum_{\pi_K(h)=\bar{h}} \mu(h) \overline{\psi(\bar{h})} = \widehat{\mu}(q^\vee \psi).$$

The local factorization in (2349) proves (2789). Because $\text{supp}(\bar{\mu}_K) = C$ and $C \cap U = \emptyset$, one has $\bar{\mu}_K(u) = 0$ for every $u \in U$. Fourier inversion on Q , followed by isolation of the principal character, gives (2790). The triangle inequality gives (2791).

If the shell were empty, theorem 10.7 would supply the subgroup $K = \text{Stab}(B)$ with $\Sigma_T(K) \leq 0$, and (2791) would contradict (2792). This proves the first sufficient condition.

Finally, Cauchy–Schwarz and Parseval on Q give

$$\begin{aligned} \left(\sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} |\widehat{\bar{\mu}_K}(\psi)| \right)^2 &\leq (|Q| - 1) \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} |\widehat{\bar{\mu}_K}(\psi)|^2 \\ &= (|Q| - 1) \left(|Q| \sum_{\bar{h} \in Q} \bar{\mu}_K(\bar{h})^2 - M_{p,a}^2 \right). \end{aligned}$$

Thus (2793) implies (2792), completing the proof. $\square$

Corollary 14.336 (Target-averaged Fourier filter and strict two-target coupling). *Retain the conditions and notation of theorem 14.335, and suppose $U \neq \emptyset$. Define the target-average character*

$$\mathcal{A}_U(\psi) := \frac{1}{|U|} \sum_{u \in U} \psi(u). \quad (2794)$$

If the shell is empty, then

$$\sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} \widehat{\mu}(q^\vee \psi) \mathcal{A}_U(\psi) = -M_{p,a}, \quad (2795)$$

and consequently

$$M_{p,a} \leq \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} \left(\prod_{i=1}^s |D_i(q^\vee \psi)| \right) |\mathcal{A}_U(\psi)|. \quad (2796)$$

Therefore the strict reverse of (2796), for every $K \leq H$ with $\Sigma_T(K) \leq 0$, forces shell occupancy.

Suppose now that the two target cosets are distinct. Put

$$\bar{p}_K := \pi_K(\bar{p}) \in Q.$$

Then

$$|\mathcal{A}_U(\psi)| = \frac{|1 + \psi(\bar{p}_K)|}{2}. \quad (2797)$$

In particular, every character with $\psi(\bar{p}_K) = -1$ is removed from the simultaneous obstruction. The sufficient condition

$$\sum_{\substack{\psi \in \widehat{H/K} \\ \psi \neq 1}} \left(\prod_{i=1}^s |D_i(q^\vee \psi)| \right) \frac{|1 + \psi(\bar{p}_K)|}{2} < M_{p,a} \quad (2798)$$

is no weaker than (2792). More generally, for

$$\nu := \bar{\mu}_K, \quad d := \bar{p}_K,$$

put

$$S(\nu) := \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} |\widehat{\nu}(\psi)|, \quad S_d(\nu) := \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} |\widehat{\nu}(\psi)| \frac{|1 + \psi(d)|}{2}.$$

Then

$$S_d(\nu) = S(\nu) \iff \nu(x+d) = \nu(x) \text{ for every } x \in Q. \quad (2799)$$

For the particular quotient $K = \text{Stab}(B)$, two distinct target cosets give the strict contraction

$$\sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} \left(\prod_{i=1}^s |D_i(q^\vee \psi)| \right) \frac{|1 + \psi(\bar{p}_K)|}{2} < \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} \prod_{i=1}^s |D_i(q^\vee \psi)|. \quad (2800)$$

This inequality holds independently of shell occupancy. If both target cosets are missed, then (2796) and (2800) give the stronger chain

$$M_{p,a} \leq \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} \left(\prod_i |D_i(q^\vee \psi)| \right) \frac{|1 + \psi(\bar{p}_K)|}{2} < \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} \prod_i |D_i(q^\vee \psi)|.$$

In particular, simultaneous failure satisfies the strict bound

$$M_{p,a} < \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1}} \prod_{i=1}^s |D_i(q^\vee \psi)|. \quad (2801)$$

Consequently, when $T \neq \emptyset$, shell occupancy follows if every $K \leq H$ with $\Sigma_T(K) \leq 0$ satisfies

$$\begin{cases} \sum_{\psi \neq 1} \prod_i |D_i(q^\vee \psi)| < M_{p,a}, & t(K) = 1, \\ \sum_{\psi \neq 1} \prod_i |D_i(q^\vee \psi)| \leq M_{p,a}, & t(K) = 2. \end{cases} \quad (2802)$$

Proof. Average (2790) over $u \in U$. This gives (2795); the triangle inequality and (2789) give (2796).

When the two target cosets are distinct, they are the images of $\overline{-1}$ and $\overline{-p^{-1}}$. Factoring the first character value from their average and using $|\psi(\bar{p}_K)| = 1$ gives (2797). Since $|\mathcal{A}_U(\psi)| \leq 1$, the filtered sum is at most the unweighted sum, proving the stated comparison of the two sufficient conditions.

It remains to prove (2800). Write $\nu := \bar{\mu}_K$ and $d := \bar{p}_K \neq 0$. Every summand on the left is its unweighted counterpart multiplied by a number in $[0, 1]$. Equality of the two sums would therefore imply

$$\widehat{\nu}(\psi) \neq 0 \implies \frac{|1 + \psi(d)|}{2} = 1 \implies \psi(d) = 1$$

for every nonprincipal character. The principal character also takes the value one at d . Fourier inversion would consequently give

$$\nu(x + d) = \nu(x) \quad (x \in Q).$$

Conversely, this periodicity forces $\widehat{\nu}(\psi) = 0$ whenever $\psi(d) \neq 1$, so every nonzero term has target factor one and the two sums are equal. This proves (2799). Thus d would stabilize $\text{supp}(\nu) = C$. But

$$\text{Stab}_Q(C) = \text{Stab}_H(B)/K = \{0\}$$

because $K = \text{Stab}_H(B)$. This contradicts $d \neq 0$ and proves (2800). Combining it with the averaged necessary bound proves (2801).

If a shell satisfying (2802) were empty, choose its full support stabilizer. When $t(K) = 1$, (2791) contradicts the first line. When $t(K) = 2$, (2800) makes the filtered sum strictly smaller than the second line's unweighted sum, contradicting (2796). $\square$

Corollary 14.337 (Quantitative target-rotation gap). *Retain the conditions and notation of theorem 14.336, and suppose the two target cosets are distinct. Put*

$$\nu := \bar{\mu}_K, \quad d := \bar{p}_K, \quad m := \text{ord}_Q(d) \geq 2.$$

Define the cyclic orbital average

$$(P_d \nu)(x) := \frac{1}{m} \sum_{j=0}^{m-1} \nu(x + jd) \quad (x \in Q) \quad (2803)$$

and the rotated Fourier mass

$$W_{\text{rot}}(\nu, d) := \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1, \psi(d) \neq 1}} |\widehat{\nu}(\psi)| = \sum_{\substack{\psi \in \widehat{Q} \\ \psi \neq 1, \psi(d) \neq 1}} \prod_{i=1}^s |D_i(q^\vee \psi)|. \quad (2804)$$

Then

$$\widehat{P_d \nu}(\psi) = \begin{cases} \widehat{\nu}(\psi), & \psi(d) = 1, \\ 0, & \psi(d) \neq 1, \end{cases} \quad W_{\text{rot}}(\nu, d) = S(\nu) - S(P_d \nu), \quad (2805)$$

and

$$\boxed{S_d(\nu) \leq S(\nu) - \left(1 - \cos \frac{\pi}{m}\right) W_{\text{rot}}(\nu, d).} \quad (2806)$$

For $K = \text{Stab}(B)$,

$$W_{\text{rot}}(\nu, d) > 0.$$

If the shell is empty, then

$$M_{p,a} \leq S_d(\nu) \leq S(\nu) - \left(1 - \cos \frac{\pi}{m}\right) W_{\text{rot}}(\nu, d). \quad (2807)$$

Consequently the strict reverse of the final inequality in (2807) forces shell occupancy.

Proof. By (2803) and the definition of the Fourier transform,

$$\widehat{P_d\nu}(\psi) = \widehat{\nu}(\psi) \frac{1}{m} \sum_{j=0}^{m-1} \psi(d)^j.$$

The geometric sum is one when $\psi(d) = 1$ and zero otherwise. This proves (2805), including its mass identity.

If $\psi(d) \neq 1$, write

$$\psi(d) = e^{2\pi i k/m}, \quad 1 \leq k \leq m-1, \quad r := \min\{k, m-k\}.$$

Then $1 \leq r \leq m/2$, and

$$\frac{|1 + \psi(d)|}{2} = \cos \frac{\pi r}{m} \leq \cos \frac{\pi}{m}.$$

Split the sum defining $S_d(\nu)$ according as $\psi(d) = 1$ or $\psi(d) \neq 1$. The first part is unchanged; the second is at most $\cos(\pi/m)W_{\text{rot}}(\nu, d)$. This gives (2806).

For the full support-stabilizer quotient, vanishing of $W_{\text{rot}}(\nu, d)$ would imply $S_d(\nu) = S(\nu)$. By (2799), d would then stabilize ν and its support. This contradicts $d \neq 0$ and the trivial support stabilizer in Q . Hence the rotated mass is positive. Finally, (2796) gives the first inequality in (2807); the second is (2806). $\square$

Corollary 14.338 (Integral orbit-variance target gap). *Retain the notation of theorem 14.337, and take*

$$K = \text{Stab}(B).$$

Then the integer-valued shell measure satisfies

$$\boxed{W_{\text{rot}}(\nu, d) \geq \sqrt{|Q| \frac{m-1}{m}}.} \quad (2808)$$

Consequently

$$\boxed{S_d(\nu) \leq S(\nu) - \left(1 - \cos \frac{\pi}{m}\right) \sqrt{|Q| \frac{m-1}{m}}.} \quad (2809)$$

In particular, the shell is occupied whenever

$$\boxed{S(\nu) < M_{p,a} + \left(1 - \cos \frac{\pi}{m}\right) \sqrt{|Q| \frac{m-1}{m}}.} \quad (2810)$$

Proof. The translation $x \mapsto x + d$ decomposes Q into orbits of cardinality m . The full support quotient has no nonzero support period, so ν is nonconstant on at least one such orbit. Let

$$n_0, \dots, n_{m-1} \in \mathbb{Z}_{\geq 0}, \quad \bar{n} := \frac{1}{m} \sum_{j=0}^{m-1} n_j,$$

be its values on that orbit. The pairwise-difference identity gives

$$\sum_{j=0}^{m-1} (n_j - \bar{n})^2 = \frac{1}{m} \sum_{0 \leq i < j < m} (n_i - n_j)^2. \quad (2811)$$

If k entries are strictly larger than the minimum entry, then $1 \leq k \leq m-1$, and each of the $k(m-k)$ cross pairs differs by at least one. Hence

$$\sum_{0 \leq i < j < m} (n_i - n_j)^2 \geq k(m-k) \geq m-1.$$

Summing over all d -orbits therefore gives

$$\sum_{x \in Q} |\nu(x) - (P_d \nu)(x)|^2 \geq \frac{m-1}{m}. \quad (2812)$$

By (2805) and Parseval,

$$\sum_{\substack{\psi \in \widehat{Q} \\ \psi(d) \neq 1}} |\widehat{\nu}(\psi)|^2 = |Q| \sum_{x \in Q} |\nu(x) - (P_d \nu)(x)|^2.$$

The ℓ^1 -norm dominates the ℓ^2 -norm, so

$$W_{\text{rot}}(\nu, d) \geq \left(\sum_{\psi(d) \neq 1} |\widehat{\nu}(\psi)|^2 \right)^{1/2} \geq \sqrt{|Q| \frac{m-1}{m}}.$$

This proves (2808). Substitution into (2806) gives (2809). If the shell were empty, (2796) would imply

$$M_{p,a} \leq S(\nu) - \left(1 - \cos \frac{\pi}{m}\right) \sqrt{|Q| \frac{m-1}{m}},$$

contradicting (2810). $\square$

Corollary 14.339 (Support-boundary target gap). *Retain the notation of theorem 14.337. Put*

$$C := \text{supp } \nu$$

and let $\mathcal{O}_d$ be the set of orbits of the translation $x \mapsto x + d$ on Q . Every $O \in \mathcal{O}_d$ has cardinality m . Define

$$b_O := |C \cap O|, \quad \mathcal{B}_d(C) := \frac{1}{m} \sum_{O \in \mathcal{O}_d} b_O(m - b_O). \quad (2813)$$

Then

$$\boxed{\sum_{x \in Q} |\nu(x) - (P_d \nu)(x)|^2 \geq \mathcal{B}_d(C)}, \quad (2814)$$

and consequently

$$\boxed{W_{\text{rot}}(\nu, d) \geq \sqrt{|Q| \mathcal{B}_d(C)}}. \quad (2815)$$

It follows that

$$\boxed{S_d(\nu) \leq S(\nu) - \left(1 - \cos \frac{\pi}{m}\right) \sqrt{|Q| \mathcal{B}_d(C)}}. \quad (2816)$$

In particular, the shell is occupied whenever

$$\boxed{S(\nu) < M_{p,a} + \left(1 - \cos \frac{\pi}{m}\right) \sqrt{|Q| \mathcal{B}_d(C)}}. \quad (2817)$$

For the full support-stabilizer quotient,

$$\mathcal{B}_d(C) \geq \frac{m-1}{m},$$

so theorem 14.338 follows from this support-boundary estimate.

Proof. Fix $O \in \mathcal{O}_d$, write the values of ν on O as $n_0, \dots, n_{m-1}$, and put $\bar{n} = m^{-1} \sum_j n_j$. The pairwise identity (2811) gives

$$\sum_{j=0}^{m-1} (n_j - \bar{n})^2 = \frac{1}{m} \sum_{0 \leq i < j < m} (n_i - n_j)^2.$$

There are exactly $b_O(m - b_O)$ unordered pairs with one entry in $C \cap O$ and the other in $O \setminus C$. On each such pair, one value is a positive integer and the other is zero, so $(n_i - n_j)^2 \geq 1$. Therefore

$$\sum_{j=0}^{m-1} (n_j - \bar{n})^2 \geq \frac{b_O(m - b_O)}{m}.$$

Summing over the d -orbits proves (2814). Parseval and (2805) give

$$\sum_{\psi(d) \neq 1} |\hat{\nu}(\psi)|^2 = |Q| \sum_{x \in Q} |\nu(x) - (P_d \nu)(x)|^2 \geq |Q| \mathcal{B}_d(C).$$

The Fourier ℓ^1 -norm dominates its ℓ^2 -norm, proving (2815). Substitution into (2806) proves (2816); the empty-shell necessary bound (2796) then proves (2817).

For the full support quotient, $d \neq 0$ cannot preserve C . Thus some d -orbit has $1 \leq b_O \leq m - 1$, and

$$b_O(m - b_O) \geq m - 1.$$

This proves the final assertion. $\square$

Corollary 14.340 (Support-cardinality boundary floor). *Retain the notation of theorem 14.339, put*

$$c := |C|, \quad r_c := c \bmod m, \quad 0 \leq r_c < m,$$

and define

$$\beta_m(c) := \begin{cases} \frac{r_c(m - r_c)}{m}, & r_c \neq 0, \\ \frac{2(m - 1)}{m}, & r_c = 0 \text{ and } C + d \neq C. \end{cases} \quad (2818)$$

Then

$$\boxed{\mathcal{B}_d(C) \geq \beta_m(c).} \quad (2819)$$

For the full support-stabilizer quotient, put

$$q := |Q|, \quad t := |U|,$$

and retain the local-support quantity $\kappa(K)$ from (418). If the shell is empty, then

$$\kappa(K) \leq c \leq q - t. \quad (2820)$$

Consequently, whenever $\kappa(K) \leq q - t$,

$$\boxed{\mathcal{B}_d(C) \geq \mathcal{B}_{q,m,\kappa,t}^{\text{Kn}} := \min_{\kappa(K) \leq h \leq q-t} \beta_m(h).} \quad (2821)$$

If $\kappa(K) > q - t$, Star-Kneser forcing already proves occupancy.

Proof. Write $b_1, \dots, b_{q/m}$ for the integers $|C \cap O|$ on the d -orbits. They satisfy

$$0 \leq b_i \leq m, \quad \sum_i b_i = c, \quad m\mathcal{B}_d(C) = \sum_i b_i(m - b_i).$$

For $1 \leq a \leq b \leq m - 1$, replacing the pair (a, b) by $(a - 1, b + 1)$ changes the final sum by

$$2(a - b - 1) < 0.$$

Repeated transfers therefore place all mass in full or empty orbits and, when $r_c \neq 0$, one orbit containing r_c points. This gives the first line of (2818). If $r_c = 0$ and $C + d \neq C$, at least two orbits are partial. Every partial orbit contributes

$$b_i(m - b_i) \geq m - 1,$$

which proves the second line and (2819).

On the full support quotient, Kneser's theorem gives

$$c \geq \kappa(K).$$

If the shell is empty, $C \cap U = \emptyset$, so $c \leq q - t$. This proves (2820) and the minimum in (2821). The remaining case is exactly the strict Star-Kneser criterion. $\square$

Theorem 14.341 (Displacement-quotient forcing for a complete target fibre). *Retain the additive finite group H , local supports A_i , carrier $B = A_1 + \cdots + A_s$, and two-target set T from theorem 10.6. Put*

$$K := \text{Stab}_H(B), \quad Q := H/K, \quad \pi : H \longrightarrow Q,$$

and define

$$C := \pi(B), \quad \bar{A}_i := \pi(A_i), \quad U := \pi(T).$$

Suppose

$$U = \{u_0, u_1\}, \quad u_0 \neq u_1, \quad d := u_1 - u_0.$$

Let

$$D := \langle d \rangle, \quad Y := Q/D, \quad \rho : Q \longrightarrow Y,$$

and write

$$\tilde{A}_i := \rho(\bar{A}_i), \quad \tilde{C} := \rho(C) = \tilde{A}_1 + \cdots + \tilde{A}_s, \quad \tilde{U} := \rho(U).$$

For

$$P := \text{Stab}_Y(\tilde{C}),$$

define the projected target surplus

$$\Sigma_D(P) := 1 + \sum_{i=1}^s \left(\left| (\tilde{A}_i + P)/P \right| - 1 \right) + \left| (\tilde{U} + P)/P \right| - |Y/P|. \quad (2822)$$

If

$$\Sigma_D(P) > 0 \quad \text{and} \quad U + D = U, \quad (2823)$$

then

$$B \cap T \neq \emptyset. \quad (2824)$$

In particular, the second condition in (2823) holds whenever $|D| = 2$.

Proof. Kneser's theorem on Y , applied with $P = \text{Stab}_Y(\tilde{C})$, gives

$$\left| \tilde{C}/P \right| \geq 1 + \sum_{i=1}^s \left(\left| (\tilde{A}_i + P)/P \right| - 1 \right). \quad (2825)$$

If $\tilde{C} \cap \tilde{U} = \emptyset$, then the P -stability of $\tilde{C}$ implies

$$\left| \tilde{C}/P \right| + \left| (\tilde{U} + P)/P \right| \leq |Y/P|.$$

Together with (2825), this would give $\Sigma_D(P) \leq 0$. Hence positive projected surplus supplies

$$c \in C, \quad u \in U, \quad \rho(c) = \rho(u).$$

Thus $c - u \in D$. If $U + D = U$, then $c \in U$, so $C \cap U \neq \emptyset$.

Choose $b \in B$ and $t \in T$ with $\pi(b) = \pi(t) \in C \cap U$. Then $t - b \in K$. Since $K = \text{Stab}_H(B)$, one has $B + K = B$, and therefore $t \in B \cap T$. This proves (2824).

Finally, if $|D| = 2$, then

$$D = \{0, d\}, \quad U = \{u_0, u_0 + d\} = u_0 + D,$$

so $U + D = U$. □

Lemma 14.342 (Projected local exponent intervals). *Retain the maps*

$$H \xrightarrow{\pi} Q \xrightarrow{\rho} Y$$

from theorem 14.341. For each prime factor $\ell_i^{e_i} \parallel a$, put

$$y_i := \rho(\pi(\bar{\ell}_i)), \quad n_i := \text{ord}_Y(y_i).$$

Then

$$\tilde{A}_i = \{ky_i : -e_i \leq k \leq e_i\}, \quad |\tilde{A}_i| = \min(2e_i + 1, n_i). \quad (2826)$$

Consequently

$$\tilde{A}_i = Y \iff \langle y_i \rangle = Y \text{ and } 2e_i + 1 \geq |Y|. \quad (2827)$$

Proof. The first set identity is the image of

$$A_i = \{-e_i \bar{\ell}_i, \dots, e_i \bar{\ell}_i\}$$

under $\rho \circ \pi$. If $2e_i + 1 < n_i$, any equality $ky_i = k'y_i$ with $-e_i \leq k, k' \leq e_i$ would make $n_i \mid k - k'$, while $|k - k'| < n_i$; hence $k = k'$. If $2e_i + 1 \geq n_i$, the consecutive interval contains a complete residue system modulo n_i , so its image is exactly $\langle y_i \rangle$. This proves (2826). Equality with all of Y occurs precisely when the cyclic subgroup is Y and the interval reaches its order, proving (2827). □

Corollary 14.343 (Single-local-factor complete-fibre forcing). *Retain the conditions of theorem 14.341, and suppose*

$$U + D = U.$$

If some prime-power factor $\ell_i^{e_i} \parallel a$ satisfies

$$\langle \rho(\pi(\bar{\ell}_i)) \rangle = Y, \quad 2e_i + 1 \geq |Y|, \quad (2828)$$

then the shell is occupied.

Proof. Theorem 14.342 gives $\tilde{A}_i = Y$. Therefore

$$\tilde{C} = Y, \quad P = \text{Stab}_Y(\tilde{C}) = Y.$$

Every nonempty subset of Y has one image in Y/P , so

$$\Sigma_D(P) = 1 + \sum_{j=1}^s (1 - 1) + 1 - 1 = 1.$$

Theorem 14.341 proves occupancy. □

Theorem 14.344 (One-factor cyclic-interval boundary formula). *Suppose*

$$a = \ell^e, \quad H = \langle \bar{\ell} \rangle, \quad n := |H|, \quad L := 2e + 1 < n,$$

and suppose the two target classes lie in H and are distinct. Identify H with $\mathbb{Z}/n\mathbb{Z}$ by $\bar{\ell}^k \leftrightarrow k$, and write

$$\bar{p} = \bar{\ell}^r, \quad 1 \leq r < n, \quad g := \gcd(n, r), \quad m := \frac{n}{g}.$$

Write

$$L = ug + v, \quad 0 \leq v < g.$$

Then the centered divisor measure is the indicator of the proper cyclic interval

$$C = \{-e, -e + 1, \dots, e\} \subset \mathbb{Z}/n\mathbb{Z},$$

its support stabilizer is trivial, and

$$\mathcal{B}_d(C) = \frac{v(u+1)(m-u-1) + (g-v)u(m-u)}{m}. \quad (2829)$$

Let

$$\chi_j(\bar{\ell}) = e^{2\pi i j/n}, \quad 0 \leq j < n.$$

The Fourier and two-target coordinates are

$$\hat{\mu}(\chi_j) = \sum_{\beta=-e}^e e^{-2\pi i j \beta/n} = \begin{cases} L, & j = 0, \\ \frac{\sin(\pi L j/n)}{\sin(\pi j/n)}, & 1 \leq j < n, \end{cases} \quad (2830)$$

$$|\mathcal{A}_U(\chi_j)| = \left| \cos \frac{\pi j r}{n} \right|. \quad (2831)$$

Thus

$$M_{p,a} = L, \quad (2832)$$

$$S(\mu) = \sum_{j=1}^{n-1} \left| \frac{\sin(\pi L j/n)}{\sin(\pi j/n)} \right|, \quad (2833)$$

$$S_d(\mu) = \sum_{j=1}^{n-1} \left| \frac{\sin(\pi L j/n)}{\sin(\pi j/n)} \right| \left| \cos \frac{\pi j r}{n} \right|. \quad (2834)$$

Equations (2829)–(2834) give a closed support-boundary and Fourier certificate for every proper one-prime-power shell.

Proof. The exponents $-e, \dots, e$ are distinct modulo n because $L < n$, so the local multiplicity function is the indicator of C . A nonzero translation cannot preserve a nonempty proper consecutive cyclic interval: its binary indicator has one entrance and one exit, whereas a nontrivial period would repeat both boundary transitions. Hence the support stabilizer is trivial.

Translation by $d = \bar{p}$ is addition by r in exponent coordinates. Its orbits are the g residue classes modulo g , each of cardinality m . A consecutive interval of length $L = ug + v$ meets exactly v residue classes in $u + 1$ points and the other $g - v$ classes in u points. Substitution into (2813) proves (2829).

The first identity in (2830) is the Fourier transform of the interval indicator. Evaluation of its finite geometric sum gives the displayed sine quotient. The two target classes differ by $\bar{p} = \bar{\ell}^r$; therefore

$$|\mathcal{A}_U(\chi_j)| = \frac{|1 + e^{2\pi i j r/n}|}{2} = \left| \cos \frac{\pi j r}{n} \right|,$$

which proves (2831). Summing the absolute values proves the three mass identities. $\square$

Theorem 14.345 (Exact support-boundary certificate for $(p, R, a) = (13, 7, 5)$). For $p = 13$, $R = 7$, and $a = 5$, let μ be the centered divisor convolution of theorem 14.260. Put

$$H = (\mathbb{Z}/7\mathbb{Z})^\times = \langle 3 \rangle \cong C_6.$$

In the ordered basis $(1, 3, 3^2, \dots, 3^5)$,

$$(\mu(1), \mu(3), \dots, \mu(3^5)) = (1, 1, 0, 0, 0, 1). \quad (2835)$$

The support stabilizer is trivial, the two targets have exponents 0 and 3, and their displacement has order $m = 2$. If

$$M := \hat{\mu}(1), \quad S := \sum_{\substack{\chi \in \hat{H} \\ \chi \neq 1}} |\hat{\mu}(\chi)|, \quad S_A := \sum_{\substack{\chi \in \hat{H} \\ \chi \neq 1}} |\hat{\mu}(\chi)| |\mathcal{A}_U(\chi)|,$$

then

$$\boxed{M = 3, \quad S = 5, \quad S_A = 0, \quad \mathcal{B}_d(\text{supp } \mu) = \frac{3}{2}.} \quad (2836)$$

Consequently the support-boundary estimate gives the strict certificate

$$\boxed{S - \left(1 - \cos \frac{\pi}{2}\right) \sqrt{6 \cdot \frac{3}{2}} = 2 < 3 = M,} \quad (2837)$$

and the shell is occupied.

Proof. The divisors of $a^2 = 25$ are 1, 5, 25, and $a^{-1} = 3 \pmod{7}$. Their centered residues are 3, 1, 5, which proves (2835). Its support $\{1, 3, 5\}$ has trivial stabilizer in H .

Let $\zeta = \zeta_6$ and let $\chi_j(3) = \zeta^j$. Equation (2835) gives

$$\widehat{\mu}(\chi_j) = 1 + \zeta^{-j} + \zeta^j.$$

Exact reduction by $\Phi_6(\zeta) = \zeta^2 - \zeta + 1 = 0$ yields

$$(\widehat{\mu}(\chi_0), \dots, \widehat{\mu}(\chi_5)) = (3, 2, 0, -1, 0, 2), \quad (2838)$$

and hence $M = 3$ and $S = 5$.

Since $p = 6 = 3^3 \pmod{7}$, the two targets differ by 3^3 . Therefore

$$|\mathcal{A}_U(\chi_j)| = \frac{|1 + (-1)^j|}{2},$$

whose values for $j = 0, \dots, 5$ are

$$(1, 0, 1, 0, 1, 0).$$

Every active nonprincipal Fourier coefficient in (2838) is consequently removed, proving $S_A = 0$.

The displacement orbits in exponent coordinates are

$$\{0, 3\}, \quad \{1, 4\}, \quad \{2, 5\}.$$

Each contains exactly one point of the support $\{0, 1, 5\}$, so

$$n = 6, \quad L = 3, \quad r = 3, \quad g = 3, \quad m = 2,$$

and (2829) gives

$$\mathcal{B}_d(\text{supp } \mu) = \frac{1}{2}(1 + 1 + 1) = \frac{3}{2}.$$

Equation (2837) now follows from theorem 14.339. It contradicts the necessary inequality for an empty shell and proves occupancy. $\square$

Theorem 14.346 (Exact displacement-quotient certificate for $(p, R, a) = (37, 19, 14)$). *For $p = 37$, $R = 19$, and $a = 14 = 2 \cdot 7$, the full residual group, the support stabilizer, and its quotient are*

$$H = (\mathbb{Z}/19\mathbb{Z})^\times, \quad K = \{1, 7, 11\}, \quad Q = H/K = \{1, 2, 4, 5, 8, 10\}, \quad (2839)$$

where every displayed member of Q denotes its K -coset. In these coordinates,

$$C = \{1, 2, 10\}, \quad U = \{1, 8\}, \quad D = \langle 8 \rangle = \{1, 8\}. \quad (2840)$$

The displacement quotient $Y = Q/D$ has the three elements

$$Y = \{\{1, 8\}, \{2, 5\}, \{4, 10\}\}. \quad (2841)$$

The projected local supports satisfy

$$\rho(\overline{A}_2) = Y, \quad \rho(\overline{A}_7) = \{D\}, \quad \tilde{C} = Y, \quad \tilde{U} = \{D\}. \quad (2842)$$

Consequently

$$P = Y, \quad \Sigma_D(P) = 1 > 0. \quad (2843)$$

The shell is therefore occupied. Its divisor certificate reconstructs

$$\boxed{\frac{4}{37} = \frac{1}{14} + \frac{1}{28} + \frac{1}{1036}}. \quad (2844)$$

Proof. The divisors of $a^2 = 196$, multiplied by $a^{-1} = 15 \pmod{19}$, give the centered support

$$\{1, 2, 3, 7, 10, 11, 13, 14, 15\} = K \dot{\cup} 2K \dot{\cup} 10K.$$

The group $H = \langle 2 \rangle$ has order 18, and the image of 2 generates the cyclic quotient Q of order six. The quotient support $\{1, 2, 10\}$, corresponding to the exponent interval $\{0, 1, 5\} \subset \mathbb{Z}/6\mathbb{Z}$, has trivial stabilizer. Hence K is exactly the full support stabilizer.

Since $37 \equiv -1 \pmod{19}$, the two target classes are

$$\overline{-1} = 8K, \quad \overline{-37^{-1}} = K.$$

Their displacement is $8K$, whose square is K , proving (2840) and (2841). The factor 2 contributes

$$\overline{A}_2 = \{10K, K, 2K\},$$

which meets all three D -cosets, while

$$\overline{A}_7 = \{K\}$$

because $7 \in K$. This proves (2842). Since the projected carrier is all of Y , its stabilizer is $P = Y$, and direct substitution in (2822) gives

$$\Sigma_D(P) = 1 + (1 - 1) + (1 - 1) + 1 - 1 = 1.$$

Theorem 14.341 proves occupancy.

For the explicit reconstruction, take $v = 14 \mid a^2$ and

$$N = pa = 518, \quad \delta = p^2v = 19166, \quad \frac{N^2}{\delta} = 14.$$

One has $\delta \equiv -N \pmod{19}$. Therefore theorem 9.1 gives

$$\frac{N + \delta}{R} = 1036, \quad \frac{N + N^2/\delta}{R} = 28,$$

which proves (2844). □

Theorem 14.347 (Residual-19 complete-fibre prime-pair family). *Let q be prime with*

$$q \equiv 7 \text{ or } 11 \pmod{19},$$

and suppose

$$p := 8q - 19$$

is prime. Put $a := 2q$. Then $4a - p = 19$, and the residual-19 shell is occupied.

If $q \equiv 7 \pmod{19}$, then

$$\boxed{\frac{4}{p} = \frac{1}{2q} + \frac{1}{2q(p+1)/19} + \frac{1}{2pq(p+1)/19}}. \quad (2845)$$

If $q \equiv 11 \pmod{19}$, then

$$\boxed{\frac{4}{p} = \frac{1}{2q} + \frac{1}{2(pq+1)/19} + \frac{1}{2pq(pq+1)/19}}. \quad (2846)$$

All displayed denominators are positive integers.

Proof. The residue classes 7 and 11 are the two nonidentity members of the order-three subgroup

$$K_0 := \{1, 7, 11\} \leq (\mathbb{Z}/19\mathbb{Z})^\times.$$

The class 2 has order 18, so

$$H = \langle \bar{2}, \bar{q} \rangle = (\mathbb{Z}/19\mathbb{Z})^\times.$$

Since q is prime and occurs to exponent one in a , its centered local support is

$$A_q = \{\bar{q}^{-1}, 1, \bar{q}\} = K_0.$$

Hence

$$B = A_2 + K_0.$$

The quotient H/K_0 is cyclic of order six, generated by the image of $\bar{2}$, and the image of A_2 is the proper exponent interval $\{-1, 0, 1\}$. It has trivial translation stabilizer, so

$$\text{Stab}_H(B) = K_0.$$

Modulo K_0 ,

$$\bar{p} = \bar{8}\bar{q} = \bar{2}^3,$$

which has order two in $Q = H/K_0$. Thus the two target images form one complete fibre of $D = \langle \bar{p} \rangle$. The quotient

$$Y = Q/D$$

is cyclic of order three, generated by the image of $\bar{2}$. Since $e_2 = 1$, one has $2e_2 + 1 = 3 = |Y|$, and theorem 14.343 proves shell occupancy.

It remains to compute the denominators. Since $p + 19 = 8q$, both identities below reduce to

$$\frac{1}{2q} + \frac{19}{2pq} = \frac{p+19}{2pq} = \frac{4}{p}.$$

If $q \equiv 7 \pmod{19}$, then $p \equiv -1 \pmod{19}$, so $19 \mid p+1$, and

$$\frac{1}{2q(p+1)/19} + \frac{1}{2pq(p+1)/19} = \frac{19}{2pq}.$$

This proves (2845).

If $q \equiv 11 \pmod{19}$, then

$$pq \equiv -1 \pmod{19},$$

so $19 \mid pq+1$, and

$$\frac{1}{2(pq+1)/19} + \frac{1}{2pq(pq+1)/19} = \frac{19}{2pq}.$$

This proves (2846). □

Theorem 14.348 (Power-of-two complete-fibre ladder). *Let $e \geq 1$, let R be an odd prime, and suppose*

$$\text{ord}_R(2) = 6e + 12. \tag{2847}$$

Put

$$m := 2e + 4, \quad \omega := 2^m \pmod{R}. \tag{2848}$$

Let q be prime with

$$q \equiv \omega \text{ or } \omega^2 \pmod{R}, \tag{2849}$$

and suppose

$$p := 2^{e+2}q - R \tag{2850}$$

is prime. Put $a := 2^e q$. Then $4a - p = R$, and the residual- R shell is occupied.

If $q \equiv \omega \pmod{R}$, then

$$\boxed{\frac{4}{p} = \frac{1}{2^e q} + \frac{1}{2^e q(p+1)/R} + \frac{1}{2^e pq(p+1)/R}}. \tag{2851}$$

If $q \equiv \omega^2 \pmod{R}$, then

$$\boxed{\frac{4}{p} = \frac{1}{2^e q} + \frac{1}{2^e(pq+1)/R} + \frac{1}{2^e pq(pq+1)/R}}. \quad (2852)$$

All displayed denominators are positive integers.

Proof. Equation (2847) says that the subgroup

$$H := \langle \bar{2} \rangle \leq (\mathbb{Z}/R\mathbb{Z})^\times$$

has order $3m$. Consequently ω has order three. Since $q \equiv \omega$ or ω^2 , one has

$$K_0 := \langle \bar{q} \rangle = \{1, \omega, \omega^2\}, \quad H = \langle \bar{2}, \bar{q} \rangle.$$

The prime q occurs to exponent one in a , and hence

$$A_q = \{\bar{q}^{-1}, 1, \bar{q}\} = K_0.$$

Thus

$$B = A_2 + K_0.$$

The quotient $Q = H/K_0$ is cyclic of order m , generated by $g := \bar{2}K_0$. The image of A_2 is the exponent interval

$$I = \{g^{-e}, \dots, 1, \dots, g^e\}.$$

It has $2e + 1 = m - 3$ elements, and its complement consists of the three consecutive powers

$$Q \setminus I = \{g^{e+1}, g^{e+2}, g^{e+3}\}.$$

Let S be the translation stabilizer of I . It also stabilizes the three-element complement, so $|S|$ divides three. If $|S| = 3$, the complement is one S -coset. Its first two elements differ by g , which would put g in S and force $S = Q$. This is impossible because $m \geq 6$. Hence S is trivial, and

$$\text{Stab}_H(B) = K_0. \quad (2853)$$

Modulo K_0 , the definition of p gives

$$\bar{p}K_0 = g^{e+2} = g^{m/2}.$$

This is the unique order-two element of Q . The two target images therefore form one complete fibre of

$$D := \langle \bar{p}K_0 \rangle.$$

The displacement quotient

$$Y := Q/D$$

is cyclic of order

$$|Y| = \frac{m}{2} = e + 2$$

and is generated by the image of g . The projected 2-factor contains $2e + 1$ consecutive powers. Since

$$2e + 1 \geq e + 2$$

for $e \geq 1$, it fills Y . The single-local-factor criterion theorem 14.343 now proves shell occupancy.

It remains to compute the denominators. From

$$p + R = 2^{e+2}q$$

one gets

$$\frac{1}{2^e q} + \frac{R}{2^e pq} = \frac{p + R}{2^e pq} = \frac{4}{p}. \quad (2854)$$

If $q \equiv \omega = 2^m \pmod{R}$, then

$$p \equiv 2^{e+2}\omega = 2^{3m/2} \equiv -1 \pmod{R}.$$

Thus $R \mid p + 1$, and

$$\frac{1}{2^e q(p+1)/R} + \frac{1}{2^e pq(p+1)/R} = \frac{R}{2^e pq}.$$

Together with (2854), this proves (2851).

If $q \equiv \omega^2 \pmod{R}$, then

$$pq \equiv 2^{e+2}\omega^4 = 2^{e+2}\omega = 2^{3m/2} \equiv -1 \pmod{R}.$$

Thus $R \mid pq + 1$, and

$$\frac{1}{2^e(pq+1)/R} + \frac{1}{2^e pq(pq+1)/R} = \frac{R}{2^e pq}.$$

This proves (2852). □

Corollary 14.349 (Unique first rung and a compact higher-rung certificate). *At $e = 1$, condition (2847) forces $R = 19$, and theorem 14.348 is exactly theorem 14.347.*

At $e = 3$, the data

$$R = 331, \quad q = 31, \quad p = 661, \quad a = 248$$

satisfy the ladder conditions and give

$$\boxed{\frac{4}{661} = \frac{1}{248} + \frac{1}{496} + \frac{1}{327856}}. \quad (2855)$$

Proof. For $e = 1$, a residual prime satisfying $\text{ord}_R(2) = 18$ divides

$$2^{18} - 1 = 3^3 \cdot 7 \cdot 19 \cdot 73.$$

The orders at 3, 7, 19, 73 are respectively 2, 3, 18, 9, so $R = 19$. Here $m = 6$ and

$$2^m \equiv 7, \quad 2^{2m} \equiv 11 \pmod{19}.$$

For $e = 3$, one has

$$2^{15} \equiv -1 \pmod{331}, \quad 2^{30} \equiv 1 \pmod{331}.$$

The proper even divisors 2, 6, 10 of 30 are excluded by direct evaluation, so $\text{ord}_{331}(2) = 30$. Moreover

$$m = 10, \quad 2^{10} \equiv 31 \pmod{331}, \quad 2^{e+2}q - R = 32 \cdot 31 - 331 = 661.$$

Trial division by the primes at most 23 proves that 661 is prime. Substitution in (2851) gives (2855). □

Theorem 14.350 (Cubic-total cofactor closure of the power-of-two ladder). *Retain e , R , m , and ω from theorem 14.348. Let $b \geq 2$ be an integer whose total residue is cubic:*

$$\bar{b} \in K_0 := \{1, \omega, \omega^2\} \subset (\mathbb{Z}/R\mathbb{Z})^\times, \quad (2856)$$

and suppose that at least one prime divisor $\ell \mid b$ satisfies

$$\bar{\ell} \in \{\omega, \omega^2\}. \quad (2857)$$

The other prime divisors of b are unrestricted modulo R . If

$$p := 2^{e+2}b - R \quad (2858)$$

is prime, then the residual- R shell at

$$a := 2^e b$$

is occupied.

More precisely, define a positive divisor $u \mid a^2$ as follows:

$$u = \begin{cases} a, & b \equiv \omega \pmod{R}, \\ 2^e, & b \equiv \omega^2 \pmod{R}, \\ a/\ell, & b \equiv 1 \pmod{R}, \quad \ell \mid b, \quad \ell \equiv \omega \pmod{R}, \\ a\ell, & b \equiv 1 \pmod{R}, \quad \ell \mid b, \quad \ell \equiv \omega^2 \pmod{R}. \end{cases} \quad (2859)$$

In the last two rows, any prime divisor with nonidentity residue may be chosen. Put

$$v := \frac{a^2 p^2}{u}. \quad (2860)$$

Then

$$R \mid ap + u, \quad R \mid ap + v, \quad (2861)$$

and

$$\boxed{\frac{4}{p} = \frac{1}{a} + \frac{1}{(ap+u)/R} + \frac{1}{(ap+v)/R}}. \quad (2862)$$

Proof. Choose a prime ℓ satisfying (2857), and write $\ell^f \parallel b$. Its centered local divisor-ratio support is

$$A_{\ell^f} = \{\bar{\ell}^{-f}, \dots, 1, \dots, \bar{\ell}^f\}.$$

Since $f \geq 1$, the exponents $-1, 0, 1$ occur and

$$A_{\ell^f} = K_0.$$

Every other prime-power support contains its centered exponent zero. The local 2-support of $a = 2^e b$ contains the $2e + 1$ powers coming from 2^e , even when b is even. Consequently the full centered support B_{full} contains the embedded carrier

$$B_0 := A_2 + K_0.$$

The support and stabilizer calculation in theorem 14.348 gives

$$\text{Stab}_H(B_0) = K_0, \quad H/K_0 \cong C_m.$$

Because $\bar{b} \in K_0$, equation (2858) gives

$$\bar{p}K_0 = g^{e+2} = g^{m/2}.$$

Thus the targets form one complete order-two fibre. The displacement quotient has order $e + 2$, whereas the projected 2-factor contains $2e + 1$ consecutive powers and fills that quotient. Hence theorem 14.343 produces a target representation in B_0 . Since $B_0 \subseteq B_{\text{full}}$, the full shell is occupied.

It remains to prove the explicit certificate. Since $\text{ord}_R(2) = 3m$ and m is even,

$$-1 \equiv 2^{3m/2} \pmod{R}.$$

Consequently

$$-2^{e+2} \equiv 2^{3m/2+e+2} = 2^{2m} = \omega^2 \pmod{R}. \quad (2863)$$

Since $p \equiv 2^{e+2}b \pmod{R}$, it follows that

$$-p \equiv \omega^2 b \pmod{R}.$$

Each row of (2859) now gives

$$\frac{u}{a} \equiv \omega^2 b \equiv -p \pmod{R}. \quad (2864)$$

Indeed, the four ratios u/a are, respectively,

$$1, \quad b^{-1}, \quad \ell^{-1}, \quad \ell,$$

and their stated residue hypotheses give the required value. The first three divisibility assertions $u \mid a^2$ are immediate. In the fourth row, $\ell \mid a$, so $a\ell \mid a^2$. Thus v in (2860) is a positive integer.

Equation (2864) gives $R \mid ap + u$. Since $u \mid a^2$ and $R \nmid a$, the residue of u is invertible modulo R , and

$$v = \frac{(ap)^2}{u} \equiv \frac{(ap)^2}{-ap} = -ap \pmod{R}.$$

This proves (2861). Finally,

$$ap + v = ap + \frac{(ap)^2}{u} = \frac{ap(ap + u)}{u},$$

and therefore

$$\frac{1}{(ap + u)/R} + \frac{1}{(ap + v)/R} = \frac{R}{ap + u} + \frac{Ru}{ap(ap + u)} = \frac{R}{ap}.$$

Adding $1/a$ and using $p + R = 4a$ proves (2862). $\square$

Corollary 14.351 (All three cubic cofactor classes at $R = 19$). *The three composite cofactors*

$$49 = 7^2, \quad 1141 = 7 \cdot 163, \quad 2629 = 11 \cdot 239$$

represent, respectively, the residue classes

$$11, \quad 1, \quad 7 \pmod{19}.$$

They give the three exact decompositions

$$\frac{4}{373} = \frac{1}{98} + \frac{1}{1924} + \frac{1}{35164948}, \quad (2865)$$

$$\frac{4}{9109} = \frac{1}{2282} + \frac{1}{1094056} + \frac{1}{69760292728}, \quad (2866)$$

$$\frac{4}{21013} = \frac{1}{5258} + \frac{1}{5815348} + \frac{1}{122197907524}. \quad (2867)$$

Proof. The corresponding choices in (2859) are

$$u = 2, \quad u = 2282/7 = 326, \quad u = 5258.$$

The exact checker cited below verifies the three factorizations, the primality of 373, 9109, and 21013, every divisibility in (2861), and the three rational identities. The companion Lean certificate proves the three primality statements and the three identities independently. $\square$

Corollary 14.352 (Mixed prime residues in all three cubic total classes). *At $R = 19$, the cofactors*

$$220 = 2^2 \cdot 5 \cdot 11, \quad 286 = 2 \cdot 11 \cdot 13, \quad 330 = 2 \cdot 3 \cdot 5 \cdot 11$$

have total residues

$$11, \quad 1, \quad 7 \pmod{19},$$

respectively. Each contains the nonidentity cubic prime residue 11, while its remaining prime residues need not lie in $\{1, 7, 11\}$. They give

$$\frac{4}{1741} = \frac{1}{440} + \frac{1}{40318} + \frac{1}{15442600360}, \quad (2868)$$

$$\frac{4}{2269} = \frac{1}{572} + \frac{1}{68640} + \frac{1}{14158560}, \quad (2869)$$

$$\frac{4}{2621} = \frac{1}{660} + \frac{1}{91080} + \frac{1}{238720680}. \quad (2870)$$

Proof. For the three cofactors, the choices in (2859) are

$$u = 2, \quad u = 572 \cdot 11 = 6292, \quad u = 660.$$

The exact checker verifies the displayed factorizations, all three total residues, the cubic witness residue, the embedded support inclusion, the primality of 1741, 2269, and 2621, and the three denominator identities. The companion Lean certificate proves the three primality statements and rational identities independently. $\square$

Theorem 14.353 (Congruence exclusion for the power-of-two ladder). *Let $e \geq 1$, let R be prime, and suppose*

$$\text{ord}_R(2) = 6e + 12. \quad (2871)$$

For every integer b , put

$$p = 2^{e+2}b - R. \quad (2872)$$

Then

$$R \bmod 8 \in \{1, 3, 5\}, \quad p \bmod 8 \in \{7, 5, 3\}, \quad (2873)$$

where the entries in the second set are the negatives modulo 8 of the corresponding entries in the first. In particular,

$$p \not\equiv 1 \pmod{8}. \quad (2874)$$

Proof. The prime R is odd because the order of 2 modulo R is defined. Suppose $R \equiv 7 \pmod{8}$. The supplementary law for the quadratic character of 2 gives

$$\left(\frac{2}{R}\right) = 1;$$

see [20, Theorem 95]. Euler's criterion then gives

$$2^{(R-1)/2} \equiv 1 \pmod{R}.$$

Therefore

$$\text{ord}_R(2) \mid \frac{R-1}{2}.$$

The integer $(R-1)/2$ is odd when $R \equiv 7 \pmod{8}$, whereas the order in (2871) is even. This contradiction excludes $R \equiv 7 \pmod{8}$, leaving

$$R \bmod 8 \in \{1, 3, 5\}.$$

Since $e \geq 1$, one has $8 \mid 2^{e+2}b$. Equation (2872) therefore gives

$$p \equiv -R \pmod{8},$$

which proves (2873) and (2874). $\square$

Corollary 14.354 (Existing congruence coverage of the power-of-two ladder). *Every prime p produced under the conditions of theorem 14.350 belongs to one of the previously certified classes*

$$p \equiv 3 \pmod{4}, \quad p \equiv 5 \pmod{12}, \quad p \equiv 13 \pmod{24}. \quad (2875)$$

Consequently, the ladder supplies new support-theoretic structure and explicit divisor certificates inside known classes, while adding no prime congruence class to the coverage already established in theorems 14.48 and 14.93.

Proof. By theorem 14.353, the residue of p modulo 8 is 3, 5, or 7. The residues 3 and 7 give $p \equiv 3 \pmod{4}$.

Now suppose $p \equiv 5 \pmod{8}$. A prime in this class has residue 5 or 13 modulo 24: the third lift, $21 \pmod{24}$, is divisible by 3. The first lift is $5 \pmod{12}$, and the second is $13 \pmod{24}$. The explicit formulas in theorem 14.48 cover the first two classes in (2875), and theorem 14.93 covers the third. $\square$

Theorem 14.355 (Exact target-coupled certificate for $(p, R, a) = (37, 11, 12)$). *For $p = 37$, $R = 11$, and $a = 12 = 2^2 \cdot 3$, let μ be the centered divisor convolution of theorem 14.260. Then*

$$H = (\mathbb{Z}/11\mathbb{Z})^\times = \langle 2 \rangle \cong C_{10}, \quad \text{Stab}(\text{supp } \mu) = \{1\},$$

and the two target classes are distinct. If

$$M := \widehat{\mu}(1), \quad S := \sum_{\substack{\chi \in \widehat{H} \\ \chi \neq 1}} |\widehat{\mu}(\chi)|,$$

and

$$S_A := \sum_{\substack{\chi \in \widehat{H} \\ \chi \neq 1}} |\widehat{\mu}(\chi)| \left| \frac{\chi(\overline{-1}) + \chi(\overline{-p^{-1}})}{2} \right|,$$

then

$$\boxed{M = 15, \quad S = 15, \quad S_A = 3 + 4\sqrt{5} < 15.} \quad (2876)$$

Thus the strict unweighted test is inconclusive, while the target-averaged Fourier test proves that the shell is occupied. The strict gap is

$$M - S_A = 4(3 - \sqrt{5}) > 0. \quad (2877)$$

Proof. Put $g = 2 \bmod 11$, so that $3 = g^8$. In the ordered basis $(1, g, \dots, g^9)$, direct convolution of

$$\beta_2 \in \{-2, -1, 0, 1, 2\}, \quad \beta_3 \in \{-1, 0, 1\}$$

gives

$$(\mu(1), \mu(g), \dots, \mu(g^9)) = (3, 2, 2, 1, 1, 0, 1, 1, 2, 2). \quad (2878)$$

Hence

$$\text{supp } \mu = H \setminus \{g^5\}, \quad \text{Stab}(\text{supp } \mu) = \{1\}.$$

Since $p = g^2$,

$$\overline{-1} = g^5, \quad \overline{-p^{-1}} = g^3.$$

Let $\zeta = e^{2\pi i/10}$ and $\chi_j(g) = \zeta^j$. The local factors are

$$D_{2,j} = \sum_{\beta=-2}^2 \zeta^{-j\beta}, \quad D_{3,j} = \sum_{\beta=-1}^1 \zeta^{-8j\beta}.$$

Writing $s = \sqrt{5}$, exact reduction in $\mathbb{Q}[\zeta]/(\Phi_{10})$ gives

j	$\widehat{\mu}(\chi_j)$	$ (\chi_j(g^5) + \chi_j(g^3))/2 $
0	15	1
1, 9	$3 + s$	$(1 + s)/4$
2, 4, 6, 8	0	irrelevant
3, 7	$3 - s$	$(s - 1)/4$
5	3	1.

(2879)

All displayed nonzero transform values are positive because $2 < s < 3$. Therefore

$$S = 2(3 + s) + 2(3 - s) + 3 = 15,$$

whereas

$$\begin{aligned} S_A &= 2(3 + s) \frac{1 + s}{4} + 2(3 - s) \frac{s - 1}{4} + 3 \\ &= 3 + 4s. \end{aligned}$$

The inequality $s < 3$ gives

$$3 + 4s < 15, \quad 15 - (3 + 4s) = 4(3 - s) > 0.$$

An empty shell would contradict (2796). This proves shell occupancy and both identities. $\square$

Exact computation 14.356 (Exact and bounded target-filter certificates). The exact cyclotomic checker and its receipt

```
verification/verify_p37_R11_target_averaged_character_certificate.py,
verification/verify_p37_R11_target_averaged_character_certificate_output.txt
```

recompute the centered multiplicity measure, both targets, every local factor, every row of (2879), and the two masses in (2876) inside $\mathbb{Q}[\zeta_{10}]/(\Phi_{10})$. The Lean 4.32 source and receipt

```
verification/P37R11TargetAveragedCertificate.lean,
verification/P37R11TargetAveragedCertificate_output.txt
```

certify the radical arithmetic and the strict positive gap without a proof placeholder or added axiom. The exact C_6 checker and its receipt

```
verification/verify_p13_R7_support_boundary_certificate.py,
verification/verify_p13_R7_support_boundary_certificate_output.txt
```

verify the centered support, the six cyclotomic Fourier coefficients in $\mathbb{Q}[\zeta_6]/(\Phi_6)$, the two-target filter, all three displacement orbits, and the strict support-boundary certificate in (2837).

The one-factor interval checker and its receipt

```
verification/verify_one_factor_cyclic_interval_boundary.py,
verification/verify_one_factor_cyclic_interval_boundary_output.txt
```

verify the support stabilizer, orbit-count distribution, and boundary formula in 679480 interval/displacement triples for every cyclic order through 160. The Lean 4.32 source and receipt

```
verification/SupportBoundaryCardinalityCertificate.lean,
verification/SupportBoundaryCardinalityCertificate_output.txt
```

prove the strict boundary-mass transfer identity, the lower bound $b(m - b) \geq m - 1$ for every partial orbit, and the rational $(13, 7, 5)$ boundary arithmetic. The formal source contains no proof placeholder or added axiom.

The displacement-quotient checker and its receipt

```
verification/verify_displacement_quotient_target_forcing.py,
verification/verify_displacement_quotient_target_forcing_output.txt
```

verify the complete-target-fibre implication in 1159599 exhaustive cyclic assertions through ambient order eight, including 894371 positive-surplus cases. They independently recover every group, support, quotient, projected local support, period, and surplus in the $(13, 7, 5)$ and $(37, 19, 14)$ certificates. The Lean 4.32 source and receipt

```
verification/DisplacementQuotientTargetCertificate.lean,
verification/DisplacementQuotientTargetCertificate_output.txt
```

prove that a hit in the projected complete target fibre pulls back to an actual carrier-target hit. The formal source contains no proof placeholder or added axiom.

The residual-19 family checker and its receipt

```
verification/verify_residual_nineteen_complete_fibre_family.py,
verification/verify_residual_nineteen_complete_fibre_family_output.txt
```

verify the local support, full stabilizer, two quotient stages, complete target fibre, projected local supports, and exact unit-fraction identity for every prime pair with $q \leq 200000$. There are 226 such

pairs: 115 in the residue-7 branch and 111 in the residue-11 branch. The first members are

$$\frac{4}{37} = \frac{1}{14} + \frac{1}{28} + \frac{1}{1036}, \quad (2880)$$

$$\frac{4}{4933} = \frac{1}{1238} + \frac{1}{321424} + \frac{1}{981476862448}. \quad (2881)$$

The Lean 4.32 source and receipt

```
verification/ResidualNineteenCompleteFibreFamilyCertificate.lean,
verification/ResidualNineteenCompleteFibreFamilyCertificate_output.txt
```

prove both rational denominator identities and the $p = 37$ instance. The formal source contains no proof placeholder or added axiom.

The power-of-two ladder checker and its receipt

```
verification/verify_power_two_complete_fibre_ladder.py,
verification/verify_power_two_complete_fibre_ladder_output.txt
```

verify 1600 abstract cyclic assertions, 154 exact residual-group assertions, and 36 exact arithmetic-certificate assertions. Through $e = 8$, the order condition finds the eleven residual rungs

e	R
1	19
2	241
3	331
4	37, 109
5	5419
6	97, 673
7	87211
8	61, 1321.

The checker reconstructs the full subgroup, support, stabilizer, target fibre, displacement quotient, and denominator identity at every tested prime-pair certificate. In particular, it verifies (2855) exactly. The Lean 4.32 source and receipt

```
verification/PowerTwoCompleteFibreLadderCertificate.lean,
verification/PowerTwoCompleteFibreLadderCertificate_output.txt
```

prove the two rational denominator schemas, the midpoint and interval bounds

$$e + 2 = \frac{2e + 4}{2}, \quad e + 2 \leq 2e + 1, \quad 3(2e + 4) = 6e + 12,$$

and the exact $4/661$ identity. The formal source contains no proof placeholder or added axiom.

The cubic-cofactor extension checker and its receipt

```
verification/verify_power_two_cubic_cofactor_ladder.py,
verification/verify_power_two_cubic_cofactor_ladder_output.txt
```

verify 400 abstract cubic-support assertions, one exact composite-cofactor support template on each of the eleven rungs through $e = 8$, and 19 prime arithmetic certificates. Six of the $R = 19$ certificates contain prime factors outside K_0 , including three with an additional 2-factor. Every certificate reconstructs the full cofactor support, embedded cubic carrier, carrier stabilizer, complete target fibre, explicit divisor u , complementary divisor v , denominator integrality, and rational identity. The checker also verifies at every rung and certificate that

$$R \bmod 8 \in \{1, 3, 5\}, \quad p \bmod 8 \in \{7, 5, 3\},$$

in agreement with theorem 14.353. The Lean 4.32 source and receipt

```
verification/PowerTwoCubicCofactorCertificate.lean,  
verification/PowerTwoCubicCofactorCertificate_output.txt
```

prove the general divisor-split identity, the exponent arithmetic behind $-2^{e+2} \equiv \omega^2$, the divisor-exponent bound, ten exact rational certificates, and nine primality statements. The formal source contains no proof placeholder or added axiom.

The separate bounded diagnostic

```
verification/verify_target_averaged_filter_effectiveness.py,  
verification/verify_target_averaged_filter_effectiveness_output.txt,  
verification/verify_target_averaged_filter_effectiveness_output_160dps.txt
```

evaluates the full support-stabilizer quotient for every residual shell of every prime $p \equiv 1 \pmod{12}$, $p \leq 150$. Of the 120 shells having at least one target coset, 94 are empty and 26 are occupied. The old strict test certifies 10 shells; the target-averaged strict test certifies 15, with five occupied shells certified by the latter and not the former. Exactly 103 of the scanned quotients have two distinct target cosets, and all 103 exhibit the strict contraction in (2800). The integral orbit bound (2808) and the phase-gap inequality pass in all 103 cases. The support-boundary variance inequality (2814) also passes in all 103 cases. The support-cardinality and Star–Kneser interval floors also pass in all applicable cases. Among the 15 proper one-factor two-target shells, the closed interval formula (2829) passes in every case. Only $(13, 7, 5)$ is certified there by the support-boundary or exact target-filter test. The resulting quotient-size/order certificate proves occupancy for two shells, exactly the two already caught by the nonstrict unweighted criterion in this bounded population. The support-boundary certificate proves occupancy for three shells and adds exactly $(13, 7, 5)$, which is caught by neither of those two earlier criteria. The complete-target-fibre projected criterion proves two shells: $(13, 7, 5)$ and $(37, 19, 14)$. The latter is missed by the uniform, nonstrict, and support-boundary criteria, so it is one new certificate beyond the preceding hierarchy. Independent runs at 100 and 160 decimal digits give the same classification. No empty shell violates the averaged necessary inequality. These counts are a bounded numerical audit; the strict-contraction, integral orbit-gap, support-boundary, and complete-target-fibre statements are proved above.

Exact computation 14.357 (Lean certificate for the averaged two-target filter). The Lean 4.32 source and captured receipt

```
verification/TwoTargetAveragedFourierCertificate.lean,  
verification/TwoTargetAveragedFourierCertificate_output.txt
```

prove the finite complex two-target triangle bound, its division by two, and the paired-phase filter

$$M \leq \sum_j |c_j| \frac{|1 + r_j|}{2}$$

whenever the two missed-target phases differ by the unit phase r_j . The certificate contains no `sorry`, `admit`, or added axiom.

Exact computation 14.358 (Lean certificate for strict target averaging). The Lean 4.32 source and captured receipt

```
verification/StrictTargetAveragingCertificate.lean,  
verification/StrictTargetAveragingCertificate_output.txt
```

prove the finite weighted strict-contraction step used in (2800): nonnegative weights multiplied by factors at most one have a strictly smaller total whenever one positive weight receives a factor strictly below one. The support-period argument in the proof of theorem 14.336 supplies that active rotated character. The certificate contains no `sorry`, `admit`, or added axiom.

Exact computation 14.359 (Lean certificate for the quantitative target gap). The Lean 4.32 source and captured receipt

```
verification/TargetRotationGapCertificate.lean,
verification/TargetRotationGapCertificate_output.txt
```

prove the finite weighted inequality underlying (2806) and prove that a positive rotated mass and a coefficient strictly below one force strict total contraction. They also certify the final contradiction obtained from any uniform lower bound for the rotated mass, as used in (2810). The certificate contains no `sorry`, `admit`, or added axiom.

Exact computation 14.360 (Lean certificate for the quotient obstruction). The Lean 4.32 source and captured receipt

```
verification/TwoTargetQuotientCertificate.lean,
verification/TwoTargetQuotientCertificate_output.txt
```

formally certify the two nonnegative defect summands and their exact sum, the finite principal/nonprincipal decomposition at a missed target, the resulting principal-mass lower bound, the incompatibility of that lower bound with strict Cauchy–Schwarz/Parseval domination, and the quantitative one-third binary-sheet positivity gap. It also certifies that a power-of-two quotient order different from one and two is at least four. The certificate uses no proof placeholder or added axiom. The quotient-pushforward and annihilator-character identities that supply its inputs are checked independently by theorem 14.293.

Theorem 14.361 (Three-origin Hadamard cone and the count–amplitude rapidity defect). *Let*

$$\Phi(s, m) := \left(\frac{s + m}{2}, \frac{s - m}{2} \right). \quad (2882)$$

Then

$$\Phi : \mathbb{R}_{\geq 0}^2 \xrightarrow{\sim} \{(t, x) \in \mathbb{R}^2 : t \geq 0, t^2 - x^2 \geq 0\} \quad (2883)$$

is a bijection, with inverse

$$s = t + x, \quad m = t - x, \quad (2884)$$

and satisfies

$$t^2 - x^2 = sm. \quad (2885)$$

For the weighted three-origin vector

$$(\alpha_0, \alpha_1, \alpha_2) = (\alpha_2, \alpha_1, \alpha_2),$$

put

$$s_{\text{amp}} := \alpha_0 + \alpha_2 = 2\alpha_2, \quad m_{\text{amp}} := \alpha_1. \quad (2886)$$

Then

$$\Phi(s_{\text{amp}}, m_{\text{amp}}) = (\lambda_+, \lambda_-). \quad (2887)$$

For the origin-count vector

$$(m_0, m_1, m_2) = (m_2, m_1, m_2),$$

put

$$s_{\text{cnt}} := m_0 + m_2 = 2m_2, \quad m_{\text{cnt}} := m_1 \quad (2888)$$

and

$$(\kappa_+, \kappa_-) := \Phi(s_{\text{cnt}}, m_{\text{cnt}}) = \left(m_2 + \frac{m_1}{2}, m_2 - \frac{m_1}{2} \right). \quad (2889)$$

Thus the count and amplitude vectors are carried by the same Hadamard map.

Assume both target orbit types are occupied, and define their mean orbit weights by

$$\bar{w}_{02} := \frac{\alpha_2}{m_2}, \quad \bar{w}_{11} := \frac{\alpha_1}{m_1}. \quad (2890)$$

The count rapidity and amplitude rapidity are

$$\vartheta_{\text{cnt}} := \frac{1}{2} \log\left(\frac{2m_2}{m_1}\right), \quad \vartheta_{\text{amp}} := \frac{1}{2} \log\left(\frac{2\alpha_2}{\alpha_1}\right), \quad (2891)$$

and obey the exact defect identity

$$\vartheta_{\text{amp}} - \vartheta_{\text{cnt}} = \frac{1}{2} \log\left(\frac{\bar{w}_{02}}{\bar{w}_{11}}\right). \quad (2892)$$

For the 12289 shell,

$$(m_0, m_1, m_2) = (3, 6, 3), \quad (\kappa_+, \kappa_-) = (6, 0), \quad \vartheta_{\text{cnt}} = 0. \quad (2893)$$

Its amplitude rapidity and its entire rapidity defect are

$$\vartheta_{\text{amp}} = \vartheta_{\text{amp}} - \vartheta_{\text{cnt}} = \frac{1}{2} \log\left(\frac{5176199252561022635}{251615052171246869483}\right) < 0. \quad (2894)$$

Proof. If $s, m \geq 0$, then

$$t = \frac{s+m}{2} \geq 0, \quad t+x = s \geq 0, \quad t-x = m \geq 0.$$

Hence $t \geq |x|$. Conversely, $t \geq |x|$ makes the two coordinates in (2884) nonnegative. Direct multiplication gives

$$t^2 - x^2 = \frac{(s+m)^2 - (s-m)^2}{4} = sm.$$

This proves (2883)–(2885).

The exterior-origin equality $\alpha_0 = \alpha_2$ and (2376) give

$$\Phi(2\alpha_2, \alpha_1) = \left(\alpha_2 + \frac{\alpha_1}{2}, \alpha_2 - \frac{\alpha_1}{2}\right) = (\lambda_+, \lambda_-).$$

The count identity follows by the same calculation with (α_2, α_1) replaced by (m_2, m_1) .

Since α_2 is the sum of the m_2 exterior-orbit weights and $\alpha_1/2$ is the sum of the $m_1/2$ middle-orbit weights,

$$\alpha_2 = m_2 \bar{w}_{02}, \quad \alpha_1 = m_1 \bar{w}_{11}.$$

Therefore

$$\frac{2\alpha_2}{\alpha_1} = \frac{2m_2}{m_1} \frac{\bar{w}_{02}}{\bar{w}_{11}},$$

and logarithms prove (2892).

For 12289, substitution of $(m_0, m_1, m_2) = (3, 6, 3)$ proves (2893). Equation (2579) then proves (2894). $\square$

Proposition 14.362 (Exact $N = 3$ rank-one refinement of a singleton modular channel). *Let $N = 3$, $A = \{a\}$, and write $R_c := \frac{1}{3}v_c v_c^* = P_{-c}$. Then*

$$\Pi_A M_3 Q_A = \widehat{\bigoplus_{b \in C_3 \setminus \{a\}} R_a M_3 R_b}, \quad (2895)$$

$$Q_A M_3 \Pi_A = \widehat{\bigoplus_{b \in C_3 \setminus \{a\}} R_b M_3 R_a}. \quad (2896)$$

Every summand is one-dimensional. For the cyclic shift $Ue_j = e_{j+1}$, conjugation by U acts on these lines by

$$\text{Ad}_U(f_a f_b^*) = \omega_3^{b-a} f_a f_b^*, \quad (2897)$$

$$\text{Ad}_U(f_b f_a^*) = \omega_3^{a-b} f_b f_a^*, \quad (2898)$$

where $f_c = v_c/\sqrt{3}$. The ordered pair (Π_A, Q_A) determines each two-dimensional channel in (2895) and (2896). The specified cyclic shift resolves it into its two character lines. Identification with one blade line additionally requires the choice of one complementary character b .

Proof. Fourier orthogonality gives

$$\Pi_A = R_a, \quad Q_A = \sum_{b \neq a} R_b.$$

Distributing the two products proves (2895) and (2896). Each R_c has rank one, so every displayed summand is one-dimensional. Finally, $U f_c = \omega_3^{-c} f_c$, and hence

$$U(f_a f_b^*) U^{-1} = \omega_3^{-a} \omega_3^b f_a f_b^*.$$

This proves (2897); transposition gives (2898). $\square$

The Fourier carriage, the ordered $N = 2$ channel factors, the blade source–range products, and the $N = 3$ character-line decomposition are checked exactly in `verification/verify_fourier_blade_channels.py`. The captured receipt is `verification/verify_fourier_blade_channels_output.txt`.

Remark 14.363 (Outer support and internal modular sign). The set A records supported universes relative to τ . The signs $\pm \log \lambda$ record the two directed off-diagonal channels for that fixed support projection. A change of A and a change between the two off-diagonal channels are separate operations.

All support subsets for $N = 1, 2, 3$, the four channel actions, and the two antiunitary C_3 -transport laws are checked over the exact cyclotomic field $\mathbb{Q}(\omega_3)$ in `verification/verify_support_indexed_modular_family.py`. The captured receipt is `verification/verify_support_indexed_modular_family_output.txt`.

For the exact test values

$$\rho_1 = (1), \quad \rho_2 = \text{diag}(2, 3), \quad \rho_3 = \text{diag}(5, 7, 11),$$

the modular spectra are respectively

$$\text{Spec}(\Delta_1) = \{1\},$$

$$\text{Spec}(\Delta_2) = \{1, 1, 2/3, 3/2\},$$

$$\text{Spec}(\Delta_3) = \{1, 1, 1, 5/7, 5/11, 7/5, 7/11, 11/5, 11/7\}.$$

These identities and Proposition 14.2 are checked exactly by `verification/verify_sectorial_modular_ladder.py`; its captured output is `verification/verify_sectorial_modular_ladder_output.txt`.

Proposition 14.364 (Consecutive-prime block continuation). *Let p_j denote the j -th prime. Define the selected family*

$$\rho_1^{\text{pb}} = (1), \quad \rho_n^{\text{pb}} = \text{diag}(p_{n(n-1)/2}, p_{n(n-1)/2+1}, \dots, p_{n(n+1)/2-1}), \quad n \geq 2. \quad (2899)$$

Then the first three members are the test values displayed above, and the algebraic spectrum of the associated Hilbert–Schmidt modular operator is the multiset

$$\text{Spec}_{\text{alg}}(\Delta_n^{\text{pb}}) = \left\{ \frac{p_a}{p_b} : a, b \in I_n \right\}, \quad I_n = \left\{ \frac{n(n-1)}{2}, \dots, \frac{n(n+1)}{2} - 1 \right\}. \quad (2900)$$

The eigenvalue 1 has multiplicity n . Every off-diagonal eigenvalue is simple, transposition sends p_a/p_b to p_b/p_a , and

$$\det(\Delta_n^{\text{pb}}) = 1. \quad (2901)$$

Proof. For $n = 2$, the index set in (2899) is $I_2 = \{1, 2\}$; for $n = 3$, it is $I_3 = \{3, 4, 5\}$. These give $\text{diag}(2, 3)$ and $\text{diag}(5, 7, 11)$, respectively. Equation (1060) gives

$$\Delta_n^{\text{pb}}(E_{ab}) = \frac{p_a}{p_b} E_{ab} \quad (a, b \in I_n),$$

which proves (2900). Unique factorization shows that $p_a/p_b = p_c/p_d$ with $a \neq b$ and $c \neq d$ forces $a = c$ and $b = d$. The diagonal units give the n copies of 1. Transposition exchanges E_{ab} with E_{ba} . Finally,

$$\prod_{a,b \in I_n} \frac{p_a}{p_b} = \frac{\prod_{a \in I_n} p_a^n}{\prod_{b \in I_n} p_b^n} = 1.$$

□

Corollary 14.365 (The fourth prime block). *For*

$$\rho_4^{\text{pb}} = \text{diag}(13, 17, 19, 23),$$

the modular ratio array is

$$\left(\frac{p_a}{p_b}\right)_{a,b=1}^4 = \begin{pmatrix} 1 & 13/17 & 13/19 & 13/23 \\ 17/13 & 1 & 17/19 & 17/23 \\ 19/13 & 19/17 & 1 & 19/23 \\ 23/13 & 23/17 & 23/19 & 1 \end{pmatrix}. \quad (2902)$$

Consequently,

$$\begin{aligned} \text{Spec}_{\text{alg}}(\Delta_4^{\text{pb}}) = \{ & 1, 1, 1, 1, 13/17, 13/19, 13/23, 17/13, 17/19, 17/23, \\ & 19/13, 19/17, 19/23, 23/13, 23/17, 23/19 \}. \end{aligned} \quad (2903)$$

Its characteristic polynomial and trace are

$$\chi_{\Delta_4^{\text{pb}}}(x) = (x-1)^4 \prod_{1 \leq a < b \leq 4} \left(x - \frac{p_a}{p_b}\right) \left(x - \frac{p_b}{p_a}\right), \quad (2904)$$

$$\text{Tr}(\Delta_4^{\text{pb}}) = (13 + 17 + 19 + 23) \left(\frac{1}{13} + \frac{1}{17} + \frac{1}{19} + \frac{1}{23}\right) = \frac{1612224}{96577}. \quad (2905)$$

If the ordered eigenlines are labelled by the four Times (M, N, T, E) , the diagonal units E_{aa} are the four fixed channels and the units E_{ab} , $a \neq b$, are the twelve directed edges of the complete graph on those labels. The modular Hamiltonian satisfies

$$K_4^{\text{pb}}(E_{ab}) = \log\left(\frac{p_b}{p_a}\right) E_{ab}, \quad K_4^{\text{pb}}(E_{ba}) = -\log\left(\frac{p_b}{p_a}\right) E_{ba}. \quad (2906)$$

Proof. Here $I_4 = \{6, 7, 8, 9\}$, whose primes are $(13, 17, 19, 23)$. Proposition 14.364 gives (2902)–(2904). The trace is $\sum_{a,b} p_a/p_b = (\sum_a p_a)(\sum_b p_b^{-1})$, which gives (2905). Equation (1061) proves (2906). □

The prime-block family in (2899) is an explicit extension of the three recorded test matrices. Tomita–Takesaki theory permits every positive definite density matrix and therefore does not select the prime blocks. An identification of the four prime eigenlines with the vertices of the previously constructed Times simplex requires an explicit intertwiner preserving the structures under comparison. The block indices, all ratio multiplicities, reciprocal pairs, determinants, and the fourth trace are checked over $\mathbb{Q}$ in `verification/verify_prime_block_modular_spectra.py`. Its captured receipt is `verification/verify_prime_block_modular_spectra_output.txt`.

The direct sum in Theorem 14.4 records three modular generators. The rectangular operators, the six path tensors, and the six D_3 -intertwined modular Hamiltonians of theorem 14.9 supply explicit inter-block maps. Several von Neumann algebras on one Hilbert space with a common cyclic and separating vector give another proved route to joint modular geometry. Under the half-sided

modular-intersection hypotheses stated by Summers, three modular groups generate a faithful unitary representation of $\mathrm{SO}^\uparrow(1, 2)$ [32, Theorem 5.3]. A morphism from the finite linking carrier to a carrier satisfying those half-sided inclusion and intersection hypotheses remains open.

The plural in Bost–Connes–Marcolli systems is structural. Ha and Paugam attach quantum statistical systems to BCM data, recover the original rational system, construct the number-field system, and construct conductor-dependent character systems [44]. For a number field F , the number-field Hamiltonian acts on an ideal-labelled basis by

$$H_F \delta_a = \log N(\mathfrak{a}) \delta_a, \quad \mathrm{Tr}(e^{-\beta H_F}) = \zeta_F(\beta), \quad \beta > 1. \quad (2907)$$

On a conductor-restricted ideal carrier, all ray-class projections P_a , all character multipliers U_χ , and the norm Hamiltonian commute. Consequently the positive partial-zeta traces and every character-twined trace occur simultaneously:

$$Z_a(\beta) = \mathrm{Tr}(P_a e^{-\beta H}), \quad \mathrm{Tr}(U_\chi e^{-\beta H}) = \sum_a \chi(a) Z_a(\beta) = L(\beta, \chi). \quad (2908)$$

Finite Fourier inversion recovers the Z_a from the complete family of twined traces. This is one Hamiltonian carrying several simultaneous observables; the nontrivial character traces need not be positive thermal partition functions.

Theorem 14.366 (Simultaneous BCM realization of the cyclic-cubic factors). *Let $F/\mathbb{Q}$ be cyclic cubic, with characters $1, \chi, \bar{\chi}$. Choose the Ha–Paugam representations giving*

$$\mathrm{Tr}(e^{-\beta H_{\mathbb{Q}}}) = \zeta(\beta), \quad \mathrm{Tr}(U_\chi e^{-\beta H_\chi}) = L(\beta, \chi), \quad \mathrm{Tr}(U_{\bar{\chi}} e^{-\beta H_{\bar{\chi}}}) = L(\beta, \bar{\chi}). \quad (2909)$$

On

$$\mathcal{H}_{\mathrm{Art}} = \mathcal{H}_{\mathbb{Q}} \otimes \mathcal{H}_\chi \otimes \mathcal{H}_{\bar{\chi}}, \quad (2910)$$

put

$$H_{\mathrm{Art}} = H_{\mathbb{Q}} \otimes I \otimes I + I \otimes H_\chi \otimes I + I \otimes I \otimes H_{\bar{\chi}}, \quad (2911)$$

$$U_{\mathrm{Art}} = I \otimes U_\chi \otimes U_{\bar{\chi}}. \quad (2912)$$

Then, for $\beta > 1$,

$$\mathrm{Tr}(U_{\mathrm{Art}} e^{-\beta H_{\mathrm{Art}}}) = \zeta(\beta) L(\beta, \chi) L(\beta, \bar{\chi}) = \zeta_F(\beta). \quad (2913)$$

Let $(\mathcal{H}_F, H_F)$ be the number-field BCM representation from (2907). The block carrier

$$\mathcal{H}_{\mathrm{all}} = \mathcal{H}_F \oplus \mathcal{H}_{\mathrm{Art}}, \quad H_{\mathrm{all}} = H_F \oplus H_{\mathrm{Art}} \quad (2914)$$

has canonical inclusions $\iota_F, \iota_{\mathrm{Art}}$ and projections $\pi_F, \pi_{\mathrm{Art}}$ satisfying

$$\pi_i \iota_j = \delta_{ij} I, \quad \iota_F \pi_F + \iota_{\mathrm{Art}} \pi_{\mathrm{Art}} = I. \quad (2915)$$

The two block functionals

$$\Phi_F(A) = \mathrm{Tr}_{\mathcal{H}_F}(\pi_F A \iota_F), \quad (2916)$$

$$\Phi_{\mathrm{Art}}(A) = \mathrm{Tr}_{\mathcal{H}_{\mathrm{Art}}}(U_{\mathrm{Art}} \pi_{\mathrm{Art}} A \iota_{\mathrm{Art}}) \quad (2917)$$

therefore obey

$$\Phi_F(e^{-\beta H_{\mathrm{all}}}) = \Phi_{\mathrm{Art}}(e^{-\beta H_{\mathrm{all}}}) = \zeta_F(\beta), \quad \beta > 1. \quad (2918)$$

If the underlying untwisted blocks are placed in faithful normal standard forms, their Tomita operators, modular operators, and modular Hamiltonians are the corresponding finite direct sums; for product states the modular operator on $\mathcal{H}_{\mathrm{Art}}$ is the tensor product of the three factor modular operators.

Proof. The three summands in (2911) act on different tensor factors and strongly commute. Functional calculus gives

$$e^{-\beta H_{\text{Art}}} = e^{-\beta H_{\mathbb{Q}}} \otimes e^{-\beta H_{\chi}} \otimes e^{-\beta H_{\bar{\chi}}}. \quad (2919)$$

Each factor in (2909) is trace class after its bounded character insertion. Multiplicativity of the trace on tensor products gives the first equality in (2913); abelian Artin factorization gives the second. The inclusion–projection identities in (2915) give (2918) by restriction to the two central blocks. Finally, the Tomita rule acts blockwise on a finite direct sum, while the Tomita operator of a product standard form is the tensor product of the factor Tomita operators. Polar decomposition gives the assertions for J and Δ , and functional calculus gives the assertions for the modular Hamiltonians. The simultaneous ray-class, character, direct-sum, and determinant constructions are also proved in the integrated arithmetic-time source [43]. $\square$

Theorem 14.367 (Exact character-sector D_3 carrier and charge conjugation). *Let χ be a primitive cubic Dirichlet character and let $\bar{\chi}$ be its complex conjugate. On $\mathcal{H} = \ell^2(\mathbb{N}_{\geq 1})$, with basis $(\delta_n)_{n \geq 1}$, put*

$$H\delta_n = (\log n)\delta_n, \quad U_{\chi}\delta_n = \chi(n)\delta_n, \quad \kappa\left(\sum_n a_n \delta_n\right) = \sum_n \overline{a_n} \delta_n.$$

Let $V_{\chi} = \mathbb{C}d_0 \oplus \mathbb{C}d_+ \oplus \mathbb{C}d_-$, and in the ordered basis (d_0, d_+, d_-) define

$$R = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (2920)$$

Write P_0, P_+, P_- for the three coordinate projections and set

$$\mathcal{H}_{\text{sec}} = V_{\chi} \otimes \mathcal{H}, \quad H_{\text{sec}} = I_3 \otimes H, \quad (2921)$$

$$D_{\chi} = P_0 \otimes I + P_+ \otimes U_{\chi} + P_- \otimes U_{\bar{\chi}}, \quad \mathcal{C}_{\text{sec}} = J \otimes \kappa. \quad (2922)$$

Then $\mathcal{C}_{\text{sec}}$ is an antiunitary involution and

$$R^3 = J^2 = I_3, \quad J R J = R^{-1}, \quad (2923)$$

$$\mathcal{C}_{\text{sec}} H_{\text{sec}} \mathcal{C}_{\text{sec}}^{-1} = H_{\text{sec}}, \quad \mathcal{C}_{\text{sec}} (R \otimes I) \mathcal{C}_{\text{sec}}^{-1} = R^{-1} \otimes I, \quad (2924)$$

$$\mathcal{C}_{\text{sec}} D_{\chi} \mathcal{C}_{\text{sec}}^{-1} = D_{\chi}. \quad (2925)$$

For $\beta > 1$,

$$\text{Tr}_{\mathcal{H}_{\text{sec}}} (D_{\chi} e^{-\beta H_{\text{sec}}}) = \zeta(\beta) + L(\beta, \chi) + L(\beta, \bar{\chi}). \quad (2926)$$

There is also an explicit D_3 -equivariant label map

$$T : \mathbb{C} \otimes_{\mathbb{Q}} V_{\text{F}} \longrightarrow V_{\chi}, \quad T(e_0) = d_0, \quad T(e_1) = d_+, \quad T(e_2) = d_-, \quad (2927)$$

which satisfies

$$T \circ r = R \circ T, \quad T \circ j = J \circ T. \quad (2928)$$

After scalar extension, $\tilde{T} = I_{\mathbb{C}[\varepsilon]} \otimes T$ is an isomorphism of the corresponding supported D_3 -modules and

$$\tilde{T}(\mathbf{f}) = \varepsilon \otimes (d_0 + d_+ + d_-). \quad (2929)$$

Finally, on $\mathcal{H}^{\otimes 3}$, let Σ_{23} exchange the second and third tensor factors, let $\kappa^{(3)}$ be coefficientwise complex conjugation in the product basis, and put

$$\mathcal{C}_{\text{Art}} := \Sigma_{23} \kappa^{(3)}. \quad (2930)$$

For the common-carrier realization of (2911)–(2912),

$$\mathcal{C}_{\text{Art}}^2 = I, \quad \mathcal{C}_{\text{Art}} H_{\text{Art}} \mathcal{C}_{\text{Art}}^{-1} = H_{\text{Art}}, \quad \mathcal{C}_{\text{Art}} U_{\text{Art}} \mathcal{C}_{\text{Art}}^{-1} = U_{\text{Art}}. \quad (2931)$$

Proof. Coordinatewise conjugation gives

$$\kappa^2 = I, \quad \kappa H \kappa = H, \quad \kappa U_\chi \kappa = U_{\bar{\chi}}. \quad (2932)$$

Direct multiplication of the two matrices in (2920) proves (2923). Since $JP_0J = P_0$, $JP_+J = P_-$, and $JP_-J = P_+$, equation (2932) gives

$$\begin{aligned} \mathcal{C}_{\text{sec}}(P_+ \otimes U_\chi) \mathcal{C}_{\text{sec}}^{-1} &= P_- \otimes U_{\bar{\chi}}, \\ \mathcal{C}_{\text{sec}}(P_- \otimes U_{\bar{\chi}}) \mathcal{C}_{\text{sec}}^{-1} &= P_+ \otimes U_\chi. \end{aligned}$$

This proves (2924) and (2925). Taking the trace on the three coordinate blocks proves (2926).

Equations (2928) follow by applying both sides to e_0, e_1, e_2 . The map T sends a basis to a basis, hence is invertible. Tensoring with $\mathbb{C}[\varepsilon]$ preserves invertibility and the displayed D_3 -action, proving (2929).

The maps Σ_{23} and $\kappa^{(3)}$ commute and are involutions, so $\mathcal{C}_{\text{Art}}^2 = I$. Conjugation by $\kappa^{(3)}$ sends $I \otimes U_\chi \otimes U_{\bar{\chi}}$ to $I \otimes U_{\bar{\chi}} \otimes U_\chi$; conjugation by Σ_{23} returns $I \otimes U_\chi \otimes U_{\bar{\chi}}$. The three summands of H_{Art} are fixed as a sum because $\kappa H \kappa = H$ and Σ_{23} exchanges its second and third summands. This proves (2931). $\square$

The map $\tilde{T}$ proves an exact equivalence of the displayed supported D_3 -representations. A morphism intertwining the polynomial map, the arithmetic divisor shell, and the BCM algebra actions would provide a stronger bridge; its defining equations remain an open calculation.

15. THE FIRST EXACT MAP FROM A BUSY-BEAVER EXTREMUM TO THE FINITE SHELL CARRIERS

Definition 15.1 (Busy-Beaver source map). For $n \geq 1$, let $\mathcal{H}_n$ be the finite set of halting n -state, two-symbol Turing machines started on the blank tape. For $M \in \mathcal{H}_n$, let

$t(M)$ = the number of executed steps, $w(M)$ = the number of final tape symbols equal to (1), and let $s(M)$ be the number of tape cells scanned before halting. Define

$$\mathbf{B}_n : \mathcal{H}_n \longrightarrow \mathbb{N}^3, \quad \mathbf{B}_n(M) = (t(M), w(M), s(M)). \quad (2933)$$

Then

$$S(n) = \max_{M \in \mathcal{H}_n} t(M), \quad \Sigma(n) = \max_{M \in \mathcal{H}_n} w(M), \quad \text{Space}(n) = \max_{M \in \mathcal{H}_n} s(M).$$

Theorem 15.2 (Four-to-five Busy-Beaver transport and obstruction to the literal state shift). *Put*

$$u_4 := S(3) = 21, \quad R_4 := S(4) = 107, \quad v_5 := S(5) = 47,176,870.$$

Then

$$4u_4^2 v_5 \equiv 1 \pmod{R_4}, \quad -u_4 v_5 \equiv 11^2 \pmod{R_4}. \quad (2934)$$

The two roots of

$$x^2 \equiv -u_4 v_5 \pmod{R_4}$$

are 11 and 96. Hence

$$R_5 := 11 \quad (2935)$$

is the unique root in $0 < x < R_4$ satisfying $x \equiv 3 \pmod{4}$.

The five-state space extremum

$$p_5 := \text{Space}(5) = 12289$$

is prime, and

$$a_5 := \frac{p_5 + R_5}{4} = 3075 = 3 \cdot 5^2 \cdot 41, \quad 4a_5 - p_5 = R_5. \quad (2936)$$

Consequently

$$(S(3), S(4), S(5)) \longmapsto R_5 = 11, \quad (\text{Space}(5), R_5) \longmapsto (p_5, a_5, R_5) = (12289, 3075, 11) \quad (2937)$$

is an exact adjacent-state transport into a residual divisor shell.

The runtime $S(5)$ cannot itself be a residual $4a - p$ with p odd, because

$$S(5) \equiv 2 \pmod{4}, \quad 4a - p \equiv 1 \text{ or } 3 \pmod{4}. \quad (2938)$$

The literal next-state copy of (2934) also fails on the odd part

$$m_5 := \frac{S(5)}{2} = 23,588,435 = 5 \cdot 13 \cdot 17 \cdot 21347.$$

Indeed, there is no integer X satisfying both

$$4S(4)^2 X \equiv 1 \pmod{m_5}, \quad -S(4)X \in \{y^2 : y \in \mathbb{Z}/m_5\mathbb{Z}\}. \quad (2939)$$

Thus the verified five-state extension is the space-to-shell chain in (2937); the two congruences in (2939) do not define a recurrence for an unknown sixth runtime.

Proof. Reduction of v_5 modulo 107 gives $v_5 \equiv 35$. Hence

$$4u_4^2 v_5 \equiv 4 \cdot 21^2 \cdot 35 \equiv 1 \pmod{107}.$$

Moreover

$$-u_4 v_5 \equiv -21 \cdot 35 \equiv 14 \equiv 11^2 \pmod{107}.$$

Since 107 is prime and $11 \not\equiv 0 \pmod{107}$, the quadratic has exactly the two roots 11 and $-11 \equiv 96$. Of their least positive representatives, only 11 is congruent to 3 modulo four.

For completeness, a Pocklington certificate for p_5 uses

$$p_5 - 1 = 12288 = 2^{12} \cdot 3$$

and the witness 11:

$$11^{12288} \equiv 1, \quad 11^{6144} \equiv 12288, \quad 11^{4096} \equiv 6240 \pmod{12289},$$

while

$$\gcd(12287, 12289) = 1, \quad \gcd(6239, 12289) = 1.$$

Pocklington's criterion proves that 12289 is prime. Direct calculation then gives

$$\frac{12289 + 11}{4} = 3075 = 3 \cdot 5^2 \cdot 41,$$

which proves (2936) and (2937).

The residue statement in (2938) follows from $S(5) = 47,176,870 \equiv 2 \pmod{4}$. If p is odd, then $4a - p$ is odd, proving the first obstruction.

Finally, suppose that X satisfied (2939). Reduction of its first congruence modulo $5 \mid m_5$ gives

$$4 \cdot 107^2 X \equiv X \equiv 1 \pmod{5}.$$

The second condition would then require

$$-107X \equiv 3 \pmod{5}$$

to be a square. The squares modulo five are 0, 1, 4, which is a contradiction.

The script and captured receipt

```
verification/verify_busy_beaver_successor_shell.py,
verification/verify_busy_beaver_successor_shell_output.txt
```

check every integer factorization, congruence, root, primality-certificate residue, and shell identity used above. $\square$

Theorem 15.3 (Exact five-state space-to-shell carrier). *There is a machine $M_5^* \in \mathcal{H}_5$ such that*

$$\mathbf{B}_5(M_5^*) = (47, 176, 870, 4098, 12289) = (S(5), \Sigma(5), \text{Space}(5)). \quad (2940)$$

Put

$$p := s(M_5^*) = 12289, \quad R := 11, \quad a := \frac{p+R}{4} = 3075 = 3 \cdot 5^2 \cdot 41.$$

The adjacent four-state data satisfy

$$p = S(4)^2 + 840, \quad p \equiv -\text{Space}(4) \pmod{S(4)}, \quad (2941)$$

because $S(4) = 107$ and $\text{Space}(4) = 16$.

Let $H = C_{10}$, and define

$$\mu_{\text{BB}5}(h) := \# \{(\beta_3, \beta_5, \beta_{41}) \in [-1, 1]_{\mathbb{Z}} \times [-2, 2]_{\mathbb{Z}} \times [-1, 1]_{\mathbb{Z}} : 8\beta_3 + 4\beta_5 + 3\beta_{41} = h \pmod{10}\}. \quad (2942)$$

Then

$$\mu_{\text{BB}5} = (3, 6, 3, 6, 3, 6, 3, 6, 3, 6) \in \mathbb{Z}_{\geq 0}^{C_{10}}. \quad (2943)$$

This vector is the binary-survivor case of theorem 14.294: its principal and binary Fourier coefficients are 45 and -15 , respectively, and the one-third amplitude gap forces every residue coefficient to be positive. For

$$q : C_{10} \longrightarrow C_2, \quad q(h) = h \pmod{2},$$

let

$$\text{desc}_q : \text{im}(q^*) \longrightarrow \mathbb{C}^{C_2}, \quad \text{desc}_q(q^*g) = g.$$

The exact descent and convolution operator are

$$\nu := \text{desc}_q(\mu_{\text{BB}5}) = \begin{pmatrix} 3 \\ 6 \end{pmatrix}, \quad \mathbf{L}_\nu = \begin{pmatrix} 3 & 6 \\ 6 & 3 \end{pmatrix}. \quad (2944)$$

They satisfy

$$\text{Spec}(\mathbf{L}_\nu) = \{9, -3\}, \quad \text{gap}(\mathbf{L}_\nu) := 9 - (-3) = 12, \quad (2945)$$

Put

$$D_\nu := U_0^* \mathbf{L}_\nu U_0 = \text{diag}(9, -3), \quad X_{+,-} := U_0 E_{12} U_0^*, \quad X_{-,+} := U_0 E_{21} U_0^*.$$

Then

$$\text{ad}_{\mathbf{L}_\nu}(X_{+,-}) = 12X_{+,-}, \quad \text{ad}_{\mathbf{L}_\nu}(X_{-,+}) = -12X_{-,+}, \quad (2946)$$

$$\text{ad}_{D_\nu}(E_{12}) = 12E_{12}, \quad \text{ad}_{D_\nu}(E_{21}) = -12E_{21}. \quad (2947)$$

The finite shell and cover cardinalities are

$$|[-1, 1]_{\mathbb{Z}} \times [-2, 2]_{\mathbb{Z}} \times [-1, 1]_{\mathbb{Z}}| = 45, \quad |\mathcal{X}_{11,p}| = 12, \quad |\tilde{\mathcal{X}}_{11,p}| = 36, \quad |\tilde{\mathcal{X}}_{11,p}/D_3| = 6. \quad (2948)$$

In particular,

$$\text{gap}(\mathbf{L}_\nu) = \Omega_{11}(s(M_5^*)) = 12. \quad (2949)$$

Thus the maps

$$M_5^* \xrightarrow{s} 12289 \xrightarrow{(2942)} \mu_{\text{BB}5} \xrightarrow{\text{desc}_q} \nu \xrightarrow{\text{left convolution}} \mathbf{L}_\nu \xrightarrow{\text{ad}} \text{ad}_{\mathbf{L}_\nu} \quad (2950)$$

form an exact typed chain from the five-state machine carrier to the finite shell, quotient, Fourier, modular, and ordered-blade carriers.

Proof. Equation (2940) is the formally verified result cited above. Equation (2941) is the direct calculation

$$12289 = 107^2 + 840, \quad 12289 \equiv 91 \equiv -16 \pmod{107}.$$

For $p = 12289$, the shell factorization and additive logarithms established in theorem 10.27 are

$$a = 3 \cdot 5^2 \cdot 41, \quad (\log_2 3, \log_2 5, \log_2 41) = (8, 4, 3) \pmod{10}.$$

Therefore (2942) is precisely the divisor convolution of theorem 14.260, and that theorem gives (2943). Its entries are constant on the fibres of q , so the quotient criterion in theorem 14.262 gives (2944). The same theorem gives the spectrum, the convolution spectral gap, and the two ordered Fourier-carried blade actions in (2945)–(2947).

The exponent-box cardinality is $3 \cdot 5 \cdot 3 = 45$. The exact shell count is $\Omega_{11}(12289) = 12$ by theorem 10.27. The free-cover formulas in theorem 14.27 give

$$|\tilde{\mathcal{X}}_{11,p}| = 3\Omega_{11}(p) = 36, \quad |\tilde{\mathcal{X}}_{11,p}/D_3| = \frac{\Omega_{11}(p)}{2} = 6.$$

This proves (2948). Substitution of $s(M_5^*) = 12289$ proves (2949) and the displayed chain. $\square$

Definition 15.4 (Trajectory metric and scanned-support measure). Let (M) be a deterministic single-tape Turing machine which halts on the blank tape after $(T=t(M))$ transitions. Write

$$c_0, c_1, \dots, c_T$$

for its ordered configurations. The trajectory graph Γ_M has vertex set $\{c_0, \dots, c_T\}$ and one unit-length edge $\{c_{k-1}, c_k\}$ for each $1 \leq k \leq T$. Its path metric is denoted by d_M .

Let $X_M \subset \mathbb{Z}$ be the set of tape-cell indices scanned during the computation, so $s(M) = |X_M|$. The associated unit-cell complex is

$$K_M := \bigcup_{j \in X_M} [j, j+1] \subset \mathbb{R},$$

and its one-dimensional Lebesgue measure is denoted by $\text{vol}_1(K_M)$.

Theorem 15.5 (Exact operation-length and tape-volume carrier). *For every deterministic halting single-tape machine M , the ordered configurations $c_0, \dots, c_T$ are pairwise distinct and*

$$d_M(c_0, c_T) = \mathcal{H}^1(|\Gamma_M|) = t(M), \quad (2951)$$

Nearest-neighbour head motion makes X_M an integer interval, and hence

$$\text{vol}_1(K_M) = s(M). \quad (2952)$$

Consequently the five-state extremal machine carries the exact metric triple

$$\left(d_{M_5^*}(c_0, c_T), \mathcal{H}^1(|\Gamma_{M_5^*}|), \text{vol}_1(K_{M_5^*})\right) = (47, 176, 870, 47, 176, 870, 12289), \quad (2953)$$

with

$$\gcd(47, 176, 870, 12289) = 1, \quad \frac{d_{M_5^*}(c_0, c_T)}{\text{vol}_1(K_{M_5^*})} = \frac{47, 176, 870}{12289}. \quad (2954)$$

The preceding space-to-shell carrier therefore factors through the tape volume:

$$\text{gap}(\mathbf{L}_\nu) = \Omega_{11}(\text{vol}_1(K_{M_5^*})) = 12. \quad (2955)$$

Proof. Suppose $c_i = c_j$ for some $0 \leq i < j \leq T$. If $j = T$, then the machine has already reached its halting configuration at time $i < T$, contrary to the definition of T . If $j < T$, determinism gives

$$c_{i+r} = c_{j+r} \quad (0 \leq r \leq T-j).$$

Taking $r = T - j$ places the halting configuration c_T at the earlier time $i + T - j < T$, again a contradiction. Thus all trajectory vertices are distinct. The trajectory graph is the path with $T + 1$

ordered vertices. Its endpoint distance and the one-dimensional Hausdorff measure of its metric realization both equal its number of unit edges, namely $T = t(M)$.

If X_M has one element, (2952) is immediate. Otherwise put $a = \min X_M < b = \max X_M$. Nearest-neighbour motion forces the head to scan every integer index between a and b . Therefore

$$X_M = [a, b]_{\mathbb{Z}}, \quad K_M = [a, b + 1],$$

and

$$\text{vol}_1(K_M) = b - a + 1 = |X_M| = s(M).$$

Substitution of (2940) proves (2953). The Euclidean algorithm gives $\gcd(47,176,870,12289) = 1$, proving (2954). Finally, (2952) and (2949) give (2955). $\square$

Theorem 15.6 (Exact lifted Pauli-geometry carrier of the shell gap). *Let*

$$\sigma_z := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \widehat{D}_\nu := D_\nu - \frac{\text{tr}(D_\nu)}{2} I_2.$$

Then the central and traceless terms of the Fourier-carried convolution are

$$D_\nu = 3I_2 + 6\sigma_z, \quad \widehat{D}_\nu = 6\sigma_z, \quad \text{ad}_{D_\nu} = \text{ad}_{\widehat{D}_\nu}. \quad (2956)$$

On $\mathfrak{su}(2)$, write a Hermitian control as $H = h_x\sigma_x + h_y\sigma_y + h_z\sigma_z$ and put

$$F_P(H) := \sqrt{h_x^2 + h_y^2 + h_z^2}.$$

This Pauli metric is bi-invariant. For the maximal torus

$$T_z := \{\exp(-i\theta\sigma_z) : \theta \in \mathbb{R}\}$$

and its covering map

$$\pi_z : \mathbb{R} \longrightarrow T_z, \quad \pi_z(\theta) = \exp(-i\theta\sigma_z),$$

consider the lifted path and its projection

$$\tilde{\gamma}_\nu(t) = 6t, \quad \gamma_\nu(t) = \pi_z(6t) = \exp(-it\widehat{D}_\nu), \quad 0 \leq t \leq 1. \quad (2957)$$

Their lengths are

$$L_{\mathbb{R}}(\tilde{\gamma}_\nu) = L_{F_P}(\gamma_\nu) = 6 = \frac{1}{2} \text{gap}(\mathbf{L}_\nu) = \frac{1}{2} \Omega_{11}(\text{vol}_1(K_{M_5^*})). \quad (2958)$$

The endpoint fibre and the endpoint distance are

$$\pi_z^{-1}(\gamma_\nu(1)) = 6 + 2\pi\mathbb{Z}, \quad (2959)$$

$$d_{F_P}(I_2, \gamma_\nu(1)) = \min_{k \in \mathbb{Z}} |6 - 2\pi k| = 2\pi - 6. \quad (2960)$$

Finally, let

$$\mathcal{G}_1^\pm := \{\exp(-i\alpha\sigma) : \sigma \in \{\pm\sigma_x, \pm\sigma_y, \pm\sigma_z\}, 0 \leq \alpha \leq 1\}.$$

Define its corresponding Hermitian set by

$$\mathcal{H}_1^\pm := \{0\} \cup \{\alpha\sigma : \sigma \in \{\pm\sigma_x, \pm\sigma_y, \pm\sigma_z\}, 0 < \alpha \leq 1\}.$$

The exponential map

$$\mathcal{H}_1^\pm \longrightarrow \mathcal{G}_1^\pm, \quad H \longmapsto \exp(-iH)$$

is bijective, $\mathcal{G}_1^\pm$ is exactly universal on $SU(2)$, and the local metric $F(V, H) := F_P(H)$ is $\mathcal{G}_1^\pm$ -bounding. The endpoint has exact gate count

$$m_{\mathcal{G}_1^\pm}(\gamma_\nu(1)) = 1, \quad d_{F_P}(I_2, \gamma_\nu(1)) \leq m_{\mathcal{G}_1^\pm}(\gamma_\nu(1)), \quad (2961)$$

where the inequality is the one-qubit instance of Nielsen's geometric gate-count bound [34].

Proof. Since $D_\nu = \text{diag}(9, -3)$, its trace is 6, and direct subtraction gives

$$\widehat{D}_\nu = \text{diag}(6, -6) = 6\sigma_z.$$

The scalar matrix $3I_2$ commutes with every matrix, which proves (2956). The adjoint action of $SU(2)$ rotates the Pauli coefficient vector $(h_x, h_y, h_z) \in \mathbb{R}^3$; it therefore preserves F_P . This proves bi-invariance.

The adapted Hamiltonian of γ_ν is the constant $6\sigma_z$. Hence

$$L_{F_P}(\gamma_\nu) = \int_0^1 F_P(6\sigma_z) dt = 6.$$

The lifted path in $\mathbb{R}$ has the same length. Equation (2958) now follows from (2955).

The kernel of π_z is $2\pi\mathbb{Z}$, which proves (2959). The bi-invariant Pauli distance on T_z is the least absolute value among the lifts. The elementary bounds $3 < \pi < 7/2$ give

$$0 < 2\pi - 6 < 1,$$

so the least lift is $6 - 2\pi$, proving (2960).

Put $\delta := 2\pi - 6$. Then

$$\gamma_\nu(1) = \exp(-i6\sigma_z) = \exp(-i\delta(-\sigma_z)).$$

Because $0 < \delta < 1$, this endpoint belongs to $\mathcal{G}_1^\pm$. It differs from the identity, so its exact gate count is one.

For $0 < \alpha \leq 1$,

$$\exp(-i\alpha\sigma) = \cos(\alpha)I_2 - i\sin(\alpha)\sigma.$$

Since $1 < \pi$, the trace determines α uniquely on this interval, and the Pauli coefficient then determines σ . The identity has the unique preimage 0. This proves that the displayed exponential map from $\mathcal{H}_1^\pm$ to $\mathcal{G}_1^\pm$ is bijective.

Euler-angle factorization generates every element of $SU(2)$ from rotations about two Pauli axes. Each signed rotation angle is a finite sum of signed terms whose absolute values lie in $[0, 1]$, so $\mathcal{G}_1^\pm$ is exactly universal. Finally,

$$F(V, \alpha\sigma) = F_P(\alpha\sigma) = \alpha \leq 1$$

for every nonzero element $\alpha\sigma \in \mathcal{H}_1^\pm$, while $F(V, 0) = 0$. Thus F is $\mathcal{G}_1^\pm$ -bounding. Nielsen's lower-bound theorem [34, §Minimal curves and lower bounds, the theorem labelled `thm:lower-bounds`] now gives the final inequality in (2961). $\square$

Theorem 15.7 (Exact finite-field operator-volume carrier). *Work over $\mathbb{F}_{107}$. Put*

$$\epsilon_6(m) := 5 - \mathbf{1}_{2|m} + \mathbf{1}_{3|m} - 5\mathbf{1}_{6|m}, \quad P(q) := \prod_{m \geq 1} (1 - q^m)^{\epsilon_6(m)},$$

and define

$$T = q^{-1}P, \quad F = -2\eta(6\tau)^4T, \quad J = T + 5 + \frac{72}{T}, \quad f_6 = -q \frac{d}{dq} \log T.$$

For $n \geq 1$, write

$$T_n = [q^n]T, \quad F_n = [q^n]F, \quad J_n = [q^n]J, \quad C_n = [q^n]f_6,$$

and set

$$\widehat{\Phi}(n) := (n \bmod 107, T_n, F_n, J_n, C_n) \in \mathbb{F}_{107}^5.$$

For the ordered block $B = (744, 745, 746, 747, 748)$, let R_B have rows $\widehat{\Phi}(n)$, $n \in B$, in the displayed order. For an integer h , let R_{B+h} have rows $\widehat{\Phi}(744+h), \dots, \widehat{\Phi}(748+h)$, and put

$$A_h := R_{B+h}R_B^{-1}.$$

The exact coefficient recurrence gives

$$R_B = \begin{pmatrix} 102 & 34 & 19 & 16 & 67 \\ 103 & 56 & 19 & 32 & 6 \\ 104 & 105 & 6 & 96 & 47 \\ 105 & 83 & 77 & 1 & 48 \\ 106 & 62 & 12 & 58 & 58 \end{pmatrix}, \quad \det R_B = 8. \quad (2962)$$

The unit and Coxeter shifts are

$$A_1 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 72 & 40 & 4 & 32 & 46 \end{pmatrix}, \quad \det A_1 = 72, \quad (2963)$$

$$A_6 = \begin{pmatrix} 62 & 30 & 43 & 97 & 102 \\ 3 & 12 & 64 & 68 & 35 \\ 54 & 74 & 14 & 80 & 85 \\ 13 & 76 & 44 & 102 & 40 \\ 10 & 47 & 88 & 31 & 73 \end{pmatrix}, \quad \det A_6 = 40, \quad \text{tr } A_6 = 49. \quad (2964)$$

Their commutator satisfies

$$\det[A_1, A_6] = 15, \quad \text{tr}[A_1, A_6] = 0. \quad (2965)$$

Order the twenty-five matrices

$$\mathcal{B}_{1,6} := (A_1^i A_6^j)_{0 \leq i,j \leq 4}$$

lexicographically in (i, j) , and vectorize each matrix in row order. Define its oriented algebraic volume by

$$\text{Vol}_{107}^{\text{or}}(A_1, A_6) := \det(\text{vec}(A_1^i A_6^j))_{0 \leq i,j \leq 4}. \quad (2966)$$

Then

$$\boxed{\text{Vol}_{107}^{\text{or}}(A_1, A_6) = 32 = \Sigma(5) \pmod{107}.} \quad (2967)$$

In particular,

$$\mathbb{F}_{107} \langle A_1, A_6 \rangle = M_5(\mathbb{F}_{107}). \quad (2968)$$

The volume in (2966) is invariant under simultaneous change of basis. Explicitly, for $Q \in GL_5(\mathbb{F}_{107})$, the transformation is

$$(A_1, A_6) \mapsto (QA_1Q^{-1}, QA_6Q^{-1}).$$

Consequently the five-state extremal machine has the exact score carrier

$$M_5^* \xrightarrow{w} \Sigma(5) = 4098 \xrightarrow{\text{mod } 107} 32 = \text{Vol}_{107}^{\text{or}}(A_1, A_6), \quad (2969)$$

in addition to the runtime-length and scanned-support-volume carriers in theorem 15.5.

Proof. Write $P(q) = \sum_{n \geq 0} a_n q^n$, $a_0 = 1$. Finite multiplication of the factors $(1 - q^m)^{\epsilon_6(m)}$ through degree 755 determines the coefficients required for all rows from 744 through 754. If $P(q)^{-1} = \sum_{n \geq 0} b_n q^n$, then

$$b_0 = 1, \quad b_n = - \sum_{k=1}^n a_k b_{n-k}.$$

The remaining coefficient channels are obtained from

$$J_n = T_n + 72b_{n-1}, \quad F = -2\eta(6\tau)^4 T,$$

and

$$C_n = 5\sigma_1(n) - 2\sigma_1(n/2) + 3\sigma_1(n/3) - 30\sigma_1(n/6),$$

where $\sigma_1(x) = 0$ for nonintegral x . These exact recurrences give (2962). Finite-field elimination gives $\det R_B = 8$, hence R_B is invertible. Multiplication of $R_{B+1}R_B^{-1}$ and $R_{B+6}R_B^{-1}$ gives (2963) and (2964). Direct multiplication gives (2965).

The determinant in (2966) is 32. It is nonzero, so the twenty-five elements of $\mathcal{B}_{1,6}$ are linearly independent in the 25-dimensional vector space $M_5(\mathbb{F}_{107})$. They therefore form a basis, proving (2968). The verified score value $\Sigma(5) = 4098$ has residue 32 modulo 107, which proves (2967) and (2969).

For the final invariance statement, simultaneous conjugation acts on $M_5(\mathbb{F}_{107})$ through the linear map

$$X \longmapsto QXQ^{-1}.$$

In vector coordinates its matrix is $Q^{-T} \otimes Q$, whose determinant is

$$\det(Q^{-T})^5 \det(Q)^5 = \det(Q)^{-5} \det(Q)^5 = 1.$$

Thus simultaneous conjugation preserves the oriented determinant in (2966).

The independent checker

`verification/verify_bb107_operator_volume.py`

reconstructs R_B, A_1, A_6 from the coefficient products and verifies all displayed determinants. Its twenty-line receipt is

`verification/verify_bb107_operator_volume_output.txt.`

□

Theorem 15.8 (Exact support–glue–Fricke transport). *Let*

$$V_{840} := \{r \in (\mathbb{Z}/840\mathbb{Z})^\times : r \equiv 1 \pmod{24}\},$$

and define

$$\beta(r) := \left(\frac{1 - \left(\frac{r}{5}\right)}{2}, \frac{1 - \left(\frac{r}{7}\right)}{2} \right) \in \mathbb{F}_2^2.$$

Then β is a surjective group homomorphism with

$$\ker \beta = S_{840} = \{1, 121, 169, 289, 361, 529\} = \langle 289 \rangle \cong C_6. \quad (2970)$$

Its four fibres $F_\eta := \beta^{-1}(\eta)$, $\eta \in \mathbb{F}_2^2$, have cardinality six. Each augmentation lattice

$$A(F_\eta) := \ker \left(\mathbb{Z}[F_\eta] \xrightarrow{\Sigma} \mathbb{Z} \right)$$

is an A_5 root lattice.

Put

$$D_4 := \{x \in \mathbb{Z}^4 : x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod{2}\}.$$

The four vectors

$$0, \quad v = (1, 0, 0, 0), \quad s = \frac{1}{2}(1, 1, 1, 1), \quad c = \frac{1}{2}(-1, 1, 1, 1)$$

represent the four classes in $D_4^\vee/D_4$. Their discriminant form satisfies

$$q(0) = 0, \quad q(v) = q(s) = q(c) = 1 \pmod{2\mathbb{Z}}, \quad v + s = c. \quad (2971)$$

The orthogonal transformations

$$R = \text{diag}(-1, 1, 1, 1), \quad H = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \quad (2972)$$

preserve D_4 and induce

$$R : (s \ c), \quad H : (v \ s)$$

on its three nonzero discriminant classes. Consequently these two maps generate the full triality group S_3 , and all six markings

$$\iota : \mathbb{F}_2^2 \xrightarrow{\sim} D_4^\vee / D_4$$

are related by explicit lattice isometries.

After fixing one such marking, set

$$R_{\text{ES}} := \left(\bigoplus_{\eta \in \mathbb{F}_2^2} A(F_\eta) \right) \oplus D_4 \cong A_5^4 D_4. \quad (2973)$$

The root frame has rank 24, and each irreducible component has Coxeter number six. In its discriminant group

$$D_{R_{\text{ES}}} \cong (\mathbb{Z}/6\mathbb{Z})^4 \oplus \mathbb{F}_2^2,$$

let H_{old} be generated by

$$\begin{aligned} h_1 &= (1, 3, 1, 1; 0, 0), \\ h_2 &= (1, 4, 2, 3; 1, 1), \\ h_3 &= (3, 0, 3, 0; 0, 1), \end{aligned}$$

with respective orders 6, 6, 2. Let

$$C_3^{\text{gl}} := \text{span}_{\mathbb{F}_3} \{(1, 1, 1, 0), (1, 2, 0, 1)\} \subset \mathbb{F}_3^4$$

and let $C_2^{\text{gl}} \subset \mathbb{F}_2^4 \oplus \mathbb{F}_2^2$ be the eight-element set

u	y	u	y
0000	00	1111	00
1100	10	0011	10
1010	01	0101	01
1001	11	0110	11.

Define the code presentation

$$H_{\text{code}} := \left\{ ((3u_1 + 2z_1, \dots, 3u_4 + 2z_4), y) : (u, y) \in C_2^{\text{gl}}, z \in C_3^{\text{gl}} \right\}. \quad (2974)$$

The two labeled subsets are distinct, and the exact transporter

$$P(x_1, x_2, x_3, x_4; y) := (x_3, x_4, x_1, x_2; y) \quad (2975)$$

satisfies

$$P(H_{\text{old}}) = H_{\text{code}}. \quad (2976)$$

Within

$$(\{\pm 1\}^4 \rtimes S_4) \times GL_2(\mathbb{F}_2),$$

exactly 48 transformations carry H_{old} to H_{code} . Both groups are isotropic of order 72; their nonzero glue-coset minima are exactly 4 and 6. Each therefore determines an even unimodular rank-twenty-four overlattice with root system $A_5^4 D_4$, hence the Niemeier lattice of that root type [9, 11, 12]. The umbral Niemeier table assigns this root system lambency 6 [13].

Let $\mathcal{F}_6 := \bigoplus_{d|6} \mathbb{Z}[d]$, and define the multiplicative eta map

$$\mathcal{E} : \mathcal{F}_6 \longrightarrow \mathcal{M}(\mathbb{H})^\times, \quad \mathcal{E} \left(\sum_{d|6} a_d [d] \right) := \prod_{d|6} \eta(d\tau)^{a_d}. \quad (2977)$$

For a Coxeter element of A_5 ,

$$\det(xI - c_{A_5}) = \frac{x^6 - 1}{x - 1}.$$

For a Coxeter element of D_4 ,

$$\det(xI - c_{D_4}) = (x^2 - x + 1)(x + 1)^2 = \frac{(x^2 - 1)(x^6 - 1)}{(x - 1)(x^3 - 1)}.$$

Thus the Coxeter frame exponent of $A_5^4 D_4$ is

$$\pi_{\text{cox}} = -5[1] + [2] - [3] + 5[6]. \quad (2978)$$

Its reciprocal eta image is exactly

$$T(\tau) = \mathcal{E}(-\pi_{\text{cox}}) = \frac{\eta(\tau)^5 \eta(3\tau)}{\eta(2\tau) \eta(6\tau)^5}. \quad (2979)$$

It is a Hauptmodul for $\Gamma_0(6)$, with initial expansion

$$T = q^{-1} - 5 + 6q + 4q^2 - 3q^3 - 12q^4 - 8q^5 + O(q^6).$$

For the Fricke involution $W_6\tau = -1/(6\tau)$,

$$T(W_6\tau) = \frac{72}{T(\tau)}. \quad (2980)$$

Consequently

$$J = T + 5 + \frac{72}{T}, \quad Y = T - \frac{72}{T} \quad (2981)$$

satisfy

$$W_6 J = J, \quad W_6 Y = -Y, \quad Y^2 = (J - 5)^2 - 288, \quad dJ = Y \frac{dT}{T}. \quad (2982)$$

Define

$$U(\tau) := \left(\frac{\eta(2\tau)\eta(3\tau)}{\eta(\tau)\eta(6\tau)} \right)^{12}. \quad (2983)$$

Then

$$U = J + 12 \quad (2984)$$

and

$$J = q^{-1} + 78q + 364q^2 + 1365q^3 + 4380q^4 + 12520q^5 + O(q^6). \quad (2985)$$

These are the initial coefficients of the Monster-class 6B McKay–Thompson Hauptmodul recorded by Borcherds [14]. The coefficient carrier in theorem 15.7 therefore factors through the exact sequence of constructions

$$(V_{840}, \beta) \mapsto (R_{\text{ES}}, H_{\text{code}}) \mapsto \pi_{\text{cox}} \xrightarrow{\mathcal{E}} T \mapsto (J, Y, F, f_6) \mapsto (R_B, A_1, A_6). \quad (2986)$$

Every displayed arrow is specified by the formulas above or by theorem 15.7.

Proof. The Chinese remainder theorem gives

$$V_{840} \cong (\mathbb{Z}/5\mathbb{Z})^\times \times (\mathbb{Z}/7\mathbb{Z})^\times \cong C_4 \times C_6.$$

The two Legendre characters are surjective, proving that β is surjective and that its four fibres have size six. Direct multiplication gives

$$(289^j)_{j=0}^5 = (1, 289, 361, 169, 121, 529) \pmod{840},$$

which proves (2970). If a fibre is ordered by this cyclic action, the basis

$$e_j - e_{j+1}, \quad 0 \leq j \leq 4,$$

of its augmentation lattice has Gram matrix

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}.$$

This is the A_5 Cartan matrix. Translation in the fibre permutes the six coordinate vectors, and replacing 289 by its inverse reverses the displayed basis. Both operations are lattice isometries.

The dual of D_4 consists of the integral and half-integral vectors whose coordinates have a common fractional part. The four representatives in the theorem are pairwise inequivalent modulo D_4 . Their squared norms are 0, 1, 1, 1, and $v + s - c = (2, 0, 0, 0)$ lies in D_4 , proving (2971). Direct multiplication gives

$$R^T R = H^T H = I_4.$$

Each map sends the root basis

$$e_1 - e_2, quade_2 - e_3, quade_3 - e_4, quade_3 + e_4$$

into D_4 . Hence both preserve D_4 . Their actions on v, s, c are the displayed transpositions, which generate S_3 . This proves the assertion about markings and establishes (2973).

The orders of h_1, h_2, h_3 are 6, 6, 2. Suppose

$$rh_1 + sh_2 + th_3 = 0, \quad r, s \in \mathbb{Z}/6\mathbb{Z}, \quad t \in \mathbb{Z}/2\mathbb{Z}.$$

The final two coordinates give

$$(s, s + t) = (0, 0) \in \mathbb{F}_2^2.$$

Hence $s \equiv 0 \pmod{2}$ and $t = 0$. The first and third coordinates then give $r + s = 0$ and $r + 2s = 0$ modulo six. Thus $s = 0$ and $r = 0$, proving $|H_{\text{old}}| = 6 \cdot 6 \cdot 2 = 72$.

The binary set C_2^{gl} is generated by

$$(1111; 00), \quad (1100; 10), \quad (1010; 01),$$

so it is a three-dimensional binary subspace and has order eight. A word of C_3^{gl} has the form

$$(\alpha + \beta, \alpha + 2\beta, \alpha, \beta).$$

The two generators are independent. For $(\alpha, \beta) \neq (0, 0)$, exactly one displayed coordinate vanishes. Hence $|C_3^{\text{gl}}| = 9$, and each nonzero word has Hamming weight three. Coordinatewise Chinese remaindering $(u, z) \mapsto 3u + 2z \pmod{6}$ is bijective, so $|H_{\text{code}}| = 8 \cdot 9 = 72$.

The images of the earlier generators under (2975) have the code decompositions

$$P(h_1) = (1, 1, 1, 3; 00) = 3(1111; 00) + 2(2, 2, 2, 0; 00),$$

$$P(h_2) = (2, 3, 1, 4; 11) = 3(0110; 11) + 2(1, 0, 2, 2; 00),$$

$$P(h_3) = (3, 0, 3, 0; 01) = 3(1010; 01).$$

Here $(2, 2, 2, 0) = 2(1, 1, 1, 0)$ and $(1, 0, 2, 2) = 2(1, 1, 1, 0) + 2(1, 2, 0, 1)$ in $\mathbb{F}_3^4$. Thus $P(H_{\text{old}}) \subseteq H_{\text{code}}$; equality follows from the two orders. Exhausting the $2^4 \cdot 24 \cdot 6 = 2304$ elements of

$$(\{\pm 1\}^4 \rtimes S_4) \times GL_2(\mathbb{F}_2)$$

gives exactly 48 transporters.

For $k = 3u + 2z \pmod{6}$,

$$\frac{5k^2}{6} \equiv \frac{3}{2}u + \frac{4}{3}z^2 \pmod{2\mathbb{Z}}.$$

The nonzero binary words have either weight four with trivial D_4 -class or weight two with nontrivial D_4 -class. Their quadratic contributions are respectively 6 and $3 + 1 = 4$. A nonzero ternary word contributes $(4/3) \cdot 3 = 4$. Hence the code glue is isotropic; the transporter proves the same statement

for the earlier presentation. The minimum norm in the A_5 discriminant class k is $k(6-k)/6$, and a nonzero D_4 discriminant class has minimum one. The six possible binary/ternary support patterns give

$\text{wt}(u)$	$\text{wt}(z)$	$\text{wt}(\text{supp } u \cap \text{supp } z)$	minimum
2	0	0	$2(3/2) + 1 = 4$
4	0	0	$4(3/2) = 6$
0	3	0	$3(4/3) = 4$
2	3	1	$(3/2) + 2(4/3) + (5/6) + 1 = 6$
2	3	2	$(4/3) + 2(5/6) + 1 = 4$
4	3	3	$(3/2) + 3(5/6) = 4.$

Therefore the nonzero minima are precisely four and six. Since

$$|D_{R_{\text{ES}}}| = 6^4 \cdot 4 = 5184 = 72^2,$$

the glue is maximal isotropic. The associated overlattice is even and unimodular. Its nontrivial glue cosets contain no norm-two vector, so its root system remains $A_5^4 D_4$. The Niemeier classification supplies the final lattice identification.

The two Coxeter characteristic-polynomial identities give (2978). The eta map is a group homomorphism because addition of exponent vectors is multiplication of eta products. Applying it to $-\pi_{\text{cox}}$ proves (2979). Its exponent vector at $d = 1, 2, 3, 6$ is

$$(r_1, r_2, r_3, r_6) = (5, -1, 1, -5).$$

The weight and the two eta congruence sums are

$$\frac{1}{2} \sum_{d|6} r_d = 0, \quad \sum_{d|6} d r_d = -24, \quad \sum_{d|6} \frac{6}{d} r_d = 24.$$

The character is trivial because

$$\prod_{d|6} d^{r_d} = 2^{-6} 3^{-4} = 72^{-2}.$$

For the four cusp denominators $d = 1, 2, 3, 6$, the exact order formula

$$\text{ord}_{1/d}(T) = \frac{6}{24} \sum_{\delta|6} \frac{\gcd(d, \delta)^2 r_\delta}{\gcd(d, 6/d) d \delta}$$

gives

$$(1, 0, 0, -1).$$

Thus T has a unique simple pole on $X_0(6)$. Moreover

$$[SL_2(\mathbb{Z}) : \Gamma_0(6)] = 6 \left(1 + \frac{1}{2}\right) \left(1 + \frac{1}{3}\right) = 12.$$

There are four cusps and no elliptic points, so

$$g(X_0(6)) = 1 + \frac{12}{12} - \frac{4}{2} = 0.$$

It follows that T is a Hauptmodul.

The eta transformation formula

$$\eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau)$$

sends the exponent vector to

$$(r_6, r_3, r_2, r_1) = (-5, 1, -1, 5) = -(r_1, r_2, r_3, r_6).$$

The positive scalar is determined by

$$\prod_{d|6} \left(\frac{6}{d}\right)^{r_d/2} = 6^{5/2} 3^{-1/2} 2^{1/2} = 72.$$

This proves (2980). Substitution and differentiation then give every identity in (2982).

The exponent vector of U at $d = 1, 2, 3, 6$ is

$$(-12, 12, 12, -12).$$

Its weight is zero, both eta congruence sums are -24 , its character is trivial, and the same cusp-order formula gives

$$(\text{ord}_{1/d}(U))_{d=1,2,3,6} = (-1, 1, 1, -1).$$

Reversal of the exponent vector fixes it, so $W_6 U = U$. The involution W_6 exchanges the cusp denominators $1, 6$ and $2, 3$. Hence U has one simple pole on $X_0(6)/\langle W_6 \rangle$. The action $T \mapsto 72/T$ shows that the fixed function field is

$$\mathbb{C}(T)^{\langle W_6 \rangle} = \mathbb{C}\left(T + \frac{72}{T}\right) = \mathbb{C}(J).$$

Both U and J have principal part q^{-1} at the pole of this quotient curve. Their difference is therefore constant. Their constant terms are respectively 12 and 0 , proving (2984).

Formal inversion of the displayed expansion of T gives

$$\frac{72}{T} = 72q + 360q^2 + 1368q^3 + 4392q^4 + 12528q^5 + O(q^6).$$

Adding this to $T + 5$ proves (2985).

The checker

`verification/verify_support_fricke_chain.py`

reconstructs the finite fibres, the A_5 Gram matrix, the D_4 triality maps, both glue presentations and their transporter, the Coxeter frame exponents, the cusp orders, and the Fricke scalar. Its receipt is

`verification/verify_support_fricke_chain_output.txt`.

□

Remark 15.9 (Scope of the first complexity map). The counting measure of each finite set in (2948) is fixed by the preceding theorems, and theorem 15.5 supplies the machine-side metric carrier. The circuit-geometric comparison additionally specifies a gate set, a target unitary, and a right-invariant cost metric. Evaluation of a holographic volume additionally specifies a boundary state, its bulk dual, and a regularized maximal slice. The theorem above supplies the exact finite-field operator-volume carrier. Independently, theorem 9.35 supplies a Euclidean rank-volume carrier, and theorem 9.43 realizes its $n = 4$ case in the Fable raw-cell coordinates. A holographic comparison requires an explicit map from these finite carriers to the bulk geometric data.

REFERENCES

- [1] C. Elsholtz and T. Tao, *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, J. Aust. Math. Soc. **94** (2013), 50–105; arXiv:1107.1010.
- [2] Reddit user u/CivQ17, comment containing the discriminant-five identity and root instruction, in *Erdős–Straus Conjecture / Umbral Moonshine Project*, r/LLMmathematics, 13 May 2026. https://www.reddit.com/r/LLMmathematics/comments/1szs5wf/erdosstrauss_conjecture_umbral_moonshine_project/olmxr22/.
- [3] Reddit user u/UmbrellaCorp_HR, *Three theorems for constructing inversive circle packings over arbitrary algebraic number fields, and deriving generalized Machin formula with arguments over the chosen field/fields and/or towers of extensions in the general case of multiple fields*, r/LLMmathematics, 29 July 2026. https://www.reddit.com/r/LLMmathematics/comments/1va0hr5/three_theorems_for_constructing_inversive_circle/.

- [4] Reddit user u/JGPTech, *Claude used my pipeline to find a counterexample to the Jacobian conjecture. 2026 is so wild.*, r/mathematics, 24 July 2026. https://www.reddit.com/r/mathematics/comments/1v5sbya/claude_used_my_pipeline_to_find_a_counterexample/.
- [5] J. Poplett (GitHub user JGPTech), *Local Fidelity, Global Alias: A Pipeline Proof of the Three-Dimensional Jacobian Counterexample*, JGPTech Research Note and exact code archive, 24 July 2026; blind-search and generator package at revision 2fe1901840ef3aa28232930e1aec8d8bbb61281a. <https://github.com/JGPTech/Fun/tree/2fe1901840ef3aa28232930e1aec8d8bbb61281a/Jacobian>.
- [6] T. Tao, *A digestion of the Jacobian conjecture counterexample*, What's New, 21 July 2026. <https://terrytao.wordpress.com/2026/07/21/a-digestion-of-the-jacobian-conjecture-counterexample/>.
- [7] L. Alpöge, public announcement of the explicit three-dimensional Jacobian counterexample, crediting A. Mathew for the question and Claude Fable 5 for the construction, 20 July 2026. https://x.com/_alpoge_/status/2079028340955197566.
- [8] The Clankers, *BB(107) rigidity reports and full residual sweep*, local research corpus, 2026; source files BEAVERSHINE/bb107_rigidity_report.txt, BEAVERSHINE/13/3/reports/bb107_rigidity_report.txt, and BEAVERSHINE/3/bb107_sweep_full.csv.
- [9] The Clankers, *Residual Divisor Shells, Support-Homogeneous Signs, and the $A_5^4 D_4$ Niemeier Refinement of the Modulo-840 Erdős–Straus Datum*, technical note and exact finite checker, 2026; local source files E-Sumbr1/residual_support_niemeier_refinement.tex and umbralsply/verify_residual_support_niemeier.py.
- [10] The Clankers, *The Level-Six Eta Quotient, Fricke Sheet, and 107-Shell Coefficient Transport*, technical note, 2026; local source file BEAVERSHINE/3/bb107_full_note.tex.
- [11] G. Chenevier and J. Lannes, *Automorphic Forms and Even Unimodular Lattices: Kneser Neighbors of Niemeier Lattices*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 69, Springer, 2019.
- [12] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, third ed., Springer, 1999.
- [13] M. C. N. Cheng, J. F. R. Duncan, and J. A. Harvey, *Umbral Moonshine and the Niemeier Lattices*, Research in the Mathematical Sciences **1** (2014), article 3. <https://arxiv.org/abs/1307.5793>.
- [14] R. E. Borcherds, *Modular Moonshine III*, Duke Mathematical Journal **93** (1998), no. 1, 129–154. <https://math.berkeley.edu/~reb/papers/modular3/modular3.pdf>.
- [15] The bbchallenge Collaboration, J. Blanchard, D. Briggs, K. Deka, N. Fenner, Y. Forster, et al., *Determination of the fifth Busy Beaver value*, arXiv:2509.12337, 2025. <https://arxiv.org/abs/2509.12337>.
- [16] The Clankers, *A Star–Kneser Positivity Criterion for Residual Erdős–Straus Shells*, May 2026, BEAVERSHINE/8/2/star_kneser_erdos_straus.tex.
- [17] The Clankers, *Integrated Completed Tesserine Sign Algebra, Circular Completion, and Blade-One Positivity for the 107-Residual Erdős–Straus Shell*, Zenodo preprint draft, May 2026, BEAVERSHINE/11/completed_supersignum_tesserine_blade1_integrated_paper.tex.
- [18] The Clankers, *A Note on the Globalization Semiring $G(\mathbb{Z})$* , March 2026, globalization_note.tex.
- [19] The Clankers, *A Secondary Note on Split-Zero Globalization*, April 2026, globalizationnte/2/split_zero_secondary_note.tex.
- [20] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, sixth edition, revised by D. R. Heath-Brown and J. H. Silverman, Oxford University Press, Oxford, 2008, especially Theorem 95.
- [21] Kokuno Yumeto, Miroku Akagi, and Maya Sakuyah, *The Z_3 Center Symmetry: Arithmetic Lattices, Automorphic Spectral Triples and QCD*, Zenodo, 2025. <https://doi.org/10.5281/zenodo.16936041>.
- [22] M. Kneser, *Abschätzung der asymptotischen Dichte von Summenmengen*, Mathematische Zeitschrift **58** (1953), 459–484.
- [23] P. Sarnak, *Letter to J. Lagarias about Integral Apollonian Packings*, June 2007.
- [24] J. Kocik, *The Koide Lepton Mass Formula and Geometry of Circles*, arXiv:1201.2067v1, 2012. <https://arxiv.org/abs/1201.2067>.
- [25] T. Ponweiser, *Computer Algebra and Analysis: Complex Variables Visualized*, Diploma thesis, Research Institute for Symbolic Computation, Johannes Kepler University Linz, revised 17 June 2014.
- [26] D. Tong, *Line Operators in the Standard Model*, Journal of High Energy Physics **07** (2017), article 104. [https://doi.org/10.1007/JHEP07\(2017\)104](https://doi.org/10.1007/JHEP07(2017)104).
- [27] S. Groot Nibbelink and P. K. S. Vaudrevange, *T-duality orbifolds of heterotic Narain compactifications*, Journal of High Energy Physics **04** (2017), article 030. [https://doi.org/10.1007/JHEP04\(2017\)030](https://doi.org/10.1007/JHEP04(2017)030).
- [28] F. Beye, T. Kobayashi, and S. Kuwakino, *Gauge Symmetries in Heterotic Asymmetric Orbifolds*, Nuclear Physics B **875** (2013), 599–620. <https://doi.org/10.1016/j.nuclphysb.2013.07.018>.
- [29] J. C. Baez, *Klein’s Quartic Curve*, 23 May 2013. <https://math.ucr.edu/home/baez/klein.html>.
- [30] A. Connes and C. Consani, *Knots, Primes and Class Field Theory*, arXiv:2501.06560v1, 2025. <https://arxiv.org/abs/2501.06560>.

- [31] The Clankers, *Sectorial modular data for a three-character gauge system*, local research corpus, 2025; source files `supporting_sources/sectorial_modular_data/1.tex`, `supporting_sources/sectorial_modular_data/2.tex`, and `supporting_sources/sectorial_modular_data/3.tex`.
- [32] S. J. Summers, *Tomita–Takesaki Modular Theory*, arXiv:math-ph/0511034v1, 2005. <https://arxiv.org/abs/math-ph/0511034>.
- [33] M. Takesaki, *Theory of Operator Algebras II*, Springer, Berlin, 2003.
- [34] M. A. Nielsen, *A geometric approach to quantum circuit lower bounds*, arXiv:quant-ph/0502070, 2005. <https://arxiv.org/abs/quant-ph/0502070>.
- [35] M. A. Nielsen, M. R. Dowling, M. Gu, and A. C. Doherty, *Quantum Computation as Geometry*, arXiv:quant-ph/0603161, 2006. <https://arxiv.org/abs/quant-ph/0603161>.
- [36] X. Pennec, P. Fillard, and N. Ayache, *A Riemannian Framework for Tensor Computing*, International Journal of Computer Vision **66** (2006), 41–66. <https://doi.org/10.1007/s11263-005-3222-z>.
- [37] D. Stanford and L. Susskind, *Complexity and Shock Wave Geometries*, arXiv:1406.2678, 2014. <https://arxiv.org/abs/1406.2678>.
- [38] J. Maldacena and L. Susskind, *Cool horizons for entangled black holes*, Fortschritte der Physik **61** (2013), 781–811. <https://arxiv.org/abs/1306.0533>.
- [39] R. Emparan and H. S. Reall, *Black Holes in Higher Dimensions*, Living Reviews in Relativity **11** (2008), article 6. <https://arxiv.org/abs/0801.3471>.
- [40] I. Bena, E. J. Martinec, S. D. Mathur, and N. P. Warner, *Fuzzballs and Microstate Geometries: Black-Hole Structure in String Theory*, arXiv:2204.13113, 2022. <https://arxiv.org/abs/2204.13113>.
- [41] The Clankers, *Corrected Wick-Rotated Spline-Wavelet Note*, local research corpus, 2026; source file `globalizationnte/shefy/Newfolder/corrected_wick_rotated_spline_wavelet_note.tex`.
- [42] M. Gerbershagen, J. Hernandez, M. Khramtsov, and M. Knysh, *Holographic dual of Bures metric and subregion complexity*, arXiv:2412.08707, 2024. <https://arxiv.org/abs/2412.08707>.
- [43] The Clankers, *Arithmetic Time, CPT Spectral Packets*, version 35 integrated replacement, local research corpus, 2026; source file `Soc/Arithmetic_Time_CPT_Spectral_Packets_v35_integrated_replacement.tex`, especially the BCM gauge-sector, Artin determinant, character-Hamiltonian, and direct-sum modular decomposition theorems.
- [44] E. Ha and F. Paugam, *Bost–Connes–Marcolli systems for Shimura varieties. I. Definitions and formal analytic properties*, arXiv:math/0507101v1, 2005. <https://arxiv.org/abs/math/0507101>.
- [45] T. T. Takahashi, H. Matsufuru, Y. Nemoto, and H. Suganuma, *Three-Quark Potential in $SU(3)$ Lattice QCD*, Physical Review Letters **86** (2001), 18–21. <https://doi.org/10.1103/PhysRevLett.86.18>.
- [46] J. M. Cornwall, *The Baryon Wilson Loop Area Law in QCD*, Physical Review D **54** (1996), 6527–6536. <https://doi.org/10.1103/PhysRevD.54.6527>.